

V. B. Berestetskii E. M. Lifshitz L. P. Pitaevskii

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Introduction

§ 1. Uncertainty relations in the relativistic domain

The quantum theory presented in volume III of this course is essentially nonrelativistic and cannot be applied to phenomena involving motion at speeds that are not small compared with the speed of light. At first sight, one might expect to obtain a relativistic theory by a more or less direct generalization of the machinery of nonrelativistic quantum mechanics. Closer examination shows, however, that a logically complete relativistic theory cannot be constructed without new physical principles.

Let us recall some of the physical ideas underlying nonrelativistic quantum mechanics (III, § 1). Measurement has a fundamental role in that theory. It means an interaction between a quantum system and a “classical object” (an “apparatus”), as a result of which the quantum system acquires definite values of particular dynamical variables, such as coordinates or velocities. We also saw that quantum mechanics severely restricts the simultaneous possession of different dynamical variables by an electron¹. Thus the uncertainties Δq and Δp with which coordinate and momentum can coexist are related by $\Delta q \Delta p \sim \hbar^2$; the more accurately one quantity is measured, the less accurately the other can be measured at the same time.

It is essential, however, that each dynamical variable of the electron, considered separately, could be measured with arbitrarily high accuracy in an arbitrarily short interval of time. This circumstance is fundamental to the whole of nonrelativistic quantum mechanics. It alone makes possible the concept of a wave function, which is basic to the formalism of the theory. The physical meaning of $\psi(q)$ is that its squared modulus gives the probability that a measurement performed at the given instant will yield one or another value of the electron coordinate. A necessary premise for introducing such a probability is plainly the possibility, in principle, of measuring the coordinate with unlimited accuracy and speed. Without that possibility, the probability would have no subject and would lose its physical meaning.

The existence of a limiting speed, the speed of light c , places new fundamental restrictions on the measurement of physical quantities (*L. D. Landau, R. Peierls, 1930*).

In volume III, § 44, the relation

$$(v' - v) \Delta p \Delta t \sim \hbar, \quad (1.1)$$

was obtained between the uncertainty Δp in a measurement of the electron momentum and the duration Δt of the measurement process itself; v and v' are the electron velocities before and after the measurement. According to this relation, measuring the momentum accurately in a short time (small Δp together with small Δt) is possible only at the cost of a sufficiently large change in velocity caused by the measurement itself. In nonrelativistic theory, this expresses the impossibility of repeating a momentum measurement after

¹As in volume III, § 1, we speak briefly of an electron while meaning any quantum system.

²Ordinary units are used in this section.

a short interval, but does not affect the possibility in principle of a single momentum measurement of unlimited accuracy, since $v' - v$ may be made arbitrarily large.

A limiting speed changes the situation completely. Neither $v' - v$ nor the velocities themselves can now exceed c (more precisely, $2c$). Replacing $v' - v$ in (1.1) by c , we obtain

$$\Delta p \Delta t \sim \hbar/c, \quad (1.2)$$

which sets the best accuracy in momentum measurement that is attainable in principle for a given measurement time Δt . Thus relativistic theory makes an arbitrarily rapid and accurate momentum measurement fundamentally impossible. Exact momentum measurement, $\Delta p \rightarrow 0$, is possible only in the limit of an infinitely long measurement.

There are reasons to expect that the measurability of the electron coordinate by itself is also altered. In the mathematical formalism, this appears as an incompatibility between exact coordinate measurement and the requirement that the energy of a free particle be positive. We shall see that a complete system of eigenfunctions of the relativistic wave equation for a free particle contains not only solutions with the “proper” time dependence, but also solutions of “negative frequency.” In general, these functions also enter the expansion of a wave packet representing an electron localized in a small region of space.

As will be shown, negative-frequency wave functions are connected with the existence of antiparticles, the positrons. Their occurrence in the wave-packet expansion expresses the generally unavoidable creation of electron-positron pairs during a measurement of the electron coordinates. The production of new particles, uncontrolled by the measurement process itself, deprives a measurement of the electron coordinates of meaning.

In the electron rest frame, the least uncertainty in measuring its coordinates is

$$\Delta q \sim \hbar/mc. \quad (1.3)$$

This value, which is also the only one allowed by dimensional considerations, corresponds to a momentum uncertainty $\Delta p \sim mc$ and hence to the minimum threshold energy for pair creation.

In a reference frame in which the electron has energy ε , equation (1.3) is replaced by

$$\Delta q \sim c\hbar/\varepsilon. \quad (1.4)$$

In particular, in the limiting ultrarelativistic case, energy and momentum obey $\varepsilon \approx cp$, and consequently

$$\Delta q \sim \hbar/p, \quad (1.5)$$

so the uncertainty Δq equals the particle’s de Broglie wavelength³.

For photons the ultrarelativistic case always applies, and therefore (1.5) is valid. It follows that photon coordinates are meaningful only when the characteristic dimensions are large compared with the wavelength. This is precisely the “classical” limiting case of geometrical optics, in which light may be said to propagate along definite paths,

³We mean measurements for which every possible experimental outcome permits a conclusion about the state of the electron. Thus we exclude coordinate measurements made by collisions when, during the observation time, the relevant outcome does not occur with probability 1. A particle’s deflection in such a case permits a conclusion about the electron’s location, but the absence of a deflection permits no conclusion at all.

the rays. In the quantum case, when the wavelength cannot be regarded as small, the notion of photon coordinates has no subject. We shall see below (see § 4) that within the mathematical formalism, the unmeasurability of photon coordinates already appears in the impossibility of constructing from the photon wave function a quantity that could serve as a probability density and meet the necessary requirements of relativistic invariance.

The preceding considerations naturally suggest that a future theory will abandon any description of the time course of particle-interaction processes. Such a theory will show that these processes have no exactly definable characteristics, even within ordinary quantum-mechanical accuracy. A description in time would then prove as illusory as the classical trajectories of nonrelativistic quantum mechanics. The only observable quantities would be properties such as the momenta and polarizations of free particles: the initial particles entering the interaction and the final particles produced by it (*L. D. Landau, R. Peierls, 1930*).

A characteristic problem in relativistic quantum theory is to determine probability amplitudes for transitions connecting specified initial and final states of a particle system, that is, its states at $t \rightarrow \mp\infty$. The collection of transition amplitudes between all possible states forms the *scattering matrix*, or *S-matrix*. This matrix contains all physically observable information about particle-interaction processes (*W. Heisenberg, 1938*).

At present, a complete and logically closed relativistic quantum theory does not yet exist. We shall see that the existing theory introduces new physical features into the description of particle states, giving that description some aspects of field theory (see § 10). Nevertheless, it is built to a considerable extent on the pattern and concepts of ordinary quantum mechanics. This construction has been successful in quantum electrodynamics. Its lack of complete logical closure is manifested by divergent expressions that arise when its mathematical apparatus is applied directly, although definite procedures exist for removing those divergences. These procedures still retain much of the character of semi-empirical prescriptions, and confidence in the results obtained from them rests ultimately on their excellent agreement with experiment rather than on the internal consistency and logical coherence of the theory's basic principles.

Photon

§2. Quantization of the free electromagnetic field

To treat the electromagnetic field as a quantum object, it is useful to begin with a classical description in which the field is specified by a discrete, though infinite, sequence of variables. Such a description allows the ordinary apparatus of quantum mechanics to be applied directly. By contrast, representing the field through potentials assigned at every point in space amounts essentially to a description in terms of a continuous set of variables.

Let $\mathbf{A}(\mathbf{r}, t)$ be the vector potential of a free electromagnetic field, satisfying the “transversality condition”

$$\nabla \cdot \mathbf{A} = 0. \quad (2.1)$$

The scalar potential is then $\Phi = 0$, while the fields $\mathbf{E}$ and $\mathbf{H}$ are

$$\mathbf{E} = -\dot{\mathbf{A}}, \quad \mathbf{H} = \nabla \times \mathbf{A}. \quad (2.2)$$

Maxwell’s equations reduce to the wave equation for $\mathbf{A}$:

$$\Delta \mathbf{A} - \frac{\partial^2 \mathbf{A}}{\partial t^2} = 0. \quad (2.3)$$

As is known (see II, § 52), in classical electrodynamics a description by a discrete sequence of variables is obtained by considering the field within some large but finite volume of space V ¹. We briefly recall this construction, omitting the computational details.

Within a finite volume the field may be expanded in travelling plane waves, so that its potential takes the form

$$\mathbf{A} = \sum_{\mathbf{k}} \left(\mathbf{a}_{\mathbf{k}} e^{i\mathbf{k}\mathbf{r}} + \mathbf{a}_{\mathbf{k}}^* e^{-i\mathbf{k}\mathbf{r}} \right), \quad (2.4)$$

where the coefficients $\mathbf{a}_{\mathbf{k}}$ depend on time according to

$$\mathbf{a}_{\mathbf{k}} \sim e^{-i\omega t}, \quad \omega = |\mathbf{k}|. \quad (2.5)$$

By virtue of condition (2.1), the complex vectors $\mathbf{a}_{\mathbf{k}}$ are orthogonal to the corresponding wave vectors: $\mathbf{a}_{\mathbf{k}} \mathbf{k} = 0$.

The summation in (2.4) extends over an infinite discrete set of values of the wave vector (that is, of its three components k_x, k_y, k_z). Passage to integration over a continuous distribution is effected by using

$$\frac{d^3 k}{(2\pi)^3}$$

¹To avoid burdening the formulas with superfluous factors, we shall set $V = 1$.

for the number of possible values of $\mathbf{k}$ belonging to the element $d^3k = dk_x dk_y dk_z$ of volume in $\mathbf{k}$ -space.

The vectors $\mathbf{a}_{\mathbf{k}}$ determine the field in the given volume completely. They may therefore be regarded as a discrete collection of classical “field variables.” To determine how to pass to the quantum theory, however, we must first transform these variables further, so that the field equations acquire a form analogous to the canonical (Hamilton) equations of classical mechanics. The canonical field variables are defined by

$$\begin{aligned}\mathbf{Q}_{\mathbf{k}} &= \frac{1}{\sqrt{4\pi}} (\mathbf{a}_{\mathbf{k}} + \mathbf{a}_{\mathbf{k}}^*), \\ \mathbf{P}_{\mathbf{k}} &= -\frac{i\omega}{\sqrt{4\pi}} (\mathbf{a}_{\mathbf{k}} - \mathbf{a}_{\mathbf{k}}^*) = \dot{\mathbf{Q}}_{\mathbf{k}}\end{aligned}\tag{2.6}$$

(these quantities are plainly real). In terms of the canonical variables the vector potential is

$$\mathbf{A} = \sqrt{4\pi} \sum_{\mathbf{k}} \left(\mathbf{Q}_{\mathbf{k}} \cos \mathbf{k}\mathbf{r} - \frac{1}{\omega} \mathbf{P}_{\mathbf{k}} \sin \mathbf{k}\mathbf{r} \right).\tag{2.7}$$

To obtain the Hamiltonian H , we must calculate the total field energy

$$\frac{1}{8\pi} \int (\mathbf{E}^2 + \mathbf{H}^2) d^3x,$$

expressing it in terms of $\mathbf{Q}_{\mathbf{k}}$ and $\mathbf{P}_{\mathbf{k}}$. On substituting expansion (2.7) for $\mathbf{A}$, evaluating $\mathbf{E}$ and $\mathbf{H}$ from (2.2), and carrying out the integration, we find

$$H = \frac{1}{2} \sum_{\mathbf{k}} (\mathbf{P}_{\mathbf{k}}^2 + \omega^2 \mathbf{Q}_{\mathbf{k}}^2).$$

Both $\mathbf{P}_{\mathbf{k}}$ and $\mathbf{Q}_{\mathbf{k}}$ are perpendicular to the wave vector $\mathbf{k}$ and thus have two independent components apiece. Their direction fixes the polarization direction of the corresponding wave. Denoting the two components of $\mathbf{Q}_{\mathbf{k}}$ and $\mathbf{P}_{\mathbf{k}}$ (in the plane normal to $\mathbf{k}$) by $Q_{\mathbf{k}\alpha}$ and $P_{\mathbf{k}\alpha}$, where $\alpha = 1, 2$, we rewrite the Hamiltonian as

$$H = \sum_{\mathbf{k}\alpha} \frac{1}{2} (P_{\mathbf{k}\alpha}^2 + \omega^2 Q_{\mathbf{k}\alpha}^2).\tag{2.8}$$

The Hamiltonian has therefore separated into a sum of independent terms, each involving only one pair $Q_{\mathbf{k}\alpha}$, $P_{\mathbf{k}\alpha}$. Each term corresponds to a travelling wave of specified wave vector and polarization, and has the form of the Hamiltonian of a one-dimensional harmonic oscillator. The resulting expansion is consequently called a decomposition of the field into *oscillators*.

We now quantize the free electromagnetic field. The classical formulation just described makes the transition to the quantum theory immediate. The canonical variables—the generalized coordinates $Q_{\mathbf{k}\alpha}$ and generalized momenta $P_{\mathbf{k}\alpha}$ —must now be treated as operators obeying the commutation rule

$$\hat{P}_{\mathbf{k}\alpha} \hat{Q}_{\mathbf{k}\alpha} - \hat{Q}_{\mathbf{k}\alpha} \hat{P}_{\mathbf{k}\alpha} = -i\tag{2.9}$$

(operators bearing different $\mathbf{k}\alpha$ commute with one another). Along with them, the potential $\mathbf{A}$ and, by (2.2), the field strengths $\mathbf{E}$ and $\mathbf{H}$ also become Hermitian operators.

A consistent definition of the field Hamiltonian requires evaluation of the integral

$$\hat{H} = \frac{1}{8\pi} \int \left(\hat{\mathbf{E}}^2 + \hat{\mathbf{H}}^2 \right) d^3x, \quad (2.10)$$

with $\hat{\mathbf{E}}$ and $\hat{\mathbf{H}}$ expressed through $\hat{P}_{\mathbf{k}\alpha}$ and $\hat{Q}_{\mathbf{k}\alpha}$. Their noncommutativity does not in fact enter here: the products $\hat{Q}_{\mathbf{k}\alpha}\hat{P}_{\mathbf{k}\alpha}$ occur multiplied by $\cos \mathbf{k}\mathbf{r} \cdot \sin \mathbf{k}\mathbf{r}$, whose integral over the whole volume vanishes. The Hamiltonian is thus

$$\hat{H} = \sum_{\mathbf{k}\alpha} \frac{1}{2} \left(\hat{P}_{\mathbf{k}\alpha}^2 + \omega^2 \hat{Q}_{\mathbf{k}\alpha}^2 \right), \quad (2.11)$$

which agrees exactly with the classical Hamiltonian, as one would naturally expect.

No special calculation is needed to find the eigenvalues of this Hamiltonian, since the problem reduces to the familiar energy levels of linear oscillators (see III, § 23). We may therefore write the field energy levels at once as

$$E = \sum_{\mathbf{k}\alpha} \left(N_{\mathbf{k}\alpha} + \frac{1}{2} \right) \omega, \quad (2.12)$$

where the $N_{\mathbf{k}\alpha}$ are integers.

We shall return to this formula in the next section. For the moment, let us write the matrix elements of $\hat{Q}_{\mathbf{k}\alpha}$,

which follow directly from the familiar expressions for the coordinate matrix elements of an oscillator (see III, § 23). The nonzero elements are

$$\langle N_{\mathbf{k}\alpha} | \hat{Q}_{\mathbf{k}\alpha} | N_{\mathbf{k}\alpha} - 1 \rangle = \langle N_{\mathbf{k}\alpha} - 1 | \hat{Q}_{\mathbf{k}\alpha} | N_{\mathbf{k}\alpha} \rangle = \sqrt{\frac{N_{\mathbf{k}\alpha}}{2\omega}}. \quad (2.13)$$

The matrix elements of $\hat{P}_{\mathbf{k}\alpha} = \dot{\hat{Q}}_{\mathbf{k}\alpha}$ differ from those of $\hat{Q}_{\mathbf{k}\alpha}$ only by a factor $\pm i\omega$.

For subsequent calculations it is more convenient to replace $\hat{Q}_{\mathbf{k}\alpha}$ and $\hat{P}_{\mathbf{k}\alpha}$ by the linear combinations $\omega\hat{Q}_{\mathbf{k}\alpha} \pm i\hat{P}_{\mathbf{k}\alpha}$, whose matrix elements are nonzero only for the transitions $N_{\mathbf{k}\alpha} \rightarrow N_{\mathbf{k}\alpha} \pm 1$. We accordingly introduce the operators

$$\hat{c}_{\mathbf{k}\alpha} = \frac{1}{\sqrt{2\omega}} \left(\omega\hat{Q}_{\mathbf{k}\alpha} + i\hat{P}_{\mathbf{k}\alpha} \right), \quad \hat{c}_{\mathbf{k}\alpha}^+ = \frac{1}{\sqrt{2\omega}} \left(\omega\hat{Q}_{\mathbf{k}\alpha} - i\hat{P}_{\mathbf{k}\alpha} \right) \quad (2.14)$$

(apart from a factor $\sqrt{2\pi/\omega}$, the classical quantities $c_{\mathbf{k}\alpha}$, $c_{\mathbf{k}\alpha}^*$ coincide with the coefficients $a_{\mathbf{k}\alpha}$, $a_{\mathbf{k}\alpha}^*$ in expansion (2.4)). Their matrix elements are

$$\langle N_{\mathbf{k}\alpha} - 1 | \hat{c}_{\mathbf{k}\alpha} | N_{\mathbf{k}\alpha} \rangle = \langle N_{\mathbf{k}\alpha} | \hat{c}_{\mathbf{k}\alpha}^+ | N_{\mathbf{k}\alpha} - 1 \rangle = \sqrt{N_{\mathbf{k}\alpha}}. \quad (2.15)$$

The commutation rule for $\hat{c}_{\mathbf{k}\alpha}$ and $\hat{c}_{\mathbf{k}\alpha}^+$ follows from definition (2.14) and rule (2.9):

$$\hat{c}_{\mathbf{k}\alpha}\hat{c}_{\mathbf{k}\alpha}^+ - \hat{c}_{\mathbf{k}\alpha}^+\hat{c}_{\mathbf{k}\alpha} = 1. \quad (2.16)$$

For the vector potential we return to an expansion of the form (2.4), now with operator-valued coefficients. Write it as

$$\hat{\mathbf{A}} = \sum_{\mathbf{k}\alpha} (\hat{c}_{\mathbf{k}\alpha} \mathbf{A}_{\mathbf{k}\alpha} + \hat{c}_{\mathbf{k}\alpha}^{\dagger} \mathbf{A}_{\mathbf{k}\alpha}^*), \quad (2.17)$$

where

$$\mathbf{A}_{\mathbf{k}\alpha} = \sqrt{4\pi} \frac{\mathbf{e}^{(\alpha)}}{\sqrt{2\omega}} e^{i\mathbf{k}\mathbf{r}}. \quad (2.18)$$

Here $\mathbf{e}^{(\alpha)}$ denotes a unit vector along the polarization direction of an oscillator. These vectors are perpendicular to the wave vector $\mathbf{k}$, and two independent polarizations exist for each $\mathbf{k}$.

Similarly, for the operators $\hat{\mathbf{E}}$ and $\hat{\mathbf{H}}$ we write

$$\hat{\mathbf{E}} = \sum_{\mathbf{k}\alpha} (\hat{c}_{\mathbf{k}\alpha} \mathbf{E}_{\mathbf{k}\alpha} + \hat{c}_{\mathbf{k}\alpha}^{\dagger} \mathbf{E}_{\mathbf{k}\alpha}^*), \quad \hat{\mathbf{H}} = \sum_{\mathbf{k}\alpha} (\hat{c}_{\mathbf{k}\alpha} \mathbf{H}_{\mathbf{k}\alpha} + \hat{c}_{\mathbf{k}\alpha}^{\dagger} \mathbf{H}_{\mathbf{k}\alpha}^*), \quad (2.19)$$

with

$$\mathbf{E}_{\mathbf{k}\alpha} = i\omega \mathbf{A}_{\mathbf{k}\alpha}, \quad \mathbf{H}_{\mathbf{k}\alpha} = \mathbf{n} \times \mathbf{E}_{\mathbf{k}\alpha} \quad (\mathbf{n} = \mathbf{k}/\omega). \quad (2.20)$$

The vectors $\mathbf{A}_{\mathbf{k}\alpha}$ are mutually orthogonal in the sense that

$$\int \mathbf{A}_{\mathbf{k}\alpha} \mathbf{A}_{\mathbf{k}'\alpha'}^* d^3x = \frac{2\pi}{\omega} \delta_{\alpha\alpha'} \delta_{\mathbf{k}\mathbf{k}'}. \quad (2.21)$$

Indeed, if $\mathbf{A}_{\mathbf{k}\alpha}$ and $\mathbf{A}_{\mathbf{k}'\alpha'}^*$ have different wave vectors, their product contains $e^{i(\mathbf{k}-\mathbf{k}')\mathbf{r}}$, which integrates to zero over the volume. If only their polarizations differ, then $\mathbf{e}^{(\alpha)} \mathbf{e}^{(\alpha')*} = 0$, since the two independent polarization directions are mutually orthogonal. Corresponding relations hold for $\mathbf{E}_{\mathbf{k}\alpha}$ and $\mathbf{H}_{\mathbf{k}\alpha}$. Their normalization is conveniently written as

$$\frac{1}{4\pi} \int (\mathbf{E}_{\mathbf{k}\alpha} \mathbf{E}_{\mathbf{k}'\alpha'}^* + \mathbf{H}_{\mathbf{k}\alpha} \mathbf{H}_{\mathbf{k}'\alpha'}^*) d^3x = \omega \delta_{\mathbf{k}\mathbf{k}'} \delta_{\alpha\alpha'}. \quad (2.22)$$

Substitution of operators (2.19) into (2.10), followed by integration using (2.22), gives the field Hamiltonian in terms of $\hat{c}$ and $\hat{c}^{\dagger}$:

$$\hat{H} = \sum_{\mathbf{k}\alpha} \frac{1}{2} \omega (\hat{c}_{\mathbf{k}\alpha} \hat{c}_{\mathbf{k}\alpha}^{\dagger} + \hat{c}_{\mathbf{k}\alpha}^{\dagger} \hat{c}_{\mathbf{k}\alpha}). \quad (2.23)$$

In the representation under consideration (with the matrix elements (2.15) of $\hat{c}$ and $\hat{c}^{\dagger}$), this operator is diagonal, and its eigenvalues naturally coincide with (2.12).

In classical theory the field momentum is defined by the integral

$$\mathbf{P} = \frac{1}{4\pi} \int \mathbf{E} \times \mathbf{H} d^3x.$$

On passing to the quantum theory, we replace $\mathbf{E}$ and $\mathbf{H}$ by operators (2.19) and readily obtain

$$\hat{\mathbf{P}} = \sum_{\mathbf{k}\alpha} \frac{1}{2} (\hat{P}_{\mathbf{k}\alpha}^2 + \omega^2 \hat{Q}_{\mathbf{k}\alpha}^2) \mathbf{n}, \quad (2.24)$$

in agreement with the familiar classical relation between the energy and momentum of plane waves. The eigenvalues of this operator are

$$\mathbf{P} = \sum_{\mathbf{k}\alpha} \mathbf{k} \left(N_{\mathbf{k}\alpha} + \frac{1}{2} \right). \quad (2.25)$$

The operator representation furnished by matrix elements (2.15) is the “occupation-number representation”: it describes the state of the system (the field) by specifying the quantum numbers $N_{\mathbf{k}\alpha}$ (the *occupation numbers*). In this representation, field operators (2.19) (and hence Hamiltonian (2.11)) act on the wave function of the system expressed as a function

of the numbers $N_{\mathbf{k}\alpha}$; denote it by $\Phi(N_{\mathbf{k}\alpha}, t)$. The field operators (2.19) have no explicit time dependence. This is the Schrödinger representation of operators customarily used in nonrelativistic quantum mechanics. The state $\Phi(N_{\mathbf{k}\alpha}, t)$ is instead time-dependent, and its dependence is governed by the Schrödinger equation

$$i \frac{\partial \Phi}{\partial t} = \hat{H} \Phi.$$

This description of the field is relativistically invariant in essence, since it rests on the invariant Maxwell equations. The invariance is not displayed explicitly, however, chiefly because the space coordinates and time enter the description in a highly asymmetric fashion.

In relativistic theory it is expedient to cast the description into a form that appears more invariant. For this purpose one uses the so-called Heisenberg representation, in which the explicit time dependence is transferred to the operators themselves (see III, § 13). Time and the coordinates then enter the field operators on an equal footing, while the state Φ becomes a function of the occupation numbers alone.

For $\hat{\mathbf{A}}$, passage to the Heisenberg representation consists in replacing, in every term of the sum in (2.17), (2.18), the factor $e^{i\mathbf{k}\mathbf{r}}$ by $e^{i(\mathbf{k}\mathbf{r} - \omega t)}$. Thus $\mathbf{A}_{\mathbf{k}\alpha}$ is to mean the time-dependent function

$$\mathbf{A}_{\mathbf{k}\alpha} = \sqrt{4\pi} \frac{\mathbf{e}^{(\alpha)}}{\sqrt{2\omega}} e^{-i(\omega t - \mathbf{k}\mathbf{r})}. \quad (2.26)$$

This can be checked at once by noting that a matrix element of a Heisenberg operator for a transition $i \rightarrow f$ must contain the factor $\exp\{-i(E_i - E_f)t\}$, where E_i and E_f are the energies of the initial and final states (see III, § 13). For a transition in which $N_{\mathbf{k}}$ decreases or increases by 1, this factor becomes $e^{-i\omega t}$ or $e^{i\omega t}$, respectively. The replacement just described satisfies this requirement.

In what follows, for both the electromagnetic field and particle fields, the Heisenberg representation of operators will always be understood.

§3. Photons

We now turn to a discussion of the field-quantization formulas obtained above.

First of all, formula (2.12) for the field energy presents the following difficulty. The lowest energy level of the field is attained when the quantum numbers $N_{\mathbf{k}\alpha}$ of all the

oscillators vanish (this state is called the *vacuum state of the electromagnetic field*). Yet even in this state every oscillator has a nonzero “zero-point energy” $\omega/2$. Summation over the entire infinite set of oscillators therefore gives an infinite result. We thus encounter one of the “divergences” arising from the fact that the present theory is not logically complete.

As long as only the eigenvalues of the field energy are at issue, this difficulty can be removed simply by omitting the zero-point energy, that is, by writing the field energy and momentum as²:

$$E = \sum_{\mathbf{k}\alpha} N_{\mathbf{k}\alpha} \omega, \quad \mathbf{P} = \sum_{\mathbf{k}\alpha} N_{\mathbf{k}\alpha} \mathbf{k}. \quad (3.1)$$

These formulas allow us to introduce a concept fundamental to all of quantum electrodynamics: *light quanta*, or *photons*³. More precisely, the free electromagnetic field may be regarded as an assembly of particles, each having energy $\omega (= \hbar\omega)$ and momentum $\mathbf{k} (= \hbar\omega/c)$. The relation between a photon’s energy and momentum is just the one required by relativistic mechanics for particles of zero rest mass moving at the speed of light. The occupation numbers $N_{\mathbf{k}\alpha}$ now have the meaning of the numbers of photons with specified momenta $\mathbf{k}$ and polarizations $\mathbf{e}^{(\alpha)}$. A photon’s polarization property is analogous to the spin concept for other particles (the special features of the photon in this respect will be considered below, in § 6).

It is readily seen that the mathematical formalism developed in the preceding section fully agrees with the interpretation of an electromagnetic field as an assembly of photons; it is precisely the machinery of so-called second quantization applied to a photon system⁴. In this method (see III, § 64), the occupation numbers of the states serve as the independent variables, and operators act on functions of these numbers. The principal part is played by particle “annihilation” and “creation” operators, which respectively decrease or increase an occupation number by one. The operators $\hat{c}_{\mathbf{k}\alpha}$ and $\hat{c}_{\mathbf{k}\alpha}^+$ are exactly of this kind: $\hat{c}_{\mathbf{k}\alpha}$ annihilates a photon in state $\mathbf{k}\alpha$, while $\hat{c}_{\mathbf{k}\alpha}^+$ creates a photon in that state.

The commutation rule (2.16) is the rule for particles obeying Bose statistics. Hence photons are bosons, as could have been anticipated: any number of photons is allowed in a given state (we shall return in § 5 to the significance of this fact).

The plane waves $\mathbf{A}_{\mathbf{k}\alpha}$ (2.26), which occur in the operator $\hat{\mathbf{A}}$ (2.17) as coefficients of the photon annihilation operators, can be interpreted as wave functions of photons with definite momenta $\mathbf{k}$ and polarizations $\mathbf{e}^{(\alpha)}$. This interpretation corresponds to expanding the ψ -operator in a series of wave functions for the stationary states of a particle in the nonrelativistic second-quantization formalism (unlike that formalism, however, expansion (2.17) contains both particle annihilation and particle creation operators; the

²This omission can be carried out formally without inconsistency if products of operators in (2.10) are understood to be “normal” products, i.e. products in which the operators $\hat{c}^+$ always stand to the left of the operators $\hat{c}$. Formula (2.23) then takes the form

$$\hat{H} = \sum_{\mathbf{k}\alpha} (\omega \hat{c}_{\mathbf{k}\alpha}^+ \hat{c}_{\mathbf{k}\alpha}).$$

³The photon concept was first introduced by Einstein (A. Einstein, 1905).

⁴The second-quantization method was developed for radiation theory by Dirac (P. A. M. Dirac, 1927).

meaning of this difference will become clear later; see § 12).

The wave function (2.26) is normalized by the condition

$$\int \frac{1}{4\pi} (|\mathbf{E}_{\mathbf{k}\alpha}|^2 + |\mathbf{H}_{\mathbf{k}\alpha}|^2) d^3x = \omega. \quad (3.2)$$

This is normalization to one photon in the volume $V = 1$. Indeed, the integral on the left-hand side is the quantum-mechanical expectation value of the photon energy in the state described by the given wave function⁵. The right-hand side of (3.2), in turn, is the energy of one photon.

Maxwell's equations serve as the photon's "Schrödinger equation." In the present case (for a potential $\mathbf{A}(\mathbf{r}, t)$ satisfying condition (2.1)), this is the wave equation

$$\frac{\partial^2 \mathbf{A}}{\partial t^2} - \Delta \mathbf{A} = 0.$$

In the general case of arbitrary stationary states, the photon "wave functions" are complex solutions of this equation whose time dependence is supplied by the factor $e^{-i\omega t}$.

In speaking of a photon wave function, we emphasize once more that it must by no means be regarded as the probability amplitude for a photon's spatial localization—in contrast to the basic meaning of a wave function in nonrelativistic quantum mechanics. The reason is that (as pointed out in § 1) the very concept of photon coordinates has no physical meaning. We shall return to the mathematical aspect of this circumstance at the end of the next section.

The components of the spatial Fourier expansion of $\mathbf{A}(\mathbf{r}, t)$ form the photon wave function in the momentum representation; write it as $\mathbf{A}(\mathbf{k}, t) = \mathbf{A}(\mathbf{k})e^{-i\omega t}$. Thus, for a state of definite momentum $\mathbf{k}$ and polarization $\mathbf{e}^{(\alpha)}$, the momentum-representation wave function is simply the coefficient of the exponential factor in (2.26):

$$\mathbf{A}_{\mathbf{k}\alpha}(\mathbf{k}', \alpha') = \sqrt{4\pi} \frac{\mathbf{e}^{(\alpha)}}{\sqrt{2\omega}} \delta_{\mathbf{k}'\mathbf{k}} \delta_{\alpha\alpha'}. \quad (3.3)$$

Because the momentum of a free particle is measurable, the momentum-representation wave function has a deeper physical meaning than its coordinate-representation counterpart: it permits calculation of the probabilities $w_{\mathbf{k}\alpha}$ of the various possible values of momentum and polarization for a photon in the specified state. By the general rules of quantum mechanics, $w_{\mathbf{k}\alpha}$ is given by the squared modulus of the coefficients in the expansion of $\mathbf{A}(\mathbf{k}')$ in wave functions of states with definite $\mathbf{k}$ and $\mathbf{e}^{(\alpha)}$:

$$w_{\mathbf{k}\alpha} \propto \left| \sum_{\mathbf{k}'\alpha'} \mathbf{A}_{\mathbf{k}\alpha}^*(\mathbf{k}', \alpha') \mathbf{A}(\mathbf{k}') \right|^2$$

(the proportionality factor depends on the normalization convention for the functions). Substitution of (3.3) gives

$$w_{\mathbf{k}\alpha} \propto \left| \mathbf{e}^{(\alpha)} \mathbf{A}(\mathbf{k}) \right|^2. \quad (3.4)$$

⁵Notice that the factor $1/(4\pi)$ in the integral (3.2) is twice the usual factor $1/(8\pi)$ in (2.10). Ultimately, this difference is connected with the fact that the vectors $\mathbf{E}_{\mathbf{k}\alpha}$ and $\mathbf{H}_{\mathbf{k}\alpha}$ are complex, whereas the field operators $\hat{\mathbf{E}}$ and $\hat{\mathbf{H}}$ are Hermitian.

After summing over both polarizations, we obtain the probability that the photon has momentum $\mathbf{k}$:

$$w_{\mathbf{k}} \propto |\mathbf{A}(\mathbf{k})|^2. \quad (3.5)$$

§4. Gauge invariance

As is well known, the choice of field potentials in classical electrodynamics is not unique: the components of the 4-potential A_μ may undergo an arbitrary *gauge* (or *gradient*) transformation of the form

$$A_\mu \rightarrow A_\mu + \partial_\mu \chi, \quad (4.1)$$

where χ is an arbitrary function of the coordinates and time (see II, § 18).

For a plane wave, if we restrict ourselves to transformations that leave the form of the potential unchanged (its proportionality to the factor $\exp(-ik_\mu x^\mu)$), the ambiguity reduces to the freedom to add to the wave amplitude any 4-vector proportional to k^μ .

The nonuniqueness of the potential is of course retained in quantum theory, whether applied to field operators or to photon wave functions. Without specifying a choice of potentials in advance, one must replace (2.17) by the analogous expansion of the operator 4-potential

$$\hat{A}^\mu = \sum_{\mathbf{k}\alpha} (\hat{c}_{\mathbf{k}\alpha} A_{\mathbf{k}\alpha}^\mu + \hat{c}_{\mathbf{k}\alpha}^\dagger A_{\mathbf{k}\alpha}^{\mu*}), \quad (4.2)$$

where the wave functions $A_{\mathbf{k}\alpha}^\mu$ are 4-vectors of the form

$$A_{\mathbf{k}}^\mu = \sqrt{4\pi} \frac{e^\mu}{\sqrt{2\omega}} e^{-ik_\nu x^\nu}, \quad e_\mu e^{\mu*} = -1,$$

or, in abbreviated notation with the four-dimensional vector indices suppressed,

$$A_{\mathbf{k}} = \sqrt{4\pi} \frac{e}{\sqrt{2\omega}} e^{-ikx}, \quad ee^* = -1. \quad (4.3)$$

Here $k_\mu = (\omega, \mathbf{k})$ is the 4-momentum (so that $kx = \omega t - \mathbf{k}\mathbf{r}$), and e is a unit polarization 4-vector⁶.

If the gauge transformations are restricted to those that do not alter the coordinate and time dependence of (4.3), they consist in the replacement

$$e_\mu \rightarrow e_\mu + \chi k_\mu, \quad (4.4)$$

where $\chi = \chi(k^\mu)$ is arbitrary. Transverse polarization means that a gauge can always be chosen in which the 4-vector e has the form

$$e^\mu = (0, \mathbf{e}), \quad \mathbf{e}\mathbf{k} = 0 \quad (4.5)$$

⁶Expression (4.3) is not fully relativistically covariant in its 4-vector form. This is connected with the noninvariant character of the normalization to a finite volume $V = 1$ adopted here. The point is not fundamental, however, and is amply offset by the convenience of this normalization. We shall see that it leads simply and automatically to real physical quantities in the required invariant form.

(we shall call this a *three-dimensionally transverse gauge*). In invariant four-dimensional form, the requirement is the condition of *four-dimensional transversality*

$$ek = 0. \quad (4.6)$$

Notice that this condition, like the normalization $ee^* = -1$, is unaffected by transformation (4.4) because $k^2 = 0$. Conversely, the vanishing square of a particle's 4-momentum means that its mass vanishes. This reveals the connection between gauge invariance and the zero photon mass (other aspects of this connection will be discussed in § 14).

No measurable physical quantity may change under a gauge transformation of the photon wave functions participating in a process. This requirement of *gauge invariance* has an even greater role in quantum electrodynamics than in classical theory. Numerous examples will show that, alongside relativistic invariance, it is a powerful heuristic principle.

Gauge invariance of the theory is in turn closely related to conservation of electric charge; this aspect will be considered in § 43.

We already mentioned in the preceding section that the coordinate-space wave function of a photon cannot be interpreted as the probability amplitude for its spatial localization. Mathematically, this is expressed by the impossibility of constructing from the wave function a quantity that could play the role of a probability density even by its formal properties alone. Such a quantity would have to be an essentially positive bilinear combination of the wave function A_μ and its complex conjugate. It would also have to satisfy definite requirements of relativistic covariance: it would have to be the time component of a 4-vector. Indeed, the continuity equation expressing conservation of particle number has the four-dimensional form of a vanishing divergence of the 4-current, whose time component in the present case is the probability density for localizing the particle (see II, § 29). On the other hand, gauge invariance would allow the 4-vector A_μ to enter the current only through the antisymmetric tensor $F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu = -i(k_\mu A_\nu - k_\nu A_\mu)$. The 4-current would therefore have to be constructed bilinearly from $F_{\mu\nu}$ and $F_{\mu\nu}^*$, together with components of k_μ . But no such 4-vector can be constructed at all: every expression satisfying these conditions, for example $k^\lambda F_{\mu\nu}^* F_\lambda{}^\nu$, vanishes by transversality, $k^\lambda F_{\nu\lambda} = 0$. Moreover, such an expression would not be essentially positive, since it contains odd powers of the components of k_μ .

§ 5. The electromagnetic field in quantum theory

Describing the field as an assembly of photons is the only description fully adequate to the physical meaning of the electromagnetic field in quantum theory. It replaces the classical description in terms of field strengths. Within the mathematical machinery of the photon picture, these strengths appear as second-quantized operators.

It is well known that the properties of a quantum system approach classical behavior when the quantum numbers specifying its stationary states are large. For the free electromagnetic field in a given volume, this means that the quantum numbers of the oscillators, namely the photon numbers $N_{\mathbf{k}\alpha}$, must be large. In this connection, the fact that photons obey Bose statistics has a deep significance. In the mathematical formalism,

the relation between Bose statistics and the properties of the classical field appears in the commutation rules for $\hat{c}_{\mathbf{k}\alpha}$ and $\hat{c}_{\mathbf{k}\alpha}^+$. When $N_{\mathbf{k}\alpha}$ is large, so are the matrix elements of these operators, and the unit on the right-hand side of commutation relation (2.16) may be neglected. The result is

$$\hat{c}_{\mathbf{k}\alpha}\hat{c}_{\mathbf{k}\alpha}^+ \simeq \hat{c}_{\mathbf{k}\alpha}^+\hat{c}_{\mathbf{k}\alpha},$$

so the operators pass into the mutually commuting classical quantities $c_{\mathbf{k}\alpha}$ and $c_{\mathbf{k}\alpha}^*$ that determine the classical field strengths.

The condition for a quasiclassical field still requires refinement. If every $N_{\mathbf{k}\alpha}$ were large, summation over all states $\mathbf{k}\alpha$ would necessarily give an infinite field energy, making the condition meaningless.

A physically meaningful formulation of the quasiclassical condition is based on field values averaged over a short time interval Δt . Suppose the classical electric field $\mathbf{E}$ (or magnetic field $\mathbf{H}$) is expanded as a Fourier integral in time. When it is averaged over Δt , an appreciable contribution to the mean $\bar{\mathbf{E}}$ comes only from Fourier components whose frequencies obey $\omega\Delta t \lesssim 1$; otherwise, the oscillating factor $e^{-i\omega t}$ nearly averages to zero. Thus, in determining the condition for the averaged field to be quasiclassical, one need consider only quantum oscillators with $\omega \lesssim 1/\Delta t$. It is enough to require large quantum numbers for these oscillators.

The number of oscillators with frequencies between zero and $\omega \sim 1/\Delta t$, per unit volume $V = 1$, is of the order ⁷

$$\left(\frac{\omega}{c}\right)^3 \sim \frac{1}{(c\Delta t)^3}. \quad (5.1)$$

The total field energy per unit volume is $\sim \bar{\mathbf{E}}^2$. Dividing it by the number of oscillators and by a representative energy of a single photon, $\sim \hbar\omega$, gives the order of magnitude of the photon numbers:

$$N_{\mathbf{k}} \sim \frac{\bar{\mathbf{E}}^2 c^3}{\hbar\omega^4}.$$

Requiring this number to be large gives the inequality

$$|\bar{\mathbf{E}}| \gg \frac{\sqrt{\hbar c}}{(c\Delta t)^2}. \quad (5.2)$$

This is the required condition under which a field averaged over intervals Δt may be treated classically. The field must be sufficiently strong, and increasingly so as the averaging interval Δt becomes shorter. For time-dependent fields, this interval must of course not exceed the periods over which the field changes appreciably. Consequently, sufficiently weak time-dependent fields cannot in any event be quasiclassical. Only for static fields, constant in time, may one set $\Delta t \rightarrow \infty$, so that the right-hand side of (5.2) vanishes. A static field is therefore always classical.

It was already pointed out that the classical expressions for an electromagnetic field as a superposition of plane waves must be regarded as operators in quantum theory. The

⁷Ordinary units are used in this section.

physical meaning of these operators is, however, very limited. A physically meaningful field operator would have to give zero field values in the photon-vacuum state. Yet the mean value of the squared-field operator $\hat{\mathbf{E}}^2$ in a normal state, which is proportional to the zero-point energy of the field, is infinite. Here “mean value” denotes the quantum-mechanical mean, that is, the corresponding diagonal matrix element of the operator. This cannot be avoided even by a formal deletion procedure, as can be done for the field energy: here one would have to accomplish it by some reasonable modification of the operators $\hat{\mathbf{E}}$ and $\hat{\mathbf{H}}$ themselves, rather than of their squares, and that is impossible.

§ 6. Angular momentum and parity of the photon

Like any particle, a photon may have a definite angular momentum. To determine the properties of this quantity for the photon, we first recall how the mathematical apparatus of quantum mechanics relates the properties of a particle's wave function to its angular momentum.

The particle angular momentum $\mathbf{j}$ is composed of the orbital angular momentum $\mathbf{l}$ and the intrinsic angular momentum, or spin, $\mathbf{s}$. The wave function of a particle of spin s is a symmetric spinor of rank $2s$: it is a collection of $2s + 1$ components that transform into one another according to a definite rule under rotations of the coordinate system. Orbital angular momentum is connected instead with the coordinate dependence of the wave functions. States of orbital angular momentum l have wave-function components expressible linearly through spherical functions of order l .

A consistent distinction between spin and orbital angular momentum therefore requires the “spin” and “coordinate” properties of the wave functions to be independent: at a specified time, the coordinate dependence of the spinor components must not be restricted by any additional condition.

In the momentum representation, coordinate dependence is replaced by dependence on the momentum $\mathbf{k}$. The photon wave function in a three-dimensionally transverse gauge is the vector $\mathbf{A}(\mathbf{k})$. A vector is equivalent to a spinor of rank two, and in this sense one might assign spin 1 to the photon. But this vector wave function obeys the transversality condition $\mathbf{kA}(\mathbf{k}) = 0$, an additional restriction on the momentum dependence of $\mathbf{A}(\mathbf{k})$. Its components can consequently no longer be specified independently and arbitrarily at the same time, which makes separation of orbital angular momentum and spin impossible.

Nor can photon spin be defined as the angular momentum of a particle at rest, since no rest frame exists for a photon moving at the speed of light.

Thus only the total angular momentum of the photon can be discussed. It is clear in advance that this total angular momentum can take only integer values. This follows already from the absence of spinors of odd rank among the quantities characterizing a photon.

As with any particle, a photon state is also characterized by its parity, which is connected with the behavior of the wave function under inversion of the coordinate system (see III, § 30). In the momentum representation, reversal of the coordinates corresponds to reversal of every component of $\mathbf{k}$. On a scalar function $\varphi(\mathbf{k})$, the inversion operator $\hat{P}$ acts only by this reversal: $\hat{P}\varphi(\mathbf{k}) = \varphi(-\mathbf{k})$. For a vector function $\mathbf{A}(\mathbf{k})$, one must also account for the fact that reversing the directions of the axes reverses every

vector component. Hence⁸

$$\hat{P}\mathbf{A}(\mathbf{k}) = -\mathbf{A}(-\mathbf{k}). \quad (6.1)$$

Although separating photon angular momentum into orbital angular momentum and spin has no physical meaning, it is nevertheless convenient to introduce “spin” $\mathbf{s}$ and “orbital angular momentum” $\mathbf{l}$ formally as auxiliary concepts expressing how the wave function transforms under rotations. The value $s = 1$ corresponds to the vector character of the wave function, while l is the order of the spherical functions entering it. Here we mean wave functions of states with definite photon angular momentum, which are spherical waves for a free particle. In particular, l determines the parity of the photon state:

$$P = (-1)^{l+1} \quad (6.2)$$

In the same formal sense, the angular-momentum operator $\hat{\mathbf{j}}$ may be represented as the sum $\hat{\mathbf{s}} + \hat{\mathbf{l}}$. As is known, the angular-momentum operator is connected with the operator of an infinitesimal rotation of the coordinate system; here it is connected with the action of that operator on a vector field. In the sum $\hat{\mathbf{s}} + \hat{\mathbf{l}}$, the operator $\hat{\mathbf{s}}$ acts on the vector index, transforming the different vector components into one another. The operator $\hat{\mathbf{l}}$ acts on those components as functions of momentum (or coordinates).

Let us count the states of a given energy that are possible for a specified value j of the photon angular momentum, disregarding the trivial $(2j + 1)$ -fold degeneracy in its directions.

If $\mathbf{l}$ and $\mathbf{s}$ were independent, this count would simply amount to finding the number of ways in which the vector-model rules allow $\mathbf{l}$ and $\mathbf{s}$ to be added so as to obtain the required value j . For a particle of spin $s = 1$ and specified nonzero j , this would give three states with the following l values and parities:

$$l = j, \quad P = (-1)^{l+1} = (-1)^{j+1}; \quad l = j \pm 1, \quad P = (-1)^{l+1} = (-1)^j.$$

For $j = 0$, there would be only one state, with $l = 1$ and parity $P = +1$.

This count, however, does not include the transversality condition on $\mathbf{A}$; all three of its components were treated as independent. The number of states corresponding to a longitudinal vector must therefore be subtracted. Such a vector can be written $\mathbf{k}\varphi(\mathbf{k})$, showing that, by their transformation properties under rotations, its three components are equivalent to the single scalar φ ⁹. The superfluous state incompatible with transversality may consequently be said to correspond to a particle state with a scalar wave function, a spinor of rank 0, and hence with “spin 0”¹⁰. The angular momentum j of this state is

⁸We define the parity of a state by the action of inversion on the polar vector $\mathbf{A}$, or equivalently on the electric vector $\mathbf{E} = i\omega\mathbf{A}$. Its sign differs from the action on the axial vector $\mathbf{H} = i(\mathbf{k} \times \mathbf{A})$, since inversion does not reverse such a vector:

$$\hat{P}\mathbf{H}(\mathbf{k}) = \mathbf{H}(-\mathbf{k}).$$

⁹Indeed, a statement about how a quantity transforms under rotation concerns the transformation at a given point, that is, at a specified $\mathbf{k}$. Under such a transformation, $\mathbf{k}\varphi(\mathbf{k})$ does not change at all and therefore behaves as a scalar.

¹⁰We emphasize again that no state of any real particle is meant here. The count is formal and, mathematically, amounts to classifying the entire set of quantities that transform into one another according to the irreducible representations of the rotation group.

therefore the order of the spherical functions entering φ . Its parity as a photon state is determined by the action of inversion on the vector function $\mathbf{k}\varphi$:

$$\widehat{P}(\mathbf{k}\varphi) = -(-\mathbf{k})\varphi(-\mathbf{k}) = (-1)^j \mathbf{k}\varphi(\mathbf{k}),$$

and is thus $(-1)^j$. One state must therefore be subtracted from the number obtained above with parity $(-1)^j$: two states when $j \neq 0$, and one when $j = 0$.

The final result is that for every nonzero photon angular momentum j , there is one even state and one odd state. For $j = 0$, there are no states at all. A photon therefore cannot have zero angular momentum, and j takes only the values $1, 2, 3, \dots$. The impossibility of $j = 0$ is also evident directly: a wave function of zero angular momentum would have to be spherically symmetric, which is plainly impossible for a transverse wave.

A standard terminology is used for the different photon states. A photon of angular momentum j and parity $(-1)^j$ is called an *electric 2^j -pole* photon (or an *Ej* photon), and one of parity $(-1)^{j+1}$ is called a *magnetic 2^j -pole* photon (or an *Mj* photon). Thus an electric dipole photon is an odd state with $j = 1$, an electric quadrupole photon is an even state with $j = 2$, and a magnetic dipole photon is an even state with $j = 1$ ¹¹.

§7. Spherical photon waves

Having determined the possible values of photon angular momentum, we must now find the corresponding wave functions.¹²

First consider the formal problem of finding vector functions that are eigenfunctions of the operators $\widehat{\mathbf{j}}^2$ and $\widehat{j}_z$. At this stage we neither decide which of these functions enter the photon wave functions of interest nor impose transversality.

We seek the functions in the momentum representation. The coordinate operator in this representation is $\widehat{\mathbf{r}} = i\partial/\partial\mathbf{k}$ (see III, (15.12)). The orbital-angular-momentum operator is

$$\widehat{\mathbf{l}} = \widehat{\mathbf{r}} \times \mathbf{k} = -i \left(\mathbf{k} \times \frac{\partial}{\partial \mathbf{k}} \right),$$

and thus differs from the angular-momentum operator in the coordinate representation only by replacing $\mathbf{r}$ with $\mathbf{k}$. The formal solution of the problem is therefore the same in both representations.

Denote the required eigenfunctions by $\mathbf{Y}_{jm}$ and call them vector spherical harmonics. They must obey

$$\widehat{\mathbf{j}}^2 \mathbf{Y}_{jm} = j(j+1) \mathbf{Y}_{jm}, \quad \widehat{j}_z \mathbf{Y}_{jm} = m \mathbf{Y}_{jm} \quad (7.1)$$

(the z axis is a specified direction in space). We shall show that every function of the form $\mathbf{a}Y_{jm}$ has this property, where $\mathbf{a}$ is any vector constructed from the unit vector $\mathbf{n} = \mathbf{k}/\omega$,

¹¹These names agree with the terminology of classical radiation theory. We shall see below (§ 46, 47) that the emission of electric- and magnetic-type photons is determined by the corresponding electric and magnetic moments of the system of charges.

¹²This question was first considered by Heitler (W. Heitler, 1936). The form of the solution presented here is due to V. B. Berestetskii (1947).

and Y_{jm} is an ordinary scalar spherical function. We use throughout the definition from III, § 28:

$$Y_{lm}(\mathbf{n}) = (-1)^{(m+|m|)/2} i^l \sqrt{\frac{(2l+1)(l-|m|)!}{4\pi(l+|m|)!}} P_l^{|m|}(\cos \theta) e^{im\varphi} \quad (7.2)$$

(θ and φ are the spherical angles specifying the direction $\mathbf{n}$).¹³ Recall the commutation rule III, (29.4):

$$\{\widehat{l}_i, a_k\}_- = i e_{ikl} a_l.$$

The right-hand side can be written $(-\widehat{s}_i a_k)$, where $\widehat{\mathbf{s}}$ is the spin-1 operator; its action on a vector function is precisely defined by $\widehat{s}_i a_k = -i e_{ikl} a_l$ (see III, § 57, problem 2). Hence

$$\widehat{l}_i a_k - a_k \widehat{l}_i = -\widehat{s}_i a_k.$$

Using this equality, we find

$$\widehat{j}_i a_k = (\widehat{l}_i + \widehat{s}_i) a_k = a_k \widehat{l}_i.$$

Consequently,

$$\widehat{\mathbf{j}}^2(\mathbf{a} Y_{jm}) = \mathbf{a} \widehat{\mathbf{l}}^2 Y_{jm}, \quad \widehat{j}_z(\mathbf{a} Y_{jm}) = \mathbf{a} \widehat{l}_z Y_{jm}.$$

But Y_{jm} is an eigenfunction of $\widehat{\mathbf{l}}^2$ and $\widehat{l}_z$ with eigenvalues $j(j+1)$ and m , respectively, and equations (7.1) follow.

Three essentially different types of vector spherical harmonic result from choosing $\mathbf{a}$ to be one of the following vectors:¹⁴

$$\frac{\nabla_{\mathbf{n}}}{\sqrt{j(j+1)}}, \quad \frac{\mathbf{n} \times \nabla_{\mathbf{n}}}{\sqrt{j(j+1)}}, \quad \mathbf{n}. \quad (7.3)$$

We therefore define the vector spherical harmonics by

$$\begin{aligned} \mathbf{Y}_{jm}^{(e)} &= \frac{1}{\sqrt{j(j+1)}} \nabla_{\mathbf{n}} Y_{jm}, & P &= (-1)^j, \\ \mathbf{Y}_{jm}^{(m)} &= \mathbf{n} \times \mathbf{Y}_{jm}^{(e)}, & P &= (-1)^{j+1}, \\ \mathbf{Y}_{jm}^{(l)} &= \mathbf{n} Y_{jm}, & P &= (-1)^j. \end{aligned} \quad (7.4)$$

¹³For later reference, we record the value at $\theta = 0$, when $\mathbf{n}$ lies along the z axis:

$$Y_{lm}(\mathbf{n}_z) = i^l \sqrt{\frac{2l+1}{4\pi}} \delta_{m0}. \quad (7.2a)$$

¹⁴The operator $\nabla_{\mathbf{n}} = |\mathbf{k}| \nabla_{\mathbf{k}}$ acts on functions that depend only on the direction $\mathbf{n}$. In spherical coordinates it has only two components:

$$\nabla_{\mathbf{n}} = \left(\frac{\partial}{\partial \theta}, \frac{1}{\sin \theta} \frac{\partial}{\partial \varphi} \right).$$

The operator denoted below by $\Delta_{\mathbf{n}}$ is the angular part of the Laplacian:

$$\Delta_{\mathbf{n}} = \frac{1}{\sin \theta} \frac{\partial}{\partial \theta} \sin \theta \frac{\partial}{\partial \theta} + \frac{1}{\sin^2 \theta} \frac{\partial^2}{\partial \varphi^2}.$$

The parity P is shown alongside each vector. The three types are mutually orthogonal; $\mathbf{Y}_{jm}^{(l)}$ is longitudinal, while $\mathbf{Y}_{jm}^{(e)}$ and $\mathbf{Y}_{jm}^{(m)}$ are transverse to $\mathbf{n}$.

The vector spherical harmonics can be expressed through scalar spherical functions. The harmonics $\mathbf{Y}_{jm}^{(m)}$ contain spherical functions of only one order, $l = j$, whereas $\mathbf{Y}_{jm}^{(e)}$ and $\mathbf{Y}_{jm}^{(l)}$ contain orders $l = j \pm 1$. This is evident on comparing their parities in (7.4) with the parity $(-1)^{l+1}$ of a vector field expressed through spherical functions of order l .

Within each type, the vector spherical harmonics are mutually orthogonal and normalized by

$$\int \mathbf{Y}_{jm} \mathbf{Y}_{j'm'}^* d\omega = \delta_{jj'} \delta_{mm'}. \quad (7.5)$$

For $\mathbf{Y}_{jm}^{(l)}$, this follows immediately from the normalization of Y_{jm} . For $\mathbf{Y}_{jm}^{(e)}$, the normalization integral is

$$\frac{1}{\sqrt{j(j+1)j'(j'+1)}} \int \nabla_{\mathbf{n}} Y_{jm} \nabla_{\mathbf{n}} Y_{j'm'}^* d\omega = -\frac{1}{\sqrt{j(j+1)j'(j'+1)}} \int Y_{j'm'}^* \Delta_{\mathbf{n}} Y_{jm} d\omega,$$

and since $\Delta_{\mathbf{n}} Y_{jm} = -j(j+1)Y_{jm}$, equation (7.5) follows. The normalization of $\mathbf{Y}_{jm}^{(m)}$ reduces to the same integral.

The harmonics (7.4) could also have been obtained without the direct verification of (7.1), solely from general transformation arguments. In the preceding section, such arguments showed that a vector function $\mathbf{n}\varphi$ has an angular momentum j equal to the order of the spherical functions entering φ . Taking $\varphi = Y_{jm}$ also fixes the projection m , giving $\mathbf{Y}_{jm}^{(l)}$ immediately. The transformation argument of § 6 is unchanged if $\mathbf{n}$ in $\mathbf{n}\varphi$ is replaced by $\nabla_{\mathbf{n}}$ or $\mathbf{n} \times \nabla_{\mathbf{n}}$, producing the other two types.

We now return to photon wave functions. For an electric-type photon (E_j), the vector $\mathbf{A}(\mathbf{k})$ must have parity $(-1)^j$. Both $\mathbf{Y}_{jm}^{(e)}$ and $\mathbf{Y}_{jm}^{(l)}$ have this parity, but only the former satisfies transversality. For a magnetic-type photon (M_j), $\mathbf{A}(\mathbf{k})$ must have parity $(-1)^{j+1}$, which belongs only to $\mathbf{Y}_{jm}^{(m)}$. Thus the wave functions of a photon with specified angular momentum j , projection m , and energy ω are

$$\mathbf{A}_{\omega jm}(\mathbf{k}) = \frac{4\pi^2}{\omega^{3/2}} \delta(|\mathbf{k}| - \omega) \mathbf{Y}_{jm}(\mathbf{n}), \quad (7.6)$$

where $\mathbf{Y}_{jm}$ is $\mathbf{Y}_{jm}^{(e)}$ for an electric-type photon and $\mathbf{Y}_{jm}^{(m)}$ for a magnetic-type photon. The factor $\delta(|\mathbf{k}| - \omega)$ imposes the specified energy. The normalization is

$$\frac{1}{(2\pi)^4} \int \omega \omega' \mathbf{A}_{\omega' j'm'}^*(\mathbf{k}) \mathbf{A}_{\omega jm}(\mathbf{k}) d^3k = \omega \delta(\omega' - \omega) \delta_{jj'} \delta_{mm'}. \quad (7.7)$$

For coordinate-representation wave functions, condition (7.7) is equivalent to¹⁵

$$\begin{aligned} \frac{1}{4\pi} \int \{ \mathbf{E}_{\omega' j'm'}^*(\mathbf{r}) \mathbf{E}_{\omega jm}(\mathbf{r}) + \mathbf{H}_{\omega' j'm'}^*(\mathbf{r}) \mathbf{H}_{\omega jm}(\mathbf{r}) \} d^3x \\ = \omega \delta(\omega' - \omega) \delta_{jj'} \delta_{mm'}. \end{aligned} \quad (7.8)$$

¹⁵This condition is of the same type as (2.22). The factor $\delta(\omega' - \omega)$ occurs on the right because the field, a spherical wave, is considered throughout infinite space rather than in a finite volume $V = 1$.

Indeed, expressed through the potentials, the integral on the left is

$$\frac{1}{2\pi} \int \mathbf{A}_{\omega'j'm'}^*(\mathbf{r}) \mathbf{A}_{\omega jm}(\mathbf{r}) \omega' \omega d^3x.$$

Substitute

$$\begin{aligned} \mathbf{A}_{\omega jm}(\mathbf{r}) &= \int \mathbf{A}_{\omega jm}(\mathbf{k}) e^{i\mathbf{k}\mathbf{r}} \frac{d^3k}{(2\pi)^3}, \\ \mathbf{A}_{\omega'j'm'}^*(\mathbf{r}) &= \int \mathbf{A}_{\omega'j'm'}^*(\mathbf{k}') e^{-i\mathbf{k}'\mathbf{r}} \frac{d^3k'}{(2\pi)^3}. \end{aligned} \quad (7.9)$$

Integration over d^3x then gives $(2\pi)^3 \delta(\mathbf{k}' - \mathbf{k})$, which is removed by the d^3k' integration, reducing the integral to (7.7).

Up to this point we have assumed a transverse gauge, in which the scalar potential is $\Phi = 0$. Other gauges for a spherical wave may, however, be more convenient in some applications.

An admissible transformation of the potentials in momentum space is

$$\mathbf{A} \rightarrow \mathbf{A} + \mathbf{n}f(\mathbf{k}), \quad \Phi \rightarrow \Phi + f(\mathbf{k}),$$

where $f(\mathbf{k})$ is arbitrary. Here choose it so that the new potentials are expressed through the same spherical functions and retain a definite parity. For an electric-type photon, these requirements restrict the potentials to

$$\begin{aligned} \mathbf{A}_{\omega jm}^{(e)}(\mathbf{k}) &= \frac{4\pi^2}{\omega^{3/2}} \delta(|\mathbf{k}| - \omega) \left(\mathbf{Y}_{jm}^{(e)} + CnY_{jm} \right), \\ \Phi_{\omega jm}^{(e)}(\mathbf{k}) &= \frac{4\pi^2}{\omega^{3/2}} \delta(|\mathbf{k}| - \omega) CY_{jm}, \end{aligned} \quad (7.10)$$

where C is arbitrary. For a magnetic-type photon, adding such a term to $\mathbf{A}_{\omega jm}^{(m)}(\mathbf{k})$ would destroy definite parity; under the same requirements, the choice (7.6) is therefore unique.

By (3.5) and (7.6), the probability that a photon of definite angular momentum and parity will be detected moving in direction $\mathbf{n}$ within the solid-angle element do is

$$w(\mathbf{n}) do = |\mathbf{Y}_{jm}^{(e)}|^2 do. \quad (7.11)$$

This expression was written for an E -type photon. Since $|\mathbf{Y}_{jm}^{(m)}|^2 = |\mathbf{Y}_{jm}^{(e)}|^2$, however, the distributions $w(\mathbf{n})$ are identical for both types.

The modulus squared $|\mathbf{Y}_{jm}^{(e)}|^2$ is independent of the azimuth φ , because the factors $e^{\pm im\varphi}$ in the spherical functions cancel. Thus $w(\mathbf{n})$ is symmetric about the z axis. Moreover, each vector spherical harmonic has a definite parity, so its modulus squared is even under inversion, or under $\theta \rightarrow \pi - \theta$. Hence the expansion of $w(\theta)$ in Legendre polynomials contains only even orders. Determining its coefficients reduces to evaluating integrals of products of three spherical functions and then summing over components. Formulas from III, §§ 107–108 perform both operations and give

$$\begin{aligned} w(\theta) &= (-1)^{m+1} \frac{(2j+1)^2}{4\pi} \sum_{n=0}^{\infty} (4n+1) \begin{pmatrix} j & j & 2n \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} j & j & 2n \\ m & -m & 0 \end{pmatrix} \\ &\quad \times \begin{Bmatrix} j & j & 1 \\ j & j & 2n \end{Bmatrix} P_{2n}(\cos \theta). \end{aligned} \quad (7.12)$$

Finally, we give the components of the vector spherical harmonics as expansions in scalar spherical functions. We use the “spherical components” of a vector defined in III, § 107. For a vector $\mathbf{f}$, they are

$$f_0 = if_z, \quad f_{+1} = -\frac{i}{\sqrt{2}}(f_x + if_y), \quad f_{-1} = \frac{i}{\sqrt{2}}(f_x - if_y). \quad (7.13)$$

Introduce the “circular unit vectors”

$$\mathbf{e}^{(0)} = i\mathbf{e}^{(z)}, \quad \mathbf{e}^{(+1)} = -\frac{i}{\sqrt{2}}(\mathbf{e}^{(x)} + i\mathbf{e}^{(y)}), \quad \mathbf{e}^{(-1)} = \frac{i}{\sqrt{2}}(\mathbf{e}^{(x)} - i\mathbf{e}^{(y)}) \quad (7.14)$$

($\mathbf{e}^{(x,y,z)}$ are unit vectors along the x , y , and z axes). Then

$$\mathbf{f} = \sum_{\lambda} (-1)^{1-\lambda} f_{-\lambda} \mathbf{e}^{(\lambda)}, \quad f_{\lambda} = (-1)^{1-\lambda} \mathbf{f} \mathbf{e}^{(-\lambda)*} = \mathbf{f} \mathbf{e}^{(\lambda)}. \quad (7.15)$$

The spherical components of the vector spherical harmonics are expressed through spherical functions by means of $3j$ symbols:

$$\begin{aligned} (-1)^{j+m+\lambda+1} (\mathbf{Y}_{jm}^{(e)})_{\lambda} &= -\sqrt{j} \begin{pmatrix} j+1 & 1 & j \\ m+\lambda & -\lambda & -m \end{pmatrix} Y_{j+1,m+\lambda} \\ &\quad + \sqrt{j+1} \begin{pmatrix} j-1 & 1 & j \\ m+\lambda & -\lambda & -m \end{pmatrix} Y_{j-1,m+\lambda}, \\ (-1)^{j+m+\lambda+1} (\mathbf{Y}_{jm}^{(m)})_{\lambda} &= -\sqrt{2j+1} \begin{pmatrix} j & 1 & j \\ m+\lambda & -\lambda & -m \end{pmatrix} Y_{j,m+\lambda}, \\ (-1)^{j+m+\lambda+1} (\mathbf{Y}_{jm}^{(l)})_{\lambda} &= \sqrt{j+1} \begin{pmatrix} j+1 & 1 & j \\ m+\lambda & -\lambda & -m \end{pmatrix} Y_{j+1,m+\lambda} \\ &\quad + \sqrt{j} \begin{pmatrix} j-1 & 1 & j \\ m+\lambda & -\lambda & -m \end{pmatrix} Y_{j-1,m+\lambda}. \end{aligned} \quad (7.16)$$

These formulas are derived as follows. Each of the three vector spherical harmonics has the form $\mathbf{Y}_{jm} = \mathbf{a} Y_{jm}$, where $\mathbf{a}$ is one of the vectors in (7.3). Therefore

$$\mathbf{Y}_{jm} = \sum_{lm'} \langle lm' | \mathbf{a} | jm \rangle Y_{lm'},$$

and the problem reduces to finding matrix elements of $\mathbf{a}$ between eigenfunctions of orbital angular momentum. According to (107.6) (see III),

$$\langle lm' | a_{\lambda} | jm \rangle = i(-1)^{j_{\max}-m'} \begin{pmatrix} l & 1 & j \\ -m' & \lambda & m \end{pmatrix} \langle l || \mathbf{a} || j \rangle,$$

where $j_{\max}$ is the greater of l and j . It is therefore enough to know the nonzero reduced matrix elements $\langle l || \mathbf{a} || j \rangle$. They are

$$\begin{aligned} \langle l-1 || \mathbf{n} || l \rangle &= \langle l || \mathbf{n} || l-1 \rangle^* = i\sqrt{l}, \\ \langle l || \mathbf{\nabla}_{\mathbf{n}} || l-1 \rangle &= i(l-1)\sqrt{l}, \\ \langle l-1 || \mathbf{\nabla}_{\mathbf{n}} || l \rangle &= i(l+1)\sqrt{l}, \\ \langle l || \mathbf{n} \times \mathbf{\nabla}_{\mathbf{n}} || l \rangle &= i\sqrt{l(l+1)(2l+1)}. \end{aligned} \quad (7.17)$$

§ 8. Photon polarization

For a photon, the polarization vector $\mathbf{e}$ serves as the “spin part” of the wave function (subject to the qualifications concerning the notion of photon spin stated in § 6).

The various possibilities for photon polarization are the same as the possible polarization types of a classical electromagnetic wave (see II, § 48).

An arbitrary polarization $\mathbf{e}$ can be represented as a superposition of two mutually orthogonal polarizations $\mathbf{e}^{(1)}$ and $\mathbf{e}^{(2)}$, chosen in some definite manner ($\mathbf{e}^{(1)}\mathbf{e}^{(2)*} = 0$). In the expansion

$$\mathbf{e} = e_1\mathbf{e}^{(1)} + e_2\mathbf{e}^{(2)} \quad (8.1)$$

the squared moduli of e_1 and e_2 give the probabilities that the photon has polarization $\mathbf{e}^{(1)}$ or $\mathbf{e}^{(2)}$.

The two chosen polarizations may be taken as mutually perpendicular linear polarizations. Alternatively, an arbitrary polarization can be resolved into two circular polarizations rotating in opposite senses. We denote the right- and left-circular polarization vectors by $\mathbf{e}^{(+1)}$ and $\mathbf{e}^{(-1)}$, respectively; in a coordinate system $\xi\eta\zeta$ whose ζ axis lies along the photon direction $\mathbf{n} = \mathbf{k}/\omega$,

$$\mathbf{e}^{(+1)} = -\frac{i}{\sqrt{2}}(\mathbf{e}^{(\xi)} + i\mathbf{e}^{(\eta)}), \quad \mathbf{e}^{(-1)} = \frac{i}{\sqrt{2}}(\mathbf{e}^{(\xi)} - i\mathbf{e}^{(\eta)}). \quad (8.2)$$

The availability of two distinct photon polarizations (at a specified momentum) means, equivalently, that every momentum eigenvalue is doubly degenerate. This fact is closely connected with the vanishing of the photon mass.

A freely moving particle of nonzero mass always has a rest frame. Evidently, it is in that frame that the particle’s own symmetry properties become manifest. One must then consider symmetry under every possible rotation about the center (that is, under the entire spherical-symmetry group). The particle’s symmetry with respect to this group is characterized by its spin s , which fixes the degeneracy multiplicity (the number $2s + 1$ of distinct wave functions transformed into one another). In particular, a particle with a vector wave function (three components) has spin 1.

A zero-mass particle, by contrast, has no rest frame: it travels at the speed of light in every reference frame. Such a particle always singles out a direction in space, namely the direction of its momentum vector $\mathbf{k}$ (the ζ axis). Hence there is no symmetry under the full group of three-dimensional rotations; only axial symmetry about the distinguished axis remains.

With axial symmetry, the only conserved quantity is the particle’s *helicity*, the projection of its angular momentum on the ζ axis; denote it by λ .¹⁶ If symmetry under reflections in planes containing the ζ axis is required as well, states differing by the sign of λ are mutually degenerate; consequently, $\lambda \neq 0$ gives a twofold degeneracy.¹⁷ A photon state of specified momentum $\mathbf{k}$ belongs to one of these types of doubly degenerate states. Its “spin” wave function is a vector $\mathbf{e}$ in the $\xi\eta$ plane; the two components of this

¹⁶This is to be distinguished from the projection m of angular momentum on a specified direction in space (the z axis), discussed in the preceding section.

¹⁷The electronic terms of a diatomic molecule are classified in the same way (see III, § 78).

vector transform into each other under all rotations about the ζ axis and under reflections in planes containing that axis.

The different polarization cases correspond in a definite way to the possible helicity values. This correspondence follows from the formulas III, (57.9), which relate the components of a vector wave function to those of the equivalent second-rank spinor.¹⁸ Projections $\lambda = +1$ and -1 correspond to vectors $\mathbf{e}$ for which only $e_\xi - ie_\eta$ or $e_\xi + ie_\eta$, respectively, is nonzero; thus $\mathbf{e} = \mathbf{e}^{(+1)}$ or $\mathbf{e} = \mathbf{e}^{(-1)}$. In other words, $\lambda = +1$ and -1 correspond to right- and left-circular photon polarization (the same result will be obtained in § 16 by directly calculating the eigenfunctions of the spin-projection operator). The photon angular momentum projected on its direction of motion can therefore take only the two values ± 1 ; zero cannot occur.

The state of a photon with specified momentum and polarization is a pure state (in the sense explained in III, § 14); it is described by a wave function and supplies a complete quantum-mechanical specification of the particle's state. "Mixed" photon states are possible as well; they correspond to a less complete description, given only by a density matrix rather than by a wave function.

Consider a photon state that is mixed with respect to polarization but has a definite momentum $\mathbf{k}$. In such a state (called a state of *partial polarization*) a "coordinate" wave function exists.

The photon polarization density matrix is a second-rank tensor $\rho_{\alpha\beta}$ in the plane perpendicular to $\mathbf{n}$ (the $\xi\eta$ plane; the indices α, β take only two values). This tensor is Hermitian:

$$\rho_{\alpha\beta} = \rho_{\beta\alpha}^*, \quad (8.3)$$

and is normalized by

$$\rho_{\alpha\alpha} = \rho_{11} + \rho_{22} = 1. \quad (8.4)$$

By (8.3), the diagonal components ρ_{11} and ρ_{22} are real, and (8.4) determines either one from the other. The component ρ_{12} is complex, while $\rho_{21} = \rho_{12}^*$. Thus the density matrix is specified by three real parameters in all.

Once the polarization density matrix is known, the probability that the photon has any particular polarization $\mathbf{e}$ can be found. It is given by the "projection" of the tensor $\rho_{\alpha\beta}$ on the direction of $\mathbf{e}$, namely

$$\rho_{\alpha\beta} e_\alpha^* e_\beta. \quad (8.5)$$

Thus ρ_{11} and ρ_{22} are the probabilities of linear polarization along the ξ and η axes. Projection onto the vectors (8.2) gives the probabilities of the two circular polarizations:

$$\frac{1}{2} [1 \pm i(\rho_{12} - \rho_{21})]. \quad (8.6)$$

Both in form and in content, the properties of $\rho_{\alpha\beta}$ agree with those of the tensor $J_{\alpha\beta}$ used for partially polarized light in the classical theory (see II, § 50). We recall some of these properties here.

¹⁸Recall that contravariant spinor components correspond to the components of a wave function regarded as probability amplitudes for the various values of the particle's angular-momentum projection, which are precisely the quantities at issue here.

For a pure state of definite polarization $\mathbf{e}$, the tensor $\rho_{\alpha\beta}$ reduces to products of components of $\mathbf{e}$:

$$\rho_{\alpha\beta} = e_\alpha e_\beta^*. \quad (8.7)$$

Its determinant is then $|\rho_{\alpha\beta}| = 0$. In the opposite case of an unpolarized photon, all polarization directions are equally probable, so that

$$\rho_{\alpha\beta} = \delta_{\alpha\beta}/2, \quad (8.8)$$

and $|\rho_{\alpha\beta}| = 1/4$.

In the general case, partial polarization is conveniently described by three real Stokes parameters ξ_1, ξ_2, ξ_3 ,¹⁹ in terms of which the density matrix is

$$\rho_{\alpha\beta} = \frac{1}{2} \begin{pmatrix} 1 + \xi_3 & \xi_1 - i\xi_2 \\ \xi_1 + i\xi_2 & 1 - \xi_3 \end{pmatrix}. \quad (8.9)$$

Each parameter ranges from -1 to $+1$. In the unpolarized state $\xi_1 = \xi_2 = \xi_3 = 0$; for a fully polarized photon, $\xi_1^2 + \xi_2^2 + \xi_3^2 = 1$.

The parameter ξ_3 characterizes linear polarization along the ξ or η axis; the probability of photon polarization along these axes is, respectively, $(1 + \xi_3)/2$ or $(1 - \xi_3)/2$. Thus $\xi_3 = +1$ or -1 represents complete polarization in the corresponding direction.

The parameter ξ_1 characterizes linear polarization along directions making an angle $\varphi = \pi/4$ or $\varphi = -\pi/4$ with the ξ axis. The respective probabilities of photon polarization along these directions are $(1 + \xi_1)/2$ and $(1 - \xi_1)/2$; this follows immediately by projecting $\rho_{\alpha\beta}$ on the directions $\mathbf{e} = (1, \pm 1)/\sqrt{2}$.

Finally, ξ_2 is the degree of circular polarization; by (8.6), the probabilities that the photon has right- or left-circular polarization are $(1 + \xi_2)/2$ and $(1 - \xi_2)/2$, respectively. Since the two polarizations correspond to helicities $\lambda = \pm 1$, it follows that, in the general case, ξ_2 is the mean photon helicity. For a pure state with polarization $\mathbf{e}$, note also that

$$\xi_2 = i(\mathbf{e} \times \mathbf{e}^*) \cdot \mathbf{n}. \quad (8.10)$$

Recall (see II, § 50) that ξ_2 and $\sqrt{\xi_1^2 + \xi_3^2}$ are invariant under Lorentz transformations.

We shall also need below the behavior of the Stokes parameters under time reversal. It is readily seen that they are invariant under this transformation. This property clearly does not depend on the nature of the polarization state, so it is enough to verify it for a pure state. In quantum mechanics, time reversal corresponds to replacing the wave function by its complex conjugate (see III, § 18). For a plane-polarized wave this gives the replacement²⁰

$$\mathbf{k} \rightarrow -\mathbf{k}, \quad \mathbf{e} \rightarrow -\mathbf{e}^*. \quad (8.11)$$

¹⁹The notation for these parameters must not be confused with that for the ξ axis.

²⁰The additional sign change of $\mathbf{e}$ arises because time reversal changes the sign of the electromagnetic vector potential. The scalar potential does not change sign; hence time reversal acts on the 4-vector e as

$$(e_0, \mathbf{e}) \rightarrow (e_0^*, -\mathbf{e}^*). \quad (8.11a)$$

Under this transformation, the symmetric part of the density matrix, $(1/2)(e_\alpha e_\beta^* + e_\beta e_\alpha^*)$, and therefore ξ_1 and ξ_3 , remain unchanged. The invariance of ξ_2 under the same transformation follows from (8.10); it is also immediate from the meaning of ξ_2 as the mean helicity. Indeed, helicity is the projection of the angular momentum $\mathbf{j}$ on the direction $\mathbf{n}$, namely the product $\mathbf{j}\mathbf{n}$, and time reversal reverses both vectors.

In subsequent calculations we shall require the photon density matrix written in four-dimensional form, that is, as a 4-tensor $\rho_{\mu\nu}$. For a polarized photon described by the 4-vector e_μ , this tensor is naturally defined by

$$\rho_{\mu\nu} = e_\mu e_\nu^*. \quad (8.12)$$

In the three-dimensionally transverse gauge $e = (0, \mathbf{e})$, if one spatial coordinate axis is taken along $\mathbf{n}$, the nonzero components of this 4-tensor coincide with (8.7).

For an unpolarized photon, the three-dimensionally transverse gauge gives a tensor $\rho_{\mu\nu}$ whose components are

$$\rho_{ik} = \frac{1}{2}(\delta_{ik} - n_i n_k), \quad \rho_{0i} = \rho_{i0} = \rho_{00} = 0 \quad (8.13)$$

(if one axis coincides with $\mathbf{n}$, this reduces to (8.8)). Direct use of $\rho_{\mu\nu}$ in this three-dimensional form is inconvenient, however. We may instead make a gauge transformation; for the density matrix it has the form

$$\rho_{\mu\nu} \rightarrow \rho_{\mu\nu} + \chi_\mu k_\nu + \chi_\nu k_\mu, \quad (8.14)$$

where the χ_μ are arbitrary functions. Setting

$$\chi_0 = -1/(4\omega), \quad \chi_i = k_i/(4\omega^2),$$

we obtain, in place of (8.13), the simple four-dimensional expression

$$\rho_{\mu\nu} = -g_{\mu\nu}/2. \quad (8.15)$$

The four-dimensional representation of the density matrix for a partially polarized photon is readily constructed by first rewriting the two-dimensional tensor (8.9) in three-dimensional form:

$$\begin{aligned} \rho_{ik} = & (1/2)(e_i^{(1)} e_k^{(1)} + e_i^{(2)} e_k^{(2)}) + (\xi_1/2)(e_i^{(1)} e_k^{(2)} + e_i^{(2)} e_k^{(1)}) - \\ & (i\xi_2/2)(e_i^{(1)} e_k^{(2)} - e_i^{(2)} e_k^{(1)}) + (\xi_3/2)(e_i^{(1)} e_k^{(1)} - e_i^{(2)} e_k^{(2)}), \end{aligned}$$

where $\mathbf{e}^{(1)}, \mathbf{e}^{(2)}$ are unit vectors along the ξ and η axes. The required generalization is obtained by replacing these 3-vectors with real spacelike unit 4-vectors $e^{(1)}, e^{(2)}$ that are mutually orthogonal and orthogonal to the photon 4-momentum k :

$$e^{(1)2} = e^{(2)2} = -1, \quad e^{(1)} e^{(2)} = 0, \quad e^{(1)} k = e^{(2)} k = 0. \quad (8.16)$$

In the three-dimensionally transverse gauge, $e^{(1)} = (0, \mathbf{e}^{(1)})$, $e^{(2)} = (0, \mathbf{e}^{(2)})$. Thus the four-dimensional photon density matrix is

$$\begin{aligned} \rho_{\mu\nu} = & (1/2)(e_\mu^{(1)} e_\nu^{(1)} + e_\mu^{(2)} e_\nu^{(2)}) + (\xi_1/2)(e_\mu^{(1)} e_\nu^{(2)} + e_\mu^{(2)} e_\nu^{(1)}) - \\ & (i\xi_2/2)(e_\mu^{(1)} e_\nu^{(2)} - e_\mu^{(2)} e_\nu^{(1)}) + (\xi_3/2)(e_\mu^{(1)} e_\nu^{(1)} - e_\mu^{(2)} e_\nu^{(2)}). \end{aligned} \quad (8.17)$$

The most convenient specific choice of the 4-vectors $e^{(1)}$, $e^{(2)}$ depends on the conditions of the problem under consideration.

It must be borne in mind that (8.16) does not fix $e_\mu^{(1)}$ and $e_\mu^{(2)}$ uniquely. If a 4-vector e_μ satisfies these conditions, then so does every 4-vector $e_\mu + \chi k_\mu$ (because $k^2 = 0$). This nonuniqueness is connected with the gauge nonuniqueness of the density matrix.

The first term in (8.17) represents the unpolarized state. It could therefore be replaced, in accordance with (8.15), by $-g_{\mu\nu}/2$. Such a replacement is again equivalent to a gauge transformation. In operations with 4-tensors of the form (8.17), expanded in terms of two independent 4-vectors, the following formal device is useful. Writing (8.17) as

$$\rho_{\mu\nu} = \sum_{a,b=1}^3 \rho^{(ab)} e_\mu^{(a)} e_\nu^{(b)},$$

represent the coefficients $\rho^{(ab)}$ as a two-row matrix,

$$\rho = \begin{pmatrix} \rho^{(11)} & \rho^{(12)} \\ \rho^{(21)} & \rho^{(22)} \end{pmatrix}.$$

Like any Hermitian two-row matrix, it may be expanded in four independent two-row matrices: the Pauli matrices σ_x , σ_y , σ_z , and the unit matrix 1:

$$\rho = (1/2)(1 + \xi \sigma), \quad \xi = (\xi_1, \xi_2, \xi_3), \quad (8.18)$$

as follows directly by comparison with (8.17), using the familiar expressions (18.5) for the Pauli matrices. (Combining the three quantities ξ_1 , ξ_2 , ξ_3 into a “vector” ξ is, of course, purely formal and serves only to make the notation compact.)

PROBLEM

Write the photon density matrix in the representation whose coordinate “axes” are the circular unit vectors (8.2).

SOLUTION. Components of the tensor $\rho'_{\alpha\beta}$ in the new axes ($\alpha, \beta = \pm 1$) are obtained by projecting the tensor (8.9) onto the unit vectors (8.2):

$$\rho'_{11} = \rho_{\alpha\beta} e_\alpha^{(+1)*} e_\beta^{(+1)}, \quad \rho'_{-1-1} = \rho_{\alpha\beta} e_\alpha^{(-1)*} e_\beta^{(-1)},$$

$$\rho' = \frac{1}{2} \begin{pmatrix} 1 + \xi_2 & -\xi_3 + i\xi_1 \\ -\xi_3 - i\xi_1 & 1 - \xi_2 \end{pmatrix}.$$

§9. A system of two photons

Reasoning analogous to that used in § 6 also permits the number of possible states to be counted in the more complicated case of a two-photon system (*L. Landau*, 1948).

We consider the photons in their center-of-momentum frame; their momenta are $\mathbf{k}_1 = -\mathbf{k}_2 \equiv \mathbf{k}$.²¹ In the momentum representation, the wave function of the two-photon system may be written as a three-dimensional second-rank tensor $A_{ik}(\mathbf{n})$, formed bilinearly from the components of the vector wave functions of the two photons; each tensor index belongs to one photon ($\mathbf{n}$ is the unit vector along $\mathbf{k}$). The transversality of each photon is expressed by the orthogonality of A_{ik} to $\mathbf{n}$:

$$A_{il}n_l = 0, \quad A_{lk}n_l = 0. \quad (9.1)$$

Interchanging the photons means interchanging the indices of A_{ik} while simultaneously reversing $\mathbf{n}$. Since photons obey Bose statistics,

$$A_{ik}(-\mathbf{n}) = A_{ki}(\mathbf{n}). \quad (9.2)$$

In general, the tensor A_{ik} is not symmetric in its indices. Separate it into symmetric (s_{ik}) and antisymmetric (a_{ik}) parts: $A_{ik} = s_{ik} + a_{ik}$. Each part must separately satisfy (9.2), as well as the orthogonality conditions (9.1). It follows that

$$s_{ik}(-\mathbf{n}) = s_{ik}(\mathbf{n}), \quad (9.3)$$

$$a_{ik}(-\mathbf{n}) = -a_{ik}(\mathbf{n}). \quad (9.4)$$

Inversion of the coordinate system does not itself change the signs of the components of a second-rank tensor, but it does reverse $\mathbf{n}$. Equation (9.3) therefore shows that the wave function s_{ik} is invariant under inversion and thus describes even-parity states of the photon system; the wave function a_{ik} describes odd-parity states.

An antisymmetric second-rank tensor is equivalent (dual) to an axial vector $\mathbf{a}$ whose components are related to those of the tensor by $a_i = \frac{1}{2}e_{ikl}a_{kl}$, where e_{ikl} is the antisymmetric unit tensor (see II, § 6). Orthogonality of a_{kl} to $\mathbf{n}$ means that $\mathbf{a}$ and $\mathbf{n}$ are parallel.²² Hence we may write $\mathbf{a} = \varphi(\mathbf{n})$, where φ is a scalar. According to (9.4), $\mathbf{a}(-\mathbf{n}) = -\mathbf{a}(\mathbf{n})$; consequently,

$$\varphi(-\mathbf{n}) = \varphi(\mathbf{n}).$$

This equality means that the scalar φ can be constructed linearly only from spherical functions of even order L , including order zero.

Thus, with respect to rotations, the antisymmetric tensor a_{ik} transforms like a single scalar (cf. the note on p. 34). Assigning "spin" 0 to this scalar, we find that the angular momentum of the state is $J = L$. The tensor a_{ik} therefore gives odd-parity two-photon states with even J .

We now turn to the symmetric tensor s_{ik} . Since it is even under reversal of $\mathbf{n}$, it describes even-parity states of the photon system. The same fact implies that all

²¹Such a reference frame always exists except when the two photons travel parallel to one another in the same direction. For those photons, the total momentum $\mathbf{k}_1 + \mathbf{k}_2$ and total energy $\omega_1 + \omega_2$ obey the same relation as for a single photon, and hence there is no frame in which $\mathbf{k}_1 + \mathbf{k}_2 = 0$.

²²Indeed, $a_{ik} = e_{ikl}a_l$, and the orthogonality condition gives

$$a_{ik}n_k = e_{ikl}a_l n_k = (\mathbf{n} \times \mathbf{a})_i = 0.$$

components s_{ik} are expressed through spherical functions of even order L , including $L = 0$. As is well known, an arbitrary symmetric second-rank tensor s_{ik} decomposes into a scalar (s_{ii}) and a traceless symmetric tensor (s'_{ik}), with $s'_{ii} = 0$.

The scalar s_{ii} is associated with “spin” 0, so the angular momentum of its states is $J = L$ and is therefore even. The tensor s'_{ik} is associated with “spin” 2 (see III, § 57). Combining this “spin” with the even “orbital angular momentum” L by the angular-momentum addition rule, we find three states for each specified even $J \neq 0$ (with $L = J \pm 2, J$), and two states for each odd $J \neq 1$ (with $L = J \pm 1$). The exceptions are $J = 0$, with one state ($L = 2$), and $J = 1$, likewise with one state ($L = 2$).

These counts do not yet incorporate the condition that s_{ik} be orthogonal to $\mathbf{n}$. We must therefore subtract the number of states corresponding to a symmetric second-rank tensor “parallel” to $\mathbf{n}$. Such a tensor, denoted by s''_{ik} , may be written

$$s''_{ik} = n_i b_k + n_k b_i,$$

where $\mathbf{b}$ is a vector. Equation (9.3) requires this vector to obey $\mathbf{b}(-\mathbf{n}) = -\mathbf{b}(\mathbf{n})$. Thus the tensor s''_{ik} responsible for the “extra” states is equivalent to an odd vector. That vector must consequently be expressed only through spherical functions of odd orders L . Noting also that a vector is associated with “spin” 1, we find two states for every even $J \neq 0$ (with $L = J \pm 1$), and one state for every odd J (with $L = J$); the special case $J = 0$ has one state ($L = 1$). Combining these results gives the following table, which lists the numbers of possible even- and odd-parity states of a two-photon system (with zero total momentum) for the different values of the total angular momentum J :

J	Even	Odd
0	1	1
1	—	—
$2k$	2	1
$2k + 1$	1	—

(9.5)

(k is a nonzero positive integer). We see that odd-parity states are absent for odd J , while $J = 1$ is altogether impossible.²³

The two-photon wave function A_{ik} determines the correlation between their polarizations. The probability that the two photons simultaneously have specified polarizations $\mathbf{e}_1$ and $\mathbf{e}_2$ is proportional to

$$A_{ik} e_{1i}^* e_{2k}^*.$$

In other words, when the polarization $\mathbf{e}_1$ of one photon is specified, the polarization $\mathbf{e}_2$ of the other is proportional to

$$e_{2k} \propto A_{ik} e_{1i}^*. \quad (9.6)$$

In odd-parity states of the system, A_{ik} coincides with the antisymmetric tensor a_{ik} . Then

$$\mathbf{e}_2 \mathbf{e}_1^* \propto a_{ik} e_{1i}^* e_{1k}^* = 0,$$

²³For another derivation of these results, see problem 1 to § 69.

so the polarizations of the two photons are mutually orthogonal. For linear polarization this means that their directions are perpendicular; for circular polarizations, their senses of rotation are opposite.

The even-parity state with $J = 0$ is described by a symmetric tensor reducible to the scalar form

$$s_{ik} = \text{const} \cdot (\delta_{ik} - n_i n_k).$$

It follows from (9.6) that $\mathbf{e}_1 = \mathbf{e}_2^*$. For linear polarization their directions are therefore parallel; for circular polarizations their senses of rotation are again opposite. The latter fact is evident: when $J = 0$, the sum of the projections of the photon angular momenta on the same direction $\mathbf{k}$ must in any event vanish (their projections on the opposite directions $\mathbf{k}_1$ and $\mathbf{k}_2$, that is, their helicities, are then equal).

CHAPTER II

Bosons

§ 10. Wave equation for spin-zero particles

Chapter I showed how a quantum description of the free electromagnetic field can be constructed by starting from the known properties of the field in the classical limit and using the concepts of ordinary quantum mechanics. The resulting picture of the field as a system of photons contains many features that also carry over to the relativistic quantum description of particles.

The electromagnetic field is a system with infinitely many degrees of freedom. It has no conservation law for the number of particles (photons), and its possible states include states with any number of particles.¹ In general, however, systems of any particles must possess the same property in a relativistic theory. Conservation of particle number in nonrelativistic theory is connected with conservation of mass: the sum of the masses (rest masses) of the particles does not change when they interact; for example, conservation of the sum of the masses in a system of electrons also means that their number remains fixed. Relativistic mechanics, by contrast, has no law of mass conservation; only the total energy of the system, including the particles' rest energy, must be conserved. The particle number therefore need not be conserved, and every relativistic particle theory consequently has to be a theory of systems with infinitely many degrees of freedom. In other words, such a particle theory assumes the character of a field theory.

The apparatus of second quantization is the appropriate mathematical means of describing systems with variable particle number (see III, §§ 64, 65). In the quantum description of the electromagnetic field, the 4-potential $\hat{A}$ acts as the second-quantized operator. It is expressed through the (coordinate) wave functions of individual particles (photons) and their creation and annihilation operators. A quantized wave-function operator plays an analogous role in describing a system of particles. Its construction first requires the form of the wave function of one free particle and the equation obeyed by that function.

The auxiliary nature of the concept of a free-particle field should be emphasized. Real particles interact, and the purpose of the theory is to study these interactions. Every interaction, however, reduces to a collision, before and after which the system may be regarded as an assemblage of free particles. Section 1 noted that these are the only measurable objects. We therefore use free-particle fields as a means of describing initial and final states.

We begin the relativistic description of free particles with particles of spin 0. The mathematical simplicity of this case makes the central ideas and characteristic features of the description especially transparent.

The state of a free particle without spin can be specified completely by its momentum

¹In reality, of course, the photon number changes only through various interaction processes.

$\mathbf{p}$ alone. The particle energy ε^2 then obeys $\varepsilon^2 = \mathbf{p}^2 + m^2$, where m is the particle mass, or, in four-dimensional form,

$$p^2 = m^2. \quad (10.1)$$

As is known, the conservation laws for momentum and energy are connected with the homogeneity of space and time, that is, with symmetry under an arbitrary parallel displacement of the 4-coordinate system. In quantum theory, this symmetry requires the wave function of a particle with definite 4-momentum to acquire only a phase factor of unit modulus under such a transformation of the 4-coordinates. Only an exponential function with an exponent linear in the 4-coordinates satisfies this requirement. Thus the wave function of a free-particle state with specified 4-momentum $p^\mu = (\varepsilon, \mathbf{p})$ must be a plane wave:

$$\text{const} \cdot e^{-i p x}, \quad p x = \varepsilon t - \mathbf{p} \mathbf{r} \quad (10.2)$$

(the choice of sign in the exponent is itself conventional in relativistic theory; it has been made to agree with the nonrelativistic case).

The wave equation must admit (10.2) as particular solutions for every 4-vector p satisfying (10.1). It must be linear, as required by the superposition principle: any linear combination of functions (10.2) also describes a possible particle state and must therefore also be a solution. Finally, its order should be as low as possible; an equation of higher order would introduce extraneous solutions.

Spin is the angular momentum of a particle in the reference frame in which it is at rest. If the particle spin is s , its wave function in the rest frame is a three-dimensional spinor of rank $2s$. To describe the particle in an arbitrary reference frame, however, its wave function must be expressed in terms of four-dimensional quantities.

In the rest frame, a spin-zero particle is described by a three-dimensional scalar. Such a scalar can nevertheless have different four-dimensional "origins": it may be a four-dimensional scalar ψ , or it may be the fourth component of a timelike 4-vector ψ_μ whose only nonzero component in the rest frame is ψ_0 .³

For a free particle, the only operator that can enter the wave equation is the 4-momentum operator $\hat{p}$. Its components are the differentiation operators with respect to the coordinates and time:

$$\hat{p}^\mu = i \partial^\mu = \left(i \frac{\partial}{\partial t}, -i \nabla \right). \quad (10.3)$$

The wave equation must be a differential relation between ψ and ψ_μ , effected by the operator $\hat{p}$. This relation must, of course, be expressed in relativistically invariant form. The required relations are

$$m \psi_\mu = \hat{p}_\mu \psi, \quad \hat{p}^\mu \psi_\mu = m \psi, \quad (10.4)$$

where m is a dimensional constant characterizing the particle.⁴

²We denote the energy of an individual particle by ε , to distinguish it from the energy E of a system of particles.

³Or, similarly, it may be the time component of a higher-rank 4-tensor; that case, however, would lead to equations of higher order.

⁴The constants m have been inserted in (10.4) so that ψ_μ and ψ have the same dimension. Introducing different constants m_1 and m_2 into the two equations would serve no purpose, since they could always be made equal by redefining either ψ or ψ_μ .

Substitution of ψ_μ from the first equation into the second gives

$$(\hat{p}^2 - m^2)\psi = 0 \quad (10.5)$$

(*O. Klein, V. A. Fock, 1926; W. Gordon, 1927*). Written out explicitly, this equation is

$$-\partial_\mu \partial^\mu \psi = \left(-\frac{\partial^2}{\partial t^2} + \Delta \right) \psi = m^2 \psi. \quad (10.6)$$

Putting the plane wave (10.2) into this equation yields $p^2 = m^2$, which shows that m is the particle mass. Notice that the form of (10.5) is already evident from the fact that $\hat{p}^2$ is the only scalar operator constructible from $\hat{p}$ (for the same reason, every component of the wave function of a particle of any spin satisfies the same equation, as we shall repeatedly see below).

Thus, in essence, a spin-zero particle is described by just one four-dimensional scalar ψ , satisfying the second-order equation (10.5). In the first-order equations (10.4), the wave function is the collection of quantities ψ and ψ_μ , with the 4-vector ψ_μ reducing to the 4-gradient of the scalar ψ . In the rest frame, the particle wave function is independent of the spatial coordinates, and the spatial components of the 4-vector ψ_μ therefore vanish, as they should.

For second quantization it is useful to express the energy and momentum of the particle as spatial integrals of combinations bilinear in ψ and ψ^* , which act as spatial densities of these quantities. In other words, we must find the energy-momentum tensor $T_{\mu\nu}$ corresponding to (10.5). In terms of this tensor, conservation of energy and momentum is written

$$\partial_\mu T_\nu^\mu = 0. \quad (10.7)$$

Following the general rules of field theory (see II, § 32), we formulate a variational principle from which (10.5) follows. Such a principle must require the “action integral”

$$S = \int L d^4x \quad (10.8)$$

of some real 4-scalar L , the density of the field’s Lagrange function, to be minimal.⁵ From the scalar ψ and the operator ∂^μ , one can form the real bilinear scalar expression

$$L = \partial_\mu \psi^* \cdot \partial^\mu \psi - m^2 \psi^* \psi, \quad (10.9)$$

where m is a dimensional constant. Treating ψ and ψ^* as independent variables describing the field (the field’s “generalized coordinates” q), one readily sees that the Lagrange equations

$$\frac{\partial}{\partial x^\mu} \frac{\partial L}{\partial q_{,\mu}} = \frac{\partial L}{\partial q} \quad (10.10)$$

($q_{,\mu} \equiv \partial_\mu q$) coincide with (10.5) for ψ and ψ^* , with m being the particle mass. Notice also that (10.9) has been assigned an overall sign such that the square of the time

⁵The corresponding second-quantized operator $\hat{L}$ is called the *field Lagrangian*. To simplify the terminology, we shall use this term for both the “quantized” and the “unquantized” density of the Lagrange function.

derivative, $|\partial\psi/\partial t|^2$, enters L with a plus sign; otherwise the action could not have a minimum (cf. II, § 27). The choice of an overall numerical coefficient in L is conventional and affects only the normalization factor in ψ .

The energy–momentum tensor is now calculated from

$$T_\mu^\nu = \sum_q q_{,\mu} \frac{\partial L}{\partial q_{,\nu}} - L \delta_\mu^\nu \quad (10.11)$$

(summation over all q). Substitution of (10.9) gives

$$T_{\mu\nu} = \partial_\mu \psi^* \cdot \partial_\nu \psi + \partial_\nu \psi^* \cdot \partial_\mu \psi - L g_{\mu\nu} \quad (10.12)$$

(these quantities are real, as required, because L is real). In particular,

$$T_{00} = 2 \frac{\partial \psi^*}{\partial t} \frac{\partial \psi}{\partial t} - L = \frac{\partial \psi^*}{\partial t} \frac{\partial \psi}{\partial t} + \nabla \psi^* \cdot \nabla \psi + m^2 \psi^* \psi, \quad (10.13)$$

$$T_{i0} = \frac{\partial \psi^*}{\partial t} \frac{\partial \psi}{\partial x^i} + \frac{\partial \psi^*}{\partial x^i} \frac{\partial \psi}{\partial t}. \quad (10.14)$$

The field 4-momentum is given by the integral

$$P_\mu = \int T_{\mu 0} d^3x, \quad (10.15)$$

so T_{00} and T_{i0} serve as the energy and momentum densities. Notice that T_{00} is positive definite.

Formula (10.13) can be used to normalize the wave function. A plane wave normalized to one particle in a volume $V = 1$ is

$$\psi_p = \frac{1}{\sqrt{2\varepsilon}} e^{-ipx}. \quad (10.16)$$

Indeed, this function gives $T_{00} = \varepsilon$, so that the total energy in the volume $V = 1$ equals the energy of one particle.

The angular momentum, whose conservation is associated with the isotropy of space, can likewise be expressed as a spatial integral; that representation will not be needed below.

Finally, besides the conservation laws directly associated with space-time symmetry, equations (10.4) admit one further conservation law. It is readily verified from (10.4), together with the corresponding equations for ψ^* , that

$$\partial_\mu j^\mu = 0, \quad (10.17)$$

where

$$j_\mu = m(\psi^* \psi_\mu + \psi_\mu^* \psi) = i[\psi^* \partial_\mu \psi - (\partial_\mu \psi^*) \psi]. \quad (10.18)$$

It follows that j^μ plays the role of a current-density 4-vector. Equation (10.17) is the continuity equation expressing conservation of the quantity

$$Q = \int j_0 d^3x, \quad (10.19)$$

where

$$j_0 = j^0 = i \left(\psi^* \frac{\partial \psi}{\partial t} - \frac{\partial \psi^*}{\partial t} \psi \right). \quad (10.20)$$

Note that j_0 is not positive definite. This fact alone shows that, in general, it cannot be interpreted as a probability density for the particle's spatial localization. The meaning of the conservation law expressed by (10.17) will become clear in the next section.

§ 11. Particles and antiparticles

Following the general procedure of second quantization, we must expand an arbitrary wave function in eigenfunctions of a complete set of possible free-particle states, for example in the plane waves ψ_p :

$$\psi = \sum_{\mathbf{p}} a_{\mathbf{p}} \psi_p, \quad \psi^* = \sum_{\mathbf{p}} a_{\mathbf{p}}^* \psi_p^*.$$

The coefficients $a_{\mathbf{p}}$, $a_{\mathbf{p}}^*$ would then have to be understood as the operators $\hat{a}_{\mathbf{p}}$, $\hat{a}_{\mathbf{p}}^+$ that annihilate and create particles in the corresponding states.⁶

At this point, however, we immediately encounter a new circumstance of principle, absent in nonrelativistic theory. For a plane wave satisfying (10.5), the energy ε at a given momentum $\mathbf{p}$ is constrained only by $\varepsilon^2 = \mathbf{p}^2 + m^2$ and can therefore take the two values $\pm\sqrt{\mathbf{p}^2 + m^2}$. Yet only positive values of ε can have the physical meaning of the energy of a free particle. The negative values cannot simply be discarded: the general solution of the wave equation is a superposition of all its independent particular solutions. This circumstance shows that some modification is needed in the interpretation of the expansion coefficients of ψ and ψ^* upon second quantization.

Write the expansion as

$$\psi = \sum_{\mathbf{p}} \frac{1}{\sqrt{2\varepsilon}} a_{\mathbf{p}}^{(+)} e^{i(\mathbf{p}\mathbf{r} - \varepsilon t)} + \sum_{\mathbf{p}} \frac{1}{\sqrt{2\varepsilon}} a_{\mathbf{p}}^{(-)} e^{i(\mathbf{p}\mathbf{r} + \varepsilon t)}, \quad (11.1)$$

where the first sum contains plane waves of positive frequency, normalized according to (10.16), and the second contains waves of negative “frequency”; throughout, ε denotes the positive quantity $\varepsilon = +\sqrt{\mathbf{p}^2 + m^2}$. In second quantization, the coefficients $a_{\mathbf{p}}^{(+)}$ in the first sum are replaced in the usual manner by the particle annihilation operators $\hat{a}_{\mathbf{p}}$. In the second sum, however, we note that when matrix elements are subsequently formed, the time dependence of its terms corresponds not to annihilation but to creation of particles; the factor $e^{i\varepsilon t} = (e^{-i\varepsilon t})^*$ represents one additional particle of energy ε in the final state (cf. the end of § 2). Accordingly, we replace $a_{\mathbf{p}}^{(-)}$ by operators $\hat{b}_{-\mathbf{p}}^+$ that create particles of another kind. Also changing the summation variable in the second sum of (11.1) from $\mathbf{p}$ to $-\mathbf{p}$, so that its exponential factor becomes $e^{-i(\mathbf{p}\mathbf{r} - \varepsilon t)}$, we obtain the ψ -operators in the form

$$\hat{\psi} = \sum_{\mathbf{p}} \frac{1}{\sqrt{2\varepsilon}} \left(\hat{a}_{\mathbf{p}} e^{-i\mathbf{p}\mathbf{x}} + \hat{b}_{\mathbf{p}}^+ e^{i\mathbf{p}\mathbf{x}} \right), \quad \hat{\psi}^+ = \sum_{\mathbf{p}} \frac{1}{\sqrt{2\varepsilon}} \left(\hat{a}_{\mathbf{p}}^+ e^{i\mathbf{p}\mathbf{x}} + \hat{b}_{\mathbf{p}} e^{-i\mathbf{p}\mathbf{x}} \right). \quad (11.2)$$

⁶We attach the 4-momentum index p to the ψ -functions; below, functions of “negative frequency” will be denoted by ψ_{-p} . The operators $\hat{a}$, $\hat{a}^+$ instead carry the three-dimensional momentum index $\mathbf{p}$, which fully specifies the state of a real particle.

Thus all operators $\hat{a}_{\mathbf{p}}, \hat{b}_{\mathbf{p}}$ multiply functions having the “correct” time dependence, proportional to $e^{-i\varepsilon t}$, while $\hat{a}_{\mathbf{p}}^+, \hat{b}_{\mathbf{p}}^+$ multiply their complex conjugates. In accordance with the general rules, this makes it possible to interpret $\hat{a}_{\mathbf{p}}, \hat{b}_{\mathbf{p}}$ as annihilation operators and $\hat{a}_{\mathbf{p}}^+, \hat{b}_{\mathbf{p}}^+$ as creation operators for particles with momenta $\mathbf{p}$ and energies ε .

We are led to two kinds of particles that occur together and on an equal footing. They are called *particles* and *antiparticles* (the reason for these names will become clear below). In the second-quantized formalism, one kind is represented by $\hat{a}_{\mathbf{p}}, \hat{a}_{\mathbf{p}}^+$ and the other by $\hat{b}_{\mathbf{p}}, \hat{b}_{\mathbf{p}}^+$. Since the operators for both kinds occur in the same ψ -operator, the two kinds have equal masses.

The same conclusions can also be reached directly from the requirements of relativistic invariance.

In mathematical terms, Lorentz transformations are rotations of the four-dimensional coordinate system that change the direction of the time axis. Together with purely spatial rotations, which leave the time axis unchanged, they form a transformation group called the *Lorentz group*.⁷ All these transformations share the property that they do not carry the t axis outside the corresponding sheet of the light cone; this expresses the physical principle that signal propagation has a limiting speed.

From a purely mathematical standpoint, simultaneous reversal of all four coordinates, a *four-dimensional inversion*, is also a rotation: the determinant of this transformation is +1, just as for every other rotational transformation. It carries the time axis from one sheet of the light cone to the other. Although this makes the transformation physically unrealizable as a change of reference frame, mathematically the distinction amounts only to the fact that, owing to the pseudo-Euclidean metric, such a rotation cannot be performed without allowing the coordinates to become complex along the way.

It is natural to suppose that this distinction is immaterial when four-dimensional invariance is at issue. Every expression invariant under Lorentz transformations should then also be invariant under 4-inversion. The precise statement of this requirement for a scalar ψ -operator will be given in § 13. Here we merely note that it necessarily requires terms with both signs before ε in the exponents to occur simultaneously in the ψ -operators, since $t \rightarrow -t$ changes precisely this sign.

Return to (11.2) and establish the permutation relations among $\hat{a}_{\mathbf{p}}, \hat{a}_{\mathbf{p}}^+$, and among $\hat{b}_{\mathbf{p}}, \hat{b}_{\mathbf{p}}^+$. For photons this was done for $\hat{c}_{\mathbf{p}}, \hat{c}_{\mathbf{p}}^+$ by analogy with oscillators, and thus ultimately from the properties of the electromagnetic field in the classical limit. No such analogy is now available. To determine whether the operators obey Bose or Fermi commutation rules, we can be guided only by the form of the Hamiltonian constructed from them.

This Hamiltonian is obtained (see III, § 64) by substituting $\hat{\psi}$ and $\hat{\psi}^+$ for ψ and ψ^* in

⁷The set of all three-dimensional spatial rotations is itself a group and is a subgroup of the Lorentz group. The set of Lorentz transformations by itself, however, is not a group: the result of successive Lorentz transformations may reduce to a purely spatial rotation.

the integral $\int T_{00} d^3x$.⁸ We thus find

$$\hat{H} = \sum_{\mathbf{p}} \varepsilon \left(\hat{a}_{\mathbf{p}}^+ \hat{a}_{\mathbf{p}} + \hat{b}_{\mathbf{p}} \hat{b}_{\mathbf{p}}^+ \right). \quad (11.3)$$

A sensible result for the eigenvalues of this Hamiltonian is obtained only if the operators obey the Bose commutation rules:

$$\{\hat{a}_{\mathbf{p}}, \hat{a}_{\mathbf{p}}^+\}_- = \{\hat{b}_{\mathbf{p}}, \hat{b}_{\mathbf{p}}^+\}_- = 1 \quad (11.4)$$

(all other operator pairs commute; in particular, every particle operator $\hat{a}_{\mathbf{p}}, \hat{a}_{\mathbf{p}}^+$ commutes with every antiparticle operator $\hat{b}_{\mathbf{p}}, \hat{b}_{\mathbf{p}}^+$). Indeed, in that case

$$\hat{H} = \sum_{\mathbf{p}} \varepsilon \left(\hat{a}_{\mathbf{p}}^+ \hat{a}_{\mathbf{p}} + \hat{b}_{\mathbf{p}} \hat{b}_{\mathbf{p}}^+ + 1 \right).$$

The eigenvalues of $\hat{a}_{\mathbf{p}}^+ \hat{a}_{\mathbf{p}}$ and $\hat{b}_{\mathbf{p}} \hat{b}_{\mathbf{p}}^+$ are the positive integers $N_{\mathbf{p}}$ and $\bar{N}_{\mathbf{p}}$, the numbers of particles and antiparticles. The infinite additive constant $\sum \varepsilon$, the “vacuum energy,” may again simply be omitted:

$$E = \sum_{\mathbf{p}} \varepsilon \left(N_{\mathbf{p}} + \bar{N}_{\mathbf{p}} \right) \quad (11.5)$$

(cf. (3.1) and the note thereto). This expression is positive definite and agrees with the conception of two kinds of particles that actually exist. Similarly, the total momentum of the particle system is

$$\mathbf{P} = \sum_{\mathbf{p}} \mathbf{p} \left(N_{\mathbf{p}} + \bar{N}_{\mathbf{p}} \right). \quad (11.6)$$

Had we adopted Fermi permutation relations in place of (11.4), with anticommutators rather than commutators, we would have obtained

$$\hat{H} = \sum_{\mathbf{p}} \varepsilon \left(\hat{a}_{\mathbf{p}}^+ \hat{a}_{\mathbf{p}} - \hat{b}_{\mathbf{p}} \hat{b}_{\mathbf{p}}^+ + 1 \right)$$

and, instead of (11.5), the physically meaningless expression $\sum \varepsilon (N_{\mathbf{p}} - \bar{N}_{\mathbf{p}})$. This expression is not positive definite and therefore cannot represent the energy of a system of free particles.

Thus spin-zero particles are bosons.

Next consider the integral Q in (10.19). Replacing ψ and ψ^* in j^0 by the operators $\hat{\psi}$ and $\hat{\psi}^+$ and carrying out the integration gives

$$\hat{Q} = \sum_{\mathbf{p}} \left(\hat{a}_{\mathbf{p}}^+ \hat{a}_{\mathbf{p}} - \hat{b}_{\mathbf{p}} \hat{b}_{\mathbf{p}}^+ \right) = \sum_{\mathbf{p}} \left(\hat{a}_{\mathbf{p}}^+ \hat{a}_{\mathbf{p}} - \hat{b}_{\mathbf{p}} \hat{b}_{\mathbf{p}}^+ - 1 \right). \quad (11.7)$$

⁸In nonrelativistic theory the conjugate operator $\hat{\psi}^+$ is placed to the left of $\hat{\psi}$. Here the order is immaterial, because interchanging $\hat{\psi}^+$ and $\hat{\psi}$ would merely interchange the equivalent operators $\hat{a}_{\mathbf{p}}$ and $\hat{b}_{\mathbf{p}}$. Once one order or the other has been selected, however, the same convention must always be followed.

The eigenvalues of this operator, after subtraction of the inessential additive constant $\sum 1$, are

$$Q = \sum_{\mathbf{p}} (N_{\mathbf{p}} - \bar{N}_{\mathbf{p}}), \quad (11.8)$$

that is, the differences between the total numbers of particles and antiparticles.

As long as free particles are being considered without any interaction between them, the meaning of conservation of Q , like that of conservation of the total energy and momentum (11.5, 11.6), remains largely conditional: in fact, not only this sum but every $N_{\mathbf{p}}$ and $\bar{N}_{\mathbf{p}}$ is separately conserved. Whether Q remains conserved when the particles interact depends on the nature of the interaction. If Q is conserved, meaning that $\hat{Q}$ commutes with the interaction Hamiltonian, then (11.8) shows how this law restricts possible changes in particle number: only pairs “particle + antiparticle” can appear or disappear.

If a particle is electrically charged, its antiparticle must have charge of the opposite sign. If the two had charges of the same sign, creation or annihilation of a pair would contradict the strict law of nature requiring conservation of total electric charge. We shall see below, in § 32, how this opposition of charges arises automatically in the theory when particles interact with the electromagnetic field.

The quantity Q is sometimes called the *charge* of the field of the particles in question. For electrically charged particles, Q specifies, in particular, the total electric charge of the system in units of the elementary charge e . It should be emphasized, however, that particles and antiparticles may be electrically neutral.

We have thus seen how the relativistic relation between energy and momentum, specifically the two-valued root of $\varepsilon^2 = \mathbf{p}^2 + m^2$, together with the requirements of relativistic invariance, introduces a new classification principle for particles into quantum theory: the possible existence of pairs of distinct particles, a particle and its antiparticle, related to one another in the manner described above. This remarkable prediction was first made by Dirac in 1930 for particles of spin 1/2, before the first antiparticle, the positron, was actually discovered.⁹

§ 12. Genuinely neutral particles

In carrying out the second quantization of the wave function (11.1), the coefficients $a_{\mathbf{p}}^{(+)}$ and $a_{\mathbf{p}}^{(-)}$ were treated as operators belonging to different particles. This need not be so, however: as a special case, the annihilation and creation operators entering $\hat{\psi}$ may refer to the same particles (as they did for photons; cf. (2.17)). Denoting these operators in this case by $\hat{c}_{\mathbf{p}}$ and $\hat{c}_{\mathbf{p}}^+$, we write the ψ -operator as

$$\hat{\psi} = \sum_{\mathbf{p}} \frac{1}{\sqrt{2\varepsilon}} \left(\hat{c}_{\mathbf{p}} e^{-i p x} + \hat{c}_{\mathbf{p}}^+ e^{i p x} \right). \quad (12.1)$$

The field described by this operator corresponds to a system of identical particles that may be said to “coincide with their antiparticles.”

⁹Weisskopf and Pauli extended the concept of antiparticles to bosons (V. Weisskopf, W. Pauli, 1934).

The operator (12.1) is Hermitian ($\widehat{\psi}^+ = \widehat{\psi}$). In this sense such a field has half as many “degrees of freedom” as a complex field, whose operators $\widehat{\psi}$ and $\widehat{\psi}^+$ do not coincide.

For this reason, the field Lagrangian expressed through the Hermitian operator $\widehat{\psi}$ must contain an extra factor 1/2 relative to (10.9)¹⁰

$$\widehat{L} = (1/2) \left(\partial_\mu \widehat{\psi} \cdot \partial^\mu \widehat{\psi} - m^2 \widehat{\psi}^2 \right). \quad (12.2)$$

The corresponding energy–momentum tensor is

$$\widehat{T}_{\mu\nu} = \partial_\mu \widehat{\psi} \cdot \partial_\nu \widehat{\psi} - \widehat{L} g_{\mu\nu}, \quad (12.3)$$

and hence the energy-density operator is

$$\widehat{T}_{00} = \left(\frac{\partial \widehat{\psi}}{\partial t} \right)^2 - \widehat{L} = \frac{1}{2} \left[\left(\frac{\partial \widehat{\psi}}{\partial t} \right)^2 + (\nabla \widehat{\psi})^2 + m^2 \widehat{\psi}^2 \right]. \quad (12.4)$$

Substitution of (12.1) into the integral $\int \widehat{T}_{00} d^3x$ gives the field Hamiltonian

$$\widehat{H} = \frac{1}{2} \sum_{\mathbf{p}} \varepsilon \left(\widehat{c}_{\mathbf{p}}^+ \widehat{c}_{\mathbf{p}} + \widehat{c}_{\mathbf{p}} \widehat{c}_{\mathbf{p}}^+ \right). \quad (12.5)$$

This again shows why Bose quantization is required:

$$\{\widehat{c}_{\mathbf{p}}, \widehat{c}_{\mathbf{p}}^+\}_- = 1, \quad (12.6)$$

and the energy eigenvalues (again with the additive constant removed) are

$$E = \sum_{\mathbf{p}} \varepsilon_{\mathbf{p}} N_{\mathbf{p}}. \quad (12.7)$$

Fermi quantization, by contrast, would give the meaningless result of an E independent of $N_{\mathbf{p}}$.

The “charge” Q of the field under consideration is zero. This follows already from the fact that Q must reverse sign when particles are replaced by antiparticles, whereas here the two coincide. Consequently, no current-density 4-vector exists either. Indeed, the expression

$$\widehat{j}_\mu = i \left[\widehat{\psi}^+ \partial_\mu \widehat{\psi} - (\partial_\mu \widehat{\psi}^+) \widehat{\psi} \right] \quad (12.8)$$

for the operator of the conserved 4-vector $\widehat{j}$ vanishes when $\widehat{\psi} = \widehat{\psi}^+$ (the vector $\widehat{\psi} \partial_\mu \widehat{\psi}$, taken by itself, is not conserved). This in turn means that there is no special conservation law restricting possible changes in particle number. Such particles are plainly electrically neutral in any event.

Particles of this kind are called *genuinely neutral*, in distinction to electrically neutral particles that possess antiparticles. Whereas the latter can annihilate (turning into photons) only in pairs, genuinely neutral particles can annihilate one at a time.

¹⁰This is analogous to the extra factor 1/2 in the electromagnetic-field energy-density operator (2.10), expressed through the Hermitian operators $\widehat{\mathbf{E}}$ and $\widehat{\mathbf{H}}$, as compared with the photon energy density (3.2), expressed through its complex wave function; cf. the note on p. 26.

The structure of the ψ -operator (12.1) is the same as that of the electromagnetic-field operators (2.17)–(2.20). In this sense the photons themselves may also be called genuinely neutral particles. For the electromagnetic field, the Hermiticity of the operators was tied to the reality of the field strengths as physical quantities measurable in the classical limit. No such connection exists for particle ψ -operators, since no directly measurable quantities correspond to them at all.

The absence of a conserved current 4-vector is a general property of genuinely neutral particles and is unrelated to zero spin (photons, for example, also have this property). Physically, it expresses the absence of corresponding restrictions on changes in particle number. Formally, however, there is a direct connection between the absence of a conserved current and the reality of the field—that is, the Hermiticity of the operator $\hat{\psi}$.

The Lagrangian of a complex field

$$\widehat{L} = \partial_\mu \hat{\psi}^+ \cdot \partial^\mu \hat{\psi} - m^2 \hat{\psi}^+ \hat{\psi} \quad (12.9)$$

is invariant when the ψ -operator is multiplied by an arbitrary phase factor, that is, under the transformations

$$\hat{\psi} \rightarrow e^{i\alpha} \hat{\psi}, \quad \hat{\psi}^+ \rightarrow e^{-i\alpha} \hat{\psi}^+ \quad (12.10)$$

(called *gauge transformations*). In particular, the Lagrangian is unchanged under the infinitesimal gauge transformation

$$\hat{\psi} \rightarrow \hat{\psi} + i\delta\alpha \cdot \hat{\psi}, \quad \hat{\psi}^+ \rightarrow \hat{\psi}^+ - i\delta\alpha \cdot \hat{\psi}^+. \quad (12.11)$$

Under an infinitesimal change of the “generalized coordinates” q , the Lagrangian changes by

$$\begin{aligned} \delta\widehat{L} &= \sum \left(\frac{\partial\widehat{L}}{\partial q} \delta q + \frac{\partial\widehat{L}}{\partial q_{,\mu}} \right) = \\ &= \sum \left(\frac{\partial\widehat{L}}{\partial q} - \frac{\partial}{\partial x^\mu} \frac{\partial\widehat{L}}{\partial q_{,\mu}} \right) \delta q + \sum \frac{\partial}{\partial x^\mu} \left(\frac{\partial\widehat{L}}{\partial q_{,\mu}} \delta q \right) \end{aligned}$$

(the sum is over all q). The first term vanishes by virtue of the “equations of motion” (the Lagrange equations). Taking the “coordinates” q to be the operators $\hat{\psi}$ and $\hat{\psi}^+$, and setting

$$\delta\hat{\psi} = i\delta\alpha \cdot \hat{\psi}, \quad \delta\hat{\psi}^+ = -i\delta\alpha \cdot \hat{\psi}^+,$$

we obtain

$$\delta\widehat{L} = i\delta\alpha \frac{\partial}{\partial x^\mu} \left(\hat{\psi} \frac{\partial\widehat{L}}{\partial\hat{\psi}_{,\mu}} - \hat{\psi}^+ \frac{\partial\widehat{L}}{\partial\hat{\psi}^+_{,\mu}} \right).$$

It follows that invariance of the Lagrangian ($\delta\widehat{L} = 0$) is equivalent to the continuity equation ($\partial_\mu \hat{j}^\mu = 0$) for the 4-vector

$$\hat{j}^\mu = i \left(\hat{\psi}^+ \frac{\partial\widehat{L}}{\partial\hat{\psi}^+_{,\mu}} - \hat{\psi} \frac{\partial\widehat{L}}{\partial\hat{\psi}_{,\mu}} \right). \quad (12.12)$$

It is readily checked that for the Lagrangian (12.9), this formula gives the current (12.8).

Thus, in the theory's mathematical formalism, the existence of a conserved current is linked to invariance of the Lagrangian under gauge transformations (W. Pauli, 1941). The Lagrangian (12.2) of a genuinely neutral field has no such symmetry.

§ 13. The transformations C , P , and T

Unlike 4-inversion, three-dimensional (spatial) inversion cannot be reduced to rotations of the 4-dimensional coordinate frame: the determinant of this transformation is -1 , not $+1$. Consequently, relativistic invariance by itself does not determine the symmetry properties of particles under inversion (the P transformation)¹¹.

For a scalar wave function, inversion acts according to

$$\widehat{P}\psi(t, \mathbf{r}) = \pm\psi(t, -\mathbf{r}), \quad (13.1)$$

where the plus and minus signs on the right correspond, respectively, to a true scalar and a pseudoscalar.

Thus two aspects of the wave function's behavior under inversion must be distinguished. The first concerns its coordinate dependence. This was the only aspect considered in nonrelativistic quantum mechanics; it gives the parity of the state, which we shall now call its *orbital parity*, and characterizes the symmetry of the particle's motion. A state has a definite orbital parity, $+1$ or -1 , if

$$\psi(t, -\mathbf{r}) = \pm\psi(t, \mathbf{r}).$$

The second aspect is the wave function's behavior at a given point when the coordinate axes are inverted; this point may conveniently be regarded as the origin. It gives rise to the *intrinsic parity of the particle*. For a spin-zero particle, the two signs in (13.1) correspond to intrinsic parities $+1$ and -1 . The total parity of a system of particles is the product of their intrinsic parities and the orbital parity of their relative motion.

The "internal" symmetry properties of different particles can of course manifest themselves only in processes that transform one particle into another. In nonrelativistic quantum mechanics, the analogue of intrinsic parity is the parity of a composite system's bound state, such as a nucleus. Relativistic theory draws no fundamental distinction between composite and elementary particles, so this intrinsic parity is no different from that of particles treated as elementary in nonrelativistic theory. In the nonrelativistic domain these latter particles behave as immutable objects; their internal symmetry properties are then unobservable, and considering them would have no physical content.

In second quantization, intrinsic parity is expressed through the behavior of the ψ operators under inversion. Scalar and pseudoscalar fields obey

$$P : \widehat{\psi}(t, \mathbf{r}) \rightarrow \pm\widehat{\psi}(t, -\mathbf{r}). \quad (13.2)$$

¹¹The Lorentz group supplemented by spatial inversion is called the *extended Lorentz group*. The original group, which does not contain P , is correspondingly called the *proper Lorentz group*. The extended group comprises all transformations that leave the t axis within the corresponding halves of the light cone.

The action of inversion on a ψ operator must itself be formulated as a definite transformation of particle annihilation and creation operators, chosen so that it produces (13.2). This transformation is readily seen to be

$$P : \hat{a}_{\mathbf{p}} \rightarrow \pm \hat{a}_{-\mathbf{p}}, \quad \hat{b}_{\mathbf{p}} \rightarrow \pm \hat{b}_{-\mathbf{p}} \quad (13.3)$$

(with the same rule for the adjoint operators). Indeed, make this replacement in

$$\hat{\psi}(t, \mathbf{r}) = \sum_{\mathbf{p}} \frac{1}{\sqrt{2\varepsilon}} \left(\hat{a}_{\mathbf{p}} e^{-i\omega t + i\mathbf{p}\mathbf{r}} + \hat{b}_{\mathbf{p}}^{\dagger} e^{i\omega t - i\mathbf{p}\mathbf{r}} \right) \quad (13.4)$$

and then relabel the summation variable, $\mathbf{p} \rightarrow -\mathbf{p}$; the result is $\pm \hat{\psi}(t, -\mathbf{r})$. Hence, if $\hat{\psi}^P(t, \mathbf{r})$ denotes the operator obtained by applying (13.3), then

$$\hat{\psi}^P(t, \mathbf{r}) = \pm \hat{\psi}(t, -\mathbf{r}). \quad (13.5)$$

The form of (13.3) is entirely natural: inversion reverses the polar vector $\mathbf{p}$, replacing particles of momentum $\mathbf{p}$ by particles of momentum $-\mathbf{p}$.

In (13.3), both $\hat{a}_{\mathbf{p}}$ and $\hat{b}_{\mathbf{p}}$ transform either with the upper signs or with the lower signs. Within second quantization this expresses the equality of the intrinsic parities of a spin-zero particle and its antiparticle. That equality is already clear from the fact that the same scalar or pseudoscalar wave functions describe the particle and antiparticle.

Relativistic theory also has a symmetry under a transformation with no nonrelativistic counterpart, called *charge conjugation* (the C transformation). Interchange every pair of operators $\hat{a}_{\mathbf{p}}$ and $\hat{b}_{\mathbf{p}}$:

$$C : \hat{a}_{\mathbf{p}} \rightarrow \hat{b}_{\mathbf{p}}, \quad \hat{b}_{\mathbf{p}} \rightarrow \hat{a}_{\mathbf{p}} \quad (13.6)$$

(that is, exchange particles and antiparticles). The operator $\hat{\psi}$ then becomes its charge-conjugate $\hat{\psi}^C$, with

$$\hat{\psi}^C(t, \mathbf{r}) = \hat{\psi}^{\dagger}(t, \mathbf{r}). \quad (13.7)$$

This equation displays the symmetry with which the concepts of particle and antiparticle enter the theory.

There is an inessential formal freedom in the definition of charge conjugation. Its meaning is unchanged if an arbitrary phase factor is introduced into (13.6):

$$\hat{a}_{\mathbf{p}} \rightarrow e^{i\alpha} \hat{b}_{\mathbf{p}}, \quad \hat{b}_{\mathbf{p}} \rightarrow e^{-i\alpha} \hat{a}_{\mathbf{p}}.$$

One would then have

$$\hat{\psi} \rightarrow e^{i\alpha} \hat{\psi}^{\dagger}, \quad \hat{\psi}^{\dagger} \rightarrow e^{-i\alpha} \hat{\psi},$$

while applying the transformation twice would still give the identity, $\hat{\psi} \rightarrow \hat{\psi}$. All such definitions are equivalent. Since multiplying the ψ operators by a phase does not alter their properties (compare the end of the preceding section), one may simply redefine $\hat{\psi}$ as $\hat{\psi} e^{i\alpha/2}$ and thereby return to (13.6), (13.7).

Since charge conjugation replaces a particle by a distinct antiparticle, in general it supplies no new characteristic of a particle or of a particle system as such.

Systems containing equal numbers of particles and antiparticles are an exception. The operator $\hat{C}$ maps such a system into itself, so it possesses eigenstates with eigenvalues

$C = \pm 1$ (because $\widehat{C}^2 = 1$). To describe charge symmetry, the particle and antiparticle may here be regarded as two different “charge states” of one particle, distinguished by a charge quantum number $Q = \pm 1$. The system’s wave function is a product of an orbital function and a “charge” function, and it must be symmetric under simultaneous interchange of all variables, both coordinate and charge, for any pair of particles. The symmetry of the “charge” function then determines the charge parity of the system (see the problem)¹².

The notion of charge parity, arising naturally for “genuinely neutral” systems, applies in full also to genuinely neutral “elementary” particles. In second quantization it is expressed by

$$\widehat{\psi}^C = \pm \widehat{\psi}; \quad (13.8)$$

the plus and minus signs characterize charge-even and charge-odd particles, respectively.

It was stated in § 11 that relativistic invariance must include invariance under 4-inversion. For the operator of a field scalar under 4-rotations, this requires

$$\widehat{\psi}(t, \mathbf{r}) = \widehat{\psi}(-t, -\mathbf{r})$$

with the plus sign invariably on the right. In terms of the transformations of $\widehat{a}_{\mathbf{p}}$ and $\widehat{b}_{\mathbf{p}}$, changing $\widehat{\psi}(t, \mathbf{r})$ into $\widehat{\psi}(-t, -\mathbf{r})$ is achieved by interchanging in (13.4) the coefficients of e^{-ipx} and e^{ipx} , namely by

$$\widehat{a}_{\mathbf{p}} \rightarrow \widehat{b}_{-\mathbf{p}}^+, \quad \widehat{b}_{\mathbf{p}} \rightarrow \widehat{a}_{-\mathbf{p}}^+ \quad (13.9)$$

Because it replaces a operators by b operators, this transformation exchanges particles and antiparticles. Thus relativistic theory naturally requires invariance under a transformation in which spatial inversion (P) and time reversal (T) are accompanied by charge conjugation (C). This statement is called the *CPT theorem*¹³.

It is important, however, to stress that although the arguments presented here and in § 11, 12 appear to extend naturally the ideas of ordinary quantum mechanics and classical relativity, the results obtained go beyond both theories in form (the ψ operators contain particle creation and annihilation operators simultaneously) and in substance (particles and antiparticles). They therefore cannot be regarded as consequences of logic alone. They embody new physical principles whose validity can be judged only by experiment.

Let $\widehat{\psi}^{CPT}(t, \mathbf{r})$ denote the operator (13.4) after transformation (13.9). Then

$$\widehat{\psi}^{CPT}(t, \mathbf{r}) = \widehat{\psi}(-t, -\mathbf{r}). \quad (13.10)$$

By formulating 4-inversion as (13.9), we also obtain the transformation rule for time reversal of the ψ operator: together with CP (called *combined inversion*), it must yield (13.9). Equations (13.3) and (13.6) therefore give

$$T : \widehat{a}_{\mathbf{p}} \rightarrow \pm \widehat{a}_{-\mathbf{p}}^+, \quad \widehat{b}_{\mathbf{p}} \rightarrow \pm \widehat{b}_{-\mathbf{p}}^+ \quad (13.11)$$

¹²This argument concerns spin-zero particles. The construction generalizes directly to other cases; see, for example, the problem in § 27.

¹³It was formulated by Lüders (G. Lüders, 1954) and Pauli (W. Pauli, 1955).

(the two signs correspond to those in (13.3)). This transformation also has a natural meaning. Time reversal not only turns motion with momentum $\mathbf{p}$ into motion with momentum $-\mathbf{p}$, but also interchanges initial and final states in matrix elements. It consequently replaces annihilation operators for particles of momentum $\mathbf{p}$ by creation operators for particles of momentum $-\mathbf{p}$. Apply (13.11) to (13.4) and relabel the summation variable, $\mathbf{p} \rightarrow -\mathbf{p}$; one obtains¹⁴

$$\widehat{\psi}^T(t, \mathbf{r}) = \pm \widehat{\psi}^+(-t, \mathbf{r}). \quad (13.12)$$

This equation is analogous to the usual rule for time reversal in quantum mechanics. If a state is described by $\psi(t, \mathbf{r})$, its time-reversed state is described by $\psi^*(-t, \mathbf{r})$. Passing to the complex-conjugate function is necessary to restore the “correct” time dependence disrupted by changing the sign of t (*E. P. Wigner*, 1932).

Because T and hence CPT interchange initial and final states, eigenstates and eigenvalues are not meaningful for these transformations. They therefore add no characteristics of particles as such. Their consequences for scattering processes will be discussed in § 69, 71.

We now examine how the current 4-vector operator $\widehat{j}^\mu$ of (12.8) transforms under C , P , and T . Transformation (13.2), together with $(\partial_0, \partial_i) \rightarrow (\partial_0, -\partial_i)$, gives

$$P : (\widehat{j}^0, \widehat{\mathbf{j}})_{t, \mathbf{r}} \rightarrow (\widehat{j}^0, -\widehat{\mathbf{j}})_{t, -\mathbf{r}}, \quad (13.13)$$

as required for a true 4-vector. Transformation (13.7) would simply give

$$C : (\widehat{j}^0, \widehat{\mathbf{j}})_{t, \mathbf{r}} \rightarrow (-\widehat{j}^0, -\widehat{\mathbf{j}})_{t, \mathbf{r}}, \quad (13.14)$$

if $\widehat{\psi}$ and $\widehat{\psi}^+$ commuted. Their noncommutativity comes only from that of $\widehat{a}_{\mathbf{p}}$ and $\widehat{a}_{\mathbf{p}}^+$ (or $\widehat{b}_{\mathbf{p}}$ and $\widehat{b}_{\mathbf{p}}^+$ for equal $\mathbf{p}$). By the commutation rules (11.4), however, interchanging these operators produces only terms independent of the occupation numbers, and hence independent of the field state. Discarding these terms as in (11.5), (11.6), we recover (13.14), whose meaning is natural: charge conjugation changes the sign of every component of the 4-current when it replaces particles by antiparticles.

Since time reversal transposes initial and final states, its action on an operator product reverses the order of the factors. Thus

$$(\widehat{\psi}^+ \partial_\mu \widehat{\psi})^T = (\partial_\mu \widehat{\psi})^T (\widehat{\psi}^+)^T.$$

Here this fact is immaterial: commutativity of the ψ operators, in the sense just specified, allows the original order to be restored without affecting the result. Also noting that $(\partial_0, \partial_i) \rightarrow (-\partial_0, \partial_i)$ under time reversal, we find

$$T : (\widehat{j}^0, \widehat{\mathbf{j}})_{t, \mathbf{r}} \rightarrow (\widehat{j}^0, -\widehat{\mathbf{j}})_{-t, \mathbf{r}}. \quad (13.15)$$

The three-dimensional vector $\mathbf{j}$ reverses sign, in agreement with its classical meaning.

¹⁴If T is defined independently of the other transformations, its phase factor has the same freedom as that of C . Requiring CPT symmetry leaves the phase factor arbitrary in only one of the two transformations, C or T .

Finally, under CPT ,

$$CPT : (\hat{j}^0, \hat{\mathbf{j}})_{t,\mathbf{r}} \rightarrow (-\hat{j}^0, -\hat{\mathbf{j}})_{-t,-\mathbf{r}}, \quad (13.16)$$

as appropriate to 4-inversion. We emphasize that since 4-inversion reduces to a rotation of the 4-dimensional coordinate frame, no distinction between true and pseudo 4-tensors of any rank exists with respect to it.

So far the particles have been understood to be free. Parity quantum numbers acquire real significance only for interacting particles, where they entail selection rules that allow or forbid particular processes. Such significance is possible only for conserved characteristics: eigenvalues of operators commuting with the Hamiltonian of the interacting particles.

Relativistic invariance requires the CPT operator in any event to commute with the Hamiltonian. Experiment shows that electromagnetic and strong interactions are separately invariant under C and T (and also P), so the associated parity quantum numbers are conserved in these interactions. In the weak interaction, these conservation laws are violated¹⁵.

Anticipating a later result, the interaction operator for charged particles and the electromagnetic field is the product of the 4-vector operators $\hat{A}$ and $\hat{j}$. Charge conjugation reverses $\hat{j}$; invariance of the electromagnetic interaction under this transformation therefore requires $\hat{A}$ to reverse as well. In other words, photons are charge-odd particles.

This behavior of the operators $\hat{A}$ agrees with the properties of the 4-potential in classical theory. Indeed, the transformations

$$\begin{aligned} C : (\hat{A}_0, \hat{\mathbf{A}}) &\rightarrow (-\hat{A}_0, -\hat{\mathbf{A}})_{t,\mathbf{r}}, \\ P : (\hat{A}_0, \hat{\mathbf{A}}) &\rightarrow (\hat{A}_0, -\hat{\mathbf{A}})_{t,-\mathbf{r}}, \\ CPT : (\hat{A}_0, \hat{\mathbf{A}}) &\rightarrow (-\hat{A}_0, -\hat{\mathbf{A}})_{-t,-\mathbf{r}}, \end{aligned}$$

imply

$$T : (\hat{A}_0, \hat{\mathbf{A}}) \rightarrow (\hat{A}_0, -\hat{\mathbf{A}})_{-t,\mathbf{r}},$$

which is precisely the classical time-reversal rule for the electromagnetic potentials.

The demand for CPT invariance places no restrictions on the properties of a particle considered alone. It does, however, impose relations between particle and antiparticle properties. Foremost among these is equality of their masses; this is already evident from the connection, discussed in § 11, between 4-inversion and the very origin of the concepts of particle and antiparticle.

Furthermore, CPT invariance implies that the proportionality coefficients relating the electric and magnetic moment vectors to the spin vector differ between a particle and its antiparticle only in sign. The magnetic moment changes sign under C and T , and, being an axial vector, is invariant under P . Hence CPT , while turning a particle into its antiparticle, leaves the sign of the magnetic moment unchanged, whereas the spin vector reverses. The same conclusion holds for the electric moment: it is unchanged by

¹⁵The possibility that parity might not be conserved in weak interactions was first proposed by *Lee and Yang* (*T. D. Lee, C. N. Yang*, 1956). Earlier, *Dirac* (1949) had expressed the more general idea that physical laws need not be P - and T -invariant.

time reversal and reverses under charge conjugation and, as a polar vector, under spatial inversion.

The requirements of P or T invariance, whenever they hold, already constrain the properties of each individual particle: they forbid it to have an electric dipole moment. For a stationary elementary particle, the only vector constructible from its ψ operators is its spin operator vector. This vector is even under P and odd under T , so it can determine a magnetic moment but not an electric one. Notice that either P invariance or T invariance alone suffices for this prohibition.

PROBLEM

Find the charge parity and spatial parity of a system consisting of a spin-zero particle and its antiparticle, when their relative motion has orbital angular momentum l .

SOLUTION. Interchanging the particle coordinates is equivalent to inversion about the center of mass and therefore multiplies the orbital function by $(-1)^l$. Interchanging the charge variables is equivalent to charge conjugation and multiplies the “charge” factor in the wave function by the required value C . The condition $C(-1)^l = 1$ gives

$$C = (-1)^l.$$

The spatial parity P of the system is the product of the orbital parity and the intrinsic parities of the two particles. Since the particle and antiparticle have equal intrinsic parities here, P equals the orbital parity: $P = (-1)^l$.

§ 14. Wave equation for a spin-1 particle

In its rest frame, a spin-1 particle is described by a three-component wave function, namely a three-dimensional vector (such a particle is often called a *vector particle*). As four-dimensional quantities, these components may originate either as the three spatial components of a spacelike 4-vector ψ^μ , or as the mixed components of an antisymmetric rank-two 4-tensor $\psi^{\mu\nu}$. In the rest frame, the temporal component (ψ^0) of the former and the spatial components (ψ^{ik}) of the latter vanish¹⁶.

The wave equation, a differential relation between ψ^μ and $\psi^{\mu\nu}$, is specified by the relations

$$i\psi_{\mu\nu} = \widehat{p}_\mu\psi_\nu - \widehat{p}_\nu\psi_\mu, \quad (14.1)$$

$$im^2\psi_\mu = \widehat{p}^\nu\psi_{\mu\nu}, \quad (14.2)$$

where $\widehat{p} = i\partial$ (A. Proca, 1936). Apply $\widehat{p}^\mu$ to both sides of (14.2). The antisymmetry of $\psi_{\mu\nu}$ then gives

$$\widehat{p}^\mu\psi_\mu = 0. \quad (14.3)$$

¹⁶Anticipating a later result, the combined 4-vector ψ_μ and 4-tensor $\psi^{\nu\mu}$ correspond to the set of four-dimensional rank-two spinors $\xi^{\alpha\beta}$, $\eta_{\dot{\alpha}\dot{\beta}}$, and $\zeta^{\dot{\alpha}\dot{\beta}}$. Here $\xi^{\alpha\beta}$ and $\eta_{\dot{\alpha}\dot{\beta}}$ are symmetric spinors transformed into one another by inversion (see § 19).

The quantity $\psi_{\mu\nu}$ can be eliminated from (14.1), (14.2) by substituting the first equation into the second. With (14.3), the result is

$$(\hat{p}^2 - m^2)\psi_\mu = 0, \quad (14.4)$$

which again shows (compare § 10) that m is the particle mass. Thus a free spin-1 particle can be described using only one 4-vector ψ^μ . Its components obey the second-order equation (14.4) together with the supplementary condition (14.3), which removes from ψ^μ the part belonging to spin 0.

In the rest frame, ψ_μ is independent of the spatial coordinates, and hence $\hat{p}^0\psi_0 = 0$. At the same time $\hat{p}^0\psi_0 = m\psi_0$, so $\psi_0 = 0$ in that frame, as required. The components ψ_{ik} vanish along with ψ_0 .

A spin-1 particle may have either of two intrinsic parities, according as ψ^μ is a polar vector or a pseudovector. In the former case

$$\hat{P}\psi^\mu = (\psi^0, -\psi^i),$$

whereas in the latter

$$\hat{P}\psi^\mu = (-\psi^0, \psi^i).$$

Equations (14.1), (14.2) follow from a variational principle with the Lagrangian

$$\begin{aligned} L = & (1/2)\psi_{\mu\nu}\psi^{\mu\nu*} - (1/2)\psi^{\mu\nu*}(\partial_\mu\psi_\nu - \partial_\nu\psi_\mu) \\ & - (1/2)\psi^{\mu\nu}(\partial_\mu\psi_\nu^* - \partial_\nu\psi_\mu^*) + m^2\psi_\mu\psi^{\mu*}. \end{aligned} \quad (14.5)$$

Its independent generalized coordinates are ψ_μ , ψ_μ^* , $\psi_{\mu\nu}$, and $\psi_{\mu\nu}^*$ ¹⁷.

For finding the energy-momentum tensor, formula (10.11) is not entirely convenient here: it would produce a nonsymmetric tensor requiring subsequent symmetrization. Instead one may use

$$\frac{1}{2}T_{\mu\nu}\sqrt{-g} = -\frac{\partial}{\partial x^\lambda} \frac{\partial\sqrt{-g}L}{\partial g_{,\lambda}^{\mu\nu}} + \frac{\partial\sqrt{-g}L}{\partial g^{\mu\nu}}, \quad (14.6)$$

where L is to be written in a form valid for arbitrary curvilinear coordinates (see II, § 94). If L contains only the components of the metric tensor $g_{\mu\nu}$ itself, and not their coordinate derivatives, the formula simplifies to

$$T_{\mu\nu} = \frac{2}{\sqrt{-g}} \frac{\partial\sqrt{-g}L}{\partial g^{\mu\nu}} = 2 \frac{\partial L}{\partial g^{\mu\nu}} - g_{\mu\nu}L$$

(recall that $d \ln g = -g_{\mu\nu}dg^{\mu\nu}$).

The differentiation in (14.6) is not performed with respect to ψ_μ and $\psi_{\mu\nu}$. Consequently, these quantities need not be treated as independent when that formula is used: relation (14.1) can be applied at once, and (14.5) can be rewritten as

$$L = (-1/2)\psi_{\mu\nu}\psi_{\lambda\rho}^*g^{\mu\lambda}g^{\nu\rho} + m^2\psi_\mu\psi_\nu^*g^{\mu\nu}. \quad (14.7)$$

¹⁷Had the variation been performed only with respect to ψ_μ , with $\psi_{\mu\nu}$ assumed from the outset to be expressed through ψ_μ by (14.1), equation (14.3) would have had to be introduced as a supplementary condition unrelated to the variational principle.

It then follows that

$$T_{\mu\nu} = -\psi_{\mu\lambda}\psi_{\nu}^{\lambda*} - \psi_{\mu\lambda}^*\psi_{\nu}^{\lambda} + m^2(\psi_{\mu}^*\psi_{\nu} + \psi_{\nu}^*\psi_{\mu}) + g_{\mu\nu} \left((1/2)\psi_{\lambda\rho}\psi^{\lambda\rho*} - m^2\psi_{\lambda}^*\psi^{\lambda} \right). \quad (14.8)$$

In particular, the energy density is the positive-definite expression

$$T_{00} = (1/2)\psi_{ik}\psi_{ik}^* + \psi_{0i}\psi_{0i}^* + m^2(\psi_0\psi_0^* + \psi_i\psi_i^*). \quad (14.9)$$

The conserved 4-vector of current density is

$$j^{\mu} = i(\psi^{\mu\nu*}\psi_{\nu} - \psi^{\mu\nu}\psi_{\nu}^*). \quad (14.10)$$

It can be obtained from (12.12) by differentiating the Lagrangian (14.5) with respect to the derivatives $\partial_{\mu}\psi_{\nu}$. In particular,

$$j^0 = i(\psi^{0k*}\psi_k - \psi^{0k}\psi_k^*) \quad (14.11)$$

and this quantity is not positive definite.

A plane wave normalized to one particle in the volume $V = 1$ is

$$\psi_{\mu} = \frac{1}{\sqrt{2\varepsilon}}u_{\mu}e^{-ipx}, \quad u_{\mu}u^{\mu*} = -1, \quad (14.12)$$

where u_{μ} is a unit polarization 4-vector satisfying, by virtue of (14.3), the 4-transversality condition

$$u_{\mu}p^{\mu} = 0. \quad (14.13)$$

Indeed, substitution of (14.12) in (14.9) and (14.11) gives

$$T_{00} = -2\varepsilon^2\psi_{\mu}\psi^{\mu*} = \varepsilon, \quad j^0 = 1.$$

Unlike a photon, a massive vector particle has three independent polarization directions. Their corresponding amplitudes are given in (16.21).

For partially polarized vector particles, the polarization density matrix is defined so that in a pure state it reduces to the product

$$\rho_{\mu\nu} = u_{\mu}u_{\nu}^*$$

(in analogy with (8.7) for photons). According to (14.12), (14.13), it obeys

$$p^{\mu}\rho_{\mu\nu} = 0, \quad \rho^{\mu}_{\mu} = -1. \quad (14.14)$$

For unpolarized particles, $\rho_{\mu\nu}$ must have the form $ag_{\mu\nu} + bp_{\mu}p_{\nu}$. Determining a and b from (14.14) yields

$$\rho_{\mu\nu} = -\frac{1}{3} \left(g_{\mu\nu} - \frac{p_{\mu}p_{\nu}}{m^2} \right). \quad (14.15)$$

The vector-particle field is quantized in the same way as the scalar field, so there is no need to repeat the entire argument. The ψ -operators of the vector field are

$$\begin{aligned}\hat{\psi}_\mu &= \sum_{\mathbf{p}\alpha} \frac{1}{\sqrt{2\varepsilon}} \left(\hat{a}_{\mathbf{p}\alpha} u_\mu^{(\alpha)} e^{-ipx} + \hat{b}_{\mathbf{p}\alpha}^+ u_\mu^{(\alpha)*} e^{ipx} \right), \\ \hat{\psi}_\mu^+ &= \sum_{\mathbf{p}\alpha} \frac{1}{\sqrt{2\varepsilon}} \left(\hat{a}_{\mathbf{p}\alpha}^+ u_\mu^{(\alpha)*} e^{ipx} + \hat{b}_{\mathbf{p}\alpha} u_\mu^{(\alpha)} e^{-ipx} \right),\end{aligned}\quad (14.16)$$

where the index α labels the three independent polarizations.

The positive definiteness of (14.9) for T_{00} and the indefiniteness of j^0 in (14.11) lead, just as in the scalar case, to Bose quantization.

There is a close connection between the properties of a genuinely neutral vector field and those of the electromagnetic field. A neutral vector field is described by a Hermitian ψ -operator:

$$\hat{\psi}_\mu = \sum_{\mathbf{p}\alpha} \frac{1}{\sqrt{2\varepsilon}} \left(\hat{c}_{\mathbf{p}\alpha} u_\mu^{(\alpha)} e^{-ipx} + \hat{c}_{\mathbf{p}\alpha}^+ u_\mu^{(\alpha)*} e^{ipx} \right). \quad (14.17)$$

The Lagrangian of this field is

$$\hat{L} = \frac{1}{4} \hat{\psi}_{\mu\nu} \hat{\psi}^{\mu\nu} - \frac{1}{2} \hat{\psi}^{\mu\nu} (\partial_\mu \hat{\psi}_\nu - \partial_\nu \hat{\psi}_\mu) + \frac{1}{2} m^2 \hat{\psi}_\mu \hat{\psi}^\mu. \quad (14.18)$$

The electromagnetic field is obtained by taking $m = 0$. The 4-vector ψ^μ then becomes the 4-potential A^μ , while the 4-tensor $\psi^{\mu\nu}$ becomes the field tensor $F^{\mu\nu}$, related to the potential by definition (14.1). Equation (14.2) becomes $\partial^\nu \psi_{\mu\nu} = 0$, which is the second pair of Maxwell equations. Condition (14.3) no longer follows from it, and is therefore no longer compulsory. Since the supplementary condition is absent, $\hat{\psi}_\mu$ and $\hat{\psi}_{\mu\nu}$ need not be regarded as independent “coordinates” in the Lagrangian,

and (14.18) reduces to

$$\hat{L} = -\frac{1}{4} \hat{\psi}_{\mu\nu} \hat{\psi}^{\mu\nu} \quad (14.19)$$

in agreement with the familiar classical expression for the electromagnetic-field Lagrangian. This Lagrangian, together with the tensor $\hat{\psi}_{\mu\nu}$, is invariant under an arbitrary gauge transformation of the “potentials” $\hat{\psi}_\mu$. The connection of this fact with zero mass is evident: because of the term $m^2 \hat{\psi}_\mu \hat{\psi}^\mu$, the Lagrangian (14.18) does not possess that invariance.

§ 15. The wave equation for particles of higher integral spin

The wave equations (14.3, 14.4) follow directly once the particle’s mass and spin are specified. In practice, therefore, the Lagrangian is useful less for deriving these equations than for constructing expressions for the field energy, momentum, and charge.

For this purpose, as already noted, (14.7) may be used in place of (14.5), and it can be transformed further. Using (14.1), rewrite (14.7) as

$$\begin{aligned}L &= -(\partial_\mu \psi_\nu^*)(\partial^\mu \psi^\nu) + (\partial_\nu \psi_\mu^*)(\partial^\mu \psi^\nu) + m^2 \psi_\mu \psi^{\mu*} = \\ &= -(\partial_\mu \psi_\nu^*)(\partial^\mu \psi^\nu) + m^2 \psi_\mu^* \psi^\mu + \partial_\nu (\psi_\mu^* \partial^\mu \psi^\nu) - \psi_\mu^* \partial^\mu \partial_\nu \psi^\nu.\end{aligned}$$

The final term vanishes by (14.3), while the preceding term is a total derivative. Omitting it leaves the Lagrangian

$$L' = -(\partial_\mu \psi_\nu^*)(\partial^\mu \psi^\nu) + m^2 \psi_\mu^* \psi^\mu. \quad (15.1)$$

Its structure is the same as that of the spin-zero Lagrangian (10.9), except that the scalar ψ has been replaced by the 4-vector ψ_μ and the overall sign has changed. The sign change occurs because ψ_μ is a spacelike vector, so that $\psi_\mu \psi^{\mu*} < 0$, whereas $\psi \psi^* > 0$ for a scalar particle.

Constructing the energy-momentum 4-tensor and the current 4-vector from (15.1) gives expressions with the same form as (10.12) and (10.18) for a scalar field:

$$T_{\mu\nu} = -\partial_\mu \psi^{\lambda*} \cdot \partial_\nu \psi_\lambda - \partial_\nu \psi^{\lambda*} \cdot \partial_\mu \psi_\lambda - L' g_{\mu\nu}, \quad (15.2)$$

$$j_\mu = -i [\psi_\lambda^* \partial_\mu \psi^\lambda - (\partial_\mu \psi_\lambda^*) \psi^\lambda]. \quad (15.3)$$

Their differences from (14.8) and (14.10) likewise reduce to total derivatives. But, as stressed earlier, the local values of these quantities have no deep physical significance. Only the volume integrals P_μ (10.15) and Q (10.19) matter; they are identical for the two choices of $T_{\mu\nu}$ and j_μ .

This description extends directly to particles of arbitrary integral spin. The wave function of a spin- s particle is an irreducible 4-tensor of rank s : it is symmetric in all indices and has zero contraction over any pair,

$$\psi_{\dots\mu\dots\nu\dots} = \psi_{\dots\nu\dots\mu\dots}, \quad \psi_{\dots\mu\dots}^{\mu\dots} = 0. \quad (15.4)$$

This tensor must also satisfy the 4-transversality condition

$$\widehat{p}^\mu \psi_{\mu\dots} = 0, \quad (15.5)$$

and each of its components must obey the second-order equation

$$(\widehat{p}^2 - m^2) \psi_{\dots} = 0. \quad (15.6)$$

In the rest frame, (15.5) makes every component of the 4-tensor with a zero among its indices vanish. In other words, in the rest frame—that is, in the nonrelativistic limit—the wave function reduces, as expected, to an irreducible 3-tensor of rank s , having $2s + 1$ independent components.

For a field of spin- s particles, the Lagrangian, energy-momentum tensor, and current vector differ from (15.1)–(15.3) only by replacing ψ_λ with $\psi_{\lambda\mu\dots}$.

A normalized plane wave is

$$\psi^{\mu\nu\dots} = \frac{1}{\sqrt{2\varepsilon}} u^{\mu\nu\dots} e^{-ipx}, \quad u_{\mu\nu\dots}^* u^{\mu\nu\dots} = -1, \quad (15.7)$$

and its amplitude satisfies

$$u^{\mu\nu\dots} p_\mu = 0. \quad (15.8)$$

There are $2s + 1$ independent polarization states.

The field is quantized by the evident generalization of the spin-zero or spin-one cases.

The construction just given is entirely adequate for its purpose: describing a field of free particles. The situation is different if one asks for a description of the interaction between the particles and an electromagnetic field. That interaction would have to enter a Lagrangian from which every equation could be derived without separately imposing supplementary conditions. In fact, such a description of the interaction proves applicable only to electrons, which have spin $1/2$ (see § 32). For other spins, the question would therefore be of methodological interest only.

For every integral or half-integral spin $s > 1$, it is impossible to formulate a variational principle using only a single tensor or spinor function whose rank corresponds to the given spin. Lower-rank tensor or spinor quantities must also be introduced as auxiliaries. The Lagrangian is then chosen so that the field equations for free particles following from the variational principle make these auxiliary quantities vanish automatically¹⁸.

§ 16. Helicity states of a particle

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In relativistic theory, the orbital angular momentum $\mathbf{l}$ and spin $\mathbf{s}$ of a moving particle are not conserved separately. Only the total angular momentum $\mathbf{j} = \mathbf{l} + \mathbf{s}$ is conserved. The projection of the spin on any prescribed direction (the z axis) is therefore not conserved either, so this quantity cannot be used to enumerate the polarization (spin) states of a moving particle.

The projection of the spin on the momentum direction is conserved, however: since $\mathbf{l} = \mathbf{r} \times \mathbf{p}$, the product $\mathbf{s}\mathbf{n}$ coincides with the conserved product $\mathbf{j}\mathbf{n}$ ($\mathbf{n} = \mathbf{p}/|\mathbf{p}|$). This quantity is called *helicity*²⁰ (we already considered it for the photon in § 8). We denote its eigenvalues by λ ($\lambda = -s, \dots, +s$), and call particle states with definite λ helicity states. Let $\psi_{\mathbf{p}\lambda}$ be the wave function (a plane wave) describing a state with definite $\mathbf{p}$ and λ , and let $u^{(\lambda)}(\mathbf{p})$ be its amplitude; to keep the notation brief, we suppress the component indices of this function (for integral spin these are 4-tensor indices).

We saw in the preceding sections that a relativistic description of particles with nonzero integral spin requires a wave function with more than $2s + 1$ components. The number of independent components nevertheless remains $2s + 1$; the “extra” components are removed by imposing supplementary conditions under which these components vanish in the rest frame (in the next chapter we shall see the same for half-integral s).

According to the transformation formulas for angular momentum (see II, § 14), helicity is invariant under Lorentz transformations that leave unchanged the direction $\mathbf{p}$ on which the angular momentum is projected. The number λ thus retains its meaning as a quantum number under such transformations. To study the symmetry properties of helicity states, we may therefore use a reference frame in which $|\mathbf{p}| \ll m$ (in the limit, the rest frame). Then $\psi_{\mathbf{p}\lambda}$ reduces to a nonrelativistic $(2s + 1)$ -component wave function. Denote its amplitude by $w^{(\lambda)}(\mathbf{n})$, writing as its argument the direction $\mathbf{n} = \mathbf{p}/|\mathbf{p}|$ along

¹⁸See Fierz M., *Pauli W.*//Proc. Roy. Soc. — 1939. — V. A 173. — P. 211. The program described here was carried out in that paper for particles of spin $3/2$ and 2 .

¹⁹The contents of this section apply to particles of any spin, whether integral or half-integral.

²⁰In the English-language literature: helicity.

which the angular momentum is quantized. The amplitude $w^{(\lambda)}$ is an eigenfunction of the operator $\widehat{\mathbf{n}}\widehat{\mathbf{s}}$:

$$\widehat{\mathbf{n}}\widehat{\mathbf{s}} w^{(\lambda)}(\mathbf{n}) = \lambda w^{(\lambda)}(\mathbf{n}). \quad (16.1)$$

In the spinor representation, $w^{(\lambda)}$ is a contravariant symmetric spinor of rank $2s$; by the correspondence formulas (57.2) (see III), its components may also be indexed by the associated values of the spin projection σ on a fixed z axis.²¹

In the momentum representation, the wave functions of the states under consideration essentially coincide with the amplitudes $u^{(\lambda)}(\mathbf{p})$. More precisely,

$$\psi_{\mathbf{p}\lambda}(\mathbf{k}) = u^{(\lambda)}(\mathbf{k})\delta^{(2)}(\boldsymbol{\nu} - \mathbf{n}) = u^{(\lambda)}(\mathbf{p})\delta^{(2)}(\boldsymbol{\nu} - \mathbf{n}), \quad (16.2)$$

where the momentum as an independent variable is denoted by $\mathbf{k}$, as distinct from its eigenvalue $\mathbf{p}$, and $\boldsymbol{\nu} = \mathbf{k}/|\mathbf{k}|$, as distinct from $\mathbf{n} = \mathbf{p}/|\mathbf{p}|$.²² In the nonrelativistic limit,

$$\psi_{\mathbf{n}\lambda}(\boldsymbol{\nu}) = w^{(\lambda)}(\boldsymbol{\nu})\delta^{(2)}(\boldsymbol{\nu} - \mathbf{n}) = w^{(\lambda)}(\mathbf{n})\delta^{(2)}(\boldsymbol{\nu} - \mathbf{n}). \quad (16.3)$$

Written more fully, this expression would be

$$\psi_{\mathbf{n}\lambda}(\boldsymbol{\nu}, \sigma) = w_{\sigma}^{(\lambda)}(\boldsymbol{\nu})\delta^{(2)}(\boldsymbol{\nu} - \mathbf{n}),$$

where the discrete independent variable σ has also been displayed explicitly.

The helicity operator $\widehat{\mathbf{s}}\widehat{\mathbf{n}}$ commutes with $\widehat{j}_z$ and $\widehat{\mathbf{j}}^2$. Indeed, the angular-momentum operator is associated with an infinitesimal rotation of the coordinate system, while the scalar product of two vectors is invariant under every rotation. There consequently exist stationary states in which the particle has definite values of the angular momentum j , its projection $j_z = m$, and the helicity λ simultaneously. We shall call these *spherical helicity states*.

Let us determine the wave functions of these states in the momentum representation. This may be done directly by analogy with the formulas obtained in III, § 103 for the wave functions of a symmetric top. Those formulas were derived there from the transformation of wave functions under finite rotations (see III, § 58). The latter formulas, in turn, depend only on rotational symmetry and therefore apply to momentum-space functions just as they do to coordinate-space functions.

Alongside the space-fixed coordinate system xyz (relative to which the functions $\psi_{jm\lambda}$ are written), introduce a “moving” system $\xi\eta\zeta$ whose ζ axis lies along $\boldsymbol{\nu}$. Without repeating the corresponding reasoning (cf. the derivation of formula (103.8) in III), we write

$$\psi_{jm\lambda}(\mathbf{k}) = \psi_{j\lambda}^{(0)} D_{\lambda m}^{(j)}(\boldsymbol{\nu}),$$

²¹This reasoning (as well as the enumeration of the possible values of λ) applies to particles of nonzero mass. A massless particle has no rest frame, and its helicity can take only the two values $\lambda = \pm s$. This is connected with the circumstance already mentioned in § 8: the states of such a particle are classified by their behavior under the axial-symmetry group, which permits only twofold degeneracy of the levels. In terms of the wave equation, this means that in the limit $m \rightarrow 0$ the system of equations for a particle of spin s separates into independent equations corresponding to massless particles of spins $s, s-1, \dots$. Thus, for the photon $\lambda = \pm 1$, and the corresponding $w^{(\lambda)}$ are the three-dimensional vectors $\mathbf{e}^{(\pm 1)}$ of (8.2).

²²Here the delta function $\delta^{(2)}$ is defined so that $\int \delta^{(2)}(\boldsymbol{\nu} - \mathbf{n}) d\omega_{\boldsymbol{\nu}} = 1$. In (16.2), and in the analogous case below (see (16.4)), the delta function fixing the prescribed energy has been omitted.

where $\psi_{j\lambda}^{(0)}$ is the wave function in the “moving” coordinate system and describes a state with a definite ζ projection of the angular momentum, $j_\zeta = \lambda$; in the momentum representation this function plainly coincides with the amplitude $u^{(\lambda)}$. The normalized wave function (see below) is

$$\psi_{jm\lambda}(\mathbf{k}) = \sqrt{\frac{2j+1}{4\pi}} D_{\lambda m}^{(j)}(\mathbf{v}) u^{(\lambda)}(\mathbf{k}). \quad (16.4)$$

There remains, however, a phase-choice question associated with the following ambiguity. A rotation of the coordinate system $\xi\eta\zeta$ relative to xyz is specified by three Euler angles α , β , and γ , whereas the direction $\mathbf{v}$, on which alone the particle wave function can depend, is specified by only two spherical angles, $\alpha \equiv \varphi$ and $\beta \equiv \theta$. A convention must therefore be adopted for γ . We put $\gamma = 0$, thereby defining $D_{\lambda m}^{(j)}(\mathbf{v})$ by

$$D_{\lambda m}^{(j)}(\mathbf{v}) = D_{\lambda m}^{(j)}(\varphi, \theta, 0) = e^{im\varphi} d_{\lambda m}^{(j)}(\theta). \quad (16.5)$$

By (58.21) (see III), the functions (16.5) satisfy the orthogonality and normalization conditions

$$\int D_{\lambda_1 m_1}^{(j_1)*}(\mathbf{v}) D_{\lambda_2 m_2}^{(j_2)}(\mathbf{v}) \frac{d\mathbf{v}}{4\pi} = \frac{1}{2j_1+1} \delta_{j_1 j_2} \delta_{m_1 m_2} \quad (16.6)$$

($d\mathbf{v} = \sin\theta d\theta d\varphi$). Orthogonality of the functions $\psi_{jm\lambda}$ with respect to λ is supplied by the factor $u^{(\lambda)}$. Thus the functions $\psi_{jm\lambda}$ are orthogonal, as required, in all the indices $jm\lambda$, and with the coefficient chosen in (16.4) they obey the normalization

$$\int |\psi_{jm\lambda}|^2 d\mathbf{v} = 1. \quad (16.7)$$

Here the amplitudes $u^{(\lambda)}$ are assumed to be normalized to unity: $u^{(\lambda)} u^{(\lambda)*} = 1$.

Consider the behavior of the helicity-state wave functions under coordinate inversion. The scalar product of the polar vector $\mathbf{v}$ and the axial vector $\mathbf{j}$ is a pseudoscalar. It is therefore clear that inversion carries a state of helicity λ into one of helicity $-\lambda$; only the phase factors in these transformations remain to be found.

Under inversion, $\mathbf{v} \rightarrow -\mathbf{v}$. The vector $\mathbf{v}$ is specified by the two angles φ and θ , and $\mathbf{v} \rightarrow -\mathbf{v}$ is effected by $\varphi \rightarrow \varphi + \pi$, $\theta \rightarrow \pi - \theta$. This fixes the ζ axis, but leaves undetermined the positions of the ξ and η axes, which also depend on the third Euler angle γ ; transforming only θ and φ does not in this sense distinguish inversion of the coordinate system from a rotation of the ζ axis. In terms of all three Euler angles, inversion is the transformation

$$\alpha \equiv \varphi \rightarrow \varphi + \pi, \quad \beta \equiv \theta \rightarrow \pi - \theta, \quad \gamma \rightarrow \pi - \gamma. \quad (16.8)$$

Consequently, if $D_{\lambda m}^{(j)}(\mathbf{v})$ is defined by (16.5) (that is, with $\gamma = 0$), and $\mathbf{v} \rightarrow -\mathbf{v}$ is understood as the result of inversion, then

$$D_{\lambda m}^{(j)}(-\mathbf{v}) = D_{\lambda m}^{(j)}(\varphi + \pi, \pi - \theta, \pi). \quad (16.9)$$

Using formulas (58.9), (58.16), and (58.18) (see III), we therefore find

$$\begin{aligned} D_{\lambda m}^{(j)}(-\mathbf{v}) &= e^{i\lambda\pi} d_{\lambda m}^{(j)}(\pi - \theta) e^{im(\varphi + \pi)} = \\ &= (-1)^{j-\lambda} e^{im\varphi} d_{-\lambda, m}^{(j)}(\theta) = (-1)^{j-\lambda} D_{-\lambda, m}^{(j)}(\varphi, \theta, 0), \end{aligned}$$

or

$$D_{\lambda m}^{(j)}(-\mathbf{v}) = (-1)^{j-\lambda} D_{-\lambda, m}^{(j)}(\mathbf{v}) \quad (16.10)$$

($j - \lambda$ is an integer).

An analogous formula for the spinor $w^{(\lambda)}$ may be obtained by noting that its components $w_{\sigma}^{(\lambda)}$ coincide, apart from a factor, with the functions

$$w_{\sigma}^{(\lambda)}(\mathbf{v}) \propto D_{\lambda \sigma}^{(s)}(\mathbf{v})^*. \quad (16.11)$$

Indeed, apply the transformation formula (58.7) (see III) to the spin eigenfunctions and require its ζ projection to have the definite value λ (that is, replace $\psi_{jm'}$ on the right-hand side of (58.7) by $\delta_{m'\lambda}$). We then find that $D_{\lambda \sigma}^{(s)}(\mathbf{v})$ are the spin wave functions corresponding to the definite values σ and λ of the spin's z and ζ projections. The set of these functions ($\sigma = -s, \dots, +s$) forms, by the correspondence formulas (57.6) (see III), a covariant spinor of rank $2s$. The components of the contravariant spinor (to which the components $w_{\sigma}^{(\lambda)}$ correspond by formulas (57.2) in III) transform as the complex conjugates of the components of a covariant spinor of the same rank. From (16.10) and (16.11),

$$w^{(\lambda)}(-\mathbf{v}) = (-1)^{s-\lambda} w^{(-\lambda)}(\mathbf{v}) \quad (16.12)$$

($s - \lambda$ is an integer). Applied to $w^{(\lambda)}$, however, inversion consists not only in replacing $\mathbf{v}$ by $-\mathbf{v}$, but also in multiplying by an overall phase factor (the particle's "intrinsic parity"), which we denote by η :

$$\hat{P}w^{(\lambda)}(\mathbf{v}) = \eta w^{(\lambda)}(-\mathbf{v}) = \eta(-1)^{s-\lambda} w^{(-\lambda)}(\mathbf{v}). \quad (16.13)$$

For the relativistic amplitude $u^{(\lambda)}(\mathbf{k})$ this transformation reads

$$\hat{P}u^{(\lambda)}(\mathbf{k}) = \eta \beta u^{(\lambda)}(-\mathbf{k}) = \eta(-1)^{s-\lambda} u^{(-\lambda)}(\mathbf{k}), \quad (16.14)$$

where β is a matrix that is the identity on those components of $u^{(\lambda)}$ which remain in the limit $|\mathbf{p}| \rightarrow 0$. What matters is that this matrix does not depend on the quantum numbers of the state; in this sense the difference between (16.13) and (16.14) is inessential.²³

Applying (16.14) to (16.2), we obtain the transformation law for the wave functions of the states $|\mathbf{n}\lambda\rangle$:

$$\hat{P}\psi_{\mathbf{n}\lambda}(\mathbf{v}) = \eta(-1)^{s-\lambda} \psi_{-\mathbf{n}, -\lambda}(\mathbf{v}). \quad (16.15)$$

For the spherical helicity states, use of (16.10) and (16.12) gives the transformation law

$$\hat{P}\psi_{jm\lambda}(\mathbf{v}) = \eta(-1)^{j-s} \psi_{jm, -\lambda}(\mathbf{v}). \quad (16.16)$$

The states ψ_{jm0} transform into themselves according to (16.16), and thus have definite parity. If $\lambda \neq 0$, only superpositions of states with opposite helicities have definite parity:

$$\psi_{jm|\lambda|}^{(\pm)} = \frac{1}{\sqrt{2}} (\psi_{jm\lambda} \pm \psi_{jm, -\lambda}). \quad (16.17)$$

²³For $s = 1$, for example, the amplitudes $u^{(\lambda)}$ are the 4-vectors (16.22); β is then the complete identity matrix in the 4-vector indices: $\beta_{\mu\nu} = \delta_{\mu\nu}$. For $s = 1/2$ (as we shall see in the next chapter), $u^{(\lambda)}$ is a bispinor; the phase factor is then $\eta = i$, while β is the Dirac matrix γ^0 (see (21.10)).

Under inversion they transform into themselves according to

$$\widehat{P}\psi_{jm|\lambda|}^{(\pm)}(\mathbf{v}) = \pm\eta(-1)^{j-s}\psi_{jm|\lambda|}^{(\pm)}(\mathbf{v}). \quad (16.18)$$

Notice that in this section we have classified the states of a free particle with prescribed angular momentum by using only conserved quantities and without invoking orbital angular momentum (used, for example, in § 6 and 7 to classify photon states).

As an example, consider spin 1. In the rest frame the amplitudes $u^{(\lambda)}$ (4-vectors) reduce to the three-dimensional vectors $\mathbf{e}^{(\lambda)}$, which here serve as the amplitudes $w^{(\lambda)}$. The action of the spin-1 operator on a vector function $\mathbf{e}$ is given by

$$(\widehat{s}_i \mathbf{e})_k = -ie_{ikl}e_l \quad (16.19)$$

(see III, § 57, problem 2). Equation (16.1) therefore becomes

$$i\mathbf{n} \times \mathbf{e}^{(\lambda)} = \lambda \mathbf{e}^{(\lambda)}. \quad (16.20)$$

Its solutions (in the coordinate system $\xi\eta\zeta$, with the ζ axis along $\mathbf{n}$) coincide with the circular unit vectors (7.14)²⁴:

$$\mathbf{e}^{(0)} = i(0, 0, 1), \quad \mathbf{e}^{(\pm 1)} = \mp \frac{i}{\sqrt{2}}(1, \pm i, 0). \quad (16.21)$$

In a reference frame in which the particle has momentum $\mathbf{p}$, the amplitudes of the helicity states are the 4-vectors

$$u^{(0)\mu} = \left(\frac{|\mathbf{p}|}{m}, \frac{\varepsilon}{m} \mathbf{e}^{(0)} \right), \quad u^{(\pm 1)\mu} = (0, \mathbf{e}^{(\pm 1)}). \quad (16.22)$$

If $\mathbf{e}$ is a polar vector, then $\eta = -1$. The functions (16.17) (which for $s = 1$ are three-dimensional vectors) then have the parities

$$\begin{aligned} \psi_{jm|\lambda|}^{(+)} : \quad P &= (-1)^j, \\ \psi_{jm|\lambda|}^{(-)} : \quad P &= (-1)^{j+1}, \\ \psi_{jm0} : \quad P &= (-1)^j. \end{aligned}$$

Comparison with the definition of the vector spherical harmonics (7.4) shows that these functions are identical, up to phase factors, respectively to $\mathbf{Y}_{jm}^{(e)}$, $\mathbf{Y}_{jm}^{(m)}$, and $\mathbf{Y}_{jm}^{(l)}$. Determining the phase factors (for example, by comparing the values at $\theta = 0$), we obtain

$$\begin{aligned} \mathbf{Y}_{jm}^{(e)} &= i^{j-1} \sqrt{\frac{2j+1}{8\pi}} \left(\mathbf{e}^{(1)} D_{1m}^{(j)} + \mathbf{e}^{(-1)} D_{-1m}^{(j)} \right), \\ \mathbf{Y}_{jm}^{(m)} &= i^{j-1} \sqrt{\frac{2j+1}{8\pi}} \left(\mathbf{e}^{(1)'} D_{1m}^{(j)} + \mathbf{e}^{(-1)'} D_{-1m}^{(j)} \right), \\ \mathbf{Y}_{jm}^{(l)} &= i^{j-1} \sqrt{\frac{2j+1}{4\pi}} \mathbf{e}^{(0)} D_{0m}^{(j)}. \end{aligned} \quad (16.23)$$

²⁴The choice of phase factors is fixed by requiring that the matrix elements of the spin operators calculated with the eigenfunctions (16.21) agree with the general definitions in III, § 27 and 107.

(j is an integer!); $\mathbf{e}^{(\lambda)'} = \mathbf{n} \times \mathbf{e}^{(\lambda)}$ are the circular unit vectors along the axes $\xi'\eta'\zeta$, obtained by rotating $\xi\eta\zeta$ through 90° about the ζ axis.

The last of formulas (16.23) is equivalent to expression (58.23) (see III) for $d_{0m}^{(j)}(\theta)$. The first (or second) formula gives a simple expression for $d_{\pm 1m}^{(j)}$. We have

$$i^{j-1} \sqrt{\frac{2j+1}{8\pi}} D_{\pm 1m}^{(j)} = \mathbf{Y}_{jm}^{(c)} \mathbf{e}^{(\pm 1)*} = \frac{1}{\sqrt{j(j+1)}} \mathbf{e}^{(\pm 1)*} \nabla Y_{jm}.$$

The scalar product on the right is expanded in the $\xi\eta\zeta$ system, where

$$\left(\frac{\partial}{\partial \xi}, \frac{\partial}{\partial \eta} \right) \rightarrow \left(\frac{\partial}{\partial \theta}, \frac{1}{\sin \theta} \frac{\partial}{\partial \varphi} \right).$$

Recalling the definition (7.2) of Y_{jm} and definition (16.5), we finally obtain

$$d_{\pm 1m}^{(j)}(\theta) = (-1)^{m+1} \sqrt{\frac{(j-m)!}{(j+m)!j(j+1)}} \left(\pm \frac{\partial}{\partial \theta} + \frac{m}{\sin \theta} \right) P_j^m(\cos \theta), \quad m \geq 0. \quad (16.24)$$

Fermions

§ 17. Four-dimensional spinors

In nonrelativistic theory, a particle of arbitrary spin s is described by a quantity with $2s + 1$ components—a symmetric spinor of rank $2s$. Mathematically, such quantities furnish irreducible representations of the spatial-rotation group.

In relativistic theory, this group is only a subgroup of the larger group of four-dimensional rotations, the Lorentz group. It is consequently necessary to develop the theory of four-dimensional spinors (4-spinors), namely quantities furnishing irreducible representations of the Lorentz group. This theory is presented in §§ 17–19. Sections 17 and 18 deal only with the proper Lorentz group, which does not include spatial inversion; inversion itself is considered in § 19.

The theory of 4-spinors is constructed by analogy with that of three-dimensional spinors (B. L. van der Waerden, 1929; G. E. Uhlenbeck, O. Laporte, 1931).

A spinor ξ^α is a two-component quantity ($\alpha = 1, 2$). When these are components of the wave function of a spin-1/2 particle, ξ^1 and ξ^2 correspond respectively to the eigenvalues $+1/2$ and $-1/2$ of the spin projection on the z -axis. Under any transformation belonging to the (proper) Lorentz group, the two quantities ξ^1 and ξ^2 are transformed into linear combinations of one another:

$$\xi^{1'} = \alpha \xi^1 + \beta \xi^2, \quad \xi^{2'} = \gamma \xi^1 + \delta \xi^2, \quad (17.1)$$

The coefficients α , β , γ , and δ are definite functions of the rotation angles of the four-dimensional coordinate system, subject to the condition

$$\alpha\delta - \beta\gamma = 1, \quad (17.2)$$

that is, the determinant of the binary transformation (17.1) is unity, just as are the determinants of coordinate transformations in the Lorentz group.

Condition (17.2) makes the bilinear form $\xi^1 \Xi^2 - \xi^2 \Xi^1$ invariant under (17.1), where ξ^α and Ξ^α are two spinors. This form corresponds to a spin-zero particle “composed” of two spin-1/2 particles. To write invariants of this kind naturally, one introduces, together with the “contravariant” components ξ^α , the “covariant” components ξ_α . Conversion between them is performed with the “metric spinor” $g_{\alpha\beta}$ ¹:

$$\xi_\alpha = g_{\alpha\beta} \xi^\beta, \quad (17.3)$$

where

$$g = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}, \quad (17.4)$$

¹ Spinor indices will be denoted by the first letters of the Greek alphabet: $\alpha, \beta, \gamma, \dots$

and hence

$$\xi_1 = \xi^2, \quad \xi_2 = -\xi^1. \quad (17.5)$$

The invariant $\xi^1 \Xi^2 - \xi^2 \Xi^1$ can then be written as the scalar product $\xi^\alpha \Xi_\alpha$. Notice also that $\xi^\alpha \Xi_\alpha = -\xi_\alpha \Xi^\alpha$.

The properties listed so far formally coincide with those of three-dimensional spinors. A difference appears, however, when complex-conjugate spinors are considered.

In nonrelativistic theory the sum

$$\psi^1 \psi^{1*} + \psi^2 \psi^{2*}, \quad (17.6)$$

which determines the probability density for locating particles in space, had to be a scalar. This required the components $\psi^{\alpha*}$ to transform as covariant spinor components; in other words, transformation (17.1) had to be unitary ($\alpha = \delta^*$, $\beta = -\gamma^*$). In relativistic theory, by contrast, the particle density is not a scalar: it is the time component of a 4-vector. The stated requirement therefore disappears, and the transformation coefficients are now constrained only by (17.2), with no additional conditions. The four complex quantities α , β , γ , and δ , subject solely to (17.2), are equivalent to $8 - 2 = 6$ real parameters. This agrees with the number of angles specifying a rotation of a four-dimensional coordinate system, since there are six coordinate planes.

Thus the two mutually complex-conjugate binary transformations are essentially different, and relativistic theory consequently has two types of spinor. A special notation is used to distinguish them: spinors transformed by formulas complex-conjugate to (17.1) carry numeral indices marked with dots above them (*dotted indices*). Accordingly, by definition,

$$\eta^{\dot{\alpha}} \sim \xi^{\alpha*}, \quad (17.7)$$

where $\sim$ means “transforms as.” In explicit form, the transformation formulas for the “dotted” spinor are

$$\eta^{\dot{1}'} = \alpha^* \eta^{\dot{1}} + \beta^* \eta^{\dot{2}}, \quad \eta^{\dot{2}'} = \gamma^* \eta^{\dot{1}} + \delta^* \eta^{\dot{2}}. \quad (17.8)$$

Dotted indices are lowered and raised in the same way as undotted ones:

$$\eta_{\dot{1}} = \eta^{\dot{2}}, \quad \eta_{\dot{2}} = -\eta^{\dot{1}}. \quad (17.9)$$

With respect to spatial rotations, 4-spinors behave in the same way as 3-spinors. For the latter we know that $\psi_\alpha^* \sim \psi^\alpha$. Definition (17.7) therefore implies that, under rotations, the 4-spinor $\eta_{\dot{\alpha}}$ behaves as a contravariant 3-spinor ψ^α . The covariant components $\eta_{\dot{1}}$ and $\eta_{\dot{2}}$ thus correspond to the spin-projection eigenvalues $1/2$ and $-1/2$, respectively.

Spinors of higher rank are defined as collections of quantities that transform like products of components of several first-rank spinors. A higher-rank spinor may have both dotted and undotted indices. For example, there are three types of second-rank spinor:

$$\xi^{\alpha\beta} \sim \xi^\alpha \Xi^\beta, \quad \zeta^{\alpha\dot{\beta}} \sim \xi^\alpha \eta^{\dot{\beta}}, \quad \eta^{\dot{\alpha}\dot{\beta}} \sim \eta^{\dot{\alpha}} \eta^{\dot{\beta}}.$$

Consequently, specifying only the total spinor rank does not determine the type uniquely. Where necessary, we shall therefore give the rank as a pair (k, l) : the numbers of undotted and dotted indices.

Transformations (17.1) and (17.8) are algebraically independent. There is consequently no need to prescribe an ordering of dotted and undotted indices; in this sense, for example, the spinors $\zeta^{\alpha\dot{\beta}}$ and $\zeta^{\dot{\beta}\alpha}$ are the same.

For a spinor equation to be invariant, its two sides must contain equal numbers of undotted indices and equal numbers of dotted indices. Otherwise the equality necessarily fails after a change of reference frame. It must also be remembered that complex conjugation replaces dotted indices by undotted ones and vice versa. Thus the relation $\eta^{\dot{\alpha}\dot{\beta}} = (\xi^{\alpha\beta})^*$ between two spinors is invariant.

Spinors, or products of them, can be contracted only over pairs of indices of the same kind: either two dotted indices or two undotted indices. Summation over one dotted and one undotted index is not invariant. Hence, from the spinor

$$\zeta^{\alpha_1 \alpha_2 \dots \alpha_k \dot{\beta}_1 \dot{\beta}_2 \dots \dot{\beta}_l}, \quad (17.10)$$

which is symmetric in all its k undotted indices and in all its l dotted indices, no spinor of lower rank can be formed (recall that simplification over a pair of indices in which the spinor is symmetric gives zero). This means that no smaller set of linear combinations can be formed from the quantities (17.10) such that the combinations transform among themselves under every group transformation. In other words, symmetric 4-spinors furnish irreducible representations of the proper Lorentz group. Each irreducible representation is specified by a pair (k, l) .

Since each spinor index has two possible values, there are $k + 1$ essentially distinct strings $\alpha_1 \alpha_2 \dots \alpha_k$ in (17.10), containing respectively $0, 1, 2, \dots, k$ occurrences of 1 and $k, k - 1, \dots, 0$ occurrences of 2. Likewise there are $l + 1$ strings $\dot{\beta}_1 \dot{\beta}_2 \dots \dot{\beta}_l$. A symmetric spinor of rank (k, l) therefore has a total of $(k + 1)(l + 1)$ independent components. This is the dimension of the irreducible representation it furnishes.

§ 18. Relation between spinors and 4-vectors

A spinor $\zeta^{\alpha\dot{\beta}}$ with one dotted and one undotted index has $2 \cdot 2 = 4$ independent components, precisely the number possessed by a 4-vector. It is therefore clear that the spinor and the 4-vector furnish the same irreducible representation of the proper Lorentz group, so a definite correspondence must exist between their components.

To establish this correspondence, we begin with its three-dimensional analogue, bearing in mind that 3- and 4-spinors must behave alike under purely spatial rotations.

For a three-dimensional spinor $\psi^{\alpha\beta}$, the correspondence formulas (see III, § 57) may be written as

$$\begin{aligned} a_x &= \frac{1}{2}(\psi^{22} - \psi^{11}) = \frac{1}{2}(\psi_1^2 + \psi_2^1), \\ a_y &= -\frac{i}{2}(\psi^{22} + \psi^{11}) = \frac{i}{2}(\psi_2^1 - \psi_1^2), \\ a_z &= \frac{1}{2}(\psi^{12} + \psi^{21}) = \frac{1}{2}(\psi_1^1 - \psi_2^2), \end{aligned}$$

where a_x , a_y , and a_z are the components of a certain three-dimensional vector **a**. In passing to four dimensions, the components ψ_β^α must be replaced by $\zeta^{\alpha\dot{\beta}}$, while a_x , a_y , and a_z are to be understood as the contravariant components a^1 , a^2 , and a^3 of a 4-vector.

The form of the fourth component, a^0 , follows at once from the circumstance noted in § 17: quantity (17.6) must transform as a^0 . Thus $a^0 \sim \zeta^{11} + \zeta^{22}$; the proportionality coefficient is chosen so that the scalar $\zeta_{\alpha\beta}\zeta^{\alpha\beta}$ equals the scalar $2a_\mu a^\mu = 2a^2$.

The resulting correspondence formulas are

$$\begin{aligned} a^1 &= \frac{1}{2}(\zeta^{12} + \zeta^{21}), & a^2 &= \frac{i}{2}(\zeta^{12} - \zeta^{21}), \\ a^3 &= \frac{1}{2}(\zeta^{11} - \zeta^{22}), & a^0 &= \frac{1}{2}(\zeta^{11} + \zeta^{22}). \end{aligned} \quad (18.1)$$

The inverse formulas are

$$\begin{aligned} \zeta^{11} &= \zeta_{22} = a^3 + a^0, & \zeta^{22} &= \zeta_{11} = a^0 - a^3, \\ \zeta^{12} &= -\zeta_{21} = a^1 - ia^2, & \zeta^{21} &= -\zeta_{12} = a^1 + ia^2. \end{aligned} \quad (18.2)$$

They give

$$\zeta_{\alpha\beta}\zeta^{\alpha\beta} = 2a^2. \quad (18.3)$$

We also note that

$$\zeta_{\alpha\beta}\zeta^{\gamma\beta} = \delta_\alpha^\gamma a^2. \quad (18.4)$$

The last equality follows because the second-rank spinor $\zeta_{\alpha\beta}\zeta^{\gamma\beta}$ is antisymmetric in $\alpha\gamma$ and is therefore proportional to the metric spinor.

The correspondence between $\zeta^{\alpha\beta}$ and a 4-vector is a special case of a general rule: every symmetric spinor of rank (k, k) is equivalent to a symmetric irreducible 4-tensor of rank k , where irreducibility here means that it vanishes upon simplification over any pair of indices.

The relation between a spinor and a 4-vector can be written compactly using the two-row Pauli matrices²:

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}. \quad (18.5)$$

Let ζ symbolically denote the matrix of the quantities $\zeta^{\alpha\beta}$ with upper indices, the first being undotted. Formulas (18.2) then take the form

$$\zeta = \mathbf{a}\sigma + a^0 \quad (18.6)$$

(the second term is, of course, understood to be the product of a^0 and the unit matrix). The inverse formulas are

$$\mathbf{a} = \frac{1}{2} \text{Tr}(\zeta\sigma), \quad a^0 = \frac{1}{2} \text{Tr} \zeta. \quad (18.7)$$

Formulas (18.6) and (18.7) allow the transformation laws of a 4-vector and a spinor to be related. The spinor transformation law can thereby be expressed in terms of the rotation parameters of the four-dimensional coordinate system.

²To simplify the notation, operators (matrices) acting on spin variables will be denoted by letters without hats.

Write the transformation of ξ^α as

$$\xi^{\alpha'} = (B\xi)^\alpha, \quad B = \begin{pmatrix} \alpha & \beta \\ \gamma & \delta \end{pmatrix}, \quad (18.8)$$

where B is the two-row matrix formed from the coefficients of the binary transformation. The dotted spinor then transforms according to

$$\eta^{\dot{\beta}'} = (B^* \eta)^{\dot{\beta}} = (\eta B^+)^{\dot{\beta}}, \quad (18.9)$$

and the transformation of the second-rank spinor $\zeta^{\alpha\dot{\beta}} \sim \xi^\alpha \eta^{\dot{\beta}}$ can be written symbolically as $\zeta' = B\zeta B^{+3}$. For an infinitesimal transformation $B = 1 + \lambda$, where λ is a small matrix, retention of terms through first order gives

$$\zeta' = \zeta + (\lambda\zeta + \zeta\lambda^+). \quad (18.10)$$

First consider a Lorentz transformation to a reference frame moving with infinitesimal velocity $\delta\mathbf{V}$, with no change in the directions of the spatial axes. The 4-vector $a^\mu = (a^0, \mathbf{a})$ then transforms as

$$\mathbf{a}' = \mathbf{a} - a^0 \delta\mathbf{V}, \quad a^{0'} = a^0 - \mathbf{a} \delta\mathbf{V}. \quad (18.11)$$

We now use (18.7). On the one hand, the transformation of a^0 may be written

$$a^{0'} = a^0 - \mathbf{a} \delta\mathbf{V} = a^0 - \frac{1}{2} \text{Tr}(\zeta \sigma \delta\mathbf{V}),$$

and, on the other hand,

$$a^{0'} = \frac{1}{2} \text{Tr} \zeta' = a^0 + \frac{1}{2} \text{Tr}(\lambda\zeta + \zeta\lambda^+) = a^0 + \frac{1}{2} \text{Tr} \zeta(\lambda + \lambda^+).$$

These expressions must agree identically, that is, for arbitrary ζ . It follows that

$$\lambda + \lambda^+ = -\sigma \delta\mathbf{V}.$$

Applying the same procedure to the transformation of $\mathbf{a}$ gives

$$\sigma\lambda + \lambda^+\sigma = -\delta\mathbf{V}.$$

Regarded as equations for λ , these relations have the solution

$$\lambda = \lambda^+ = -\frac{1}{2} \sigma \delta\mathbf{V}.$$

Thus the infinitesimal Lorentz transformation of ξ^α is effected by the matrix

$$B = 1 - \frac{1}{2}(\sigma \mathbf{n}) \delta V, \quad (18.12)$$

³For the covariant components,

$$\xi'_\alpha = (\tilde{B}^{-1} \xi)_\alpha = (\xi B^{-1})_\alpha, \quad \eta'_{\dot{\alpha}} = (\eta B^{*-1})_{\dot{\alpha}} \quad (18.8a)$$

(so that the product of two spinors $\xi_\alpha \Xi^\alpha$ remains invariant).

where $\mathbf{n}$ is a unit vector in the direction of the velocity $\delta\mathbf{V}$. The transformation for a finite velocity $\mathbf{V}$ is now readily obtained. Recall that, geometrically, a Lorentz transformation is a rotation of the four-dimensional coordinate system in the $t\mathbf{n}$ plane through an angle φ , related to $\mathbf{V}$ by $\tanh \varphi = V$ ⁴. An infinitesimal transformation corresponds to the angle $\delta\varphi = \delta V$, and a finite rotation through φ is obtained by repeating the rotation through $\delta\varphi$ a total of $\varphi/\delta\varphi$ times. Raising the operator (18.12) to the power $\varphi/\delta\varphi$ and then taking the limit $\delta\varphi \rightarrow 0$, we find

$$B = e^{-\frac{\varphi}{2}\mathbf{n}\sigma}. \quad (18.13)$$

The mathematical content of this operator becomes clear from the properties of the Pauli matrices: every even power of $\mathbf{n}\sigma$ is 1, whereas every odd power is $\mathbf{n}\sigma$. Since the expansion of cosh contains the even powers of its argument and that of sinh the odd powers, the final result is

$$B = \cosh \frac{\varphi}{2} - \mathbf{n}\sigma \sinh \frac{\varphi}{2}, \quad \tanh \varphi = V. \quad (18.14)$$

The Lorentz-transformation matrices B are thus Hermitian: $B = B^\dagger$.

We next consider an infinitesimal rotation of the spatial coordinate system. The three-dimensional vector $\mathbf{a}$ transforms as

$$\mathbf{a}' = \mathbf{a} - \delta\boldsymbol{\theta} \times \mathbf{a}, \quad (18.15)$$

where $\delta\boldsymbol{\theta}$ is the vector of the infinitesimal rotation angle. The corresponding spinor transformation could be found in the same way. This is unnecessary, however, because 4-spinors and 3-spinors behave alike under spatial rotations, while for the latter the transformation is already known from the general relation between the spin operator and the generator of an infinitesimal rotation:

$$B = 1 + \frac{i}{2}\boldsymbol{\sigma}\delta\boldsymbol{\theta}. \quad (18.16)$$

Passage to a finite rotation through an angle θ proceeds as in the passage from (18.12) to (18.14):

$$B = \exp\left(\frac{i\theta}{2}\mathbf{n}\sigma\right) = \cos \frac{\theta}{2} + i\mathbf{n}\sigma \sin \frac{\theta}{2}, \quad (18.17)$$

where $\mathbf{n}$ is a unit vector along the rotation axis. This matrix is unitary ($B^\dagger = B^{-1}$), as is required for a spatial rotation.

§ 19. Inversion of spinors

In presenting the three-dimensional theory of spinors in Volume III, we did not examine how they behave under spatial inversion, since in nonrelativistic theory this would yield no new physical results. We now consider the question in order to clarify the subsequent discussion of inversion properties of 4-spinors.

⁴Recall that the metric is pseudo-Euclidean in planes containing the time axis.

Inversion does not reverse an axial vector such as the spin vector. Its projection s_z is therefore unchanged. It follows that inversion can transform each component of a spinor ψ^α only into itself; hence

$$\psi^\alpha \rightarrow P\psi^\alpha, \quad (19.1)$$

where P is a constant coefficient. Applying inversion twice returns the coordinate frame to its initial position. For spinors, however, this return can be understood in two different ways: as a rotation through 0° or through 360° . These alternatives are not equivalent for spinors, since ψ^α changes sign under a 360° rotation. There are consequently two possible conceptions of inversion. In one,

$$P^2 = 1, \quad P = \pm 1, \quad (19.2)$$

whereas in the other,

$$P^2 = -1, \quad P = \pm i. \quad (19.3)$$

It is essential that inversion be defined in the same way for every spinor. Different spinors cannot be allowed to behave differently under inversion, according to (19.2) or (19.3), because then a scalar or pseudoscalar could not be formed from every pair of spinors. If ψ^α transformed according to (19.2) and φ^α according to (19.3), inversion would multiply $\psi^\alpha \varphi_\alpha$ by $\pm i$ rather than leaving it unchanged or merely reversing its sign.

It should also be stressed that, under either definition of inversion, assigning a particular parity P to a spinor has no absolute meaning. Spinors change sign under a rotation through 2π , which can always be performed together with inversion. The “relative parity” of two spinors does have an absolute meaning: it is defined as the parity of the scalar $\psi^\alpha \varphi_\alpha$ formed from them. Since a 2π rotation reverses all spinors simultaneously, the associated ambiguity does not affect the parity of this scalar.

We turn now to four-dimensional spinors.

First observe that inversion reverses only three coordinates, (x, y, z) , out of the four (t, x, y, z) . It therefore commutes with spatial rotations but not with transformations that rotate the t axis. If $\widehat{L}$ is the Lorentz transformation to a frame moving with velocity $\mathbf{V}$, then $\widehat{P}\widehat{L} = \widehat{L}'\widehat{P}$, where $\widehat{L}'$ is the transformation to the frame moving with velocity $-\mathbf{V}$.

It follows that under inversion the components of a 4-spinor ξ^α cannot transform among themselves. If inversion of ξ^α were still given by (19.1), and thus represented by a matrix proportional to the identity, it would commute with every Lorentz transformation. This is certainly impossible, since $\widehat{L}$ and $\widehat{L}'$ certainly do not act identically on ξ^α .

Thus inversion must transform the components of ξ^α into other quantities. The only possibilities are the components of another spinor $\eta^{\dot{\alpha}}$ whose transformation properties differ from those of ξ^α . Since inversion leaves the z projection of spin unchanged, as noted above, ξ^1 and ξ^2 can transform only into η_1 and η_2 , which correspond to the same values $s_z = 1/2$ and $s_z = -1/2$. If inversion is understood as an operation whose square is 1, its action may be defined by

$$\xi^\alpha \rightarrow \eta_{\dot{\alpha}}, \quad \eta_{\dot{\alpha}} \rightarrow \xi^\alpha. \quad (19.4)$$

For the covariant components ξ_α and contravariant components $\eta^{\dot{\alpha}}$ these transformations carry the opposite sign:

$$\xi_\alpha \rightarrow -\eta^{\dot{\alpha}}, \quad \eta^{\dot{\alpha}} \rightarrow -\xi_\alpha, \quad (19.4a)$$

because lowering and raising the same index introduce opposite signs; see (17.5) and (17.9)⁵. If inversion is instead understood so that $P^2 = -1$, its action is defined by

$$\xi^\alpha \rightarrow i\eta_{\dot{\alpha}}, \quad \eta_{\dot{\alpha}} \rightarrow i\xi^\alpha \quad (19.5)$$

or, equivalently,

$$\xi_\alpha \rightarrow -i\eta^{\dot{\alpha}}, \quad \eta^{\dot{\alpha}} \rightarrow -i\xi_\alpha. \quad (19.5a)$$

The two definitions of inversion differ in one respect. With the second definition, the complex-conjugate spinors transform by the same rule: if $\Xi_\alpha = \eta_{\dot{\alpha}}^*$ and $H^{\dot{\alpha}} = \xi^{\alpha*}$, then (19.5) gives $\Xi_\alpha \rightarrow -iH^{\dot{\alpha}}$ and $H^{\dot{\alpha}} \rightarrow -i\Xi_\alpha$, the same rule as for ξ_α and $\eta^{\dot{\alpha}}$. Definition (19.4), by contrast, would give $\Xi_\alpha \rightarrow H^{\dot{\alpha}}$ and $H^{\dot{\alpha}} \rightarrow \Xi_\alpha$, with the sign opposite to the transformation of ξ_α and $\eta^{\dot{\alpha}}$. We shall return in § 27 to possible physical aspects of this distinction.

Below, for definiteness, we shall always mean definition (19.5).

Under the rotation subgroup, ξ^α and $\eta_{\dot{\alpha}}$ transform alike, as we know. Forming the combinations

$$\xi^\alpha \pm \eta_{\dot{\alpha}}, \quad (19.6)$$

would give quantities transforming under inversion according to (19.1), with $P = \pm i$. These combinations, however, do not behave as spinors under every transformation of the Lorentz group.

Thus, including inversion in the symmetry group requires the pair $(\xi^\alpha, \eta_{\dot{\alpha}})$ to be considered together; such a pair is called a first-rank *bispinor*. Its four components realize one irreducible representation of the extended Lorentz group.

The scalar product of two bispinors $(\xi^\alpha, \eta_{\dot{\alpha}})$ and $(\Xi^\alpha, H_{\dot{\alpha}})$ can be formed in two ways. The quantity

$$\xi^\alpha \Xi_\alpha + \eta_{\dot{\alpha}} H^{\dot{\alpha}} \quad (19.7)$$

is wholly unchanged by inversion and is therefore a true scalar. The quantity

$$\xi^\alpha \Xi_\alpha - \eta_{\dot{\alpha}} H^{\dot{\alpha}} \quad (19.8)$$

is likewise invariant under rotations of the 4-dimensional coordinate frame, but reverses sign under inversion; it is therefore a pseudoscalar.

A second-rank spinor $\zeta^{\alpha\dot{\beta}}$ can also be defined in two ways. Defining it by

$$\zeta^{\alpha\dot{\beta}} \sim \xi^\alpha H^{\dot{\beta}} + \Xi^\alpha \eta^{\dot{\beta}}, \quad (19.9)$$

gives quantities that transform under inversion according to

$$\zeta^{\alpha\dot{\beta}} \rightarrow \zeta_{\dot{\alpha}\beta}. \quad (19.10)$$

⁵Definition (19.4) is of course conventional in a certain sense, because ξ^α and $\eta_{\dot{\alpha}}$ are independent quantities. For example, introduce in place of $\eta_{\dot{\alpha}}$ a new spinor $\eta'_{\dot{\alpha}} = e^{i\delta} \eta_{\dot{\alpha}}$. The equivalent definition replacing (19.4) is then

$$\xi^\alpha \rightarrow e^{-i\delta} \eta'_{\dot{\alpha}}, \quad \eta'_{\dot{\alpha}} \rightarrow e^{i\delta} \xi^\alpha.$$

The equivalent 4-vector a^μ then transforms, in accordance with (18.1), as $(a^0, \mathbf{a}) \rightarrow (a^0, -\mathbf{a})$. Thus it is a true 4-vector, while the three-dimensional vector $\mathbf{a}$ is polar.

Alternatively, $\zeta^{\alpha\beta}$ may be defined by

$$\zeta^{\alpha\beta} \sim \xi^\alpha H^\beta - \Xi^\alpha \eta^\beta. \quad (19.11)$$

Then⁶

$$\zeta^{\alpha\beta} \rightarrow -\zeta_{\alpha\beta}. \quad (19.12)$$

The corresponding 4-vector transforms under inversion as $(a^0, \mathbf{a}) \rightarrow (-a^0, \mathbf{a})$ and is therefore a 4-pseudovector; its three-dimensional vector $\mathbf{a}$ is axial.

Symmetric second-rank spinors with indices of the same type are defined by

$$\xi^{\alpha\beta} \sim \xi^\alpha \Xi^\beta + \xi^\beta \Xi^\alpha, \quad \eta_{\dot{\alpha}\dot{\beta}} \sim \eta_{\dot{\alpha}} H_{\dot{\beta}} + \eta_{\dot{\beta}} H_{\dot{\alpha}}. \quad (19.13)$$

Under inversion they turn into one another:

$$\xi^{\alpha\beta} \rightarrow -\eta_{\dot{\alpha}\dot{\beta}}. \quad (19.14)$$

The pair $(\xi^{\alpha\beta}, \eta_{\dot{\alpha}\dot{\beta}})$ is a second-rank bispinor. It has $3 + 3 = 6$ independent components. An antisymmetric second-rank 4-tensor $a^{\mu\nu}$ has the same number. A definite correspondence must therefore exist between the two; they realize equivalent irreducible representations of the extended Lorentz group.

Under the proper Lorentz group, $\xi^{\alpha\beta}$ and $\eta_{\dot{\alpha}\dot{\beta}}$ transform independently. The components of $a^{\mu\nu}$ can accordingly be divided into two sets of quantities, each of which transforms only within itself under every rotation of the 4-dimensional coordinate frame. This division is made as follows.

Introduce a three-dimensional polar vector $\mathbf{p}$ and a three-dimensional axial vector $\mathbf{a}$, related to the components of $a^{\mu\nu}$ by

$$a^{\mu\nu} = \begin{pmatrix} 0 & p_x & p_y & p_z \\ -p_x & 0 & -a_z & a_y \\ -p_y & a_z & 0 & -a_x \\ -p_z & -a_y & a_x & 0 \end{pmatrix} \equiv (\mathbf{p}, \mathbf{a}) \quad (19.15)$$

(we shall use $(\mathbf{p}, \mathbf{a})$ as a concise notation for listing the components of such a tensor). Then $a_{\mu\nu} = (-\mathbf{p}, \mathbf{a})$, and of the two quantities

$$\mathbf{a}^2 - \mathbf{p}^2 = \frac{1}{2} a_{\mu\nu} a^{\mu\nu}, \quad \mathbf{a}\mathbf{p} = \frac{1}{8} e_{\mu\nu\rho\sigma} a^{\mu\nu} a^{\rho\sigma}$$

the first is a scalar and the second a pseudoscalar; both are equally invariant under the proper Lorentz group. The squares of the three-dimensional vectors $\mathbf{f}^\pm = \mathbf{p} \pm i\mathbf{a}$ are invariant along with them. Thus every rotation in 4-space acts on $\mathbf{f}^\pm$ like a “rotation” in three-dimensional space, generally through complex angles: the six rotation angles in

⁶We emphasize that the transformation laws (19.10) and (19.12), which differ by the sign on the right, are by no means equivalent, because both sides of each contain components of the same spinor (compare the note on p. 92).

4-space correspond to three complex “rotation angles” of the three-dimensional frame. Spatial inversion reverses $\mathbf{p}$ but not $\mathbf{a}$, and consequently interchanges $\mathbf{f}^+$ and $-\mathbf{f}^-$. The components of these vectors constitute the required two sets formed from the components of $a^{\mu\nu}$.

The correspondence between the components of $a^{\mu\nu}$ and those of $\xi^{\alpha\beta}$ and $\eta_{\dot{\alpha}\dot{\beta}}$ is now also evident. Since spatial rotations form a subgroup of the Lorentz group, the relations between the spinor components and those of the three-dimensional vector must be the same as for three-dimensional spinors:

$$\begin{aligned} f_x^+ &= \frac{1}{2}(\xi^{22} - \xi^{11}), & f_y^+ &= \frac{i}{2}(\xi^{22} + \xi^{11}), & f_z^+ &= \xi^{12}; \\ f_x^- &= \frac{1}{2}(\eta_{\dot{2}\dot{2}} - \eta_{\dot{1}\dot{1}}), & f_y^- &= \frac{i}{2}(\eta_{\dot{2}\dot{2}} + \eta_{\dot{1}\dot{1}}), & f_z^- &= \eta_{\dot{1}\dot{2}}. \end{aligned} \quad (19.16)$$

PROBLEM

Establish the general correspondence between spinors of even rank and 4-tensors.

SOLUTION. All spinors with even $k + l$ realize single-valued irreducible representations of the extended Lorentz group and are therefore equivalent to 4-tensors realizing the same representations⁷.

A spinor of rank (k, k) transforms under inversion according to

$$\zeta^{\alpha\beta\dots\dot{\gamma}\dot{\delta}\dots} \rightarrow \pm \zeta_{\dot{\alpha}\dot{\beta}\dots\gamma\delta\dots}. \quad (1)$$

Such a spinor is equivalent to a symmetric irreducible 4-tensor of rank k , which is a true tensor or pseudotensor according to the sign in (1).

Spinors of ranks (k, l) and (l, k) , which together form a bispinor, transform under inversion as

$$\underbrace{\zeta^{\alpha\beta\dots}}_k \underbrace{\dot{\gamma}\dot{\delta}\dots}_l \rightarrow (-1)^{(k-l)/2} \underbrace{\chi_{\dot{\alpha}\dot{\beta}\dots}}_k \underbrace{\gamma\delta\dots}_l. \quad (2)$$

For $l = k + 2$, the bispinor is equivalent to an irreducible 4-tensor $a_{[\mu\nu]\rho\sigma\dots}$ of rank $k + 2$, antisymmetric in the indices $[\mu\nu]$ and symmetric in all its other indices. Irreducibility means that contraction over any pair of indices gives zero and that forming the dual over any three indices gives zero, that is, $e^{\lambda\mu\nu\sigma} a_{[\mu\nu]\rho\sigma\dots} = 0$. The latter condition says that the cyclic sum over three indices— $\mu\nu$ and any one of the others—vanishes.

For $l = k + 4$, the bispinor is equivalent to an irreducible 4-tensor $a_{[\lambda\mu][\nu\rho]\sigma\tau\dots}$ of rank $k + 4$ with the following properties: it is antisymmetric in each pair $[\lambda\mu]$ and $[\nu\rho]$, symmetric in all remaining indices, symmetric under interchange of the pair $[\lambda\mu]$ with $[\nu\rho]$, zero upon contraction over any pair of indices, and zero upon forming the dual over any three indices.

⁷Spinors of odd rank realize double-valued representations of the group: a spatial rotation through 360° reverses their sign, so every group element corresponds to two matrices of opposite sign.

In general, for $l = k + 2n$, the bispinor is equivalent to an irreducible 4-tensor of rank $k + 2n$, antisymmetric in n pairs of indices and symmetric in the remaining k indices. Tensors antisymmetric in larger sets of indices—triples, quadruples, and so forth—do not occur in this classification for an evident reason. A third-rank antisymmetric tensor is equivalent (dual) to a pseudovector, while a fourth-rank antisymmetric tensor reduces to a scalar, being proportional to the unit pseudotensor $e^{\lambda\mu\nu\rho}$. Antisymmetry in still more indices is impossible in 4-space.

§ 20. The Dirac equation in spinor form

In its rest frame, a spin-1/2 particle is described by a two-component wave function, a 3-spinor. By its “four-dimensional origin,” this may be either an undotted or a dotted 4-spinor. Both kinds of 4-spinor enter the description of the particle in an arbitrary reference frame; denote them by ξ^α and $\eta_{\dot{\alpha}}$ ⁸.

For a free particle, the only operator that can occur in the wave equation is, as already stated in § 10, the 4-momentum operator $\widehat{p}_\mu = i\partial_\mu$. In spinor notation this 4-vector corresponds to the operator-valued spinor $\widehat{p}_{\alpha\dot{\beta}}$, with

$$\begin{aligned}\widehat{p}^{\dot{1}1} &\equiv \widehat{p}_{22} = \widehat{p}_0 + \widehat{p}_z, & \widehat{p}^{2\dot{2}} &\equiv \widehat{p}_{11} = \widehat{p}_0 - \widehat{p}_z, \\ \widehat{p}^{\dot{1}2} &= -\widehat{p}_{2\dot{1}} = \widehat{p}_x - i\widehat{p}_y, & \widehat{p}^{2\dot{1}} &= -\widehat{p}_{1\dot{2}} = \widehat{p}_x + i\widehat{p}_y.\end{aligned}\quad (20.1)$$

The wave equation is a linear differential relation among the spinor components, effected by $\widehat{p}_{\alpha\dot{\beta}}$. Relativistic invariance fixes the following system:

$$\widehat{p}^{\alpha\dot{\beta}}\eta_{\dot{\beta}} = m\xi^\alpha, \quad \widehat{p}_{\dot{\beta}\alpha}\xi^\alpha = m\eta_{\dot{\beta}}, \quad (20.2)$$

where m is a dimensionful constant. It would serve no purpose to use different constants m_1 and m_2 in the two equations, or to change the sign preceding m : a suitable redefinition of either ξ^α or $\eta_{\dot{\alpha}}$ would in every case restore the equations to the form above.

Eliminate one of the two spinors from (20.2) by substituting $\eta_{\dot{\beta}}$ from the second equation into the first:

$$\widehat{p}^{\alpha\dot{\beta}}\eta_{\dot{\beta}} = \frac{1}{m}\widehat{p}^{\alpha\dot{\beta}}\widehat{p}_{\gamma\dot{\beta}}\xi^\gamma = m\xi^\alpha.$$

According to (18.4), however, $\widehat{p}^{\alpha\dot{\beta}}\widehat{p}_{\gamma\dot{\beta}} = \widehat{p}^2\delta^\alpha_\gamma$, and therefore

$$(\widehat{p}^2 - m^2)\xi^\gamma = 0. \quad (20.3)$$

It follows that m is the mass of the particle.

The need for a mass in the wave equation requires the simultaneous consideration of two spinors, ξ^α and $\eta_{\dot{\alpha}}$: with either one alone, no relativistically invariant equation containing a dimensionful parameter can be formed. The wave equation is thereby automatically invariant under spatial inversion if the wave-function transformation is defined by

⁸A first-rank three-dimensional spinor may also “originate” from 4-spinors of higher odd rank which, in the rest frame, become antisymmetric in one or more pairs of indices. Such possibilities, however, would lead to equations of higher order (cf. the note on p. 52).

$$P : \quad \xi^\alpha \rightarrow i\eta_{\dot{\alpha}}, \quad \eta_{\dot{\alpha}} \rightarrow i\xi^\alpha. \quad (20.4)$$

It is readily seen that this replacement, accompanied by the replacement $\widehat{p}^{\dot{\alpha}\beta} \rightarrow \widehat{p}_{\alpha\dot{\beta}}$ evident from (20.1), interchanges the two equations (20.2). Two spinors which pass into one another under inversion together form a four-component quantity, a bispinor.

The relativistic wave equation represented by the system (20.2) is called the *Dirac equation*; it was established by *Dirac* in 1928. For its further study and use, we shall examine several forms in which it can be expressed.

Using (18.6), equations (20.2) become

$$(\widehat{p}_0 + \widehat{\mathbf{p}}\boldsymbol{\sigma})\eta = m\xi, \quad (\widehat{p}_0 - \widehat{\mathbf{p}}\boldsymbol{\sigma})\xi = m\eta. \quad (20.5)$$

Here ξ and η denote the two-component quantities

$$\xi = \begin{pmatrix} \xi^1 \\ \xi^2 \end{pmatrix}, \quad \eta = \begin{pmatrix} \eta_1 \\ \eta_2 \end{pmatrix} \quad (20.6)$$

(the first with upper indices and the second with lower indices). Here and below, multiplication of a matrix σ by any two-component quantity f always means ordinary matrix multiplication:

$$(\sigma f)_\alpha = \sigma_{\alpha\beta} f_\beta. \quad (20.7)$$

Writing f as the vertical column $\begin{pmatrix} f_1 \\ f_2 \end{pmatrix}$ corresponds to multiplying each row of σ by the column f .

For convenient reference, we once more write out the Pauli matrices,

$$\sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad (20.8)$$

and recall their principal properties:

$$\sigma_i \sigma_k + \sigma_k \sigma_i = 2\delta_{ik}, \quad \sigma_i \sigma_k = i e_{ikl} \sigma_l + \delta_{ik} \quad (20.9)$$

(see III, § 55).

We shall also write the wave equation satisfied by the complex-conjugate wave function, composed of the spinors

$$\xi^* = \left(\xi^{1*}, \xi^{2*} \right), \quad \eta^* = \left(\eta_1^*, \eta_2^* \right). \quad (20.10)$$

Every operator $\widehat{p}_\mu$ contains the factor i , so $\widehat{p}_\mu^* = -\widehat{p}_\mu$. On taking the complex conjugate of both sides of (20.5), one must also use the Hermiticity of the σ matrices ($\sigma^* = \overline{\sigma}$):

$$(\sigma f)_\alpha^* = \sigma_{\alpha\beta}^* f_\beta^* = f_\beta^* \sigma_{\beta\alpha} = (f^* \sigma)_\alpha.$$

The resulting equations are

$$\eta^* (\widehat{p}_0 + \widehat{\mathbf{p}}\boldsymbol{\sigma}) = -m\xi^*, \quad \xi^* (\widehat{p}_0 - \widehat{\mathbf{p}}\boldsymbol{\sigma}) = -m\eta^*. \quad (20.11)$$

In this notation the operators $\widehat{p}_\mu$ are understood conventionally to act on the function placed to their left. Writing ξ^* and η^* as horizontal rows matches the matrix multiplication in these equations: the row f^* is multiplied by the columns of the matrices σ ,

$$(f^* \sigma)_\alpha = f_\beta^* \sigma_{\beta\alpha}. \quad (20.12)$$

The inversion transformation of ξ^* and η^* is defined as the complex conjugate of (20.4):

$$P : \quad \xi^{\alpha*} \rightarrow -i\eta_{\dot{\alpha}}^*, \quad \eta_{\dot{\alpha}}^* \rightarrow -i\xi^{\alpha*}. \quad (20.13)$$

§ 21. Symmetric form of the Dirac equation

The spinor form of the Dirac equation is the most natural one in that it displays its relativistic invariance directly. In applications, however, other representations of the wave equation may be more convenient; these result from choosing a different set of four independent components of the wave function.

Let the four-component wave function be denoted by ψ (with components ψ_i , $i = 1, 2, 3, 4$). In the spinor representation it is a bispinor:

$$\psi = \begin{pmatrix} \xi \\ \eta \end{pmatrix}. \quad (21.1)$$

But any linear combinations of the components of the spinors ξ and η may equally well be chosen as the independent components of ψ ⁹. We shall restrict the admissible linear transformations only by requiring them to be unitary; such transformations leave unchanged the bilinear forms constructed from ψ and ψ^* (see § 28).

For an arbitrary choice of the components of ψ , the Dirac equation can in general be written as

$$\widehat{p}_\mu \gamma_{ik}^\mu \psi_k = m \psi_i,$$

where γ^μ ($\mu = 0, 1, 2, 3$) are certain four-by-four matrices (the *Dirac matrices*). We shall usually suppress the matrix indices and write this equation symbolically as

$$(\gamma \widehat{p} - m) \psi = 0, \quad (21.2)$$

where

$$\gamma \widehat{p} \equiv \gamma^\mu \widehat{p}_\mu = \widehat{p}_0 \gamma^0 - \widehat{\mathbf{p}} \boldsymbol{\gamma} = i \gamma^0 \frac{\partial}{\partial t} + i \boldsymbol{\gamma} \nabla, \quad \boldsymbol{\gamma} = (\gamma^1, \gamma^2, \gamma^3).$$

In the spinor form of the equation, with the components of ψ given by (21.1), the corresponding matrices are¹⁰

$$\gamma^0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \boldsymbol{\gamma} = \begin{pmatrix} 0 & -\boldsymbol{\sigma} \\ \boldsymbol{\sigma} & 0 \end{pmatrix}, \quad (21.3)$$

⁹For brevity, we shall call the four-component quantity ψ a bispinor even in its nonspinor representations.

¹⁰Here and below, four-by-four matrices are written compactly in terms of two-by-two matrices: each symbol in (21.3) represents a two-by-two matrix.

as is readily seen by writing equations (20.5) in the form

$$\begin{pmatrix} 0 & \widehat{p}_0 + \widehat{\mathbf{p}}\boldsymbol{\sigma} \\ \widehat{p}_0 - \widehat{\mathbf{p}}\boldsymbol{\sigma} & 0 \end{pmatrix} \begin{pmatrix} \xi \\ \eta \end{pmatrix} = m \begin{pmatrix} \xi \\ \eta \end{pmatrix}$$

and comparing with (21.2).

In general, the matrices γ need obey only the conditions that ensure the equality $\widehat{p}^2 = m^2$. To find these conditions, multiply equation (21.2) on the left by $\gamma\widehat{p}$. This gives

$$(\gamma^\mu \widehat{p}_\mu) (\gamma^\nu \widehat{p}_\nu) \psi = m (\widehat{p}_\mu \gamma^\mu) \psi = m^2 \psi.$$

Since $\widehat{p}_\mu \widehat{p}_\nu$ is a symmetric tensor (all the operators $\widehat{p}_\mu$ commute), the equality may be rewritten

$$\frac{1}{2} \widehat{p}_\mu \widehat{p}_\nu (\gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu) \psi = m^2 \psi,$$

which shows that

$$\gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu = 2g^{\mu\nu}. \quad (21.4)$$

Thus every pair of distinct matrices γ^μ anticommutes, while their squares are

$$(\gamma^1)^2 = (\gamma^2)^2 = (\gamma^3)^2 = -1, \quad (\gamma^0)^2 = 1. \quad (21.5)$$

Under an arbitrary unitary transformation of the components of ψ ($\psi' = U\psi$, where U is a unitary four-by-four matrix), the matrices γ transform according to

$$\gamma' = U\gamma U^{-1} = U\gamma U^+ \quad (21.6)$$

(so that the equation $(\gamma\widehat{p} - m)\psi = 0$ becomes $(\gamma'\widehat{p} - m)\psi' = 0$). The commutation relations (21.4) of course remain unchanged.

The matrix γ^0 in (21.3) is Hermitian, whereas the matrices $\boldsymbol{\gamma}$ are anti-Hermitian. These properties are retained under every unitary transformation (21.6), and consequently we shall always have¹¹:

$$\gamma^+ = -\boldsymbol{\gamma}, \quad \gamma^{0+} = \gamma^0. \quad (21.7)$$

We also write the equation for the complex-conjugate function ψ^* . Taking the complex conjugate of equation (21.2) and using (21.7), we obtain

$$(\widehat{p}_0 \widetilde{\gamma}^0 - \widetilde{\mathbf{p}}\widetilde{\boldsymbol{\gamma}} - m)\psi^* = 0.$$

Move ψ^* to the left by means of $\widetilde{\gamma}^\mu \psi^* = \psi^* \gamma^\mu$, and then multiply the equation on the right by γ^0 . Noting that $\boldsymbol{\gamma}\gamma^0 = -\gamma^0\boldsymbol{\gamma}$ and introducing the new bispinor

$$\bar{\psi} = \psi^* \gamma^0, \quad \psi^* = \bar{\psi} \gamma^0, \quad (21.8)$$

we find

$$\bar{\psi}(\gamma\widehat{p} + m) = 0. \quad (21.9)$$

¹¹These equalities can be combined in the form

$$\gamma^{\lambda+} = \gamma^0 \gamma^\lambda \gamma^0. \quad (21.7 \text{ a})$$

As in (20.11), the operator $\widehat{p}$ is understood here to act on the function to its left. The function $\bar{\psi}$ is called the *Dirac-conjugate* (or *relativistically conjugate*) of ψ . The purpose of the factor γ^0 in its definition is that, in the spinor representation, it interchanges the spinors ξ^* and η^* so that in $\bar{\psi} = (\eta^*, \xi^*)$ the undotted spinor comes first and the dotted one second, just as in ψ . This is precisely why $\bar{\psi}$ is a more natural “partner” of ψ than ψ^* when, for example, they occur together in various bilinear combinations (see § 28).

The inversion transformation of the wave function may be written

$$P: \quad \psi \rightarrow i\gamma^0\psi, \quad \bar{\psi} \rightarrow -i\bar{\psi}\gamma^0. \quad (21.10)$$

When ψ is in the spinor representation, the matrix γ^0 interchanges the components ξ and η , as inversion requires. The invariance of the Dirac equation under transformation (21.10) is also evident directly in the general case. Replacing $\widehat{p} \rightarrow -\widehat{p}$ in equation (21.2) and simultaneously replacing $\psi \rightarrow i\gamma^0\psi$, we obtain

$$(\widehat{p}_0\gamma^0 + \widehat{\mathbf{p}}\boldsymbol{\gamma} - m)\gamma^0\psi = 0.$$

Multiplying this equation on the left by γ^0 and using the anticommutation of γ^0 and $\boldsymbol{\gamma}$ returns us to the original equation.

Multiply $(\gamma\widehat{p} - m)\psi = 0$ on the left by $\bar{\psi}$, multiply $\bar{\psi}(\gamma\widehat{p} + m) = 0$ on the right by ψ , and add the results. We get

$$\bar{\psi}\gamma^\mu(\widehat{p}_\mu\psi) + (\widehat{p}_\mu\bar{\psi})\gamma^\mu\psi = \widehat{p}_\mu(\bar{\psi}\gamma^\mu\psi) = 0,$$

where the parentheses indicate the function on which the operator $\widehat{p}$ acts. The resulting equality has the form of the continuity equation $\partial_\mu j^\mu = 0$, so that

$$j^\mu = \bar{\psi}\gamma^\mu\psi = (\psi^*\psi, \psi^*\gamma^0\boldsymbol{\gamma}\psi) \quad (21.11)$$

is the 4-vector of particle current density. Its time component $j^0 = \psi^*\psi$ is positive definite.

The Dirac equation may be put in a form solved for the time derivative:

$$i\frac{\partial\psi}{\partial t} = \widehat{H}\psi, \quad (21.12)$$

where $\widehat{H}$ is the particle Hamiltonian¹². It is enough to multiply equation (21.2) on the left by γ^0 . The Hamiltonian is then

$$\widehat{H} = \boldsymbol{\alpha}\widehat{\mathbf{p}} + \beta m, \quad (21.13)$$

where the customary notation for the matrices occurring here has been introduced:

$$\boldsymbol{\alpha} = \gamma^0\boldsymbol{\gamma}, \quad \beta = \gamma^0. \quad (21.14)$$

¹²For a spin-zero particle the wave equation could not be written in this form: equation (10.5) for the scalar ψ is second order in time, while the first-order system (10.4) for the five-component quantity (ψ, ψ_μ) contains time derivatives of only some of the components.

We note that

$$\alpha_i \alpha_k + \alpha_k \alpha_i = 2\delta_{ik}, \quad \beta \alpha + \alpha \beta = 0, \quad \beta^2 = 1, \quad (21.15)$$

that is, all the matrices α and β anticommute pairwise, and their squares equal 1; all of them are Hermitian. In the spinor representation,

$$\alpha = \begin{pmatrix} \sigma & 0 \\ 0 & -\sigma \end{pmatrix}, \quad \beta = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}. \quad (21.16)$$

In the low-velocity limit, a particle must be described by only one two-component spinor, as in nonrelativistic theory. Indeed, taking the limit $\mathbf{p} \rightarrow 0$, $\varepsilon \rightarrow m$ in equations (20.5), we obtain $\xi = \eta$: the two spinors making up the bispinor coincide. Here, however, a shortcoming of the spinor form of the Dirac equation becomes apparent. All four components of ψ remain nonzero in the limiting process, although only two of them are actually independent. It is more convenient to use a representation of ψ in which two components tend to zero in this limit.

Accordingly, replace ξ and η by their linear combinations φ and χ :

$$\psi = \begin{pmatrix} \varphi \\ \chi \end{pmatrix}, \quad \varphi = \frac{1}{\sqrt{2}}(\xi + \eta), \quad \chi = \frac{1}{\sqrt{2}}(\xi - \eta). \quad (21.17)$$

For a particle at rest, $\chi = 0$. We shall call this representation of ψ the *standard representation*. Under inversion, φ and χ transform separately according to

$$P: \quad \varphi \rightarrow i\varphi, \quad \chi \rightarrow -i\chi. \quad (21.18)$$

The equations for φ and χ follow by adding and subtracting equations (20.5):

$$\widehat{p}_0 \varphi - \widehat{\mathbf{p}} \boldsymbol{\sigma} \chi = m\varphi, \quad -\widehat{p}_0 \chi + \widehat{\mathbf{p}} \boldsymbol{\sigma} \varphi = m\chi. \quad (21.19)$$

It follows that the matrices in the standard representation are

$$\gamma^0 \equiv \beta = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \quad \boldsymbol{\gamma} = \begin{pmatrix} 0 & \boldsymbol{\sigma} \\ -\boldsymbol{\sigma} & 0 \end{pmatrix}, \quad \alpha = \begin{pmatrix} 0 & \boldsymbol{\sigma} \\ \boldsymbol{\sigma} & 0 \end{pmatrix}. \quad (21.20)$$

Because (21.17) combines the first components of ξ and η separately from their second components, in the standard representation, just as in the spinor representation, the components ψ_1 and ψ_3 correspond to the eigenvalue $+1/2$ of the spin projection, while ψ_2 and ψ_4 correspond to $-1/2$. In both representations, therefore, the matrix $\frac{1}{2}\boldsymbol{\Sigma}$, where

$$\boldsymbol{\Sigma} = \begin{pmatrix} \boldsymbol{\sigma} & 0 \\ 0 & \boldsymbol{\sigma} \end{pmatrix}, \quad (21.21)$$

is the three-dimensional spin operator: when $\frac{1}{2}\Sigma_z$ acts on a bispinor containing only the components ψ_1, ψ_3 , or only ψ_2, ψ_4 , the bispinor is multiplied by $+1/2$ or $-1/2$, respectively. In an arbitrary representation this matrix may be written

$$\boldsymbol{\Sigma} = -\boldsymbol{\alpha} \boldsymbol{\gamma}^5 = -\frac{i}{2}(\boldsymbol{\alpha} \times \boldsymbol{\alpha}) \quad (21.22)$$

(for the definition of $\boldsymbol{\gamma}^5$, see (22.14) below).

PROBLEM

1. Find the transformation laws of the wave function under an infinitesimal Lorentz transformation and an infinitesimal spatial rotation.

SOLUTION. In the spinor representation, under an infinitesimal Lorentz transformation ψ obeys

$$\xi' = \left(1 - \frac{1}{2}\sigma\delta\mathbf{V}\right)\xi, \quad \eta' = \left(1 + \frac{1}{2}\sigma\delta\mathbf{V}\right)\eta$$

(see (18.8), (18.8 a), (18.12)). The two formulas may be combined as

$$\psi' = \left(1 - \frac{1}{2}\alpha\delta\mathbf{V}\right)\psi. \quad (1)$$

Similarly, the transformation law under an infinitesimal rotation is

$$\psi' = \left(1 + \frac{i}{2}\Sigma\delta\theta\right)\psi. \quad (2)$$

Written in this way, the formulas hold in any representation of ψ , provided α and Σ are understood to be the matrices in that same representation.

It is readily verified that the matrices α and Σ form the components of an antisymmetric “matrix 4-tensor”

$$\sigma^{\mu\nu} = \frac{1}{2}(\gamma^\mu\gamma^\nu - \gamma^\nu\gamma^\mu) = (\alpha, i\Sigma)$$

(the components are listed according to the rule (19.15)). Introduce also the infinitesimal antisymmetric tensor $\delta\varepsilon^{\mu\nu} = (\delta\mathbf{V}, \delta\theta)$. Then

$$\sigma^{\mu\nu}\delta\varepsilon_{\mu\nu} = 2i\Sigma\delta\theta - 2\gamma\delta\mathbf{V}$$

and formulas (1), (2) may both be written in the unified form

$$\psi' = \left(1 + \frac{1}{4}\sigma^{\mu\nu}\delta\varepsilon_{\mu\nu}\right)\psi. \quad (3)$$

2. Write the Dirac equation in a representation in which it contains no imaginary coefficients (*E. Majorana*, 1937).

SOLUTION. In the standard representation, in the equation

$$\left(\frac{\partial}{\partial t} + \alpha_x \frac{\partial}{\partial x} + \alpha_y \frac{\partial}{\partial y} + \alpha_z \frac{\partial}{\partial z} + i\beta\right)\psi = 0$$

only the matrices α_y and $i\beta$ are imaginary. This imaginary character can be removed by making a transformation $\psi' = U\psi$ that interchanges the imaginary matrix α_y with the real matrix β . For this purpose set

$$U = \frac{1}{\sqrt{2}}(\alpha_y + \beta) = U^{-1}.$$

Then

$$\alpha'_x = U\alpha_x U = -\alpha_x, \quad \alpha'_y = \beta, \quad \alpha'_z = -\alpha_z, \quad \beta' = \alpha_y,$$

and the Dirac equation becomes

$$\left(\frac{\partial}{\partial t} - \alpha_x \frac{\partial}{\partial x} + \beta \frac{\partial}{\partial y} - \alpha_z \frac{\partial}{\partial z} + im\alpha_y \right) \psi' = 0,$$

in which every coefficient is real.

§ 22. Algebra of the Dirac Matrices

Calculations involving the Dirac equation make extensive use of the γ matrices without requiring their explicit form in any particular representation. The rules for manipulating these matrices follow entirely from the anticommutation relations

$$\gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu = 2g^{\mu\nu}, \quad \mu, \nu = 0, 1, 2, 3, \quad (22.1)$$

which express all their general properties.

Here we collect several formulas and rules from the algebra of the γ matrices that are useful in calculations.

The “scalar product” of the γ matrices with themselves is $g_{\mu\nu} \gamma^\mu \gamma^\nu = 4$. For conciseness, in analogy with covariant components of 4-vectors, we introduce the notation $\gamma_\mu = g_{\mu\nu} \gamma^\nu$. Then

$$\gamma_\mu \gamma^\mu = 4. \quad (22.2)$$

If γ_μ and γ^μ are separated by one or more γ factors, repeated use of (22.1) brings them into neighboring positions, after which the sum over μ is evaluated with (22.2). In this way one obtains

$$\begin{aligned} \gamma_\mu \gamma^\nu \gamma^\mu &= -2\gamma^\nu, \\ \gamma_\mu \gamma^\lambda \gamma^\nu \gamma^\mu &= 4g^{\lambda\nu}, \\ \gamma_\mu \gamma^\lambda \gamma^\nu \gamma^\rho \gamma^\mu &= -2\gamma^\rho \gamma^\nu \gamma^\lambda, \\ \gamma_\mu \gamma^\lambda \gamma^\nu \gamma^\rho \gamma^\sigma \gamma^\mu &= 2(\gamma^\sigma \gamma^\lambda \gamma^\nu \gamma^\rho + \gamma^\rho \gamma^\nu \gamma^\lambda \gamma^\sigma). \end{aligned} \quad (22.3)$$

Usually the factors $\gamma^\mu, \dots$ occur together with different 4-vectors in the form of “scalar products”¹³

$$\gamma a \equiv \gamma^\mu a_\mu. \quad (22.4)$$

For these products, formulas (22.1) become

$$(a\gamma)(b\gamma) + (b\gamma)(a\gamma) = 2(ab), \quad (a\gamma)(a\gamma) = a^2 \quad (22.5)$$

and formulas (22.3) take the form

$$\begin{aligned} \gamma_\mu (a\gamma) \gamma^\mu &= -2(a\gamma), \\ \gamma_\mu (a\gamma)(b\gamma) \gamma^\mu &= 4(ab), \\ \gamma_\mu (a\gamma)(b\gamma)(c\gamma) \gamma^\mu &= -2(c\gamma)(b\gamma)(a\gamma), \\ \gamma_\mu (a\gamma)(b\gamma)(c\gamma)(d\gamma) \gamma^\mu &= 2[(d\gamma)(a\gamma)(b\gamma)(c\gamma) + (c\gamma)(b\gamma)(a\gamma)(d\gamma)]. \end{aligned} \quad (22.6)$$

¹³In this edition we use no special symbol for such a product. Letters carrying a hat or a slash are frequently used for it in the literature.

A frequently used operation is the trace of a product containing some number of γ matrices. Consider the quantities

$$T^{\mu_1 \mu_2 \dots \mu_n} \equiv \frac{1}{4} \text{Tr}(\gamma^{\mu_1} \gamma^{\mu_2} \dots \gamma^{\mu_n}). \quad (22.7)$$

The familiar property of the trace of a matrix product makes this tensor symmetric under cyclic permutations of the indices $\mu_1 \mu_2 \dots \mu_n$.

Because the γ matrices have the same form in every reference frame, the quantities T are likewise independent of the frame chosen. They therefore form a tensor expressible solely in terms of the metric tensor $g_{\mu\nu}$, which has the same property.

Only tensors of even rank can be constructed from the rank-two tensor $g_{\mu\nu}$. It follows at once that the trace of a product containing any odd number of γ factors vanishes. In particular, every individual γ matrix has zero trace¹⁴:

$$\text{Tr} \gamma^\mu = 0. \quad (22.8)$$

The trace of the identity matrix of order four (implicitly present on the right-hand side of the anticommutation relation (22.1)) equals 4. Taking the trace of both sides of (22.1) consequently gives

$$T^{\mu\nu} = g^{\mu\nu}. \quad (22.9)$$

For the trace of a product of four matrices one has

$$T^{\lambda\mu\nu\rho} = g^{\lambda\mu} g^{\nu\rho} - g^{\lambda\nu} g^{\mu\rho} + g^{\lambda\rho} g^{\mu\nu}. \quad (22.10)$$

For example, this formula can be derived by moving the factor γ^λ to the right through $\text{Tr}(\gamma^\lambda \gamma^\mu \gamma^\nu \gamma^\rho)$ with the aid of (22.1). Each interchange produces one of the terms appearing in (22.10):

$$T^{\lambda\mu\nu\rho} = 2g^{\lambda\mu} T^{\nu\rho} - T^{\mu\lambda\nu\rho} = 2g^{\lambda\mu} g^{\nu\rho} - T^{\mu\lambda\nu\rho}$$

and so on. Once all interchanges have been made, the right-hand side retains $-T^{\mu\nu\rho\lambda} = -T^{\lambda\mu\nu\rho}$, which is transferred to the left. The same procedure reduces the trace of a product of six γ matrices to traces of products of four factors, and so forth. Thus,

$$\begin{aligned} T^{\lambda\mu\nu\rho\sigma\tau} &= g^{\lambda\mu} T^{\nu\rho\sigma\tau} - g^{\lambda\nu} T^{\mu\rho\sigma\tau} + g^{\lambda\rho} T^{\mu\nu\sigma\tau} - \\ &\quad - g^{\lambda\sigma} T^{\mu\nu\rho\tau} + g^{\lambda\tau} T^{\mu\nu\rho\sigma}. \end{aligned} \quad (22.11)$$

We note that every trace $T^{\lambda\mu\dots}$ is real and can be nonzero only when each of the matrices $\gamma^0, \gamma^1, \dots$ occurs an even number of times in the product; both statements are evident from the formulas just obtained. It then follows readily that reversing the order of all factors leaves the trace unchanged:

$$T^{\lambda\mu\dots\rho\sigma} = T^{\sigma\rho\dots\mu\lambda}. \quad (22.12)$$

¹⁴The trace of a matrix is invariant under transformations $\gamma = U\gamma U^{-1}$. Equation (22.8) is therefore also evident from the explicit matrices in (21.3).

As already mentioned, γ factors usually appear in scalar products with various 4-vectors. In such cases formulas (22.9) and (22.10), for example, mean that

$$\begin{aligned} \frac{1}{4} \text{Tr}(a\gamma)(b\gamma) &= ab, \\ \frac{1}{4} \text{Tr}(a\gamma)(b\gamma)(c\gamma)(d\gamma) &= (ab)(cd) - (ac)(bd) + (ad)(bc). \end{aligned} \quad (22.13)$$

The product $\gamma^0\gamma^1\gamma^2\gamma^3$ has a special role. It is conventionally denoted by

$$\gamma^5 = -i\gamma^0\gamma^1\gamma^2\gamma^3. \quad (22.14)$$

It is easy to see that

$$\gamma^5\gamma^\mu + \gamma^\mu\gamma^5 = 0, \quad (\gamma^5)^2 = 1, \quad (22.15)$$

so that γ^5 anticommutes with every γ^μ . With respect to the matrices α and β , the corresponding rules are

$$\alpha\gamma^5 - \gamma^5\alpha = 0, \quad \beta\gamma^5 + \gamma^5\beta = 0 \quad (22.16)$$

(commutation with α follows because $\alpha = \gamma^0\gamma$ is a product of two γ^μ matrices).

The matrix γ^5 is Hermitian. Indeed,

$$\gamma^{5+} = i\gamma^{3+}\gamma^{2+}\gamma^{1+}\gamma^{0+} = -i\gamma^3\gamma^2\gamma^1\gamma^0,$$

and since the sequence 3210 is reduced to 0123 by an even number of interchanges,

$$\gamma^{5+} = \gamma^5. \quad (22.17)$$

For completeness, its form in two particular representations is

$$\begin{aligned} \text{spinor} \quad \gamma^5 &= \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}; \\ \text{standard} \quad \gamma^5 &= \begin{pmatrix} 0 & -1 \\ -1 & 0 \end{pmatrix}. \end{aligned} \quad (22.18)$$

The trace of γ^5 is zero:

$$\text{Tr} \gamma^5 = 0 \quad (22.19)$$

(this is also seen directly from (22.18)). The traces of products $\gamma^5\gamma^\mu\gamma^\nu$ vanish as well. For products of γ^5 with four γ^μ factors, however,

$$\frac{1}{4} \text{Tr} \gamma^5\gamma^\lambda\gamma^\mu\gamma^\nu\gamma^\rho = ie^{\lambda\mu\nu\rho}. \quad (22.20)$$

Another useful formula is

$$\gamma N = i\gamma^5(\gamma a)(\gamma b)(\gamma c), \quad N^\lambda = e^{\lambda\mu\nu\rho} a_\mu b_\nu c_\rho, \quad (22.21)$$

valid for mutually orthogonal 4-vectors a , b , and c : $ab = ac = bc = 0$.

In some situations (in problems involving nonrelativistic particles) one must evaluate traces of products in which γ^0 and the three-dimensional “vector” $\boldsymbol{\gamma}$ enter separately. Only products containing an even number of γ^0 and $\boldsymbol{\gamma}$ factors can have a nonzero trace. All γ^0 factors then reduce to 1, while traces of products with two and four $\boldsymbol{\gamma}$ factors are given by

$$\begin{aligned}\frac{1}{4} \text{Tr}(\mathbf{a}\boldsymbol{\gamma})(\mathbf{b}\boldsymbol{\gamma}) &= -\mathbf{a}\mathbf{b}, \\ \frac{1}{4} \text{Tr}(\mathbf{a}\boldsymbol{\gamma})(\mathbf{b}\boldsymbol{\gamma})(\mathbf{c}\boldsymbol{\gamma})(\mathbf{d}\boldsymbol{\gamma}) &= (\mathbf{a}\mathbf{b})(\mathbf{c}\mathbf{d}) - (\mathbf{a}\mathbf{c})(\mathbf{b}\mathbf{d}) + (\mathbf{a}\mathbf{d})(\mathbf{b}\mathbf{c}).\end{aligned}\quad (22.22)$$

§ 23. Plane waves

The state of a free particle with definite momentum and energy is described by a plane wave, which we write as

$$\psi_p = \frac{1}{\sqrt{2\varepsilon}} u_p e^{-ipx}. \quad (23.1)$$

The index p specifies the value of the 4-momentum; the wave amplitude u_p is a bispinor with a particular normalization.

When we subsequently carry out second quantization, we shall need, in addition to the wave functions (23.1), functions of “negative frequency.” As explained in § 11, these arise in relativistic theory from the two-valued root $\pm\sqrt{\mathbf{p}^2 + m^2}$. As in § 11, ε will everywhere mean the positive quantity $\varepsilon = +\sqrt{\mathbf{p}^2 + m^2}$, so that the “negative frequency” is $-\varepsilon$. Reversing the sign of $\mathbf{p}$ as well gives a function naturally denoted by ψ_{-p} :

$$\psi_{-p} = \frac{1}{\sqrt{2\varepsilon}} u_{-p} e^{ipx}. \quad (23.2)$$

The meaning of these functions will become clear in § 26. Below, formulas for ψ_p and ψ_{-p} will be written in parallel.

The components of the bispinor amplitudes u_p and u_{-p} satisfy the systems of algebraic equations

$$(\gamma p - m)u_p = 0, \quad (\gamma p + m)u_{-p} = 0, \quad (23.3)$$

obtained by substituting (23.1), (23.2) into the Dirac equation (which amounts to replacing the operator $\hat{p}$ in that equation by $\pm p$)¹⁵. The relation $p^2 = m^2$ is the compatibility condition for each system. We shall always normalize the bispinor amplitudes by the invariant conditions

$$\bar{u}_p u_p = 2m, \quad \bar{u}_{-p} u_{-p} = -2m \quad (23.4)$$

(the bar over a letter denotes, as everywhere, Dirac conjugation: $\bar{u} = u^* \gamma^0$). Multiplying equations (23.3) on the left by $\bar{u}_{\pm p}$ gives $(\bar{u}_{\pm p} \gamma u_{\pm p})p = 2m^2 = 2p^2$, and therefore

$$\bar{u}_p \gamma u_p = \bar{u}_{-p} \gamma u_{-p} = 2p \quad (23.5)$$

¹⁵We also note the analogous systems obtained from the Dirac equation (21.9) for the complex-conjugate function:

$$\bar{u}_p (\gamma p - m) = 0, \quad \bar{u}_{-p} (\gamma p + m) = 0. \quad (23.3 \text{ a})$$

Notice that formulas for u_{-p} are obtained from those for u_p by changing the sign of m .

The 4-vector of current density is

$$j = \bar{\psi}_{\pm p} \gamma \psi_{\pm p} = \frac{1}{2\varepsilon} \bar{u}_{\pm p} \gamma u_{\pm p} = \frac{p}{\varepsilon}, \quad (23.6)$$

that is, $j^\mu = (1, \mathbf{v})$, where $\mathbf{v} = \mathbf{p}/\varepsilon$ is the particle velocity. Hence the functions ψ_p are normalized to “one particle in a volume $V = 1$ ”.

By virtue of equations (23.3), the components of the wave amplitude obey certain relations whose explicit form naturally depends on the chosen representation of ψ . Let us determine them in the standard representation.

For a plane wave, equations (21.19) give

$$(\varepsilon - m)\varphi - \mathbf{p}\boldsymbol{\sigma}\chi = 0, \quad (\varepsilon + m)\chi - \mathbf{p}\boldsymbol{\sigma}\varphi = 0. \quad (23.7)$$

These equalities give the relation between φ and χ in two equivalent forms:

$$\varphi = \frac{\mathbf{p}\boldsymbol{\sigma}}{\varepsilon - m}\chi, \quad \chi = \frac{\mathbf{p}\boldsymbol{\sigma}}{\varepsilon + m}\varphi \quad (23.8)$$

(their equivalence is immediate: multiply the first on the left by $\mathbf{p}\boldsymbol{\sigma}/(\varepsilon + m)$ and use $(\mathbf{p}\boldsymbol{\sigma})^2 = \mathbf{p}^2$ and $\varepsilon^2 - m^2 = \mathbf{p}^2$ to obtain the second). The common factor in φ and χ is chosen to satisfy the normalization condition (23.4). We then obtain the following expressions for u_p (and similarly for u_{-p}):

$$u_p = \begin{pmatrix} \sqrt{\varepsilon + m} w \\ \sqrt{\varepsilon - m}(\mathbf{n}\boldsymbol{\sigma})w \end{pmatrix}, \quad u_{-p} = \begin{pmatrix} \sqrt{\varepsilon - m}(\mathbf{n}\boldsymbol{\sigma})w' \\ \sqrt{\varepsilon + m} w' \end{pmatrix} \quad (23.9)$$

(the second formula follows from the first by changing the sign before m and relabeling $w \rightarrow (\mathbf{n}\boldsymbol{\sigma})w'$). Here $\mathbf{n}$ is the unit vector along $\mathbf{p}$, while w is an arbitrary two-component quantity subject only to the normalization condition

$$w^* w = 1. \quad (23.10)$$

For $\bar{u} = u^* \gamma^0$ (with γ^0 from (21.20)), we have

$$\begin{aligned} \bar{u}_p &= (\sqrt{\varepsilon + m} w^*, -\sqrt{\varepsilon - m} w^*(\mathbf{n}\boldsymbol{\sigma})), \\ \bar{u}_{-p} &= (\sqrt{\varepsilon - m} w'^*(\mathbf{n}\boldsymbol{\sigma}), -\sqrt{\varepsilon + m} w'^*) \end{aligned} \quad (23.11)$$

and multiplication confirms that $\bar{u}_{\pm p} u_{\pm p} = \pm 2m$.

In the rest frame, where $\varepsilon = m$, we have

$$u_p = \sqrt{2m} \begin{pmatrix} w \\ 0 \end{pmatrix}, \quad u_{-p} = \sqrt{2m} \begin{pmatrix} 0 \\ w' \end{pmatrix}, \quad (23.12)$$

so w is the 3-spinor to which the amplitude of each wave reduces in the nonrelativistic limit. Note that in the rest frame it is the first two components of the bispinor u_{-p} , rather than the second two, that vanish. This property of solutions of the Dirac equation with

“negative frequencies” is evident: setting $\mathbf{p} = 0$ in (23.7) and replacing ε by $-m$ gives $\varphi = 0$ ¹⁶.

The amplitude of a plane wave contains one arbitrary two-component quantity. In other words, for a specified momentum there are two distinct independent states, corresponding to the two possible values of the spin projection. The projection of the spin on an arbitrary axis z , however, cannot have a definite value. Indeed, the Hamiltonian of a particle with specified $\mathbf{p}$ (the matrix $H = \alpha\mathbf{p} + \beta m$) does not commute with $\Sigma_z = -i\alpha_x\alpha_y$. Nevertheless, in accordance with the general statements made in § 16, the helicity λ is conserved: this is the spin projection along $\mathbf{p}$, and the Hamiltonian commutes with the matrix $\mathbf{n}\Sigma$.

Helicity states correspond to plane waves for which the 3-spinor $w = w^{(\lambda)}(\mathbf{n})$ is an eigenfunction of the operator $\mathbf{n}\sigma$:

$$\frac{1}{2}(\mathbf{n}\sigma)w^{(\lambda)} = \lambda w^{(\lambda)}. \quad (23.13)$$

These spinors are explicitly

$$w^{(\lambda=1/2)} = \begin{pmatrix} e^{-i\varphi/2} \cos \frac{\theta}{2} \\ e^{i\varphi/2} \sin \frac{\theta}{2} \end{pmatrix}, \quad w^{(\lambda=-1/2)} = \begin{pmatrix} -e^{-i\varphi/2} \sin \frac{\theta}{2} \\ e^{i\varphi/2} \cos \frac{\theta}{2} \end{pmatrix}, \quad (23.14)$$

where θ and φ are the polar and azimuthal angles of the direction $\mathbf{n}$ relative to fixed axes xyz ¹⁷.

Another possible choice of two independent states for a free particle with specified $\mathbf{p}$, simpler though less transparent, corresponds to the two values of the z -projection of spin in the rest frame; denote it by σ . The corresponding spinors are

$$w^{(\sigma=1/2)} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad w^{(\sigma=-1/2)} = \begin{pmatrix} 0 \\ 1 \end{pmatrix}. \quad (23.15)$$

As the two linearly independent solutions of “negative frequency,” we choose plane waves in which the 3-spinors are

$$w^{(\sigma)'} = -\sigma_y w^{(-\sigma)} = 2\sigma i w^{(\sigma)} \quad (23.16)$$

(the significance of this choice will become clear in § 26).

One can find a representation of a plane wave in which, in every reference frame and not only in the rest frame, it has only two components corresponding to definite values of the same physical characteristic, the rest-frame spin projection (*L. Foldy, S. A. Wouthuysen, 1950*).

Starting from the amplitude u_p in the standard representation (23.9), seek a unitary transformation to such a representation in the form

$$u'_p = U u_p, \quad U = e^{W\gamma^n},$$

¹⁶In the spinor representation one instead has $\xi = -\eta$, in place of the relation $\xi = \eta$ that holds in the rest frame for solutions of “positive frequency.”

¹⁷The solution of equations (23.13) may be multiplied by an arbitrary phase factor; this freedom is associated with an arbitrary rotation about the direction $\mathbf{n}$.

where W is real. Since $\gamma^+ = -\gamma$, this automatically gives $U^+ = U^{-1}$. Expanding in a series and using $(\gamma \mathbf{n})^2 = -1$, write U as

$$U = \cos W + \gamma \mathbf{n} \sin W$$

(compare the transition from (18.13) to (18.14)). The condition that the second two components of the transformed amplitude u'_p vanish gives

$$\tan W = \frac{|\mathbf{p}|}{m + \varepsilon},$$

and hence

$$U = \frac{m + \varepsilon + (\gamma \mathbf{n})|\mathbf{p}|}{\sqrt{2\varepsilon(\varepsilon + m)}}.$$

In the new representation,

$$u'_p = \sqrt{2\varepsilon} \begin{pmatrix} w \\ 0 \end{pmatrix}. \quad (23.17)$$

The particle Hamiltonian in this representation is

$$\hat{H}' = U(\alpha \mathbf{p} + \beta m)U^{-1} = \beta \varepsilon \quad (23.18)$$

(all the matrices β , α , and γ are in the standard representation). This Hamiltonian commutes with the matrix

$$\Sigma = -\alpha \gamma^5 = \begin{pmatrix} \sigma & 0 \\ 0 & \sigma \end{pmatrix},$$

which in the new representation is the operator of the conserved quantity, the spin in the rest frame.

§ 24. Spherical waves

The wave functions of free-particle states (for spin 1/2) with definite values of the angular momentum j are spinor spherical waves. We shall determine their form, first recalling the analogous formulas of nonrelativistic theory.

The nonrelativistic wave function is a 3-spinor $\psi = \begin{pmatrix} \psi^1 \\ \psi^2 \end{pmatrix}$. For a state with definite energy ε (and hence also a definite momentum magnitude p ¹⁸), orbital angular momentum l , total angular momentum j , and its projection m , the wave function has the form

$$\psi = R_{pl}(r)\Omega_{jlm}(\theta, \varphi). \quad (24.1)$$

Its angular part Ω_{jlm} consists of three-dimensional spinors whose components (for the

¹⁸In this section, p denotes $|\mathbf{p}|$.

two values $j = l \pm 1/2$ possible at a given l) are

$$\begin{aligned}\Omega_{l+1/2, l, m} &= \begin{pmatrix} \sqrt{\frac{j+m}{2j}} Y_{l, m-1/2} \\ \sqrt{\frac{j-m}{2j}} Y_{l, m+1/2} \end{pmatrix}, \\ \Omega_{l-1/2, l, m} &= \begin{pmatrix} -\sqrt{\frac{j-m+1}{2j+2}} Y_{l, m-1/2} \\ \sqrt{\frac{j+m+1}{2j+2}} Y_{l, m+1/2} \end{pmatrix}\end{aligned}\quad (24.2)$$

(see III, § 106, problem). We shall call Ω_{jlm} *spherical spinors*. Their normalization is

$$\int \Omega_{jlm}^* \Omega_{j'l'm'} d\omega = \delta_{jj'} \delta_{ll'} \delta_{mm'}. \quad (24.3)$$

The radial functions R_{pl} , on the other hand, are a common factor in both components of the spinor ψ and are given by formula

(33.10) (see III):

$$R_{pl} = \sqrt{\frac{2\pi p}{r}} J_{l+1/2}(pr). \quad (24.4)$$

They obey the normalization condition

$$\int_0^\infty r^2 R_{p'l} R_{pl} dr = 2\pi \delta(p' - p). \quad (24.5)$$

Returning to the relativistic case, we first recall that spin and orbital angular momentum are not separately conserved for a moving particle: neither $\hat{\mathbf{s}}$ nor $\hat{\mathbf{l}}$, taken individually, commutes with the Hamiltonian. The parity of a state nevertheless remains conserved (for a free particle). Thus the quantum number l no longer specifies a definite orbital-angular-momentum value, but it does determine the parity of the state, as shown below.

We take the required wave function (a bispinor) in the standard representation: $\psi = \begin{pmatrix} \varphi \\ \chi \end{pmatrix}$. Under rotations, φ and χ transform as 3-spinors. Their angular dependence is therefore described by the same spherical spinors Ω_{jlm} . Let $\varphi \propto \Omega_{jlm}$, where l is one definite member of the pair $j + 1/2$ and $j - 1/2$. Under inversion, $\varphi(\mathbf{r}) \rightarrow i\varphi(-\mathbf{r})$ (see (21.18)); since $\Omega_{jlm}(-\mathbf{n}) = (-1)^l \Omega_{jlm}(\mathbf{n})$, it follows that

$$\varphi(\mathbf{r}) \rightarrow i(-1)^l \varphi(\mathbf{r}).$$

The components of χ , however, transform under inversion according to $\chi(\mathbf{r}) \rightarrow -i\chi(-\mathbf{r})$. For the state to have a definite parity (that is, for every component to acquire the same factor under inversion), the angular dependence of χ must consequently be given by the spherical spinor $\Omega_{j'l'm}$ with the other possible value of l . Since the two values differ by 1, $(-1)^{l'} = -(-1)^l$.

The radial dependences of φ and χ are in turn determined by the same functions R_{pl} and $R_{p'l'}$, with l and l' matching the orders of the spherical functions entering

Ω_{jlm} . This follows because every component of ψ satisfies the second-order equation $(\widehat{\mathbf{p}}^2 - m^2)\psi = 0$, which, at a specified value of $|\mathbf{p}|$, takes the form

$$(\Delta + \mathbf{p}^2)\psi = 0,$$

formally identical to the nonrelativistic Schrödinger equation for a free particle.

Hence

$$\varphi = AR_{pl}\Omega_{jlm}, \quad \chi = BR_{pl'}\Omega_{jl'm}, \quad (24.6)$$

and only the constant coefficients A and B remain to be found. For this purpose, consider the distant region, where a spherical wave may be treated as a plane wave. By the asymptotic formula (33.12) (see III),

$$R_{pl} \approx \frac{1}{ir} \left\{ e^{i\left(pr - \frac{\pi l}{2}\right)} - e^{-i\left(pr - \frac{\pi l}{2}\right)} \right\}, \quad (24.7)$$

so that φ is the difference of two plane waves propagating in the directions $\pm \mathbf{n}$ ($\mathbf{n} = \mathbf{r}/r$). For each of them, equation (23.8) gives

$$\chi = \frac{p}{\varepsilon + m} (\pm \mathbf{n} \boldsymbol{\sigma}) \varphi.$$

The preceding discussion and formulas (24.6) show that $(\mathbf{n} \boldsymbol{\sigma})\Omega_{jlm} = a\Omega_{jl'm}$, where a is a constant. It can be found readily by comparing the two sides at $m = 1/2$ when $\mathbf{n}$ points along the z axis. Using (7.2a), we obtain

$$(\mathbf{n} \boldsymbol{\sigma})\Omega_{jlm} = i^{l'-l}\Omega_{jl'm}. \quad (24.8)$$

Combining these formulas and comparing the result with (24.6), we find

$$B = -\frac{p}{\varepsilon + m} A.$$

Finally, A is fixed by the overall normalization of ψ . With the condition

$$\int \psi_{pjlm}^* \psi_{p'j'l'm'} d^3x = 2\pi \delta_{jj'} \delta_{ll'} \delta_{mm'} \delta(p - p'), \quad (24.9)$$

the final result is

$$\psi_{pjlm} = \frac{1}{\sqrt{2\varepsilon}} \begin{pmatrix} \sqrt{\varepsilon + m} R_{pl} \Omega_{jlm} \\ -\sqrt{\varepsilon - m} R_{pl'} \Omega_{jl'm} \end{pmatrix}, \quad l' = 2j - l. \quad (24.10)$$

Thus, for specified j and m (and energy ε), there are two states distinguished by their parity. The latter is fixed uniquely by l , which takes the values $j \pm 1/2$: under inversion, the bispinor (24.10) is multiplied by $i(-1)^l$. Its components, however, contain spherical functions of both orders l and l' , expressing the absence of a definite orbital-angular-momentum value.

As $r \rightarrow \infty$, the spherical waves (24.7) can be treated in every small region of space as plane waves with momentum $\mathbf{p} = \pm p\mathbf{n}$. It is therefore clear that the momentum-representation wave functions differ from (24.10) mainly by

the absence of radial factors and by interpreting $\mathbf{n}$ as the momentum direction.

To pass directly to the momentum representation, one performs the Fourier expansion

$$\psi(\mathbf{p}') = \int \psi(\mathbf{r}) e^{-i\mathbf{p}'\mathbf{r}} d^3x, \quad (24.11)$$

The integral is evaluated with the expansion of a plane wave in spherical functions (see III, (34.3)):

$$e^{i\mathbf{p}\mathbf{r}} = \frac{2\pi}{p} \sum_{l=0}^{\infty} \sum_{m=-l}^l i^l R_{pl}(r) Y_{lm}^*\left(\frac{\mathbf{p}}{p}\right) Y_{lm}\left(\frac{\mathbf{r}}{r}\right). \quad (24.12)$$

Expanding the factor $e^{-i\mathbf{p}'\mathbf{r}}$ in (24.11) in this way and using (24.5), the Fourier components of the function

$$\psi(\mathbf{r}) = R_{pl}(r) \Omega_{jlm}\left(\frac{\mathbf{r}}{r}\right)$$

are

$$\psi(\mathbf{p}') = \frac{(2\pi)^2}{p} \delta(p' - p) i^{-l} Y_{lm'}\left(\frac{\mathbf{p}'}{p'}\right) \int \Omega_{jlm}\left(\frac{\mathbf{r}}{r}\right) Y_{lm'}^*\left(\frac{\mathbf{r}}{r}\right) d\Omega.$$

The integral here equals the coefficients of the spherical functions in the definition of the spherical spinors (24.2). Together with the factor $Y_{lm'}(\mathbf{p}'/p')$, these coefficients again form the same spherical spinor, now with argument $\mathbf{p}'/p'$:

$$\psi(\mathbf{p}') = \frac{(2\pi)^2}{p} \delta(p' - p) i^{-l} \Omega_{jlm}\left(\frac{\mathbf{p}'}{p'}\right).$$

Applying this result to the bispinor wave function (24.10), we obtain its momentum representation:

$$\psi_{pjlm}(\mathbf{p}') = \delta(p' - p) \frac{(2\pi)^2}{p\sqrt{2\varepsilon}} \begin{pmatrix} \sqrt{\varepsilon + m} i^{-l} \Omega_{jlm}(\mathbf{p}'/p') \\ -\sqrt{\varepsilon - m} i^{-l'} \Omega_{jl'm}(\mathbf{p}'/p') \end{pmatrix}. \quad (24.13)$$

The states $|pjlm\rangle$ coincide with the states $|pjm|\lambda\rangle$ considered in § 16 (where $|\lambda| = 1/2$): both have definite values of pjm and parity. The spherical spinors Ω_{jlm} can therefore be expressed through the functions $D_m^{(j)}$ (both being functions of $\mathbf{p}/p$). As $p \rightarrow 0$, the wave functions (24.13) reduce to the 3-spinors Ω_{jlm} , whose parity is $P = \eta(-1)^l$ (where $\eta = i$ is the “intrinsic parity” of the spinor). Comparison with the results of § 16 yields

$$\Omega_{jlm} = i^l \sqrt{\frac{2j+1}{8\pi}} \left(w^{(-1/2)} D_{-1/2,m}^{(j)} \pm w^{(1/2)} D_{1/2,m}^{(j)} \right) \quad (24.14)$$

(for $l = j \mp 1/2$, where $w^{(\lambda)}$ are the 3-spinors (23.14)).

§ 25. The relation between spin and statistics

Second quantization of the field of spin-1/2 particles (the spinor field) is carried out in the same way as for the scalar field in § 11.

Without repeating the entire argument, we write down the field operators at once, in a form fully analogous to formulas (11.2):

$$\begin{aligned}\hat{\psi} &= \sum_{\mathbf{p}\sigma} \frac{1}{\sqrt{2\varepsilon}} (\hat{a}_{\mathbf{p}\sigma} u_{\mathbf{p}\sigma} e^{-ipx} + \hat{b}_{\mathbf{p}\sigma}^+ u_{-\mathbf{p}-\sigma} e^{ipx}), \\ \hat{\bar{\psi}} &\equiv \hat{\psi}^+ \gamma^0 = \sum_{\mathbf{p}\sigma} \frac{1}{\sqrt{2\varepsilon}} (\hat{a}_{\mathbf{p}\sigma}^+ \bar{u}_{\mathbf{p}\sigma} e^{ipx} + \hat{b}_{\mathbf{p}\sigma} \bar{u}_{-\mathbf{p}-\sigma} e^{-ipx});\end{aligned}\quad (25.1)$$

the summation extends over every momentum $\mathbf{p}$ and over $\sigma = \pm 1/2$. The antiparticle annihilation operators $\hat{b}_{\mathbf{p}\sigma}$ (like the particle annihilation operators $\hat{a}_{\mathbf{p}\sigma}$) occur as coefficients of functions whose coordinate dependence ($e^{i\mathbf{p}\mathbf{r}}$) corresponds to a state of momentum $\mathbf{p}$ ¹⁹.

To calculate the Hamiltonian of the spinor field, there is no need to determine its energy-momentum tensor (as was done for the scalar field), because here a particle Hamiltonian exists and allows the wave equation, the Dirac equation (21.12), to be written. The mean energy of a particle in a state with wave function ψ is the integral

$$\int \psi^* \hat{H} \psi d^3x = i \int \psi^* \frac{\partial \psi}{\partial t} d^3x = i \int \bar{\psi} \gamma^0 \frac{\partial \psi}{\partial t} d^3x. \quad (25.2)$$

Notice that the “energy density” (the integrand) is not positive definite in this case.

Replacing ψ and $\bar{\psi}$ in (25.2) by the corresponding field operators, and using the mutual orthogonality of wave functions with different $\mathbf{p}$ or σ , together with the relation $\bar{u}_{\pm p\sigma} \gamma^0 u_{\pm p\sigma} = 2\varepsilon$ for the wave amplitudes, we obtain the field Hamiltonian

$$\hat{H} = \sum_{\mathbf{p}\sigma} \varepsilon (\hat{a}_{\mathbf{p}\sigma}^+ \hat{a}_{\mathbf{p}\sigma} - \hat{b}_{\mathbf{p}\sigma} \hat{b}_{\mathbf{p}\sigma}^+). \quad (25.3)$$

It follows that Fermi quantization must be used here:

$$\{\hat{a}_{\mathbf{p}\sigma}, \hat{a}_{\mathbf{p}\sigma}^+\}_+ = 1, \quad \{\hat{b}_{\mathbf{p}\sigma}, \hat{b}_{\mathbf{p}\sigma}^+\}_+ = 1, \quad (25.4)$$

while every other pair among the operators $\hat{a}$, $\hat{a}^+$, $\hat{b}$, and $\hat{b}^+$ anticommutes (see III, § 65). Indeed, the Hamiltonian (25.3) can then be rewritten as

$$\hat{H} = \sum_{\mathbf{p}\sigma} \varepsilon (\hat{a}_{\mathbf{p}\sigma}^+ \hat{a}_{\mathbf{p}\sigma} + \hat{b}_{\mathbf{p}\sigma}^+ \hat{b}_{\mathbf{p}\sigma} - 1),$$

and its energy eigenvalues (after subtracting, as always, the infinite additive constant) are

$$E = \sum_{\mathbf{p}\sigma} \varepsilon (N_{\mathbf{p}\sigma} + \bar{N}_{\mathbf{p}\sigma}), \quad (25.5)$$

and are therefore positive definite, as required. Bose quantization applied to (25.3), on the other hand, would give the meaningless, non-positive-definite eigenvalues

$$\sum_{\mathbf{p}\sigma} \varepsilon (N_{\mathbf{p}\sigma} - \bar{N}_{\mathbf{p}\sigma})$$

¹⁹Both kinds of function also correspond to the same value of the rest-frame spin projection σ : for the functions $\bar{\psi}_{-p, -\sigma}$ this will be shown in § 26; see (26.10).

The analogous expression

$$\mathbf{P} = \sum_{\mathbf{p}\sigma} \mathbf{p}(N_{\mathbf{p}\sigma} + \bar{N}_{\mathbf{p}\sigma}) \quad (25.6)$$

for the momentum of the system is obtained from the eigenvalues of the operator $\int \hat{\psi}^+ \hat{\mathbf{p}} \hat{\psi} d^3x$.

The 4-current operator is

$$\hat{j}^\mu = \hat{\psi} \gamma^\mu \hat{\psi}, \quad (25.7)$$

and the operator of the field “charge” is

$$\hat{Q} = \int \hat{\psi} \gamma^0 \hat{\psi} d^3x = \sum_{\mathbf{p}\sigma} (\hat{a}_{\mathbf{p}\sigma}^+ \hat{a}_{\mathbf{p}\sigma} + \hat{b}_{\mathbf{p}\sigma} \hat{b}_{\mathbf{p}\sigma}^+) = \sum_{\mathbf{p}\sigma} (\hat{a}_{\mathbf{p}\sigma}^+ \hat{a}_{\mathbf{p}\sigma} - \hat{b}_{\mathbf{p}\sigma}^+ \hat{b}_{\mathbf{p}\sigma} + 1), \quad (25.8)$$

with eigenvalues

$$Q = \sum_{\mathbf{p}\sigma} (N_{\mathbf{p}\sigma} - \bar{N}_{\mathbf{p}\sigma}) \quad (25.9)$$

We are thus led once more to particles and antiparticles, to which everything said about them in § 11 applies.

Particles of spin 0, however, are bosons, whereas particles of spin 1/2 are fermions. Tracing the formal origin of this distinction shows that it arises from the different character of the “energy-density” expressions for scalar and spinor fields. In the first case the expression is positive definite, so both terms ($\hat{a}^+ \hat{a}$ and $\hat{b} \hat{b}^+$) enter the Hamiltonian (11.3) with a plus sign. To keep the energy eigenvalues positive, $\hat{b} \hat{b}^+$ must then be replaced by $\hat{b}^+ \hat{b}$ without a sign change, that is, according to the Bose commutation rule. For a spinor field, by contrast, the “energy density” is not positive definite; consequently the term $\hat{b} \hat{b}^+$ has a minus sign in the Hamiltonian (25.3), and obtaining positive eigenvalues requires the replacement of $\hat{b} \hat{b}^+$ by $\hat{b}^+ \hat{b}$ to be accompanied by a sign change, that is, to follow the Fermi commutation rule.

On the other hand, the form of the energy density is directly connected with the transformation properties of the wave function and with the demands of relativistic invariance. In this sense, the relation between a particle’s spin and its statistics may likewise be regarded as a direct consequence of those demands.

The fact that spin-1/2 particles are fermions also yields the general statement that all particles of half-integer spin are fermions and particles of integer spin are bosons (including the result for a spin-0 particle proved in § 11)²⁰.

This becomes evident by imagining a particle of spin s as being “composed” of $2s$ spin-1/2 particles. For half-integer s , the number $2s$ is odd; for integer s , it is even. A “composite” particle containing an even number of fermions is a boson, whereas one containing an odd number is a fermion²¹.

²⁰The origin of the relation between a particle’s spin and the statistics it obeys was clarified by *Pauli* (W. Pauli, 1940).

²¹This reasoning assumes that all particles with the same spin obey the same statistics, independently of how they are “composed.” That this is indeed so follows from a similar argument. If spin-0 fermions existed, for example, a spin-0 fermion and a spin-1/2 fermion could form a spin-1/2 particle that would be a boson, contradicting the general result proved for spin 1/2.

If a system contains particles of different kinds, separate creation and annihilation operators must be introduced for each kind. Operators belonging to different bosons, or to bosons and fermions, commute with one another. Operators belonging to different fermions could, within nonrelativistic theory, be taken either to commute or to anticommute (III, § 65). Relativistic theory permits mutual transformations of particles, however, and therefore the creation and annihilation operators of different fermions must be taken to anticommute, just as do the operators belonging to different states of the same fermions.

PROBLEM

Find the Lagrangian of the spinor field.

SOLUTION. The Lagrangian function corresponding to the Dirac equation is given by the real scalar expression

$$L = \frac{i}{2} (\bar{\psi} \gamma^\mu \partial_\mu \psi - \partial_\mu \bar{\psi} \cdot \gamma^\mu \psi) - m \bar{\psi} \psi. \quad (1)$$

Taking the components of ψ and $\bar{\psi}$ as the “generalized coordinates” q , one readily verifies that the corresponding Lagrange equations (10.10) coincide with the Dirac equations for $\bar{\psi}$ and ψ . The overall sign of the Lagrangian (as well as its overall coefficient) is conventional in this case. Since L is linear in the derivatives of ψ and $\bar{\psi}$, the action $S = \int L d^4x$ cannot have either a minimum or a maximum. The condition $\delta S = 0$ here determines only a stationary point, not an extremum of the integral.

The Lagrangian of the spinor field is obtained by replacing ψ in (1) with the operator $\hat{\psi}$. Applying formula (12.12) to this Lagrangian gives the current operator (25.7).

§ 26. Charge conjugation and time reversal of spinors

The factors $\psi_{p\sigma} = u_{p\sigma} e^{-ipx}$ that multiply the operators $\hat{a}_{p\sigma}$ in (25.1) are the wave functions of free particles (which we shall call “electrons”) with momenta $\mathbf{p}$ and polarizations σ :

$$\psi_{p\sigma}^{(e)} = \psi_{p\sigma}.$$

The factors $\bar{\psi}_{-p-\sigma}$ multiplying the operators $\hat{b}_{p\sigma}$ must instead be regarded as the wave functions of positrons with the same $\mathbf{p}$ and σ . It will turn out, however, that the electron and positron functions are expressed in different bispinor representations. This follows from the different transformation properties of ψ and $\bar{\psi}$ and from the different systems of equations satisfied by their components. To remove this defect, the components of $\bar{\psi}_{-p-\sigma}$ must undergo a certain unitary transformation such that the new four-component function satisfies the same equation as $\psi_{p\sigma}$ ²². This is the function that we shall call the positron wave function, with momentum $\mathbf{p}$ and polarization σ . Denoting the required unitary-transformation matrix by U_C , we write

$$\psi_{p\sigma}^{(p)} = U_C \bar{\psi}_{-p-\sigma}. \quad (26.1)$$

²²For particles of spin 0 this issue did not arise at all: the scalar functions ψ and ψ^* satisfy the same equation, and ψ_{-p}^* simply coincides with ψ_p .

The operation C that produces this function from $\psi_{-p-\sigma}$ is called *charge conjugation* of the wave function (*H. A. Kramers, 1937*). The concept is of course not restricted to plane waves. For any function ψ there is a “charge-conjugate” function

$$\widehat{C}\psi(t, \mathbf{r}) = U_C \widetilde{\psi}(t, \mathbf{r}), \quad (26.2)$$

which transforms like ψ and satisfies the same equation.

The properties of U_C follow from this definition. If ψ is a solution of the Dirac equation $(\gamma \widehat{p} - m)\psi = 0$, then $\widetilde{\psi}$ satisfies

$$\widetilde{\psi}(\gamma \widehat{p} + m) = 0, \quad \text{or} \quad (\widetilde{\gamma} \widehat{p} + m)\widetilde{\psi} = 0.$$

Multiplication on the left by U_C gives

$$U_C \widetilde{\gamma} \widehat{p} \widetilde{\psi} + m U_C \widetilde{\psi} = 0,$$

We require $U_C \widetilde{\psi}$ to satisfy the same equation as ψ :

$$(\gamma \widehat{p} - m)U_C \widetilde{\psi} = 0.$$

Comparison gives the following “commutation relation” between U_C and the matrices γ^μ ²³:

$$U_C \widetilde{\gamma}^\mu = -\gamma^\mu U_C. \quad (26.3)$$

In what follows, we assume that the wave functions are given in the spinor or standard representation; the general case of an arbitrary representation will be considered only at the end of this section. In these representations,

$$\begin{aligned} \gamma^{0,2} &= \widetilde{\gamma}^{0,2}, & \gamma^{1,3} &= -\widetilde{\gamma}^{1,3}, \\ (\gamma^{0,1,3})^* &= \gamma^{0,1,3}, & \gamma^{2*} &= -\gamma^2. \end{aligned} \quad (26.4)$$

The conditions (26.3) are then satisfied by $U_C = \eta_C \gamma^2 \gamma^0$, where η_C is an arbitrary constant. The requirement $\widehat{C}^2 = 1$ gives $|\eta_C|^2 = 1$, so U_C is fixed only up to a phase factor. We shall choose $\eta_C = 1$ from now on, giving

$$U_C = \gamma^2 \gamma^0 = -\alpha_y. \quad (26.5)$$

Since also $\widetilde{\psi} = \psi^* \gamma^0 = \widetilde{\gamma}^0 \psi^* = \gamma^0 \psi^*$, the action of $\widehat{C}$ can be written

$$\widehat{C}\psi = \gamma^2 \gamma^0 \widetilde{\psi} = \gamma^2 \psi^*. \quad (26.6)$$

Explicitly, transformation (26.6) in the spinor representation is

$$C : \quad \xi^\alpha \rightarrow -i\eta^{\dot{\alpha}*}, \quad \eta_{\dot{\alpha}} \rightarrow -i\xi_{\alpha}^*. \quad (26.7a)$$

²³We also note the resulting equality

$$U_C \widetilde{\gamma}^5 = \gamma^5 U_C. \quad (26.3a)$$

Equivalently,

$$C : \quad \xi_\alpha \rightarrow -i\eta_{\dot{\alpha}}^*, \quad \eta^{\dot{\alpha}} \rightarrow -i\xi^{\alpha*}. \quad (26.7b)$$

Charge conjugation of the plane waves $\psi_{\pm p\sigma}$ is readily performed using their explicit expressions (23.9) and the standard-representation matrix

$$U_C = \begin{pmatrix} 0 & -\sigma_y \\ -\sigma_y & 0 \end{pmatrix}. \quad (26.8)$$

Noting that

$$\sigma_y \sigma^* = -\sigma \sigma_y,$$

and defining $w^{(\sigma)'} as in (23.16), we obtain$

$$U_C \tilde{u}_{-p-\sigma} = u_{p\sigma}, \quad U_C u_{-p-\sigma} = \tilde{u}_{p\sigma}. \quad (26.9)$$

Thus

$$\hat{C}\psi_{-p-\sigma} = \psi_{p\sigma}, \quad (26.10)$$

so the functions $\psi_{-p-\sigma}$ that occur in the ψ -operators (25.1) together with $\hat{b}_{p\sigma}$ do indeed correspond to particle states of momentum $\mathbf{p}$ and polarization σ . We also see that electron and positron states are described by the same functions:

$$\psi_{p\sigma}^{(e)} = \psi_{p\sigma}^{(p)} = \psi_{p\sigma}.$$

This is natural, since the functions $\psi_{p\sigma}$ contain information only about the particle's momentum and polarization.

Time reversal may be considered in an analogous way. Reversing the sign of time must be accompanied by complex conjugation of the wave function. To obtain the resulting time-reversed wave function ($\hat{T}\psi$) in the same representation as the original ψ , the components of ψ^* (or $\tilde{\psi}$) must also undergo a unitary transformation. In analogy with (26.2), we therefore represent the action of $\hat{T}$ on ψ as

$$\hat{T}\psi(t, \mathbf{r}) = U_T \tilde{\psi}(-t, \mathbf{r}), \quad (26.11)$$

where U_T is unitary.

We again write the Dirac equation satisfied by ψ :

$$\left(i\gamma^0 \frac{\partial}{\partial t} + i\boldsymbol{\gamma} \boldsymbol{\nabla} - m \right) \psi(t, \mathbf{r}) = 0,$$

and the equation for $\tilde{\psi}$:

$$\left(i\tilde{\gamma}^0 \frac{\partial}{\partial t} + i\tilde{\boldsymbol{\gamma}} \boldsymbol{\nabla} + m \right) \tilde{\psi}(t, \mathbf{r}) = 0.$$

In the latter equation, replace t by $-t$ and multiply from the left by $-U_T$:

$$\left(iU_T \tilde{\gamma}^0 \frac{\partial}{\partial t} - iU_T \tilde{\boldsymbol{\gamma}} \boldsymbol{\nabla} \right) \tilde{\psi}(-t, \mathbf{r}) - mU_T \tilde{\psi}(-t, \mathbf{r}) = 0.$$

We require $U_T \tilde{\psi}(-t, \mathbf{r})$ to satisfy the same equation as $\psi(t, \mathbf{r})$:

$$\left(i\gamma^0 \frac{\partial}{\partial t} + i\boldsymbol{\gamma} \nabla \right) U_T \tilde{\psi}(-t, \mathbf{r}) - m U_T \tilde{\psi}(-t, \mathbf{r}) = 0.$$

Comparison of the two equations shows that U_T must obey

$$U_T \tilde{\gamma}^0 = \gamma^0 U_T, \quad U_T \tilde{\boldsymbol{\gamma}} = -\boldsymbol{\gamma} U_T. \quad (26.12)$$

In the spinor and standard representations, these conditions are satisfied by²⁴

$$U_T = i\gamma^3 \gamma^1 \gamma^0. \quad (26.13)$$

The action of $\hat{T}$ is consequently

$$\hat{T}\psi(t, \mathbf{r}) = i\gamma^3 \gamma^1 \gamma^0 \tilde{\psi}(-t, \mathbf{r}) = i\gamma^3 \gamma^1 \psi^*(-t, \mathbf{r}). \quad (26.14)$$

Explicitly, this transformation in the spinor representation is

$$T : \quad \xi^\alpha \rightarrow -i\xi_\alpha^*, \quad \eta_{\dot{\alpha}} \rightarrow i\eta^{\dot{\alpha}*} \quad (26.15a)$$

or

$$T : \quad \xi_\alpha \rightarrow i\xi^{\alpha*}, \quad \eta^{\dot{\alpha}} \rightarrow -i\eta_{\dot{\alpha}}^*. \quad (26.15b)$$

In the standard representation,

$$U_T = \begin{pmatrix} \sigma_y & 0 \\ 0 & -\sigma_y \end{pmatrix}. \quad (26.16)$$

We now find the result of applying all three operations P , T , and C to ψ . Successively,

$$\hat{T}\psi(t, \mathbf{r}) = -i\gamma^1 \gamma^3 \psi^*(-t, \mathbf{r}),$$

$$\hat{P}\hat{T}\psi(t, \mathbf{r}) = i\gamma^0 (\hat{T}\psi) = \gamma^0 \gamma^1 \gamma^3 \psi^*(-t, -\mathbf{r}),$$

$$\hat{C}\hat{P}\hat{T}\psi(t, \mathbf{r}) = \gamma^2 (\gamma^0 \gamma^1 \gamma^3 \psi^*)^* = \gamma^2 \gamma^0 \gamma^1 \gamma^3 \psi(-t, -\mathbf{r}),$$

or

$$\hat{C}\hat{P}\hat{T}\psi(t, \mathbf{r}) = i\gamma^5 \psi(-t, -\mathbf{r}). \quad (26.17)$$

In the spinor representation,

$$CPT : \quad \xi^\alpha \rightarrow -i\xi^\alpha, \quad \eta_{\dot{\alpha}} \rightarrow i\eta_{\dot{\alpha}}, \quad (26.18)$$

as can also be checked directly from the transformation rules (20.4), (26.7), and (26.15)²⁵.

The expressions above for U_C and U_T assumed the spinor or standard representation of ψ . We finally determine which of their properties persist in an arbitrary representation.

²⁴The choice of phase factor in (26.13) is related to that in (26.5) for the reasons stated below in the note on p. 124.

²⁵The notation $\hat{C}\hat{P}\hat{T}$ means that the operators act from right to left. The overall sign in (26.17), (26.18) depends on their order because $\hat{T}$ does not commute with $\hat{C}$ or $\hat{P}$ in their action on a bispinor.

If ψ undergoes a unitary transformation,

$$\psi' = U\psi, \quad \gamma' = U\gamma U^{-1}, \quad \bar{\psi}' = \psi'^* \gamma^{0'} = \bar{\psi} U^+ = \bar{\psi} \widetilde{U}^{-1}, \quad (26.19)$$

then, in the new representation,

$$(\widehat{C}\psi)' = U(\widehat{C}\psi) = UU_C \widetilde{\psi} = UU_C (\widetilde{\psi' U}) = UU_C \widetilde{U} \widetilde{\psi'}.$$

Comparing this with the definition of U'_C in the new representation, $((\widehat{C}\psi)' = U'_C \widetilde{\psi'})$, we find

$$U'_C = UU_C \widetilde{U}. \quad (26.20)$$

Transformation (26.20) agrees with the transformation of the γ matrices only when U is real. Thus expression (26.5) is valid only in representations obtained from the spinor or standard representation by a real transformation.

The matrix (26.5) is unitary, and transposition changes its sign:

$$U_C U_C^+ = 1, \quad \widetilde{U}_C = -U_C. \quad (26.21)$$

These properties are invariant under (26.20) and therefore hold in every representation. The matrix (26.5) is also Hermitian ($U_C = U_C^+$), but in general this property is not preserved by (26.20).

All these statements, including (26.21), apply equally to U_T .

In second-quantized theory, the transformations C , P , and T of the ψ -operators must be formulated as transformation rules for particle creation and annihilation operators. These rules may be established (as in § 13 for spin-0 particles) by requiring that the transformed ψ -operators admit the representation

$$\begin{aligned} \widehat{\psi}^C(t, \mathbf{r}) &= U_C \widetilde{\widehat{\psi}}(t, \mathbf{r}), \\ \widehat{\psi}^P(t, \mathbf{r}) &= i\gamma^0 \widehat{\psi}(t, -\mathbf{r}), \\ \widehat{\psi}^T(t, \mathbf{r}) &= U_T \widetilde{\widehat{\psi}}(-t, \mathbf{r}). \end{aligned} \quad (26.22)$$

PROBLEM

Find the charge-conjugation operator in the Majorana representation (see problem 2, § 21).

SOLUTION. The matrix U'_C in the Majorana representation is obtained from the standard-representation matrix $U_C = -\alpha_y$ by transformation (26.20), with $U = (\alpha_y + \beta)/\sqrt{2}$, and equals $U'_C = \alpha_y$ (here α_y and β denote the standard-representation matrices). Using primes for quantities in the Majorana representation, we have $\widehat{C}\psi' = U'_C(\psi'^* \beta')$; since $\beta' = \alpha_y$,

$$\widehat{C}\psi' = \alpha_y(\psi'^* \alpha_y) = \alpha_y \widetilde{\alpha_y} \psi'^* = \psi'^*,$$

so charge conjugation is equivalent to complex conjugation.

§ 27. Intrinsic symmetry of particles and antiparticles

In its rest frame, the wave function of a spin-1/2 particle reduces to a single 3-spinor (which we denote by Φ^α). The concept of a particle's intrinsic parity is associated with the behavior of this spinor under inversion. However, as already noted in § 19, although the two possible transformation laws for 3-spinors ($\Phi^\alpha \rightarrow \pm i\Phi^\alpha$) are not equivalent to each other, assigning a definite parity to a spinor has no absolute meaning. It is therefore likewise meaningless to speak of the intrinsic parity of a spin-1/2 particle by itself. One may, however, speak of the relative intrinsic parity of two such particles.

From two three-dimensional spinors $\Phi^{(e)}$ and $\Phi^{(p)}$ one can form the scalar $\Phi_\alpha^{(e)} \Phi^{(p)\alpha}$. If this is a true scalar, the particles described by these spinors are said to have the same parity; if it is a pseudoscalar, the particles are said to have opposite intrinsic parities.

We shall show that the intrinsic parities of a particle and its antiparticle (of spin 1/2) are opposite (V. B. Berestetskii, 1948).

To this end, observe that applying the operation C from (26.7) to both sides of the P transformation (19.5), which in the spinor representation is

$$P : \quad \xi^\alpha \rightarrow i\eta_{\dot{\alpha}}, \quad \eta_{\dot{\alpha}} \rightarrow i\xi^\alpha, \quad (27.1)$$

gives

$$\eta^{c\dot{\alpha}*} \rightarrow i\xi_\alpha^{c*}, \quad \xi_\alpha^{c*} \rightarrow i\eta_{\dot{\alpha}}^{c*},$$

where the superscript c marks the components of the bispinor $\psi^C = \begin{pmatrix} \xi^c \\ \eta^c \end{pmatrix}$ that is charge-conjugate to the bispinor $\psi = \begin{pmatrix} \xi \\ \eta \end{pmatrix}$. Taking the complex conjugate and raising or lowering the indices as required, we find

$$P : \quad \eta_\alpha^c \rightarrow i\xi^{c\alpha}, \quad \xi^{c\alpha} \rightarrow i\eta_\alpha^c. \quad (27.2)$$

Thus charge-conjugate bispinors transform under inversion according to the same law.

Let $\psi^{(e)}$ be the wave function of the particle (the electron), and let $\psi^{(p)}$ be the wave function of the antiparticle (the positron). The latter is the bispinor charge-conjugate to some negative-frequency solution of the Dirac equation. In the rest frame, each of them reduces to a certain 3-spinor:

$$\xi^{(e)\alpha} = \eta_{\dot{\alpha}}^{(e)} = \Phi^{(e)\alpha}, \quad \xi^{(p)\alpha} = \eta_{\dot{\alpha}}^{(p)} = \Phi^{(p)\alpha}.$$

According to (27.1) and (27.2), these spinors transform under inversion by the rule

$$\Phi^\alpha \rightarrow i\Phi^\alpha, \quad (27.3)$$

which is the same for $\Phi^{(e)}$ and $\Phi^{(p)}$. Their product $\Phi^{(e)}\Phi^{(p)}$, however, changes sign, proving the assertion.

A particle that coincides with its antiparticle is called genuinely neutral (see § 12). The field operator $\hat{\psi}$ for such particles satisfies

$$\hat{\psi}(t, \mathbf{r}) = \hat{\psi}^C(t, \mathbf{r}).$$

For spin-1/2 particles, this gives the following conditions in the spinor representation²⁶:

$$\widehat{\xi}^\alpha = -i\widehat{\eta}^{\dot{\alpha}+}, \quad \widehat{\eta}_{\dot{\alpha}} = -i\widehat{\xi}_{\alpha}^+ \quad (27.4)$$

Like any relations expressing physical properties, these conditions are invariant under the CPT transformation²⁷. It is readily verified that they are in fact invariant not only under CPT , but under each of the three transformations separately.

In § 19 we agreed to define spinor inversion as the transformation for which $\widehat{P}^2 = -1$, and we have used that definition up to this point. It is easy to see that the result obtained above for the relative parity of particles and antiparticles is independent, as it must be, of how inversion is defined.

If inversion is defined by the condition $\widehat{P}^2 = 1$, then (27.1) is replaced by

$$P: \quad \xi^\alpha \rightarrow \eta_{\dot{\alpha}}, \quad \eta_{\dot{\alpha}} \rightarrow \xi^\alpha. \quad (27.5)$$

The charge-conjugate function then transforms according to

$$\xi^{c\alpha} \rightarrow -\eta_{\dot{\alpha}}^c, \quad \eta_{\dot{\alpha}}^c \rightarrow -\xi^{c\alpha},$$

which differs from (27.5) by a sign. Accordingly, the three-dimensional spinors Φ transform as

$$\Phi^{(e)\alpha} \rightarrow \Phi^{(e)\alpha}, \quad \Phi^{(p)\alpha} \rightarrow -\Phi^{(p)\alpha},$$

so that the product $\Phi^{(e)}\Phi^{(p)}$ remains a pseudoscalar.

The only possible difference between the physical consequences of the two definitions of inversion is that, with definition (27.5), the condition of genuine neutrality of the field would not be invariant under this transformation (or under CP): it would change the relative sign of the two sides of the equalities (27.4). No genuinely neutral spin-1/2 particles are known, and at present one cannot say whether this distinction between the two definitions of inversion has any real physical meaning²⁸.

PROBLEM

Find the charge parity of positronium (the hydrogen-like system consisting of an electron and a positron).

SOLUTION. The wave function of two fermions must be antisymmetric under the simultaneous interchange of their coordinates, spins, and charge variables (compare the problem in § 13). Interchanging the first multiplies the function by $(-1)^l$, interchanging the second by $(-1)^{l+S}$ (where $S = 0$ or 1 is the total spin of the system), and interchanging the third by the required quantity C . The condition $(-1)^l(-1)^{l+S}C = -1$ therefore gives

$$C = (-1)^{l+S}.$$

Since the intrinsic parities of the electron and positron are opposite, the spatial parity of the system is $P = (-1)^{l+1}$. Its combined parity is $CP = (-1)^{S+1}$.

²⁶In the Majorana representation, genuine neutrality simply means that the operator $\widehat{\psi}$ is Hermitian (see the problem in § 26).

²⁷More precisely, in this case the CPT transformation must be defined so as to leave relations of the type (27.4) invariant. This is achieved by the corresponding choice of phase factor in the definition of the matrix U_T (see the note on p. 121).

²⁸The incomplete equivalence of the two definitions of inversion was noted by *Racah* (*G. Racah*, 1937).

§28. Bilinear forms

Consider the transformation properties of the various bilinear forms that can be constructed from the components of the functions ψ and ψ^* . Such forms are of broad importance in quantum mechanics; the four-vector of current density (21.11) is one example.

Since ψ and ψ^* each have four components, they can be used to construct $4 \cdot 4 = 16$ independent bilinear combinations. Their classification according to transformation properties follows directly from the ways of multiplying two arbitrary bispinors listed in §19 (here the two bispinors are ψ and ψ^*). Specifically, one can form a scalar (denoted by S), a pseudoscalar (P), a mixed spinor of rank two equivalent to a true four-vector V^μ (four independent quantities), a mixed spinor of rank two equivalent to an axial four-vector A^μ (four quantities), and a rank-two bispinor equivalent to an antisymmetric four-tensor $T^{\mu\nu}$ (six quantities).

In symmetric form, valid in any representation of ψ , these combinations are

$$\begin{aligned} S &= \bar{\psi}\psi, & P &= i\bar{\psi}\gamma^5\psi, \\ V^\mu &= \bar{\psi}\gamma^\mu\psi, & A^\mu &= \bar{\psi}\gamma^\mu\gamma^5\psi, & T^{\mu\nu} &= i\bar{\psi}\sigma^{\mu\nu}\psi. \end{aligned} \quad (28.1)$$

Here

$$\sigma^{\mu\nu} = \frac{1}{2}(\gamma^\mu\gamma^\nu - \gamma^\nu\gamma^\mu) = (\boldsymbol{\alpha}, i\boldsymbol{\Sigma}) \quad (28.2)$$

(the components in (28.2) are ordered as in (19.15))²⁹. All the expressions written above are real.

That S and P are respectively a scalar and a pseudoscalar follows immediately from their spinor-representation forms:

$$S = \xi^*\eta + \eta^*\xi, \quad P = i(\xi^*\eta - \eta^*\xi),$$

which are precisely of the forms (19.7) and (19.8). The vector character of V^μ then follows from the Dirac equation. Multiplying $\hat{p}_\mu\gamma^\mu\psi = m\psi$ on the left by $\bar{\psi}$ gives

$$(\bar{\psi}\hat{p}_\mu\gamma^\mu\psi) = m\bar{\psi}\psi;$$

since the right-hand side is a scalar, the expression on the left must also be a scalar.

The rule for constructing the quantities (28.1) is clear: they are formed as though the matrices γ^μ constituted a four-vector, γ^5

were a pseudoscalar, and the factors $\bar{\psi}$ and ψ on either side together formed a scalar³⁰. The absence of bilinear forms having the character of a symmetric four-tensor, which is evident in the spinor representation, also follows from this rule: because the symmetric combination $\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2g^{\mu\nu}$, any such form would reduce to a scalar.

²⁹Under a unitary transformation of ψ (a change of representation),

$$\psi \rightarrow U\psi, \quad \gamma \rightarrow U\gamma U^{-1}, \quad \bar{\psi} \rightarrow \bar{\psi}U^{-1},$$

and the invariance of the bilinear forms under such transformations is evident.

³⁰The “pseudoscalar character” of γ^5 itself conforms to these rules, since

$$\gamma^5 = \frac{i}{24}e_{\lambda\mu\nu\rho}\gamma^\lambda\gamma^\mu\gamma^\nu\gamma^\rho.$$

Second-quantized bilinear forms are obtained by replacing the ψ functions in (28.1) with ψ operators. For greater generality, suppose that the two ψ operators refer to fields of different particles, distinguished by the indices a and b . We determine how these operator forms transform under charge conjugation. Noting that ³¹

$$\widehat{\psi}^C = U_C \widetilde{\widehat{\psi}}, \quad \widehat{\psi}^C = U_C^+ \widehat{\psi}, \quad (28.3)$$

and using (26.3) and (26.21), we have

$$\begin{aligned} \widehat{\psi}_a^C \widehat{\psi}_b^C &= \widehat{\psi}_a U_C^* U_C \widetilde{\widehat{\psi}}_b = -\widehat{\psi}_a U_C^+ U_C \widehat{\psi}_b = -\widehat{\psi}_a \widehat{\psi}_b, \\ \widehat{\psi}_a^C \gamma^\mu \widehat{\psi}_b^C &= \widehat{\psi}_a U_C^* \gamma^\mu U_C \widetilde{\widehat{\psi}}_b = -\widehat{\psi}_a U_C^+ \gamma^\mu U_C \widehat{\psi}_b = \widehat{\psi}_a \gamma^\mu \widehat{\psi}_b. \end{aligned}$$

When the operators are returned to their original order ($\widehat{\psi}$ to the left of $\widehat{\psi}$), the sign of the product changes by the Fermi commutation rules (25.4). In addition, terms independent of the state of the field appear; these are omitted, as in the analogous derivations in § 13. We thus obtain

$$\widehat{\psi}_a^C \widehat{\psi}_b^C = \widehat{\psi}_b \widehat{\psi}_a, \quad \widehat{\psi}_a^C \gamma^\mu \widehat{\psi}_b^C = -\widehat{\psi}_b \gamma^\mu \widehat{\psi}_a.$$

Transforming the remaining forms in the same way, we find that under charge conjugation ³²

$$\begin{aligned} C : \quad \widehat{S}_{ab} &\rightarrow \widehat{S}_{ba}, & \widehat{P}_{ab} &\rightarrow \widehat{P}_{ba}, & \widehat{V}_{ab}^\mu &\rightarrow -\widehat{V}_{ba}^\mu, \\ \widehat{A}_{ab}^\mu &\rightarrow \widehat{A}_{ba}^\mu, & \widehat{T}_{ab}^{\mu\nu} &\rightarrow -\widehat{T}_{ba}^{\mu\nu}. \end{aligned} \quad (28.4)$$

The behavior of the same forms under time reversal is found similarly. Here one must remember (see § 13) that this operation entails a reversal of operator order, so that, for example,

$$(\widehat{\psi}_a \widehat{\psi}_b)^T = \widehat{\psi}_b^T \widehat{\psi}_a^T.$$

Substitution of

$$\widehat{\psi}^T = U_T \widetilde{\widehat{\psi}}, \quad \widehat{\psi}^T = -U_T^+ \widehat{\psi} \quad (28.5)$$

gives

$$(\widehat{\psi}_a \widehat{\psi}_b)^T = -\widehat{\psi}_b \widetilde{U_T U_T^+} \widehat{\psi}_a = \widehat{\psi}_b U_T U_T^+ \widehat{\psi}_a = \widehat{\psi}_b \widehat{\psi}_a.$$

Treating the other forms in the same manner, we obtain

$$\begin{aligned} T : \quad \widehat{S}_{ab} &\rightarrow \widehat{S}_{ba}, & \widehat{P}_{ab} &\rightarrow -\widehat{P}_{ba}, & (\widehat{V}^0, \widehat{\mathbf{V}})_{ab} &\rightarrow (\widehat{V}^0, -\widehat{\mathbf{V}})_{ba}, \\ (\widehat{A}^0, \widehat{\mathbf{A}})_{ab} &\rightarrow (\widehat{A}^0, -\widehat{\mathbf{A}})_{ba}, & \widehat{T}_{ab}^{\mu\nu} &= (\widehat{\mathbf{p}}, \widehat{\mathbf{a}})_{ab} &\rightarrow (\widehat{\mathbf{p}}, -\widehat{\mathbf{a}})_{ba} \end{aligned} \quad (28.6)$$

³¹To obtain the second equality from the first, write

$$\widehat{\psi}^C = [U_C^* (\widehat{\psi} \gamma^{0*})] \gamma^0 = \widetilde{\gamma^0} U_C^* \gamma^0 \widehat{\psi} = -\widetilde{\gamma^0} U_C^+ \gamma^0 \widehat{\psi} = \widetilde{\gamma^0} \gamma^{0*} U_C^+ \widehat{\psi} = U_C^+ \widehat{\psi}$$

(we used (26.3), (26.21), and the Hermiticity of γ^0).

³²Notice that for bilinear forms constructed from ψ functions rather than ψ operators, transformations (28.4) would have the opposite sign, since restoring the original order of the factors $\widetilde{\psi}$ and ψ would not be accompanied by a sign change.

($\widehat{\mathbf{p}}$ and $\widehat{\mathbf{a}}$ are the three-dimensional vectors equivalent to the components of $\widehat{T}^{\mu\nu}$ according to (19.15)).

Under spatial inversion, on the other hand, the tensor character of the quantities gives³³

$$\begin{aligned} P : \quad \widehat{S}_{ab} &\rightarrow \widehat{S}_{ab}, & \widehat{P}_{ab} &\rightarrow -\widehat{P}_{ab}, & (\widehat{V}^0, \widehat{\mathbf{V}})_{ab} &\rightarrow (\widehat{V}^0, -\widehat{\mathbf{V}})_{ab}, \\ (\widehat{A}^0, \widehat{\mathbf{A}})_{ab} &\rightarrow (-\widehat{A}^0, \widehat{\mathbf{A}})_{ab}, & \widehat{T}_{ab}^{\mu\nu} = (\widehat{\mathbf{p}}, \widehat{\mathbf{a}})_{ab} &\rightarrow (-\widehat{\mathbf{p}}, \widehat{\mathbf{a}})_{ab}. \end{aligned} \quad (28.7)$$

Finally, applying all three operations together leaves every $\widehat{S}_{ab}$, $\widehat{P}_{ab}$, and $\widehat{T}_{ab}^{\mu\nu}$ unchanged and reverses the sign of every $\widehat{V}_{ab}^{\mu}$ and $\widehat{A}_{ab}^{\mu}$. This agrees precisely with the meaning of the transformation as four-dimensional inversion: since four-dimensional inversion is equivalent to a rotation of the four-dimensional coordinate system, it makes no distinction between true tensors and pseudotensors of any rank.

Consider pairwise products of bilinear forms made from four different functions ψ^a , ψ^b , ψ^c , and ψ^d . Different results arise depending on which functions are paired. Nevertheless, any such product can be reduced to products of bilinear forms having fixed pairs of factors (*W. Pauli, M. Fierz*, 1936). We derive the relation underlying this reduction.

Consider the set of four-by-four matrices

$$1, \quad \gamma^5, \quad \gamma^\mu, \quad i\gamma^\mu\gamma^5, \quad i\sigma^{\mu\nu} \quad (28.8)$$

(1 denotes the identity matrix). Number these 16 ($= 1 + 1 + 4 + 4 + 6$) matrices in some definite order and denote them by γ^A ($A = 1, \dots, 16$); denote the same matrices with lowered four-tensor indices (μ, ν) by γ_A . They have the properties

$$\text{Tr } \gamma^A = 0 \quad (\gamma^A \neq 1), \quad \gamma^A \gamma_A = 1, \quad \frac{1}{4} \text{Tr } \gamma^A \gamma_B = \delta_B^A. \quad (28.9)$$

The last of these properties shows that the matrices γ^A are linearly independent. Their number equals the number ($4 \cdot 4$) of elements of a four-by-four matrix, so the γ^A form a complete system in which an arbitrary four-by-four matrix Γ can be expanded:

$$\Gamma = \sum_A c_A \gamma^A, \quad c_A = \frac{1}{4} \text{Tr } \gamma_A \Gamma, \quad (28.10)$$

or, explicitly displaying the matrix indices ($i, k = 1, 2, 3, 4$),

$$\Gamma_{ik} = \frac{1}{4} \sum_A \Gamma_{lm} \gamma_{ml}^A \gamma_{Aik}.$$

³³To avoid misunderstanding, recall that the transformations T and P also require a change of the function arguments. If the left-hand sides are functions of $x = (t, \mathbf{r})$, the right-hand sides (the transformed forms in (28.6) and (28.7)) are respectively functions of

$$x^T = (-t, \mathbf{r}), \quad x^P = (t, -\mathbf{r}).$$

Taking, in particular, a matrix Γ with only one nonzero element (Γ_{lm}), we obtain the required relation, the “completeness condition”:

$$\delta_{il}\delta_{km} = \frac{1}{4} \sum_A \gamma_{Aik} \gamma_{ml}^A. \quad (28.11)$$

Multiplying both sides of this equality by $\bar{\psi}_i^a \psi_k^b \bar{\psi}_m^c \psi_l^d$, we have

$$(\bar{\psi}^a \psi^d)(\bar{\psi}^c \psi^b) = \frac{1}{4} \sum_A (\bar{\psi}^a \gamma_A \psi^b)(\bar{\psi}^c \gamma^A \psi^d). \quad (28.12)$$

This is one equality of the type described above: it reduces a product of two scalar bilinear forms to products of forms constructed from other pairs of factors³⁴.

Other equalities of this type can be obtained from (28.12) by making the replacements

$$\psi^d \rightarrow \gamma^B \psi^d, \quad \psi^b \rightarrow \gamma^C \psi^b$$

and using the expansion (see the problem)

$$\gamma^A \gamma^B = \sum_R c_R \gamma^R, \quad c_R = \frac{1}{4} \text{Tr } \gamma^A \gamma^B \gamma_R.$$

For later reference, we also give the analogue of (28.11) for two-by-two matrices. A complete system of linearly independent two-by-two matrices σ^A ($A = 1, \dots, 4$) is

$$1, \quad \sigma_x, \quad \sigma_y, \quad \sigma_z. \quad (28.13)$$

For these matrices,

$$\text{Tr } \sigma^A = 0 \quad (\sigma^A \neq 1), \quad \frac{1}{2} \text{Tr } \sigma^A \sigma^B = \delta_{AB}. \quad (28.14)$$

The completeness condition is

$$\delta_{\alpha\gamma} \delta_{\beta\delta} = (1/2) \sum_A \sigma_{\alpha\beta}^A \sigma_{\delta\gamma}^A = (1/2) \sigma_{\alpha\beta} \sigma_{\delta\gamma} + (1/2) \delta_{\alpha\beta} \delta_{\delta\gamma} \quad (28.15)$$

($\alpha, \beta, \dots = 1, 2$), or equivalently,

$$\sigma_{\alpha\beta} \sigma_{\delta\gamma} = -(1/2) \sigma_{\alpha\gamma} \sigma_{\delta\beta} + (3/2) \delta_{\alpha\gamma} \delta_{\delta\beta}. \quad (28.16)$$

³⁴To avoid misunderstanding, recall that the forms here are constructed from ψ functions. For forms constructed from anticommuting ψ operators, the sign of the transformation would be reversed.

PROBLEM

Derive formulae analogous to (28.12) for scalar products of two bilinear forms P , V , A , and T .

SOLUTION. Introduce the notation

$$\begin{aligned} J_S &= (\bar{\psi}^a \psi^b)(\bar{\psi}^c \psi^d), & J_P &= (\bar{\psi}^a \gamma^5 \psi^b)(\bar{\psi}^c \gamma^5 \psi^d), \\ J_V &= (\bar{\psi}^a \gamma^\mu \psi^b)(\bar{\psi}^c \gamma_\mu \psi^d), & J_A &= (\bar{\psi}^a i \gamma^\mu \gamma^5 \psi^b)(\bar{\psi}^c i \gamma_\mu \gamma^5 \psi^d), \\ J_T &= (\bar{\psi}^a i \sigma^{\mu\nu} \psi^b)(\bar{\psi}^c i \sigma_{\mu\nu} \psi^d). \end{aligned}$$

Use the same letters with primes for the corresponding products in which ψ^b and ψ^d have been interchanged. The procedure described in the text gives

$$\begin{aligned} 4J'_S &= J_S + J_V + J_T + J_A + J_P, \\ 4J'_V &= 4J_S - 2J_V + 2J_A - 4J_P, \\ 4J'_T &= 6J_S - 2J_T + 6J_P, \\ 4J'_A &= 4J_S + 2J_V - 2J_A - 4J_P, \\ 4J'_P &= J_S - J_V + J_T - J_A + J_P \end{aligned}$$

(the first line follows from (28.12)).

§ 29. Polarization density matrix

The coordinate dependence of the wave function ψ describing free motion with momentum $\mathbf{p}$ (a plane wave) reduces to the common factor $e^{i\mathbf{p}\mathbf{r}}$, while the amplitude u_p plays the role of the spin wave function. In such a pure state, the particle is completely polarized (see III, § 59). In nonrelativistic theory, this means that the particle's spin has a definite direction in space (more precisely, there is a direction along which the spin projection has the definite value $+1/2$). In relativistic theory, such a characterization of the state in an arbitrary reference frame is impossible because the spin vector is not conserved, as already noted in § 23. Purity of the state means only that the spin has a definite direction in the particle's rest frame.

In a state of partial polarization there is no definite amplitude, but only a *polarization density matrix* ρ_{ik} ($i, k = 1, 2, 3, 4$ are bispinor indices). We define this matrix so that, for a pure state, it reduces to the products

$$\rho_{ik} = u_{pi} \bar{u}_{pk}. \quad (29.1)$$

Accordingly, ρ is normalized by

$$\text{Tr } \rho = 2m \quad (29.2)$$

(compare (23.4)).

For a pure state, the mean spin is determined by

$$\bar{\mathbf{s}} = \frac{1}{2} \int \psi^* \boldsymbol{\Sigma} \psi d^3x = \frac{1}{4\varepsilon} u_p^+ \boldsymbol{\Sigma} u_p = \frac{1}{4\varepsilon} \bar{u}_p \gamma^0 \boldsymbol{\Sigma} u_p. \quad (29.3)$$

The corresponding expression for a partially polarized state is

$$\bar{\mathbf{s}} = \frac{1}{4\varepsilon} \text{Tr}(\rho \gamma^0 \Sigma) = \frac{1}{4\varepsilon} \text{Tr}(\rho \gamma^5 \gamma). \quad (29.4)$$

The amplitudes u_p and $\bar{u}_p$ satisfy the algebraic systems

$$(\gamma p - m)u_p = 0, \quad \bar{u}_p(\gamma p - m) = 0.$$

The matrix (29.1) therefore satisfies

$$(\gamma p - m)\rho = 0, \quad \rho(\gamma p - m) = 0. \quad (29.5)$$

The density matrix of a general state mixed with respect to spin must obey the same linear equations (compare the analogous argument in III, § 14).

For a free particle in its rest frame, nonrelativistic theory applies. In that theory, a state of partial polarization is fully specified by three parameters, the components of the mean-spin vector $\bar{\mathbf{s}}$ (see III, § 59). The same parameters will plainly determine the polarization state after any Lorentz transformation, and hence also for a moving particle.

Denote twice the mean spin vector in the rest frame by ζ (for a pure state $|\zeta| = 1$, and for a mixed state $|\zeta| < 1$). For a four-dimensional description of the polarization state, it is convenient to introduce a 4-vector a^μ whose spatial part in the rest frame is ζ . Since ζ is an axial vector, a^μ is a 4-pseudovector. In the rest frame this 4-vector is orthogonal to the 4-momentum (there $a^\mu = (0, \zeta)$ and $p^\mu = (m, 0)$), and therefore in any reference frame

$$a^\mu p_\mu = 0. \quad (29.6)$$

In an arbitrary reference frame, one also has

$$a_\mu a^\mu = -\zeta^2. \quad (29.7)$$

The components of a^μ in a frame where the particle moves with velocity $\mathbf{v} = \mathbf{p}/\varepsilon$ are obtained by a Lorentz transformation from the rest frame and equal

$$a^0 = \frac{|\mathbf{p}|}{m} \zeta_\parallel, \quad \mathbf{a}_\perp = \zeta_\perp, \quad a_\parallel = \frac{\varepsilon}{m} \zeta_\parallel, \quad (29.8)$$

where $\parallel$ and $\perp$ denote components of ζ and $\mathbf{a}$ parallel and perpendicular to the direction of $\mathbf{p}$ ³⁵. These formulas can be written in vector form:

$$\mathbf{a} = \zeta + \frac{\mathbf{p}(\zeta \mathbf{p})}{m(\varepsilon + m)}, \quad a^0 = \frac{\mathbf{a} \mathbf{p}}{\varepsilon} = \frac{\mathbf{p} \zeta}{m}, \quad \mathbf{a}^2 = \zeta^2 + \frac{(\mathbf{p} \zeta)^2}{m^2}. \quad (29.9)$$

³⁵By their transformation properties, the components of the mean-spin vector $\bar{\mathbf{s}}$ (like those of any angular momentum) are the spatial components of an antisymmetric tensor $S^{\lambda\mu}$ in relativistic mechanics. The 4-vector a^λ is related to this tensor by

$$S^{\lambda\mu} = \frac{1}{2m} \epsilon^{\lambda\mu\nu\rho} a_\nu p_\rho, \quad a^\lambda = -\frac{2}{m} \epsilon^{\lambda\mu\nu\rho} S_{\mu\nu} p_\rho.$$

We emphasize that in an arbitrary reference frame, the spatial part of a^λ by no means coincides with $2\bar{\mathbf{s}}$. It is readily seen that $2\bar{\mathbf{s}}_\parallel = \frac{1}{m}(a_\parallel \varepsilon - a^0 |\mathbf{p}|) = \zeta_\parallel$, $2\bar{\mathbf{s}}_\perp = \frac{\varepsilon}{m} \mathbf{a}_\perp = \frac{\varepsilon}{m} \zeta_\perp$.

Consider first an unpolarized state ($\zeta = 0$). In this case, the density matrix can contain only the 4-momentum p as a parameter. The unique matrix of this kind satisfying (29.5) is

$$\rho = \frac{1}{2}(\gamma p + m) \quad (29.10)$$

(*I. E. Tamm*, 1930; *H. B. G. Casimir*, 1933). The constant coefficient has been chosen in accordance with the normalization (29.2).

For a general partially polarized state ($\zeta \neq 0$), seek the density matrix in the form

$$\rho = \frac{1}{4m}(\gamma p + m)\rho'(\gamma p + m), \quad (29.11)$$

which satisfies (29.5) automatically. When $\zeta \neq 0$, the auxiliary matrix ρ' must reduce to the identity; since

$$(\gamma p + m)^2 = 2m(\gamma p + m),$$

equation (29.11) then coincides with (29.10). Furthermore, ρ' must contain the 4-vector a linearly as a parameter, and hence must have the form

$$\rho' = 1 - A\gamma^5(\gamma a); \quad (29.12)$$

the second term contains the scalar product of the pseudovector a and the “matrix 4-pseudovector” $\gamma^5\gamma$. To determine A , write the density matrix in the rest frame:

$$\rho = \frac{m}{4}(1 + \gamma^0)(1 + A\gamma^5\gamma\zeta)(1 + \gamma^0) = \frac{m}{2}(1 + \gamma^0)(1 + A\gamma^5\gamma\zeta),$$

and calculate the mean spin from (29.4). Using the rules listed in § 22, we readily find that the only nonzero term in the required trace is

$$2\bar{s} = \frac{1}{2m} \text{Tr}(\rho\gamma^5\gamma) = -\frac{A}{4} \text{Tr}((\gamma\zeta)\gamma) = A\zeta.$$

Equating this expression to ζ gives $A = 1$. The final expression for ρ follows by substituting (29.12) into (29.11) and interchanging the factors ρ' and $(\gamma p + m)$. Since a and p are orthogonal, γp anticommutes with γa :

$$(\gamma a)(\gamma p) = 2ap - (\gamma p)(\gamma a) = -(\gamma p)(\gamma a),$$

and consequently commutes with $\gamma^5(\gamma a)$.

Thus the density matrix of a partially polarized electron is

$$\rho = \frac{1}{2}(\gamma p + m)[1 - \gamma^5(\gamma a)] \quad (29.13)$$

(*L. Michel*, *A. S. Wightman*, 1955). If ρ is known, the 4-vector a that characterizes the state (and hence also ζ) can be found from

$$a^\mu = \frac{1}{2m} \text{Tr}(\rho\gamma^5\gamma^\mu). \quad (29.14)$$

The formulas for the positron density matrix are analogous to those for the electron. If the positron (with 4-momentum p) were described by the positron amplitude $u_p^{(\text{pos})}$ and

by a density matrix $\rho^{(\text{pos})}$ defined from this amplitude, there would be no difference at all from the electron case, and $\rho^{(\text{pos})}$ would be given by the same formula (29.13). In actual calculations of cross sections for scattering processes involving positrons, however, one deals (as we shall see below) not with $u_p^{(\text{pos})}$ but with the “negative-frequency” amplitudes u_{-p} . The polarization density matrix, which we denote by $\rho^{(-)}$, must accordingly be defined so that for a pure state it reduces to $u_{-p} i \bar{u}_{-p} k$.

According to (26.1), the positron amplitude is $u_p^{(\text{pos})} = U_C \tilde{u}_{-p}$. Conversely,

$$u_{-p} = U_C \tilde{u}_p^{(\text{pos})}, \quad \tilde{u}_{-p} = U_C^+ u_p^{(\text{pos})} = \tilde{u}_p^{(\text{pos})} U_C^*$$

(compare (28.3)). If

$$\rho_{ik}^{(-)} = u_{-p} i \bar{u}_{-p} k, \quad \rho_{ik}^{(\text{pos})} = u_{pi}^{(\text{pos})} \bar{u}_{pk}^{(\text{pos})},$$

then these formulas give

$$\rho^{(-)} = U_C \tilde{\rho}^{(\text{pos})} U_C^*. \quad (29.15)$$

Substitution of (29.13) for $\rho^{(\text{pos})}$, followed by simple transformations using (26.3) and (26.21), gives

$$\rho^{(-)} = \frac{1}{2} (\gamma p - m) [1 - \gamma^5 (\gamma a)]. \quad (29.16)$$

In particular, for an unpolarized state,

$$\rho^{(-)} = \frac{1}{2} (\gamma p - m). \quad (29.17)$$

Henceforth, by positron density matrices we shall mean the matrices $\rho^{(-)}$ and shall omit their index $(-)$; in practice, the matrices $\rho^{(\text{pos})}$ are not needed.

In various calculations we shall often need to average over spin states expressions of the form $\bar{u} F u (\equiv \bar{u}_i F_{ik} u_k)$, where F is some four-row matrix and u is the bispinor amplitude of a state with definite 4-momentum p . Such an average is equivalent to replacing the products $u_k \bar{u}_i$ by the density matrix ρ_{ik} of a partially polarized state.

In particular, a complete average over the two independent spin states is equivalent to passing to an unpolarized state. By (29.10),

$$\frac{1}{2} \sum_{\text{pol}} \bar{u}_p F u_p = \frac{1}{2} \text{Tr}(\gamma p + m) F. \quad (29.18)$$

Similarly, for negative-frequency wave functions,

$$\frac{1}{2} \sum_{\text{pol}} \bar{u}_{-p} F u_{-p} = \frac{1}{2} \text{Tr}(\gamma p - m) F. \quad (29.19)$$

If the spin states are summed rather than averaged, the result is twice as large.

We now trace how the density matrix (29.13) approaches its nonrelativistic form. For this purpose, pass to the electron rest frame. In the standard representation of the

wave functions, the amplitudes u_p become two-component objects in this frame; the density matrix must correspondingly become two-row. Indeed, in the rest frame,

$$\rho = \frac{m}{2}(\gamma^0 + 1)(1 + \gamma^5 \boldsymbol{\gamma} \boldsymbol{\zeta}),$$

and the matrix expressions (21.20) and (22.18) give

$$\rho = \begin{pmatrix} \rho_{\text{nr}} & 0 \\ 0 & 0 \end{pmatrix}, \quad \rho_{\text{nr}} = m(1 + \boldsymbol{\sigma} \boldsymbol{\zeta}) \quad (29.20)$$

(the zeros denote two-row zero matrices). If the density matrix is given the normalization to unity customary in nonrelativistic theory ($\text{Tr } \rho_{\text{nr}} = 1$) instead of normalization to $2m$, this expression must be divided by $2m$, yielding, in agreement with formula (59.6) (see III),

$$\frac{1}{2}(1 + \boldsymbol{\sigma} \boldsymbol{\zeta}).$$

The nonrelativistic limit of the positron density matrix is obtained similarly:

$$\rho = \begin{pmatrix} 0 & 0 \\ 0 & \rho_{\text{nr}} \end{pmatrix}, \quad \rho_{\text{nr}} = -m(1 + \boldsymbol{\sigma} \boldsymbol{\zeta}).$$

Finally, consider a simplified expression for the density matrix in the ultrarelativistic case. Setting $|\mathbf{p}| \approx \varepsilon$ in (29.8), thereby neglecting quantities of relative order $(m/\varepsilon)^2$, substituting the result into (29.13) or (29.16), and choosing the direction of $\mathbf{p}$ as the x axis, we write

$$\rho = \frac{1}{2}[\varepsilon(\gamma^0 - \gamma^1) \pm m] \left[1 - \gamma^5 \left(\frac{\varepsilon}{m}(\gamma^0 - \gamma^1)\zeta_{\parallel} - \boldsymbol{\zeta}_{\perp} \boldsymbol{\gamma}_{\perp} \right) \right],$$

where the upper sign applies to the electron and the lower sign to the positron. On expanding the product, the leading terms cancel, while the next-order terms give

$$\rho = \frac{1}{2}\varepsilon(\gamma^0 - \gamma^1)[1 + \gamma^5(\pm\zeta_{\parallel} + \boldsymbol{\zeta}_{\perp} \boldsymbol{\gamma}_{\perp})]$$

or, writing $\varepsilon(\gamma^0 - \gamma^1)$ as γp ,

$$\rho = \frac{1}{2}(\gamma p)[1 + \gamma^5(\pm\zeta_{\parallel} + \boldsymbol{\zeta}_{\perp} \boldsymbol{\gamma}_{\perp})]. \quad (29.21)$$

This is the required density matrix in the ultrarelativistic case. Notice that all components of the polarization vector $\boldsymbol{\zeta}$ enter on an equal footing, as terms of the same order. Recall that $\zeta_{\parallel}$ is the component of this vector parallel to the particle momentum when $\zeta_{\parallel} > 0$ and antiparallel when $\zeta_{\parallel} < 0$. In particular, for a particle in a helicity state, $\zeta_{\parallel} = 2\lambda = \pm 1$; the density matrix then has the especially simple form

$$\rho = \frac{1}{2}(\gamma p)(1 \pm 2\lambda\gamma^5), \quad (29.22)$$

which, as expected, coincides with the density matrix of a neutrino or antineutrino, a particle of zero mass and definite helicity (see § 30).

§ 30. Two-component fermions

We saw in § 20 that the need to describe a spin-1/2 particle by two spinors, ξ and η , is connected with the particle's mass. This reason disappears when the mass is zero. The wave equation for such a particle can be constructed with only one spinor, say the dotted spinor η :

$$\widehat{p}^{\alpha\dot{\beta}}\eta_{\dot{\beta}} = 0, \quad (30.1)$$

or, equivalently,

$$(\widehat{p}_0 + \widehat{\mathbf{p}}\boldsymbol{\sigma})\eta = 0. \quad (30.2)$$

It was also noted in § 20 that a wave equation containing a mass m is automatically symmetric under inversion, with transformation (20.4). This symmetry is lost when the particle is described by a single spinor. It is not required, however, since inversion symmetry is not a universal property of nature.

For a particle with $m = 0$, energy and momentum are related by $\varepsilon = |\mathbf{p}|$. Hence, for a plane wave $\eta_p \propto e^{-ipx}$, (30.2) gives

$$(\mathbf{n}\boldsymbol{\sigma})\eta_p = -\eta_p, \quad (30.3)$$

where $\mathbf{n}$ is the unit vector along $\mathbf{p}$. The same equation,

$$(\mathbf{n}\boldsymbol{\sigma})\eta_{-p} = -\eta_{-p} \quad (30.4)$$

holds for the wave of “negative frequency,” $\eta_{-p} \propto e^{ipx}$. The second-quantized ψ operator is

$$\widehat{\eta} = \sum_{\mathbf{p}} (\eta_p \widehat{a}_{\mathbf{p}} + \eta_{-p} \widehat{b}_{\mathbf{p}}^{\dagger}), \quad \widehat{\eta}^{\dagger} = \sum_{\mathbf{p}} (\eta_p^* \widehat{a}_{\mathbf{p}}^{\dagger} + \eta_{-p}^* \widehat{b}_{\mathbf{p}}). \quad (30.5)$$

It follows in the usual way that η_{-p}^* are the wave functions of the antiparticle.

The definition of the operators $\widehat{p}^{\alpha\dot{\beta}}$ in (20.1) shows that $\widehat{p}^{\alpha\dot{\beta}*} = -\widehat{p}^{\dot{\alpha}\beta}$. The complex-conjugate spinor η^* therefore satisfies $\widehat{p}^{\alpha\dot{\beta}}\eta_{\dot{\beta}}^* = 0$, or equivalently

$$\widehat{p}_{\dot{\alpha}\beta}\eta^{\dot{\beta}*} = 0.$$

Write $\eta^{\dot{\beta}*} = \xi^{\beta}$, thereby expressing the fact that complex conjugation changes a dotted spinor into an undotted one. The antiparticle wave functions thus satisfy

$$\widehat{p}_{\dot{\alpha}\beta}\xi^{\beta} = 0, \quad (30.6)$$

or

$$(\widehat{p}_0 - \widehat{\mathbf{p}}\boldsymbol{\sigma})\xi = 0. \quad (30.7)$$

For a plane wave this gives

$$(\mathbf{n}\boldsymbol{\sigma})\xi_p = \xi_p. \quad (30.8)$$

But $1/2(\mathbf{n}\boldsymbol{\sigma})$ is the operator for the projection of spin on the direction of motion. Equations (30.3) and (30.8) therefore mean that a particle state of definite momentum is automatically a helicity state: the spin projection along the direction of motion has a

definite value. If the particle's spin is opposite to its momentum, giving helicity $-1/2$, the antiparticle's spin is directed along its momentum, giving helicity $+1/2$.

Neutrinos occurring in nature may be particles with these properties. By convention, the particle of helicity $-1/2$ is called a neutrino and that of helicity $+1/2$ an antineutrino³⁶.

In connection with the absence of spin-direction degeneracy for neutrino states, recall the observation in § 8 that a particle of zero mass has only axial symmetry about the direction of its momentum. For a genuinely neutral particle, the photon, this symmetry includes both rotations about the axis and reflections in planes through it. For the neutrino there is no reflection symmetry; only the group of rotations about the axis remains, preserving the projection of angular momentum on the axis without reversing its sign. Reflection symmetry exists only if particle and antiparticle are interchanged at the same time.

It should also be noted that compulsory longitudinal polarization means that for a neutrino the spin cannot be separated at all from the orbital angular momentum, just as for a photon the fields are necessarily transverse; see § 6.

Only four bilinear combinations can be formed from a single spinor η , or ξ ; together they make up the 4-vector

$$j^\mu = (\eta^* \eta, \eta^* \sigma \eta). \quad (30.9)$$

It is readily checked from the equations

$$(\widehat{p}_0 + \widehat{\mathbf{p}} \sigma) \eta = 0, \quad \eta^* (\widehat{p}_0 - \widehat{\mathbf{p}} \sigma) = 0$$

that the continuity equation $\partial_\mu j^\mu = 0$ holds. Thus j^μ serves as the particle-current density 4-vector.

It is convenient to normalize plane neutrino waves in a manner analogous to that used in § 23 for massive particles:

$$\eta_p = \frac{1}{\sqrt{2\varepsilon}} u_p e^{-ipx}, \quad \eta_{-p} = \frac{1}{\sqrt{2\varepsilon}} u_{-p} e^{ipx}, \quad (30.10)$$

where the spinor amplitudes obey the invariant normalization

$$u_{\pm p}^* (1, \sigma) u_{\pm p} = 2(\varepsilon, \mathbf{p}). \quad (30.11)$$

The particle density and its current density are then $j^0 = 1$ and $\mathbf{j} = \mathbf{p}/\varepsilon = \mathbf{n}$.

Since a free neutrino of specified momentum is always fully polarized, there is no notion of a spin-mixed state in this case. It can nevertheless be convenient to introduce a two-row polarization “density matrix,” defined simply as the second-rank spinor

$$\rho_{\alpha\beta} = u_\alpha u_\beta^* \quad (30.12)$$

³⁶The existence of the neutrino was predicted theoretically by Pauli to explain the properties of β decay (1931). Equation (30.1) was first considered by Weyl (H. Weyl, 1929). A neutrino theory based on these equations was formulated in 1957 by L. D. Landau, and by Lee, Yang, and Salam (T. D. Lee, C. N. Yang, A. Salam).

Experiment has not yet settled conclusively whether the neutrino mass is zero. Below, we shall use the term “neutrino” conventionally for a particle described by (30.3).

(with $\text{Tr } \rho = 2\varepsilon$). Its expression follows by observing that it must satisfy

$$(\varepsilon + \mathbf{p}\boldsymbol{\sigma})\rho = \rho(\varepsilon + \mathbf{p}\boldsymbol{\sigma}) = 0.$$

It follows that

$$\rho = \varepsilon - \mathbf{p}\boldsymbol{\sigma}. \quad (30.13)$$

In studying various interaction processes, neutrinos may occur together with other, massive particles of spin 1/2, which are consequently described by four-component wave functions. It is then convenient to keep the notation uniform by formally introducing a “bispinor” wave function for the neutrino as well, although two of its components vanish: $\psi = \begin{pmatrix} 0 \\ \eta \end{pmatrix}$. In general, however, this form of ψ is not preserved on passing to another, nonspinor representation. The difficulty can be avoided by noting that in the spinor representation the identities

$$\frac{1 + \gamma^5}{2} \begin{pmatrix} \xi \\ \eta \end{pmatrix} = \begin{pmatrix} 0 \\ \eta \end{pmatrix}, \quad (\eta^* \ \xi^*) \frac{1 - \gamma^5}{2} = (\eta^* \ 0),$$

hold, where ξ is an arbitrary “ballast” spinor that drops out of the result; the matrix γ^5 is given in (22.18). The neutrino’s genuinely “two-component” character is therefore maintained when it is described by a four-component ψ in any representation, provided ψ is understood to be a solution of the Dirac equation with $m = 0$,

$$(\gamma p)\psi = 0, \quad (30.14)$$

subject to the supplementary condition $1/2 (1 + \gamma^5)\psi = \psi$, or

$$\gamma^5\psi = \psi. \quad (30.15)$$

This condition can be incorporated by agreeing to make the following replacement in every formula containing ψ and $\bar{\psi}$:

$$\psi \rightarrow \frac{1 + \gamma^5}{2}\psi, \quad \bar{\psi} \rightarrow \bar{\psi} \frac{1 - \gamma^5}{2}. \quad (30.16)$$

For example, the current-density 4-vector becomes, upon making (30.16) in $\bar{\psi}\gamma^\mu\psi$,

$$j^\mu = \frac{1}{4}\bar{\psi}(1 - \gamma^5)\gamma^\mu(1 + \gamma^5)\psi = \frac{1}{2}\bar{\psi}\gamma^\mu(1 + \gamma^5)\psi. \quad (30.17)$$

By the same rule, the four-row neutrino density matrix must be written

$$\rho = \frac{1}{4}(1 + \gamma^5)(\gamma p)(1 - \gamma^5) = \frac{1}{2}(1 + \gamma^5)(\gamma p). \quad (30.18)$$

In the spinor representation it reduces, as it should, to the two-row matrix (30.13):

$$\rho = \begin{pmatrix} 0 & 0 \\ \varepsilon - \boldsymbol{\sigma}\mathbf{p} & 0 \end{pmatrix}.$$

The analogous formulas for the antineutrino differ from those above by reversing the sign before γ^5 .

The neutrino is electrically neutral. A neutrino with the properties described above is not, however, genuinely neutral. In this connection, note that the “neutrino field” described by a two-component spinor is equivalent, in the number of particle states available to it, though certainly not in its other physical properties, to a genuinely neutral field described by a four-component bispinor. In place of particle and antiparticle states with definite helicities, there would be the same number of states of one particle with the two possible helicities, and inversion symmetry would hold automatically. The vanishing mass of such a “four-component” neutrino would, however, have an “accidental” character, so to speak, since it would not follow from the symmetry properties of its wave equation, which also permits a nonzero mass. Including the various interactions of such a particle would therefore automatically generate a rest mass that might be small but would not be exactly zero.

§ 31. Wave equation for a spin-3/2 particle

In its rest frame, a spin-3/2 particle is described by a symmetric third-rank 3-spinor, with $2s + 1 = 4$ independent components. Accordingly, in an arbitrary reference frame its description may involve the 4-spinors $\xi^{\alpha\beta\dot{\gamma}}$, $\eta_{\dot{\alpha}\beta\gamma}$ and $\zeta^{\alpha\beta\gamma}$, $\chi_{\dot{\alpha}\beta\dot{\gamma}}$. Each is symmetric in all indices of the same kind, dotted or undotted; inversion interchanges the spinors within the first pair and within the second pair.

For the 4-spinors $\xi^{\alpha\beta\dot{\gamma}}$ and $\eta_{\dot{\alpha}\beta\gamma}$ to become 3-spinors symmetric in all three indices in the rest frame, they must obey

$$\widehat{p}^{\dot{\alpha}\beta}\eta_{\dot{\alpha}\beta\gamma} = 0, \quad \widehat{p}_{\alpha\dot{\beta}}\xi^{\alpha\beta\dot{\gamma}} = 0. \quad (31.1)$$

Indeed, in the rest frame,

$$\widehat{p}^{\dot{\alpha}\beta} \rightarrow \widehat{p}_0\delta_{\alpha}^{\beta} = m\delta_{\alpha}^{\beta}$$

as follows from (20.1). Conditions (31.1) therefore give

$$\delta_{\alpha}^{\beta}\eta'^{\alpha}_{\beta\gamma} = 0, \quad \delta_{\alpha}^{\beta}\xi'^{\alpha}_{\beta\gamma} = 0,$$

where primed letters denote the corresponding 3-spinors. In other words, these spinors vanish upon simplification over the indices $\alpha\beta$. This means that they are symmetric in those indices, and hence in all three indices.

The differential relation between the spinors ξ and η is established by

$$\widehat{p}^{\delta\dot{\gamma}}\eta_{\alpha\dot{\delta}}^{\beta} = m\xi_{\alpha}^{\beta\dot{\gamma}}, \quad \widehat{p}_{\delta\dot{\gamma}}\xi_{\alpha}^{\beta\dot{\gamma}} = m\eta_{\alpha\dot{\delta}}^{\beta}. \quad (31.2)$$

The left-hand sides of these equations are symmetric in $\dot{\beta}, \dot{\gamma}$ or in α, δ , respectively. This symmetry is ensured by (31.1), by virtue of which they vanish upon simplification over all the indices. In the rest frame, equations (31.2) make the 3-spinors ξ' and η' identical. Eliminating either η or ξ from (31.2), we find that every component of both spinors satisfies the second-order equation

$$(\widehat{p}^2 - m^2)\xi^{\alpha\beta\dot{\gamma}} = 0. \quad (31.3)$$

Equations (31.1) and (31.2) together form the complete system of wave equations for a spin-3/2 particle³⁷. Adding the spinors ζ and χ would introduce nothing new. They are constructed according to

$$m\zeta^{\alpha\beta\gamma} = \widehat{p}^{\alpha\delta}\eta_{\delta}^{\beta\gamma}, \quad m\chi_{\dot{\alpha}\dot{\beta}\dot{\gamma}} = \widehat{p}_{\dot{\alpha}\delta}\xi_{\dot{\beta}\dot{\gamma}}^{\delta}.$$

The equations for spin-3/2 particles also admit another formulation, which uses the vector aspects of spinor properties (W. Rarita, J. Schwinger, 1941; A. S. Davydov, I. E. Tamm, 1942). A single four-dimensional vector index μ is associated with the pair of spinor indices $\alpha\dot{\beta}$. The components of the third-rank spinor $\xi^{\alpha\dot{\beta}\dot{\gamma}}$ can therefore be put into correspondence with components of the “mixed” quantities $\psi_{\mu}^{\dot{\gamma}}$, which carry one vector and one spinor index. Similarly, the spinor $\eta^{\dot{\beta}\alpha\gamma}$ corresponds to the quantities ψ_{μ}^{γ} , while the two spinors together correspond to a “vector” bispinor ψ_{μ} ; its bispinor index is not displayed. The wave equation then takes the form of a “Dirac equation” for each vector component ψ_{μ} :

$$(\gamma^{\mu}\widehat{p} - m)\psi_{\mu} = 0 \quad (31.4)$$

with the additional condition

$$\gamma^{\mu}\psi_{\mu} = 0. \quad (31.5)$$

Using the spinor-representation expressions for the matrices γ^{μ} and the formulas relating spinor and vector components (18.6), (18.7), one readily verifies that (31.2) is contained in (31.4), while condition (31.5) is equivalent to symmetry of the spinors $\xi^{\alpha\dot{\beta}\dot{\gamma}}$ and $\eta^{\dot{\beta}\alpha\gamma}$ in $\dot{\beta}\dot{\gamma}$ or $\beta\gamma$. Multiplication of (31.4) by γ^{μ} gives, in view of (31.5),

$$\gamma^{\mu}\gamma^{\nu}\widehat{p}_{\nu}\psi_{\mu} = 0,$$

or, on using the commutation rules for the matrices γ^{μ} ,

$$2g^{\mu\nu}\widehat{p}_{\nu}\psi_{\mu} - \gamma^{\nu}\widehat{p}_{\nu}\gamma^{\mu}\psi_{\mu} = 0. \quad (31.6)$$

The second term again vanishes by (31.5), while the first yields

$$\widehat{p}^{\mu}\psi_{\mu} = 0. \quad (31.7)$$

It is readily seen that this condition, which follows automatically from (31.4) and (31.5), is equivalent to (31.1).

Finally, a further formulation of the wave equation introduces quantities ψ_{ikl} ($i, k, l = 1, 2, 3, 4$) symmetric in their three bispinor indices (V. Bargmann, E. P. Wigner, 1948). Together these quantities are equivalent to the combined components of all four spinors ξ, η, ζ , and χ . The wave equation is written as a system of “Dirac equations”:

$$\widehat{p}_{\mu}\gamma_{im}^{\mu}\psi_{mkl} = m\psi_{ikl}. \quad (31.8)$$

These equations already leave the required number, four, of independent components ψ_{ikl} , so no additional conditions need be imposed. Indeed, in the rest frame (31.8) reduces to

$$\gamma_{im}^0\psi_{mkl} = \psi_{ikl},$$

³⁷For a Lagrangian formulation of these equations, see the paper by Fierz and Pauli cited on p. 76.

which, in the standard representation, makes all components with $i, k, l = 3, 4$ vanish; thus ψ_{ikl} reduces to the components of a third-rank 3-spinor.

The results above generalize directly to particles of any half-integral spin s . In a description by equations of the form (31.4), (31.5), the wave function is a symmetric 4-tensor of rank $(2s - 1)/2$ carrying one bispinor index. In a description by equations of the form (31.8), the wave function instead carries $2s$ bispinor indices and is symmetric in them.

A Particle in an External Field

§ 32. The Dirac equation for an electron in an external field

In essence, the wave equations for free particles express only those properties that follow from the general requirements of space-time symmetry. The physical processes involving particles, by contrast, depend on the properties of their interactions.

Electromagnetic interactions between particles can be described in relativistic quantum theory by extending the method used for this purpose in classical theory and in non-relativistic quantum theory.

This method, however, describes the electromagnetic interactions only of particles that do not undergo strong interactions. Electrons (and positrons) belong to this class, and hence the existing theory encompasses the whole extensive domain of electron quantum electrodynamics. The unstable particles known as muons likewise do not undergo strong interactions; they are governed by the same quantum electrodynamics for phenomena occurring over times short compared with their lifetime (which is determined by weak interactions).

In this chapter we consider the part of quantum electrodynamics that lies within a one-particle theory. These are problems in which the number of particles remains fixed and the interaction can be introduced through an external electromagnetic field. Besides the conditions under which the external field may be regarded as prescribed, the range of validity of such a theory is also restricted by conditions associated with the so-called radiative corrections.

The wave equation of an electron in a prescribed external field can be obtained in the same way as in non-relativistic theory (see III, § 111). Let $A^\mu = (\Phi, \mathbf{A})$ be the 4-potential of the external electromagnetic field ($\mathbf{A}$ and Φ are respectively the vector and scalar potentials). The required equation follows by replacing the 4-momentum operator $\hat{p}$ in the Dirac equation by $\hat{p} - eA$, where e is the charge of the particle¹:

$$[\gamma(\hat{p} - eA) - m]\psi = 0. \quad (32.1)$$

The Hamiltonian corresponding to this equation is obtained from (21.13) by the same replacement:

$$\hat{H} = \alpha(\hat{\mathbf{p}} - e\mathbf{A}) + \beta m + e\Phi. \quad (32.2)$$

Gauge invariance of the Dirac equation under a transformation of the electromagnetic potentials means that its form is unchanged if, together with $A \rightarrow A + i\hat{p}\chi$ (where χ is

¹The charge is understood to include its sign; thus, for an electron, $e = -|e|$.

an arbitrary function), the wave function is transformed according to²

$$\psi \rightarrow \psi e^{ie\chi} \quad (32.3)$$

(compare the analogous transformation of the Schrödinger equation in vol. III, § 111).

The current density expressed through the wave function is given, just as in the absence of an external field, by (21.11), $j = \bar{\psi}\gamma\psi$. If the derivation of (21.11) is repeated using (32.1) and equation (32.4) below, it is readily seen that the external field cancels; the continuity equation therefore remains valid with the same expression for the current.

We now apply charge conjugation to (32.1). For this purpose, write

$$\bar{\psi}[\gamma(\hat{p} + eA) + m] = 0, \quad (32.4)$$

which follows from (32.1) by complex conjugation in the same manner as equation (21.9) was obtained earlier (recall here that the 4-vector A is real). Rewriting this equation as

$$[\bar{\gamma}(\hat{p} + eA) + m]\bar{\psi} = 0,$$

multiplying it on the left by the matrix U_C , and using (26.3), we find

$$[\gamma(\hat{p} + eA) - m](\hat{C}\psi) = 0. \quad (32.5)$$

Thus the charge-conjugate wave function obeys an equation that differs from the original one by a reversal of the charge sign. On the other hand, charge conjugation changes particles into antiparticles. It follows that, when the particles carry electric charge, the electron and positron automatically have charges of opposite sign.

The first-order equation (32.1) can be converted into a second-order equation by applying the operator $\gamma(\hat{p} - eA) + m$ to it:

$$[\gamma^\mu\gamma^\nu(\hat{p}_\mu - eA_\mu)(\hat{p}_\nu - eA_\nu) - m^2]\psi = 0.$$

We replace the product $\gamma^\mu\gamma^\nu$ by

$$\gamma^\mu\gamma^\nu = \frac{1}{2}(\gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu) + \frac{1}{2}(\gamma^\mu\gamma^\nu - \gamma^\nu\gamma^\mu) = g^{\mu\nu} + \sigma^{\mu\nu},$$

where $\sigma^{\mu\nu}$ is the antisymmetric «matrix 4-tensor» (28.2). In multiplication by $\sigma^{\mu\nu}$ we may antisymmetrize, that is, make the replacement

$$\begin{aligned} (\hat{p}_\mu - eA_\mu)(\hat{p}_\nu - eA_\nu) &\rightarrow \frac{1}{2}\{(\hat{p}_\mu - eA_\mu)(\hat{p}_\nu - eA_\nu)\} - \\ &= \frac{1}{2}e(-A_\mu\hat{p}_\nu + \hat{p}_\nu A_\mu - \hat{p}_\mu A_\nu + A_\nu\hat{p}_\mu) \\ &= \frac{1}{2}ie(\partial_\nu A_\mu - \partial_\mu A_\nu) = -\frac{ie}{2}F_{\mu\nu} \end{aligned}$$

($F_{\mu\nu} = \partial_\mu A_\nu - \partial_\nu A_\mu$ is the electromagnetic-field tensor). The resulting second-order equation is

$$\left[(\hat{p} - eA)^2 - m^2 - \frac{i}{2}eF_{\mu\nu}\sigma^{\mu\nu} \right] \psi = 0. \quad (32.6)$$

²Transformation (32.3) with a function $\chi(t, \mathbf{r})$ is sometimes called a «local gauge transformation», in contrast to the «global gauge transformation» (12.10), whose phase α is constant.

The product $F_{\mu\nu}\sigma^{\mu\nu}$ can be put in three-dimensional form by expressing it through the components

$$\sigma^{\mu\nu} = (\boldsymbol{\alpha}, i\boldsymbol{\Sigma}), \quad F^{\mu\nu} = (-\mathbf{E}, \mathbf{H}).$$

Then

$$[(\hat{p} - eA)^2 - m^2 + e\boldsymbol{\Sigma}\mathbf{H} - ie\boldsymbol{\alpha}\mathbf{E}]\psi = 0, \quad (32.7)$$

or, in ordinary units,

$$\left[\left(\frac{i\hbar}{c} \frac{\partial}{\partial t} - \frac{e}{c} \Phi \right)^2 - \left(i\hbar \boldsymbol{\nabla} + \frac{e}{c} \mathbf{A} \right)^2 - m^2 c^2 + \frac{e\hbar}{c} \boldsymbol{\Sigma}\mathbf{H} - i \frac{e\hbar}{c} \boldsymbol{\alpha}\mathbf{E} \right] \psi = 0. \quad (32.7a)$$

The terms involving the fields $\mathbf{E}$ and $\mathbf{H}$ appear in these equations because the particle has spin; we return to them in the next section.

The solutions of the second-order equation naturally include some extraneous ones that do not satisfy the original first-order equation (32.1) (they solve

equation (32.1) with the sign before m reversed). In particular cases the required solutions are usually easy to select. A systematic selection procedure is as follows: if φ is any solution of the second-order equation, then a solution of the correct first-order equation is

$$\psi = [\gamma(\hat{p} - eA) + m]\varphi. \quad (32.8)$$

Indeed, multiplying this equality by $\gamma(\hat{p} - eA) - m$ shows that its right-hand side vanishes when φ satisfies (32.6).

It should be stressed that introducing an external field into a relativistic wave equation by replacing $\hat{p}$ with $\hat{p} - eA$ is not self-evident. In making this prescription we have, in effect, invoked an additional principle: the stated replacement must be carried out in first-order equations. It is precisely for this reason that the extra terms in (32.6) arise; they would not have appeared had the replacement been made directly in the second-order equation.

The stationary solutions of the Dirac equation in an external field may include states in both continuous and discrete spectra. As in non-relativistic theory, continuous-spectrum states describe unbounded motion: the particle can reach infinity, where it may be treated as free. The free-particle Hamiltonian has eigenvalues $\pm\sqrt{\mathbf{p}^2 + m^2}$, so the continuous energy spectrum clearly occupies $\varepsilon \geq m$ and $\varepsilon \leq -m$. For $-m < \varepsilon < m$, by contrast, the particle cannot be at infinity; its motion is bound and the state belongs to the discrete spectrum.

As for free particles, wave functions of «positive frequency» ($\varepsilon > 0$) and «negative frequency» ($\varepsilon < 0$) enter the second-quantized description in a definite manner. For particles in an external field, this description is naturally generalized by replacing the plane waves in (25.1) by normalized eigenfunctions $\psi_n^{(+)}$ and $\psi_n^{(-)}$ of the Dirac equation, associated respectively with positive frequencies $\varepsilon_n^{(+)}$ and negative frequencies $-\varepsilon_n^{(-)}$:

$$\begin{aligned} \hat{\psi} &= \sum_n \left\{ \hat{a}_n \psi_n^{(+)} \exp(-i\varepsilon_n^{(+)} t) + \hat{b}_n^+ \psi_n^{(-)} \exp(i\varepsilon_n^{(-)} t) \right\}, \\ \hat{\bar{\psi}} &= \sum_n \left\{ \hat{a}_n^+ \bar{\psi}_n^{(+)} \exp(i\varepsilon_n^{(+)} t) + \hat{b}_n \bar{\psi}_n^{(-)} \exp(-i\varepsilon_n^{(-)} t) \right\}. \end{aligned} \quad (32.9)$$

One must bear in mind that, as a potential well is deepened, energy levels may cross $\varepsilon = 0$, changing from positive to negative (or, for a potential of the opposite sign, from negative to positive). Continuity nevertheless requires that such levels continue to be counted as electron levels, not positron levels. In other words, the electron states are all those that, when the field is switched off infinitely slowly, join the positive boundary of the continuous spectrum ($\varepsilon = m$).

Although, as already noted, the Dirac equation for an electron in an external field makes it possible to solve a broad class of quantum-electrodynamic problems, the external-field concept within a one-particle problem still has a restricted range in relativistic theory. This restriction results from the spontaneous creation of electron-positron pairs in sufficiently strong fields (see below, § 35, 36).

We shall not discuss in this book how an external field is introduced into the wave equations of particles whose spin differs from $1/2$, since the question has no direct physical meaning: real particles with such spins are hadrons, and wave equations cannot describe their electromagnetic interactions. It should also be noted in this connection that these equations can produce physically inconsistent results. Thus, for particles of spin 0, the wave equation in a sufficiently deep potential well has complex energy levels (with imaginary parts of either sign). For particles of spin $3/2$, the wave equation violates causality, as is manifested by solutions that propagate faster than light.

PROBLEM

Determine the energy levels of an electron in a constant magnetic field.

SOLUTION. The vector potential is $A_x = A_z = 0$, $A_y = Hx$ (the field $\mathbf{H}$ is directed along the z axis). Besides the energy, the components p_y and p_z of the generalized momentum are conserved.

We use the second-order equation for the auxiliary function φ (see (32.8)) and take φ to be an eigenfunction of Σ_z (with eigenvalue $\sigma = \pm 1$), as well as of the operators $\hat{p}_y$ and $\hat{p}_z$. The equation for φ is

$$\left\{ -\frac{d^2}{dx^2} + (eHx - p_y)^2 - eH\sigma \right\} \varphi = (\varepsilon^2 - m^2 - p_z^2) \varphi.$$

This equation has the same form as the Schrödinger equation for a linear oscillator. The eigenvalues ε are given by

$$\varepsilon^2 - m^2 - p_z^2 = |e|H(2n + 1) - eH\sigma, \quad n = 0, 1, 2, \dots$$

(compare III, § 112). Note that the wave function ψ , which must be obtained from φ using (32.8), is not an eigenfunction of Σ_z ; this is consistent with the fact that spin is not a conserved quantity for a moving particle.

§ 33. Expansion in powers of $1/c$

3

³Conventional units are used throughout this section.

We saw in § 21 that, in the nonrelativistic limit ($v \rightarrow 0$), two components (χ) of the bispinor $\psi = \begin{pmatrix} \varphi \\ \chi \end{pmatrix}$ vanish. Thus $\chi \ll \varphi$ when the electron velocity is small. This allows an approximate equation involving only the two-component quantity φ to be obtained by formally expanding the wave function in powers of $1/c$.

We begin with the Dirac equation for an electron in an external field, written as

$$i\hbar \frac{\partial \psi}{\partial t} = \left\{ c\alpha \left(\hat{\mathbf{p}} - \frac{e}{c} \mathbf{A} \right) + \beta mc^2 + e\Phi \right\} \psi. \quad (33.1)$$

The relativistic energy of the particle includes its rest energy mc^2 . To pass to the nonrelativistic approximation, we remove this term by replacing ψ with a function ψ' defined by

$$\psi = \psi' e^{-imc^2 t/\hbar}.$$

It then follows that

$$\left(i\hbar \frac{\partial}{\partial t} + mc^2 \right) \psi' = \left\{ c\alpha \left(\hat{\mathbf{p}} - \frac{e}{c} \mathbf{A} \right) + \beta mc^2 + e\Phi \right\} \psi'.$$

Writing ψ' as $\psi' = \begin{pmatrix} \varphi' \\ \chi' \end{pmatrix}$ gives the pair of equations

$$\left(i\hbar \frac{\partial}{\partial t} - e\Phi \right) \varphi' = c\sigma \left(\hat{\mathbf{p}} - \frac{e}{c} \mathbf{A} \right) \chi', \quad (33.2)$$

$$\left(i\hbar \frac{\partial}{\partial t} - e\Phi + 2mc^2 \right) \chi' = c\sigma \left(\hat{\mathbf{p}} - \frac{e}{c} \mathbf{A} \right) \varphi' \quad (33.3)$$

(below, the primes on φ and χ will be omitted; this cannot cause confusion because only the transformed function ψ' is used in this section).

At first order we retain only $2mc^2\chi$ on the left side of (33.3), obtaining

$$\chi = \frac{1}{2mc} \sigma \left(\hat{\mathbf{p}} - \frac{e}{c} \mathbf{A} \right) \varphi \quad (33.4)$$

(notice that $\chi \sim \varphi/c$). Substitution in (33.2) yields

$$\left(i\hbar \frac{\partial}{\partial t} - e\Phi \right) \varphi = \frac{1}{2m} \left(\sigma \left(\hat{\mathbf{p}} - \frac{e}{c} \mathbf{A} \right) \right)^2 \varphi.$$

The Pauli matrices obey

$$(\sigma \mathbf{a})(\sigma \mathbf{b}) = \mathbf{a} \cdot \mathbf{b} + i\sigma(\mathbf{a} \times \mathbf{b}), \quad (33.5)$$

where $\mathbf{a}$ and $\mathbf{b}$ are arbitrary vectors (see (20.9)). Here $\mathbf{a} = \mathbf{b} = \hat{\mathbf{p}} - \frac{e}{c} \mathbf{A}$, but the cross product does not vanish because $\hat{\mathbf{p}}$ and $\mathbf{A}$ do not commute:

$$\left[\left(\hat{\mathbf{p}} - \frac{e}{c} \mathbf{A} \right) \times \left(\hat{\mathbf{p}} - \frac{e}{c} \mathbf{A} \right) \right] \varphi = i \frac{e\hbar}{c} \{ \mathbf{A} \times \nabla + \nabla \times \mathbf{A} \} \varphi = i \frac{e\hbar}{c} \nabla \times \mathbf{A} \varphi.$$

Consequently,

$$\left(\boldsymbol{\sigma} \left(\hat{\mathbf{p}} - \frac{e}{c}\mathbf{A}\right)\right)^2 = \left(\hat{\mathbf{p}} - \frac{e}{c}\mathbf{A}\right)^2 - \frac{e\hbar}{c}\boldsymbol{\sigma}\mathbf{H} \quad (33.6)$$

(where $\mathbf{H} = \nabla \times \mathbf{A}$ is the magnetic field), and the equation for φ becomes

$$i\hbar \frac{\partial \varphi}{\partial t} = \hat{H}\varphi = \left[\frac{1}{2m} \left(\hat{\mathbf{p}} - \frac{e}{c}\mathbf{A}\right)^2 + e\Phi - \frac{e\hbar}{2mc}\boldsymbol{\sigma}\mathbf{H} \right] \varphi. \quad (33.7)$$

This is the *Pauli equation*. It differs from the nonrelativistic Schrödinger equation through the last term in the Hamiltonian, which has the form of the potential energy of a magnetic dipole in an external field (compare III, § 111). Hence, to first order in $1/c$, the electron behaves as a particle carrying, in addition to its charge, a magnetic moment

$$\boldsymbol{\mu} = \frac{e}{mc}\hbar\mathbf{s}. \quad (33.8)$$

The gyromagnetic ratio e/mc is twice the value that a magnetic moment associated with orbital motion would have⁴.

The density is $\rho = \psi^*\psi = \varphi^*\varphi + \chi^*\chi$. In the first approximation the second term must be omitted, leaving $\rho = |\varphi|^2$, as required for the Schrödinger equation.

The current density, on the other hand, is

$$\mathbf{j} = c\psi^*\boldsymbol{\alpha}\psi = c(\varphi^*\boldsymbol{\sigma}\chi + \chi^*\boldsymbol{\sigma}\varphi).$$

Using (33.4), we substitute

$$\chi = \frac{1}{2mc}\boldsymbol{\sigma} \left(-i\hbar\nabla - \frac{e}{c}\mathbf{A}\right)\varphi, \quad \chi^* = \frac{1}{2mc} \left(i\hbar\nabla - \frac{e}{c}\mathbf{A}\right)\varphi^*\boldsymbol{\sigma},$$

and transform the products containing two factors $\boldsymbol{\sigma}$ by using (33.5) in the form

$$(\boldsymbol{\sigma}\mathbf{a})\boldsymbol{\sigma} = \mathbf{a} + i(\boldsymbol{\sigma} \times \mathbf{a}), \quad \boldsymbol{\sigma}(\boldsymbol{\sigma}\mathbf{a}) = \mathbf{a} + i(\mathbf{a} \times \boldsymbol{\sigma}). \quad (33.9)$$

The result is

$$\mathbf{j} = \frac{i\hbar}{2m}(\varphi\nabla\varphi^* - \varphi^*\nabla\varphi) - \frac{e}{mc}\mathbf{A}\varphi^*\varphi + \frac{\hbar}{2m}\nabla \times (\varphi^*\boldsymbol{\sigma}\varphi), \quad (33.10)$$

in agreement with the expression (115.4) (see III) from nonrelativistic theory.

We now obtain the second approximation by extending the expansion through terms of order $1/c^2$ ⁵. We assume that the external field is purely electric ($\mathbf{A} = 0$).

First observe that, when terms of order $1/c^2$ are retained, the density is

$$\rho = |\varphi|^2 + |\chi|^2 = |\varphi|^2 + \frac{\hbar^2}{4m^2c^2}|\boldsymbol{\sigma}\nabla\varphi|^2.$$

This is not the Schrödinger expression. To obtain, in the second approximation, a wave equation analogous to the Schrödinger equation, we must replace φ by another

⁴Dirac obtained this remarkable result in 1928. Pauli had introduced the two-component wave function satisfying (33.7) in 1927, before Dirac discovered his equation.

⁵The method followed below is due to V. B. Berestetskii and L. D. Landau (1949).

two-component function φ_{Sch} for which the time-independent integral has the form $\int |\varphi_{\text{Sch}}|^2 d^3x$, as a Schrödinger equation requires.

To find the necessary transformation, write

$$\int \varphi_{\text{Sch}}^* \varphi_{\text{Sch}} d^3x = \int \left\{ \varphi^* \varphi + \frac{\hbar^2}{4m^2c^2} (\nabla \varphi^* \cdot \sigma)(\sigma \nabla \varphi) \right\} d^3x$$

and integrate by parts:

$$\int (\nabla \varphi^* \cdot \sigma)(\sigma \nabla \varphi) d^3x = - \int \varphi^* (\sigma \nabla)(\sigma \nabla) \varphi d^3x = - \int \varphi^* \Delta \varphi d^3x$$

(or equivalently with φ and φ^* interchanged). It follows that

$$\int \varphi_{\text{Sch}}^* \varphi_{\text{Sch}} d^3x = \int \left\{ \varphi^* \varphi - \frac{\hbar^2}{8m^2c^2} (\varphi^* \Delta \varphi + \varphi \Delta \varphi^*) \right\} d^3x,$$

and hence

$$\varphi_{\text{Sch}} = \left(1 + \frac{\hat{\mathbf{p}}^2}{8m^2c^2} \right) \varphi, \quad \varphi = \left(1 - \frac{\hat{\mathbf{p}}^2}{8m^2c^2} \right) \varphi_{\text{Sch}}. \quad (33.11)$$

For compactness, take the state to be stationary and replace the operator $i\hbar\partial/\partial t$ by the energy ε (with the rest energy subtracted). At the next order beyond (33.4), equation (33.3) gives

$$\chi = \frac{1}{2mc} \left(1 - \frac{\varepsilon - e\Phi}{2mc^2} \right) (\sigma \hat{\mathbf{p}}) \varphi.$$

This expression is to be inserted in (33.2), after which φ is replaced by φ_{Sch} according to (33.11), with terms beyond order $1/c^2$ discarded throughout. A direct calculation then gives the equation $\varepsilon \varphi_{\text{Sch}} = \hat{H} \varphi_{\text{Sch}}$, with Hamiltonian

$$\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m} + e\Phi - \frac{\hat{\mathbf{p}}^4}{8m^3c^2} + \frac{e}{4m^2c^2} \left\{ (\sigma \hat{\mathbf{p}}) \Phi (\sigma \hat{\mathbf{p}}) - \frac{1}{2} (\hat{\mathbf{p}}^2 \Phi + \Phi \hat{\mathbf{p}}^2) \right\}.$$

The expression in braces is transformed with the identities

$$(\sigma \hat{\mathbf{p}}) \Phi (\sigma \hat{\mathbf{p}}) = \Phi \hat{\mathbf{p}}^2 + (\sigma \hat{\mathbf{p}} \Phi) (\sigma \hat{\mathbf{p}}) = \Phi \hat{\mathbf{p}}^2 + i\hbar (\sigma \mathbf{E}) (\sigma \hat{\mathbf{p}}),$$

$$\hat{\mathbf{p}}^2 \Phi - \Phi \hat{\mathbf{p}}^2 = -\hbar^2 \Delta \Phi + 2i\hbar \mathbf{E} \hat{\mathbf{p}},$$

where $\mathbf{E} = -\nabla \Phi$ is the electric field. The final Hamiltonian is

$$\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m} + e\Phi - \frac{\hat{\mathbf{p}}^4}{8m^3c^2} - \frac{e\hbar}{4m^2c^2} \sigma (\mathbf{E} \times \hat{\mathbf{p}}) - \frac{e\hbar^2}{8m^2c^2} \nabla \cdot \mathbf{E}. \quad (33.12)$$

The last three terms are the required corrections of order $1/c^2$. The first comes from the relativistic momentum dependence of the kinetic energy, as follows by expanding $c\sqrt{p^2 + m^2c^2} - mc^2$. The second may be called the energy of the *spin-orbit interaction*;

it is the interaction energy of a moving magnetic moment with an electric field⁶. The final term is nonzero only at points occupied by the charges producing the external field; for the Coulomb field of a point charge Ze , for example, $\Delta\Phi = -4\pi Ze\delta(\mathbf{r})$ (C. G. Darwin, 1928).

If the electric field is centrally symmetric, then

$$\mathbf{E} = -\frac{\mathbf{r}}{r} \frac{d\Phi}{dr},$$

and the spin-orbit interaction operator can be written

$$\frac{e\hbar}{4m^2c^2r} \boldsymbol{\sigma}(\mathbf{r} \times \hat{\mathbf{p}}) \frac{d\Phi}{dr} = \frac{\hbar^2}{2m^2c^2r} \frac{dU}{dr} \hat{\mathbf{L}} \cdot \hat{\mathbf{S}}. \quad (33.13)$$

Here $\hat{\mathbf{L}}$ is the orbital-angular-momentum operator, $\hat{\mathbf{S}} = \frac{1}{2}\boldsymbol{\sigma}$ is the electron spin operator, and $U = e\Phi$ is the electron's potential energy in the field.

§ 34. Fine structure of the hydrogen-atom levels

We shall determine the relativistic corrections to the energy levels of the hydrogen atom, that is, of an electron in the Coulomb field of a stationary nucleus⁷. The electron velocity in a hydrogen atom satisfies $v/c \sim \alpha \ll 1$. The desired corrections can therefore be calculated by perturbation theory: one averages the relativistic terms in the approximate Hamiltonian (33.12) over the unperturbed state, using the nonrelativistic wave function. To make the result somewhat more general, take the nuclear charge to be Ze , while still assuming $Z\alpha \ll 1$.

The electric field strength of the nucleus is $\mathbf{E} = Ze\mathbf{r}/r^3$, and its potential obeys $\Delta\Phi = -4\pi Ze\delta(\mathbf{r})$. Substitution in the last three terms of (33.12), with account taken of the negative electron charge, gives the perturbation operator

$$\hat{V} = -\frac{\hat{\mathbf{p}}^4}{8m^3} + \frac{Z\alpha}{2r^3m^2} \hat{\mathbf{L}} \cdot \hat{\mathbf{S}} + \frac{Z\alpha\pi}{2m^2} \delta(\mathbf{r}). \quad (34.1)$$

According to the nonrelativistic Schrödinger equation,

$$\hat{\mathbf{p}}^2\psi = 2m \left(\varepsilon_0 + \frac{Z\alpha}{r} \right) \psi$$

where $\varepsilon_0 = -mZ^2\alpha^2/2n^2$ is the unperturbed level and n is the principal quantum number. Hence the mean value is

$$\overline{\mathbf{p}^4} = 4m^2 \overline{\left(\varepsilon_0 + \frac{Z\alpha}{r} \right)^2}.$$

⁶Introducing the magnetic moment (33.8) and the velocity $\mathbf{v} = \mathbf{p}/m$, this energy takes the form $-\frac{1}{2c}\boldsymbol{\mu} \cdot (\mathbf{E} \times \mathbf{v})$. At first sight the result may seem unnatural: in the frame moving with the particle, a magnetic field $\mathbf{H} = \frac{1}{c}(\mathbf{E} \times \mathbf{v})$ appears, in which the magnetic moment should have the energy $-\boldsymbol{\mu}\mathbf{H}$. The factor 1/2 (the "Thomas half," L. Thomas, 1926) actually follows from the general requirements of relativistic invariance together with the distinctive properties of the electron as a spinor particle, including its particular gyromagnetic ratio (see § 41).

⁷The effect of nuclear motion on these corrections is of higher order in the small quantities and will not concern us here.

This quantity, as well as the mean value of the second term in (34.1), is evaluated from the formulas (see III, § 36)

$$\overline{r^{-1}} = \frac{m\alpha Z}{n^2}, \quad \overline{r^{-2}} = \frac{(m\alpha Z)^2}{n^3(l+1/2)}, \quad \overline{r^{-3}} = \frac{(m\alpha Z)^3}{n^3l(l+1/2)(l+1)} \quad (34.2)$$

where the last formula applies for $l \neq 0$. The eigenvalue is

$$\mathbf{ls} = \begin{cases} \frac{1}{2} [j(j+1) - l(l+1) - 3/4], & l \neq 0, \\ 0, & l = 0. \end{cases}$$

Finally, the third term is averaged using

$$\psi(0) = \begin{cases} \frac{1}{\sqrt{\pi}} \left(\frac{Z\alpha m}{n} \right)^{3/2}, & l = 0, \\ 0, & l \neq 0. \end{cases} \quad (34.3)$$

Straightforward calculation with the formulas above yields, for every j and l ,

$$\Delta\varepsilon = -\frac{m(Z\alpha)^4}{2n^3} \left(\frac{1}{j+1/2} - \frac{3}{4n} \right). \quad (34.4)$$

Formula (34.4) is the required relativistic correction to the energy of a hydrogen level, the fine-structure energy⁸. Recall that nonrelativistic theory has both degeneracy with respect to spin direction and Coulomb degeneracy with respect to l . Fine structure, or the spin-orbit interaction, removes this degeneracy, but not completely. Levels with the same n and j but different $l = j \pm 1/2$ remain mutually twofold degenerate. The only nondegenerate levels are those with the largest possible value at fixed n , namely $j = j_{\max} = l_{\max} + 1/2 = n - 1/2$. The hydrogen levels, including fine structure, therefore occur in the sequence

$$\begin{array}{l} 1s_{1/2}; \\ \underbrace{2s_{1/2}, 2p_{1/2}}, \quad 2p_{3/2}; \\ \underbrace{3s_{1/2}, 3p_{1/2}}, \quad \underbrace{3p_{3/2}, 3d_{3/2}}, \quad 3d_{5/2}. \\ \dots \quad \dots \end{array}$$

A level with principal quantum number n splits into n fine-structure components.

Recall also that in nonrelativistic mechanics the “accidental” degeneracy of the energy levels in a Coulomb field is connected with a conservation law specific to that field. The conserved quantity is $\mathbf{A}$, whose operator is

$$\widehat{\mathbf{A}} = \frac{\mathbf{r}}{r} + \frac{1}{2m\alpha} \left\{ \widehat{\mathbf{l}} \times \widehat{\mathbf{p}} - \widehat{\mathbf{p}} \times \widehat{\mathbf{l}} \right\}$$

⁸This formula, as well as the more accurate formula (36.10), was obtained by Sommerfeld (A. Sommerfeld) from the old Bohr theory even before quantum mechanics was created.

(see III, (36.30)). The twofold degeneracy that remains in the relativistic case is likewise associated with a special conservation law: the Dirac-equation Hamiltonian $\hat{H} = \alpha \hat{\mathbf{p}} + \beta m - e^2/r$ commutes with the operator

$$\hat{I} = \frac{\mathbf{r}}{r} \cdot \boldsymbol{\Sigma} + \frac{i}{m\alpha} \beta (\boldsymbol{\Sigma} \hat{\mathbf{l}} + 1) \gamma^5 (\hat{H} - m\beta)$$

(M. H. Johnson, B. A. Lippmann, 1950). In the nonrelativistic limit, this operator becomes $\hat{I} \rightarrow \boldsymbol{\Sigma} \hat{\mathbf{A}}$.

We shall see later (§ 123) that this residual degeneracy is removed by the so-called radiative corrections (the *Lamb shift*), which are not included in the one-electron Dirac equation.

Anticipating that discussion, we note here that these corrections are of order $mZ^4\alpha^5 \ln(1/\alpha)$. The second-order correction from the spin-orbit interaction would instead be of order $m(Z\alpha)^6$, so its ratio to the radiative corrections is of order $Z^2\alpha/\ln(1/\alpha)$. For hydrogen ($Z = 1$) this ratio is certainly small, and an exact solution of the Dirac equation is therefore not meaningful in this case. Such a problem may, however, be meaningful for electron energy levels in the field of a nucleus with large Z (see § 36).

§ 35. Motion in a spherically symmetric field

Consider the motion of an electron in a spherically symmetric electric field.

Angular momentum and parity, relative to the center of the field chosen as the coordinate origin, are conserved during motion in a central field. Everything said in § 24 about the angular dependence of free-particle spherical waves therefore applies here as well; only the radial functions change. Accordingly, we seek the stationary-state wave function, in the standard representation, in the form

$$\psi = \begin{pmatrix} \varphi \\ \chi \end{pmatrix} = \begin{pmatrix} f(r)\Omega_{jlm} \\ (-1)^{\frac{1+l-l'}{2}} g(r)\Omega_{jl'm} \end{pmatrix}, \quad (35.1)$$

where $l = j \pm 1/2$, $l' = 2j - l$. The power of -1 has been introduced to simplify the subsequent formulas.

In the standard representation, the Dirac equation gives the following system for φ and χ :

$$(\varepsilon - m - U)\varphi = (\boldsymbol{\sigma} \hat{\mathbf{p}})\chi, \quad (\varepsilon + m - U)\chi = (\boldsymbol{\sigma} \hat{\mathbf{p}})\varphi, \quad (35.2)$$

where $U(r) = e\Phi(r)$ is the potential energy of the electron in the field. Substituting (35.1) into these equations reduces the calculation to evaluation of their right-hand sides.

Expressing the spherical spinor $\Omega_{jl'm}$ in terms of Ω_{jlm} according to

$$\Omega_{jl'm} = i^{l-l'} \left(\boldsymbol{\sigma} \frac{\mathbf{r}}{r} \right) \Omega_{jlm}$$

(see (24.8)), we write

$$(\boldsymbol{\sigma} \hat{\mathbf{p}})\chi = -i(\boldsymbol{\sigma} \hat{\mathbf{p}})(\boldsymbol{\sigma} \mathbf{r}) \frac{g}{r} \Omega_{jlm}.$$

Transforming the product $(\sigma\hat{\mathbf{p}})(\sigma\mathbf{r})$ by means of (33.5), and then carrying out the vector operations, gives

$$\begin{aligned} (\sigma\hat{\mathbf{p}})\chi &= -i\{\hat{\mathbf{p}}\mathbf{r} + i\sigma(\hat{\mathbf{p}} \times \mathbf{r})\} \frac{g}{r} \Omega_{jlm} = \\ &= \{-\nabla \cdot \mathbf{r} - (\mathbf{r} \nabla) - \sigma(\mathbf{r} \times \hat{\mathbf{p}})\} \frac{g}{r} \Omega_{jlm} = -\left\{g' + \frac{2}{r}g + \frac{g}{r}\sigma\hat{\mathbf{l}}\right\} \Omega_{jlm}, \end{aligned}$$

where $\hat{\mathbf{l}} = \mathbf{r} \times \hat{\mathbf{p}}$ is the orbital-angular-momentum operator, and a prime denotes differentiation with respect to r . The eigenvalues of the product $\sigma\hat{\mathbf{l}} = 2\hat{\mathbf{l}}s$ are

$$2\hat{\mathbf{l}}s = \hat{\mathbf{j}}^2 - \hat{\mathbf{l}}^2 - \hat{\mathbf{s}}^2 = j(j+1) - l(l+1) - \frac{3}{4} = \begin{cases} j-1/2, & l = j-1/2, \\ -j-3/2, & l = j+1/2. \end{cases}$$

It is convenient to introduce a common notation for both cases $l = j \pm 1/2$:

$$\kappa = \begin{cases} -(j+1/2) = -(l+1), & j = l+1/2, \\ +(j+1/2) = l, & j = l-1/2. \end{cases} \quad (35.3)$$

The number κ assumes every integral value except zero; positive values correspond to $j = l-1/2$, and negative values to $j = l+1/2$. We then have $\hat{\mathbf{l}}\sigma = -(1+\kappa)$, and hence

$$(\sigma\hat{\mathbf{p}})\chi = -\left(g' + \frac{1-\kappa}{r}g\right) \Omega_{jlm}.$$

Upon substitution of this expression into the first equation (35.2), the spherical spinor Ω_{jlm} cancels from both sides. Treating the second equation in the same way, we obtain the following system for the radial functions:

$$\begin{aligned} f' + \frac{1+\kappa}{r}f - (\varepsilon + m - U)g &= 0, \\ g' + \frac{1-\kappa}{r}g + (\varepsilon - m - U)f &= 0, \end{aligned} \quad (35.4)$$

or

$$\begin{aligned} (fr)' + \frac{\kappa}{r}(fr) - (\varepsilon + m - U)gr &= 0, \\ (gr)' - \frac{\kappa}{r}(gr) + (\varepsilon - m - U)fr &= 0. \end{aligned} \quad (35.5)$$

Let us examine the behavior of f and g at small distances, assuming that $U(r)$ increases faster than $1/r$ as $r \rightarrow 0$. In the small- r region, equations (35.4) become

$$f' + Ug = 0, \quad g' - Uf = 0.$$

They have real solutions of the form

$$f = \text{const} \cdot \sin\left(\int U dr + \delta\right), \quad g = \text{const} \cdot \cos\left(\int U dr + \delta\right), \quad (35.6)$$

where δ is an arbitrary constant. As $r \rightarrow 0$, these functions oscillate without tending to any limit. It is readily seen that this situation corresponds to the particle's "fall" to the center in nonrelativistic theory.

First note that in this case the small-distance region imposes no restriction on the choice of solution: an oscillatory function has no boundary condition at $r = 0$, and the constant δ remains arbitrary. Correct behavior of the wave function at large r can be obtained at any ε by a suitable choice of δ . This ambiguity can be removed by regarding the potential singular at $r = 0$ as the $r_0 \rightarrow 0$ limit of a potential "cut off" at some r_0 , that is, one equal to $U(r)$ for $r > r_0$ and to $U(r_0)$ for $r < r_0$. For finite r_0 there is, of course, a definite system of energy levels. The ground-state energy, however, tends to $-\infty$ as $r_0 \rightarrow 0$.

In nonrelativistic theory this is precisely what is meant by "fall" to the center, since a particle on a deep level is localized in a small region around $r = 0$. In relativistic theory such a situation is altogether inadmissible because it signifies instability of the system against spontaneous creation of electron-positron pairs. Indeed, whereas creation of such a pair in vacuum requires energy greater than $2m$, less energy suffices in a field. If there is a bound electron state of energy $\varepsilon < m$, a pair can be created at an energy cost of only $\varepsilon + m < 2m$, producing a free positron and an electron in the bound state. If the bound-state level has energy $\varepsilon < -m$, the field can create positrons, of energy $-\varepsilon > m$, spontaneously and without receiving energy from an external source. In the field under consideration, as $r_0 \rightarrow 0$ there are infinitely many such "anomalous" levels with $\varepsilon < -m$. Potentials $\Phi(r)$ that increase faster than $1/r$ as $r \rightarrow 0$ therefore cannot be considered at all in Dirac theory. We emphasize that this statement applies to potentials of either sign. "Fall" occurs, of course, only for attraction, but the sign of $U = e\Phi$ also depends on the sign of the charge. Electron levels behave anomalously for one sign and positron levels for the other; in the latter case the field creates free electrons.

We next consider the large-distance behavior of the wave functions. If $U(r)$ decreases sufficiently rapidly as $r \rightarrow \infty$, the field may be neglected entirely in the equations when the asymptotic form of the wave

functions at large distances is determined. For $\varepsilon > m$, in the continuous-spectrum region, we then recover the equation of free motion. The asymptotic form of the wave functions, which are spherical waves, differs from that of a free particle only through the appearance of additional "phase shifts" whose values are determined by the field at short distances⁹. These shifts depend on j and l , or equivalently on the number κ introduced above, and also, of course, on the energy ε . Denoting them by δ_κ and using the free spherical wave (24.7), we can write the required asymptotic formula immediately:

$$\psi \approx \frac{2}{r} \frac{1}{\sqrt{2\varepsilon}} \begin{pmatrix} \sqrt{\varepsilon + m} \Omega_{jlm} \sin \left(pr - \frac{\pi l}{2} + \delta_\kappa \right) \\ -\sqrt{\varepsilon - m} \Omega_{jl'm} \sin \left(pr - \frac{\pi l'}{2} + \delta_\kappa \right) \end{pmatrix}, \quad (35.7)$$

⁹Cf. III, § 33. As in nonrelativistic theory, $U(r)$ must decrease faster than $1/r$. The case $U \sim 1/r$ will be considered separately in § 36.

or, taking account of definition (35.1),

$$\begin{aligned} f &= \frac{\sqrt{2}}{r} \sqrt{\frac{\varepsilon + m}{\varepsilon}} \sin \left(pr - \frac{\pi l}{2} + \delta_{\kappa} \right), \\ g &= \frac{\sqrt{2}}{r} \sqrt{\frac{\varepsilon - m}{\varepsilon}} \cos \left(pr - \frac{\pi l}{2} + \delta_{\kappa} \right), \end{aligned} \quad (35.8)$$

where $p = \sqrt{\varepsilon^2 - m^2}$. The overall coefficient here corresponds to normalization of the radial functions according to (24.5).

The wave functions of the discrete spectrum ($\varepsilon < m$), on the other hand, decay exponentially as $r \rightarrow \infty$ according to

$$f = -\sqrt{\frac{m + \varepsilon}{m - \varepsilon}} g = \frac{A_0}{r} \exp \left(-r \sqrt{m^2 - \varepsilon^2} \right), \quad (35.9)$$

where A_0 is a constant.

As in nonrelativistic theory, the phase shifts δ_{κ} , or more precisely the quantities $e^{2i\delta_{\kappa}} - 1$, determine the scattering amplitude in the given field; this will be discussed in more detail in § 37. We shall not investigate the analytic properties of these quantities here (cf. III, § 128). We note only that, as a function of energy, $e^{2i\delta_{\kappa}}$ still has poles at points corresponding to the particle's bound-state levels. The residue of $e^{2i\delta_{\kappa}}$ at such a pole is related in a definite way to the coefficient in the asymptotic expression for the corresponding discrete-spectrum wave function. We shall find this relation, which generalizes the nonrelativistic formula (128.17) (see III). The required calculation is entirely analogous to that carried out in III, § 128.

Differentiate equations (35.5) with respect to the energy:

$$\begin{aligned} \left(\frac{\partial r f}{\partial \varepsilon} \right)' + \frac{\kappa}{r} \frac{\partial r f}{\partial \varepsilon} - (\varepsilon + m - U) \frac{\partial r g}{\partial \varepsilon} &= r g, \\ \left(\frac{\partial r g}{\partial \varepsilon} \right)' - \frac{\kappa}{r} \frac{\partial r g}{\partial \varepsilon} + (\varepsilon - m - U) \frac{\partial r f}{\partial \varepsilon} &= -r f. \end{aligned}$$

Multiply these two equations respectively by rg and $-rf$, and the two equations (35.5) respectively by $-rg$ and rf ; then add all four equations term by term. After cancellation, this gives

$$\frac{\partial}{\partial r} \left[r^2 \left(g \frac{\partial f}{\partial \varepsilon} - f \frac{\partial g}{\partial \varepsilon} \right) \right] = r^2 (f^2 + g^2).$$

Integrating this equality over r , we obtain

$$r^2 \left(g \frac{\partial f}{\partial \varepsilon} - f \frac{\partial g}{\partial \varepsilon} \right) = \int_0^r (f^2 + g^2) r^2 dr,$$

and then pass to the limit $r \rightarrow \infty$. By the normalization condition, the integral on the right tends to unity. On the left we use the relation between f and g in the asymptotic region,

$$rg = \frac{(rf)'}{\varepsilon + m},$$

which follows from (35.5) when the terms containing U and $1/r$ are neglected. The result is

$$\frac{1}{\varepsilon + m} \left[(rf)' \frac{\partial rf}{\partial \varepsilon} - rf \left(\frac{\partial rf}{\partial \varepsilon} \right)' \right] = 1. \quad (35.10)$$

This formula differs from its nonrelativistic analogue, for the function χ , only in the coefficient $\varepsilon + m$ in place of $2m$. There is therefore no need to repeat the subsequent calculation, and we give at once the final expression valid near $\varepsilon = \varepsilon_0$, where ε_0 is an energy level:

$$e^{2i\delta_\kappa} = (-1)^l \frac{2A_0^2}{\varepsilon - \varepsilon_0} \sqrt{\frac{m - \varepsilon_0}{m + \varepsilon_0}}, \quad (35.11)$$

where A_0 is the coefficient in the asymptotic expression (35.9).

PROBLEM

Find the limiting form of the wave function at small r in a field $U \sim r^{-s}$, $s < 1$.

SOLUTION. For a free particle at small r , $f \sim r^l$ and $g \sim r^{l'}$. Thus $f \gg g$ when $l < l'$, while $f \ll g$ when $l > l'$. We assume (as the result will confirm) that the same relation holds in the field under consideration. For $l < l'$, that is, $l = j - 1/2$ and $\kappa = l + 1$, the term containing g can be omitted from the first equation (35.4), and hence $f \sim r^l$ as before. The second equation then gives $g \sim rfU$, that is, $g \sim r^{l+1-s} = r^{l'-s}$. The case $l > l'$ is treated similarly. We finally obtain

$$\begin{aligned} \text{for } l < l' : \quad f &\sim r^l, & g &\sim r^{l'-s}; \\ \text{for } l > l' : \quad f &\sim r^{l-s}, & g &\sim r^{l'}. \end{aligned}$$

§ 36. Motion in a Coulomb field

We begin the study of motion in the especially important Coulomb field by examining the behavior of the wave functions at short distances. For definiteness, consider an attractive field: $U = -Z\alpha/r$ ¹⁰.

At small r , the terms containing $\varepsilon \pm m$ may be dropped from equations (35.5); then

$$\begin{aligned} (fr)' + \frac{\kappa}{r} fr - \frac{Z\alpha}{r} gr &= 0, \\ (gr)' - \frac{\kappa}{r} gr + \frac{Z\alpha}{r} fr &= 0. \end{aligned}$$

The functions fr and gr enter each of these equations symmetrically. We therefore seek both as the same powers of r : $fr = ar^\gamma$, $gr = br^\gamma$. Substitution gives

$$a(\gamma + \kappa) - bZ\alpha = 0, \quad aZ\alpha + b(\gamma - \kappa) = 0,$$

¹⁰In conventional units, $U = -Ze^2/r$. On passing to relativistic units, e^2 is replaced by the dimensionless quantity α .

whence

$$\gamma^2 = \kappa^2 - (Z\alpha)^2. \quad (36.1)$$

Let $(Z\alpha)^2 < \kappa^2$. Then γ is real, and its positive value must be selected: the corresponding solution either remains finite at $r = 0$ or diverges less rapidly than the other. This selection can be justified by considering a potential cut off (as explained in the preceding section) at some small r_0 , and then taking the limit $r_0 \rightarrow 0$ (compare the analogous argument in III, § 35). Thus,

$$f = \frac{Z\alpha}{\gamma + \kappa} g = \text{const} \cdot r^{-1+\gamma}, \quad \gamma = \sqrt{\kappa^2 - Z^2\alpha^2} = \sqrt{(j + 1/2)^2 - Z^2\alpha^2}. \quad (36.2)$$

Although the wave function may become infinite at $r = 0$ (when $\gamma < 1$), the integral of $|\psi|^2$ remains, of course, convergent. If $(Z\alpha)^2 > \kappa^2$, both values of γ in (36.2) are imaginary. The corresponding solutions oscillate as $r \rightarrow 0$ (like $r^{-1} \cos(|\gamma| \ln r)$), again describing the inadmissible relativistic situation of “fall” to the center discussed above. Since $\kappa^2 \geq 1$, a pure Coulomb field can therefore be treated in Dirac theory only for $Z\alpha < 1$, that is, $Z < 137$.

We pause to describe qualitatively what occurs for $Z > 137$. To avoid an ambiguity in the boundary condition at $r = 0$, one must again use a potential cut off at some distance r_0 (I. Ya. Pomeranchuk and Ya. A. Smorodinsky, 1945). This has not only a formal but also a direct physical meaning. A charge $Z > 137$ can in fact be concentrated only in a finite-radius “superheavy” nucleus. Let us therefore see how the level positions change as Z increases at fixed r_0 .

In the “uncut” Coulomb field, the energy ε_1 of the lowest level vanishes at $Z\alpha = 1$, and the curve $\varepsilon_1(Z)$ terminates: for $Z\alpha > 1$, the level ε_1 becomes imaginary (see (36.10)). In the “cut-off” field, at a fixed $r_0 \neq 0$, the level ε_1 passes through zero only at some $Z\alpha > 1$. But the value $\varepsilon_1 = 0$ has no special physical status, and for $r_0 \neq 0$ it is not distinguished formally either: the curve $\varepsilon_1(Z)$ does not terminate there. As Z is increased further, the levels continue to descend, and at a certain “critical” value $Z = Z_c(r_0)$ the energy ε_1 reaches the boundary $(-m)$ of the lower continuum. As explained in the preceding section, this means that the energy required to create a free positron has fallen to zero. Hence the critical value Z_c is the greatest charge that a “bare” nucleus of the specified r_0 can possess.

For $Z > Z_c$, the level has $\varepsilon_1 < -m$, and creation of two electron–positron pairs becomes energetically favorable. The positrons escape to infinity, carrying away kinetic energy $2(|\varepsilon_1| - m)$, while the two electrons occupy the level ε_1 . The result is an “ion” with a filled K shell and charge $Z_{\text{eff}} = Z - 2$ (S. S. Gershtein and Ya. B. Zeldovich, 1969). This system is stable for $Z > Z_c$ until Z reaches values at which the next level arrives at the boundary $-m$ ¹¹.

Finally, even for a point charge, radiative corrections distort the potential at short distances. Their inclusion, however, produces only corrections of order α to the value of $Z_c\alpha$.

¹¹For example, if the nuclear charge is distributed uniformly over a sphere of radius $r_0 = 1.2 \cdot 10^{-12}$ cm, the critical value is $Z_c = 170$, and the next level reaches the boundary $-m$ at $Z = 185$ (V. S. Popov, 1970). For a detailed account of the quantitative theory, see the review by Ya. B. Zeldovich and V. S. Popov (Usp. Fiz. Nauk — 1971. — Vol. 105. — P. 403).

We now turn to the exact solution of the wave equation (G. Darwin, 1928; W. Gordon, 1928).

Discrete spectrum ($\varepsilon < m$). Seek the functions f and g in the form

$$\begin{aligned} f &= \sqrt{m + \varepsilon} e^{-\rho/2} \rho^{\gamma-1} (Q_1 + Q_2), \\ g &= -\sqrt{m - \varepsilon} e^{-\rho/2} \rho^{\gamma-1} (Q_1 - Q_2), \end{aligned} \quad (36.3)$$

where

$$\rho = 2\lambda r, \quad \lambda = \sqrt{m^2 - \varepsilon^2}, \quad \gamma = \sqrt{\kappa^2 - Z^2 \alpha^2}. \quad (36.4)$$

This form is natural in view of the already known behavior of the functions as $\rho \rightarrow 0$, (36.2), and their exponential decay ($\sim e^{-\rho/2}$) as $\rho \rightarrow \infty$. Since the first relation (35.9) must also hold for the Coulomb field when $\rho \rightarrow \infty$, we expect $Q_1 \gg Q_2$ in this limit.

Substitution of (36.3) in (35.4) gives

$$\begin{aligned} \rho(Q_1 + Q_2)' + (\gamma + \kappa)(Q_1 + Q_2) - \rho Q_2 + Z\alpha \sqrt{\frac{m - \varepsilon}{m + \varepsilon}} (Q_1 - Q_2) &= 0, \\ \rho(Q_1 - Q_2)' + (\gamma - \kappa)(Q_1 - Q_2) + \rho Q_2 - Z\alpha \sqrt{\frac{m + \varepsilon}{m - \varepsilon}} (Q_1 + Q_2) &= 0 \end{aligned}$$

(a prime denotes differentiation with respect to ρ). Their sum and difference yield

$$\begin{aligned} \rho Q_1' + \left(\gamma - \frac{Z\alpha\varepsilon}{\lambda} \right) Q_1 + \left(\kappa - \frac{Z\alpha m}{\lambda} \right) Q_2 &= 0, \\ \rho Q_2' + \left(\gamma + \frac{Z\alpha\varepsilon}{\lambda} - \rho \right) Q_2 + \left(\kappa + \frac{Z\alpha m}{\lambda} \right) Q_1 &= 0, \end{aligned} \quad (36.5)$$

or, on eliminating Q_1 or Q_2 ,

$$\begin{aligned} \rho Q_1'' + (2\gamma + 1 - \rho) Q_1' - \left(\gamma - \frac{Z\alpha\varepsilon}{\lambda} \right) Q_1 &= 0, \\ \rho Q_2'' + (2\gamma + 1 - \rho) Q_2' - \left(\gamma + 1 - \frac{Z\alpha\varepsilon}{\lambda} \right) Q_2 &= 0 \end{aligned}$$

(where $\gamma^2 - (Z\alpha\varepsilon/\lambda)^2 = \kappa^2 - (Z\alpha m/\lambda)^2$ must be used). The solutions finite at $\rho = 0$ are

$$\begin{aligned} Q_1 &= AF \left(\gamma - \frac{Z\alpha\varepsilon}{\lambda}, 2\gamma + 1, \rho \right), \\ Q_2 &= BF \left(\gamma + 1 - \frac{Z\alpha\varepsilon}{\lambda}, 2\gamma + 1, \rho \right), \end{aligned} \quad (36.6)$$

where $F(\alpha, \beta, z)$ is the confluent hypergeometric function. Setting $\rho = 0$ in either equation (36.5), we find the relation between the constants A and B :

$$B = -\frac{\gamma - Z\alpha\varepsilon/\lambda}{\kappa - Z\alpha m/\lambda} A. \quad (36.7)$$

Both hypergeometric functions in (36.6) must reduce to polynomials (otherwise they grow as e^ρ when $\rho \rightarrow \infty$, and the entire wave function grows with them as $e^{\rho/2}$). The

function $F(\alpha, \beta, z)$ reduces to a polynomial when the parameter α is a negative integer or zero. Write

$$\gamma - \frac{Z\alpha\varepsilon}{\lambda} = -n_r. \quad (36.8)$$

For $n_r = 1, 2, \dots$, both hypergeometric functions reduce to polynomials. If $n_r = 0$, only one of them does. But $n_r = 0$ means $\gamma = Z\alpha\varepsilon/\lambda$, and then, as is readily verified, $Z\alpha m/\lambda = |\kappa|$. If $\kappa < 0$, the coefficient B in (36.7) vanishes, so that $Q_2 = 0$ and the required condition is not violated. If $\kappa > 0$, however, $B = A$, and for $n_r = 0$ the function Q_2 remains divergent. The permitted values of the quantum number n_r are therefore

$$n_r = \begin{cases} 0, 1, 2, \dots & \text{for } \kappa < 0, \\ 1, 2, 3, \dots & \text{for } \kappa > 0. \end{cases} \quad (36.9)$$

Definition (36.8) now gives the following expression for the discrete energy levels:

$$\frac{\varepsilon}{m} = \left[1 + \frac{(Z\alpha)^2}{\left(\sqrt{\kappa^2 - (Z\alpha)^2} + n_r \right)^2} \right]^{-1/2}. \quad (36.10)$$

In particular, the energy of the ground level $1s_{1/2}$ ($|\kappa| = 1$, $n_r = 0$) is

$$\varepsilon_1 = m\sqrt{1 - (Z\alpha)^2}.$$

For $Z\alpha \ll 1$, the first terms in the expansion of (36.10) give

$$\frac{\varepsilon}{m} - 1 = -\frac{(Z\alpha)^2}{2(|\kappa| + n_r)^2} \left\{ 1 + \frac{(Z\alpha)^2}{|\kappa| + n_r} \left[\frac{1}{|\kappa|} - \frac{3}{4(|\kappa| + n_r)} \right] \right\}.$$

Putting $n_r + |\kappa| = n$ ($n = 1, 2, \dots$) and noting that $|\kappa| = j + 1/2$, we recover formula (34.4), obtained earlier by perturbation theory. As noted at the end of § 34, the subsequent terms of this expansion have no meaning, since they are certainly masked by radiative corrections. The exact expression (36.10), however, remains meaningful when $Z\alpha \sim 1$.

The twofold degeneracy of the levels revealed by the approximate formula (34.4) is also retained in the exact expression: because only $|\kappa|$ enters it, levels with different l and the same j still coincide.

It remains to determine the overall normalization coefficient A in the wave function. As usual, a discrete-spectrum wave function must satisfy $\int |\psi|^2 d^3x = 1$; for f and g this condition is

$$\int_0^\infty (f^2 + g^2) r^2 dr = 1.$$

The simplest way to find A is from the asymptotic form of the functions as $r \rightarrow \infty$. The asymptotic formula

$$F(-n_r, 2\gamma + 1, \rho) \approx \frac{\Gamma(2\gamma + 1)}{\Gamma(n_r + 2\gamma + 1)} (-\rho)^{n_r}$$

(see III, (d. 14)) gives

$$f \approx (-1)^{n_r} A \sqrt{m + \varepsilon} \frac{\Gamma(2\gamma + 1)}{\Gamma(n_r + 2\gamma + 1)} e^{-\lambda r} (2\lambda r)^{\gamma + n_r - 1}.$$

Comparison with (36.22), derived below, determines A . Collecting the resulting formulas, we write the final expressions for the normalized wave functions:

$$\left. \begin{matrix} f \\ g \end{matrix} \right\} = \frac{\pm (2\lambda)^{3/2}}{\Gamma(2\gamma + 1)} \left[\frac{(m \pm \varepsilon)\Gamma(2\gamma + n_r + 1)}{4m(Z\alpha m/\lambda)(Z\alpha m/\lambda - \kappa)n_r!} \right]^{1/2} (2\lambda r)^{\gamma-1} e^{-\lambda r} \times \quad (36.11)$$

$$\times \left\{ \left(\frac{Z\alpha m}{\lambda} - \kappa \right) F(-n_r, 2\gamma + 1, 2\lambda r) + n_r F(1 - n_r, 2\gamma + 1, 2\lambda r) \right\}.$$

(the upper signs refer to f , and the lower signs to g).

Continuous spectrum ($\varepsilon > m$). There is no need to solve the wave equation anew for states in the continuous spectrum. Their wave functions follow from the discrete-spectrum functions by the replacements¹²

$$\sqrt{m - \varepsilon} \rightarrow -i\sqrt{\varepsilon - m}, \quad \lambda \rightarrow -ip, \quad -n_r \rightarrow \gamma - i\frac{Z\alpha\varepsilon}{p} \quad (36.12)$$

(for the choice of sign in analytically continuing the root $\sqrt{m - \varepsilon}$, see III, § 128). The functions must, however, be normalized again.

Making these replacements in (36.11), write f and g as

$$\left. \begin{matrix} f \\ g \end{matrix} \right\} = \frac{\sqrt{\varepsilon + m}}{i\sqrt{\varepsilon - m}} \left\{ A' e^{ipr} (2pr)^{\gamma-1} \times \right.$$

$$\times \left[e^{i\xi} F(\gamma - i\nu, 2\gamma + 1, -2ipr) \mp e^{-i\xi} F(\gamma + 1 - i\nu, 2\gamma + 1, -2ipr) \right],$$

where A' is a new normalization constant and

$$\nu = \frac{Z\alpha\varepsilon}{p}, \quad e^{-2i\xi} = \frac{\gamma - i\nu}{\kappa - i\nu m/\varepsilon} \quad (36.13)$$

(ξ is real, since $\gamma^2 + (Z\alpha\varepsilon/p)^2 = \kappa^2 + (Z\alpha m/p)^2$).

The known formula

$$F(\alpha, \beta, z) = e^z F(\beta - \alpha, \beta, -z)$$

(see III, (d. 10)) gives

$$F(\gamma + 1 - i\nu, 2\gamma + 1, -2ipr) = e^{-2ipr} F(\gamma + i\nu, 2\gamma + 1, 2ipr) =$$

$$= e^{-2ipr} F^*(\gamma - i\nu, 2\gamma + 1, -2ipr),$$

and therefore

$$\left. \begin{matrix} f \\ g \end{matrix} \right\} = 2iA'\sqrt{\varepsilon \pm m}(2pr)^{\gamma-1} \frac{\text{Im}}{\text{Re}} \left\{ e^{i(pr+\xi)} F(\gamma - i\nu, 2\gamma + 1, -2ipr) \right\}. \quad (36.14)$$

The normalization coefficient A' is fixed by comparing the asymptotic form of this function with the general expression (35.7) for a normalized spherical wave. We give

¹²Below in this section, p denotes $|\mathbf{p}| = \sqrt{\varepsilon^2 - m^2}$.

directly the resulting wave functions of the continuous spectrum (and then verify them)¹³:

$$\left. \begin{aligned} f \\ g \end{aligned} \right\} = 2^{3/2} \sqrt{\frac{m \pm \varepsilon}{\varepsilon}} e^{\pi \nu / 2} \frac{|\Gamma(\gamma + 1 + i\nu)|}{\Gamma(2\gamma + 1)} \frac{(2pr)^\gamma}{r} \times \\ \times \left. \begin{aligned} \text{Im} \\ \text{Re} \end{aligned} \right\} \left\{ e^{i(pr+\xi)} F(\gamma - i\nu, 2\gamma + 1, -2ipr) \right\}. \quad (36.15)$$

The asymptotic expression for this function follows from formula III, (d. 14), of which only the first term is relevant here (the second decreases with a higher power of $1/r$):

$$\left. \begin{aligned} f \\ g \end{aligned} \right\} = \frac{\sqrt{2}}{r} \sqrt{\frac{\varepsilon \pm m}{\varepsilon}} \frac{\sin}{\cos} \left\{ pr - \frac{\pi l}{2} + \delta_\kappa + \nu \ln 2pr - \frac{\pi l}{2} \right\}, \quad (36.16)$$

where

$$\delta_\kappa = \xi - \arg \Gamma(\gamma + 1 + i\nu) - \frac{\pi \gamma}{2} + \frac{\pi l}{2}, \quad (36.17)$$

or

$$e^{2i\delta_\kappa} = \frac{\kappa - i\nu m/\varepsilon}{\gamma - i\nu} \frac{\Gamma(\gamma + 1 - i\nu)}{\Gamma(\gamma + 1 + i\nu)} e^{i\pi(l-\gamma)}. \quad (36.18)$$

For later reference, note the phases in the ultrarelativistic case ($\varepsilon \gg m$, $\nu \approx Z\alpha$):

$$e^{2i\delta_\kappa} = \frac{\kappa}{\gamma - iZ\alpha} \frac{\Gamma(\gamma + 1 - iZ\alpha)}{\Gamma(\gamma + 1 + iZ\alpha)} e^{i\pi(l-\gamma)}. \quad (36.19)$$

Expression (36.16) differs from (35.8) only by the logarithmic term in the argument of the trigonometric function. As with the Schrödinger equation, the slow decrease of the Coulomb potential distorts the wave phase, making it a slowly varying function of r .

On analytic continuation into the region $\varepsilon < m$, expression (36.18) becomes

$$e^{2i\delta_\kappa} = \frac{\kappa - Z\alpha m/\lambda}{\gamma - Z\alpha\varepsilon/\lambda} \frac{\Gamma(\gamma + 1 - Z\alpha\varepsilon/\lambda)}{\Gamma(\gamma + 1 + Z\alpha\varepsilon/\lambda)} e^{i\pi(l-\gamma)}. \quad (36.20)$$

It has poles at the points where $\gamma + 1 - Z\alpha\varepsilon/\lambda = 1 - n_r$, $n_r = 1, 2, \dots$ (the poles of the Γ function in the numerator), and also at $\gamma - Z\alpha\varepsilon/\lambda = -n_r = 0$ (provided $\kappa < 0$); as expected, these points coincide with the discrete energy levels.

Near any pole with $n_r \neq 0$,

$$e^{2i\delta_\kappa} \approx \frac{(Z\alpha m/\lambda - \kappa) e^{i\pi(l-\gamma)}}{n_r! \Gamma(2\gamma + 1 + n_r)} \Gamma\left(\gamma + 1 - \frac{Z\alpha\varepsilon}{\lambda}\right).$$

The form of the Γ function near its pole follows from the known formula $\Gamma(z)\Gamma(1-z) = \pi/\sin \pi z$:

$$\begin{aligned} \Gamma\left(\gamma + 1 - \frac{Z\alpha\varepsilon}{\lambda}\right) &\approx \frac{\pi}{\Gamma(n_r) \sin \pi(\gamma + 1 - Z\alpha\varepsilon/\lambda)}, \\ \sin \pi\left(\gamma + 1 - \frac{Z\alpha\varepsilon}{\lambda}\right) &\approx \pi \cos \pi n_r \frac{d}{d\varepsilon} \left(\frac{Z\alpha\varepsilon}{\lambda}\right) (\varepsilon - \varepsilon_0) = \\ &= (-1)^{n_r} \frac{\pi Z\alpha m^2}{\lambda^3} (\varepsilon - \varepsilon_0) \end{aligned}$$

¹³The wave functions in a repulsive field follow by reversing the sign before $Z\alpha$, that is, by reversing the sign of ν .

(ε_0 is an energy level). Thus¹⁴,

$$e^{2i\delta_\kappa} \approx (-1)^{l+n_r} \frac{e^{-i\pi\gamma}(Z\alpha m/\lambda - \kappa)}{n_r! \Gamma(2\gamma + 1 + n_r)} \frac{\lambda^3}{Z\alpha m^2} \frac{1}{\varepsilon - \varepsilon_0}. \quad (36.21)$$

At the end of the preceding section, formula (35.11) was obtained, relating the residue of $e^{2i\delta_\kappa}$ at its pole to the coefficient in the asymptotic wave function of the corresponding bound state. For a Coulomb field, however, this formula must be modified slightly because (36.16) contains the sum $\delta_\kappa + \nu \ln(2pr)$ instead of the constant phase shift δ_κ used in (35.7). The left side of (35.11) must therefore contain, in place of $e^{2i\delta_\kappa}$,

$$\exp[2i\delta_\kappa + 2i\nu \ln(2pr)] \rightarrow e^{2i\delta_\kappa} (2i\lambda r)^{2(n_r+\gamma)}.$$

Using (36.21) and determining from (35.11) the coefficient A_0 (which is now a power function of r), we obtain the asymptotic form of the normalized discrete-spectrum wave function:

$$f = \left[\frac{(Z\alpha m/\lambda - \kappa)(m + \varepsilon)\lambda^2}{2n_r! Z\alpha m^2 \Gamma(2\gamma + 1 + n_r)} \right]^{1/2} \frac{e^{-\lambda r}}{r} (2\lambda r)^{n_r+\gamma}. \quad (36.22)$$

This formula was already used to determine the coefficient in (36.11).

§ 37. Scattering in a spherically symmetric field

Write the asymptotic expression for the wave function of a particle scattered by the field of a stationary force center as¹⁵

$$\psi = u_{\varepsilon\mathbf{p}} e^{ipz} + u'_{\varepsilon\mathbf{p}'} e^{ip'r}/r. \quad (37.1)$$

Here $u_{\varepsilon\mathbf{p}}$ is the bispinor amplitude of the incident plane wave. The bispinor $u'_{\varepsilon\mathbf{p}'}$ is a function of the scattering direction $\mathbf{n}'$. For each specified value of $\mathbf{n}'$, its form, though of course not its normalization, coincides with the bispinor amplitude of a plane wave propagating along $\mathbf{n}'$.

We saw in § 24 that the bispinor amplitude of a plane wave is fixed completely by a two-component quantity, the 3-spinor w , which is the nonrelativistic wave function in the particle's rest frame. The flux density is also expressed through this spinor: it is proportional to w^*w , with a proportionality coefficient depending only on the energy ε and consequently identical for the incident and scattered particles. The scattering cross section is therefore $d\sigma = (w'^*w'/w^*w)do$, or, if the incident wave is normalized by $w^*w = 1$ as in § 24,

$$d\sigma = w'^*w' do.$$

Introduce the scattering operator f by the definition

$$w' = \widehat{f}w. \quad (37.2)$$

Since w and w' have two components, the operator so defined is analogous to the operator scattering amplitude used in nonrelativistic scattering theory with spin (see III, § 140).

¹⁴It is readily verified that this formula remains valid for $n_r = 0$ as well.

¹⁵In §§ 37 and 38, p denotes $|\mathbf{p}|$, while ε and $\mathbf{p}$ are written separately as indices on an amplitude.

The formulas derived there, which express the operator in terms of phase shifts of the wave functions in the scattering field, can therefore be transferred directly to the present problem. Only the phases must be relabeled, expressing the shifts δ_l^+ and δ_l^- introduced in III, § 140 in terms of the phase shift δ_κ that appears in the relativistic formula (35.7). Recall that δ_l^+ and δ_l^- referred to states of orbital angular momentum l and total angular momentum $j = l + 1/2$ and $j = l - 1/2$, respectively. By definition (35.3), $\kappa = -l - 1$ for $j = l + 1/2$, while $\kappa = l$ for $j = l - 1/2$. We must therefore make the replacements

$$\delta_l^+ \rightarrow \delta_{-(l+1)}, \quad \delta_l^- \rightarrow \delta_l$$

and remember that the index on δ now specifies the value of κ . We thus obtain

$$\hat{f} = A + B \boldsymbol{\nu} \boldsymbol{\sigma}, \quad (37.3)$$

$$A = \frac{1}{2ip} \sum_{l=0}^{\infty} [(l+1)(e^{2i\delta_{-l-1}} - 1) + l(e^{2i\delta_l} - 1)] P_l(\cos \theta), \quad (37.4)$$

$$B = \frac{1}{2p} \sum_{l=1}^{\infty} (e^{2i\delta_{-l-1}} - e^{2i\delta_l}) P_l^1(\cos \theta), \quad (37.5)$$

where $\boldsymbol{\nu}$ is a unit vector in the direction of $\mathbf{n} \times \mathbf{n}'$.

Since w is the spinor wave function in the rest frame, the polarization properties of the scattering are described in terms of $\hat{f}$ by the same formulas as in III, § 140.

For a Coulomb field, both functions $A(\theta)$ and $B(\theta)$ can be expressed through a single function. We briefly indicate the calculation¹⁶.

For a Coulomb field, the phases δ_κ are given by (36.18), which we write in the form

$$e^{2i\delta_\kappa} = - \left(\kappa - i \frac{Ze^2m}{p} \right) \frac{\kappa}{|\kappa|} C_\kappa, \quad (37.6)$$

$$C_\kappa = - \frac{\Gamma(\gamma - i\nu)}{\Gamma(\gamma + 1 + i\nu)} e^{i\pi(|\kappa| - \gamma)}$$

(note that $e^{i\pi l} = e^{i\pi\kappa}$ for $\kappa > 0$, while $e^{i\pi l} = -e^{i\pi\kappa}$ for $\kappa < 0$). In terms of the quantities introduced in this way, the series (37.4) and (37.5) can be written

$$A(\theta) = \frac{1}{p} G(\theta) - i \frac{Ze^2m}{p^2} F(\theta), \quad (37.7)$$

$$B(\theta) = -\frac{i}{p} \tan \frac{\theta}{2} G(\theta) + \frac{Ze^2m}{p^2} \cot \frac{\theta}{2} F(\theta),$$

where

$$G(\theta) = \frac{i}{2} \sum_{l=1}^{\infty} l^2 C_l (P_l + P_{l-1}), \quad F(\theta) = \frac{i}{2} \sum_{l=1}^{\infty} l C_l (P_l - P_{l-1}). \quad (37.8)$$

¹⁶Gluckstern R. L., Lin S. R.//J. Math. Phys. — 1964. — Vol. 5. — P. 1594.

In transforming the series for $B(\theta)$, the following recurrence relations between Legendre polynomials have been used:

$$P_l^1 + P_{l-1}^1 = \cot \frac{\theta}{2} \cdot l(P_l - P_{l-1}), \quad (37.9)$$

$$P_l^1 - P_{l-1}^1 = \tan \frac{\theta}{2} \cdot l(P_l + P_{l-1}). \quad (37.10)$$

On the other hand, by the identity

$$(1 + \cos \theta) \frac{d}{d \cos \theta} [P_l(\cos \theta) - P_{l-1}(\cos \theta)] = l[P_l(\cos \theta) + P_{l-1}(\cos \theta)] \quad (37.11)$$

the functions $F(\theta)$ and $G(\theta)$ are related by

$$G = (1 - \cos \theta) \frac{dF}{d \cos \theta} = -\cot \frac{\theta}{2} \frac{dF}{d\theta}. \quad (37.12)$$

Thus $A(\theta)$ and $B(\theta)$ are both expressed through the single function $F(\theta)$ ¹⁷.

§ 38. Scattering in the ultrarelativistic case

We consider separately scattering in the ultrarelativistic case ($\varepsilon \gg m$). In the first approximation, the mass m is neglected altogether in the wave equation. It is then convenient to use the spinor representation $\psi = \begin{pmatrix} \xi \\ \eta \end{pmatrix}$, because at $m = 0$ the equations for ξ and η separate:

$$-i\sigma \nabla \xi = (\varepsilon - U)\xi, \quad -i\sigma \nabla \eta = -(\varepsilon - U)\eta \quad (38.1)$$

and acquire the “neutrino” form; see § 30.

An electron in a helicity state polarized along $\mathbf{p}$ has the wave function $\psi = \begin{pmatrix} \xi \\ 0 \end{pmatrix}$, while polarization opposite to $\mathbf{p}$ corresponds to $\psi = \begin{pmatrix} 0 \\ \eta \end{pmatrix}$. Since the equations for ξ and η are independent, this property plainly remains unchanged in scattering. In other words, helicity is conserved when ultrarelativistic electrons are scattered. Symmetry considerations, namely longitudinal polarization, show that scattering of particles in helicity states has no azimuthal asymmetry. One can also state that the scattering cross section of electrons in helicity states does not depend on the sign of the helicity. This follows because a central field is invariant under inversion, while inversion reverses the helicity sign.

In the ultrarelativistic case, formulas (37.3)–(37.5) can be simplified substantially (*D. R. Yennie, D. G. Ravenhall, R. N. Wilson, 1954*).

Let the incident electron be polarized, say, along its direction of motion $\mathbf{n}$. For a plane wave with a definite value of $\mathbf{n}\sigma$, the spinor $\xi (= (\varphi + \chi)/\sqrt{2})$ is proportional to the same 3-spinor w that appeared in the standard representation of the wave. The relation between the spinor amplitudes of the incident and scattered waves in the new representation is therefore still effected by the same operator $\hat{f}$.

¹⁷The function $F(\theta)$ cannot be expressed in closed form in terms of elementary functions. It can, however, be written as a definite double integral; see the paper cited above.

As a result of scattering, the polarization vector turns together with the momentum and acquires the direction $\mathbf{n}'$. The action of $\hat{f}$ on the electron spin wave function consequently reduces to a rotation of the spin through the angle θ , the angle between $\mathbf{n}$ and $\mathbf{n}'$, about the axis $\mathbf{v}$. Such a rotation is in turn equivalent to rotating the coordinate system about the same axis in the opposite sense, through the angle $-\theta$. It follows that, apart from a coefficient, $\hat{f}$ must coincide with the operator that transforms the wave function under this change of coordinate system, namely operator (18.17) with θ replaced by $-\theta$.

Comparison of (37.3) and (18.17) gives

$$\frac{B}{A} = -i \tan \frac{\theta}{2}. \quad (38.2)$$

Thus, in the ultrarelativistic limit,

$$\hat{f} = A(\theta) \left[1 - i \tan \frac{\theta}{2} \cdot \mathbf{v} \boldsymbol{\sigma} \right]. \quad (38.3)$$

The expression (37.4) for $A(\theta)$ can also be simplified by using the relation between the phases δ_{κ} and $\delta_{-\kappa}$ that arises in the same limit. To derive it, note that after the terms containing m are deleted, equations (35.4) for f and g are invariant under the replacement

$$\kappa \rightarrow -\kappa, \quad f \rightarrow g, \quad g \rightarrow -f,$$

which changes no parameter of either the particle or the field. We must therefore have $f_{\kappa}/g_{\kappa} = -g_{-\kappa}/f_{-\kappa}$. Substitution of the asymptotic expressions then gives

$$\begin{aligned} \tan \left(pr - \frac{l\pi}{2} + \delta_{\kappa} \right) &= -\cot \left(pr - \frac{l'\pi}{2} + \delta_{-\kappa} \right), \\ \delta_{\kappa} &= \delta_{-\kappa} - (l' - l) \frac{\pi}{2} + \left(n + \frac{1}{2} \right) \pi, \end{aligned}$$

whence

$$e^{2i\delta_{\kappa}} = e^{2i\delta_{-\kappa}}. \quad (38.4)$$

Using this relation, and changing the summation index l to $l - 1$ in the first term of the sum in (37.4), we find

$$A(\theta) = \frac{1}{2ip} \sum_{l=1}^{\infty} l(e^{2i\delta_l} - 1) [P_l(\cos \theta) + P_{l-1}(\cos \theta)]. \quad (38.5)$$

Equation (38.2) implies $\text{Re}(AB^*) = 0$. Thus, in the approximation under consideration, the cross section is independent of the initial particle polarization, and an unpolarized beam remains unpolarized after scattering (see formulas III, (140.8)–(140.10)). We also note that as $\theta \rightarrow \pi$, the expression (38.5) for $A(\theta)$ tends to zero as $(\pi - \theta)^2$; recall that $P_l(-1) = (-1)^l$. The cross section tends to zero together with it:

$$\frac{d\sigma}{d\omega} = |A|^2 + |B|^2 = \frac{|A(\theta)|^2}{\cos^2(\theta/2)}. \quad (38.6)$$

These properties naturally disappear in the subsequent approximations in the small quantity m/ε . In particular, analysis shows that as $\theta \rightarrow \pi$, the cross section tends to a limit proportional to $(m/\varepsilon)^2$.

For a Coulomb field in the ultrarelativistic case, the phases δ_κ are independent of energy, as can be seen from (36.9)¹⁸. Therefore, in a pure Coulomb field the scattering cross section for $\varepsilon \gg m$ has the form

$$d\sigma = \frac{\tau(\theta)}{\varepsilon^2} d\omega, \quad (38.7)$$

where τ is a function of the angle alone.

§ 39. System of continuous-spectrum wave functions for scattering in a Coulomb field

Later (see §§ 95 and 96) we shall consider various inelastic processes occurring when ultrarelativistic electrons are scattered in the field of a heavy nucleus, with $Z\alpha \sim 1$. To calculate the corresponding matrix elements, we shall need wave functions whose asymptotic form as $r \rightarrow \infty$ consists of a plane wave and a spherical wave.

We shall see that in the ultrarelativistic case, where the electron energy $\varepsilon \gg m$, the main role in scattering is played by momentum transfers from the electron to the nucleus of magnitude $q = |\mathbf{p}' - \mathbf{p}| \sim m$. These values of q correspond to impact parameters $\rho \sim 1/q \sim 1/m$, while the electron is deflected through angles¹⁹

$$\theta \sim \frac{q}{p} \sim \frac{m}{\varepsilon}. \quad (39.1)$$

In terms of the coordinates r , the distance from the center, and $z = r \cos \theta$, this specifies the region

$$\rho = r \sin \theta \sim 1/m, \quad p(r - z) = pr(1 - \cos \theta) \sim 1. \quad (39.2)$$

Here $r \sim \varepsilon/m^2$, so this is a large-distance region.

Write the Dirac equation as

$$(\varepsilon - U - m\beta + i\alpha\nabla)\psi = 0, \quad U = -Z\alpha/r. \quad (39.3)$$

To transform it into a second-order equation, apply the operator $(\varepsilon - U + m\beta - i\alpha\nabla)$ to (39.3):

$$(\Delta + p^2 - 2\varepsilon U)\psi = (-i\alpha\nabla U - U^2)\psi. \quad (39.4)$$

In the region under consideration $r \gg Z\alpha/\varepsilon$, and hence $U \ll \varepsilon$. To first approximation the right-hand side of (39.4) may be neglected. The remaining equation,

$$\left(\Delta + p^2 + \frac{2\varepsilon Z\alpha}{r} \right) \psi = 0, \quad (39.5)$$

¹⁸This can also be seen directly from equations (38.1), since for a Coulomb field the substitution $\mathbf{r} \rightarrow \mathbf{r}'/\varepsilon$ eliminates the energy ε from the equations altogether.

¹⁹In this section, p denotes $|\mathbf{p}|$.

has the same form as the nonrelativistic Schrödinger equation in a Coulomb field,

$$\left(\frac{1}{2m} \Delta + \frac{p^2}{2m} + \frac{Z\alpha}{r} \right) \psi = 0, \quad (39.5a)$$

differing from it only by an evident relabeling of the parameters: the “potential energy” has an extra factor ε/m . We can therefore write down immediately the solution with the required asymptotic form (see III, § 136).

Thus the wave function which asymptotically contains a plane wave proportional to $e^{i\mathbf{p}\mathbf{r}}$ and an outgoing spherical wave is

$$\begin{aligned} \psi_{\varepsilon\mathbf{p}}^{(+)} &= C \frac{u_{\varepsilon\mathbf{p}}}{\sqrt{2\varepsilon}} e^{i\mathbf{p}\mathbf{r}} F\left(\frac{iZ\alpha\varepsilon}{p}, 1, i(pr - \mathbf{p}\mathbf{r})\right), \\ C &= e^{\pi Z\alpha\varepsilon/(2p)} \Gamma\left(1 - i\frac{Z\alpha\varepsilon}{p}\right), \end{aligned} \quad (39.6)$$

where F is the confluent hypergeometric function, and $u_{\varepsilon\mathbf{p}}$ is the constant bispinor amplitude of the plane wave, normalized by our convention (23.4):

$$\bar{u}_{\varepsilon\mathbf{p}} u_{\varepsilon\mathbf{p}} = 2m. \quad (39.7)$$

The normalization of (39.6) is such that the plane wave in its asymptotic expression has the usual form

$$\frac{u_{\varepsilon\mathbf{p}}}{\sqrt{2\varepsilon}} e^{i\mathbf{p}\mathbf{r}},$$

corresponding to one particle in unit volume. Since $p \approx \varepsilon$ in the ultrarelativistic case, we may set $Z\alpha\varepsilon/p \approx Z\alpha$ in (39.6):

$$\begin{aligned} \psi_{\varepsilon\mathbf{p}}^{(+)} &= C \frac{u_{\varepsilon\mathbf{p}}}{\sqrt{2\varepsilon}} e^{i\mathbf{p}\mathbf{r}} F(iZ\alpha, 1, i(pr - \mathbf{p}\mathbf{r})), \\ C &= e^{\pi Z\alpha/2} \Gamma(1 - iZ\alpha). \end{aligned} \quad (39.8)$$

Although the distances under consideration are so large that $pr \gg 1$, the hypergeometric function in (39.8) cannot be replaced by its asymptotic expression. Its argument is not pr but $pr(1 - \cos\theta)$, which is not assumed to be large²⁰.

Applications also require the next approximation to ψ , which has a spinor structure different from that of (39.8), where the spinor dependence reduces to the factor $u_{\varepsilon\mathbf{p}}$. To calculate it, write ψ as

$$\psi = \frac{C}{\sqrt{2\varepsilon}} e^{i\mathbf{p}\mathbf{r}} (u_{\varepsilon\mathbf{p}} F + \varphi).$$

We now retain the term linear in U on the right-hand side of (39.4), obtaining for φ

$$(\Delta + 2i\mathbf{p}\nabla - 2\varepsilon U)\varphi = -iu_{\varepsilon\mathbf{p}}(\boldsymbol{\alpha}\nabla U)F. \quad (39.9)$$

Its solution can be found by noting that F satisfies

$$(\Delta + 2i\mathbf{p}\nabla - 2\varepsilon U)F = 0,$$

²⁰In III, § 135, we were interested in arbitrarily large r , so such a replacement was possible at every angle θ .

as follows directly upon substitution of (39.6) in (39.5). Applying ∇ to this equation gives

$$(\Delta + 2i\mathbf{p}\nabla - 2\varepsilon U)\nabla F = 2\varepsilon F\nabla U.$$

Comparison with (39.9) yields

$$\varphi = -\frac{i}{2\varepsilon}(\alpha\nabla)u_{\varepsilon\mathbf{p}}F.$$

We now write the final expression for $\psi^{(+)}$, and for the analogous function $\psi^{(-)}$ whose asymptotic expression contains an incoming spherical wave:

$$\begin{aligned}\psi_{\varepsilon\mathbf{p}}^{(+)} &= \frac{C}{\sqrt{2\varepsilon}}e^{i\mathbf{p}\mathbf{r}}\left(1 - \frac{i\alpha}{2\varepsilon}\nabla\right)F(iZ\alpha, 1, i(pr - \mathbf{p}\mathbf{r}))u_{\varepsilon\mathbf{p}}, \\ \psi_{\varepsilon\mathbf{p}}^{(-)} &= \frac{C^*}{\sqrt{2\varepsilon}}e^{i\mathbf{p}\mathbf{r}}\left(1 - \frac{i\alpha}{2\varepsilon}\nabla\right)F(-iZ\alpha, 1, -i(pr + \mathbf{p}\mathbf{r}))u_{\varepsilon\mathbf{p}}, \\ C &= e^{\pi Z\alpha/2}\Gamma(1 - iZ\alpha)\end{aligned}\quad (39.10)$$

(W. H. Furry, 1934). We also write the analogous “negative-frequency” functions $\psi_{-\varepsilon-\mathbf{p}}$, needed in processes involving positrons. They follow from $\psi_{\varepsilon\mathbf{p}}$ by the replacements $\mathbf{p} \rightarrow -\mathbf{p}$, $\varepsilon \rightarrow -\varepsilon$, while $p = |\mathbf{p}|$ remains unchanged. Because of the last fact, the parameter $iZ\alpha$ of the hypergeometric function changes sign, as can be seen from the original expression (39.6), where it appears as $iZ\alpha\varepsilon/p$. We consequently obtain

$$\begin{aligned}\psi_{-\varepsilon-\mathbf{p}}^{(+)} &= \frac{C}{\sqrt{2\varepsilon}}e^{-i\mathbf{p}\mathbf{r}}\left(1 + \frac{i\alpha}{2\varepsilon}\nabla\right)F(-iZ\alpha, 1, i(pr + \mathbf{p}\mathbf{r}))u_{-\varepsilon-\mathbf{p}}, \\ \psi_{-\varepsilon-\mathbf{p}}^{(-)} &= \frac{C^*}{\sqrt{2\varepsilon}}e^{-i\mathbf{p}\mathbf{r}}\left(1 + \frac{i\alpha}{2\varepsilon}\nabla\right)F(iZ\alpha, 1, -i(pr - \mathbf{p}\mathbf{r}))u_{-\varepsilon-\mathbf{p}}, \\ C &= e^{-\pi Z\alpha/2}\Gamma(1 + iZ\alpha).\end{aligned}\quad (39.11)$$

One further comment must be made about this calculation. The asymptotic condition imposed above is by no means sufficient by itself to select a unique solution of the wave equation. This is clear, for example, because any outgoing Coulomb spherical wave can always be added to ψ without violating that condition. By writing the solution of (39.5) in the form (39.6), we tacitly selected the solution finite at $r = 0$. Such a requirement was necessary in III, §§ 135 and 136, where solutions of the exact Schrödinger equation, valid throughout space, were considered²¹. In the present case, however, equation (39.5) applies only at large distances, so the selection just made requires additional justification.

That justification is provided by the fact that large impact parameters $\rho = r \sin \theta$ correspond to large orbital angular momenta l and small scattering angles θ . For $\rho \sim 1/m$,

$$l \sim \rho p \sim \rho\varepsilon \sim \varepsilon/m \gg 1,$$

while θ can be estimated semiclassically:

$$\theta \sim \frac{1}{p} \int \frac{dU}{dr} dt \sim \frac{U'(\rho)\rho}{p} \sim \frac{m}{\varepsilon} \ll 1.$$

²¹In the method of solution presented in III, § 135, this condition was ensured by choosing a particular integral of the form (135.1) instead of the general sum of integrals with different values of β_1 and β_2 .

Thus, in the region of r and θ under consideration, the spherical-wave expansion of ψ mainly contains waves with these large values of l . A spherical wave with large l necessarily decreases to small values as it approaches the origin through distances $r \ll l/\varepsilon$ that are “classically inaccessible” because of the centrifugal barrier. If the solution of (39.5) is therefore “matched” to the solution of the exact equation (39.4) at short distances $r \sim r_1$, where $l/\varepsilon \gg r_1 \gg Z\alpha/\varepsilon$, the boundary condition on the solution of (39.5) requires it to be small. This justifies the choice made above.

PROBLEM

For an attractive Coulomb field with $Z\alpha \ll 1$, find the correction, of relative order $Z\alpha$, to the nonrelativistic discrete-spectrum wave function.

SOLUTION. The electron velocity in a bound state is $v \sim Z\alpha$. Thus, for $Z\alpha \ll 1$, the zeroth-approximation wave function is nonrelativistic:

$$\psi = u\psi_{\text{nr}},$$

where ψ_{nr} satisfies the nonrelativistic Schrödinger equation and u is a bispinor of the form $u = \begin{pmatrix} w \\ 0 \end{pmatrix}$, with w the spinor describing the electron's polarization state. In the next approximation write $\psi = u\psi_{\text{nr}} + \psi^{(1)}$. Substitution in (39.4) gives the following equation for $\psi^{(1)}$:

$$\left(\frac{1}{2m} \Delta - |\varepsilon_n| + \frac{Z\alpha}{r} \right) \psi^{(1)} = i \frac{Z\alpha}{2m} \left(\nabla \frac{1}{r} \right) (\alpha u) \psi_{\text{nr}},$$

where ε_n is the nonrelativistic discrete energy level. Terms of relative order $(Z\alpha)^2$ have been omitted here; one must remember that in the nonrelativistic case the characteristic distances are of the order of the Bohr radius, $r \sim 1/mZ\alpha$. The solution of this equation is $\psi^{(1)} = -\frac{i}{2m} \alpha u \nabla \psi_{\text{nr}}$, and hence

$$\psi = \left(1 - \frac{i}{2m} \alpha \nabla \right) u \psi_{\text{nr}}.$$

§ 40. An electron in the field of a plane electromagnetic wave

For an electron moving in the field of a plane electromagnetic wave, the Dirac equation admits an exact solution (D. M. Volkov, 1937).

The field of a plane wave with wave 4-vector k ($k^2 = 0$) depends on the 4-coordinates only through the combination $\varphi = kx$. Thus the 4-potential is

$$A^\mu = A^\mu(\varphi), \quad (40.1)$$

and obeys the Lorenz gauge condition

$$\partial_\mu A^\mu = k_\mu A^{\mu'} = 0$$

(the prime denotes differentiation with respect to φ). A constant term in A is immaterial, so the prime may be dropped in this condition, giving

$$kA = 0. \quad (40.2)$$

We start from the second-order equation (32.6), for which the field tensor is

$$F_{\mu\nu} = k_\mu A'_\nu - k_\nu A'_\mu. \quad (40.3)$$

On expanding the square $(i\partial - eA)^2$, one must also use (40.2), by which $\partial_\mu(A^\mu\psi) = A^\mu\partial_\mu\psi$. This gives

$$[-\partial^2 - 2ie(A\partial) + e^2A^2 - m^2 - ie(\gamma k)(\gamma A')]\psi = 0 \quad (40.4)$$

($\partial^2 = \partial_\mu\partial^\mu$).

We seek a solution in the form

$$\psi = e^{-ipx}F(\varphi), \quad (40.5)$$

where p is a constant 4-vector. Adding to p any vector of the form $\text{const} \cdot k$ leaves the form of ψ unchanged (only a corresponding redefinition of $F(\varphi)$ is needed). We may therefore impose one further condition on p without loss of generality. Let

$$p^2 = m^2. \quad (40.6)$$

When the field is removed, the quantum numbers p^μ then become the components of the free particle's 4-momentum. In the presence of the field, the meaning of the components of p is clearest in a special reference frame chosen so that $A_0 = 0$. In this frame let $\mathbf{A}$ point along the x^1 axis and $\mathbf{k}$ along the x^3 axis (the wave's electric field is then directed along x^1 , its magnetic field along x^2 , and its direction of propagation along x^3). Equation (40.5) is then an eigenfunction of the operators

$$\widehat{p}_1 = i\frac{\partial}{\partial x^1}, \quad \widehat{p}_2 = i\frac{\partial}{\partial x^2}, \quad \widehat{p}_0 - \widehat{p}_3 = i\left(\frac{\partial}{\partial x^0} - \frac{\partial}{\partial x^3}\right)$$

with eigenvalues p_1 , p_2 , and $p_0 - p_3$ (as is readily seen, these operators commute with the Hamiltonian of the Dirac equation). Thus, in this frame, p^1 and p^2 are the components of the generalized momentum along the x^1 and x^2 axes, while $p^0 - p^3$ is the difference between the total energy and the generalized-momentum component along the x^3 axis.

Upon substituting (40.5) into (40.4), we note that

$$\partial^\mu F = k^\mu F', \quad \partial_\mu\partial^\mu F = k^2 F'' = 0,$$

and obtain the following equation for $F(\varphi)$:

$$2i(kp)F' + [-2e(pA) + e^2A^2 - ie(\gamma k)(\gamma A')]F = 0.$$

Its formal solution is

$$F = \exp\left\{-i\int_0^{kx}\left[\frac{e}{(kp)}(pA) - \frac{e^2}{2(kp)}A^2\right]d\varphi + \frac{e(\gamma k)(\gamma A)}{2(kp)}\right\}\frac{u}{\sqrt{2p_0}},$$

where $u/\sqrt{2p_0}$ is an arbitrary constant bispinor (its explicit form is discussed below).

Every power of $(\gamma k)(\gamma A)$ above the first vanishes, since

$$\begin{aligned} (\gamma k)(\gamma A)(\gamma k)(\gamma A) &= -(\gamma k)(\gamma k)(\gamma A)(\gamma A) + 2(kA)(\gamma k)(\gamma A) \\ &= -k^2 A^2 = 0. \end{aligned}$$

Hence we may replace

$$\exp \frac{e(\gamma k)(\gamma A)}{2(kp)} = 1 + \frac{e}{2(kp)}(\gamma k)(\gamma A),$$

and ψ takes the form

$$\psi_p = \left[1 + \frac{e}{2(kp)}(\gamma k)(\gamma A) \right] \frac{u}{\sqrt{2p_0}} e^{iS}, \quad (40.7)$$

where²²

$$S = -px - \int_0^{kx} \frac{e}{(kp)} \left[(pA) - \frac{e}{2} A^2 \right] d\varphi. \quad (40.8)$$

To determine the conditions on the constant bispinor u , suppose that the wave is switched on infinitely slowly from $t = -\infty$. Then $A \rightarrow 0$ as $kx \rightarrow -\infty$, and ψ must tend to a solution of the free Dirac equation. This requires $u = u(p)$ to obey

$$(\gamma p - m)u = 0. \quad (40.9)$$

This condition eliminates the extraneous solutions of the second-order equation. Since u is time-independent, the condition continues to hold at finite kx . Thus $u(p)$ is the bispinor amplitude of a free plane wave; we shall assume that it is normalized by the same condition (23.4), $\bar{u}u = 2m$.

The preceding argument also determines the normalization of the wave functions (40.7) directly. Switching the field on infinitely slowly leaves the normalization integral unchanged. Consequently, the functions (40.7) satisfy the same normalization condition as free plane waves:

$$\int \psi_p^* \psi_p d^3x = \int \bar{\psi}_{p'} \gamma^0 \psi_p d^3x = (2\pi)^3 \delta(\mathbf{p}' - \mathbf{p}). \quad (40.10)$$

We now find the current density associated with the functions (40.7). Using

$$\bar{\psi}_p = \frac{\bar{u}}{\sqrt{2p_0}} \left[1 + \frac{e}{2(kp)}(\gamma A)(\gamma k) \right] e^{-iS},$$

direct multiplication gives

$$j^\mu = \bar{\psi}_p \gamma^\mu \psi_p = \frac{1}{p_0} \left\{ p^\mu - eA^\mu + k^\mu \left(\frac{e(pA)}{(kp)} - \frac{e^2 A^2}{2(kp)} \right) \right\}. \quad (40.11)$$

²²The expression for S is the classical action for a particle moving in the wave field (compare II, § 47, problem 2).

If $A^\mu(\varphi)$ are periodic functions whose time average vanishes, the mean current density is

$$\overline{j^\mu} = \frac{1}{p_0} \left(p^\mu - \frac{e^2}{2(kp)} \overline{A^2} k^\mu \right). \quad (40.12)$$

We also calculate the kinetic-momentum density in the state ψ_p . The kinetic-momentum operator is the difference $\widehat{p} - eA = i\partial - eA$. A direct calculation yields

$$\begin{aligned} \psi_p^* (\widehat{p}^\mu - eA^\mu) \psi_p &= \overline{\psi}_p \gamma^0 (\widehat{p}^\mu - eA^\mu) \psi_p = \\ &= p^\mu - eA^\mu + k^\mu \left(\frac{e(pA)}{(kp)} - \frac{e^2 A^2}{2(kp)} \right) + k^\mu \frac{ie}{8(kp)p_0} F_{\lambda\nu} (u^* \sigma^{\lambda\nu} u). \end{aligned} \quad (40.13)$$

Its time average, denoted by q^μ , is

$$q^\mu = p^\mu - \frac{e^2 \overline{A^2}}{2(kp)} k^\mu. \quad (40.14)$$

Its square is

$$q^2 = m_*^2, \quad m_* = m \sqrt{1 - \frac{e^2 \overline{A^2}}{m^2}}; \quad (40.15)$$

m_* plays the role of the electron's "effective mass" in the field. Comparing (40.14) and (40.12), we find

$$\overline{j^\mu} = q^\mu / p_0. \quad (40.16)$$

Finally, when the normalization condition (40.10) is expressed in terms of the vector $\mathbf{q}$, it becomes

$$\int \psi_p^* \psi_p d^3x = (2\pi)^3 \frac{q_0}{p_0} \delta(\mathbf{q}' - \mathbf{q}) \quad (40.17)$$

(the passage from (40.10) to (40.17) is most easily made in the special reference frame specified above).

§ 41. Spin motion in an external field

The semiclassical limit of the Dirac equation is taken in the same way as in nonrelativistic theory. In the second-order equation (32.7a), put²³

$$\psi = u e^{iS/\hbar},$$

where S is a scalar and u is a slowly varying bispinor. The usual semiclassical condition is assumed: the particle momentum must change little over distances of the order of the wavelength $\hbar/|\mathbf{p}|$.

At zeroth order in $\hbar$, one obtains the ordinary classical relativistic Hamilton–Jacobi equation for the action S . All spin-dependent terms (which are proportional to $\hbar$) then drop out of the equations of motion. Spin would appear only at the next order in $\hbar$. In other words, the effect of the electron's magnetic moment on its motion is always of the

²³We initially use conventional units.

same order as quantum corrections. This is natural in view of the purely quantum origin of spin angular momentum, which is proportional to $\hbar$.

This circumstance makes it meaningful to ask how the spin of an electron behaves while the electron follows a prescribed semiclassical motion in an external field. The answer is contained in the next order in $\hbar$ of the Dirac equation. We shall instead use a different method, which is more transparent and does not appeal directly to the Dirac equation. Its advantage is that it applies to any particle, including one with an “anomalous” gyromagnetic ratio not described by the Dirac equation.

Our aim is to establish an “equation of motion” for spin during an arbitrary prescribed motion of the particle. We begin with the nonrelativistic case.

The nonrelativistic Hamiltonian of a particle in an external field is

$$\hat{H} = \hat{H}' - \mu \boldsymbol{\sigma} \cdot \mathbf{H}, \quad (41.1)$$

where $\hat{H}'$ includes all terms independent of spin (see III, § 111), and μ is the particle's magnetic moment. This form of the Hamiltonian is not tied to any particular species. For an electron, $\mu = e\hbar/2mc$ (the electron charge is $e = -|e|!$), while for nucleons μ also contains an “anomalous” part²⁴

$$\mu' = \mu - \frac{e\hbar}{2mc}. \quad (41.2)$$

By the general rules of quantum mechanics, the operator equation of spin motion follows from

$$\hat{\mathbf{s}} = \frac{i}{\hbar}(\hat{H}\hat{\mathbf{s}} - \hat{\mathbf{s}}\hat{H}) = \frac{i}{2\hbar}(\hat{H}\boldsymbol{\sigma} - \boldsymbol{\sigma}\hat{H}). \quad (41.3)$$

Substitution of (41.1) gives

$$\hat{s}_i = -\frac{i\mu}{2\hbar}H_k(\sigma_k\sigma_i - \sigma_i\sigma_k) = -\frac{\mu}{\hbar}e_{ikl}H_k\sigma_l,$$

or

$$\hat{\mathbf{s}} = \frac{2\mu}{\hbar}(\hat{\mathbf{s}} \times \mathbf{H}). \quad (41.4)$$

Average this operator equality over the state of a semiclassical wave packet moving along a specified trajectory. This amounts to replacing the spin operator by its mean value $\bar{\mathbf{s}}$ and the vector $\mathbf{H}$ by the function $\mathbf{H}(t)$ giving the magnetic field at the particle's (wave packet's) position as it moves along the prescribed trajectory. In the nonrelativistic approximation, that is, within the Pauli equation, $\hat{\mathbf{s}} = \boldsymbol{\sigma}/2$ is the particle's spin operator in its rest frame; in § 29 we denoted its mean value by $\boldsymbol{\zeta}/2$. We therefore obtain

$$\frac{d\boldsymbol{\zeta}}{dt} = \frac{2\mu}{\hbar}(\boldsymbol{\zeta} \times \mathbf{H}(t)). \quad (41.5)$$

In this form the equation is, in essence, purely classical. It states that the magnetic-moment vector precesses about the field direction with angular velocity $-2\mu H/\hbar$, while retaining its magnitude²⁵.

²⁴When radiative corrections are included, the electron magnetic moment also contains a very small “anomalous part.”

²⁵Classically, equation (41.5) follows directly from

$$d\mathbf{M}/dt = \boldsymbol{\mu} \times \mathbf{H},$$

In the same nonrelativistic case, the particle velocity $\mathbf{v}$ changes according to

$$\frac{d\mathbf{v}}{dt} = \frac{e}{mc} (\mathbf{v} \times \mathbf{H}),$$

so that $\mathbf{v}$ rotates about the direction of $\mathbf{H}$ with angular velocity $-eH/mc$. If $\mu' = 0$, then $\mu = e\hbar/2mc$, and this angular velocity equals the angular velocity $-2\mu H/\hbar$ of $\boldsymbol{\zeta}$. Thus the polarization vector keeps a constant angle with the direction of motion (we shall see below that this remains true in the relativistic case).

We now generalize equation (41.5) relativistically. A covariant description of polarization requires the 4-vector a introduced in § 29, and the equation of spin motion must determine the derivative $da/d\tau$ with respect to the proper time τ ²⁶.

The possible form of this equation can already be fixed by relativistic invariance. Its right side must be linear and homogeneous in the electromagnetic field tensor $F^{\mu\nu}$ and the 4-vector a^μ , and besides these may contain only the 4-velocity $u^\mu = p^\mu/m$. The only equation satisfying these requirements has the form

$$\frac{da^\mu}{d\tau} = \alpha F^{\mu\nu} a_\nu + \beta u^\mu F^{\nu\lambda} u_\nu a_\lambda, \quad (41.6)$$

where α and β are constants. The condition $a_\mu u^\mu = 0$ and the antisymmetry of $F^{\mu\nu}$ (so that $F^{\mu\nu} u_\mu u_\nu = 0$) make it clear that no other expression of the required type can be formed.

As $v \rightarrow 0$, this equation must reduce to (41.5). Setting $a^\mu = (0, \boldsymbol{\zeta})$, $u^\mu = (1, 0)$, and $\tau = t$, we obtain

$$\frac{d\boldsymbol{\zeta}}{dt} = \alpha (\boldsymbol{\zeta} \times \mathbf{H}).$$

Comparison with (41.5) gives $\alpha = 2\mu$.

To determine β , use $a^\mu u_\mu = 0$. Differentiating this equality with respect to τ and using the classical equation of motion of a charge in a field,

$$m \frac{du^\mu}{d\tau} = e F^{\mu\nu} u_\nu$$

(see II, § 23), we find

$$u_\mu \frac{da^\mu}{d\tau} = -a_\mu \frac{du^\mu}{d\tau} = -a_\mu \frac{e}{m} F^{\mu\nu} u_\nu = \frac{e}{m} F^{\mu\nu} u_\mu a_\nu.$$

Multiplying both sides of (41.6) by u_μ , using $u_\mu u^\mu = 1$, and cancelling the common factor $F^{\mu\nu} u_\mu a_\nu$, we obtain

$$\beta = -2 \left(\mu - \frac{e}{2m} \right) = -2\mu'.$$

The final relativistic equation of spin motion is therefore

$$\frac{da^\mu}{d\tau} = 2\mu F^{\mu\nu} a_\nu - 2\mu' u^\mu F^{\nu\lambda} u_\nu a_\lambda \quad (41.7)$$

where $\mathbf{M}$ is the angular momentum of the system, $\boldsymbol{\mu}$ its magnetic moment, and $\boldsymbol{\mu} \times \mathbf{H}$ the torque acting on it. Setting $\mathbf{M} = \frac{1}{2}\hbar\boldsymbol{\zeta}$ and $\boldsymbol{\mu} = \frac{\mu}{2s}\boldsymbol{\zeta} = \mu\boldsymbol{\zeta}$ gives (41.5).

²⁶Below we again put $c = 1$, $\hbar = 1$.

(V. Bargmann, L. Michel, V. Telegdi, 1959)²⁷.

We pass from the 4-vector a to ζ , which directly describes the particle polarization in its “instantaneous” rest frame; their relation is given by (29.7)–(29.9). First note that (41.7) automatically gives $a_\mu da^\mu/d\tau = 0$, and hence $a_\mu a^\mu = \text{const}$. Since $a_\mu a^\mu = -\zeta^2$, this yields the natural result that the magnitude of the particle polarization ζ remains unchanged during the motion.

To obtain the equation governing the direction of polarization, convert (41.7) to three-dimensional notation. Writing out the spatial components gives

$$\begin{aligned} \frac{d\mathbf{a}}{dt} = & \frac{2\mu m}{\varepsilon}(\mathbf{a} \times \mathbf{H}) + \frac{2\mu m}{\varepsilon}(\mathbf{a} \cdot \mathbf{v})\mathbf{E} - \frac{2\mu'\varepsilon}{m}\mathbf{v}(\mathbf{a} \cdot \mathbf{E}) + \\ & + \frac{2\mu'\varepsilon}{m}\mathbf{v}(\mathbf{v} \cdot (\mathbf{a} \times \mathbf{H})) + \frac{2\mu'\varepsilon}{m}\mathbf{v}(\mathbf{a} \cdot \mathbf{v})(\mathbf{v} \cdot \mathbf{E}). \end{aligned}$$

Substitute (29.9) here, using in the differentiation $\mathbf{p} = \varepsilon\mathbf{v}$, $\varepsilon^2 = \mathbf{p}^2 + m^2$, and the equations of motion

$$\frac{d\mathbf{p}}{dt} = e\mathbf{E} + e(\mathbf{v} \times \mathbf{H}), \quad \frac{d\varepsilon}{dt} = e(\mathbf{v} \cdot \mathbf{E}). \quad (41.8)$$

A straightforward, though fairly long, calculation gives²⁸:

$$\frac{d\zeta}{dt} = \frac{2\mu m + 2\mu'(\varepsilon - m)}{\varepsilon}(\zeta \times \mathbf{H}) + \frac{2\mu'\varepsilon}{\varepsilon + m}(\mathbf{v} \cdot \mathbf{H})(\mathbf{v} \times \zeta) + \frac{2\mu m + 2\mu'\varepsilon}{\varepsilon + m}(\zeta \times (\mathbf{E} \times \mathbf{v})). \quad (41.9)$$

Of particular interest is not so much the change in the absolute direction of polarization in space as its change relative to the direction of motion. Write

$$\zeta = \mathbf{n}\zeta_{\parallel} + \zeta_{\perp} \quad (41.10)$$

(where $\mathbf{n} = \mathbf{v}/v$), and form the equation for the projection $\zeta_{\parallel}$ of the polarization on the direction of motion. Calculation with (41.8) and (41.9) gives²⁹

$$\frac{d\zeta_{\parallel}}{dt} = 2\mu'(\zeta_{\perp} \cdot (\mathbf{H} \times \mathbf{n})) + \frac{2}{v} \left(\frac{\mu m^2}{\varepsilon^2} - \mu' \right) (\zeta_{\perp} \cdot \mathbf{E}). \quad (41.11)$$

Several applications of these equations are considered in the problems to this section. Here we note only that, in a purely magnetic field, the polarization of a particle without an anomalous magnetic moment maintains a constant angle with its velocity ($\zeta_{\parallel} = \text{const}$). Thus the result stated above for the nonrelativistic case is indeed general.

²⁷An equation of this kind, in another form, was first found by Ya. I. Frenkel (1926).

²⁸If, as is often done for charged particles, the gyromagnetic coefficient (Landé factor) g is defined by $\mu = g \frac{e}{2m} \cdot \frac{1}{2} (= g \frac{e\hbar}{2mc} \frac{1}{2})$, the equation takes the form

$$\begin{aligned} \frac{d\zeta}{dt} = & \frac{e}{2m} \left(g - 2 + 2\frac{m}{\varepsilon} \right) (\zeta \times \mathbf{H}) + \frac{e}{2m} (g - 2) \frac{\varepsilon}{\varepsilon + m} (\mathbf{v} \cdot \mathbf{H})(\mathbf{v} \times \zeta) + \\ & + \frac{e}{2m} \left(g - \frac{2\varepsilon}{\varepsilon + m} \right) (\zeta \times (\mathbf{E} \times \mathbf{v})). \end{aligned} \quad (41.9a)$$

²⁹A slightly shorter derivation follows by writing out the time component of (41.7).

We specify more precisely the conditions under which the equations obtained are applicable. The requirement, mentioned at the outset, that the particle momentum vary sufficiently slowly reduces to a condition that the fields $\mathbf{E}$ and $\mathbf{H}$ be sufficiently weak; in particular, the Larmor radius in a magnetic field ($\sim p/eH$) must be large compared with the particle wavelength. Strictly speaking, an additional condition is that the fields not vary too rapidly in space: the field must change little across the semiclassical wave packet. It must therefore vary little over distances of the order of the particle wavelength ($1/p$) and also over the Compton wavelength $1/m$ ³⁰.

In practical problems concerning motion in macroscopic fields, the slow-variation condition is certainly satisfied, so that only sufficient weakness is actually required.

In § 33 the first relativistic corrections were found for the Hamiltonian of an electron moving in an external field. For an electron in an electric field, the approximate Hamiltonian has the form (see (33.12))

$$\hat{H} = \hat{H}' - \frac{e}{4m} \left(\boldsymbol{\sigma} \cdot \left(\mathbf{E} \times \frac{\hat{\mathbf{p}}}{m} \right) \right), \quad \hat{\mathbf{p}} = -i\nabla, \quad (41.12)$$

where $\hat{H}'$ contains the spin-independent terms. In the present case, because the field varies slowly, the term containing derivatives of $\mathbf{E}$ (that is, $\nabla \cdot \mathbf{E}$) must be neglected in $\hat{H}'$. The small term containing $\hat{\mathbf{p}}^4$ may also be omitted, since it is unrelated to the field effects of interest here. Thus in the absence of a magnetic field, $\hat{H}'$ reduces to the nonrelativistic Hamiltonian $\hat{H}' = \hat{\mathbf{p}}^2/(2m) + e\Phi$.

Formula (41.12) can also be obtained from equation (41.9), without direct use of the Dirac equation. This will at the same time generalize it, in the semiclassical case, to particles with an anomalous magnetic moment.

To first order in the velocity $\mathbf{v}$, the equation of spin motion in an electric field follows from (41.9) as

$$\frac{d\boldsymbol{\zeta}}{dt} = (\mu + \mu')(\boldsymbol{\zeta} \times (\mathbf{E} \times \mathbf{v})) = \left(\frac{e}{2m} + 2\mu' \right) (\boldsymbol{\zeta} \times (\mathbf{E} \times \mathbf{v})).$$

If this equation is required to follow quantum mechanically by commuting the spin operator with the Hamiltonian, according to (41.3), then, as is readily checked, one must put

$$\hat{H} - \hat{H}' = \left(\mu' + \frac{e}{4m} \right) \left(\boldsymbol{\sigma} \cdot \left(\mathbf{E} \times \frac{\hat{\mathbf{p}}}{m} \right) \right). \quad (41.13)$$

This is the required expression. For $\mu' = 0$, it reduces to (41.12). Notice that the “normal” magnetic moment $e/2m$ enters with an extra factor 1/2 relative to the anomalous moment μ' ³¹.

³⁰The latter requirement follows from the condition that, in the rest frame of the wave packet, its velocity spread be small compared with c ; otherwise nonrelativistic formulas could not be used in that frame.

If the field varies too rapidly, additional terms containing spatial derivatives of the field may become important in the equations.

³¹This is the “Thomas half” mentioned in the note on p. 151. The derivation given here clearly shows its origin.

PROBLEM

1. Determine the change in the polarization direction of a particle moving in a plane perpendicular to a uniform magnetic field ($\mathbf{v} \perp \mathbf{H}$).

SOLUTION. Only the first term remains on the right side of (41.9). Thus ζ precesses about the direction of $\mathbf{H}$ (the z axis) with angular velocity

$$-\frac{2\mu m + 2\mu'(\varepsilon - m)}{\varepsilon} \mathbf{H} = -\left(\frac{e}{\varepsilon} + 2\mu'\right) \mathbf{H}.$$

The projection of ζ on the xy plane (denote it by ζ_1) rotates in that plane with the same angular velocity. The vector $\mathbf{v}$ rotates in the same plane with angular velocity $-e\mathbf{H}/\varepsilon$, as follows from the equation of motion $\dot{\mathbf{p}} = \varepsilon\dot{\mathbf{v}} = e(\mathbf{v} \times \mathbf{H})$. Hence ζ_1 turns relative to the direction of $\mathbf{v}$ with angular velocity $-2\mu'\mathbf{H}$.

2. The same question for motion along the magnetic-field direction.

SOLUTION. When $\mathbf{v}$ and $\mathbf{H}$ have the same direction, equation (41.9) becomes

$$\frac{d\zeta}{dt} = \frac{2\mu m}{\varepsilon} (\zeta \times \mathbf{H}),$$

so that ζ precesses about the common direction of $\mathbf{v}$ and $\mathbf{H}$ with angular velocity $-2\mu m H/\varepsilon$.

3. The same question for motion in a uniform electric field.

SOLUTION. Let the field $\mathbf{E}$ point along the x axis, and let the motion take place in the xy plane (so that $p_y = \text{const}$). Equation (41.9) shows that ζ precesses about the z axis with instantaneous angular velocity

$$-\left(\frac{e}{\varepsilon + m} + 2\mu'\right) E \frac{p_y}{\varepsilon}.$$

Again decompose ζ into ζ_z and $\zeta_\perp$ (in the xy plane). Then

$$\zeta_\parallel = \zeta_1 \cos \varphi, \quad \zeta_\perp \cdot \mathbf{E} = -\zeta_1 \sin \varphi \cdot \frac{v_y}{v}.$$

Equation (41.11) gives the instantaneous angular velocity with which ζ_1 rotates relative to the direction of $\mathbf{v}$:

$$\dot{\varphi} = \frac{2v_y}{v^2} \left(\frac{\mu m^2}{\varepsilon^2} - \mu' \right) E = \frac{p_y}{\varepsilon} \left(\frac{em}{p^2} - 2\mu' \right).$$

§ 42. Neutron scattering in an electric field

In collisions between neutrons and nuclei, scattering through large angles is governed by the principal interaction, the nuclear forces. For scattering through small angles, however, the interaction of the neutron magnetic moment with the electric field of the nucleus becomes important, as we shall see (J. Schwinger, 1948).

We shall treat the neutron as nonrelativistic, so that the interaction in question is described by the approximate Hamiltonian (41.13). The entire magnetic moment of an

electrically neutral particle is “anomalous”, and in this case the operator $\hat{H}'$ reduces to the kinetic-energy operator³²:

$$\hat{H} = -\frac{\hbar^2}{2m}\Delta + i\frac{\mu\hbar}{mc}\boldsymbol{\sigma}(\mathbf{E} \times \nabla). \quad (42.1)$$

Since the neutron's electromagnetic interaction is weak, the amplitude f_{em} for the scattering caused by it can be calculated in the Born approximation:

$$f_{em} = -\frac{m}{2\pi\hbar^2} \int e^{-i\mathbf{p}'\mathbf{r}/\hbar} \left(i\frac{\mu\hbar}{mc}\boldsymbol{\sigma}(\mathbf{E} \times \nabla) \right) e^{i\mathbf{p}\mathbf{r}/\hbar} d^3x$$

(see III, § 126), or

$$f_{em} = \frac{\mu}{2\pi c\hbar^2}\boldsymbol{\sigma}(\mathbf{E}_{\mathbf{q}} \times \mathbf{p}), \quad \mathbf{E}_{\mathbf{q}} = \int \mathbf{E}(\mathbf{r})e^{-i\mathbf{q}\mathbf{r}}d^3x \quad (42.2)$$

($\mathbf{p}$ and $\mathbf{p}'$ are the neutron momenta before and after scattering, and $\hbar\mathbf{q} = \mathbf{p}' - \mathbf{p}$). In the form written, the amplitude f_{em} is an operator with respect to the spin variable.

Before proceeding with the calculation, we make the following observation. Formula (42.1) was derived in the preceding section for slowly varying fields (which in effect meant neglecting Hamiltonian terms containing spatial derivatives of the field). For its application to the Coulomb field of a nucleus, the wavelength $\hbar/p$ must therefore be small compared with the distances relevant to the integral $\mathbf{E}_{\mathbf{q}}$, namely $r \sim 1/q$. Hence $\hbar q \ll p$, and the scattering angle $\theta \sim \hbar q/p \ll 1$. Thus, the required condition is fulfilled precisely for scattering through small angles.

For a Coulomb field with potential $\Phi = Ze/r$, the Fourier component of the electric field strength is

$$\mathbf{E}_{\mathbf{q}} = -i\mathbf{q}\Phi_{\mathbf{q}} = -i\mathbf{q} \cdot 4\pi Ze/q^2$$

(see II, (51.5)). Substitution in (42.2) gives

$$f_{em} = i\frac{2Ze\mu}{q^2c\hbar^3}\boldsymbol{\sigma}(\mathbf{p} \times \mathbf{p}').$$

At small scattering angles, $\hbar q \approx p\theta$ and $\mathbf{p} \times \mathbf{p}' \approx p^2\theta\mathbf{v}$, where $\mathbf{v}$ is a unit vector in the direction of $\mathbf{p} \times \mathbf{p}'$. Consequently,

$$f_{em} = i\frac{2Ze\mu}{\theta\hbar c}\boldsymbol{\sigma}\mathbf{v}.$$

The nuclear-scattering amplitude must be added to this expression. Because nuclear forces decrease rapidly with distance, this amplitude approaches a finite, energy-dependent complex value at small angles; denote it by a . The total scattering amplitude is therefore

$$f = a + i\frac{b}{\theta}\boldsymbol{\sigma}\mathbf{v}, \quad b = \frac{2Ze\mu}{c\hbar} = 2Z\alpha\frac{\mu}{e}. \quad (42.3)$$

We see that electromagnetic scattering does indeed become predominant at sufficiently small angles.

³²In this section we use ordinary units, and the letter m denotes the neutron mass.

The form of expression (42.3) is the same as that considered in Vol. III, § 140. We may therefore use the formulas derived there directly. The scattering cross-section summed over all possible final polarization states is

$$\frac{d\sigma}{d\Omega} = |a|^2 + \frac{b^2}{\theta^2} + 2b \operatorname{Im} a \cdot \nu \zeta, \quad (42.4)$$

where ζ is the initial polarization of the neutron beam (for $\mathbf{P}$, see Vol. III, § 140). If the initial state is unpolarized ($\zeta = 0$), the polarization after scattering is

$$\zeta' = \frac{2b \operatorname{Im} a \cdot \theta}{|a|^2 \theta^2 + b^2} \nu. \quad (42.5)$$

This polarization reaches its maximum at $\theta = b/|a|$, with $\zeta'_{\max} = \operatorname{Im} a/|a|$.

Radiation

§ 43. The electromagnetic interaction operator

The interaction of electrons with an electromagnetic field can generally be treated by perturbation theory. This is possible because the electromagnetic interaction is comparatively weak, as expressed by the smallness of its dimensionless “coupling constant”, the fine-structure constant $\alpha = e^2/\hbar c = 1/137$. This small parameter has a fundamental role in quantum electrodynamics.

In classical electrodynamics (see II, § 16, 28), the electromagnetic interaction is described by the term

$$-e j^\mu A_\mu \quad (43.1)$$

in the Lagrangian density of the “field+charges” system (A is the field 4-potential and j is the 4-vector of the particle-current density). The current density obeys the continuity equation

$$\partial_\mu j^\mu = 0, \quad (43.2)$$

which expresses charge conservation. Recall (see II, § 29) that the gauge invariance of the theory is closely connected with this law. Indeed, under the replacement $A_\mu \rightarrow A_\mu + \partial_\mu \chi$ (4.1), the quantity $-e j^\mu \partial_\mu \chi$ is added to the Lagrangian density (43.1). By (43.2), this quantity can be written as the 4-divergence

$$-e \partial_\mu (\chi j^\mu)$$

and consequently makes no contribution after integration over d^4x in the action $S = \int L d^4x$.

In quantum electrodynamics, the 4-vectors j and A are replaced by the corresponding second-quantized operators. The current operator is then expressed in terms of the ψ operators according to $\hat{j} = \hat{\bar{\psi}} \gamma \hat{\psi}$. The values of $\hat{\bar{\psi}}$, $\hat{\psi}$, and $\hat{A}$ at every point of space play the role of the generalized “coordinates” q in the Lagrangian

$$\int \hat{L}_{\text{int}} d^3x = -e \int (\hat{j} \hat{A}) d^3x.$$

Since the Lagrangian density depends only on the “coordinates” q themselves (and not on their derivatives with respect to x), the passage to the Hamiltonian density by formula (10.11) merely reverses the sign of the Lagrangian density¹. Thus, the electromagnetic interaction operator (the spatial integral of the interaction Hamiltonian density) is

$$\hat{V} = e \int (\hat{j} \hat{A}) d^3x. \quad (43.3)$$

¹Independently of this argument, we note that when only a first-order correction is at issue, any small correction to the Lagrangian enters the Hamiltonian with its sign reversed (see I, § 40).

The free electromagnetic-field operator is the sum

$$\hat{A} = \sum_n [\hat{c}_n A_n(x) + \hat{c}_n^* A_n^*(x)], \quad (43.4)$$

which contains the photon creation and annihilation operators for the different states (labelled by the index n). Each of these operators has matrix elements only when the corresponding occupation number N_n increases or decreases by 1 (with all other occupation numbers unchanged). The operator $\hat{A}$ therefore also has matrix elements only for transitions in which the photon number changes by 1. In other words, first-order perturbation theory permits only the emission or absorption of a single photon.

According to (2.15), the matrix elements are

$$\langle N_n - 1 | \hat{c}_n | N_n \rangle = \langle N_n | \hat{c}_n^* | N_n - 1 \rangle = \sqrt{N_n}. \quad (43.5)$$

If no photons (of type n) are present in the initial field state, then $\langle 1 | \hat{c}_n^* | 0 \rangle = 1$. The matrix element of operator (43.3) for photon emission is

$$V_{fi}(t) = e \int (j_{fi} A_n^*) d^3x, \quad (43.6)$$

where $A_n(x)$ is the wave function of the emitted photon, while j_{fi} is the matrix element of the operator $\hat{j}$ for the transition of the radiating system from its initial state i to its final state f ². The 4-vector $j_{fi}^\mu = (\rho_{fi}, \mathbf{j}_{fi})$ is called the *transition current*.

The matrix element for photon absorption is obtained similarly:

$$V_{fi}(t) = e \int (j_{fi} A_n) d^3x. \quad (43.7)$$

It differs from (43.6) only in having $A_n(x)$ in place of $A_n^*(x)$.

By displaying the argument t in V_{fi} , we emphasize that this matrix element depends on time. Separating the time factors in the wave functions, we may pass in the usual way to time-independent matrix elements:

$$V_{fi}(t) = V_{fi} e^{-i(E_i - E_f \mp \omega)t} \quad (43.8)$$

(E_i and E_f are the initial and final energies of the radiating system; the signs $\mp$ correspond to emission and absorption of a photon of frequency ω).

The wave function of a photon with definite momentum $\mathbf{k}$ and definite polarization is

$$A^\mu = \sqrt{4\pi} \frac{e^\mu}{\sqrt{2\omega}} e^{i\mathbf{k}\mathbf{r}} \quad (43.9)$$

(see (4.3); the time factor has been omitted). Substitution in (43.6) gives the matrix element for emission of such a photon in the form

$$V_{fi} = e\sqrt{4\pi} \frac{1}{\sqrt{2\omega}} e_\mu^* j_{fi}^\mu(\mathbf{k}), \quad (43.10)$$

²There is a slight inconsistency in the notation in (43.6): the indices of V_{fi} refer to states of the entire "radiating system+field", whereas those of j_{fi} refer to states of the radiating system alone.

where $j_{fi}(\mathbf{k})$ is the transition current in the momentum representation, that is, its Fourier components are

$$j_{fi}(\mathbf{k}) = \int j_{fi}(\mathbf{r}) e^{-i\mathbf{k}\mathbf{r}} d^3x. \quad (43.11)$$

The corresponding formula for photon absorption is

$$V_{fi} = e\sqrt{4\pi} \frac{1}{\sqrt{2\omega}} e_{\mu} j_{fi}^{\mu}(-\mathbf{k}). \quad (43.12)$$

In the momentum representation, the current-conservation equation takes the form of the 4-transversality condition for transition currents:

$$k_{\mu} j_{fi}^{\mu} = \omega \rho_{fi}(\mathbf{k}) - \mathbf{k} \mathbf{j}_{fi}(\mathbf{k}) = 0. \quad (43.13)$$

The formulas written in this section, in which the form of the current operator has not been specified, are general and apply to electromagnetic processes involving any charged particles. The existing theory allows the form of the current operator to be established (and hence its matrix elements to be calculated in principle) only for electrons. In applying these formulas to systems of strongly interacting particles, including nuclei, we shall instead give a semiphenomenological theory in which the transition currents are quantities taken from experiment and constrained only by the general requirements of space-time symmetry and by the continuity equation.

§ 44. Emission and absorption

In first-order perturbation theory, the probability of a transition produced by the perturbation $\hat{V}$ is given by the standard formulas (III, § 42). Let the initial and final states of the radiating system belong to the discrete spectrum³. The probability per unit time of the transition $i \rightarrow f$ accompanied by photon emission is then

$$dw = 2\pi |V_{fi}|^2 \delta(E_i - E_f - \omega) d\nu, \quad (44.1)$$

where ν denotes, symbolically, the collection of quantities characterizing the photon state and ranging continuously over their values (the photon wave function is here understood to be normalized to a delta function on the ν scale).

If the emitted photon has a specified angular momentum, the frequency ω is the only continuous variable. Integrating (44.1) over $d\nu = d\omega$ removes the delta function (fixing ω at $\omega = E_i - E_f$), and the transition probability becomes

$$w = 2\pi |V_{fi}|^2. \quad (44.2)$$

If instead photon emission with a prescribed momentum $\mathbf{k}$ is considered, then $d\nu = d^3k/(2\pi)^3 = \omega^2 d\omega d\Omega/(2\pi)^3$. The photon wave function (a plane wave) is assumed to be normalized to one photon in the volume $V = 1$, as it is throughout this

³This assumption entails, in particular, neglect of recoil: the radiator remains at rest as a whole.

book; $d\nu$ is the number of states contained in the phase-space volume Vd^3k . Hence the probability of emitting a photon with a prescribed momentum is

$$dw = 2\pi|V_{fi}|^2\delta(E_i - E_f - \omega)\frac{d^3k}{(2\pi)^3}, \quad (44.3)$$

or, after integration over $d\omega$,

$$dw = \frac{1}{4\pi^2}|V_{fi}|^2\omega^2 d\omega. \quad (44.4)$$

The matrix element V_{fi} from (43.10) must be substituted here.

In the sections that follow, these formulas will be used to calculate emission probabilities in various specific cases. Here we consider several general relations between different kinds of radiative processes.

If the initial field state already contains a nonzero number N_n of the photons in question, the transition matrix element acquires the additional factor

$$\langle N_n + 1 | c_n^+ | N_n \rangle = \sqrt{N_n + 1}, \quad (44.5)$$

so that the transition probability is multiplied by $N_n + 1$. The unity in this factor accounts for spontaneous emission, which also occurs at $N_n = 0$. The term N_n , by contrast, produces *stimulated* (or *induced*) emission: the presence of photons in the initial field state stimulates the emission of further photons of the same kind.

For the reverse change of state of the system ($f \rightarrow i$), the matrix element V_{if} differs from V_{fi} by replacing (44.5) with

$$\langle N_n - 1 | c_n | N_n \rangle = \sqrt{N_n}$$

(and by replacing the other quantities with their complex conjugates). This reverse transition is the absorption of a photon by a system that passes from the level E_f to the level E_i . The emission and absorption probabilities for a given pair of states i, f therefore obey the important relation⁴

$$\frac{w^{(\text{em})}}{w^{(\text{abs})}} = \frac{N_n + 1}{N_n} \quad (44.6)$$

(first given by A. Einstein in 1916).

We now relate the photon number to the intensity of radiation incident on the system from outside. Let

$$I_{\mathbf{ke}} d\omega do \quad (44.7)$$

be the radiation energy incident per unit time on unit area, with polarization $\mathbf{e}$, frequency in the interval $d\omega$, and wave-vector direction in the solid-angle element do . These intervals correspond to $k^2 dk do / (2\pi)^3$ field oscillators, each containing $N_{\mathbf{ke}}$ photons of the specified polarization. The same energy (44.7) is therefore obtained from the product

$$c \frac{k^2 dk do}{(2\pi)^3} N_{\mathbf{ke}} \hbar \omega = \frac{\hbar \omega^3}{8\pi^3 c^2} N_{\mathbf{ke}} d\omega do.$$

⁴For the remainder of this section we use conventional units.

This gives the required relation:

$$N_{\mathbf{ke}} = \frac{8\pi^3 c^2}{\hbar\omega^3} I_{\mathbf{ke}}. \quad (44.8)$$

Let $dw_{\mathbf{ke}}^{(\text{sp})}$ denote the probability of spontaneous emission of a photon of polarization $\mathbf{e}$ into the solid angle do ; the labels (ind) and (abs) will denote the corresponding probabilities of induced emission and absorption. Equations (44.6) and (44.8) give the following relations among them:

$$dw_{\mathbf{ke}}^{(\text{abs})} = dw_{\mathbf{ke}}^{(\text{ind})} = dw_{\mathbf{ke}}^{(\text{sp})} \frac{8\pi^3 c^2}{\hbar\omega^3} I_{\mathbf{ke}}. \quad (44.9)$$

For isotropic, unpolarized incident radiation ($I_{\mathbf{ke}}$ is independent of the directions of $\mathbf{k}$ and $\mathbf{e}$), integration of (44.9) over do and summation over $\mathbf{e}$ yield analogous relations between the total probabilities of radiative transitions for the specified pair of system states i and f :

$$w^{(\text{abs})} = w^{(\text{ind})} = w^{(\text{sp})} \frac{\pi^2 c^2}{\hbar\omega^3} I, \quad (44.10)$$

where $I = 2 \cdot 4\pi I_{\mathbf{ke}}$ is the total spectral intensity of the incident radiation.

If the states i and f of the emitting (or absorbing) system are degenerate, the total probability for emission (or absorption) of these photons is found by summing over all mutually degenerate final states and averaging over every possible initial state. Denote the degeneracies (statistical weights) of states i and f by g_i and g_f . The initial states are the states i for spontaneous and induced emission, and the states f for absorption. If all g_i or g_f initial states are taken to be equiprobable in each case, then (44.10) is replaced by

$$g_f w^{(\text{abs})} = g_i w^{(\text{ind})} = g_i w^{(\text{sp})} \frac{\pi^2 c^2}{\hbar\omega^3} I. \quad (44.11)$$

The quantities known as the *Einstein coefficients* are often used in the literature; they are defined by

$$A_{if} = w^{(\text{sp})}, \quad B_{if} = w^{(\text{ind})} \frac{c}{I}, \quad B_{fi} = w^{(\text{abs})} \frac{c}{I} \quad (44.12)$$

(I/c is the spatial spectral density of the radiation energy). They are related by

$$g_f B_{fi} = g_i B_{if} = g_i A_{if} \frac{\pi^2 c^3}{\hbar\omega^3}. \quad (44.13)$$

§ 45. Dipole radiation

We apply the formulas obtained above to photon emission by a relativistic electron in a prescribed external field. In this case, the transition current is the matrix element of the operator

$$\hat{j} = \hat{\psi} \gamma \hat{\psi},$$

in which the ψ operators are understood to be expanded in a system of wave functions for the stationary electron states in the given field (see § 32). The matrix element $\langle 0_i 1_f | \hat{j} | 1_i 0_f \rangle$ corresponds to the electron transition from state i to state f . This change in the occupation numbers is effected by the operator $\hat{a}_f^\dagger \hat{a}_i$, and the transition current is therefore

$$j_{fi}^\mu = \bar{\psi}_f \gamma^\mu \psi_i = (\psi_f^* \psi_i, \psi_f^* \alpha \psi_i), \quad (45.1)$$

where ψ_i and ψ_f are the wave functions of the initial and final electron states.

Choose the photon wave function in the three-dimensionally transverse gauge (the polarization 4-vector is $e = (0, \mathbf{e})$). The product in (43.10) is then $j_{fi} e^* = -\mathbf{j}_{fi} \mathbf{e}^*$. Substituting V_{fi} in (44.4), we obtain the following formula for the emission probability (per second) into the solid-angle element do for a photon with polarization $\mathbf{e}$:

$$dw_{\text{en}} = e^2 \frac{\omega}{2\pi} |\mathbf{e}^* \mathbf{j}_{fi}(\mathbf{k})|^2 do, \quad (45.2)$$

where

$$\mathbf{j}_{fi}(\mathbf{k}) = \int \psi_f^* \alpha \psi_i e^{-i\mathbf{k}\mathbf{r}} d^3x. \quad (45.3)$$

Summation over the photon polarizations is carried out by averaging over the directions of $\mathbf{e}$ (in the plane perpendicular to the specified direction $\mathbf{n} = \mathbf{k}/\omega$), and then multiplying the result by 2 for the two independent possibilities for transverse photon polarization⁵. This gives

$$dw_{\text{n}} = e^2 \frac{\omega}{2\pi} |\mathbf{n} \times \mathbf{j}_{fi}(\mathbf{k})|^2 do. \quad (45.4)$$

Of particular importance is the case in which the photon wavelength λ is large compared with the dimensions a of the radiating system. This situation is usually connected with particle velocities small compared with the speed of light. To the first approximation in a/λ (corresponding to dipole radiation; cf. II, § 67), the factor $e^{-i\mathbf{k}\mathbf{r}}$ in the transition current (45.3), which varies little in the region where either ψ_i or ψ_f differs appreciably from zero, may be replaced by unity. In other words, this replacement means neglecting the photon momentum in comparison with the particle momenta in the system.

In the same approximation, the integral $\mathbf{j}_{fi}(0)$ may be replaced by its nonrelativistic expression, namely the matrix element $\mathbf{v}_{fi}$ of the electron velocity evaluated with Schrödinger wave functions. In turn, this element is $\mathbf{v}_{fi} = -i\omega \mathbf{r}_{fi}$, while $e \mathbf{r}_{fi} = \mathbf{d}_{fi}$, where $\mathbf{d}$ is the dipole moment of the electron (in its orbital motion). We consequently obtain the following formula for the dipole-radiation probability:

$$dw_{\text{en}} = \frac{\omega^3}{2\pi} |\mathbf{e}^* \mathbf{d}_{fi}|^2 do \quad (45.5)$$

⁵The averaging uses the formula

$$\overline{e_i e_k^*} = \frac{1}{2} (\delta_{ik} - n_i n_k) \quad (45.4a)$$

or

$$\overline{(\mathbf{a}\mathbf{e})(\mathbf{b}\mathbf{e}^*)} = \frac{1}{2} \{ \mathbf{a}\mathbf{b} - (\mathbf{a}\mathbf{n})(\mathbf{b}\mathbf{n}) \} = \frac{1}{2} (\mathbf{a} \times \mathbf{n})(\mathbf{b} \times \mathbf{n}), \quad (45.4b)$$

where $\mathbf{a}$ and $\mathbf{b}$ are constant vectors.

(the direction $\mathbf{n}$ enters here implicitly: the vector $\mathbf{e}$ must be perpendicular to $\mathbf{n}$). Summing over polarizations gives

$$dw_{\mathbf{n}} = \frac{\omega^3}{2\pi} |\mathbf{n} \times \mathbf{d}_{fi}|^2 d\Omega. \quad (45.6)$$

Because these formulas are nonrelativistic with respect to the electron, their generalization to arbitrary electron systems is immediate: $\mathbf{d}_{fi}$ is to mean the matrix element of the total dipole moment of the system.

Integrating formula (45.6) over all directions, we find the total emission probability:

$$w = \frac{4\omega^3}{3} |\mathbf{d}_{fi}|^2, \quad (45.7)$$

or, in ordinary units,

$$w = \frac{4\omega^3}{3\hbar c^3} |\mathbf{d}_{fi}|^2. \quad (45.7a)$$

The radiation intensity I is obtained by multiplying the probability by $\hbar\omega$:

$$I = \frac{4\omega^4}{3c^3} |\mathbf{d}_{fi}|^2. \quad (45.8)$$

This formula displays a direct analogy with the classical formula (see II, (67.11)) for the intensity of dipole radiation from a system of periodically moving particles: the radiation intensity at frequency $\omega_s = s\omega$ (where ω is the frequency of the particle motion and s is an integer) is

$$I_s = \frac{4\omega_s^4}{3c^3} |\mathbf{d}_s|^2, \quad (45.9)$$

where $\mathbf{d}_s$ are the Fourier components of the dipole moment, that is, the coefficients in the expansion

$$\mathbf{d}(t) = \sum_{s=-\infty}^{\infty} \mathbf{d}_s e^{-is\omega t}. \quad (45.10)$$

The quantum formula (45.8) follows from (45.9) when these Fourier components are replaced by the matrix elements for the corresponding transitions. This rule, which expresses Bohr's *correspondence principle*, is a special case of the general correspondence between Fourier components of classical quantities and quantum matrix elements in the semiclassical case (see III, § 48). Radiation is semiclassical for transitions between states having large quantum numbers; the transition frequency $\hbar\omega = E_i - E_f$ is then small compared with the energies E_i and E_f of the radiating system. This circumstance would not, however, change the form of (45.8), which is valid for arbitrary transitions. It explains the fact—accidental in a certain sense—that the correspondence principle for the radiation intensity is valid not only in the semiclassical case but also in the general quantum case.

§ 46. Electric multipole radiation

Rather than considering the emission of a photon in a prescribed direction (that is, with prescribed momentum), we now consider emission of a photon having specified values

of the angular momentum j and of its projection m on a chosen direction z . As shown in § 6, such photons can be of two kinds, electric and magnetic; we begin with the emission of electric-type photons. The dimensions of the emitting system will again be taken as small compared with the wavelength.

It is convenient to carry out the calculation using photon wave functions in the momentum representation, representing the 4-vector $A^\mu(\mathbf{r})$ by a Fourier integral. The matrix element is then

$$V_{fi} = e \int j_{fi}^\mu(\mathbf{r}) A_\mu^*(\mathbf{r}) d^3x = e \int d^3x \cdot j_{fi}^\mu(\mathbf{r}) \int \frac{d^3k}{(2\pi)^3} A_\mu^*(\mathbf{k}) e^{-i\mathbf{k}\mathbf{r}} \quad (46.1)$$

(to simplify the notation, we omit the indices ωjm on the photon wave functions).

For an Ej photon, take the wave function from (7.10), choosing the arbitrary constant C as

$$C = -\sqrt{\frac{j+1}{j}}.$$

The purpose of this choice is to cancel from the spatial components of the wave function ($\mathbf{A}$) the terms containing spherical functions of order $j-1$ (as follows from (7.16)). The remaining terms in $\mathbf{A}$ then contain only spherical functions of order $j+1$. Their corresponding contribution to V_{fi} is of higher order in a/λ (as the calculation below will make clear) than the contribution from the component $A^0 \equiv \Phi$, which contains spherical functions of the lower order j .

We therefore set

$$A^\mu = (\Phi, \mathbf{0}), \quad \Phi = -\sqrt{\frac{j+1}{j}} \frac{4\pi^2}{\omega^{3/2}} \delta(|\mathbf{k}| - \omega) Y_{jm}(\mathbf{n})$$

($\mathbf{n} = \mathbf{k}/\omega$). Substitution in (46.1), followed by integration over $|\mathbf{k}|$, gives

$$V_{fi} = -e \sqrt{\frac{j+1}{j}} \frac{\sqrt{\omega}}{2\pi} \int d^3x \cdot \rho_{fi}(\mathbf{r}) \int d\mathbf{n} e^{-i\mathbf{k}\mathbf{r}} Y_{jm}^*(\mathbf{n}). \quad (46.2)$$

To calculate the inner integral, write the expansion (24.12) as

$$e^{i\mathbf{p}\mathbf{r}} = 4\pi \sum_{l=0}^{\infty} \sum_{m=-l}^l i^l g_l(kr) Y_{lm}^*\left(\frac{\mathbf{k}}{k}\right) Y_{lm}\left(\frac{\mathbf{r}}{r}\right), \quad (46.3)$$

where

$$g_l(kr) = \sqrt{\pi/(2kr)} J_{l+1/2}(kr) \quad (46.4)$$

(see III, (34.3))⁶. Substitution of this expansion in (46.2) gives

$$\int e^{-i\mathbf{k}\mathbf{r}} Y_{jm}^*(\mathbf{n}) d\mathbf{n} = 4\pi i^{-j} g_j(kr) Y_{jm}^*\left(\frac{\mathbf{r}}{r}\right)$$

⁶The normalization of the functions g_l is such that their asymptotic form for $kr \rightarrow \infty$ is

$$g_l(kr) \approx \frac{\sin(kr - \pi l/2)}{kr}. \quad (46.4a)$$

(all other terms vanish by orthogonality of the spherical functions). Since $a/\lambda \ll 1$, only distances satisfying $kr \ll 1$ contribute to the integral over d^3x . The functions $g_j(kr)$ may consequently be replaced by the first terms of their expansions in powers of kr ⁷:

$$g_j(kr) \approx \frac{(kr)^j}{(2j+1)!!}. \quad (46.5)$$

The result is

$$V_{fi} = (-1)^{m+1} i^j \sqrt{\frac{(2j+1)(j+1)}{\pi j}} \frac{\omega^{j+1/2}}{(2j+1)!!} e(Q_{j,-m}^{(E)})_{fi}, \quad (46.6)$$

where we have introduced

$$(Q_{jm}^{(E)})_{fi} = \sqrt{\frac{4\pi}{2j+1}} \int \rho_{fi}(\mathbf{r}) r^j Y_{jm}\left(\frac{\mathbf{r}}{r}\right) d^3x \quad (46.7)$$

(recall that $Y_{j,-m} = (-1)^{j-m} Y_{jm}^*$). By analogy with the corresponding classical quantities (II, § 41), the quantities (46.7) are called the 2^j -pole electric transition moments of the system⁸.

For an electron in an external field, $\rho_{fi} = \psi_f^* \psi_i$, so that (46.7) is calculated as a matrix element of the classical quantity

$$Q_{jm}^{(E)} = \sqrt{\frac{4\pi}{2j+1}} r^j Y_{jm}.$$

In the nonrelativistic case (with respect to the particle velocities), the transition moment can in principle be calculated in the same way for any system of N interacting particles. The transition density is then expressed through the system wave functions as

$$\rho_{fi}(\mathbf{r}) = \int \psi_f^*(\mathbf{r}_1, \dots, \mathbf{r}_N) \psi_i(\mathbf{r}_1, \dots, \mathbf{r}_N) \sum_{n=1}^N \delta(\mathbf{r} - \mathbf{r}_n) \cdot d^3x_1 \dots d^3x_N, \quad (46.8)$$

where the integration extends over the entire configuration space⁹.

The photon wave function used here is normalized in the coordinate representation to a delta function on the ω scale, as assumed in (44.2). Substitution of (46.6) into that formula gives the probability of Ej radiation¹⁰

$$w_{jm}^{(E)} = \frac{2(2j+1)(j+1)}{j[(2j+1)!!]^2} \omega^{2j+1} e^2 |(Q_{j,-m}^{(E)})_{fi}|^2. \quad (46.9)$$

⁷The power of kr is equal to the order of the function Y_{jm} that accompanies g_j . This also justifies neglecting the terms in $\mathbf{A}$ that contain spherical functions of higher order.

⁸We define the multipole moments without a factor e , consistently with the definition of the currents in this book without a charge factor.

⁹The transition probability may vanish because of approximate selection rules that hold only when electron spin-orbit interaction is neglected. A nonzero result must then be obtained with wave functions containing the relativistic correction that accounts for this interaction.

¹⁰At first sight, spatial isotropy might seem to imply that the total probability of photon emission cannot depend on m . This is not so because, for a fixed initial state of the system, the final states must differ when photons with different values of m are emitted; compare the rule (46.16) below.

For $j = 1$, in particular,

$$w_{1m}^{(E)} = \frac{4\omega^3}{3} e^2 |(Q_{1,-m}^{(E)})_{fi}|^2. \quad (46.10)$$

The quantities $Q_{1m}^{(E)}$ are related to the components of the electric dipole-moment vector by

$$eQ_{10}^{(E)} = id_z, \quad eQ_{1\pm 1}^{(E)} = \mp \frac{i}{\sqrt{2}} (d_x \pm id_y). \quad (46.11)$$

Summing (46.10) over m returns, as it should, the previously obtained formula (45.7) for the total probability of dipole radiation.

The angular distribution of multipole radiation is determined by (7.11). Normalizing it to the total emission probability w_{jm} gives

$$dw_{jm} = |\mathbf{Y}_{jm}^{(E)}(\mathbf{n})|^2 w_{jm} d\Omega = \frac{w_{jm}}{j(j+1)} |\nabla_{\mathbf{n}} Y_{jm}|^2 d\Omega. \quad (46.12)$$

For $j = 1$, in particular,

$$Y_{10} = i\sqrt{\frac{3}{4\pi}} \cos \theta, \quad Y_{1,\pm 1} = \mp i\sqrt{\frac{3}{8\pi}} \sin \theta \cdot e^{\pm i\varphi},$$

where θ, φ are the polar and azimuthal angles of the direction $\mathbf{n}$ relative to the z axis. Evaluating the gradient, we find that the angular distributions of dipole radiation for specified values of m are

$$dw_{10} = w_{10} \frac{3}{8\pi} \sin^2 \theta d\Omega, \quad dw_{1,\pm 1} = w_{1,\pm 1} \frac{3}{8\pi} \frac{1 + \cos^2 \theta}{2} d\Omega. \quad (46.13)$$

These expressions could also have been obtained directly from (45.6), first by setting, for $m = 0$, $d_x = d_y = 0$, $d_z = d$, and then, for $m = \pm 1$, $d_y = \mp id_x = d/\sqrt{2}$, $d_z = 0$.

If the characteristic dimension of the system (atom or nucleus) is a , then the electric multipole moments are in general of order $Q_{jm}^E \sim a^j$. The multipole-radiation probability is of order

$$w_{jm}^E \sim \alpha k (ka)^{2j}. \quad (46.14)$$

Increasing the multipole order by one reduces the emission probability by a factor of order $(ka)^2$.

Conservation of angular momentum and parity gives selection rules that restrict the possible changes of state of the emitting system. If the initial angular momentum of the system is J_i , then after emission of a photon with angular momentum j , the system angular momentum can take only those values J_f allowed by the angular-momentum addition rule ($\mathbf{J}_i - \mathbf{J}_f = \mathbf{j}$):

$$|J_i - J_f| \leq j \leq J_i + J_f. \quad (46.15)$$

For specified J_i and J_f , the same rule (46.15) determines the possible photon angular momenta j . Because the emission probability falls rapidly as j increases, radiation occurs mainly with the lowest allowed multipolarity.

The projections M_i and M_f of $\mathbf{J}_i$ and $\mathbf{J}_f$, together with the projection m of the photon angular momentum, satisfy the immediate consequence of the same addition rule

$$M_i - M_f = m. \quad (46.16)$$

The parities P_i and P_f of the initial and final states of the emitting system must satisfy $P_f P_\gamma = P_i$, where P_γ is the parity of the emitted photon. Since a parity can only be ± 1 , this condition may also be written

$$P_i P_f = P_\gamma. \quad (46.17)$$

For an electric-type photon, $P_\gamma = (-1)^j$; thus the parity selection rule for electric multipole radiation is

$$P_i P_f = (-1)^j. \quad (46.18)$$

The selection rules for total angular momentum and parity are exact and must hold for radiation from any system. Other, more restrictive rules may coexist with them, depending on particular structural features of the emitting systems. Such rules are necessarily only approximate to some degree; they will be considered in later sections of this chapter.

The dependence of the emission probability on the quantum numbers m , M_i , and M_f is fixed entirely by the tensor character of the multipole moments. For a given j , the quantities Q_{jm} form a spherical tensor of rank j . The dependence of its matrix elements on these quantum numbers is

$$|\langle n_f J_f M_f | Q_{j,-m} | n_i J_i M_i \rangle|^2 = \begin{pmatrix} J_f & j & J_i \\ M_f & m & -M_i \end{pmatrix}^2 |\langle n_f J_f || Q_j || n_i J_i \rangle|^2 \quad (46.19)$$

(see III, (107.6)), where n denotes symbolically the collection of the remaining quantum numbers of the system state, apart from J and M . The reduced matrix elements on the right-hand side of (46.19) are independent of m , M_i , and M_f . Substitution of this formula in (46.9) determines the required dependence, which is proportional to

$$\begin{pmatrix} J_f & j & J_i \\ M_f & m & -M_i \end{pmatrix}^2$$

(the emitter is, of course, assumed not to be in an external field, so the transition frequency ω is independent of M_i and M_f).

Summing the probability over all values of M_f for a fixed M_i gives the total probability of emission of a photon of the specified frequency from the system's initial level $n_i J_i$. By spatial isotropy, this quantity is clearly also independent of the initial value M_i . The sum is evaluated with

$$\sum_{M_f} |\langle n_f J_f M_f | Q_{j,-m} | n_i J_i M_i \rangle|^2 = \frac{1}{2J_i + 1} |\langle n_f J_f || Q_j || n_i J_i \rangle|^2 \quad (46.20)$$

(see III, (107.11)).

§ 47. Magnetic multipole radiation

The wave function of a magnetic-type photon is $A^\mu = (0, \mathbf{A})$, with $\mathbf{A}$ given by (7.6). Substitution in (46.1) gives the transition matrix element

$$V_{fi} = -e \frac{\sqrt{\omega}}{2\pi} \int d^3x \cdot \mathbf{j}_{fi}(\mathbf{r}) \int d\mathbf{o}_{\mathbf{n}} \cdot e^{-i\mathbf{k}\mathbf{r}} \mathbf{Y}_{jm}^{(M)*}(\mathbf{n}). \quad (47.1)$$

According to (7.16), the components of $\mathbf{Y}_{jm}^{(M)}$ are expressed through spherical functions of order j . Using the expansion (46.3) once more, the inner integral becomes

$$\int e^{-i\mathbf{k}\mathbf{r}} \mathbf{Y}_{jm}^{(M)*}(\mathbf{n}) d\mathbf{o}_{\mathbf{n}} = 4\pi i^{-j} g_j(kr) \mathbf{Y}_{jm}^{(M)*}\left(\frac{\mathbf{r}}{r}\right),$$

and substitution of g_j from (46.5) gives¹¹

$$V_{fi} = -ei^{-j} \frac{2\omega^{j+1/2}}{(2j+1)!!} \int \mathbf{j}_{fi}(\mathbf{r}) r^j \mathbf{Y}_{jm}^{(M)*}\left(\frac{\mathbf{r}}{r}\right) d^3x.$$

The definition (7.4) requires the substitution

$$\mathbf{Y}_{jm}^{(M)}\left(\frac{\mathbf{r}}{r}\right) = \frac{1}{\sqrt{j(j+1)}} \mathbf{r} \times \nabla Y_{jm}.$$

The integrand may then be transformed as

$$r^j \mathbf{j}_{fi} \cdot (\mathbf{r} \times \nabla Y_{jm}^*) = -(\mathbf{r} \times \mathbf{j}_{fi}) \cdot \nabla (r^j Y_{jm}^*),$$

which yields

$$V_{fi} = (-1)^m i^j \sqrt{\frac{(2j+1)(j+1)}{\pi j}} \frac{\omega^{j+1/2}}{(2j+1)!!} e(Q_{j,-m}^{(M)})_{fi}, \quad (47.2)$$

where

$$(Q_{jm}^{(M)})_{fi} = \frac{1}{j+1} \sqrt{\frac{4\pi}{2j+1}} \int (\mathbf{r} \times \mathbf{j}_{fi}) \cdot \nabla (r^j Y_{jm}) d^3x. \quad (47.3)$$

These quantities are called the 2^j -pole magnetic transition moments.

Because (47.2) and (46.6) have analogous forms, the emission probability is given by a formula differing from (46.9) only in the replacement of the electric moments by magnetic ones. Formula (46.12) for the angular distribution likewise remains valid, as was already noted in connection with (7.11).

Consider the structure of (47.3) for $j = 1$. In this case,

$$\sqrt{\frac{4\pi}{3}} r Y_{10} = iz, \quad \sqrt{\frac{4\pi}{3}} r Y_{1,\pm 1} = \mp \frac{i}{\sqrt{2}} (x \pm iy),$$

¹¹Do not confuse the current $\mathbf{j}$ with the angular momentum j .

and their gradients are simply the circular unit vectors $\mathbf{e}^{(0)}$, $\mathbf{e}^{(\pm 1)}$ of (7.14). Consequently, the quantities $e(Q_{lm}^{(M)})_{fi}$ are the spherical components of the vector

$$\boldsymbol{\mu}_{fi} = \frac{e}{2} \int \mathbf{r} \times \mathbf{j}_{fi} d^3x, \quad (47.4)$$

whose structure is analogous to that of the classical magnetic moment (see II, § 44). In conventional units, the total probability of $M1$ radiation is expressed through this quantity as

$$w = \frac{4\omega^3}{3\hbar c^3} |\boldsymbol{\mu}_{fi}|^2. \quad (47.5)$$

We now show how (47.4) is related to the usual nonrelativistic quantum expression for the magnetic-moment operator.

The transition current is (see III, § 115)

$$\mathbf{j}_{fi} = -\frac{i}{2m} (\psi_f^* \nabla \psi_i - \psi_i \nabla \psi_f^*) + \frac{\mu}{es} \nabla \times \psi_f^* \mathbf{s} \psi_i, \quad (47.6)$$

where μ is the magnetic moment of the particle and s is its spin. Hence

$$\begin{aligned} \boldsymbol{\mu}_{fi} = & -\frac{ie}{4m} \int \psi_f^* (\mathbf{r} \times \nabla) \psi_i d^3x + \frac{ie}{4m} \int \psi_i (\mathbf{r} \times \nabla) \psi_f^* d^3x + \\ & + \frac{\mu}{2s} \int \mathbf{r} \times \nabla \times \psi_f^* \mathbf{s} \psi_i d^3x. \end{aligned} \quad (47.7)$$

In the second term, write

$$\int \psi_i (\mathbf{r} \times \nabla) \psi_f^* d^3x = - \int \psi_f^* (\mathbf{r} \times \nabla) \psi_i d^3x + \int \nabla \times \mathbf{r} \psi_f^* \psi_i d^3x.$$

The last integral becomes an integral over a surface at infinity and vanishes. The first two terms in (47.7) are therefore equal. In the third term, transform the integral as follows (temporarily writing $\mathbf{F} = \psi_f^* \mathbf{s} \psi_i$):

$$\int \mathbf{r} \times (\nabla \times \mathbf{F}) d^3x = \oint \mathbf{r} \times (d\mathbf{f} \cdot \mathbf{F}) - \int (\mathbf{F} \times \nabla) \times \mathbf{r} d^3x.$$

The surface integral vanishes, while in the last integral $((\mathbf{F} \times \nabla) \times \mathbf{r}) = -\mathbf{F} \nabla \cdot \mathbf{r} + \mathbf{F} = -2\mathbf{F}$. Thus

$$\int \mathbf{r} \times \nabla \times \mathbf{F} d^3x = 2 \int \mathbf{F} d^3x.$$

The resulting expression for $\boldsymbol{\mu}_{fi}$ is

$$\boldsymbol{\mu}_{fi} = \int \psi_f^* \left(\frac{e}{2m} \widehat{\mathbf{L}} + \frac{\mu}{s} \widehat{\mathbf{s}} \right) \psi_i d^3x, \quad (47.8)$$

where $\widehat{\mathbf{L}} = -i\mathbf{r} \times \nabla$ is the orbital-angular-momentum operator of the particle. As required, $\boldsymbol{\mu}_{fi}$ is the matrix element of the operator

$$\widehat{\boldsymbol{\mu}} = \frac{e}{2m} \widehat{\mathbf{L}} + \frac{\mu}{s} \widehat{\mathbf{s}}, \quad (47.9)$$

which is the sum of the particle's orbital and intrinsic magnetic-moment operators.

The selection rules for magnetic multipole radiation are analogous to those for the electric case: the same rules (46.15), (46.16) hold for the total angular momentum, while parity obeys

$$P_i P_f = (-1)^{j+1}, \quad (47.10)$$

as follows by substituting the parity of an Mj photon, $P_\gamma = (-1)^{j+1}$, in (46.17).

§ 48. Angular distribution and polarization of radiation

The formulas derived in § 46 and 47 concerned emission of a photon with specified values of the angular momentum j and its projection m . Correspondingly, the emitting system (a nucleus, for example) was assumed, both before and after emission, to possess not only a definite angular momentum J but also a definite polarization, that is, a definite value of M .

We now consider the more general case in which a partially polarized nucleus emits radiation (its dimensions are still assumed small compared with the wavelength). The emitted photon again has a definite angular momentum j , but it may be only partially polarized. We shall determine the emission probability as a function of the photon direction $\mathbf{n}$. It must be expressed through density matrices describing the polarization states of the nucleus and photon.

As a preliminary step, consider the emission probability as a function of the direction $\mathbf{n}$ and photon helicity λ ($\lambda = \pm 1$), when the initial and final nuclei have definite values $J_i M_i$ and $J_f M_f$.

The matrix element for emission of a photon with specified jm is proportional to the matrix element of the 2^j -pole (electric or magnetic) moment of the nucleus:

$$\langle J_f M_f; jm | V | J_i M_i \rangle \propto (-1)^m \langle J_f M_f | Q_{j,-m} | J_i M_i \rangle. \quad (48.1)$$

The wave function of the emitted photon in momentum representation is proportional to $\mathbf{Y}_{jm}^{(E)}(\mathbf{n})$ or $\mathbf{Y}_{jm}^{(M)}(\mathbf{n})$. A photon whose momentum points along $\mathbf{n}$ and whose helicity is λ has a wave function proportional to the polarization vector $\mathbf{e}^{(\lambda)}$. The matrix element for emission of a photon $\mathbf{n}\lambda$ is obtained by multiplying (48.1) by the projection of the wave function of the state $|jm\rangle$ on that of the state $|\mathbf{n}\lambda\rangle$:

$$\langle J_f M_f; \mathbf{n}\lambda | V | J_i M_i \rangle \propto (-1)^m \langle J_f M_f | Q_{j,-m} | J_i M_i \rangle (\mathbf{e}^{(\lambda)*} \mathbf{Y}_{jm}).$$

According to (16.23), for photons of either type,

$$\mathbf{e}^{(\lambda)*} \mathbf{Y}_{jm}(\mathbf{n}) \propto D_{\lambda m}^{(j)}(\mathbf{n}). \quad (48.2)$$

We express the multipole-moment matrix element in the usual way through its reduced element. The resulting transition amplitude is

$$\langle J_f M_f; \mathbf{n}\lambda | V | J_i M_i \rangle \propto (-1)^{J_f - M_f + m} \begin{pmatrix} J_f & j & J_i \\ -M_f & -m & M_i \end{pmatrix} Q D_{\lambda m}^{(j)}(\mathbf{n}), \quad (48.3)$$

where Q denotes $\langle J_f || Q || J_i \rangle$.

We may now pass to the general case of mixed polarization states. By the general rules of quantum mechanics, the transition probability is proportional to the expression¹²

$$\sum_{(m)} \langle J_f M_f; \mathbf{n} \lambda | V | J_i M_i \rangle \langle J_f M'_f; \mathbf{n} \lambda' | V | J_i M'_i \rangle^* \quad (48.4)$$

$$\times \langle M_i | \rho^{(i)} | M'_i \rangle \langle M'_f | \rho^{(f)} | M_f \rangle \langle \lambda' | \rho^{(\gamma)} | \lambda \rangle,$$

where $\rho^{(i)}$, $\rho^{(f)}$, and $\rho^{(\gamma)}$ are the density matrices of the initial nucleus, final nucleus, and emitted photon. The symbol (m) under the summation sign means that the sum is taken over every twice-repeated m -type index $(M_i M'_i M_f M'_f \lambda \lambda')$. Equation (48.3) is to be substituted in (48.4).

Let $w(\mathbf{n})do$ denote the probability of emission into the solid-angle element do . The total probability, summed over all directions and all polarizations of the photon and secondary nucleus, is plainly independent of the initial polarization state of the nucleus. It is given by formulas already obtained and is not of interest here. We shall therefore normalize $w(\mathbf{n})$ to unity. This gives¹³

$$w(\mathbf{n}) = \frac{(2j+1)(2J_i+1)}{8\pi} \sum_{(m)} (-1)^{2j-M_i-M'_i} D_{\lambda m}^{(j)} D_{\lambda' m'}^{(j)*}$$

$$\times \begin{pmatrix} J_f & j & J_i \\ -M_f & -m & M_i \end{pmatrix} \begin{pmatrix} J_f & j & J_i \\ -M'_f & -m' & M'_i \end{pmatrix} \langle M_i | \rho^{(i)} | M'_i \rangle$$

$$\times \langle M'_f | \rho^{(f)} | M_f \rangle \langle \lambda' | \rho^{(\gamma)} | \lambda \rangle.$$

(the normalization will be verified below). We transform this formula by expanding the product of two D -functions in the series (110.2) (see III):

$$D_{\lambda m}^{(j)} D_{\lambda' m'}^{(j)*} = (-1)^{\lambda'+m'} D_{\lambda m}^{(j)} D_{-\lambda', -m'}^{(j)}$$

$$= (-1)^{\lambda+m} \sum_L (2L+1) \begin{pmatrix} j & j & L \\ \lambda & -\lambda' & -\lambda \end{pmatrix} \begin{pmatrix} j & j & L \\ m & -m' & -\mu \end{pmatrix} D_{\Lambda \mu}^{(L)}$$

¹²If the initial and final states of the system are described by the superpositions

$$\psi^{(i)} = \sum_n a_n \psi_n^{(i)}, \quad \psi^{(f)} = \sum_m b_m \psi_m^{(f)},$$

then the matrix element is

$$\langle f | V | i \rangle = \sum_{mn} b_m^* a_n V_{mn},$$

and its square is

$$|\langle f | V | i \rangle|^2 = \sum_{nn' mm'} V_{mn} V_{m'n'}^* a_n a_n^* b_m^* b_{m'}.$$

The mixed-state case is obtained by making the replacements

$$a_n a_n^* \rightarrow \rho_{nn'}^{(i)}, \quad b_m^* b_{m'} \rightarrow \rho_{m'm}^{(f)},$$

so that

$$|\langle f | V | i \rangle|^2 \rightarrow \sum_{nn' mm'} V_{mn} V_{m'n'}^* \rho_{nn'}^{(i)} \rho_{m'm}^{(f)}.$$

¹³When transforming the sign factor, one may use the fact that $2J_i$, $2J_f$, $2M_i$, and $2M_f$ have the same parity. Recall also that j and m are integers, while $\lambda = \pm 1$.

(the indices are $\Lambda = \lambda - \lambda'$ and $\mu = m - m'$; L is integral and $L \leq 2j$). We thus finally obtain

$$\begin{aligned}
 w(\mathbf{n}) = & \frac{(2j+1)(2J_i+1)}{8\pi} \sum_L \sum_{(m)} (-1)^{2J_i-M_i-M'_i+m+1} (2L+1) \\
 & \times \begin{pmatrix} j & j & L \\ \lambda & -\lambda' & -\Lambda \end{pmatrix} \begin{pmatrix} j & j & L \\ m & -m' & -\mu \end{pmatrix} \begin{pmatrix} J_f & j & J_i \\ -M_f & -m & M_i \end{pmatrix} \\
 & \times \begin{pmatrix} J_f & j & J_i \\ -M'_f & -m' & M'_i \end{pmatrix} D_{\Lambda\mu}^{(L)}(\mathbf{n}) \langle M_i | \rho^{(i)} | M'_i \rangle \\
 & \times \langle M'_f | \rho^{(f)} | M_f \rangle \langle \lambda' | \rho^{(\gamma)} | \lambda \rangle.
 \end{aligned} \tag{48.5}$$

As above, $\sum_{(m)}$ denotes summation over all twice-repeated m -type indices. The indices λ, λ' differ from the other m -type indices: the sum over them is not over all $2j+1$ values possible for a given j index, but only over the two values $\lambda, \lambda' = \pm 1$, corresponding to the two photon polarizations.

Formula (48.5) contains all the information needed about the angular distribution and polarization of the emitted photons, and also about the polarization of the secondary nuclei (those which have emitted a photon). The initial density matrix is understood to be specified.

Angular distribution.

The angular distribution of the photons is obtained by summing over all polarizations of the photon and secondary nucleus. Averaging over polarizations amounts to substituting the density matrices of unpolarized states:

$$\langle \lambda | \rho^{(\gamma)} | \lambda' \rangle = \frac{1}{2} \delta_{\lambda\lambda'}, \quad \langle M_f | \rho^{(f)} | M'_f \rangle = \frac{1}{2J_f+1} \delta_{M_f M'_f}, \tag{48.6}$$

after which summation reduces to multiplication by 2 for the photon or by $2J_f+1$ for the nucleus. Equivalently, summation is effected simply by the replacements

$$\langle \lambda | \rho^{(\gamma)} | \lambda' \rangle \rightarrow \delta_{\lambda\lambda'}, \quad \langle M_f | \rho^{(f)} | M'_f \rangle \rightarrow \delta_{M_f M'_f}. \tag{48.7}$$

The angular distribution is therefore

$$\begin{aligned}
 \bar{w}(\mathbf{n}) = & \frac{(2j+1)(2J_i+1)}{8\pi} \sum_L \sum_{(m)} (-1)^{m'+1} (2L+1) D_{0\mu}^{(L)}(\mathbf{n}) \\
 & \times \begin{pmatrix} j & j & L \\ \lambda & -\lambda & 0 \end{pmatrix} \begin{pmatrix} j & j & L \\ m & -m' & -\mu \end{pmatrix} \begin{pmatrix} J_f & j & J_i \\ -M_f & -m & M_i \end{pmatrix} \\
 & \times \begin{pmatrix} J_f & j & J_i \\ -M_f & -m' & M'_i \end{pmatrix} \langle M_i | \rho^{(i)} | M'_i \rangle.
 \end{aligned}$$

This formula can be simplified substantially by carrying out the sums over the m -type indices.

First note that

$$\begin{pmatrix} j & j & L \\ \lambda & -\lambda & 0 \end{pmatrix} = (-1)^L \begin{pmatrix} j & j & L \\ -\lambda & \lambda & 0 \end{pmatrix}, \quad (48.8)$$

and hence

$$\sum_{\lambda=\pm 1} \begin{pmatrix} j & j & L \\ \lambda & -\lambda & 0 \end{pmatrix} = \begin{cases} 2 \begin{pmatrix} j & j & L \\ 1 & -1 & 0 \end{pmatrix}, & \text{for even } L, \\ 0, & \text{for odd } L. \end{cases}$$

Only terms with even L remain in the sum over L ; thus it contains spherical functions $D_{0\mu}^{(L)}$ only of even order. This result could have been anticipated: conservation of parity requires the probability to be invariant under inversion, that is, under $\mathbf{n} \rightarrow -\mathbf{n}$.

Consequently,

$$\begin{aligned} \bar{w}(\mathbf{n}) &= \frac{(2j+1)(2J_i+1)}{8\pi} \sum_L (2L+1) \begin{pmatrix} j & j & L \\ 1 & -1 & 0 \end{pmatrix} D_{0\mu}^{(L)}(\mathbf{n}) \\ &\times \sum_{(m)} (-1)^{m'+1} \begin{pmatrix} j & j & L \\ m & -m' & -\mu \end{pmatrix} \begin{pmatrix} J_f & j & J_i \\ -M_f & -m & M_i \end{pmatrix} \\ &\times \begin{pmatrix} J_f & j & J_i \\ -M_f & -m' & M'_i \end{pmatrix} \langle M_i | \rho^{(i)} | M'_i \rangle. \end{aligned}$$

The normalization can readily be checked here. From

$$\int D_{0\mu}^{(L)}(\mathbf{n}) \frac{d\Omega}{4\pi} = \delta_{L0} \delta_{\mu 0},$$

only the term $L = \mu = 0$ remains after integration over directions. Using

$$\begin{aligned} \begin{pmatrix} j & j & 0 \\ m & -m & 0 \end{pmatrix} &= (-1)^{j-m} \frac{1}{\sqrt{2j+1}}, \\ \sum_{M_f m} \begin{pmatrix} J_f & j & J_i \\ -M_f & -m & M_i \end{pmatrix}^2 &= \frac{1}{2J_i+1}, \quad \text{Tr } \rho^{(i)} = 1, \end{aligned}$$

we see that the integral is unity.

The remaining sum over $mm'M_f$ in the inner sum for $\bar{w}(\mathbf{n})$ is evaluated by means of formula (108.4) (see III). The following final expression for the photon angular distribution results:

$$\begin{aligned} \bar{w}(\mathbf{n}) &= (-1)^{1+J_i+J_f} \frac{(2j+1)\sqrt{2J_i+1}}{4\pi} \sum_{\text{even } L} (-1)^L \sqrt{2L+1} \\ &\times \begin{pmatrix} j & j & L \\ 1 & -1 & 0 \end{pmatrix} \begin{Bmatrix} J_i & J_i & L \\ j & j & J_f \end{Bmatrix} \sum_{\mu} \mathcal{P}_{L\mu}^{(i)*} D_{0\mu}^{(L)}(\mathbf{n}), \end{aligned} \quad (48.9)$$

where

$$\begin{aligned}\mathcal{P}_{L\mu}^{(i)} &= i^L \sqrt{(2L+1)(2J_i+1)} \sum_{M_i M_i'} (-1)^{J_i-M_i'} \\ &\quad \times \begin{pmatrix} J_i & L & J_i \\ -M_i' & \mu & M_i \end{pmatrix} \langle M_i | \rho^{(i)} | M_i' \rangle, \\ \mathcal{P}_{L\mu}^{(i)*} &= (-1)^{L-\mu} \mathcal{P}_{L,-\mu}^{(i)}.\end{aligned}\tag{48.10}$$

The inner sum in (48.9) extends over all $|\mu| \leq L$, and the outer sum over all even L satisfying

$$L \leq 2j, \quad L \leq 2J_i \tag{48.11}$$

(these conditions follow from the triangle rule obeyed by the j indices in the $3j$ symbols appearing in (48.9) and (48.10)). The number of terms is therefore usually small. For $J_i = 0$ or $1/2$, only the term $L = 0$ remains, so the radiation is isotropic (one readily verifies that this term is $1/4\pi$, as required by normalization). For $J_i = 1, 3/2$, or for $j = 1$, the sum over L has two terms, $L = 0, 2$. If the density matrix $\rho^{(i)}$ is diagonal ($M_i = M_i'$), then $\mu = 0$ and the distribution (48.9) becomes an expansion in Legendre polynomials (by (16.5) and (58.23) (see III), the functions $D_{00}^{(L)}$ reduce to $P_L(\cos \theta)$). Finally, if

$$\langle M_i | \rho^{(i)} | M_i' \rangle = \frac{1}{2J_i + 1} \delta_{M_i M_i'},$$

so that the initial nucleus is unpolarized, all $\mathcal{P}_{L\mu}^{(i)}$ vanish except $\mathcal{P}_{00}^{(i)} = 1$ ¹⁴.

The quantities $\mathcal{P}_{L\mu}$ provide convenient characteristics of the nuclear polarization state; we shall call them *polarization moments*. Formula (48.10) defines them in terms of the density matrix $\rho_{MM'}$. Direct substitution readily verifies the inverse formula expressing this matrix through the polarization moments:

$$\rho_{MM'} = \sum_{L\mu} \sqrt{\frac{2L+1}{2J+1}} i^{-L} (-1)^{J-M'} \begin{pmatrix} J & L & J \\ -M' & \mu & M \end{pmatrix} \mathcal{P}_{L\mu}. \tag{48.12}$$

Let $f_{L\mu}$ be a spherical tensor depending on the nuclear polarization state. By the general rules (see III, (14.8)), its mean value in a state with density matrix $\rho_{MM'}$ is

$$\bar{f}_{L\mu} = \sum_{MM'} \rho_{MM'} \langle JM' | f_{L\mu} | JM \rangle. \tag{48.13}$$

¹⁴Indeed, noting that

$$\begin{pmatrix} J & 0 & J \\ -M' & 0 & M \end{pmatrix} = (-1)^{J-M} \frac{1}{\sqrt{2J+1}} \delta_{MM'},$$

we have

$$\begin{aligned}\sum_{MM'} (-1)^{J-M'} \begin{pmatrix} J & L & J \\ -M' & \mu & M \end{pmatrix} \delta_{MM'} \\ = \sqrt{2J+1} \sum_{MM'} \begin{pmatrix} J & L & J \\ -M' & \mu & M \end{pmatrix} \begin{pmatrix} J & 0 & J \\ -M' & 0 & M \end{pmatrix} \\ = \sqrt{2J+1} \delta_{L0} \delta_{\mu 0},\end{aligned}$$

and the stated result follows from definition (48.10).

Expressing the matrix elements of $f_{L\mu}$ through the reduced element $\langle J \| f_L \| J \rangle$ according to

$$\langle JM' | f_{L\mu} | JM \rangle = i^L (-1)^{J-M'} \begin{pmatrix} J & L & J \\ -M' & \mu & M \end{pmatrix} \langle J \| f_L \| J \rangle$$

and introducing the polarization moments by definition (48.10), we obtain

$$\bar{f}_{L\mu} = \frac{\langle J \| f_L \| J \rangle}{\sqrt{(2L+1)(2J+1)}} \mathcal{P}_{L\mu}. \quad (48.14)$$

Photon polarization.

For specified matrices $\rho^{(\gamma)}$ and $\rho^{(f)}$, in addition to $\rho^{(i)}$, formula (48.5) determines the probability of a transition in which the emitted photon and nucleus occupy specified polarization states. These states characterize, in substance, not the radiation process itself but the detectors that register the photon and recoil nucleus while selecting particular polarizations. A more natural formulation leaves the final state of the “nucleus+photon” system unspecified and asks for the polarization density matrix of that state when only the photon emission direction is given.

The same formula (48.5) answers this question. Written as

$$w = \bar{w}(\mathbf{n}) \sum_{(m)} \langle M_f; \mathbf{n} \lambda | \rho | M'_f; \mathbf{n} \lambda' \rangle \langle \lambda' | \rho^{(\gamma)} | \lambda \rangle \langle M'_f | \rho^{(f)} | M_f \rangle, \quad (48.15)$$

the expression $\langle M_f; \mathbf{n} \lambda | \rho | M'_f; \mathbf{n} \lambda' \rangle$ is the required density matrix, since the general rules of quantum mechanics give the probability w for transition to a prescribed state as its “projection” on the specified $\rho^{(\gamma)} \rho^{(f)}$. The factor $\bar{w}(\mathbf{n})$ was extracted in (48.15) so that this matrix has the conventional normalization

$$\sum_{\lambda M_f} \langle M_f; \mathbf{n} \lambda | \rho | M_f; \mathbf{n} \lambda \rangle = 1.$$

If only the photon polarization is wanted, we must sum over $M_f = M'_f$:

$$\langle \mathbf{n} \lambda | \rho | \mathbf{n} \lambda' \rangle = \sum_{M_f} \langle M_f; \mathbf{n} \lambda | \rho | M_f; \mathbf{n} \lambda' \rangle.$$

A calculation entirely analogous to the derivation of (48.9) gives

$$\begin{aligned} \langle \mathbf{n} \lambda | \rho | \mathbf{n} \lambda' \rangle &= (-1)^{1+J_i+J_f} \frac{(2j+1)\sqrt{2J_i+1}}{8\pi\bar{w}(\mathbf{n})} \sum_L (-i)^L \sqrt{2L+1} \\ &\times \begin{pmatrix} j & j & L \\ \lambda & -\lambda' & -\Lambda \end{pmatrix} \begin{Bmatrix} J_i & J_i & L \\ j & j & J_f \end{Bmatrix} \\ &\times \sum_{\mu} \mathcal{P}_{L\mu}^{(i)*} D_{\Lambda\mu}^{(L)}(\mathbf{n}), \end{aligned} \quad (48.16)$$

where $\Lambda = \lambda - \lambda'$, and the sum is over every integral L satisfying (48.11).

In particular, circular polarization is determined by the Stokes parameter

$$\xi_2 = \langle \mathbf{n} | \rho | \mathbf{n} \rangle - \langle \mathbf{n}, -1 | \rho | \mathbf{n}, -1 \rangle$$

(see the problem to § 8). By (48.8), all terms with even L cancel in this difference, and the formula for ξ_2 differs from (48.9) only in having a sum over odd rather than even L .

Polarization of secondary nuclei.

Finally, if only the final nuclear polarization is required, we put $\rho^{(\gamma)} \rightarrow \delta$. If we also integrate over photon directions, the density matrix of the secondary nucleus is

$$\begin{aligned} \langle M_f | \rho | M'_f \rangle &= \int \bar{w}(\mathbf{n}) \langle M_f \mathbf{n} | \rho | M'_f \mathbf{n} \rangle d\mathbf{o} \\ &= (2J_i + 1) \sum_{m M_i M'_i} (-1)^{2J_i - M_i - M'_i} \begin{pmatrix} J_f & j & J_i \\ -M_f & -m & M_i \end{pmatrix} \\ &\quad \times \begin{pmatrix} J_f & j & J_i \\ -M'_f & -m & M'_i \end{pmatrix} \langle M_i | \rho^{(i)} | M'_i \rangle. \end{aligned}$$

The polarization moments calculated from this matrix are

$$\mathcal{P}_{L\mu}^{(f)} = (-1)^{J_i + J_f + L + j} \sqrt{(2J_i + 1)(2J_f + 1)} \begin{Bmatrix} J_i & J_i & L \\ J_f & J_f & j \end{Bmatrix} \mathcal{P}_{L\mu}^{(i)}. \quad (48.17)$$

If the initial nucleus is unpolarized, the final nucleus is likewise unpolarized. There is nevertheless a correlation polarization: the nucleus is polarized after emission in a specified direction. Setting $\rho^{(i)} \rightarrow \delta / (2J_i + 1)$ (and hence $\bar{w}(\mathbf{n}) = 1/(4\pi)$) and carrying out a calculation analogous to the derivation of (48.9), we obtain the density matrix describing this polarization:

$$\begin{aligned} \langle M_f; \mathbf{n} | \rho | M'_f; \mathbf{n} \rangle &= (2j + 1) (-1)^{J_i + M'_f + 1} \sum_{\text{even } L} (2L + 1) \begin{pmatrix} j & j & L \\ 1 & -1 & 0 \end{pmatrix} \\ &\quad \times \begin{pmatrix} J_f & L & J_f \\ -M'_f & \mu & M_f \end{pmatrix} \begin{Bmatrix} J_f & J_f & L \\ j & j & J_i \end{Bmatrix} D_{0\mu}^{(L)}(\mathbf{n}). \end{aligned} \quad (48.18)$$

The corresponding polarization moments are

$$\begin{aligned} \mathcal{P}_{L\mu}^{(f)} &= i^L (-1)^{1 + J_i + J_f} (2j + 1) \sqrt{(2L + 1)(2J_f + 1)} \\ &\quad \times \begin{pmatrix} j & j & L \\ 1 & -1 & 0 \end{pmatrix} \begin{Bmatrix} J_f & J_f & L \\ j & j & J_i \end{Bmatrix} D_{0\mu}^{(L)}(\mathbf{n}). \end{aligned} \quad (48.19)$$

Only moments of even order occur (another consequence of the parity conservation noted above).

If the secondary nucleus emits in turn, its polarization produces an anisotropic photon distribution. Since the polarization moments (48.19) depend on the direction

n of the photon emitted in the first decay, a definite correlation arises between the directions of photons emitted in succession when the primary nucleus is unpolarized. Other correlation phenomena in cascade emission, such as polarization correlations, may be treated similarly.¹⁵

PROBLEM

Relate the polarization moments $\mathcal{P}_{1\mu}$ and $\mathcal{P}_{2\mu}$ to the mean values of the angular-momentum vector **J** and the quadrupole-moment tensor Q_{ik} .

SOLUTION. The reduced elements of the vector **J** and tensor Q_{ik} are determined from

$$\overline{\mathbf{J}^2} = \frac{|\langle J \| \mathbf{J} \| J \rangle|^2}{2J+1}, \quad \overline{Q_{ik}^2} = \frac{|\langle J \| Q \| J \rangle|^2}{2J+1}$$

(compare III, (107.10), (107.11)). The operator $\widehat{Q}_{ik}$ is expressed in terms of the angular-momentum operators by formula (75.2) (see III):

$$\widehat{Q}_{ik} = \frac{3Q}{2J(2J-1)} \left(\widehat{J}_i \widehat{J}_k + \widehat{J}_k \widehat{J}_i - \frac{2}{3} \widehat{\mathbf{J}}^2 \delta_{ik} \right).$$

It follows that the mean value is

$$\overline{Q_{ik}^2} = \frac{3Q^2}{2J^2(2J-1)^2} \overline{\mathbf{J}^2} (4J^2 - 3) = Q^2 \frac{3(J+1)(2J+3)}{2J(2J-1)}.$$

The reduced matrix elements are

$$\begin{aligned} \langle J \| \mathbf{J} \| J \rangle &= \sqrt{J(J+1)(2J+1)}, \\ \langle J \| Q \| J \rangle &= Q \sqrt{\frac{3(2J+1)(J+1)(2J+3)}{2J(2J-1)}}. \end{aligned}$$

It is now evident from (48.14) that the polarization moments $\mathcal{P}_{1\mu}$ coincide with the spherical components of the vector

$$\sqrt{\frac{3}{J(J+1)}} \mathbf{J},$$

while the moments $\mathcal{P}_{2\mu}$ are the spherical components of the tensor

$$\sqrt{\frac{10J(2J-1)}{3(J+1)(2J+3)}} \frac{\widehat{Q}_{ik}}{Q}.$$

¹⁵A detailed account of these questions may be found in the article by A. Z. Dolginov in the book "Gamma Rays" (USSR Academy of Sciences Press, 1961).

§ 49. Radiation from atoms. Electric type¹⁶

The energies of an atom's outer electrons (those participating in optical radiative transitions) have the rough order of magnitude $E \sim me^4/\hbar^2$, and hence the emitted wavelengths are $\lambda \sim \hbar c/E \sim \hbar^2/(\alpha me^2)$. The dimensions of the atom are $a \sim \hbar^2/me^2$. Thus atomic optical spectra generally satisfy $a/\lambda \sim \alpha \ll 1$. The ratio $v/c \sim \alpha$, where v denotes the velocities of the optical electrons, has the same order of magnitude.

Consequently, the condition prevailing in atomic optical spectra makes the probability of electric-dipole radiation (when allowed by the selection rules) much greater than the probabilities of multipole transitions¹⁷. Electric-dipole transitions therefore have the central role in atomic spectroscopy.

As already stated, such transitions obey strict selection rules for the total atomic angular momentum J and the parity P ¹⁸:

$$|J' - J| \leq 1 \leq J + J', \quad (49.1)$$

$$PP' = -1. \quad (49.2)$$

The inequality $|J' - J| \leq 1$ means that J may change only by 0 or ± 1 ; the inequality $J + J' \geq 1$ additionally forbids the transition $0 \rightarrow 0$. The initial and final states must have opposite parities¹⁹.

For emission in a transition $nJM \rightarrow n'J'M'$, the probability is determined by the corresponding matrix element of the atomic dipole moment:

$$w(nJM \rightarrow n'J'M') = \frac{4\omega^3}{3\hbar c^3} |\langle n'J'M' | d_{-m} | nJM \rangle|^2, \quad (49.3)$$

$$\omega = \omega(nJ \rightarrow n'J').$$

Summing (49.3) over all values $M' = M - m$ (for a given M), we obtain the total emission probability at the specified frequency from the atomic level nJ . Using (46.20) to perform the summation gives

$$w(nJ \rightarrow n'J') = \frac{4\omega^3}{3\hbar c^3} \frac{1}{2J+1} |\langle n'J' || d || nJ \rangle|^2. \quad (49.4)$$

The squared modulus of the reduced matrix element appearing here is sometimes called the *line strength*; it is symmetric in the initial and final states.

The observed radiation intensity is found by multiplying w by $\hbar\omega$ and by the number N_{nJ} of source atoms on the specified excited level. For example, in a gas at temperature T , this number is $N_{nJ} \propto (2J+1) \exp(-E_{nJ}/T)$; the factor $2J+1$ is the statistical weight of the level with angular momentum J .

Further conclusions about transition probabilities in atomic spectra require the character of the atomic states to be specified more fully. We shall not discuss methods

¹⁶In § 49–51 and 53–55 we use ordinary units.

¹⁷Typical dipole-transition probabilities in the optical region of atomic spectra are of order 10^8 s^{-1} .

¹⁸From now on, quantum numbers of the initial and final states will be denoted respectively by unprimed and primed letters. The letters n and n' will denote the sets of the remaining quantum numbers, apart from those displayed explicitly, that specify the states of the system.

¹⁹The parity selection rule was established by Laporte (O. Laporte, 1924).

for calculating matrix elements here, since their degree of approximation has no sharply defined theoretical character. We derive only some relations for a fairly broad category of states (particularly in light atoms) constructed according to the LS -coupling scheme (see III, § 72). Besides total angular momentum, these states are characterized by definite values of the orbital angular momentum L and spin S , both conserved in this case.

Since the dipole moment is a purely orbital quantity, its operator commutes with the spin operator; its matrix is therefore diagonal in S . With respect to L , the dipole moment obeys the same selection rules as any orbital vector (see III, § 29). Hence transitions between states constructed in the LS scheme obey the additional selection rules, besides (49.1) and (49.2),

$$S' - S = 0, \quad (49.5)$$

$$|L' - L| \leq 1 \leq L + L'. \quad (49.6)$$

We emphasize again that these rules are approximate and are violated when spin-orbit interaction is taken into account.

The rule (49.5), which forbids transitions between terms of different multiplicity, is valid not only for dipole transitions but for all transitions of electric type: electric multipole moments of every order are orbital tensors, so their matrices are diagonal in spin. Thus electric-quadrupole transitions obey the general rules

$$|J' - J| \leq 2 \leq J + J', \quad PP' = 1, \quad (49.7)$$

and, for LS coupling, the additional selection rules

$$S' - S = 0, \quad |L' - L| \leq 2 \leq L + L'. \quad (49.8)$$

The dependence of the emission probability on S , J , and J' can be determined explicitly. It follows directly from the general formulas for matrix elements of spherical tensors when angular momenta are coupled. According to formula (109.3) (see III),²⁰

$$|\langle n'L'SJ' \| d \| nLSJ \rangle|^2 = (2J+1)(2J'+1) \left\{ \begin{matrix} L' & J' & S \\ J & L & 1 \end{matrix} \right\}^2 |\langle n'L' \| d \| nL \rangle|^2. \quad (49.9)$$

Substitution in (49.4) gives

$$w(nLSJ \rightarrow n'L'SJ') = \frac{4\omega^3}{3\hbar c^3} (2J'+1) \left\{ \begin{matrix} L' & J' & S \\ J & L & 1 \end{matrix} \right\}^2 |\langle n'L' \| d \| nL \rangle|^2, \quad (49.10)$$

where $\omega = \omega(nLS \rightarrow n'L'S)$ ²¹.

A definite sum rule can be obtained for these probabilities. The squares of the $6j$ symbols obey the summation formula (see III, (108.7))

$$\sum_{J'} (2J'+1) \left\{ \begin{matrix} L' & J' & S \\ J & L & 1 \end{matrix} \right\}^2 = \frac{1}{2L+1}. \quad (49.11)$$

²⁰In the formulas of Vol. III, § 109, the “angular momenta of subsystems 1 and 2” are now to mean the orbital angular momentum and spin of the atom, whose mutual interaction is neglected. The orbital vector d_q plays the role of $f_{1q}^{(1)}$.

²¹In neglecting spin-orbit interaction when calculating the matrix elements, we also neglect the dependence of the frequencies on J and J' , that is, the fine structure of the initial and final atomic levels.

Using it in (49.10), we find

$$\sum_{J'} w(nLSJ \rightarrow n'L'SJ') = \frac{4\omega^3}{3\hbar c^3} \frac{1}{2L+1} |\langle n'L' \| d \| nL \rangle|^2. \quad (49.12)$$

This quantity is independent of the initial value of J .

Suppose that the emitting gas has a temperature much greater than the fine-structure intervals of the atomic term nSL . States with different J are then populated uniformly, so all values of J are equally probable. In that case, the probability that an atom is on a level having a specified value of J is

$$\frac{2J+1}{(2L+1)(2S+1)}, \quad (49.13)$$

the ratio of the statistical weight of that level to the total statistical weight of the term nSL . Averaging expressions (49.10), or their sums (49.12), with these probabilities amounts to multiplication by the factor (49.13); we denote this averaging by a bar over the letter. The total emission probability for all lines of the spectral multiplet (formed by every possible transition between fine-structure components of the two terms nSL and $n'L'S$) is the sum

$$\bar{w}(nLS \rightarrow n'L'S) = \sum_J \sum_{J'} \bar{w}(nLSJ \rightarrow n'L'SJ'). \quad (49.14)$$

Since, of course, $\sum_J (2J+1) = (2S+1)(2L+1)$, the resulting expression for the total probability coincides with (49.12).

The relative probability (or, equivalently, relative intensity) of an individual line is therefore

$$\frac{\bar{w}(nLSJ \rightarrow n'L'SJ')}{\bar{w}(nLS \rightarrow n'L'S)} = \frac{(2J+1)(2J'+1)}{2S+1} \left\{ \begin{matrix} L' & J' & S \\ J & L & 1 \end{matrix} \right\}^2. \quad (49.15)$$

Inspection of the numerical values given by this formula shows that the most intense lines of the multiplet are those with $\Delta J = \Delta L$ (they are called *principal lines*, as distinct from the other multiplet components, called *satellites*). The intensity of the principal lines increases with the initial value of J .

Summing the quantities (49.15) over J or over J' gives

$$\begin{aligned} \frac{\sum_{J'} \bar{w}(nLSJ \rightarrow n'L'SJ')}{\bar{w}(nLS \rightarrow n'L'S)} &= \frac{2J+1}{(2L+1)(2S+1)}, \\ \frac{\sum_J \bar{w}(nLSJ \rightarrow n'L'SJ')}{\bar{w}(nLS \rightarrow n'L'S)} &= \frac{2J'+1}{(2L'+1)(2S+1)}. \end{aligned} \quad (49.16)$$

Thus the sum of the intensities of all spectral-multiplet lines having the same initial (or final) level is proportional to the statistical weight of that initial (or final) level.

We now consider the hyperfine structure of atomic spectral lines. Recall that hyperfine splitting of atomic levels results from the interaction of the electrons with the nuclear spin when that spin is nonzero (see III, § 122). The total angular momentum $\mathbf{F}$ of the

atom, including the nucleus, is the sum of the total electronic angular momentum $\mathbf{J}$ and the nuclear angular momentum $\mathbf{I}$. Each hyperfine-structure component of the level nJ is characterized by its own value of the quantum number F .

The exact conservation of angular momentum now leads to a strict selection rule for the total angular momentum F ; for electric-dipole radiation,

$$|F' - F| \leq 1 \leq F + F'. \quad (49.17)$$

The interaction between the electrons and the nuclear spin is exceedingly weak, however, and can be neglected entirely when the matrix elements of the electric (and magnetic) moments of the atomic electron shell are calculated. The previous selection rules for the electronic angular momentum J and the electronic parity therefore remain valid. In particular, electric-dipole transitions between hyperfine components of the same term are impossible because all these levels have identical parity, whereas such transitions require states of opposite parity.

Because the dipole-moment operator commutes with the nuclear spin, the dependence of the matrix elements on I and F can be found explicitly; the calculation differs from that performed above for LS coupling only by an evident change of notation. The emission probability summed over the final values of the projection of the total angular momentum $\mathbf{F}$ is

$$w(nJIF \rightarrow n'J'IF') = \frac{4\omega^3}{3\hbar c^3} \frac{1}{2F+1} |\langle n'J'IF' \| d \| nJIF \rangle|^2, \quad (49.18)$$

$$\omega = \omega(nJ \rightarrow n'J'),$$

while the squared reduced matrix element is

$$|\langle n'J'IF' \| d \| nJIF \rangle|^2 = (2F+1)(2F'+1) \begin{Bmatrix} J' & F' & I \\ F & J & 1 \end{Bmatrix}^2 |\langle n'J' \| d \| nJ \rangle|^2. \quad (49.19)$$

PROBLEM

Most lines in the spectra of alkali metals can be described as transitions of a single outer (optical) electron in the self-consistent field of the atomic core, which forms a closed configuration; the atomic state is constructed according to the LS -coupling scheme. Under these assumptions, determine the relative intensities of the fine-structure components of the spectral lines.

SOLUTION. The total angular momenta L and $S = 1/2$ of the atom coincide with the orbital angular momentum and spin of the optical electron. The parity of the state is consequently $(-1)^L$ (the closed configuration of the atomic core has positive parity). The parity selection rule therefore forbids a dipole transition with $L' = L$, leaving only transitions with $L' - L = \pm 1$. By the ΔJ selection rule, transitions between the components of the doublet levels n, L and $n', L - 1$ produce only three lines (Fig. 1).

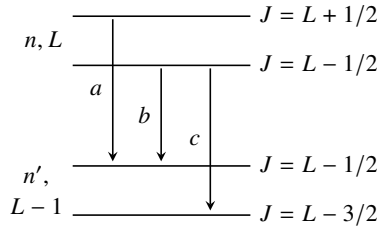


Fig. 1

Their relative intensities, denoted by a , b , and c , are more easily obtained from the rules (49.16) than directly from formula (49.15). Taking ratios of the total intensities of lines with one or the other common initial (or final) level gives two equations:

$$\frac{b+c}{a} = \frac{2L}{2L+2}, \quad \frac{a+b}{c} = \frac{2L}{2L-2},$$

and hence

$$a : b : c = [(L+1)(2L-1)] : 1 : [(L-1)(2L+1)].$$

For $L = 1$, the lower level is not split, line c is absent, and $a/b = 2$.

§ 50. Radiation from atoms. Magnetic type

The order of magnitude of an atom's magnetic moment is given by the Bohr magneton: $\mu \sim e\hbar/mc$. This estimate differs by a factor α from the order of magnitude of the electric dipole moment, $d \sim ea \sim \hbar^2/(me)$ (since $v/c \sim \alpha$ as well, $\mu \sim dv/c$, as expected). It follows that the probability of magnetic-dipole ($M1$) radiation from an atom is approximately α^2 times the probability of electric-dipole radiation at the same frequency. Magnetic radiation therefore has a significant role only in transitions forbidden by the selection rules for the electric type.

For electric-quadrupole ($E2$) radiation, the order of magnitude of its probability relative to that of $M1$ radiation is

$$\frac{E2}{M1} \sim \frac{(ea^2)^2\omega^2/c^2}{\mu^2} \sim \frac{a^4m^2\omega^2}{\hbar^2} \sim \left(\frac{\Delta E}{E}\right)^2 \quad (50.1)$$

(the quadrupole moment is $\sim ea^2$, $E \sim \hbar^2/ma^2$ is the atomic energy, and ΔE is the change in energy during the transition). We see that at ordinary atomic frequencies, that is, when $\Delta E \sim E$, the probabilities of $E2$ and $M1$ radiation have the same order of magnitude (provided, of course, that both are allowed by the selection rules). If $\Delta E \ll E$, as for transitions between fine-structure components of the same term, $M1$ radiation is more probable than $E2$ radiation.

Magnetic-dipole transitions obey the strict selection rules

$$|J' - J| \leq 1 \leq J + J', \quad (50.2)$$

$$PP' = 1. \quad (50.3)$$

In the case of LS coupling, additional selection rules arise that are even more restrictive than in the electric case. This is connected with a specific property of the atomic magnetic moment resulting from the identity of all the particles in the system (the electrons). Namely, the atomic magnetic-moment operator is expressed in terms of the operators of its total orbital and spin angular momenta:

$$\hat{\mu} = -\mu_0(\hat{\mathbf{L}} + 2\hat{\mathbf{S}}) = -\mu_0(\hat{\mathbf{J}} + \hat{\mathbf{S}}), \quad (50.4)$$

where $\mu_0 = |e|\hbar/(2mc)$ is the Bohr magneton (see III, § 113). Because total angular momentum is conserved, the operator $\hat{\mathbf{J}}$ has no matrix elements off-diagonal in energy at all; thus, for radiation theory it is sufficient to write $\hat{\mu} = -\mu_0\hat{\mathbf{S}}^{22}$.

When spin-orbit interaction is neglected, each of the angular momenta L and S is conserved separately. Hence the spin operator is diagonal in all the quantum numbers nSL that characterize the unsplit term. For any transition to occur at all, the number J must therefore change. We thus have the selection rules

$$n' = n, \quad S' = S, \quad L' = L, \quad J' - J = \pm 1, \quad (50.5)$$

that is, transitions are possible only between fine-structure components of the same term.

The radiation probability in this case can be calculated completely. Changing the notation in formula (49.10) accordingly, we find

$$w(nLSJ \rightarrow nLSJ') = \frac{4\omega^3\mu_0^2}{3\hbar c^3} (2J' + 1) \begin{Bmatrix} S & J' & L \\ J & S & 1 \end{Bmatrix}^2 |\langle S||S||S \rangle|^2.$$

The reduced matrix element of the spin with respect to its own eigenfunctions that enters here is given by

$$\langle S||S||S \rangle = \sqrt{S(S+1)(2S+1)} \quad (50.6)$$

(see III, (29.13)). The required $6j$ symbol is

$$\begin{Bmatrix} S & J-1 & L \\ J & S & 1 \end{Bmatrix}^2 = \frac{(L+S+J+1)(L+S-J+1)(L-S+J)(S-L+J)}{S(2S+1)(2S+2)(2J-1)2J(2J+1)} \quad (50.7)$$

(see the table in III, § 108). The result is

$$\begin{aligned} w(nLSJ \rightarrow nLS, J-1) &= \frac{2J+1}{2J-1} w(nLS, J-1 \rightarrow nLSJ) = \\ &= \frac{\omega^3\mu_0^2}{3\hbar c^3 (2J+1)J} (L+S+J+1)(L+S-J+1) \times \\ &\quad \times (J+S-L)(J+L-S). \end{aligned} \quad (50.8)$$

Transitions between hyperfine-structure components of the same level (their frequencies lie in the radio-wave region) cannot occur as electric-dipole transitions at all, since all these components have the same parity. The $E2$ and $M1$ transitions occur without a change of parity. Because the hyperfine-structure intervals are very small, however, $E2$ radiation is improbable in comparison with $M1$ radiation (cf. (50.1)), and the transitions in question therefore take place as magnetic-dipole transitions.

²²Exceptions occur when the electronic angular momentum $\mathbf{J}$ of the atom is not conserved: when hyperfine structure is included, in the presence of an external field, and so forth (see the problems).

PROBLEM

1. Find the probability of an $M1$ transition between hyperfine-structure components of the same level.

SOLUTION. The transition probability is given by formulas (49.18) and (49.19), in which the diagonal reduced matrix element of the magnetic moment, $\langle nJ || \mu || nJ \rangle$, now occurs. Its value may be written down at once by observing that the total (unreduced) matrix element $\langle nJM | \mu_z | nJM \rangle$ determines precisely the splitting of this level in the Zeeman effect (see III, § 113) and is equal to $-\mu_0 g M$, where g is the Landé factor. The reduced matrix element (see III, (29.7)) is

$$\begin{aligned} \langle nJ || \mu || nJ \rangle &= \frac{1}{M} \sqrt{J(J+1)(2J+1)} \langle nJM | \mu_z | nJM \rangle = \\ &= -\mu_0 g \sqrt{J(J+1)(2J+1)}. \end{aligned}$$

Consequently, the required probability is²³:

$$\begin{aligned} w(nJIF \rightarrow nJI, F-1) &= \frac{2F+1}{2F-1} w(nJI, F-1 \rightarrow nJIF) = \\ &= \frac{\omega^3 \mu_0^2 g^2}{3\hbar c^3 (2F+1)F} (J+I+F+1)(J+I-F+1) \\ &\quad \times (F+J-I)(F-J+I). \end{aligned}$$

This expression differs from (50.8) only by the evident change of notation and the additional factor g^2 .

PROBLEM

2. Find the probability of an $M1$ transition between Zeeman components of the same atomic level.

SOLUTION. This is a transition $M \rightarrow M-1$ with the values of nJ unchanged; its frequency (see below, (51.3)) is $\hbar\omega = \mu_0 g H$ (g is the Landé factor). The matrix element of the spherical component μ_{-1} of the vector μ is

$$\begin{aligned} |\langle nJ, M-1 | \mu_{-1} | nJM \rangle| &= \sqrt{\frac{(J-M+1)(J+M)}{2J(J+1)(2J+1)}} \langle nJ || \mu || nJ \rangle = \\ &= -\mu_0 g \sqrt{\frac{1}{2} (J-M+1)(J+M)} \end{aligned}$$

²³An interesting example is the transition between the hyperfine-structure components of the ground level of the hydrogen atom ($1s_{1/2}$), which is strictly forbidden not only as $E1$ but also as $E2$ (the latter by the rule forbidding a quadrupole transition with $J+J'=1$). This transition has frequency $\omega = 2\pi \cdot 1.42 \cdot 10^9 \text{ s}^{-1}$ (wavelength $\lambda = 21 \text{ cm}$). Setting $g = 2$, $I = 1/2$, $J = 1/2$, $F = 1$, and $F' = 0$, we obtain

$$w = \frac{4\omega^3 \mu_0^2}{3\hbar c^3} = 2.85 \cdot 10^{-15} \text{ s}^{-1}.$$

(see III, (27.12) and the preceding problem). The transition probability is

$$w = \frac{4\omega^3}{3\hbar c^3} |\langle nJ, M-1 | \mu_{-1} | nJM \rangle|^2 = \frac{2\mu_0^5 H^3}{3\hbar^4 c^3} (J-M+1)(J+M).$$

§ 51. Atomic radiation. The Zeeman and Stark effects

In an external magnetic field H , assumed weak, every atomic level of total angular momentum J splits into $2J+1$ levels,

$$E_M = E^{(0)} + \mu_0 g M H, \quad (51.1)$$

where $E^{(0)}$ is the unperturbed level, μ_0 is the Bohr magneton, g is the Landé factor, and M is the projection of the angular momentum J on the field direction (see III, § 113). Thus, the degeneracy with respect to angular-momentum direction is removed completely.

The spectral lines arising from transitions between two split levels undergo a corresponding splitting. The number of line components is determined by the selection rule for M , according to which dipole radiation requires

$$m = M - M' = 0, \pm 1. \quad (51.2)$$

In addition to this rule, transitions with $M = M' = 0$ are forbidden when $J' = J$. This follows directly from the general expressions (29.7) (see III) for the matrix elements of an arbitrary vector.

The components produced by transitions with $m = 0, \pm 1$ are called the π and σ components, respectively. Their frequencies are

$$\begin{aligned} \hbar\omega_\pi &= \hbar\omega^{(0)} + \mu_0 H(g - g')M, \\ \hbar\omega_\sigma &= \hbar\omega^{(0)} + \mu_0 H[gM - g'(M \pm 1)]. \end{aligned} \quad (51.3)$$

In the special case $g = g'$, we have

$$\hbar\omega_\pi = \hbar\omega^{(0)}, \quad \hbar\omega_\sigma = \hbar\omega^{(0)} \mp \mu_0 g H, \quad (51.4)$$

independently of M . In other words, the line then splits into a triplet consisting of an undisplaced π component and two σ components placed symmetrically on its two sides (the so-called *normal Zeeman effect*).

The total radiation probability, summed over all directions, is proportional to the squared modulus $|\langle n'J'M' | d_{-m} | nJM \rangle|^2$. Hence, by formula (46.19) with $j = 1$, the relative radiation probability of each Zeeman component of the spectral line is

$$\begin{pmatrix} J' & 1 & J \\ M' & m & -M \end{pmatrix}^2. \quad (51.5)$$

In the special case of the normal Zeeman effect there are only three components, each arising from transitions out of all initial values of M for a specified m . Since

$$\sum_{MM'} \begin{pmatrix} J' & 1 & J \\ M' & m & -M \end{pmatrix}^2 = \frac{1}{3} \quad (51.6)$$

(see III, (106.12)), the radiation probabilities of all three components are equal in this case.

Of greater interest, however, is the relative intensity of the Zeeman components when they are observed in a specified direction relative to the direction of the magnetic field applied to the source. According to (45.5), the probability of radiation, and therefore the line intensity, in a given direction $\mathbf{n}$ is proportional to $\sum |\mathbf{e}^* \mathbf{d}_{fi}|^2$, where the sum is taken over the two independent polarizations $\mathbf{e}$ possible for this $\mathbf{n}$.

For observation along the field (the z axis), this sum is

$$|(d_x)_{fi}|^2 + |(d_y)_{fi}|^2.$$

On passing to spherical components, we obtain

$$|(d_1)_{fi}|^2 + |(d_{-1})_{fi}|^2.$$

This means that only the two σ components ($m = \pm 1$) are observed in the longitudinal direction, along the field. Their intensities are proportional to

$$\begin{pmatrix} J' & 1 & J \\ M \mp 1 & \pm 1 & -M \end{pmatrix}^2. \quad (51.7)$$

Because they have definite values of the angular-momentum projection m along the direction of propagation, these lines have right ($m = 1$) and left ($m = -1$) circular polarization (see § 8).

For observation in a direction perpendicular to the field (let it be the x axis), the intensity is proportional to the sum

$$|(d_z)_{fi}|^2 + |(d_y)_{fi}|^2 = |(d_0)_{fi}|^2 + \frac{1}{2}\{|(d_1)_{fi}|^2 + |(d_{-1})_{fi}|^2\}.$$

Thus, in the transverse direction one observes two σ components and one π component, with intensities proportional respectively to

$$\frac{1}{2} \begin{pmatrix} J' & 1 & J \\ M \mp 1 & \pm 1 & -M \end{pmatrix}^2 \quad \text{and} \quad \begin{pmatrix} J' & 1 & J \\ M & 0 & -M \end{pmatrix}^2 \quad (51.8)$$

(the intensities of the σ components are half those for longitudinal observation). Here the π component is linearly polarized along the z axis, while the σ components observed in this direction are linearly polarized along the y axis.

We note that the relative intensities of the Zeeman components are determined entirely by the initial and final values of J and M , independently of all other characteristics of the levels.

The selection rules forbid electric-dipole transitions between Zeeman components of the same level, since they all have the same parity. For the same reason given at the end of the preceding section for transitions between components of a level's hyperfine structure, these transitions occur as magnetic-dipole transitions. By the selection rule for M , transitions occur only between adjacent components ($M' - M = \pm 1$)²⁴.

²⁴These transitions usually have frequencies in the centimetre range and are observed in absorption and stimulated emission (electron paramagnetic resonance): the absorbing atoms are placed in a strong constant magnetic field, which produces the Zeeman splitting, and a weak radio-frequency field at the resonant frequency.

The splitting of atomic levels in a weak electric field (the *Stark effect*), unlike their splitting in a magnetic field, does not remove the degeneracy with respect to angular-momentum direction completely. All levels other than those with $M = 0$ remain doubly degenerate: each corresponds to two states with angular-momentum projections M and $-M$.

The relative intensities of the Stark components of a spectral line are calculated in the same way as described above for the Zeeman effect²⁵. It must be borne in mind that transitions $M \rightarrow M$ and $-M \rightarrow -M$ contribute to the intensity of a π component (for $M \neq 0$), whereas transitions $M \rightarrow M \pm 1$ and $-M \rightarrow -(M \pm 1)$ contribute to the intensities of the σ components. Thus, for example, in transverse observation of the effect, the intensities of the π components are proportional to

$$2 \begin{pmatrix} J' & 1 & J \\ M & 0 & -M \end{pmatrix}^2,$$

while the intensities of the σ components are proportional to the sums

$$\begin{aligned} \frac{1}{2} \begin{pmatrix} J' & 1 & J \\ M \pm 1 & \mp 1 & -M \end{pmatrix}^2 + \frac{1}{2} \begin{pmatrix} J' & 1 & J \\ -M \mp 1 & \pm 1 & M \end{pmatrix}^2 \\ = \begin{pmatrix} J' & 1 & J \\ M \pm 1 & \mp 1 & -M \end{pmatrix}^2. \end{aligned}$$

(Recall that when the signs of all the numbers in the second row are reversed, the $3j$ symbols can at most change sign; their squares are unchanged.)

Strictly speaking, in an external field, however weak, the total angular momentum $\mathbf{J}$ ceases to be conserved; in a uniform field only conservation of its projection M remains exact. Consequently, conservation of angular momentum is no longer an absolute requirement for radiative transitions in a weak field, and lines forbidden by the usual selection rules can appear in the spectra.

Calculation of the intensities of these lines reduces to calculating corrections to the dipole-moment matrix, which in turn requires the corrections to the wavefunctions of the stationary states. In the first approximation of perturbation theory in the weak external field, the wavefunction acquires “admixture” of states connected to the original state by nonzero matrix elements of the perturbation ($-\mathbf{E}\mathbf{d}$ in an electric field): the addition of a certain state ψ_2 to the state ψ_1 is

$$-\frac{\mathbf{E}\mathbf{d}_{21}}{E_1 - E_2}\psi_2.$$

As a result, the matrix element of the “forbidden” transition acquires the term

$$-\frac{(\mathbf{E}\mathbf{d}_{21})\mathbf{d}_{32}}{E_1 - E_2},$$

which is nonzero if transitions from the “intermediate” state 2 to the initial and final states 1 and 3 are allowed.

²⁵Here we mean the quadratic Stark effect, which occurs in all atoms except hydrogen (see III, § 76). The field is assumed so weak that the level splitting it produces is small even in comparison with the fine-structure intervals.

§ 52. Radiation from atoms. The hydrogen atom

The hydrogen atom is the sole case in which transition matrix elements can be evaluated completely in analytic form (W. Gordon, 1929).

The parity of a hydrogen-atom state is $(-1)^l$, and is thus fixed uniquely by the electron's orbital angular momentum (recall that the number l , as the quantity determining the parity of a state, retains its meaning for the exact relativistic wave functions as well, that is, when spin-orbit interaction is included). The parity selection rule therefore strictly forbids electric-dipole transitions in which l is unchanged; only transitions $l \rightarrow l \pm 1$ are possible. Changes in the principal quantum number n , by contrast, are unrestricted.

The dipole moment of the hydrogen atom reduces to the electron radius vector: $\mathbf{d} = e\mathbf{r}$. Since the electron wave function in hydrogen is a product of an angular part and a radial function R_{nl} , the reduced matrix elements of the radius vector can likewise be written as a product:

$$\langle n', l-1 || r || nl \rangle = \langle l-1 || \boldsymbol{\nu} || l \rangle \int_0^\infty R_{n', l-1} R_{nl} r^3 dr,$$

where $\langle l-1 || \boldsymbol{\nu} || l \rangle$ is the reduced matrix element of the unit vector $\boldsymbol{\nu}$ in the direction of $\mathbf{r}$. These elements are

$$\langle l-1 || \boldsymbol{\nu} || l \rangle = \langle l || \boldsymbol{\nu} || l-1 \rangle^* = i\sqrt{l}$$

(see III, (29.14)). Consequently,

$$\langle n', l-1 || r || nl \rangle = -\langle nl || r || n', l-1 \rangle = i\sqrt{l} \int_0^\infty R_{n', l-1} R_{nl} r^3 dr. \quad (52.1)$$

The nonrelativistic radial functions for the discrete spectrum of hydrogen are given by formula (36.13) (see III)²⁶:

$$R_{nl} = \frac{2}{n^{l+2}(2l+1)!} \sqrt{\frac{(n+l)!}{(n-l-1)!}} (2r)^l e^{-r/n} \times \\ \times F\left(-n+l+1, 2l+2, \frac{2r}{n}\right). \quad (52.2)$$

The integral (52.1), containing a product of two confluent hypergeometric functions, is

²⁶Atomic units are used in this section. In ordinary units, the coordinate matrix elements written below must be multiplied by $\hbar^2/(me^2)$ (or by $\hbar^2/(mZe^2)$ for a hydrogen-like ion of atomic number Z).

evaluated by means of the formulas given in III, § f²⁷. The calculation gives

$$\begin{aligned}
 \langle n', l-1 \| r \| nl \rangle &= i\sqrt{l} \frac{(-1)^{n'-1}}{4(2l-1)!} \sqrt{\frac{(n+l)!(n'+l-1)!}{(n-l-1)!(n'-l)!}} \\
 &\times \frac{(4nn')^{l+1} (n-n')^{n+n'-2l-2}}{(n+n')^{n+n'}} \\
 &\times \left\{ F\left(-n+l+1, -n'+l, 2l, -\frac{4nn'}{(n-n')^2}\right) \right. \\
 &\quad \left. - \left(\frac{n-n'}{n+n'}\right)^2 F\left(-n+l-1, -n'+l, 2l, -\frac{4nn'}{(n-n')^2}\right) \right\}.
 \end{aligned} \tag{52.3}$$

Here $F(\alpha, \beta, \gamma, z)$ denotes the hypergeometric function. Because the parameters α and β are negative integers (or zero) in the present case, these functions reduce to polynomials²⁸.

For reference, we give the expressions obtained from (52.3) in several special cases (the value of l is denoted by the spectroscopic symbol $s, p, d, \dots$):

$$\begin{aligned}
 |\langle 1s \| r \| np \rangle|^2 &= \frac{2^8 n^7 (n-1)^{2n-5}}{(n+1)^{2n+5}}, \\
 |\langle 2s \| r \| np \rangle|^2 &= \frac{2^{17} n^7 (n^2-1)(n-2)^{2n-6}}{(n+2)^{2n+6}}, \\
 |\langle 2p \| r \| nd \rangle|^2 &= \frac{2^{19} n^9 (n^2-1)(n-2)^{2n-7}}{3(n+2)^{2n+7}}, \\
 |\langle 2p \| r \| ns \rangle|^2 &= \frac{2^{15} n^9 (n-2)^{2n-6}}{3(n+2)^{2n+6}}.
 \end{aligned} \tag{52.4}$$

Formula (52.3) cannot be used for transitions in which the principal quantum number n does not change (transitions between the fine-structure components of a level). In this case ($n = n'$), we carry out the integration by starting from the representation of the radial functions in terms of generalized Laguerre polynomials:

$$R_{nl} = -\frac{2}{n^2} \sqrt{\frac{(n-l-1)!}{[(n+l)!]^3}} e^{-r/n} \left(\frac{2r}{n}\right)^l L_{n+l}^{2l+1} \left(\frac{2r}{n}\right). \tag{52.5}$$

In the integral

$$\int_0^\infty R_{n,l-1} R_{nl} r^3 dr \propto \int_0^\infty e^{-\rho} \rho^{2l+2} L_{n+l}^{2l+1}(\rho) L_{n+l-1}^{2l-1}(\rho) d\rho$$

we replace one polynomial by its expression in terms of the generating function (see III, § d):

$$L_{n+l}^{2l+1}(\rho) = -\frac{(n+l)!}{(n-l-1)!} e^\rho \rho^{-2l-1} \left(\frac{d}{d\rho}\right)^{n-l-1} e^{-\rho} \rho^{n+l}.$$

²⁷In the notation introduced there, the integral to be evaluated is $J_{2l+2}^{12}(-n+l+1, -n'+l)$. The evaluation uses formulas (f.12)–(f.16).

²⁸Numerical tables of matrix elements and transition probabilities for hydrogen may be found in: H. Bethe and E. Salpeter, *Quantum Mechanics of One- and Two-Electron Atoms*, Moscow: IL, 1960.

After integrating by parts $n - l - 1$ times, we obtain an integral of the form

$$\int_0^\infty e^{-\rho} \rho^{n+l} \left(\frac{d}{d\rho} \right)^{n-l-1} \rho L_{n+l-1}^{2l-1}(\rho) d\rho,$$

in which the Laguerre polynomial is replaced by its explicit expression according to the formula

$$L_n^m(\rho) = (-1)^m n! \sum_{k=0}^{n-m} \binom{n}{m+k} \frac{(-\rho)^k}{k!}.$$

Once the differentiations have been performed, only three terms remain in the sum, and the integration is then elementary. The calculation gives the simple result

$$\langle n, l-1 \| r \| nl \rangle = i\sqrt{l} \cdot \frac{3}{2} n \sqrt{n^2 - l^2}. \quad (52.6)$$

The integral

$$\int_0^\infty R_{n', l-1} R_{nl} r^3 dr = \int_0^\infty \chi_{n', l-1}(r) \chi_{nl}(r) dr$$

(where $\chi_{nl} = r R_{nl}$) is a coefficient in the expansion of the function $r \chi_{nl}$ in the orthogonal system of functions $\chi_{n', l-1}$ ($n' = 1, 2, \dots$). The sum of the squared moduli of these coefficients equals the integral of the square of the function being expanded²⁹. Hence

$$\sum_{n'} |\langle n', l-1 \| r \| nl \rangle|^2 = l \int_0^\infty r^2 \chi_{nl}^2 dr. \quad (52.7)$$

Using the known expression for the mean square r^2 in the state nl (see III, (36.16)), we obtain the sum rule

$$\sum_{n'} |\langle n', l-1 \| r \| nl \rangle|^2 = l \frac{n^2}{2} [5n^2 + 1 - 3l(l+1)]. \quad (52.8)$$

For fixed n, l and large n' , the matrix element for a transition $nl \rightarrow n'l'$ decreases according to

$$|\langle n'l' \| r \| nl \rangle|^2 \propto \frac{3}{n'^3}, \quad (52.9)$$

as follows both from the special expressions (52.4) and from the general formula (52.3). This result is entirely natural: at large n' , the Coulomb energy levels $E' = -1/2n'^2$ form a quasi-continuous sequence, and the probability of a transition to any level in an interval dE' is proportional to the density of levels there, which is itself $\propto n'^{-3}$.

The Stark effect in hydrogen is known to have a special character (see III, § 77): the splitting is proportional to the first power of the electric field. The field is assumed not to be strong (as required for perturbation theory to apply), yet at the same time strong enough that the level splitting greatly exceeds the fine-structure intervals. Under these conditions the magnitude of the angular momentum is not conserved in general, and

²⁹The sum is over states in both the discrete and continuous spectra.

the levels must be classified by the parabolic quantum numbers n_1, n_2, m . The last of these, the magnetic quantum number m , continues to specify the projection of the orbital angular momentum on the z axis (the direction of the field); under the present conditions, in which spin-orbit interaction is neglected, this projection is conserved. It therefore obeys the usual selection rule

$$m' - m = 0, \pm 1. \quad (52.10)$$

There are no restrictions on changes of n_1 and n_2 .

The dipole-moment matrix elements in parabolic coordinates can also be evaluated analytically. The resulting formulas are, however, very cumbersome, and we shall not give them here³⁰.

PROBLEM

1. Find the Stark splitting of the hydrogen levels when the splitting is small compared with the fine-structure intervals (but large compared with the Lamb shift).

SOLUTION. Under these conditions, the unperturbed levels with $l = j \pm 1/2$ remain twofold degenerate, and the Stark splitting consequently remains linear in the field. The value of the splitting Δ is determined from the secular equation

$$\begin{vmatrix} -\Delta & -E(d_z)_{12} \\ -E(d_z)_{21} & -\Delta \end{vmatrix} = 0, \quad \Delta = \pm E|(d_z)_{12}|$$

(the indices 1 and 2 refer to the states with $l = j \pm 1/2$ and a specified magnetic quantum number m ; the perturbation $V = -Ed_z$ is diagonal in m and has no matrix elements diagonal in l). The matrix element of the orbital quantity d_z is evaluated using formulas (29.7) and (109.3) (see III), which give

$$\langle j, l-1, m | d_z | jlm \rangle = \frac{m}{\sqrt{j(j+1)(2j+1)}} \langle j, l-1 || d || j, l \rangle,$$

$$\langle j, l-1 || d || j, l \rangle = -(2j+1) \begin{Bmatrix} l-1 & j & 1/2 \\ j & l & 1 \end{Bmatrix} \langle l-1 || d || l \rangle,$$

where $l = j + 1/2$ must be used; the quantity $\langle l-1 || d || l \rangle$ is taken from (52.6). The result is

$$\Delta = \pm \frac{3}{4} \sqrt{n^2 - (j+1/2)^2} \frac{nm}{j(j+1)} E.$$

2. Determine the probability for photon emission in the transition between the hydrogen states $2s_{1/2} \rightarrow 1s_{1/2}$ (G. Breit, E. Teller, 1940).

SOLUTION. The process in question is strictly forbidden for an $E1$ transition by parity and for an $E2$ transition by rule (46.15). We must therefore calculate the probability of an $M1$ transition, which is given by formula (47.5). In the present case ($l = 0$), however, the magnetic moment is a purely spin quantity, and, when spin-orbit interaction is neglected,

³⁰For these formulas and the corresponding numerical tables, see the book by H. Bethe and E. Salpeter cited above.

its matrix element vanishes because orbital wave functions with different principal quantum numbers are mutually orthogonal. Thus the Pauli-equation approximation is insufficient for obtaining a nonzero answer, and one must start from the full Dirac equation.

In the standard representation of the wave functions, the transition current is³¹

$$\mathbf{j}_{fi} = \psi_f^* \boldsymbol{\alpha} \psi_i = \varphi_f^* \boldsymbol{\sigma} \chi_i + \chi_f^* \boldsymbol{\sigma} \varphi_i.$$

According to (35.1), (24.2), and (24.8), the wave functions of states with $l = 0$, $j = 1/2$ have the form

$$\psi = \begin{pmatrix} \varphi \\ \chi \end{pmatrix} = \frac{1}{4\pi} \begin{pmatrix} f(r)w(m) \\ -ig(r)(\boldsymbol{\sigma}\mathbf{n})w(m) \end{pmatrix},$$

where $\mathbf{n} = \mathbf{r}/r$, and $w(m)$ is a real unit 3-spinor corresponding to the value m of the spin projection. Hence

$$\mathbf{j}_{fi} = \frac{1}{4\pi i} \{f_f g_i w_f \boldsymbol{\sigma}(\boldsymbol{\sigma}\mathbf{n}) w_i - g_i f_i w_f(\boldsymbol{\sigma}\mathbf{n}) \boldsymbol{\sigma} w_i\}.$$

Substitution of this expression in (47.4), followed by integration over the directions of $\mathbf{n}$, gives

$$\boldsymbol{\mu}_{fi} = -\frac{e}{6i} w_f(\boldsymbol{\sigma} \times \boldsymbol{\sigma}) w_i I = -\frac{e}{3} w_f \boldsymbol{\sigma} w_i I$$

(by the Pauli-matrix commutation relation $\boldsymbol{\sigma} \times \boldsymbol{\sigma} = 2i\boldsymbol{\sigma}$); here

$$I = \int_0^\infty (f_f g_i + f_i g_f) r^3 dr. \quad (1)$$

The photon-emission probability (47.5), summed over the values of m_f , is

$$w = \frac{4e^2 \omega^3}{27} w_i \boldsymbol{\sigma}^2 w_i I^2 = \frac{4e^2 \omega^3}{9} I^2. \quad (2)$$

From (35.4), with $\varkappa = -1$, we have

$$g = \frac{f'}{\varepsilon + m + \alpha/r} \approx \frac{f'}{2m} - \left(\varepsilon - m + \frac{\alpha}{r} \right) \frac{R'}{4m^2};$$

in the second term the exact function f has been replaced by the nonrelativistic radial function R . If the approximation is restricted to $g = R'/2m$, the integral

$$I = \frac{1}{2m} \int_0^\infty (R_f R_i)' r^3 dr = -\frac{3}{2m} \int_0^\infty R_f R_i r^2 dr = 0 \quad (3)$$

by orthogonality of R_f and R_i . In the next approximation, with (3) taken into account,

$$\begin{aligned} I &= \frac{1}{2m} \int_0^\infty (f_f f_i)' r^3 dr \\ &+ \frac{1}{4m^2} \int_0^\infty \left\{ R_f' R_i (\varepsilon_i - \varepsilon_f) - \frac{\alpha}{r} (R_f R_i)' \right\} r^3 dr. \end{aligned} \quad (4)$$

³¹Relativistic units are used in this problem.

Since orthogonality of the exact functions ψ_i and ψ_f gives (for $\kappa_i = \kappa_f$)

$$\int_0^\infty (f_i f_f + g_i g_f) r^2 dr = 0,$$

the first term in (4), after integration by parts, can be written as

$$-\frac{3}{2m} \int_0^\infty f_f f_i r^2 dr = \frac{3}{2m} \int_0^\infty g_f g_i r^2 dr \approx \frac{3}{8m^3} \int_0^\infty R'_f R'_i r^2 dr.$$

Evaluation of the integral with the functions

$$R_f = 2(m\alpha)^{3/2} e^{-m\alpha r}, \quad R_i = \frac{(m\alpha)^{3/2}}{\sqrt{2}} \left(1 - \frac{m\alpha r}{2}\right) e^{-m\alpha r/2}$$

(see III, § 36) and the energy difference

$$\omega = \varepsilon_i - \varepsilon_f = \frac{m\alpha^2}{2} \left(1 - \frac{1}{2^2}\right) = \frac{3}{8} m\alpha^2$$

gives $I = 2^{3/2} \alpha^2 / (9m)$. Hence the transition probability in ordinary units is

$$w = \frac{2^5 \alpha^5 \hbar^2 \omega^3}{3^6 m^2 c^4} = \frac{mc^2 \alpha^{11}}{2^4 \cdot 3^3 \hbar} = 5.6 \cdot 10^{-6} \text{ s}^{-1}.$$

The corresponding lifetime of the $2s_{1/2}$ state is very long; in practice, radiative decay through the simultaneous emission of two photons is much more probable (see the note on p. 263).

§ 53. Radiation from diatomic molecules. Electronic spectra

The distinctive features of molecular spectra arise primarily from the division of a molecule's energy into electronic, vibrational, and rotational parts, each successive part being small in comparison with the preceding one. The level structure of diatomic molecules was studied in detail in vol. III, chap. XI. Here we shall determine the resulting form of the spectrum and calculate the intensities of its lines³².

We begin with the general case in which a transition changes the electronic state of the molecule (and, generally, its vibrational and rotational states as well). The frequencies of these transitions lie in the visible and ultraviolet regions of the spectrum. Their totality is called the molecule's *electronic spectrum*. We shall always have electric-dipole transitions in mind; transitions of other types are generally of little importance in molecular spectroscopy.

As for dipole transitions in any system, the selection rule for the total angular momentum J of the molecule is

$$|J' - J| \leq 1 \leq J + J'. \quad (53.1)$$

³²The discussion below is based on the material in vol. III, §§ 78, 82–88. To avoid burdening the text, we shall not refer continually to those sections.

The strict selection rule for the parity of the system here corresponds to a selection rule for the sign of the level. Recall that, in the terminology adopted in molecular spectroscopy, states whose wavefunctions retain or reverse their sign under inversion (reversal of the coordinates of the electrons and nuclei) are called positive or negative, respectively. Thus the strict rule is

$$+ \rightarrow -, \quad - \rightarrow +. \quad (53.2)$$

If the molecule consists of identical atoms (with nuclei of the same isotope), its levels are also classified with respect to interchange of the nuclear coordinates: symmetric (s) levels, whose wavefunctions retain their sign under this transformation, and antisymmetric (a) levels, whose functions reverse sign. Since this transformation does not affect the electronic dipole-moment operator at all, its matrix elements are nonzero only for transitions that preserve this symmetry³³:

$$s \rightarrow s, \quad a \rightarrow a. \quad (53.3)$$

This rule is not absolutely strict, however. The possession by a level of a given symmetry property is connected with the molecule having one particular value of the total nuclear spin I . Because the interaction of the nuclear spins with the electrons is extremely weak, I is conserved to high accuracy, but not strictly. When this interaction is included, I has no definite value, the symmetry property (s or a) is not conserved, and the selection rule (53.3) is lost.

The electronic terms of a molecule made of identical atoms are also characterized by their parity (g or u), that is, by the behavior of the wavefunctions when the signs of the electron coordinates (measured from the molecular center) are reversed while the nuclear coordinates remain fixed. This property of the electronic term is closely connected with the nuclear symmetry and the sign of the rotational levels belonging to that term. Levels belonging to an even (g) electronic term may have the characteristics $s+$ or $a-$, whereas those belonging to an odd term may have $s-$ or $a+$. Rules (53.2) or (53.3) therefore also imply

$$g \rightarrow u, \quad u \rightarrow g. \quad (53.4)$$

As an approximate rule, (53.4) remains valid for molecules made of different isotopes of the same element. Since the nuclear charges are equal, if we consider the electronic term with fixed nuclei, we have a system of electrons in an electric field possessing a center of symmetry (at the point bisecting the internuclear distance). The symmetry of the electronic wavefunction under inversion through this point determines the parity of the term. Since the electric dipole-moment vector reverses sign under this transformation, rule (53.4) follows. A rule derived in this way is approximate because the nuclei have had to be treated as fixed. It is therefore violated when the interaction between the electronic state and molecular rotation is taken into account.

The subsequent selection rules depend on particular assumptions about the relative strengths of the various interactions in the molecule (that is, on its coupling case). These rules can consequently be only approximate.

³³This rule plainly applies also to transitions of any multipolarity.

Most electronic terms of diatomic molecules belong to coupling cases *a* or *b*. Both are characterized by coupling of the orbital angular momentum to the axis (the electric interaction of the two atoms in the molecule) that is strong in comparison with all other interactions. There consequently exist quantum numbers Λ and S (the projection of the electronic orbital angular momentum on the molecular axis and the total electron spin). The operator of the orbital quantity, the electronic dipole moment, commutes with the spin operator, so that

$$S' - S = 0 \quad (\text{cases } a, b). \quad (53.5)$$

The change in Λ obeys the selection rule

$$\Lambda' - \Lambda = 0, \pm 1 \quad (\text{cases } a, b), \quad (53.6)$$

and transitions between states with $\Lambda = 0$ (Σ terms) obey the additional rule

$$\Sigma^+ \rightarrow \Sigma^+, \quad \Sigma^- \rightarrow \Sigma^- \quad (\text{cases } a, b) \quad (53.7)$$

(recall that the states Σ^+ and Σ^- differ in their behavior under reflection in a plane through the molecular axis). Rules (53.6) and (53.7) are obtained by considering the molecule in a coordinate system rigidly attached to the nuclei (see III, § 87); rule (53.6) is analogous to the selection rule for the magnetic quantum number in atoms.

Coupling cases *a* and *b* differ in the relation between the energy of the spin–axis interaction and the rotational energy (the separations of the rotational levels). The former energy is greater than the latter in case *a*, but much smaller in case *b*. We now consider the two cases separately.

Case *a*. Here there is a quantum number Σ , the projection of the total spin on the molecular axis (and with it the number $\Omega = \Sigma + \Lambda$, the projection of the total angular momentum). If both the initial and final states belong to case *a*, then

$$\Sigma' - \Sigma = 0 \quad (\text{case } a) \quad (53.8)$$

(as follows from the already mentioned commutation of the dipole moment with the spin). Equations (53.6) and (53.8) imply³⁴

$$\Omega' - \Omega = 0, \pm 1. \quad (53.9)$$

If $\Omega = \Omega' = 0$, transitions with $J' = J$ are forbidden in addition to the general rule (53.1)³⁵:

$$J' - J = \pm 1 \quad \text{for } \Omega = \Omega' = 0 \quad (\text{case } a). \quad (53.10)$$

Consider transitions between any two specified vibrational levels belonging to two different electronic terms (of case *a*). When the fine structure of an electronic term is

³⁴This rule also remains valid in case *c* (where coupling of the orbital angular momentum to the axis is weak in comparison with spin–orbit coupling), in which the numbers Λ and Σ do not exist separately.

³⁵This rule is analogous to the prohibition of transitions with $J = J'$ when $M = M' = 0$ in atoms (see § 5), where, however, it could be relevant only in the presence of an external field. Here the rule follows directly from formula (53.12) below; the $3j$ symbol $\begin{pmatrix} J & 1 & J \\ 0 & 0 & 0 \end{pmatrix}$ vanishes because the sum $J + J + 1$ is odd.

taken into account, each of these levels splits into several components; by rule (53.5), their number $(2S + 1)$ must be the same for both levels. According to rule (53.8), each component of one level combines with only one component of the other, having the same value of Σ .

Now take one pair of levels with the same Σ ; their values of Ω and Ω' may differ (together with Λ and Λ') by 0 or ± 1 . When rotation is included, each splits into a sequence of levels distinguished by the numbers J and J' , which take the values $J \geq |\Omega|$ and $J' \geq |\Omega'|$. The dependence of the transition probabilities on these numbers can be found in general form (H. Hönl, R. London, 1925).

The matrix element for the transition $n\Lambda\Omega JM_J \rightarrow n'\Lambda'\Omega'J'M'_J$ (where n denotes the remaining characteristics of the electronic term, apart from Ω and Λ) is

$$\begin{aligned} & |\langle n'\Lambda'\Omega'J'M'_J | d_q | n\Lambda\Omega JM_J \rangle| = \\ & = \sqrt{(2J+1)(2J'+1)} \begin{pmatrix} J' & 1 & J \\ -\Omega' & q & \Omega \end{pmatrix} \begin{pmatrix} J' & 1 & J \\ -M'_J & q & M_J \end{pmatrix} |\langle n'\Lambda' | \bar{d}_{q'} | n\Lambda \rangle|, \end{aligned} \quad (53.11)$$

where d_q and $\bar{d}_{q'}$ are the spherical components of the dipole-moment vector in the fixed coordinate system xyz and in the “moving” system $\xi\eta\zeta$, respectively, with the ζ axis along the molecular axis (this formula follows with the aid of formula (110.6), see III). The matrix elements $\langle n'\Lambda' | \bar{d}_{q'} | n\Lambda \rangle$ are independent of the rotational quantum numbers J, J' and depend only on the characteristics of the electronic terms (in the present case they are also independent of Σ ³⁶); the indices $\Omega' = \Lambda' + \Sigma$ and $\Omega = \Lambda + \Sigma$ have therefore been omitted from the notation for the matrix element.

The probability of the transition $n\Lambda\Omega J \rightarrow n'\Lambda'\Omega'J'$ is proportional to the square of the matrix element (53.11), summed over M'_J . By formula (106.12) (see III),

$$\sum_{M'_J} \begin{pmatrix} J' & 1 & J \\ -M'_J & q & M_J \end{pmatrix}^2 = \frac{1}{2J+1},$$

and hence

$$\begin{aligned} w(n\Lambda\Omega J \rightarrow n'\Lambda'\Omega'J') &= (2J'+1) \begin{pmatrix} J' & 1 & J \\ -\Omega' & \Omega' - \Omega & \Omega \end{pmatrix}^2 \\ &\times B(n', n; \Lambda', \Lambda), \end{aligned} \quad (53.12)$$

where the coefficients B are independent of J, J' (we of course neglect the relatively small difference among the frequencies of transitions with different J, J')³⁷.

Summing (53.12) over J' gives simply $B(n', n; \Lambda', \Lambda)$, by the orthogonality property of the $3j$ symbols (see III, (106.13)). In other words, the total probability of transition from a rotational level J of the state Ω to all levels J' of the state Ω' is independent of J .

³⁶This may be verified in the same way as at the beginning of III, § 29 for a scalar quantity f . Here the vector operator $\mathbf{d}$ commutes with the operator of the vector $\mathbf{S}$, which is conserved in the zeroth approximation, while Σ is the projection of $\mathbf{S}$ on the ζ axis in the rotating coordinate system in which the commutation condition for $\mathbf{d}$ and $\mathbf{S}$ must be considered.

³⁷When Λ doubling is included, each rotational level J considered here splits further into two levels, one positive and the other negative. Thus, in place of one transition $J \rightarrow J'$, rule (53.2) gives two transitions: from the positive (negative) component of level J to the negative (positive) component of level J' . The probabilities of these transitions are equal.

Case *b*. In this case, besides the total angular momentum J , there is also a quantum number K , the angular momentum of the molecule excluding its spin. The selection rules for this number coincide with the general selection rules for any orbital vector quantity (such as the electric dipole moment):

$$|K' - K| \leq 1 \leq K + K' \quad (\text{case } b) \quad (53.13)$$

with the additional prohibition of the transition $K = K'$ when $\Lambda = \Lambda' = 0$ (analogously to (53.10)):

$$K' - K = \pm 1 \quad \text{for } \Lambda = \Lambda' = 0. \quad (53.14)$$

Consider transitions between rotational components of specified vibrational levels of two electronic states belonging to case *b*. Their transition probabilities are given by the same formulas (53.12), with K, Λ written in place of J, Ω . When the fine structure is included (for $S \neq 0$), each rotational level K splits into $2S + 1$ components: $J = |K - S|, \dots, K + S$, so that a multiplet appears in place of a single line $K \rightarrow K'$. Since this case involves the addition of the free angular momenta K and S (which are not coupled to the molecular axis), the formulas for the relative probabilities of the various multiplet lines coincide with the analogous formulas (49.15) for fine-structure components of atomic spectra, where the angular momenta L and S play the same role (in the case of LS coupling).

We have thus considered the selection rules that determine the possible spectral lines in all the principal cases encountered in diatomic molecules.

The set of lines from transitions between the rotational components of two given electronic-vibrational levels constitutes what spectroscopy calls a *band*: because the rotational intervals are small, the lines in a band lie very close together. Their frequencies are given by the differences

$$\hbar\omega_{J,J'} = \text{const} + BJ(J+1) - B'J'(J'+1), \quad (53.15)$$

where B, B' are the rotational constants in the two electronic states (to avoid needless complications, we assume singlet electronic terms). For $J' = J, J \pm 1$, formula (53.15) is represented graphically (Fig. 2) by three branches (parabolas), whose points at integer J give the frequencies (the arrangement of the branches in Fig. 2 corresponds to $B' < B$; for $B' > B$ they open toward smaller ω , and the curve for $J' = J - 1$ lies uppermost)³⁸. As is clear from the figure, the presence of a turning branch causes the lines to crowd toward a particular limiting position (the band head).

In discussing line intensities, we must also mention the distinctive alternation of intensities in certain bands of the electronic spectra of molecules made of atoms of the same isotope (W. Heisenberg, F. Hund, 1927). The symmetry requirements associated with the nuclear spins cause rotational components with even and odd values of K in electronic Σ terms to have opposite nuclear symmetry and, accordingly, different nuclear statistical weights g_s and g_a (see III, § 86). By rule (53.14), only $J' = J \pm 1$ is allowed for transitions between two different Σ terms; by (53.4), one of the Σ terms must be

³⁸The series of lines corresponding to transitions with $J' = J + 1, J, J - 1$ are called the *P*, *Q*, and *R* branches, respectively.

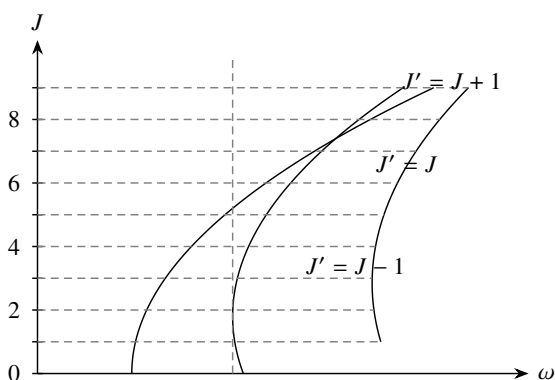


Figure 1

even and the other odd. Consequently, for a specified value of $J' - J$, transitions at successive values of J occur alternately between pairs of symmetric levels and pairs of antisymmetric levels (as illustrated in Fig. 3 for the states Σ_u^+ and Σ_g^+). On the other hand, the observed line intensity is proportional to the number of molecules in the given initial state and therefore to its statistical weight. Hence the intensities of successive lines ($J = 0, 1, 2, \dots$) will alternate between larger and smaller values, proportional in turn to g_s and g_a (apart from the monotonic variation predicted by formulas (53.12))³⁹.

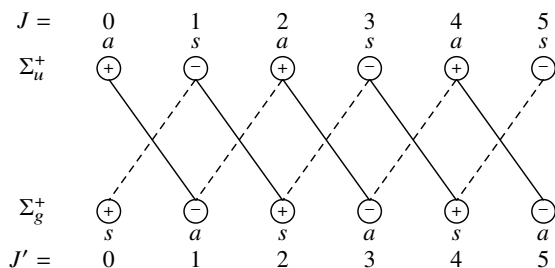


Figure 2

There are no strict selection rules for the change of the vibrational quantum number in transitions between two different electronic terms. There is, however, a rule (the Franck–Condon principle) that predicts the most probable change of vibrational state. It rests on the quasiclassical character of the nuclear motion resulting from the large nuclear masses (compare the discussion of predissociation in III, § 90)⁴⁰.

In the integral determining the matrix element of the transition between the vibrational states E and E' of the electronic terms $U(r)$ and $U'(r)$, the main contribution comes from the neighborhood of the point $r = r_0$ at which

$$U(r_0) - U'(r_0) = E - E' \quad (53.16)$$

³⁹Here it is assumed that all states with different values of the total nuclear spin are uniformly populated.

⁴⁰Strictly speaking, the vibrational quantum number must also be sufficiently large.

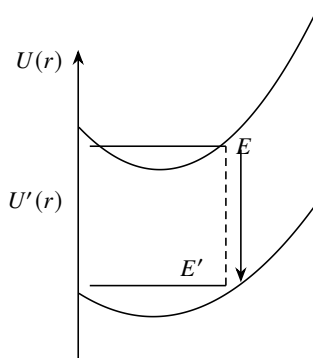


Figure 3

(that is, the momenta of the relative nuclear motion are the same in both states: $p = p'$). For a specified E , the transition probability (as a function of the final energy E') is the greater, the smaller each of the differences $E - U$ and $E' - U'$. It is greatest when

$$E - U(r_0) = E' - U'(r_0) = 0, \quad (53.17)$$

that is, when the “transition point” r_0 (a root of equation (53.16)) coincides with a classical turning point of the nuclei (Fig. 4 illustrates graphically this relation between E and the most probable E'). In pictorial terms, the transition is most probable near a point where the nuclei stop and in whose neighborhood they consequently spend more time.

§ 54. Radiation from diatomic molecules. Vibrational and rotational spectra

The selection rules and transition-probability formulas listed in the preceding section remain valid for transitions in which the molecule’s electronic state does not change⁴¹. We shall consider here only a few features particular to these transitions.

First of all, selection rule (53.4) forbids dipole transitions without a change of electronic state in molecules made of identical atoms: such a transition would leave the parity of the electronic term unchanged. As follows from § 53, this prohibition can be violated only when the interaction of the nuclear spins with the electrons is taken into account; in molecules composed of different isotopes of one element, the effect of rotation on the electronic state is already sufficient to violate it.

The calculation of dipole-moment matrix elements reduces (by the formulas of § 87 (see III)) to calculating them in a coordinate system that rotates with the molecule. In this system the molecular wave function is the product of the electron wave function at a specified separation r of the nuclei and the wave function for the vibrational motion of the nuclei in the effective field $U(r)$ of the electrons and nuclei. If the influence

⁴¹Transitions that alter the vibrational state (and, with it, the rotational state) constitute what is called the molecule’s *vibrational spectrum*; it lies in the near-infrared region (wavelengths below $20\ \mu\text{m}$). Transitions that alter only the rotational state constitute the *rotational spectrum*, which lies in the far-infrared region (wavelengths above $20\ \mu\text{m}$).

of nuclear motion on the electronic state is neglected completely, the initial and final electron wave functions for the transitions under consideration are identical. Integration over the electron coordinates therefore leaves in the matrix element simply the mean dipole moment $\bar{\mathbf{d}}$ of the molecule (directed, evidently, along its axis), as a function of the separation r . Since the vibrations are small, $\bar{\mathbf{d}}(r)$ can be expanded in powers of the vibrational coordinate $q = r - r_0$. For transitions involving a change of vibrational state, the constant term of this expansion drops out of the matrix element because the wave functions for vibrational motion in the same field $U(q)$ are orthogonal; the term proportional to q remains. If the vibrations are treated as harmonic, the familiar properties of the linear oscillator (see III, § 23) show that the matrix elements are nonzero only for transitions between adjacent vibrational states. Thus the vibrational quantum number ν obeys the selection rule

$$\nu' - \nu = \pm 1. \quad (54.1)$$

This rule is violated, however, when vibrational anharmonicity and the subsequent terms in the expansion of $\bar{\mathbf{d}}(q)$ are included.

For a purely rotational transition (with no change of vibrational state either), the matrix element of the dipole-moment projection on the moving axis ζ may be replaced simply by the mean molecular dipole moment $\bar{\mathbf{d}} = \bar{\mathbf{d}}(0)$ ⁴². The resulting formula for the probability of the transition $J \rightarrow J - 1$ is

$$w(nJ \rightarrow n, J - 1) = \frac{4\omega^3}{3\hbar c^3} \bar{\mathbf{d}}^2 \frac{J^2 - \Omega^2}{J(2J + 1)}, \quad (54.2)$$

which gives not only relative probabilities, as does (53.12), but their absolute values as well. (Formula (54.2) is written for case *a*; in case *b*, K, Λ must replace J, Ω .)

The frequencies of purely rotational transitions are the differences between the rotational energies $BJ(J + 1)$ and are therefore

$$\hbar\omega_{J,J-1} = 2BJ. \quad (54.3)$$

Successive lines are separated by the same interval, $2B$.

§ 55. Nuclear radiation

For nuclear γ radiation, the dimensions of the system (the nuclear radius R) are generally small compared with the photon wavelength. The separations between nuclear levels, however, and hence the energy of a γ quantum, are usually small compared with the energy per nucleon in the nucleus. The quantity R/λ is therefore not directly related to the velocity v/c of the nucleons in the nucleus and is, in general, much smaller. Correspondingly, the probability of Ml radiation is usually greater than that of $E, l + 1$ radiation (compare the beginning of § 50).

The general selection rules for the total angular momentum ("spin") of the nucleus and for parity are the same as for radiation by any system. A distinctive feature of

⁴²For a molecule made of identical atoms, $\bar{\mathbf{d}} = 0$, as is evident from symmetry.

nuclear radiation is the prevalence of transitions of higher multipolarity. Unlike atoms, whose radiation is ordinarily electric dipole, nuclei at low energies make such transitions comparatively rarely because the selection rules forbid them.

If a radiative nuclear transition can be treated as a one-particle transition — a change in the state of one nucleon with the state of the nuclear “core” unchanged — additional selection rules apply to the angular momentum of that nucleon. Such “one-particle” selection rules, however, are obeyed with very low accuracy.

Selection rules involving isospin are specific to the nucleus. Recall that the projection T_3 of the isospin is already fixed by the mass number and nuclear charge:

$$T_3 = \frac{1}{2}(Z - N) = Z - \frac{A}{2}.$$

At a specified value of T_3 , the magnitude of the isospin may take any value $T \geq |T_3|$. A selection rule for T in radiative transitions arises because the operators of the electric and magnetic moments of the nucleus, when expressed through the nucleon isospin operators, are sums of a scalar and the x_3 component of a vector in isospin space (see III, § 116). Their matrix elements are consequently nonzero only if

$$T' - T = 0, \pm 1. \quad (55.1)$$

By itself, however, this rule imposes no significant restriction on transitions in light nuclei (the only nuclei for which conservation of isospin is accurate enough to be meaningful): in fact, the low-lying levels of these nuclei include no levels with $T > 1$.

For $E1$ transitions there is an additional rule, because the isoscalar part of the electric dipole moment drops out and its operator reduces to the x_3 component of an isospin vector (see III, § 116). Thus, when $T_3 = 0$, transitions with $\Delta T = 0$ are additionally forbidden. In other words, in nuclei having equal numbers of neutrons and protons ($N = Z$, $A = 2Z$), $E1$ transitions are possible only when

$$T' - T = \pm 1 \quad (T_3 = 0). \quad (55.2)$$

The accuracy of this rule naturally depends on how accurately the isospin of the nucleus is conserved.

The recoil of the nuclear core during the motion of individual nucleons also affects the probability of nuclear $E1$ transitions. As a result of this effect, protons contribute to the dipole moment with an effective charge $e(1 - Z/A)$ rather than e , while neutrons contribute with a charge $-eZ/A$ rather than zero (see III, § 118). The reduction of the proton's effective charge produces some suppression of the $E1$ transition probability.

The energy levels of nonspherical nuclei have rotational structure. Such nuclei therefore have a characteristic rotational structure in their γ -radiation spectra.

The symmetry of the field in which the nucleons move in a “stationary” nonspherical (axial) nucleus is the same as that of the field in which the electrons move in a “stationary” diatomic molecule composed of identical atoms (the point group $C_{\infty h}$). The symmetry properties of the levels of a nonspherical nucleus, and with them the selection rules for matrix elements, are consequently analogous to those for the levels of a diatomic molecule (see III, § 119). In particular, just as in a diatomic molecule made of identical atoms,

electric-dipole transitions within the same rotational band (that is, without a change of the internal nuclear state) are forbidden; compare § 54. Such transitions therefore occur as $E2$ or $M1$ transitions. In the first case the total nuclear angular momentum J may change by 2 or 1, and in the second by 1.

According to (46.9), the quadrupole-transition probability summed over the values of the projection M' of the total angular momentum of the nucleus in the final state is

$$w_{E2} = \frac{\omega^5}{15\hbar c^5} \sum_{M'} \left| \langle J' \Omega M' | Q_{2,-m}^{(e)} | J \Omega M \rangle \right|^2$$

(J is the total angular momentum of the nucleus, Ω its projection on the nuclear axis, and $m = M - M'$). Formula (110.8) (see III) expresses this sum in terms of the squares of specified quantities: the diagonal (in the internal nuclear state) transition quadrupole moments $\bar{Q}_{2\lambda}$, defined relative to the nuclear coordinate axes $\xi\eta\zeta$. Here $\lambda = \Omega - \Omega'$, so in the present case ($\Omega' = \Omega$) only the component $\bar{Q}_{20}$ occurs. The quantity called simply the quadrupole moment of the nucleus is defined by

$$eQ_0 = e \int \rho_{ii} (2\zeta^2 - \xi^2 - \eta^2) d\xi d\eta d\zeta = -2e(\bar{Q}_{20})_{ii}.$$

It follows that

$$w_{E2}(\Omega J \rightarrow \Omega J') = \frac{\omega^5}{60\hbar c^5} Q_0^2 (2j' + 1) \begin{pmatrix} J' & 2 & J \\ -\Omega & 0 & \Omega \end{pmatrix}^2. \quad (55.3)$$

In expanded form,

$$\begin{aligned} w_{E2}(\Omega J \rightarrow \Omega, J-1) &= \frac{\omega^5}{20\hbar c^5} Q_0^2 \frac{\Omega^2 (J^2 - \Omega^2)}{(J-1)J(J+1)(2J+1)}, \\ w_{E2}(\Omega J \rightarrow \Omega, J-2) &= \frac{\omega^5}{40\hbar c^5} Q_0^2 \frac{(J^2 - \Omega^2)[(J-1)^2 - \Omega^2]}{(J-1)J(2J-1)(2J+1)}. \end{aligned}$$

The following qualification must, however, be made concerning these formulas. They use matrix elements calculated with wave functions of the form

$$\psi_{J\Omega M} = \text{const} \cdot \chi_{\Omega} D_{\Omega M}^{(J)}(\mathbf{n})$$

(χ is the wave function of the internal nuclear state). These functions correspond to specified values, both in magnitude and sign, of the angular-momentum projection on the ζ axis. In a nucleus, however, the states have only a specified parity and magnitude of the angular-momentum projection (the latter is normally what is meant by Ω). Thus, when $\Omega \neq 0$, the initial and final wave functions should instead be combinations of the form

$$\frac{1}{\sqrt{2}}(\psi_{J\Omega M} \pm \psi_{J, -\Omega, M}), \quad \frac{1}{\sqrt{2}}(\psi_{J'\Omega M'} \pm \psi_{J', -\Omega, M'}).$$

The products of the first terms and of the second terms give the previous value of the quadrupole-moment matrix element. The “cross” products, on the other hand,

give nonzero integrals when $2\Omega \leq l^{43}$. Strictly speaking, formula (55.3) is therefore inapplicable for $\Omega = 1/2, 1$; in these cases the transition probability contains an additional term that cannot be expressed through the mean quadrupole moment⁴⁴.

By a derivation analogous to that of (55.3), the probability of an $M1$ transition is

$$\begin{aligned} w_{M1}(\Omega J \rightarrow \Omega, J-1) &= \frac{4\omega^3}{3\hbar c^3} \mu^2 (2J-1) \begin{pmatrix} J-1 & 1 & J \\ -\Omega & 0 & \Omega \end{pmatrix}^2 \\ &= \frac{4\omega^3}{3\hbar c^3} \mu^2 \frac{J^2 - \Omega^2}{J(2J+1)}, \end{aligned} \quad (55.4)$$

where μ is the magnetic moment of the nucleus (this formula is inapplicable for $\Omega = 1/2$).

§ 56. Photoelectric effect. Nonrelativistic case

In §§ 49–52 we considered radiative transitions (emission or absorption of a photon) between atomic levels belonging to the discrete spectrum. The photoelectric effect differs from such a process of photon absorption only in that the final state belongs to the continuous spectrum.

For a hydrogen atom, or a hydrogen-like ion (with nuclear charge $Z \ll 137$), the photoelectric cross section can be calculated analytically in closed form.

In the initial state there is an electron in a discrete level $\varepsilon_i = -I$ (I is the ionization potential of the atom) and a photon of definite momentum $\mathbf{k}$. In the final state the electron has momentum $\mathbf{p}$ (and energy $\varepsilon_f = \varepsilon$). Since $\mathbf{p}$ ranges over a continuous set of values, the photoelectric cross section is

$$d\sigma = 2\pi |V_{fi}|^2 \delta(-I + \omega - \varepsilon) \frac{d^3 p}{(2\pi)^3} \quad (56.1)$$

(cf. (44.3)), where the wave function of the final electron state is assumed to be normalized to one particle in a volume $V = 1$. The photon wave function is normalized in the same way as before; in passing from the probability dw to the cross section $d\sigma$, it must therefore be divided by the photon flux density (equal to $c/V = c$), but in relativistic units this does not alter the form of (56.1).

As in (45.2), choose the three-dimensionally transverse gauge for the photon. Then

$$V_{fi} = -e\mathbf{A}\mathbf{j}_{fi} = -e\sqrt{4\pi} \frac{1}{\sqrt{2\omega}} M_{fi},$$

where

$$M_{fi} = \int \psi'^*(\alpha\mathbf{e}) e^{i\mathbf{k}\mathbf{r}} \psi d^3x \quad (56.2)$$

⁴³For matrix elements of 2^l -pole moments, the integrands contain products of the form

$$D_{-\Omega M}^{(J')*} D_{q' q}^{(l)} D_{\Omega M}^{(J)}.$$

The angular integral is nonzero for $q' = -2\Omega$, whereas q' ranges only from $-l$ to l ; hence $2\Omega \leq l$.

⁴⁴In practice this term gives an appreciable correction only for $\Omega = 1/2$, when the coupling between rotation and the internal state of the nucleus is especially strong (see III, § 119).

($\psi \equiv \psi_i$ and $\psi' \equiv \psi_f$ are the initial and final electron wave functions). Replacing $d^3p \rightarrow \mathbf{p}^2 d|\mathbf{p}| d\omega = \varepsilon |\mathbf{p}| d\varepsilon d\omega$ in (56.1) and integrating the δ -function with respect to $d\varepsilon$, we rewrite this formula as

$$d\sigma = e^2 \frac{\varepsilon |\mathbf{p}|}{2\pi\omega} |M_{fi}|^2 d\omega. \quad (56.3)$$

We shall carry out the calculation in two cases, distinguished by the photon energy: for $\omega \gg I$ and for $\omega \ll m$. Since $I \sim me^4 Z^2 \ll m$, these two regions overlap in part (when $I \ll \omega \ll m$), so their examination gives, in essence, a complete description of the photoelectric effect.

We begin with the case

$$\omega \ll m. \quad (56.4)$$

Here the electron velocity is small in both the initial and final states, and the problem is therefore wholly nonrelativistic as regards the electron. Accordingly, in (56.2) replace α by the nonrelativistic velocity operator $\hat{\mathbf{v}} = -i\nabla/m$ (cf. § 45). We may also pass to the dipole approximation, putting $e^{i\mathbf{k}\mathbf{r}} \approx 1$, that is, neglecting the photon momentum in comparison with the electron momentum. We then have

$$d\sigma = e^2 \frac{m |\mathbf{p}|}{2\pi\omega} |\mathbf{e}\mathbf{v}_{fi}|^2 d\omega, \quad \mathbf{v}_{fi} = -\frac{i}{m} \int \psi'^* \nabla \psi \cdot d^3x. \quad (56.5)$$

Consider the photoelectric effect from the ground level of a hydrogen atom (or a hydrogen-like ion). Then

$$\psi = \frac{(Ze^2m)^{3/2}}{\sqrt{\pi}} e^{-Ze^2mr} \quad (56.6)$$

(in ordinary units $me^2 \rightarrow 1/a_0$, where $a_0 = \hbar^2/me^2$ is the Bohr radius).

For ψ' , on the other hand, we must use the wave function whose asymptotic form contains a plane wave ($e^{i\mathbf{p}\mathbf{r}}$) together with an incoming spherical wave (see III, § 136, where this function was denoted by $\psi_{\mathbf{p}}^{(-)}$). By the selection rule for l , a transition from an s -state is possible only to a p -state (in the dipole case). Hence, in the expansion⁴⁵

$$\psi_{\mathbf{p}}^{(-)} = \frac{1}{2p} \sum_{l=0}^{\infty} i^l (2l+1) e^{-i\delta_l} R_{pl}(r) P_l(\mathbf{n}\mathbf{n}_1) \quad (56.7)$$

($\mathbf{n} = \mathbf{p}/p$, $\mathbf{n}_1 = \mathbf{r}/r$), it is sufficient to retain only the term with $l = 1$. Dropping inessential phase factors, we obtain

$$\psi' = \frac{3}{2p} (\mathbf{n}\mathbf{n}_1) R_{p1}(r). \quad (56.8)$$

With the functions ψ and ψ' from (56.6) and (56.8), we have

$$\begin{aligned} \mathbf{e}\mathbf{v}_{fi} &= \frac{3(Ze^2m)^{5/2}}{2\sqrt{\pi}mp} \iint (\mathbf{n}\mathbf{n}_1)(\mathbf{n}_1\mathbf{e}) e^{-Ze^2mr} R_{p1}(r) d\omega_1 \cdot r^2 dr = \\ &= \frac{\sqrt{2\pi}(Ze^2m)^{5/2}}{pm} \int_0^\infty r^2 e^{-Ze^2mr} R_{p1}(r) dr. \end{aligned}$$

⁴⁵Below in this section, p denotes $|\mathbf{p}|$.

According to (36.18) and (36.24) (see III), the radial function (in the units used here) is

$$R_{pl} = \frac{\sqrt{8\pi} Z e^2 m}{3} \sqrt{\frac{1 + \nu^2}{\nu(1 - e^{-2\pi\nu})}} pr e^{-ipr} F(2 + i\nu, 4, 2ipr),$$

where

$$\nu = \frac{Ze^2 m}{p} \left(= \frac{Ze^2}{\hbar v} \right). \quad (56.9)$$

The integral required here is evaluated with the aid of the formula

$$\int_0^\infty e^{-\lambda z} z^{\gamma-1} F(\alpha, \gamma, kz) dz = \Gamma(\gamma) \lambda^{\alpha-\gamma} (\lambda - k)^{-\alpha}$$

(see III, (f. 3)). Noting also that

$$\left(\frac{\nu + i}{\nu - i} \right)^{i\nu} = e^{-2\nu \operatorname{arccot} \nu},$$

we obtain

$$\mathbf{e} \nu_{fi} = \frac{2^{7/2} \pi \nu^3 (\mathbf{ne})}{\sqrt{\rho m} (1 + \nu^2)^{3/2}} \frac{e^{-2\nu \operatorname{arccot} \nu}}{\sqrt{1 - e^{-2\pi\nu}}}.$$

The ionization energy from the ground level of a hydrogen atom (or hydrogen-like ion) is $I = Z^2 e^4 m / 2$. Therefore

$$\omega = \frac{p^2}{2m} + I = \frac{p^2}{2m} (1 + \nu^2). \quad (56.10)$$

Taking this relation into account, the final expression for the photoelectric cross section with emission of an electron into the solid-angle element do is

$$d\sigma = 2^7 \pi \alpha a^2 \left(\frac{I}{\hbar \omega} \right)^4 \frac{e^{-4\nu \operatorname{arccot} \nu}}{1 - e^{-2\pi\nu}} (\mathbf{ne})^2 do, \quad (56.11)$$

where $a = \hbar^2 / (mZe^2) = a_0 / Z$ (here and below, ordinary units are used). Notice that the angular distribution of the photoelectrons is determined by the factor $(\mathbf{ne})^2$. It is greatest in directions parallel to the polarization of the incident photons, and vanishes in directions perpendicular to $\mathbf{e}$, including the direction of incidence. For unpolarized photons, (56.11) must be averaged over the directions of $\mathbf{e}$, which amounts to the replacement

$$(\mathbf{ne})^2 \rightarrow \frac{1}{2} [\mathbf{n}_0 \mathbf{n}]^2,$$

where $\mathbf{n}_0 = \mathbf{k} / k$ (see (45.46)).

Integration of (56.11) over the angles gives the total photoelectric cross section:

$$\sigma = \frac{2^9 \pi^2}{3} \alpha a^2 \left(\frac{I}{\hbar \omega} \right)^4 \frac{e^{-4\nu \operatorname{arccot} \nu}}{1 - e^{-2\pi\nu}} \quad (56.12)$$

(*M. Stobbe*, 1930).

The limiting value of σ as $\hbar\omega \rightarrow I$ (that is, $\nu \rightarrow \infty$) is

$$\sigma = \frac{2^9 \pi^2}{3e^4} \alpha a^2 = \frac{2^9 \pi^2}{3e^4} \frac{\alpha a_0^2}{Z^2} = 0.23 \frac{a_0^2}{Z^2} \quad (56.13)$$

(in the denominator, $e = 2.71 \dots$!). As is required for a reaction producing charged particles (see III, § 147), the photoelectric cross section tends to a constant limit near its threshold.

The case $\hbar\omega \gg I$ (while still $\hbar\omega \ll mc^2$) corresponds to the Born approximation ($\nu = Ze^2/(\hbar\nu) \ll 1$). Formula (56.12) then becomes

$$\sigma = \frac{2^8 \pi}{3} \alpha a_0^2 Z^5 \left(\frac{I_0}{\hbar\omega} \right)^{7/2} \quad (56.14)$$

($I_0 = e^4 m / (2\hbar^2)$ is the ionization energy of the hydrogen atom).

The inverse of the photoelectric effect is radiative recombination of an electron with a stationary ion. The cross section for this process (σ_{rec}) can be found from the photoelectric cross section (σ_{ph}) by means of the principle of detailed balance (III, § 144). According to this principle, the cross sections of the processes $i \rightarrow f$ and $f \rightarrow i$ (with two particles in each of the states i and f) are related by

$$g_i p_i^2 \sigma_{i \rightarrow f} = g_f p_f^2 \sigma_{f \rightarrow i},$$

where p_i, p_f are the relative-motion momenta of the particles, and g_i, g_f are the spin statistical weights of states i and f . Taking account also of the fact that $g = 2$ for a photon (two polarization directions), while the statistical weight of a free electron and ion is equal to that of the ground state of the hydrogen atom, we obtain for this state

$$\sigma_{\text{rec}} = \sigma_{\text{ph}} \frac{2k^2}{p^2} \quad (56.15)$$

($\mathbf{p} = m\mathbf{v}$ is the momentum of the incident electron, and $\mathbf{k}$ is the momentum of the emitted photon).

PROBLEM

1. Derive (56.14) by applying the Born approximation directly in the nonrelativistic case.

SOLUTION. In the Born approximation, ψ' in (56.5) must simply be taken as the plane wave $\psi' = e^{i\mathbf{p}\mathbf{r}}$, while ψ remains the function (56.6). Then

$$\mathbf{v}_{fi} = \mathbf{v}_{if} = \frac{1}{m} \int \psi \hat{\mathbf{p}} \psi' d^3x = \frac{\mathbf{p}}{m} \frac{(Ze^2 m)^{3/2}}{\sqrt{\pi}} (e^{-Ze^2 m r})_{\mathbf{p}}.$$

The Fourier component is given by (57.66), and hence

$$\mathbf{v}_{fi} \approx 8\sqrt{\pi} p^{-3} m^{3/2} (Ze^2)^{5/2} \mathbf{n}.$$

Substitution in (56.5), followed by integration over do , gives (56.14) (to the required accuracy, $p^2/(2m) \approx \omega$ here).

2. Find the total cross section for radiative recombination of a fast nonrelativistic electron ($I \ll mv^2 \ll mc^2$) with a nucleus (of charge $Z \ll 137$).

SOLUTION. The capture cross section into the K shell (principal quantum number $n = 1$) follows by substituting (56.14) into (56.15):

$$\sigma_1^{\text{rec}} = \frac{2^7 \pi}{3} Z^5 \alpha^3 a_0^2 \left(\frac{I_0}{\varepsilon} \right)^{5/2}$$

($\varepsilon = mv^2/2$ is the energy of the incident electron; $\hbar\omega \approx \varepsilon$). Of the other states of the atom being formed, only the s -states are important: in calculating the matrix element in the Born approximation, the values of the bound-state wave function at small r are relevant (as will be seen from the calculations in § 57), and for $l > 0$ these values are small compared with those of the functions having $l = 0$; it is sufficient here to retain the first two terms in the expansion of ψ in powers of r . For states with $l = 0$ and arbitrary n , these terms are

$$\psi = \frac{1}{\sqrt{\pi a^3 n^{3/2}}} \left(1 - \frac{r}{a} \right),$$

that is, they contain n only through the overall factor $n^{-3/2}$ (the displayed expression follows by expanding (36.13) (see III)). Consequently, the total recombination cross section is

$$\sigma^{\text{rec}} = \sum_{n=1}^{\infty} \sigma_n^{\text{rec}} = \sigma_1^{\text{rec}} \sum_{n=1}^{\infty} \frac{1}{n^3} = \zeta(3) \sigma_1^{\text{rec}}$$

(the value of the zeta function is $\zeta(3) = 1.202$).

§ 57. Photoelectric effect. Relativistic case

We turn to the case

$$\omega \gg I. \quad (57.1)$$

Here also $\varepsilon = \omega - I \gg I$, and the effect of the nuclear Coulomb field on the photoelectron wave function (ψ') can therefore be treated by perturbation theory. Write ψ' as

$$\psi' = \frac{1}{\sqrt{2\varepsilon}} \left(u' e^{i\mathbf{p}\mathbf{r}} + \psi^{(1)} \right). \quad (57.2)$$

The photoelectron may be relativistic; for this reason the unperturbed function in (57.2) is written as the relativistic plane wave (23.1).

Although the electron is nonrelativistic in the initial state, its wave function ψ must nevertheless include a relativistic correction ($\sim Ze^2$), for reasons that will become clear below. This function is given by (see the problem to § 39)

$$\psi = \left(1 - \frac{i}{2m} \gamma^0 \boldsymbol{\gamma} \nabla \right) \frac{u}{\sqrt{2m}} \psi_{\text{nr}}, \quad (57.3)$$

where ψ_{nr} is the nonrelativistic bound-state function (56.6), and u is the bispinor amplitude of an electron at rest, normalized by our convention $\bar{u}u = 2m$.

Substitute the functions (57.2), (57.3) into the matrix element (56.2)⁴⁶:

$$M_{fi} = \frac{1}{2\sqrt{m\varepsilon}} \int \left\{ \bar{u}'(\gamma\mathbf{e}) \left[\left(1 - \frac{i}{2m} \gamma^0 \gamma \nabla \right) u \psi_{\text{nr}} \right] e^{-i(\mathbf{p}-\mathbf{k})\mathbf{r}} + \bar{\psi}^{(1)}(\gamma\mathbf{e}) e^{i\mathbf{k}\mathbf{r}} u \psi_{\text{nr}} \right\} d^3x. \quad (57.4)$$

Since we seek the first term in the expansion of this quantity in Ze^2 , in the second term within the braces we may replace ψ_{nr} simply by the constant $(Ze^2m)^{3/2}/\sqrt{\pi}$. The first term would vanish under the same replacement (when $\mathbf{p} - \mathbf{k} \neq 0$). This is precisely why the first relativistic correction, proportional to Ze^2 , must also be retained in ψ : when $v \sim 1$, this correction contributes to the cross section in the same order as the next term in the expansion of ψ_{nr} in Ze^2 .

In the first term of (57.4), integrate by parts so that the operator ∇ acts on the exponential factor rather than on ψ_{nr} . The result is

$$M_{fi} = \frac{(Ze^2m)^{3/2}}{2(\pi m\varepsilon)^{1/2}} \left\{ \bar{u}'(\gamma\mathbf{e}) \left[1 + \frac{1}{2m} \gamma^0 \gamma(\mathbf{p} - \mathbf{k}) \right] u \left(e^{-Ze^2mr} \right)_{\mathbf{p}-\mathbf{k}} + \bar{\psi}_{-\mathbf{k}}^{(1)}(\gamma\mathbf{e}) u \right\}, \quad (57.5)$$

where a vector subscript denotes a spatial Fourier component. To accuracy through the term $\sim Ze^{247}$

$$\left(e^{-Ze^2mr} \right)_{\mathbf{p}-\mathbf{k}} = \frac{8\pi Ze^2m}{(\mathbf{p} - \mathbf{k})^4}. \quad (57.6)$$

To calculate the Fourier component $\psi_{\mathbf{k}}^{(1)}$, write the equation satisfied by $\psi^{(1)}$:

$$(\gamma^0\varepsilon + i\gamma\nabla - m)\psi^{(1)} = e(\gamma^\mu A_\mu)u'e^{i\mathbf{p}\mathbf{r}} = -\frac{Ze^2}{r}\gamma^0u'e^{i\mathbf{p}\mathbf{r}}$$

⁴⁶The function (57.3) was obtained for distances $r \sim 1/(mZe^2)$, at which its correction term is of relative order Ze^2 . For the ground state (and indeed for all s -states), however, formula (57.3) is valid at every r , since the derivative of the purely exponential function (56.6), and therefore also the correction term in (57.3), is always proportional to Ze^2 . This makes it possible to use (57.3) in the present problem, where small r are important. (When $v \sim 1$, the important distances are $r \sim 1/m$.)

⁴⁷Taking the Fourier component of both sides of

$$(\Delta - \lambda^2) \frac{e^{-\lambda r}}{r} = -4\pi\delta(\mathbf{r}),$$

we obtain

$$\left(\frac{e^{-\lambda r}}{r} \right)_{\mathbf{q}} = \frac{4\pi}{\mathbf{q}^2 + \lambda^2}. \quad (57.6a)$$

Differentiation of this expression with respect to the parameter λ gives

$$\left(e^{-\lambda r} \right)_{\mathbf{q}} = \frac{8\pi\lambda}{(\mathbf{q}^2 + \lambda^2)^2}. \quad (57.6b)$$

(it follows by substituting (57.2) into (32.1)). Apply the operator $(\gamma^0 \varepsilon + i\boldsymbol{\gamma} \nabla + m)$ to both sides of this equation. We obtain

$$(\Delta + \mathbf{p}^2)\psi^{(1)} = -Ze^2(\gamma^0 \varepsilon + i\boldsymbol{\gamma} \nabla + m)(\gamma^0 u') \frac{1}{r} e^{i\mathbf{p}\mathbf{r}}.$$

Multiply this equation by $e^{-i\mathbf{k}\mathbf{r}}$ and integrate over d^3x , performing the usual integrations by parts in the terms containing Δ and ∇ :

$$\begin{aligned} (\mathbf{p}^2 - \mathbf{k}^2)\psi_{\mathbf{k}}^{(1)} &= -Ze^2(\gamma^0 \varepsilon - \boldsymbol{\gamma} \mathbf{k} + m)(\gamma^0 u') \left(\frac{1}{r} \right)_{\mathbf{k}-\mathbf{p}} = \\ &= -Ze^2(2\varepsilon\gamma^0 - \boldsymbol{\gamma}(\mathbf{k} - \mathbf{p}))(\gamma^0 u') \frac{4\pi}{(\mathbf{k} - \mathbf{p})^2}. \end{aligned}$$

In the last line we have used the equation obeyed by the amplitude u' :

$$(\varepsilon\gamma^0 - \mathbf{p}\boldsymbol{\gamma} - m)u' = 0, \quad \text{or} \quad (\varepsilon\gamma^0 + \mathbf{p}\boldsymbol{\gamma} - m)\gamma^0 u' = 0.$$

It follows that

$$\bar{\psi}_{-\mathbf{k}}^{(1)} = \psi_{\mathbf{k}}^{(1)*} \gamma^0 = 4\pi Ze^2 \bar{u}' \frac{2\varepsilon\gamma^0 + \boldsymbol{\gamma}(\mathbf{k} - \mathbf{p})}{(\mathbf{k}^2 - \mathbf{p}^2)(\mathbf{k} - \mathbf{p})^2} \gamma^0. \quad (57.7)$$

Substituting (57.6), (57.7) into the matrix element (57.5), we put it in the form

$$M_{fi} = \frac{4\pi^{1/2}(Ze^2m)^{5/2}}{(\varepsilon m)^{1/2}(\mathbf{k} - \mathbf{p})^2} \bar{u}' Au,$$

where

$$\begin{aligned} A &= a(\boldsymbol{\gamma} \mathbf{e}) + (\boldsymbol{\gamma} \mathbf{e})\gamma^0(\boldsymbol{\gamma} \mathbf{b}) + (\boldsymbol{\gamma} \mathbf{c})\gamma^0(\boldsymbol{\gamma} \mathbf{e}), \\ a &= \frac{1}{(\mathbf{k} - \mathbf{p})^2} + \frac{\varepsilon}{m} \frac{1}{\mathbf{k}^2 - \mathbf{p}^2}, \quad \mathbf{b} = -\frac{\mathbf{k} - \mathbf{p}}{2m(\mathbf{k} - \mathbf{p})^2}, \quad \mathbf{c} = \frac{\mathbf{k} - \mathbf{p}}{2m(\mathbf{k}^2 - \mathbf{p}^2)}. \end{aligned}$$

The cross section is

$$d\sigma = \frac{8e^2(Ze^2m)^5|\mathbf{p}|}{\omega(\mathbf{k} - \mathbf{p})^4m} (\bar{u}' Au)(\bar{u} \bar{A} u') do,$$

where $\bar{A} = \gamma^0 A^+ \gamma^0$ (see § 65). This expression must still be summed over the final spin directions and averaged over the initial ones. These operations are carried out by the rules described below in § 65, using the polarization density matrices of the initial and final states:

$$\rho = \frac{m}{2}(\gamma^0 + 1), \quad \rho' = \frac{1}{2}(\gamma^0 \varepsilon - \boldsymbol{\gamma} \mathbf{p} + m)$$

(in the initial state $\mathbf{p} = 0$, $\varepsilon = m$). They give

$$d\sigma = \frac{16e^2(Ze^2m)^5|\mathbf{p}|}{m\omega(\mathbf{k} - \mathbf{p})^4} \text{Tr}(\rho' A \rho \bar{A}) do.$$

Evaluation of the trace (using formulas (22.22)) is a purely algebraic operation and gives

$$\begin{aligned} \text{Tr}(\rho' A \rho \bar{A}) &= \frac{m}{\varepsilon + m} \left[a\mathbf{p} - (\mathbf{b} - \mathbf{c})(\varepsilon + m) \right]^2 + \\ &\quad + 4m(\mathbf{b}\mathbf{e})[(\varepsilon + m)(\mathbf{c}\mathbf{e}) + a(\mathbf{p}\mathbf{e})] \end{aligned}$$

(the vector $\mathbf{e}$ is assumed real, corresponding to linear photon polarization).

Put the photoelectric cross-section formula into its final form by introducing the polar angle θ and azimuth φ of the direction of $\mathbf{p}$. The direction of $\mathbf{k}$ is chosen as the z axis, and the plane of $\mathbf{k}$ and $\mathbf{e}$ as the xz plane (so that $\mathbf{pe} = |\mathbf{p}| \cos \varphi \sin \theta$). When $\omega \gg I$, energy conservation can be written as $\varepsilon - m = \omega$ (rather than $\varepsilon - m = \omega - I$). It is readily verified that

$$\mathbf{k}^2 - \mathbf{p}^2 = -2m(\varepsilon - m), \quad (\mathbf{k} - \mathbf{p})^2 = 2\varepsilon(\varepsilon - m)(1 - v \cos \theta),$$

where $v = p/\varepsilon$ is the photoelectron velocity. Simple transformations then give

$$\begin{aligned} d\sigma = Z^5 \alpha^4 r_e^2 \frac{v^3 (1 - v^2)^3 \sin^2 \theta}{(1 - \sqrt{1 - v^2})^5 (1 - v \cos \theta)^4} \times \\ \times \left\{ \frac{(1 - \sqrt{1 - v^2})^2}{2(1 - v^2)^{3/2}} (1 - v \cos \theta) + \right. \\ \left. + \left[2 - \frac{(1 - \sqrt{1 - v^2})(1 - v \cos \theta)}{1 - v^2} \right] \cos^2 \varphi \right\} d\theta, \end{aligned} \quad (57.8)$$

where $r_e = e^2/m$.

In the ultrarelativistic case ($\varepsilon \gg m$), the photoelectric cross section has a sharp maximum at small angles ($\theta \sim \sqrt{1 - v^2}$); thus the electrons are emitted mainly in the direction of the incident photon. Near the maximum, write

$$1 - v \cos \theta \approx \frac{1}{2} [(1 - v^2) + \theta^2].$$

The leading terms in (57.8) then give

$$d\sigma \approx 4Z^5 \alpha^4 r_e^2 \frac{(1 - v^2)^{3/2}}{(1 - v^2 + \theta^2)^3} d\theta d\varphi. \quad (57.9)$$

An elementary, though fairly lengthy, integration of (57.8) over the angles yields the following formula for the total cross section (*F. Sauter*, 1931):

$$\begin{aligned} \sigma = 2\pi Z^5 \alpha^4 r_e^2 \frac{(\gamma^2 - 1)^{3/2}}{(\gamma - 1)^5} \times \\ \times \left\{ \frac{3}{4} + \frac{\gamma(\gamma - 2)}{\gamma + 1} \left(1 - \frac{1}{2\gamma\sqrt{\gamma^2 - 1}} \ln \frac{\gamma + \sqrt{\gamma^2 - 1}}{\gamma - \sqrt{\gamma^2 - 1}} \right) \right\}, \end{aligned} \quad (57.10)$$

where, for brevity, the ‘‘Lorentz factor’’ has been introduced:

$$\gamma = \frac{1}{\sqrt{1 - v^2}} = \frac{\varepsilon}{m} \approx \frac{m + \omega}{m}. \quad (57.11)$$

In the ultrarelativistic case this formula reduces to the simple expression

$$\sigma = 2\pi Z^5 \alpha^4 r_e^2 / \gamma. \quad (57.12)$$

In the case $I \ll \omega \ll m$, on the other hand, taking the limit of small $\gamma - 1$ in (57.10) gives the result (56.14), already obtained above.

§ 58. Photodisintegration of the deuteron

The deuteron is distinguished by its small binding energy compared with the depth of the potential well. Because of this, reactions involving a deuteron can be described using the binding energy alone, without detailed knowledge of the nuclear forces (see III, § 133). The wavelengths of the colliding particles are assumed to be large relative to the range a of the nuclear forces.

This also applies when a deuteron is broken up by γ quanta satisfying $ka \ll 1$. We shall assume in addition that $pa \ll 1$, where $\mathbf{p}$ is the momentum of the relative motion of the released neutron and proton (this condition is stronger than the preceding one⁴⁸).

We begin with the nonrelativistic photoelectric cross section (56.5), integrated over directions:

$$\sigma = \frac{e^2 p}{2\pi\omega} \frac{M}{2} \frac{4\pi}{3} |(\mathbf{v}_p)_{fi}|^2.$$

Here $\mathbf{p}$ is the relative-motion momentum of the proton and neutron⁴⁹, while the mass m in (56.5) has been replaced by their reduced mass $M/2$ (M is the nucleon mass). The matrix element is taken for the proton velocity $\mathbf{v}_p$, since only the proton interacts with the photon. On expressing $\mathbf{v}_p$ through the momentum $\mathbf{p}$ ($\mathbf{v}_p = \mathbf{v}/2 = \mathbf{p}/M$), we obtain

$$\sigma^{(e)} = \frac{e^2 p}{3M\omega} |\mathbf{p}_{fi}|^2. \quad (58.1)$$

The superscript e indicates that this formula describes electric-dipole transitions: $e\mathbf{p}/M = e\mathbf{v}_p = \dot{\mathbf{d}}$, and hence $e\mathbf{p}_{fi}/M = i\omega\mathbf{d}_{fi}$.

The normalized wavefunction of the initial (ground) state of the deuteron is

$$\psi = \sqrt{\frac{\kappa}{2\pi}} \frac{e^{-\kappa r}}{r}, \quad \kappa = \sqrt{MI}, \quad (58.2)$$

where $I = 2.23$ MeV is the binding energy (see III, § 133)⁵⁰. For the final state, however, we may use a free-motion wavefunction, namely a plane wave,

$$\psi' = e^{i\mathbf{p}\mathbf{r}}. \quad (58.3)$$

The reason is that the “deuteron size” $1/\kappa$ is treated in this theory as large in comparison with the effective interaction range a . The proton-neutron interaction must therefore be retained only in S states; it may be neglected in states with $l \neq 0$, whose wavefunctions are small at short distances. Meanwhile, the selection rules prohibit

⁴⁸The photon energy at which $pa \approx 1$ ($a = 1.5 \cdot 10^{-13}$ cm) is 15 MeV.

⁴⁹In this section, p denotes $|\mathbf{p}|$.

⁵⁰A correction due to the finite value of a can be introduced into this function. It amounts to replacing the normalization coefficient in (58.2) by

$$\sqrt{\frac{\kappa}{2\pi(1 - a\kappa)}}$$

(see III, (133.13)). The factor $1/(1 - a\kappa)$ then also appears in the cross-section formulas. In fact, this correction is not especially small: for the deuteron ground state, $a\kappa \approx 0.4$.

The deuteron ground state is a 3S_1 state with a small admixture of the 3D_1 state caused by tensor nuclear forces (see III, § 117). We shall neglect this admixture, and with it the tensor forces.

electric-dipole transitions between two S states (the ground state and an S state of the continuous spectrum). This is what permits us here to disregard the nucleon interaction in the final state.

Integration by parts gives the matrix element

$$\begin{aligned}\mathbf{p}_{fi} &= -i\sqrt{\frac{\kappa}{2\pi}} \int e^{-i\mathbf{p}\mathbf{r}} \nabla \frac{e^{-\kappa r}}{r} d^3x = \sqrt{\frac{\kappa}{2\pi}} \mathbf{p} \left(\frac{e^{-\kappa r}}{r} \right)_p \\ &= \sqrt{\frac{\kappa}{2\pi}} \frac{4\pi\mathbf{p}}{p^2 + \kappa^2}\end{aligned}$$

(see the note on p. 246).

Also using the equality

$$\frac{1}{M}(\kappa^2 + p^2) = I + \frac{p^2}{M} = \omega,$$

which expresses energy conservation, we arrive at the photodisintegration cross section (in ordinary units)

$$\sigma^{(e)} = \frac{8\pi}{3} \alpha \frac{\hbar^2}{M} \frac{\sqrt{I}(\hbar\omega - I)^{3/2}}{(\hbar\omega)^3}. \quad (58.4)$$

(*H. A. Bethe, R. Peierls, 1935*). It reaches its maximum at $\hbar\omega = 2I$ and tends to zero both as $\hbar\omega \rightarrow I$ and as $\hbar\omega \rightarrow \infty$.

The electric-dipole photon absorption represented by (58.4) does not, however, furnish the leading contribution to the cross section near the photoelectric threshold ($\hbar\omega$ close to I). In that region the principal effect must arise from transitions to an S state, which electric-dipole absorption does not contain. Such transitions are also absent from electric-quadrupole absorption: although parity does not forbid them in this case, the orbital-angular-momentum selection rule does (recall that tensor forces have been neglected, so that L and S are conserved separately). To calculate photodisintegration near threshold, we must therefore consider magnetic-dipole absorption, whose selection rules allow transitions between S states (*E. Fermi, 1935*).

Replacing the electric moment in (58.1) by the magnetic moment gives

$$\sigma^{(m)} = \frac{1}{3} \omega M p |\boldsymbol{\mu}_{fi}|^2. \quad (58.5)$$

The magnetic moment of the orbital motion contributes nothing to $\boldsymbol{\mu}_{fi}$, because the orbital angular momentum $\mathbf{L}$ has no matrix elements for transitions between S states. The spin magnetic moment is

$$\boldsymbol{\mu} = 2\mu_p \mathbf{s}_p + 2\mu_n \mathbf{s}_n = 2(\mu_p - \mu_n) \mathbf{s}_p + 2\mu_n \mathbf{S},$$

where $\mathbf{S} = \mathbf{s}_p + \mathbf{s}_n$, and μ_p, μ_n are the magnetic moments of the proton and neutron. When tensor nuclear forces are neglected, the total spin is conserved and its operator produces no transitions. Hence

$$\boldsymbol{\mu}_{fi} = 2(\mathbf{s}_p)_{fi}(\mu_p - \mu_n).$$

In the same approximation (without tensor forces), the spin and coordinate variables separate. The matrix element, like the wavefunctions, therefore factors into spin and coordinate parts:

$$\mu_{fi} = 2(\mu_p - \mu_n) \langle s_p S' M' | s_p | s_p S M \rangle \int \psi'^*(\mathbf{r}) \psi(\mathbf{r}) d^3x.$$

The presence of spin-spin nuclear forces nevertheless means that the wave equation for the coordinate functions $\psi(\mathbf{r})$ contains the spin value S as a parameter. If $S' = S$, then $\psi'(r)$ and $\psi(r)$ are eigenfunctions of the same operator and are consequently orthogonal. It follows that photodisintegration from the initial 3S state can lead only to a 1S state in the continuous spectrum. The square $|\mu_{fi}|^2$ in (58.5) must of course be averaged over the projections M of the initial spin S . The calculation thus reduces to the quantity

$$\frac{1}{2S+1} \sum_M |\langle s_p S' M' | s_p | s_p S M \rangle|^2,$$

where $s_p = s_n = 1/2$, $S = 1$, and $S' = 0$. The general formulas for matrix elements under angular-momentum coupling give

$$\begin{aligned} & \frac{1}{(2S+1)(2S'+1)} |\langle s_p S' \| s_p \| s_p S \rangle|^2 \\ &= \left\{ \begin{matrix} s_p & S' & s_n \\ S & s_p & 1 \end{matrix} \right\}^2 |\langle s_p \| s_p \| s_p \rangle|^2 = \frac{1}{6} |\langle s_p \| s_p \| s_p \rangle|^2 \end{aligned}$$

(formulas III, (107.11) and (109.3) have been used). The reduced matrix element is

$$\langle s_p \| s_p \| s_p \rangle = \sqrt{s_p(s_p+1)(2s_p+1)} = \sqrt{3/2}.$$

Formula (58.5) therefore becomes

$$\sigma^{(m)} = \frac{1}{3} \omega M p (\mu_p - \mu_n)^2 \left| \int \psi'^* \psi d^3x \right|^2. \quad (58.6)$$

The initial function ψ is given by (58.2). The final function is

$$\psi' = \frac{1}{2p} R_{p0}(r).$$

This is the first ($l = 0$) term in expansion (56.7) of the function that asymptotically contains a plane wave and an incoming spherical wave; an inessential phase factor has been omitted. Since the integration is over the region outside the range of the nuclear forces, the radial function is

$$R_{p0}(r) = 2 \frac{\sin(pr + \delta)}{r}.$$

The phase δ is related to the energy of the virtual level ($I_1 = 0.067$ MeV) of the proton+neutron system with $S = 0$:

$$\cot \delta = \frac{\kappa_1}{p}, \quad \kappa_1 = \sqrt{M I_1}$$

(see III, § 133). We now have

$$\begin{aligned} \int \psi'^* \psi d^3x &= (2\pi)^{3/2} \frac{\sqrt{\kappa}}{p\pi} \operatorname{Im} \int e^{-\kappa r + i p r} e^{i\delta} dr \\ &= (2\pi)^{3/2} \frac{\sqrt{\kappa}}{p\pi} \operatorname{Im} \frac{e^{i\delta}}{\kappa - ip}. \end{aligned}$$

Straightforward algebra then yields the following photodisintegration cross section (in ordinary units):

$$\sigma^{(m)} = \frac{8\pi}{3\hbar c} (\mu_p - \mu_n)^2 \frac{\sqrt{I(\hbar\omega - I)} (\sqrt{I} + \sqrt{I_1})^2}{\hbar\omega(\hbar\omega - I + I_1)}. \quad (58.7)$$

As $\hbar\omega \rightarrow I$, this cross section vanishes as $\sqrt{\hbar\omega - I}$, in accordance with the general threshold behavior of reaction cross sections (see III, § 147).

The inverse of photodisintegration is the radiative capture of a proton by a neutron. The capture cross section (σ_c) follows from the photoelectric cross section (σ_{ph}) by detailed balance (compare the derivation of (56.15)). The combined spin statistical weight of the neutron and proton is $2 \cdot 2 = 4$. The corresponding weight of the deuteron (in the state with $S = 1$) and photon is $3 \cdot 2 = 6$. Therefore

$$\sigma_c = \frac{3}{2} \frac{(\hbar\omega)^2}{c^2 p^2} \sigma_{ph} = \frac{3(\hbar\omega)^2}{2Mc^2(\hbar\omega - I)} \sigma_{ph}. \quad (58.8)$$

Scattering of Light

§ 59. The scattering tensor

Scattering of a photon by an electron system (for definiteness, we shall speak of an atom) consists in absorption of the initial photon $\mathbf{k}$ accompanied by emission of another photon $\mathbf{k}'$. The atom may remain either in its initial energy level or in another discrete level. In the first case the photon frequency is unchanged (*Rayleigh*, or *unshifted*, *scattering*); in the second it changes by

$$\omega' - \omega = E_1 - E_2, \quad (59.1)$$

where E_1 and E_2 are the initial and final atomic energies (*combination*, or *shifted*, *scattering*).¹ If the initial atomic state is the ground state, then $E_2 > E_1$ in combination scattering, so that $\omega' < \omega$: the frequency decreases (the *Stokes* case). For scattering by an excited atom, both the Stokes and the *anti-Stokes* ($\omega' > \omega$) cases are possible.

Since the electromagnetic-perturbation operator has no matrix elements for transitions in which two photon occupation numbers change simultaneously, scattering first appears in second-order perturbation theory. It must be regarded as proceeding through intermediate states of two possible types:

I. The photon $\mathbf{k}$ is absorbed and the atom passes to one of its possible states E_n ; in the subsequent transition to the final state, the photon $\mathbf{k}'$ is emitted.

II. The photon $\mathbf{k}'$ is emitted and the atom passes to a state E_n ; in the transition to the final state, the photon $\mathbf{k}$ is absorbed.

The role of the matrix element for this process is played by the sum (see vol. III, (43.7))

$$V_{21} = \sum_n' \left(\frac{V'_{2n} V_{n1}}{\mathcal{E}_1 - \mathcal{E}_n^I} + \frac{V_{2n} V'_{n1}}{\mathcal{E}_1 - \mathcal{E}_n^{II}} \right), \quad (59.2)$$

where the initial energy of the “atom+photons” system is $\mathcal{E}_1 = E_1 + \omega$, while the intermediate-state energies are

$$\mathcal{E}_n^I = E_n, \quad \mathcal{E}_n^{II} = E_n + \omega + \omega';$$

V denotes matrix elements for absorption of the photon $\mathbf{k}$, and V' those for emission of the photon $\mathbf{k}'$; the initial state is excluded from the sum over n (as indicated by the prime). The scattering cross section is

$$d\sigma = 2\pi |V_{21}|^2 \frac{\omega'^2 d\omega'}{(2\pi)^3}, \quad (59.3)$$

¹In this chapter, quantities referring to the initial and final states of the scattering system carry the indices 1 and 2.

where do' is the solid-angle element for the directions of $\mathbf{k}'$. The light energy dI' scattered per second into do' is related to the incident-light intensity I (energy-flux density) by

$$dI' = I \frac{\omega'}{\omega} d\sigma.$$

We suppose that the wavelengths of the initial and final photons are large compared with the size a of the scattering system, and accordingly treat every transition in the dipole approximation. If the photon states are described by plane waves, this approximation amounts to replacing the factors $e^{i\mathbf{k}\mathbf{r}}$ by unity. The photon wave functions (in the three-dimensionally transverse gauge) are then

$$\mathbf{A}_{\mathbf{e}\omega} = \sqrt{4\pi} \frac{\mathbf{e}}{\sqrt{2\omega}} e^{-i\omega t}, \quad \mathbf{A}_{\mathbf{e}'\omega'} = \sqrt{4\pi} \frac{\mathbf{e}'}{\sqrt{2\omega'}} e^{-i\omega' t}.$$

Under these conditions the electromagnetic interaction operator can be written

$$\hat{V} = -\hat{\mathbf{d}}\hat{\mathbf{E}}, \quad (59.4)$$

where $\hat{\mathbf{E}} = -\hat{\mathbf{A}}$ is the field-strength operator and $\hat{\mathbf{d}}$ the atomic dipole-moment operator (as in the classical expression for the energy of a small system in an electric field; see vol. II, § 42). Its matrix elements are

$$V_{n1} = -i\sqrt{2\pi\omega} (\mathbf{e}\mathbf{d}_{n1}), \quad V'_{2n} = i\sqrt{2\pi\omega'} (\mathbf{e}'^*\mathbf{d}_{2n}).$$

Substitution in (59.2), (59.3) gives the scattering cross section (in ordinary units)²:

$$d\sigma = \left| \sum_n \left\{ \frac{(\mathbf{d}_{2n}\mathbf{e}'^*)(\mathbf{d}_{n1}\mathbf{e})}{\omega_{n1} - \omega - i0} + \frac{(\mathbf{d}_{2n}\mathbf{e})(\mathbf{d}_{n1}\mathbf{e}'^*)}{\omega_{n1} + \omega - i0} \right\} \right|^2 \frac{\omega\omega'^3}{\hbar^2 c^4} do', \quad (59.5)$$

$$\hbar\omega_{n1} = E_n - E_1, \quad \omega' - \omega = \omega_{12}.$$

The sum extends over all possible atomic states, including continuum states (states 1 and 2 drop out automatically because $d_{11} = d_{22} = 0$). The infinitesimal imaginary additions in the denominators implement the usual pole-bypass rule of perturbation theory (see vol. III, § 43): an infinitesimal negative imaginary part is added to the intermediate-state energies E_n being summed. The prescription matters when poles of (59.5), regarded as a function of E_n , lie in the continuum (if state 1 is the atomic ground state, for example, $\hbar\omega$ must then exceed the ionization threshold).³

Introduce the notation (ordinary units)⁴

$$(c_{ik})_{21} = \frac{1}{\hbar} \sum_n \left[\frac{(d_i)_{2n}(d_k)_{n1}}{\omega_{n1} - \omega - i0} + \frac{(d_k)_{2n}(d_i)_{n1}}{\omega_{n1} + \omega - i0} \right] \quad (59.6)$$

²This formula was first obtained by Kramers and Heisenberg (*H. A. Kramers, W. Heisenberg, 1925*), before quantum mechanics was developed.

³For a molecule, the dissociation threshold into atoms plays the role of the ionization threshold here.

⁴Most of the results presented in §§ 59–61 are due to Placzek (*G. Placzek, 1931–1933*).

($i, k = x, y, z$ are three-dimensional vector indices). Formula (59.5) then becomes

$$d\sigma = \frac{\omega(\omega + \omega_{12})^3}{c^4} |(c_{ik})_{21} e_i'^* e_k|^2 d\omega'. \quad (59.7)$$

The notation (59.6) is justified because the sum can indeed be represented as a matrix element of a tensor. This is seen most easily by introducing a vector $\mathbf{b}$ whose operator satisfies

$$i\dot{\mathbf{b}} + \omega\mathbf{b} = \dot{\mathbf{d}}.$$

Its matrix elements are

$$\mathbf{b}_{n1} = \frac{\mathbf{d}_{n1}}{\omega - \omega_{n1}}, \quad \mathbf{b}_{2n} = \frac{\mathbf{d}_{2n}}{\omega + \omega_{n2}},$$

so that

$$(c_{ik})_{21} = (b_k d_i - d_i b_k)_{21}. \quad (59.8)$$

We shall call the matrix elements $(c_{ik})_{21}$ the *light-scattering tensor*.

It follows that the selection rules for scattering coincide with those for matrix elements of an arbitrary second-rank tensor. In particular, if the system has a center of symmetry (so its states can

be classified by parity), transitions are possible only between states of the same parity, including transitions with no change of state. This is the reverse of the parity selection rule for electric-dipole radiation; there is an alternating prohibition: transitions allowed in radiation are forbidden in scattering, and those allowed in scattering are forbidden in radiation.

Decompose c_{ik} into irreducible parts:

$$c_{ik} = c^0 \delta_{ik} + c_{ik}^s + c_{ik}^a, \quad (59.9)$$

where

$$c^0 = \frac{1}{3} c_{ii}, \quad c_{ik}^s = \frac{1}{2} (c_{ik} + c_{ki}) - c^0 \delta_{ik}, \quad c_{ik}^a = \frac{1}{2} (c_{ik} - c_{ki}) \quad (59.10)$$

are, respectively, a scalar, a symmetric traceless tensor, and an antisymmetric tensor. Their matrix elements are

$$(c^0)_{21} = \frac{1}{3} \sum_n \frac{\omega_{n1} + \omega_{n2}}{(\omega_{n1} - \omega)(\omega_{n2} + \omega)} (\mathbf{d}_{2n} \mathbf{d}_{n1}), \quad (59.11)$$

$$(c_{ik}^s)_{21} = \frac{1}{2} \sum_n \frac{\omega_{n1} + \omega_{n2}}{(\omega_{n1} - \omega)(\omega_{n2} + \omega)} [(d_i)_{2n} (d_k)_{n1} + (d_k)_{2n} (d_i)_{n1}] - (c^0)_{21} \delta_{ik}, \quad (59.12)$$

$$(c_{ik}^a)_{21} = \frac{2\omega + \omega_{12}}{2} \sum_n \frac{(d_i)_{2n} (d_k)_{n1} - (d_k)_{2n} (d_i)_{n1}}{(\omega_{n1} - \omega)(\omega_{n2} + \omega)} \quad (59.13)$$

(the pole-bypass signs have been omitted for brevity).

Consider some properties of the scattering tensor in the limiting cases of low and high photon frequencies.⁵

For unshifted scattering ($\omega_{12} = 0$), the antisymmetric part vanishes as $\omega \rightarrow 0$ because of the factor ω preceding the sum in (59.13). The scalar and symmetric parts, however, approach finite limits. The cross section at low ω is therefore proportional to ω^4 .

In the opposite case, where ω is large compared with all frequencies ω_{n1} and ω_{n2} relevant in (59.6) (while the wavelength, of course, still satisfies $\gg a$), the classical formulas must be recovered. The first term in the expansion in powers of $1/\omega$ is

$$\frac{1}{\omega} \sum_n [(d_k)_{2n}(d_i)_{n1} - (d_i)_{2n}(d_k)_{n1}] = \frac{1}{\omega} (d_k d_i - d_i d_k)_{21}$$

and vanishes because d_i and d_k commute. The next term is

$$\begin{aligned} (c_{ik})_{21} &= \frac{1}{\omega^2} \sum_n \omega_{n1} [(d_i)_{2n}(d_k)_{n1} - (d_k)_{2n}(d_i)_{n1}] \\ &= \frac{1}{i\omega^2} (\dot{d}_i d_k - d_i \dot{d}_k)_{21}. \end{aligned}$$

Using $\mathbf{d} = \sum \mathbf{e} \mathbf{r}$ (the sum is over all atomic electrons) and the momentum–coordinate commutation rules, we obtain

$$(c_{ik})_{11} = -\frac{Ze^2}{m\omega^2} \delta_{ik}, \quad (c_{ik})_{21} = 0, \quad (59.14)$$

where Z is the total number of electrons in the system and m the electron mass. Thus at high frequencies only the scalar part remains, and scattering occurs without a change in the state of the system (it is entirely coherent; see below). The cross section is then

$$d\sigma = r_e^2 Z^2 |\mathbf{e}'^* \mathbf{e}|^2 d\omega', \quad (59.15)$$

where $r_e = e^2/m$. Summing over final-photon polarizations gives

$$d\sigma = r_e^2 Z^2 \{1 - (\mathbf{e}\mathbf{e}')^2\} d\omega' = r_e^2 Z^2 \sin^2 \theta \cdot d\omega', \quad (59.16)$$

which is indeed the classical Thomson formula (see vol. II, (80.7); θ is the angle between the scattering direction and the incident photon's polarization vector).

Consider scattering by N identical atoms occupying a volume whose dimensions are small compared with the wavelength. The scattering tensor of the collection is the sum of the tensors of the individual atoms. We must, however, take account of the fact that the wave functions used to calculate dipole-moment matrix elements for several identical atoms considered together cannot simply be regarded as identical. A wave function is intrinsically defined only up to an arbitrary phase factor, and each atom has its own. The scattering cross section must therefore be averaged independently over the phase factors of the atoms.

⁵The resonant case, in which ω is close to one of the frequencies ω_{n1} or ω_{n2} , will be considered in § 63.

The tensor $(c_{ik})_{21}$ of each atom contains the factor $e^{i(\varphi_1 - \varphi_2)}$, where φ_1 and φ_2 are the phases of the initial- and final-state wave functions. In shifted scattering states 1 and 2 differ, so this factor is not unity. In the squared modulus

$$\left| e_i^* e_k \sum (c_{ik})_{21} \right|^2$$

(the sum is over all N atoms), products of terms belonging to different atoms contain phase factors that vanish under independent averaging over the atomic phases; only the squared moduli of the individual terms remain. Thus the total cross section for N atoms is N times the single-atom cross section: the scattering is incoherent.

If the initial and final atomic states coincide, the factors $e^{i(\varphi_1 - \varphi_2)} = 1$. The scattering amplitude of the collection is then N times the single-atom amplitude, and the cross section is correspondingly larger by N^2 : the scattering is coherent.⁶ Thus, if the atomic energy level is nondegenerate, unshifted scattering is completely coherent. If it is degenerate, there is also incoherent unshifted scattering due to transitions between distinct mutually degenerate states. The latter is a purely quantum effect: in classical theory, scattering without a frequency change is always coherent.

The coherent-scattering tensor is the diagonal matrix element $(c_{ik})_{11}$; denote it by α_{ik} (suppressing the index that should indicate the atomic state). According to (59.6),

$$\alpha_{ik}(\omega) = (c_{ik})_{11} = \sum_n \left[\frac{(d_i)_{1n}(d_k)_{n1}}{\omega_{n1} - \omega - i0} + \frac{(d_k)_{1n}(d_i)_{n1}}{\omega_{n1} + \omega - i0} \right]. \quad (59.17)$$

This can also be written

$$\alpha_{ik}(\omega) = \frac{e^2}{m\omega^2} \left\{ -Z\delta_{ik} + \frac{1}{m} \sum_n \left[\frac{(p_i)_{1n}(p_k)_{n1}}{\omega_{n1} - \omega - i0} + \frac{(p_k)_{1n}(p_i)_{n1}}{\omega_{n1} + \omega - i0} \right] \right\}, \quad (59.18)$$

where the limiting expression (59.14) has been separated out. Here $\mathbf{p}$ is the total momentum of the atomic electrons. The equivalence follows at once from

$$\frac{e}{m} \mathbf{p}_{1n} = i\omega_{1n} \mathbf{d}_{1n},$$

together with the relations used in deriving (59.14).

If neither $E_1 + \omega$ nor $E_1 - \omega$ coincides with an atomic energy level E_n , including in the continuum, the $i0$ terms may be omitted. Since $\mathbf{p}_{1n}^* = \mathbf{p}_{n1}$, the tensor is then Hermitian⁷:

$$\alpha_{ik} = \alpha_{ki}^*. \quad (59.19)$$

Its scalar and symmetric parts are therefore real, while its antisymmetric part is imaginary. Notice that the antisymmetric part necessarily vanishes if the atom is in a nondegenerate

⁶The factor Z^2 in (59.15), (59.16) has the same origin: the cross section for coherent scattering by the Z electrons of one atom is Z^2 times that for one electron.

⁷This result follows from neglecting the natural line width, and hence the possibility of absorption of the incident light (see § 62).

state: the wave function of such a state is real, and hence so are the diagonal matrix elements.

The tensor α_{ik} is related to the polarizability of an atom in an external electric field. To establish the relation, calculate the correction to the mean dipole moment when the system is placed in the field

$$\frac{1}{2} (\mathbf{E}e^{-i\omega t} + \mathbf{E}^*e^{i\omega t}). \quad (59.20)$$

We use the standard perturbation-theory formula (see vol. III, § 40): if the perturbation is

$$\widehat{V} = \widehat{F}e^{-i\omega t} + \widehat{F}^+e^{i\omega t},$$

the first-order correction to the diagonal matrix elements of a quantity f is

$$f_{11}^{(1)}(t) = - \sum_n \left\{ \left[\frac{f_{1n}^{(0)} F_{n1}}{\omega_{n1} - \omega - i0} + \frac{f_{n1}^{(0)} F_{1n}}{\omega_{n1} + \omega + i0} \right] e^{-i\omega t} + \left[\frac{f_{1n}^{(0)} F_{1n}^*}{\omega_{n1} + \omega - i0} + \frac{f_{n1}^{(0)} F_{n1}^*}{\omega_{n1} - \omega + i0} \right] e^{i\omega t} \right\}$$

(the perturbation must be regarded as switched on infinitely slowly from $t = -\infty$, so ω is understood as $\omega + i0$ in the first term and as $\omega - i0$ in the second; the imaginary additions in the denominators have been written accordingly).

Here $\widehat{F} = -\mathbf{d}\mathbf{E}/2$, and the correction to the diagonal dipole-moment matrix element is

$$\mathbf{d}_{11}^{(1)} = \frac{1}{2} (\bar{\mathbf{d}}e^{-i\omega t} + \bar{\mathbf{d}}^*e^{i\omega t}), \quad (59.21)$$

where $\bar{\mathbf{d}}$ has components

$$\bar{d}_i = \alpha_{ik}^{(\text{pol})} E_k, \quad (59.22)$$

and the expression for $\alpha_{ik}^{(\text{pol})}(\omega)$ differs from (59.17) for α_{ik} by the reversed sign of the imaginary addition in the denominator of the second term. By definition, $\alpha_{ik}^{(\text{pol})}(\omega)$ is the tensor of atomic polarizability in a field of frequency ω . At frequencies for which the imaginary additions may be omitted and α_{ik} is Hermitian, α_{ik} and $\alpha_{ik}^{(\text{pol})}$ simply coincide. In particular, at $\omega = 0$, (59.22) becomes formula (76.4) of vol. III, and the static-polarizability tensor (vol. III, (76.5)) coincides with $\alpha_{ik}(0)$ from (59.17). Notice also that if state 1 is the ground state,⁸ all $\omega_{n1} > 0$; the bypass prescription in the first term of (59.17) then matters only for $\omega > 0$, and that in the second only for $\omega < 0$. Hence

$$\alpha_{ik}(\omega) = \alpha_{ik}^{(\text{pol})}(|\omega|). \quad (59.23)$$

The scattering formulas implicitly take $\omega > 0$; the tensor α_{ik} then coincides with the polarizability tensor.

⁸Only this case, which will be understood below, permits a fully rigorous treatment because excited states have finite lifetimes (see § 62).

We shall also need the photon scattering amplitude f . As usual in perturbation theory, apart from a normalization factor it is the negative of the matrix element (59.2). Choosing the factor so that (59.7) becomes $d\sigma = |f|^2 d\omega'$, we find for elastic scattering

$$f = \omega^2 \alpha_{ik} e_i^* e_k. \quad (59.24)$$

By the optical theorem (see (71.10) below), the imaginary part of the forward-scattering amplitude (with neither momentum nor polarization changed) determines the total cross section σ_t of all possible elastic and inelastic processes for the given initial photon state:

$$\sigma_t = \frac{4\pi}{\omega} \operatorname{Im}(\omega^2 \alpha_{ik} e_i^* e_k) = 4\pi\omega \frac{\alpha_{ik} - \alpha_{ki}^*}{2i} e_i^* e_k. \quad (59.25)$$

Thus the total cross section is determined by the anti-Hermitian part of the scattering tensor.

Formula (59.25) has a simple classical meaning. The electric field $\mathbf{E}$ does work on the system of charges at the rate $\sum e \mathbf{v} \mathbf{E} = \mathbf{E} \mathbf{d}$. Writing the field as in (59.20), the dipole moment as in (59.21), (59.22), and averaging this work over time gives

$$\frac{1}{2} \omega |E|^2 e_i^* e_k \frac{\alpha_{ik} - \alpha_{ki}^*}{2i}$$

($\mathbf{E} = \mathbf{e}E$). On the other hand, if $\mathbf{E}$ is the incident-light field, its mean energy-flux density is $|E|^2/(8\pi)$, and the energy absorbed by the atom is

$$\frac{1}{8\pi} |E|^2 \sigma_t.$$

Equating these expressions gives (59.25).

If the ground-state angular momentum is $J = 0$, spherical symmetry gives $\alpha_{ik} = \alpha \delta_{ik}$. Then

$$\sigma_t = 4\pi\omega \operatorname{Im} \alpha. \quad (59.26)$$

For a system with angular momentum, the same relation holds for quantities averaged over its directions in space (see § 60).

Above the atomic ionization threshold, the principal contribution to σ_t is ionization, namely photon absorption in the photoelectric effect. The scattering cross section is of higher order in e^2 (compare, for example, (56.13) and (59.16)).

Below the ionization threshold, but away from resonance with a discrete atomic excitation frequency, the cross section reduces to the scattering cross section; it, and hence the imaginary part of the amplitude, is of higher order than the real part. Neglecting the imaginary part again gives (59.19). Near resonance the cross section increases; that case will be considered in § 63.

Alongside scattering, the two-photon processes appearing in second-order perturbation theory include *double emission*, the simultaneous emission of two quanta by an atom.

Its probability differs from (59.5) only by the substitutions $\omega \rightarrow -\omega$, $\mathbf{e} \rightarrow \mathbf{e}^*$ (emission rather than absorption of the photon ω), and by the additional factor

$$\frac{d^3 k}{(2\pi)^3} = \frac{\omega^2 d\omega d\Omega}{(2\pi)^3},$$

the number of quantum states of the emitted photon in the specified intervals of frequency ω and direction $\mathbf{k}$; the second photon frequency is fixed by $\omega + \omega' = \omega_{12}$. The emission probability per unit time is therefore⁹

$$dw = |(b_{ik})_{21} e_i'^* e_k^*|^2 \frac{\omega^3 \omega'^3}{(2\pi)^3 c^6 \hbar^2} d\omega d\omega' d\omega, \quad (59.27)$$

where

$$(b_{ik})_{21} = \sum_n \left[\frac{(d_i)_{2n}(d_k)_{n1}}{\omega_{n1} + \omega - i0} + \frac{(d_k)_{2n}(d_i)_{n1}}{\omega_{n1} + \omega' - i0} \right]$$

differs from $(c_{ik})_{21}$ in (59.6) only in the sign before ω . Summing over photon polarizations and integrating over their emission directions¹⁰ gives

$$dw = \frac{8\omega^3 \omega'^3}{9\pi \hbar^2 c^6} |(b_{ik})_{21}|^2 d\omega. \quad (59.28)$$

The probability of emitting two photons ω and ω' is usually very small compared with that of emitting one photon of frequency $\omega + \omega'$. Exceptions occur when selection rules forbid the latter process but allow the former. Examples are transitions between two $J = 0$ states, for which every one-photon emission process is strictly forbidden, and the transition from the first excited state of hydrogen ($2s_{1/2}$) to the ground state ($1s_{1/2}$), which is forbidden for both $E1$ and $M1$ radiation (see problem 2, § 52).¹¹

If the atom is in an incident field of photons $\omega, \mathbf{k}$, then in addition to spontaneous double emission, whose probability is (59.27), stimulated double emission is possible: the field induces emission of another identical photon and, with it, a photon $\omega', \mathbf{k}'$. Its probability differs from the spontaneous probability by the factor $N_{\mathbf{k}\mathbf{e}}$, the number density of incident photons with specified $\mathbf{k}, \mathbf{e}$. The incident photon-flux density is

$$dI = c N_{\mathbf{k}\mathbf{e}} \frac{d^3 k}{(2\pi)^3} = N_{\mathbf{k}\mathbf{e}} \frac{\omega^2}{8\pi^3 c^2} d\omega d\Omega.$$

Solving for $N_{\mathbf{k}\mathbf{e}}$ and dividing the process probability by dI gives its cross section

$$d\sigma = \frac{\omega \omega'^3}{\hbar^2 c^4} |(b_{ik})_{12} e_i'^* e_k^*|^2 d\omega'. \quad (59.29)$$

Similarly, if the atom is in a field of photons $\omega', \mathbf{k}'$, incidence of a photon $\omega, \mathbf{k}$ produces *stimulated combination scattering*, whose cross section is proportional to the number density of the $\omega', \mathbf{k}'$ photons.

Calculation of $(c_{ik})_{12}$ or $(b_{ik})_{12}$ for particular atoms requires sums of the form

$$(M_{ik}^{(2)})_{21} = \sum_n \frac{(d_i)_{2n}(d_k)_{n1}}{E_n - E - i0}, \quad (59.30)$$

⁹Here and below in this section, ordinary units are used.

¹⁰This operation reduces to complete averaging over the directions of $\mathbf{e}$ according to $\overline{e_i e_k^*} = \delta_{ik}/3$, followed by multiplication by $2 \cdot 2 \cdot 4\pi \cdot 4\pi$.

¹¹The lifetime of the $2s_{1/2}$ level due to double emission is 0.15 s.

where E takes the values $E_1 \pm \hbar\omega$ or $E_1 \pm \hbar\omega'$. For brevity, consider a hydrogen atom. Write (59.30) as the integral

$$(M_{ik}^{(2)})_{21} = \int \psi_2^*(\mathbf{r}) d_i G(\mathbf{r}, \mathbf{r}'; E) d'_k \psi_1(\mathbf{r}') d^3x d^3x', \quad (59.31)$$

where

$$G(\mathbf{r}, \mathbf{r}'; E) = \sum_n \frac{\psi_n(\mathbf{r}) \psi_n^*(\mathbf{r}')}{E_n - E - i0}. \quad (59.32)$$

Act on G with $\hat{H} - E$, where $\hat{H}$ is the atomic Hamiltonian. Since $\hat{H}\psi_n = E_n\psi_n$,

$$(\hat{H} - E)G = \sum_n \psi_n(\mathbf{r}) \psi_n^*(\mathbf{r}').$$

By completeness, the sum is $\delta(\mathbf{r} - \mathbf{r}')$. Thus G satisfies

$$(\hat{H} - E)G(\mathbf{r}, \mathbf{r}'; E) = \delta(\mathbf{r} - \mathbf{r}'), \quad (59.33)$$

and is the Green function of the Schrödinger equation (the bypass prescription in (59.32) selects the required solution). The calculation of (59.30) is thereby reduced to finding the atomic Green function. An exact solution of (59.33), however, is possible only when exact solutions of the homogeneous Schrödinger equation are known, in practice only for hydrogen.¹²

PROBLEM

Calculate the probability for elastic scattering of a nonrelativistic electron by an almost monochromatic standing light wave (*P. L. Kapitza, P. A. M. Dirac, 1933*).

SOLUTION. A standing wave may be regarded as a collection of photons with momenta $\mathbf{k}$ and $-\mathbf{k}$ and identical polarizations. Electron scattering may be regarded as absorption of a photon of momentum $\mathbf{k}$ followed by stimulated emission of a photon of momentum $-\mathbf{k}$. The electron momentum $\mathbf{p}$ thereby increases by $2\hbar\mathbf{k}$, turning through an angle θ without changing its magnitude:

$|\mathbf{p}| \sin(\theta/2) = \hbar\omega/c$. The process probability follows from the Thomson-scattering cross section (59.15),

$$d\sigma = r_e^2 |\mathbf{e}^* \cdot \mathbf{e}|^2 d\omega' = r_e^2 d\omega',$$

by multiplying it by the flux density of photons of momentum $\mathbf{k}$ and the number of photons of momentum $-\mathbf{k}$.

The flux density of photons with frequencies in $d\omega$ is

$$\frac{cU_\omega d\omega}{2\hbar\omega},$$

where $U_\omega d\omega$ is the standing-wave energy density in the spectral interval $d\omega$ (the factor 1/2 accounts for the equal division of the wave energy between photons moving in

¹²See *Hostler L./J. Math. Phys.* — 1964. — Vol. 5. — P. 591. For an application of this Green function to the scattering amplitude of hydrogen, see *Granovsky Ya. I./ZhETF.* — 1969. — Vol. 56. — P. 605.

opposite directions). The momenta $\mathbf{k}$ of all photons forming the standing wave are parallel to a specified direction $\mathbf{n}$ (the “direction” of the standing wave). Thus the energy density as a function of photon frequency and direction $\mathbf{n}'$ is $U_{\omega\mathbf{n}'} = U_{\omega}\delta^{(2)}(\mathbf{n}' - \mathbf{n})$. Accordingly, the number of photons with momentum $-\mathbf{k}$ is (cf. (44.8))

$$\int N_{-\mathbf{k}} d\omega' = \frac{8\pi^3 c^3}{\hbar\omega^3} \frac{U_{\omega}}{2}.$$

The electron scattering probability per second is therefore

$$w = \frac{2\pi^3 e^4}{m^2 \hbar^2 \omega^4} \int U_{\omega}^2 d\omega.$$

The factor ω^{-4} has been taken outside the integral because the degree of nonmonochromaticity $\Delta\omega$ is assumed small. At fixed total intensity, the integral is inversely proportional to $\Delta\omega$.

§ 60. Scattering by freely oriented systems

If an atomic energy level is nondegenerate, the polarizability and the coherent-scattering intensity are determined by the same tensor $\alpha_{ik} \equiv (c_{ik})_{11}$. If the level is degenerate, the observed values of these quantities are obtained by averaging over all states belonging to that level. The polarizability must be defined as the mean

$$\alpha_{ik} = \overline{(c_{ik})_{11}}.$$

The observed scattering intensity, on the other hand, is determined by averages of the products

$$\overline{(c_{ik})_{11}(c_{lm})_{11}}.$$

The connection between polarizability and scattering is therefore less direct.

Notice that although each quantity $(c_{ik})_{11}$ may be complex, their mean values are real (we suppose that there is no absorption and that α_{ik} is Hermitian). Indeed, in carrying out the average one may choose the set of independent wave functions belonging to the degenerate level arbitrarily, and they can always all be chosen real.

For free atoms or molecules (those not in an external field), level degeneracy is usually associated with an angular momentum whose orientation in space is free. Let the initial state in the scattering have angular momentum J_1 , and the final state J_2 . As usual, the scattering cross section must be averaged over all values of the projection M_1 and summed over the values of M_2 . After the first average the cross section no longer depends on M_2 , so the subsequent summation reduces to multiplication by $(2J_2 + 1)$. The averaged scattering cross section is thus

$$d\overline{\sigma} = \omega\omega'^3 c_{iklm}^{(21)} e_i'^* e_k e_l' e_m^* d\omega', \quad (60.1)$$

where

$$c_{iklm}^{(21)} = \frac{1}{2J_1 + 1} \sum_{M_1 M_2} (c_{ik})_{21} (c_{lm})_{21}^* = (2J_2 + 1) \overline{(c_{ik})_{21} (c_{lm})_{21}^*}, \quad (60.2)$$

and the bar with the index 1 denotes averaging over M_1 .

For unshifted scattering, states 1 and 2 belong to the same energy level ($\omega_{12} = 0$). If only coherent scattering is meant, states 1 and 2 must coincide completely, so that $M_1 = M_2$. The sum over M_2 , and with it the factor $2J_2 + 1$ in (60.2), then disappears:

$$c_{iklm}^{(\text{coh})} = \overline{(c_{ik})_{11}(c_{lm})_{11}^*}^1. \quad (60.3)$$

The result of the averaging can be written without a detailed calculation. Averaging over M_1 is equivalent to averaging over all orientations of the system, after which the mean can be expressed only in terms of the unit tensor δ_{ik} . Only products of components belonging separately to the scalar, symmetric, and antisymmetric parts of the scattering tensor can have nonzero means: no expression constructed from the unit tensor can have the symmetry properties required by mixed products of these parts. Consequently,

$$c_{iklm}^{(21)} = G_{21}^0 \delta_{ik} \delta_{lm} + c_{iklm}^{(21)s} + c_{iklm}^{(21)a}, \quad (60.4)$$

where

$$\begin{aligned} G_{21}^0 &= (2J_2 + 1) \overline{(c^0)_{21}}^1, \\ c_{iklm}^{(21)s} &= (2J_2 + 1) \overline{(c_{ik}^s)_{21}(c_{lm}^s)_{21}^*}^1, \\ c_{iklm}^{(21)a} &= (2J_2 + 1) \overline{(c_{ik}^a)_{21}(c_{lm}^a)_{21}^*}^1. \end{aligned} \quad (60.5)$$

In other words, the cross section (and hence the intensity) for scattering by a freely oriented system separates into the sum of three

independent parts, which we shall call *scalar*, *symmetric*, and *antisymmetric scattering*.

Each of the three terms in (60.4) is expressed through just one independent quantity. The scalar scattering is expressed through G_{21}^0 , while for the symmetric and antisymmetric scattering we have

$$\begin{aligned} c_{iklm}^{(21)s} &= \frac{1}{10} G_{21}^s \left(\delta_{il} \delta_{km} + \delta_{im} \delta_{kl} - \frac{2}{3} \delta_{ik} \delta_{lm} \right), \\ G_{21}^s &= (2J_2 + 1) \overline{(c_{ik}^s)_{21}(c_{ik}^s)_{21}^*}^1; \\ c_{iklm}^{(21)a} &= \frac{1}{6} G_{21}^a (\delta_{il} \delta_{km} - \delta_{im} \delta_{kl}), \\ G_{21}^a &= (2J_2 + 1) \overline{(c_{ik}^a)_{21}(c_{ik}^a)_{21}^*}^1 \end{aligned} \quad (60.6)$$

(the combination of unit tensors is constructed from the symmetry properties, after which the overall coefficient is found by contracting over the pairs of indices il and km).

Substitution of (60.4)–(60.6) into (60.1) gives the following expression for the scattering cross section:

$$d\bar{\sigma} = \omega \omega'^3 \left\{ G_{21}^0 |\mathbf{e}'^* \mathbf{e}|^2 + \frac{1}{10} G_{21}^s \left(1 + |\mathbf{e}' \mathbf{e}|^2 - \frac{2}{3} |\mathbf{e}'^* \mathbf{e}|^2 \right) + \frac{1}{6} G_{21}^a (1 - |\mathbf{e}' \mathbf{e}|^2) \right\} d\omega'. \quad (60.7)$$

This formula gives explicitly the angular dependence and polarization properties of the scattering.

The total scattering cross section over all directions, summed over the final-photon polarizations and averaged over the polarizations and incident directions of the initial photon, follows directly from (60.1). We note that

$$\overline{e_i^* e_k} = \delta_{ik}/3$$

when the average is taken over both polarizations and directions of photon propagation (summing over them instead gives a result larger by a factor $2 \cdot 4\pi$). We obtain

$$\overline{\sigma} = \frac{8\pi}{9} \omega \omega'^3 c_{ikik}^{(21)} = \frac{8\pi}{9} \omega \omega'^3 (3G_{21}^0 + G_{21}^s + G_{21}^a). \quad (60.8)$$

It was stated above that the selection rules for scattering coincide with those for matrix elements of an arbitrary second-rank tensor. In view of the division of the scattering intensity into three independent parts, it is useful to formulate these rules separately for each part.

The selection rules for symmetric scattering coincide with those for electric-quadrupole radiation, since the latter is also determined by an irreducible symmetric tensor (the quadrupole-moment tensor). For antisymmetric scattering they coincide with those for magnetic-dipole radiation, since both are determined by an axial vector (recall that an antisymmetric tensor is equivalent, or dual, to an axial vector).¹³ There is, however, one difference. Diagonal matrix elements, which in radiation give mean values of electric or magnetic moments (and do not correspond to radiative transitions), are important in scattering: they belong to coherent scattering.

For scalar scattering, the selection rules coincide with those for matrix elements of a scalar quantity. Transitions are therefore possible only between states of the same symmetry. In particular, the total angular momentum J and its projection M must have the same values (and diagonal matrix elements in M are independent of the value of M ; see vol. III, (29.3)). For unshifted scattering, states 1 and 2 must consequently coincide completely, not merely in energy but also in M , so unshifted scalar scattering is entirely coherent. Conversely, since every state is in any event combined with itself in scalar scattering, coherent scattering always contains a scalar part.

Just as the scattering cross section was averaged above, the polarizability tensor of a system freely oriented in space must also be averaged over the directions of the angular momentum J_1 . The averaging is immediate:

$$\alpha_{ik} \equiv \overline{(c_{ik})_{11}} = \overline{(c^0)_{11}} \delta_{ik}.$$

The symmetric and antisymmetric parts of the scattering tensor vanish upon averaging, since δ_{ik} is the only isotropic second-rank tensor.

It was noted above that diagonal matrix elements of a scalar do not depend on M_1 . The averaging sign over $(c^0)_{11}$ may therefore be omitted altogether (and $(c^0)_{11}$ may be calculated for any value of M_1), giving the polarizability

$$\alpha_{ik} \equiv (c^0)_{11} \delta_{ik}. \quad (60.9)$$

¹³This statement concerns, of course, the selection rules associated with symmetry, not the specific form of the axial vector in radiation: the magnetic-moment vector contains a spin part, whereas scattering involves matrix elements of quantities of orbital (coordinate) origin.

For the same reason, the averaging sign may also be omitted in G_{11}^0 , which determines the scalar part of coherent scattering:

$$G_{11}^0 = \overline{|(c^0)_{11}|^2}^1 = (c^0)_{11}^2 \quad (60.10)$$

(the factor $2J_2 + 1$ is omitted in accordance with (60.3)). Thus there is a simple relation between the mean polarizability and the scalar part of coherent scattering. Both are determined by the quantity

$$(c^0)_{11} = \frac{2}{3} \sum_n \frac{\omega_{n1}}{\omega_{n1}^2 - \omega^2} |\mathbf{d}_{n1}|^2. \quad (60.11)$$

PROBLEM

1. Find the angular distribution and degree of depolarization in the scattering of linearly polarized light.

SOLUTION. Let θ be the angle between the scattering direction $\mathbf{n}'$ and the polarization direction $\mathbf{e}$ of the incident light. The scattered light contains two independent components, polarized in the plane $\mathbf{n}'\mathbf{e}$ (intensity I_1) and perpendicular to it (intensity I_2); the degree of depolarization is the ratio I_2/I_1 . The intensities I_1 and I_2 follow from (60.7), with $\mathbf{e}'$ directed appropriately.

In scalar scattering the light remains completely polarized in the same plane ($I_2 = 0$), and the angular intensity distribution is

$$I = \frac{3}{2} \sin^2 \theta.$$

(Here and below, the expressions for $I = I_1 + I_2$ are normalized to have mean value 1 when averaged over directions.) For symmetric scattering,

$$I = \frac{3}{20}(6 + \sin^2 \theta), \quad \frac{I_2}{I_1} = \frac{3}{3 + \sin^2 \theta}.$$

For antisymmetric scattering,

$$I = \frac{3}{4}(1 + \cos^2 \theta), \quad \frac{I_2}{I_1} = \frac{1}{\cos^2 \theta}.$$

2. Find the same quantities for the scattering of unpolarized light.

SOLUTION. The transition in (60.7) to unpolarized incident light is effected by the replacement

$$e_i e_k^* \longrightarrow \frac{1}{2}(\delta_{ik} - n_i n_k),$$

which represents averaging over the polarization directions $\mathbf{e}$ at a fixed incident direction $\mathbf{n}$. The scattered light is partially polarized. By symmetry, its two independent components are linearly polarized in the scattering plane $\mathbf{nn}'$ (intensity $I_{\parallel}$) and perpendicular to it (intensity $I_{\perp}$). Denote by θ the scattering angle, that is, the angle between $\mathbf{n}$ and $\mathbf{n}'$.

For scalar scattering,

$$I = I_{\perp} + I_{\parallel} = \frac{3}{4}(1 + \cos^2 \theta), \quad \frac{I_{\parallel}}{I_{\perp}} = \cos^2 \theta;$$

for symmetric scattering,

$$I = \frac{3}{40}(13 + \cos^2 \theta), \quad \frac{I_{\parallel}}{I_{\perp}} = \frac{6 + \cos^2 \theta}{7};$$

and for antisymmetric scattering,

$$I = \frac{3}{8}(2 + \sin^2 \theta), \quad \frac{I_{\parallel}}{I_{\perp}} = 1 + \sin^2 \theta.$$

3. For the scattering of circularly polarized light, determine the reversal coefficient (the ratio of the intensity of the component circularly polarized in the “reversed” sense to that of the component polarized in the “proper” sense).

SOLUTION. With circularly polarized incident light, the angular distribution and the degree of depolarization (the ratio $I_{\parallel}/I_{\perp}$) are the same as for the scattering of unpolarized light.

Let the polarization vector of the incident light have components $\mathbf{e} = (1, i, 0)/\sqrt{2}$ (in a coordinate system whose xz plane is the scattering plane and whose z axis is along $\mathbf{n}$). The polarization vectors of the “reversed” and “proper” circularly polarized components of the scattered light are then, respectively,

$$\mathbf{e}' = \frac{1}{\sqrt{2}}(\cos \theta, -i, -\sin \theta) \quad \text{and} \quad \mathbf{e}' = \frac{1}{\sqrt{2}}(\cos \theta, i, -\sin \theta).$$

Calculating the intensity from (60.7), we find the reversal coefficients P for the three types of scattering:

$$P^0 = \tan^4 \frac{\theta}{2}, \quad P^s = \frac{13 + \cos^2 \theta + 10 \cos \theta}{13 + \cos^2 \theta - 10 \cos \theta}, \quad P^a = \frac{1 - \cos^4(\theta/2)}{1 - \sin^4(\theta/2)}$$

(θ is the scattering angle).

4. Calculate the cross section for scattering of a low-frequency photon by a hydrogen atom in its ground state.

SOLUTION. A low-frequency photon can scatter only elastically. Since the orbital angular momentum in the ground state of the hydrogen atom is $L = 0$, the selection rules show that only scalar scattering occurs when spin-orbit coupling is neglected. The static polarizability of the atom (in ordinary units) is

$$\alpha = \frac{9}{2} \left(\frac{\hbar^2}{me^2} \right)^3$$

(see vol. III, § 76, problem 4). Substitution in (60.8) gives the required cross section:

$$\sigma_t = 54\pi \left(\frac{\omega}{c} \right)^4 \left(\frac{\hbar^2}{me^2} \right)^6.$$

5. Calculate the cross section for elastic scattering of γ radiation by a deuteron (*H. A. Bethe, R. Peierls, 1935*).

SOLUTION. The wave functions of the deuteron ground state and of its continuum states (a dissociated deuteron) are

$$\psi_0 = \sqrt{\frac{\kappa}{2\pi}} \frac{e^{-\kappa r}}{r}, \quad \psi_{\mathbf{p}} = e^{i\mathbf{p}\mathbf{r}}, \quad \kappa = \sqrt{MI}$$

(see (58.2), (58.3)). The matrix element of the dipole moment, equal to

$$\mathbf{d}_{\mathbf{p}0} = -\frac{ie\mathbf{p}p_0}{M\omega_{p0}},$$

was calculated in § 58:

$$\mathbf{d}_{\mathbf{p}0} = -\frac{4\pi ie}{M\omega_{p0}} \sqrt{\frac{\kappa}{2\pi}} \frac{\mathbf{p}}{\kappa^2 + p^2},$$

where the frequencies are $\omega_{p0} = (\mathbf{p}^2 + \kappa^2)/M$. The polarizability tensor is

$$\alpha_{ik} = \left\{ \frac{2}{3} \int \frac{\omega_{p0}}{\omega_{p0}^2 - \omega^2} |\mathbf{d}_{\mathbf{p}0}|^2 \frac{d^3 p}{(2\pi)^3} - \frac{e^2}{2M\omega^2} \right\} \delta_{ik}.$$

The first term is associated with virtual excitation of the deuteron's internal degrees of freedom; it is written in the form (60.11). The second term is associated with the action of the wave field on the translational motion of the deuteron as a whole. Since this motion is quasiclassical, the corresponding part of the scattering tensor is given by (59.14), with the deuteron mass $2M$ used for m .

The calculation of α_{ik} reduces to the integral

$$J = \frac{1}{2} \int_{-\infty}^{\infty} \frac{z^4 dz}{(z^2 + 1)^3 [(z^2 + 1)^2 - \gamma^2]}, \quad z = \frac{p}{\kappa}, \quad \gamma = \frac{M\omega}{\kappa^2} = \frac{\omega}{I}.$$

We have

$$J = \frac{1}{8} \frac{d}{d\lambda} \left(\frac{1}{\lambda} \frac{dJ_0}{d\lambda} \right) \Big|_{\lambda=1}, \quad J_0 = \frac{1}{2} \int_{-\infty}^{\infty} \frac{z^4 dz}{(z^2 + \lambda^2) [(z^2 + 1)^2 - \gamma^2]}.$$

For $\gamma < 1$, the integrand has poles in the upper half-plane of the complex variable z at $i\lambda$, $i\sqrt{1+\gamma}$, and $i\sqrt{1-\gamma}$; the integral J_0 is evaluated from the residues at these poles. The result is

$$J = \frac{\pi}{2} \left\{ \frac{(1+\gamma)^{3/2}}{2\gamma^4} + \frac{(1-\gamma)^{3/2}}{2\gamma^4} - \left(\frac{3}{8\gamma^2} + \frac{1}{\gamma^4} \right) \right\}.$$

By (60.8) and (60.10), the total scattering cross section is expressed in terms of α_{ik} and is (in ordinary units)

$$\sigma = \frac{8\pi}{3} \left(\frac{e^2}{Mc^2} \right)^2 \left| -1 - \frac{4}{3\gamma^2} + \frac{2}{3\gamma^2} [(1+\gamma)^{3/2} + (1-\gamma)^{3/2}] \right|^2 \quad \text{for } \gamma = \frac{\hbar\omega}{I} < 1.$$

For $\gamma > 1$ (above the deuteron-dissociation threshold), the scattering amplitude is obtained by analytic continuation of the amplitude for $\gamma < 1$; it acquires an imaginary part, which must be positive (in accordance with the bypass prescription in (59.17)):

$$\sigma = \frac{8\pi}{3} \left(\frac{e^2}{Mc^2} \right)^2 \left| -1 - \frac{4}{3\gamma^2} + \frac{2}{3\gamma^2}(\gamma + 1)^{3/2} + i \frac{2}{3\gamma^2}(\gamma - 1)^{3/2} \right|^2 \quad \text{for } \gamma > 1.$$

For $\gamma \gg 1$ this gives $\sigma = (8\pi/3)(e^2/Mc^2)^2$, as expected for nonrelativistic scattering by a free proton.

The angular distribution of the radiation is

$$d\sigma = \sigma \frac{3}{4} (1 + \cos^2 \theta) \frac{d\theta}{4\pi},$$

where θ is the scattering angle. Defining the scattering amplitude according to (59.24), we have

$$\text{Im } f(0) = \frac{2e^2}{3Mc^2} \frac{(\gamma - 1)^{3/2}}{\gamma^2}, \quad \gamma > 1.$$

By the optical theorem (59.26), this quantity equals $\omega\sigma_t/(4\pi)$, where σ_t is the total photodissociation cross section (58.4). Recall that the elastic-scattering cross section is of higher order ($\sim e^4$) than the dissociation cross section ($\sim e^2$; see (58.4)), so that σ_t coincides with the dissociation cross section. For the same reason, in the approximation used here the scattering amplitude for $\gamma < 1$ (below the dissociation threshold) is real.

§ 61. Scattering by molecules

The distinctive features of molecular scattering arise from the same molecular properties on which the general theory of molecular spectra rests: the electronic state can be considered separately for fixed nuclei, and the nuclear motion can then be considered in the resulting effective electronic field.

Let the incident-light frequency ω be smaller than the energy ω_e of the first electronic excitation. The electronic terms then cannot be excited in the scattering. The scattering is either unshifted or shifted through excitation of rotational or vibrational levels.

Assume also that the ground electronic term of the molecule is nondegenerate (and has no fine structure). In other words, both the total electron spin and the projection of the total electron orbital angular momentum on the molecular axis (for a symmetric-top molecule) are assumed to vanish. For a diatomic molecule, this means that the ground electronic term must be $^1\Sigma$. These conditions are known to hold for the ground states of most molecules¹⁴.

Finally, we shall suppose that ω is large in comparison with the intervals of the nuclear (rotational and vibrational) structure of the ground term, and that $\omega_e - \omega$ bears the same relation to the nuclear structure of the excited electronic term. Thus the

¹⁴The results given below may nevertheless remain valid, to a certain accuracy, when the degeneracy of the ground electronic term is associated with nonzero spin and the spin-orbit interaction is weak (so that the fine structure it produces can be neglected). In this approximation, states with different spin directions do not combine and, in this sense, behave as nondegenerate states. An example is the O_2 molecule, whose ground term is $^3\Sigma$.

incident-light frequency must lie sufficiently far from resonances. These are precisely the conditions that allow the nuclear motion initially to be set aside when the scattering tensor is calculated, so that the problem is first treated for a specified nuclear configuration.

In this fixed-configuration problem, the scattering tensor coincides with the polarizability tensor $\alpha_{ik} = (\alpha_{ik})_{11}$ and can in principle be calculated from the general formula (59.17), with the sum extending over all excited electronic terms. The quantities α_{ik} obtained in this way are functions of the nuclear-configuration coordinates q (the energies and wavefunctions of the electronic terms depend parametrically on these coordinates). Since the state is nondegenerate, the tensor $\alpha_{ik}(q)$ is real and therefore symmetric.

The tensor $\alpha_{ik}(q)$ is the electronic polarizability of the molecule in the specified nuclear configuration. To solve the actual scattering problem, the nuclear motion in the initial and final states must still be included. Let $\psi_{s_1}(q)$ and $\psi_{s_2}(q)$ be the nuclear wavefunctions of these states, so that s_1, s_2 are sets of vibrational and rotational quantum numbers. The required scattering tensor is the matrix element of $\alpha_{ik}(q)$ evaluated with these functions:

$$\langle s_2 | \alpha_{ik} | s_1 \rangle = \int \psi_{s_2}^*(q) \alpha_{ik} \psi_{s_1} dq. \quad (61.1)$$

Because $\alpha_{ik}(q)$ is symmetric, the tensor (61.1) is also symmetric, whether s_1 and s_2 coincide or differ. Under the conditions considered here, the antisymmetric part is therefore absent from both unshifted and shifted scattering. Only scalar and symmetric parts remain.

The scalar part of the polarizability, $\alpha^0(q)$, is independent of the molecular orientation and depends only on the internal arrangement of its atoms. Denote the full set of vibrational quantum numbers by v , and the set of rotational numbers other than the magnetic number m by r . Then

$$\langle v_2 r_2 m_2 | \alpha^0 | v_1 r_1 m_1 \rangle = \langle v_2 | \alpha^0 | v_1 \rangle \delta_{r_1 r_2} \delta_{m_1 m_2}. \quad (61.2)$$

Diagonality in r, m is a general property of every scalar. What is specific to (61.2) is that here the matrix elements do not depend on these numbers at all. Scalar scattering thus occurs only for purely vibrational transitions and is independent of the rotational state.

Symmetric scattering is determined by matrix elements of the tensor α_{ik}^s . Its components in the fixed coordinate system xyz are expressed through the components $\bar{\alpha}_{i'k'}^s$ in the molecule-fixed system $\xi\eta\zeta$ as

$$\alpha_{ik}^s = \sum_{i'k'} \bar{\alpha}_{i'k'}^s D_{i'i} D_{k'k}, \quad (61.3)$$

where $D_{i'i}$ are the direction cosines of the new axes with respect to the old ones. The quantities $\bar{\alpha}_{i'k'}^s$ do not depend on the molecular orientation, while $D_{i'i}$ do not depend on its internal coordinates. Hence

$$\langle v_2 r_2 m_2 | \alpha_{ik}^s | v_1 r_1 m_1 \rangle = \sum_{i'k'} \langle v_2 | \bar{\alpha}_{i'k'}^s | v_1 \rangle \langle r_2 m_2 | D_{i'i} D_{k'k} | r_1 m_1 \rangle.$$

As is readily verified, the sum over $r_2 m_2, ik$ of the squared moduli of these quantities

is¹⁵:

$$\sum_{r_2 m_2} \sum_{ik} |\langle v_2 r_2 m_2 | \alpha_{ik}^s | v_1 r_1 m_1 \rangle|^2 = \sum_{i'k'} |\langle v_2 | \bar{\alpha}_{i'k'}^s | v_1 \rangle|^2. \quad (61.4)$$

Thus the total intensity of scattering from a given vibrational-rotational level $v_1 r_1$ to all rotational levels of the vibrational state v_2 does not depend on r_1 .

For symmetric-top molecules one can go further and determine the dependence of the scattering intensity on the rotational quantum numbers for each transition $v_1 r_1 \rightarrow v_2 r_2$. In this case the numbers r are the angular momentum J and its projection k on the molecular axis. Replace the Cartesian components α_{ik}^s by the corresponding second-rank spherical tensor, whose components will be denoted by α_λ ($\lambda = 0, \pm 1, \pm 2$). According to formula (110.7) (see III), the squared moduli of its matrix elements are

$$\begin{aligned} |\langle v_2 J_2 k_2 m_2 | \alpha_\lambda | v_1 J_1 k_1 m_1 \rangle|^2 &= \\ &= (2J_1 + 1)(2J_2 + 1) \begin{pmatrix} J_2 & 2 & J_1 \\ -k_2 & \lambda' & k_1 \end{pmatrix}^2 \begin{pmatrix} J_2 & 2 & J_1 \\ -m_2 & \lambda & m_1 \end{pmatrix}^2 |\langle v_2 | \bar{\alpha}_{\lambda'} | v_1 \rangle|^2, \end{aligned}$$

where $\bar{\alpha}_{\lambda'}(q)$ is the spherical polarizability tensor referred to axes fixed in the molecule, and $\lambda' = k_2 - k_1$. Summing over m_2 and $\lambda = m_2 - m_1$ for a specified m_1 , we obtain (compare III, (110.8))

$$\begin{aligned} \sum_{m_2 \lambda} |\langle v_2 J_2 k_2 m_2 | \alpha_\lambda | v_1 J_1 k_1 m_1 \rangle|^2 &= \\ &= (2J_2 + 1) \begin{pmatrix} J_2 & 2 & J_1 \\ -k_2 & \lambda' & k_1 \end{pmatrix}^2 |\langle v_2 | \bar{\alpha}_{\lambda'} | v_1 \rangle|^2. \end{aligned} \quad (61.5)$$

This quantity determines the scattering intensity for the vibrational-rotational transition $v_1 J_1 k_1 \rightarrow v_2 J_2 k_2$. Since the matrix elements $\langle v_2 | \bar{\alpha}_{\lambda'} | v_1 \rangle$ do not depend on the molecular rotation, this formula gives the intensity dependence on both J_1, J_2 and k_1, k_2 . Notice that only one spherical component of the polarizability tensor enters the right-hand side of (61.5).

Summing (61.5) over J_2 and k_2 gives¹⁶:

$$\sum_{\lambda} \sum_{J_2 k_2 m_2} |\langle v_2 J_2 k_2 m_2 | \alpha_\lambda | v_1 J_1 k_1 m_1 \rangle|^2 = \sum_{\lambda'} |\langle v_2 | \bar{\alpha}_{\lambda'} | v_1 \rangle|^2,$$

¹⁵In transforming the sum, one uses the equality expressing the unitarity of the matrix $D_{i'i}$:

$$\begin{aligned} \sum_{ik} \sum_{r_2 m_2} \langle r_1 m_1 | D_{i'i} D_{k'k} | r_2 m_2 \rangle \langle r_2 m_2 | D_{i''i} D_{k''k} | r_1 m_1 \rangle \\ = \left\langle r_1 m_1 \left| \sum_{ik} D_{i'i} D_{i''i} D_{k'k} D_{k''k} \right| r_1 m_1 \right\rangle \\ = \langle r_1 m_1 | \delta_{i'i''} \delta_{k'k''} | r_1 m_1 \rangle = \delta_{i'i''} \delta_{k'k''}. \end{aligned}$$

¹⁶For the sum over J_2 with k_1 and λ' specified (and hence also $k_2 = k_1 + \lambda'$), one has

$$\sum_{J_2} (2J_2 + 1) \begin{pmatrix} J_2 & 2 & J_1 \\ -k_2 & \lambda' & k_1 \end{pmatrix}^2 = 1$$

by (106.13) (see III). One then sums over k_2 , or equivalently over $\lambda' = k_2 - k_1$, for the specified k_1 .

so that the sum rule (61.4) is recovered.

A rotor, that is, a linear molecule (in particular a diatomic molecule), is a special case of the symmetric top. The projection of the angular momentum on the axis of such a molecule vanishes (in a nondegenerate electronic state with zero electronic orbital angular momentum)¹⁷. Consequently, one must set $k_1 = k_2 = 0$ in (61.5) for this case.

Finally, consider the selection rules for vibrational Raman scattering, together with the analogous question for the vibrational emission (or absorption) spectra of a molecule¹⁸.

For scattering, the problem is to find the conditions under which the matrix elements of the tensor $\alpha_{ik}(q)$ evaluated with the vibrational wavefunctions $\psi_v(q)$ are nonzero. The scalar α^0 (for scalar scattering) and the irreducible symmetric tensor α_{ik}^s (for symmetric scattering) must be considered separately. In emission (or absorption), the analogous role is played by matrix elements of the vector $\mathbf{d}(q)$, the molecular dipole moment averaged over the electronic state at a specified nuclear configuration (this was already noted in § 54 for diatomic molecules).

The vibrations of a polyatomic molecule are classified by symmetry types, that is, by irreducible representations of the corresponding point group: D_a , where a labels the representation (see III, § 100). These representations also determine the symmetry of the wavefunctions of the molecular vibrational states (see III, § 101). The symmetry of the first vibrational state (quantum number $v_a = 1$) is that of the representation D_a assigned to the vibrational mode. For higher states ($v_a > 1$), the symmetry is given by the representation $[D_a^{v_a}]$, the symmetric product of D_a with itself v_a times. Finally, the symmetry of states in which distinct vibrations a and b are excited simultaneously is given by the direct product $[D_a^{v_a}] \times [D_b^{v_b}]$ ¹⁹. The procedure for obtaining the selection rules for various quantities (scalar, vector, and tensor) from their symmetry types is described in vol. III, § 97.

Selection rules based on molecular symmetry are exact. Alongside them there are approximate rules associated with the harmonic-vibration assumption and with expansion of the functions $\alpha_{ik}(q)$ or $\mathbf{d}(q)$ in powers of the vibrational coordinates q . They follow from the familiar selection rule for a harmonic oscillator: matrix elements of its coordinate q are nonzero only for transitions in which the vibrational quantum number changes by $\Delta v = \pm 1$.

§ 62. Natural width of spectral lines

Until now, in studying the emission and scattering of light, we have treated all levels of the system (an atom, for example) as strictly discrete. An excited level, however, has a finite lifetime because it can radiatively decay. By the general principles of quantum mechanics, the level consequently becomes quasidiscrete and acquires a finite (small) width (see III, § 134); it is written as $E - i\Gamma/2$, where $\Gamma (= \Gamma/\hbar)$ is the total probability per second of all possible “decay” processes of the state.

¹⁷Effects associated with the interaction between molecular vibration and rotation are not considered here (see III, § 104).

¹⁸These spectra lie in the infrared region and are usually observed in absorption.

¹⁹The symmetry properties of vibrational wavefunctions do not, of course, depend on the particular form of the vibrational potential energy. In particular, they are independent of the assumption of harmonic vibrations made in vol. III, § 101.

Let us examine how this circumstance affects emission (V. Weisskopf, E. Wigner, 1930). It is clear in advance that the finite level width prevents the emitted light from being strictly monochromatic: its frequencies are spread over an interval $\Delta\omega \sim \Gamma (= \Gamma/\hbar)$. Since the initial state of the emitting system has a finite lifetime, it is more natural here to seek the total probability of emitting a photon of a specified frequency than the probability per unit time. We first calculate this probability for a transition of an atom from an excited level

$$E_1 - \frac{i}{2}\Gamma_1$$

to a ground level E_2 having an infinite lifetime and therefore remaining strictly discrete.

Let Ψ be the wavefunction of the atom and the photon field, and let $\hat{H} = \hat{H}^{(0)} + \hat{V}$ be the Hamiltonian of this system, with $\hat{V}$ the interaction operator between the atom and the photon field. We seek a solution of the Schrödinger equation

$$i\frac{\partial\Psi}{\partial t} = (\hat{H}^{(0)} + \hat{V})\Psi \quad (62.1)$$

as an expansion in the wavefunctions of the unperturbed states:

$$\Psi = \sum_{\nu} a_{\nu}(t)\Psi_{\nu}^{(0)} = \sum_{\nu} a_{\nu}(t)e^{-i\mathcal{E}_{\nu}t}\psi_{\nu}^{(0)}. \quad (62.2)$$

The coefficients $a_{\nu}(t)$ then satisfy

$$i\frac{\partial a_{\nu}}{\partial t} = \sum_{\nu'} \langle \nu|V|\nu' \rangle a_{\nu'} \exp\{i(\mathcal{E}_{\nu} - \mathcal{E}_{\nu'})t\}. \quad (62.3)$$

Let $|\nu\rangle$ be a state of energy $\mathcal{E}_{\nu} = E_2 + \omega$ in which the atom is in its ground level E_2 and there is one quantum of specified frequency ω ; denote this state by $|\omega 2\rangle$. At the initial instant, the system is in the state $|1\rangle$, with the atom excited to level E_1 and no photons present. Thus at $t = 0$,

$$a_1 = 1, \quad a_{\nu'} = 0 \quad \text{for } |\nu'\rangle \neq |1\rangle. \quad (62.4)$$

With this initial condition, the solution of (62.3), for suitably normalized wavefunctions, gives the probability by time t for the atomic transition $1 \rightarrow 2$ accompanied by emission of a photon in the frequency interval $d\omega$:

$$|a_{\omega 2}(t)|^2 d\omega.$$

We require the probability as $t \rightarrow \infty$:

$$dw = |a_{\omega 2}(\infty)|^2 d\omega. \quad (62.5)$$

To clarify this formulation, recall how the usual emission probability per second for a transition $1 \rightarrow 2$ is found when the level width is neglected. Equation (62.3) is solved by replacing, in the first approximation, every $a_{\nu'}(t)$ on its right-hand side by the values (62.4). The resulting solution is then considered at large t (compare III, § 42). We can now make the meaning of this procedure more precise: it applies to times short

in comparison with the lifetime of the excited level. Here large t means a time large compared with the period $1/(E_1 - E_2)$, but still small compared with $1/\Gamma_1$.

In the present problem, where times comparable with $1/\Gamma_1$ are considered, the function $a_1(t)$ decreases as

$$a_1(t) = e^{-\Gamma_1 t/2}. \quad (62.6)$$

The functions $a_{\nu'}(t)$ for states $|\nu'\rangle$ that can be produced when the atom radiates increase with time. If the level E_1 can radiate to several atomic levels besides E_2 , there are many increasing functions $a_{\nu'}(t)$; each corresponds to a state in which the atom occupies one of its levels and one photon of the appropriate energy is present. Even so, only one term remains on the right-hand side of (62.3), namely the term with $|\nu'\rangle = |1\rangle$. Indeed, matrix elements can be nonzero only for transitions in which the number of photons of one particular energy changes by one; they therefore vanish identically between states each containing one photon, but of different energies.

We consequently have the following equation for $a_{\omega 2}(t)$:

$$\begin{aligned} i \frac{da_{\omega 2}}{dt} &= \langle \omega 2 | V | 1 \rangle e^{i(E_2 + \omega - E_1)t} a_1 = \\ &= \langle \omega 2 | V | 1 \rangle \exp \left\{ i(\omega - \omega_{12})t - \frac{\Gamma_1}{2}t \right\}, \end{aligned} \quad (62.7)$$

where $\omega_{12} = E_1 - E_2$. Integrating with $a_{\omega 2}(0) = 0$ gives

$$a_{\omega 2} = \langle \omega 2 | V | 1 \rangle \frac{1 - \exp\{i(\omega - \omega_{12})t - \Gamma_1 t/2\}}{\omega - \omega_{12} + i\Gamma_1/2}. \quad (62.8)$$

It follows that the probability dw in (62.5) is

$$dw = |\langle \omega 2 | V | 1 \rangle|^2 \frac{d\omega}{(\omega - \omega_{12})^2 + \Gamma_1^2/4}.$$

Since the width $\Gamma_1 \ll \omega_{12}$, one may put $\omega = \omega_{12}$ in the factor $|\langle \omega 2 | V | 1 \rangle|^2$. Then $2\pi|\langle \omega 2 | V | 1 \rangle|^2$ is the usual probability per second for emission of a photon of frequency ω_{12} and of the other characteristics besides frequency (direction of motion and polarization), which have so far been suppressed for simplicity. The dependence of the probability on these characteristics is completely determined by the factor $|\langle \omega 2 | V | 1 \rangle|^2$. Thus allowing for the level width does not change the polarization properties or the angular distribution of the radiation.

The sum

$$\Gamma_{1 \rightarrow 2} = 2\pi \sum |\langle \omega 2 | V | 1 \rangle|^2, \quad (62.9)$$

taken over photon polarizations and directions of motion, is the total usual emission probability. At the same time, it is the part of the width of level E_1 (the partial width) associated with the transition $1 \rightarrow 2$, as distinct from the total width Γ_1 , which is composed of contributions from every possible mode of “decay” of this quasistationary state²⁰.

²⁰Formulas (62.6) and (62.9) can of course also be obtained by solving for $a_1(t)$ an equation analogous to (62.7).

Transitions to continuous-spectrum states that produce a finite level width need not involve photon emission. Highly excited (x-ray) levels may decay by emitting an electron and leaving a positive ion in its ground state (the Auger effect).

Summing the probability dw in the same way gives the final expression for the frequency distribution of the emitted light:

$$dw = w_t \frac{\Gamma_1}{2\pi} \frac{d\omega}{(\omega_{12} - \omega)^2 + \Gamma_1^2/4}, \quad (62.10)$$

where $w_t = \Gamma_{1 \rightarrow 2}/\Gamma_1$ is the total relative probability of the transition $1 \rightarrow 2$. This is a dispersion-type distribution. The spectral-line shape described by (62.10) belongs to an isolated atom at rest and is called *natural*²¹.

Now let the atomic level E_2 also be excited, with a finite width Γ_2 . For simplicity, suppose that this width is associated with a transition of the atom to the ground state E_0 with emission of one photon (the final result, formula (62.12), does not depend on this assumption). The decay of state 1 can then be regarded as a two-photon emission process of the kind studied in § 59. Before the finite lifetime of state 2 is included, the matrix element of this process is

$$\langle \omega \omega' 0 | V^{(2)} | 1 \rangle = \frac{\langle \omega \omega' 0 | V | \omega 2 \rangle \langle \omega 2 | V | 1 \rangle}{E_0 - E_2 + \omega' + i0} \quad (62.11)$$

(in (59.2), the label of state 2 has been changed to 0, and the sum over n retains only the term corresponding to the atom being in state 2, which is resonantly large when ω' is close to $E_2 - E_0$). If the finite lifetime of state 2 is now included, the only change in (62.11) is the replacement $E_2 \rightarrow E_2 - i\Gamma_2/2$, so that

$$\langle \omega \omega' 0 | V^{(2)} | 1 \rangle = \frac{\langle \omega \omega' 0 | V | \omega 2 \rangle \langle \omega 2 | V | 1 \rangle}{E_0 - E_2 + \omega' + i\Gamma_2/2}.$$

Substitution of this matrix element in the equation for $a_{\omega \omega' 0}(t)$ (which differs from (62.7) only in its notation) gives, by a derivation entirely analogous to that of (62.8),

$$a_{\omega \omega' 0}(\infty) = \frac{\langle \omega \omega' 0 | V | \omega 2 \rangle \langle \omega 2 | V | 1 \rangle}{(\omega' - \omega_{20} + i\Gamma_2/2)(\omega + \omega' - \omega_{10} + i\Gamma_1/2)}.$$

The probability of emitting photons ω and ω' is

$$\begin{aligned} dw &= |a_{\omega \omega' 0}(\infty)|^2 d\omega d\omega' = \\ &= \frac{\Gamma_{1 \rightarrow 2}}{2\pi} \frac{\Gamma_{2 \rightarrow 0}}{2\pi} \frac{d\omega d\omega'}{[(\omega' - \omega_{20})^2 + \Gamma_2^2/4][(\omega + \omega' - \omega_{10})^2 + \Gamma_1^2/4]}. \end{aligned} \quad (62.12)$$

As expected, this expression has sharp maxima at $\omega' \approx \omega_{20}$ and $\omega \approx \omega_{12}$.

The required line shape for the transition $1 \rightarrow 2$ follows by integrating (62.12) over $d\omega'$; the integration may be extended from $-\infty$ to $+\infty$. The simplest evaluation closes the integration path by an infinitely distant semicircle in the upper half-plane of the complex variable ω' . The result is determined by the sum of the residues of the integrand at the poles

$$\omega' = \omega_{20} + i\Gamma_2/2 \quad \text{and} \quad \omega' = \omega_{10} - \omega + i\Gamma_1/2.$$

²¹This is to be distinguished from broadening caused by interactions of the atom with other atoms (collision broadening), or by the presence in the source of atoms moving at different velocities (Doppler broadening).

We thereby obtain

$$dw = w_t \frac{\Gamma_1 + \Gamma_2}{2\pi} \frac{d\omega}{(\omega - \omega_{12})^2 + (\Gamma_1 + \Gamma_2)^2/4}, \quad (62.13)$$

where $w_t = \Gamma_{1 \rightarrow 2} \Gamma_{2 \rightarrow 0} / (\Gamma_1 \Gamma_2)$ is the total probability of the double transition $1 \rightarrow 2 \rightarrow 0$ ²².

The line shape (62.13) differs from (62.10) only by the replacement of Γ_1 with $\Gamma_1 + \Gamma_2$: the line width is the sum of the widths of the initial and final states.

In general, the line width is not equal to the probability $\Gamma_{1 \rightarrow 2}$ of the transition $1 \rightarrow 2$ itself, and hence is not proportional to the line intensity (as it would be in classical theory). Since $\Gamma_1 + \Gamma_2 > \Gamma_{1 \rightarrow 2}$, a line can have a large width while its intensity is comparatively small.

§ 63. Resonance fluorescence

Allowance for the finite width of levels in the problem of light scattering is essential when the frequency ω of the incident light is close to one of the “intermediate” frequencies ω_{n1} or ω_{2n} —the phenomenon known as *resonance fluorescence* (V. Weisskopf, 1931).

Consider unshifted scattering by a system (an atom, for example) in its ground state, so that the initial and final levels coincide and are strictly discrete. Let the light frequency be close to some frequency ω_{n1} , with n an excited and therefore quasidiscrete level.

This question could be solved by the method set out in the preceding section. That is unnecessary, however, because the problem is entirely analogous to the problem of nonrelativistic resonance scattering through a quasidiscrete level considered in Vol. III, § 134. According to the result obtained there, the scattering amplitude must contain the pole factor

$$\left[\omega - \left(E_n - \frac{i}{2} \Gamma_n - E_1 \right) \right]^{-1}.$$

On the other hand, when $|\omega - \omega_{n1}| \gg \Gamma_n$, the formula must reduce to the nonresonant formula (59.5). It follows that the required scattering cross section is obtained simply by replacing E_n with $E_n - i\Gamma_n/2$ in (59.5), while retaining only the resonant terms in the sum over n :

$$d\sigma = \frac{|\sum M_n (\mathbf{d}_{2n} \mathbf{e}^*) (\mathbf{d}_{n1} \mathbf{e})|^2}{(\omega_{n1} - \omega)^2 + \Gamma_n^2/4} \omega^4 d\omega'. \quad (63.1)$$

The sum extends over all states (with different angular-momentum projections M_n) belonging to the resonant level E_n ; states 1 and 2 belong to the same (ground) level but may have different values of M_1 and M_2 .

The cross section (63.1) is largest at $\omega = \omega_{n1}$. In order of magnitude, its value at the maximum is $\sigma_{\max} \sim \omega^4 d^4 / \Gamma_n^2$. Since the probability of the spontaneous transition $n \rightarrow 1$, and hence the width Γ_n , is of order $\omega^3 d^2$, this value is

$$\sigma_{\max} \sim \omega^{-2} \sim \lambda^2, \quad (63.2)$$

²²In more complicated cases, w_t is the total probability of all cascades that begin with the transition $1 \rightarrow 2$ and terminate at level 0.

that is, of the order of the square of the light wavelength and independent of the fine-structure constant, in place of the typical values $\sim r_e^2$ outside the resonance region.

We emphasize that, because the atom is in a strictly discrete (ground) level both before and after scattering, the frequencies of the primary and secondary photons coincide exactly. Therefore, under irradiation by monochromatic light, the scattered light is monochromatic as well. If instead the incident light has a spectral intensity distribution $I(\omega)$ and $I(\omega)$ varies little over the width Γ_n , the intensity of the scattered light is proportional to

$$\frac{I(\omega_{n1})d\omega}{(\omega - \omega_{n1})^2 + \Gamma_n^2/4}. \quad (63.3)$$

In other words, the scattering line has the same shape as the natural line shape for spontaneous emission from the level E_n .

The scattering tensor corresponding to the cross section (63.1) is

$$(c_{ik})_{21} = \frac{\sum M_n (d_i)_{2n} (d_k)_{n1}}{\omega_{n1} - \omega - i\Gamma_n/2}. \quad (63.4)$$

In particular, the polarizability tensor is

$$\alpha_{ik} = (c_{ik})_{11} = \frac{\sum M_n (d_i)_{1n} (d_k)_{n1}}{\omega_{n1} - \omega - i\Gamma_n/2}. \quad (63.5)$$

Let us note at once that adding an imaginary part to the energy levels of the intermediate excited states makes the polarizability tensor non-Hermitian even at frequencies below the ionization threshold. The tensor acquires an imaginary part directly connected with the absorption of light.

After absorbing a quantum, the atom will sooner or later return to the ground state by emitting one or more photons. From this viewpoint, therefore, the absorption cross section is simply the total cross section σ_t for all possible scattering processes²³. On the other hand, according to formula (59.25), which expresses the optical theorem, this cross section is determined by the anti-Hermitian part of the polarizability tensor.

Substitution of the tensor α_{ik} from (63.5) into (59.25) gives the following formula for the absorption cross section of a photon whose frequency ω is close to ω_{n1} :

$$\sigma^{(\text{abs})} = 4\pi^2 \sum_{M_n} |\mathbf{d}_{n1}\mathbf{e}|^2 \omega \frac{\Gamma_n/2}{\pi[(\omega - \omega_{n1})^2 + \Gamma_n^2/4]}. \quad (63.6)$$

In the limit $\Gamma_n \rightarrow 0$, the last factor in this formula tends to the delta function $\delta(\omega - \omega_{n1})$, in agreement with the fact that in this case only a photon of a precisely specified frequency can be absorbed. Let light with spectral and angular energy-flux density $I_{\mathbf{ke}}$ fall on the atom (cf. (44.7)). The number flux density of photons is then $I_{\mathbf{ke}} d\omega d\Omega/\omega$, and the absorption probability is

$$dW^{(\text{abs})} = \sigma^{(\text{abs})} \frac{I_{\mathbf{ke}}}{\omega} d\omega d\Omega. \quad (63.7)$$

²³We emphasize that this concerns absorption by a system in a stable ground state. Since the duration of an experiment is finite, the formulation of the question for an excited state would be different.

If $I_{\mathbf{ke}}(\omega)$ varies little over the width Γ_n , integration over frequencies gives

$$dw^{(\text{abs})} = 4\pi^2 \sum_{M_n} |\mathbf{d}_{n1}\mathbf{e}|^2 I_{\mathbf{ke}}(\omega_{n1}) d\omega do.$$

On the other hand, noting that, according to (45.5),

$$dw^{(\text{sp})} = \frac{\omega^3}{2\pi} \sum_{M_n} |\mathbf{d}_{1n}\mathbf{e}^*|^2 do = \frac{\omega^3}{2\pi} \sum_{M_n} |\mathbf{d}_{n1}\mathbf{e}|^2 do$$

is the probability of spontaneous emission of a photon with frequency ω_{n1} , we recover formula (44.9).

The Scattering Matrix

§ 64. The scattering amplitude

The general collision problem is to determine, from a specified initial state of a system (some collection of free particles), the probabilities of its various possible final states (other collections of free particles). If $|i\rangle$ denotes the initial state, the result of the collision may be represented by the superposition

$$\sum_f |f\rangle \langle f|S|i\rangle, \quad (64.1)$$

where the sum extends over the possible final states $|f\rangle$. The coefficients $\langle f|S|i\rangle$, abbreviated S_{fi} , form the *scattering matrix*, or *S-matrix*.¹ The squares $|S_{fi}|^2$ give the transition probabilities to specified states $|f\rangle$.

Without interactions between the particles, the system's state would not change, corresponding to a unit *S*-matrix (no scattering). It is convenient always to separate out this unit matrix and write

$$S_{fi} = \delta_{fi} + i(2\pi)^4 \delta^{(4)}(P_f - P_i) T_{fi}, \quad (64.2)$$

where T_{fi} is a new matrix. In the second term the four-dimensional delta function expressing conservation of 4-momentum has been separated out (P_i and P_f are the sums of the 4-momenta of all particles in the initial and final states); the remaining factors are included for later convenience. For off-diagonal matrix elements the first term in (64.2) vanishes, and for a transition $i \rightarrow f$,

$$S_{fi} = i(2\pi)^4 \delta^{(4)}(P_f - P_i) T_{fi}. \quad (64.3)$$

We shall call the matrix elements T_{fi} left after extracting the delta function the *scattering amplitudes*.

On forming $|S_{fi}|^2$, the square of a delta function occurs. It is understood as follows. The delta function arises from

$$\delta^{(4)}(P_f - P_i) = \frac{1}{(2\pi)^4} \int e^{i(P_f - P_i)x} d^4x. \quad (64.4)$$

Evaluate another identical integral at $P_f = P_i$ (as enforced by the first delta function),

¹The letter is taken from the English word scattering or the German Streuung.

over a large but finite volume V and time interval t ; it gives $Vt/(2\pi)^4$.² Thus

$$|S_{fi}|^2 = (2\pi)^4 \delta^{(4)}(P_f - P_i) |T_{fi}|^2 Vt.$$

Division by t gives the transition probability per unit time:

$$w_{i \rightarrow f} = (2\pi)^4 \delta^{(4)}(P_f - P_i) |T_{fi}|^2 V. \quad (64.5)$$

Each free initial or final particle is described by its own wave function, a plane wave with amplitude u (a bispinor for an electron, a 4-vector for a photon, and so on). The scattering amplitude has the structure

$$T_{fi} = u_1^* u_2^* \dots Q u_1 u_2 \dots, \quad (64.6)$$

where the amplitudes of the final-particle wave functions stand on the left and those of the initial particles on the right; Q is a matrix in the component indices of all these amplitudes.

The most important cases have only one or two particles initially. The first is a decay and the second a two-particle collision.

First consider decay into any number of particles with momenta $\mathbf{p}'_a$ in the momentum-space element $\prod_a d^3 p'_a$ (the index a labels final particles, so $\sum_a \mathbf{p}'_a = \mathbf{P}_f$). The number of states in this element and normalization volume V ³ is

$$\prod_a \frac{V d^3 p'_a}{(2\pi)^3}.$$

Multiplication of (64.5) by this quantity gives

$$dw = (2\pi)^4 \delta^{(4)}(P_f - P_i) |T_{fi}|^2 V \prod_a \frac{V d^3 p'_a}{(2\pi)^3}. \quad (64.7)$$

The wave functions used in the matrix element must be normalized to one particle in volume V . For an electron this is the plane wave (23.1), for a spin-1 particle (14.12), and for a photon (4.3). Each contains $1/\sqrt{2\varepsilon V}$, where ε is the particle energy. It is convenient below to write all particle wave functions without these factors and include them in the probability instead. Thus an electron plane wave is

$$\psi = u e^{-i p x}, \quad \bar{u} u = 2m, \quad (64.8)$$

and a photon wave is

$$A = \sqrt{4\pi} \mathbf{e} e^{-i k x}, \quad \mathbf{e} \mathbf{e}^* = -1, \quad \mathbf{e} \mathbf{k} = 0. \quad (64.9)$$

²Equivalently, first evaluate the integral over each coordinate in (64.4) between finite limits and then send the limits to infinity using (vol. III, (42.4))

$$\lim_{\xi \rightarrow \infty} \frac{\sin^2 \alpha \xi}{\xi^2} = \pi \delta(\alpha).$$

³For clarity in this section, we shall not set the normalization volume equal to unity.

Denote the amplitude calculated with these functions by M_{fi} , to distinguish it from T_{fi} . Evidently,

$$T_{fi} = \frac{M_{fi}}{(2\varepsilon_1 V \dots 2\varepsilon'_1 V \dots)^{1/2}}, \quad (64.10)$$

with one factor $\sqrt{2\varepsilon V}$ for every initial or final particle.

For a decay, (64.7) consequently becomes

$$dw = (2\pi)^4 \delta^{(4)}(P_f - P_i) |M_{fi}|^2 \frac{1}{2\varepsilon} \prod_a \frac{d^3 p'_a}{(2\pi)^3 2\varepsilon'_a}, \quad (64.11)$$

where ε is the energy of the decaying particle; as required, the normalization volume has disappeared.⁴

For a two-particle decay, we can eliminate the delta functions from (64.11). In the rest frame of the decaying particle, $\mathbf{p}'_1 = -\mathbf{p}'_2 \equiv \mathbf{p}'$ and $\varepsilon'_1 + \varepsilon'_2 = m$, so

$$dw = \frac{1}{(2\pi)^2} |M_{fi}|^2 \frac{1}{2m} \frac{1}{4\varepsilon'_1 \varepsilon'_2} \delta(\mathbf{p}'_1 + \mathbf{p}'_2) \delta(\varepsilon'_1 + \varepsilon'_2 - m) d^3 p'_1 d^3 p'_2.$$

The first delta function removes the $d^3 p'_2$ integration, while

$$d^3 p' = \mathbf{p}'^2 d|\mathbf{p}'| d\phi' = |\mathbf{p}'| d\phi' \frac{\varepsilon'_1 \varepsilon'_2 d(\varepsilon'_1 + \varepsilon'_2)}{\varepsilon'_1 + \varepsilon'_2} \quad (64.12)$$

(this follows at once from $\varepsilon_1'^2 - m_1'^2 = \varepsilon_2'^2 - m_2'^2 = \mathbf{p}'^2$). Integration over $d(\varepsilon'_1 + \varepsilon'_2)$ removes the second delta function and yields

$$dw = \frac{1}{32\pi^2 m^2} |M_{fi}|^2 |\mathbf{p}'| d\phi'. \quad (64.13)$$

Now consider collision of two particles with momenta $\mathbf{p}_1, \mathbf{p}_2$ and energies $\varepsilon_1, \varepsilon_2$, producing any number of particles with momenta $\mathbf{p}'_a$. Instead of (64.11),

$$dw = (2\pi)^4 \delta^{(4)}(P_f - P_i) |M_{fi}|^2 \frac{1}{4\varepsilon_1 \varepsilon_2 V} \prod_a \frac{d^3 p'_a}{(2\pi)^3 2\varepsilon'_a}.$$

Here the quantity of interest is the cross section $d\sigma$, not the probability. A Lorentz-invariant cross section is obtained by dividing dw by

$$j = \frac{I}{V \varepsilon_1 \varepsilon_2}, \quad (64.14)$$

where I is the 4-scalar

$$I = \sqrt{(p_1 p_2)^2 - m_1^2 m_2^2} \quad (64.15)$$

⁴If the final state contains N identical particles, integration over their momenta must include a factor $1/N!$ to account for the identity of states differing only by permutations of these particles.

(see vol. II, § 12).⁵ In the center-of-momentum frame ($\mathbf{p}_1 = -\mathbf{p}_2 \equiv \mathbf{p}$),

$$I = |\mathbf{p}|(\varepsilon_1 + \varepsilon_2), \quad (64.16)$$

and hence

$$j = \frac{|\mathbf{p}|}{V} \left(\frac{1}{\varepsilon_1} + \frac{1}{\varepsilon_2} \right) = \frac{v_1 + v_2}{V}, \quad (64.17)$$

which agrees with the usual collision-flux density (v_1, v_2 are the speeds).⁶ Therefore

$$d\sigma = (2\pi)^4 \delta^{(4)}(P_f - P_i) |M_{fi}|^2 \frac{1}{4I} \prod_a \frac{d^3 p'_a}{(2\pi)^3 2\varepsilon'_a}. \quad (64.18)$$

For a two-particle final state, eliminate the delta function in the center-of-momentum frame. Let $\varepsilon = \varepsilon_1 + \varepsilon_2 = \varepsilon'_1 + \varepsilon'_2$ be the total energy, and $\mathbf{p}_1 = -\mathbf{p}_2 \equiv \mathbf{p}$, $\mathbf{p}'_1 = -\mathbf{p}'_2 \equiv \mathbf{p}'$. The same elimination as in deriving (64.13) gives

$$d\sigma = \frac{1}{64\pi^2} |M_{fi}|^2 \frac{|\mathbf{p}'|}{|\mathbf{p}| \varepsilon^2} d\omega' \quad (64.19)$$

($|\mathbf{p}'| = |\mathbf{p}|$ for elastic scattering without a change in particle species).

Introduce the invariant

$$\begin{aligned} t &\equiv (p_1 - p'_1)^2 = m_1^2 + m_1'^2 - 2(p_1 p'_1) = \\ &= m_1^2 + m_1'^2 - 2\varepsilon_1 \varepsilon'_1 + 2|\mathbf{p}_1||\mathbf{p}'_1| \cos \theta, \end{aligned} \quad (64.20)$$

where θ is the angle between $\mathbf{p}_1$ and $\mathbf{p}'_1$. In the center-of-momentum frame, $|\mathbf{p}_1| = |\mathbf{p}|$ and $|\mathbf{p}'_1| = |\mathbf{p}'|$ depend only on ε , and at fixed ε ,

$$dt = 2|\mathbf{p}||\mathbf{p}'| d \cos \theta. \quad (64.21)$$

Thus in (64.19),

$$d\omega' = -d\varphi d \cos \theta = \frac{d\varphi d(-t)}{2|\mathbf{p}||\mathbf{p}'|},$$

where φ is the azimuth of $\mathbf{p}'_1$ about $\mathbf{p}_1$.⁷ Hence

$$d\sigma = \frac{1}{64\pi} |M_{fi}|^2 \frac{dt}{I^2} \frac{d\varphi}{2\pi} \quad (64.22)$$

⁵For later use, another form is

$$I^2 = \frac{1}{4} [s - (m_1 + m_2)^2] [s - (m_1 - m_2)^2], \quad (64.15a)$$

where $s = (p_1 + p_2)^2$.

⁶In an arbitrary frame,

$$j = (1/V) \sqrt{(\mathbf{v}_1 - \mathbf{v}_2)^2 - (\mathbf{v}_1 \times \mathbf{v}_2)^2}.$$

This reduces to the usual flux density whenever $\mathbf{v}_1 \parallel \mathbf{v}_2$: $j = |\mathbf{v}_1 - \mathbf{v}_2|/V$.

⁷Since the correct sign of the differential is evident in such cases, below we shall write dt instead of $d(-t)$, and similarly.

(we have again used (64.16) to introduce I). The azimuth φ , and hence (64.22), is invariant under Lorentz transformations that do not change the direction of relative motion. If the cross section is azimuth-independent,

$$d\sigma = \frac{1}{64\pi} |M_{fi}|^2 \frac{dt}{I^2}. \quad (64.23)$$

If one colliding particle is sufficiently heavy and its state is unchanged, its role reduces to that of a stationary source of a constant field in which the other particle scatters. Since energy, but not momentum, is conserved in a constant field, write

$$S_{fi} = i \cdot 2\pi\delta(E_f - E_i)T_{fi}. \quad (64.24)$$

In $|S_{fi}|^2$, the square of the one-dimensional delta function means

$$[\delta(E_f - E_i)]^2 \rightarrow \frac{1}{2\pi} \delta(E_f - E_i)t.$$

Passing from T_{fi} to M_{fi} as in deriving (64.11), the probability for one particle scattering in a constant field and producing any number of final particles is

$$dw = 2\pi\delta(E_f - \varepsilon) |M_{fi}|^2 \frac{1}{2\varepsilon V} \prod_a \frac{d^3 p'_a}{(2\pi)^3 2\varepsilon'_a}.$$

Here $\varepsilon = E_i$ is the initial-particle energy, while $\mathbf{p}'_a, \varepsilon'_a$ are the final momenta and energies. Dividing by the flux density $j = v/V$, with $v = |\mathbf{p}|/\varepsilon$, gives

$$d\sigma = 2\pi\delta(E_f - \varepsilon) |M_{fi}|^2 \frac{1}{2|\mathbf{p}|} \prod_a \frac{d^3 p'_a}{(2\pi)^3 2\varepsilon'_a}. \quad (64.25)$$

For elastic scattering the final state contains one particle with the same momentum magnitude and energy. Replacing $d^3 p' \rightarrow p'^2 dp' |do'| = |\mathbf{p}'| \varepsilon' d\varepsilon' do'$ and eliminating $\delta(\varepsilon' - \varepsilon)$ by integration over $d\varepsilon'$ gives

$$d\sigma = \frac{1}{16\pi^2} |M_{fi}|^2 do'. \quad (64.26)$$

Finally, if the external field depends on time (for example, the field of particles in prescribed motion), the S -matrix also lacks an energy delta function. Then $S_{fi} = iT_{fi}$, and after passing from T_{fi} to M_{fi} by (64.10), the probability, for example, that the field produces a specified collection of particles is

$$dw = |M_{fi}|^2 \prod_a \frac{d^3 p'_a}{(2\pi)^3 2\varepsilon'_a}. \quad (64.27)$$

§ 65. Reactions involving polarized particles

In this section we use simple examples to show how the polarization states of particles taking part in a reaction are included in calculating its cross section.

Suppose there is one electron in each of the initial and final states. The scattering amplitude then has the form

$$M_{fi} = \bar{u}' Au \quad (\equiv \bar{u}'_i A_{ik} u_k), \quad (65.1)$$

where u, u' are the bispinor amplitudes of the initial and final electrons, and A is a matrix (depending on the momenta and polarizations of any other particles taking part in the reaction).

The scattering cross section is proportional to $|M_{fi}|^2$. We have

$$(\bar{u}' Au)^* = u'^* \gamma^{0*} A^* u_k^* = u^* A^+ \gamma^{0+} u',$$

or

$$(\bar{u}' Au)^* = \bar{u} \bar{A} u', \quad (65.2)$$

where⁸

$$\bar{A} = \gamma^0 A^+ \gamma^0.$$

Thus

$$|M_{fi}|^2 = (\bar{u}' Au)(\bar{u} \bar{A} u') = u'_i \bar{u}'_k A_{kl} u_l \bar{u}_m \bar{A}_{mi}. \quad (65.3)$$

If the initial electron is in a mixed (partially polarized) state with density matrix ρ , and the final electron is to be detected in a specified polarization state ρ' , replace the products of bispinor components by

$$u'_i \bar{u}'_k \rightarrow \rho'_{ik}, \quad u_l \bar{u}_m \rightarrow \rho_{lm}.$$

Then

$$|M_{fi}|^2 = \text{Tr}(\rho' A \rho \bar{A}). \quad (65.4)$$

The density matrices ρ, ρ' are given by (29.13):

$$\rho = \frac{1}{2} (\gamma p + m) [1 - \gamma^5 (\gamma a)] \quad (65.5)$$

(and similarly for ρ').

For an unpolarized initial electron,

$$\rho = \frac{1}{2} (\gamma p + m). \quad (65.6)$$

Substitution of this expression is equivalent to averaging over the electron polarizations. If the cross section is required with arbitrary final-electron polarization, one must also put $\rho' = (\gamma p' + m)/2$ and double the result; this is equivalent to summing over the final-electron polarizations. Hence

$$\frac{1}{2} \sum_{\text{pol}} |M_{fi}|^2 = \frac{1}{2} \text{Tr}\{(\gamma p' + m) A (\gamma p + m) \bar{A}\}, \quad (65.7)$$

⁸For later use in forming $\bar{A}$, note the readily verified relations

$$\overline{\gamma^\mu} = \gamma^\mu, \quad \overline{\gamma^\mu \gamma^\nu \dots \gamma^\rho} = \gamma^\rho \dots \gamma^\nu \gamma^\mu, \quad \overline{\gamma^5} = -\gamma^5, \quad \overline{\gamma^5 \gamma^\mu} = \gamma^5 \gamma^\mu. \quad (65.2a)$$

where $\sum_{\text{pol}}$ denotes summation over initial and final polarizations, and the factor $1/2$ converts one of the sums into an average.

The density matrix ρ' in (65.4) is an auxiliary concept that essentially describes the detector, which selects a particular final-electron polarization, rather than the scattering process itself. We may instead ask for the polarization state into which the scattering process itself places the electron. If $\rho^{(f)}$ is its density matrix, the probability of detecting the electron in state ρ' is obtained by projecting $\rho^{(f)}$ onto ρ' , that is, by forming $\text{Tr}(\rho^{(f)}\rho')$. The corresponding cross section, and hence $|M_{fi}|^2$, is proportional to the same quantity. Comparison with (65.4) gives

$$\rho^{(f)} \sim A\rho\bar{A}. \quad (65.8)$$

Since $\rho^{(f)}$ must have the form (65.5) with some 4-vector $a^{(f)}$, the problem reduces to determining this vector. Formula (29.14) could be used, but the procedure below is still simpler.

We saw in § 29 that the components of a are expressed through the components of the 3-vector ζ , the mean doubled electron spin in its rest frame. Electron polarization states are completely specified by these vectors, so it is useful to express the cross section in terms of them as well. Evidently, $|M_{fi}|^2$ is linear in each of the vectors ζ, ζ' belonging to the initial and final electrons. As a function of ζ' it has the form

$$|M_{fi}|^2 = \alpha + \beta\zeta', \quad (65.9)$$

where α, β are themselves linear functions of ζ .

The vector ζ' in (65.9) is the specified final-electron polarization selected by the detector. The vector $\zeta^{(f)}$ corresponding to $\rho^{(f)}$ is found as follows. From the preceding discussion,

$$|M_{fi}|^2 \sim \text{Tr}(\rho'\rho^{(f)}).$$

This invariant may be calculated in any frame. In the final electron's rest frame, (29.20) gives

$$\rho'\rho^{(f)} \sim (1 + \sigma\zeta')(1 + \sigma\zeta^{(f)}).$$

Therefore

$$|M_{fi}|^2 \sim 1 + \zeta'\zeta^{(f)},$$

and comparison with (65.9) gives

$$\zeta^{(f)} = \beta/\alpha. \quad (65.10)$$

Thus calculating the cross section as a function of the parameter ζ' also determines the polarization $\zeta^{(f)}$.

More complicated cases, with more than one initial or final electron, are treated analogously by the same scheme.

For example, with two electrons in both the initial and final states, the amplitude is

$$M_{fi} = (\bar{u}'_1 A u_1)(\bar{u}'_2 B u_2) + (\bar{u}'_2 C u_1)(\bar{u}'_1 D u_2),$$

where u_1, u_2 are the initial and u'_1, u'_2 the final bispinor amplitudes. Forming $|M_{fi}|^2$ produces terms of the form

$$|\overline{u'_1} A u_1|^2 |\overline{u'_2} B u_2|^2$$

and terms of the form

$$(\overline{u'_1} A u_1)(\overline{u'_2} B u_2)(\overline{u'_2} C u_1)^* (\overline{u'_1} D u_2)^*.$$

The first reduce to products of two traces of the form (65.4), and the second to traces of the form

$$\text{Tr}(\rho'_1 A \rho_1 \overline{C} \rho'_2 B \rho_2 \overline{D}).$$

Positrons are described by “negative-frequency” amplitudes $u(-p)$. For reactions involving positrons, the only change is to use density matrices differing from (65.5), (65.6) by reversal of the sign before m (cf. (29.16), (29.17)).

We now turn to the polarization states of photons participating in the reaction.

The polarization of each initial photon enters the scattering amplitude linearly as a 4-vector e , and that of each final photon as e^* . In both cases the cross section, that is $|M_{fi}|^2$, contains the 4-tensor $e_\mu e_\nu^*$. For an arbitrary partially polarized state this tensor must

be replaced by the four-dimensional density matrix, the 4-tensor $\rho_{\mu\nu}$:

$$e_\mu e_\nu^* \rightarrow \rho_{\mu\nu}. \quad (65.11)$$

In particular, for an unpolarized photon, (8.15) gives

$$\rho_{\mu\nu} = -g_{\mu\nu}/2. \quad (65.12)$$

Thus averaging over photon polarizations reduces to contraction in $|M_{fi}|^2$ over the corresponding pair of tensor indices $\mu\nu$.⁹

To sum, rather than average, over photon polarizations, replace $e_\mu e_\nu^*$ by twice this expression:

$$e_\mu e_\nu^* \rightarrow -g_{\mu\nu}. \quad (65.13)$$

The density matrix of a polarized photon is given by (8.17). The choice of the 4-vectors $e^{(1)}, e^{(2)}$ occurring there is generally dictated by the particular problem. Sometimes they may be tied to specified spatial directions in a particular frame; in other cases it is more convenient to relate them to characteristic 4-vectors in the problem, namely particle 4-momenta.

In (8.17), photon polarization is described by the Stokes parameters forming the “vector” $\xi = (\xi_1, \xi_2, \xi_3)$. As for the electron, the polarization $\xi^{(f)}$ of the final photon itself must be distinguished from the polarization ξ' selected by the detector. If the squared amplitude is known as a function of ξ' ,

$$|M_{fi}|^2 = \alpha + \beta \xi',$$

then $\xi^{(f)} = \beta/\alpha$, in analogy with (65.10).

⁹Expression (65.12) formally reduces averaging over the two physically possible photon polarizations to averaging over four independent directions of the 4-vector e .

§ 66. Kinematic invariants

Let us consider some kinematic relations for scattering processes in which both the initial and the final state contain just two particles. We have in mind relations that follow from the general conservation laws alone and are therefore valid irrespective of the nature of the particles and of the laws governing their interaction.

We write four-momentum conservation in a general form that does not specify in advance which momenta belong to the initial particles and which to the final ones:

$$q_1 + q_2 + q_3 + q_4 = 0. \quad (66.1)$$

Here the $+q_a$ are four-momentum vectors; two correspond to the incident particles, while the other two correspond to the scattered particles, whose momenta are $-q_a$. In other words, two of the q_a have time component $q_a^0 > 0$, and two have $q_a^0 < 0$.

Charge conservation must hold along with four-momentum conservation. Charge here may mean not only electric charge, but also other conserved quantities whose signs are opposite for particles and antiparticles.

When the types of particles participating in the process are specified, the squares of the four-vectors q_a are the specified squared particle masses ($q_a^2 = m_a^2$). Three distinct reactions result, depending on the ranges of the time components q_a^0 and on the charges. We write these three processes as

$$\begin{aligned} \text{I.} \quad & 1 + 2 \rightarrow 3 + 4, \\ \text{II.} \quad & 1 + \bar{3} \rightarrow \bar{2} + 4, \\ \text{III.} \quad & 1 + \bar{4} \rightarrow \bar{2} + 3. \end{aligned} \quad (66.2)$$

Here a numeral denotes the particle number, and a bar over it distinguishes an antiparticle from a particle. Passing from one reaction to another—that is, moving a particle from one side of a formula to the other—corresponds to changing the sign of the relevant time component q_a^0 and also the sign of the charge, or equivalently to replacing a particle by its antiparticle. (The reverse reactions are, of course, also possible alongside the processes (66.2).)

The three processes (66.2) are described as the three *crossed channels* of a single generalized reaction.

Here are several examples. If particles 1 and 3 are electrons and particles 2 and 4 are photons, channel I is the scattering of a photon by an electron; because the photon is truly neutral, channel III is the same as I. Channel II is the conversion of an electron–positron pair into two photons. If all four particles are electrons, channel I is electron–electron scattering, whereas channels II and III are positron–electron scattering. If particles 1 and 3 are electrons and particles 2 and 4 are muons, channel I is $e-\mu$ scattering, channel III is $e-\bar{\mu}$ scattering, and channel II is the conversion of an $e\bar{e}$ pair into a $\mu\bar{\mu}$ pair.

Invariant quantities formed from the four-momenta play a special role in the study of scattering processes. The invariant scattering amplitudes are functions of these quantities (see § 70).

Two independent invariant quantities can be formed from the four four-momenta. Indeed, by (66.1), only three of the four-vectors q_a are independent; take them to be q_1 ,

q_2 , and q_3 . Six invariants can be formed from them: the three squares q_1^2, q_2^2, q_3^2 and the three products $q_1 q_2, q_1 q_3, q_2 q_3$. The first three, however, are the specified squared masses, while the latter three obey one relation following from the equality¹⁰

$$(q_1 + q_2 + q_3)^2 = q_4^2 = m_4^2.$$

For greater symmetry, however, it is convenient to consider not two but three invariants, chosen as follows:

$$\begin{aligned} s &= (q_1 + q_2)^2 = (q_3 + q_4)^2, \\ t &= (q_1 + q_3)^2 = (q_2 + q_4)^2, \\ u &= (q_1 + q_4)^2 = (q_2 + q_3)^2. \end{aligned} \quad (66.3)$$

As is readily seen, they satisfy

$$s + t + u = h, \quad (66.4)$$

where

$$h = m_1^2 + m_2^2 + m_3^2 + m_4^2. \quad (66.5)$$

In the principal channel (I), the invariant s has a simple physical meaning. It is the square of the total energy of the colliding particles (1 and 2) in their center-of-momentum frame (when $\mathbf{p}_1 + \mathbf{p}_2 = 0$: $s = (\varepsilon_1 + \varepsilon_2)^2$). In channel II the invariant t plays the analogous role, and in channel III the invariant u does so. Accordingly, channels I, II, and III are often called the s -, t -, and u -channels.

It is straightforward to express the invariants s , t , and u in terms of the energies and momenta of the colliding particles in each channel. Consider the s -channel. In the center-of-momentum frame of particles 1 and 2, the time and space components of the four-vectors q_a are

$$\begin{aligned} q_1 &= p_1 = (\varepsilon_1, \mathbf{p}_s), & q_2 &= p_2 = (\varepsilon_2, -\mathbf{p}_s), \\ q_3 &= -p_3 = (-\varepsilon_3, -\mathbf{p}'_s), & q_4 &= -p_4 = (-\varepsilon_4, \mathbf{p}'_s) \end{aligned} \quad (66.6)$$

(the subscript s on $\mathbf{p}_s$ and $\mathbf{p}'_s$ serves as a reminder that these momenta refer to the reaction in the s -channel). Then

$$s = \varepsilon_s^2, \quad \varepsilon_s = \varepsilon_1 + \varepsilon_2 = \varepsilon_3 + \varepsilon_4; \quad (66.7)$$

$$\begin{aligned} 4s\mathbf{p}_s^2 &= [s - (m_1 + m_2)^2][s - (m_1 - m_2)^2], \\ 4s\mathbf{p}'_s{}^2 &= [s - (m_3 + m_4)^2][s - (m_3 - m_4)^2]; \end{aligned} \quad (66.8)$$

$$\begin{aligned} 2t &= h - s + 4\mathbf{p}_s\mathbf{p}'_s - \frac{1}{s}(m_1^2 - m_2^2)(m_3^2 - m_4^2), \\ 2u &= h - s - 4\mathbf{p}_s\mathbf{p}'_s + \frac{1}{s}(m_1^2 - m_2^2)(m_3^2 - m_4^2). \end{aligned} \quad (66.9)$$

¹⁰In the general case, when $n > 4$ particles participate in a reaction, the number of functionally independent invariant variables is $3n - 10$. Indeed, there are altogether $4n$ quantities, the components of the n four-momenta q_a . They obey n functional relations $q_a^2 = m_a^2$ and four more supplied by the conservation law $\sum q_a = 0$. Arbitrary values may be assigned to six quantities, equal to the number of parameters specifying a general Lorentz transformation (a general rotation in four dimensions). Hence the number of independent invariant variables is $4n - n - 4 - 6 = 3n - 10$.

For elastic scattering ($m_1 = m_3$, $m_2 = m_4$), we have $|\mathbf{p}_s| = |\mathbf{p}'_s|$, and hence $\varepsilon_1 = \varepsilon_3$, $\varepsilon_2 = \varepsilon_4$. In place of (66.9), this gives the simpler formulas

$$\begin{aligned} t &= -(\mathbf{p}_s - \mathbf{p}'_s)^2 = -2\mathbf{p}_s^2(1 - \cos \theta_s), \\ u &= -2\mathbf{p}_s^2(1 + \cos \theta_s) + (\varepsilon_1 - \varepsilon_2)^2, \end{aligned} \quad (66.10)$$

where θ_s is the angle between $\mathbf{p}_s$ and $\mathbf{p}'_s$. Notice that the invariant $-t$ is then the square of the (three-dimensional) momentum transferred in the collision.

The corresponding formulas for the other channels follow by a simple change of notation. To pass to the t -channel, make the substitutions $s \leftrightarrow t$, $2 \leftrightarrow 3$ in (66.6)–(66.10); to pass to the u -channel, make the substitutions $s \leftrightarrow u$, $2 \leftrightarrow 4$.

§ 67. Physical regions

When scattering amplitudes are regarded as functions of the independent variables s, t, u (subject only to $s + t + u = h$), one must distinguish physically admissible and inadmissible regions of their values. Values that can correspond to a physical scattering process must satisfy conditions following from 4-momentum conservation and from the fact that the square of each 4-vector q_a has the specified value $q_a^2 = m_a^2$.

The product of two 4-momenta satisfies

$$p_a p_b \geq m_a m_b. \quad (67.1)$$

Consequently,

$$(q_a + q_b)^2 = (p_a + p_b)^2 \geq (m_a + m_b)^2,$$

if $q_a = p_a, q_b = p_b$ (or $q_a = -p_a, q_b = -p_b$), whereas

$$(q_a + q_b)^2 = (p_a - p_b)^2 \leq (m_a - m_b)^2,$$

if $q_a = p_a, q_b = -p_b$.

Thus for a reaction in the s -channel,

$$\begin{aligned} (m_1 + m_2)^2 &\leq s \leq (m_3 + m_4)^2, \\ (m_1 - m_3)^2 &\geq t \leq (m_2 - m_4)^2, \\ (m_1 - m_4)^2 &\geq u \leq (m_2 - m_3)^2 \end{aligned} \quad (67.2)$$

(with analogous inequalities in the t - and u -channels).

To find the remaining conditions, form the 4-vector L dual to the product of any three of the q_a , say

$$L_\lambda = e_{\lambda\mu\nu\rho} q_1^\mu q_2^\nu q_3^\rho. \quad (67.3)$$

In the rest frame of one particle, for example particle 1, $q_1 = (q_1^0, 0)$. Then L has only spatial components: $L_i = e_{ikl} q_1^0 q_2^k q_3^l$. Thus L is spacelike, and $L^2 \leq 0$ in every frame. Expanding L^2 gives

$$\begin{vmatrix} q_1^2 & q_1 q_2 & q_1 q_3 \\ q_2 q_1 & q_2^2 & q_2 q_3 \\ q_3 q_1 & q_3 q_2 & q_3^2 \end{vmatrix} \geq 0. \quad (67.4)$$

In terms of s, t, u , this has the same form in every channel:

$$stu \geq as + bt + cu, \quad (67.5)$$

where

$$\begin{aligned} ah &= (m_1^2 m_2^2 - m_3^2 m_4^2)(m_1^2 + m_2^2 - m_3^2 - m_4^2), \\ bh &= (m_1^2 m_3^2 - m_2^2 m_4^2)(m_1^2 + m_3^2 - m_2^2 - m_4^2), \\ ch &= (m_1^2 m_4^2 - m_2^2 m_3^2)(m_1^2 + m_4^2 - m_2^2 - m_3^2) \end{aligned} \quad (67.6)$$

(*T. W. B. Kibble*, 1960).

The ranges of s, t, u are conveniently displayed with triangular coordinates in a plane (the Mandelstam plane; *S. Mandelstam*, 1958). Its coordinate axes are three lines whose intersections form an equilateral triangle. The coordinates s, t, u are measured perpendicular to these lines, with the positive directions pointing into the triangle as shown by the arrows in Fig. 5. Thus every point corresponds to values s, t, u represented, with their appropriate signs, by

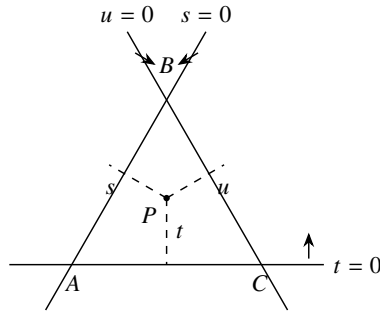


Fig. 5

the lengths of the perpendiculars dropped to the three axes. The condition $s+t+u = h$ is then ensured by a familiar geometrical theorem, if the altitude of the equilateral triangle is h .¹¹

Consider the important case in which the principal (s) channel is elastic scattering; the particle masses are then equal in pairs:

$$m_1 = m_3 \equiv m, \quad m_2 = m_4 \equiv \mu. \quad (67.7)$$

Let $m > \mu$. In (67.5),

$$h = 2(m^2 + \mu^2), \quad a = c = 0, \quad b = (m^2 - \mu^2)^2,$$

and hence

$$sut \geq (m^2 - \mu^2)^2 t. \quad (67.8)$$

The boundary consists of the line $t = 0$ and the hyperbola

$$su = (m^2 - \mu^2)^2, \quad (67.9)$$

¹¹Joining P in Fig. 5 to the three vertices of ABC divides the triangle into three triangles of altitudes s, t, u . Equating the sum of their areas to that of ABC gives the required equality. It is proved similarly when P lies outside ABC .

whose two branches lie in the sectors $u < 0, s < 0$ and $s > 0, u > 0$; the axes $s = 0, u = 0$ are its asymptotes. Equivalently, (67.8) may be written

$$t > 0, \quad su > (m^2 - \mu^2)^2$$

or

$$t < 0, \quad su < (m^2 - \mu^2)^2.$$

In addition, (67.2) requires $s > (m + \mu)^2$ in the s -channel and $u > (m + \mu)^2$ in the u -channel; the other inequalities then hold automatically. The *physical regions* of channels I, II, III (s, t, u) are therefore the shaded regions in Fig. 6.

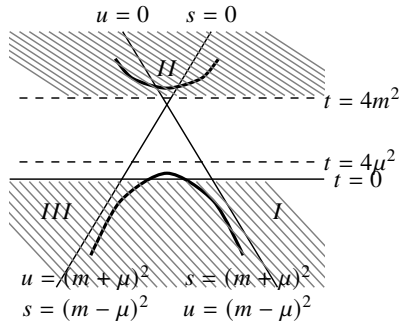


Fig. 6

If $\mu = 0$ (particles 2 and 4 are photons), the lower branch of the hyperbola touches the axis $t = 0$, and the physical regions are as shown in Fig. 7.

If $m = \mu$, the boundaries (67.8) degenerate into the coordinate axes, and the physical regions are the three sectors shown in Fig. 8.

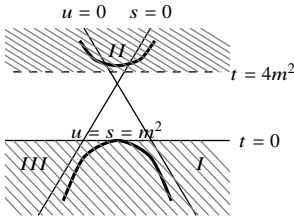


Fig. 7

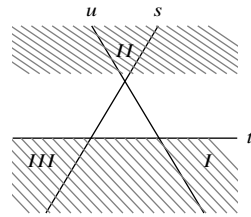


Fig. 8

For four distinct masses, the equation

$$stu = as + bt + cu \quad (67.10)$$

defines a cubic curve whose branches bound the physical regions of the three channels, as in Fig. 9.



Fig. 9

Let

$$m_1 \geq m_2 \geq m_3 \geq m_4.$$

Then

$$a \geq b \geq c, \quad a > 0, \quad b > 0.$$

The curve (67.10) intersects the coordinate axes at points on the line

$$as + bt + cu = 0$$

(the dashed lines in Fig. 9). Its form depends on the sign of c as shown. For $c < 0$, the physical region of the

u -channel occupies part of the coordinate triangle; s, t, u can then all be positive. The three coordinate axes are asymptotes of the corresponding branches. Conditions (67.2) generally add nothing to the boundaries fixed by (67.10): their equality lines do not cross the shaded physical regions, although some touch their boundaries at extreme values of s, t, u .

If one mass exceeds the sum of the other three, $m_1 > m_2 + m_3 + m_4$, a fourth, decay channel is possible in addition to I, II, III:

$$\text{IV. } 1 \rightarrow \bar{2} + 3 + 4. \quad (67.11)$$

In the rest frame of the decaying particle,

$$q_1 = (m_1, 0), \quad q_2 = (-\varepsilon_2, -\mathbf{p}_2), \quad q_3 = (-\varepsilon_3, -\mathbf{p}_3), \quad q_4 = (-\varepsilon_4, -\mathbf{p}_4),$$

$$\varepsilon_2 + \varepsilon_3 + \varepsilon_4 = m_1, \quad \mathbf{p}_2 + \mathbf{p}_3 + \mathbf{p}_4 = 0.$$

The invariants are

$$\begin{aligned} s &= m_1^2 + m_2^2 - 2m_1\varepsilon_2, \\ t &= m_1^2 + m_3^2 - 2m_1\varepsilon_3, \\ u &= m_1^2 + m_4^2 - 2m_1\varepsilon_4. \end{aligned} \quad (67.12)$$

Equation (67.1) now gives

$$\begin{aligned} (m_3 + m_4)^2 &\leq s \leq (m_1 - m_2)^2, \\ (m_2 + m_4)^2 &\leq t \leq (m_1 - m_3)^2, \\ (m_2 + m_3)^2 &\leq u \leq (m_1 - m_4)^2. \end{aligned} \quad (67.13)$$

All three invariants are positive, so the physical region of the decay channel lies inside the coordinate triangle.

PROBLEM

1. Find the physical regions for three equal masses: $m_1 \equiv m, m_2 = m_3 = m_4 \equiv \mu$ (for example, $K + \pi \rightarrow \pi + \pi$).

SOLUTION. Equation (67.10) becomes

$$stu = \mu^2(m^2 - \mu^2)^2, \quad (1)$$

with

$$s + t + u = 3\mu^2 + m^2.$$

Regions I, II, III are bounded by curves of identical form (for I, $s > 0, t < 0, u < 0$, and similarly for II and III). If $m > 3\mu$, (1) also has a branch, a closed curve with $s, t, u > 0$, bounding the region of channel IV (Fig. 10).

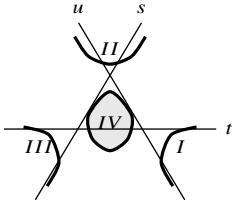


Fig. 10

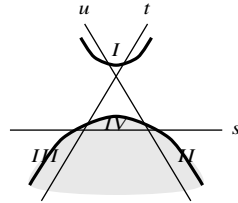


Fig. 11

2. Find the same regions for $m_1 \equiv m, m_2 \equiv \mu, m_3 = m_4 = 0, m > \mu$ (for example, $\mu + \nu \rightarrow e + \nu$).

SOLUTION. Condition (67.5) becomes

$$stu \geq m^2 \mu^2 s,$$

with $s + t + u = m^2 + \mu^2$. The physical regions are bounded by the axis $s = 0$ and the two branches of $tu = m^2 \mu^2$ (Fig. 11).

3. Find the same regions for $m_1 = m_3 \equiv m, m_2 = 0, m_4 = \mu$, with $m > 2\mu$ (for example, $\rho + \gamma \rightarrow \rho + \pi^0$).

SOLUTION. The boundary equation (67.10) becomes

$$\begin{aligned} stu &= a(s + u) + bt, & ah &= m^2 \mu^4, \\ bh &= m^4(2m^2 - \mu^2), & h &= 2m^2 + \mu^2. \end{aligned}$$

Eliminating u gives

$$t^2 + \left(\frac{b-a}{s} + s - h \right) t + \frac{ah}{s} = 0.$$

At fixed s this is a quadratic in t . For $s > (m + \mu)^2$ (the s -channel region), each s has two negative values of t . At $s = (m + \mu)^2$ the roots merge at $t = -m\mu^2/(m + \mu)$. The s -channel boundary is shown in Fig. 12. Its lower branch approaches $u = 0$ asymptotically, and the upper branch crosses this axis at $t = \mu^4/(\mu^2 - m^2)$.

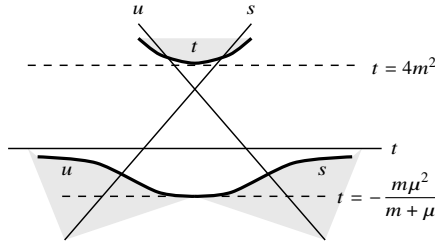


Fig. 12

The u -channel region is symmetric to the s -channel region, while the t -channel region is positioned as shown.

§ 68. Expansion in partial-wave amplitudes

An essential step in the analysis of a reaction of the form

$$a + b \rightarrow c + d \quad (68.1)$$

is the expansion of the scattering amplitude in partial-wave amplitudes, each corresponding (at a specified total energy ε) to a definite value of the total angular momentum J of the particles in their center-of-momentum frame¹².

In other words, these partial-wave amplitudes are elements of the S -matrix in the angular-momentum representation:

$$\langle \varepsilon J' M' | S | \varepsilon J M \rangle.$$

Since the angular momentum J and its projection M on a specified z axis are conserved, the S -matrix is diagonal in these quantum numbers (as it is in the energy ε). By spatial isotropy, moreover, the diagonal elements are independent of the value of M . At specified J , M , and ε , the scattering matrix remains a matrix with respect to the spin quantum numbers; we write its elements more briefly as

$$\langle \varepsilon J M \lambda' | S | \varepsilon J M \lambda \rangle \equiv \langle \lambda' | S^J(\varepsilon) | \lambda \rangle, \quad (68.2)$$

where λ and λ' denote sets of spin quantum numbers. The natural choice for these numbers here is the particle helicities. Recall that helicity (unlike the projection of spin on an arbitrary spatial axis) is conserved for a free particle and also commutes with both its momentum and its angular momentum (see § 16). Helicity may therefore be used in both the momentum and the angular-momentum representations of the scattering matrix.

We shall call the elements of the S -matrix with helicity indices *helicity scattering amplitudes*; thus λ and λ' will denote the sets of helicities of the initial and final particles: $\lambda = (\lambda_a, \lambda_b)$, $\lambda' = (\lambda_c, \lambda_d)$.

In the momentum representation, the scattering-matrix elements are defined with respect to the states $|\varepsilon \mathbf{n} \lambda\rangle$ ($\mathbf{n} = \mathbf{p}/|\mathbf{p}|$ is the direction of the relative-motion momentum

¹²Most of the results presented in § 68 and 69 are due to Jacob and Wick (*M. Jacob, G. C. Wick, 1959*).

in the center-of-momentum frame), and in the angular-momentum representation with respect to the states $|\varepsilon JM\lambda\rangle$. The two are related by the expansions

$$|JM\lambda\rangle = \int |\mathbf{n}\lambda\rangle \langle \mathbf{n}\lambda | JM\lambda \rangle d\mathbf{n}, \quad (68.3)$$

where the integration is over the directions $\mathbf{n}$ (for brevity, the energy ε will be omitted from the state symbols). Since this transformation is unitary (see Vol. III, § 12), the coefficients of the inverse transformation are

$$\langle JM\lambda | \mathbf{n}\lambda \rangle = \langle \mathbf{n}\lambda | JM\lambda \rangle^*. \quad (68.4)$$

By the general rule for matrix transformations, the same coefficients determine the relation between the S -matrix elements in the two representations:

$$\langle \mathbf{n}'\lambda' | S | \mathbf{n}\lambda \rangle = \sum_{JM} \langle \mathbf{n}'\lambda' | JM\lambda' \rangle \langle JM\lambda' | S | JM\lambda \rangle \langle JM\lambda | \mathbf{n}\lambda \rangle. \quad (68.5)$$

The expansion coefficients in (68.3) are readily found from the results of § 16.

Let the wave functions of all the states be expressed in the momentum representation, that is, as functions of the momentum direction (at a specified energy). To distinguish this direction as an independent variable from the direction $\mathbf{n}$ serving as a quantum number of the state, denote it by $\mathbf{v}$. In this representation the wave function has the form (16.2)

$$\psi_{\mathbf{n}\lambda}(\mathbf{v}) = u^{(\lambda)} \delta^{(2)}(\mathbf{v} - \mathbf{n}). \quad (68.6)$$

Upon substitution of (68.6), the expansion (68.3) reduces to a single term:

$$\psi_{JM\lambda} = \langle \mathbf{v}\lambda | JM\lambda \rangle u^{(\lambda)}. \quad (68.7)$$

The helicity λ_a or λ_b of each of the two particles is defined as the projection of its spin on the direction of its own momentum. If the particle momenta are $\mathbf{p}_a \equiv \mathbf{p}$, $\mathbf{p}_b = -\mathbf{p}$, this direction is $\mathbf{n}$ for the first particle and $-\mathbf{n}$ for the second. If the system is now regarded as a single particle with helicity Λ in the direction $\mathbf{n}$, then $\Lambda = \lambda_a - \lambda_b$. According to (16.4), its wave function (in the momentum representation) can be written as

$$\psi_{JM\lambda}(\mathbf{v}) = u^{(\lambda)} D_{\Lambda M}^{(J)}(\mathbf{v}) \sqrt{\frac{2J+1}{4\pi}}. \quad (68.8)$$

Comparison of (68.7) and (68.8), followed by changing the name of the variable $\mathbf{v}$ to $\mathbf{n}$, gives the required coefficients:

$$\langle \mathbf{n}\lambda | JM\lambda \rangle = \sqrt{\frac{2J+1}{4\pi}} D_{\Lambda M}^{(J)}(\mathbf{n}). \quad (68.9)$$

Substitution of these coefficients into (68.5) gives

$$\langle \mathbf{n}'\lambda' | S | \mathbf{n}\lambda \rangle = \sum_{JM} \frac{2J+1}{4\pi} D_{\Lambda'M}^{(J)}(\mathbf{n}') D_{\Lambda M}^{(J)*}(\mathbf{n}) \langle \lambda' | S^J | \lambda \rangle, \quad (68.10)$$

$$\Lambda = \lambda_a - \lambda_b, \quad \Lambda' = \lambda_c - \lambda_d,$$

where the abbreviated notation (68.2) has been used. Choose the direction $\mathbf{n}$ as the z axis; then

$$D_{\Lambda M}^{(J)}(\mathbf{n}) = \delta_{\Lambda M}$$

and (68.10) becomes

$$\langle \mathbf{n}' \lambda' | S | \mathbf{n} \lambda \rangle = \sum_J \frac{2J+1}{4\pi} D_{\Lambda' \Lambda}^{(J)}(\mathbf{n}') \langle \lambda' | S^J | \lambda \rangle. \quad (68.11)$$

We see that the expansion in partial-wave amplitudes has the functions $D_{\Lambda' \Lambda}^{(J)}$ as its coefficients. For a reaction of the form (68.1), it is convenient to define the scattering amplitude f so that the cross section (in the center-of-momentum frame) is

$$d\sigma = |\langle \mathbf{n}' \lambda' | f | \mathbf{n} \lambda \rangle|^2 d\Omega' \quad (68.12)$$

(comparison with (64.19) relates this amplitude to the matrix element M_{fi}). We write its expansion in partial-wave amplitudes as

$$\langle \mathbf{n}' \lambda' | f | \mathbf{n} \lambda \rangle = \sum_{JM} (2J+1) D_{\Lambda' M}^{(J)}(\mathbf{n}') D_{\Lambda M}^{(J)*}(\mathbf{n}) \langle \lambda' | f^J | \lambda \rangle, \quad (68.13)$$

or, choosing the z axis along $\mathbf{n}$,

$$\langle \mathbf{n}' \lambda' | f | \mathbf{n} \lambda \rangle = \sum_J (2J+1) D_{\Lambda' \Lambda}^{(J)}(\mathbf{n}') \langle \lambda' | f^J | \lambda \rangle. \quad (68.14)$$

This formula generalizes the usual partial-wave expansion for the scattering of spinless particles (see Vol. III, (123.14)). Since $D_{00}^{(L)} = P_L(\cos \theta)$, when all spins vanish, (68.14) reduces to the expansion in Legendre polynomials

$$f(\theta) = \sum_L (2L+1) f_L P_L(\cos \theta).$$

The cross section (68.12) applies when every particle has a definite helicity. If the particles are in mixed polarization states, the cross section is obtained by averaging the product

$$\langle \lambda_c \lambda_d | f | \lambda_a \lambda_b \rangle \langle \lambda'_c \lambda'_d | f | \lambda'_a \lambda'_b \rangle^*$$

with the polarization density matrices of the particles,

$$\langle \lambda_a | \rho^{(a)} | \lambda'_a \rangle \langle \lambda_b | \rho^{(b)} | \lambda'_b \rangle \langle \lambda'_c | \rho^{(c)} | \lambda_c \rangle \langle \lambda'_d | \rho^{(d)} | \lambda_d \rangle$$

(see the note on p. 204). Thus, for a reaction between unpolarized particles a, b producing likewise unpolarized particles c, d , we obtain

$$d\sigma = \frac{d\Omega}{(2s_a+1)(2s_b+1)} \sum_{(\lambda)} \sum_{JJ'} (2J+1)(2J'+1) \langle \lambda_c \lambda_d | f^J | \lambda_a \lambda_b \rangle \times \quad (68.15)$$

$$\times \langle \lambda_c \lambda_d | f^{J'} | \lambda_a \lambda_b \rangle^* D_{\Lambda' \Lambda}^{(J)}(\mathbf{n}') D_{\Lambda' \Lambda}^{(J')*}(\mathbf{n}')$$

(the z axis is directed along $\mathbf{n}$, and $\sum_{(\lambda)}$ denotes summation over $\lambda_a \lambda_b \lambda_c \lambda_d$). Replacing the function $D_{\Lambda' \Lambda}^{(J')*}$ by means of formula (58.19) (see Vol. III) and then using the expansion (110.2) (see Vol. III), we finally obtain

$$\begin{aligned} d\sigma = \frac{d\sigma}{(2s_a + 1)(2s_b + 1)} \sum_{(\lambda)JJ'} (-1)^{\Lambda - \Lambda'} (2J + 1)(2J' + 1) \times \\ \times \langle \lambda_c \lambda_d | f^J | \lambda_a \lambda_b \rangle \langle \lambda_c \lambda_d | f^{J'} | \lambda_a \lambda_b \rangle^* \sum_L (2L + 1) \times \\ \times \begin{pmatrix} J & J' & L \\ \Lambda & -\Lambda & 0 \end{pmatrix} \begin{pmatrix} J & J' & L \\ \Lambda' & -\Lambda' & 0 \end{pmatrix} P_L(\cos \theta) \end{aligned} \quad (68.16)$$

(θ is the angle between $\mathbf{n}'$ and the z axis); the sum over L extends over all integer values arising from the vector addition of J and J' .

The expansion of the scattering amplitude in partial-wave amplitudes fully accounts for all properties of the angular distribution of scattering that follow from symmetry under spatial rotations. It does not, however, explicitly incorporate the properties associated with symmetry under spatial inversion. P invariance (when possessed by the interaction) leads to definite relations among the various helicity amplitudes (see § 69 below).

§ 69. Symmetry of helicity scattering amplitudes

The requirements imposed by symmetry under P , C , and T transformations (provided, of course, that the particle interaction in question actually has the corresponding symmetry) produce relations between different helicity scattering amplitudes and thereby reduce the number of independent amplitudes.¹³

To establish these relations, we first determine the symmetry properties of the helicity states of a two-particle system.

Consider the particles in their center-of-momentum frame. One has momentum $\mathbf{p}_1 \equiv \mathbf{p}$ and helicity λ_1 relative to $\mathbf{p}$, while the other has momentum $\mathbf{p}_2 = -\mathbf{p}$ and helicity λ_2 relative to $-\mathbf{p}$. If instead both helicities are defined relative to the same direction $\mathbf{p}$, they are λ_1 and $-\lambda_2$. The particles are correspondingly described by plane waves with amplitudes $u_{\mathbf{p}}^{(\lambda_1)}$ and $u_{\mathbf{p}}^{(-\lambda_2)}$. The two-particle system is described by the multicomponent function $u_{\mathbf{p}}^{(\lambda_1 \lambda_2)}$ formed from products of these amplitudes.

Regarding the system as a single particle of helicity $\Lambda = \lambda_1 - \lambda_2$ in the direction $\mathbf{n} = \mathbf{p}/|\mathbf{p}|$, we may write its momentum-representation wave function, as a function of $\mathbf{n}$, for a state with specified $J, M, \lambda_1, \lambda_2$ and total energy ε :

$$\psi_{JM\lambda_1\lambda_2} = u_{\mathbf{p}}^{(\lambda_1\lambda_2)} D_{\Lambda M}^{(J)}(\mathbf{n}) \sqrt{\frac{2J+1}{4\pi}}, \quad \Lambda = \lambda_1 - \lambda_2. \quad (69.1)$$

(Cf. (68.8).) Since Λ is the projection of the total angular momentum on $\mathbf{p}$,

$$|\Lambda| \leq J. \quad (69.2)$$

¹³The number of independent amplitudes does not, of course, depend on the particular representation of the matrix S^J and remains the same for any choice of spin variables.

According to (16.14), under inversion

$$\begin{aligned}\widehat{P}u^{(\lambda_1\lambda_2)}(\mathbf{n}) &= \eta_1\eta_2 u^{(\lambda_1\lambda_2)}(-\mathbf{n}) = \\ &= \eta_1\eta_2(-1)^{s_1+s_2-\lambda_1+\lambda_2} u^{(-\lambda_1,-\lambda_2)}(\mathbf{n}),\end{aligned}\quad (69.3)$$

where η_1, η_2 are the intrinsic parities of the particles. Using (16.10) as well, we find the transformation law for (69.1):

$$\widehat{P}\psi_{JM\lambda_1\lambda_2} = \eta_1\eta_2(-1)^{s_1+s_2-J}\psi_{JM,-\lambda_1,-\lambda_2}. \quad (69.4)$$

For identical particles, their exchange symmetry must also be considered. Exchanging the particles exchanges their momenta and spins. The definition of (69.1) treats the particles asymmetrically: both angular momenta are projected on $\mathbf{p}_1 = \mathbf{p}$, the momentum of the first particle. After exchange this direction is replaced by $\mathbf{p}_2 = -\mathbf{p}$; the projections of $\mathbf{j}_1, \mathbf{j}_2$ on it are $-\lambda_1, \lambda_2$, instead of $\lambda_1, -\lambda_2$ on $\mathbf{p}$. Hence the exchange operator $\widehat{P}_{12}$ acts as

$$\widehat{P}_{12}\psi_{JM\lambda_1\lambda_2} = u_{\mathbf{p}}^{(-\lambda_1,-\lambda_2)}(-\mathbf{n})D_{\Lambda M}^{(J)}(-\mathbf{n})\sqrt{\frac{2J+1}{4\pi}},$$

where $\Lambda = \lambda_1 - \lambda_2$ as before. Equations (69.3), (16.10) give

$$\widehat{P}_{12}\psi_{JM\lambda_1\lambda_2} = (-1)^{2s-J}\psi_{JM\lambda_2\lambda_1}, \quad (69.5)$$

where $s_1 = s_2 \equiv s$.

For identical particles, only states symmetric under exchange for bosons or antisymmetric for fermions are allowed. Since the former have integral and the latter half-integral s , the allowed helicity states in both cases may be written

$$\left[1 + (-1)^{2s}\widehat{P}_{12}\right]\psi_{JM\lambda_1\lambda_2},$$

or, using (69.5),

$$\psi_{JM\lambda_1\lambda_2} + (-1)^J\psi_{JM\lambda_2\lambda_1}. \quad (69.6)$$

Remarkably, this combination has the same form for bosons and fermions.

For a particle–antiparticle system, exchange is described by the same formula (69.5). Unlike identical particles, however, both exchange symmetries are allowed, giving the two combinations

$$\psi^\pm = \psi_{JM\lambda_1\lambda_2} \pm (-1)^J\psi_{JM\lambda_2\lambda_1}. \quad (69.7)$$

These states have definite charge parity C . Charge conjugation can be represented as complete exchange of all variables, spin and charge, followed by reverse exchange of the spin variables (helicities). The first operation must give the same result as exchange in a system of two identical particles. Thus the upper sign in (69.7), which is also the sign in the identical-particle state (69.6), is charge-even, while the lower sign is charge-odd:

$$\widehat{C}\psi^\pm = \pm\psi^\pm.$$

Finally consider time reversal. The rest-frame wave function of a particle with spin s and projection σ transforms as

$$\widehat{T}\psi_{s\sigma} = (-1)^{s-\sigma}\psi_{s,-\sigma}$$

(see vol. III, (60.2)). In its transformation properties, the two-particle center-of-momentum wave function may likewise be regarded as that of a “particle at rest” with angular momentum J and projection M . The helicities λ_1, λ_2 do not change: time reversal reverses both momentum and angular momentum, so the products $\mathbf{j}\mathbf{p}$ remain unchanged. Hence

$$\widehat{T}\psi_{JM\lambda_1\lambda_2} = (-1)^{J-M}\psi_{J,-M,\lambda_1\lambda_2}. \quad (69.8)$$

We can now write the symmetry relations for the helicity amplitudes directly.

If the interaction is P -invariant, then for the reaction

$$a + b \rightarrow c + d$$

the transition amplitudes, at fixed J, ε , for

$$|\lambda_a\lambda_b\rangle \rightarrow |\lambda_c\lambda_d\rangle \quad \text{and} \quad \widehat{P}|\lambda_a\lambda_b\rangle \rightarrow \widehat{P}|\lambda_c\lambda_d\rangle$$

must coincide. Equation (69.4) therefore gives

$$\begin{aligned} \langle \lambda_c\lambda_d | S^J | \lambda_a\lambda_b \rangle &= \frac{\eta_c\eta_d}{\eta_a\eta_b} (-1)^{s_c+s_d-s_a-s_b} \\ &\quad \langle -\lambda_c, -\lambda_d | S^J | -\lambda_a, -\lambda_b \rangle. \end{aligned} \quad (69.9)$$

If states of definite parity are used instead, the combinations

$$\frac{1}{\sqrt{2}} \left(\psi_{JM\lambda_1\lambda_2} \pm \widehat{P}\psi_{JM\lambda_1\lambda_2} \right),$$

where $\lambda_1\lambda_2$ denotes either $\lambda_a\lambda_b$ or $\lambda_c\lambda_d$, then amplitudes for parity-changing transitions vanish.

Time reversal transforms each state by (69.8) and also interchanges initial and final states. Thus T -invariance gives

$$\langle \lambda_c\lambda_d | S^J(\varepsilon) | \lambda_a\lambda_b \rangle = \langle \lambda_a\lambda_b | S^J(\varepsilon) | \lambda_c\lambda_d \rangle. \quad (69.10)$$

These amplitudes generally belong to different processes, the direct and inverse reactions. Only in elastic scattering do the processes coincide, and (69.10) then relates helicity amplitudes of the same reaction.

In elastic scattering of two identical particles, exchange symmetry further reduces the number of different amplitudes. At fixed J , only states symmetric or only states anti-symmetric in λ_1, λ_2 occur. Conservation of angular momentum therefore automatically conserves the symmetry under exchange of helicities.

The situation is similar for elastic particle-antiparticle scattering, or conversion of one such pair into another, $a + \bar{a} \rightarrow b + \bar{b}$. At fixed J , both symmetric and antisymmetric states in λ_1, λ_2 exist, but they have different

charge parities. If the interaction is C -invariant, so that charge parity is conserved, transitions between states of different symmetry in λ_1, λ_2 are forbidden.¹⁴ This differs

¹⁴A similar prohibition may follow from isotopic invariance for nonidentical particles. To the accuracy of that invariance, transitions between states of different symmetry in λ_1, λ_2 are forbidden in neutron-proton scattering.

from identical particles, for which one of the two symmetries is absent altogether at each fixed J . In a particle–antiparticle system, only transitions between different symmetries are forbidden; both kinds of states exist for every J .

Because CPT invariance is universal, T -invariance also implies CP -invariance. The latter equates the amplitudes of two reactions related by replacing every particle with its antiparticle and reversing the helicities, with $\lambda_{\bar{a}} = -\lambda_a, \dots$ ¹⁵

$$\langle \lambda_c \lambda_d | S^J | \lambda_a \lambda_b \rangle = \langle \lambda_{\bar{c}} \lambda_{\bar{d}} | S^J | \lambda_{\bar{a}} \lambda_{\bar{b}} \rangle. \quad (69.11)$$

The number of independent amplitudes is the same in all crossed channels of one generalized reaction, so any channel may be used to determine it. Thus elastic scattering $a + b \rightarrow a + b$ and annihilation $a + \bar{a} \rightarrow b + \bar{b}$ are described by the same number of independent amplitudes. The restrictions imposed by T -invariance in the first process are equivalent to those imposed by C -invariance in the second.

Consider also the decay of one particle into two, $a \rightarrow b + c$. In the center-of-momentum frame, the rest frame of a , $\mathbf{p}_b = -\mathbf{p}_c$. Taking the scalar product of $\mathbf{j}_a = \mathbf{j}_b + \mathbf{j}_c$ with $\mathbf{p}_b$ gives

$$\lambda_a = \lambda_b - \lambda_c \quad (69.12)$$

(the helicity λ_a of the primary particle is defined as the projection of its spin on the momentum direction of one secondary particle). This relation may be viewed as a consequence of an additional symmetry of the process, axial symmetry about the direction of $\mathbf{p}_b, \mathbf{p}_c$. If $s_a < s_b + s_c$, (69.12) reduces the allowed sets $\lambda_a, \lambda_b, \lambda_c$, and hence the number of independent decay helicity amplitudes. Here the total angular momentum is fixed at $J = s_a$.

P -invariance in decay is expressed by

$$\langle \lambda_b \lambda_c | S^J | \lambda_a \rangle = \frac{\eta_b \eta_c}{\eta_a} (-1)^{s_a - s_b - s_c} \langle -\lambda_b, -\lambda_c | S^J | -\lambda_a \rangle \quad (69.13)$$

(we have also used the one-particle wave-function transformation law (16.16)).

If the primary particle is truly neutral, conservation of C parity may impose further restrictions. Three cases must be distinguished. If the decay products are also truly neutral, $C_a = C_b C_c$; this condition either forbids the decay entirely or is satisfied without imposing further restrictions. If b, c are distinct particles, C -invariance relates the amplitudes of different processes, $a \rightarrow b + c$ and $a \rightarrow \bar{b} + \bar{c}$. Finally, in $a \rightarrow b + \bar{b}$, fixed charge parity C and total angular momentum $J = s_a$ allow states of only one symmetry in the helicities, symmetric or antisymmetric according to the parity of J and the sign of C .

CP -invariance equates the decay amplitudes $a \rightarrow b + c$ and $\bar{a} \rightarrow \bar{b} + \bar{c}$:

$$\langle \lambda_b \lambda_c | S^J | \lambda_a \rangle = \langle \lambda_{\bar{b}} \lambda_{\bar{c}} | S^J | \lambda_{\bar{a}} \rangle \quad (69.14)$$

with $\lambda_{\bar{a}} = -\lambda_a, \dots$, and hence equates the particle and antiparticle decay probabilities. If several decay channels are possible, the equality holds separately for each. This

¹⁵Since these amplitudes belong to different reactions and therefore cannot interfere, the phase factor in (69.11) has no meaning and may be set equal to 1. Only the equality of cross sections following from (69.11) has physical meaning.

conclusion assumes CP -invariance, which is not universal. Only CPT -invariance is universal; by itself it gives merely

$$\langle \lambda_b \lambda_c | S^J | \lambda_a \rangle = \langle \lambda_{\bar{a}} | S^J | \lambda_{\bar{b}} \lambda_{\bar{c}} \rangle,$$

whose right side belongs to the inverse-decay process. We shall see in § 71 that CPT -invariance together with unitarity nevertheless gives a more restricted relation between particle and antiparticle decay probabilities.

PROBLEM

1. Use (69.6) to obtain the classification of possible states of a two-photon system.

SOLUTION. Here $\lambda_1, \lambda_2 = \pm 1$. For even J ($J > 0$), (69.6) allows three states symmetric in $\lambda_1 \lambda_2$:

$$\text{a) } \psi_{JM11}, \quad \text{b) } \psi_{JM,-1,-1}, \quad \text{c) } \psi_{JM,1,-1} + \psi_{JM,-1,1}.$$

For odd J ($J > 1$), one antisymmetric state is allowed:

$$\text{d) } \psi_{JM,1,-1} - \psi_{JM,-1,1}.$$

States c) and d) also have definite parity, namely $+1$: by (69.4),

$$\widehat{P}(\psi_{JM,1,-1} \pm \psi_{JM,-1,1}) = \pm(-1)^J (\psi_{JM,1,-1} \pm \psi_{JM,-1,1});$$

the factor $\pm(-1)^J = 1$, since the upper sign belongs to even and the lower to odd J . States a) and b) do not separately have definite parity, but the combinations

$$\text{a') } \psi_{JM11} + \psi_{JM,-1,-1}, \quad \text{b') } \psi_{JM11} - \psi_{JM,-1,-1}$$

are even and odd states. At $J = 0$, the condition $|\lambda_1 - \lambda_2| \leq J$ permits only $\lambda_1 = \lambda_2$, so state c) drops out and only the even and odd states a'), b') remain. At $J = 1$, the only state allowed for odd J , state d), is forbidden because $\Lambda = 2 > J$. We thus obtain the table of allowed states (9.5).

2. In the nonrelativistic approximation, the total angular momentum J is formed by adding spin S and orbital angular momentum L . For a two-particle system, find the relation between $|JLSM\rangle$ and $|JM\lambda_1\lambda_2\rangle$.

SOLUTION. The rule for constructing wave functions when angular momenta are added gives

$$\psi_{JLSM} = \sum \{ \psi_{s_1\sigma_1} \psi_{s_2\sigma_2} \langle \sigma_1 \sigma_2 | SM_S \rangle \} \psi_{LM_L} \langle M_L M_S | JM \rangle. \quad (1)$$

Here $\psi_{s\sigma}$ are eigenfunctions of spin s with projection σ on a fixed z axis, and ψ_{LM_L} are the corresponding functions for orbital angular momentum L and projection M_L . The braces describe addition of s_1, s_2 to form S , after which S, L are added to form J ; the sum is over all magnetic indices. Express all functions in momentum space as functions of the direction $\mathbf{n}$ of $\mathbf{p} \equiv \mathbf{p}_1$. By vol. III, (58.7), express the spin functions through helicity-state functions $\psi_{\mathbf{n}\lambda}$:

$$\psi_{s_1\sigma_1} = \sum_{\lambda_1} D_{\lambda_1\sigma_1}^{(s_1)}(\mathbf{n}) \psi_{\mathbf{n}\lambda_1},$$

$$\psi_{s_2\sigma_2} = \sum_{\lambda_2} D_{-\lambda_2\sigma_2}^{(s_2)}(\mathbf{n}) \psi_{\mathbf{n}, -\lambda_2}.$$

For ψ_{LM_L} ,

$$\psi_{LM_L} = Y_{LM_L}(\mathbf{n}) = i^L \sqrt{\frac{2L+1}{4\pi}} D_{0M_L}^{(L)}(\mathbf{n})$$

(using vol. III, (58.25), and definition (16.5)). Substitution in (1), followed twice by the expansion vol. III, (110.1) and use of the orthogonality of Clebsch–Gordan coefficients (vol. III, (106.13)), gives

$$\psi_{JLSM} = \sum_{\lambda_1\lambda_2} \psi_{JM\lambda_1\lambda_2} \langle JM\lambda_1\lambda_2 | JLSM \rangle, \quad (2)$$

where

$$\psi_{JM\lambda_1\lambda_2} = \psi_{\mathbf{n}\lambda_1} \psi_{\mathbf{n}, -\lambda_2} D_{\Lambda M}^{(J)}(\mathbf{n}) \sqrt{\frac{2J+1}{4\pi}}, \quad \Lambda = \lambda_1 - \lambda_2,$$

and

$$\begin{aligned} \langle JM\lambda_1\lambda_2 | JLSM \rangle &= (-i)^L (-1)^{s_1-s_2+S} \sqrt{(2L+1)(2S+1)} \\ &\quad \begin{pmatrix} s_1 & s_2 & S \\ \lambda_1 & -\lambda_2 & -\Lambda \end{pmatrix} \begin{pmatrix} L & S & J \\ 0 & \Lambda & -\Lambda \end{pmatrix}. \end{aligned} \quad (3)$$

By unitarity of (2),

$$\langle JLSM | JM\lambda_1\lambda_2 \rangle = \langle JM\lambda_1\lambda_2 | JLSM \rangle^*.$$

§ 70. Invariant amplitudes

Helicity amplitudes use a particular reference frame, namely the center-of-momentum frame. When scattering amplitudes are calculated by invariant perturbation theory (and also when their general analytic properties are studied), however, it is convenient to write them in a manifestly invariant form.

If the particles participating in the reaction have no spin, the scattering amplitude depends only on invariant products of their four-momenta. For a reaction of the form

$$a + b \rightarrow c + d \quad (70.1)$$

any two of the quantities s , t , and u defined in § 66 may be chosen as these invariants. The scattering amplitude then reduces to a single function, $M_{fi} = f(s, t)$.

If the particles have spin, there are, in addition to the kinematic invariants s , t , and u , invariants that can be formed from the particle wave amplitudes (bispinors, four-tensors, and so on). The scattering amplitudes must then have the form

$$M_{fi} = \sum_n f_n(s, t) F_n, \quad (70.2)$$

where the F_n are invariants depending linearly on the wave amplitudes of all participating particles (and also on their four-momenta). The coefficients $f_n(s, t)$ are called *invariant amplitudes*.

If the wave amplitudes are chosen to describe particles of definite helicity, the invariants acquire definite values $F_n = F_n(\lambda_i, \lambda_f)$. The helicity scattering amplitudes are then homogeneous linear combinations of the invariant amplitudes f_n . It follows that the number of independent functions $f_n(s, t)$ equals the number of independent helicity amplitudes. Since the latter number is readily found (as explained in § 69), the construction of the invariants F_n is thereby simplified: their required number is known in advance.

Let us consider some examples. In every example we assume that the interaction is T - and P -invariant; the latter property means that the invariants F_n must be true scalars rather than pseudoscalars.

Scattering of a spin-0 particle by a spin-1/2 particle.

To count the invariants—or, equivalently, the independent helicity amplitudes—note that the total number of elements of the matrix S^J (that is, the number of different sets of values $\lambda_1, \lambda_2, \lambda'_1, \lambda'_2$) is 4 in this case ($\lambda_1 = \lambda'_1 = 0, \lambda_2, \lambda'_2 = \pm 1/2$). Taking P invariance into account reduces the number of independent elements to two, after which T invariance produces no further reduction.

Two independent invariants may be chosen as

$$F_1 = \bar{u}'u, \quad F_2 = \bar{u}'(\gamma K)u. \quad (70.3)$$

Here $u = u(p)$ and $u' = u(p')$ are the bispinor amplitudes of the initial and final fermions; $K = k + k'$, where k and k' are the four-momenta of the initial and final bosons¹⁶.

The T invariance of the quantities (70.3) becomes evident on noting that the products $\bar{u}'u$ and $\bar{u}'\gamma^\mu u$ transform under time reversal by the same law (28.6) as the operators $\widehat{\bar{\psi}\psi}$ and $\widehat{\bar{\psi}\gamma^\mu\psi}$ of which they are matrix elements: the product $\bar{u}'u$ is invariant by itself, whereas the four-vector $\bar{u}'\gamma u$ transforms as

$$\bar{u}'\gamma^0 u \rightarrow \bar{u}'\gamma^0 u, \quad \bar{u}'\gamma u \rightarrow -\bar{u}'\gamma u.$$

The four-momenta transform in the same way, $(K^0, \mathbf{K}) \rightarrow (k^0, -\mathbf{K})$, and consequently the scalar product $F_2 = K_\mu (\bar{u}'\gamma^\mu u)$ is invariant.

Elastic scattering of two identical spin-1/2 particles.

To count the independent helicity amplitudes, it is convenient to begin with the linear combinations of helicity states

$$\psi_{1g} = \psi_{++} + \psi_{--}, \quad \psi_{2g} = \psi_{++} - \psi_{--},$$

¹⁶At first sight, one might also form an invariant of the type $\bar{u}'\sigma_{\mu\nu}k^\mu k'^\nu u$ (the matrices $\sigma_{\mu\nu}$ are defined in (28.2)). It is readily shown, however, that this invariant reduces to those in (70.3) when the conservation law $k' = p + k - p'$ and the equations

$$(\gamma p)u = mu, \quad \bar{u}'(\gamma p') = m\bar{u}',$$

satisfied by the bispinor amplitudes are used.

$$\psi_{3g} = \psi_{+-} + \psi_{-+}, \quad \psi_u = \psi_{+-} - \psi_{-+},$$

where the subscripts $+$ and $-$ indicate the helicities ($\pm 1/2$) of the two particles. The states $1g$, $2g$, and $3g$ are even, while the state u is odd under particle interchange. The transitions $g \leftrightarrow u$ are therefore forbidden, leaving $16 - 6 = 10$ matrix elements after interchange symmetry is taken into account. Under the inversion P , the functions ψ_{1g} , ψ_{3g} have parity opposite to that of ψ_{2g} ; forbidding transitions between them reduces the number of independent amplitudes to six. Finally, T invariance makes the amplitudes of the transitions $1g \rightarrow 3g$ and $3g \rightarrow 1g$ equal, so that only five independent amplitudes remain. Five independent invariants may be chosen as

$$\begin{aligned} F_1 &= (\bar{u}'_1 u_1)(\bar{u}'_2 u_2), & F_2 &= (\bar{u}'_1 \gamma^5 u_1)(\bar{u}'_2 \gamma^5 u_2), \\ F_3 &= (\bar{u}'_1 \gamma^\mu u_1)(\bar{u}'_2 \gamma_\mu u_2), & F_4 &= (\bar{u}'_1 \gamma^\mu \gamma^5 u_1)(\bar{u}'_2 \gamma_\mu \gamma^5 u_2), \\ F_5 &= (\bar{u}'_1 \sigma^{\mu\nu} u_1)(\bar{u}'_2 \sigma_{\mu\nu} u_2), \end{aligned} \quad (70.4)$$

where u_1, u_2 are the bispinor amplitudes of the initial particles, and u'_1, u'_2 those of the final particles. Interchanging the initial (or final) particles produces no new invariants: the resulting invariants can be expressed through the original ones (see the problem to § 28). But expression (70.2) with the F_n from (70.4) does not explicitly incorporate the requirement that interchanging two identical fermions must change the sign of the scattering amplitude. An expression satisfying this requirement can be written as

$$M_{fi} = [(\bar{u}'_1 u_1)(\bar{u}'_2 u_2)f_1(t, u) - (\bar{u}'_2 u_1)(\bar{u}'_1 u_2)f_1(u, t)] + \dots \quad (70.5)$$

When p'_1 and p'_2 (or p_1 and p_2) are interchanged, the kinematic invariants transform as $s \rightarrow s, t \rightarrow u, u \rightarrow t$, so the stated requirement is fulfilled automatically.

Elastic scattering of a photon by particles of spin 0 and $1/2$.

It is useful to express the amplitudes of these processes in terms of unit spacelike four-vectors $e^{(1)}, e^{(2)}$ satisfying

$$\begin{aligned} e^{(1)2} &= e^{(2)2} = -1, & e^{(1)} e^{(2)} &= 0, \\ e^{(1)} k &= e^{(2)} k = 0, & e^{(1)} k' &= e^{(2)} k' = 0 \end{aligned} \quad (70.6)$$

(for each of the two photons these four-vectors may serve as the four-unit vectors used in an invariant description of its polarization properties; see § 8).

Let k and k' be the initial and final photon four-momenta, and let p and p' be the corresponding four-momenta of the scattering particle. Consider the four-vectors

$$P^\lambda = p^\lambda + p'^\lambda - K^\lambda \frac{pK + p'K}{K^2}, \quad N^\lambda = e^{\lambda\mu\nu\rho} P_\mu q_\nu K_\rho, \quad (70.7)$$

where

$$K = k + k', \quad q = p - p' = k' - k.$$

They are manifestly mutually orthogonal. They are also orthogonal to the four-vectors K, q , and hence to k, k' . Since they are orthogonal to the timelike four-vector K

($K^2 = 2kk' > 0$), they are themselves spacelike (indeed, in a frame where $\mathbf{K} = 0$, the relation $KP = 0$ gives $P_0 = 0$, and therefore $P^2 < 0$). Normalizing P and N , that is, forming

$$e^{(1)\lambda} = \frac{N^\lambda}{\sqrt{-N^2}}, \quad e^{(2)\lambda} = \frac{P^\lambda}{\sqrt{-P^2}}, \quad (70.8)$$

gives a pair of four-vectors with all the required properties. Notice that $e^{(2)}$ is a true vector, whereas $e^{(1)}$ is a pseudovector. Write the photon scattering amplitude as

$$M_{fi} = F^{\lambda\mu} e_\lambda'^* e_\mu, \quad (70.9)$$

with the polarization four-vectors e and e' of the initial and final photons displayed explicitly.

Photon helicity takes only two values (± 1). Hence, for the scattering of a photon by a spin-0 particle, the number of independent helicity amplitudes is the same as for the mutual scattering of particles of spin 0 and 1/2, namely 2. The tensor $F^{\lambda\mu}$ in (70.9) must be constructed only from the particle four-momenta. It can be written as

$$F^{\lambda\mu} = f_1 e^{(1)\lambda} e^{(1)\mu} + f_2 e^{(2)\lambda} e^{(2)\mu}, \quad (70.10)$$

where f_1, f_2 are invariant amplitudes. Note that $F^{\lambda\mu}$ cannot contain a term involving the product $e^{(1)\lambda} e^{(2)\mu}$, because this product is a pseudotensor and would give a pseudoscalar upon substitution into (70.9).

Finally, consider the scattering of a photon by a spin-1/2 particle. To count the independent helicity amplitudes, note that the total number of elements of the matrix S^J is 16 in this case (the helicity of each of the two initial and two final particles takes two values). The requirement of P invariance reduces this number to 8, and the requirement of T invariance then reduces it to 6.

In this case, write the tensor $F_{\lambda\mu}$ as

$$\begin{aligned} F_{\lambda\mu} = & G_0 (e_\lambda^{(1)} e_\mu^{(1)} + e_\lambda^{(2)} e_\mu^{(2)}) + G_1 (e_\lambda^{(1)} e_\mu^{(2)} + e_\lambda^{(2)} e_\mu^{(1)}) + \\ & + G_2 (e_\lambda^{(1)} e_\mu^{(2)} - e_\lambda^{(2)} e_\mu^{(1)}) + G_3 (e_\lambda^{(1)} e_\mu^{(1)} - e_\lambda^{(2)} e_\mu^{(2)}), \end{aligned} \quad (70.11)$$

where G_0, G_3 are true scalars, while G_1, G_2 are pseudoscalars. Both kinds are bilinear in the fermion bispinor amplitudes $\bar{u}(p')$ and $u(p)$, that is, they have the form

$$G_n = \bar{u}(p') Q_n u(p). \quad (70.12)$$

The general form of the matrices (in bispinor indices) Q_n is

$$\begin{aligned} Q_0 = & f_1 + f_2(\gamma K), & Q_1 = & \gamma^5 [f_3 + f_4(\gamma K)], \\ Q_2 = & \gamma^5 [f_5 + f_6(\gamma K)], & Q_3 = & f_7 + f_8(\gamma K), \end{aligned} \quad (70.13)$$

where $K = k + k'$. The coefficients $f_1, \dots, f_8$ are invariant amplitudes, whose number here is 8 (instead of the required 6) because the requirement of T invariance has not yet been imposed.

Time reversal interchanges the initial and final particle four-momenta and also changes the signs of their spatial components:

$$(k_0, \mathbf{k}) \leftrightarrow (k'_0, -\mathbf{k}'), \quad (p_0, \mathbf{p}) \leftrightarrow (p'_0, -\mathbf{p}'); \quad (70.14)$$

the photon polarization four-vectors transform according to

$$(e_0, \mathbf{e}) \leftrightarrow (e'^*_0, -\mathbf{e}'^*) \quad (70.15)$$

(cf. (8.11a)), so that

$$(e'^*_0 e_0, e'^*_i e_0, e'^*_i e_k) \rightarrow (e'^*_0 e_0, -e'^*_0 e_i, e'^*_k e_i).$$

By this last transformation, the condition that the scattering amplitude (70.9) be invariant is equivalent to the requirement

$$(F_{00}, F_{i0}, F_{ik}) \rightarrow (F_{00}, -F_{0i}, F_{ki}).$$

On the other hand, the substitutions (70.14) give

$$(K_0, \mathbf{K}) \rightarrow (K_0, -\mathbf{K}), \quad (q_0, \mathbf{q}) \rightarrow (-q_0, \mathbf{q}),$$

$$(P_0, \mathbf{P}) \rightarrow (P_0, -\mathbf{P}), \quad (N_0, \mathbf{N}) \rightarrow (N_0, -\mathbf{N}),$$

and hence

$$(e^{(1,2)}_0, \mathbf{e}^{(1,2)}) \rightarrow (e^{(1,2)}_0, -\mathbf{e}^{(1,2)}). \quad (70.16)$$

It follows from (70.11) that

$$G_{0,1,3} \rightarrow G_{0,1,3}, \quad G_2 \rightarrow -G_2.$$

But under time reversal

$$\bar{u}' \gamma^5 u \rightarrow -\bar{u}' \gamma^5 u, \quad \bar{u}' \gamma^5 (\gamma K) u \rightarrow \bar{u}' \gamma^5 (\gamma K) u,$$

as follows from the transformation laws for the pseudoscalar and pseudovector bilinear forms in (28.6). It is therefore evident from (70.12) and (70.13) that T invariance of the scattering amplitude requires

$$f_3 = f_6 = 0. \quad (70.17)$$

§ 71. The unitarity condition

The scattering matrix must be unitary: $\hat{S}\hat{S}^\dagger = 1$, or in matrix elements,

$$(SS^\dagger)_{fi} = \sum_n S_{fn} S_{in}^* = \delta_{fi}, \quad (71.1)$$

where n labels all possible intermediate states.¹⁷ This is the most general property of the S -matrix and

¹⁷The meaning of δ_{fi} in (71.1) depends, of course, on the particular choice of quantum numbers and on the normalization of the system's wave functions. It must be defined so that $\sum_f \delta_{if} = 1$.

ensures preservation of the normalization and orthogonality of states in a reaction (cf. vol. III, §§ 125, 144). In particular, the diagonal elements of (71.1) state simply that the sum of transition probabilities from a given initial state to every possible final state is unity:

$$\sum_n |S_{ni}|^2 = 1.$$

Substitution of (64.2) into (71.1) gives

$$\begin{aligned} T_{fi} - T_{if}^* &= i(2\pi)^4 \sum_n \delta^{(4)}(P_f - P_n) T_{fn} T_{in}^* = \\ &= i(2\pi)^4 \sum_n \delta^{(4)}(P_f - P_n) T_{nf}^* T_{ni}. \end{aligned} \quad (71.2)$$

The two equivalent forms on the right follow by writing unitarity as $\hat{S}\hat{S}^+ = 1$ or $\hat{S}^+\hat{S} = 1$, with the factors in opposite orders.

The left side is linear, whereas the right side is quadratic, in the matrix elements T . Thus, if the interaction contains a small parameter, as does the electromagnetic interaction, the two sides are respectively of first and second order. In the first approximation the right side may be neglected, giving

$$T_{fi} = T_{if}^*, \quad (71.3)$$

so the matrix T is Hermitian.

To make (71.2) more explicit, the sum over n must be specified. Consider a two-particle collision for which conservation laws permit only elastic scattering; all intermediate states in (71.2) are then two-particle states. Summation over them means integration over the intermediate momenta $\mathbf{p}_1'', \mathbf{p}_2''$ and summation over the spin quantum numbers, such as helicities, of both particles, denoted collectively by λ'' :

$$\sum_n \rightarrow \int \frac{V^2 d^3 p_1'' d^3 p_2''}{(2\pi)^6} \sum_{\lambda''}.$$

Eliminating the delta functions as in § 64 gives the two-particle unitarity condition

$$T_{fi} - T_{if}^* = \frac{iV^2}{(2\pi)^2} \sum_{\lambda''} \frac{|\mathbf{p}|}{\varepsilon} \int T_{fn} T_{in}^* \varepsilon_1'' \varepsilon_2'' d\omega'',$$

where $\mathbf{p}$ and ε are the momentum and total energy in the center-of-momentum frame. On passing from T_{fi} to M_{fi} by (64.10), the normalization volume disappears:

$$M_{fi} - M_{if}^* = \frac{i}{(4\pi)^2} \sum_{\lambda''} \frac{|\mathbf{p}|}{\varepsilon} \int M_{fn} M_{in}^* d\omega''. \quad (71.4)$$

Define the elastic-scattering amplitude by

$$d\sigma = |\langle \mathbf{n}' \lambda' | f | \mathbf{n} \lambda \rangle|^2 d\omega' \quad (71.5)$$

($\mathbf{n}, \mathbf{n}'$ are the initial and final momentum directions, and λ, λ' the initial and final spin quantum numbers). Comparison with (64.19) gives

$$\langle \mathbf{n}' \lambda' | f | \mathbf{n} \lambda \rangle = \frac{1}{8\pi\epsilon} M_{fi}, \quad (71.6)$$

and (71.4) becomes

$$\begin{aligned} \langle \mathbf{n}' \lambda' | f | \mathbf{n} \lambda \rangle - \langle \mathbf{n} \lambda | f | \mathbf{n}' \lambda' \rangle^* &= \\ &= \frac{i|\mathbf{p}|}{2\pi} \sum_{\lambda''} \int \langle \mathbf{n}' \lambda' | f | \mathbf{n}'' \lambda'' \rangle \langle \mathbf{n} \lambda | f | \mathbf{n}'' \lambda'' \rangle^* d\omega'', \end{aligned} \quad (71.7)$$

which generalizes the familiar nonrelativistic formula (see vol. III, (125.8)).

The forward elastic-scattering amplitude is the diagonal matrix element T_{ii} for which the final particle state coincides with the initial state.¹⁸ For this amplitude, (71.2) gives

$$2 \operatorname{Im} T_{ii} = (2\pi)^4 \sum_n |T_{in}|^2 \delta^{(4)}(P_i - P_n). \quad (71.8)$$

The right side differs only by a factor from the total cross section σ_t of all possible scattering processes from the initial state i . Indeed, summing (64.5) over f and dividing by the flux density j gives

$$\sigma_t = \frac{(2\pi)^4 V}{j} \sum_n |T_{in}|^2 \delta^{(4)}(P_i - P_n),$$

so that

$$\frac{2V}{j} \operatorname{Im} T_{ii} = \sigma_t.$$

Replacing $T_{ii} = M_{ii}/(2\epsilon_1 V \cdot 2\epsilon_2 V)$ and using j from (64.17) removes the normalization volume:

$$\operatorname{Im} M_{ii} = 2|\mathbf{p}|\epsilon\sigma_t. \quad (71.9)$$

This is the content of the *optical theorem*. In terms of the elastic amplitude (71.6), it takes its usual form

$$\operatorname{Im} \langle \mathbf{n} \lambda | f | \mathbf{n} \lambda \rangle = \frac{|\mathbf{p}|}{4\pi} \sigma_t \quad (71.10)$$

(cf. vol. III, (142.10)).

If the S -matrix is given in the angular-momentum representation, as partial amplitudes, its diagonality in J allows the unitarity condition to be written separately for each J .

If only elastic scattering is possible,

$$\sum_{\lambda''} \langle \lambda' | S^J | \lambda'' \rangle \langle \lambda | S^J | \lambda'' \rangle^* = \delta_{\lambda\lambda'}. \quad (71.11)$$

By T -invariance, the elastic-scattering matrix is symmetric (cf. (69.10)) and can therefore be diagonalized. Unitarity then requires its diagonal elements to have unit modulus; they are conventionally written

$$S_n^J = \exp(2i\delta_{Jn}), \quad (71.12)$$

¹⁸This means an element of T , not of S : the diagonal element is taken after the unit matrix has been removed from S .

where the real quantities δ_{Jn} are functions of the energy, and n labels the diagonal elements for fixed J . In general, if the number N of independent amplitudes exceeds the rank of the square matrix S^J , the coefficients of the transformation diagonalizing S^J depend on J, E ; together with its principal values they contain independent quantities equivalent to the original N . If N equals the rank, and hence the number of principal values, the diagonalization coefficients are universal. The diagonalizing states are then states of definite parity, though no longer of definite helicity.

In terms of the partial amplitudes $\langle \lambda' | f^J | \lambda \rangle$, (71.11) is

$$\begin{aligned} \langle \lambda' | f^J | \lambda \rangle - \langle \lambda | f^J | \lambda' \rangle^* &= \\ &= 2i|\mathbf{p}| \sum_{\lambda''} \langle \lambda' | f^J | \lambda'' \rangle \langle \lambda | f^J | \lambda'' \rangle^*, \end{aligned} \quad (71.13)$$

as follows by substituting (68.13) in (71.7) and using orthonormality of the D functions. Under T -invariance the matrix $\langle \lambda' | f^J | \lambda \rangle$ is symmetric, and (71.13) becomes

$$\text{Im} \langle \lambda' | f^J | \lambda \rangle = |\mathbf{p}| \langle \lambda' | f^J f^{J*} | \lambda \rangle. \quad (71.14)$$

If the matrix is diagonalized, its diagonal elements are

$$f_n^J = \frac{1}{2i|\mathbf{p}|} (\exp(2i\delta_{Jn}) - 1) = \frac{1}{|\mathbf{p}|} \exp(i\delta_{Jn}) \sin \delta_{Jn}. \quad (71.15)$$

Finally, consider some consequences of unitarity together with CPT -invariance. The latter gives

$$T_{fi} = T_{\bar{i}\bar{f}}, \quad (71.16)$$

where $\bar{i}, \bar{f}$ differ from i, f by replacing every particle by its antiparticle and reversing angular-momentum vectors while leaving momenta unchanged. In particular,

$$T_{ii} = T_{\bar{i}\bar{i}}.$$

It follows from (71.8) or (71.9) that the total cross sections for all possible processes from specified initial states are the same for reactions of particles and antiparticles.

In particular, the total decay probabilities, and hence lifetimes, are the same for a particle and its antiparticle. Together with equality of their masses (§ 11), these results are among the most important consequences of CPT -invariance. Recall from the end of § 69 that the same assertion separately for each decay channel also requires CP -invariance.

PROBLEM

Starting from unitarity, find the relation between the phases of the partial amplitudes for pion photoproduction on nucleons ($\gamma + N \rightarrow \pi + N$) and elastic pion-nucleon scattering ($\pi + N \rightarrow \pi + N$). Take into account that πN scattering is due to the strong interaction, whereas photoproduction and γN scattering are electromagnetic.

SOLUTION. Denote the partial amplitudes by

$$\langle \pi N | S | \gamma N \rangle = S_{\pi\gamma}, \quad \langle \gamma N | S | \gamma N \rangle = S_{\gamma\gamma}, \quad \langle \pi N | S | \pi N \rangle = S_{\pi\pi}$$

(the J and helicity indices are suppressed). Photoproduction is first order and γN scattering second order in the charge e ; hence $S_{\pi\gamma} \sim e$, $S_{\gamma\gamma} - 1 \sim e^2$. The amplitude $S_{\pi\pi}$ contains no small parameter. To terms of order e , (71.1) gives

$$S_{\pi\gamma}S_{\gamma\gamma}^* + S_{\pi\pi}S_{\gamma\pi}^* \approx S_{\pi\gamma} + S_{\pi\pi}S_{\gamma\pi}^* = 0, \quad (1)$$

$$S_{\pi\gamma}S_{\pi\gamma}^* + S_{\pi\pi}S_{\pi\pi}^* \approx S_{\pi\pi}S_{\pi\pi}^* = 1 \quad (2)$$

(the 1 on the right of (2) is the unit matrix in spin variables). By T -invariance, $S_{\pi\pi}$ is symmetric and $S_{\gamma\pi} = S_{\pi\gamma}$. Choose $S_{\pi\pi}$ diagonal, with respect to pion states of definite parity. Equation (2) then requires its diagonal elements to have the form $e^{2i\delta_\pi}$, with different constants δ_π . Equation (1) gives for each element of $S_{\pi\gamma}$

$$S_{\pi\gamma}/S_{\pi\gamma}^* = -e^{2i\delta_\pi},$$

and hence

$$S_{\pi\gamma} = \pm |S_{\pi\gamma}| e^{i\delta_\pi}.$$

Thus the phase of the partial photoproduction amplitude into a state of definite parity is determined by the phase of elastic πN scattering.

Invariant Perturbation Theory

§ 72. The chronological product

The probabilities of processes in collisions of particles whose interaction may be regarded as small are calculated by perturbation theory. In its usual form for nonrelativistic quantum mechanics, however, this formalism has the disadvantage that relativistic invariance is not explicit. Although its application to relativistic problems ultimately gives invariant results, the noninvariant form of the intermediate formulas greatly complicates the calculations. This chapter develops a consistently relativistic perturbation theory free of this defect; it was constructed by Feynman (*R. P. Feynman*, 1948–1949).

With a second-quantized description in mind, denote by Φ the wave function of the system in the representation of occupation numbers for the various free-particle states. The Hamiltonian is $\hat{H} = \hat{H}_0 + \hat{V}$, where $\hat{V}$ is the interaction operator. Let Φ_n be eigenfunctions of the unperturbed Hamiltonian, each corresponding to specified values of all occupation numbers. An arbitrary Φ is expanded as $\Phi = \sum C_n \Phi_n$. The exact wave equation

$$i \frac{\partial \Phi}{\partial t} = (\hat{H}_0 + \hat{V})\Phi \quad (72.1)$$

then becomes a system of equations for C_n :

$$i \dot{C}_n = \sum_m V_{nm} \exp[i(E_n - E_m)t] C_m, \quad (72.2)$$

where V_{nm} are the time-independent matrix elements of $\hat{V}$, and E_n are the energy levels of the unperturbed system (cf. vol. III, § 40).

By definition, $\hat{V}$ has no explicit time dependence. The quantities

$$V_{nm}(t) = V_{nm} \exp[i(E_n - E_m)t] \quad (72.3)$$

may be regarded as matrix elements of the time-dependent operator

$$\hat{V}(t) = \exp(i\hat{H}_0 t) \hat{V} \exp(-i\hat{H}_0 t). \quad (72.4)$$

This is the operator in the *interaction representation*, as distinct from the original time-independent Schrödinger operator $\hat{V}$.¹ Denoting the wave function in this new representation again by Φ , equations (72.2) take the symbolic form

$$i \dot{\Phi} = \hat{V}(t)\Phi. \quad (72.5)$$

¹Notice that the definition (72.4) contains the unperturbed Hamiltonian $\hat{H}_0$. It therefore differs from the *Heisenberg representation*, in which

$$\hat{V}^H(t) = \exp(i\hat{H}t) \hat{V} \exp(-i\hat{H}t)$$

(see vol. III, § 13, and § 102 below).

In this representation, the wave function changes only through the perturbation, and therefore describes processes caused by the particle interaction.

If $\Phi(t)$, $\Phi(t + \delta t)$ are its values at two infinitesimally close times, (72.5) gives

$$\Phi(t + \delta t) = [1 - i\delta t \hat{V}(t)]\Phi(t) = \exp[-i\delta t \cdot \hat{V}(t)]\Phi(t).$$

Thus its value at an arbitrary time t_f is related to that at an initial time $t_i < t_f$ by

$$\Phi(t_f) = \left\{ \prod_i^f \exp[-i\delta t_\alpha \cdot \hat{V}(t_\alpha)] \right\} \Phi(t_i), \quad (72.6)$$

where $\prod$ denotes the limiting product over all infinitesimal intervals δt_α between t_i, t_f . If $V(t)$ were an ordinary function, the limit would reduce to

$$\exp \left\{ -i \int_{t_i}^{t_f} V(t) dt \right\}.$$

This reduction depends on commutativity of factors belonging to different times when the product in (72.6) is replaced by a sum in the exponential. The operators $\hat{V}(t)$ do not in general commute, so no reduction to an ordinary integral is possible.

Write (72.6) symbolically as

$$\Phi(t_f) = T \exp \left\{ -i \int_{t_i}^{t_f} \hat{V}(t) dt \right\} \Phi(t_i), \quad (72.7)$$

where T is the symbol of *chronological ordering*, specifying the chronological sequence of the times in the successive factors of (72.6). In particular, taking $t_i \rightarrow -\infty$, $t_f \rightarrow +\infty$ gives

$$\Phi(+\infty) = \hat{S}\Phi(-\infty), \quad (72.8)$$

where

$$\hat{S} = T \exp \left\{ -i \int_{-\infty}^{\infty} \hat{V}(t) dt \right\}. \quad (72.9)$$

The usefulness of (72.7)–(72.9), a formally exact solution of the wave equation, is that it immediately gives the perturbation expansion

$$\hat{S} = \sum_{k=0}^{\infty} \frac{(-i)^k}{k!} \int_{-\infty}^{\infty} dt_1 \int_{-\infty}^{\infty} dt_2 \dots \int_{-\infty}^{\infty} dt_k \cdot T \{ \hat{V}(t_1) \hat{V}(t_2) \dots \hat{V}(t_k) \}. \quad (72.10)$$

Here the k th power of the integral is written as a k -fold integral, and T directs that, in every region of values of $t_1, t_2, \dots, t_k$, the operators be placed chronologically from right to left in order of increasing t .²

It follows from (72.8) that if the system was in state Φ_i , a particular collection of free particles, before the collision, then the probability amplitude for transition to state Φ_f , another collection of free particles, is S_{fi} . These elements constitute the S -matrix.

²The derivation of the rules of relativistic perturbation theory from (72.10) is due to Dayson (*F. Dayson*, 1949).

The electromagnetic interaction operator was given in § 43:

$$\hat{V} = e \int (\hat{j}\hat{A}) d^3x. \quad (72.11)$$

Substitution in (72.9) gives

$$\hat{S} = \text{T exp} \left\{ -ie \int (\hat{j}\hat{A}) d^4x \right\}. \quad (72.12)$$

The operator (72.12) is relativistically invariant. This follows from the scalar character of the integrand, the invariant integration over d^4x , and the invariance of chronological ordering. The last point requires explanation.

The order of two times t_1, t_2 , that is, the sign of $t_2 - t_1$, is frame-independent if the corresponding world points x_1, x_2 are separated by a timelike interval, $(x_2 - x_1)^2 > 0$. Chronological ordering is then automatically invariant. If $(x_2 - x_1)^2 < 0$, a spacelike interval, different frames may have either $t_2 > t_1$ or $t_2 < t_1$.³ But no causal relation can exist between events at such points. The operators of two physical quantities referring to them therefore cannot fail to commute: noncommutativity physically means that the quantities cannot be measured jointly, which presupposes a physical connection between the measurements. Chronological ordering remains invariant in this case as well. A Lorentz transformation may reverse the time order, but commutativity allows the factors to be restored to chronological order.⁴

The definition of the S -matrix given here automatically satisfies unitarity. Write $\hat{S}$ as the chronological product in (72.6). Since $\hat{V}$ is Hermitian, $\hat{S}^+$ is the product of the same factors $\exp(i\delta t_\alpha \cdot \hat{V}(t_\alpha))$

in reverse chronological order, with the opposite sign in the exponent. In multiplying $\hat{S}$ by $\hat{S}^+$, all factors cancel in pairs.

Thus unitarity of $\hat{S}$ is ensured here by Hermiticity of the Hamiltonian. The requirement of unitarity is actually more general than the assumptions underlying this theory: it must also hold in a quantum-mechanical description that does not use the concepts of a Hamiltonian and wave functions.

§ 73. Feynman diagrams for electron scattering

We shall use specific examples to show how scattering-matrix elements are calculated. These examples will facilitate the subsequent formulation of the general rules of invariant perturbation theory.

The current operator $\hat{j}$ contains a product of two electron ψ operators. First-order perturbation theory could therefore give processes involving only three particles in the

³Instead of timelike and spacelike intervals, one often speaks briefly of regions inside and outside the light cone: all points x separated from x' by $(x - x')^2 > 0$ lie inside the double cone with vertex x' , while points separated by $(x - x')^2 < 0$ lie outside it.

⁴For the product $\hat{V}(t_1)\hat{V}(t_2)\dots$, this statement requires qualification. Since $\hat{V}$ itself is not gauge invariant (it changes with $\hat{A}_\mu$), factors $\hat{V}(t_1), \hat{V}(t_2), \dots$ that commute in one gauge may not commute in another. The statement must therefore be understood as the possibility of choosing a gauge in which $\hat{V}(t_1), \hat{V}(t_2)$ commute outside the light cone. This qualification does not affect the invariance of the S -matrix: scattering amplitudes are physical quantities and cannot depend on the gauge. Formally, this follows from the gauge invariance of the action integral noted in § 43.

initial and final states: two electrons from $\hat{j}$ and one photon from $\hat{A}$. Such processes between free particles are impossible, however, because energy-momentum conservation forbids them. If p_1, p_2 are the electron 4-momenta and k the photon 4-momentum, conservation would require $k = p_2 - p_1$ or $k = p_2 + p_1$. These equalities are impossible because $k^2 = 0$ for a photon, whereas $(p_2 \pm p_1)^2$ is necessarily nonzero. Evaluating this invariant in the rest frame of one electron gives

$$(p_2 \pm p_1)^2 = 2(m^2 \pm p_1 p_2) = 2(m^2 \pm \varepsilon_1 \varepsilon_2 \mp \mathbf{p}_1 \mathbf{p}_2) = 2m(m \pm \varepsilon_2).$$

Since $\varepsilon_2 > m$,

$$(p_2 + p_1)^2 > 0, \quad (p_2 - p_1)^2 < 0. \quad (73.1)$$

The first nonvanishing off-diagonal elements of the S -matrix can therefore occur only in second order. All such processes are contained in the second-order operator obtained by expanding (72.12):

$$\hat{S}^{(2)} = -\frac{e^2}{2!} \iint d^4x d^4x' \cdot T(\hat{j}^\mu(x) \hat{A}_\mu(x) \hat{j}^\nu(x') \hat{A}_\nu(x')).$$

Since electron and photon operators commute with one another, this chronological product may

be separated into two:

$$\hat{S}^{(2)} = -\frac{e^2}{2!} \iint d^4x d^4x' \cdot T(\hat{j}^\mu(x) \hat{j}^\nu(x')) T(\hat{A}_\mu(x) \hat{A}_\nu(x')). \quad (73.2)$$

As a first example, consider elastic scattering of two electrons. Initially their 4-momenta are p_1, p_2 , and finally p_3, p_4 . All electrons are also understood to be in specified spin states; spin-variable indices are suppressed throughout.

Since neither state contains photons, the required matrix element of the photon chronological product is the diagonal element $\langle 0 | \dots | 0 \rangle$, where $|0\rangle$ denotes the photon vacuum. This vacuum expectation value is, for each $\mu\nu$, a function of the coordinates x, x' . Homogeneity of 4-space permits dependence only on $x - x'$. The tensor

$$D_{\mu\nu}(x - x') = i \langle 0 | T \hat{A}_\mu(x) \hat{A}_\nu(x') | 0 \rangle \quad (73.3)$$

is called the *photon propagation function*, or *photon propagator*. Its explicit form will be found in § 76.

For the electron chronological product we require

$$\langle 34 | T \hat{j}^\mu(x) \hat{j}^\nu(x') | 12 \rangle, \quad (73.4)$$

where $|12\rangle, |34\rangle$ denote the corresponding two-electron states. It may also be written as a vacuum expectation value using

$$\langle 2 | \hat{F} | 1 \rangle = \langle 0 | \hat{a}_2 \hat{F} \hat{a}_1^+ | 0 \rangle,$$

where $\hat{F}$ is arbitrary and $\hat{a}_1^+, \hat{a}_2$ respectively create the first and annihilate the second electron. Thus instead of (73.4) we may calculate

$$\langle 0 | \hat{a}_3 \hat{a}_4 T(\hat{j}^\mu(x) \hat{j}^\nu(x')) \hat{a}_2^+ \hat{a}_1^+ | 0 \rangle \quad (73.5)$$

(the indices abbreviate $p_1, p_2, \dots$).

Each current is $\hat{j} = \hat{\bar{\psi}}\gamma\hat{\psi}$, and each ψ operator is

$$\hat{\psi} = \sum_{\mathbf{p}} (\hat{a}_{\mathbf{p}}\psi_{\mathbf{p}} + \hat{b}_{\mathbf{p}}^+\psi_{-\mathbf{p}}), \quad \hat{\bar{\psi}} = \sum_{\mathbf{p}} (\hat{a}_{\mathbf{p}}^+\bar{\psi}_{\mathbf{p}} + \hat{b}_{\mathbf{p}}\bar{\psi}_{-\mathbf{p}}) \quad (73.6)$$

(the second terms contain positron operators, which do not contribute here). The product $\hat{j}^\mu(x)\hat{j}^\nu(x')$

is a sum of terms each containing two $\hat{a}_{\mathbf{p}}$ and two $\hat{a}_{\mathbf{p}}^+$. They must annihilate electrons 1,2 and create electrons 3,4; they are therefore $\hat{a}_1, \hat{a}_2, \hat{a}_3^+, \hat{a}_4^+$. These are said to be *contracted* with the external operators $\hat{a}_1^+, \hat{a}_2^+, \hat{a}_3, \hat{a}_4$ in (73.5), and cancel by

$$\langle 0|\hat{a}_{\mathbf{p}}\hat{a}_{\mathbf{p}}^+|0\rangle = 1. \quad (73.7)$$

Depending on which ψ operators supply $\hat{a}_1, \hat{a}_2, \hat{a}_3^+, \hat{a}_4^+$, (73.5) contains four terms:

$$\begin{aligned} (73.5) = & \hat{a}_3\hat{a}_4(\hat{\bar{\psi}}\gamma^\mu\hat{\psi})(\hat{\bar{\psi}}'\gamma^\nu\hat{\psi}')\hat{a}_2^+\hat{a}_1^+ \\ & + \hat{a}_3\hat{a}_4(\hat{\bar{\psi}}\gamma^\mu\hat{\psi})(\hat{\bar{\psi}}'\gamma^\nu\hat{\psi}')\hat{a}_2^+\hat{a}_1^+ \\ & + \hat{a}_3\hat{a}_4(\hat{\bar{\psi}}\gamma^\mu\hat{\psi})(\hat{\bar{\psi}}'\gamma^\nu\hat{\psi}')\hat{a}_2^+\hat{a}_1^+ \\ & + \hat{a}_3\hat{a}_4(\hat{\bar{\psi}}\gamma^\mu\hat{\psi})(\hat{\bar{\psi}}'\gamma^\nu\hat{\psi}')\hat{a}_2^+\hat{a}_1^+, \end{aligned} \quad (73.8)$$

where $\psi = \psi(x)$, $\psi' = \psi(x')$, and arcs join contracted operators, those supplying a pair $\hat{a}, \hat{a}^+$ for cancellation by (73.7). In each term, successive interchanges bring conjugate operators into adjacent pairs. Since all these operators anticommute (states 1,2,3,4 are distinct),⁵ (73.4) is

$$\begin{aligned} \langle 34|T\hat{j}^\mu(x)\hat{j}^\nu(x')|12\rangle = & (\bar{\psi}_4\gamma^\mu\psi_2)(\bar{\psi}_3'\gamma^\nu\psi_1') + (\bar{\psi}_3\gamma^\mu\psi_1)(\bar{\psi}_4'\gamma^\nu\psi_2') \\ & - (\bar{\psi}_3\gamma^\mu\psi_2)(\bar{\psi}_4'\gamma^\nu\psi_1') - (\bar{\psi}_4\gamma^\mu\psi_1)(\bar{\psi}_3'\gamma^\nu\psi_2'). \end{aligned} \quad (73.9)$$

The overall sign is conventional and depends on the ordering of the external electron operators in (73.5), as is generally true for identical-fermion scattering. The relative signs are independent of that ordering.

The two terms in the first and second lines of (73.9) differ only by simultaneous interchange of μ, ν and x, x' . This interchange also leaves (73.3) unchanged, since the ordering of its factors is in any case fixed by T . After multiplication of (73.3) by (73.9) and integration over $d^4x d^4x'$, the four terms therefore give equal results in pairs, and

$$\begin{aligned} S_{fi} = & ie^2 \iint d^4x d^4x' \cdot D_{\mu\nu}(x-x') \{ (\bar{\psi}_4\gamma^\mu\psi_2)(\bar{\psi}_3'\gamma^\nu\psi_1') \\ & - (\bar{\psi}_4\gamma^\mu\psi_1)(\bar{\psi}_3'\gamma^\nu\psi_2') \} \end{aligned} \quad (73.10)$$

⁵Because of this anticommutativity, the operators $\hat{j}(x), \hat{j}(x')$ may here be regarded as commuting when the matrix element is evaluated, and the chronological-ordering symbol may be omitted.

(notice that the factor $1/2!$ has disappeared).

The electron wave functions are the plane waves (64.8), so the expression in braces is

$$\begin{aligned} \{ \dots \} &= (\bar{u}_4 \gamma^\mu u_2)(\bar{u}_3 \gamma^\nu u_1) e^{-i(p_2-p_4)x-i(p_1-p_3)x'} - \\ &\quad - (\bar{u}_4 \gamma^\mu u_1)(\bar{u}_3 \gamma^\nu u_2) e^{-i(p_1-p_4)x-i(p_2-p_3)x'} = \\ &= \{ (\bar{u}_4 \gamma^\mu u_2)(\bar{u}_3 \gamma^\nu u_1) e^{-i[(p_2-p_4)+(p_3-p_1)]\xi/2} - \\ &\quad - (\bar{u}_4 \gamma^\mu u_1)(\bar{u}_3 \gamma^\nu u_2) e^{-i[(p_1-p_4)+(p_3-p_2)]\xi/2} \} e^{-i(p_1+p_2-p_3-p_4)X}, \end{aligned}$$

where $X = (x + x')/2$, $\xi = x - x'$. Replace integration over $d^4x d^4x'$ by integration over $d^4\xi d^4X$. The d^4X integral gives a delta function, enforcing $p_1 + p_2 = p_3 + p_4$. Passing from S to M as in § 64 gives the scattering amplitude

$$M_{fi} = e^2 \{ (\bar{u}_4 \gamma^\mu u_2) D_{\mu\nu}(p_4 - p_2)(\bar{u}_3 \gamma^\nu u_1) - (\bar{u}_4 \gamma^\mu u_1) D_{\mu\nu}(p_4 - p_1)(\bar{u}_3 \gamma^\nu u_2) \}. \quad (73.11)$$

Here the momentum-representation photon propagator has been introduced:

$$D_{\mu\nu}(k) = \int D_{\mu\nu}(\xi) e^{ik\xi} d^4\xi. \quad (73.12)$$

Each of the two terms in (73.11) may be represented symbolically by a *Feynman diagram*. The first is

$$e^2 (\bar{u}_4 \gamma^\mu u_2) D_{\mu\nu}(k) (\bar{u}_3 \gamma^\nu u_1) = \begin{array}{c} p_3 \swarrow \quad \nwarrow p_1 \\ \quad \quad \quad \downarrow k \\ p_4 \swarrow \quad \nwarrow p_2 \end{array} \quad (73.13)$$

Each intersection of lines, or *vertex*, supplies a factor γ . Incoming solid lines directed toward a vertex represent initial electrons and supply the bispinor amplitudes u ; outgoing solid lines directed away represent final electrons and supply $\bar{u}$. When a diagram is read, these factors are written from left to right in the order encountered along solid lines against the arrows. The two vertices are joined by a dashed line representing a *virtual* photon emitted at one vertex and absorbed at the other; it supplies $-iD_{\mu\nu}(k)$. Its 4-momentum is fixed by conservation at a vertex: here $k = p_1 - p_3 = p_4 - p_2$. The whole diagram also supplies $(-ie)^2$, the exponent being the number of vertices, and in this form contributes to iM_{fi} . Similarly, the second term is

$$e^2 (\bar{u}_4 \gamma^\mu u_1) D_{\mu\nu}(k') (\bar{u}_3 \gamma^\nu u_2) = \begin{array}{c} p_4 \swarrow \quad \nwarrow p_1 \\ \quad \quad \quad \downarrow k' \\ p_3 \swarrow \quad \nwarrow p_2 \end{array} \quad (73.14)$$

where $k' = p_1 - p_4 = p_3 - p_2$. The diagram may be read starting from either the p_3 or p_4 end; the results coincide because $D_{\mu\nu}$ is symmetric. The virtual-photon line may

likewise be oriented either way: reversing it merely changes the sign of k , which is immaterial because $D_{\mu\nu}(k)$ is even (see § 76).

Lines representing initial and final particles are called *external lines*, or *free ends*. Diagrams (73.13), (73.14) differ by exchange of the two free electron ends p_3, p_4 . Exchanging two fermions reverses the sign of a diagram, corresponding to the opposite signs of the two terms in (73.11).

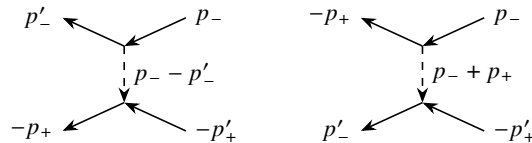
We shall always use Feynman diagrams in this momentum representation. They can also be associated with the coordinate-representation terms of (73.10): electron amplitudes are then coordinate wave functions, propagators are in coordinate space, and each vertex corresponds to an integration variable (x or x'). Factors belonging to lines meeting at a vertex are evaluated at that variable.

Now consider electron–positron scattering. Denote the initial momenta by p_-, p_+ and the final momenta by p'_-, p'_+ .

Positron creation and annihilation operators occur in (73.6) together with electron annihilation and creation operators, respectively. Previously, $\hat{\psi}$ annihilated both initial particles and $\hat{\bar{\psi}}$ created both final ones; here their roles are opposite for the electron and positron. Thus the initial positron is described by $\bar{u}(-p_+)$ and the final positron by $u(-p'_+)$, both with reversed 4-momentum. Allowing for this difference gives⁶

$$M_{fi} = -e^2 (\bar{u}(p'_-) \gamma^\mu u(p_-)) D_{\mu\nu}(p_- - p'_-) (\bar{u}(-p_+) \gamma^\nu u(-p'_+)) + e^2 (\bar{u}(-p_+) \gamma^\mu u(p_-)) D_{\mu\nu}(p_- + p_+) (\bar{u}(p'_-) \gamma^\nu u(-p'_+)). \quad (73.15)$$

The two terms are represented by



$$\quad (73.16)$$

The diagram rules change only for positrons. Incoming solid lines still supply u and outgoing lines $\bar{u}$, but incoming lines now represent final positrons and outgoing lines initial positrons, with all positron momenta reversed.

The diagrams in (73.16) have different characters. In the first, initial and final electron lines meet at one vertex, and initial and final positron lines at the other. In the second, electron and positron lines meet at each vertex: the pair appears to annihilate into a virtual photon at the upper vertex and to be created from it at the lower.

This difference is reflected in the virtual photons. In the first, the scattering-type diagram, the virtual-photon 4-momentum is the difference of two electron or positron

⁶For scattering of nonidentical particles, the overall sign of the amplitude is fixed. In (73.5), the external operators must be arranged so that both electron operators stand at the edges,

$$\langle 0 | a' b' \dots b^+ a^+ | 0 \rangle$$

(or both in the middle); this ensures the same sign for the initial and final vacuum states. The overall sign can also be checked from the nonrelativistic limit: in § 81 we shall see that the second term in (73.15) tends to zero, while the first tends to the Born amplitude for Rutherford scattering.

4-momenta, so $k^2 < 0$. In the annihilation diagram, $k' = p_- + p_+$, hence $k'^2 > 0$. For a virtual photon always $k^2 \neq 0$, unlike a real photon, for which $k^2 = 0$.

If the colliding particles are neither identical nor particle and antiparticle, say an electron and a muon, only one diagram occurs:

$$(73.17)$$

An annihilation- or exchange-type diagram is impossible. Analytically, write the current as

$$\hat{j} = \hat{j}^{(e)} + \hat{j}^{(\mu)} = (\bar{\hat{\psi}}^{(e)} \gamma \hat{\psi}^{(e)}) + (\bar{\hat{\psi}}^{(\mu)} \gamma \hat{\psi}^{(\mu)})$$

and in $\hat{j}^\mu(x) \hat{j}^\nu(x')$ take matrix elements of the terms producing the required annihilations and creations.

Return to the first-order processes forbidden by 4-momentum conservation. Matrix elements of

$$\hat{S}^{(1)} = -ie \int \hat{j}(x) \hat{A}(x) d^4x \quad (73.18)$$

describe creation or annihilation at one point x of three real particles, two electrons and a photon. They result from contracting $\hat{\psi}(x), \bar{\hat{\psi}}(x)$ at the same point and, for photon emission, contain integrals such as

$$S_{fi} = -ie \int \bar{\psi}_2(x) \psi_1(x) (\gamma A^*(x)) d^4x,$$

which vanish because the integrand contains $\exp[-i(p_1 - p_2 - k)x]$ with a nonzero exponent. Diagrammatically, diagrams with three free ends vanish:

$$(73.19)$$

For the same reason, second-order processes involving six particles in the initial and final states are impossible. Their S_{fi} integrals over $d^4x d^4x'$ would factor into two vanishing integrals, each over a product of three wave functions at one point. Equivalently, their diagrams separate into two independent diagrams of the form (73.19).

§74. Feynman diagrams for photon scattering

Consider another second-order effect, the scattering of a photon by an electron (the *Compton effect*). Let the photon and electron have 4-momenta k_1 and p_1 in the initial state and k_2 and p_2 in the final state (with specified polarizations as well, which we omit for brevity).

The photon matrix element is

$$\langle 2 | T \hat{A}_\mu(x) \hat{A}_\nu(x') | 1 \rangle = \langle 0 | \hat{c}_2 T \hat{A}_\mu(x) \hat{A}_\nu(x') \hat{c}_1^\dagger | 0 \rangle, \quad (74.1)$$

where

$$\hat{A} = \sum_{\mathbf{k}} (\hat{c}_{\mathbf{k}} A_{\mathbf{k}} + \hat{c}_{\mathbf{k}}^\dagger A_{\mathbf{k}}^*).$$

Contracting the external and internal operators gives

$$\begin{aligned} (74.1) &= \hat{c}_2 A_\mu A_\nu' \hat{c}_1^\dagger + \hat{c}_2 A_\mu A_\nu' \hat{c}_1^\dagger \\ &= A_{2\mu}^* A_{1\nu}' + A_{1\mu} A_{2\nu}'. \end{aligned} \quad (74.2)$$

(here the commutativity of $\hat{c}_1$ and $\hat{c}_2^\dagger$ has been used; for the same reason the symbol T may be omitted in this case). The electron matrix element is

$$\langle 2 | T \hat{j}^\mu(x) \hat{j}^\nu(x') | 1 \rangle = \langle 0 | \hat{a}_2 T(\hat{\bar{\psi}} \gamma^\mu \hat{\psi})(\hat{\bar{\psi}}' \gamma^\nu \hat{\psi}') \hat{a}_1^\dagger | 0 \rangle. \quad (74.3)$$

Four ψ -operators occur in it. Only two are used to annihilate electron 1 and create electron 2; that is, they are contracted with $\hat{a}_1^\dagger$ and $\hat{a}_2$. These may be $\hat{\bar{\psi}}'$, $\hat{\psi}$, or $\hat{\psi}'$, $\hat{\bar{\psi}}$ (but not $\hat{\bar{\psi}}$, $\hat{\psi}$, or $\hat{\bar{\psi}}'$, $\hat{\psi}'$: the creation and annihilation at the same point x or x' of two real electrons together with one real photon gives zero). Performing the contraction in the two possible ways produces two terms in the matrix element (74.3). Write them first under the assumption $t > t'$:

$$\begin{aligned} (74.3) &= \hat{a}_2 (\hat{\bar{\psi}} \gamma^\mu \hat{\psi}) (\hat{\bar{\psi}}' \gamma^\nu \hat{\psi}') \hat{a}_1^\dagger \\ &\quad + \hat{a}_2 (\hat{\bar{\psi}} \gamma^\mu \hat{\psi}') (\hat{\bar{\psi}}' \gamma^\nu \hat{\psi}) \hat{a}_1^\dagger. \end{aligned} \quad (74.4)$$

The operators contracted in the first term give

$$\hat{a}_2 \hat{\bar{\psi}} \rightarrow \hat{a}_2 \hat{a}_2^\dagger \bar{\psi}_2, \quad \hat{\psi}' \hat{a}_1^\dagger \rightarrow \hat{a}_1 \hat{a}_1^\dagger \psi'_1.$$

Since the operators $\hat{a}_2 \hat{a}_2^\dagger$ and $\hat{a}_1 \hat{a}_1^\dagger$ are diagonal and stand at the ends of the product, they are replaced by their vacuum mean values, namely unity. To make the analogous transformation in the second term of (74.4), one must first move $\hat{a}_2^\dagger$ to the left and $\hat{a}_1$ to the right. This is done with the commutation rules for $\hat{a}_{\mathbf{p}}$ and $\hat{a}_{\mathbf{p}}^\dagger$, which imply

$$\begin{aligned} \{\hat{a}_{\mathbf{p}}, \hat{\bar{\psi}}\}_+ &= \{\hat{a}_{\mathbf{p}}^\dagger, \hat{\bar{\psi}}\}_+ = 0, \\ \{\hat{a}_{\mathbf{p}}, \hat{\bar{\psi}}'\}_+ &= \bar{\psi}_{\mathbf{p}}, \quad \{\hat{a}_{\mathbf{p}}^\dagger, \hat{\psi}\}_+ = \psi_{\mathbf{p}}. \end{aligned} \quad (74.5)$$

Consequently, (74.4) becomes

$$\langle 0 | (\bar{\psi}_2 \gamma^\mu \hat{\psi}) (\hat{\bar{\psi}}' \gamma^\nu \psi'_1) - (\bar{\psi}_2 \gamma^\mu \psi_1) (\bar{\psi}_2' \gamma^\nu \hat{\psi}') | 0 \rangle, \quad t > t' \quad (74.6)$$

(only the operator factors, of course, are averaged). Similarly, for $t < t'$ we obtain an expression differing by the interchange of the prime and the indices μ, ν :

$$\langle 0 | -(\hat{\bar{\psi}}' \gamma^\nu \psi'_1) (\bar{\psi}_2 \gamma^\mu \hat{\psi}) + (\bar{\psi}_2' \gamma^\nu \hat{\psi}') (\bar{\psi}_2 \gamma^\mu \psi_1) | 0 \rangle, \quad t < t'. \quad (74.7)$$

The two expressions can be written in a single form by introducing the chronological product of ψ -operators through the definition

$$T \hat{\psi}_i(x) \hat{\bar{\psi}}_k(x') = \begin{cases} \hat{\psi}_i(x) \hat{\bar{\psi}}_k(x'), & t' < t, \\ -\hat{\bar{\psi}}_k(x') \hat{\psi}_i(x), & t' > t \end{cases} \quad (74.8)$$

(i and k are bispinor indices). The first and second terms of (74.6) and (74.7) can then be written together as

$$\bar{\psi}_2 \gamma^\mu \langle 0 | T \hat{\psi} \cdot \hat{\bar{\psi}}' | 0 \rangle \gamma^\nu \psi'_1 + \bar{\psi}'_2 \gamma^\nu \langle 0 | T \hat{\psi}' \cdot \hat{\bar{\psi}} | 0 \rangle \gamma^\mu \psi_1 \quad (74.9)$$

(here $\psi \cdot \bar{\psi}$ denotes the matrix $\psi_i \bar{\psi}_k$).

Notice that in the definition (74.8), which arose naturally here, the operator products for $t < t'$ and $t > t'$ carry different signs. This distinguishes it from the definition of the T-product used for the operators $\hat{A}$ and $\hat{j}$. The difference arises because the fermion operators $\hat{\psi}$ and $\hat{\bar{\psi}}$ anticommute outside the light cone (unlike the commuting boson operators $\hat{A}$ and the bilinear operators $\hat{j} = \hat{\bar{\psi}} \gamma \hat{\psi}$)⁷. This ensures the relativistic invariance of the definition (74.8) (a formal proof of the commutation rules for the ψ -operators will be given in § 75)⁸.

Define the *electron propagation function*, or *electron propagator*, to be the second-rank bispinor $G_{ik}(x - x')$ given by

$$G_{ik}(x - x') = -i \langle 0 | T \hat{\psi}_i(x) \hat{\bar{\psi}}_k(x') | 0 \rangle. \quad (74.10)$$

The electron matrix element then takes the form

$$\langle 2 | T \hat{j}^\mu(x) \hat{j}^\nu(x') | 1 \rangle = i \bar{\psi}_2 \gamma^\mu G \gamma^\nu \psi'_1 + i \bar{\psi}'_2 \gamma^\nu G \gamma^\mu \psi_1. \quad (74.11)$$

After multiplication by the photon matrix element (74.1) and integration over $d^4x d^4x'$, the two terms in (74.11) give identical results, yielding

$$S_{fi} = -ie^2 \iint d^4x d^4x' \bar{\psi}_2(x) \gamma^\mu G(x - x') \gamma^\nu \psi_1(x') \times \\ \times \{A_{2\mu}^*(x) A_{1\nu}(x') + A_{2\nu}^*(x') A_{1\mu}(x)\}. \quad (74.12)$$

Substituting the plane waves (64.8), (64.9) for the electron and photon wave functions and extracting the delta function as for (73.10), we finally obtain the scattering amplitude

$$M_{fi} = -4\pi e^2 \bar{u}_2 \{ (\gamma e_2^*) G(p_1 + k_1) (\gamma e_1) + (\gamma e_1) G(p_1 - k_2) (\gamma e_2^*) \} u_1, \quad (74.13)$$

⁷Recall that the ψ -operators themselves do not correspond to any measurable physical quantities and therefore need not commute outside the light cone.

⁸The T-product of any number of ψ -operators can be defined similarly. It is the product of all these operators arranged from right to left in order of increasing time, with a sign determined by the parity of the permutation required to obtain this order from the order displayed under the T-product. With this definition, the sign of the T-product changes when any two ψ -operators are interchanged; for example,

$$T \hat{\psi}_i(x) \hat{\bar{\psi}}_k(x') = -T \hat{\bar{\psi}}_k(x') \hat{\psi}_i(x).$$

where e_1 and e_2 are the photon polarization 4-vectors, and $G(p)$ is the electron propagator in the momentum representation.

The two terms in this expression are represented by the following Feynman diagrams:

$$\begin{aligned}
 4\pi e^2 \bar{u}_2 (\gamma e_2^*) G(f) (\gamma e_1) u_1 = & \quad \begin{array}{c} \text{Diagram 1: A vertex where an incoming solid line } p_1 \text{ and an incoming dashed line } k_1 \text{ meet. A solid line } p_2 \text{ and a dashed line } k_2 \text{ emerge. The internal solid line is labeled } f = p_1 + k_1. \end{array} \\
 4\pi e^2 \bar{u}_2 (\gamma e_1) G(f') (\gamma e_2^*) u_1 = & \quad \begin{array}{c} \text{Diagram 2: A vertex where an incoming solid line } p_1 \text{ and an incoming dashed line } k_2 \text{ meet. A solid line } p_2 \text{ and a dashed line } k_1 \text{ emerge. The internal solid line is labeled } f' = p_1 - k_2. \end{array}
 \end{aligned} \tag{74.14}$$

The dashed free ends of the diagrams represent real photons. An incoming line (the initial photon) supplies a factor $\sqrt{4\pi}e$, and an outgoing line (the final photon) a factor $\sqrt{4\pi}e^*$, where e is the polarization 4-vector. In the first diagram the initial photon is absorbed together with the initial electron, while the final photon is emitted together with the final electron. In the second diagram, emission of the final photon occurs together with annihilation of the initial electron, and absorption of the initial photon with creation of the final electron.

The internal solid line joining the two vertices represents a virtual electron, whose 4-momentum is determined by 4-momentum conservation at the vertices. This line supplies a factor $iG(f)$. Unlike the 4-momentum of a real particle, the square of a virtual electron's 4-momentum is not m^2 . Considering the invariant f^2 , for example in the electron rest frame, readily gives

$$f^2 = (p_1 + k_1)^2 > m^2, \quad f'^2 = (p_1 - k_2)^2 < m^2. \tag{74.15}$$

§ 75. The electron propagator

The propagation functions, or propagators, introduced in the preceding sections play a central role in the apparatus of quantum electrodynamics. The photon propagator $D_{\mu\nu}$ becomes the principal quantity characterizing the interaction of two electrons. This role is clearly displayed by its place in the electron-scattering amplitude, where $D_{\mu\nu}$ is multiplied by the transition currents of the two particles. A similar role is played by the electron propagator in the interaction between an electron and a photon.

We now turn to the actual calculation of the propagators, beginning with the electron case.

Apply the operator $\gamma \hat{p} - m$, where $\hat{p}_\mu = i\partial_\mu$, to the function

$$G_{ik}(x - x') = -i \langle 0 | T \hat{\psi}_i(x) \hat{\bar{\psi}}_k(x') | 0 \rangle \tag{75.1}$$

(i and k are bispinor indices). Since $\hat{\psi}(x)$ satisfies the Dirac equation $(\gamma\hat{p} - m)\hat{\psi}(x) = 0$, the result is zero at every x except where $t = t'$. Indeed, $G(x - x')$ approaches different limits as $t \rightarrow t' + 0$ and $t \rightarrow t' - 0$. By the definition (74.8), these limits are respectively

$$-i\langle 0|\hat{\psi}_i(\mathbf{r}, t)\hat{\bar{\psi}}_k(\mathbf{r}', t)|0\rangle \quad \text{and} \quad +i\langle 0|\hat{\bar{\psi}}_k(\mathbf{r}', t)\hat{\psi}_i(\mathbf{r}, t)|0\rangle.$$

As we shall see, they do not coincide on the light cone. The derivative $\partial G/\partial t$ therefore acquires an additional delta-function term:

$$\begin{aligned} \frac{\partial G}{\partial t} = & -i\left\langle 0\left|T\frac{\partial\hat{\psi}_i(x)}{\partial t}\hat{\bar{\psi}}_k(x')\right|0\right\rangle \\ & + \delta(t - t')(G|_{t \rightarrow t' + 0} - G|_{t \rightarrow t' - 0}). \end{aligned} \quad (75.2)$$

Noting that the time derivative enters $\gamma\hat{p} - m$ as $i\gamma^0\partial/\partial t$, we thus have

$$(\gamma\hat{p} - m)_{ik}G_{kl}(x - x') = \delta(t - t')\gamma_{ik}^0\langle 0|\{\hat{\psi}_k(\mathbf{r}, t), \hat{\bar{\psi}}_l(\mathbf{r}', t)\}_+|0\rangle. \quad (75.3)$$

Let us calculate the anticommutator in this expression. Multiplying the operators $\hat{\psi}(\mathbf{r}, t)$ and $\hat{\bar{\psi}}(\mathbf{r}', t)$ (see (73.6)) and applying the interchange rules for the fermion operators $\hat{a}_{\mathbf{p}}$ and $\hat{b}_{\mathbf{p}}$, we find

$$\{\hat{\psi}_i(\mathbf{r}, t), \hat{\bar{\psi}}_k(\mathbf{r}', t)\}_+ = \sum_{\mathbf{p}} [\psi_{\mathbf{p}i}(\mathbf{r})\bar{\psi}_{\mathbf{p}k}(\mathbf{r}') + \psi_{-\mathbf{p}i}(\mathbf{r})\bar{\psi}_{-\mathbf{p}k}(\mathbf{r}')]. \quad (75.4)$$

Here $\psi_{\pm\mathbf{p}}(\mathbf{r})$ are the wave functions without the time factor (as in §§ 73 and 74, their polarization indices are suppressed for brevity). The collection of all $\psi_{\pm\mathbf{p}}(\mathbf{r})$, the eigenfunctions of the electron Hamiltonian, forms a complete normalized system; hence, by the general properties of such systems (compare III, (5.12)),

$$\sum_{\mathbf{p}} [\psi_{\mathbf{p}i}(\mathbf{r})\psi_{\mathbf{p}k}^*(\mathbf{r}') + \psi_{-\mathbf{p}i}(\mathbf{r})\psi_{-\mathbf{p}k}^*(\mathbf{r}')] = \delta_{ik}\delta(\mathbf{r} - \mathbf{r}'). \quad (75.5)$$

The sum on the right-hand side of (75.4) differs from this one by the replacement of ψ_k^* with $(\psi^*\gamma^0)_k$ and is therefore equal to $\gamma_{ik}^0\delta(\mathbf{r} - \mathbf{r}')$. Thus

$$\{\hat{\psi}_i(\mathbf{r}, t), \hat{\bar{\psi}}_k(\mathbf{r}', t)\}_+ = \delta(\mathbf{r} - \mathbf{r}')\gamma_{ik}^0. \quad (75.6)$$

In particular, this formula implies the statement already mentioned in § 74 that $\hat{\psi}$ and $\hat{\bar{\psi}}$ anticommute outside the light cone. If $(x - x')^2 < 0$, a reference frame always exists in which $t = t'$; when $\mathbf{r} \neq \mathbf{r}'$ as well, the anticommutator (75.6) is indeed zero.

Substitution of (75.6) into (75.3), with the bispinor indices suppressed, finally gives⁹

$$(\gamma\hat{p} - m)G(x - x') = \delta^{(4)}(x - x'). \quad (75.7)$$

⁹Written explicitly with bispinor indices,

$$(\gamma\hat{p} - m)_{il}G_{lk}(x - x') = \delta^{(4)}(x - x')\delta_{ik}. \quad (75.7a)$$

Thus the electron propagator satisfies the Dirac equation with a delta function on the right-hand side. In other words, it is a *Green function* for the Dirac equation.

In what follows we shall need not $G(\xi)$ itself, where $\xi = x - x'$, but its Fourier components

$$G(p) = \int G(\xi) e^{ip\xi} d^4\xi. \quad (75.8)$$

(the propagator in the momentum representation). Taking the Fourier component of both sides of (75.7), we find that $G(p)$ satisfies the system of algebraic equations

$$(\gamma p - m)G(p) = 1. \quad (75.9)$$

The solution of this system is

$$G(p) = \frac{\gamma p + m}{p^2 - m^2}. \quad (75.10)$$

The four components of the 4-vector p in $G(p)$ are independent variables; they are not constrained by $p^2 = p_0^2 - \mathbf{p}^2 = m^2$. Writing the denominator of (75.10) as $p_0^2 - (\mathbf{p}^2 + m^2)$ shows that, as a function of p_0 at fixed $\mathbf{p}$, $G(p)$ has two poles: $p_0 = \pm\varepsilon$, where $\varepsilon = \sqrt{\mathbf{p}^2 + m^2}$. Integration over dp_0 in the integral

$$\begin{aligned} G(\xi) &= \frac{1}{(2\pi)^4} \int e^{-ip\xi} G(p) d^4p \\ &= \frac{1}{(2\pi)^4} \int d^3p e^{i\mathbf{p}\xi} \int dp_0 e^{-ip_0\tau} G(p) \end{aligned} \quad (75.11)$$

where $\tau = t - t'$, consequently raises the question of how the poles are to be bypassed; until this prescription is specified, (75.10) is still essentially undefined.

To settle this question, return to the original definition (75.1). Substitute the expansions (73.6) for the ψ -operators, observing that only the following vacuum mean values of products of creation and annihilation operators are nonzero:

$$\langle 0 | \hat{a}_{\mathbf{p}} \hat{a}_{\mathbf{p}}^+ | 0 \rangle = 1, \quad \langle 0 | \hat{b}_{\mathbf{p}} \hat{b}_{\mathbf{p}}^+ | 0 \rangle = 1.$$

(Because the vacuum contains no particles, a particle must first be created by $\hat{a}_{\mathbf{p}}^+$ or $\hat{b}_{\mathbf{p}}^+$ before it can be annihilated by $\hat{a}_{\mathbf{p}}$ or $\hat{b}_{\mathbf{p}}$.) We obtain

$$\begin{aligned} G_{ik}(x - x') &= -i \sum_{\mathbf{p}} \psi_{\mathbf{p}i}(\mathbf{r}, t) \bar{\psi}_{\mathbf{p}k}(\mathbf{r}', t') \\ &= -i \sum_{\mathbf{p}} e^{-i\varepsilon(t-t')} \psi_{\mathbf{p}i}(\mathbf{r}) \bar{\psi}_{\mathbf{p}k}(\mathbf{r}'), \quad t - t' > 0; \end{aligned} \quad (75.12)$$

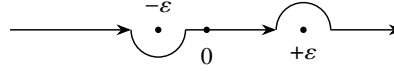
$$\begin{aligned} G_{ik}(x - x') &= i \sum_{\mathbf{p}} \bar{\psi}_{-\mathbf{p}k}(\mathbf{r}', t') \psi_{-\mathbf{p}i}(\mathbf{r}, t) \\ &= i \sum_{\mathbf{p}} e^{i\varepsilon(t-t')} \psi_{-\mathbf{p}i}(\mathbf{r}) \bar{\psi}_{-\mathbf{p}k}(\mathbf{r}'), \quad t - t' < 0. \end{aligned}$$

(For $t > t'$ only the electron terms contribute to G , and for $t < t'$ only the positron terms.)

Replace the sum over $\mathbf{p}$ conceptually by an integral over d^3p and compare (75.12) with (75.11). The integral

$$\int e^{-ip_0\tau} G(p) dp_0 \quad (75.13)$$

must have the phase factor $e^{-i\varepsilon\tau}$ for $\tau > 0$ and $e^{i\varepsilon\tau}$ for $\tau < 0$. This condition is met if the poles $p_0 = \varepsilon$ and $p_0 = -\varepsilon$ are bypassed above and below, respectively, in the complex p_0 plane:



$$(75.14)$$

Indeed, for $\tau > 0$ the integration path is closed by a semicircle at infinity in the lower half-plane, so the value of (75.13) is given by the residue at the pole $p_0 = +\varepsilon$. For $\tau < 0$ the contour is closed in the upper half-plane and the integral is determined by the residue at $p_0 = -\varepsilon$. The required result follows in both cases.

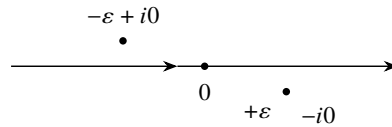
This bypass prescription, the *Feynman prescription*, can be stated in another way: integration is performed everywhere along the real axis itself, but the particle mass m is assigned an infinitesimal negative imaginary part:

$$m \rightarrow m - i0. \quad (75.15)$$

Indeed, then

$$\varepsilon \rightarrow \sqrt{\mathbf{p}^2 + (m - i0)^2} = \sqrt{\mathbf{p}^2 + m^2 - i0} = \varepsilon - i0.$$

In other words, the poles $p_0 = \pm\varepsilon$ are displaced below and above the real axis:



$$(75.16)$$

so that integration along this axis becomes equivalent to integration along the path in (75.14)¹⁰. With the prescription (75.15), the propagator (75.10) may be written

$$G(p) = \frac{\gamma p + m}{p^2 - m^2 + i0}. \quad (75.17)$$

The integration rule for a displaced pole is expressed by

$$\frac{1}{x + i0} = \mathcal{P} \frac{1}{x} - i\pi\delta(x). \quad (75.18)$$

This is to be understood in the sense that multiplication by a function $f(x)$ and integration gives

$$\int_{-\infty}^{\infty} \frac{f(x)}{x + i0} dx = \mathcal{P} \int_{-\infty}^{\infty} \frac{f(x)}{x} dx - i\pi f(0), \quad (75.19)$$

¹⁰It is useful to note that the pole-displacement prescription corresponds to an infinitesimal damping of $G(x - x')$ with $|\tau|$, where $\tau = t - t'$. Indeed, if the values of p_0 at the displaced poles are written as $-(\varepsilon - i\delta)$ and $(\varepsilon - i\delta)$, where $\delta \rightarrow +0$, the time factor in (75.13) is $\exp(-i\varepsilon|\tau| - \delta|\tau|)$.

where the crossed integral sign, or the symbol P , denotes the principal value.

The Green function (75.10) is the product of the bispinor factor $\gamma p + m$ and the scalar

$$G^{(0)}(p) = \frac{1}{p^2 - m^2}. \quad (75.20)$$

The corresponding coordinate function $G^{(0)}(\xi)$ is evidently a solution of

$$(\hat{p}^2 - m^2)G^{(0)}(x - x') = \delta^{(4)}(x - x'), \quad (75.21)$$

that is, a Green function for the equation $(\hat{p}^2 - m^2)\psi = 0$. In this sense, $G^{(0)}(x - x')$ may be called the propagator for scalar particles. A direct calculation similar to the one above readily shows that the scalar-field propagation function is expressed in terms of the ψ -operators (11.2) by

$$G^{(0)}(x - x') = -i\langle 0 | T \hat{\psi}(x) \hat{\psi}^+(x') | 0 \rangle, \quad (75.22)$$

in analogy with the definition (75.1). Here the chronological product is defined, as for all boson operators, by

$$T \hat{\psi}(x) \hat{\psi}^+(x') = \begin{cases} \hat{\psi}(x) \hat{\psi}^+(x'), & t > t', \\ \hat{\psi}^+(x') \hat{\psi}(x), & t < t' \end{cases} \quad (75.23)$$

with the same signs for $t > t'$ and $t < t'$.

§ 76. The photon propagator

So far, in § 43 and 74, we have needed the explicit form of the electromagnetic-field operators $\hat{\mathbf{A}}$ only when calculating matrix elements involving a change in the number of real photons. For this purpose it was sufficient to use the representation given in § 2, in which the free-field potentials are expanded in transverse plane waves.

This representation by itself, however, does not describe an arbitrary field completely. This is evident already from the fact that the scattering diagrams (73.13), (73.14) must also include the Coulomb interaction between the electrons. That interaction is described by the scalar potential Φ and plainly cannot be reduced to the exchange of transverse virtual photons alone (which are described by a vector potential subject to $\nabla \cdot \mathbf{A} = 0$)¹¹.

Thus we still have, in substance, no complete definition of the operators $\hat{\mathbf{A}}$; without one, the photon propagator cannot be calculated directly from

$$D_{\mu\nu}(x - x') = i\langle 0 | T A_\mu(x) A_\nu(x') | 0 \rangle. \quad (76.1)$$

On the other hand, the gauge ambiguity of the potentials largely deprives the operators that would have to be introduced for an exhaustive quantization of the electromagnetic field of physical meaning.

¹¹When $\nabla \cdot \mathbf{A} = 0$, Maxwell's equations give the following equations for $\mathbf{A}$ and Φ :

$$\square \mathbf{A} = -4\pi \mathbf{j} + \nabla \frac{\partial \Phi}{\partial t}, \quad \Delta \Phi = -4\pi \rho.$$

In this gauge, Φ obeys the static Poisson equation (compare formula (76.13) for D_{00} in the same gauge).

These difficulties are merely formal rather than physical. They can be avoided by using certain general properties of the propagator that follow directly from relativistic and gauge invariance.

The most general form of a rank-two 4-tensor that depends only on the 4-vector $\xi = x - x'$ is

$$D_{\mu\nu}(\xi) = g_{\mu\nu}D(\xi^2) - \partial_\mu\partial_\nu D^{(l)}(\xi^2), \quad (76.2)$$

where D and $D^{(l)}$ are scalar functions of the invariant ξ^2 ¹². Notice that the tensor is automatically symmetric.

Accordingly, in the momentum representation we have

$$D_{\mu\nu}(k) = D(k^2)g_{\mu\nu} + k_\mu k_\nu D^{(l)}(k^2), \quad (76.3)$$

where $D(k^2)$ and $D^{(l)}(k^2)$ are the Fourier components of $D(\xi^2)$ and $D^{(l)}(\xi^2)$.

In physical quantities—the scattering amplitudes—the photon propagator is multiplied by the transition currents of two electrons, and therefore occurs in combinations such as $j_{21}^\mu D_{\mu\nu} j_{43}^\nu$ (see, for example, (73.13)). Current conservation ($\partial_\mu j^\mu = 0$), however, implies that its matrix elements $j_{21}^\mu = \bar{\psi}_2 \gamma^\mu \psi_1$ obey the 4-transversality condition

$$k_\mu (j^\mu)_{21} = 0, \quad (76.4)$$

where $k = p_2 - p_1$ (compare (43.13)). It is consequently clear that no physical result changes under the replacement

$$D_{\mu\nu} \rightarrow D_{\mu\nu} + \chi_\mu k_\nu + \chi_\nu k_\mu, \quad (76.5)$$

where the χ_μ may be arbitrary functions of $\mathbf{k}$ and k_0 . This freedom in the choice of $D_{\mu\nu}$ corresponds to the freedom in the gauge of the field potentials.

An arbitrary gauge transformation (76.5) may spoil the relativistically invariant form of $D_{\mu\nu}$ assumed in (76.3), if the quantities χ_μ do not make up a 4-vector. Yet even if we restrict ourselves to relativistically invariant forms of the propagator, the choice of the function $D^{(l)}(k^2)$ in (76.3) remains arbitrary. It cannot affect physical results and may be fixed for convenience (L. D. Landau, A. A. Abrikosov, I. M. Khalatnikov, 1954).

The determination of the propagator thus reduces to finding just one gauge-invariant function, $D(k^2)$. For a prescribed value of k^2 , choose the z axis along $\mathbf{k}$. The transformations (76.5) then leave the components $D_{xx} = D_{yy} = D(k^2)$ unchanged. It is therefore enough to calculate the single component D_{xx} , using any gauge for the potentials.

We use the gauge in which $\nabla \cdot \hat{\mathbf{A}} = 0$ and the operator $\hat{\mathbf{A}}$ is given by the expansions (2.17), (2.18):

$$\hat{\mathbf{A}} = \sum_{\mathbf{k}\alpha} \sqrt{\frac{2\pi}{\omega}} \left(\hat{c}_{\mathbf{k}\alpha} \mathbf{e}^{(\alpha)} e^{-ikx} + \hat{c}_{\mathbf{k}\alpha}^+ \mathbf{e}^{(\alpha)*} e^{ikx} \right), \quad \omega = |\mathbf{k}| \quad (76.6)$$

¹²These functions are different in the three ranges of their argument that cannot be transformed into one another by Lorentz transformations: outside the light cone ($\xi^2 < 0$), in its future interior ($\xi^2 > 0$, $\xi_0 > 0$), and in its past interior ($\xi^2 > 0$, $\xi_0 < 0$).

(the index $\alpha = 1, 2$ labels the polarizations). Of all the vacuum expectation values of products of the operators $\widehat{c}$ and $\widehat{c}^+$, the only nonzero ones are

$$\langle 0 | \widehat{c}_{\mathbf{k}\alpha} \widehat{c}_{\mathbf{k}\alpha}^+ | 0 \rangle = 1.$$

The definition (76.1) therefore gives

$$D_{ik}(\xi) = \frac{1}{(2\pi)^3} \int \frac{2\pi i d^3 k}{\omega} \left(\sum_{\alpha} e_i^{(\alpha)} e_k^{(\alpha)*} \right) e^{-i\omega|\tau| + i\mathbf{k}\xi} \quad (76.7)$$

(i and k are three-dimensional vector indices; we have replaced the sum over $\mathbf{k}$ by integration with the measure $d^3 k / (2\pi)^3$). The absolute value of the difference $\tau = t - t'$ in the exponent results from the time ordering of the operator product in (76.1).

Equation (76.7) shows that the integrand without the factor $e^{i\mathbf{k}\xi}$ is the component in the three-dimensional Fourier expansion of $D_{ik}(\mathbf{r}, t)$. For $D_{xx} = -D$ it is

$$\frac{2\pi i}{\omega} e^{-i\omega|\tau|} \sum_{\alpha} |e_x^{(\alpha)}|^2 = \frac{2\pi i}{\omega} e^{-i\omega|\tau|}.$$

To find $D_{xx}(k^2)$, it remains to expand this function as a Fourier integral over time. The required expansion is

$$\frac{2\pi i}{\omega} e^{-i\omega|\tau|} = -\frac{1}{2\pi} \int_{-\infty}^{\infty} \frac{4\pi}{k_0^2 - \mathbf{k}^2 + i0} e^{-ik_0\tau} dk_0.$$

As explained in the preceding section, this integration entails passing below the pole $k_0 = +|\mathbf{k}| = +\omega$ and above the pole $k_0 = -|\mathbf{k}| = -\omega$. For $\tau > 0$ the integral is determined by the residue at $k_0 = +\omega$, and for $\tau < 0$ by the residue at $k_0 = -\omega$.

We finally obtain

$$D(k^2) = \frac{4\pi}{k^2 + i0}. \quad (76.8)$$

The occurrence of $+i0$ in the denominator, obtained automatically in the preceding derivation, agrees with rule (75.15): $i0$ is subtracted from the (zero) photon mass. It follows from (76.8) that the corresponding coordinate-space function $D(\xi^2)$ satisfies

$$-\partial_{\mu} \partial^{\mu} D(x - x') = 4\pi \delta^{(4)}(x - x'), \quad (76.9)$$

so it is a Green function of the wave equation.

We shall normally set $D^{(l)} = 0$, thus using the propagator in the form

$$D_{\mu\nu} = g_{\mu\nu} D(k^2) = \frac{4\pi}{k^2 + i0} g_{\mu\nu} \quad (76.10)$$

(the *Feynman gauge*).

We also give other gauge choices that can offer advantages in some applications.

Putting $D^{(l)} = -D/k^2$, we obtain the propagator

$$D_{\mu\nu} = \frac{4\pi}{k^2} \left(g_{\mu\nu} - \frac{k_{\mu} k_{\nu}}{k^2} \right) \quad (76.11)$$

(the *Landau gauge*). In this case $D_{\mu\nu}k^\nu = 0$. This choice is analogous to the Lorenz gauge for the potentials ($A_\mu k^\mu = 0$).

The gauge of the potentials defined by the three-dimensional condition $\nabla \cdot \mathbf{A} = 0$ has as its analogue the propagator gauge specified by

$$D_{il}k^l = 0, \quad D_{0l}k^l = 0.$$

Together with $D_{xx} = -D = -4\pi/k^2$, these conditions give

$$D_{il} = \frac{4\pi}{\omega^2 - \mathbf{k}^2} \left(\delta_{il} - \frac{k_i k_l}{\mathbf{k}^2} \right). \quad (76.12)$$

To obtain this D_{il} , transformation (76.5) must be applied to propagator (76.10), with

$$\chi_0 = -\frac{4\pi\omega}{2(\omega^2 - \mathbf{k}^2)\mathbf{k}^2}, \quad \chi_i = -\frac{4\pi k_i}{2(\omega^2 - \mathbf{k}^2)\mathbf{k}^2}.$$

The remaining components $D_{\mu\nu}$ then become

$$D_{00} = -4\pi/\mathbf{k}^2, \quad D_{0i} = 0. \quad (76.13)$$

This gauge is called the Coulomb gauge (E. Salpeter, 1952). Notice that D_{00} is here the Fourier component of the Coulomb potential.

Finally, the propagator gauge analogous to the potential gauge $\Phi = 0$ is

$$D_{il} = \frac{4\pi}{\omega^2 - \mathbf{k}^2} \left(\delta_{il} - \frac{k_i k_l}{\omega^2} \right), \quad D_{0i} = D_{00} = 0. \quad (76.14)$$

This form is convenient in nonrelativistic problems (I. E. Dzyaloshinsky, L. P. Pitaevsky, 1959).

All the expressions written above refer to the momentum representation of the propagator. In some cases it is convenient to use a mixed frequency–coordinate representation, namely the function

$$D_{\mu\nu}(\omega, \mathbf{r}) = \int D_{\mu\nu}(\omega, \mathbf{k}) e^{i\mathbf{k}\mathbf{r}} \frac{d^3k}{(2\pi)^3}. \quad (76.15)$$

In the Feynman gauge (76.10),

$$D_{\mu\nu}(\omega, \mathbf{r}) = g_{\mu\nu} D(\omega, \mathbf{r}),$$

where

$$D(\omega, \mathbf{r}) = 4\pi \int \frac{e^{i\mathbf{k}\mathbf{r}}}{\omega^2 - \mathbf{k}^2 + i0} \frac{d^3k}{(2\pi)^3} = -\frac{i}{\pi r} \int_0^\infty \frac{e^{ikr} - e^{-ikr}}{\omega^2 - k^2 + i0} k dk,$$

or, after replacing k by $-k$ in the second term of the integrand,

$$D(\omega, \mathbf{r}) = -\frac{i}{\pi r} \int_{-\infty}^\infty \frac{e^{ikr} k dk}{\omega^2 - k^2 + i0}.$$

The last integral is evaluated by closing the contour with an infinitely distant semicircle in the upper half-plane of the complex k plane. It then reduces to the residue at the pole $k = |\omega| + i0$. The result is

$$D(\omega, \mathbf{r}) = -e^{i|\omega|r}/r. \quad (76.16)$$

We make the following observation concerning this expression. The process represented by diagrams (73.13), (73.14) can be pictured as electron 2 being scattered in the field produced by electron 1 (or conversely). Function (76.16) corresponds to the usual “retarded” potential $\propto e^{i\omega r}$ (see II, (64.1), (64.2)) only when $\omega > 0$. The sign of ω , however, depends on the arbitrary choice of direction for the arrow k in the diagram. This property of $D(\omega, \mathbf{r})$ means that, in quantum electrodynamics, the source of the field must be taken to be the particle that gives up energy, i.e. emits the virtual photon.

We conclude with the propagator for spin-1 particles of nonzero mass. In this case there is no gauge freedom, and the choice of propagator is unique.

Substitution of the ψ operators (14.16) into the definition

$$G_{\mu\nu} = -i\langle 0 | T \psi_\mu(x) \psi_\nu^\dagger(x') | 0 \rangle, \quad (76.17)$$

gives an expression differing from (76.7) only in that the polarization sum in its integrand is replaced by

$$\sum_{\alpha} u_{\mu}^{(\alpha)} u_{\nu}^{(\alpha)*}.$$

Summation over polarizations is equivalent to averaging and then multiplying by 3, the number of independent polarizations. The average gives the density matrix (14.15) for unpolarized particles. Hence the vector-particle propagator is

$$G_{\mu\nu}(p) = -\frac{1}{p^2 - m^2 + i0} \left(g_{\mu\nu} - \frac{p_{\mu} p_{\nu}}{m^2} \right). \quad (76.18)$$

Notice the analogous structure of propagators (75.17) and (76.18): their denominator is the difference $p^2 - m^2$, while the numerator is, up to a factor, the density matrix of unpolarized particles having the specified spin.

§ 77. General rules of the diagrammatic method

The calculations of scattering-matrix elements carried out in § 73 and 74 for several simple cases contain all the essential features of the general method. Their appropriate generalization readily gives rules for calculating matrix elements at an arbitrary order of perturbation theory.

As noted above, the matrix element of the scattering operator $\hat{S}$ between any initial and final states equals the vacuum average of the operator obtained by multiplying $\hat{S}$ on the right by the creation operators for all initial particles and on the left by the annihilation operators for all final particles.

After this reduction, the S -matrix element at order n of perturbation theory takes the

form

$$\begin{aligned} \langle f | S^{(n)} | i \rangle &= \frac{1}{n!} \langle 0 | \dots b_{2f} b_{1f} \dots a_{1f} \dots c_{1f} \times \\ &\quad \times \int d^4 x_1 \dots d^4 x_n T \left\{ (\bar{\psi}_1 (-ie\gamma \hat{A}_1) \psi_1) \dots (\bar{\psi}_n (-ie\gamma \hat{A}_n) \psi_n) \right\} \times \\ &\quad \times c_{1i}^+ \dots a_{1i}^+ \dots b_{1i}^+ | 0 \rangle. \end{aligned} \quad (77.1)$$

(The indices $1i, 2i, \dots$ enumerate the initial particles, separately for positrons, electrons, and photons; $1f, 2f, \dots$ enumerate the final particles. The indices $1, 2, \dots$ on $\hat{\psi}$ and $\hat{A}$ mean that $\hat{\psi}_1 = \hat{\psi}(x_1), \dots$) The operators $\hat{\psi}$ and $\hat{A}$ appearing here are

linear combinations of creation and annihilation operators for the corresponding particles in different states. Hence the matrix elements become vacuum averages of products of particle creation and annihilation operators and of their linear combinations. Such averages are evaluated by the following statements, which constitute *Wick's theorem* (G. C. Wick, 1950).

1. The vacuum average of a product containing any number of bosonic operators $\hat{c}^+$ and $\hat{c}$ is the sum of the products of every possible set of pairwise averages (contractions) of these operators. Within each pair, the factors must retain their order in the original product.

2. For fermionic operators $\hat{a}^+, \hat{a}, \hat{b}^+, \text{ and } \hat{b}$ (whether for like or unlike particles), the sole change is that each term enters the sum with a plus or minus sign according as the number of interchanges of fermionic operators needed to place all contracted pairs next to one another is even or odd.

Clearly, an average can be nonzero only when every factor $\hat{a}, \hat{b}$, or $\hat{c}$ in the product is accompanied by a factor $\hat{a}^+, \hat{b}^+$, or $\hat{c}^+$, respectively. Moreover, only operator pairs $(\hat{a}, \hat{a}^+), \dots$ belonging to the same state are to be contracted, and only when $\hat{a}^+, \dots$ stands to the right of $\hat{a}, \dots$: the particle is first created and is then annihilated (whereas $\langle 0 | \hat{a}^+ \hat{a} | 0 \rangle = 0, \dots$).

When every pair $(\hat{a}, \hat{a}^+), \dots$ occurs in the product only once, Wick's theorem is immediate: the average reduces to a single product of pair averages. It is equally immediate when all annihilation operators stand to the right of the creation operators in the product (a product of this kind is called *normal*); its average vanishes. These facts make it easy to prove Wick's theorem by complete induction in the general case where the same operator pair occurs several times, say k times, in the product.

Consider an average $\langle 0 | \dots \hat{c} \hat{c}^+ \dots | 0 \rangle$ in which a pair of bosonic operators occurs k times (the subsequent reasoning for fermionic operators is entirely analogous). Interchanging the factors $\hat{c}$ and $\hat{c}^+$ in one pair and using their commutation rule gives

$$\langle 0 | \dots \hat{c} \hat{c}^+ \dots | 0 \rangle = \langle 0 | \dots \hat{c}^+ \hat{c} \dots | 0 \rangle + \langle 0 | \dots 1 \dots | 0 \rangle. \quad (77.2)$$

The average $\langle 0 | \dots 1 \dots | 0 \rangle$ contains $k - 1$ pairs, so Wick's theorem is assumed to hold for it. On the other hand,

expanding $\langle 0 | \dots \hat{c} \hat{c}^+ \dots | 0 \rangle$ by Wick's theorem makes it differ from the corresponding expansion of $\langle 0 | \dots \hat{c}^+ \hat{c} \dots | 0 \rangle$ precisely by the term

$$\langle 0 | \dots 1 \dots | 0 \rangle \langle 0 | \hat{c} \hat{c}^+ | 0 \rangle = \langle 0 | \dots 1 \dots | 0 \rangle$$

(the analogous term in the expansion of $\langle 0 | \dots \hat{c}^+ \hat{c} \dots | 0 \rangle$ is $\langle 0 | \dots 1 \dots | 0 \rangle \times \langle 0 | \hat{c}^+ \hat{c} | 0 \rangle = 0$). Equation (77.2) therefore shows that, if Wick's theorem holds for the matrix element $\langle 0 | \dots \hat{c}^+ \hat{c} \dots | 0 \rangle$, it continues to hold after $\hat{c}$ and $\hat{c}^+$ are interchanged. Since the theorem certainly holds for one particular ordering of the factors, namely normal ordering, it follows that it holds for every ordering.

Since Wick's theorem holds for products of $\hat{a}$, $\hat{b}$, $\dots$, it also holds for arbitrary products containing their linear combinations $\hat{\psi}$, $\tilde{\hat{\psi}}$, and $\hat{A}$ alongside $\hat{a}$, $\hat{b}$, $\dots$ themselves. Applying the theorem to the matrix element (77.1) expresses it as a sum of terms, each a product of certain pairwise averages. Some of these are contractions of $\hat{\psi}$, $\tilde{\hat{\psi}}$, and $\hat{A}$ with "external" operators—the creation operators of initial particles or the annihilation operators of final particles. They are related to the wave functions of the initial and final particles by

$$\begin{aligned} \langle 0 | \hat{A} \hat{c}_{\mathbf{p}}^+ | 0 \rangle &= A_p, & \langle 0 | \hat{c}_{\mathbf{p}} \hat{A} | 0 \rangle &= A_p^*, \\ \langle 0 | \hat{\psi} \hat{a}_{\mathbf{p}}^+ | 0 \rangle &= \psi_p, & \langle 0 | \hat{a}_{\mathbf{p}} \tilde{\hat{\psi}} | 0 \rangle &= \tilde{\psi}_p^*, \\ \langle 0 | \hat{b}_{\mathbf{p}} \hat{\psi} | 0 \rangle &= \psi_{-p}, & \langle 0 | \tilde{\hat{\psi}} \hat{b}_{\mathbf{p}}^+ | 0 \rangle &= \tilde{\psi}_{-p}. \end{aligned} \quad (77.3)$$

Here A_p and ψ_p are the photon and electron wave functions with momenta $\mathbf{p}$ (as in § 73 and 74, polarization indices have been suppressed for brevity). There are also contractions of the "internal" operators under the T -product signs. Because application of Wick's theorem preserves the sequence of the factors within every contracted pair, these contractions preserve the chronological ordering of the operators and may therefore be replaced by the corresponding propagators¹³.

Every term in the sum obtained by expanding the matrix element according to Wick's theorem is represented by a particular Feynman diagram. A diagram of order n has n vertices, each associated with one integration variable, namely one of the 4-vectors $x_1, x_2, \dots$. Three rays meet at every vertex: two solid (electron) rays and one dashed (photon) ray. They correspond to the electron operators $\hat{\psi}$, $\tilde{\hat{\psi}}$ and the photon operator $\hat{A}$, all evaluated at the same variable x . The operator $\hat{\psi}$ corresponds to a line entering the vertex, while $\tilde{\hat{\psi}}$ corresponds to one leaving it.

For illustration, consider several examples of the correspondence between terms in a third-order matrix element and diagrams. Omitting the integral sign, the operator and T signs, and the factors $-ie\gamma$, and suppressing the arguments of the operators, we write

¹³The last statement requires the following qualification. In proving Wick's theorem we used the commutation rules for $\hat{c}$ and $\hat{c}^+$, which are meaningful only for real ("transverse") photons. The "external" operators $\hat{c}_i^+$ and $\hat{c}_f$ do, of course, represent precisely such initial and final photons. As stated in § 76, however, the operators $\hat{A}$ under the T -product signs describe more than transverse

photons. The situation is the same as in the calculation of $D_{\mu\nu}$ in § 76. Relativistic and gauge invariance make it sufficient to prove the theorem for those products—that is, for those components of the tensor $\langle 0 | T A_\mu A_\nu \dots | 0 \rangle$ —which are determined by the transverse parts of the potentials. This proves it for arbitrary products as well.

these terms symbolically as

$$\begin{aligned}
 \text{a} \quad & \begin{array}{c} \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \end{array} = \begin{array}{c} \text{---} \\ \text{---} \text{---} \text{---} \\ \text{---} \end{array} \\
 \text{b} \quad & \begin{array}{c} \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \end{array} = \begin{array}{c} \text{---} \\ \text{---} \text{---} \text{---} \\ \text{---} \end{array} \\
 \text{c} \quad & \begin{array}{c} \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \end{array} = \begin{array}{c} \text{---} \\ \text{---} \text{---} \text{---} \\ \text{---} \end{array} \\
 \text{d} \quad & \begin{array}{c} \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \\ \text{---} \text{---} \text{---} \end{array} = \begin{array}{c} \text{---} \\ \text{---} \text{---} \text{---} \\ \text{---} \end{array}
 \end{aligned} \tag{77.4}$$

For clarity, electron and photon contractions have been drawn as solid and dashed arcs, respectively, just as in the diagram. The arrows on the electron contractions point from $\hat{\bar{\psi}}$ to $\hat{\psi}$, in agreement with their direction in the diagrams. Direction has no significance for internal photon contractions; this is also expressed by the evenness of the photon propagator as a function of $x - x'$.

Among the terms obtained in this way are equivalent ones that differ solely in a permutation of vertex numbers—the assignment of the variables $x_1, x_2, \dots$ to the vertices, which amounts merely to renaming integration variables. There are $n!$ such permutations. This number cancels the factor $1/n!$

in (77.1), after which diagrams differing by a vertex permutation need no longer be included. We already encountered this in § 73 and 74. For example, the following two second-order diagrams are equivalent:

$$\begin{aligned}
 & \begin{array}{c} \text{---} \\ \text{---} \text{---} \text{---} \\ \text{---} \end{array} = \left\{ \begin{array}{c} \text{---} \\ \text{---} \text{---} \text{---} \\ \text{---} \end{array} \right. \\
 & \begin{array}{c} \text{---} \\ \text{---} \text{---} \text{---} \\ \text{---} \end{array} = \left\{ \begin{array}{c} \text{---} \\ \text{---} \text{---} \text{---} \\ \text{---} \end{array} \right.
 \end{aligned} \tag{77.5}$$

Only the internal contractions are shown in (77.4) and (77.5); these correspond to the internal lines of the diagrams, representing virtual electrons and photons. The operators left free are contracted with particular external operators, thereby assigning particular initial and final particles to the free ends of the diagrams. Thus $\hat{\bar{\psi}}$, contracted with $\hat{a}_f$ or $\hat{b}_i^+$, gives the line of a final electron or an initial positron; $\hat{\psi}$, contracted with $\hat{a}_i^+$ or $\hat{b}_f$, gives the line of an initial electron or a final positron. A free operator $\hat{A}$, contracted with $\hat{c}_i^+$ or $\hat{c}_f$, may represent either an initial or a final photon. Consequently, several diagrams

in the process. The case where an external electromagnetic field occurs in the problem is handled similarly; this is the field produced by “passive” particles whose state is unchanged by the process under consideration.

Let $A^{(e)}(x)$ be the 4-potential of the external field. It enters the interaction Lagrangian together with the photon operator $\hat{A}$ as the

sum $\hat{A} + A^{(e)}$, which multiplies the current operator $\hat{j}$. Since $A^{(e)}$ contains no operators, it cannot form contractions with other operators. In other words, an external field is represented in Feynman diagrams only by external lines.

Write $A^{(e)}$ as a Fourier integral:

$$\begin{aligned} A^{(e)}(x) &= \int A^{(e)}(q) e^{-iqx} \frac{d^4q}{(2\pi)^4}, \\ A^{(e)}(q) &= \int A^{(e)}(x) e^{iqx} d^4x. \end{aligned} \quad (77.6)$$

In momentum-space expressions for matrix elements, the 4-vector q occurs alongside the 4-momenta of the other external lines, which represent real particles. Every external-field line supplies a factor $A^{(e)}(q)$ and is to be regarded as “incoming”, in accordance with the sign of the exponent in e^{-iqx} with which $A^{(e)}(q)$ enters the Fourier integral. (An “outgoing” line would instead supply $A^{(e)*}(q)$.) If 4-momentum conservation does not uniquely fix the 4-momenta of all external-field lines when the 4-momenta of every real particle are specified, the remaining “free” values of q are integrated over with measure $d^4q/(2\pi)^4$, just like all other line 4-momenta not fixed by the diagram.

If the external field is time independent, then

$$A^{(e)}(q) = 2\pi\delta(q^0)A^{(e)}(\mathbf{q}), \quad (77.7)$$

where $A^{(e)}(\mathbf{q})$ is the three-dimensional Fourier component:

$$A^{(e)}(\mathbf{q}) = \int A^{(e)}(\mathbf{r}) e^{-i\mathbf{q}\mathbf{r}} d^3x. \quad (77.8)$$

In this case an external line supplies the factor $A^{(e)}(\mathbf{q})$ and is assigned the 4-momentum $q^\mu = (0, \mathbf{q})$. Conservation then makes the energies of the electron lines meeting it and the field line at the vertex equal. Every three-dimensional momentum $\mathbf{p}$ of an internal line that remains undetermined is integrated with measure $d^3p/(2\pi)^3$. The amplitude M_{fi} calculated in this manner determines, for example, the scattering cross section (64.25).

We now summarize the final rules of the *diagrammatic method* for constructing the momentum-space expression for a scattering amplitude, or more precisely for iM_{fi} .

1. Order n in perturbation theory is represented by diagrams with n vertices. At each vertex meet one incoming and one outgoing electron (solid) line and one photon (dashed) line. The amplitude for a scattering process contains every diagram whose number of free ends (external lines) equals the total number of initial and final particles.

2. Every external incoming solid line supplies the amplitude $u(p)$ of an initial electron or $u(-p)$ of a final positron, where p is the particle’s 4-momentum. Every outgoing solid line supplies the amplitude $\bar{u}(p)$ of a final electron or $\bar{u}(-p)$ of an initial positron.

3. Every vertex supplies the 4-vector $-ie\gamma^\mu$.
 4. Every external incoming dashed line supplies the initial-photon amplitude $\sqrt{4\pi}e_\mu$, and every outgoing dashed line supplies the final-photon amplitude $\sqrt{4\pi}e_\mu^*$, where e is the polarization 4-vector. The vector index μ is the same as the index of γ^μ at the corresponding vertex, producing the scalar product γe or γe^* .
 5. Every internal solid line supplies a factor $iG(p)$, while every internal dashed line supplies $-iD_{\mu\nu}(p)$. The tensor indices $\mu\nu$ agree with those of the matrices γ^μ and γ^ν at the vertices joined by the dashed line.
 6. The arrows retain one direction along every continuous succession of electron lines. The bispinor-index order along these lines corresponds to writing the matrices from left to right while moving opposite to the arrows. A closed electron loop corresponds to the trace of the product of matrices encountered along it.
 7. At every vertex, the 4-momenta of the lines meeting there obey conservation: the sum of the momenta on incoming lines equals the sum on outgoing lines. The momenta at free ends are prescribed quantities satisfying overall conservation, and a positron line is assigned momentum $-p$. Integrate over the internal-line momenta that remain unfixed after applying conservation at every vertex, with measure $d^4p/(2\pi)^4$.
 8. An incoming free end representing an external field supplies a factor $A^{(e)}(q)$; the 4-vector q is connected to the 4-momenta of the other lines by momentum conservation at the vertex. For a time-independent field, the free end supplies $A^{(e)}(\mathbf{q})$, and every three-dimensional internal-line momentum left unfixed is integrated with measure $d^3p/(2\pi)^3$.
 9. Every closed electron loop in a diagram contributes an additional factor -1 to the expression for iM_{fi} . The same factor is contributed by every pair of external positron ends when they are the beginning and the end of a single continuous sequence of solid lines. If several electrons or positrons occur among either the initial or the final particles, diagrams differing by an odd permutation of pairs of identical particles—and hence of their corresponding external ends—must have opposite relative signs.
- To make the last rule more precise, diagrams having the same solid lines must in any event have the same signs; that is, they would become identical after all photon lines were removed. Recall also that, in the presence of identical fermions, the overall sign of the amplitude is conventional.

§ 78. Crossing invariance

Representing the scattering amplitudes M_{fi} by Feynman integrals reveals a remarkable symmetry, as follows.

Without changing its direction, any incoming external line of a Feynman diagram may represent either an initial particle or a final antiparticle; any outgoing line may represent either a final particle or an initial antiparticle. Passing from particle to antiparticle simultaneously changes the meaning of the 4-momentum p assigned to the line: $p = p_e$ for a particle, say an electron, and $p = -p_p$ for a positron. The assigned polarization also changes. An incoming line carries the wave amplitude u and an outgoing line u^* ; $u = u_e$ for an electron, but $u = u_p^*$ for a positron. Replacing u by u^* reverses the particle's spin projection, or helicity.

For a photon, a genuinely neutral particle, changing the meaning of an external line merely interchanges emission and absorption. A photon line of momentum k represents either absorption with $k_{\text{abs}} = k$, or emission with $k_{\text{em}} = -k$ and the opposite helicity.

Changing the meaning of the external lines in this way amounts to passing from one crossed reaction channel to the others. Hence the same amplitude, as a function of the momenta at the diagrams' free ends, describes every reaction channel¹⁴. Only the interpretation of its arguments changes: replacing a particle by an antiparticle means substituting $p_i \rightarrow -p_f$, where p_i is the initial-particle 4-momentum in one channel and p_f the final-particle 4-momentum in another. This property of a scattering amplitude is called *crossing symmetry*, or *crossing invariance*.

For the invariant amplitudes introduced in § 70 as functions of kinematic invariants, the functions are the same in every channel, but their arguments range over a different physical region in each. Thus Feynman integrals define invariant amplitudes as analytic functions; values in different physical regions are analytic continuations of the function specified in one region. Since their integrands contain singularities, the invariant amplitudes have singularities determined by the integral expressions, including the pole-bypass prescription. Once calculated from Feynman integrals in one channel, their analytic continuation to other channels automatically incorporates these singularities.

Crossing invariance is more than the scattering-matrix properties implied by general space-time symmetry. Those requirements equate amplitudes for processes related by interchanging initial and final states and replacing all particles by antiparticles, with every particle momentum $\mathbf{p}$ unchanged and the signs of the angular-momentum projections reversed. This is *CPT* invariance¹⁵. Crossing invariance permits this transformation not only for all particles together, but for any one particle separately.

§79. Virtual particles

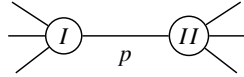
The internal lines of Feynman diagrams play a role in invariant perturbation theory analogous to that of intermediate states in the "ordinary" theory. The character of these states, however, differs in the two theories. In the ordinary theory, three-dimensional momentum is conserved in intermediate states but energy is not conserved; in this sense they are called *virtual states*. In the invariant theory, by contrast, momentum and energy enter on an equal footing: every component of 4-momentum is conserved in intermediate states. This follows because the *S*-matrix elements are integrated over both the coordinates and time, thereby making the theory invariant. What fails in an intermediate state is the relation between energy and momentum that holds for real particles, expressed by $p^2 = m^2$. In this sense one speaks of intermediate *virtual particles*. The relation between the momentum and energy of a virtual particle is arbitrary: it is whatever 4-momentum conservation at the vertices requires.

Consider a diagram made up of two parts, *I* and *II*, joined by one line. Leaving their

¹⁴If a channel is forbidden by 4-momentum conservation, the delta function occurring as a common factor in (64.5) automatically makes its transition probability zero.

¹⁵Formally passing between these reactions by reversing all 4-momenta in the Feynman diagrams corresponds to *CPT* as four-dimensional inversion.

internal structure aside, represent the diagram schematically as


(79.1)

(the lines shown may be either solid or dashed). By overall conservation, the sums of the 4-momenta of the external lines of parts *I* and *II* are equal. Conservation at every vertex makes the 4-momentum p of the internal line joining *I* and *II* equal to this same quantity. Thus this momentum is fixed uniquely and is not integrated over in the matrix element.

Depending on the reaction channel, p^2 may be positive or negative. There is always a channel in which $p^2 > 0$ ¹⁶. The virtual particle then has formal properties entirely analogous to those of a real particle with a real mass $M = \sqrt{p^2}$. A rest frame can be introduced for it, its spin can be defined, and so forth.

The tensor structure of the photon propagator (76.11) is the same as the density matrix of an unpolarized massive spin-1 particle:

$$\rho_{\mu\nu} = -\frac{1}{3} \left(g_{\mu\nu} - \frac{p_\mu p_\nu}{m^2} \right)$$

(see (14.15)). On the other hand, the propagator, being quadratic in the field operators, plays for a virtual particle a role analogous to the density matrix of a real particle. A virtual photon must therefore be assigned spin 1, like a real photon.

Unlike a real photon, which has two independent polarizations, a virtual photon as a finite-mass “particle” may have all three polarizations.

The electron propagation function has

$$G \propto \gamma p + m.$$

Here m is the mass of a real electron, whereas the “mass” of the virtual particle is $M = \sqrt{p^2}$. Writing

$$\gamma p + m = \frac{M + m}{2M} (\gamma p + M) + \frac{M - m}{2M} (\gamma p - M), \quad (79.2)$$

we see that the first term corresponds to the density matrix of a mass- M , spin-1/2 particle, while the second corresponds to the density matrix of the analogous “antiparticle” (compare (29.10) and (29.17)). Since a particle and its antiparticle have different intrinsic parities (see § 27), a virtual electron must be assigned the same spin 1/2, but cannot be assigned a definite parity.

A characteristic feature of the diagram (79.1) is that it can be cut into two mutually unconnected parts by intersecting only one internal line¹⁷. In this case that line represents a *one-particle* intermediate state, containing only one virtual particle. The scattering amplitude for such a diagram contains the characteristic factor, which is not integrated over,

$$\frac{1}{p^2 - m^2 + i0},$$

¹⁶For example, if energetically allowed, take the channel in which all free ends of part *I* represent initial particles and those of part *II* final particles. Then $p = P_i$, the sum of the 4-momenta of all initial particles, and in the center-of-momentum frame $p = (P_i^0, 0)$, so $p^2 > 0$.

¹⁷Almost all processes in their first nonvanishing approximation have diagrams with this property.

arising from the internal line p . Here m is the electron mass if the line is electronic, and $m = 0$ if it is photonic. Thus the scattering amplitude has a pole at those values of p for which the virtual particle would become physical, $p^2 = m^2$. This resembles the poles of a nonrelativistic quantum-mechanical scattering amplitude at energies corresponding to bound states of the colliding-particle system (see III, § 128).

Consider (79.1) in the reaction channel where all right-hand free ends represent initial particles and all left-hand ends final particles, so that $p^2 > 0$. The initial-particle system may then be said to turn into a single virtual particle in the intermediate state. This is possible only if the transformation obeys the necessary conservation laws apart from 4-momentum conservation: conservation of angular momentum,

charge, charge parity, and so on. This is the necessary condition for the appearance of what are called *pole diagrams*. If present in one channel, crossing invariance ensures that these diagrams also occur in the other reaction channels.

For example, these conservation laws do not prevent the formation of a virtual electron through $e + \gamma \rightarrow e$. This possibility corresponds to a pole of the Compton-effect amplitude and therefore also of the amplitude in the other channel of the same reaction, two-photon annihilation of an electron pair. Formation of a virtual photon through $e^- + e^+ \rightarrow \gamma$ corresponds to a pole in electron-positron scattering and hence also in electron-electron scattering. Two photons, however, can produce neither a virtual electron nor a virtual photon: $\gamma + \gamma \rightarrow e$ is forbidden by charge and angular-momentum conservation, while $\gamma + \gamma \rightarrow \gamma$ is forbidden by charge-parity conservation. Accordingly, the photon-photon scattering amplitude cannot contain pole diagrams.

The pole singularities of scattering amplitudes have here been traced from Feynman integrals, but their origin is actually more general and is not tied to perturbation theory. We shall show that they already follow from the unitarity condition (71.2).

Suppose that among the intermediate states n in (71.2) there is a one-particle state. Its contribution is

$$(T_{fi} - T_{if}^*)^{(\text{one-particle})} = i(2\pi)^4 \sum_{\lambda} \int \delta^{(4)}(P_f - p) T_{fn} T_{in}^* \frac{V d^3 p}{(2\pi)^3},$$

where p and λ are the 4-momentum and helicity of the intermediate particle. Replace the integration over $d^3 p$ by one over $d^4 p$, in the region $p^0 \equiv \varepsilon > 0$, according to

$$d^3 p \rightarrow 2\varepsilon \delta(p^2 - M^2) d^4 p$$

where M is the mass of the intermediate particle. The integration removes the delta function $\delta^{(4)}(P_f - p)$. Then changing from T_{fi} to M_{fi} according to (64.10), we find

$$(M_{fi} - M_{if}^*)^{(\text{one-particle})} = 2\pi i \delta(p^2 - M^2) \sum_{\lambda} M_{fn} M_{in}^*. \quad (79.3)$$

Assuming T and P invariance, we have, up to a phase factor, $M_{if} = M_{f'\nu'}$, where i' and f' differ from i and f only in the signs of the particle helicities, at unchanged momenta. Adding (79.3) to the analogous equation for $M_{f'\nu'} - M_{i'\nu'}^*$ gives

$$\text{Im } \overline{M}_{fi}^{(\text{one-particle})} = -\pi \delta(p^2 - M^2) R, \quad (79.4)$$

where

$$\overline{M}_{fi} = M_{fi} + M_{f'i'}, \quad R = - \sum_{\lambda} (M_{fn} M_{in}^* + M_{f'n} M_{i'n}^*).$$

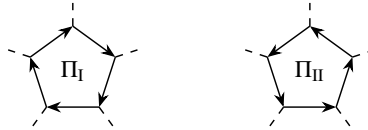
It follows that $\overline{M}_{fi}$, as an analytic function of $p^2 = P_i^2 = P_f^2$, has a pole at $p^2 = M^2$. According to (75.18), its pole part is

$$\overline{M}_{fi}^{(\text{one-particle})} = \frac{R}{p^2 - M^2 + i0}. \quad (79.5)$$

Real transitions to a one-particle state are possible only when $P_i^2 = P_f^2$ equals M^2 . Thus we have indeed obtained the scattering-amplitude structure corresponding to a diagram of the form (79.1).

Finally, consider an important property of diagrams containing closed electron loops. It follows readily by applying the notion of charge parity to a virtual photon: a virtual photon, like a real one, must be assigned a definite negative charge parity¹⁸.

If a diagram contains a closed loop with $N > 2$ vertices, the amplitude for the process must also include another diagram differing only in the direction in which the loop is traversed. For $N = 2$, a direction of traversal plainly has no meaning. “Cut out” the loops along the dashed lines leading to them. This gives two loops, Π_I and Π_{II} :


(79.6)

They may be regarded as diagrams defining the amplitude for transforming one collection of photons, real or virtual, into another; N is then the sum of the numbers of initial and final photons. Charge-parity conservation forbids an even number of photons from turning into an odd number. Therefore, for odd N , the sum of the expressions corresponding to the loops (79.6) must vanish. Hence the combined contribution to the scattering amplitude of the two diagrams containing these loops as components also vanishes. This is *Furry's theorem* (W. H. Furry, 1937).

Thus, in constructing the amplitude of any process, diagrams containing loops with an odd number of vertices need not be considered at all.

Let us trace the origin of this mutual cancellation more closely. For specified photon-line momenta $k_1, k_2, \dots, k_N$, a closed electron loop corresponds to the expression

$$\int d^4p \cdot \text{Tr}[(\gamma e_1)G(p)(\gamma e_2)G(p+k_1)\dots], \quad (79.7)$$

where $p, p+k_1, \dots$ are the electron-line momenta, which remain incompletely determined after the conservation laws at the vertices have been applied. Perform charge conjugation on all matrices γ^μ and G , replacing them by $U_C^{-1}\gamma^\mu U_C$ and $U_C^{-1}GU_C$. This leaves (79.7)

¹⁸This follows from the same considerations concerning the electromagnetic-interaction operator acting at each vertex that were given in § 13 for a real photon.

unchanged because the trace of a matrix product is invariant under such a transformation. On the other hand, by (26.3),

$$U_C^{-1} \gamma^\mu U_C = -\tilde{\gamma}^\mu, \quad (79.8)$$

and therefore

$$U_C^{-1} G(p) U_C = \frac{-p\tilde{\gamma} + m}{p^2 - m^2} = \tilde{G}(-p). \quad (79.9)$$

Replacing $G(p)$ by the transposed matrix with the sign of p reversed plainly changes the direction in which the loop is traversed, reversing every arrow. Thus the transformation takes one loop into the other and produces a factor $(-1)^N$ from the substitution (79.8) at every vertex. Hence

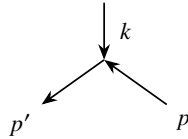
$$\Pi_I = (-1)^N \Pi_{II}, \quad (79.10)$$

so the two loop contributions are equal for even N and opposite in sign for odd N .

Interaction of Electrons

§ 80. Electron scattering in an external field

Elastic scattering of an electron in a constant external field is the simplest process that already occurs in first-order perturbation theory (the first Born approximation). It is represented by a diagram with one vertex,


(80.1)

where p and p' are the initial and final electron four-momenta, and $q = p' - p$. Since the electron energy is conserved in scattering by a constant field ($\varepsilon = \varepsilon'$), $q = (0, \mathbf{q})$ ¹.

The corresponding scattering amplitude is

$$M_{fi} = -e\bar{u}(p')[\gamma A^{(e)}(\mathbf{q})]u(p), \quad (80.2)$$

where $A^{(e)}(\mathbf{q})$ is the spatial Fourier component of the external field. According to (64.26), the scattering cross section is

$$d\sigma = \frac{1}{16\pi^2} |M_{fi}|^2 d\Omega'. \quad (80.3)$$

For an electrostatic field $A^{(e)} = (A_0^{(e)}, 0)$, and hence

$$M_{fi} = -e\bar{u}(p')\gamma^0 u(p)A_0^{(e)}(\mathbf{q}) = -eu^*(p')u(p)A_0^{(e)}(\mathbf{q}). \quad (80.4)$$

In the nonrelativistic case, the plane-wave bispinor amplitudes $u(p)$ reduce to nonrelativistic (two-component) amplitudes. For scattering without a change of polarization, this is a quantity independent of $\mathbf{p}$; under our normalization convention, $u^*u = 2m$. Taking this into account gives

$$d\sigma = \left| \frac{m}{2\pi} U(\mathbf{q}) \right|^2 d\Omega',$$

where $U(\mathbf{q}) = eA_0^{(e)}(\mathbf{q})$ is the Fourier component of the electron's potential energy in the field; this expression agrees with the familiar Born formula (Vol. III, (126.7)).

¹ In the case of an external field, such a diagram is of course not forbidden by four-momentum conservation (as was diagram (73.19) with a real photon): unlike the square of the four-momentum of a real photon, q^2 need not vanish; the component with the required q is selected automatically from the Fourier integral representing the external field.

In the general relativistic case, the scattering cross section for unpolarized electrons is obtained by averaging $|M_{fi}|^2$ over the initial polarizations and summing over the final ones, that is, by forming

$$\frac{1}{2} \sum_{\text{pol}} |M_{fi}|^2,$$

where the sum is over the spin directions of the initial and final electrons; the factor $1/2$ converts one of these summations into an average. The rules set out in § 65 give

$$\begin{aligned} \frac{1}{2} \sum_{\text{pol}} |M_{fi}|^2 &= 2 \text{Tr} \rho' (\gamma A_0^{(e)}) \rho (\gamma A_0^{(e)*}) = \\ &= \frac{1}{2} |A_0^{(e)}(\mathbf{q})|^2 \text{Tr}(m + \gamma p') \gamma^0 (m + \gamma p) \gamma^0. \end{aligned}$$

To evaluate the trace, note that $\gamma^0 (\gamma p) \gamma^0 = \gamma \tilde{p}$, where $\tilde{p} = (\varepsilon, -\mathbf{p})$, and therefore

$$\begin{aligned} \frac{1}{4} \text{Tr}(m + \gamma p') \gamma^0 (m + \gamma p) \gamma^0 &= \frac{1}{4} \text{Tr}(m + \gamma p') (m + \gamma \tilde{p}) = \\ &= m^2 + p' \tilde{p} = \varepsilon^2 + m^2 + \mathbf{p} \mathbf{p}' = 2\varepsilon^2 - \mathbf{q}^2/2. \end{aligned}$$

It follows that the cross section is

$$d\sigma = \frac{e^2 |A_0^{(e)}(\mathbf{q})|^2}{4\pi^2} \varepsilon^2 \left(1 - \frac{\mathbf{q}^2}{4\varepsilon^2}\right) d\sigma'. \quad (80.5)$$

For the field produced by a static charge distribution of density $\rho(\mathbf{r})$, we have

$$A_0^{(e)}(\mathbf{q}) = \frac{4\pi\rho(\mathbf{q})}{\mathbf{q}^2}, \quad (80.6)$$

where $\rho(\mathbf{q})$ is the Fourier transform of the distribution $\rho(\mathbf{r})$ (the form factor). In particular, for the Coulomb field of a point charge Ze , $\rho(\mathbf{q}) = Ze$. The scattering cross section is then

$$d\sigma = d\sigma' \frac{4(Ze^2)^2 \varepsilon^2}{\mathbf{q}^4} \left(1 - \frac{\mathbf{q}^2}{4\varepsilon^2}\right) \quad (80.7)$$

(*N. F. Mott*, 1929). The square

$$\mathbf{q}^2 = 4\mathbf{p}^2 \sin^2(\theta/2),$$

where θ is the scattering angle. The expression preceding the parentheses may therefore be called, from its angular dependence, the Rutherford cross section:

$$d\sigma_R = d\sigma' \frac{4(Ze^2)^2 \varepsilon^2}{\mathbf{q}^4} = d\sigma' \frac{4(Ze^2)^2 \varepsilon^2}{4\mathbf{p}^4} \sin^{-4} \frac{\theta}{2} \quad (80.8)$$

(in the nonrelativistic limit, the factor $\varepsilon^2/\mathbf{p}^4 \rightarrow 1/(m^2v^4)$). Thus²,

$$d\sigma = d\sigma_R \left(1 - v^2 \sin^2 \frac{\theta}{2}\right). \quad (80.9)$$

Notice that in the ultrarelativistic case the angular distribution differs from the nonrelativistic one by a strong suppression of backward scattering (as $\theta \rightarrow \pi$: $d\sigma/d\sigma_R \rightarrow m^2/\varepsilon^2$).

For small-angle scattering in the ultrarelativistic case, (80.7) gives

$$d\sigma = \frac{4(Ze^2)^2}{\varepsilon^4 \theta^4} d\theta'. \quad (80.10)$$

Although this formula was obtained in the Born approximation (that is, under the assumption $Ze^2 \ll 1$), it remains valid (for angles $\theta \lesssim m/\varepsilon$) also when $Ze^2 \sim 1$. This can be verified using the ultrarelativistic wave function $\psi_{\mathbf{p}}^{(+)}$ (39.10), which is exact in Ze^2 . This solution, valid in the region (39.2), naturally remains valid in the asymptotic region at arbitrarily large r . There,

$$F \propto 1 + \text{const} \cdot e^{i(pr - \mathbf{p}\mathbf{r})}, \quad \frac{\alpha \nabla F}{\varepsilon} \sim 1 - \cos \theta \sim \theta^2 \ll 1,$$

so the correction term remains small, as expected. The wave function of the form $e^{ipr}F$, being identical in form to the nonrelativistic function (with the evident change of parameters), has the same asymptotic form; the Rutherford expression therefore results for the cross section as well.

The cross section for scattering of electrons with arbitrary polarization could be calculated by the general rules using the density matrix (29.13). In the present case, however,

the result can be obtained less laboriously by writing the bispinor amplitudes $u(p')$ and $u(p)$ in the form (23.9); multiplication gives

$$u^*(p')u(p) = w'^* \{ \varepsilon + m + (\varepsilon - m)(\mathbf{n}'\boldsymbol{\sigma})(\mathbf{n}\boldsymbol{\sigma}) \} w,$$

or, using formula (33.5),

$$u^*(p')u(p) = w'^* \hat{f} w, \quad (80.11)$$

where³

$$\begin{aligned} \hat{f} &= A + B\boldsymbol{\nu}\boldsymbol{\sigma}, \\ A &= (\varepsilon + m) + (\varepsilon - m) \cos \theta, \quad B = -i(\varepsilon - m) \sin \theta, \\ \boldsymbol{\nu} &= \frac{\mathbf{n} \times \mathbf{n}'}{\sin \theta}. \end{aligned} \quad (80.12)$$

²The difference between $d\sigma$ and $d\sigma_R$ expressed by this formula is specific to particles of spin 1/2. For the scattering of spin-0 particles (if their motion in an electromagnetic field were described by a wave equation), one would obtain $d\sigma = d\sigma_R$. At first sight it may seem strange that the factor expressing this purely quantum effect contains no $\hbar$. It must be remembered, however, that the condition for applicability of the Born approximation ($e^2/(\hbar v) \ll 1$) is opposite to the condition for quasiclassical motion in a Coulomb field; consequently, the classical limit cannot be taken in formula (80.9).

³The definitions of $\hat{f}$ here and in § 37, § 38 differ by an overall factor.

The two-component quantity (three-spinor) w is the nonrelativistic spin wave function of the electron. Partially polarized states are therefore introduced by replacing the products $w_\alpha w_\beta^*$ (α, β are spinor indices) by the nonrelativistic two-by-two density matrix $\rho_{\alpha\beta}$. Thus one must replace

$$|M_{fi}|^2 \rightarrow e^2 |A_0^{(e)}(\mathbf{q})|^2 \text{Tr} \rho (A - B\mathbf{v}\boldsymbol{\sigma}) \rho' (A + B\mathbf{v}\boldsymbol{\sigma}),$$

where

$$\rho = \frac{1}{2}(1 + \boldsymbol{\sigma}\boldsymbol{\zeta}), \quad \rho' = \frac{1}{2}(1 + \boldsymbol{\sigma}\boldsymbol{\zeta}'),$$

and $\boldsymbol{\zeta}$ and $\boldsymbol{\zeta}'$ are the initial polarization vector and the final polarization vector selected by the detector. Evaluation of the trace gives

$$d\sigma = d\sigma_0 \left\{ 1 + \frac{(A^2 - |B|^2)(\boldsymbol{\zeta}\boldsymbol{\zeta}') + 2|B|^2(\mathbf{v}\boldsymbol{\zeta})(\mathbf{v}\boldsymbol{\zeta}') + 2A|B|\mathbf{v}(\boldsymbol{\zeta} \times \boldsymbol{\zeta}')}{A^2 + |B|^2} \right\}, \quad (80.13)$$

where $d\sigma_0$ is the cross section for scattering of unpolarized electrons.

Writing the expression in braces in (80.13) as $\{1 + \boldsymbol{\zeta}^{(f)}\boldsymbol{\zeta}'\}$, we find the polarization of the final electron itself (as distinct from the detected polarization $\boldsymbol{\zeta}'$; see § 65)⁴:

$$\boldsymbol{\zeta}^{(f)} = \frac{(A^2 - |B|^2)\boldsymbol{\zeta} + 2|B|^2(\mathbf{v}\boldsymbol{\zeta})\mathbf{v} + 2A|B|(\mathbf{v} \times \boldsymbol{\zeta})}{A^2 + |B|^2}. \quad (80.14)$$

We see that the scattered electrons are polarized only if the incident electrons are polarized. This is a general property of the first Born approximation (cf. Vol. III, § 140).

In the nonrelativistic case ($\varepsilon \rightarrow m$), (80.14) gives $\boldsymbol{\zeta}^{(f)} = \boldsymbol{\zeta}$, that is, the electron retains its polarization upon scattering (a natural consequence of neglecting the spin-orbit interaction).

In the opposite, ultrarelativistic case, we have

$$A = \varepsilon(1 + \cos \theta), \quad B = -i\varepsilon \sin \theta$$

(in agreement with the general formula (38.2)).

If the incident electron then has a definite helicity ($\boldsymbol{\zeta} = 2\lambda\mathbf{n}$, $\lambda = \pm 1/2$), straightforward simplification of (80.14) gives

$$\boldsymbol{\zeta}^{(f)} = 2\lambda\mathbf{n}'.$$

In other words, after scattering the electron remains in a helicity state and retains its previous helicity value λ .

As already explained in § 38, this property is connected with the fact that, when the mass is neglected, the Dirac equation in the spinor representation separates into two independent equations for the functions ξ and η . The result has a more general significance as well, because the current

$$\hat{j} = (\xi^*\xi + \eta^*\eta, \xi^*\boldsymbol{\sigma}\xi - \eta^*\boldsymbol{\sigma}\eta),$$

⁴Formula (80.14) corresponds to the formula found in Problem 1 (see Vol. III, § 140) and follows from it when A is real and B imaginary.

and hence the electromagnetic perturbation operator $\widehat{V} = e\widehat{j}\widehat{A}$, contain no terms mixing ξ and η and therefore have no matrix elements for transitions between ξ and η states. It follows that if an ultrarelativistic electron has a definite helicity (that is, either ξ or η is nonzero), then this helicity is conserved in interaction processes in the approximation in which the electron mass is neglected completely.

§ 81. Scattering of electrons and positrons on an electron

Let us consider the scattering of an electron on an electron: two electrons with 4-momenta p_1, p_2 collide, acquiring 4-momenta p'_1, p'_2 . Conservation of 4-momentum is expressed by the equality

$$p_1 + p_2 = p'_1 + p'_2. \quad (81.1)$$

Below we shall use the kinematic invariants introduced in § 66, defined as

$$\begin{aligned} s &= (p_1 + p_2)^2 = 2(m^2 + p_1 p_2), \\ t &= (p_1 - p'_1)^2 = 2(m^2 - p_1 p'_1), \\ u &= (p_1 - p'_2)^2 = 2(m^2 - p_1 p'_2), \\ s + t + u &= 4m^2. \end{aligned} \quad (81.2)$$

The process under consideration is depicted by two Feynman diagrams (73.13), (73.14), and its amplitude equals⁵

$$M_{fi} = 4\pi e^2 \left\{ \frac{1}{t} (\bar{u}'_2 \gamma^\mu u_2) (\bar{u}'_1 \gamma_\mu u_1) - \frac{1}{u} (\bar{u}'_1 \gamma^\nu u_2) (\bar{u}'_2 \gamma_\nu u_1) \right\}. \quad (81.3)$$

According to the rules stated in § 65 for the states of initial and final particles, described by the polarisation density matrices $\rho_1, \rho'_1, \dots$, we replace

$$\begin{aligned} |M_{fi}|^2 &\rightarrow 16\pi^2 \alpha^2 e^4 \left\{ \frac{1}{t^2} \text{Sp}(\rho'_2 \gamma^\mu \rho_2 \gamma^\nu) \text{Sp}(\rho'_1 \gamma_\mu \rho_1 \gamma_\nu) + \right. \\ &\quad + \frac{1}{u^2} \text{Sp}(\rho'_1 \gamma^\mu \rho_2 \gamma^\nu) \text{Sp}(\rho'_2 \gamma_\mu \rho_1 \gamma_\nu) - \\ &\quad - \frac{1}{tu} \text{Sp}(\rho'_2 \gamma^\mu \rho_2 \gamma^\nu \rho'_1 \gamma_\mu \rho_1 \gamma_\nu) - \\ &\quad \left. - \frac{1}{tu} \text{Sp}(\rho'_1 \gamma^\mu \rho_2 \gamma^\nu \rho'_2 \gamma_\mu \rho_1 \gamma_\nu) \right\}. \end{aligned} \quad (81.4)$$

For the scattering of unpolarised electrons (not concerning ourselves with their polarisation after scattering) we must set for all density matrices $\rho = \frac{1}{2}(\gamma p + m)$, multiplying the result by $2 \cdot 2 = 4$ (averaging over the polarisations of the two initial and summing over the polarisations of the two final electrons). The scattering cross

⁵This form of M_{fi} is in agreement with the general expression (70.5). In the first non-vanishing approximation of perturbation theory only one of the five invariant amplitudes differs from zero: $f_3(t, u) = 4\pi e^2/t$.

section is determined by formula (64.23), in which one must set, according to (64.15a), $I^2 = \frac{1}{4}s(s - 4m^2)$. Let us represent the cross section in the form

$$d\sigma = dt \frac{4\pi\alpha^2}{s(s - 4m^2)} \{f(t, u) + g(t, u) + f(u, t) + g(u, t)\}, \quad (81.5)$$

$$f(t, u) = \frac{1}{16t^2} \text{Sp}[(\gamma p'_2 + m)\gamma^\mu(\gamma p_2 + m)\gamma^\nu] \times \\ \times \text{Sp}[(\gamma p'_1 + m)\gamma_\mu(\gamma p_1 + m)\gamma_\nu], \quad (\text{IX.1})$$

$$g(t, u) = -\frac{1}{16tu} \text{Sp}[(\gamma p'_2 + m)\gamma^\mu(\gamma p_2 + m)\gamma^\nu(\gamma p'_1 + m)\gamma_\mu(\gamma p_1 + m)\gamma_\nu]. \quad (\text{IX.2})$$

In $f(t, u)$ the traces are first computed (using (22.9), (22.10)), and then the summation over μ and ν is carried out⁶; in $g(t, u)$ the summation over μ and ν is first carried out (using formula (22.6)).

As a result we obtain

$$f(t, u) = \frac{2}{t^2} [(p_1 p_2)^2 + (p_1 p'_2)^2 + 2m^2(m^2 - p_1 p'_1)],$$

$$g(t, u) = \frac{2}{tu} (p_1 p_2 - 2m^2)(p_1 p_2),$$

or, expressing the functions f and g through the invariants (81.2),

$$f(t, u) = \frac{1}{t^2} \left[\frac{s^2 + u^2}{2} + 4m^2(t - m^2) \right], \quad (81.6)$$

$$g(t, u) = g(u, t) = \frac{2}{tu} \left(\frac{s}{2} - m^2 \right) \left(\frac{s}{2} - 3m^2 \right).$$

Thus, the cross section

$$d\sigma = r_e^2 \frac{4\pi m^2 dt}{s(s - 4m^2)} \left\{ \frac{1}{t^2} \left[\frac{s^2 + u^2}{2} + 4m^2(t - m^2) \right] + \right. \\ \left. + \frac{1}{u^2} \left[\frac{s^2 + t^2}{2} + 4m^2(u - m^2) \right] + \frac{4}{tu} \left(\frac{s}{2} - m^2 \right) \left(\frac{s}{2} - 3m^2 \right) \right\}, \quad (81.7)$$

where $r_e = e^2/m$.

Let us apply this formula in the centre-of-inertia system. Here

$$s = 4\varepsilon^2, \quad t = -4\mathbf{p}^2 \sin^2 \frac{\theta}{2}, \quad u = -4\mathbf{p}^2 \cos^2 \frac{\theta}{2}, \quad (81.8)$$

$$-dt = -2\mathbf{p}^2 d \cos \theta = \frac{\mathbf{p}^2}{\pi} d\phi$$

⁶Let us note for future reference the formula

$$\frac{1}{4} \text{Sp}(\gamma p_1 + m)\gamma^\mu(\gamma p_2 + m)\gamma^\nu = g^{\mu\nu}(m^2 - p_1 p_2) + p_1^\mu p_2^\nu + p_1^\nu p_2^\mu.$$

($|\mathbf{p}|$, ε are the magnitude of the momentum and the energy of the electrons, which do not change upon scattering; θ is the scattering angle). In the non-relativistic case ($\varepsilon \approx m$)⁷ we obtain

$$\begin{aligned} d\sigma &= r_e^2 \frac{2\pi m^4 dt}{\mathbf{p}^2} \left(\frac{1}{t^2} + \frac{1}{u^2} - \frac{1}{tu} \right) = \\ &= \left(\frac{e^2}{mv^2} \right)^2 \left(\frac{1}{\sin^4 \frac{\theta}{2}} + \frac{1}{\cos^4 \frac{\theta}{2}} - \frac{1}{\sin^2 \frac{\theta}{2} \cos^2 \frac{\theta}{2}} \right) d\theta = \\ &= \left(\frac{e^2}{mv^2} \right)^2 \frac{4(1 + 3 \cos^2 \theta)}{\sin^4 \theta} d\theta \quad (\text{n. r.}) \end{aligned} \quad (81.9)$$

(where $\mathbf{v} = 2\mathbf{p}/m$ is the relative velocity of the electrons) which agrees with non-relativistic theory (see III, § 137).

In the general case of arbitrary velocities formula (81.7) after substitution of (81.8) and simple transformations can be brought to the form

$$d\sigma = r_e^2 \frac{m^2(\varepsilon^2 + \mathbf{p}^2)^2}{4\mathbf{p}^4 \varepsilon^2} \left[\frac{4}{\sin^4 \theta} - \frac{3}{\sin^2 \theta} + \left(\frac{\mathbf{p}^2}{\varepsilon^2 + \mathbf{p}^2} \right)^2 \left(1 + \frac{4}{\sin^2 \theta} \right) \right] d\theta \quad (81.10)$$

(Ch. Möller, 1932). In the ultrarelativistic case ($\mathbf{p}^2 \approx \varepsilon^2$)

$$d\sigma = r_e^2 \frac{m^2}{\varepsilon^2} \frac{(3 + \cos^2 \theta)^2}{4 \sin^4 \theta} d\theta \quad (\text{u. r.}) \quad (81.11)$$

In the laboratory system of reference, in which one of the electrons (say, the second) was at rest before the collision, we express the cross section through the quantity

$$\Delta = \frac{\varepsilon_1 - \varepsilon'_1}{m} = \frac{\varepsilon'_2 - m}{m} \quad (81.12)$$

—the energy (in units of m) transferred by the first (incident) electron to the second⁸. The invariants

$$\begin{aligned} s &= 2m(m + \varepsilon_1), \quad t = -2m^2 \Delta, \\ u &= -2m(\varepsilon_1 - m - m\Delta). \end{aligned} \quad (81.13)$$

Substitution of these expressions in (81.7) leads to the following formula for the distribution over energies of the secondary electrons (or, as one says, δ -electrons), arising in the scattering of fast primary electrons:

$$d\sigma = 2\pi r_e^2 \frac{d\Delta}{\gamma^2 - 1} \left\{ \frac{(\gamma - 1)^2 \gamma^2}{\Delta^2 (\gamma - 1 - \Delta)^2} - \frac{2\gamma^2 + 2\gamma - 1}{\Delta(\gamma - 1 - \Delta)} + 1 \right\}, \quad (81.14)$$

where $\gamma = \varepsilon_1/m$; $m\Delta$ and $m(\gamma - 1 - \Delta)$ are the kinetic energies of the two electrons after the collision; the identity of both particles manifests itself here in the symmetry

⁷The velocity v is assumed to be small ($v \ll 1$), but such that the condition of applicability of perturbation theory is still satisfied: $e^2/v (= e^2/(\hbar v)) \ll 1$.

⁸The kinematic relations for elastic collisions in different systems of reference see in II, § 13.

of the formula with respect to these quantities. If one agrees to call the recoil electron the one which has the smaller energy, then Δ ranges from 0 to $(\gamma - 1)/2$. For small Δ formula (81.14) takes the form

$$d\sigma = 2\pi r_e^2 \frac{\gamma^2}{\gamma^2 - 1} \frac{d\Delta}{\Delta^2} = \frac{2\pi r_e^2}{\varepsilon_1^2} \frac{d\Delta}{\Delta^2}, \quad \Delta \ll \gamma - 1. \quad (81.15)$$

Let us note that this formula, expressed through the velocity of the incident electron ($v_1 = |\mathbf{p}_1|/\varepsilon_1$), preserves its form in the transition to the non-relativistic case. It is natural therefore that it agrees in form with the result of non-relativistic theory (cf. III, (148.17)).

Let us now consider the scattering of a positron on an electron (*H. Bhabha*, 1936). This is another cross channel of the same generalised reaction, to which the scattering of an electron on an electron pertains. If p_- , p_+ are the initial, and p'_- , p'_+ the final 4-momenta of the electron and positron, then the transition from one case to the other is accomplished by the substitution

$$p_1 \rightarrow -p'_+, \quad p_2 \rightarrow p_-, \quad p'_1 \rightarrow -p_+, \quad p'_2 \rightarrow p'_-.$$

The kinematic invariants (81.2) thereby acquire the following meaning:

$$s = (p_- - p'_+)^2, \quad t = (p_+ - p'_-)^2, \quad u = (p_- - p_+)^2. \quad (81.16)$$

If the ee -scattering was the s -channel, then the $\bar{e}e$ -scattering is the u -channel of the reaction. The square of the scattering amplitude, expressed through s , t , u , remains the same, while in the denominator of formula (81.5) one must replace $s \rightarrow u$. Thus, for the cross section of scattering of the positron on the electron we obtain instead of (81.7)

$$d\sigma = r_e^2 \frac{4\pi m^2 dt}{u(u - 4m^2)} \left\{ \frac{1}{t^2} \left[\frac{s^2 + u^2}{2} + 4m^2(t - m^2) \right] + \right. \\ \left. + \frac{1}{u^2} \left[\frac{s^2 + t^2}{2} + 4m^2(u - m^2) \right] + \frac{4}{tu} \left(\frac{s}{2} - m^2 \right) \left(\frac{s}{2} - 3m^2 \right) \right\}. \quad (81.17)$$

In the centre-of-inertia system the values of the invariants s , t , u differ from (81.8) by the permutation of s and u :

$$s = -4\mathbf{p}^2 \cos^2 \frac{\theta}{2}, \quad t = -4\mathbf{p}^2 \sin^2 \frac{\theta}{2}, \quad u = 4\varepsilon^2. \quad (81.18)$$

In the non-relativistic limit formula (81.17) reduces to the Rutherford formula

$$d\sigma = \left(\frac{e^2}{mv^2} \right)^2 \frac{d\omega}{\sin^4(\theta/2)} \quad (\text{n. r.}), \quad (81.19)$$

where $\mathbf{v} = 2\mathbf{p}/m$ is the relative velocity of the electron and positron. It is obtained from the first term in the curly brackets in (81.17), originating from the diagram of the “scattering” type (see § 73). The contributions of the “annihilation” diagram (the second term in (81.17)) and of its interference with the scattering diagram (the third term) vanish in the non-relativistic limit⁹.

⁹The passage to the non-relativistic limit in the scattering and annihilation parts of the scattering amplitude see below (83.4) and (83.20). The annihilation term (83.20) contains the factor $1/c^2$ and therefore vanishes in this limit.

In the general case of arbitrary velocities the contributions of all three terms in (81.17) are of the same order of magnitude (only in the region

of small angles the first term predominates thanks to the factor $t^{-2} \propto \sin^{-4}(\theta/2)$). After the reduction of similar terms one can represent the cross section of positron scattering on the electron (in the centre-of-inertia system) in the form

$$d\sigma = d\sigma \frac{r_e^2}{16} \frac{m^2}{\varepsilon^2} \left\{ \frac{(\varepsilon^2 + \mathbf{p}^2)^2}{\mathbf{p}^2} \frac{1}{\sin^4(\theta/2)} - \frac{8\varepsilon^4 - m^4}{\mathbf{p}^2 \varepsilon^2} \frac{1}{\sin^2(\theta/2)} + \right. \\ \left. + \frac{12\varepsilon^4 + m^4}{\varepsilon^4} - \frac{4\mathbf{p}^2(\varepsilon^2 + \mathbf{p}^2)}{\varepsilon^4} \sin^2 \frac{\theta}{2} + \frac{4\mathbf{p}^4}{\varepsilon^4} \sin^2 \frac{\theta}{2} \right\}. \quad (81.20)$$

The symmetry with respect to the substitution $\theta \rightarrow \pi - \theta$, characteristic of the scattering of identical particles, is, of course, absent in the scattering of the positron on the electron. In the ultrarelativistic limit expression (81.20) differs from the electron-electron scattering cross section by the factor $\cos^4(\theta/2)$ alone:

$$d\sigma_{\bar{e}e} = \cos^4 \frac{\theta}{2} d\sigma_{ee} \quad (\text{u. r.}) \quad (81.21)$$

In the laboratory system, in which one of the particles (say, the electron) was initially at rest, we again introduce the quantity

$$\Delta = \frac{\varepsilon_+ - \varepsilon'_+}{m} = \frac{\varepsilon'_- - m}{m}, \quad (81.22)$$

i.e. the energy transferred by the positron to the electron. Analogously to (81.13) we now have

$$s = -2m(\varepsilon_+ - m - m\Delta), \quad t = -2m^2\Delta, \quad u = 2m(m + \varepsilon_+).$$

Substituting these expressions in (81.17), after simple transformations we obtain the following formula for the distribution of secondary electrons by energy:

$$d\sigma = 2\pi r_e^2 \frac{d\Delta}{\gamma^2 - 1} \left\{ \frac{\gamma^2}{\Delta^2} - \frac{2\gamma^2 + 4\gamma + 1}{\gamma + 1} \frac{1}{\Delta} + \frac{3\gamma^2 + 6\gamma + 4}{(\gamma + 1)^2} - \frac{2\gamma}{(\gamma + 1)^2} \Delta + \frac{1}{(\gamma + 1)^2} \Delta^2 \right\}, \quad (81.23)$$

where $\gamma = \varepsilon_+/m$; Δ ranges from 0 to $\gamma - 1$. For $\Delta \ll \gamma - 1$ from (81.23) one obtains the same formula (81.15), as for the scattering of electrons as well.

Polarisation effects in the scattering of electrons or positrons are computed by the general rules set out in § 65. In any general case the computations lead to cumbersome formulas. Here we limit ourselves to only a few remarks¹⁰.

In the approximation being considered (first, non-vanishing approximation of perturbation theory) there are absent in the cross section terms linear in the polarisation vectors of the initial or final particles. As in the non-relativistic theory (see III, § 140), such terms are forbidden by the requirements stemming from the hermiticity of the scattering matrix. Therefore the scattering cross section does not change if only one of the colliding

¹⁰More detailed information on this topic can be found in the review article *McMaster W. H* Rev. Mod. Phys. 1961. V. 133. P. 8.

particles is polarised, and the scattering of unpolarised particles does not lead to their polarisation.

These same requirements also forbid the correlation terms in the cross section containing products of polarisations of three of the four participating (initial and final) particles in the process. The cross section contains, however, double and quadruple correlation terms. In the scattering of non-identical particles (electron and positron, electron and muon) in the non-relativistic limit these terms vanish, since the spin-orbit interaction is absent. In the collision of identical particles correlation terms arise already in the non-relativistic case thanks to the exchange effect.

Problems

1. Determine the cross section of scattering of polarised electrons in the non-relativistic case.

Solution. In the non-relativistic case the bispinor amplitudes in the standard representation become two-component, and the density matrices become two-row matrices (29.20). In the scattering amplitude (81.3) there remain only the terms with $\mu = \nu = 0$, containing the diagonal (in the standard representation) matrices γ^0 . Instead of (81.4) we shall have

$$\sum_{\text{polar}} |M_{fi}|^2 = 16\pi^2 \alpha^2 e^4 \cdot 4m^4 \left\{ \left(\frac{1}{t^2} + \frac{1}{u^2} \right) \text{Sp}(1 + \sigma \zeta_1) \text{Sp}(1 + \sigma \zeta_2) - \frac{2}{tu} \text{Sp}(1 + \sigma \zeta_1) \text{Sp}(1 + \sigma \zeta_2) \right\} = 16\pi^2 \alpha^2 e^4 \cdot 4m^4 \left[\frac{1}{t^2} + \frac{1}{u^2} - \frac{1}{tu} (1 + \zeta_1 \zeta_2) \right].$$

(summation over the polarisations of the final electrons only). Hence the cross section of scattering

$$d\sigma = d\sigma_0 \left[1 - \zeta'_1 \zeta_2 \frac{\sin^2 \theta}{1 + 3 \cos^2 \theta} \right].$$

Whence for the vector of polarisation of the scattered electron we have

$$\zeta_1^{(f)} = -\zeta_2 \frac{2 \cos \theta (1 - \cos \theta)}{1 + 3 \cos^2 \theta}.$$

From this, for the probability of reversal of the direction of the spin of a completely polarised electron on scattering by an unpolarised electron

$$\frac{d\sigma}{d\sigma_0} = \frac{(1 - \cos \theta)^2}{2(1 + 3 \cos^2 \theta)}.$$

2. In the non-relativistic case, determine the cross section for the given initial and final polarisations ζ_1 and ζ'_1 and also as in Problem 1 we obtain

$$d\sigma = \frac{1}{2} d\sigma_0 \left[1 + \zeta_1 \zeta'_1 \frac{2 \cos \theta (1 + \cos \theta)}{1 + 3 \cos^2 \theta} \right].$$

Setting $\zeta_1 \zeta'_1 = -1$, we find from this the probability of spin reversal:

$$\frac{d\sigma}{d\sigma_0} = \frac{(1 - \cos \theta)^2}{2(1 + 3 \cos^2 \theta)}.$$

3. In the non-relativistic case, determine the probability of reversal of the direction of the spin of a completely polarised electron on scattering by an unpolarised electron.

Solution. Analogously we find the cross section for the given initial polarisations ζ_1 and ζ'_1 :

$$d\sigma = \frac{1}{2}d\sigma_0 \left[1 + \zeta_1 \zeta'_1 \frac{2 \cos \theta (1 + \cos \theta)}{1 + 3 \cos^2 \theta} \right].$$

Setting $\zeta_1 \zeta'_1 = -1$, we find from this the probability of spin reversal:

$$\frac{d\sigma}{d\sigma_0} = \frac{(1 - \cos \theta)^2}{2(1 + 3 \cos^2 \theta)}.$$

4. Determine the ratio of scattering cross sections of helical electrons with parallel and antiparallel spins in the ultrarelativistic case.

Solution. In (81.4) one must set, according to (29.22),

$$\rho_1 = \frac{1}{2}(\gamma p_1)(1 - 2\lambda_1 \gamma^5), \quad \rho_2 = \frac{1}{2}(\gamma p_2)(1 - 2\lambda_2 \gamma^5);$$

$$\rho'_1 = \frac{1}{2}\gamma p'_1, \quad \rho'_2 = \frac{1}{2}\gamma p'_2,$$

where $\lambda_1, \lambda_2 = \pm 1/2$. The computation of traces is carried out by the formulas cited in § 22; in particular,

$$\begin{aligned} & \text{Sp}[\gamma^5(\gamma a)\gamma^\mu(\gamma b)\gamma^\nu] \text{Sp}[\gamma^5(\gamma c)\gamma_\mu(\gamma d)\gamma_\nu] = \\ & = i^2(e^{\mu\lambda\kappa\sigma}a_\kappa b_\lambda)(e_{\mu\nu\rho\tau}c^\rho d^\tau) = 2(\delta_\nu^\lambda \delta_\rho^\kappa \delta_\tau^\sigma - \delta_\rho^\lambda \delta_\tau^\kappa \delta_\nu^\sigma)a_\kappa b_\lambda c^\rho d^\tau = 2(ac)(bd) - 2(ad)(bc). \end{aligned}$$

From this, for the ratio of cross sections one obtains the result coinciding with the formula found in Problem 4 of III, § 137 (taking $\mathbf{p}^2 \approx \varepsilon^2$), we find the desired ratio of cross sections:

$$\frac{d\sigma_{\uparrow\uparrow}}{d\sigma_{\uparrow\downarrow}} = \frac{1}{8}(1 + 6 \cos^2 \theta + \cos^4 \theta). \quad (1)$$

This ratio is minimal (1/8) at $\theta = \pi/2$.

Since the momenta of the colliding electrons (in the centre-of-inertia system) are mutually opposite, equal helicities ($\lambda_1 = \lambda_2$) correspond to antiparallel spins, and different helicities ($\lambda_1 = -\lambda_2$) correspond to parallel spins. Substituting s, t, u from (81.8) (taking $\mathbf{p}^2 \approx \varepsilon^2$), we find for the desired ratio of cross sections

$$\frac{d\sigma_{\uparrow\uparrow}}{d\sigma_{\uparrow\downarrow}} = \frac{1}{8}(1 + 6 \cos^2 \theta + \cos^4 \theta).$$

This ratio is minimal (1/8) at $\theta = \pi/2$.

5. The same for the scattering of positrons on electrons.

Solution. Instead of (81.4) one must compute

$$\begin{aligned} |M_{fi}|^2 \rightarrow & 16\pi^2 e^4 \left\{ \frac{1}{t^2} \text{Sp}(\rho'_- \gamma^\mu \rho_- \gamma^\nu) \text{Sp}(\rho_+ \gamma_\mu \rho'_+ \gamma_\nu) - \right. \\ & \left. - \frac{1}{tu} \text{Sp}(\rho'_- \gamma^\mu \rho_- \gamma^\nu \rho_+ \gamma_\mu \rho'_+ \gamma_\nu) + \dots \right\} \end{aligned}$$

(the remaining terms are obtained from those written by the permutation ρ_+ and ρ'_-). The density matrices:

$$\rho_- = \frac{1}{2}(\gamma p_-)(1 - 2\lambda_- \gamma^5), \quad \rho_+ = \frac{1}{2}(\gamma p_+)(1 + 2\lambda_+ \gamma^5),$$

$$\rho'_- = \frac{1}{2}\gamma p'_-, \quad \rho'_+ = \frac{1}{2}\gamma p'_+,$$

where $\lambda_+, \lambda_- = \pm 1/2$ (note that for the positron, as for the electron, $\lambda = 1/2$ denotes a spin directed along the momentum). The computation gives

$$\frac{d\sigma}{dt} \propto \left(\frac{s^2 + u^2}{t^2} + \frac{s^2 + t^2}{u^2} + \frac{2s^2}{tu} \right) - 4\lambda_+ \lambda_- \left(\frac{s^2 - u^2}{t^2} + \frac{s^2 - t^2}{u^2} + \frac{2s^2}{tu} \right).$$

From this for the ratio of cross sections one obtains a result coinciding with the formula of Problem 4.

6. Determine the cross section of scattering of muons on electrons.

Solution. The process is described by only one diagram (73.17). Instead of (81.5) we have

$$d\sigma = \frac{\pi e^4 dt}{(p_e p_\mu)^2 - m^2 \mu^2} f(t, u), \quad (1)$$

$$f(t, u) = \frac{1}{16t^2} \text{Sp}[(\gamma p'_\mu + \mu)\gamma^\lambda(\gamma p_\mu + \mu)\gamma^\nu] \text{Sp}[(\gamma p'_e + m)\gamma_\lambda(\gamma p_e + m)\gamma_\nu]$$

(p_e, p_μ and p'_e, p'_μ are the initial and final 4-momenta of the electron and muon; m, μ are their masses). The invariants:

$$s = (p_e + p_\mu)^2 = m^2 + \mu^2 + 2p_e p_\mu,$$

$$t = (p_e - p'_e)^2 = 2(m^2 - p_e p'_e) = 2(\mu^2 - p_\mu p'_\mu),$$

$$u = (p_e - p'_\mu)^2 = m^2 + \mu^2 - 2p_e p'_\mu,$$

$$s + t + u = 2(m^2 + \mu^2).$$

The computation leads to the result

$$f = \frac{1}{t^2} \left\{ \frac{s^2 + u^2}{2} + \frac{1}{2}(m^2 + \mu^2)(2t - m^2 - \mu^2) \right\} = \quad (\text{IX.3})$$

$$= \frac{1}{t^2} \left\{ \frac{s^2 + u^2}{2} + (m^2 + \mu^2)(2t - m^2 - \mu^2) \right\}. \quad (2)$$

Formulas (1) and (2) solve the problem posed. In the centre-of-inertia system

$$d\sigma = \frac{e^4 do}{8(\varepsilon_e + \varepsilon_\mu)^2 \mathbf{p}^4 \sin^4(\theta/2)} \times \\ \times [(\varepsilon_e \varepsilon_\mu + \mathbf{p}^2)^2 + (\varepsilon_e \varepsilon_\mu + \mathbf{p}^2 \cos \theta)^2 - 2(m^2 + \mu^2) \mathbf{p}^2 \sin^2(\theta/2)],$$

where $do = 2\pi \sin \theta d\theta$; $\varepsilon_e, \varepsilon_\mu$ are the energies of the electron and muon; $\mathbf{p}^2 = \varepsilon_e^2 - m^2 = \varepsilon_\mu^2 - \mu^2$.

When $\mathbf{p}^2 \ll \mu^2$ we return to formula (80.9) for the scattering of an electron on a stationary Coulomb centre.

7. Determine the ratio of cross sections of mutual scattering of helical electrons and muons with parallel and antiparallel spins in the ultrarelativistic case ($\varepsilon_\mu \gg \mu$, $\varepsilon_e \gg m$).

Solution. Analogously to Problem 4 we find

$$\frac{d\sigma_{\uparrow\downarrow}}{d\sigma_{\uparrow\uparrow}} = \cos^4 \frac{\theta}{2}$$

(θ is the scattering angle in the centre-of-inertia system).

8. Determine the cross section of conversion of an electron pair into a muon pair (B. B. Berestetskii, I. Ya. Pomeranchuk, 1955).

Solution. This is another cross-channel of the reaction, which pertains to μe -scattering. In this channel:

$$s = (p_e - \bar{p}_\mu)^2, \quad t = (p_\mu - \bar{p}_\mu)^2, \quad u = (p_e - p_\mu)^2,$$

where $p_e, \bar{p}_e$ are the 4-momenta of the electron and positron, and $p_\mu, \bar{p}_\mu$ those of the muon and antimuon. The reaction threshold corresponds to the energy of the electron pair, equal (in the centre-of-inertia system) to 2μ , so that $t > 4\mu^2$ must hold. In the laboratory system, in which the electron before the collision rests, the positron has energy ε_+ , $t = 2m(\varepsilon_+ + m) \approx 2m\varepsilon_+$, so that $\varepsilon_+ > \varepsilon_{\text{th}}$, where the threshold energy $\varepsilon_{\text{th}} = 2\mu^2/m$. In the laboratory system in which the electron before the collision rests the positron has energy ε_+ ; $t = 2m(\varepsilon_+ + m)$.

The differential cross section (instead of (1), in Problem 6):

$$d\sigma = \frac{4\pi e^4 ds}{[s - (m + \mu)^2][s - (m - \mu)^2]} f(t, u), \quad (1)$$

$$f(t, u) = \frac{1}{16t^2} \text{Sp}[(\gamma p'_\mu + \mu)\gamma^\lambda(\gamma p_\mu + \mu)\gamma^\nu] \text{Sp}[(\gamma p'_e + m)\gamma_\lambda(\gamma p_e + m)\gamma_\nu].$$

At a given t , the quantity s ranges between limits determined by the equations $su \approx \mu^2$, $s + t + u \approx 2\mu^2$, i.e.

$$\mu^2 - \frac{t}{2} - \frac{1}{2}\sqrt{t(t - 4\mu^2)} \leq s \leq \mu^2 - \frac{t}{2} + \frac{1}{2}\sqrt{t(t - 4\mu^2)}.$$

Elementary integration yields the result:

$$\sigma = \frac{4\pi}{3} r_e^2 \frac{m^2}{t} \sqrt{1 - \frac{4\mu^2}{t}} \left(1 + \frac{2\mu^2}{t}\right), \quad r_e = \frac{e^2}{m}. \quad (1)$$

(in the laboratory system $t = 2m\varepsilon_+$). This formula is inapplicable in the immediate vicinity of the threshold: when $\varepsilon_+ - \varepsilon_{\text{th}} \sim \mu e^4$, the emerging muons cannot be considered free particles (taking into account the Coulomb interaction between them, the cross section will not tend to zero but to a constant; see III, § 147).

The cross section (1) is maximal at $\varepsilon_+ = 1.7\varepsilon_{\text{th}}$. Its maximum value is approximately 20 times smaller than the cross section of two-photon annihilation at the same energy.

§ 82. Ionization losses of fast particles

Let us consider the collisions of a fast relativistic particle with an atom, accompanied by excitation and ionisation of the latter. In the non-relativistic case such inelastic collisions were considered in III, §§ 144–150; here we give a relativistic generalisation of the results obtained there (*H. A. Bethe*, 1933).

The velocity of the incident particle is assumed to be large in comparison with the velocities of the atomic electrons (thereby $Z\alpha \ll 1$, i.e. the atomic number is not too large). This condition ensures the applicability of the Born approximation to the process being considered.

The solution of the problem is somewhat different depending on whether the fast particle is a light (electron, positron) or heavy (meson, proton, α -particle and so on) one. We shall consider here the latter case, which is simpler.

Let $p = (\varepsilon, \mathbf{p})$ and $p' = (\varepsilon', \mathbf{p}')$ be the initial and final 4-momenta of the fast particle in the laboratory system of reference, in which the atom before the collision was at rest; the difference $q = p' - p$ gives the energy and momentum transferred by the particle to the atom. Let us divide the whole range of possible momentum transfers to the atom into two regions:

$$\text{I) } \frac{\mathbf{q}^2}{m} \ll m, \quad \text{II) } \frac{\mathbf{q}^2}{m} \gg I, \quad (82.1)$$

where m is the mass of the electron, I is some average atomic energy (the ionisation potential of the atom). Both regions overlap each other when $I \ll \mathbf{q}^2/m \ll m$; this circumstance will permit making a precise matching of the results obtained for each of the regions. We shall speak of the values of $\mathbf{q}$ in the first and in the second regions respectively as small and large momentum transfers.

Small momentum transfers. In this region the atomic electrons can be considered non-relativistic both in the initial and in the final states of the atom. The amplitude of the process is given by the expression

$$M_{fi}^{(n)} = e^2 J_{n0}^\mu (-q) J_{p'p}^\nu (q) D_{\mu\nu}(q), \quad (82.2)$$

where J_{n0} is the 4-current of transition of the atom from the initial (0) to the final (n) state, $J_{p'p}$ is the 4-current of transition of the fast particle; these currents, which replace here the expressions $(\bar{u}'\gamma u)$ that would stand, for example, in the amplitude of scattering of two “elementary” particles—the electron and the muon (73.17) (cf. also (139.3)). The transition currents are taken in the momentum representation (see (43.11)). The cross section of the process in the laboratory system of reference:

$$d\sigma_n = 2\pi\delta(\varepsilon - \varepsilon' - \omega_{n0}) \left| M_{fi}^{(n)} \right|^2 \frac{d^3 p'}{2|\mathbf{p}|2\varepsilon'(2\pi)^3}, \quad (82.3)$$

where $\omega_{n0} = E_n - E_0$ is the frequency of transition between states of the atom. The final state may correspond both to a discrete and to a continuous spectrum; the first case corresponds to excitation, and the second to ionisation of the atom.

The photon propagator is conveniently chosen in the present case in the Coulomb gauge (76.14), in which only its spatial components are different from zero:

$$D_{ik}(q) = -\frac{4\pi}{\omega^2 - \mathbf{q}^2} \left(\delta_{ik} - \frac{q_i q_k}{\mathbf{q}^2} \right). \quad (82.4)$$

Then in (82.2) only the spatial components of the 4-currents of transition are needed.

The atomic transition current $\mathbf{J}_{n0}(\mathbf{q})$ in the present case is the Fourier component of the usual non-relativistic expression:

$$\mathbf{J}_{n0}(\mathbf{q}) = \frac{i}{2m} \int e^{-i\mathbf{q}\mathbf{r}} (\psi_0 \nabla \psi_n^* - \psi_n^* \nabla \psi_0) d^3x, \quad (82.5)$$

where ψ_0, ψ_n are the atomic wave functions (with the summation sign over the atomic electrons, which for brevity of notation we omit here and below, to be understood).

Prointegrating over the first member by parts, we may rewrite this expression in the form of the matrix element:

$$\mathbf{J}_{n0}(\mathbf{q}) = \frac{1}{2} (\mathbf{v} e^{-i\mathbf{q}\mathbf{r}} + e^{-i\mathbf{q}\mathbf{r}} \mathbf{v})_{n0}, \quad (82.6)$$

where $\hat{\mathbf{v}} = -\frac{i}{m} \nabla$ is the operator of the velocity of the electron.

As regards the current of transition of the scattered particle, in view of the relative smallness of its recoil ($|\mathbf{q}| \ll |\mathbf{p}|$) it can simply be replaced by the diagonal element

$$\mathbf{J}_{pp}(0) = 2\mathbf{p}z, \quad (82.7)$$

corresponding to a classical rectilinear motion (cf. below (99.5)); here is also introduced a factor z that takes into account the possible difference of the charge of the particle (ze) from the electron charge.

The smallness of $\mathbf{q}$ means also the smallness of the deflection angle of the particle ϑ . Furthermore, the longitudinal (with respect to $\mathbf{p}$) and transverse components of $\mathbf{q}$ are

$$-q_{\parallel} \approx \frac{dp}{d\varepsilon} \omega_{n0} = \frac{\omega_{n0}}{v}, \quad q_{\perp} \approx |\mathbf{p}| \vartheta, \quad (82.8)$$

so that $\mathbf{q}\mathbf{p} \approx -\varepsilon\omega_{n0}$.

Substitution of ((82.4)–(82.8)) in (82.2) gives, taking into account that

$$M_{fi}^{(n)} = -\frac{4\pi ze^2}{q^2} \left\langle n \left| \frac{\varepsilon}{\omega_{n0}} (\mathbf{q}\mathbf{v} e^{-i\mathbf{q}\mathbf{r}} + e^{-i\mathbf{q}\mathbf{r}} \mathbf{q}\mathbf{v}) + (\mathbf{p}\mathbf{v} e^{-i\mathbf{q}\mathbf{r}} + e^{-i\mathbf{q}\mathbf{r}} \mathbf{p}\mathbf{v}) \right| 0 \right\rangle.$$

In the first member we note that

$$\mathbf{q}\hat{\mathbf{v}}f + f\mathbf{q}\hat{\mathbf{v}} = 2i\hat{f},$$

where $f \equiv e^{-i\mathbf{q}\mathbf{r}}$ (see III, § 149); therefore the matrix element of this operator coincides with the matrix element $2i(\hat{f})_{n0} = 2\omega_{n0}f_{n0}$. In the second member it suffices to replace, in view of the smallness of $\mathbf{q}$, $e^{-i\mathbf{q}\mathbf{r}}$ by unity. Then

$$M_{fi}^{(n)} = -\frac{8\pi ze^2}{q^2} \{ \varepsilon (e^{-i\mathbf{q}\mathbf{r}})_{n0} - i\mathbf{p}\mathbf{r}_{n0}\omega_{n0} \}.$$

The square of the modulus of this expression:

$$\begin{aligned} |M_{fi}^{(n)}|^2 = \frac{64\pi^2 (ze)^2}{(q^2)^2} & \left\{ \varepsilon^2 |(e^{-i\mathbf{q}\mathbf{r}})_{n0}|^2 + 2(\mathbf{q}\mathbf{r}_{n0})(\mathbf{p}\mathbf{r}_{n0})\varepsilon\omega_{n0} + \right. \\ & \left. + (\mathbf{p}\mathbf{r}_{n0})^2 \omega_{n0}^2 \right\} \end{aligned} \quad (82.9)$$

(in the second member it is assumed that $e^{-i\mathbf{q}\mathbf{r}} \approx 1 - i\mathbf{q}\mathbf{r}$; in the first member this cannot be done for the reason which will be clarified below—see remark on p. 378).

The energy losses of a fast particle as a result of its inelastic collisions with atoms¹¹ are determined by the quantity

$$\kappa = \sum_n \int \omega_{n0} d\sigma_n = \frac{1}{16\pi^2} \sum_n \int \omega_{n0} \left| M_{fi}^{(n)} \right|^2 d\vartheta', \quad (82.10)$$

where the summation is carried out over all possible final states of the atom, and the integration over the directions of the scattered particle; we shall call this quantity the *effective retardation* (the ratio κ/ε is called the *cross section of energy losses*).

The integration in (82.10) can be carried out in two stages: as an averaging over the direction of $\mathbf{p}'$ relative to $\mathbf{p}$ and then integration over $d\vartheta' \approx 2\pi\vartheta d\vartheta$, where ϑ is the small angle of deflection. The first operation replaces $\mathbf{q}\mathbf{r}_{n0}$ by

$$\mathbf{q}\mathbf{r}_{n0} \rightarrow q_{\parallel} x_{n0} = -\frac{\omega_{n0}}{v} x_{n0},$$

where x_{n0} is the matrix element of one of the Cartesian coordinates of the atomic electrons¹². The integration over ϑ can be replaced by integration over q^2 , noting that

$$-q^2 = -\omega_{n0}^2 + \mathbf{q}^2 \approx -\omega_{n0}^2 + \frac{\omega_{n0}^2}{v^2} + \mathbf{p}^2 \vartheta^2 = \frac{\omega_{n0}^2 M^2}{\mathbf{p}^2} + \mathbf{p}^2 \vartheta^2 \quad (82.11)$$

and therefore $2\vartheta d\vartheta = d|q^2|/\mathbf{p}^2$ (M is the mass of the fast particle). As a result we obtain

$$\kappa = 4\pi(z e)^2 \sum_n \int \left\{ |(e^{-i\mathbf{q}\mathbf{r}})_{n0}|^2 \frac{\omega_{n0}}{v^2} - \omega_{n0}^3 |x_{n0}|^2 \left| \frac{M^2}{\mathbf{p}^2} + \frac{1}{v^2} \right| \right\} \frac{d|q^2|}{|q^2|^2}. \quad (82.12)$$

The lower limit of integration over q^2 :

$$|q^2|_{\min} = \frac{M^2}{\mathbf{p}^2} \omega_{n0}^2. \quad (82.13)$$

As the upper limit we shall choose a certain value $|q^2|_1$ such that

$$I \ll \frac{|q^2|_1}{m} \ll m, \quad (82.14)$$

i.e. lying in the region of overlap of regions I and II (see (82.1)).

The integration and summation in (82.12) is accomplished similarly to what was done in III, § 149 for the non-relativistic case. Let us divide the whole interval of integration into two parts still: a) from $|q^2|_{\min}$ to $|q^2|_0$ and b) from $|q^2|_0$ to $|q^2|_1$, where the value $|q^2|_0$ is such that

$$\frac{IM}{|\mathbf{p}|} \ll \sqrt{|q^2|_0} \ll m\alpha \quad (82.15)$$

¹¹These losses are frequently called ionisation losses, although they are connected not only with ionisation but also with excitation of atoms.

¹²It is immaterial which one: after the summation over the direction of the momentum, the matrix element x_{n0} arising below no longer depends on the direction of the axis x .

(the quantity $m\alpha$ is of the order of the magnitudes of the momenta of the atomic electrons). In region a) one can expand $e^{-i\mathbf{q}\mathbf{r}} \approx 1 - i\mathbf{q}\mathbf{r}$, and the contribution of this region to κ takes the form

$$4\pi(ze)^2 \sum_n \int_{|q^2|_{\min}}^{|q^2|_0} \left\{ \frac{1}{v^2} \omega_{n0} |x_{n0}|^2 \frac{1}{|q^2|} - \frac{M^2}{\mathbf{p}^2} \omega_{n0}^3 |x_{n0}|^2 \frac{1}{|q^2|^2} \right\} d|q^2| \approx$$

$$\approx \frac{4\pi(ze)^2}{v^2} \sum_n \omega_{n0} |x_{n0}|^2 \left[\ln \frac{|q^2|_0 \mathbf{p}^2}{M^2 \omega_{n0}^2} - v^2 \right].$$

(The integration in the second member can be extended to infinity.)

The summation is accomplished with the help of the formula

$$\sum_n \omega_{n0} |x_{n0}|^2 = \frac{Z}{2m}, \quad (82.16)$$

where Z is the number of electrons in the atom (see III, (149.10)). The result can be represented in the form

$$\frac{2\pi(ze)^2 Z}{mv^2} \left[\ln \frac{|q^2|_0 \mathbf{p}^2}{M^2 I^2} - v^2 \right], \quad (82.17)$$

where I is a certain average atomic energy, defined by the formula

$$\ln I = \frac{\sum_n \omega_{n0} |x_{n0}|^2 \ln \omega_{n0}}{\sum_n \omega_{n0} |x_{n0}|^2} = \frac{2m}{Z} \sum_n \omega_{n0} |x_{n0}|^2 \ln \omega_{n0}. \quad (82.18)$$

In region b) we have, according to (82.11), $|q^2| \approx \mathbf{p}^2 \vartheta^2$, i.e. $|q^2|$ does not depend on the number n of the final state of the atom; n also does not affect the limits of integration. Therefore the summation over n in (82.12) can be carried out under the integral sign. In the first member it is accomplished by the formula

$$\sum_n |(e^{-i\mathbf{q}\mathbf{r}})_{n0}|^2 \omega_{n0} = \frac{Z}{2m} |q^2| \quad (82.19)$$

(see III, (149.5)), and the integral from it is equal to ¹³

$$\frac{2\pi Z(ze)^2}{mv^2} \ln \frac{|q^2|_1}{|q^2|_0}.$$

The integral from the second member in (82.12) in this region gives a negligible contribution to κ .

Combining the last formula with (82.17), we find the contribution to κ from the whole region of small momentum transfers:

$$\frac{2\pi Z(ze)^2}{mv^2} \left[\ln \frac{|q^2|_1 \mathbf{p}^2}{M^2 I^2} - v^2 \right]. \quad (82.20)$$

¹³The logarithmic divergence of the integral at the upper limit is precisely the reason why in the first member in (82.12) it was not possible to expand $e^{-i\mathbf{q}\mathbf{r}}$ in powers of $\mathbf{q}$.

Large momentum transfers. Let us turn to collisions with momentum transfer large in comparison with the momentum of the atomic electrons ($\mathbf{q}^2 \gg mI$). In this region one can, obviously, neglect the binding of the electrons in the atom, i.e. consider their scattering free. Correspondingly, the collision of the fast particle with the atom will represent in this case the elastic scattering on each of the Z atomic electrons individually. Due to the large velocity of the fast particle the atomic electrons can be considered initially at rest.

Let us denote by $m\Delta$ the energy transferred by the fast particle to the atomic electron, and let $d\sigma_\Delta$ be the cross section of elastic scattering with such transfer. The differential effective retardation on each atom will then be

$$d\kappa = Zm\Delta d\sigma_\Delta. \quad (82.21)$$

The maximum energy that can be transferred to the resting electron by a collision with a particle of mass $M \gg m$ is

$$m\Delta_{\max} = \frac{2m\mathbf{p}^2}{m^2 + M^2 + 2m\varepsilon} \approx \frac{2m\mathbf{p}^2}{M^2 + 2m\varepsilon}, \quad (82.22)$$

where ε and $\mathbf{p}$ are the energy and momentum of the incident particle (see II, (13.13)). We shall assume further that the energy ε although it can be ultrarelativistic ($\varepsilon \gg M$), is at the same time

$$\varepsilon \ll \frac{M^2}{m}. \quad (82.23)$$

Then the maximal transferred energy

$$m\Delta_{\max} \approx \frac{2m\mathbf{p}^2}{M^2} = 2mv^2\gamma^2, \quad \gamma = \frac{1}{\sqrt{1-v^2}} \quad (82.24)$$

remains small in comparison with the original kinetic energy of the incident particle ($m\Delta_{\max} \ll \varepsilon - M$). Correspondingly the momentum transfer $\mathbf{q}$ also remains always small in comparison with the initial momentum of the particle $\mathbf{p}$. This circumstance permits one to consider the motion of the latter not changing during the collision, i.e. to consider the incident particle as infinitely heavy. Then the scattering cross section is obtained simply by transformation of the cross section of scattering of the electron on a stationary Coulomb centre (80.7) to the laboratory system of reference in which the electron

was initially at rest. This is easy to do, noting that in the indicated approximation

$$-q^2 \approx \mathbf{q}^2 = 4\mathbf{p}^2 \sin^2 \frac{\theta}{2}, \quad d\sigma' = \frac{\pi d|q^2|}{\mathbf{p}^2},$$

and the relative velocity v is the same in both systems. Formula (80.7) takes the form

$$d\sigma = \frac{4\pi(ze)^2}{v^2} \left(1 - \frac{|q^2|}{4m^2\gamma^2} \right) \frac{d|q^2|}{|q^2|^2}.$$

The energy transfer Δ is expressed through the same invariant q^2 : according

to $-q^2 = 2m^2\Delta$. Therefore we have¹⁴

$$d\sigma_\Delta = \frac{2\pi(ze)^2}{mv^2} \left(1 - v^2 \frac{\Delta}{\Delta_{\max}} \right) \frac{d\Delta}{\Delta^2}. \quad (82.25)$$

The contribution to the effective retardation from the region of momentum transfers under consideration is obtained by integrating (82.21) from the previously introduced boundary $|q^2|_1$ to $|q^2|_{\max} = 2m^2\Delta_{\max}$. It equals

$$\frac{2\pi(ze)^2 Z}{mv^2} \left(\ln \frac{2\Delta_{\max} m^2}{|q^2|_1} - v^2 \right). \quad (82.26)$$

Finally, adding the contributions (82.20) and (82.25), we obtain the following result for the total ionisation losses of a fast heavy particle:

$$\kappa = \frac{4\pi Z(ze)^2}{mv^2} \left(\ln \frac{2mv^2}{I(1 - v^2/c^2)} - \frac{v^2}{c^2} \right) \quad (82.27)$$

(in ordinary units). In the non-relativistic limit from this one recovers the previous formula (150.10) (see III):

$$\kappa = \frac{4\pi Z(ze)^2}{mv^2} \ln \frac{2mv}{I}, \quad (82.28)$$

and in the ultrarelativistic case

$$\kappa = \frac{4\pi Z(ze)^2}{mc^2} \left(\ln \frac{2mc^2}{I(1 - v^2/c^2)} - 1 \right). \quad (82.29)$$

The retardation depends only on the velocity (but not on the mass) of the fast particle. The decrease of the retardation with increasing velocity of the fast particle, according to (82.27), is replaced in the ultrarelativistic region by a slow (logarithmic) growth.

Problems

1. Determine the effective retardation of a relativistic electron.

Solution. For the region of small momentum transfers the result is the same as the expression (82.20). For the region of large transfers, instead of (82.24), one should use formula (81.14), taking into account the exchange effects. Integrating $\Delta d\sigma_\Delta$ from $d\Delta$ over $|q^2|_1/2m^2$ to $(\gamma - 1)/2$ and combining with (82.20), we find

$$\kappa = \frac{2\pi Ze^4}{mv^2} \left[\ln \frac{m^2(\gamma^2 - 1)(\gamma - 1)}{2I^2} - \left(\frac{2}{\gamma} - \frac{1}{\gamma^2} \right) \ln 2 + \frac{1}{\gamma^2} + \frac{(\gamma - 1)^2}{8\gamma^2} \right]. \quad (1)$$

In the non-relativistic case from this one recovers the formula from Problem to § 149 in III, and in the ultrarelativistic ($\gamma \gg 1$)

$$\kappa = \frac{2\pi Ze^4}{mc^2} \left(\ln \frac{m^2 c^4 \gamma^3}{2I^2} + \frac{1}{8} \right). \quad (2)$$

¹⁴In this formula, of course, specific effects of strong interactions are not taken into account, if the heavy particle is a hadron. These effects (the hadronic form factor), however, become important only for $|q^2| \propto 1/M^2$, and for the condition (82.22) such momentum transfers are excluded.

2. The same for the positron.

Solution. For $d\sigma_\Delta$ in the region of large transfers one should use (81.23), with the upper limit of integration over Δ being $\gamma - 1$. The answer in the ultrarelativistic case:

$$\kappa = \frac{2\pi Ze^4}{mc^2} \left(\ln \frac{2m^2 c^4 \gamma^3}{I^2} - \frac{23}{12} \right).$$

§ 83. Breit equation

As is well known, in classical electrodynamics a system of interacting particles can be described with the help of a Lagrangian function dependent only on the coordinates and velocities of the particles themselves and correct to terms $\sim 1/c^2$ (see II, § 65). This circumstance is connected with the fact that radiation appears only as an effect of order $1/c^2$.

In quantum theory this situation corresponds to the possibility of description of the system by a Schrödinger equation, with perturbation terms of second order. For an electron moving in an external electromagnetic field, such an equation was established in § 33. Now we turn to deriving an analogous equation describing a system of two interacting particles.

We shall proceed from the relativistic expression for the amplitude of scattering of two particles. In the non-relativistic approximation it goes over to the ordinary Born scattering amplitude, proportional to the Fourier component of the potential of electrostatic interaction of two charges. Computing the amplitude to accuracy of terms of second order in $1/c^2$, we shall be able to establish the form of the potential corresponding to it.

Let us suppose first that the two particles are different, with masses m_1 and m_2 (say, electron and muon). Then the scattering

is depicted by a single diagram

To it there corresponds the amplitude

$$M_{fi} = e^2 (\bar{u}'_1 \gamma^\mu u_1) D_{\mu\nu}(q) (\bar{u}'_2 \gamma^\nu u_2), \quad q = p'_1 - p_1 = p_2 - p'_2 \quad (83.1)$$

(here it is assumed that the charges of the particles are of the same sign; in the opposite case e^2 is replaced by $-e^2$).

Further calculations are substantially simplified if we choose the photon propagator $D_{\mu\nu}$ not in the ordinary but in the Coulomb gauge (76.12), (76.13)¹⁵:

$$D_{00} = -\frac{4\pi}{\mathbf{q}^2}, \quad D_{0i} = 0, \quad D_{ik} = \frac{4\pi}{\mathbf{q}^2 - \omega^2/c^2} \left(\delta_{ik} - \frac{q_i q_k}{\mathbf{q}^2} \right). \quad (83.2)$$

Then the amplitude of scattering

$$M_{fi} = e^2 \{ (\bar{u}'_1 \gamma^0 u_1) (\bar{u}'_2 \gamma^0 u_2) D_{00} + (\bar{u}'_1 \gamma^k u_1) (\bar{u}'_2 \gamma^k u_2) D_{ik} \}. \quad (83.3)$$

¹⁵In this paragraph we write out the factors c in all intermediate formulas, and in the final formulas also $\hbar$.

To the neglect of all terms containing the factor $1/c$, the second member in the curly brackets drops out entirely, and the first gives

$$M_{fi} = -2m_1 \cdot 2m_2 \begin{pmatrix} w_1^{(0)'} \\ w_1^{(0)} \end{pmatrix} \begin{pmatrix} w_2^{(0)'} \\ w_2^{(0)} \end{pmatrix} U(\mathbf{q}), \quad (83.4)$$

where

$$U(\mathbf{q}) = \frac{4\pi e^2}{\mathbf{q}^2}, \quad (83.5)$$

and by $w_1^{(0)}$, $w_2^{(0)}$, ... are denoted the 3-spinors of the non-relativistic plane waves introduced in §23 (two-component amplitudes). The function $U(\mathbf{q})$ represents the Fourier component of the potential energy of the Coulomb interaction: $U(r) = e^2/r$.

In the next (by $1/c$) approximation the “Schrödinger” wave function of the free particle φ_{Schr} (normalised by the integral $\int |\varphi_{\text{Schr}}|^2 d^3x$) satisfies the equation

$$\hat{H}^{(0)} \varphi_{\text{Schr}} = (\varepsilon - mc^2) \varphi_{\text{Schr}}, \quad \hat{H}^{(0)} = \frac{\hat{\mathbf{p}}^2}{2m} - \frac{\hat{\mathbf{p}}^4}{8m^3 c^2}, \quad \hat{\mathbf{p}} = -i\hbar \nabla, \quad (83.6)$$

in which the next-order term of the relativistic expansion for the kinetic energy has been taken into account. The amplitude of the (spinor) part of such a plane wave we denote by w (when $1/c \rightarrow 0$ it goes over to $w^{(0)}$). It is precisely through these amplitudes that the desired scattering amplitude should be expressed so that from its form one could determine the “Schrödinger” interaction potential in the approximation being considered.

In accordance with formula (33.11) the bispinor amplitude of the free particle u is expressed through the “Schrödinger” amplitude w —with the accuracy required here—in the form

$$u = \sqrt{2m} \begin{pmatrix} \left(1 - \frac{\mathbf{p}^2}{8m^2 c^2}\right) w \\ \frac{\boldsymbol{\sigma} \mathbf{p}}{2mc} w \end{pmatrix}. \quad (83.7)$$

With the help of this formula we find

$$\begin{aligned} \bar{u}'_1 \gamma^0 u_1 &= u'^*_1 \alpha_1 u_1 = 2m_1 \left(1 - \frac{\mathbf{p}_1'^2 + \mathbf{p}_1^2}{8m_1^2 c^2}\right) w'^*_1 w_1 + \\ &+ \frac{1}{2m_1 c^2} w'^*_1 (\boldsymbol{\sigma} \mathbf{p}'_1) (\boldsymbol{\sigma} \mathbf{p}_1) w_1 = 2m_1 w'^*_1 \left\{1 - \frac{\mathbf{q}^2}{8m_1^2 c^2} + \frac{i\boldsymbol{\sigma} [\mathbf{q} \mathbf{p}_1]}{4m_1^2 c^2}\right\} w_1, \\ \bar{u}'_1 \gamma u_1 &= u'^*_1 \alpha_1 u_1 = \frac{1}{c} w'^*_1 \{\boldsymbol{\sigma} (\boldsymbol{\sigma} \mathbf{p}_1) + (\boldsymbol{\sigma} \mathbf{p}'_1) \boldsymbol{\sigma}\} w_1 = \\ &= \frac{1}{c} w'^*_1 \{i[\boldsymbol{\sigma} \mathbf{q}] + 2\mathbf{p}_1 + \mathbf{q}\} w_1 \end{aligned}$$

(where $\mathbf{q} = \mathbf{p}'_1 - \mathbf{p}_1 = \mathbf{p}_2 - \mathbf{p}'_2$). Analogous expressions for $(\bar{u}'_2 \gamma^0 u_2)$ and $(\bar{u}'_2 \gamma u_2)$ differ by the replacement of indices 1 by 2 and correspondingly the replacement of q by $-q$.

Substituting these expressions in (83.3). Since the product $(\bar{u}'_1 \gamma u_1)(\bar{u}'_2 \gamma u_2)$ already contains the factor $1/c^2$, in it one can neglect the term ω^2/c^2 in the denominator of D_{ik} . As a result we obtain the scattering amplitude in the form

$$M_{fi} = -2m_1 \cdot 2m_2 (w_1'^* w_2'^* U(\mathbf{p}_1, \mathbf{p}_2, \mathbf{q}) w_1 w_2), \quad (83.8)$$

where

$$\begin{aligned} U(\mathbf{p}_1, \mathbf{p}_2, \mathbf{q}) = 4\pi e^2 \left\{ \frac{1}{\mathbf{q}^2} - \frac{1}{8m_1^2 c^2} - \frac{1}{8m_2^2 c^2} + \frac{(\mathbf{q}\mathbf{p}_1)(\mathbf{q}\mathbf{p}_2)}{m_1 m_2 c^2 \mathbf{q}^4} - \frac{\mathbf{p}_1 \mathbf{p}_2}{m_1 m_2 c^2 \mathbf{q}^2} + \right. \\ \left. + \frac{i\sigma_1[\mathbf{q}\mathbf{p}_1]}{4m_1^2 c^2 \mathbf{q}^2} - \frac{i\sigma_1[\mathbf{q}\mathbf{p}_2]}{2m_1 m_2 c^2 \mathbf{q}^2} - \frac{i\sigma_2[\mathbf{q}\mathbf{p}_2]}{4m_2^2 c^2 \mathbf{q}^2} + \frac{i\sigma_2[\mathbf{q}\mathbf{p}_1]}{2m_1 m_2 c^2 \mathbf{q}^2} + \right. \\ \left. + \frac{(\sigma_1 \mathbf{q})(\sigma_2 \mathbf{q})}{4m_1 m_2 c^2 \mathbf{q}^2} - \frac{\sigma_1 \sigma_2}{4m_1 m_2 c^2} \right\} \end{aligned} \quad (83.9)$$

(indices 1, 2 of the Pauli matrices indicate on which spinor indices they act: σ_1 acts on w_1 , and σ_2 on w_2).

The function $U(\mathbf{p}_1, \mathbf{p}_2, \mathbf{q})$ is the operator of particle interaction in the momentum representation. It is related to the operator $\hat{U}(\hat{\mathbf{p}}_1, \hat{\mathbf{p}}_2, \mathbf{r})$ in the coordinate representation by the formula

$$\int e^{-i(\mathbf{p}'_1 \mathbf{r}_1 + \mathbf{p}'_2 \mathbf{r}_2)} \hat{U}(\hat{\mathbf{p}}_1, \hat{\mathbf{p}}_2, \mathbf{r}) e^{i(\mathbf{p}_1 \mathbf{r}_1 + \mathbf{p}_2 \mathbf{r}_2)} d^3 x_1 d^3 x_2 = \quad (IX.4)$$

$$= (2\pi)^3 \delta(\mathbf{p}_1 + \mathbf{p}_2 - \mathbf{p}'_1 - \mathbf{p}'_2) U(\mathbf{p}_1, \mathbf{p}_2, \mathbf{q}). \quad (83.10)$$

If the operator $\hat{U}$ represents simply a function $U(\mathbf{r})$ ($\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2$) that does not depend on $\mathbf{p}_1$ and $\mathbf{p}_2$, then formula (83.10) reduces to the ordinary definition of the Fourier components:

$$\int e^{-i\mathbf{q}\mathbf{r}} U(\mathbf{r}) d^3 x = U(\mathbf{q}).$$

From this it is clear that to find $\hat{U}(\hat{\mathbf{p}}_1, \hat{\mathbf{p}}_2, \mathbf{r})$ one must compute the integral

$$\int e^{i\mathbf{q}\mathbf{r}} U(\mathbf{p}_1, \mathbf{p}_2, \mathbf{q}) \frac{d^3 q}{(2\pi)^3}$$

and then replace $\mathbf{p}_1, \mathbf{p}_2$ by operators $\hat{\mathbf{p}}_1 = -i\nabla_1, \hat{\mathbf{p}}_2 = -i\nabla_2$, placing them to the right of all other factors.

The necessary integrals are computed by differentiating the formula

$$\int e^{i\mathbf{q}\mathbf{r}} \frac{4\pi}{\mathbf{q}^2} \frac{d^3 q}{(2\pi)^3} = \frac{1}{r}. \quad (83.11)$$

Thus, taking the gradient we find

$$\int e^{i\mathbf{q}\mathbf{r}} \cdot \frac{4\pi \mathbf{q}}{\mathbf{q}^2} \frac{d^3 q}{(2\pi)^3} = -i\nabla \frac{1}{r} = \frac{i\mathbf{r}}{r^3}. \quad (83.12)$$

Further ($\mathbf{a}$, $\mathbf{b}$ are constant vectors)

$$\int \frac{4\pi(\mathbf{a}\mathbf{q})(\mathbf{b}\mathbf{q})}{q^4} e^{i\mathbf{q}\mathbf{r}} \frac{d^3q}{(2\pi)^3} = \frac{i}{2} \left(\mathbf{a} \frac{\partial}{\partial \mathbf{r}} \right) \int e^{i\mathbf{q}\mathbf{r}} \left(\mathbf{b} \frac{\partial}{\partial \mathbf{q}} \right) \frac{1}{q^2} \frac{d^3q}{(2\pi)^3}; \quad (83.13)$$

the resulting integral after integration by parts reduces to (83.12) and gives

$$\int \frac{4\pi(\mathbf{a}\mathbf{q})(\mathbf{b}\mathbf{q})}{q^4} e^{i\mathbf{q}\mathbf{r}} \frac{d^3q}{(2\pi)^3} = \frac{1}{2} (\mathbf{a}\nabla) \frac{\mathbf{b}\mathbf{r}}{r} = \frac{1}{2r} \left[\mathbf{a}\mathbf{b} - \frac{(\mathbf{a}\mathbf{r})(\mathbf{b}\mathbf{r})}{r^2} \right].$$

Finally,

$$\int \frac{4\pi(\mathbf{a}\mathbf{q})(\mathbf{b}\mathbf{q})}{q^2} e^{i\mathbf{q}\mathbf{r}} \frac{d^3q}{(2\pi)^3} = -(\mathbf{a}\nabla)(\mathbf{b}\nabla) \frac{1}{r}.$$

In expanding the derivatives one must bear in mind that this expression contains the δ -function $\delta(\mathbf{r})$. To separate it we note that after averaging over the directions of $\mathbf{r}$:

$$-(\mathbf{a}\nabla)(\mathbf{b}\nabla) \frac{1}{r} = -\frac{1}{3} (\mathbf{a}\mathbf{b}) \Delta \frac{1}{r} = \frac{4\pi}{3} (\mathbf{a}\mathbf{b}) \delta(\mathbf{r}).$$

Expanding now the derivatives in the ordinary way, we find

$$\int \frac{4\pi(\mathbf{a}\mathbf{q})(\mathbf{b}\mathbf{q})}{q^2} e^{i\mathbf{q}\mathbf{r}} \frac{d^3q}{(2\pi)^3} = \frac{1}{r^3} \left\{ \mathbf{a}\mathbf{b} - 3 \frac{(\mathbf{a}\mathbf{r})(\mathbf{b}\mathbf{r})}{r^2} \right\} + \frac{4\pi}{3} (\mathbf{a}\mathbf{b}) \delta(\mathbf{r}) \quad (83.14)$$

(on averaging over the directions of $\mathbf{r}$ the first term on the right vanishes and there remains only the term with the δ -function).

With the help of these formulas we obtain the following final expression for the operator of particle interaction:

$$\begin{aligned} \hat{U}(\hat{\mathbf{p}}_1, \hat{\mathbf{p}}_2, \mathbf{r}) &= \frac{e^2}{r} - \frac{\pi e^2 \hbar^2}{2c^2} \left(\frac{1}{m_1^2} + \frac{1}{m_2^2} \right) \delta(\mathbf{r}) - \\ &- \frac{e^2}{2m_1 m_2 c^2 r} \left[\hat{\mathbf{p}}_1 \hat{\mathbf{p}}_2 + \frac{\mathbf{r}(\mathbf{r} \hat{\mathbf{p}}_1) \hat{\mathbf{p}}_2}{r^2} \right] - \frac{e^2 \hbar}{4m_1^2 c^2 r^3} [\mathbf{r} \hat{\mathbf{p}}_1] \sigma_1 + \frac{e^2 \hbar}{4m_2^2 c^2 r^3} [\mathbf{r} \hat{\mathbf{p}}_2] \sigma_2 - \\ &- \frac{e^2}{2m_1 m_2 c^2 r} \{ -(\sigma_1 + 2\sigma_2) [\mathbf{r} \hat{\mathbf{p}}_1] + (\sigma_2 + 2\sigma_1) [\mathbf{r} \hat{\mathbf{p}}_2] \} + \\ &+ \frac{e^2 \hbar^2}{4m_1 m_2 c^2} \left\{ \frac{\sigma_1 \sigma_2}{r^3} - 3 \frac{(\sigma_1 \mathbf{r})(\sigma_2 \mathbf{r})}{r^5} - \frac{8\pi}{3} \sigma_1 \sigma_2 \delta(\mathbf{r}) \right\}. \end{aligned} \quad (83.15)$$

The full Hamiltonian of the system of two particles in this approximation

$$\hat{H} = \hat{H}_1^{(0)} + \hat{H}_2^{(0)} + \hat{U}, \quad (83.16)$$

where $\hat{H}^{(0)}$ are the Hamiltonians of free particles (83.6).

Two electrons. If both particles are identical (two electrons), then in the scattering amplitude there appears a second term, depicted by the “exchange” diagram

Computing its contribution to the interaction operator, however, is not necessary. The point is that the description of a system of identical particles by the Schrödinger equation

can be accomplished with the help of the same interaction operator as for non-identical particles, if one takes care of the proper symmetrisation of the

wave functions. In particular, in the consideration of the scattering of identical particles such symmetrisation automatically takes into account the contributions to the amplitude corresponding to both Feynman diagrams.

Thus, the Hamiltonian of a system of two electrons is obtained from formulas (83.15), (83.16), if one simply puts $m_1 = m_2$ ¹⁶:

$$\hat{H} = \frac{1}{2m}(\hat{\mathbf{p}}_1^2 + \hat{\mathbf{p}}_2^2) - \frac{1}{8m^3c^2}(\hat{\mathbf{p}}_1^4 + \hat{\mathbf{p}}_2^4) + \hat{U}(\hat{\mathbf{p}}_1, \hat{\mathbf{p}}_2, \mathbf{r}), \quad (\text{IX.5})$$

$$\begin{aligned} \hat{U}(\hat{\mathbf{p}}_1, \hat{\mathbf{p}}_2, \mathbf{r}) = & \frac{e^2}{r} - \pi \left(\frac{e\hbar}{mc} \right)^2 \delta(\mathbf{r}) - \frac{e^2}{2m^2c^2r} \left(\hat{\mathbf{p}}_1 \hat{\mathbf{p}}_2 + \frac{\mathbf{r}(\mathbf{r}\hat{\mathbf{p}}_1)\hat{\mathbf{p}}_2}{r^2} \right) + \\ & + \frac{e^2\hbar}{4m^2c^2r^3} \{ -(\boldsymbol{\sigma}_1 + 2\boldsymbol{\sigma}_2)[\mathbf{r}\hat{\mathbf{p}}_1] + (\boldsymbol{\sigma}_2 + 2\boldsymbol{\sigma}_1)[\mathbf{r}\hat{\mathbf{p}}_2] \} + \\ & + \frac{1}{4} \left(\frac{e\hbar}{mc} \right)^2 \frac{1}{r^3} \left\{ \frac{\boldsymbol{\sigma}_1 \boldsymbol{\sigma}_2}{r^3} - 3 \frac{(\boldsymbol{\sigma}_1 \mathbf{r})(\boldsymbol{\sigma}_2 \mathbf{r})}{r^5} - \frac{8\pi}{3} \boldsymbol{\sigma}_1 \boldsymbol{\sigma}_2 \delta(\mathbf{r}) \right\}. \end{aligned} \quad (83.17)$$

Let us note that the presence of terms with $\delta(\mathbf{r})$ does not, of course, signify the existence of a special strong interaction. The integral magnitude of all correction terms is the same, and in the spirit of the expansion performed they must all be considered small in comparison with the first term—the Coulomb interaction.

The various groups of terms in the interaction operator (83.17) have different character. The terms of the first row in $\hat{U}$ have a purely orbital origin. In the second row stand the terms, linear in the spin operators of the particles; they correspond to the *spin-orbit interaction*. Finally, the terms of the third row, quadratic in the spin operators, describe the *spin-spin interaction*¹⁷.

Electron and positron. The system of electron and positron requires special consideration. The scattering amplitude in this case is composed of two terms:

$$\begin{aligned} M_{fi} = & -e^2 [\bar{u}(p'_-) \gamma^\mu u(p_-)] D_{\mu\nu}(p_- - p'_-) [\bar{u}(-p_+) \gamma^\nu u(-p'_+)] + \\ & + e^2 [\bar{u}(-p_+) \gamma^\mu u(p_-)] D_{\mu\nu}(p_- + p_+) [\bar{u}(p'_-) \gamma^\nu u(-p'_+)] \end{aligned} \quad (83.18)$$

(the first corresponds to the scattering diagram, and the second to the annihilation diagram). Since the wave function of the “electron +

positron” system should not be antisymmetric, both terms give independent contributions to the operator of interaction.

The first term (whose structure coincides with the structure of the scattering amplitude (83.1)) leads, naturally, to an operator differing from (83.17) only by the overall sign.

¹⁶The wave equation with Hamiltonian (83.17) was first established by *Breit* (*G. Breit*, 1929), and its consistent quantum-mechanical derivation was given by *L. D. Landau* (1932).

¹⁷This interaction was mentioned in III, § 72 in connection with the fine structure of atomic levels, and the spin-spin interaction of electrons with the nucleus was treated in III, § 121, in connection with the superhyperfine structure of levels. In particular, formula (121.9) (see III) corresponds to the δ -function term in the spin-spin interaction operator.

Let us turn to the transformation of the second term.

We use here the photon propagator in the ordinary gauge:

$$D_{\mu\nu} = \frac{4\pi}{k^2} g_{\mu\nu} = \frac{4\pi}{\omega^2/c^2 - \mathbf{k}^2} g_{\mu\nu}. \quad (83.19)$$

In the present case $k = p_+ + p_-$, and since the particles are “almost non-relativistic”, we have

$$\frac{\omega^2}{c^2} \equiv \frac{(\varepsilon_+ + \varepsilon_-)^2}{c^2} \approx 4m^2 c^2 \gg (\mathbf{p}_+ + \mathbf{p}_-)^2 \equiv \mathbf{k}^2.$$

Therefore for the photon propagator it suffices to write

$$D_{\mu\nu} \approx \frac{\pi}{m^2 c^2} g_{\mu\nu}.$$

This already contains the factor $1/c^2$. Therefore the amplitudes $u(p)$ can be taken in the zeroth approximation:

$$u(p_-) = \sqrt{2m} \begin{pmatrix} w_-^{(0)} \\ 0 \end{pmatrix}, \quad u(-p_+) = \sqrt{2m} \begin{pmatrix} 0 \\ w_+^{(0)} \end{pmatrix},$$

where $w_-^{(0)}$, $w_+^{(0)}$ are the 3-spinors figuring in (23.12) (below we omit the indices (0) on them). With these amplitudes

$$\bar{u}(-p_+) \gamma^0 u(p_-) = u^*(-p_+) u(p_-) = 0,$$

$$\bar{u}(-p_+) \gamma u(p_-) = u^*(-p_+) \alpha u(p_-) = 2m(w_-^* \sigma w_-),$$

where $w_-^{(0)}$, $w_+^{(0)}$ are the 3-spinors figuring in (23.12). With these amplitudes

$$M_{fi}^{(\text{ann})} = -e^2 \frac{\pi}{m^2 c^2} (2m)^2 (w_-^* \sigma w_-) (w_+^{*\prime} \sigma w_+'). \quad (83.20)$$

From this, however, one cannot yet draw direct conclusions about the form of the operator of interaction. First, the scattering amplitude should be brought to the form in which the electronic (w_- and w_-') and positronic (w_+ and w_+') spinors are grouped together. This is achieved with the help of the formula

$$(w_-^* \sigma w_-) (w_+^{*\prime} \sigma w_+') = \frac{3}{2} (w_-^{*\prime} w_-) (w_+^* w_+') - \frac{1}{2} (w_-^{*\prime} \sigma w_-) (w_+^* \sigma w_+'), \quad (83.22)$$

which itself follows from (28.16).

Finally, expressing w and w' through w_+ and w_+' , according to (83.21), we find, as is easy to verify,

$$(w_-^* w_+') = (w_+^{*\prime} w_+), \quad (w_-^* \sigma w_+') = -(w_+^{*\prime} \sigma w_+). \quad (83.23)$$

Substituting (83.23) in (83.22) and then in (83.20), we obtain the final expression for the annihilation part of the scattering amplitude

$$M_{fi}^{(\text{ann})} = -4m^2 \left\{ w_-^{*\prime} w_+^{*\prime} \left[\frac{\pi e^2}{2m^2 c^2} (3 + \sigma_+ \sigma_-) \right] w_- w_+ \right\} \quad (\text{IX.6})$$

(the matrices σ_- and σ_+ act respectively on w_- and w_+). The expression in the square brackets represents the interaction operator in the momentum representation. The corresponding coordinate operator

$$\hat{U}^{(\text{ann})}(\mathbf{r}) = \frac{\pi\hbar^2 e^2}{2m^2 c^2} (3 + \sigma_+ \sigma_-) \delta(\mathbf{r}), \quad \mathbf{r} = \mathbf{r}_- - \mathbf{r}_+ \quad (83.24)$$

(Pirenne, 1947; B. B. Berestetskii and L. D. Landau, 1949). The full interaction operator of the electron and positron is

$$-\hat{U} + \hat{U}^{(\text{ann})}$$

with $\hat{U}$ from (83.17).

§ 84. Positronium

The formulas obtained in the preceding paragraph can be applied to positronium—a hydrogen-like system of electron and positron.

In the centre-of-inertia system the operators of momenta of the electron and positron: $\hat{\mathbf{p}}_- = -\hat{\mathbf{p}}_+ \equiv \hat{\mathbf{p}}$, where $\hat{\mathbf{p}} = -i\hbar\nabla$ is the operator of the momentum of relative motion corresponding to the relative radius vector $\mathbf{r} = \mathbf{r}_- - \mathbf{r}_+$. The full Hamiltonian of positronium¹⁸

$$\begin{aligned} \hat{H} &= \frac{\hat{\mathbf{p}}^2}{m} - \frac{e^2}{r} + \hat{V}_1 + \hat{V}_2 + \hat{V}_3, \\ \hat{V}_1 &= \frac{\hat{\mathbf{p}}^4}{4m^3 c^2} + 4\pi\mu_0^2 \delta(\mathbf{r}) - \frac{e^2}{2m^2 c^2 r} \left\{ \hat{\mathbf{p}}^2 + \frac{\mathbf{r}(\mathbf{r}\hat{\mathbf{p}})\hat{\mathbf{p}}}{r^2} \right\}, \\ \hat{V}_2 &= 6\mu_0^2 \frac{1}{r^3} \hat{\mathbf{I}}\hat{\mathbf{S}}, \\ \hat{V}_3 &= 6\mu_0^2 \frac{1}{r^3} \left\{ \frac{(\hat{\mathbf{S}}\mathbf{r})(\hat{\mathbf{S}}\mathbf{r})}{r^2} - \frac{1}{3} \hat{\mathbf{S}}^2 \right\} + 4\pi\mu_0^2 \left(\frac{7}{3} \hat{\mathbf{S}}^2 - 2 \right) \delta(\mathbf{r}). \end{aligned} \quad (84.1)$$

Here $\mu_0 = e\hbar/(2mc)$ is the Bohr magneton, $\hat{\mathbf{I}} = [\mathbf{r}\hat{\mathbf{p}}]$ is the operator of orbital angular momentum, $\hat{\mathbf{S}} = (\sigma_+ + \sigma_-)/2$ is the operator of total spin of the system (its square $\hat{\mathbf{S}}^2 = \frac{1}{2}(3 + \sigma_+ \sigma_-)$). In $\hat{V}_1$ are included all correction terms of purely orbital character; $\hat{V}_2$ is the spin-orbit interaction; $\hat{V}_3$ includes the spin-spin and “annihilation” interactions.

The “unperturbed” Hamiltonian

$$\hat{H} = \frac{\hat{\mathbf{p}}^2}{m} - \frac{e^2}{r}$$

differs, naturally, from the Hamiltonian of the hydrogen atom only by the replacement of the electron mass by the reduced mass $m/2$. The energy levels of positronium are therefore half (in absolute value) of the levels of the hydrogen atom:

$$E = -\frac{me^4}{4\hbar^2 n^2} \quad (84.2)$$

¹⁸In ordinary units.

(n is the principal quantum number).

The remaining terms in (84.1) lead to a splitting of the levels (84.2)—the appearance of fine structure. The arising levels are classified first of all by the values of the total angular momentum j . We also see that the spin operators of the particles enter in the Hamiltonian (84.1) only in the form of the sum $\hat{\mathbf{S}}$. This means that the Hamiltonian commutes with the operator of the square of the total spin $\hat{\mathbf{S}}^2$, i.e. the total spin continues to be conserved also in the approximation under consideration (second in $1/c$). Therefore the energy levels of positronium can also be classified by the total spin, taking the values $S = 0$ and $S = 1$. The levels with spin 0 are called levels of *parapositronium*, and the levels with spin 1—levels of *orthopositronium*.

Let us emphasise that the conservation of total spin in positronium is in fact an exact law, not connected with this or that approximation by $1/c$; it follows from the

CP -invariance of electromagnetic interactions. Positronium represents a truly neutral system, and therefore its states are characterised by definite charge and combined parities. The latter is equal to $(-1)^{S+1}$ (see Problem to § 27); since S can take only the two values, 0 and 1, the conservation of combined parity is equivalent to conservation of total spin.

For $S = 0$ the total moment j coincides with the orbital. For spin $S = 1$ and given j the number l takes the values $j, j \pm 1$, correspondingly each level (n, j) of orthopositronium splits, generally speaking, into three levels. Since the two values $l = j$ and $l = j \pm 1$ correspond to different parities, the Hamiltonian has no matrix elements connecting these states. However the operator of perturbation (the first member in $\hat{V}_3$), generally speaking, has non-diagonal matrix elements connecting states with $l = j + 1$ and $l = j - 1$, so that when this number l loses, of course, its strict meaning of the orbital angular momentum.

Specific properties are possessed by the Zeeman effect in positronium (*B. B. Berestetskii, I. Ya. Pomeranchuk, 1949*). The orbital magnetic moment of positronium is always zero: since in positronium $[\mathbf{r}_+ \hat{\mathbf{p}}_+] = [\mathbf{r}_- \hat{\mathbf{p}}_-]$, the operator

$$\hat{\mu}_l = \mu_0([\mathbf{r}_+ \hat{\mathbf{p}}_+] - [\mathbf{r}_- \hat{\mathbf{p}}_-]) = 0.$$

The operator of the spin magnetic moment

$$\hat{\mu}_s = \mu_0(\boldsymbol{\sigma}_+ - \boldsymbol{\sigma}_-) \quad (84.3)$$

is not proportional to the operator of total spin $\hat{\mathbf{S}} = \frac{1}{2}(\boldsymbol{\sigma}_+ + \boldsymbol{\sigma}_-)$, and the operators $\hat{\mathbf{S}}^2$ and $\hat{\boldsymbol{\mu}}^2$ are not commutative. Therefore the states with definite values of the total spin S and its projection S_z are not eigenstates of the magnetic moment.

The states with given S and S_z are described by the spin functions χ_{SS_z} of the form

$$\begin{aligned} \chi_{11} &= \alpha_+ \alpha_-, & \chi_{1,-1} &= \beta_+ \beta_-, \\ \chi_{10} &= \frac{1}{\sqrt{2}}(\alpha_+ \beta_- + \alpha_- \beta_+), \\ \chi_{00} &= \frac{1}{\sqrt{2}}(\alpha_+ \beta_- - \alpha_- \beta_+), \end{aligned} \quad (84.4)$$

where α and β are the spin functions of one particle, corresponding to the spin projections $+1/2$ and $-1/2$ (the indices “+” or “-” indicate whether the function refers to the positron

or the electron). Of these the first two (χ_{11} and $\chi_{1,-1}$) are also eigenfunctions of the operator μ_z , with eigenvalue $\mu_z = 0$. The func-

tions χ_{10} and χ_{00} , however, are not eigenfunctions of μ_z ; such are the combinations

$$\frac{1}{\sqrt{2}}(\chi_{10} + \chi_{00}) = \alpha_+\beta_-, \quad \frac{1}{\sqrt{2}}(\chi_{10} - \chi_{00}) = \alpha_-\beta_+.$$

It is easy to see that the only non-zero matrix elements of the matrix $\langle S'S'_z|\mu_z|SS_z\rangle$, computed with the functions (84.4), are the elements

$$\langle 00|\mu_z|10\rangle = \langle 10|\mu_z|00\rangle = 2\mu_0. \quad (84.6)$$

In weak magnetic fields (when $\mu_0 H \ll \Delta$, where Δ is the difference between the energies of levels with $S = 0$ and $S = 1$) the initial approximation for computing the Zeeman splitting are the states with definite values of the total spin. In the first approximation this splitting is given by the average value of the energy of perturbation

$$\hat{V}_H = -\hat{\mu}_z H.$$

But all diagonal matrix elements of the operator $\hat{\mu}_z$, computed with the functions (84.4), are zero. Thus, in weak fields the linear Zeeman effect in positronium is absent.

In the limiting case of strong fields ($\mu_0 H \gg \Delta$) one can neglect the spin interaction, leading to the establishment of definite values of S . The components of the split level will in this case correspond to states with definite values of $\mu_z = \pm 2\mu_0$ (described by the functions (84.5)), and the magnitude of their shift is $\pm 2\mu_0 H$.

Problems

1. Determine the fine structure of the levels of parapositronium (*B. B. Berestetskii*, 1949)¹⁹.

Solution. The desired energy of level splitting is given by the average values of the correction terms in the Hamiltonian (84.1), computed with the wave functions of the unperturbed states with $j = l$ (for $S = 0$ the contribution that differs from zero arises only from $\hat{V}_1$ and the second member in $\hat{V}_3$).

The unperturbed wave functions (we denote them by ψ) satisfy the Schrödinger equation

$$\hat{\mathbf{p}}^2 \psi = -\Delta \psi = \left(E + \frac{1}{r}\right) \psi, \quad E = -\frac{1}{4n^2}.$$

Therefore

$$\begin{aligned} \hat{\mathbf{p}}^4 \psi &= \hat{\mathbf{p}}^2 \left(E + \frac{1}{r}\right) \psi = \left(E + \frac{1}{r}\right)^2 \psi - \psi \Delta \frac{1}{r} - 2 \left(\nabla \frac{1}{r}\right) (\nabla \psi) = \\ &= \left(E + \frac{1}{r}\right)^2 \psi + 4\pi \delta(\mathbf{r}) \psi + \frac{2}{r^2} \frac{\partial \psi}{\partial r} \end{aligned}$$

¹⁹The computation of the fine structure of orthopositronium see *Sokolov A. A., Tsitovich V. N.* JETP. 1953. V. 24. P. 253.

The average value

$$\overline{\mathbf{p}^4} = \overline{\left(E + \frac{1}{r}\right)^2} + 4\pi|\psi(0)|^2 + \int_0^\infty \int \frac{\partial|\psi|^2}{\partial r} dr d\Omega.$$

The last integral is non-zero only when $\psi(0) \neq 0$, i.e. only for $l = 0$, and the wave functions of S -states are spherically symmetric; the integral equals $-4\pi|\psi(0)|^2$ and cancels with the second term.

Introducing the operator of orbital angular momentum $\hat{\mathbf{l}} = [\mathbf{r}\hat{\mathbf{p}}]$ and writing

$$-\hat{\mathbf{p}}^2 = \frac{\partial^2}{\partial r^2} + \frac{2}{r} \frac{\partial}{\partial r} - \frac{\hat{\mathbf{l}}^2}{r^2}, \quad -\hat{\mathbf{p}}^2\psi = -\left(E + \frac{1}{r}\right)\psi.$$

From this, for the other mean value we need:

$$\begin{aligned} \int \psi^* \frac{r}{r^3} (\mathbf{r}\hat{\mathbf{p}})\hat{\mathbf{p}}\psi d^3x &= - \int \psi^* \frac{1}{r} \frac{\partial^2\psi}{\partial r^2} d^3x = \\ &= \frac{1}{r} \left(E + \frac{1}{r}\right) - 4\pi|\psi(0)|^2 - l(l+1)\overline{r^{-3}}. \end{aligned}$$

(for $l = 0$ the last member is absent).

According to the well-known formulas of the theory of the hydrogen atom (see III, (36.14), (36.16)) with the replacement of the electron mass m by $m/2$ we have

$$|\psi(0)|^2 = \frac{1}{8\pi n^3} \delta_{l0}, \quad \overline{r^{-2}} = \frac{1}{2n^3(2l+1)}, \quad \overline{r^{-3}} = \frac{1}{4n^3 l(l+1)(2l+1)} \quad (l \neq 0).$$

With the help of these formulas we finally obtain for the desired energy levels of parapositronium

$$E_{nl} = -\frac{1}{4n^2} - \alpha^2 \frac{me^4}{\hbar^2} \frac{1}{2n^3} \left(\frac{1}{2l+1} - \frac{11}{32n} \right).$$

2. Determine the difference of energies of the ground states ($n = 1, l = 0$) of ortho- and parapositronium.

Solution. The dependence of energy on the total spin S at $l = 0$ is contained only in the average value of the second member in $\hat{V}_3$ (the first member vanishes on averaging in a spherically-symmetric S -state)²⁰. The ground level of orthopositronium (3S_1) lies above the ground level of parapositronium (1S_0) by the amount

$$E(^3S_1) - E(^1S_0) = \frac{7}{12} \alpha^2 \frac{me^4}{\hbar^2} = 8.2 \cdot 10^{-4} \text{ eV}.$$

²⁰The averaging over angles should be carried out before integration over r , as is seen from the manner of computation of the integral (83.14), leading to the first member in $\hat{V}_3$.

§ 85. Interaction of atoms at large distances

Between two neutral atoms, situated at large (in comparison with their own dimensions) distances r , there act forces of attraction. The ordinary quantum-mechanical calculation of these forces (see III, § 89) becomes, however, inapplicable at too large distances. The point is that in this calculation only the electrostatic interaction is considered, i.e. retardation effects are not taken into account. This consideration is valid only so long as the distance r remains small in comparison with the characteristic wavelengths λ_0 in the spectra of the interacting atoms. In this paragraph we shall carry out a calculation free from such a restriction.

We shall proceed in approximately the same way as in § 83, i.e. we shall compute in the first non-vanishing approximation the amplitude of elastic scattering (without change of internal state) of two different atoms. The obtained expression will be compared with the amplitude that would arise in describing the interaction of atoms by a potential energy $U(r)$.

In the latter case the first non-vanishing element of the S -matrix describing the given process would be the element of the first approximation of perturbation theory:

$$S_{fi} = -i \int \psi_1'^* \psi_2'^* U(r) \psi_1 \psi_2 d^3 x_1 d^3 x_2 \times \quad (\text{IX.7})$$

$$\times \int \exp\{-i(\varepsilon_1 + \varepsilon_2 - \varepsilon_1' - \varepsilon_2')t\} dt. \quad (85.1)$$

Here ψ_1, ψ_2 and ψ_1', ψ_2' are the time-independent parts of the wave functions (plane waves) describing the translational motion of the atoms with initial and final momenta; $\varepsilon_1, \varepsilon_2$ and $\varepsilon_1', \varepsilon_2'$ are the kinetic energies of this motion; the coordinates $\mathbf{r}_1$ and $\mathbf{r}_2$ of the atoms as a whole may be understood as the coordinates of their nuclei, and the distance $r = |\mathbf{r}_1 - \mathbf{r}_2|$. The temporal integral gives, as usual, the δ -function expressing the law of conservation of energy.

The actual computation of the elastic scattering amplitude can be split, under the assumptions made, into two stages. First we average the S -operator over the wave functions of the unchanged (ground) states of both atoms (for given coordinates of their nuclei $\mathbf{r}_1$ and $\mathbf{r}_2$), and also over the photon vacuum—in the initial and final states of the process there are no photons. As a result we obtain a quantity which is a function of the distance between the nuclei; let us denote it by $\langle S(r) \rangle$ ²¹. To find the desired matrix element of transition, one must then compute the integral

$$S_{fi} = \int \psi_1'^* \psi_2'^* \langle S(r) \rangle \psi_1 \psi_2 d^3 x_1 d^3 x_2. \quad (85.3)$$

Comparing with (85.2), we see that if one obtains for $\langle S(r) \rangle$ the expression

$$\langle S(r) \rangle = -itU(r),$$

then $U(r)$ will be the desired energy of interaction of the atoms.

²¹Instead of a more cumbersome notation for the diagonal matrix element with indication of the states of the atom and photon field.

Since we have to deal in this case with collisions not of elementary particles but of complex systems (atoms, which in the intermediate states can be excited), the usual formal rules of diagram technique are here directly inapplicable, and we start from the initial expression for the S -operator in the form of the expansion (72.10).

For the interaction of atoms only the components of the fields with frequencies of the order of atomic (and smaller) are important. The corresponding wavelengths are large in comparison with atomic dimensions. Therefore the operator of electromagnetic interaction may be taken in the form

$$\hat{V} = -\hat{\mathbf{E}}(\mathbf{r}_1)\hat{\mathbf{d}}_1 - \hat{\mathbf{E}}(\mathbf{r}_2)\hat{\mathbf{d}}_2, \quad (85.4)$$

where $\hat{\mathbf{d}}_1$, $\hat{\mathbf{d}}_2$ are the operators of the dipole moments of the atoms (the Heisenberg operators, depending on time are meant), and $\hat{\mathbf{E}}(\mathbf{r})$ is the operator of the electric field, which is evaluated at the points of location of the corresponding atoms.

As is well known, the average values of the dipole moment of the atom in its stationary states are zero (see III, § 75). From this it follows that the non-zero scattering amplitude will appear only in the fourth approximation of perturbation theory, i.e. as the matrix element of the operator

$$\hat{S}^{(4)} = \frac{(-i)^4}{4!} \int dt_1 \dots \int dt_4 \cdot T\{\hat{V}(t_1)\hat{V}(t_2)\hat{V}(t_3)\hat{V}(t_4)\}. \quad (85.5)$$

Indeed, in lower orders each member in the products of the operators $\hat{V}$ will contain at most one of the operators $\hat{\mathbf{d}}_1$ or $\hat{\mathbf{d}}_2$ to the first power and upon averaging over the state of the corresponding atom will vanish.

We average the operator (85.5) over the photon vacuum. By Wick's theorem the average of the product of four operators of the field $\hat{\mathbf{E}}$ reduces to the sum of products of their pairwise averages (contractions). The splitting into pairs can be accomplished in three ways, which can be depicted by diagrams:

$$(85.6)$$

where the dashed lines depict the contractions, and the numbers correspond to the arguments t_1, t_2, t_3, t_4 . Furthermore, each point may correspond to the spatial coordinates $\mathbf{r}_1$ or $\mathbf{r}_2$ (since at each point one of the summands in the operator $\hat{V}$ of the operators $\hat{\mathbf{d}}_1$ or $\hat{\mathbf{d}}_2$ enters to the first power and upon averaging over the state of each atom will vanish). Obviously, in the points connected by lines there should stand different $\mathbf{r}_1$ and $\mathbf{r}_2$. In the opposite case the diagram (i.e. the corresponding member in the matrix element) would reduce to a product of independent functions of $\mathbf{r}_1$ and of $\mathbf{r}_2$, instead of being a function of the difference $\mathbf{r}_1 - \mathbf{r}_2$; such terms have no relation to scattering²². In accordance with these conditions one can assign to the four points of the diagrams the arguments $\mathbf{r}_1$ and $\mathbf{r}_2$ in four ways. Taking also the commutativity of the operators $\hat{\mathbf{d}}_1$ and $\hat{\mathbf{d}}_2$, and averaging over the state of each of the atoms, into account, we find that all the resulting $3 \cdot 4 = 12$ members are identical (they differ only in the labelling of the variables of integration).

²²They give corrections not of interest to us here to the self-energies of each of the atoms.

As a result we obtain

$$\begin{aligned} \langle S(r) \rangle = & \frac{1}{2} \int dt_1 \dots \int dt_4 \cdot \langle T(E_i(\mathbf{r}_1, t_1) E_k(\mathbf{r}_2, t_2)) \rangle \times \\ & \times \langle T(E_l(\mathbf{r}_2, t_3) E_m(\mathbf{r}_1, t_4)) \rangle \times \\ & \times \langle T(d_{1i}(t_1) d_{1m}(t_4)) \rangle \langle T(d_{2k}(t_2) d_{2l}(t_3)) \rangle, \end{aligned} \quad (85.7)$$

where $i, k, \dots$ are three-dimensional vector indices.

For computing the quantities

$$D_{ik}^E(x_1 - x_2) = \langle T(E_i(x_1) E_k(x_2)) \rangle \quad (85.8)$$

we use the gauge of the potentials in which the scalar potential $\Phi = 0$. Then $\hat{\mathbf{E}} = -\partial \hat{\mathbf{A}} / \partial t$, and we have

$$D_{ik}^E(x_1 - x_2) = \frac{\partial^2}{\partial t_1 \partial t_2} \langle T(A_i(x_1) A_k(x_2)) \rangle = i \frac{\partial^2}{\partial t^2} D_{ik}(x),$$

where $x = x_1 - x_2$, and $D_{ik}(x)$ is the photon propagator in the given gauge²³.

It will be more convenient below to use the propagator $D_{ik}(\omega, \mathbf{r})$ in the mixed ω , $\mathbf{r}$ -representation, which is related to $D_{ik}(t, \mathbf{r})$ by the relation

$$D_{ik}(t, \mathbf{r}) = \int D_{ik}(\omega, \mathbf{r}) e^{-i\omega t} \frac{d\omega}{2\pi}.$$

With this

$$D_{ik}^E(t, \mathbf{r}) = -i \int \omega^2 D_{ik}(\omega, \mathbf{r}) e^{-i\omega t} \frac{d\omega}{2\pi}. \quad (85.9)$$

The quantities

$$\alpha_{ik}(t_1 - t_2) = i \langle T(d_i(t_1) d_k(t_2)) \rangle \quad (85.10)$$

we expand into the Fourier integral

$$\alpha_{ik}(t) = \int e^{-i\omega t} \alpha_{ik}(\omega) \frac{d\omega}{2\pi}.$$

Setting for convenience $t_2 = 0$, $t_1 = t$, we write down the T-product by definition

$$\alpha_{ik}(\omega) = \int_{-\infty}^{\infty} e^{i\omega t} \alpha_{ik}(t) dt = \quad (IX.8)$$

$$= i \int_{-\infty}^0 e^{i\omega t} \langle d_k(0) d_i(t) \rangle dt + i \int_0^{\infty} e^{i\omega t} \langle d_i(t) d_k(0) \rangle dt. \quad (85.11)$$

The averages entering here (over the ground state of the atom) are expressed through the matrix elements of the dipole moment:

$$\langle d_k(0) d_i(t) \rangle = \sum_n (d_k)_{0n} (d_i)_{n0} e^{i\omega_{n0} t},$$

²³The first derivative $\frac{\partial}{\partial t} D_{ik}(t)$ has a finite jump at $t = 0$. Therefore the second derivative, i.e. the function $D_{ik}^E(t)$, contains also a δ -function term ($\sim \delta^{(4)}(x_2 - x_1)$). This term, however, is zero for all $\mathbf{r}_1 \neq \mathbf{r}_2$ and does not concern us here.

$$\langle d_i(t) d_k(o) \rangle = \sum_n (d_i)_{0n} (d_k)_{n0} e^{-i\omega_{n0}t}.$$

For the convergence of the integrals in (85.11) one must understand ω in the first of them as $\omega - i0$, and in the second as $\omega + i0$. Carrying out the integration, we obtain

$$\alpha_{ik}(\omega) = \sum_n \left\{ \frac{(d_i)_{0n} (d_k)_{n0}}{\omega_{n0} - \omega - i0} + \frac{(d_k)_{0n} (d_i)_{n0}}{\omega_{n0} + \omega + i0} \right\}. \quad (85.12)$$

If the ground state is an S -state, then this tensor reduces to a scalar: $\alpha_{ik} = \alpha \delta_{ik}$, where

$$\alpha(\omega) = \frac{1}{3} \sum_n |\mathbf{d}_{0n}|^2 \left(\frac{1}{\omega_{n0} - \omega - i0} + \frac{1}{\omega_{n0} + \omega + i0} \right). \quad (85.13)$$

If the atom possesses angular momentum, then the same result is obtained after averaging over its directions, which is to be understood (for our purposes, the interaction of atoms averaged over their mutual orientations).

Comparing (85.12) with expression (59.17), we see that $\alpha_{ik}(\omega)$ coincides with the tensor of coherent scattering of a photon of frequency ω on the atom. According to (59.23) $\alpha(\omega)$ at $\omega > 0$ coincides with the polarisability of the atom. The values of $\alpha(\omega)$ at $\omega < 0$ are expressed through the values at $\omega > 0$ with the help of the obvious from (85.13) relation $\alpha(-\omega) = \alpha(\omega)$.

Substituting the obtained expressions in (85.7), we find

$$\begin{aligned} \langle S(r) \rangle = & \frac{1}{2} \int dt_1 \dots dt_4 \frac{d\Omega_1}{2\pi} \frac{d\Omega_2}{2\pi} \frac{d\omega_1}{2\pi} \frac{d\omega_2}{2\pi} \times \\ & \times \alpha_1(\Omega_1) \alpha_2(\Omega_2) \omega_1^2 D_{ik}(\omega_1, \mathbf{r}) \omega_2^2 D_{ik}(\omega_2, \mathbf{r}) \times \\ & \times \exp\{-i\omega_1(t_1 - t_2) - i\omega_2(t_3 - t_4) - i\Omega_1(t_1 - t_4) - i\Omega_2(t_2 - t_3)\} \end{aligned} \quad (\text{IX.9})$$

($\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2$; we have also taken into account the evenness of the function $D_{ik}(\omega, \mathbf{r})$ in $\mathbf{r}$).

Integration over three of the times gives δ -functions (by virtue of which $-\Omega_1 = \Omega_2 = \omega_2 = \omega_1$), and over the fourth—the factor t :

$$\langle S(r) \rangle = -itU(r),$$

where

$$U(r) = \frac{i}{2} \int_{-\infty}^{\infty} \omega^4 \alpha_1(\omega) \alpha_2(\omega) [D_{ik}(\omega, \mathbf{r})]^2 \frac{d\omega}{2\pi}. \quad (85.14)$$

This formula determines the energy of interaction of two atoms at any distances large in comparison with the atomic dimensions a . It remains to find and substitute into it the explicit expression for the function $D_{ik}(\omega, \mathbf{r})$.

Comparing expressions (76.14) and (76.8) with one another, we find that

$$D_{ik}(\omega, \mathbf{k}) = \left(\delta_{ik} - \frac{k_i k_k}{\omega^2} \right) D(\omega, \mathbf{k}),$$

where D is given by formula (76.8). In the $\omega, \mathbf{r}$ -representation this relation is expressed, evidently, by the equality

$$D_{ik}(\omega, \mathbf{r}) = - \left(\delta_{ik} + \frac{1}{\omega^2} \frac{\partial^2}{\partial x_i \partial x_k} \right) D(\omega, \mathbf{r}). \quad (85.15)$$

Substituting here $D(\omega, \mathbf{r})$ from (76.16) and performing the differentiation, we find

$$D_{ik}(\omega, \mathbf{r}) = \left[\delta_{ik} \left(1 + \frac{i}{|\omega|r} - \frac{1}{\omega^2 r^2} \right) + \frac{x_i x_k}{r^2} \left(\frac{3}{\omega^2 r^2} - \frac{3i}{|\omega|r} - 1 \right) \right] \frac{e^{i|\omega|r}}{r}. \quad (85.16)$$

Finally, substituting this expression in (85.14), after simple transformations taking into account the evenness of the function $\alpha(\omega)$ we obtain the following final expression for the energy of interaction of atoms:

$$U(r) = \frac{i}{\pi r^2} \int_0^\infty \omega^4 \alpha_1(\omega) \alpha_2(\omega) e^{2i\omega r} \left[1 + \frac{2i}{\omega r} - \frac{5}{(\omega r)^2} - \frac{6i}{(\omega r)^3} + \frac{3}{(\omega r)^4} \right] d\omega. \quad (85.17)$$

This general expression can be simplified in the limiting cases of “small” ($a \ll r \ll \lambda_0$) and “large” ($r \gg \lambda_0$) distances.

When $r \ll \lambda_0$, in the integral the values $\omega \sim \omega_0$ are important (see below), where $\omega_0 \sim c/\lambda_0$ are atomic frequencies; therefore $\omega r \ll 1$. In this case one can retain in the square brackets only the last member and replace the exponential by unity. Writing the integral from $-\infty$ to ∞ (with the aim of further transformation), we find

$$U(r) = \frac{3i}{2\pi r^6} \int_{-\infty}^\infty \alpha_1(\omega) \alpha_1(\omega) d\omega. \quad (85.18)$$

As expected, we have obtained for the interaction the law $1/r^6$. The integral in this formula is easily

computed, after substituting into it $\alpha(\omega)$ from (85.13), by closing the contour of integration by an infinitely distant semicircle in the lower half-plane of the complex variable ω ; at this the integral is determined by the residues of the subintegral expression in the poles $\omega = \omega_{n0} \sim \omega_0$. Assuming for simplicity of notation that both atoms are identical, we obtain (in ordinary units):

$$U(r) = -\frac{2}{3r^6} \sum_{n,n'} \frac{|\mathbf{d}_{0n}|^2 |\mathbf{d}_{0n'}|^2}{\hbar(\omega_{n0} + \omega_{n'0})}, \quad (85.19)$$

which coincides with the London formula (see III, § 89, Problem).

In the opposite limiting case of large distances, $r \gg \lambda_0$, in the integral the values $\omega \lesssim c/r \ll \omega_0$ are important; when $\omega \gtrsim \omega_0$ the integral is rapidly quenched by the oscillating factor $\exp(2i\omega r)$. Therefore one can replace $\alpha_1(\omega)$ and $\alpha_2(\omega)$ by their static values $\alpha_1(0)$ and $\alpha_2(0)$. After this the integration is carried out elementarily (for ensuring convergence one should replace $r \rightarrow r + i0$ in the exponent). As a result we finally obtain (in ordinary units):

$$U(r) = -\frac{23}{4\pi} \frac{\hbar c \alpha_1(0) \alpha_2(0)}{r^7} \quad (85.20)$$

(H. B. G. Casimir, D. Polder, 1948)²⁴.

²⁴In the outlined derivation we partially followed I. E. Dzyaloshinskii (1956).

Interaction of Electrons with Photons

§ 86. Scattering of a photon by an electron

Conservation of 4-momentum in the scattering of a photon by a free electron (*Compton effect*) is expressed by the equation

$$p + k = p' + k', \quad (\text{X.1})$$

where p and k are the 4-momenta of the electron and photon before the collision, and p' and k' are their 4-momenta after the collision. The kinematic invariants introduced in § 66 are:

$$\begin{aligned} s &= (p + k)^2 = (p' + k')^2 = m^2 + 2pk = m^2 + 2p'k', \\ t &= (p - p')^2 = (k' - k)^2 = 2(m^2 - pp') = -2kk', \\ u &= (p - k')^2 = (p' - k)^2 = m^2 - 2pk' = m^2 - 2p'k, \\ s + t + u &= 2m^2. \end{aligned} \quad (\text{X.2})$$

The process under consideration is represented by the two Feynman diagrams (74.14), and its amplitude is

$$M_{fi} = -4\pi e^2 e'^*_{\mu} e_{\nu} (\bar{u}' Q^{\mu\nu} u), \quad (\text{X.3})$$

where

$$Q^{\mu\nu} = \frac{1}{s - m^2} \gamma^{\mu} (\gamma p + \gamma k + m) \gamma^{\nu} - \frac{1}{u - m^2} \gamma^{\nu} (\gamma p - \gamma k' + m) \gamma^{\mu}. \quad (\text{X.4})$$

Here e, e' are the polarisation 4-vectors of the initial and final photons; u, u' are the bispinor amplitudes of the initial and final electrons.

According to the rules set out in § 65, for arbitrary polarisation states of the particles, $|M_{fi}|^2$ is replaced by

$$|M_{fi}|^2 \rightarrow 16\pi^2 e^4 \text{Sp} \left\{ \rho^{(e)'}_{\lambda\mu} Q^{(\gamma)'} Q^{\mu\rho} \rho^{(e)}_{\nu\sigma} Q^{\lambda\sigma} \right\}. \quad (\text{X.5})$$

Here $\rho^{(e)}, \rho^{(e)'}$ are the density matrices of the initial and final electrons, $\rho^{(\gamma)}, \rho^{(\gamma)'}$ are the same for the photons; the photonic (tensor) indices are written explicitly, while the electronic (bispinor) indices are implied; the Sp sign refers specifically to the latter indices. The same indices carry the hermitian conjugation sign in the definition $\bar{Q}_{\mu\nu} = \gamma^0 Q^+_{\mu\nu} \gamma^0$.

Let us consider the scattering of an unpolarised photon by an unpolarised electron, without concern for the polarisations after scattering. Averaging over the polarisations of all particles is achieved by means of the density matrices:

$$\rho^{(\gamma)}_{\lambda\mu} = \rho^{(\gamma)'}_{\lambda\mu} = -\frac{1}{2} g_{\lambda\mu}, \quad \rho^{(e)} = \frac{1}{2} (\gamma p + m), \quad \rho^{(e)'} = \frac{1}{2} (\gamma p' + m);$$

the transition to summation over polarisations of the final-state particles is accomplished by an additional factor of $2 \cdot 2 = 4$.

By formula (64.23) (in which one must now set $I^2 = \frac{1}{4}(s - m^2)^2$ —see (64.15a)) we obtain for the cross section

$$d\sigma = \frac{\pi e^4}{4} \frac{dt}{(s - m^2)^2} \text{Sp} \left\{ (\gamma p' + m) Q^{\lambda\mu} (\gamma p + m) \tilde{Q}_{\lambda\mu} \right\}.$$

With the aid of formula (65.2a) we find that $\tilde{Q}_{\mu\lambda} = Q_{\lambda\mu}$. Separating the terms that transform into one another under the substitution $k \leftrightarrow -k'$ (and correspondingly $s \leftrightarrow u$), we write the cross section in the form

$$d\sigma = dt \frac{\pi e^4}{(s - m^2)^2} [f(s, u) + g(s, u) + f(u, s) + g(u, s)],$$

where we have introduced the notation

$$f(s, u) = \frac{1}{4(s - m^2)^2} \times \\ \times \text{Sp} \{ (\gamma p' + m) \gamma^\mu (\gamma p + \gamma k + m) \gamma^\nu (\gamma p + m) \gamma_\mu (\gamma p + \gamma k + m) \gamma_\nu \},$$

$$g(s, u) = \frac{1}{4(s - m^2)(u - m^2)} \times \\ \times \text{Sp} \{ (\gamma p' + m) \gamma^\mu (\gamma p + \gamma k + m) \gamma^\nu (\gamma p + m) \gamma_\mu (\gamma p - \gamma k' + m) \gamma_\nu \}$$

(in these expressions we bear in mind in advance that the result will depend only on the invariant quantities).

Summation over μ and ν is carried out with the aid of formulas (22.6); discarding the terms with an odd number of γ factors, we obtain

$$f(s, u) = \frac{1}{(s - m^2)^2} \text{Sp} \{ (\gamma p') (\gamma p + \gamma k) (\gamma p) (\gamma p + \gamma k) + \\ + 4m^2 (\gamma p + \gamma k) (\gamma k - \gamma p') + m^2 (\gamma p) (\gamma p') + 4m^4 \}.$$

Computing the trace with the aid of formulas (22.13) and expressing all quantities through the invariants s, u , we find after straightforward manipulations

$$f(s, u) = \frac{2}{(s - m^2)^2} \{ 4m^2 - (s - m^2)(u - m^2) + 2m^2(s - m^2) \}.$$

In an analogous fashion, g is computed:

$$g(s, u) = \frac{2m^2}{(s - m^2)(u - m^2)} \{ 4m^2 + (s - m^2) + (u - m^2) \}.$$

As a result, for the cross section we obtain

$$d\sigma = 8\pi r_e^2 \frac{m^2 dt}{(s - m^2)^2} \left\{ \left(\frac{m^2}{s - m^2} + \frac{m^2}{u - m^2} \right)^2 + \right. \\ \left. + \left(\frac{m^2}{s - m^2} + \frac{m^2}{u - m^2} \right) - \frac{1}{4} \left(\frac{s - m^2}{u - m^2} + \frac{u - m^2}{s - m^2} \right) \right\}, \quad (\text{X.6})$$

where $r_e = e^2/m$. This formula expresses the cross section through invariant quantities. With its help it is easy to express the cross section through the collision parameters in any particular reference frame.

Let us do this for the laboratory frame, in which the electron before the collision is at rest: $p = (m, \mathbf{0})$. Here

$$s - m^2 = 2m\omega, \quad u - m^2 = -2m\omega'. \quad (\text{X.7})$$

Writing the 4-momentum conservation equation in the form $p + k - k' = p'$ and squaring, we obtain

$$pk - pk' - kk' = 0,$$

whence (in the laboratory frame)

$$m(\omega - \omega') - \omega\omega'(1 - \cos \vartheta) = 0,$$

where ϑ is the photon scattering angle. This equation determines the relation between the change in the photon energy and the scattering angle:

$$\frac{1}{\omega'} - \frac{1}{\omega} = \frac{1}{m}(1 - \cos \vartheta) = 0. \quad (\text{X.8})$$

Invariant t :

$$t = -2kk' = -2\omega\omega'(1 - \cos \vartheta) = 0.$$

At a given energy ω we find (using (86.8))

$$dt = 2\omega'^2 d \cos \vartheta = \frac{1}{\pi} \omega'^2 d\vartheta', \quad d\vartheta' = 2\pi \sin \vartheta d\vartheta.$$

Substitution of the expressions written in (86.6) into the following formula for the scattering cross section in the laboratory frame gives:

$$d\sigma = \frac{r_e^2}{2} \left(\frac{\omega'}{\omega} \right)^2 \left(\frac{\omega'}{\omega} + \frac{\omega}{\omega'} - \sin^2 \vartheta \right) d\vartheta' \quad (\text{X.9})$$

(*O. Klein, Y. Nishina, 1929; I. E. Tamm, 1930*).

Since the angle ϑ is uniquely related to ω' by the relation (86.8), the cross section can also be expressed through the energy of the scattered photon ω' :

$$d\sigma = \pi r_e^2 \frac{m d\omega'}{\omega^2} \left[\frac{\omega}{\omega'} + \frac{\omega'}{\omega} + \left(\frac{m}{\omega'} - \frac{m}{\omega} \right)^2 - 2m \left(\frac{1}{\omega'} - \frac{1}{\omega} \right) \right], \quad (\text{X.10})$$

where ω' varies within the limits

$$\frac{\omega}{1 + 2\omega/m} \leq \omega' \leq \omega. \quad (\text{X.11})$$

When $\omega \ll m$, one can set $\omega' \approx \omega$ in (86.9), and one obtains, as expected, the classical non-relativistic Thomson formula

$$d\sigma = \frac{1}{2} r_e^2 (1 + \cos^2 \vartheta) d\vartheta' \quad (\text{X.12})$$

(cf. II, (78.7)).

For computing the total cross section let us return to formula (86.6) (the invariants s , t , and u entering it take values satisfying the inequalities

$$s \geq m^2, \quad t \leq 0, \quad us \leq m^4. \quad (\text{X.13})$$

These were already obtained in § 67 (corresponding to the physical region—I in Fig. 7, p. 299). It is easy to verify them directly, writing the expressions for the invariants in the centre-of-inertia frame. Here $\mathbf{p} + \mathbf{k} = 0$, and the energies of the electron ε and photon ω are related by $\varepsilon = \sqrt{\omega^2 + m^2}$. The invariants:

$$\begin{aligned} s &= (\varepsilon + \omega)^2 = m^2 + 2\omega(\omega + \varepsilon), \\ u &= m^2 - 2\omega(\varepsilon + \omega \cos \theta), \\ t &= -2\omega^2(1 - \cos \theta), \end{aligned} \quad (\text{X.14})$$

where θ is the scattering angle (the angle between $\mathbf{p}$ and $\mathbf{p}'$ or $\mathbf{k}$ and $\mathbf{k}'$). The three inequalities (86.13) follow from the conditions: $\omega \geq 0$, $-1 \leq \cos \theta \leq 1$.

At given s (given energy of the particles) the integration over t can be replaced by integration over $u = 2m^2 - s - t$ in the interval

$$m^4/s \geq u \geq 2m^2 - s.$$

Introducing, instead of s , u , the quantities

$$x = \frac{s - m^2}{m^2}, \quad y = \frac{m^2 - u}{m^2}, \quad (\text{X.15})$$

we obtain

$$\sigma = \frac{8\pi r_e^2}{x^2} \int_{x/(x+1)}^x \left[\left(\frac{1}{x} - \frac{1}{y} \right)^2 + \frac{1}{x} - \frac{1}{y} + \frac{1}{4} \left(\frac{x}{y} + \frac{y}{x} \right) \right] dy$$

and after elementary integration

$$\sigma = 2\pi r_e^2 \frac{1}{x} \left\{ \left(1 - \frac{4}{x} - \frac{8}{x^2} \right) \ln(1+x) + \frac{1}{2} + \frac{8}{x} - \frac{1}{2(1+x)^2} \right\}. \quad (\text{X.16})$$

The leading terms of the expansion for $x \ll 1$ (non-relativistic case) give

$$\sigma = \frac{8\pi r_e^2}{3} (1 - x) \quad (\text{n.r.}). \quad (\text{X.17})$$

The first term is the classical Thomson cross section. In the opposite, ultrarelativistic case $x \gg 1$, expansion of formula (86.16) gives

$$\sigma = 2\pi r_e^2 \frac{1}{x} \left(\ln x + \frac{1}{2} \right) \quad (\text{u.r.}). \quad (\text{X.18})$$

In the laboratory frame

$$x = 2\omega/m, \quad (\text{X.19})$$

so that formulas (86.16)–(86.18) directly give the dependence of the scattering cross section on the energy of the photon incident on a stationary electron as a function of ω/m .

We note that in the ultrarelativistic case the cross section decreases with increasing energy both in the laboratory frame ($\sigma \propto \omega^{-1} \ln \omega$) and in the centre-of-inertia frame ($x \approx 4\omega^2/m^2$, $\sigma \propto \omega^{-2} \ln \omega$). The angular distribution of the scattering in the ultrarelativistic case is entirely different in the two reference frames.

In the laboratory frame the differential cross section has a sharp maximum in the forward direction. In a narrow cone $\vartheta \lesssim \sqrt{m/\omega}$ we have $\omega' \sim \omega$ and the cross section $d\sigma/d\omega' \sim r_e^2$ (reaching the value r_e^2 at $\vartheta \rightarrow 0$). Outside this cone the cross section drops off, and in the region $\vartheta^2 \gg m/\omega$ (where $\omega' \approx m/(1 - \cos \vartheta)$) we have

$$\frac{d\sigma}{d\omega'} = \frac{r_e^2}{2} \frac{m}{\omega(1 - \cos \vartheta)},$$

i.e. the cross section decreases by a factor $\sim \omega/m$.

In the centre-of-inertia frame the differential cross section has a maximum in the backward direction. When $\pi - \theta \ll 1$ we have from (86.14)

$$\frac{s - m^2}{m^2} \approx \frac{4\omega^2}{m^2}, \quad \frac{m^2 - u}{m^2} \approx 1 + \frac{\omega^2}{m^2} (\pi - \theta)^2.$$

The leading term in the cross section (86.6) is

$$d\sigma \approx 8\pi r_e^2 \frac{m^2 dt}{4(s - m^2)(m^2 - u)};$$

whence

$$d\sigma = \frac{r_e^2}{2} \frac{d\omega'}{1 + (\pi - \theta)^2 \omega^2/m^2}. \quad (\text{X.20})$$

The cross section $d\sigma/d\omega' \sim r_e^2$ in a narrow cone $\pi - \theta \lesssim m/\omega$, and outside it the cross section decreases by a factor of order ω^2/m^2 .

§ 87. Scattering of a photon by an electron. Polarisation effects

Let us return to the original formulas of the preceding section and show how the calculations should be carried out taking into account the polarisations of the initial and final photons and electrons.

The photon density matrix is expressed, according to (8.17), with the aid of a pair of unit 4-vectors $e^{(1)}, e^{(2)}$, satisfying the conditions (8.16). In the present case one can choose these vectors in a unified manner for both photons. These are the 4-vectors introduced in § 70¹

$$e^{(1)} = \frac{N}{\sqrt{-N^2}}, \quad e^{(2)} = \frac{P}{\sqrt{-P^2}}, \quad (\text{X.21})$$

¹An alternative method of calculation consists in choosing from the outset a specific reference frame (e.g. the laboratory frame), and for each photon taking as $e^{(1)}, e^{(2)}$ two purely spatial-like ($\mathbf{e} = (0, \mathbf{e})$) vectors, orthogonal to the momentum of the photon and to each other. In that case, however, all the calculations will be carried out in three-dimensional form, and the result will not have an invariant appearance.

where

$$\begin{aligned} P^\lambda &= (p^\lambda + p'^\lambda) - K^\lambda \frac{pK + p'K}{K^2}, \quad N^\lambda = e^{\lambda\mu\nu\rho} P_\mu q_\nu K_\rho, \\ K^\lambda &= k^\lambda + k'^\lambda, \quad q^\lambda = k^\lambda - k'^\lambda = p^\lambda - p'^\lambda. \end{aligned} \quad (\text{X.22})$$

The quantities $Q^{\mu\nu}$ in (86.5) are given by formula (86.4). They can be regarded as components of a 4-tensor (in the sense that the 4-tensor formed after constructing the quantity $\bar{u}' Q^{\mu\nu} u$, as one says, “in bra-kets”). All the components of the 4-tensor can be exhausted by projecting it onto four mutually orthogonal 4-vectors, for example those defined above: P, N, q, K . Since the tensors $\rho_{\mu\nu}^{(\gamma)'}$, $\rho_{\mu\nu}^{(\gamma)}$ contain only components along P and N , in practice we need only the components of $Q_{\mu\nu}$ along these two 4-vectors.

In other words, it suffices to seek in $Q_{\mu\nu}$ only terms of the form

$$\begin{aligned} Q_{\mu\nu} &= Q_0 (e_\mu^{(1)} e_\nu^{(1)} + e_\mu^{(2)} e_\nu^{(2)}) + Q_1 (e_\mu^{(1)} e_\nu^{(2)} + e_\mu^{(2)} e_\nu^{(1)}) - \\ &\quad - iQ_2 (e_\mu^{(1)} e_\nu^{(2)} - e_\mu^{(2)} e_\nu^{(1)}) + Q_3 (e_\mu^{(1)} e_\nu^{(1)} - e_\mu^{(2)} e_\nu^{(2)}); \end{aligned} \quad (\text{X.23})$$

the remaining terms upon substitution in (86.5) would vanish anyway. The quantities Q_0 and Q_3 are scalars—in the same sense as $Q_{\mu\nu}$ is a 4-tensor; they therefore contain γ matrices only in “invariant” combinations: γK and so on. In this same sense Q_1 and Q_2 are pseudoscalars (N is a pseudovector!) and must therefore contain the matrix γ^5 .

By direct projection of the tensor $Q_{\mu\nu}$ we find

$$Q_0 = \frac{1}{2} Q^{\mu\nu} (e_\mu^{(1)} e_\nu^{(1)} + e_\mu^{(2)} e_\nu^{(2)})$$

and so on. For the calculations it is convenient first to express $Q_{\mu\nu}$ through the mutually orthogonal 4-vectors P, N, q, K :

$$Q^{\mu\nu} = \gamma^\mu \frac{\gamma P/2 + m}{s - m^2} \gamma^\nu + \gamma^\nu \frac{\gamma P/2 + m}{u - m^2} \gamma^\mu - \frac{1}{t} [\gamma^\mu (\gamma K) \gamma^\nu - \gamma^\nu (\gamma K) \gamma^\mu].$$

The further calculation reduces to purely algebraic manipulations with the aid of the formulas given in § 22. Furthermore, one can make in $Q^{\mu\nu}$ replacements that will not affect the result upon further formation of the product $\bar{u}' Q^{\mu\nu} u$. For example, since

$$\begin{aligned} \bar{u}' (\gamma p + \gamma p') u &= 2m \bar{u}' u, \\ \bar{u}' \gamma^5 (\gamma q) u &= \bar{u}' [\gamma^5 (\gamma p) + (\gamma p') \gamma^5] u = 2m \bar{u}' \gamma^5 u, \end{aligned}$$

one can replace in $Q^{\mu\nu}$ the terms

$$\gamma p + \gamma p' \rightarrow 2m, \quad \gamma^5 (\gamma q) \rightarrow 2m \gamma^5. \quad (\text{X.24})$$

Omitting the details of the calculations, we present the result²:

$$\begin{aligned} Q_0 &= -ma_+, \quad Q_1 = \frac{1}{2} a_+ \gamma^5 (\gamma K), \\ Q_2 &= -ma_+ \gamma^5, \quad Q_3 = ma_+ + \frac{1}{2} a_- (\gamma K), \end{aligned} \quad (\text{X.25})$$

²Expression (87.3) with the values (87.5) has the form of (70.11) (70.13), established in § 70 from general considerations. In addition to the equalities $f_1 = f_2 = 0$, which follow from T -invariance, here it also turns out that there is only one independent amplitude (f_+). This is a property of the Born approximation of perturbation theory under consideration, and it would vanish in higher approximations.

where

$$a_{\pm} = \frac{1}{s - m^2} \pm \frac{1}{u - m^2}.$$

For the further calculations it is convenient to apply to $Q_{\mu\nu}$ the same formal device that was described in § 8 for the photon density matrix: the four components of the tensor (87.3) in the directions $e^{(1)}, e^{(2)}$ are combined into a two-row matrix Q , which is then expanded in the Pauli matrices. Analogously to formula (8.18) we obtain

$$Q = Q_0 + \mathbf{Q}\sigma, \quad \mathbf{Q} = (Q_1, Q_2, Q_3). \quad (\text{X.26})$$

As regards the tensor $\tilde{Q}_{\mu\nu} = \gamma^0 Q_{\mu\nu}^+ \gamma^0$ appearing in (86.5), using (87.3) and (87.5) it is easy to verify (with the aid of the rules ((65.2a))) that its components are obtained from the components of $Q_{\mu\nu}$ by the replacement of the quantities $Q_0, Q_1, \dots$ by $\bar{Q}_0, \bar{Q}_1, \dots$, where

$$\bar{Q}_0 = Q_0, \quad \bar{Q}_1 = -Q_1, \quad \bar{Q}_2 = -Q_2, \quad \bar{Q}_3 = Q_3, \quad (\text{X.27})$$

and a simultaneous permutation of the indices $\mu\nu$ ³. In matrix form this means that

$$\bar{Q} = \bar{Q}_0 + \bar{\mathbf{Q}}\sigma. \quad (\text{X.28})$$

Let us now clarify the meaning of the 4-vectors $e^{(1)}, e^{(2)}$ in relation to the polarisation of the photons. For each photon the independent directions of polarisation will be determined by the transverse (with respect to the photon momentum $\mathbf{k}$) components of the 3-vectors $e^{(1)}, e^{(2)}$ ⁴. It is easy to see that, both in the centre-of-inertia frame and in the laboratory frame (the rest frame of the initial electron), the vector $\mathbf{P}$ lies in the plane $\mathbf{k}\mathbf{k}'$, and the vector $\mathbf{N}$ is perpendicular to this plane. Therefore the direction $e^{(1)}$ has the meaning of polarisation perpendicular to the scattering plane, and the direction $e^{(2)}$ is polarisation in the scattering plane. We note also that for the initial photon the x axis is the same for both photons and perpendicular to the scattering plane:

$$x \parallel [\mathbf{k}\mathbf{k}'],$$

while the y axes lie in the scattering plane:

$$y \parallel [\mathbf{k}[\mathbf{k}\mathbf{k}']], \quad y' \parallel [\mathbf{k}'[\mathbf{k}\mathbf{k}']].$$

For the initial photon the relevant vectors are $\mathbf{N}, -\mathbf{P}_{\perp}, \mathbf{k}$, and for the final photon—the vectors $\mathbf{N}, \mathbf{P}'_{\perp}$ ($\mathbf{P}_{\perp}, \mathbf{P}'_{\perp}$ are the components of $\mathbf{P}, \mathbf{P}'$ perpendicular to $\mathbf{k}$ and $\mathbf{k}'$ respectively). A change in the sign of $e^{(2)}$ in the photon density matrix (8.17) is equivalent to changing the sign of ξ_1 and ξ_2 . Therefore the photon density matrices of the initial and final photons, referred to the 4-orts $e^{(1)}, e^{(2)}$, will be

$$\begin{aligned} \rho^{(\gamma)} &= \frac{1}{2}(1 + \xi\sigma), \quad \xi = (\xi_1, \xi_2, \xi_3); \\ \rho^{(\gamma)'} &= \frac{1}{2}(1 + \xi'\sigma), \quad \xi' = (-\xi'_1, -\xi'_2, \xi'_3). \end{aligned} \quad (\text{X.29})$$

³For the matrix $Q_{\mu\nu}$ in the original form (86.4) we would simply have $\tilde{Q}_{\mu\nu} = Q_{\nu\mu}$. This property is, however, lost in the result of transformations including replacements of the type (87.4) and so on.

⁴The longitudinal components of e (as well as the time components of the 4-vectors e) can simply be ignored: their inessentiality is ensured by gauge invariance.

Now the tensor trace

$$\rho_{\lambda\mu}^{(\gamma)'} Q^{\mu\nu} \rho_{\nu\rho}^{(\gamma)} \bar{Q}^{\rho\lambda}$$

is computed as the trace of a matrix product of the matrices (87.6)–(87.9) with the aid of (33.5). The result is

$$\begin{aligned} |M_{fi}|^2 = & 8\pi^2 e^4 \text{Sp}_1 \{ (\rho^{(e)'} Q_0 \rho^{(e)} \bar{Q}_0 + \rho^{(e)'} \mathbf{Q} \rho^{(e)} \bar{\mathbf{Q}}) + \\ & + (\xi + \xi') (\rho^{(e)'} Q_0 \rho^{(e)} \bar{\mathbf{Q}} + \rho^{(e)'} \mathbf{Q} \rho^{(e)} \bar{Q}_0) - i(\xi - \xi') [\rho^{(e)'} Q_0 \rho^{(e)} \bar{\mathbf{Q}}] + \\ & + (\xi \xi') (\rho^{(e)'} Q_0 \bar{Q}_0 \rho^{(e)} - \rho^{(e)'} \mathbf{Q} \rho^{(e)} \bar{\mathbf{Q}}) + \rho^{(e)'} (\xi' \mathbf{Q}) \rho^{(e)} (\xi \bar{\mathbf{Q}}) + \\ & + \rho^{(e)'} (\xi \mathbf{Q}) \rho^{(e)} (\xi' \bar{\mathbf{Q}}) - i[\xi \xi'] (\rho^{(e)'} Q_0 \rho^{(e)} \bar{\mathbf{Q}} - \rho^{(e)'} \mathbf{Q} \rho^{(e)} \bar{Q}_0) \}. \end{aligned} \quad (\text{X.30})$$

Scattering on unpolarised electrons. Let us compute the complete cross section for scattering of polarised photons by an unpolarised electron, summed over the polarisations of the final electron. For this one must set in (87.10)

$$\rho^{(e)} = \frac{1}{2}(\gamma p + m), \quad \rho^{(e)'} = \frac{1}{2}(\gamma p' + m)$$

and double the result, which must be substituted in place of $|M_{fi}|^2$ in the formula for the cross section (64.22)

$$d\sigma = \frac{1}{32\pi^2} \frac{dt d\varphi}{(s - m^2)^2} |M_{fi}|^2$$

(φ is the azimuth in the centre-of-inertia frame or in the laboratory frame). The computation of the remaining traces leads to the following final result (in the notation of (86.15)):

$$\begin{aligned} d\sigma = & \frac{1}{2} d\bar{\sigma} + 2r_e^2 \frac{dy d\varphi}{x^2} \left\{ (\xi_3 + \xi'_3) \left[-\left(\frac{1}{x} - \frac{1}{y}\right)^2 - \left(\frac{1}{x} - \frac{1}{y}\right) \right] + \right. \\ & + \xi_1 \xi'_1 \left(\frac{1}{x} - \frac{1}{y} + \frac{1}{2} \right) + \xi_2 \xi'_2 \cdot \frac{1}{4} \left(\frac{x}{y} + \frac{y}{x} \right) \left(1 + \frac{2}{x} - \frac{2}{y} \right) + \\ & \left. + \xi_3 \xi'_3 \left[\left(\frac{1}{x} - \frac{1}{y} \right)^2 + \left(\frac{1}{x} - \frac{1}{y} \right) + \frac{1}{2} \right] \right\}, \end{aligned} \quad (\text{X.31})$$

where $d\bar{\sigma}$ is the cross section for scattering of unpolarised photons, given by formula (86.9); the factor $\frac{1}{2}$ is connected with the fact that in (87.11) there is no summation over polarisations of the final photon.

In the laboratory frame formula (87.11) takes the form

$$d\sigma = \frac{r_e^2}{4} \left(\frac{\omega'}{\omega} \right)^2 d\omega' \{ F_0 + F_3(\xi_3 + \xi'_3) + F_{11}\xi_1\xi'_1 + F_{22}\xi_2\xi'_2 + F_{33}\xi_3\xi'_3 \}, \quad (\text{X.32})$$

$$d\omega' = \sin \vartheta d\vartheta d\varphi,$$

where

$$\begin{aligned} F_0 = & \frac{\omega}{\omega'} + \frac{\omega'}{\omega} - \sin^2 \vartheta, \quad F_3 = \sin^2 \vartheta, \\ F_{11} = & 2 \cos \vartheta, \quad F_{22} = \left(\frac{\omega}{\omega'} + \frac{\omega'}{\omega} \right) \cos \vartheta, \quad F_{33} = 1 + \cos^2 \vartheta \end{aligned} \quad (\text{X.33})$$

(*U. Fano*, 1949). We note that although expression (87.12) does not contain an explicit dependence on the azimuth of the scattering plane φ , there is an implicit dependence, since the Stokes parameters ξ_1, ξ_2, ξ_3 are defined relative to the x, y, z axes associated with the scattering plane. Recall that the x axis is the same for both photons and perpendicular to the scattering plane:

$$x \parallel [\mathbf{k}\mathbf{k}'],$$

while the y axes lie in the scattering plane:

$$y \parallel [\mathbf{k}[\mathbf{k}\mathbf{k}']], \quad y' \parallel [\mathbf{k}'[\mathbf{k}\mathbf{k}']].$$

Taking the sum of the cross sections differing by the sign of ξ' (i.e. setting $\xi' = 0$ and doubling the result), we obtain the complete (summed over polarisations of the final photon) cross section for scattering of a polarised photon on an unpolarised electron. Denoting it by $d\sigma(\xi)$, we have

$$d\sigma(\xi) = \frac{1}{2} r_e^2 \left(\frac{\omega'}{\omega} \right)^2 F d\omega', \quad (\text{X.34})$$

where

$$F = F_0 + \xi_3 F_3 = \frac{\omega'}{\omega} + \frac{\omega}{\omega'} - (1 - \xi_3) \sin^2 \vartheta. \quad (\text{X.35})$$

We see that the cross section for scattering of photons polarised perpendicular to the scattering plane ($\xi_3 = 1$) is greater than for photons polarised in the scattering plane ($\xi_3 = -1$). For circular polarisation the cross section does not depend on ξ_1 . Therefore the scattering cross section coincides with the cross section for unpolarised photons if there is no linear polarisation relative to the x or y axes ($\xi_3 = 0$) or even if there is polarisation relative to directions making 45° with these axes.

The cross section for scattering of unpolarised photons with detection of the polarised final

photon possesses analogous properties. This cross section (denote it by $d\sigma(\xi')$) is obtained from formula (87.12) by setting $\xi = 0$:

$$d\sigma(\xi') = \frac{1}{4} r_e^2 \left(\frac{\omega'}{\omega} \right)^2 F' d\omega', \quad F' = F_0 + \xi'_3 F_3. \quad (\text{X.36})$$

From formula (87.12) one can also find the polarisation of the secondary photon as such; the parameters of this polarisation are denoted $\xi^{(f)}$ in distinction from the detectable polarisation ξ' . According to the rules set out in § 65, the quantities $\xi_i^{(f)}$ are the ratios of the coefficients at ξ'_i to the term not containing ξ' :

$$\xi_1^{(f)} = \frac{F_{11}}{F} \xi_1, \quad \xi_2^{(f)} = \frac{F_{22}}{F} \xi_2, \quad \xi_3^{(f)} = \frac{F_3 + F_{33}\xi_3}{F}. \quad (\text{X.37})$$

In particular, when an unpolarised photon is scattered,

$$\xi_1^{(f)} = 0, \quad \xi_2^{(f)} = 0, \quad \xi_3^{(f)} = \frac{\sin^2 \vartheta}{\omega/\omega' + \omega'/\omega - \sin^2 \vartheta}. \quad (\text{X.38})$$

Here $\xi_3^{(f)} > 0$, i.e. the secondary photon is polarised perpendicular to the scattering plane. Circular polarisation of the secondary photon arises only if the primary photon is circularly polarised: $\xi_2^{(f)} \neq 0$ only when $\xi_2 \neq 0$.

Consider the case when the incident photon is completely linearly polarised ($\xi_2 = 0$, $\xi_1^2 + \xi_3^2 = 1$), and let us find the cross section for scattering in which a linearly polarised secondary photon is also detected. Expressing the Stokes parameters ξ_i and ξ_i' through the components of the polarisation vectors $\mathbf{e}$ and $\mathbf{e}'$, we obtain the following expression for the cross section of scattering:

$$d\sigma = \frac{r_e^2}{4} \left(\frac{\omega'}{\omega} \right)^2 \left(\frac{\omega}{\omega'} + \frac{\omega'}{\omega} - 2 + 4 \cos^2 \Theta \right) d\omega', \quad (\text{X.39})$$

where Θ is the angle between the polarisation directions of the incident and scattered photons⁵.

According to this formula, the cross section behaves substantially differently when the polarisations $\mathbf{e}$ and $\mathbf{e}'$ are mutually perpendicular and when they lie in one plane. Distinguishing these two cases by the subscripts $\perp$ and $\parallel$, we have in the non-relativistic limit ($\omega \ll m$, $\omega' \approx \omega$)

$$d\sigma_{\perp} = 0, \quad d\sigma_{\parallel} = r_e^2 \cos^2 \Theta d\omega' \quad (\text{X.40})$$

in agreement with the classical formulas. In the opposite, ultrarelativistic case $\omega \gg m$, $\omega' \approx m/(1 - \cos \vartheta)$. Here one must distinguish the regions of large and small angles (ω'/ω large or small):

$$\begin{aligned} d\sigma_{\perp} = d\sigma_{\parallel} &= \frac{1}{4} r_e^2 \frac{\omega'}{\omega} d\omega' = \frac{1}{4} r_e^2 \frac{m d\omega'}{\omega(1 - \cos \vartheta)}, \quad \vartheta^2 \gg \frac{m}{\omega}, \\ d\sigma_{\perp} &= 0, \quad d\sigma_{\parallel} = r_e^2 \cos^2 \Theta d\omega', \quad \vartheta^2 \ll \frac{m}{\omega}. \end{aligned} \quad (\text{X.41})$$

We see that in the region of very small scattering angles the cross section coincides with the classical one. The equality $d\sigma_{\perp} \approx d\sigma_{\parallel}$ at not too small angles does not mean that the scattered radiation in this region is unpolarised; we stress, however, that this conclusion refers specifically to a linearly polarised incident photon: from (87.17) it is seen that for a circularly polarised photon in the ultrarelativistic case $\xi_2^{(f)} \approx \cos \vartheta \cdot \xi_2$.

Scattering on polarised electrons. For polarised electrons the computation of the traces in formula (87.10) becomes very cumbersome, though it does not present any difficulties in principle. We give here some final results of such a calculation⁶.

In the general case the cross section depends both on the polarisation parameters of the initial and final photons ξ and ξ' , and on the polarisations of the initial and final

⁵Formula (87.19) could have been obtained more simply from the outset, by setting in the scattering amplitude (86.3) $e = (0, \mathbf{e})$, $e' = (0, \mathbf{e}')$ and computing the square of the amplitude in three-dimensional form (i.e. separating the time and spatial components of the 4-vectors). Averaging $\cos^2 \Theta = (\mathbf{e}\mathbf{e}')^2$ over the directions of $\mathbf{e}$ and $\mathbf{e}'$ (with the aid of (45.4a)) and doubling the cross section (the transition to summation over $\mathbf{e}'$), we return, of course, to (86.9).

⁶More detailed information can be found in the review articles: Tolhoek *H. A. Rev. Mod. Phys.* 1956. V. 28. P. 277; McMaster W. H. *Rev. Mod. Phys.* 1961. V. 33. P. 8.

electrons, characterised by the vectors ζ and ζ' . The dependence of the cross section on each of these parameters is linear. The cross section has the form

$$d\sigma = \frac{1}{2} d\sigma(\xi, \xi') + \frac{r_e^2}{8} \left(\frac{\omega'}{\omega} \right)^2 d\omega' \{ f \zeta \xi_2 + f' \zeta' \xi_2' + g \zeta' \xi_2 + g' \zeta \xi_2' + G_{ik} \zeta_i \zeta_k' + \dots \}. \quad (\text{X.42})$$

Here $d\sigma(\xi, \xi')$ is the cross section (87.12). Listed are all the terms containing products of two polarisation parameters; omitted are

the terms containing products of three or four parameters; these terms are inessential if we are interested only in correlations between the polarisations of two particles; they vanish when the polarisation parameters of the other two particles are set equal to zero. We give the values of some of the coefficients in the laboratory frame:

$$\begin{aligned} \mathbf{f} &= -\frac{1 - \cos \vartheta}{m} (\mathbf{k} \cos \vartheta + \mathbf{k}'), \\ \mathbf{f}' &= -\frac{1 - \cos \vartheta}{m} (\mathbf{k} + \mathbf{k}' \cos \vartheta), \\ \mathbf{g} &= -\frac{1 - \cos \vartheta}{m} \left[(\mathbf{k} \cos \vartheta + \mathbf{k}') - (1 + \cos \vartheta) \frac{\omega + \omega'}{\omega - \omega' + 2m} (\mathbf{k} - \mathbf{k}') \right], \\ \mathbf{g}' &= -\frac{1 - \cos \vartheta}{m} \left[(\mathbf{k} + \mathbf{k}' \cos \vartheta) - (1 + \cos \vartheta) \frac{\omega + \omega'}{\omega - \omega' + 2m} (\mathbf{k} - \mathbf{k}') \right]. \end{aligned}$$

In the cross section (87.22) there is no term of the form $\mathbf{G}\zeta$; this means that the polarisation of the electron does not affect the complete (summed over ξ' and ζ') cross section for scattering of polarised photons. There is also no term of the form $\mathbf{G}'\zeta'$; this means that upon scattering of unpolarised photons the recoil electron is not polarised.

We see also that the terms bilinear in the polarisations of the electron and photon contain only the parameters ξ_2, ξ_2' corresponding to circular polarisation of the photon. The polarisation vectors of the electrons enter in the form of scalar products $f\zeta, \dots$, containing only the projections of these vectors onto the scattering plane. Therefore, for example, the cross section for scattering of a polarised photon by a polarised electron is

$$d\sigma(\xi, \zeta) = d\sigma(\xi) + \frac{1}{2} r_e^2 \left(\frac{\omega'}{\omega} \right)^2 \xi_2 \mathbf{f} \zeta d\omega' \quad (\text{X.24})$$

and differs from $d\sigma(\xi)$ only when the photon has circular polarisation and the electron has a non-zero projection of its mean spin on the scattering plane. For the same reason the recoil electron is polarised only when the photon possesses circular polarisation; the vector of the resulting electron polarisation lies in the scattering plane:

$$\zeta^{(f)} = \frac{1}{F} \xi_2 \mathbf{g}. \quad (\text{X.25})$$

Symmetry relations. In conclusion we point out that the qualitative properties of polarisation effects in the scattering of photons by electrons follow already from the general requirements of symmetry.

The parameter ξ_2 of circular polarisation is a pseudoscalar (see § 8). Therefore, by virtue of P -invariance, terms of the form $\propto \xi_2$ (or $\propto \xi_2'$) could appear in the cross section only as the product of ξ_2 with some other pseudoscalar that can be constructed from the vectors at our disposal $\mathbf{k}$ and $\mathbf{k}'$ ⁷. But from two polar vectors one cannot construct a pseudoscalar. It follows from this that the indicated terms cannot appear in the cross section.

The parameters of linear polarisation ξ_1 and ξ_2 are related to the components of the two-dimensional (in the plane perpendicular to $\mathbf{k}$) symmetric tensor

$$S_{\alpha\beta} = \frac{1}{2}(\rho_{\alpha\beta}^{(\gamma)} + \rho_{\beta\alpha}^{(\gamma)}) = \frac{1}{2} \begin{pmatrix} 1 + \xi_3 & \xi_1 \\ \xi_1 & 1 - \xi_3 \end{pmatrix}. \quad (\text{X.26})$$

In the present case one of the polarisation axes is chosen along the vector $\mathbf{v} = [\mathbf{k}\mathbf{k}']$, and the other lies in the plane $\mathbf{k}\mathbf{k}'$ (along the vector $[\mathbf{k}\mathbf{v}]$ or $[\mathbf{k}'\mathbf{v}]$ for one or the other photon). Terms $\propto \xi_1$ could appear in the cross section only as products of $S_{\alpha\beta}v_\alpha[\mathbf{k}'\mathbf{v}]_\beta$ (or, what is the same, $S_{\alpha\beta}v_\alpha k'_\beta$) and so on. But since $\mathbf{v}$ is an axial vector and $\mathbf{k}$ is a polar vector, and $S_{\alpha\beta}$ is a true tensor, such products are not invariant under inversion. Therefore terms $\propto \xi_1$ (or $\propto \xi_1'$) in the cross section are also excluded.

Terms in the cross section proportional to the electron polarisation ζ are not forbidden by parity, since such terms could be formed as products of two axial vectors: $\zeta\mathbf{v}$. However, they must be absent as a consequence of the hermiticity of the scattering matrix in the considered approximation (cf. § 71).

By virtue of this hermiticity, the square of the modulus of the scattering amplitude (and with it the cross section) does not change upon permutation of the initial and final states. At the same time the cross section must be invariant with respect to time reversal—permutation of the initial and final states with the simultaneous reversal of the sign of the momentum and angular momentum vectors of all particles (the Stokes parameters ξ_1, ξ_2, ξ_3 do not change—see § 8). Combining both requirements, we conclude that in the considered approximation the cross section must not change upon the simultaneous reversal of the sign of all momenta and angular momenta without the permutation of

initial and final states, i.e. under the transformation

$$\mathbf{k} \rightarrow -\mathbf{k}', \quad \mathbf{k}' \rightarrow -\mathbf{k}, \quad \zeta \rightarrow -\zeta', \quad \zeta' \rightarrow -\zeta \quad (\text{X.27})$$

and unchanged parameters ξ, ξ' .

Transformation (87.26) changes the sign of the products $\zeta\mathbf{v}$, and therefore such terms cannot appear in the cross section. We stress, however, that this prohibition is not a consequence of strict symmetry requirements and may be violated in the following approximations of perturbation theory.

We note also that parity forbids terms of the type $\xi_1\xi_3$ and $\xi_2\xi_3$ in the double correlation between the polarisations of the photons with one another, and imposes no restrictions on the form of the correlation of the photons with the electrons. However, all

⁷We consider the process in the laboratory frame, where $\mathbf{p} = 0$, $\mathbf{p}' = \mathbf{k} - \mathbf{k}'$. It is clear that the interesting consequences of the requirements of symmetry (the presence or absence of given terms in the cross section) do not depend on the choice of reference frame.

the terms of the type $\xi_1\xi_2$, $\xi_1\zeta$, $\xi_3\zeta$ are forbidden in the first Born approximation relative to the transformation (87.26). Thus, the terms of the type $\xi_1\xi_2$ and $\xi_1\zeta$ can be formed (from the point of view of obeying parity) as scalars, for example $\xi'_1(S_{\alpha\beta}k'_\alpha\nu_\beta)$ and $(S_{\alpha\beta}k'_\alpha\nu_\beta)(\zeta\mathbf{k})$; these combinations, however, change sign under the transformation (87.26).

The allowed correlation terms of the type $\xi_2\zeta$ can be formed only as products of the type $\xi_2(\zeta\mathbf{k})$; the electron polarisation vectors enter them only in the form of projections on the scattering plane.

Finally, a number of relations among the coefficients in the allowed terms arises from the requirements of cross-universality. The channels of the reaction, differing by the permutation of the initial and final photons, correspond to the same process—scattering of a photon by an electron. Therefore the square of the modulus of the amplitude, and with it the cross section, must be invariant with respect to the transformation expressing the transition from one of these channels to the other:

$$k \leftrightarrow -k', \quad e \leftrightarrow e'^* \quad (\text{X.28})$$

at unchanged momenta and polarisations of the electrons. In three-dimensional form this means the replacements:

$$\omega \leftrightarrow -\omega', \quad \mathbf{k} \leftrightarrow -\mathbf{k}', \quad \xi_1 \leftrightarrow \xi'_1, \quad \xi_2 \leftrightarrow -\xi'_2, \quad \xi_3 \leftrightarrow \xi'_3.$$

The change of sign of the parameter ξ_2 is obvious from the expression $\xi_2 = i[\mathbf{e}\mathbf{e}^*]\mathbf{n}$, in which the vector $[\mathbf{e}\mathbf{e}^*]$ changes sign under the substitution $\mathbf{e} \leftrightarrow -\mathbf{e}^*$, while the vector $\mathbf{n} = \mathbf{k}/\omega$ under $\mathbf{k} \leftrightarrow -\mathbf{k}$, $\omega \leftrightarrow -\omega$ does not change. The transformation (87.27), without affecting the momenta of the electrons, leaves the laboratory frame. Therefore the cross section (87.22) must not change its form under this transformation; formulas (87.12), (87.22), (87.23) indeed satisfy this condition.

§ 88. Two-photon annihilation of an electron-positron pair

The annihilation of an electron and a positron (4-momenta p_- and p_+) with the production of two photons (k_1 and k_2) corresponds to the two diagrams:

They differ from the diagrams for scattering of a photon by an electron by the substitutions

$$p \rightarrow p_-, \quad p' \rightarrow -p_+, \quad k \rightarrow -k_1, \quad k' \rightarrow k_2. \quad (\text{X.2})$$

Both processes are two crossed channels of one and the same (generalised) reaction. After the substitutions (88.2) the kinematic invariants (86.2) acquire the following meaning:

$$\begin{aligned} s &= (p_- - k_1)^2, \\ t &= (p_- + p_+)^2 = (k_1 + k_2)^2, \\ u &= (p_- - k_2)^2. \end{aligned} \quad (\text{X.3})$$

If the scattering of the photon was the s -channel, then annihilation is the t -channel.

The square $|M_{fi}|^2$ for annihilation (averaged over polarisations of the electrons and summed over polarisations of the photons) will, when expressed through the invariants s ,

u , coincide with the analogous quantity for scattering with a change only of the meaning of the invariants⁸. In the formula for the cross section (64.23), one must now replace $s \leftrightarrow t$, and for the quantity I we now have, according to (64.15a),

$$I^2 = \frac{1}{4}t(t - 4m^2).$$

Performing the corresponding changes in formula (86.6), we obtain the annihilation cross section:

$$d\sigma = 8\pi r_e^2 \frac{m^2 ds}{t(t - 4m^2)} \left\{ \left(\frac{m^2}{s - m^2} + \frac{m^2}{u - m^2} \right)^2 + \left(\frac{m^2}{s - m^2} + \frac{m^2}{u - m^2} \right) - \frac{1}{4} \left(\frac{s - m^2}{u - m^2} + \frac{u - m^2}{s - m^2} \right) \right\}. \quad (\text{X.4})$$

The physical region of the annihilation channel is region II in Fig. 7 (see p. 299). At given t (given energy in the centre

of inertia) the interval of variation of s is determined by the equation of the boundary $su = m^4$. Together with the relation $s + t + u = 2m^2$ this gives

$$\frac{t}{2} - \frac{1}{2}\sqrt{t(t - 4m^2)} \leq s - m^2 \leq -\frac{t}{2} + \frac{1}{2}\sqrt{t(t - 4m^2)}. \quad (\text{X.5})$$

Integration of expression (88.4) is elementary; the result must be divided by 2 as well, taking into account the identity of the two final-state particles (photons). Thus we obtain

$$\sigma = \frac{\pi r_e^2}{2\tau^2(\tau - 1)} \left[\left(\tau^2 + \tau - \frac{1}{2} \right) \ln \frac{\sqrt{\tau} + \sqrt{\tau - 1}}{\sqrt{\tau} - \sqrt{\tau - 1}} - (\tau + 1)\sqrt{\tau(\tau - 1)} \right], \quad (\text{X.6})$$

where $\tau = t/4m^2$ (*P. A. M. Dirac*, 1930).

In the non-relativistic limit ($\tau \rightarrow 1$) we find

$$\sigma = \frac{\pi r_e^2}{2\sqrt{\tau - 1}} \quad (\text{n.r.}). \quad (\text{X.7})$$

In the ultrarelativistic case ($\tau \rightarrow \infty$)

$$\sigma = \frac{\pi r_e^2}{2\tau} (\ln 4\tau - 1) \quad (\text{u.r.}). \quad (\text{X.8})$$

In the laboratory frame, in which one of the particles (say, the electron) before the collision is at rest, the invariant τ is

$$\tau = \frac{1}{2}(1 + \gamma), \quad \gamma = \frac{\varepsilon_+}{m}. \quad (\text{X.9})$$

Formulas (88.6)–(88.8) give the dependence of the total cross section on the energy of the positron in flight:

$$\sigma = \frac{\pi r_e^2}{4} \left[\frac{\gamma^2 + 4\gamma + 1}{\gamma^2 - 1} \ln \left(\gamma + \sqrt{\gamma^2 - 1} \right) - \frac{\gamma + 3}{\sqrt{\gamma^2 - 1}} \right]. \quad (\text{X.10})$$

⁸Here it is taken into account that the photons and electrons have the same number (two) of independent polarisations, and therefore the inessential factor by which $|M_{fi}|^2$ is averaged is the same.

In particular, in the non-relativistic limit⁹

$$\sigma = \pi r_e^2 / v_+ \quad (\text{n.r.}), \quad (\text{X.11})$$

where v_+ is the velocity of the positron.

In the centre-of-inertia frame, electron, positron, and both photons have equal energies $\varepsilon = \omega$. The invariants:

$$\begin{aligned} m^2 - s &= 2\varepsilon(\varepsilon - |\mathbf{p}| \cos \theta), & m^2 - u &= 2\varepsilon(\varepsilon + |\mathbf{p}| \cos \theta), \\ t &= 4\varepsilon^2, \end{aligned} \quad (\text{X.12})$$

where θ is the angle between the momenta of the electron and one of the photons.

Substituting (88.12) into (88.4), we obtain the angular distribution of the annihilation photons:

$$d\sigma = \frac{r_e^2 m^2}{4\varepsilon |\mathbf{p}|} \left[\frac{\varepsilon^2 + \mathbf{p}^2 (1 + \sin^2 \theta)}{\varepsilon^2 - \mathbf{p}^2 \cos^2 \theta} - \frac{2\mathbf{p}^4 \sin^4 \theta}{(\varepsilon^2 - \mathbf{p}^2 \cos^2 \theta)^2} \right] d\theta. \quad (\text{X.13})$$

In the ultrarelativistic case it has symmetric maxima in the directions $\theta = 0$ and $\theta = \pi$. Near $\theta = 0$

$$d\sigma \approx \frac{r_e^2 m^2 d\theta}{2\varepsilon^2 (\theta^2 + m^2/\varepsilon^2)} \quad (\text{u.r.}). \quad (\text{X.14})$$

The total cross section is obtained from (88.6):

$$\sigma = \pi r_e^2 \frac{1 - v^2}{4v} \left[\frac{3 - v^4}{v} \ln \frac{1 + v}{1 - v} - 2(2 - v^2) \right], \quad (\text{X.15})$$

where $v = |\mathbf{p}|/\varepsilon = \sqrt{\varepsilon^2 - m^2}/\varepsilon$ is the velocity of the colliding particles.

We shall not consider here the details of the polarisation effects in the annihilation. Let us note only a few qualitative properties for large and small velocities v of the colliding particles. We shall consider the process in the centre-of-inertia frame.

In the limit $v \rightarrow 0$ only the state with orbital angular momentum $l = 0$ contributes to the cross section. But the S -state of the system “electron + positron” has negative parity (see the problem to § 27). In odd states of the two-photon system their polarisations are mutually orthogonal (see § 9). The two-photon system must therefore possess the same property, and the annihilation photons must be polarised perpendicular to each other in the non-relativistic case.

If the electron and positron are polarised, and also in the non-relativistic case, one can assert that their annihilation is possible only for anti-parallel spins. Indeed, since the annihilation occurs from the S -state, the total angular momentum of the system coincides with the total spin, which equals 1 for parallel spins. But a system of two photons cannot have a total angular momentum equal to 1 (see § 9).

In the ultrarelativistic limit ($v \rightarrow 1$) the annihilation of a longitudinally polarised (helical) electron and positron is possible only for different signs of their helicities¹⁰. In

⁹This formula is, however, inapplicable when $v_+ \lesssim \alpha$, and one cannot neglect the Coulomb interaction of the pair components (cf. the end of § 94).

¹⁰Since the directions of the momenta of the particles in the centre-of-inertia frame are opposite, different signs of the helicities correspond to parallel spins.

this limit the helical particles behave like neutrinos (see the end of § 80), and therefore the annihilating electron and positron must be analogous to the neutrino and antineutrino, whence the statement made follows.

The annihilation of an electron and a positron with equal helicities arises in the ultrarelativistic case only upon taking into account terms containing m . In order of magnitude the amplitude of this process differs from the amplitude of annihilation of the pair with anti-parallel spins by a factor m/ε ; the cross section differs by the factor $(m/\varepsilon)^2$.

Problem

Find the cross section of electron-pair production upon collision of two photons (*G. Breit, J. A. Wheeler, 1934*).

Solution. This process is the reverse of two-photon annihilation of an electron-positron pair. The squared amplitudes of both processes are the same, and their relation to the cross section differs only in that now $I^2 = (k_1 k_2)^2 = t^2/4$. Therefore

$$d\sigma_{\text{pr}} = v^2 d\sigma_{\text{ann}}.$$

In the centre-of-inertia frame ($t = 4\varepsilon^2 = 4\omega^2$),

v = velocity of the pair components,

$$d\sigma_{\text{pr}} = v^2 d\sigma_{\text{ann}},$$

where the total cross section in the centre of inertia is

$$\sigma_{\text{pr}} = \frac{\pi r_e^2}{2} (1 - v^2) \left\{ (3 - v^4) \ln \frac{1 + v}{1 - v} - 2v(2 - v^2) \right\}. \quad (1)$$

In an arbitrary reference frame K , in which the two photons k_1 and k_2 are directed towards each other, we have (from the invariance of $k_1 k_2$)

$$\omega_1 \omega_2 = \omega^2.$$

Since the photon energies and the pair component energies coincide in the centre-of-inertia frame, $\omega = \varepsilon = m/\sqrt{1 - v^2}$. Therefore for the transition to the frame K one must put in (1)

$$v = \sqrt{1 - \frac{m^2}{\omega_1 \omega_2}}.$$

§ 89. Annihilation of positronium

By virtue of conservation of momentum the annihilation of the electron and positron in positronium must be accompanied by the emission of at least two photons. Such a decay is possible, however (principally) only for parapositronium; and for orthopositronium it

is three-photon annihilation. In § 9 it was shown that the total angular momentum of a system of two photons cannot equal 1. Therefore an orthopositronium atom, which is in a state 3S_1 , cannot decay into two photons. Moreover, since in the 3S_1 state positronium constitutes a charge-odd system (see the problem to § 27), by the Furry theorem (see § 79) its decay into any even number of photons is also impossible. On the contrary, in the 1S_0 state positronium is charge-even, and therefore the decay of parapositronium into any odd number of photons is forbidden.

The principal process determining the lifetime of positronium is, thus, two-photon annihilation in the case of parapositronium and three-photon annihilation in the case of orthopositronium (*I. Ya. Pomeranchuk*, 1948). The probability of decay can be related to the cross section of annihilation of a free pair.

The momenta of the electron and positron in positronium are $\sim me^2/\hbar$, i.e. small in comparison with mc . Therefore in computing the probability of annihilation one can go to the limit of two particles at rest in the origin of coordinates. Let $\bar{\sigma}_{2\gamma}$ be the cross section of two-photon annihilation of a free pair, averaged over spin directions of both particles. In the non-relativistic limit, according to (88.11)¹¹,

$$\bar{\sigma}_{2\gamma} = \pi \left(\frac{e^2}{mc^2} \right)^2 \frac{c}{v}, \quad (\text{X.16})$$

where v is the relative velocity of the particles. We obtain the probability of annihilation $\bar{w}_{2\gamma}$ by multiplying $\bar{\sigma}_{2\gamma}$ by the flux density $v|\psi(0)|^2$. Here $\psi(r)$ is the wave function of the ground state of positronium normalised to 1:

$$\psi = \frac{1}{\sqrt{\pi a^3}} e^{-r/a}, \quad a = \frac{2\hbar^2}{me^2} \quad (\text{X.17})$$

(the Bohr radius of positronium a is twice the Bohr radius of the hydrogen atom because the reduced mass is half that). This probability, however, corresponds to the averaging over spin states of the initial state. But among the four possible spin states of the two-particle system, only one (with total spin 0) is capable of two-photon annihilation. Therefore the average probability of decay $\bar{w}_{2\gamma}$ is related to the probability of decay of parapositronium w_0 by the relation $\bar{w}_{2\gamma} = w_0/4$. Thus,

$$w_0 = 4|\psi(0)|^2 (v\bar{\sigma}_{2\gamma})_{v \rightarrow 0}. \quad (\text{X.18})$$

Substituting the values of the quantities from (89.1) and (89.2), we obtain for the lifetime of parapositronium

$$\tau_0 = \frac{2\hbar}{mc^2\alpha^5} = 1.23 \cdot 10^{-10} \text{ s}. \quad (\text{X.19})$$

We draw attention to the fact that the level width $\Gamma_0 = \hbar/\tau_0$ is small compared with the energy:

$$|E_{\text{gr}}| = \frac{me^4}{4\hbar^2} = mc^2 \frac{\alpha^2}{4}.$$

¹¹Formulas (89.1) and (89.7) are written in ordinary units.

It is precisely this circumstance that allows one to treat positronium as a quasi-stationary system.

In an analogous manner we find that the probability of decay of orthopositronium is related to the spin-averaged cross section of three-photon annihilation of a free pair by the relation

$$w_1 = \frac{4}{3} \bar{w}_{3\gamma} = \frac{4}{3} |\psi(0)|^2 (v \bar{\sigma}_{3\gamma})_{v \rightarrow 0} \quad (\text{X.20})$$

($\frac{3}{4}$ is the relative statistical weight of the state with spin 1). Looking ahead, we note that

$$\bar{\sigma}_{3\gamma} = \frac{4(\pi^2 - 9)}{3v} \alpha \left(\frac{e^2}{mc^2} \right)^2. \quad (\text{X.21})$$

Therefore the lifetime of orthopositronium is

$$\tau_1 = \frac{9\pi}{2(\pi^2 - 9)} \frac{\hbar}{mc^2 \alpha^6} = 1.4 \cdot 10^{-7} \text{ s}. \quad (\text{X.22})$$

The inequality $\Gamma_1 \ll |E_{\text{gr}}|$ is in this case satisfied, of course, even better than for parapositronium.

Let us compute the cross section of three-photon annihilation of a free pair (A. Ore, J. L. Powell, 1949).

According to (64.18) the cross section of the process under consideration in the centre-of-inertia frame is expressed through the squared amplitude by the formula

$$d\sigma_{3\gamma} = \frac{(2\pi)^4 |M_{fi}|^2}{4I} \delta(\mathbf{k}_1 + \mathbf{k}_2 + \mathbf{k}_3) \delta(\omega_1 + \omega_2 + \omega_3 - 2m) \times \\ \times \frac{d^3 k_1 d^3 k_2 d^3 k_3}{(2\pi)^9 2\omega_1 \cdot 2\omega_2 \cdot 2\omega_3}, \quad (\text{X.23})$$

where, according to (64.16), $I = 2m \frac{m}{2} v = m^2 v$, where v is the relative velocity of the positron and electron (which we assume to be small); $\mathbf{k}_1, \mathbf{k}_2, \mathbf{k}_3$ and $\omega_1, \omega_2, \omega_3$ are the wave vectors and frequencies of the emitted photons; the δ -functions express the laws of conservation of energy and momentum. By virtue of these laws the three frequencies $\omega_1, \omega_2, \omega_3$ must be representable as the side lengths of a triangle with perimeter $2m$.

Three-photon annihilation corresponds to the diagram

and five more diagrams obtained by permutation of the photons k_1, k_2, k_3 . The corresponding amplitude is written in the form

$$M_{fi} = (4\pi)^{3/2} e_{\lambda}^{(3)*} e_{\mu}^{(2)*} e_{\nu}^{(1)*} \bar{u}(-p_+) Q^{\lambda\mu\nu} u(p_-), \quad (\text{X.24})$$

where

$$Q^{\lambda\mu\nu} = \sum_{\text{perm}} \gamma^{\lambda} G(k_3 - p_+) \gamma^{\mu} G(p_- - k_1) \gamma^{\nu}, \quad (\text{X.25})$$

where the sum is taken over all permutations of the photon numbers 1, 2, 3 together with the simultaneously corresponding permutations of the tensor indices $\lambda\mu\nu$. The squared

amplitude, averaged over the polarisations of the electron and positron and summed over the polarisations of the photons:

$$\frac{1}{4} \sum_{\text{polar}} |M_{fi}|^2 = (4\pi)^3 \text{Sp} \left\{ \rho_+ Q^{\lambda\mu\nu} \rho_- \tilde{Q}_{\lambda\mu\nu} \right\}, \quad (\text{X.26})$$

where

$$\rho_- = \frac{1}{2}(\gamma p_- + m), \quad \rho_+ = \frac{1}{2}(\gamma p_+ - m).$$

The matrices $\tilde{Q}^{\lambda\mu\nu}$ differ from the matrices $Q^{\lambda\mu\nu}$ by the reversal of the order of factors in each term of the sum. In the case of interest to us, where the velocities of the electron and positron are small, one can set their 3-momenta $\mathbf{p}_-$ and $\mathbf{p}_+$ equal to zero, i.e. set $p_- = p_+ = (m, 0)$. Then the electron Green's functions become

$$G(p_- - k_1) = \frac{\gamma p_- - \gamma k_1 + m}{(p_- - k_1)^2 - m^2} \approx \frac{-\gamma k_1 + m(\gamma^0 + 1)}{-2m\omega_1}$$

and so on, and the density matrices reduce to

$$\rho_{\pm} = \frac{m}{2}(\gamma^0 \pm 1).$$

Upon multiplying out the terms in (89.11) there arises a large number of terms. However, the number of terms requiring computation can be considerably reduced by making full use of the symmetry with respect to the permutations of the photons. It suffices to multiply out the six terms in $Q^{\lambda\mu\nu}$ (89.10) with any one term in $\tilde{Q}^{\lambda\mu\nu}$. In the remaining five traces one can also identify certain parts that transform into one another under different permutations of the photons. The products of traces of 4-vectors p , k_1 , k_2 , k_3 arising upon expanding the traces are expressed through the frequencies ω_1 , ω_2 , ω_3 . Since $p = (m, 0)$, we have $pk_1 = m\omega_1, \dots$. The products $k_1k_2, \dots$ are determined from the equation of 4-momentum conservation: $2p = k_1 + k_2 + k_3$; rewriting this equation in the form $2p - k_3 = k_1 + k_2$ and squaring, we obtain

$$k_1k_2 = 2m(m - \omega_3), \dots \quad (\text{X.27})$$

As a result, after a rather lengthy calculation one obtains

$$\frac{1}{4} \sum_{\text{polar}} |M_{fi}|^2 = (4\pi)^3 e^6 \cdot 16 \left[\left(\frac{m - \omega_1}{\omega_2\omega_3} \right)^2 + \left(\frac{m - \omega_2}{\omega_1\omega_3} \right)^2 + \left(\frac{m - \omega_3}{\omega_1\omega_2} \right)^2 \right].$$

Substituting this expression into (89.8), we find the differential cross section of three-photon annihilation:

$$\begin{aligned} d\bar{\sigma}_{3\gamma} = & \frac{e^6}{\pi^2 m^2 v} \left[\left(\frac{m - \omega_1}{\omega_2\omega_3} \right)^2 + \left(\frac{m - \omega_2}{\omega_1\omega_3} \right)^2 + \left(\frac{m - \omega_3}{\omega_1\omega_2} \right)^2 \right] \times \\ & \times \delta(\mathbf{k}_1 + \mathbf{k}_2 + \mathbf{k}_3) \delta(\omega_1 + \omega_2 + \omega_3 - 2m) \frac{d^3k_1 d^3k_2 d^3k_3}{\omega_1\omega_2\omega_3}. \end{aligned} \quad (\text{X.28})$$

Here we must still eliminate the δ -functions. The first of them is removed by integration over d^3k_3 , after which we replace the remaining differentials:

$$d^3k_1 d^3k_2 \rightarrow 4\pi\omega_1^2 d\omega_1 \cdot 2\pi\omega_2^2 d(\cos\theta_{12}) d\omega_2,$$

where θ_{12} is the angle between $\mathbf{k}_1$ and $\mathbf{k}_2$; it is understood that the integration over the directions of $\mathbf{k}_1$ and the azimuth of $\mathbf{k}_2$ relative to $\mathbf{k}_1$ has already been performed. Differentiating the equation

$$\omega_3 = \sqrt{\omega_1^2 + \omega_2^2 + 2\omega_1\omega_2 \cos\theta_{12}},$$

we find

$$d \cos \theta_{12} = \frac{\omega_3}{\omega_1 \omega_2} d\omega_3.$$

By integrating over ω_3 we eliminate the second δ -function. As a result we obtain the cross section of annihilation with the production of photons having given energies in the form

$$d\bar{\sigma}_{3\gamma} = \frac{1}{6} \frac{8e^6}{vm^2} \left\{ \left(\frac{m - \omega_1}{\omega_2 \omega_3} \right)^2 + \left(\frac{m - \omega_2}{\omega_1 \omega_3} \right)^2 + \left(\frac{m - \omega_3}{\omega_1 \omega_2} \right)^2 \right\} d\omega_1 d\omega_2 \quad (\text{X.29})$$

(with a view to subsequent integration over the frequencies, we have introduced the factor $\frac{1}{6}$ taking into account the identity of the photons (cf. the footnote on p. 286)).

Each of the frequencies $\omega_1, \omega_2, \omega_3$ can take values between 0 and m (the value m is attained by two of the frequencies when the third equals zero). At given ω_1 the frequency ω_2 varies between $m - \omega_1$ and m . Integrating (89.14) over ω_2 in these limits, we obtain the spectral distribution of the photons of the decay:

$$d\bar{\sigma}_{3\gamma} = \frac{8e^6}{3vm^3} F(\omega_1) d\omega_1,$$

$$F(\omega_1) = \frac{\omega_1(m - \omega_1)}{(2m - \omega_1)^2} + \frac{2m - \omega_1}{\omega_1} + \left[\frac{2m(m - \omega_1)}{\omega_1^2} - \frac{2m(m - \omega_1)^2}{(2m - \omega_1)^3} \right] \ln \frac{m - \omega_1}{m}.$$

The function F monotonically increases from zero at $\omega_1 = 0$ to 1 at $\omega_1 = m$; in Fig. 14 its graph is shown.

The total annihilation cross section is obtained by integrating (89.14) over both frequencies:

$$\bar{\sigma}_{3\gamma} = \frac{4e^6}{3vm^2} \cdot 3 \int_0^m \int_{m-\omega_1}^m \frac{(\omega_1 + \omega_2 - m)^2}{\omega_1^2 \omega_2^2} d\omega_1 d\omega_2.$$

The double integral here equals $(\pi^2 - 9)/3$, and we arrive at the formula (89.6) given above.

§ 90. Magnetobremssstrahlung

According to classical theory (see II, § 74) an ultrarelativistic electron moving in a constant magnetic field H emits a quasi-continuous spectrum with maximum at a frequency of order

$$\omega \sim \omega_0 \left(\frac{\varepsilon}{m} \right)^3, \quad (\text{X.30})$$

where

$$\omega_0 = \frac{v|e|H}{|\mathbf{p}|} \approx \frac{|e|H}{\varepsilon} \quad (\text{X.31})$$

is the frequency of revolution of an electron with energy ε in a circular orbit (in the plane perpendicular to the field)¹². We shall assume that the longitudinal (along $\mathbf{H}$) component of the electron velocity is zero; this can always be achieved by choosing the appropriate reference frame.

Quantum effects in magnetobremssstrahlung radiation have a twofold origin: quantisation of the motion of the electron and quantum recoil upon emission of a photon. The latter is determined by the ratio $\hbar\omega_0/\varepsilon$, and the condition for applicability of classical theory requires its smallness. Since

$$\frac{\hbar\omega_0}{\varepsilon} = \frac{H}{H_0} \left(\frac{m}{\varepsilon} \right)^2, \quad (\text{X.32})$$

where $H_0 = m^2/(|e|\hbar)$ ($= m^2 c^3/(|e|\hbar)$) $= 4.4 \cdot 10^{13}$ G. In the classical region $\chi \sim \hbar\omega/\varepsilon \ll 1$. When $\chi \gtrsim 1$ the energy of the emitted photon $\hbar\omega \sim \varepsilon$, and moreover when $\chi \gg 1$ (as we shall see below) the essential region of the spectrum extends to frequencies at which the energy of the electron after emission is

$$\varepsilon' \sim m \frac{H_0}{H} \ll \varepsilon. \quad (\text{X.33})$$

In order for the electron to remain ultrarelativistic, the field must satisfy the condition

$$\frac{H}{H_0} \ll 1. \quad (\text{X.34})$$

As regards the quantisation of the motion itself, it is characterised by the ratio $\hbar\omega_0/\varepsilon$. Since

$$\frac{\hbar\omega_0}{\varepsilon} = \frac{H}{H_0} \left(\frac{m}{\varepsilon} \right)^2,$$

by virtue of (90.5) $\hbar\omega_0 \ll \varepsilon$, i.e. the motion of the electron is quasi-classical irrespective of the value of χ . In other words, one can neglect the non-commutativity of the operators of dynamic variables of the electron with one another (quantities $\approx \hbar\omega_0/\varepsilon$), while at the same time taking into account their non-commutativity with the operators of the photon field (quantities $\sim \hbar\omega/\varepsilon$)¹³.

¹²In this section we set $c = 1$, but retain the factors $\hbar$.

¹³The complete solution of the quantum problem of magnetobremssstrahlung radiation was given by *N. P. Klepikov* (1954), and the first quantum correction to the classical formula by *A. A. Sokolov, N. P. Klepikov* and *I. M. Ternov* (1952). The derivation presented in this section, which makes explicit use of quasi-classicality of the motion, is due to *V. N. Baier* and *V. M. Katkov* (1967). An analogous method was used earlier by *Schwinger* (*J. Schwinger*, 1967) to obtain the first quantum correction to the intensity of radiation.

The quasi-classical wave functions of the stationary states of an electron in the external field can be symbolically represented in the form

$$\psi = (2\hat{H})^{-1/2} u(\hat{p}) \exp\left(-\frac{i}{\hbar}\hat{H}t\right) \varphi(\mathbf{r}), \quad (\text{X.35})$$

where $\varphi(\mathbf{r}) \sim \exp(iS/\hbar)$ are the quasi-classical wave functions of a spinless particle ($S(\mathbf{r})$ is its classical action); $u(\hat{p})$ is an operator bispinor

$$u(\hat{p}) = \begin{pmatrix} (\hat{H} + m)^{1/2} w \\ (\hat{H} + m)^{-1/2} (\boldsymbol{\sigma} \hat{\mathbf{p}}) w \end{pmatrix},$$

obtained from the bispinor amplitude of the plane wave $u(p)$ (23.9) by the replacement of $\mathbf{p}$ and ε by operators¹⁴

$$\hat{\mathbf{p}} = \hat{\mathbf{P}} - e\mathbf{A} = -i\hbar\nabla - e\mathbf{A}, \quad \hat{H} = (\hat{\mathbf{p}}^2 + m^2)^{1/2},$$

$\mathbf{P}$ being the generalised momentum of the particle in the field with vector potential $\mathbf{A}(\mathbf{r})$; the ordering of operator factors in ψ is inessential, since their non-commutativity is neglected; the spin state of the electron is determined by the 3-spinor w .

For computing the probability of emission of a photon it is convenient in the quasi-classical case to start not from the final formula of perturbation theory (44.3), but from the formula in which the integration over time has not yet been performed. For the complete (over all time) differential probability we have¹⁵

$$dw = \sum_f |a_{fi}|^2 \frac{d^3k}{(2\pi)^3}, \quad a_{fi} = \int_{-\infty}^{\infty} V_{fi}(t) dt \quad (\text{X.36})$$

(cf. III, (41.2)); the summation is over the final states of the electron.

Using (90.6), we write the matrix element for the emission of a photon ω , $\mathbf{k}$ in the form

$$\begin{aligned} V_{fi}(t) = & -\frac{e\sqrt{4\pi}}{\sqrt{2\hbar\omega}} \int \varphi_f^* \exp\left(\frac{i}{\hbar}\hat{H}t\right) \frac{u^\dagger(\hat{p}')}{(2\hat{H}')^{1/2}} (\mathbf{e}^* \boldsymbol{\alpha}) \frac{u(\hat{p})}{(2\hat{H})^{1/2}} \times \\ & \times e^{i\omega t - i\mathbf{k}\mathbf{r}} \exp\left(-\frac{i}{\hbar}\hat{H}t\right) \varphi_i d^3x, \end{aligned}$$

where the operators in the square brackets act to the left; the field of the photon is chosen in three-dimensional transverse gauge. The factors $\exp(\pm i\hat{H}t/\hbar)$ convert the operators standing between them into operators that are explicitly time-dependent, in the Heisenberg representation. We write $V_{fi}(t)$ in the form

$$V_{fi}(t) = e\sqrt{\frac{2\pi}{\hbar\omega}} \langle f | Q(t) | i \rangle e^{i\omega t}, \quad (\text{X.37})$$

¹⁴In this section (in contrast to Ch. IV) the generalised momentum is denoted by the capital letter $\mathbf{P}$; the symbol $\mathbf{p}$ is used for the ordinary (kinetic) momentum.

¹⁵Substituting $V_{fi}(t) = V_{fi} \exp(i\omega_{fi}t)$, we obtain $a_{fi} = 2\pi V_{fi} \delta(\omega_{fi})$. Bearing in mind that the square of the δ -function must be understood as $|\delta(\omega)|^2 \rightarrow (t/2\pi) \delta(\omega)$, where t is the total observation time (cf. the derivation of formula (64.5)), we obtain from (90.7) the probability per unit time (44.3).

where $Q(t)$ denotes the Heisenberg operator

$$\hat{Q}(t) = \frac{u_f^\dagger(\hat{p}')}{(2H')^{1/2}} (\alpha \mathbf{e}^*) e^{-i\mathbf{k}\hat{\mathbf{r}}(t)} \frac{u_i(\hat{p})}{(2H)^{1/2}}, \quad (\text{X.38})$$

and the matrix element is taken with respect to the functions φ_i, φ_f .

The summation in (90.7) is over all final wave functions φ_f ; it is carried out with the aid of the completeness relation

$$\sum_f \varphi_f^*(\mathbf{r}') \varphi_f(\mathbf{r}) = \delta(\mathbf{r}' - \mathbf{r}),$$

expressing the completeness of the system of functions φ_f . As a result,

$$dw = \frac{e^2}{\hbar\omega} \frac{d^3k}{4\pi^2} \iint dt_1 dt_2 \cdot e^{i\omega(t_1-t_2)} \langle i|Q^+(t_2)Q(t_1)|i\rangle. \quad (\text{X.39})$$

If the integration is over a sufficiently large time interval, one can introduce instead of t_1, t_2 the new variables

$$\tau = t_2 - t_1, \quad t = \frac{t_1 + t_2}{2}$$

and in the integral over dt regard the sub-integral expression as the probability of emission per unit time. Multiplying it by $\hbar\omega$, we obtain the intensity

$$dI = \frac{e^2}{4\pi^2} d^3k \int e^{-i\omega\tau} \langle i|Q^+\left(t + \frac{\tau}{2}\right) Q\left(t - \frac{\tau}{2}\right) |i\rangle d\tau. \quad (\text{X.40})$$

An ultrarelativistic electron radiates into a narrow cone at angles $\vartheta \sim m/\varepsilon$ about the direction of its velocity $\mathbf{v}$. Therefore radiation in a given direction $\mathbf{n} = \mathbf{k}/\omega$ is formed on that segment of the trajectory on which $\mathbf{v}$ turns through an angle $\sim m/\varepsilon$. This segment is traversed in a time τ such that $\tau|\dot{\mathbf{v}}| \approx \tau\omega_0 \sim m/\varepsilon \ll 1$. It is precisely this region that gives the principal contribution to the integral over τ . Therefore in the further calculations we shall systematically expand all quantities in powers of $\omega_0\tau$. In doing this, however, it may be necessary to retain more than just the leading term of the expansion because of cancellations resulting from the fact that $1 - \mathbf{n}\mathbf{v} \sim \vartheta^2 \sim (m/\varepsilon)^2$.

If we bring the operator $\hat{Q}^+\hat{Q}$ to the form of a product of commutative (to the required accuracy) operators, then taking the diagonal matrix element $\langle i|\dots|i\rangle$ reduces to replacing these operators by the classical values (functions of time) of the corresponding quantities. This goal is achieved as follows.

In the expression for $\hat{Q}(t)$ (90.8) we factor out the operator $\exp(-i\mathbf{k}\hat{\mathbf{r}}(t))$ to the left and rewrite $\hat{Q}(t)$ in the form

$$\hat{Q}(t) = \exp[-i\mathbf{k}\hat{\mathbf{r}}(t)] \hat{R}(t), \quad \hat{R}(t) = \frac{u_f^\dagger(\hat{p}')}{(2H')^{1/2}} (\alpha \mathbf{e}^*) \frac{u_i(\hat{p})}{(2H)^{1/2}}, \quad (\text{X.41})$$

where $\hat{H}' = \hat{H} - \hbar\omega$, $\hat{\mathbf{p}}' = \hat{\mathbf{p}} - \hbar\mathbf{k}$.

Now,

$$\hat{Q}_2^+ \hat{Q}_1 = \hat{R}_2 \exp(i\mathbf{k}\hat{\mathbf{r}}_2) \exp(-i\mathbf{k}\hat{\mathbf{r}}_1) \hat{R}_1 \quad (\text{X.42})$$

(here and below the subscripts 1 and 2 label values of the quantities at the times $t_1 = t - \tau/2$ and $t_2 = t + \tau/2$). It remains to compute the product of the two non-commutative operators $\exp(i\mathbf{k}\hat{\mathbf{r}}_2)$ and $\exp(-i\mathbf{k}\hat{\mathbf{r}}_1)$. This product can itself already be considered commutative with the remaining factors. Denoting

$$\hat{L}(\tau) = \exp(-i\omega\tau) \exp(i\mathbf{k}\hat{\mathbf{r}}_2) \exp(-i\mathbf{k}\hat{\mathbf{r}}_1); \quad (\text{X.43})$$

this is precisely the combination of operators that enters (90.10). By the meaning of the operator $\exp(i\hat{H}\tau/\hbar)$ as the time-shift operator we have

$$\exp(i\mathbf{k}\hat{\mathbf{r}}_2) = \exp\left(i\hat{H}\frac{\tau}{\hbar}\right) \exp(i\mathbf{k}\hat{\mathbf{r}}_1) \exp\left(-i\hat{H}\frac{\tau}{\hbar}\right).$$

Substituting this expression into (90.14) and noting that $\exp(i\mathbf{k}\hat{\mathbf{r}}_1)$ is the operator of momentum-space shift, we transform L to the form

$$\hat{L}(\tau) = \exp\left\{i\left[\hat{H} - \hbar\omega\right]\frac{\tau}{\hbar}\right\} \exp\left\{-i\hat{H}(\hat{\mathbf{p}}_1 - \hbar\mathbf{k})\frac{\tau}{\hbar}\right\}. \quad (\text{X.44})$$

Differentiating (90.15) with respect to τ and noting that $\exp(i\mathbf{k}\hat{\mathbf{r}}_1)$ is the momentum-space shift operator, we transform L to the form¹⁶

$$i\hbar \frac{d\hat{L}}{d\tau} = \frac{\varepsilon}{\varepsilon'} \hbar(\omega - \mathbf{v}_2\mathbf{k})L. \quad (\text{X.45})$$

This equation must be solved with the obvious initial condition $L(0) = 1$. Noting that

$$\int_0^\tau \mathbf{v}_2 d\tau = \mathbf{r}_2 - \mathbf{r}_1,$$

we obtain

$$L(\tau) = \exp\left\{i \frac{\varepsilon}{\varepsilon'} (\mathbf{k}\mathbf{r}_2 - \mathbf{k}\mathbf{r}_1 - \omega\tau)\right\}. \quad (\text{X.46})$$

Up to this point we have not used the specific form of the electron trajectory. Let us now express $\mathbf{r}_2 - \mathbf{r}_1$ through $\mathbf{p}_1$ with the aid of the equations of motion of the electron in the plane perpendicular to the magnetic field $\mathbf{H}$ (see II, § 21):

$$\mathbf{r}_2 - \mathbf{r}_1 = \frac{\mathbf{p}_1}{eH} \sin \frac{eH\tau}{\varepsilon} + \frac{[\mathbf{p}_1, \mathbf{H}]}{|eH|} \left(1 - \cos \frac{eH\tau}{\varepsilon}\right).$$

Expanding in powers of $\omega_0\tau$ and retaining terms up to the third order, and noting that $1 - \mathbf{nv} \sim \vartheta^2 \sim (m/\varepsilon)^2$, we obtain

$$\mathbf{k}(\mathbf{r}_2 - \mathbf{r}_1) - \omega\tau \approx \omega\tau \left\{ -\frac{i\omega\tau\varepsilon}{\varepsilon'} \left(1 - \mathbf{vn} + \frac{\tau^2}{24}\omega_0^2\right) \right\}. \quad (\text{X.47})$$

(In the last term we have set $\mathbf{nv}_1 \approx 1$.)

¹⁶By virtue of energy conservation the Heisenberg energy operators $\hat{H}(\hat{\mathbf{p}}_1)$ and $\hat{H}(\hat{\mathbf{p}}_2)$ coincide, and therefore in such cases we write the argument of $\hat{H}$ without a subscript. But, of course, $\hat{H}(\hat{\mathbf{p}}_1 - \hbar\mathbf{k})$ does not coincide with $\hat{H}(\hat{\mathbf{p}}_2 - \hbar\mathbf{k})$.

Let us now transform the remaining factors in (90.13). By direct expansion of the products in $R(t)$ (with the matrix α from (21.20)) we find

$$R(t) = w_f^\dagger \mathbf{e}^* (A + i[\mathbf{B}\sigma]) w_i,$$

$$\mathbf{A} = \frac{\mathbf{p}}{2} \left(\frac{1}{\varepsilon} + \frac{1}{\varepsilon'} \right) = \frac{\varepsilon + \varepsilon'}{2\varepsilon'} \mathbf{v}, \quad (\text{X.48})$$

$$\mathbf{B} = \frac{1}{2} \left(\frac{\mathbf{p}}{\varepsilon + m} - \frac{\mathbf{p}'}{\varepsilon' + m} \right) \approx \frac{\hbar\omega}{2\varepsilon'} \left(\mathbf{n} - \mathbf{v} + \mathbf{v} \frac{m}{\varepsilon} \right),$$

where $p'(t) = p(t) - \hbar\mathbf{k}$; terms of higher orders in m/ε are omitted. Thus, finally we have

$$\exp(-i\omega\tau) \langle i | Q_2^\dagger Q_1 | i \rangle = R_2^* R_1 L(\tau), \quad (\text{X.49})$$

$$R_2^* R_1 = \text{Sp} \frac{1 + \zeta\sigma}{2} ((\mathbf{A}_2 - i[\mathbf{B}_2\sigma])\mathbf{e}) \frac{1 + \zeta'\sigma}{2} ((\mathbf{A}_1 - i[\mathbf{B}_1\sigma])\mathbf{e}^*).$$

The factors $(1 + \zeta\sigma)/2$ are the two-row polarisation density matrices of the initial and final electrons.

Let us consider the intensity of the radiation, summed over polarisations of the photon and the final electron and averaged over polarisations of the initial electron. As a result of the indicated operations we obtain after a simple calculation¹⁷

$$\frac{1}{2} \sum_{\text{polar}} R_2^* R_1 = \frac{\varepsilon^2 + \varepsilon'^2}{2\varepsilon'^2} (\mathbf{v}_1 \mathbf{v}_2 - 1) + \frac{1}{2} \left(\frac{\hbar\omega}{\varepsilon'} \right)^2 \left(\frac{m}{\varepsilon} \right)^2.$$

To the required accuracy

$$\mathbf{v}_1 \mathbf{v}_2 = \mathbf{v}^2 - \frac{\tau^2}{4} \dot{\mathbf{v}}^2 + \frac{\tau}{4} \mathbf{v} \ddot{\mathbf{v}} = 1 - \frac{m^2}{\varepsilon^2} - \frac{1}{2} \omega_0^2 \tau^2.$$

Substituting these expressions into (90.21), and then into (90.10), we obtain

$$dI = -\frac{e^2}{4\pi^2} \omega^2 d\omega d\mathbf{o}_{\mathbf{n}} \times \int_{-\infty}^{\infty} \left(\frac{m^2}{\varepsilon\varepsilon'} + \frac{\varepsilon^2 + \varepsilon'^2}{4\varepsilon'^2} \omega_0^2 \tau^2 \right) \exp \left\{ -\frac{i\omega\tau\varepsilon}{\varepsilon'} \left(1 - \mathbf{v}\mathbf{n} + \frac{\tau^2}{24} \omega_0^2 \right) \right\} d\tau. \quad (\text{X.50})$$

This formula gives the spectral and angular distribution of the intensity of the radiation. For finding the spectral distribution of the radiation the product is integrated over $d\mathbf{o}_{\mathbf{n}}$. Choosing $\mathbf{v}$ as the polar axis with angle ϑ between $\mathbf{n}$ and $\mathbf{v}$, we have

$$\mathbf{n}\mathbf{v} = v \cos \vartheta, \quad d\mathbf{o}_{\mathbf{n}} = \sin \vartheta d\vartheta d\varphi,$$

¹⁷Here we have also used the following circumstance. Upon summation over $\mathbf{e}$: $\sum_{\mathbf{e}} = (\mathbf{v}_1 \mathbf{e})(\mathbf{v}_2 \mathbf{e}^*) = \mathbf{v}_1 \mathbf{v}_2 - (\mathbf{v}_1 \mathbf{n})(\mathbf{v}_2 \mathbf{n})$. But upon substitution of (90.21) into (90.10) one can integrate by parts, noting that $(\mathbf{v}_1 \mathbf{n}) \exp(-\frac{i\varepsilon}{\varepsilon'} \mathbf{k}\mathbf{r}_1) = \frac{i\varepsilon'}{\varepsilon\omega} \frac{d}{dt_1} \exp(-\frac{i\varepsilon}{\varepsilon'} \mathbf{k}\mathbf{r}_1)$ and analogously for $\mathbf{v}_2 \mathbf{n}$. As a result, we find that for the purpose of further integration one can replace $\mathbf{v}_1 \mathbf{n}$ and $\mathbf{v}_2 \mathbf{n}$ by unity.

and the integral

$$\int \exp\left(\frac{i\omega\tau\varepsilon}{\varepsilon'} \mathbf{n}\mathbf{v}\right) d\mathbf{o}_n = \frac{2\pi\varepsilon'}{i\omega\tau\varepsilon v} \left\{ \exp\left(\frac{i\omega\tau\varepsilon v}{\varepsilon'}\right) - \exp\left(-\frac{i\omega\tau\varepsilon v}{\varepsilon'}\right) \right\}.$$

Upon substitution of this expression into (90.22) we obtain two terms with exponents of different order of magnitude. The exponent of the second term turns out to be much larger, since it contains the factor $1 + v \approx 2$ instead of the small factor $1 - v \approx m^2/(2\varepsilon^2)$ in the first term. Shifting the contour of integration over τ to the lower half-plane of the complex variable τ , one can make the second term small and neglect it. After this one can again shift the contour of integration back to the real axis. From the conclusion it is clear, however, that the pole appearing in the sub-integral expression at $\tau = 0$ must be bypassed from below. Thus,

$$\frac{dI}{d\omega} = \frac{ie^2\omega}{2\pi} \int_{-\infty}^{\infty} \left(\frac{m^2}{\varepsilon^2\tau} + \frac{\varepsilon^2 + \varepsilon'^2}{4\varepsilon'^2} \omega_0^2 \tau^2 \right) \exp\left\{ -\frac{i\omega\tau\varepsilon}{\varepsilon'} \left(1 - v + \frac{\tau^2}{24} \omega_0^2 \right) \right\} d\tau, \quad (\text{X.51})$$

where the contour of integration is chosen as indicated above. Using the integral representation of the Airy function Φ (see III, § b), it is not difficult to show that the first term reduces to an integral of the Airy function, and the second to its derivative. We finally obtain

$$\frac{dI}{d\omega} = -\frac{e^2 m^2 \omega}{\sqrt{\pi} \varepsilon^2} \left[\int_x^{\infty} \Phi(\xi) d\xi + \left(\frac{2}{x} + \frac{\hbar\omega}{\varepsilon} \chi x^{1/2} \right) \Phi'(x) \right], \quad (\text{X.52})$$

$$x = \left(\frac{\hbar\omega}{\varepsilon' \chi} \right)^{2/3} = \frac{m^2}{\varepsilon^2} \left(\frac{\varepsilon\omega}{\varepsilon' \omega_0} \right)^{2/3}, \quad (\text{X.53})$$

(A. I. Nikishov, V. I. Ritus, 1967). The maximum in the frequency distribution lies at $x \sim 1$; when $\chi \ll 1$ it follows from this that (90.1), and when $\chi \gg 1$ —(90.4). In the classical limiting case we have $\hbar\omega \ll \varepsilon$, so that $\varepsilon' \approx \varepsilon$, $x \approx (\omega/\omega_0)^{2/3} (m/\varepsilon)^2$; the second term in the square brackets is small and (90.23) goes over into the classical formula (74.13) (see II).

The quantity I_{cl} is the classical total intensity of the radiation (cf. II, (74.2)).

For computing the total intensity of the radiation, expression (90.23) must be integrated over ω from 0 to ε . Passing to integration over x , noting that

$$\hbar\omega = \varepsilon \left(1 - \frac{1}{1 + \chi x^{3/2}} \right),$$

and therefore x varies from 0 to ∞ . Carrying out in the first term of (90.23) two integrations by parts, we obtain

$$I = -\frac{e^2 m^2 \chi^2}{2\sqrt{\pi} \hbar^2} \int_0^{\infty} \frac{4 + 5\chi x^{3/2} + 4\chi^2 x^3}{(1 + \chi x^{3/2})^4} \Phi'(x) x dx. \quad (\text{X.54})$$

In Fig. 16 the graph of the function $I(\chi)/I_{cl}$ is shown.

When $\chi \ll 1$, in the integral the essential region is $x \sim 1$. Expanding the sub-integral expression in χ and integrating this expansion with the aid of the formula

$$\int_0^\infty x^\nu \Phi'(x) dx = -\frac{1}{2\sqrt{\pi}} 3^{(\nu-1)/6} \Gamma\left(\frac{\nu}{3} + 1\right) \Gamma\left(\frac{\nu}{3} + \frac{1}{3}\right),$$

we obtain

$$I = I_{cl} \left(1 - \frac{55\sqrt{3}}{16} \chi + 48\chi^2 - \dots \right). \quad (\text{X.55})$$

When $\chi \gg 1$, in the integral the essential region is that in which $\chi x^{3/2} \sim 1$, i.e. $x \ll 1$. In the first approximation one can therefore replace $\Phi'(x)$ by $\Phi'(0) = -3^{1/6} \Gamma(2/3)/(2\sqrt{\pi})$, after which integration gives

$$I \approx \frac{32\Gamma(2/3)e^2 m^2}{243\hbar^2} (3\chi)^{2/3} = 0.37 \frac{e^2 m^2}{\hbar^2} \left(\frac{H\varepsilon}{H_0 m} \right)^{2/3}. \quad (\text{X.56})$$

Magnetobremssstrahlung radiation leads to the appearance of polarisation of the moving electrons (*A. A. Sokolov, I. M. Ternov,*

1963). To examine this question one must find the probability of a radiative transition with reversal of the spin direction. Setting in (90.21) $\zeta_i = -\zeta_f \equiv \zeta$, $|\zeta| = 1$, we obtain

$$\begin{aligned} R_2^* R_1 &= (\mathbf{B}_1 \mathbf{B}_2) - (\mathbf{e}^* \mathbf{B}_2)(\mathbf{e} \mathbf{B}_2) - \\ &- (\mathbf{e}^* [\mathbf{B}_1 \zeta])(\mathbf{e} [\mathbf{B}_2 \zeta]) - i(\zeta \mathbf{e}^*)(\mathbf{e} [\mathbf{B}_1 \mathbf{B}_2]). \end{aligned}$$

Summation over the polarisations of the photon gives after straightforward transformations

$$\sum_{\mathbf{e}} R_2^* R_1 = (\mathbf{B}_1 \mathbf{B}_2)(1 - (\zeta \mathbf{n})^2) + (\zeta \mathbf{n})(\mathbf{n} \mathbf{B}_2)(\zeta \mathbf{B}_1) - i(\zeta - \mathbf{n}(\mathbf{n} \zeta))[\mathbf{B}_1 \mathbf{B}_2]. \quad (\text{X.57})$$

We shall assume that $\chi \ll 1$, and seek only the leading term of the expansion of the probability in powers of $\hbar$. Since $\mathbf{B}$ (from (90.20)) already contains $\hbar^2$, all remaining quantities (including those in the exponent in (90.18)) can be replaced by ε .

Expanding

$$\begin{aligned} \mathbf{B}_1 &= \frac{\omega}{2\varepsilon} \left(\mathbf{n} - \mathbf{v} + \frac{\tau}{2} \dot{\mathbf{v}} + \mathbf{v} \frac{m}{\varepsilon} \right), \\ \mathbf{B}_2 &= \frac{\omega}{2\varepsilon} \left(\mathbf{n} - \mathbf{v} - \frac{\tau}{2} \dot{\mathbf{v}} + \mathbf{v} \frac{m}{\varepsilon} \right), \\ \mathbf{r}_2 - \mathbf{r}_1 &= \tau \mathbf{v} + \frac{\tau^3}{24} \ddot{\mathbf{v}} \end{aligned}$$

and substituting (90.28) into (90.21) and then into (90.10), we find the differential probability of the transition per unit time ($dw = dI/\hbar\omega$). It is integrated with the aid of formula

$$\int f(k_\mu) e^{-ikx} \frac{d^3 k}{\omega} = -f(i\partial_\mu) \frac{4\pi}{x_0^2 - \mathbf{x}^2}, \quad (\text{X.58})$$

where in the present case

$$x_0 = \tau, \quad \mathbf{x} = \mathbf{r}_2 - \mathbf{r}_1, \quad x^2 = x_0^2 - \mathbf{x}^2 = \tau^2 \left(\frac{m^2}{\varepsilon^2} + \frac{\tau^2 \omega_0^2}{12} \right).$$

The computation leads to the result

$$w = \frac{5\sqrt{3}\alpha}{16} \frac{\hbar^2}{m^2} \left(\frac{\varepsilon}{m} \right)^5 \omega_0^3 \left(1 - \frac{2}{9} \zeta_{\parallel}^2 - \frac{8\sqrt{3}}{15} \frac{\varepsilon}{|e|} \zeta_{\perp} \right), \quad (\text{X.59})$$

where $\zeta_{\parallel} = \boldsymbol{\zeta} \mathbf{v}$, $\zeta_{\perp} = \boldsymbol{\zeta} \mathbf{H}/H$. This formula is valid both for electrons ($e < 0$) and for positrons ($e > 0$).

The probability (90.30) does not depend on the sign of the longitudinal polarisation $\zeta_{\parallel}$, but does depend on the sign of $\zeta_{\perp}$. Therefore the polarisation arising as a result of the radiation is transverse. For electrons the probability of transition from the state with spin “along the field” ($\zeta_{\perp} = 1$) to the state with spin “against the field” is greater than the probability of the reverse transition. Therefore the radiative polarisation of electrons is directed against the field, and its degree in the stationary state (at $\zeta_{\parallel} = 0$) is

$$\frac{w(\zeta_{\perp} = -1) - w(\zeta_{\perp} = 1)}{w(\zeta_{\perp} = -1) + w(\zeta_{\perp} = 1)} = \frac{8\sqrt{3}}{15} = 0.92.$$

Positrons are polarised (to the same degree) in the direction of the field.

§ 91. Pair production by a photon in a magnetic field

The production of an electron-positron pair by a photon in a magnetic field and magnetobremssstrahlung radiation are two crossed channels of one and the same reaction. Therefore the amplitude M_{fi} of the process of pair production is obtained from the amplitude of bremsstrahlung radiation simply by the substitutions

$$\varepsilon, \mathbf{p} \rightarrow -\varepsilon_+, -\mathbf{p}_+; \quad \varepsilon', \mathbf{p}' \rightarrow \varepsilon_-, -\mathbf{p}_-; \quad \omega, \mathbf{k} \rightarrow -\omega, -\mathbf{k} \quad (\text{X.60})$$

(here $\varepsilon_-, \mathbf{p}_-$ and $\varepsilon_+, \mathbf{p}_+$ are the energies and momenta of the electron and positron in the pair; $\varepsilon, \mathbf{p}$ and $\varepsilon', \mathbf{p}'$ are the initial and final energies and momenta of the electron in bremsstrahlung radiation)¹⁸. In terms of angles and absolute values the transformation of momenta is

$$|\mathbf{p}| \rightarrow |\mathbf{p}_+|, \quad |\mathbf{p}'| \rightarrow |\mathbf{p}_-|, \quad \theta \rightarrow \pi - \theta_+, \quad \theta' \rightarrow \theta_-, \quad \varphi \rightarrow \varphi - \pi, \quad (\text{X.61})$$

where $\theta_{\pm}$ are the angles between the momenta of the electron and positron and $\mathbf{k}$, and φ is the angle between the planes $\mathbf{k}, \mathbf{p}_+$ and $\mathbf{k}, \mathbf{p}_-$. In the case of bremsstrahlung radiation the cross section is expressed

through the amplitude by the formula¹⁹

$$d\sigma = |M_{fi}|^2 \frac{1}{8|\mathbf{p}|\varepsilon'\omega} \delta(\varepsilon - \varepsilon' - \omega) \frac{d^3 p' d^3 k}{(2\pi)^5} \quad (\text{X.62})$$

¹⁸This circumstance is, however, clear in advance: the axial vector of the arising polarisation can only be directed along the sole axial vector $\mathbf{H}$ figuring in the problem.

¹⁹In this section we again set not only $c = 1$, but also $\hbar = 1$.

(see (64.25)); the δ -function is removed by integration over ε' . Noting that

$$d^3 p' = |\mathbf{p}'| \varepsilon' d\varepsilon' d\mathbf{o}', \quad d^3 k = \omega^2 d\omega d\mathbf{o}_k,$$

we must simply replace

$$\delta(\varepsilon - \varepsilon' - \omega) d^3 p' d^3 k \rightarrow \omega^2 |\mathbf{p}'| \varepsilon' d\mathbf{o}_k d\mathbf{o}' d\omega.$$

Then

$$d\sigma = |M_{fi}|^2 \frac{\omega |\mathbf{p}'|}{8(2\pi)^5 |\mathbf{p}|} d\mathbf{o}_k d\mathbf{o}' d\omega. \quad (\text{X.63})$$

In the case of pair production by a photon the cross section is expressed through the amplitude as

$$d\sigma = |M_{fi}|^2 \frac{1}{8\omega\varepsilon_+\varepsilon_-} \delta(\omega - \varepsilon_+ - \varepsilon_-) \frac{d^3 p_+ d^3 p_-}{(2\pi)^5}$$

or, after elimination of the δ -function:

$$d\sigma = |M_{fi}|^2 \frac{|\mathbf{p}_+||\mathbf{p}_-|}{8(2\pi)^5 \omega} d\mathbf{o}_+ d\mathbf{o}_- d\varepsilon_+. \quad (\text{X.64})$$

Comparing with (91.4), we see that to obtain the cross section of pair production from the cross section of bremsstrahlung radiation one must carry out the substitution (91.1) and multiply the latter by

$$\frac{\mathbf{p}_+^2 d\varepsilon_+}{\omega^2 d\omega} \quad (\text{X.65})$$

and replace $d\mathbf{o}' d\mathbf{o}_k$ by $d\mathbf{o}_+ d\mathbf{o}_-$.

In the ultrarelativistic case ($\omega \gg m$)²⁰ this can be done in the formulas obtained in the preceding section. It is assumed that both particles of the pair are ultrarelativistic; it is easy to verify that in such a case all the approximations used in § 90 remain valid.

In particular, the probability of pair production by an unpolarised photon, summed over the spin projections of the electron and positron and integrated over the emission directions of the electron, is obtained by performing the substitution (91.1) in the formula (90.22) (more precisely, in the expression for dI/ω), where $d^3 k = \omega^2 d\omega d\mathbf{o}_n$ is replaced by $d^3 p_+$:

$$\begin{aligned} d\omega = & \frac{e^2}{4\pi^2} \frac{d^3 p_+}{\omega} \int_{-\infty}^{\infty} \left(\frac{m^2}{\varepsilon_+ \varepsilon_-} - \frac{\varepsilon_+^2 + \varepsilon_-^2}{4\varepsilon_-^2} \omega_{0+}^2 \tau^2 \right) \times \\ & \times \exp \left\{ i \frac{\omega \tau \varepsilon_+}{\varepsilon_-} \left(1 - \mathbf{n} \mathbf{v}_+ + \frac{\tau^2}{24} \omega_{0+}^2 \right) \right\} d\tau, \end{aligned} \quad (\text{X.66})$$

where $\omega_{0+} = |e|H/\varepsilon_+$; $\mathbf{n}$ is the unit vector in the direction of the photon momentum, lying in the plane perpendicular to the magnetic field. The integration is performed in the same way as in § 90, since (expression (91.7)) depends only on the angle between $\mathbf{n}$

²⁰More precisely, one should have $\omega \sin \vartheta \gg m$, where ϑ is the angle between $\mathbf{k}$ and $\mathbf{H}$; when $\vartheta = 0$ pairs are not produced at all. Below we set $\vartheta = \pi/2$.

and $\mathbf{v}_+$ —one can integrate indifferently over do_+ or over $do_{\mathbf{n}}$. The result can therefore be obtained directly by analogy with (90.23):

$$dw = \frac{m^2 e^2 d\varepsilon_+}{\sqrt{\pi} \omega^2} \left[\int_x^\infty \Phi(\xi) d\xi + \left(\frac{2}{x} - \kappa x^{1/2} \right) \Phi'(x) \right], \quad (\text{X.67})$$

where now

$$x = \left(\frac{m^3 \omega}{|e| H \varepsilon_+ \varepsilon_-} \right)^{2/3}, \quad \kappa = \frac{|e| H \omega}{m^3} \left(= \frac{\hbar^2 |e| H \omega}{m^3 c^5} \right). \quad (\text{X.68})$$

The total probability of pair production per unit time is obtained by integrating (91.8) over ε_+ (and, in view of the obvious symmetry with respect to ε_+ and $\varepsilon_- = \omega - \varepsilon_+$, it suffices to integrate from 0 to $\omega/2$ and then double the result). Performing the change of the integration variable from ε_+ to x and integrating the first term in (91.8) twice by parts, we obtain

$$w = -\frac{|e|^3 H}{m \kappa \sqrt{\pi}} \int_{(4/\kappa)^{2/3}}^\infty \frac{2(x^{3/2} + 1/\kappa)}{x^{11/4}(x^{3/2} - 4/\kappa)^{1/2}} \Phi'(x) dx \quad (\text{X.69})$$

(A. I. Nikishov, V. I. Ritus, 1967).

In the limiting case of weak fields ($\kappa \ll 1$) the essential values of x in the integral (91.10) are those near the lower limit. Since these values are large, one can use the asymptotic expression for the Airy function

$$\Phi(x) \approx \frac{1}{2x^{1/4}} \exp\left(-\frac{2}{3}x^{3/2}\right)$$

(see III, § b). Introducing the variable $y = x^{3/2} - 4/\kappa$ and setting $y = 0$ everywhere possible, we obtain in the result

$$w = \frac{3^{3/2} |e|^3 H}{2^{5/2} m} \exp\left(-\frac{8}{3\kappa}\right), \quad \kappa \ll 1. \quad (\text{X.70})$$

The exponential decrease of the probability when $\kappa \rightarrow 0$ corresponds to the impossibility of pair production in the classical limit.

In the opposite limiting case of strong fields ($\kappa \gg 1$) only the second term is essential in the integral (91.10), and it is determined by the region of values of x in which $x^{3/2} \sim 1/\kappa \ll 1$. In this region one can replace $\Phi'(x)$ by its value

$$\Phi'(0) = -\frac{3^{1/6} \Gamma(2/3)}{2\pi^{1/2}}.$$

Using the value of the integral

$$\int_1^\infty y^{-\nu} (y-1)^{\mu-1} dy = \frac{\Gamma(\nu-\mu) \Gamma(\mu)}{\Gamma(\nu)},$$

we obtain

$$w = \frac{3^{1/6} \cdot 5 \Gamma^2(2/3)}{2^{4/3} \cdot 7 \pi^{1/2} \Gamma(7/6)} \frac{|e|^3 H}{m \kappa^{1/3}} = 0.38 \frac{|e|^3 H}{m \kappa^{1/3}}, \quad \kappa \gg 1. \quad (\text{X.71})$$

The function $m w(x)/|e|^3 H$ has a maximum of 0.11 at $\kappa \approx 11$.

§ 92. Bremsstrahlung of an electron on a nucleus. Non-relativistic case

This and several of the following sections are devoted to the important phenomenon of *bremsstrahlung*, the radiation accompanying the collision of particles. We begin with the non-relativistic case, i.e. we consider the radiation produced during scattering of an electron in the Coulomb field of a stationary centre (*A. Sommerfeld*, 1931).

We start from formula (45.5) for the probability of dipole radiation

$$dw = \frac{\omega^3}{2\pi} |\mathbf{e}^* \mathbf{d}_{fi}|^2 d\mathbf{k}. \quad (\text{X.72})$$

In the present case the initial and final states of the electron belong to the continuous spectrum, and the photon frequency is

$$\omega = \frac{1}{2m} (\mathbf{p}^2 - \mathbf{p}'^2), \quad (\text{X.73})$$

where $\mathbf{p} = m\mathbf{v}$ and $\mathbf{p}' = m\mathbf{v}'$ are the initial and final momenta of the electron. If the initial and final electron wave functions are normalised to one particle in a volume $V = 1$, then expression

(92.1), multiplied by $d^3 p' / (2\pi)^3$ and divided by the density of the incident flux $v/V = v$, gives the cross section $d\sigma_{\mathbf{k}\mathbf{p}'}$ of the radiation of a photon $\mathbf{k}$ in the solid angle $d\mathbf{k}$ with the scattering of the electron into the interval of final-state momenta $d^3 p'$. Replacing the matrix element of the dipole moment $\mathbf{d} = e\mathbf{r}$ by the matrix element of momentum according to

$$\mathbf{d}_{fi} = -\frac{1}{i\omega} \frac{e}{m} \mathbf{p}_{fi},$$

we write the expression for the cross section in the form

$$d\sigma_{\mathbf{k}\mathbf{p}'} = \frac{\omega e^2}{(2\pi)^4 m p} |\mathbf{e}^* \mathbf{p}_{fi}|^2 d\mathbf{k} d^3 p', \quad (\text{X.74})$$

where

$$\mathbf{p}_{fi} = \int \psi_f^* \hat{\mathbf{p}} \psi_i d^3 x = -i \int \psi_f^* \nabla \psi_i d^3 x. \quad (\text{X.75})$$

The functions ψ_i and ψ_f must be taken as exact wave functions in the Coulomb field of attraction, those functions which asymptotically contain a plane wave and a spherical wave; in ψ_f the spherical wave must be converging, and in ψ_i —diverging (see III, § 136). These functions have the form

$$\begin{aligned} \psi_i &= A_i e^{i\mathbf{p}\mathbf{r}} F(i\nu, 1, i(pr - \mathbf{p}\mathbf{r})), & \nu &= \frac{Ze^2 m}{p}, \\ \psi_f &= A_f e^{i\mathbf{p}'\mathbf{r}} F(-i\nu', 1, -i(p'r + \mathbf{p}'\mathbf{r})), & \nu' &= \frac{Ze^2 m}{p'}, \end{aligned} \quad (\text{X.76})$$

with the normalisation coefficients

$$A_i = e^{\pi\nu/2} \Gamma(1 - i\nu), \quad A_f = e^{\pi\nu'/2} \Gamma(1 + i\nu'). \quad (\text{X.77})$$

Noting that

$$\nabla F(iv, 1, i(pr - \mathbf{p}\mathbf{r})) = i \left(p \frac{\mathbf{r}}{r} - \mathbf{p} \right) F' = - \left(\frac{\partial F}{\partial \mathbf{p}} \right),$$

we write $\nabla \psi_i$ in the form

$$\nabla \psi_i = i \mathbf{p} \psi_i - A_i e^{i \mathbf{p} \mathbf{r}} \frac{p}{r} \left(\frac{\partial F}{\partial \mathbf{p}} \right).$$

Upon multiplying by ψ_f^* and integrating, the first term vanishes by virtue of the orthogonality of ψ_i and ψ_f . Therefore for the matrix element of momentum we have

$$\mathbf{p}_{fi} = i A_i A_f p \frac{\partial J}{\partial \mathbf{p}}, \quad (\text{X.78})$$

where

$$J = \int \frac{e^{-i \mathbf{q} \mathbf{r}}}{r} F(iv', 1, i(p'r + \mathbf{p}'\mathbf{r})) F(iv, 1, i(pr - \mathbf{p}\mathbf{r})) d^3x, \quad \mathbf{q} = \mathbf{p}' - \mathbf{p}. \quad (\text{X.79})$$

We have taken $\partial/\partial \mathbf{p}$ out from under the integral sign, bearing in mind that upon differentiating J the quantities $\nu, \nu', \mathbf{q}$ must be treated as independent parameters and only after the differentiation should one express ν and $\mathbf{q}$ through $\mathbf{p}$.

The integral is computed by replacing each of the degenerate hypergeometric functions by its expression in the form of a contour integral. We give here only the result:

$$\begin{aligned} J &= BF(iv', iv, 1, z), \\ B &= 4\pi e^{-\pi\nu} (-\mathbf{q}^2 - 2\mathbf{q}\mathbf{p})^{-i\nu} (\mathbf{q}^2 - 2\mathbf{q}\mathbf{p}')^{-i\nu'} (\mathbf{q}^2)^{i\nu+i\nu'-1}, \\ z &= 2 \frac{\mathbf{q}^2(p\mathbf{p}' + \mathbf{p}\mathbf{p}') - 2(\mathbf{q}\mathbf{p})(\mathbf{q}\mathbf{p}')}{(\mathbf{q}^2 - 2\mathbf{q}\mathbf{p}')(\mathbf{q}^2 + 2\mathbf{q}\mathbf{p})}. \end{aligned} \quad (\text{X.80})$$

Here $F(iv', iv, 1, z)$ is the complete hypergeometric function.

After the differentiation in (92.7) one can set $\mathbf{q} = \mathbf{p}' - \mathbf{p}$; then

$$z = -2 \frac{p\mathbf{p}' - \mathbf{p}\mathbf{p}'}{(p - p')^2}, \quad \mathbf{q}^2 = (p - p')^2(1 - z) \quad (\text{X.81})$$

($z < 0$). We note also that

$$-\mathbf{q}^2 - 2\mathbf{q}\mathbf{p} = \mathbf{q}^2 - 2\mathbf{q}\mathbf{p}' = \mathbf{p}^2 - \mathbf{p}'^2 > 0.$$

As a result we find for the matrix element the following final expression:

$$\begin{aligned} \mathbf{p}_{fi} &= A_i A_f \frac{8\pi i e^{-\pi\nu}}{(p - p')^3(p + p')} \left(\frac{p + p'}{p - p'} \right)^{-i(\nu+\nu')} \times \\ &\times (1 - z)^{i(\nu+\nu')-1} [i\nu p \mathbf{q} F'(z) + (1 - z) F'(z)(p'\mathbf{p} - p\mathbf{p}')], \end{aligned} \quad (\text{X.82})$$

where for brevity we have denoted

$$F(z) = F(iv', iv, 1, z). \quad (\text{X.83})$$

The cross section is obtained by substitution of (92.10) into (92.3), but the general formula is very cumbersome and difficult to survey. Therefore we proceed directly to computing the spectral distribution of the radiation, i.e. we integrate the cross section over the directions of the photon and the final electron.

Integration over $do_{\mathbf{k}}$ and summation over the polarisations of the photon reduces to averaging over all directions of $\mathbf{e}$ and multiplication by $2 \cdot 4\pi$, i.e. to the replacement

$$e_i e_k^* do_{\mathbf{k}} \rightarrow \frac{8\pi}{3} \delta_{ik}.$$

After this the cross section becomes

$$d\sigma_{\mathbf{p}'} = \frac{\omega e^2 p'}{6\pi^3 p} |\mathbf{p}_{fi}|^2 d\omega do_{\mathbf{p}'}. \quad (\text{X.84})$$

The square $|\mathbf{p}_{fi}|^2$ is computed using (92.9)–(92.11), and taking into account that

$$|\Gamma(1 - i\nu)|^2 = \frac{\pi\nu}{\sinh \pi\nu}.$$

We obtain

$$|\mathbf{p}_{fi}|^2 = \frac{2^8 \pi^4 (Ze^2)^2 m^2}{(p + p')^2 (p - p')^4 (1 - e^{-2\pi\nu})(e^{2\pi\nu} - 1)} \times \\ \times \left\{ \frac{\nu\nu'}{1 - z} |F|^2 - z|F'|^2 + i \frac{\nu + \nu'}{2} \frac{z}{1 - z} (FF'^* - F^*F') \right\}. \quad (\text{X.85})$$

For integrating the cross section (92.12), $do_{\mathbf{p}'} = 2\pi \sin \vartheta d\vartheta$ where ϑ is the scattering angle, we pass from ϑ to the variable

$$z = -\frac{2pp'}{(p - p')^2} (1 - \cos \vartheta), \quad do_{\mathbf{p}'} \rightarrow \frac{\pi(p - p')^2}{pp'} dz.$$

To take the integral over dz , we transform the expression in curly brackets in (92.13). According to the differential equation for hypergeometric functions (see III, (e. 2)) we have

$$z(1 - z)F'' + [1 - (1 + i\nu + i\nu')z]F' + \nu\nu'F = 0, \\ z(1 - z)F''^* + [1 - (1 - i\nu - i\nu')z]F'^* + \nu\nu'F^* = 0.$$

Multiplying these two equations by F^* and F respectively and adding, we obtain

$$(1 - z) \left[\frac{d}{dz} z(F'F^* + F'^*F) - 2z|F'|^2 + \frac{i(\nu + \nu')z}{1 - z} (F'^*F + F^*F') + \frac{2\nu\nu'}{1 - z} |F|^2 \right] = 0.$$

From this one sees that the expression in curly brackets in (92.13) equals

$$\{\dots\} = -\frac{1}{2} \frac{d}{dz} z(F'F^* + F'^*F) \quad (\text{X.86})$$

and integrates directly.

Collecting the obtained formulas, we find the final expression for the cross section of bremsstrahlung radiation in the frequency interval $d\omega$ ²¹:

$$d\sigma_\omega = \frac{64\pi^2}{3} Z^2 \alpha r_e^2 \frac{m^2 c^2}{(p - p')^2} \frac{p'}{p} \frac{1}{(1 - e^{-2\pi\nu'})(e^{2\pi\nu} - 1)} \times \left(-\frac{d}{d\xi} |F(\xi)|^2 \right) \frac{d\omega}{\omega}, \quad (\text{X.87})$$

where

$$\nu = \frac{Z\alpha mc}{p} = \frac{Ze^2}{\hbar v}, \quad \nu' = \frac{Ze^2}{\hbar v'}, \quad p' = \sqrt{p^2 - 2m\hbar\omega},$$

$$F(\xi) = F(i\nu', i\nu, 1, \xi), \quad \xi = -\frac{4pp'}{(p - p')^2}.$$

Let us consider the limiting case when both velocities ν and ν' are so large that $\nu \ll 1$, $\nu' \ll 1$ (but still, of course, $\nu \ll 1$, so that $Z\alpha \ll \nu \ll 1$; this is possible only for small Z). For computing the derivative $F'(\xi)$ in this case we use the formula

$$\frac{d}{dz} F(\alpha, \beta, \gamma, z) = \frac{\alpha\beta}{\gamma} F(\alpha + 1, \beta + 1, \gamma + 1, z),$$

which is easily obtained by direct differentiation of the hypergeometric series. We have

$$F'(\xi) = i\nu \cdot i\nu' F(1, 1, 2, \xi) = \frac{i\nu\nu'}{\xi} \ln(1 - \xi)$$

(the last equality follows from direct comparison of the corresponding series). For the function $F(\xi)$ itself we have simply

$$F(\xi) \approx F(0, 0, 1, \xi) = 1.$$

As a result, from (92.15) we find

$$d\sigma_\omega = \frac{16}{3} Z^2 \alpha r_e^2 \frac{c^2}{v^2} \ln \frac{v + v'}{v - v'} \frac{d\omega}{\omega}, \quad \frac{Ze^2}{\hbar v} \ll 1, \quad \frac{Ze^2}{\hbar v'} \ll 1. \quad (\text{X.88})$$

The smallness of ν and ν' is precisely the condition for applicability of the Born approximation in the case of Coulomb interaction. Therefore formula (92.16) itself can be more simply obtained directly with the aid of perturbation theory (see Problem 1).

Now let the fast ($\nu \ll 1$) electron lose a considerable fraction of its energy to radiation, so that $\nu' \ll \nu$ and ν' may not be small. Then

$$-\xi \approx 4p'/p = 4\nu/\nu' \ll 1, \quad F(\xi) \approx F(i\nu', 0, 1, \xi) = 1,$$

$$F'(\xi) = -\nu\nu' F(1 + i\nu', 1, 2, \xi) \approx -\nu\nu',$$

and the cross section

$$d\sigma_\omega = \frac{64\pi^3}{3} Z^3 \alpha^2 r_e^2 \left(\frac{c}{v} \right)^3 \frac{1}{1 - \exp(-2\pi Ze^2/\hbar v')} \frac{d\omega}{\omega}, \quad (\text{X.89})$$

²¹Formulas (92.15) and (92.25) are written in ordinary units.

$$\frac{Ze^2}{\hbar v} \ll 1, \quad \frac{Ze^2}{\hbar v'} \gtrsim 1.$$

When $v' \ll 1$ this formula gives the same limiting expression

$$d\sigma_\omega = \frac{32}{3} Z^2 \alpha r_e^2 \frac{c^2 v'}{v^3} \frac{d\omega}{\omega} \quad (\text{X.90})$$

as formula (92.16) when $v' \ll v$. Therefore formulas (92.16) and (92.17) together cover (for $v \ll 1$) the entire range of values of v' .

When $\omega \rightarrow \omega_0$ (where $\hbar\omega_0 = mv^2/2$) the velocity $v' \rightarrow 0$ and $v' \rightarrow \infty$. In this limit (92.17) gives

$$d\sigma_\omega = \frac{128\pi}{3} Z^3 \alpha^2 r_e^2 \left(\frac{c}{v}\right)^3 \frac{\hbar d\omega}{mv^2}. \quad (\text{X.91})$$

Thus, $d\sigma_\omega/d\omega$ tends to a finite limit as $\omega \rightarrow \omega_0$. This circumstance can be justified in general terms by arguments analogous to those set out in vol. III, § 147. Physically it is related to the fact that the frequency $\omega = \omega_0$ is the boundary of a continuous bremsstrahlung spectrum only. The electron can also radiate at the frequency $\omega > \omega_0$, passing into a bound state. But strongly excited bound states in the Coulomb field are so similar in their properties to states near the boundary of the free spectrum that the continuous spectrum boundary is not a physically distinguished point.

Let us pass to the case when both parameters $v, v' \gg 1$. In this case the motion of both the initial and the final electrons is quasi-classical. We shall assume that $p^2/(2m) \sim \hbar\omega$; then we shall need the asymptotic expression for the function $F(\xi)$ when $v, v' \rightarrow \infty$ and $\xi \sim 1$ (the more precise condition will be formulated below, see (92.24)).

For obtaining this expression we start from the integral representation of the hypergeometric function (e. 3) (see III), which we write in the form

$$F(i\eta v', iv', 1, \xi) = \frac{e^{-\pi\eta v'}}{2\pi i} \oint e^{v' f(t)} \frac{dt}{t}, \quad (\text{X.92})$$

where we have introduced the notation

$$\eta = v/v', \quad 0 < \eta < 1, \quad (\text{X.93})$$

so that

$$\xi = -\frac{4\eta}{(1-\eta)^2}. \quad (\text{X.94})$$

As integration contour we choose the path shown in Fig. 17, passing along a segment of the real axis and going around the points $t = 0$ and $t = 1$ ²².

When $v, v' \gg 1$ the value of the sub-integral expression on the lower part of this contour is small and can be neglected: upon going around the point $t = 0$ from above downwards the sub-integral expression is multiplied by the small factor $\exp(-2\pi\eta v')$,

²²For the hypergeometric function $F(\alpha, \beta, \gamma, \xi)$ the contour must be chosen so that upon going around it the function $V(t) = e^t t^{\gamma-\beta-1} (t-1)^{1-\alpha}$ returns to its original value. When γ is integer (in the present case $\gamma = 1$) the indicated contour satisfies this condition.

while upon going around the point $t = 1$ from below upwards it is multiplied by $\exp(2\pi\eta\nu')$. The integral

$$E = \frac{e^{-\pi\eta\nu'}}{2\pi i} \int e^{\nu' f(t)} \frac{dt}{t}, \quad f(t) = i \ln \frac{t^\eta}{(1-t)^\eta (1-t\xi)^{i\nu'}},$$

is evaluated by the saddle-point method. The saddle point t_0 is determined by the condition $f'(t_0) = 0$, whence $t_0 = (1-\eta)/2$. At this point, however, f'' also vanishes, as does the derivative $f'''(t_0)$, so that one must write

$$f(t) \approx f(t_0) + \frac{ia}{3} \tau^3, \quad \tau = t - t_0,$$

where

$$f(t_0) = 2\pi\eta i + i(1+\eta) \ln \frac{1-\eta}{1+\eta}, \quad a = \frac{1}{2} f'''(t_0) = \frac{16\eta}{(1-\eta^2)^2}.$$

The pre-exponential factor $1/t$ in the sub-integral expression is written as

$$\frac{1}{t} \approx \frac{1}{t_0} - \frac{\tau}{t_0^2}$$

(one cannot restrict oneself to the term $1/t_0$ alone, as this would lead to the vanishing of the derivative $d|F(\xi)|^2/d\xi$ appearing in (92.15)). Thus, we find, after an obvious substitution, in the integrals,

$$F \approx \frac{1}{2\pi i t_0 (\alpha\nu')^{1/3}} \exp\{-\pi\eta\nu' + \nu' f(t_0)\} \times \\ \times \left[- \int_{-\infty}^{\infty} e^{ix^3/3} dx + \frac{i}{t_0 (\alpha\nu')^{1/3}} \int_{-\infty}^{\infty} x e^{ix^3/3} dx \right]. \quad (\text{X.95})$$

The integrals here are respectively

$$2 \int_0^{\infty} \cos \frac{x^3}{3} dx = \frac{2\pi}{3^{2/3} \Gamma(2/3)}, \quad 2 \int_0^{\infty} x \sin \frac{x^3}{3} dx = 3^{1/6} \Gamma(2/3).$$

In an analogous fashion the derivative $F'(\xi)$ is computed; according to (92.19) it is given by an integral differing from (92.21) only by the

replacement of the pre-exponential factor $1/it$ by $\nu'/(l-\xi t)$. After this straightforward calculation we arrive at the result

$$-\frac{d}{d\xi} |F(\xi)|^2 = \frac{(1-\eta)^2 e^{2\pi\nu}}{4\sqrt{3}\pi\nu}.$$

Finally, substituting this expression into formula (92.15), we find, to the required accuracy, the following simple result:

$$d\sigma_\omega = \frac{16\pi}{3^{3/2}} Z^2 \alpha r_e^2 \frac{m^2 c^2}{p^2} \frac{d\omega}{\omega}. \quad (\text{X.96})$$

The condition for applicability of this formula, i.e. the condition for applicability of the asymptotic expression (92.23), consists in requiring the smallness of the second term

in comparison with the first: $(1 - \eta)v \gg 1$, or, after expressing the parameters of the hypergeometric function through the physical quantities:

$$\omega \gg \frac{v}{Ze^2} \frac{mv^2}{2}. \quad (\text{X.97})$$

Condition (92.24) coincides with that defining the “high-frequency limit” in the classical radiation in the Coulomb field of attraction, and the quantity $\hbar\omega d\sigma_\omega$ from (92.23) coincides with the expression (70.22) (see II) for the “effective retardation” in this limit. This result needs some discussion. It might seem that for the applicability of the classical radiation formula one needs, besides the quasi-classicality of the motion, also the smallness of the energy of the quantum compared to the energy of the electron, i.e. the condition $\hbar\omega \ll mv^2/2$, which was not assumed in the derivation of (92.23). In reality, however, the value $\hbar\omega$ need not be small compared with the energy of the electron at infinity, but rather compared with its kinetic energy on that segment of the trajectory where the radiation occurs in the main. This energy is much larger than the initial one, because of the acceleration of the electron by the field of the ion.

Indeed, the radiation of high frequencies occurs mainly at small distances from the ion, where

$$v(r)/r \sim \omega. \quad (\text{X.98})$$

(We have denoted by $v(r)$ the velocity of the electron at distance r from the ion, in contrast to the velocity v at infinity.) Taking into account that here $Ze^2/r \sim mv^2(r)$, we find that the kinetic energy on the segment where the radiation is produced is

$$\frac{mv^2(r)}{2} \sim \frac{m}{2} \left(\frac{\omega Ze^2}{m} \right)^{2/3} \gg \frac{mv^2}{2}.$$

Therefore the radiation of a quantum with energy of order $mv^2/2$ does not change the motion on the radiation segment, and no additional condition of smallness of $\hbar\omega$ is required.

We note also that the motion on the segment (92.25) for given angular momentum $\hbar l$ does not depend on the initial energy. Correspondingly, the energy and $d\sigma_\omega$ from (92.23) can be obtained by multiplying the probability of radiation $d\mathcal{E}_\omega/\hbar\omega$ by $2\pi\rho d\rho$ (ρ is the impact parameter) and integrating over all ρ . Since in the quasi-classical case

$$\rho d\rho = \hbar^2 l dl / (m^2 v^2),$$

this leads to the dependence $d\sigma_\omega \sim 1/v^2$, i.e. to (92.23). The above argument clarifies why it is the initial (and not the final) velocity of the electron that enters this formula.

In order to go over to the classical formulas in the entire region $(1 - \eta)v \sim 1$, $v \gg 1$, one would have to find the asymptotics of the hypergeometric function near the saddle point t_0 at closer range; we shall not pursue this here in view of the evident nature of the final result.

All the formulas written above refer to the Coulomb field of attraction. The cross section for radiation in the field of repulsion is obtained from (92.15) by the substitution:

$\nu \rightarrow -\nu$, $\nu' \rightarrow -\nu'$. In this case, in particular, the limiting Born formula (92.16) does not change at all. In the limit $\nu \ll 1$, $\nu' \rightarrow \infty$ we obtain instead of (92.18)

$$d\sigma_\omega = \frac{128\pi}{3} Z^3 \alpha^2 r_e^2 \left(\frac{c}{v}\right)^3 \exp\left(-\frac{\sqrt{2mc^2} \pi Z \alpha}{\sqrt{\hbar(\omega_0 - \omega)}}\right) \frac{\hbar d\omega}{mv^2}, \quad (\text{X.99})$$

i.e. the differential cross section tends to zero exponentially as $\omega \rightarrow \omega_0$. This result is again natural, since in the field of repulsion there are no bound states and the frequency $\omega = \omega_0$ is the true boundary of the radiation spectrum.

Problems

1. Find in the Born approximation the cross section of bremsstrahlung radiation in the non-relativistic collision of two particles with different ratios e/m .

Solution. The dipole moment of two particles with charges e_1 , e_2 and masses m_1 , m_2 in the system of their centre of inertia is

$$\mathbf{d} = \mu \left(\frac{e_1}{m_1} - \frac{e_2}{m_2} \right) \mathbf{r},$$

where $\mu = m_1 m_2 / (m_1 + m_2)$, $\mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2$. Hence

$$\dot{\mathbf{d}} = \left(\frac{e_1}{m_1} - \frac{e_2}{m_2} \right) \mu \dot{\mathbf{r}} = - \left(\frac{e_1}{m_1} - \frac{e_2}{m_2} \right) \frac{e_1 e_2}{r} \nabla \frac{1}{r}.$$

The matrix element

$$\mathbf{d}_{\mathbf{p}'\mathbf{p}} = -\frac{1}{i\omega^2} (\dot{\mathbf{d}})_{\mathbf{p}'\mathbf{p}}, \quad \omega = \frac{p^2 - p'^2}{2\mu}$$

($\mathbf{p} = \mu \mathbf{v}$, $\mathbf{p}' = \mu \mathbf{v}'$ are the momenta of relative motion) is computed using plane waves²³

$$\psi_{\mathbf{p}} = e^{i\mathbf{p}\mathbf{r}}, \quad \psi_{\mathbf{p}'} = e^{i\mathbf{p}'\mathbf{r}}$$

with the aid of the formula

$$\left(\nabla \frac{1}{r} \right)_{\mathbf{p}'\mathbf{p}} = \frac{4\pi i \mathbf{q}}{q^2}, \quad \mathbf{q} = \mathbf{p}' - \mathbf{p}.$$

As a result we find

$$d\sigma_{\mathbf{k}\mathbf{p}'} = \frac{e_1^2 e_2^2}{\pi^2} \left(\frac{e_1}{m_1} - \frac{e_2}{m_2} \right)^2 \frac{v'}{v} \frac{\mu^2}{q^4} (\mathbf{e}\mathbf{q})(\mathbf{e}^*\mathbf{q}) \frac{d\omega}{\omega} d\mathbf{o}_{\mathbf{k}} d\mathbf{o}_{\mathbf{p}'}.$$

After summation over the polarisations, the angular distribution of the radiation is described by a factor $\sin^2 \Theta$, where Θ is the angle between the photon $\mathbf{k}$ and the vector $\mathbf{q}$, lying in the scattering plane (see (45.4a)). After integration over the photon directions

$$d\sigma_{\vartheta\omega} = \frac{16}{3} e_1^2 e_2^2 \left(\frac{e_1}{m_1} - \frac{e_2}{m_2} \right)^2 \frac{v'}{v} \frac{d\omega}{\omega} \frac{\sin \theta d\theta}{v^2 + v'^2 - 2vv' \cos \theta'},$$

²³The replacement of two particles with the reduced mass is admissible, of course, only in the non-relativistic case.

where θ is the scattering angle. Finally, integrating over θ ,

$$d\sigma_{\omega} = \frac{16}{3} e_1^2 e_2^2 \left(\frac{e_1}{m_1} - \frac{e_2}{m_2} \right)^2 \frac{1}{v^2} \ln \frac{v + v'}{v - v'} \frac{d\omega}{\omega}.$$

For radiation in the field of a stationary Coulomb centre this formula coincides with (92.16).

2. Find in the Born approximation the cross section of bremsstrahlung radiation in the non-relativistic collision of two electrons²⁴.

Solution. Dipole radiation in this case is absent, and one must consider quadrupole radiation. In classical theory the spectral distribution of the total intensity of quadrupole radiation is given by the formula

$$I_{\omega} = (1/90) |\hat{D}_{ik})_{\omega}|^2,$$

where $D_{ik} = \sum e(3x_i x_k - r^2 \delta_{ik})$ is the tensor of the quadrupole moment of the system of charges²⁵. For two electrons in the system of their centre of inertia

$$D_{ik} = e(3x_i x_k - r^2 \delta_{ik}), \quad \mathbf{r} = \mathbf{r}_1 - \mathbf{r}_2.$$

In the transition to quantum theory the Fourier components must be replaced by matrix elements (cf. what was said in §45 on dipole radiation), and with the appropriate normalisation of the wave functions (plane waves) one obtains, after division by the photon energy ω , the cross section of radiation with the scattering of the electrons into the interval of final states $d^3 p'$:

$$d\sigma_{\mathbf{p}'} = \frac{1}{90\omega} |(\hat{D}_{ik})_{\mathbf{p}\mathbf{p}'}|^2 \frac{d^3 p'}{v(2\pi)^3}.$$

$v = 2p/m$ is the initial velocity of relative motion; the emitted frequency $\omega = (p^2 - p'^2)/m$.

The operator $\hat{D}_{ik}$ is computed by threefold commutation of D_{ik} with the Hamiltonian

$$\hat{H} = \frac{\hat{\mathbf{p}}^2}{m} + \frac{e^2}{r}$$

and is²⁶

$$\hat{\hat{D}}_{ik} = \frac{2e^3}{m} \left[6 \left(\frac{x_i}{r^3} \hat{p}_k + \hat{p}_k \frac{x_i}{r^3} \right) + 6 \left(\frac{x_k}{r^3} \hat{p}_i + \hat{p}_i \frac{x_k}{r^3} \right) - 9 \left(\frac{x_i x_k x_l}{r^5} \hat{p}_l + \hat{p}_l \frac{x_i x_k x_l}{r^5} \right) - \delta_{ik} \left(\frac{x_l}{r^3} \hat{p}_l + \hat{p}_l \frac{x_l}{r^3} \right) \right]$$

Taking into account the identity of both particles (electrons) the matrix elements are computed using the wave functions

$$\psi_{\mathbf{p}} = \frac{1}{\sqrt{2}} (e^{i\mathbf{p}\mathbf{r}} \pm e^{-i\mathbf{p}\mathbf{r}}), \quad \psi_{\mathbf{p}'} = \frac{1}{\sqrt{2}} (e^{i\mathbf{p}'\mathbf{r}} \pm e^{-i\mathbf{p}'\mathbf{r}}),$$

²⁴The velocity of the collision satisfies the condition $\alpha \ll e^2/(\hbar v) \ll 1$.

²⁵This formula is obtained from (71.5) (see II), just as (67.11) (see II) is obtained from (67.8) (see II).

²⁶This expression is analogous to the classical formula $\ddot{D}_{ik} = \frac{4e^3}{m} \left[6 \frac{x_i x_k}{r^3} \hat{p}_k + 6 \frac{x_k}{r^3} \hat{p}_i - 9 \frac{x_i x_k}{r^5} \mathbf{p}\mathbf{r} - \frac{1}{r^3} \delta_{ik} \mathbf{p}\mathbf{r} \right]$, which would be obtained by differentiating D_{ik} using the classical equation of motion $\frac{m}{2} \ddot{\mathbf{r}} = \frac{e^2}{r^3} \mathbf{r}$.

where the signs “+” and “-” correspond respectively to total spins of the electrons 0 and 1 (the permutation of the electrons corresponds to the substitution $\mathbf{r} \rightarrow -\mathbf{r}$).

The cumbersome calculation leads to the following formula for the spectral distribution of the radiation:

$$d\sigma_{\omega} = \frac{4}{10} \alpha r_e^2 \left\{ 17 - \frac{3x^2}{(2-x)^2} + \frac{12(2-x)^4 - 7(2-x)^2 x^2 - 3x^4}{(2-x)^3 \sqrt{1-x}} \operatorname{Arcth} \frac{1}{\sqrt{x}} \right\} \sqrt{1-x} dx,$$

where $x = \omega/\varepsilon$, and $\varepsilon = p^2/m$ is the initial energy of relative motion of the electrons; the cross section is averaged over the values of the total spin. The effective retardation is

$$\kappa_{\text{eff}} = \hbar \int_0^{\varepsilon} \omega d\sigma_{\omega} = 8.1 \alpha r_e^2 \varepsilon$$

(B. K. Fedyushin, 1952).

3. Determine the energy of radiation arising upon emission of a non-relativistic electron by a nucleus in the s -state.

Solution. The wave function of the emitted electron—a diverging spherical s -wave, normalised to unit full flux:

$$\psi_i = \frac{1}{\sqrt{4\pi v}} \frac{e^{ipr}}{r}$$

(see III, (33.14)). As the wave function of the final (after emission of the photon) state of the electron we take a plane wave

$$\psi_f = e^{i\mathbf{p}'\mathbf{r}}.$$

The matrix element of the transition

$$\begin{aligned} \mathbf{p}_{fi} &= (\mathbf{p}_{fi})^* = \left(\int \psi_f^* \hat{\mathbf{p}} \psi_i d^3x \right)^* = \frac{\mathbf{p}'}{\sqrt{4\pi v}} \int e^{-i\mathbf{p}'\mathbf{r} + ipr} \frac{d^3x}{r} \\ &= \sqrt{\frac{\pi}{v}} \frac{\mathbf{p}'}{p^2 - p'^2} = \sqrt{\frac{\pi}{v}} \frac{v'}{v \omega} \end{aligned}$$

(the integral is evaluated according to (57.6a)). The energy of the radiation is obtained from formula (45.8), multiplied by $d^3p'/(2\pi)^3$ and integrated over the directions of $\mathbf{p}'$. As a result we obtain the spectral distribution of the radiated energy

$$dE_{\omega} = \frac{2e^2 v'^3}{3\pi v} d\omega.$$

When $\omega \rightarrow 0$ the final velocity of the electron $v' \rightarrow v$, and this formula coincides, as it should, with the non-relativistic limit of the classical result (see the problem in II, § 69). The total radiated energy (in ordinary units)

$$E = \frac{4}{15\pi} \alpha \left(\frac{v}{c} \right)^2 \varepsilon,$$

where $\varepsilon = mv^2/2$ is the initial energy of the electron.

4. Determine the energy of radiation arising upon the reflection of a non-relativistic electron from an infinitely high “potential wall”.

Solution. Let the electron move normally to the wall. Although the photon may be emitted in any direction, in the non-relativistic case the photon impulse is small in comparison with the electron impulse, and one can consider that the reflected electron also moves normally to the plane.

Let the wall be at $x = 0$, and the electron be on the side $x > 0$. The wave functions of the stationary states of one-dimensional motion, normalised to $\delta(p/2\pi)$ (see III, § 21):

$$\psi_i = 2 \sin px, \quad \psi_f = 2 \sin p'x.$$

The matrix element of the operator $\hat{p} = \hat{p}_x$:

$$p_{fi} = -4i \int_0^\infty \sin p'x \frac{d}{dx} \sin px \, dx = -\frac{4ipp'}{p^2 - p'^2},$$

(the integrals of this kind are to be understood as the limit when $\delta \rightarrow +0$ of the value obtained by introducing in the sub-integral expression the factor $e^{-\delta x}$).

The energy radiated upon single reflection of the electron is obtained from (45.8) by multiplication by $d\omega/v'$ and division by $v/2$ (the flux density of the running wave in the initial wave function ψ_i):

$$dE_\omega = \frac{4\omega^2 e^2}{3m^2} |p_{fi}|^2 \frac{2\pi d\omega}{vv'} = \frac{8}{3\pi} e^2 v v' d\omega. \quad (1)$$

At small frequencies ($\omega \ll \varepsilon = mv^2/2$) we have $v' \approx v$ and (1) goes over into the classical formula (69.5) (see II), which must be integrated over angles and divided by the change in velocity $\Delta v = 2v$ upon reflection; and it should be, since from the reflection from the wall the condition of smallness of the collision time (69.1) (see II) is always satisfied.

The quantum formula (1) allows, however, also to find the total radiated energy:

$$E = \int \frac{dE_\omega}{d\omega} d\omega = \frac{16}{9\pi} \alpha \varepsilon \frac{v^2}{c^2}$$

(in ordinary units).

5. Determine the energy of bremsstrahlung radiation arising upon scattering of a slow electron on an atom.

Solution. When the condition $pa \ll 1$ (where a is the atomic dimensions) is satisfied, the scattering on the atom is isotropic and does not depend on the energy of the electron (see III, § 132). The wave functions of the initial and final states of the electron are written in the form

$$\psi_i = e^{i\mathbf{p}\mathbf{r}} + f \frac{e^{ipr}}{r}, \quad \psi_f = e^{i\mathbf{p}'\mathbf{r}} + f \frac{e^{-ip'r}}{r},$$

where f is a real constant scattering amplitude. These expressions refer to the asymptotic region of distances $r \gg a$, which in the present case are just the essential ones:

$r \sim 1/p \gg a$. The computed matrix element from these functions is

$$\mathbf{p}_{fi} = \frac{2\pi f}{\omega}(\mathbf{v} - \mathbf{v}')$$

(the integrals are evaluated as in Problem 3). Substituting this expression into (92.12), we obtain the cross section of radiation with scattering of the electron in the direction $\mathbf{p}'$ (ordinary units):

$$d\sigma_{\omega\mathbf{p}'} = \frac{2\alpha p'}{3\pi pc^2}(\mathbf{v} - \mathbf{v}')^2 d\sigma_{\text{el}} \frac{d\omega}{\omega}, \quad (1)$$

where $d\sigma_{\text{el}} = f^2 d\sigma_{\mathbf{p}'}$ is the differential cross section of elastic scattering. When $\hbar\omega \ll p^2/(2m)$ one can set $p \approx p'$, and then this formula goes over, in the non-relativistic limit, into the formula for radiation of soft photons (see § 98)²⁷.

Integrating (1) over the directions of $\mathbf{p}'$, we get

$$d\sigma_{\omega} = \frac{2\alpha p'}{3\pi pc^2}(v^2 + v'^2)\sigma_{\text{el}} \frac{d\omega}{\omega}. \quad (2)$$

where $\sigma_{\text{el}} = 4\pi f^2$ is the total elastic scattering cross section. Finally, multiplying by $\hbar\omega$ and integrating over ω from 0 to $p^2/(2m) = \varepsilon$, we obtain the “effective retardation”

$$\kappa_{\text{eff}} = \int \hbar\omega d\sigma_{\omega} = \frac{32}{45\pi}\alpha\sigma_{\text{el}}\varepsilon \left(\frac{v}{c}\right)^2. \quad (3)$$

§ 93. Bremsstrahlung of an electron on a nucleus. Relativistic case

Let us turn to the bremsstrahlung radiation of an electron on a nucleus in the case of relativistic electron velocities²⁸. Here we shall again assume the Born approximation condition to be satisfied, i.e. for both the initial (v) and the final (v') velocities of the electron: $Ze^2/\hbar v \ll 1$, $Ze^2/\hbar v' \ll 1$. At the same time the charge of the nucleus must not be too large: $Z\alpha \ll 1$.

As in the preceding section, we neglect the recoil of the nucleus, so that it serves merely as the source of the external field (for the justification of such a neglect see § 97).

According to (91.4) the bremsstrahlung cross section is expressed through the amplitude by the formula

$$d\sigma = |M_{fi}|^2 \frac{\omega|\mathbf{p}'|}{8(2\pi)^5|\mathbf{p}|} d\mathbf{o}_{\mathbf{k}} d\mathbf{o}' d\omega. \quad (\text{X.100})$$

In the first non-vanishing approximation the matrix element M_{fi} corresponds to two diagrams:

The free end q corresponds to the external field, so that $q = p' - p + k$ is the 4-vector of the momentum transfer to the nucleus. The neglect of the recoil means that the time component $q^0 = 0$.

²⁷The fact that “factorisation” of the cross section (i.e. the extraction of the factor σ_{el}) has occurred in the present case for arbitrary ω is, in a certain sense, accidental and is related to the independence of the scattering amplitude from energy.

²⁸The greater part of the results presented below was obtained by *Bethe* and *Heitler* (*H. A. Bethe, W. Heitler*, 1934) and independently by *Sauter* (*F. Sauter*, 1934).

According to the diagrams (93.2) we have

$$M_{fi} = -e^2 A_0^{(e)}(\mathbf{q}) \sqrt{4\pi} e_\nu^* \bar{u}' \left(\gamma^\mu \frac{\gamma f' + m}{f'^2 - m^2} \gamma^\nu + \gamma^\nu \frac{\gamma f + m}{f^2 - m^2} \gamma^\mu \right) u. \quad (\text{X.3})$$

The intermediate 4-momenta $f = p - k$, $f' = p' + k$; we introduce the notation:

$$f^2 - m^2 = -2kp \equiv -2\kappa\omega, \quad f'^2 - m^2 = -2kp' \equiv -2\kappa'\omega. \quad (\text{X.4})$$

$A_0^{(e)}$ is the scalar potential of the external field; for a purely Coulomb field

$$A_0^{(e)}(\mathbf{q}) = \frac{4\pi Ze}{\mathbf{q}^2}. \quad (\text{X.5})$$

Substituting into (93.1), we have for the cross section

$$d\sigma = \frac{Z^2 e^6}{4\pi^2} \frac{|\mathbf{p}'| \omega}{|\mathbf{p}| \mathbf{q}^4} e_\nu^* e_\mu (\bar{u}' Q^\mu u) (\bar{u} \tilde{Q}^\nu u') d\mathbf{o}_\mathbf{k} d\mathbf{o}' d\omega, \quad (\text{X.6})$$

where

$$Q^\mu = \gamma^\mu \frac{\gamma f' + m}{2\omega\kappa'} \gamma^0 - \gamma^0 \frac{\gamma f + m}{2\omega\kappa} \gamma^\mu, \\ \tilde{Q}^\nu = \gamma^0 \frac{\gamma f' + m}{2\omega\kappa'} \gamma^\nu - \gamma^\nu \frac{\gamma f + m}{2\omega\kappa} \gamma^0.$$

Without considering polarisation effects, we average the cross section over the spin directions of the initial electron and sum over the polarisations of the final electron and the photon. This reduces to the replacement

$$e_\nu^* e_\mu (\bar{u}' Q^\mu u) (\bar{u} \tilde{Q}^\nu u') \rightarrow -\frac{1}{2} \text{Sp } Q_\mu (\gamma p + m) \tilde{Q}^\mu (\gamma p' + m).$$

Computation of the trace is carried out by the standard formulas (see § 22). A certain simplification of the algebra is achieved by using the identity

$$\gamma^0 (\gamma p) \gamma^0 = \gamma \tilde{p},$$

where $\tilde{p} = (\varepsilon, -\mathbf{p})$, if $p = (\varepsilon, \mathbf{p})$. Furthermore, the number of terms requiring computation can be reduced by using the symmetry with respect to the substitution $p \leftrightarrow p'$, $k \rightarrow -k$, $q \rightarrow -q$ (this substitution leads only to a cyclic permutation of the factors in the products of matrices and therefore does not change the trace).

As a result one obtains the following expression for the differential cross section of bremsstrahlung with the emission of a photon of given frequency and direction, and with the emission of the secondary electron in a given direction²⁹:

$$d\sigma = \frac{Z^2 \alpha r_e^2}{4\pi^2} \frac{p' m^4}{p q^4} \frac{d\omega}{\omega} d\mathbf{o}_\mathbf{k} d\mathbf{o}' \times \\ \times \left\{ \frac{q^2}{\kappa^2 m^2} (2\varepsilon^2 + 2\varepsilon'^2 - q^2) + \left(q^2 \left(\frac{1}{\kappa} - \frac{1}{\kappa'} \right)^2 - \left(\frac{\varepsilon}{\kappa} - \frac{\varepsilon'}{\kappa'} \right)^2 \right) + \right. \\ \left. + \frac{2\omega q^2}{m^2} \left(\frac{1}{\kappa} - \frac{1}{\kappa'} \right) - \frac{2\omega^2}{m^2} \left(\frac{\varepsilon^2}{\kappa'^2} - \frac{\varepsilon'^2}{\kappa^2} \right) \right\}, \quad (\text{X.7})$$

²⁹Here and below in this section p , p' , q denote the absolute values of the three-dimensional vectors: $p = |\mathbf{p}|$, $p' = |\mathbf{p}'|$, $q = |\mathbf{q}|$.

where

$$\kappa = \varepsilon - \mathbf{n}\mathbf{p}, \quad \kappa' = \varepsilon' - \mathbf{n}\mathbf{p}' \quad (\mathbf{n} = \mathbf{k}/\omega), \quad \mathbf{q} = \mathbf{p}' + \mathbf{k} - \mathbf{p}.$$

By straightforward transformations this formula can be given a somewhat more convenient form for investigation:

$$\begin{aligned} d\sigma = & \frac{Z^2 \alpha r_e^2}{2\pi^2} \frac{d\omega}{\omega} \frac{p'}{p} \frac{m^2}{q^4} \sin \theta d\theta \sin \theta' d\theta' d\varphi \times \\ & \times \left\{ \frac{p'^2}{\kappa'^2} (4\varepsilon^2 - q^2) \sin^2 \theta' + \frac{p^2}{\kappa^2} (4\varepsilon'^2 - q^2) \sin^2 \theta + \right. \\ & + \frac{2\omega^2}{\kappa\kappa'} (p^2 \sin^2 \theta + p'^2 \sin^2 \theta') - \\ & \left. - \frac{2pp'}{\kappa\kappa'} (2\varepsilon^2 + 2\varepsilon'^2 - q^2) \sin \theta \sin \theta' \cos \varphi \right\}, \end{aligned} \quad (\text{X.8})$$

where

$$\kappa = \varepsilon - p \cos \theta, \quad \kappa' = \varepsilon' - p' \cos \theta',$$

$$q^2 = p^2 + p'^2 + \omega^2 - 2p\omega \cos \theta + 2p'\omega \cos \theta' - 2pp'(\cos \theta \cos \theta' + \sin \theta \sin \theta' \cos \varphi),$$

θ and θ' are the angles between $\mathbf{k}$ and, respectively, $\mathbf{p}$ and $\mathbf{p}'$, φ the angle between the planes $\mathbf{k}, \mathbf{p}$ and $\mathbf{k}, \mathbf{p}'$.

Integration of (93.8) over the directions of the photon and secondary electron is quite cumbersome. It leads to the following formula for the spectral distribution of the radiation³⁰:

$$\begin{aligned} d\sigma_\omega = & Z^2 \alpha r_e^2 \frac{d\omega}{\omega} \frac{p'}{p} \left\{ \frac{4}{3} - 2\varepsilon\varepsilon' \frac{p^2 + p'^2}{p^2 p'^2} + \right. \\ & + m^2 \left(l \frac{\varepsilon'}{p^3} + l' \frac{\varepsilon}{p'^3} - \frac{ll'}{pp'} \right) + L \left[\frac{8\varepsilon\varepsilon'}{3pp'} + \frac{\omega^2}{p^3 p'^3} (\varepsilon^2 \varepsilon'^2 + p^2 p'^2 + m^2 \varepsilon\varepsilon') \right] + \\ & \left. + \frac{m^2 \omega}{2pp'} \left(l \frac{\varepsilon\varepsilon' + p^2}{p^3} - l' \frac{\varepsilon\varepsilon' + p'^2}{p'^3} + \frac{2\omega\varepsilon\varepsilon'}{p^2 p'^2} \right) \right\}, \end{aligned} \quad (\text{X.9})$$

where

$$L = \ln \frac{\varepsilon\varepsilon' + pp' - m^2}{\varepsilon\varepsilon' - pp' - m^2}, \quad l = \ln \frac{\varepsilon + p}{m}, \quad l' = \ln \frac{\varepsilon' + p'}{m}.$$

We recall that the permissible values of frequencies in the obtained formulas are restricted only by the condition imposed on the final velocity of the electron ($Ze^2/v' \ll 1$): the electron must not lose almost all its energy. When $\omega \rightarrow 0$ the cross section of radiation diverges as $d\omega/\omega$; this is a manifestation of the general rule discussed in § 98.

In the non-relativistic limit ($p \ll m$) the photon impulse is small compared to the electron impulse, so that $q^2 \approx (\mathbf{p}' - \mathbf{p})^2$. Setting in (93.8) $\varepsilon = \varepsilon' = m$, and neglecting all p, p', ω compared to m , we obtain

$$d\sigma = \frac{2}{\pi} Z^2 \alpha r_e^2 \frac{d\omega}{\omega} \frac{p'}{p} \frac{m}{q^4} \sin \theta d\theta \sin \theta' d\theta' d\varphi \times (p^2 \sin^2 \theta + p'^2 \sin^2 \theta' - 2pp' \sin \theta \sin \theta' \cos \varphi),$$

³⁰Integration over the directions of the secondary electron alone can also be performed in analytic form—see *Gluckstern R. L., Hull M. H.* Phys. Rev. 1953. V. 90. P. 1030.

or

$$d\sigma = \frac{Z^2 \alpha^3}{\pi^2} \frac{p'}{p} [\mathbf{nq}]^2 \frac{d\mathbf{q} d\mathbf{q}' d\omega}{q^4 \omega}, \quad (\text{X.10})$$

in agreement with the formula obtained in the Born approximation in Problem 1 of § 92. Correspondingly, for the spectral distribution of the radiation one also obtains the already known result (92.16).

In the ultrarelativistic case, when both the initial and final energies of the electron are large ($\varepsilon, \varepsilon' \gg m$), the angular distribution of the photon and secondary electrons has a very specific character. At small angles θ, θ' the quantities $\varkappa, \varkappa'$ appearing in the denominators of formula (93.8) are

$$\varkappa \approx \frac{\varepsilon}{2} \left(\frac{m^2}{\varepsilon^2} + \theta^2 \right), \quad \varkappa' \approx \frac{\varepsilon'}{2} \left(\frac{m^2}{\varepsilon'^2} + \theta'^2 \right) \quad (\text{X.11})$$

and in the region $\theta \lesssim m/\varepsilon$ become very small. In this region the magnitude of the vector $\mathbf{q}$ ($q \sim m$) is also small. Thus, in the ultrarelativistic case the photon and the secondary electron fly forward in a narrow cone with opening angle $\sim m/\varepsilon$.

A quantitative formula for the angular distribution in the ultrarelativistic case is easily obtained from (93.8) by substituting $\varkappa, \varkappa'$ from (93.11), replacing everywhere else p, p' by $\varepsilon, \varepsilon'$ and neglecting q^2 compared to ε^2 . Introducing convenient notation:

$$\delta = \frac{\varepsilon}{m} \theta, \quad \delta' = \frac{\varepsilon'}{m} \theta', \quad (\text{X.12})$$

we write this formula in the form

$$\begin{aligned} d\sigma = & \frac{8}{\pi} Z^2 \alpha r_e^2 \frac{\varepsilon' m^4 d\omega}{\varepsilon q^4 \omega} \delta d\delta \cdot \delta' d\delta' d\varphi \times \\ & \times \left\{ \frac{\delta^2}{(1 + \delta^2)^2} + \frac{\delta'^2}{(1 + \delta'^2)^2} + \frac{\omega^2}{2\varepsilon\varepsilon'} \frac{\delta^2 + \delta'^2}{(1 + \delta^2)(1 + \delta'^2)} - \right. \\ & \left. - \left(\frac{\varepsilon'}{\varepsilon} + \frac{\varepsilon}{\varepsilon'} \right) \frac{\delta\delta' \cos \varphi}{(1 + \delta^2)(1 + \delta'^2)} \right\}. \end{aligned} \quad (\text{X.13})$$

Writing $q^2 = [\mathbf{nq}]^2 + (\mathbf{nq})^2$ ($\mathbf{n} = \mathbf{k}/\omega$), it is easy to find that for small angles

$$\frac{q^2}{m^2} = (\delta^2 + \delta'^2 - 2\delta\delta' \cos \varphi) + m^2 \left(\frac{1 + \delta^2}{2\varepsilon} - \frac{1 + \delta'^2}{2\varepsilon'} \right)^2. \quad (\text{X.14})$$

When $\delta \sim \delta' \sim 1$ the second term here is small in comparison with the first. These terms become comparable in the region of even smaller angles, where $\delta \sim m/\varepsilon$. Although q here becomes especially small ($q \sim m^2/\varepsilon \ll m$), the integral contribution of this region to the cross section is still small by the factor m^2/ε^2 (as it is easy to see) compared to the contribution of the whole region $\delta \lesssim 1$. However, q can also reach values $\sim m^2/\varepsilon$ when $\delta \sim \delta' \sim 1$, if in addition

$$|\delta - \delta'| \lesssim \frac{m}{\varepsilon}, \quad \varphi \lesssim \frac{m}{\varepsilon}. \quad (\text{X.15})$$

The contribution of this region is of the same order as the total integral cross section (or even constitutes the main part of it—see below).

Integration of formula (93.13) over φ and over δ' gives the angular distribution of the photon (of given frequency) independently of the directions of the secondary electrons³¹:

$$d\sigma = 8Z^2 \alpha r_e^2 \frac{d\omega}{\omega} \frac{\varepsilon'}{\varepsilon} \frac{\delta \cdot d\delta}{(1 + \delta^2)^2} \times \\ \times \left\{ \left[\frac{\varepsilon}{\varepsilon'} + \frac{\varepsilon'}{\varepsilon} - \frac{4\delta^2}{(1 + \delta^2)^2} \right] \ln \frac{2\varepsilon\varepsilon'}{m\omega} - \right. \\ \left. - \frac{1}{2} \left[\frac{\varepsilon}{\varepsilon'} + \frac{\varepsilon'}{\varepsilon} + 2 - \frac{16\delta^2}{(1 + \delta^2)^2} \right] \right\}. \quad (\text{X.16})$$

Integrating over δ , we find the spectral distribution of the radiation in the ultrarelativistic case:

$$d\sigma_\omega = 4Z^2 \alpha r_e^2 \frac{d\omega}{\omega} \frac{\varepsilon'}{\varepsilon} \left[\left(\frac{\varepsilon}{\varepsilon'} + \frac{\varepsilon'}{\varepsilon} - \frac{2}{3} \right) \left(\ln \frac{2\varepsilon\varepsilon'}{m\omega} - \frac{1}{2} \right) \right]. \quad (\text{X.17})$$

We draw attention to the presence in (93.16) and (93.17) of a logarithm of a large quantity (even at $\omega \sim \varepsilon$ the ratio $\varepsilon\varepsilon'/(m\omega) \sim \varepsilon/m \gg 1$). If it is so large that its logarithm is also large, then in the formulas named, the terms containing the logarithm become dominant. We note that this logarithm arises from integration over the region (93.15)³². Thus, in the logarithmic approximation (i.e. neglecting terms not containing the large logarithm), the secondary electron flies at an angle $\sim (m/\varepsilon)^2$ to the direction of incidence.

Finally, we give the limiting formula for the region near the hard boundary of the spectrum, when the ultrarelativistic electron radiates almost all its energy: $\omega \approx \varepsilon \gg \varepsilon'$. From (93.9) one easily finds

$$d\sigma_\omega = 2Z^2 \alpha r_e^2 \frac{d\omega}{\varepsilon} \left\{ \frac{\varepsilon'^2}{p'^2} \ln \frac{\varepsilon' + p'}{m} - \frac{m^2 \varepsilon'^2}{4p'^3} \ln^2 \frac{\varepsilon' + p'}{m} + \frac{\varepsilon' + p'}{3pp'} \right\}. \quad (\text{X.18})$$

Formulas (93.17) and (93.18) cover the entire range of values of ω for an ultrarelativistic initial electron; when $\omega \approx \varepsilon \gg \varepsilon' \gg m$ these formulas coincide. If the secondary electron is non-relativistic ($p' \ll m$), then

$$d\sigma_\omega = 2Z^2 \alpha r_e^2 \frac{\sqrt{2m(\varepsilon - \omega)}}{m} \frac{d\omega}{\varepsilon}. \quad (\text{X.19})$$

Polarisation effects. Polarisation effects in bremsstrahlung can be investigated by the same general method as was done in § 87 for the Compton effect. The question of

³¹First one integrates over φ (in the limits from 0 to 2π). The integration over δ' is conveniently replaced by integration over the difference $|\Delta| = |\delta' - \delta|$, splitting its range into two parts: from 0 to some Δ_0 and from Δ_0 to ∞ , where Δ_0 satisfies the inequalities $m/\varepsilon \ll \Delta_0 \ll 1$. In each domain corresponding simplifications in the sub-integral expression of (93.13) are possible.

³²This is easy to verify by considering the domain of integration in which φ and $\Delta = \delta' - \delta$ satisfy conditions: $m/\varepsilon \ll \Delta_0 \ll 1$, $\varphi \ll 1$. In this domain $q^2/m^2 \approx \Delta^2 + \varphi^2 \delta^2$, and the expression in curly brackets in (93.13) becomes the expression proportional to φ^2 or Δ^2 (when $\varphi = 0$ and $\Delta = 0$ it vanishes). The integrals $\int \frac{\varphi^2 d\varphi d\Delta}{(\Delta^2 + \delta^2 \varphi^2)^2}$ or $\int \frac{\Delta^2 d\varphi d\Delta}{(\Delta^2 + \delta^2 \varphi^2)^2}$ diverge logarithmically; their “cutting off” occurs at the boundaries of the indicated domain of values of the variables.

the choice of 4-vectors $e^{(1)}, e^{(2)}$ in this case is particularly simple. Since it suffices in practice to consider the process in one specific reference frame (the rest frame of the nucleus), it is sufficient to set

$$e^{(1)} = (0, \mathbf{e}^{(1)}), \quad e^{(2)} = (0, \mathbf{e}^{(2)}),$$

where $\mathbf{e}^{(1)}, \mathbf{e}^{(2)}$ are unit vectors perpendicular to $\mathbf{k}$, one of which lies in the plane $\mathbf{k}\mathbf{p}$ and the other is perpendicular to it.

We shall not give here the rather cumbersome calculations themselves, but merely indicate some qualitative properties of the polarisation effects.

These properties can be obtained with the aid of various symmetry relations, similarly to what was done in § 87 for the Compton effect.

The theory expounded corresponds to the first Born approximation. In this approximation the cross section cannot contain a term proportional only to the vector of polarisation of one of the initial (ζ) or the final (ζ') electron. The absence of a term $\propto \zeta$ means that the complete (summed over photon polarisations and the polarisation of the secondary electron) bremsstrahlung cross section does not depend on the polarisation of the incident electron.

Of the terms proportional to only the photon polarisation parameters (ξ_1, ξ_2, ξ_3), the one $\propto \xi_2$ is absent. This means that upon radiation by an unpolarised electron the photon does not possess circular polarisation. Here there is, however, a difference from the analogous result for the Compton effect. In the latter case such terms were forbidden by the impossibility of constructing a pseudoscalar from the only two independent vectors at our disposal $\mathbf{k}$ and $\mathbf{k}'$. In the case of bremsstrahlung there are three independent momenta ($\mathbf{p}, \mathbf{p}', \mathbf{k}$), which suffice for the construction of a pseudoscalar ($\mathbf{k}[\mathbf{p}\mathbf{p}']$). The term of the form $\xi_2'(\mathbf{k}[\mathbf{p}\mathbf{p}'])$ does not contradict spatial parity and, strictly speaking, differs from zero. However, it is not invariant under the transformation (87.26) and therefore is absent in the first Born approximation.

The existence of the pseudoscalar ($\mathbf{k}[\mathbf{p}\mathbf{p}']$) leads also to the fact that along with the term $\propto \xi_3$ there also appears in the cross section a term $\propto \xi_1$ (in contrast to the situation in the Compton effect). The term of the form

$$S_{\alpha\beta} \nu_{\alpha}(\mathbf{k}[\mathbf{p}\mathbf{p}'])$$

(where $\nu = [\mathbf{k}\mathbf{p}]$), invariant under spatial inversion as well as under the transformation (87.26), is non-zero.

When also the direction of the secondary electron is detected and recorded, the cross section contains the terms $\propto \xi_1$ (allowed in the cross section also for the Compton effect) and also $\propto \xi_1'$, disappearing upon integration over all directions of $\mathbf{p}'$.

The degree of linear polarisation does not depend on the polarisation state of the incident electron: the correlation terms of the form $\xi_1'\zeta$ and $\xi_3\zeta$ are forbidden in the first Born approximation. However, the term $\xi_2'\zeta$ is allowed, so that upon radiation by a polarised electron the photon possesses circular polarisation (*Ya. B. Zeldovich, 1952*).

Screening. The formulas obtained above are derived for a purely Coulomb field. If, however, one is dealing with radiation not from a "bare" nucleus but from an atom, then the screening of the nuclear charge by the atomic electrons, leading to a reduction of the

cross section, must be taken into account. For this one must introduce into the potential of the external field $A_0^{(e)}(\mathbf{q})$ the atomic form factor $F(q)$ (see III, § 139). According to (139.2) (see III) this is achieved by the replacement of Z by $Z - F(q)$. We determine under what conditions the screening is substantial.

A definite value of q in the form factor corresponds to distances $r \sim 1/q$ in the spatial distribution of the electronic charge in the atom. The form factor approaches the value Z (complete screening) when $q \lesssim 1/a$, where a is the dimensions of the atom.

On the other hand, in the ultrarelativistic case a substantial contribution to the bremsstrahlung cross section arises, as we have seen above, already from the region of values of q close to the smallest value which q can have at given initial and final energies of the electron. In the ultrarelativistic case

$$q_{\min} = p - p' - \omega = \sqrt{\varepsilon^2 - m^2} - \sqrt{\varepsilon'^2 - m^2} - (\varepsilon - \varepsilon') \approx \frac{m^2 \omega}{2\varepsilon \varepsilon'}. \quad (\text{X.20})$$

Screening is substantial if $q_{\min} \lesssim 1/a$ or

$$\frac{\varepsilon \varepsilon'}{m \omega} \gtrsim \frac{a}{1/m} = \frac{a m}{1}. \quad (\text{X.21})$$

This condition is always satisfied at sufficiently large energies of the incident electron.

If $q_{\min} \ll 1/a$ ("complete screening"), then with logarithmic accuracy one can immediately write the answer for the spectral distribution of the radiation. Indeed, under the sign of the logarithm in (93.17) just the left-hand side of inequality (93.21) stands, the logarithm that leads to it. Upon observing the inequality the integral over dq , giving rise to this logarithm, is cut off at the value of order of the right-hand side of inequality. According to the Thomas–Fermi model $a \sim a_0 Z^{-1/3}$, where $a_0 \sim 1/(me^2)$ is the Bohr radius (see III, § 70); then $am \sim 1/(\alpha Z^{1/3})$. Thus, under complete screening the logarithm in (93.17) should be replaced by $\ln(1/(\alpha Z^{1/3}))$.

The ratio $\kappa_{\text{rad}}/\varepsilon$ is also called the cross section of energy loss to radiation. For large ε it grows logarithmically. This growth is eliminated, however, upon inclusion of screening. Under complete screening $\kappa_{\text{rad}}/\varepsilon$ tends to a constant limit $\approx 4Z^2 \alpha r_e^2 \ln(1/(\alpha Z^{1/3}))$.

When an atom is bombarded, some radiation is produced not only on the nucleus but also on the electrons. We shall see below (see § 97) that in the ultrarelativistic case the cross section of the radiation of an electron on an electron differs from the cross section of radiation on the nucleus only by the absence of the factor Z^2 . Therefore the presence of Z atomic electrons can approximately be taken into account by the replacement of Z^2 by $Z(Z+1)$.

Energy loss. The energy loss of the electron to radiation is characterised by the "effective retardation":

$$\kappa_{\text{rad}} = \int_0^{\varepsilon-m} \omega d\sigma_\omega. \quad (\text{X.22})$$

Computation of the integral of $d\sigma_\omega$ from (93.17) leads to the following result³³:

$$\kappa_{\text{rad}} = Z^2 \alpha r_e^2 \varepsilon \left\{ \frac{12\varepsilon^2 + 4m^2}{3\varepsilon p} \ln \frac{\varepsilon + p}{m} - \frac{(8\varepsilon + 6p)m^2}{3\varepsilon p^2} \ln^2 \frac{\varepsilon + p}{m} - \frac{4}{3} + \frac{2m^2}{\varepsilon p} F\left(\frac{2p(\varepsilon + p)}{m^2}\right) \right\}, \quad (\text{X.23})$$

where $F(\xi)$ is the Spence function, defined according to (131.19). In the non-relativistic case formula (93.23) goes over into

$$\kappa_{\text{rad}} = \frac{16}{3} Z^2 \alpha r_e^2 m \quad (\text{n.r.}) \quad (\text{X.24})$$

(one uses $F(\xi) \approx \xi$ at $\xi \ll 1$ —see (131.23)). This expression can, of course, be obtained also by direct integration of the non-relativistic Born formula (92.16). In the ultrarelativistic case

$$\kappa_{\text{rad}} = 4Z^2 \alpha r_e^2 \varepsilon \left(\ln \frac{2\varepsilon}{m} - \frac{1}{3} \right) \quad (\text{u.r.}) \quad (\text{X.25})$$

(when $\xi \gg 1$ we have $F(\xi) \approx \frac{1}{2} \ln^2 \xi$ —see (131.20); both terms with the square of the logarithm in (93.23) can then be omitted).

When passing through a medium containing N atoms per unit volume, a fast electron loses on average its energy to bremsstrahlung at a rate $-d\varepsilon/dx = N\kappa_{\text{rad}}$. Hence the energy decreases exponentially along the path, $\varepsilon = \varepsilon_0 e^{-x/l_{\text{rad}}}$, at distances of order

$$l_{\text{rad}} \sim \frac{\varepsilon}{N\kappa_{\text{rad}}} \sim \left[Z^2 \alpha N r_e^2 \ln \frac{1}{\alpha Z^{1/3}} \right]^{-1}; \quad (\text{X.26})$$

this length is called the *radiation length*.

Coherence length. Formula (93.20) admits also another, more general interpretation: for the applicability of the obtained formulas it is necessary that the external field in which the electron moves change little (in the direction of motion) over distances

$$l_{\text{coh}} \sim \frac{1}{q_{\text{min}}} \sim \frac{\varepsilon \varepsilon'}{m^2 \omega} \left(= \frac{\varepsilon \varepsilon'}{c^2 m^2 \omega} \right); \quad (\text{X.27})$$

this length is called the *formation length* of the radiation or the *coherence length*³⁴. The value (93.27), obtained in the Born approximation, has in fact (for ultrarelativistic particles) a completely general character—it is easy to obtain it also in the opposite limiting case of quasi-classical motion. Indeed, from formula (90.22)³⁵ it is immediately clear that for radiation at small angles to the direction of motion the essential times are

$$\tau \sim \frac{\varepsilon'}{\varepsilon \omega (1 - v)} \sim \frac{\varepsilon \varepsilon'}{m^2 \omega},$$

³³Though formula (93.17) is inapplicable close to the upper limit, this is inessential because of the convergence of the integral.

³⁴The considerations presented are due to *M. L. Ter-Mikaelyan* (1953).

³⁵The derivation of formula (90.22) is based only on the smallness of the curvature of the trajectory and in this sense is not connected with the fact that in § 90 a specifically magnetic field was considered.

i.e. a segment of trajectory of length $c\tau \sim l_{\text{coh}}$.

At a given frequency ω the coherence length grows with increase of the electron energy. Meanwhile the formulas obtained for bremsstrahlung on a single isolated atom can be valid for radiation during passage through a medium only under the condition that on the coherence length no repeated scattering of the photon or scattering of the electron occurs. The first condition means that one must have $l_{\text{coh}} \ll l_{\text{rad}}$. But already significantly earlier the second condition is violated—over a path $\sim l_{\text{rad}}$ there arises multiple scattering of the electron on nuclei of atoms of the medium.

For a quantitative formulation of the condition let us return to formula (90.22) and the expression in the exponent of the exponential; carrying out the integration over time, we write it in the form

$$-i \frac{\omega \varepsilon}{\varepsilon'} \int_{t_1}^{t_1+\tau} (1 - \mathbf{n} \mathbf{v}) dt \approx -\frac{i\omega \varepsilon}{\varepsilon'} (1 - \nu) \tau + \frac{1}{2} \int_{t_1}^{t_1+\tau} \theta^2 dt, \quad (\text{X.28})$$

where θ is the small angle between $\mathbf{v}$ and $\mathbf{n}$, associated with scattering on nuclei. In Coulomb scattering the angle θ changes by small increments, so that the variation of θ with time has the character of a slow “diffusion in angles”. The mean square deviation of the electron over a path $t - t_1 (= c(t - t_1))$

$$\overline{\theta^2} \sim (t - t_1) / l_{\text{Coul}},$$

where l_{Coul} is the mean free path with respect to Coulomb collisions. For this mean free path we have

$$\frac{1}{l_{\text{Coul}}} \sim \frac{NZ^2 e^4}{\varepsilon^2} \ln \frac{\chi_{\text{max}}}{\chi_{\text{min}}},$$

where χ_{min} and χ_{max} are the minimum and maximum scattering angles in a single collision for which the scattering can still be considered Rutherford (cf. X, § 41)³⁶. The first of them is determined by the atomic dimensions a , at which the field of the nucleus is screened: $\chi_{\text{min}} \sim 1/(pa)$. Large scattering angles are bounded (for an ultrarelativistic electron) by distances of order the nuclear radius R : $\chi_{\text{max}} \approx 1/(pR)$. Putting $R \approx 1.5 \cdot 10^{-13} Z^{1/3} \text{ cm} \sim r_e Z^{1/3}$, we obtain

$$l_{\text{Coul}} \sim \frac{\varepsilon^2}{NZ^2 e^4 \ln(1/(\alpha Z^{1/3}))} \sim \frac{\alpha \varepsilon^2}{m^2} l_{\text{rad}}. \quad (\text{X.29})$$

The second term in (93.28), accumulated over the time $\tau \sim l_{\text{coh}}$, is now estimated as

$$\overline{\theta^2} \tau \sim \frac{\omega \varepsilon l_{\text{coh}}^2}{\varepsilon' l_{\text{Coul}}} \sim \frac{l_{\text{coh}}}{\alpha l_{\text{rad}}}.$$

For the applicability of the bremsstrahlung formulas obtained without account of multiple scattering, this term must be small compared with unity. From this we find the condition

$$l_{\text{coh}} \ll \alpha l_{\text{rad}}, \quad (\text{X.30})$$

stronger than the condition $l_{\text{coh}} \ll l_{\text{rad}}$ (*L. D. Landau, I. Ya. Pomeranchuk, 1953*).

³⁶Recall that the mean free path is defined by the transport cross section $\sigma_{\text{tr}} = \int (1 - \cos \chi) d\sigma(\chi)$. For scattering of ultrarelativistic electrons on a Coulomb centre the cross section $d\sigma(\chi)$ is given by formula (80.10).

§ 94. Pair production by a photon in the field of a nucleus

The production of an electron-positron pair in the collision of a photon with a nucleus ($Z + \gamma \rightarrow Z + e^- + e^+$) and bremsstrahlung in the collision of an electron with a nucleus ($Z + e^- \rightarrow Z + e^+ \gamma$) are two crossed channels of one and the same reaction. In § 91 the rules were already formulated by which the formulas are transformed in passing from the second of these cases to the first. In the present case, applying these rules to formula (93.8), we obtain the following expression for the differential cross section of pair production by an unpolarised photon, averaged over the polarisations of the pair components³⁷:

$$\begin{aligned}
 d\sigma = & \frac{Z^2 \alpha r_e^2 m^2 p_+ p_-}{2\pi \omega^3 q^4} \sin \theta_+ d\theta_+ \sin \theta_- d\theta_- d\varphi \times \\
 & \times \left\{ \frac{p_+^2}{\kappa_+^2} (4\varepsilon_-^2 - q^2) \sin^2 \theta_+ + \frac{p_-^2}{\kappa_-^2} (4\varepsilon_+^2 - q^2) \sin^2 \theta_- - \right. \\
 & - \frac{2\omega^2}{\kappa_+ \kappa_-} (p_+^2 \sin^2 \theta_+ + p_-^2 \sin^2 \theta_-) - \\
 & \left. - \frac{2p_+ p_-}{\kappa_+ \kappa_-} (2\varepsilon_+^2 + 2\varepsilon_-^2 - q^2) \sin \theta_+ \sin \theta_- \cos \varphi \right\},
 \end{aligned} \tag{X.31}$$

$$\kappa_{\pm} = \varepsilon_{\pm} - p_{\pm} \cos \theta_{\pm}, \quad q^2 = (\mathbf{p}_+ + \mathbf{p}_- - \mathbf{k})^2, \quad \varepsilon_+ + \varepsilon_- = \omega$$

(H. A. Bethe, W. Heitler, 1934).

By the same transformation we obtain from (93.9) the distribution of the pair components in energy:

$$\begin{aligned}
 d\sigma_{\varepsilon_+} = & Z^2 \alpha r_e^2 \frac{p_+ p_-}{\omega^3} d\varepsilon_+ \left\{ -\frac{4}{3} - 2\varepsilon_+ \varepsilon_- \frac{p_+^2 + p_-^2}{p_+^2 p_-^2} + \right. \\
 & + m^2 \left(l_- \frac{\varepsilon_+}{p_-^2} + l_+ \frac{\varepsilon_-}{p_+^2} - \frac{l_+ l_-}{p_+ p_-} \right) + \\
 & + L \left[-\frac{8\varepsilon_+ \varepsilon_-}{3p_+ p_-} + \frac{\omega^3}{p_+^3 p_-^3} (\varepsilon_+^2 \varepsilon_-^2 + p_+^2 p_-^2 - m^2 \varepsilon_+ \varepsilon_-) - \right. \\
 & \left. \left. - \frac{m^2 \omega}{2p_+ p_-} \left(l_+ \frac{\varepsilon_+ \varepsilon_- - p_+^2}{p_+^3} + l_- \frac{\varepsilon_+ \varepsilon_- - p_-^2}{p_-^3} \right) \right] \right\},
 \end{aligned} \tag{X.32}$$

$$L = \ln \frac{\varepsilon_+ \varepsilon_- + p_+ p_- + m^2}{\varepsilon_+ \varepsilon_- - p_+ p_- + m^2}, \quad l_{\pm} = \ln \frac{\varepsilon_{\pm} + p_{\pm}}{\varepsilon_{\pm} - p_{\pm}}.$$

In the ultrarelativistic case ($\varepsilon_{\pm} \gg m$) the electron and positron emerge at angles $\theta_{\pm} \sim m/\varepsilon_{\pm}$ to the direction of the incident photon. The angular distribution is given by a

³⁷On polarisation effects in pair production by photons see the same literature as indicated in § 93 for bremsstrahlung.

formula analogous to (93.13):

$$d\sigma = \frac{8}{\pi} Z^2 \alpha r_e^2 \frac{m^2 \varepsilon_+ \varepsilon_-}{\omega^3 q^4} d\varepsilon_+ \times \\ \times \left\{ -\frac{\delta_+^2}{(1 + \delta_+^2)^2} - \frac{\delta_-^2}{(1 + \delta_-^2)^2} + \frac{\omega^2}{2\varepsilon_+ \varepsilon_-} \frac{\delta_+^2 + \delta_-^2}{(1 + \delta_+^2)(1 + \delta_-^2)} + \right. \\ \left. + \left(\frac{\varepsilon_+}{\varepsilon_-} + \frac{\varepsilon_-}{\varepsilon_+} \right) \frac{\delta_+ \delta_- \cos \varphi}{(1 + \delta_+^2)(1 + \delta_-^2)} \right\} \delta_+ \delta_- d\delta_+ d\delta_- d\varphi, \quad (\text{X.33})$$

where

$$\frac{q^2}{m^2} = \delta_+^2 + \delta_-^2 - 2\delta_+ \delta_- \cos \varphi + m^2 \left(\frac{1 + \delta_+^2}{2\varepsilon_+} + \frac{1 + \delta_-^2}{2\varepsilon_-} \right)^2. \quad (\text{X.34})$$

The energy distribution in this case:

$$d\sigma = 4Z^2 \alpha r_e^2 \frac{d\varepsilon_+}{\omega^3} \left(\varepsilon_+^2 + \varepsilon_-^2 + \frac{2}{3} \varepsilon_+ \varepsilon_- \right) \left(\ln \frac{2\varepsilon_+ \varepsilon_-}{m\omega} - \frac{1}{2} \right) \quad (\text{u.r.}), \quad (\text{X.35})$$

Integration of this expression over ε_+ (in the limits from m to ω) gives the total cross section of pair production by a photon of given energy³⁸:

$$\sigma = \frac{28}{9} Z^2 \alpha r_e^2 \left(\ln \frac{2\omega}{m} - \frac{109}{42} \right), \quad \omega \gg m. \quad (\text{X.36})$$

As for bremsstrahlung, the logarithmic term in the cross section in the ultrarelativistic case originates from the region of values $q \sim m^2/\varepsilon$. To this there now correspond angles for which

$$|\delta_+ - \delta_-| \lesssim \frac{m}{\varepsilon}, \quad |\pi - \varphi| \lesssim \frac{m}{\varepsilon}$$

(instead of $\varphi \lesssim m/\varepsilon$ in (93.15)). Thus, in the logarithmic approximation the directions of the electron and positron make small angles inversely proportional to the particle energies with the direction of the photon, and lie almost in one plane with the latter but on opposite sides of it.

Near the reaction threshold ($\omega \rightarrow 2m$) the Born approximation is inapplicable. The derivation of a quantitative formula in this case would require exact account of the Coulomb interaction of the three charged particles present in the final state (the nucleus and the pair). The symmetry with respect to the electron (attracted

to the nucleus) and the positron (repelled from the nucleus) in this case, of course, disappears.

If

$$Z\alpha \sqrt{\frac{\omega - 2m}{\omega}} \ll 1, \quad (\text{X.37})$$

then the Born approximation is still applicable. At non-relativistic energies of the pair $\omega \approx 2m \gg p_{\pm}$, and therefore $q \approx \omega$. In (94.1) one can put everywhere $\varepsilon_{\pm} = \kappa_{\pm} = m$, $\omega = 2m$, after which this formula reduces to the expression

$$d\sigma = \frac{Z^2 \alpha r_e^2 p_+ p_-}{64\pi^2 m^6} (p_+^2 \sin^2 \theta_+ + p_-^2 \sin^2 \theta_-) d\theta_+ d\theta_- d\varepsilon_+. \quad (\text{X.38})$$

³⁸Because of the convergence of the integral at both limits, the inapplicability of formula (94.5) when $\varepsilon_{\pm} - m$ is small is inessential.

After integration over angles

$$\begin{aligned} d\sigma &= \frac{1}{6} Z^2 \alpha r_e^2 \frac{p_+ p_- (p_+^2 + p_-^2)}{m^5} d\varepsilon_+ = \\ &= \frac{2Z^2 \alpha r_e^2}{3m^3} (\omega - 2m) \sqrt{(\varepsilon_+ - m)(\varepsilon_- - m)} d\varepsilon_+. \end{aligned} \quad (\text{X.39})$$

Finally, integrating over ε_+ (in the limits from m to $\omega - m$), we obtain the total cross section

$$\sigma = \frac{\pi}{12} Z^2 \alpha r_e^2 \left(\frac{\omega - 2m}{m} \right)^3. \quad (\text{X.40})$$

If the relative velocity v_0 of the components of the produced pair is small, it is necessary to take into account their Coulomb interaction with each other (*A. D. Sakharov*, 1948). It becomes substantial when v_0 is of the order of (or less than) the velocity of a particle in the bound state of the electron and positron (positronium):

$$v_0 \lesssim \alpha. \quad (\text{X.41})$$

Let us consider the process in the centre-of-mass frame of the pair. In the diagrams depicting the process in this frame, the essential virtual momenta are $\sim m$. In other words, the essential distances between the electron and positron are $\sim 1/m$. Meanwhile the wave function of their relative motion $\psi(r)$ changes substantially only at distances $r \sim 1/(mv_0) \sim 1/(m\alpha)$, i.e. large compared with $1/m$. Therefore the account of the interaction of the particles reduces to the appearance in the transition matrix element of a factor $\psi^*(0)$. Correspondingly, the differential cross section is multiplied by $|\psi(0)|^2$, i.e. by

$$\frac{2\pi\alpha/v_0}{1 - e^{-2\pi\alpha/v_0}} \quad (\text{X.42})$$

(see III, (136.11)). The relative velocity of two particles is the velocity of one of them in the rest frame of the other. Comparing the values of the invariant $p_- \mu p_+^\mu$ in this frame and in the laboratory frame (the rest frame of the nucleus), we obtain

$$\frac{m^2}{\sqrt{1 - v_0^2}} = \varepsilon_+ \varepsilon_- - \mathbf{p}_+ \mathbf{p}_-,$$

from which v_0 can be found. If $\mathbf{p}_+$ and $\mathbf{p}_-$ are close to each other in magnitude and direction, then for v_0 we obtain the approximate formula

$$v_0^2 = \frac{p^2}{m^2} \vartheta^2 + \frac{(p_+ - p_-)^2}{\varepsilon^2}, \quad (\text{X.43})$$

applicable when $v_0 \ll 1$; $p = (p_+ + p_-)/2$, $\varepsilon = (\varepsilon_+ + \varepsilon_-)/2$, ϑ is the angle between $\mathbf{p}_+$ and $\mathbf{p}_-$.

The correction to the cross section, determined by formulas (94.12), (94.13), leads to the appearance of an anomaly in the correlation between the momenta of the produced electron and positron: a narrow maximum at $\mathbf{p}_+ \approx \mathbf{p}_-$.

§ 95. Exact theory of pair production in the ultrarelativistic case

In the two preceding sections bremsstrahlung and pair production by a photon in the relativistic region were studied on the basis of the Born approximation, for which in any case the fulfilment of the condition $Z\alpha \ll 1$ was required. In §§ 95, 96 there is presented the theory of these processes, free from this restriction, i.e. valid also when $Z\alpha \sim 1$ (H. A. Bethe, L. Maximon, 1954). It is assumed that both particles (the initial and final electron, or the components of the pair) are ultrarelativistic; their energy $\varepsilon \gg m$.

We have seen that in the ultrarelativistic case both particles fly at small angles (θ, θ' or θ_+, θ_-) to the direction of the photon: $\theta \lesssim m/\varepsilon$. This property is preserved also in the exact (in $Z\alpha$) theory, and we shall consider precisely this region of angles.

The momentum transfer to the nucleus in this region is $q \sim m$. This means that in the wave functions the essential impact parameters are $\rho \sim 1/q \sim 1/m$, i.e. “large” distances. At such distances one can use the wave function obtained in § 39. Let us present the corresponding calculations for pair production.

The cross section of pair production has a form analogous to the cross section of the photoeffect (cf. (56.1), (56.2)):

$$d\sigma = 2\pi \left| e\sqrt{4\pi} \frac{1}{\sqrt{2\omega}} M_{fi} \right|^2 \delta(\omega - \varepsilon_+ - \varepsilon_-) \frac{d^3p_+ d^3p_-}{(2\pi)^3}, \quad (\text{X.44})$$

where

$$M_{fi} = \int \psi_{\varepsilon-\mathbf{p}_-}^{(-)*}(\alpha\mathbf{e}) e^{i\mathbf{k}\mathbf{r}} \psi_{-\varepsilon_+-\mathbf{p}_+}^{(+)} d^3x. \quad (\text{X.45})$$

Here $\psi_{\varepsilon-\mathbf{p}_-}^{(-)}$ is the wave function of the electron, and $\psi_{-\varepsilon_+-\mathbf{p}_+}^{(+)}$ is the wave function with negative energy ($-\varepsilon_+$) and momentum $-\mathbf{p}_+$.

The function $\psi_{\varepsilon-\mathbf{p}_-}^{(-)}$, pertaining to a particle in the final state, must contain in its asymptotics (alongside the plane wave) a converging spherical wave; this circumstance is indicated by the upper index $(-)$ on the function. According to (39.10) such a wave function is³⁹

$$\psi_{\varepsilon-\mathbf{p}_-}^{(-)} = \frac{C^{(-)}}{\sqrt{2\varepsilon_-}} e^{i\mathbf{p}_-\mathbf{r}} \left(1 - \frac{i\alpha\nabla}{2\varepsilon_-} \right) F(-i\nu, 1, -i(p_-r + \mathbf{p}_-\mathbf{r})) u(\mathbf{p}_-), \quad (\text{X.46})$$

$$C^{(-)} = e^{\pi\nu/2} \Gamma(1 + i\nu), \quad \nu = Z\alpha.$$

The function $\psi_{-\varepsilon_+-\mathbf{p}_+}^{(+)}$ must contain in its asymptotics a diverging spherical wave (index $(+)$ above), since by its meaning it is the wave function of an “initial state with negative energy”. The wave function of the positron (formed from $\psi_{\varepsilon_+\mathbf{p}_+}^{(+)*}$) will then have a converging wave in its asymptotics, as required for a final-state particle. According to (39.11) such a function is

$$\psi_{-\varepsilon_+-\mathbf{p}_+}^{(+)} = \frac{C^{(+)}}{\sqrt{2\varepsilon_+}} e^{-i\mathbf{p}_+\mathbf{r}} \left(1 + \frac{i\alpha\nabla}{2\varepsilon_+} \right) F(-i\nu, 1, i(p_+r + \mathbf{p}_+\mathbf{r})) u(-\mathbf{p}_+), \quad (\text{X.47})$$

$$C^{(+)} = e^{-\pi\nu/2} \Gamma(1 + i\nu).$$

³⁹In this section $p_{\pm} = |\mathbf{p}_{\pm}|$, $q = |\mathbf{q}|$.

Note that the necessity of including the terms $\sim 1/\varepsilon$ in (95.3), (95.4) is connected with the matrix structure of M_{fi} (95.2). The matrix element $(\alpha)_{fi}$ is a vector with a direction close to $\mathbf{k}$. Therefore the principal term in $(\alpha\mathbf{e})_{fi}$ turns out to be small, while the correction terms are of the same order of magnitude as it.

Substituting (95.3), (95.4) in (95.2) and neglecting terms $\sim 1/(\varepsilon_+\varepsilon_-)$, we find

$$M_{fi} = \frac{N}{2\sqrt{\varepsilon_+\varepsilon_-}} u^*(\mathbf{p}_-)\{(\mathbf{e}\alpha)I + (\mathbf{e}\alpha)(\alpha\mathbf{I}_+) + (\alpha\mathbf{I}_-)(\mathbf{e}\alpha)\}u(-\mathbf{p}_+), \quad (\text{X.48})$$

where

$$N = C^{(+)}C^{(-)} = \frac{\pi\nu}{\sinh \pi\nu}, \quad (\text{X.49})$$

$$I = \int e^{-i\mathbf{q}\mathbf{r}} F_-^* F_+ d^3x, \quad (\text{X.50})$$

$$\mathbf{I}_+ = \frac{i}{2\varepsilon_+} \int e^{-i\mathbf{q}\mathbf{r}} F_-^* \nabla F_+ d^3x, \quad \mathbf{I}_- = \frac{i}{2\varepsilon_-} \int e^{-i\mathbf{q}\mathbf{r}} (\nabla F_-)^* F_+ d^3x,$$

$$\mathbf{q} = \mathbf{p}_+ + \mathbf{p}_- - \mathbf{k}$$

(F_- and F_+ denote for brevity the hypergeometric functions appearing in (95.3) and (95.4)). Let us immediately note that the integrals $I, \mathbf{I}_+, \mathbf{I}_-$ are connected by one identity: from

$$\int \nabla(e^{-i\mathbf{q}\mathbf{r}} F_-^* F_+) d^3x = 0$$

we have

$$\mathbf{q}I + 2\varepsilon_+\mathbf{I}_+ + 2\varepsilon_-\mathbf{I}_- = 0. \quad (\text{X.51})$$

The square $|M_{fi}|^2$ is averaged over the polarisations of the incident photon and summed over the spin directions of the electron and positron⁴⁰. This is accomplished by the replacement of the tensor:

$$e_i e_k^* \rightarrow \frac{1}{2}(\delta_{ik} - n_i n_k), \quad \mathbf{n} = \frac{\mathbf{k}}{\omega},$$

and of the bispinor products:

$$u_\pm \bar{u}_\pm \rightarrow 2\rho_\pm = (\varepsilon_\pm \gamma^0 - \mathbf{p}_\pm \boldsymbol{\gamma} \mp m).$$

Replacing also $\alpha = \gamma^0 \boldsymbol{\gamma}$, we find

$$|M_{fi}|^2 \rightarrow \frac{N^2}{2\varepsilon_+\varepsilon_-} \{\text{Sp } \rho_- Q \rho_+ \bar{Q} - \text{Sp } \rho_- (\mathbf{n}Q) \rho_+ (\mathbf{n}\bar{Q})\},$$

$$Q = \gamma I - \gamma^0 \boldsymbol{\gamma} (\gamma \mathbf{I}_+) - \gamma^0 (\gamma \mathbf{I}_-) \boldsymbol{\gamma},$$

$$\bar{Q} = \gamma I^* - \gamma^0 \boldsymbol{\gamma} (\gamma \mathbf{I}_+^*) - \gamma^0 (\gamma \mathbf{I}_-^*) \boldsymbol{\gamma}.$$

⁴⁰For calculations with account of the polarisation of all particles see *Olsen H., Maximon L.* Phys. Rev. 1959. V. 114. P. 887, and also the book by V. Ya. Baier, V. M. Katkov and V. S. Fadin indicated on p. 454.

Let us write out immediately the result, obtained after the appropriate approximations, for the ultrarelativistic case of interest in the region of small angles

$$\theta_{\pm} \sim m/\varepsilon \ll 1. \quad (\text{X.52})$$

We introduce auxiliary vectors:

$$\delta_{\pm} = \frac{1}{m} (\mathbf{p}_{\pm})_{\perp}, \quad \delta_{\pm} = \frac{\varepsilon_{\pm}}{m} \theta_{\pm} \quad (\text{X.53})$$

(the subscript $\perp$ denotes the component perpendicular to the direction of $\mathbf{k}$). With their help the result is written in the form

$$|M_{fi}|^2 \rightarrow \frac{N^2}{4} \left\{ \frac{m^2 \omega^2}{\varepsilon_+^2 \varepsilon_-^2} |I|^2 + 2 \left| I \frac{m \delta_+}{2 \varepsilon_+} + \mathbf{I}_+ \right|^2 + 2 \left| I \frac{m \delta_-}{2 \varepsilon_-} + \mathbf{I}_- \right|^2 \right\}. \quad (\text{X.54})$$

Here it is taken into account that $I \sim \varepsilon/(q) \cdot I_{\pm}$ (as is evident from (95.8)), and terms of higher order in m/ε are omitted.

The integrals $\mathbf{I}_{\pm}$ can be represented in the form

$$\mathbf{I}_{\pm} = i \frac{p_{\pm}}{2 \varepsilon_{\pm}} \frac{\partial J}{\partial \mathbf{p}_{\pm}},$$

$$J = \int \frac{e^{-i\mathbf{q}\mathbf{r}}}{r} F(-i\nu, 1, i(p_+r + \mathbf{p}_+\mathbf{r})) F(i\nu, 1, i(p_-r + \mathbf{p}_-\mathbf{r})) d^3x. \quad (\text{X.55})$$

The integral J is expressed through the complete hypergeometric function⁴¹:

$$J = \frac{4\pi}{q^2} \left(\frac{q^2 - 2\mathbf{p}_+\mathbf{q}}{q^2 - 2\mathbf{p}_-\mathbf{q}} \right)^{i\nu} F(-i\nu, i\nu, 1, z), \quad (\text{X.56})$$

$$z = 2 \frac{q^2(p_+p_- - \mathbf{p}_+\mathbf{p}_-) + 2(\mathbf{p}_+\mathbf{q})(\mathbf{p}_-\mathbf{q})}{(q^2 - 2\mathbf{p}_+\mathbf{q})(q^2 - 2\mathbf{p}_-\mathbf{q})}.$$

Differentiation with respect to $\mathbf{p}_{\pm}$ is to be carried out at a given value of the parameter $\mathbf{q}$, and only afterwards can one set $\mathbf{q} = \mathbf{p}_+ + \mathbf{p}_- - \mathbf{k}$. We give the result in a form in which the approximations corresponding to the ultrarelativistic case and the conditions (95.9) have already been made:

$$\mathbf{I}_{\pm} = \frac{4\pi}{q^2} \frac{\varepsilon_{\mp}}{m^2 \omega} \left(\frac{\varepsilon_{\pm} \xi_{\pm}}{\varepsilon_{\mp} \xi_{\mp}} \right)^{i\nu} \left\{ \pm \nu \mathbf{q} \xi_{\mp} F(z) + i \frac{q^2}{m^2} F'(z) (\mathbf{q} \xi_{\mp} - m \delta_{\pm}) \right\}. \quad (\text{X.57})$$

Here the following notation has been introduced:

$$\xi_{\pm} = \frac{1}{1 + \delta_{\pm}^2}, \quad z = 1 - \frac{q^2}{m^2} \xi_+ \xi_-, \quad F(z) = F(-i\nu, i\nu, 1, z) \quad (\text{X.58})$$

($F(z)$ is a real function). The integral I is then computed directly from (95.8).

⁴¹For the computation see the paper by Nordsieck indicated on p. 438.

Substituting the values of the integrals in (95.11), and then in (95.1), we obtain the desired differential cross section. The final formula:

$$\begin{aligned}
 d\sigma = & \frac{4}{\pi} \left(\frac{\pi\nu}{\sinh \pi\nu} \right)^2 Z^2 \alpha r_e^2 \frac{m^2}{q^4 \omega^3} \delta_+ d\delta_+ \cdot \delta_- d\delta_- \cdot d\varphi d\varepsilon_+ \left\{ F^2(z) \times \right. \\
 & \times [-2\varepsilon_+ \varepsilon_- (\delta_+^2 \xi_+^2 + \delta_-^2 \xi_-^2) + \omega^2 (\delta_+^2 + \delta_-^2) \xi_+ \xi_- + 2(\varepsilon_+^2 + \varepsilon_-^2) \times \\
 & \times \delta_+ \delta_- \xi_+ \xi_- \cos \varphi] + \frac{q^4}{m^4} \frac{\xi_+^2 \xi_-^2}{\nu^2} F'^2(z) [-2\varepsilon_+ \varepsilon_- (\delta_+^2 \xi_+^2 + \delta_-^2 \xi_-^2) + \\
 & \left. + \omega^2 (1 + \delta_+^2 \delta_-^2) \xi_+ \xi_- - 2(\varepsilon_+^2 + \varepsilon_-^2) \delta_+ \delta_- \xi_+ \xi_- \cos \varphi] \right\}. \quad (\text{X.59})
 \end{aligned}$$

When $\nu \rightarrow 0$

$$\pi\nu/(\sinh \pi\nu) \rightarrow 1, \quad F(z) \rightarrow 1, \quad F'(z) \approx \nu^2 \rightarrow 0.$$

Expression (95.16) reduces in this case, as it should, to the Bethe–Heitler formula (94.3), corresponding to the Born approximation. It reduces to the same formula also for arbitrary ν , if the emergence angles of the pair satisfy the conditions

$$|\delta_+ - \delta_-| \ll 1, \quad |\pi - \varphi| \ll 1.$$

Indeed, in this case $q \ll m$, so that the second term in the curly brackets in (95.16) can be dropped in view of the presence in it of the extra (compared with the first term) factor $(q/m)^4$. In the first term, however, we have (noting that $(1 - z) \sim q^2/m^2 \ll 1$)

$$F(z) \rightarrow F(1) \equiv F(-i\nu, i\nu, 1, 1) = \frac{1}{\Gamma(1 - i\nu)\Gamma(1 + i\nu)} = \frac{\sinh \pi\nu}{\pi\nu}, \quad (\text{X.60})$$

as a result of which the analogous factor before the curly brackets cancels⁴².

Let us now turn to the integration of the cross section over the directions of emergence of the pair.

We divide the integration over angles into two regions, I and II, in which respectively

$$\text{I) } 1 - z > 1 - z_1, \quad \text{II) } 1 - z < 1 - z_1,$$

where z_1 is some value for which $1 \gg 1 - z_1 \gg (m/\varepsilon)^2$.

Since in region II $1 - z \ll 1$, $q \ll m$, then, according to what was said above, $d\sigma \approx d\sigma_B \equiv d\sigma|_{\nu \rightarrow 0}$, where $d\sigma_B$ is the cross section in the Born approximation. Therefore the integral over angles:

$$d\sigma_{\varepsilon_+} \equiv \int d\sigma = \int_I d\sigma + \int_{II} d\sigma|_{\nu \rightarrow 0} = (d\sigma_{\varepsilon_+})_B + \int_I (d\sigma - d\sigma|_{\nu \rightarrow 0}), \quad (\text{X.61})$$

where $(d\sigma_{\varepsilon_+})_B$ is the Born cross section (94.5) integrated over angles.

In region I we have

$$q^2/m^2 \approx \delta_+^2 + \delta_-^2 + 2\delta_+ \delta_- \cos \varphi.$$

⁴²This value of the function can be obtained from formula (e. 7) (see II), connecting hypergeometric functions of the arguments z and $1 - z$.

From the variables δ_+ , δ_- , φ let us go over to the variables ξ_+ , ξ_- , z . By direct computation of the Jacobian of the transformation we find

$$\delta_+ d\delta_+ \cdot \delta_- d\delta_- \cdot d\varphi \rightarrow \frac{\varepsilon_+ \varepsilon_-}{8m^2} \frac{d\xi_+ d\xi_- d\varphi}{(\xi_+ \xi_-)^3 \sin \varphi},$$

where

$$1 - z = \frac{q^2}{m^2} \xi_+ \xi_- = \xi_+ + \xi_- - 2\xi_+ \xi_- + 2\sqrt{\xi_+ \xi_- (1 - \xi_+)(1 - \xi_-)} \cos \varphi.$$

Expressing from this $\cos \varphi$ and $\sin \varphi$ and substituting in (95.16), after simple algebraic transformations we obtain

$$\begin{aligned} d\sigma = A d\varepsilon_+ & \frac{2 d\xi_+ d\xi_- dz}{[z(1-z) - (1-z)(\xi_+ + \xi_- - 1)^2 - z(\xi_+ - \xi_-)^2]^{1/2}} \times \\ & \times \left\{ \frac{F^2(z)}{1-z} [(\varepsilon_+^2 + \varepsilon_-^2)z + 2\varepsilon_+ \varepsilon_- (\xi_+ - \xi_- - 1)^2] + \right. \\ & \left. + \frac{F'^2(z)}{\nu^2} [(\varepsilon_+^2 + \varepsilon_-^2)z + 2\varepsilon_+ \varepsilon_- (\xi_+ + \xi_- - 1)^2] \right\}, \\ A = & \left(\frac{\pi \nu}{\sinh \pi \nu} \right)^2 \frac{Z^2 \alpha r_e^2}{2\pi \omega^3}. \end{aligned}$$

Finally, we introduce instead of ξ_+ , ξ_- new "spherical" variables χ , ψ :

$$\begin{aligned} \xi_+ + \xi_- - 1 &= \sqrt{z} \sin \chi \cos \psi; \\ \xi_+ - \xi_- &= \sqrt{1-z} \sin \chi \sin \psi; \\ 0 &\leq \chi \leq \pi/2, \quad 0 \leq \psi \leq 2\pi; \\ 2 d\xi_+ d\xi_- &\rightarrow \sqrt{z(1-z)} \sin \chi \cos \chi d\chi d\psi. \end{aligned}$$

The indicated intervals of variation of χ and ψ correspond to the variation of ξ_+ , ξ_- from 0 to 1, i.e. δ_+ , δ_- (or, what is the same, θ_+ , θ_-) from 0 to ∞ ; the fast convergence of the integral permits such an extension of the region of variation of angles. After the transformation, the square root in the denominator reduces to $\sqrt{z(1-z)} \cos \chi$; the integration over $d\chi d\psi$ is elementary and gives

$$d\sigma = 2A \cdot 2\pi dz \left(\varepsilon_+^2 + \varepsilon_-^2 + \frac{2}{3} \varepsilon_+ \varepsilon_- \right) \left[\frac{F^2(z)}{1-z} + \frac{z}{\nu^2} F'^2(z) \right] d\varepsilon_+.$$

Here an extra factor of 2 has been introduced, taking into account the fact that the integration over z will be carried out from 0 to z_1 , whereas as the azimuth φ varies from 0 to π and from π to 2π each value of z is traversed twice.

The integration over dz is performed with the aid of formula (92.14), which for $\nu' = -\nu$ (and correspondingly real $F(z)$) takes the form

$$\frac{F^2}{1-z} + \frac{z}{\nu^2} F'^2 = \frac{1}{\nu^2} \frac{d}{dz} (z F F').$$

The integral of this expression is $z_1 F(z_1) F'(z_1) / \nu^2$. The value $z_1 F(z_1) \approx F(1)$, while the limiting expression for $F'(z_1 \rightarrow 1)$ is given by the formula⁴³

$$\frac{1}{\nu^2} F'(z) = F(1 - i\nu, 1 + i\nu, 2, z) \approx -[\ln(1 - z) + 2f(\nu)] \frac{\sinh \pi \nu}{\pi \nu},$$

where

$$f(\nu) = \frac{1}{2} [\Psi(1 + i\nu) + \Psi(1 - i\nu) - 2\Psi(1)] = \nu^2 \sum_{n=1}^{\infty} \frac{1}{n(n^2 + \nu^2)}, \quad (\text{X.62})$$

$$\Psi(z) = \Gamma'(z) / \Gamma(z).$$

Substituting all the found expressions in (95.18), we obtain the following final formula:

$$d\sigma_{\varepsilon_+} = 4Z^2 \alpha r_e^2 \left(\varepsilon_+^2 + \varepsilon_-^2 + \frac{2}{3} \varepsilon_+ \varepsilon_- \right) \left[\ln \frac{2\varepsilon_+ \varepsilon_-}{m\omega} - \frac{1}{2} - f(\alpha Z) \right] \frac{d\varepsilon_+}{\omega^3}. \quad (\text{X.63})$$

The total cross section of pair production by a photon with energy ω :

$$\sigma = \frac{28}{9} Z^2 \alpha r_e^2 \left[\ln \frac{2\omega}{m} - \frac{109}{42} - f(\alpha Z) \right]. \quad (\text{X.64})$$

We see that in these formulas the changes reduce to the subtraction from the logarithm of a universal function of the atomic number $f(\alpha Z)$. In Fig. 18 a graph of this function is given. When $\nu \ll 1$ the function $f(\nu) \approx 1.2\nu^2$.

§96. Exact theory of bremsstrahlung in the ultrarelativistic case

The matrix element for bremsstrahlung

$$M_{fi} = \int \psi_{\varepsilon' \mathbf{p}'}^{(-)*} (\alpha \mathbf{e}^*) e^{-i\mathbf{k}\mathbf{r}} \psi_{\varepsilon \mathbf{p}}^{(+)} d^3x; \quad (\text{X.65})$$

the wave functions of the initial $(\varepsilon, \mathbf{p})$ and final $(\varepsilon', \mathbf{p}')$ electrons contain in their asymptotics respectively an incoming and outgoing spherical wave. The computation of this integral is analogous to the computation of the matrix element (95.2). We shall, however, present here a different method of computing the bremsstrahlung cross section, based on the quasi-classicality of the process and not utilising the explicit form of the wave functions of the electron in the field of the nucleus; in this sense the method is not connected with the specific form of the potential of the field (*V. N. Baier, V. M. Katkov, 1968*).

In the process of bremsstrahlung the nucleus transfers to the electron and photon a momentum $\mathbf{q} = \mathbf{p}' + \mathbf{k} - \mathbf{p}$. As in the problem of pair production, one must distinguish two regions of values of the momentum transfer $q_{\perp}$, transverse with respect to $\mathbf{p}$:

$$\text{I) } m \gtrsim q_{\perp} \gg \omega m^2 / \varepsilon^2, \quad \text{II) } q_{\perp} \sim \omega m^2 / \varepsilon^2 \ll m. \quad (\text{X.66})$$

⁴³The derivation of these formulas can be found in the appendix to the paper by *Davies H., Bethe H. A., Maximon L. C. Phys. Rev. 1954. V. 93. P. 788.*

It is evident that in region I the cross section of photon emission is given by its Born value: for such $q_{\perp}$ the change in the recoil momentum of the nucleus upon radiation is inessential, as will be shown in § 98 (see the derivation of condition (98.10)). Therefore in region I the cross section of the process equals the product of the exact cross section of scattering of the electron in the field of a fixed nucleus and the probability of photon emission, which is independent of the form of the field. But according to (80.10) the scattering cross section in a Coulomb field for small angles coincides with its Born value. The same therefore applies to the cross section of the whole process in region I.

Thus, only region II requires special consideration. To small momentum transfers there corresponds the passage of the electron past the nucleus at large impact parameters: $\rho \sim 1/q_{\perp} \gtrsim \varepsilon/m^2$

$\gtrsim \varepsilon/m^2$. But at such distances the motion of the electron is certainly quasi-classical, as one can easily verify by a simple application of the usual quasi-classicality condition (46.7) (see III) to the ultrarelativistic equation (39.5).

The quasi-classicality of the motion permits applying the method already used in § 90 for magnetobremstrahlung radiation. In this case the expression (90.7) represents the probability of emission during a single passage of the electron past the nucleus.

For the function L that appeared in § 90 the formula (90.18) remains in force; the sole distinction consists in the form of the quasi-classical trajectory of the electron $\mathbf{r} = \mathbf{r}(t)$, along which the difference $\mathbf{r}_2 - \mathbf{r}_1$ is computed.

At large impact parameters the field of the nucleus can be considered weak. In the zeroth approximation the trajectory is a straight line passing at a distance ρ from the centre. In the next approximation we have the equation of motion (cf. I, § 20)

$$\frac{d\mathbf{p}}{dt} = -\frac{\rho}{r} \frac{dU}{dr},$$

where ρ is a vector in the xy plane, perpendicular to the initial momentum of the electron, and for r on the right-hand side of the equation one should take the zeroth-approximation function:

$$r \approx \sqrt{\rho^2 + v^2 t^2} \approx \sqrt{\rho^2 + t^2}.$$

Consequently,

$$\mathbf{p}(t) - \mathbf{p}_1 = -\rho \int_{t_1}^t \frac{dU}{dr} \frac{dt}{r}. \quad (\text{X.67})$$

With sufficient accuracy the velocity $\mathbf{v}(t) = \mathbf{p}(t)/\varepsilon$ (where the energy ε depends only on the magnitude, but not on the direction of $\mathbf{p}$) can be considered constant. One more integration then gives

$$\mathbf{r}(t) - \mathbf{r}_1 = \mathbf{v}_1(t - t_1) - \frac{1}{\varepsilon} \int_{t_1}^t [\mathbf{p}(t') - \mathbf{p}_1] dt'. \quad (\text{X.68})$$

Setting $t_1 = -\infty$, the quantities $\mathbf{p}_1 = \mathbf{p}(-\infty) \equiv \mathbf{p}$ and $\mathbf{v} = \mathbf{p}/\varepsilon$ will be the initial momentum and velocity of the electron.

Let us represent the probability (90.7) in the form

$$dw = |a(\rho)|^2 \frac{d^3k}{(2\pi)^3}, \quad (\text{X.69})$$

where

$$a(\rho) = e \sqrt{\frac{2\pi}{\omega}} \int_{-\infty}^{\infty} R(t) \exp \left\{ i \frac{\varepsilon}{\varepsilon'} [\omega t - \mathbf{k} \mathbf{r}(t)] \right\} dt, \quad (\text{X.70})$$

$$R(t) = \frac{u_{\varepsilon'}^* \mathbf{p}'}{\sqrt{2\varepsilon'}} (\alpha \mathbf{e}^*) \frac{u_{\varepsilon} \mathbf{p}}{\sqrt{2\varepsilon}},$$

where $\varepsilon' = \varepsilon - \omega$, $\mathbf{p}'(t) = \mathbf{p}(t) - \mathbf{k}$. The classical function $\mathbf{p}(t)$ is given by formula (96.3). If $\mathbf{p}$ is the initial momentum of the electron, then for a Coulomb field ($U = -\nu/r$, $\nu = Z\alpha$) we have

$$\mathbf{p}(t) = \mathbf{p} - \frac{\nu \rho}{\rho^2} \left[\frac{t}{\sqrt{\rho^2 + t^2}} + 1 \right],$$

$$\mathbf{r}(t) = \frac{\mathbf{p}}{\varepsilon} t - \frac{\rho}{\rho^2} \frac{\nu}{\varepsilon} \left[\sqrt{\rho^2 + t^2} + t \right].$$

Introducing the momentum transfer in classical scattering

$$\Delta = \mathbf{p}(\infty) - \mathbf{p}(-\infty) = -\frac{2\rho\nu}{\rho^2}, \quad (\text{X.71})$$

we can rewrite these formulas as

$$\mathbf{p}(t) = \mathbf{p} + \frac{1}{2} \Delta \left[\frac{t}{\sqrt{\rho^2 + t^2}} + 1 \right], \quad (\text{X.72})$$

$$\mathbf{r}(t) = \left(\mathbf{p} + \frac{1}{2} \Delta \right) \frac{t}{\varepsilon} + \frac{\Delta}{2\varepsilon} \sqrt{t^2 + \rho^2}.$$

Using now formula (90.20) for $R(t)$ and the expressions (96.8) for $\mathbf{p}(t)$ and $\mathbf{r}(t)$, one can carry out the integration over time in (96.6). It is accomplished by the introduction of the variable

$$\xi = -\frac{\varepsilon}{\varepsilon'} [\omega t - \mathbf{k} \mathbf{r}(t)]$$

instead of t and the use of the formula

$$\int_{-\infty}^{\infty} \frac{\xi e^{-i\xi} d\xi}{\sqrt{\chi^2 + \xi^2}} = 2i\chi K_1(\chi),$$

where K_1 is the MacDonald function. In the full execution of this computation, however, there is no necessity, since we require the expression $a(\rho)$ only for small values of the independent parameter Δ ($\Delta \ll m$). In this case we find

$$a(\rho) = w_f^* \mathbf{D} w_i \cdot \Delta \chi K_1(\chi), \quad (\text{X.73})$$

where

$$\chi = \rho \frac{\omega \varepsilon}{\varepsilon'} (1 - \mathbf{n} \mathbf{v}),$$

$\mathbf{n} = \mathbf{k}/\omega$, and $\mathbf{D}$ is some function of $\mathbf{p}$, ε and $\mathbf{k}$ (but not ρ), the precise form of which is inessential⁴⁴.

⁴⁴The spinors w_i and w_f can be considered constant during the integration, i.e. one can neglect the change of polarisation of the electron during its classical ultrarelativistic motion. This follows from the equations obtained in § 41.

Since in the ultrarelativistic case the photon is emitted at a small angle θ to the direction of velocity of the electron, we have

$$\chi \approx \rho \frac{\varepsilon}{\varepsilon'} \omega \left(1 - v + \frac{\theta^2}{2} \right),$$

or

$$\chi = \rho \frac{\omega m^2}{2\varepsilon\varepsilon'} (1 + \delta^2), \quad \delta = \frac{\theta\varepsilon}{m}. \quad (\text{X.74})$$

It was already mentioned that (96.5) is the probability of photon emission during a single passage of the electron past the nucleus at impact parameter ρ . The cross section of emission of a photon with given frequency and direction is obtained by multiplying this probability by $dp_x dp_y / v \approx dp_x dp_y \equiv d^2\rho$ and integrating over the impact parameters:

$$d\sigma = \frac{d^3k}{(2\pi)^3} \int |a(\rho)|^2 d^2\rho. \quad (\text{X.75})$$

One should not, however, think that this formula without the integration over $d^2\rho$ would also give the distribution of the final electrons in direction. The deflection of the electron during its motion along the classical orbit is determined uniquely by the external field and certainly does not coincide with the uncertain quantum-mechanical deflection (and the limiting value $\mathbf{p}'(\infty)$ of the classical function $\mathbf{p}'(t)$ does not therefore coincide with the real values of the momentum of the electron). To find this distribution it is necessary, therefore, to expand the wave function of the electrons in plane waves.

As is evident from (96.11), $a(\rho)$ is the amplitude of photon emission in a collision at impact parameter ρ . From the expressions (96.5), (96.6) this amplitude is determined only up to a phase factor. The latter is, obviously, $e^{-ik\rho}$ —by virtue of the presence of the time-independent term $r_\perp(\infty) = \rho$ in $\mathbf{r}(t)$, this constant factor must appear in $V_{fi}(t)$ and can be taken outside the integral sign. Since it is not an operator, it is not affected by commutation operations and, thus, the amplitude of the emission process is

$$e^{-ik\rho} a(\rho), \quad (\text{X.76})$$

where $a(\rho)$ is given by expression (96.9).

Let the electron be described as $z \rightarrow -\infty$ by a plane wave with momentum $\mathbf{p}$, directed along the z axis. This means that the wave function of the electron as $z \rightarrow -\infty$ does not depend on x and y and reduces to a constant, which can be set equal to 1.

Then the wave function of the electron that has passed through the field, as $z \rightarrow \infty$, equals⁴⁵

$$\psi(\infty) = S(\rho) = \exp \left\{ -i \int_{-\infty}^{\infty} U(x, y, z) dz \right\}. \quad (\text{X.77})$$

⁴⁵Cf. III, (131.4). We have in mind here the analogy between equation (39.5) (in which we set $p^2 \approx \varepsilon^2$) and the non-relativistic Schrödinger equation. Taking into account the difference of coefficients in these equations, it is easy to see that in our case the condition (131.1) (see III) of applicability of equation (131.4) (see III) is indeed satisfied. The fact that this formula does not apply to the region of arbitrarily large z is inessential for the same reasons as in III, § 131.

On the other hand, by the meaning of the transition amplitude (96.12) the wave function of the electron that has passed through the field and emitted a photon is

$$e^{-i\mathbf{k}\rho} a(\rho) S(\rho). \quad (\text{X.78})$$

The amplitude of the process of photon emission, in which the electron remains in a state with definite momentum $\mathbf{p}'$, is given by the corresponding Fourier component of the function (96.14), i.e.

$$a(\mathbf{q}_\perp) = \int e^{-i\mathbf{p}'\rho} e^{-i\mathbf{k}\rho} a(\rho) S(\rho) d^2\rho = \int e^{-i\mathbf{q}_\perp\rho} a(\rho) S(\rho) d^2\rho, \quad (\text{X.79})$$

where $\mathbf{q}_\perp$ is the transverse component of the momentum transfer vector to the nucleus (cf. III, (131.7)). The cross section of scattering with given value of $\mathbf{q}_\perp$ is then

$$d\sigma = |a(\mathbf{q}_\perp)|^2 \frac{d^3k}{(2\pi)^3} \frac{d^2q_\perp}{(2\pi)^2}. \quad (\text{X.80})$$

Let us now compute $S(\rho)$. In the case under consideration of a Coulomb field the integral in the exponent diverges, in correspondence with the divergence of the phase in Coulomb scattering. Therefore the integral must be taken between finite limits:

$$\int_{-R}^R U dz = -2\nu \int_0^R \frac{dz}{\sqrt{\rho^2 + z^2}} = -2\nu [\ln(R + \sqrt{R^2 + \rho^2}) - \ln \rho] \approx -2\nu \ln R + 2\nu \ln \rho$$

($R \gg \rho$). The first, constant, term is inessential, so that

$$S(\rho) = \exp(-2i\nu \ln \rho) = \rho^{-2i\nu}. \quad (\text{X.81})$$

Substituting (96.9), (96.17) in (96.15) and integrating over the directions of the vector ρ in the xy plane, we find

$$a(\mathbf{q}_\perp) \propto \nu \int_0^\infty \rho^{-2i\nu} K_1(\chi) J_1(q_\perp \rho) \rho d\rho, \quad (\text{X.82})$$

where J_1 is the Bessel function. The factors not containing $\nu = Z\alpha$ are not written out here.

We see that the dependence of the amplitude $a(\mathbf{q}_\perp)$ (and consequently also of the cross section (96.16)) on ν is contained in a separate factor. On the other hand, when $\nu \rightarrow 0$ the cross section must tend to its Born value. Therefore it is clear that the cross section will differ from the Born one only by a factor that is independent of the polarisation of the electron and does not affect the polarisation effects.

The integral (96.18) can be expressed through a hypergeometric function with the aid of the formula

$$\int_0^\infty x^{-\lambda} K_1(ax) J_1(bx) x dx = \frac{b \Gamma(2 - \lambda/2) \Gamma(1 - \lambda/2)}{2^\lambda a^{3-\lambda}} \left(1 + \frac{b^2}{a^2}\right)^{-1+\lambda/2} F\left(\frac{\lambda}{2}, 1 - \frac{\lambda}{2}, 2, \frac{b^2}{a^2 + b^2}\right)$$

This gives

$$a(\mathbf{q}_\perp) \propto \nu(1 - i\nu) \left(\frac{q}{2}\right)^{2i\nu} \Gamma^2(1 - i\nu) F(i\nu, 1 - i\nu, 2, z), \quad (\text{X.83})$$

where

$$z = 1 - \frac{m^4 \omega^2}{4q^2 \varepsilon^2 \varepsilon'^2} (1 + \delta^2)^2, \quad \delta = \frac{\varepsilon \theta}{m}; \quad (\text{X.84})$$

here it is used that in region II (see (96.2)) the component of the vector $\mathbf{q}$ parallel to $\mathbf{p}$ is

$$q_{\parallel}^2 = q^2 - q_{\perp}^2 \approx \frac{m^4 \omega^2}{4\varepsilon^2 \varepsilon'^2} (1 + \delta^2)^2. \quad (\text{X.85})$$

In this one can easily convince oneself, if one takes into account that in the indicated region the angles between the momenta $\mathbf{p}$, $\mathbf{p}'$ and $\mathbf{k}$ satisfy conditions (93.15).

The hypergeometric function in (96.19) can be reduced to the function $F(z)$ (95.15) with the aid of the formula

$$F(a, b+1, c+1, z) = \frac{c}{c-a} F(a, b, c, z) + \frac{c(1-z)}{b(a-c)} F'(a, b, c, z).$$

The final result takes then the form

$$d\sigma = d\sigma_B \frac{1}{F^2(1)} \left[F^2(z) + \frac{(1-z)^2}{v^2} F'^2(z) \right], \quad (\text{X.86})$$

where $d\sigma_B$ is the Born cross section (93.13) (*H. A. Bethe, L. Maximon, 1954*). When $q \gg m^2/\varepsilon$ we have $z \approx 1$, so that the entire coefficient

at $d\sigma_B$ tends to unity. In this sense formula (96.22), derived for region II, is automatically valid for all $q \lesssim m$. When $q \lesssim m^2/\varepsilon$ and the correction factor in (96.22) differs from unity, the vectors $\mathbf{p}$, $\mathbf{p}'$, $\mathbf{k}$ are almost coplanar and the quantities δ and δ' are almost equal to each other; this has already been taken into account in (96.22). Thus, q in expression (96.20) for z can be rewritten as

$$\frac{q^2}{m^2} = \delta^2 + \delta'^2 - 2\delta\delta' \cos \varphi + \frac{m^2 \omega^2}{4\varepsilon^2 \varepsilon'^2} (1 + \delta^2)^2, \quad (\text{X.87})$$

i.e. one can set $\delta = \delta'$ in the second term in (93.14), but not in the first term, which does not contain the small coefficient ($\sim m^2/\varepsilon^2$).

For finding the integral (over angles) cross section of radiation there is no need to carry out the integration anew, as is clear from the following considerations (*H. Olsen, 1955*). The different directions of $\mathbf{p}'$ (at given energy ε') correspond to the degeneracy of the final state of the electron. It is evident that the result of summation over states belonging to one degenerate level does not depend on the manner in which the complete set of these states is chosen. We can therefore use for the purposes of summation over directions of $\mathbf{p}'$ the system of functions $\psi_{\varepsilon' \mathbf{p}'}^{(-)}$ instead of the system $\psi_{\varepsilon' \mathbf{p}'}^{(+)}$ (necessary for computing the differential cross section), i.e. define the matrix element of bremsstrahlung as

$$M_{fi}^{\text{brem}} = \int \psi_{\varepsilon' \mathbf{p}'}^{(+)*} (\alpha \mathbf{e}^*) e^{-i\mathbf{k}\mathbf{r}} \psi_{\varepsilon \mathbf{p}}^{(+)} d^3x.$$

It is easy to verify that this integral coincides with the integral $(M_{fi}^{\text{pair}})^*$, if in the latter one replaces the parameters of the wave functions according to

$$\mathbf{p}_+, p_+, \varepsilon_+ \rightarrow -\mathbf{p}_-, -p_-, -\varepsilon_-; \quad \mathbf{p}_-, p_-, \varepsilon_- \rightarrow \mathbf{p}', p', \varepsilon'; \quad \mathbf{k} \rightarrow -\mathbf{k}$$

(and also replaces the variables of integration: $\mathbf{r} \rightarrow -\mathbf{r}$).

From this it is clear that the integral (over angles) bremsstrahlung cross section can be obtained from the integral pair-production cross section (95.20) by multiplying the latter by

$$\frac{\omega^2 d\omega}{p_+^2} \frac{d\omega}{\omega} \approx \frac{\omega^2 d\omega}{\varepsilon_+^2 \varepsilon_-}$$

(cf. (91.6)) and replacing $\varepsilon_+ \rightarrow -\varepsilon$, $\varepsilon_- \rightarrow \varepsilon'$. Thus, we find

$$d\sigma = 4Z^2 \alpha r_e^2 \frac{\varepsilon'}{\varepsilon} \left[\frac{\varepsilon}{\varepsilon'} + \frac{\varepsilon'}{\varepsilon} - \frac{2}{3} \right] \left(\ln \frac{2\varepsilon\varepsilon'}{m\omega} - \frac{1}{2} - f(\alpha Z) \right) \frac{d\omega}{\omega}. \quad (\text{X.88})$$

We see that the corrections to the Born formulas for the integral cross sections of bremsstrahlung and pair production are given by one and the same function $f(\alpha Z)$.

Formula (96.24), not connected with any restrictions on the magnitude of $Z\alpha$, permits passage to the classical limit $\hbar \rightarrow 0$, $Z\alpha \rightarrow \infty$. In this limit one must also set $\varepsilon \approx \varepsilon'$. Bearing in mind the asymptotic expression $\Psi(z) \approx \ln z$ for $|z| \rightarrow \infty$ and the value $\Psi(1) = -C$ (C is the Euler constant), we find for the effective retardation

$$\hbar\omega d\sigma = \frac{16Z^2 r_e^2 e^2}{3c} \left[\ln \frac{2\varepsilon^2}{mc\omega Ze^2} - \frac{1}{2} - C \right] d\omega. \quad (\text{X.89})$$

This expression, not containing $\hbar$, is the classical spectral distribution of the intensity of bremsstrahlung.

§97. Bremsstrahlung of an electron on an electron in the ultrarelativistic case

The bremsstrahlung of an electron on an electron is depicted by eight Feynman diagrams: four diagrams

and four “exchange” diagrams, obtained from the depicted ones by the interchange of p'_1 and p'_2 . We present here the results of the calculations for the ultrarelativistic case (*G. Altarelli, F. Buccella, 1964; V. N. Baier, V. S. Fadin, V. A. Khoze, 1966*)⁴⁶.

In the laboratory frame (the rest frame of one of the initial electrons—say, the second), the cross section of radiation integrated over the photon directions can be represented as a sum $d\sigma = d\sigma^{(1)} + d\sigma^{(2)}$, where

$$d\sigma^{(1)} = 4\alpha r_e^2 \frac{d\omega}{\omega} \frac{\varepsilon - \omega}{\varepsilon} \left[\frac{\varepsilon}{\varepsilon - \omega} + \frac{\varepsilon - \omega}{\varepsilon} - \frac{2}{3} \right] \left(\ln \frac{2\varepsilon(\varepsilon - \omega)}{m\omega} - \frac{1}{2} \right); \quad (\text{X.90})$$

$$d\sigma^{(2)} = \frac{2}{3} \alpha r_e^2 \frac{m d\omega}{\omega^2} \left\{ \left(4 - \frac{m}{\omega} + \frac{m^2}{4\omega^2} \right) \ln \frac{2\varepsilon}{m} - 2 + \frac{2m}{\omega} - \frac{5m^2}{8\omega^2} \right\} \omega \gg \frac{m}{2}; \quad (\text{X.91})$$

$$d\sigma^{(2)} = \frac{2}{3} \alpha r_e^2 \frac{d\omega}{\omega} \left\{ 8 \left(1 - \frac{m}{\omega} + \frac{\omega^2}{m^2} \right) \ln \frac{\varepsilon}{m} - \left(1 - \frac{2\omega}{m} \right) \ln \left(1 - \frac{2\omega}{m} \right) \times \right. \\ \left. \times \left[\frac{m^3}{4\omega^3} - \frac{m^2}{2\omega^2} + \frac{3m}{\omega} - \frac{4\omega}{m} \right] - \frac{m^2}{2\omega^2} + \frac{3m}{\omega} - 2 + \frac{m^2}{2\omega^2} - \frac{4\omega^2}{m^2} \right\} \omega \leq \frac{m}{2}; \quad (\text{X.92})$$

⁴⁶The corresponding calculations can be found in the book by Baier, Katkov and Fadin indicated on p. 454.

(ε is the initial energy of the first electron).

The accuracy of these formulas is up to terms of relative order m/ε . With this accuracy it turns out that the contributions to the cross section from the various diagrams do not interfere with each other, and in this sense $d\sigma^{(1)}$ and $d\sigma^{(2)}$ correspond to radiation by each of the two electrons—respectively the fast electron and the recoil electron (diagrams (97.1a) and (97.1b)).

The exchange-type diagrams give the same contribution to the cross section as the “direct” diagrams. By virtue of the identity of the electrons the total contribution of the direct and exchange diagrams must be divided by 2; therefore one can consider only the contribution of the direct diagrams and not take into account the identity of the particles. For the collision of an electron with a positron, in place of exchange diagrams there appear annihilation diagrams. Their contribution, however, turns out to be of relative order m/ε , i.e. negligible. Therefore, with the indicated accuracy, the bremsstrahlung cross sections in electron-electron and electron-positron collisions are the same.

When $\omega \gg m$ the ratio

$$\frac{d\sigma^{(2)}}{d\sigma^{(1)}} \sim \frac{m}{\omega} \ll 1,$$

i.e. the radiation by the recoil electron is small compared with the radiation by the fast electron (when this ratio reaches order m/ε , formula (97.3), of course, loses its meaning). On the contrary, when $\omega \ll m$ both parts of the cross section are nearly equal:

$$d\sigma^{(1)} = \frac{16}{3} \alpha r_e^2 \frac{d\omega}{\omega} \ln \frac{2\varepsilon^2}{m\omega}, \quad d\sigma^{(2)} = \frac{16}{3} \alpha r_e^2 \frac{d\omega}{\omega} \ln \frac{\varepsilon}{\omega}, \quad \omega \ll m. \quad (\text{X.93})$$

For the validity of formulas (97.2)–(97.5) it is necessary that at least one of the electrons after the radiation remain ultrarelativistic. In other words, the photon frequency must be sufficiently far from the hard boundary of the spectrum, i.e. from the maximum frequency $\omega_{\max}$ that can be radiated. The final energy of the electrons will be minimal, and the photon energy maximal, when both electrons move after the radiation in the direction of the photon and have equal velocities. Then from the conservation laws we have

$$\varepsilon + m = \omega_{\max} + 2\varepsilon', \quad |\mathbf{p}| = \omega_{\max} + 2|\mathbf{p}'|.$$

Eliminating from this ε' and $\mathbf{p}'$, we obtain

$$(\varepsilon + m - \omega_{\max})^2 - (|\mathbf{p}| - \omega_{\max})^2 = 4m^2,$$

whence

$$\omega_{\max} = \frac{m(\varepsilon - m)}{m + \varepsilon - |\mathbf{p}|}. \quad (\text{X.94})$$

When $\varepsilon \gg m$ we have $\omega_{\max} \approx \varepsilon$. Thus, formulas (97.2)–(97.4) are valid under the condition

$$\omega_{\max} - \omega \sim \varepsilon - \omega \gg m. \quad (\text{X.95})$$

The cross section of radiation by the fast electron (97.2) coincides exactly with the cross section of radiation of an electron on a nucleus with $Z = 1$ (formula (93.17)). This coincidence is not accidental, and its causes are elucidated by analysis of the role of recoil in the radiation process.

In deriving formula (93.17) we neglected the recoil of the stationary particle (the nucleus)—replaced it by a constant external field. This reduces to the neglect of the time component of the 4-vector of momentum transfer $q = p' - p + k$ (the recoil energy). Let us show that in the ultrarelativistic case such neglect is permissible for radiation by an electron not only on a nucleus but also on an electron.

Let us write q^2 in the form

$$-q^2 = -(\varepsilon' + \omega - \varepsilon)^2 + (p'_{\parallel} + \omega - p_{\parallel})^2 + (\mathbf{p}'_{\perp} - \mathbf{p}_{\perp})^2, \quad (\text{X.96})$$

where the subscripts denote the components of the vectors $\mathbf{p}'$ and $\mathbf{p}$ (the initial and final momenta of the electron), parallel and perpendicular to the direction of the photon $\mathbf{k}$. In the ultrarelativistic case the angles θ and θ' (between $\mathbf{k}$ and respectively $\mathbf{p}$ and $\mathbf{p}'$) are small: $\theta \lesssim m/\varepsilon$, $\theta' \lesssim m/\varepsilon'$. Therefore

$$|\mathbf{p}_{\perp}| \sim |\mathbf{p}|\theta \sim m, \quad p_{\parallel} \approx |\mathbf{p}| - \frac{p_{\perp}^2}{2|\mathbf{p}|} \approx \varepsilon - \frac{m^2}{2\varepsilon} - \frac{\mathbf{p}_{\perp}^2}{2\varepsilon}, \quad (\text{X.97})$$

and analogously for $\mathbf{p}'_{\perp}$, $p'_{\parallel}$.

Without account of recoil we have $\varepsilon' + \omega - \varepsilon = 0$, the difference $p'_{\parallel} + \omega - p_{\parallel} \sim m^2/\varepsilon$, so that

$$-q^2 \approx (\mathbf{p}'_{\perp} - \mathbf{p}_{\perp})^2 \sim m^2. \quad (\text{X.98})$$

The recoil energy (on an electron):

$$q_0 = \varepsilon' + \omega - \varepsilon \sim \mathbf{q}^2/(2m) \sim m. \quad (\text{X.99})$$

The change of $\mathbf{p}'_{\perp}$ due to the change of ε' can be neglected. Therefore the first two terms in (97.8) give the change of q^2 upon account of

recoil; let us denote it by Δq^2 . Using (97.9), we obtain

$$\Delta q^2 \approx (\varepsilon' + \omega - \varepsilon) \left(-\frac{m^2}{\varepsilon'} - \frac{\mathbf{p}_{\perp}^2}{\varepsilon'} + \frac{m^2}{\varepsilon} + \frac{\mathbf{p}_{\perp}^2}{\varepsilon} \right) \sim m^2 \cdot \frac{m}{\varepsilon}.$$

Comparing with (97.10), we see that $\Delta q^2 \ll |q^2|$, which justifies the neglect of recoil⁴⁷.

The fact that the fast particle radiates into a narrow cone (with opening angle $\sim m/\varepsilon$) in the direction of its motion permits obtaining the radiation cross section in the centre-of-mass frame by a simple recalculation of the cross section (97.2) from the laboratory frame⁴⁸.

In the centre-of-mass frame both electrons radiate equally, each in the direction of its motion (this circumstance visually explains the reason for the absence of interference between the radiation of both particles). The energy of the ultrarelativistic electron in the centre-of-mass frame is related to its energy ε in the laboratory frame by $2E^2 = m\varepsilon$, and the frequencies Ω and ω of the photon in these frames by $\omega/\varepsilon = \Omega/E$ (these equalities

⁴⁷This conclusion is, of course, all the more valid for radiation of an electron on a nucleus, for which the recoil energy $q_0 \approx \mathbf{q}^2/(2M) \sim m^2/M$, where M is the mass of the nucleus.

⁴⁸In the general case such a recalculation is impossible, since the contribution to the spectrum in a given frequency interval $d\omega$ arises from photons emitted in essentially different directions.

are easy to obtain by comparing the values of the invariants $(p_1 p_2)$ and $(p_1 k)$ in both frames). Therefore for the cross section of radiation by each of the electrons in the centre-of-mass frame we find

$$d\sigma^{(1)} = d\sigma^{(2)} = 4\alpha r_e^2 \frac{d\Omega}{\Omega} \frac{E - \Omega}{E} \left[\frac{E}{E - \Omega} + \frac{E - \Omega}{E} - \frac{2}{3} \right] \times \\ \times \left(\ln \frac{4E^2(E - \Omega)}{m^2 \Omega} - \frac{1}{2} \right). \quad (\text{X.100})$$

For the applicability of (97.12) it is also necessary that the photon frequency not be close to the boundary of the spectrum. For an ultrarelativistic particle the transformation indicated above directly gives from $\omega_{\max} \approx \varepsilon$

$$\Omega_{\max} \sim \omega_{\max} E / \varepsilon \approx E. \quad (\text{X.101})$$

Thus, in the centre-of-mass frame the electrons can radiate only half their total energy $2E$. A direct computation of $\Omega_{\max}$ is easily carried out by noting that after radiation of such a photon the electrons will move (in the same frame) with equal velocities in the direction opposite to the direction of the photon. We have

$$2E = 2E' + \Omega_{\max}, \quad 2|\mathbf{p}'| = \Omega_{\max},$$

whence

$$\Omega_{\max} = \frac{\mathbf{p}^2}{m} = E - \frac{m^2}{E}, \quad (\text{X.102})$$

and in the ultrarelativistic case we again obtain (97.13). Thus, formula (97.12) is applicable under the condition

$$\Omega_{\max} - \Omega \sim E - \Omega \gg m. \quad (\text{X.103})$$

Let us now give formulas for radiation in the centre-of-mass frame in the opposite limiting case, near the boundary of the spectrum, when⁴⁹

$$\Omega_{\max} - \Omega \ll m. \quad (\text{X.104})$$

Since in this case the recoil is very substantial, the results differ from the case of scattering on a fixed centre and turn out to be different for electron-electron and electron-positron scattering (*V. N. Baier, V. S. Fadin, V. A. Khoze, 1967*).

In the case of electron scattering on an electron, besides the squares of diagrams (97.1), a contribution to the radiation cross section near the boundary of the spectrum is also given by the products (interference terms) of direct and exchange diagrams in which one and the same initial particle radiates.

This is connected with the fact that near the boundary the final particles have close momenta and there are no reasons for the smallness of the exchange terms. The final answer for the cross section:

$$d\sigma = 2\alpha r_e^2 \frac{[E(\Omega_{\max} - \Omega)]^{1/2}}{m} \frac{d\Omega}{\Omega_{\max}}. \quad (\text{X.105})$$

⁴⁹The result obtained in the Born approximation is, of course, valid only so long as the relative velocity of the final electrons is large compared with α . In the opposite case one should take into account the interaction of the particles in the final state.

In the scattering of an electron on a positron a logarithmically large contribution to the radiation cross section comes from the squares of annihilation diagrams in which the initial particles radiate:

With non-logarithmic accuracy the squares of other diagrams are also substantial. The interference terms, however, are small. The final answer:

$$d\sigma = 2\alpha r_e^2 \frac{[E(\Omega_{\max} - \Omega)]^{1/2}}{m} \left(\ln \frac{2E}{m} + 1 \right) \frac{d\Omega}{\Omega_{\max}}. \quad (\text{X.106})$$

Thus, radiation in electron-positron scattering is logarithmically large compared with radiation in electron-electron scattering.

§ 98. Emission of soft photons in collisions

Let $d\sigma_0$ be the cross section of some process of scattering of charged particles, which may be accompanied by the emission of a definite number of photons. Alongside this process let us consider also another process, differing from the first only by the emission of one additional photon. If the frequency ω of this additional photon is sufficiently small (the corresponding conditions will be formulated below), then the cross section $d\sigma$ is in a simple way related to $d\sigma_0$.

Indeed, at small $d\sigma_0$ one can neglect the reverse influence of the emission of this quantum on the scattering process. In other words, the cross section $d\sigma$ can be represented in the form of a product of two independent factors: the cross section $d\sigma_0$ and the probability $d\omega$ of emission of one photon in the collision. The emission of a soft photon is a quasi-classical process; therefore its probability coincides with the classically computed number of emitted quanta, i.e. with the classical intensity (total energy) of the radiation dI , divided by ω ($= \hbar\omega$). Thus,

$$d\sigma = d\sigma_0 \frac{dI}{\omega}. \quad (\text{X.107})$$

Let us show that this formula can be obtained by the general rules of the diagram technique (*J. M. Jauch, F. Rohrlich*, 1954). The diagrams of the process with an additional photon are obtained from the diagrams of the basic process by adding an external photon line, “branching off” from some (external or internal) electron line, i.e. by the replacement

It is easy to see that the main role will be played by diagrams obtained by such a replacement in the external electron lines. Indeed, if p is the momentum of the external line ($p^2 = m^2$), then for small k also $(p - k)^2 \approx m^2$, i.e. the propagator $G(p - k)$ added in the diagram will be near its pole.

For the line of the initial electron p the replacement (98.2) reduces to a replacement in the amplitude of the reaction:

$$\begin{aligned} u(p) &\rightarrow e\sqrt{4\pi} G(p - k)(\gamma e^*) u(p) = \\ &= e\sqrt{4\pi} \frac{\gamma p - \gamma k + m}{(p - k)^2 - m^2} (\gamma e^*) u(p) \approx -e\sqrt{4\pi} \frac{\gamma p + m}{2(pk)} (\gamma e^*) u(p). \end{aligned}$$

Noting that

$$(\gamma p)(\gamma e^*) = 2pe^* - (\gamma e^*)(\gamma p), \quad \gamma p u(p) = mu(p),$$

we obtain the replacement rule in the form

$$u(p) \rightarrow -e\sqrt{4\pi} \frac{(pe^*)}{(pk)} u(p). \quad (\text{X.108})$$

Analogously, for the line of the final electron p' the replacement in the diagram means a replacement in the amplitude:

$$u(p') \rightarrow e\sqrt{4\pi} \bar{u}(p') \frac{(p'e^*)}{(p'k)}. \quad (\text{X.109})$$

In all remaining parts of the diagrams one can in general neglect the changes in the momenta of lines associated with the emission of the photon k . It is assumed here that the photon energy ω is in all cases small compared with the energies of all particles participating in the reaction (including compared with the energies of hard photons radiated, if there are such).

For definiteness let the cross section $d\sigma_0$ refer to the scattering of an electron on a fixed nucleus (with possible emission of hard photons). The amplitude of this process, which we shall conventionally call elastic, has the form

$$M_{fi}^{\text{el}} = \bar{u}(p') M u(p).$$

Carrying out in it the replacement (98.3) once and (98.4) another time, and combining the results, we obtain the amplitude of bremsstrahlung of the same hard photons and a soft photon k ⁵⁰:

$$M_{fi} = M_{fi}^{\text{el}} e\sqrt{4\pi} \left(\frac{(p'e^*)}{(p'k)} - \frac{(pe^*)}{(pk)} \right). \quad (\text{X.110})$$

The corresponding cross section

$$d\sigma = d\sigma_{\text{el}} \cdot 4\pi e^2 \left| \frac{(p'e)}{(p'k)} - \frac{(pe)}{(pk)} \right|^2 \frac{d^3k}{(2\pi)^3\omega}. \quad (\text{X.111})$$

Summing over the polarisations of the photon k , we obtain

$$d\sigma = -e^2 \left| \frac{p'}{(p'k)} - \frac{p}{(pk)} \right|^2 \frac{d^3k}{4\pi^2\omega} d\sigma_{\text{el}}. \quad (\text{X.112})$$

Expressed through three-dimensional quantities, this formula has the form⁵¹

$$d\sigma = \alpha \left(\frac{[\mathbf{v}'\mathbf{n}]}{1 - \mathbf{v}'\mathbf{n}} - \frac{[\mathbf{v}\mathbf{n}]}{1 - \mathbf{v}\mathbf{n}} \right)^2 \frac{d\omega d\mathbf{o}_k}{4\pi^2\omega} d\sigma_{\text{el}}, \quad (\text{X.113})$$

⁵⁰Let us draw attention to the fact that the appearance of the difference in this formula is a natural result of gauge invariance: the amplitude of the reaction must not change under the replacement of the 4-vector of polarisation $e \rightarrow e + \text{const} \cdot k$.

⁵¹For the derivation it is convenient to return to (98.6), setting $p = (\varepsilon, \varepsilon\mathbf{v})$, $pk = \varepsilon\omega(1 - \mathbf{v}\mathbf{n})$, ..., $e = (0, \mathbf{e})$ and carrying out the summation over polarisations afresh with the aid of (45.4a).

where $\mathbf{n} = \mathbf{k}/\omega$, and $\mathbf{v}$ and $\mathbf{v}'$ are the initial and final velocities of the electron. We see that the expression standing before $d\sigma_{\text{el}}$ indeed coincides (divided by ω) with the classical intensity of radiation (cf. II, (69.4)), as was asserted in formula (98.1).

The condition of applicability of the obtained formulas requires also, besides the smallness of ω compared with ε , that the momentum transfer to the nucleus q be large compared with the change $\delta\mathbf{q}$ of this quantity associated with the emission of a soft photon. We have

$$\delta\mathbf{q} = (\mathbf{p}' - \mathbf{p} - \mathbf{k}) - (\mathbf{p}' - \mathbf{p})_{\omega=0} = \delta\mathbf{p}' - \mathbf{k},$$

where

$$|\delta\mathbf{p}'| \sim \frac{\partial|\mathbf{p}'|}{\partial\varepsilon} \omega \sim \frac{\omega}{v},$$

and $|\mathbf{k}| = \omega$. In the non-relativistic case ($v \ll 1$) we therefore obtain the condition

$$\omega/|\mathbf{q}|v \ll 1. \quad (\text{X.114})$$

In scattering on a Coulomb (and in general on a slowly decreasing with distance) potential $|\mathbf{q}| \sim 1/\rho$ (ρ is the impact parameter), so that this condition can also be represented in the form $\omega\tau \ll 1$, where $\tau \sim \rho/v$ is the characteristic collision time.

In the ultrarelativistic case the photons are radiated mainly in directions close to $\mathbf{v}$ or $\mathbf{v}'$ (as is evident from one of the denominators in (98.8)). If the scattering angle θ of the electron is small, then the directions of all three vectors $\mathbf{p}$, $\mathbf{p}'$, $\mathbf{n}$ are close to each other. Then

$$|\delta\mathbf{q}| = |\delta\mathbf{p}'| - |\mathbf{k}| = \omega \left(\frac{1}{v} - 1 \right) \sim \frac{\omega m^2}{\varepsilon^2},$$

and since $|\mathbf{q}| \sim \varepsilon\theta$, we obtain the condition

$$\theta \gg \frac{\omega}{\varepsilon} \frac{m^2}{\varepsilon^2}. \quad (\text{X.115})$$

By virtue of the quasi-classical character of formulas (98.5)–(98.8) they are valid for radiation by any charged particles (not necessarily electrons, for which the derivation was carried out). In the general case, when several particles participate in the reaction, formula (98.5) must be written in the form

$$M_{fi} = M_{fi}^{\text{el}} e\sqrt{4\pi} \sum Z \left(\frac{(p'e^*)}{(p'k)} - \frac{(pe^*)}{(pk)} \right), \quad (\text{X.116})$$

where the summation is carried out over all particles (with charges Ze); correspondingly formulas (98.6)–(98.8) also change. In particular, in the non-relativistic case

$$M_{fi} = M_{fi}^{\text{el}} \frac{e\sqrt{4\pi}}{\omega} \sum Z(\mathbf{v}' - \mathbf{v})\mathbf{e}^*. \quad (\text{X.117})$$

For two particles this formula takes the form

$$M_{fi} = M_{fi}^{\text{el}} \frac{\sqrt{4\pi}}{\omega} \left(\frac{Z_1 e}{m_1} - \frac{Z_2 e}{m_2} \right) \mathbf{q}\mathbf{e}^*, \quad (\text{X.118})$$

$$\mathbf{q} = m(\mathbf{v}' - \mathbf{v}), \quad m = \frac{m_1 m_2}{m_1 + m_2},$$

where $\mathbf{v}$ and $\mathbf{v}'$ are the relative velocity of the particles before and after the collision. Integrating the square $|M_{fi}|^2$ over the photon emission directions and summing over its polarisations, we obtain from this the non-relativistic spectral distribution of the radiation in the form

$$d\sigma = d\sigma_{\text{el}} \frac{2e^2}{3\pi} \left(\frac{Z_1}{m_1} - \frac{Z_2}{m_2} \right)^2 \mathbf{q}^2 \frac{d\omega}{\omega}.$$

The obtained results generalise to the case of simultaneous emission of several soft photons. For each of the photons one adds to the amplitude M_{fi} its own factor of the type standing at M_{fi}^{el} in (98.5). In this one can easily convince oneself directly, say, on the example of two photons. The lines of both emitted photons must be added to the external electron lines, in two different sequences, i.e. the diagram with external line p is replaced by two diagrams with lines

Each of them contains the factor (the denominators of the electron propagators)

$$\frac{1}{2((pk_1) + (pk_2))} \frac{1}{2(pk_1)} \quad \text{or} \quad \frac{1}{2((pk_1) + (pk_2))} \frac{1}{2(pk_2)}.$$

Their sum is

$$\frac{1}{2(pk_1) \cdot 2(pk_2)},$$

i.e. it contains the product of two independent factors, corresponding to the first and second photon. After this the sum of all the diagrams is assembled (by virtue of gauge invariance) into the product of differences

$$\left(\frac{(p'e_1^*)}{(p'k_1)} - \frac{(pe_1^*)}{(pk_1)} \right) \left(\frac{(p'e_2^*)}{(p'k_2)} - \frac{(pe_2^*)}{(pk_2)} \right).$$

Correspondingly the factorisation of the amplitude extends to the factor and also to the cross section of the process. Thus, soft photons are emitted independently. The cross section of the process with emission of n soft photons can be represented in the form

$$d\sigma = d\sigma_{\text{el}} dw_1 dw_2 \dots dw_n, \quad (\text{X.119})$$

where $dw_1, dw_2, \dots$ are the probabilities of emission of each individual photon $dk_1, dk_2, \dots$. Upon integration of this formula over a finite interval of the variables (frequencies and directions), equal for all the quanta, a factor $1/n!$ must be introduced, taking into account the identity of the photons.

If one integrates the radiation cross section (98.1) over frequencies in some finite interval from ω_1 to ω_2 , we obtain an expression of the form

$$d\sigma \sim \alpha \ln \frac{\omega_2}{\omega_1} d\sigma_{\text{el}} \quad (\text{X.120})$$

(cf. (98.8)). It is assumed here that both frequencies are soft, so that their possible values are limited by the condition of applicability of the method. With logarithmic accuracy, however, one can set $\omega_2 \sim \varepsilon$, where ε is the initial energy of the radiating particle. The

values of ω_1 are in no way limited from below. Sending ω_1 to zero, we see that the cross section of emission of all possible soft quanta turns to infinity. Let us elucidate the meaning of this situation—the so-called *infrared catastrophe* (F. Bloch, A. Nordsieck, 1937). When

$$\alpha \ln \frac{\varepsilon}{\omega_1} \gtrsim 1 \quad (\text{X.121})$$

$d\sigma$ will be $d\sigma \gtrsim d\sigma_{\text{el}}$. But this means the inapplicability of perturbation theory—the impossibility of computing $d\sigma$ as a quantity of higher order of smallness than $d\sigma_{\text{el}}$. In other words, the parameter of smallness should in the given case be not α , but the product $\alpha \ln(\varepsilon/\omega_1)$:

Thus, the derivation of formulas (98.5), (98.6) on the basis of perturbation theory turns out to be incorrect at sufficiently small frequencies. On the other hand, the classical formula for the intensity dI (see II, (69.4)) is applicable to a greater extent, the smaller is ω . Therefore formula (98.1) remains correct, if one somewhat modifies its meaning toward greater classicality. Namely, in (98.1) it was assumed that one photon is radiated; then the energy lost by the particle to radiation coincides with ω and the “cross section of the relative energy loss” is given by the expression $\omega d\sigma/\varepsilon$, i.e.

$$d\sigma_{\text{el}} \frac{dI}{\varepsilon}. \quad (\text{X.122})$$

In reality, however, at sufficiently small ω the probability of radiation is not small, and the probability of radiation of two or more photons is not less, but greater than the probability of radiation of one photon. Under these conditions expression (98.17) remains valid, but the classical intensity dI will determine not the probability of radiation of one photon, but the mean number of radiated photons, and

$$d\bar{n} = \frac{dI}{\omega}, \quad (\text{X.123})$$

or in a finite interval of frequencies

$$\bar{n} = \int_{\omega=\omega_1}^{\omega_2} \frac{dI}{\omega}. \quad (\text{X.124})$$

Since soft photons are radiated statistically independently (this is valid in all approximations of perturbation theory), the Poisson formula can be applied to the process of multiple radiation: the probability $w(n)$ of emission of n photons is expressed through the mean number $\bar{n}$ by the formula

$$w(n) = \frac{\bar{n}^n}{n!} \exp(-\bar{n}). \quad (\text{X.125})$$

Let us represent the cross section of the scattering process with radiation of photons in the form

$$d\sigma = d\sigma_{\text{el}} \cdot w(n). \quad (\text{X.126})$$

Since $\sum w(n) = 1$, $d\sigma_{\text{el}}$ represents the total cross section of scattering accompanied by any soft radiation. This circumstance is obvious from classical considerations; in the

theory of perturbations, however, $d\sigma_{\text{el}}$ is the cross section of purely elastic scattering. But the theory of perturbations here is inapplicable. It turns out that

$d\sigma_{\text{el}}$, computed by perturbation theory as the cross section of elastic scattering, in fact takes into account the emission of any soft photons. As regards the cross section of purely elastic scattering, it is in fact equal to zero: when $\omega_1 \rightarrow 0$ the mean number $\bar{n} \rightarrow \infty$, and according to (98.20) the probability of emission of any finite number of photons goes to zero⁵².

Problems

1. Find the spectral distribution of the bremsstrahlung of soft photons in scattering of an ultrarelativistic electron on a nucleus⁵³.

Solution. Integration of formula (98.8) over $d\mathbf{o}_k$ gives

$$d\sigma = \alpha F(\xi) \frac{d\omega}{\omega} d\sigma_{\text{el}}, \quad (1)$$

where

$$F(\xi) = \frac{2}{\pi} \left[\frac{2\xi^2 + 1}{\sqrt{\xi^2 + 1}} \ln(\xi + \sqrt{\xi^2 + 1}) - 1 \right], \quad \xi = \frac{|\mathbf{p}|}{m} \sin \frac{\theta}{2} \quad (2)$$

($\mathbf{p}$ is the momentum, θ the scattering angle of the electron). In the ultrarelativistic case the essential role is played by the angular region

$$\frac{m^2\omega}{\varepsilon^2} \ll \theta \ll \frac{m}{\varepsilon} \quad (3)$$

(the lower bound is condition (98.10), the upper bound from above). At this $\xi \approx \varepsilon\theta/(2m) \ll 1$, so that $F(\xi) \approx (8/(3\pi))\xi^2$, and the elastic scattering cross section on a nucleus (see (80.10))

$$d\sigma_{\text{el}} \approx 4Z^2 r_e^2 \frac{m^2 d\omega}{\varepsilon^2 \theta^4}. \quad (4)$$

The integral

$$d\sigma_\omega = \frac{16}{3} Z^2 \alpha r_e^2 \frac{d\omega}{\omega} \int \frac{d\theta}{\theta} \quad (5)$$

diverges logarithmically; it is cut off from below at $\theta \sim m^2\omega/\varepsilon^2$, from above at $\theta \sim m/\varepsilon$, so that

$$d\sigma_\omega = \frac{16}{3} Z^2 \alpha r_e^2 \frac{d\omega}{\omega} \ln \frac{\varepsilon^2}{m\omega}$$

in agreement with the logarithmic part of formula (93.17) (in which one should set $\varepsilon \approx \varepsilon'$). Achieving non-logarithmic accuracy is possible only by going beyond the quasi-classical region.

⁵²We shall return to the discussion of this situation in § 130 in connection with the study of radiative corrections.

⁵³The formulas given below in the application of formula (98.7) are due to V. N. Baier and V. M. Galitsky (1964).

2. For the collision of two ultrarelativistic electrons, determine (in the centre-of-mass frame) the cross section of simultaneous emission of two soft photons in opposite directions at small angles to the momenta of the electrons.

Solution. Photons flying in opposite directions are emitted by different electrons. The cross section of simultaneous emission:

$$d\sigma = d\sigma_{\text{el}} \cdot \alpha F(\xi) \frac{d\omega_1}{\omega_1} \cdot \alpha F(\xi) \frac{d\omega_2}{\omega_2}, \quad \xi = \frac{\varepsilon}{m} \sin \frac{\theta}{2} \quad (6)$$

where ε is the energy of each electron, θ the scattering angle in the centre-of-mass frame. In the ultrarelativistic case the elastic scattering cross section for small angles in the centre of mass coincides with (4) (cf. (81.11)). Unlike σ (1) the cross section (6) behaves as $\theta d\theta$ when $\theta \rightarrow 0$, so that the integration over angles is convergent down to $\theta = 0$, and one should use the exact expression (2). On the other hand, the main contribution to the integral cross section comes from $\theta \sim m/\varepsilon$, so that $F(\xi)$ should also be used exactly. The result of integration of the cross section over the scattering angles:

$$d\sigma_{\omega_1\omega_2} = \frac{2}{\pi} \left[5 + \frac{7}{2} \zeta(3) \right] r_e^2 \alpha^2 \frac{d\omega_1}{\omega_1} \frac{d\omega_2}{\omega_2} = 5.9 r_e^2 \alpha^2 \frac{d\omega_1}{\omega_1} \frac{d\omega_2}{\omega_2}.$$

(ζ is the Riemann zeta function; $\zeta(3) = 1.202$).

§ 99. The method of equivalent photons

Let us compare two processes, described by the diagrams:

The circles depict conventionally the entire internal part of the diagrams. Diagram *a*) depicts the collision of a photon k ($k^2 = 0$) with some particle with 4-momentum q (and mass m ; $q^2 = m^2$). As a result of the collision a system (particle or group of particles) with total 4-momentum Q is formed. Diagram *b*) depicts the collision of the same particle q with another particle, whose 4-momentum is p , and mass M ($p^2 = M^2$). As a result of the collision this latter particle acquires 4-momentum p' and there is formed the same system Q . The second process can be considered as the collision of the particle q with a virtual photon emitted by the particle p , the momentum of which $k = p - p'$ ($k^2 < 0$). If $|k^2|$ is small,

the virtual photon differs little from a real one. It is obvious that such a situation can be encountered in collisions of very fast particles: the electromagnetic field of a charged particle moving with velocity $v \approx 1$ is almost transverse and therefore close in its properties to the field of a light wave. Under these conditions the cross section of process *b*) can be expressed through the cross section of process *a*)⁵⁴.

Thus, let us consider particle M as ultrarelativistic: its energy (in the rest frame of particle m) $\varepsilon \gg M$. If the masses of the colliding particles m and M are different, then for definiteness we shall assume that $m < M$.

⁵⁴The method described below was developed by Weizsäcker and Williams (*K. Weizsäcker, E. J. Williams*, 1934); the basic idea of this method was already put forward earlier by Fermi (*E. Fermi*, 1924).

The amplitude of process *a*) (with participation of a real photon) can be represented in the form

$$M_{fi}^{(r)} = -e\sqrt{4\pi}(e_\mu J^\mu), \quad (\text{X.127})$$

where e_μ is the 4-vector of polarisation of the photon, and J^μ the transition current, corresponding to the upper vertex (circle) of the diagram. The amplitude of process *b*)

$$M_{fi} = Ze^2 \frac{4\pi}{k^2} (j_\mu J^\mu), \quad (\text{X.128})$$

where j_μ is the transition current of particle *m* (the lower vertex of the diagram); Ze is the charge of this particle. The current J is a function of $k = Q - q$ and therefore in the two cases is different: $k^2 = 0$ in (99.2) and $k^2 \neq 0$ in (99.3). But if in the second case

$$|k^2| \ll m^2, \quad (\text{X.129})$$

then also here one can take J at $k^2 = 0$.

The change of the momentum of particle *M* upon emission of a virtual photon, $\mathbf{p} - \mathbf{p}' = \mathbf{k}$, is small compared with its original momentum $|\mathbf{p}| \approx \varepsilon$; therefore in the current *j* one can set $\mathbf{p} = \mathbf{p}'$. In other words, we consider the motion of particle *M* as rectilinear and uniform. Since such motion is quasi-classical, the corresponding current does not depend on the spin of the particle⁵⁵:

$$j_\mu = 2p_\mu. \quad (\text{X.130})$$

The condition of transversality of the current ($jk = 0$) gives now $\varepsilon\omega - p_x k_x = 0$, where the *x* axis is chosen in the direction of $\mathbf{p}$. From this

$$\omega = vk_x, \quad (\text{X.131})$$

where $v = p_x/\varepsilon$ is the velocity of particle *M*. Since

$$-\mathbf{k}^2 = -\omega^2 + k_x^2 + \mathbf{k}_\perp^2 \approx \omega^2(1 - v^2) + \mathbf{k}_\perp^2 \quad (\text{X.132})$$

($\mathbf{k}_\perp$ is the component of the vector $\mathbf{k}$ transverse to the *x* axis), condition (99.4) is equivalent to the inequality $|\mathbf{k}_\perp| \ll m$ and a considerably stronger inequality for ω : $\omega \ll m\sqrt{1 - v^2}$.

Further, from the condition of transversality of the current J ($Jk = 0$) it follows, using (99.6), that

$$J_0 = \frac{J_x}{v} + \frac{\mathbf{J}_\perp \mathbf{k}_\perp}{\omega}.$$

Therefore for the scalar product jJ we obtain

$$jJ = 2(J_0\varepsilon - J_x p_x) \approx 2 \frac{\varepsilon}{\omega} \left(\mathbf{J}_\perp \mathbf{k}_\perp + \frac{\omega M^2}{\varepsilon^2} J_x \right). \quad (\text{X.133})$$

⁵⁵In the normalisation of wave functions to one particle in unit volume the current is $j^\mu = (1, \mathbf{v})$, where v is the velocity. But we have agreed (see § 6) to omit in the wave functions the normalisation factor $1/\sqrt{2\varepsilon}$, and we arrive at expression (99.5).

The product Je in (99.2) we expand, choosing the 4-vector of polarisation of the real photon in the three-dimensional transverse gauge: $ek = -\mathbf{ek} = 0$, whence $e_x \approx -\mathbf{e}_\perp \mathbf{k}_\perp / \omega$. Then

$$je = -\mathbf{e}_\perp \left(\mathbf{J}_\perp - \frac{\mathbf{k}_\perp}{\omega} J_x \right). \quad (\text{X.134})$$

Comparing expressions (99.8) and (99.9), they will turn out to be proportional to each other if one can neglect the second terms in the brackets. Since the current J pertains to the upper vertex of the diagrams (99.1), it is not connected with the direction of $\mathbf{p}$; therefore J_x and $\mathbf{J}_\perp$ should be considered as quantities of the same order. The permissibility of the indicated neglect requires, therefore, the fulfilment of the conditions $|\mathbf{k}_\perp| \ll \omega$ and $\omega \ll \varepsilon^2 |\mathbf{k}_\perp| / M^2$; they do not contradict the conditions already imposed on $\mathbf{k}_\perp$ and ω .

Accepting that in (99.9) the photon is polarised in the plane $x\mathbf{k}$ (so that $\mathbf{e}_\perp \parallel \mathbf{k}_\perp$), and noting that by virtue of the established conditions $\mathbf{e}_\perp^2 \approx \mathbf{e}^2 = 1$, we obtain now

$$M_{fi} = M_{fi}^{(r)} \frac{Ze\sqrt{4\pi}}{-k^2} \frac{2\varepsilon}{\omega} |\mathbf{k}_\perp|. \quad (\text{X.135})$$

According to what was said above, it is assumed here that the following conditions are fulfilled

$$|\mathbf{k}_\perp| \ll \omega \ll m\gamma, \quad (\text{X.136})$$

$$\frac{\omega}{\gamma^2} \ll |\mathbf{k}_\perp| \ll m, \quad (\text{X.137})$$

where for brevity

$$\gamma = \frac{\varepsilon}{M} = \frac{1}{\sqrt{1-v^2}}.$$

From this one can find the relation between the corresponding cross sections. According to the general formula (64.18) we have (in the rest frame of particle m)

$$d\sigma_r = |M_{fi}^{(r)}|^2 (2\pi)^4 \delta^{(4)}(P_f - P_i) \frac{1}{4m\omega} d\rho_Q,$$

$$d\sigma = |M_{fi}|^2 (2\pi)^4 \delta^{(4)}(P_f - P_i) \frac{1}{4m\omega} \frac{d^3 p'}{2\varepsilon(2\pi)^3} d\rho_Q,$$

where $d\rho_Q$ are the statistical weights of the system Q . Using (99.10) and (99.7), we obtain

$$d\sigma = d\sigma_r \cdot n(\mathbf{k}) d^3 p', \quad (\text{X.138})$$

where

$$n(\mathbf{k}) = \frac{Z^2 e^2}{\pi^2} \frac{\mathbf{k}_\perp^2}{\omega(\mathbf{k}_\perp^2 + \omega^2/\gamma^2)^2}. \quad (\text{X.139})$$

Let us recall that $d\sigma_r$ is the cross section of process a), caused by a collision with a real photon, with the resting particle, and there is formed a system of particles Q in definite intervals of their momenta. The cross section $d\sigma$ refers to process b)—the formation of the same system Q in the collision of a fast particle (mass M) with the same

resting particle, with the fast particle remaining in the interval $d^3 p'$ of values of $\mathbf{p}'$. The factor $n(\mathbf{k})$ in (99.13) can be interpreted as a density of photons (in $\mathbf{k}$ -space) equivalent to the electromagnetic field of the fast particle.

Integration over $d^3 p'$ is equivalent to integration over $d^3 k = d\omega d^2 k_\perp$. Carrying out the integration over $d^2 k_\perp$, we obtain the cross section of the process in which the total energy E of the system Q lies in a given interval $dE = d\omega$ ($E - m = \varepsilon - \varepsilon' = \omega$, where ε and ε' are the initial and final energies of particle M). The integration over directions of $\mathbf{k}_\perp$ means averaging over the polarisation directions of the incident photon (together with multiplication by 2π). After this we obtain

$$d\sigma = n(\omega) d\sigma_r d\omega, \\ n(\omega) = \int n(\mathbf{k}) \cdot 2\pi k_\perp dk_\perp = \frac{2Z^2 e^2}{\pi\omega} \int \frac{k_\perp^3 dk_\perp}{(k_\perp^2 + \omega^2/\gamma^2)^2}. \quad (\text{X.140})$$

The integral over $k_\perp$ diverges for large $k_\perp$. The divergence, however, is only logarithmic. This circumstance permits (within the limits of applicability of the method described) obtaining

the answer in the logarithmic approximation: it is assumed that not only the argument of the logarithm, but the logarithm itself, is large. With such accuracy it suffices to put for the upper limit of integration $k_{\perp\text{max}} \sim m$ —the upper limit of inequality (99.12). Carrying out the integration, we obtain for the spectral distribution of equivalent photons (in ordinary units)

$$n(\omega) d\omega = \frac{2}{\pi} Z\alpha \ln \frac{\gamma mc^2}{\hbar\omega} \frac{d\omega}{\omega}. \quad (\text{X.141})$$

The adopted approximation means that the numerical coefficient in the argument of the logarithm remains undefined: the introduction of such a coefficient would mean the addition to the large logarithm of a relatively small quantity (~ 1) and would represent an excess of the permissible accuracy.

Problems

1. Find the bremsstrahlung cross section in the collision of a fast electron with a nucleus, starting from the scattering of photons on the electron.

Solution. In the system of reference K_1 in which the electron is at rest before the collision, the process can be considered as scattering on the electron of equivalent photons of the field of the nucleus⁵⁶. According to (86.10) the cross section of photon scattering by the electron in the frame K_1

$$d\sigma_{\text{scatt}}(\omega_1, \omega'_1) = \pi r_e^2 \frac{m d\omega'_1}{\varepsilon\omega_1} \left[\frac{\omega_1}{\omega'_1} + \frac{\omega'_1}{\omega_1} + \left(\frac{m}{\omega'_1} - \frac{m}{\omega_1} \right)^2 - 2m \left(\frac{1}{\omega'_1} - \frac{1}{\omega_1} \right) \right], \quad (1)$$

⁵⁶The scattering of virtual photons on the nucleus (in the rest frame of the nucleus) is excluded by the large mass of the latter: the scattering cross section tends to zero with increasing mass of the scattering particle.

where ω_1 and ω'_1 are the initial and final energies of the photon in this frame. The bremsstrahlung cross section in the frame K_1

$$d\sigma_{\text{rad}}(\omega'_1) = \int d\omega_1 \cdot n(\omega_1) d\sigma_{\text{scatt}}(\omega_1, \omega'_1), \quad (2)$$

where $n(\omega_1)$ is function (99.16). By virtue of the invariance of the cross section the transition to the frame K in which the nucleus is at rest reduces to the change of the designated variables. For the region of values probed by ω_1 at given ω'_1 , we have (cf. (4))

$$\begin{aligned} \omega \leq \omega \leq \infty & \quad \text{when} \quad \omega' > \frac{m}{2}; \\ \omega \leq \omega \leq \frac{\omega'}{m - 2\omega'} & \quad \text{when} \quad \omega' < \frac{m}{2}. \end{aligned}$$

When $\omega' < m/2$ the integration over ω gives

$$d\sigma_{\text{rad}}^{(2)} = \frac{16}{3} \alpha r_e^2 \frac{d\omega'}{\omega'} \left(1 - \frac{\omega'}{m} + \frac{\omega'^2}{m^2} \right) \ln \frac{\varepsilon}{\omega'},$$

in agreement with (97.4). If, however, $\omega' > m/2$, one must distinguish the case $\omega' \sim m$ and

$\omega' \sim \varepsilon \gg m$. In the first case we obtain

$$d\sigma_{\text{rad}}^{(2)} = \frac{2}{3} \alpha r_e^2 \frac{m d\omega'}{\omega'^2} \left(4 - \frac{m}{\omega'} + \frac{m^2}{4\omega'^2} \right) \ln \frac{\varepsilon}{m}$$

in agreement with (97.3) (in the argument of the logarithm the replacement of ε/ω' by ε/m is made with the required accuracy). In the case $\omega' \sim \varepsilon$ the method of equivalent photons is generally inapplicable to computing $d\sigma_{\text{rad}}^{(2)}$. The frequency of the virtual photon ω ranges over values starting from ω' , and at $\omega = \omega' \sim \varepsilon$, i.e. consequently, condition (99.11) is not fulfilled.

2. The same for bremsstrahlung of the electron on an electron.

Solution. In this case the virtual photon can scatter either on the fast electron, or on the recoil electron, and conversely. The scattering of virtual photons on the fast electron gives a cross section $d\sigma_{\text{rad}}^{(1)}$ coinciding with the cross section of radiation of the electron on the nucleus with $Z = 1$.

The scattering of virtual photons on the recoil electron gives a cross section of radiation

$$d\sigma_{\text{rad}}^{(2)} = \int d\omega \cdot n(\omega) d\sigma_{\text{scatt}}(\omega, \omega'),$$

with $d\sigma_{\text{scatt}}(\omega, \omega')$ from (1) (with the correspondingly changed notation of variables). For the region of values probed by ω at given ω' , we have (cf. (4))

$$\begin{aligned} \omega \leq \omega \leq \infty & \quad \text{when} \quad \omega' > \frac{m}{2}; \\ \omega \leq \omega \leq \frac{\omega'}{m - 2\omega'} & \quad \text{when} \quad \omega' < \frac{m}{2}. \end{aligned}$$

When $\omega' < m/2$ the integration over ω gives

$$d\sigma_{\text{rad}}^{(2)} = \frac{16}{3} \alpha r_e^2 \frac{d\omega'}{\omega'} \left(1 - \frac{\omega'}{m} + \frac{\omega'^2}{m^2} \right) \ln \frac{\varepsilon}{\omega'},$$

in agreement with (97.4). If $\omega' > m/2$, one must distinguish $\omega' \sim m$ and $\omega' \sim \varepsilon$. In the first case:

$$d\sigma_{\text{rad}}^{(2)} = \frac{2}{3} \alpha r_e^2 \frac{m d\omega'}{\omega'^2} \left(4 - \frac{m}{\omega'} + \frac{m^2}{4\omega'^2} \right) \ln \frac{\varepsilon}{m}$$

in agreement with (97.3). In the case $\omega' \sim \varepsilon$ the method of equivalent photons is generally inapplicable.

3. Determine the total cross section of pair production by a photon in the field of a nucleus, starting from the cross section of pair production by two photons.

Solution. The energy of the photon in the rest frame of the nucleus (system K): $\omega \gg m$. We go to the frame K_0 in which the nucleus moves towards the photon with velocity v_0 such that

$$\frac{1}{\sqrt{1-v_0^2}} = \frac{\omega}{2m}.$$

In this frame the photon energy is

$$\omega_0 = \omega \frac{1-v_0}{\sqrt{1-v_0^2}} \approx \frac{\omega}{2} \sqrt{1-v_0^2} = m.$$

The desired cross section σ in the frame K_0 is computed as the cross section of pair production at the collision of the incident photon ω_0 with equivalent photons of the nuclear field, whose energies we denote by ω' :

$$\sigma = \int \sigma_{\gamma\gamma} n(\omega') d\omega',$$

where $\sigma_{\gamma\gamma}$ is the cross section of pair production by two photons; it is given by the solution of the Problem to § 88, in which one must set

$$v = \sqrt{1 - \frac{m^2}{\omega_0 \omega'}} = \sqrt{1 - \frac{m}{\omega'}}.$$

Changing to the integration variable v instead of ω' , we obtain

$$\sigma = 2r_e^2 \alpha Z \int v \ln \left[\frac{\omega}{m} (1-v^2) \right] \left[(3-v^4) \ln \frac{1+v}{1-v} - 2v(2-v^2) \right] dv.$$

Taking into account the convergence of the integral at the upper limit, the integration extends over the whole region from the reaction threshold $\omega' = m$ ($v = 0$) to $\omega' = \infty$ ($v = 1$) and is carried out with logarithmic accuracy (i.e. the logarithm

$\ln[\omega(1 - v^2)/m]$ is replaced by its value at $v = 0$ and taken outside the integral sign). In the result we obtain

$$\sigma = \frac{28}{9} \alpha Z^2 r_e^2 \ln \frac{\omega}{m}$$

in agreement with (94.6); the formula is valid when $\ln(\omega/m) \gg 1$.

§ 100. Pair production in particle collisions

The production of an electron pair in the collision of two charged particles is described by diagrams of two types:

The two upper solid lines correspond to the colliding particles, the lower one to the pair being produced.

Let us consider in the ultrarelativistic case the collision of two heavy particles (nuclei). The change in the state of motion of the particles themselves in such a collision can be neglected, i.e. they can be considered as sources of an external field⁵⁷. To this there correspond two diagrams of the first type:

where $q^{(1)}, q^{(2)}$ are the “momenta” of the Fourier components of the fields of the two particles.

The potential $A^\mu = (A_0, \mathbf{A})$, created by a uniformly moving with velocity $\mathbf{v}$ classical particle, satisfies the equations

$$\square A_0 = -4\pi Ze \delta(\mathbf{r} - \mathbf{v}t - \mathbf{r}_0), \quad \square \mathbf{A} = -4\pi Ze \mathbf{v} \delta(\mathbf{r} - \mathbf{v}t - \mathbf{r}_0).$$

Its Fourier components:

$$A_0(\omega, \mathbf{k}) = -\frac{8\pi^2 Ze}{\omega^2 - \mathbf{k}^2} e^{-i\mathbf{k}\mathbf{r}_0} \delta(\omega - \mathbf{k}\mathbf{v})$$

and analogously for $\mathbf{A}(\omega, \mathbf{k})$. In four-dimensional form

$$A^\mu(q) = -\frac{8\pi^2 Ze}{q^2} e^{iqx_0} U^\mu \delta(Uq),$$

where U is the 4-velocity of the particle, and the 4-vector $x_0 = (0, \mathbf{r}_0)$. If nucleus 1 is at rest at the origin ($\mathbf{r}_0^{(1)} = 0$), then $\boldsymbol{\rho} \equiv \mathbf{r}_0^{(2)}$ is the impact parameter vector (in the plane perpendicular to the direction of motion of nucleus 2). This expression for $A^\mu(q)$ should be used in the analytic representation of diagrams (100.2).

The pair production cross section can be determined by the method of equivalent photons from the known cross section of pair production by a photon on a nucleus. Replacing in diagrams (100.2) the field lines $q^{(1)}$ of one of the particles by real photon lines, the set of these two diagrams becomes identical to the set of diagrams for pair production by a photon on nucleus 2. For $\varepsilon_+, \varepsilon_- \gg m$ this cross section is given by

⁵⁷The case of the collision of two light particles (electrons), the change in the motion of which cannot be neglected, is considerably more complicated. See about this the book by V. Ya. Baier, V. M. Katkov and V. S. Fadin indicated on p. 454.

formula (94.5). Multiplying it by the equivalent photon spectrum (99.16) of the first nucleus, we obtain the differential cross section for pair production in collisions:

$$d\sigma = \frac{8}{\pi} r_e^2 (Z_1 Z_2 \alpha)^2 \frac{d\varepsilon_+ d\varepsilon_-}{(\varepsilon_+ + \varepsilon_-)^4} \left(\varepsilon_+^2 + \varepsilon_-^2 + \frac{2}{3} \varepsilon_+ \varepsilon_- \right) \ln \frac{2\varepsilon_+ \varepsilon_-}{m(\varepsilon_+ + \varepsilon_-)} \ln \frac{m\gamma}{\varepsilon_+ + \varepsilon_-}, \quad (\text{X.142})$$

where $\gamma = 1/\sqrt{1 - v^2} \gg 1$. This formula applies in the region

$$m \ll \varepsilon_+, \varepsilon_- \ll m\gamma. \quad (\text{X.143})$$

The upper inequality is the applicability condition for the equivalent photon method. The region (100.4) coincides with the region of energies of the electron and positron that are essential in integrating expression (100.3).

Integrating over ε_+ or ε_- at fixed sum $\varepsilon \equiv \varepsilon_+ + \varepsilon_- (\gg m)$, the essential region lies near the upper limit; discarding terms that are non-leading in the logarithm:

$$d\sigma = \frac{56}{9\pi} r_e^2 (Z_1 Z_2 \alpha)^2 \ln \frac{\varepsilon}{m} \ln \frac{m\gamma}{\varepsilon} \frac{d\varepsilon}{\varepsilon}.$$

The integral over ε taken over region (100.4) diverges as the cube of the logarithm, while at its boundaries only as the square of the logarithm. In the logarithmic approximation ($\ln \gamma \gg 1$), region (100.4) is the principal one, and the integral can be taken from m to $m\gamma$. Using

$$\int_1^\gamma \ln \xi (\ln \gamma - \ln \xi) \frac{d\xi}{\xi} = \frac{1}{6} \ln^3 \gamma,$$

we obtain the total cross section for pair production:

$$\sigma = \frac{28}{27\pi} r_e^2 (Z_1 Z_2 \alpha)^2 \ln^3 \frac{1}{\sqrt{1 - v^2}} \quad (\text{X.144})$$

(*L. D. Landau, E. M. Lifshitz, 1934*).

Let us now consider the non-relativistic case of colliding nuclei. The change in the state of motion of the nuclei becomes essential, and the main contribution comes from diagrams of the second type in (100.1). There are four such diagrams:

and two analogous diagrams in which the virtual photon k (producing the pair) is emitted by the first rather than the second nucleus⁵⁸.

We assume that the pair energy is small compared with the kinetic energy of relative motion of the nuclei in their centre-of-mass system:

$$\varepsilon_+ + \varepsilon_- \ll \frac{1}{2} M v^2 \quad (\text{X.145})$$

(v is the initial relative velocity, $M = M_1 M_2 / (M_1 + M_2)$ is the reduced mass). Then one can neglect the back-reaction of pair production on the nuclear motion. If one removes the electron–positron line from diagrams (100.6), the remaining parts depict emission of a virtual photon of small frequency ($\omega = \varepsilon_+ + \varepsilon_-$) by colliding particles. We thus return

⁵⁸Note that pair production in the collision of two electrons corresponds to 36 diagrams: $2! \cdot 3! = 12$ diagrams of type *a*), obtained by permutations among two initial and three final electrons, plus $2 \cdot 2! \cdot 3! = 24$ diagrams of type *b*), obtained similarly from the two diagrams (100.6).

to the situation considered in § 98 for emission of a real soft photon, and can use formula (98.13) obtained there for the non-relativistic case (with the difference that in place of the amplitude $\sqrt{4\pi} e^*$ of the real photon, the propagator of the virtual photon will stand)⁵⁹.

Thus the amplitude of the entire pair production process is written as:

$$M_{fi} = M_{fi}^{\text{el}} \frac{1}{\omega} \left(\frac{Z_1 e}{M_1} - \frac{Z_2 e}{M_2} \right) q^\lambda D_{\lambda\mu}(k) [-ie(\bar{u}_- \gamma^\mu u_+)], \quad (\text{X.146})$$

where $\mathbf{q} = M(\mathbf{v}' - \mathbf{v})$, $q = (0, \mathbf{q})$.

As usual, in the non-relativistic case the photon propagator should be chosen in the Coulomb gauge (76.14). From the amplitude (100.8) we find the cross section of the process:

$$d\sigma = d\sigma_{\text{sc}} \cdot e^4 \left(\frac{Z_1}{M_1} - \frac{Z_2}{M_2} \right)^2 \frac{d^3 p_+ d^3 p_-}{2\varepsilon_+ 2\varepsilon_- (2\pi)^6} \frac{(4\pi)^2}{\omega^2 (\omega^2 - \mathbf{k}^2)^2} |\bar{u}_- \gamma \mathbf{Q} u_+|^2, \quad (\text{X.147})$$

where

$$\omega = \varepsilon_+ + \varepsilon_-, \quad \mathbf{k} = \mathbf{p}_+ + \mathbf{p}_-, \quad \mathbf{Q} = \mathbf{q} - \frac{1}{\omega^2} \mathbf{k}(\mathbf{q} \cdot \mathbf{k});$$

$d\sigma_{\text{sc}}$ is the cross section of elastic scattering of nuclei off each other (in their centre-of-mass system). It is given by the Rutherford formula:

$$d\sigma_{\text{sc}} = 4(Z_1 Z_2 e^2)^2 \frac{M^2 d\sigma}{q^4} \approx \frac{4(Z_1 Z_2 e^2)^2 dq_y dq_z}{v^2 q^4} \quad (\text{X.148})$$

(the approximate equality assumes small deflection of the nuclei from their initial direction of motion—the x -axis)⁶⁰. Substituting (100.10) into (100.9) and performing the usual summation over polarisations of the pair:

$$d\sigma = \frac{(Z_1 Z_2 e^2)^2 e^4}{v^2} \left(\frac{Z_1}{M_1} - \frac{Z_2}{M_2} \right)^2 \frac{d^3 p_+ d^3 p_-}{4\pi^2 \varepsilon_+ \varepsilon_-} \frac{dq_y dq_z}{q^4 (\omega^2 - \mathbf{k}^2)^2 \omega^2} \times \text{Sp}\{(\gamma p_- + m)(\gamma \mathbf{Q})(\gamma p_+ - m)(\gamma \mathbf{Q})\}. \quad (\text{X.149})$$

Further computation is performed in the approximation where all logarithms arising during integration are considered large quantities. We shall see that with this accuracy the main role is played by pair energies ε_+ , $\varepsilon_- \gg m$ and angles θ between $\mathbf{p}_+$ and $\mathbf{p}_-$ in the region

$$m/\varepsilon \ll \theta \ll 1. \quad (\text{X.150})$$

⁵⁹In the non-relativistic case the photon momentum is small compared with the momentum of the radiating particles ($|\delta \mathbf{p}| \sim \omega/v$), and therefore can be neglected (compared with $\delta \mathbf{p}$) even when it is not negligible compared with the photon energy. This applies all the more in the present case to the virtual photon, for which $k^2 = (p_+ + p_-)^2 > 0$, so that $|\mathbf{k}| < \omega$. Under these conditions the difference between a real and a virtual photon vanishes, which justifies the use of formula (98.13).

⁶⁰Diagrams (100.6) correspond to the Born approximation for nuclear scattering. However, since the Rutherford formula (for Coulomb interaction) is exact, the validity of the obtained results does not actually require the applicability condition of the Born approximation.

With the corresponding approximations, the trace computation in (100.11) gives:

$$\text{Sp}\{\dots\} = 4 \left[(\varepsilon_+ \varepsilon_- - \mathbf{p}_+ \cdot \mathbf{p}_-) \left(q^2 - \frac{(\mathbf{q} \cdot \mathbf{k})^2}{\omega^2} \right) + 2(\mathbf{p}_+ \cdot \mathbf{q})(\mathbf{p}_- \cdot \mathbf{q}) \right. \\ \left. + \frac{2\varepsilon_+ \varepsilon_-}{\omega^2} (\mathbf{q} \cdot \mathbf{k})^2 - \frac{2(\mathbf{q} \cdot \mathbf{k})}{\omega} (\varepsilon_+ (\mathbf{q} \cdot \mathbf{p}_-) + \varepsilon_- (\mathbf{q} \cdot \mathbf{p}_+)) \right],$$

where one can set $|\mathbf{p}_+| = \varepsilon_+$, $|\mathbf{p}_-| = \varepsilon_-$. In the denominator:

$$\omega^2 - \mathbf{k}^2 \approx \varepsilon_+ \varepsilon_- \theta^2 + \frac{m^2 (\varepsilon_+ + \varepsilon_-)^2}{\varepsilon_+ \varepsilon_-}.$$

Integrating over the directions of $\mathbf{p}_+$ and $\mathbf{p}_-$ at fixed angle θ between them:

$$d\sigma = \frac{8}{3\pi^2} (Z_1 Z_2 e^2)^2 \frac{e^4}{v^2} \left(\frac{Z_1}{M_1} - \frac{Z_2}{M_2} \right)^2 (\varepsilon_+^2 + \varepsilon_-^2) d\varepsilon_+ d\varepsilon_- \\ \times \frac{\theta^3 d\theta}{[\theta^2 + m^2 (\varepsilon_+ + \varepsilon_-)^2 / (\varepsilon_+^2 \varepsilon_-^2)]^2} \frac{dq_y dq_z}{q^2}. \quad (\text{X.151})$$

The form of the θ -dependence confirms the assumption (100.12). Integration over the last factor in (100.13) is carried out from $q_y = q_z = 0$ to $\sqrt{q_y^2 + q_z^2} \sim 1/R$, where R is a quantity of order of the nuclear radius (this value corresponds to the minimum impact parameter—see below); this integration gives:

$$\pi \ln(q_x^2 + q_y^2 + q_z^2) \Big|_{q_y=q_z=0}^{q_y=q_z=1/R} \approx 2\pi \ln \frac{1}{Rq_x}.$$

On the other hand, the total pair energy, equal to the change in the kinetic energy of the nuclei, is:

$$\varepsilon \equiv (\varepsilon_+ + \varepsilon_-) = \frac{1}{2} M(v'^2 - v^2) \approx Mv(v'_x - v_x) = vq_x,$$

so that $q_x = \varepsilon/v$. Thus we find:

$$d\sigma = \frac{16}{3\pi} (Z_1 Z_2 e^2)^2 \frac{e^4 m^2}{v^2} \left(\frac{Z_2}{M_2} - \frac{Z_1}{M_1} \right)^2 \frac{\varepsilon_+^2 + \varepsilon_-^2}{\varepsilon^4} \\ \times \ln \frac{v}{R\varepsilon} \ln \frac{\varepsilon_+ \varepsilon_-}{m(\varepsilon_+ + \varepsilon_-)} d\varepsilon_+ d\varepsilon_-,$$

and after integration over ε_+ or ε_- at fixed sum ε :

$$d\sigma = \frac{32}{9\pi} (Z_1 Z_2 e^2)^2 \frac{e^4 m^2}{v^2} \left(\frac{Z_1}{M_1} - \frac{Z_2}{M_2} \right)^2 \ln \frac{v}{R\varepsilon} \ln \frac{\varepsilon}{m} \frac{d\varepsilon}{\varepsilon}. \quad (\text{X.152})$$

The energy ε can be related to the impact parameter: $\rho \sim v/\varepsilon$ (the pair energy is of order of the frequency corresponding to the collision time). Therefore the logarithmic divergence in ε in (100.14) signifies the same divergence in impact parameters. This means that the essential ρ are large (which, incidentally, justifies the use of the

Rutherford scattering cross section (100.10) in the pure Coulomb field of the nucleus). Correspondingly, the essential energy region is: $m \ll \varepsilon \ll \nu/R$. Integrating (100.14) gives the total pair production cross section; finally (in ordinary units)⁶¹:

$$\sigma = \frac{16}{27\pi} (Z_1 Z_2 \alpha)^2 r_e^2 \frac{c^2}{v^2} \left(\frac{Z_2 m}{M_2} - \frac{Z_1 m}{M_1} \right)^2 \ln^3 \frac{\hbar \nu}{mc^2 R} \quad (\text{X.153})$$

(*E. M. Lifshitz*, 1935).

§ 101. Emission of a photon by an electron in the field of an intense electromagnetic wave

The applicability of perturbation theory to processes of interaction of an electron with the radiation field presupposes (besides the smallness of the coupling constant α) also sufficient weakness of this field. If a is the amplitude of the classical 4-potential of the electromagnetic wave field, the characteristic quantity in this sense is the dimensionless invariant ratio

$$\xi = \frac{e\sqrt{-a^2}}{m}. \quad (\text{X.154})$$

In this section we consider radiation processes arising in the interaction of an electron with the field of a strong electromagnetic wave, for which ξ can have any value. The method used is based on exact account of this interaction; the interaction of the electron with the newly emitted photons can still be treated as a small perturbation (*A. I. Nikishov*, *V. I. Ritus*, 1964).

Consider a monochromatic plane wave, for definiteness circularly polarised. Its 4-potential:

$$A = a_1 \cos \varphi + a_2 \sin \varphi, \quad \varphi = kx, \quad (\text{X.155})$$

where $k^\mu = (\omega, \mathbf{k})$ is the wave 4-vector ($k^2 = 0$), and the 4-amplitudes a_1 and a_2 are equal in magnitude and mutually orthogonal:

$$a_1^2 = a_2^2 \equiv a^2, \quad a_1 a_2 = 0.$$

We assume the potential is calibrated by the Lorentz condition, so $a_1 k = a_2 k = 0$.

The exact wave function for an electron in the field of an arbitrary plane electromagnetic wave was found in § 40 (formulae (40.7), (40.8)). We change, however, the normalisation: we require that ψ_p correspond to unit average spatial particle density — similar to how we normalise the wave functions of free particles to one particle per unit volume. Since for function (40.7) the average density is $\bar{j}_0 = q_0/p_0$, to obtain the required normalisation we multiply it by $\sqrt{p_0/q_0}$, i.e. replace in (40.7) the factor $1/\sqrt{2p_0}$ by $1/\sqrt{2q_0}$. For the wave with 4-potential (X.155) we obtain:

$$\psi_p = \left\{ 1 + \frac{e}{2(kp)} [(\gamma k)(\gamma a_1) \cos \varphi + (\gamma k)(\gamma a_2) \sin \varphi] \right\} \frac{u(p)}{\sqrt{2q_0}} \exp \left\{ -\frac{ie(a_1 p)}{(kp)} \sin \varphi + \frac{ie(a_2 p)}{(kp)} \cos \varphi \right\} \quad (\text{X.156})$$

⁶¹A numerical error was corrected by *L. B. Okun'* (1953).

where

$$q^\mu = p^\mu - \frac{e^2 a^2}{2(kp)} k^\mu. \quad (\text{X.157})$$

According to (40.14) the 4-vector q is the average kinetic 4-momentum of the electron; we shall call it the *quasi-momentum*.

The S -matrix element for transition of the electron from state ψ_p to state $\psi_{p'}$ with emission of a photon with 4-momentum $k'^\mu = (\omega', \mathbf{k}')$ and polarisation 4-vector e' :

$$S_{fi} = -ie \int \bar{\psi}_{p'}(\gamma e'^*) \psi_p \frac{e^{ik'x}}{\sqrt{2\omega'}} d^4x. \quad (\text{X.158})$$

The integrand in (X.158) is a linear combination of the quantities

$$\exp(-i\alpha_1 \sin \varphi + i\alpha_2 \cos \varphi) \cdot \{1, \cos \varphi, \sin \varphi\},$$

where

$$\alpha_1 = e \left(\frac{(a_1 p)}{(kp)} - \frac{(a_1 p')}{(kp')} \right), \quad \alpha_2 = e \left(\frac{(a_2 p)}{(kp)} - \frac{(a_2 p')}{(kp')} \right). \quad (\text{X.159})$$

Together with the factor $\exp[i(k' + q' - q)x]$ these quantities exhaust all the dependence of the integrand on x .

Expand them in Fourier series, denoting the expansion coefficients B_s, B_{1s}, B_{2s} ; for example:

$$\exp(-i\alpha_1 \sin \varphi + i\alpha_2 \cos \varphi) = \exp[-i\sqrt{\alpha_1^2 + \alpha_2^2} \sin(\varphi - \varphi_0)] = \sum_{s=-\infty}^{\infty} B_s e^{-is\varphi}.$$

These coefficients are expressed through Bessel functions:

$$\begin{aligned} B_s &= J_s(z) e^{is\varphi_0}, \\ B_{1s} &= \frac{1}{2} [J_{s+1}(z) e^{i(s+1)\varphi_0} + J_{s-1}(z) e^{i(s-1)\varphi_0}], \\ B_{2s} &= \frac{1}{2i} [J_{s+1}(z) e^{i(s+1)\varphi_0} - J_{s-1}(z) e^{i(s-1)\varphi_0}], \end{aligned} \quad (\text{X.160})$$

where $z = \sqrt{\alpha_1^2 + \alpha_2^2}$, $\cos \varphi_0 = \alpha_2/z$, $\sin \varphi_0 = \alpha_1/z$. The functions B_s, B_{1s}, B_{2s} are related by

$$\alpha_1 B_{1s} + \alpha_2 B_{2s} = s B_s, \quad (\text{X.161})$$

which is a consequence of the known recurrence relation for Bessel functions:

$$J_{s-1}(z) + J_{s+1}(z) = \frac{2s}{z} J_s(z).$$

The matrix element (X.158) thus takes the form

$$S_{fi} = \frac{1}{\sqrt{2\omega' \cdot 2q_0 \cdot 2q'_0}} \sum_s M_{fi}^{(s)} (2\pi)^4 i \delta^{(4)}(sk + q - q' - k'); \quad (\text{X.162})$$

the rather cumbersome expressions for the amplitudes $M_{fi}^{(s)}$ we shall not write out here. Thus S_{fi} is an infinite sum of terms, each of which corresponds to a conservation law:

$$sk + q = q' + k'. \quad (\text{X.163})$$

Since

$$q^2 = q'^2 = m^2(1 + \xi^2) \equiv m_*^2 \quad (\text{X.164})$$

(cf. (40.15)), and $k^2 = k'^2 = 0$, the equality (X.163) is possible only for $s \geq 1$. The s -th term of the sum describes emission of a photon k' due to absorption of s photons with 4-momenta k from the wave. From the form of the equality (X.163) it is clear that all kinematic relations for the Compton effect will apply to the processes under consideration, if we replace the electron momenta by quasi-momenta q and the incident photon momentum by the 4-vector

sk . In particular, for the frequency of the emitted photon in the system where the electron is on average at rest ($\mathbf{q} = 0$, $q_0 = m_*$), we have

$$\omega' = \frac{s\omega}{1 + (s\omega/m_*)(1 - \cos \theta)}, \quad (\text{X.165})$$

where θ is the angle between $\mathbf{k}$ and $\mathbf{k}'$ (cf. (86.8)). One can say that the frequencies ω' are harmonics of the frequency ω . In our adopted notation (§ 64) the amplitude of the s -th harmonic radiation process coincides with $M_{fi}^{(s)}$, and the expression

$$dW_s = \frac{|M_{fi}^{(s)}|^2 d^3k' d^3q'}{(2\pi)^6 \cdot 2\omega' \cdot 2q_0 \cdot 2q'_0} (2\pi)^4 \delta^{(4)}(sk + q - q' - k') \quad (\text{X.166})$$

gives the corresponding differential probability (per unit time)⁶².

The structure of the amplitudes $M_{fi}^{(s)}$ is similar to the structure of the scattering amplitude with plane waves: $\bar{u}(p') \dots u(p)$. Therefore the operations of summation over polarisations of the particles are carried out in the usual way. After summation over polarisations of the final electron and photon and averaging over polarisations of the initial electron:

$$\begin{aligned} dW_s &= \frac{e^2 m^2}{4\pi} \frac{d^3k' d^3q'}{q_0 q'_0 \omega'} \delta^{(4)}(sk + q - q' - k') \times \\ &\times \left\{ -2J_s^2(z) + \xi^2 \left(1 + \frac{(kk')^2}{2(kp)(kp')} \right) (J_{s+1}^2 + J_{s-1}^2 - 2J_s^2) \right\}. \end{aligned} \quad (\text{X.167})$$

For integration of this expression we note that due to the axial symmetry of the circularly polarised wave field, the differential probability does not depend on the overall azimuthal angle ϕ around the direction $\mathbf{k}$. Together with the δ -function, this allows integration over all variables except one; as the latter we choose the invariant quantity $u = (kk')/(kp)$. Then after integration over $d^3k' d\phi d(q'_0 + \omega')$ we have

$$\delta^{(4)}(sk + q - q' - k') \frac{d^3q' d^3k'}{q'_0 \omega'} \rightarrow \frac{2\pi du}{(1+u)^2}.$$

⁶²Note that the normalisation of the functions ψ_p to unit density corresponds to normalisation of the δ -function "on the scale $q/(2\pi)$ " (cf. (40.17), where the factor q_0/p_0 on the right side of the equality will now be absent). It is precisely for this reason that the number of final states of the electron must be measured by the element $d^3q'/(2\pi)^3$.

Indeed, in the centre-of-inertia system (the system where $s\mathbf{k} + \mathbf{q} = \mathbf{q}' + \mathbf{k}' = 0$), the indicated integration gives $2\pi|\mathbf{q}'| d\cos\theta/E_s$, where $E_s = s\omega + q_0 = \omega' + q'_0$ and θ is the angle between $\mathbf{k}$ and $\mathbf{q}'$. On the other hand, in this same system,

$$u = \frac{E_s}{q'_0 - |\mathbf{q}'| \cos\theta} - 1, \quad d\cos\theta = \frac{E_s du}{|\mathbf{q}'|(1+u)^2}.$$

The interval $-1 \leq \cos\theta \leq 1$ corresponds to the interval

$$0 \leq u \leq u_s \equiv \frac{E_s^2}{m_*^2} - 1 = \frac{2s(kp)}{m_*^2}$$

(in the transformations one should remember that $(kp) = (kq)$).

Thus the total probability of radiation per unit time:

$$W = \sum_{s=1}^{\infty} W_s = \frac{e^2 m^2}{4q_0} \sum_{s=1}^{\infty} \int_0^{u_s} \frac{du}{(1+u)^2} \left\{ -4J_s^2(z) + \right. \\ \left. + \xi^2 \left(2 + \frac{u^2}{1+u} \right) (J_{s+1}^2 + J_{s-1}^2 - 2J_s^2) \right\}, \quad (\text{X.168})$$

where⁶³

$$u = \frac{(kk')}{(kp')}, \quad u_s = \frac{2s(kp)}{m_*^2}, \quad z = \frac{2sm^2\xi}{m_*\sqrt{1+\xi^2}} \sqrt{\frac{u}{u_s} \left(1 - \frac{u}{u_s} \right)}. \quad (\text{X.169})$$

For $\xi \ll 1$ (the condition for applicability of perturbation theory) the integrand expressions in (X.168) can be expanded in powers of ξ . Thus, for the first term of the expansion W_1 (at $\xi \ll 1$):

$$W_1 = \frac{e^2 m^2 \xi^2}{4p_0} \int_0^{u_1} \left[2 + \frac{u^2}{1+u} - \frac{4u}{u_1} \left(1 - \frac{u}{u_1} \right) \right] \frac{du}{(1+u)^2} \\ = \frac{e^2 m^2 \xi^2}{4p_0} \left[\left(1 - \frac{4}{u_1} - \frac{8}{u_1^2} \right) \ln(1+u_1) + \frac{1}{2} + \frac{8}{u_1} - \frac{1}{2(1+u_1)^2} \right], \quad (\text{X.170})$$

with $u_1 \approx 2(kp)/m^2$. As expected, this result coincides with the Klein–Nishina formula for photon scattering on an electron: substituting in (X.170) $-a^2 = 4\pi/\omega$, $\xi^2 = 4\pi e^2/(m^2\omega)$ and dividing by the flux density (64.14), we return to (86.16) (the total scattering cross section does not depend on the initial polarisation of the photon).

We also give the expression for the probability of emission of the second harmonic (first term of the expansion of W_2 at $\xi \ll 1$):

$$W_2 = \frac{e^2 m^2 \xi^4}{p_0} \int_0^{u_2} \frac{du}{(1+u)^2} \frac{u}{u_2} \left(1 - \frac{u}{u_2} \right) \left[2 + \frac{u^2}{1+u} - \frac{4u}{u_2} \left(1 - \frac{u}{u_2} \right) \right]. \quad (\text{X.171})$$

In general, the main term in W_s (for not too large s) is proportional to ξ^{2s} .

⁶³The indicated value of a^2 corresponds to normalisation of the 4-potential to one photon per unit volume. For its determination one must equate ω times the energy of the classical field with the (real) 4-potential (X.155).

Now consider the opposite case: $\xi \gg 1$. The parameter ξ can be made large, for example, by decreasing the frequency ω at fixed field intensity (obviously, $\xi = eF/(m\omega)$, where F is the field amplitude). Therefore it is clear that the case $\xi \gg 1$ essentially reduces to processes in a constant uniform field, the intensities $\mathbf{E}$ and $\mathbf{H}$ of which are mutually perpendicular and equal in magnitude (such a field is called a *crossed field*). The radiation probability can be obtained by the limiting transition $\xi \rightarrow \infty$, but it is simpler to carry out the calculations directly for a constant field, taking the 4-potential in the form

$$A^\mu = a^\mu \varphi, \quad \varphi = kx, \quad ak = 0 \quad (\text{X.172})$$

(so that $F_{\mu\nu} = k_\mu a_\nu - k_\nu a_\mu = \text{const}$). The exact wave function of the electron in this field is obtained by substitution of (X.172) into (40.7), (40.8):

$$\psi_p = \left[1 + \frac{e(\gamma k)(\gamma a)}{2(kp)} \varphi \right] \frac{u(p)}{\sqrt{2p_0}} \exp \left\{ -\frac{ie(ap)}{2(kp)} \varphi^2 + \frac{ie^2 a^2}{6(kp)} \varphi^3 - ipx \right\}. \quad (\text{X.173})$$

The result obtained with this function is exact for radiation of an electron in a crossed field at any electron energy. But in the ultrarelativistic case this result applies to radiation of an electron not only in a crossed field, but in any constant uniform electromagnetic field, including a constant magnetic field (which was considered in § 90).

To formulate this assertion we note that the state of a particle in an arbitrary constant uniform field is determined by the same quantum numbers as the state of a free particle, and these quantum numbers can always be chosen so that when the field is switched off they go over into the quantum numbers of the free particle, i.e. its 4-momentum p^μ ($p^2 = m^2$). Thus the state of the particle in the field will be described by a constant 4-vector p .

The total radiation intensity, being an invariant quantity, depends only on invariants that can be formed from the constant 4-tensor $F_{\mu\nu}$ and the 4-vector p . Taking into account also that $F_{\mu\nu}$ must enter the intensity only together with the charge e , we obtain three dimensionless invariants⁶⁴:

$$\chi^2 = -\frac{e^2}{m^6} (F_{\mu\nu} p^\nu)^2, \quad f = \frac{e^2 (F_{\mu\nu})^2}{m^4}, \quad g = \frac{e^2}{m^4} e_{\lambda\mu\nu\rho} F^{\lambda\mu} F^{\nu\rho}. \quad (\text{X.174})$$

In a crossed field $f = g \equiv 0$, while in the general case all three invariants are non-zero. But if the electron is ultrarelativistic ($p_0 \gg m$), and its momentum $\mathbf{p}$ makes with the fields $\mathbf{E}$, $\mathbf{H}$ angles $\theta \gg m/p_0$, then $\chi^2 \gg f$, g (in other words, for an ultrarelativistic particle almost for all directions of $\mathbf{p}$ any constant field looks like a crossed one). If, moreover, the field intensities $|\mathbf{E}|$, $|\mathbf{H}| \ll m^2/e (= m^2 c^3/(e\hbar))$, then $|f|$, $|g| \ll 1$. Under these conditions the radiation intensity, computed for a crossed field, also applies to radiation in any constant field, expressed through the invariant χ computed for a crossed field.

The invariant χ is expressed through the field intensities $\mathbf{E}$, $\mathbf{H}$ as

$$\chi^2 = \frac{e^2}{m^6} \{ ([\mathbf{p}\mathbf{H}] + p_0 \mathbf{E})^2 - (\mathbf{p} \cdot \mathbf{E})^2 \}.$$

⁶⁴In the expression for χ one can with the same accuracy consider p as the ordinary kinematic 4-momentum of the particle.

For a constant magnetic field, χ coincides with the quantity (90.3) introduced in § 90, so that the considerations presented here give another way to obtain the results of § 90⁶⁵.

⁶⁵Detailed treatment of the theory of various processes in strong fields is given in the reviews by *A. I. Nikishov* and *V. I. Ritus* in the collection “Kvantovaya elektrodinamika yavlenii v intensivnom pole” (Trudy FIAN, M.: Nauka, 1979, T. 111).

Exact Propagators and Vertex Parts

§ 102. Field operators in the Heisenberg representation

Up to now when considering various concrete electrodynamical processes we limited ourselves to the first non-vanishing approximation of perturbation theory. We now turn to the study of effects arising when higher-order approximations are taken into account. These effects are called *radiative corrections*.

A deeper understanding of the structure of higher approximations can be achieved on the basis of a preliminary investigation of general properties possessed by the exact (i.e. not expanded in powers of e^2) scattering amplitudes. We saw (see § 72) that successive terms of the perturbation theory series are expressed through field operators in the interaction representation—operators whose time dependence is determined by the Hamiltonian of the system of free particles $\hat{H}_0$. The exact scattering amplitudes are more conveniently expressed through operators of the field not in this, but in the Heisenberg representation, in which the time dependence is determined directly by the exact Hamiltonian of the system of interacting particles $\hat{H} = \hat{H}_0 + \hat{V}$.

By the general rule for constructing Heisenberg operators we have:¹

$$\hat{\psi}(x) \equiv \hat{\psi}(t, \mathbf{r}) = e^{i\hat{H}t} \hat{\psi}(\mathbf{r}) e^{-i\hat{H}t} \quad (\text{XI.1})$$

and similarly for $\hat{\bar{\psi}}(x)$ and $\hat{A}(x)$, where $\hat{\psi}(\mathbf{r}), \dots$ are time-independent (Schrödinger, or pre-interaction) operators.

We immediately note that Heisenberg operators taken at equal times satisfy the same commutation rules as operators in the Schrödinger representation or in the interaction representation. Indeed, we have, for example,

$$\{\hat{\psi}_i(t, \mathbf{r}), \hat{\psi}_k^\dagger(t, \mathbf{r}')\}_+ = e^{i\hat{H}t} \{\hat{\psi}_i(\mathbf{r}), \hat{\psi}_k^\dagger(\mathbf{r}')\}_+ e^{-i\hat{H}t} = \gamma_{ik}^0 \delta(\mathbf{r} - \mathbf{r}') \quad (\text{XI.2})$$

(cf. (75.6)). Analogously the operators $\hat{\psi}(t, \mathbf{r})$ and $\hat{A}(t, \mathbf{r}')$ are commutative:

$$\{\hat{\psi}_i(t, \mathbf{r}), \hat{A}(t, \mathbf{r}')\}_- = 0$$

(at different times this is no longer so!).

The “equation of motion” satisfied by the Heisenberg ψ -operator can be obtained from the general formula (13.7) (see III):

$$-i \frac{\partial \hat{\psi}(x)}{\partial t} = \hat{H} \hat{\psi}(x) - \hat{\psi}(x) \hat{H}. \quad (\text{XI.3})$$

For the Hamiltonians the Schrödinger and Heisenberg representations are identical, and the Hamiltonian is expressed identically through field operators in both representations.

¹In this chapter operators with a time argument will refer to the Heisenberg representation, and operators in the interaction representation will be marked with the additional index “int”.

In this case, when computing the right side of (XI.3), one can omit the part depending only on the operator $\hat{A}(x)$ (the Hamiltonian of the free electromagnetic field), since this part is commutative with $\hat{\psi}(x)$. According to (21.13) and (43.3) we have:

$$\begin{aligned}\hat{H} &= \int \hat{\psi}^*(t, \mathbf{r}) (\alpha \hat{\mathbf{p}} + \beta m) \hat{\psi}(t, \mathbf{r}) d^3x \\ &\quad + e \int \hat{\bar{\psi}}(t, \mathbf{r}) (\gamma \hat{A}(t, \mathbf{r})) \hat{\psi}(t, \mathbf{r}) d^3x \\ &= \int \hat{\bar{\psi}}(t, \mathbf{r}) \{ \gamma \hat{p} + m + e(\gamma \hat{A}(t, \mathbf{r})) \} \hat{\psi}(t, \mathbf{r}) d^3x.\end{aligned}\quad (\text{XI.4})$$

Computing the commutator $\{\hat{H}, \hat{\psi}(t, \mathbf{r})\}_-$ with the help of (XI.2) and eliminating the δ -function by integration over d^3x , we obtain:

$$(\gamma \hat{p} - e \gamma \hat{A} - m) \hat{\psi}(t, \mathbf{r}) = 0. \quad (\text{XI.5})$$

As expected, the operator $\hat{\psi}(t, \mathbf{r})$ satisfies the equation formally coinciding with the Dirac equation.

The equation for the operator of the electromagnetic field $\hat{A}(t, \mathbf{r})$ is obviously obtained from correspondence with the classical case. In this case (large occupation numbers—see § 5) after averaging over the field state the operator equation should go over into the classical Maxwell equation for potentials (30.2) (see II). Therefore it is clear that the equation for the operator simply coincides in form with the Maxwell equation, i.e. (in arbitrary gauge):

$$\partial_\nu \partial^\nu \hat{A}^\mu(x) - \partial^\mu \partial_\nu \hat{A}^\nu(x) = -4\pi e \hat{j}^\mu(x), \quad (\text{XI.6})$$

where $\hat{j}^\mu = \hat{\bar{\psi}}(x) \gamma^\mu \hat{\psi}(x)$ is the current operator, identically satisfying the continuity equation:

$$\partial_\mu \hat{j}^\mu(x) = 0. \quad (\text{XI.7})$$

It is essential that equations (XI.6) are linear in $\hat{A}^\mu$ and $\hat{j}^\mu$, and therefore no question arises about the ordering of these operators.

Like the equations for wave functions, the system of operator equations (XI.6), (XI.7) is invariant with respect to gauge transformation:²

$$\hat{A}^\mu(x) \rightarrow \hat{A}^\mu(x) - \partial^\mu \hat{\chi}(x), \quad \hat{\psi}(x) \rightarrow \hat{\psi}(x) e^{ie\hat{\chi}}, \quad \hat{\bar{\psi}}(x) \rightarrow e^{-ie\hat{\chi}} \hat{\bar{\psi}}(x), \quad (\text{XI.8})$$

where $\hat{\chi}(x)$ is an arbitrary Hermitian operator, commuting (at one and the same moment of time) with $\hat{\psi}$.

Now let us establish the connection between operators in the Heisenberg representation and in the interaction representation. To simplify the reasoning it is convenient to make the formal assumption (not affecting the final result) that the interaction $\hat{V}(t)$ is adiabatically “switched on” from $t = -\infty$ to finite times. Then when $t \rightarrow -\infty$ both representations—the

²We emphasise that here it is precisely about Heisenberg ψ -operators. In the interaction representation the gauge transformation of electromagnetic potentials does not at all affect the ψ -operators.

Heisenberg representation and the interaction representation—simply coincide. The corresponding wave functions of the system Φ and Φ_{int} also coincide:

$$\Phi_{\text{int}}(t = -\infty) = \Phi. \quad (\text{XI.9})$$

On the other hand, the wave function in the Heisenberg representation does not depend on time at all (all time dependence is transferred to the operators), while in the interaction representation the dependence of the wave function on time is given, according to (72.7):

$$\Phi_{\text{int}}(t) = \hat{S}(t, -\infty) \Phi_{\text{int}}(-\infty), \quad (\text{XI.10})$$

where the operator is introduced:

$$\hat{S}(t_2, t_1) = T \exp \left\{ -i \int_{t_1}^{t_2} \hat{V}(t') dt' \right\} \quad (\text{XI.11})$$

with the obvious properties:

$$\hat{S}(t, t_1) \hat{S}(t_1, t_0) = \hat{S}(t, t_0), \quad \hat{S}^{-1}(t, t_1) = \hat{S}(t_1, t). \quad (\text{XI.12})$$

Comparing formulas (XI.10) and (XI.9), we find the relation:

$$\Phi_{\text{int}}(t) = \hat{S}(t, -\infty) \Phi, \quad (\text{XI.13})$$

establishing the connection between the wave functions in both representations. Correspondingly the operator transformation formula:

$$\hat{\psi}(t, \mathbf{r}) = \hat{S}^{-1}(t, -\infty) \hat{\psi}_{\text{int}}(t, \mathbf{r}) \hat{S}(t, -\infty) = \hat{S}(-\infty, t) \hat{\psi}_{\text{int}}(t, \mathbf{r}) \hat{S}(t, -\infty) \quad (\text{XI.14})$$

(the same for $\hat{\bar{\psi}}$ and $\hat{A}$).

In conclusion let us make one more general remark. We have already repeatedly indicated that in relativistic quantum field theory the physical meaning of field operators is very limited due to the infinity of zero-point fluctuations. This applies even more to operators in the Heisenberg representation, which actually contain in themselves also divergences associated with the interaction. In this chapter, §§ 102, 109 are devoted to the exposition of formal theory, in which questions of removal of these infinities are not discussed and operations with all quantities are carried out as if they were finite. The results obtained in this way have primarily heuristic value: they allow one to more deeply understand the meaning of the expansion of perturbation theory; it is also possible that they are preserved in some form in a future theory free of the present difficulties.

§ 103. The exact photon propagator

The main role in the exact theory apparatus (without expansion in powers of e^2) is played by the concepts of exact propagators.

The exact photon propagator (which we shall denote by the script letter $\mathcal{D}$) is defined by the formula:

$$\mathcal{D}_{\mu\nu}(x - x') = i \langle 0 | T \hat{A}_\mu(x) \hat{A}_\nu(x') | 0 \rangle, \quad (\text{XI.15})$$

where $\hat{A}_\mu(x)$ are Heisenberg operators, in contrast to the definition (76.1):

$$D_{\mu\nu}(x - x') = i\langle 0 | T \hat{A}_\mu^{\text{int}}(x) \hat{A}_\nu^{\text{int}}(x') | 0 \rangle, \quad (\text{XI.16})$$

in which operators in the interaction representation figured. In contrast to the exact propagator (XI.15), function (XI.16) can be called the propagator of free photons.

In view of the impossibility of exact computation of the vacuum mean value (XI.15), the definition allows one to establish some general properties of this function. This will be the topic of § 111, and for now we shall concern ourselves with the computation of $\mathcal{D}_{\mu\nu}$ by perturbation theory with the help of diagrammatic technique. For this it is necessary to express $\mathcal{D}_{\mu\nu}$ through operators in the interaction representation.

Let first $t > t'$. Using the connection between $\hat{A}(x)$ and $\hat{A}_{\text{int}}(x)$ (cf. (XI.14)), we write:

$$\begin{aligned} \mathcal{D}_{\mu\nu}(x - x') &= i\langle 0 | \hat{A}_\mu(x) \hat{A}_\nu(x') | 0 \rangle \\ &= i\langle 0 | \hat{S}(-\infty, t) \hat{A}_\mu^{\text{int}}(x) \hat{S}(t, t') \hat{A}_\nu^{\text{int}}(x') \hat{S}(t', -\infty) | 0 \rangle. \end{aligned} \quad (\text{XI.17})$$

where for brevity it is denoted:

$$\hat{S} \equiv \hat{S}(+\infty, -\infty). \quad (\text{XI.18})$$

According to (XI.12):

$$\hat{S}(t, -\infty) \hat{S}(-\infty, t') = \hat{S}(t, t'), \quad \hat{S}(-\infty, t) = \hat{S}(-\infty, +\infty) \hat{S}(+\infty, t) = \hat{S}^{-1} \hat{S}(\infty, t).$$

Then:

$$\mathcal{D}_{\mu\nu} = i\langle 0 | \hat{S}^{-1} [\hat{S}(\infty, t) \hat{A}_\mu^{\text{int}}(x) \hat{S}(t, t') \hat{A}_\nu^{\text{int}}(x') \hat{S}(t', -\infty)] | 0 \rangle.$$

Since according to the definition (XI.11) $\hat{S}(t_2, t_1)$ contains only operators at times between t_1 and t_2 , arranged in chronological order, it is obvious that altogether all operator factors in square brackets are arranged in order of decreasing times from left to right. Placing before the bracket the sign of chronologisation T , we can then arbitrarily rearrange the order of factors, since the operator T automatically establishes them in the needed order. Using this, we rewrite the expression in brackets in the form:

$$[\dots] = T [\hat{A}_\mu^{\text{int}}(x) \hat{A}_\nu^{\text{int}}(x') \hat{S}(\infty, t) \hat{S}(t, t') \hat{S}(t', -\infty)] = T [\hat{A}_\mu^{\text{int}}(x) \hat{A}_\nu^{\text{int}}(x') \hat{S}].$$

Thus:

$$\mathcal{D}_{\mu\nu}(x - x') = i\langle 0 | \hat{S}^{-1} T [\hat{A}_\mu^{\text{int}}(x) \hat{A}_\nu^{\text{int}}(x') \hat{S}] | 0 \rangle, \quad (\text{XI.19})$$

and it is easy to verify analogously that this formula is valid also for $t < t'$.

We now show that the factor $\hat{S}^{-1}$ can be taken out from under the vacuum averaging sign in the form of some phase factor. For this we recall that the Heisenberg wave function of the vacuum Φ coincides with the value $\Phi_{\text{int}}(-\infty)$ of the wave function of this same state in the interaction representation (see (XI.9)). But the vacuum is a strictly stationary state; in it no spontaneous processes of particle production are possible; with

the passage of time the vacuum remains vacuum; this means that $\Phi_{\text{int}}(+\infty)$ can differ from $\Phi_{\text{int}}(-\infty)$ only by some phase factor $e^{i\alpha}$. According to (72.8) we have:

$$\hat{S} \Phi_{\text{int}}(-\infty) = e^{i\alpha} \Phi_{\text{int}}(-\infty) = \langle 0|S|0 \rangle \Phi_{\text{int}}(-\infty). \quad (\text{XI.20})$$

Taking the complex conjugate and using the unitarity of operator $\hat{S}$, we obtain:

$$\Phi_{\text{int}}^*(-\infty) \hat{S}^{-1} = \langle 0|S|0 \rangle^{-1} \Phi_{\text{int}}^*(-\infty).$$

From this it is clear that expression (XI.19) can be rewritten in the form:

$$\mathcal{D}_{\mu\nu}(x-x') = \frac{i \langle 0|T \hat{A}_{\mu}^{\text{int}}(x) \hat{A}_{\nu}^{\text{int}}(x') S|0 \rangle}{\langle 0|S|0 \rangle}. \quad (\text{XI.21})$$

Substituting here (in both numerator and denominator) the expansion (72.10) for $\hat{S}$ and performing the averaging with the help of Wick's theorem (see § 77), we obtain an expansion of $\mathcal{D}_{\mu\nu}$ in powers of e^2 .

In the numerator of (XI.21) the averaged expressions differ from the matrix elements of the type (77.1) considered in § 77, only in that instead of “external” operators of creation or annihilation of photons, there stand the operators $\hat{A}_{\mu}^{\text{int}}(x)$ and $\hat{A}_{\nu}^{\text{int}}(x')$. Since all factors in the averaged products stand under the chronologisation sign, the pairwise contractions of these operators with “internal” operators of creation or annihilation of photons will give photon propagators $D_{\mu\nu}$. Thus the results of the averaging will be expressed as sets of diagrams with two external ends, constructed according to the rules described in § 77, with the only difference that external photon lines, like internal ones, will now correspond to propagators $D_{\mu\nu}$ (instead of amplitudes e of real photons). In the zeroth approximation, setting $\hat{S} = 1$, the numerator of (XI.21) coincides simply with $D_{\mu\nu}(x-x')$. The next non-vanishing terms will be of order $\sim e^2$. They are represented by a set of diagrams containing two external ends and two vertices:

$$(\text{XI.22})$$

The second of these diagrams consists of two unconnected parts: a dashed line (which corresponds to $-iD_{\mu\nu}$) and a closed loop. Such a decomposition of the diagram means decomposition of the corresponding analytic expression into two independent factors. Adding to diagrams (XI.22) the diagram (dashed line) of the zeroth approximation and “taking it outside the bracket”, we find that as a result, to the accuracy of terms of second order, the numerator in (XI.21) equals:

The expression $\langle 0|S|0 \rangle$ in the denominator of (XI.21) represents the amplitude of the “transition” from vacuum to vacuum. Its expansion therefore contains only diagrams without external ends. In the zeroth approximation $\langle 0|S|0 \rangle = 1$, and to accuracy of terms of second order we get:

Dividing numerator by denominator to this accuracy, we find that the curly bracket cancels and there remains:

Thus the diagram with the disconnected loop drops out of the answer. This result has a general character. Thinking about the way diagrams are constructed for the

numerator and denominator in (XI.21), it is not difficult to understand that the role of the denominator $\langle 0|S|0\rangle$ reduces to the fact that in any order of perturbation theory the exact propagator $\mathcal{D}_{\mu\nu}$ will be represented only by diagrams not containing disconnected parts.

We note that diagrams without external ends (closed loops) have no physical significance whatsoever and should not be taken into account even independently of the fact that they drop out in the formation of propagator $\mathcal{D}$. Indeed, such loops represent radiative corrections to a diagonal matrix element of the S -matrix for the vacuum-to-vacuum transition. But according to (XI.20) the sum of all these loops (together with unity of the zeroth approximation) gives only an insignificant phase factor, which cannot affect any physical results.

The transition from coordinate to momentum representation occurs in the usual way. Thus in the second approximation of perturbation theory the propagator $-i\mathcal{D}_{\mu\nu}(k)$ (which we will depict by a bold dashed line) is given by the sum:

$$(XI.23)$$

where all diagrams are computed by the usual rules (enumerated in § 77), with the only difference that external photon lines, like internal ones, are also assigned factors $-iD_{\mu\nu}(k)$.

The analytic expression of this formula gives therefore:

$$\mathcal{D}_{\mu\nu}(k) = D_{\mu\nu}(k) + ie^2 D_{\mu\lambda}(k) \int \text{Sp } \gamma^\lambda G(p+k) \gamma^\rho G(p) \frac{d^4 p}{(2\pi)^4} D_{\rho\nu}(k) \quad (XI.24)$$

(bispinor indices on matrices γ and G , as usual, are not written out).

The terms of the following approximations are constructed analogously; they are represented by sets of diagrams with two external photon ends and the needed number of vertices. Thus, terms $\sim e^4$ correspond to the following diagrams with four vertices:

$$(XI.25)$$

Four vertices also has the diagram

the upper part of which constitutes a loop formed by a single “closed on itself” electron line. Such a loop corresponds to the convolution $\hat{\psi}(x)\gamma\hat{\psi}(x)$, i.e. simply to the vacuum-average value of the current: $\langle 0|j(x)|0\rangle$. But by the very definition of the vacuum, this quantity must identically vanish, and this identity cannot, of course, be altered by any further radiative corrections to such a loop.³

Therefore altogether no diagrams with “closed on themselves” electron lines should be taken into account in any approximation.⁴

The part of the diagram (“block”) enclosed between two photon lines (external or internal) is called the *photon self-energy part*. In the general case such a block itself can be further subdivided into parts connected pairwise by one photon line, i.e. has the structure of the form:

³Although direct computation of the diagrams would lead to divergent integrals.

⁴In determining the signs one should not forget about the factor -1 contributed by a closed electron loop!

where circles denote blocks that can no longer be further subdivided in such a way; these parts are called *compact* (for example, of the four self-energy parts of fourth order (XI.25) the first three are compact).

Denote by the symbol $i\mathcal{P}_{\mu\nu}/4\pi$ the sum of all (infinite number of) compact self-energy parts; the function $\mathcal{P}_{\mu\nu}(k)$ is called the *polarisation operator*. Classifying the diagrams by the number of compact parts they contain, one can represent the exact propagator $\mathcal{D}_{\mu\nu}$ in the form of a series:

Analytically this series is written as:

$$\begin{aligned}\mathcal{D} &= D + D \frac{\mathcal{P}}{4\pi} D + D \frac{\mathcal{P}}{4\pi} D \frac{\mathcal{P}}{4\pi} D + \dots \\ &= D \left\{ 1 + \frac{\mathcal{P}}{4\pi} \left[D + D \frac{\mathcal{P}}{4\pi} D + \dots \right] \right\}\end{aligned}\quad (\text{XI.26})$$

(tensor indices omitted for brevity). But the series in square brackets again coincides with the series for $\mathcal{D}$. Therefore we have:

$$\mathcal{D}_{\mu\nu}(k) = D_{\mu\nu}(k) + D_{\mu\lambda}(k) \frac{\mathcal{P}^{\lambda\rho}(k)}{4\pi} \mathcal{D}_{\rho\nu}(k). \quad (\text{XI.27})$$

Multiplying this equality on the left by the inverse tensor $(D^{-1})^{\tau\mu}$ and on the right by $(\mathcal{D}^{-1})^{\nu\sigma}$ (and changing the notation of indices), we obtain its equivalent form:

$$\mathcal{D}_{\mu\nu}^{-1} = D_{\mu\nu}^{-1} - \frac{\mathcal{P}_{\mu\nu}}{4\pi}. \quad (\text{XI.28})$$

We emphasise that the representation of $\mathcal{D}$ in the form (XI.26) implies that from diagrams one can extract simpler blocks, which are computed by the general rules of diagrammatic technique. Combining such blocks with each other, we obtain correct expressions for the diagrams as a whole. The admissibility of such a decomposition constitutes an important (and by no means trivial) feature of diagrammatic technique. It is connected with the fact that the total numerical coefficient in a diagram does not depend on its order.

This same property allows one to use the function $\mathcal{D}$ (if it is known) for simplification of computations of radiative corrections to amplitudes of various scattering processes: instead of each time considering diagrams with corrections to internal photon lines, we can simply replace these lines by bold ones, i.e. assign to them propagators $\mathcal{D}$ (instead of D).

If the photon line corresponds to a real (not virtual) photon, i.e. if it is an external end of the diagram as a whole, then after including in it all self-energy corrections we get what is called an *effective external line*. It corresponds, in distinction from (XI.27), to replacing the multiplier D by the polarisation amplitude of the real photon:

$$e_\mu + D_{\mu\rho}(k) \frac{\mathcal{P}^{\rho\lambda}(k)}{4\pi} e_\lambda. \quad (\text{XI.29})$$

If we are talking about a line of an external field, then instead of e_μ here one should write $A_\mu^{(e)}$.

Everything said in § 76 about the tensor structure and gauge ambiguity of the approximate propagator $D_{\mu\nu}$ also applies to the exact function $\mathcal{D}_{\mu\nu}$. Remaining within the framework of relativistically invariant representations of this function, let us write its general form:

$$\mathcal{D}_{\mu\nu}(k) = \mathcal{D}(k^2) \left(g_{\mu\nu} - \frac{k_\mu k_\nu}{k^2} \right) + \mathcal{D}^{(l)}(k^2) \frac{k_\mu k_\nu}{k^2}; \quad (\text{XI.30})$$

the first term corresponds to the Landau gauge, and in the second term $\mathcal{D}^{(l)}$ is a gauge-arbitrary function. An analogous representation for the approximate propagator:

$$D_{\mu\nu}(k) = D(k^2) \left(g_{\mu\nu} - \frac{k_\mu k_\nu}{k^2} \right) + D^{(l)}(k^2) \frac{k_\mu k_\nu}{k^2}. \quad (\text{XI.31})$$

The polarisation operator, in accordance with its definition, is also a transverse tensor:⁵

$$\mathcal{P}_{\mu\nu} = \mathcal{P}(k^2) \left(g_{\mu\nu} - \frac{k_\mu k_\nu}{k^2} \right). \quad (\text{XI.32})$$

For the direct tensors (XI.30) and (XI.31) the inverse tensors have the form:

$$\mathcal{D}_{\mu\nu}^{-1}(k) = \frac{1}{\mathcal{D}} \left(g_{\mu\nu} - \frac{k_\mu k_\nu}{k^2} \right) + \frac{1}{\mathcal{D}^{(l)}} \frac{k_\mu k_\nu}{k^2}, \quad D_{\mu\nu}^{-1}(k) = \frac{1}{D} \left(g_{\mu\nu} - \frac{k_\mu k_\nu}{k^2} \right) + \frac{1}{D^{(l)}} \frac{k_\mu k_\nu}{k^2}. \quad (\text{XI.33})$$

Substituting (XI.32) and (XI.33) into (XI.28) and comparing the transverse parts, we find:

$$\frac{1}{\mathcal{D}} = \frac{1}{D} - \frac{\mathcal{P}}{4\pi}, \quad (\text{XI.34})$$

where $\mathcal{P} = \mathcal{P}(k^2)$, $D = D(k^2) = 4\pi/k^2$, or:

$$\mathcal{D}(k^2) = \frac{4\pi}{k^2(1 - \mathcal{P}(k^2)/k^2)}. \quad (\text{XI.35})$$

Thus the polarisation operator is (in contrast to the photon propagator itself) a gauge-invariant quantity.

§ 104. The photon self-energy function

For further investigation of the analytic properties of the photon propagator it will be useful to introduce, alongside the polarisation operator, one more auxiliary quantity $\Pi_{\mu\nu}(k)$, which is called the *photon self-energy function*. Namely, $i\Pi_{\mu\nu}/(4\pi)$ is defined as the sum of all (not only compact) self-energy photon parts. Depicting this sum as a square on the diagrams, we represent the exact propagator as the sum:
i.e.:

$$\mathcal{D}_{\mu\nu} = D_{\mu\nu} + D_{\mu\lambda} \frac{\Pi^{\lambda\rho}}{4\pi} D_{\rho\nu}. \quad (\text{XI.36})$$

⁵The definition of $D^{(l)}$ in this formula does not coincide with the definition in (76.3).

Expressing from this $\Pi_{\mu\nu}$ in the form:

$$\frac{1}{4\pi} \Pi_{\mu\nu} = D_{\mu\lambda}^{-1} \mathcal{D}^{\lambda\rho} D_{\rho\nu}^{-1} - D_{\mu\nu}^{-1}$$

and substituting into this equation (103.16), (103.19), and then (103.21), we get:

$$\Pi_{\mu\nu} = \Pi(k^2) \left(g_{\mu\nu} - \frac{k_\mu k_\nu}{k^2} \right), \quad \Pi = \frac{\mathcal{P}}{1 - \mathcal{P}/k^2}. \quad (\text{XI.37})$$

We see that $\Pi_{\mu\nu}$ (like $\mathcal{P}_{\mu\nu}$) is a gauge-invariant tensor.

The usefulness of the quantity $\Pi_{\mu\nu}$ is connected with its expression in the coordinate representation. It is easy to find, noting that the equality

$$\frac{1}{4\pi} \Pi_{\mu\nu}(k) = D_{\mu\lambda}^{-1} \mathcal{D}^{\lambda\rho} D_{\rho\nu}^{-1},$$

taking into account from (103.18) the transverse tensor $\mathcal{D}^{\lambda\rho} - D^{\lambda\rho}$, in the coordinate representation can be written in the form:

$$\Pi_{\mu\nu}(x-x') = \frac{1}{4\pi} (\partial_\mu \partial_\lambda - g_{\mu\lambda} \partial_\sigma \partial^\sigma) (\partial'_\nu \partial'_\rho - g_{\nu\rho} \partial'_\sigma \partial'^\sigma) \{ \mathcal{D}^{\lambda\rho}(x-x') - D^{\lambda\rho}(x-x') \}.$$

For carrying out differentiation it is necessary to substitute here:

$$\mathcal{D}^{\lambda\rho}(x-x') - D^{\lambda\rho}(x-x') = i \langle 0 | T \hat{A}^\lambda(x) \hat{A}^\rho(x') - T \hat{A}^{\text{int},\lambda}(x) \hat{A}^{\text{int},\rho}(x') | 0 \rangle. \quad (\text{XI.38})$$

We saw in § 75 that differentiation of T -products requires, generally speaking, caution due to their discontinuous character. But the averaged expression in (XI.38) is continuous, since the commutation rules for components of operators $\hat{A}^\lambda(x)$ and $\hat{A}^{\text{int}}(x)$ (taken at one and the same moment of time) are identical and the corresponding jumps cancel (cf. § 75). Therefore differentiation of the difference (XI.38) can be carried out under the T sign. According to (XI.6) (and similarly to the equation without the right-hand part for operators of the free electromagnetic field $\hat{A}_\mu^{\text{int}}(x)$) we get in the result the expression:

$$\Pi_{\mu\nu}(x-x') = 4\pi i e^2 \langle 0 | T \hat{j}_\mu(x) \hat{j}_\nu(x') | 0 \rangle. \quad (\text{XI.39})$$

This manifestly reveals the gauge invariance of $\Pi_{\mu\nu}$, since such are the current operators.

From (XI.39) one can obtain an important integral representation of this function.

In view of (XI.37) it is sufficient to consider the scalar function $\Pi = \Pi_\mu^\mu/3$. In the coordinate representation:

$$\Pi(x-x') = \frac{4\pi}{3} i e^2 \langle 0 | T \hat{j}_\mu(x) \hat{j}^\mu(x') | 0 \rangle = \frac{4\pi}{3} i e^2 \left\{ \sum_n \langle 0 | \hat{j}_\mu(x) | n \rangle \langle n | \hat{j}^\mu(x') | 0 \rangle, \quad t > t', \right. \\ \left. \sum_n \langle 0 | \hat{j}_\mu(x') | n \rangle \langle n | \hat{j}^\mu(x) | 0 \rangle, \quad t < t', \right. \quad (\text{XI.40})$$

where the symbol n numbers the states of the system “electromagnetic + electron-positron field”.⁶

⁶The current operator conserves charge, therefore the states $|n\rangle$ in (XI.40) can contain only equal numbers of electrons and positrons.

Since the current operator $\hat{j}(x)$ depends on $x^\mu = (t, \mathbf{r})$, also the matrix elements depend on x . This dependence can be established explicitly, if one chooses as states $|n\rangle$ the states with definite values of total 4-momentum of the system (see III, (15.13)). The operator of such a translation is $e^{i\mathbf{r}\hat{\mathbf{P}}}$, where $\hat{\mathbf{P}}$ is the operator of total momentum of the system (see III, (15.13)). Bearing in mind the general rule for transformation of matrix elements (see III, (12.7)), we find:

$$\langle n|\hat{j}^\mu(t, \mathbf{r})|m\rangle = \langle n|\hat{j}^\mu(0)|m\rangle e^{-i(P_m - P_n) \cdot x}. \quad (\text{XI.41})$$

We also note that the matrix $\langle n|\hat{j}_\mu(0)|m\rangle$ is Hermitian (like the matrix (XI.41) of the operator $\hat{j}^\mu(t, \mathbf{r})$ as a whole), and by virtue of the continuity equation (XI.7) satisfies the transversality condition:

$$(P_n - P_m)^\mu \langle n|\hat{j}_\mu(0)|m\rangle = 0. \quad (\text{XI.42})$$

Let us return to the computation of the function $\Pi(x - x')$. Substituting (XI.41) into (XI.40), we get:

$$\Pi(\xi) = \frac{4\pi i e^2}{3} \sum_n \langle 0|\hat{j}_\mu(0)|n\rangle \langle n|\hat{j}^\mu(0)|0\rangle e^{\mp i P_n \cdot \xi}, \quad \tau \gtrless 0, \quad (\text{XI.43})$$

where $x - x' = \xi = (\tau, \boldsymbol{\xi})$. Denote:

$$\rho(k^2) = -\frac{4\pi e^2}{3} (2\pi)^3 \sum_n \langle 0|\hat{j}_\mu(0)|n\rangle \langle 0|\hat{j}^\mu(0)|n\rangle^* \delta^{(4)}(k - P_n). \quad (\text{XI.44})$$

The summation is over all systems of real electron pairs and photons that can be produced by a virtual photon with 4-momentum $k = (\omega, \mathbf{k})$ ($k^2 > 0$), and for each of such systems—also over its internal variables (polarisations and momenta of particles in the centre-of-inertia system).⁷

As a result of such summation the function ρ can depend only on k , and by virtue of its scalar nature—only on k^2 . In particular, ρ does not depend on the direction of $\mathbf{k}$. Having in mind these properties of the function ρ , we rewrite (XI.43) in the form:

$$\Pi(\xi) = -i \int_0^\infty d\omega \int \frac{d^3 k}{(2\pi)^3} \rho(k^2) e^{i\mathbf{k} \cdot \boldsymbol{\xi} - i\omega|\tau|} = -i \int \frac{d^3 k}{(2\pi)^3} \int_0^\infty d\omega d(\mu^2) \delta(\mu^2 - k^2) \rho(\mu^2) e^{i\mathbf{k} \cdot \boldsymbol{\xi} - i\omega|\tau|}$$

The transition to the momentum representation is effected by substitution of the formula:

$$e^{-i\omega|\tau|} = 2i\omega \int_{-\infty}^\infty \frac{dk_0}{2\pi} \frac{e^{-ik_0\tau}}{k_0^2 - \omega^2 + i0} \quad (\text{XI.45})$$

(already used in § 76) and gives:

$$\Pi(k^2) = \int d(\mu^2) \int d(\omega^2) \delta(\mu^2 - k^2 - \omega^2) \frac{\rho(\mu^2)}{k_0^2 - \omega^2 + i0},$$

⁷This definition of states $|n\rangle$ is obviously identical with identifying them as states for which the matrix elements $\langle 0|\hat{j}|n\rangle$ of the charge-odd operator are non-zero.

or finally:⁸

$$\Pi(k^2) = \int_0^\infty \frac{\rho(\mu^2) d(\mu^2)}{k^2 - \mu^2 + i0}. \quad (\text{XI.46})$$

The coefficient ρ in this integral representation is called the *spectral density* of the function $\Pi(k^2)$. It possesses the properties:

$$\rho(k^2) = 0 \quad \text{for } k^2 < 0, \quad \rho(k^2) > 0 \quad \text{for } k^2 > 0. \quad (\text{XI.47})$$

Indeed, the 4-momentum k of a virtual photon that can produce a system of real particles is necessarily timelike (k^2 coincides with the square of the total energy of the particles in their centre-of-inertia system). By virtue of the transversality condition (XI.42) we have:

$$P_n^\mu \langle 0 | \hat{j}_\mu | n \rangle = 0.$$

But the 4-vector $\langle 0 | \hat{j} | n \rangle$, orthogonal to a timelike 4-vector (P_n), is spacelike, i.e.:

$$\langle 0 | \hat{j}_\mu | n \rangle \langle 0 | \hat{j}^\mu | n \rangle^* < 0,$$

and therefore, according to definition (XI.44), $\rho > 0$.

§ 105. The exact electron propagator

Similarly to the photon, the exact electron propagator is defined by the formula:

$$\mathcal{G}_{ik}(x - x') = -i \langle 0 | T \hat{\psi}_i(x) \hat{\bar{\psi}}_k(x') | 0 \rangle \quad (\text{XI.48})$$

(i, k are bispinor indices), differing from the definition (75.1) of the propagator of free particles:

$$G_{ik}(x - x') = -i \langle 0 | T \hat{\psi}_i^{\text{int}}(x) \hat{\bar{\psi}}_k^{\text{int}}(x') | 0 \rangle, \quad (\text{XI.49})$$

by the replacement of $\hat{\psi}$ -operators in the interaction representation by Heisenberg ones.

The same reasoning as in deriving (XI.21) allows one to transform the propagator $\mathcal{G}$ to the form:

$$\mathcal{G}_{ik}(x - x') = \frac{\langle 0 | T \hat{\psi}_i^{\text{int}}(x) \hat{\bar{\psi}}_k^{\text{int}}(x') S | 0 \rangle}{\langle 0 | S | 0 \rangle}. \quad (\text{XI.50})$$

Its expansion in powers of e^2 leads to the representation of the $\mathcal{G}$ -function as a set of diagrams with two external electron lines and various numbers of vertices. Here the role of the denominator in (XI.50) again reduces to the necessity of including only diagrams without isolated “vacuum loops”. Thus, to terms $\sim e^2$ we have:⁹

$$(\text{XI.51})$$

⁸Formal calculations, analogous to those carried out above, require caution in view of the presence of the already-mentioned divergences. This leads, in particular, to the appearance on the right-hand side of (XI.46) of additional divergent terms that do not have a relativistically invariant form (so-called Schwinger terms). We do not write them out, since they are eliminated in renormalisation (see § 110) and do not affect the further results.

⁹As was already explained in § 103, one should also not include here diagrams with electron lines “closed on themselves”, which would appear already in second order.

The bold solid line corresponds (in the momentum representation) to the function $i\mathcal{G}(p)$, and all thin solid and dashed lines in the diagrams on the right side of the equality correspond to the propagators of free particles iG and $-iD$ respectively.

The block enclosed between two electron lines is called the *electron self-energy part*. As in the photon case, a self-energy part is called *compact* if it cannot be further subdivided into two self-energy parts by cutting one electron line. The sum of all possible compact parts we denote by $-i\mathcal{M}_{ik}$; the function $\mathcal{M}_{ik}(p)$ is called the *mass operator*. Thus, to accuracy $\sim e^4$ we have:

$$(XI.52)$$

By summation, in exact analogy with the derivation of (XI.27), we get:

$$\mathcal{G}(p) = G(p) + G(p) \mathcal{M}(p) \mathcal{G}(p) \quad (XI.53)$$

(bispinor indices omitted), or for inverse matrices:

$$\mathcal{G}^{-1}(p) = G^{-1}(p) - \mathcal{M}(p) = \gamma p - m - \mathcal{M}(p). \quad (XI.54)$$

In § 102 it was already noted that the Heisenberg $\hat{\psi}$ -operators (in contradistinction to $\hat{\psi}$ -operators in the interaction representation) change under gauge transformation of electromagnetic potentials. Together with them also turns out to be gauge-non-invariant the exact electron propagator $\mathcal{G}$. Let us clarify the law of its gauge transformation (L. D. Landau, I. M. Khalatnikov, 1952).

It is clear in advance that the change of $\mathcal{G}$ under gauge transformation must be expressible through the very same quantity $\mathcal{D}^{(l)}$ that is added to the photon propagator under this transformation. This will become obvious if one notes that in computing $\mathcal{G}$ from the diagrams of perturbation theory each term of the series is expressed through the functions D and no other electromagnetic quantities appear in it. This circumstance can be used to simplify the derivation: one can make any particular assumptions about the properties of the arbitrary operator $\hat{\chi}$ in the transformation (XI.8), as long as the answer is expressed through $\mathcal{D}^{(l)}$.

As a result of the transformation (XI.8) the propagators $\mathcal{D}$ (XI.15) and $\mathcal{G}$ (XI.48) go over to:

$$\mathcal{D}_{\mu\nu} \rightarrow i\langle 0|T[\hat{A}_\mu(x) - \partial_\mu\hat{\chi}(x)][\hat{A}_\nu(x') - \partial'_\nu\hat{\chi}(x')]|0\rangle, \quad (XI.55)$$

$$\mathcal{G}_{ik} \rightarrow -i\langle 0|T\hat{\psi}_i(x)e^{ie\hat{\chi}(x)}e^{-ie\hat{\chi}(x')}\hat{\psi}_k(x')|0\rangle.$$

We assume now that the operators $\hat{\chi}$ are averaged independently of all remaining operator factors in the T -product; this assumption is quite natural, since by virtue of gauge invariance the “field” $\hat{\chi}$ takes no part in the interaction. We also assume that the vacuum average of $\hat{\chi}$ itself vanishes: $\langle 0|\hat{\chi}|0\rangle = 0$. Then in (XI.55) the terms containing $\hat{\chi}$ separate out and we get:

$$\mathcal{D}_{\mu\nu} \rightarrow \mathcal{D}_{\mu\nu} + i\langle 0|T\partial_\mu\hat{\chi}(x) \cdot \partial'_\nu\hat{\chi}(x')|0\rangle, \quad (XI.56)$$

$$\mathcal{G}_{ik} \rightarrow \mathcal{G}_{ik} \langle 0|T e^{ie\hat{\chi}(x)} e^{-ie\hat{\chi}(x')} |0\rangle. \quad (XI.57)$$

The further derivation is carried out for an infinitesimally small transformation; to emphasise this smallness we shall write $\delta\hat{\chi}$ instead of $\hat{\chi}$.

The transformation (XI.56) can (independently of the smallness of $\delta\hat{\chi}$) be written as:

$$\mathcal{D}_{\mu\nu} \rightarrow \mathcal{D}_{\mu\nu} + \delta\mathcal{D}_{\mu\nu}, \quad \delta\mathcal{D}_{\mu\nu} = \partial_\mu \partial'_\nu d^{(l)}(x - x'), \quad (\text{XI.58})$$

where

$$d^{(l)}(x - x') = i\langle 0 | T \delta\hat{\chi}(x) \delta\hat{\chi}(x') | 0 \rangle. \quad (\text{XI.59})$$

From this it is clear that the function $d^{(l)}$ determines the change under gauge transformation of the longitudinal part of the photon propagator $\mathcal{D}^{(l)}$.¹⁰

The assumption about the dependence of this function only on the difference $x - x'$ means, of course, a definite restriction on the properties of the operator $\delta\hat{\chi}$; in the general case a fully arbitrary gauge transformation can violate the space-time homogeneity of the propagator.

In (XI.57) we expand the exponential factors in powers of $\delta\hat{\chi}$ to the accuracy of quadratic terms:¹¹

$$\langle 0 | T e^{ie\delta\hat{\chi}(x)} e^{-ie\delta\hat{\chi}(x')} | 0 \rangle \approx 1 - \frac{1}{2}e^2 \langle 0 | \delta\hat{\chi}^2(x) + \delta\hat{\chi}^2(x') - 2T \delta\hat{\chi}(x) \delta\hat{\chi}(x') | 0 \rangle.$$

Taking into account definition (XI.59), we find therefore the following law of transformation of the electron propagator:

$$\mathcal{G} \rightarrow \mathcal{G} + \delta\mathcal{G}, \quad \delta\mathcal{G} = ie^2 \mathcal{G}(x - x') [d^{(l)}(0) - d^{(l)}(x - x')]. \quad (\text{XI.60})$$

In the momentum representation:

$$\delta\mathcal{G}(p) = ie^2 \int d^{(l)}(q) [\mathcal{G}(p) - \mathcal{G}(p - q)] \frac{d^4 q}{(2\pi)^4}. \quad (\text{XI.61})$$

Moreover the function $d^{(l)}(q)$ is connected with the change of the function $\mathcal{D}^{(l)}$ by:

$$\delta\mathcal{D}^{(l)}(q) = q^2 d^{(l)}(q). \quad (\text{XI.62})$$

For the electron propagator one could also obtain an integral representation analogous to formula (104.11). Its derivation is based on the expressions

$$\psi_{nm}(x) = \psi_{nm}(0) e^{-i(P_m - P_n)x} \quad (\text{XI.63})$$

for matrix elements of the $\hat{\psi}$ -operator, similar to those used in § 104 expressions (104.6) for matrix elements of the current. In contradistinction to the current, however, the

¹⁰The transition from (XI.56) to (XI.58) is possible if the function $d^{(l)}$ and its derivative with respect to t are continuous at $t = t'$. In the opposite case the right-hand sides of these expressions would differ by δ -functional terms (cf. the derivation of formula (75.2)). In the momentum representation this condition is equivalent to the assumption that $d^{(l)}(q)$ decreases at $|q^2| \rightarrow \infty$ faster than $1/q^2$.

¹¹If the function $f(x) = f_1(x) f_2(x)$, then its Fourier components $f(p) = \int f(x) e^{ipx} d^4 x = \iiint d^4 x \frac{d^4 q_1 d^4 q_2}{(2\pi)^8} e^{ix(p - q_1 - q_2)} f_1(q_1) f_2(q_2) = \iint \frac{d^4 q_1 d^4 q_2}{(2\pi)^4} \delta^{(4)}(p - q_1 - q_2) f_1(q_1) f_2(q_2) = \int \frac{d^4 q}{(2\pi)^4} f_1(q) f_2(p - q)$. In the transition from (XI.60) to (XI.61) it was also taken into account that $f(x = 0) = \int f(q) d^4 q / (2\pi)^4$.

$\hat{\psi}$ -operators themselves are gauge-non-invariant. Therefore the coordinate dependence of the form

(XI.63) does not have a general character, and relates only to a particular gauge, and the integral representation of the propagator based on (XI.63) has no general character. The deep physical reason for this situation is the equality to zero of the photon mass, and the resulting infrared catastrophe (see § 98). Because of this an electron in the process of interaction emits an infinite number of soft quanta, which to a considerable extent deprives the “single-particle” propagator (XI.48) of direct physical meaning.

§ 106. The vertex operator

In complex diagrams one can distinguish certain standard structural elements (blocks), of which these diagrams are composed. In §§ 103 and 105 two of such elements were in essence already considered: the photon and electron self-energy parts. Below we shall encounter yet another type — the *vertex part*.

For this purpose let us define the *vertex function*

$$K_{ik}^{\mu}(x_1, x_2, x_3) = \langle 0 | T \hat{A}^{\mu}(x_1) \hat{\psi}_i(x_2) \hat{\bar{\psi}}_k(x_3) | 0 \rangle \quad (\text{XI.64})$$

— the vacuum average of the T -product of Heisenberg operators of the electromagnetic and electron-positron fields (one factor of each type). In the interaction representation:

$$K_{ik}^{\mu}(x_1, x_2, x_3) = \frac{\langle 0 | T \hat{A}^{\text{int}, \mu}(x_1) \hat{\psi}_i^{\text{int}}(x_2) \hat{\bar{\psi}}_k^{\text{int}}(x_3) S | 0 \rangle}{\langle 0 | S | 0 \rangle}. \quad (\text{XI.65})$$

The transition to the momentum representation is carried out by the formula

$$\begin{aligned} (2\pi)^4 \delta^{(4)}(p_1 + k - p_2) K_{ik}^{\mu}(p_2, p_1; k) = \\ = \iiint K_{ik}^{\mu}(x_1, x_2, x_3) e^{-ikx_1 + ip_2x_2 - ip_1x_3} d^4x_1 d^4x_2 d^4x_3. \end{aligned} \quad (\text{XI.66})$$

In diagrammatic technique the functions K_{ik}^{μ} correspond to blocks (three-tails) of the type

$$(\text{XI.67})$$

with three (one photon and two electron) ends, whose momenta are related by the conservation law

$$p_1 + k = p_2. \quad (\text{XI.68})$$

The zeroth-order term in the expansion of (XI.65) vanishes. In first order in the coordinate representation:

$$K^{\mu} = e \int G(x_2 - x) \gamma_{\nu} G(x - x_3) \cdot D^{\nu\mu}(x_1 - x) d^4x,$$

and in the momentum representation:

$$K^{\mu}(p_2, p_1; k) = e G(p_2) \gamma_{\nu} G(p_1) \cdot D^{\nu\mu}(k), \quad (\text{XI.69})$$

corresponding to the diagram:

(XI.70)

In higher approximations new vertices arise. In third order there are the following diagrams:

(XI.71)

The first three of these can be cut into two parts, one of which is the simple vertex (XI.70), and the other a second-order self-energy part of one of the external lines.

Corrections of this kind to the function K^μ simply amount to the replacement of the free propagators G and D of the external lines by the exact propagators $\mathcal{G}$ and $\mathcal{D}$. But the fourth diagram cannot be decomposed in this manner.

Separating from the sum of all diagrams for K^μ the propagators of the three external lines, we write by definition:

$$K^\mu(p_2, p_1; k) = \{i\mathcal{G}(p_2) [-ie\Gamma_\nu(p_2, p_1; k)] i\mathcal{G}(p_1)\} \{-i\mathcal{D}^{\nu\mu}(k)\}. \quad (\text{XI.72})$$

The block that cannot be divided into parts by cutting a single (electron or photon) line of an internal diagram is called the *vertex part*. The quantity Γ^μ , equal to the sum of all vertex parts including the simple vertex γ^μ , is called the *vertex operator* (or *vertex function*).

All diagrams of the vertex operator up to fifth order are:

(XI.73)

The operator Γ has two bispinor indices and one four-vector index; it is a function of p_1, p_2 (or equivalently of p_1, k , since $p_2 = p_1 + k$). We note that not all three four-momenta p_1, p_2, k can correspond simultaneously to real particles, since the conservation law (XI.68) cannot be satisfied when $p_1^2 = p_2^2 = m^2, k^2 = 0$.

The vertex parts can be split into *irreducible* and *reducible* ones. An irreducible vertex part is one that cannot be split into simpler parts by removing a single electron or photon line; in the terminology previously introduced these are vertex parts that do not contain self-energy insertions in internal lines and do not contain vertex corrections within themselves. From the diagrams (XI.73): only *b*) and *d*) are irreducible (not counting the bare vertex *a*)); diagrams *g*), *h*), *i*) contain self-energy parts; in diagram *c*) the upper horizontal dashed line can be regarded as a correction to the upper vertex, and the lateral dashed lines in diagrams *e*) and *f*) as corrections to the lateral vertices.

In irreducible diagrams, if one replaces the internal lines by bold (exact propagator) lines and the vertices by bold points, i.e. replaces D, G by $\mathcal{D}, \mathcal{G}$ and γ by Γ , one obtains all vertex parts whatsoever. The resulting diagrams are called *skeleton diagrams*.

This decomposition of the vertex operator can be written symbolically in the form:

(XI.74)

This is an integral equation for Γ , with an infinite number of terms on the right-hand side.

Let us state the general principle of constructing exact expressions corresponding to diagrammatic blocks with an arbitrary number of external lines. Such an expression is constructed as the vacuum average of a T -product of Heisenberg operators: one factor of $\hat{\psi}(x)$ for each final electron line, one factor of $\hat{\bar{\psi}}(x)$ for each initial electron line, and one factor of $\hat{A}(x)$ for each photon line.

As an example let us consider four-electron diagrams — diagrams with four electron external lines:

(XI.75)

The corresponding function is:

$$K_{ik,lm}(x_1, x_2; x_3, x_4) = \langle 0 | T \hat{\psi}_i(x_1) \hat{\bar{\psi}}_k(x_2) \hat{\bar{\psi}}_l(x_3) \hat{\psi}_m(x_4) | 0 \rangle. \quad (\text{XI.76})$$

Its Fourier transform contains a δ -function expressing the overall conservation of four-momentum:

$$K_{ik,lm}(p_3, p_4; p_1, p_2) \sim (2\pi)^4 \delta^{(4)}(p_1 + p_2 - p_3 - p_4). \quad (\text{XI.77})$$

We write the function K in the form:

$$\begin{aligned} K_{ik,lm}(p_3, p_4; p_1, p_2) = & (2\pi)^4 \left[\delta^{(4)}(p_1 - p_3) \mathcal{G}_{il}(p_1) \mathcal{G}_{km}(p_2) \right. \\ & \left. - \delta^{(4)}(p_2 - p_3) \mathcal{G}_{im}(p_1) \mathcal{G}_{kl}(p_2) \right] \\ & + \mathcal{G}_{in}(p_3) \mathcal{G}_{kr}(p_4) \left[-i\Gamma_{nr,st}(p_3, p_4; p_1, p_2) \right] \mathcal{G}_{sl}(p_1) \mathcal{G}_{tm}(p_2). \end{aligned} \quad (\text{XI.78})$$

The first two terms in (XI.78) exclude from Γ diagrams that split into two disconnected parts, each having two external lines:

The third term: the $\mathcal{G}$ -factors on all four external lines exclude corrections to the external electron lines (self-energy insertions on external legs).

The antisymmetry of Γ (following from the properties of the T -product of fermionic $\hat{\psi}$ -operators) is expressed by:

$$\Gamma_{ik,lm}(p_3, p_4; p_1, p_2) = -\Gamma_{ki,lm}(p_4, p_3; p_1, p_2) = -\Gamma_{ik,ml}(p_3, p_4; p_2, p_1). \quad (\text{XI.79})$$

If p_1, p_2, p_3, p_4 correspond to real particles, the non-decomposing diagrams (XI.75) describe electron-electron scattering. The scattering amplitude (with the factors $\mathcal{G}$ replaced, on the external lines, by spinor amplitudes of the particles)¹² is:

$$iM_{fi} = \bar{u}_i(p_3) \bar{u}_k(p_4) \left[-i\Gamma_{ik,lm}(p_3, p_4; p_1, p_2) \right] u_l(p_1) u_m(p_2). \quad (\text{XI.80})$$

By virtue of (XI.79) this amplitude automatically possesses the required antisymmetry with respect to permutation of the electrons.

¹²We shall see later (see § 110) that in constructing amplitudes of real processes one need not include self-energy parts on the free ends of the diagrams.

§ 107. Dyson equations

The exact propagators and the vertex part are connected with each other by definite integral relations.

The concept of irreducibility of diagrams extends to any diagrams. Let us consider compact self-energy electron diagrams. Among them only one is irreducible — the second-order diagram. All vertex corrections can be assigned to one of the two vertices.¹³ Therefore the mass operator $\mathcal{M}$ is depicted by a single skeleton diagram:

(XI.81)

Analytically:¹⁴

$$\mathcal{M}(p) = \mathcal{G}^{-1}(p) - G^{-1}(p) = -ie^2 \int \gamma^\nu \mathcal{G}(p+k) \Gamma^\mu(p+k, p; k) \mathcal{D}_{\mu\nu}(k) \frac{d^4 k}{(2\pi)^4}. \quad (\text{XI.82})$$

An analogous expression can be written for the polarisation operator $\mathcal{P}$. Among the compact photon self-energy parts only one is irreducible:

(XI.83)

The corresponding analytic equality is:

$$\frac{\mathcal{P}_{\mu\nu}(k)}{4\pi} = \mathcal{D}_{\mu\nu}^{-1}(k) - D_{\mu\nu}^{-1}(k) = ie^2 \int \text{Sp} \gamma_\mu \mathcal{G}(p+k) \Gamma_\nu(p+k, p; k) \mathcal{G}(p) \frac{d^4 p}{(2\pi)^4} \quad (\text{XI.84})$$

(in (XI.82) and (XI.84) bispinor indices are omitted). Equations (XI.82) and (XI.84) are called the *Dyson equations*.

These equations can also be derived analytically. For the derivation of (XI.82), we consider

$$(\gamma \hat{p} - m)_{il} \mathcal{G}_{lk}(x - x') = -i(\gamma \hat{p} - m)_{il} \langle 0 | T \psi_l(x) \bar{\psi}_k(x') | 0 \rangle \quad (\text{XI.85})$$

($\hat{p} = i\partial$ — the differentiation operator with respect to x). We compute this expression using (XI.5), in the same way as was done in § 75 for the free-particle propagator. The result is:

$$(\gamma \hat{p} - m)_{il} \mathcal{G}_{lk}(x - x') = -ie\gamma_{il}^\nu \langle 0 | T A_\nu(x) \psi_l(x) \bar{\psi}_k(x') | 0 \rangle + \delta_{ik} \delta^{(4)}(x - x'). \quad (\text{XI.86})$$

The δ -function term here is the same as in (75.7), since the equal-time commutation relations for operators in the Heisenberg and interaction representations are the same. The first term on the right is $-ie\gamma^\nu K_{ik}^\nu(x, x, x')$, so that (dropping bispinor indices):

$$(\gamma \hat{p} - m) \mathcal{G}(x - x') = -ie\gamma^\nu K_\nu(x, x, x') + \delta^{(4)}(x - x'). \quad (\text{XI.87})$$

¹³For clarity we emphasise that although we obtain the entire required collection of diagrams by introducing corrections to only one vertex, for each specific diagram the structure of the correction block generally depends on which vertex it is assigned to.

¹⁴If in (XI.81) the exact vertex part is assigned to the left vertex, then in equation (XI.82) the factors γ and Γ are interchanged. Both forms of the equation are, of course, essentially equivalent.

We now pass to Fourier components. If we integrate the definition (XI.66) over $d^4k d^4p_2/(2\pi)^8$, we get:

$$\int K^\mu(p+k, p; k) \frac{d^4k}{(2\pi)^4} = \int K^\mu(x, x, x') e^{ip(x-x')} d^4(x-x'), \quad (\text{XI.88})$$

which shows that the integral on the left-hand side is a Fourier component of the function $K^\mu(x, x, x')$. Taking the Fourier transform of both sides of (XI.87), using the definition (XI.72), and recalling that $\gamma p - m = G^{-1}(p)$, we obtain:

$$G^{-1}(p) \mathcal{G}(p) = 1 - ie^2 \int \gamma^\nu \mathcal{G}(p+k) \Gamma^\mu(p+k, p; k) \mathcal{G}(p) \mathcal{D}_{\mu\nu}(k) \frac{d^4k}{(2\pi)^4}. \quad (\text{XI.89})$$

Multiplying the right-hand side by $\mathcal{G}^{-1}(p)$, we return to equation (XI.82).

§ 108. Ward identity

Yet another relation between the photon propagator and the vertex part, simpler than the Dyson equation, arises as a consequence of gauge invariance.

Let us perform the gauge transformation (102.8), assuming that $\chi(x) \equiv \delta\chi(x)$ is an infinitely small simple (non-operator) function of the 4-coordinates x . Then the electron propagator acquires the addition:

$$\delta\mathcal{G}(x, x') = ie\mathcal{G}(x - x') [\delta\chi(x) - \delta\chi(x')]. \quad (\text{XI.90})$$

Note that this gauge transformation violates spatial-temporal homogeneity and $\delta\mathcal{G}$ depends on x and x' separately, not just on the difference $x - x'$. We decompose $\delta\mathcal{G}$ in a Fourier integral over x and x' separately:

$$\delta\mathcal{G}(p_2, p_1) = \int \int \delta\mathcal{G}(x, x') e^{ip_2x - ip_1x'} d^4x d^4x'.$$

Substituting (XI.90) and integrating over $d^4\xi d^4x'$ (where $\xi = x - x'$), we obtain:

$$\delta\mathcal{G}(p + q, p) = ie \delta\chi(q) [\mathcal{G}(p) - \mathcal{G}(p + q)]. \quad (\text{XI.91})$$

On the other hand, under the same gauge transformation the operator $\hat{A}_\mu(x)$ receives an addition:

$$\delta A_\mu^{(e)}(x) = -\frac{\partial}{\partial x^\mu} \delta\chi, \quad (\text{XI.92})$$

which can be regarded as an infinitely small external field. In the momentum representation:

$$\delta A_\mu^{(e)}(q) = iq_\mu \delta\chi(q). \quad (\text{XI.93})$$

The quantity $\delta\mathcal{G}$ can be computed as the change of the propagator under the influence of this field. In first-order accuracy, it is represented by the single skeleton diagram:

$$(\text{XI.94})$$

Here the bold dashed line represents the effective external field line, associated with the factor (see (103.15)):

$$\delta A_\mu^{(e)}(q) + \delta A_\lambda^{(e)}(q) \frac{\mathcal{P}^{\lambda\nu}(q)}{4\pi} \mathcal{D}_{\mu\nu}(q).$$

But the 4-vector $\delta A_\lambda^{(e)}(q)$ is longitudinal (proportional to q_λ), and the tensor $\mathcal{P}^{\lambda\nu}$ is transverse. Therefore the second term vanishes, and the effective external field reduces to simply the field $\delta A^{(e)}$:

The analytical expression is:

$$\delta \mathcal{G} = e \mathcal{G}(p+q) \Gamma^\mu(p+q, p; q) \mathcal{G}(p) \cdot \delta A_\mu^{(e)}. \quad (\text{XI.95})$$

Substituting (XI.93) and comparing with (XI.91):

$$\mathcal{G}(p+q) - \mathcal{G}(p) = -\mathcal{G}(p+q) \Gamma^\mu(p+q, p; q) \mathcal{G}(p) \cdot q_\mu,$$

or, passing to inverse matrices:

$$\mathcal{G}^{-1}(p+q) - \mathcal{G}^{-1}(p) = q_\mu \Gamma^\mu(p+q, p; q) \quad (\text{XI.96})$$

(H. S. Green, 1953).

Letting $q \rightarrow 0$ and comparing the coefficients of the infinitely small q_μ on both sides, we obtain:

$$\frac{\partial}{\partial p_\mu} \mathcal{G}^{-1}(p) = \Gamma^\mu(p, p; 0). \quad (\text{XI.97})$$

This is the *Ward identity* (J. C. Ward, 1950). The derivative of $\mathcal{G}^{-1}(p)$ with respect to the momentum equals the vertex operator at zero momentum transfer.¹⁵

For the derivative of $\mathcal{G}(p)$ itself:

$$-\frac{\partial}{\partial p_\mu} i\mathcal{G}(p) = i\mathcal{G}(p) [-i\Gamma^\mu(p, p; 0)] i\mathcal{G}(p). \quad (\text{XI.98})$$

One could similarly obtain higher derivatives from higher-order terms of the expansion of $\delta\chi$ in powers of the coordinates. We shall not need such formulae.

Now let us consider the derivative $\partial\mathcal{P}(k)/\partial k_\mu$ of the polarisation operator. Unlike $\mathcal{G}(p)$, the operator $\mathcal{P}(k)$ is gauge-invariant and does not change when the fictitious external field (XI.93) is introduced. Therefore the derivative of $\mathcal{P}$ cannot be computed in the same manner. It is, however, possible to obtain a diagrammatic expression.

Consider the first diagram entering the definition of $\mathcal{P}$ — the second-order diagram:

$$(\text{XI.99})$$

The solid lines correspond to the factors $iG(p)$ and $iG(p+k)$. Differentiating with respect to k replaces the second factor by $\partial G(p+k)/\partial k$, and according to the identity (XI.98) this replacement is equivalent to adding an extra vertex on the electron line:

¹⁵In the zeroth approximation, i.e. for the free-particle propagator, this identity is obvious: $G^{-1}(p) = \gamma p - m$, and therefore $\partial G^{-1}/\partial p_\mu = \gamma^\mu$.

(XI.100)

In the first non-vanishing order the desired derivative is thus expressed through a diagram with three photon ends (“photon three-tail”). Note that this diagram by itself does *not* give the amplitude of splitting of one photon into two. The amplitude of such a process would be the sum of diagram (XI.100) and the same diagram with the reversed direction of traversal of the loop; by Furry’s theorem this sum vanishes. But diagram (XI.100) by itself is non-zero.

Similarly one can differentiate more complex diagrams, adding the vertex with $k' = 0$ to all electron lines that depend on k . There are also diagrams in which the dependence on k resides in internal photon lines, for example:

The derivative of the diagram enclosed in curly brackets is represented by introducing a new graphical element — a *fictitious three-photon vertex* (a point at which three dashed lines meet):

$$4\pi i \frac{\partial D^{-1}}{\partial k^\nu} = 2ik_\mu \equiv v_\mu. \quad (\text{XI.101})$$

Now any diagram depending on k can be differentiated by adding the vertices v_μ or γ_μ along all lines carrying momentum k . Computing further:

$$-\frac{1}{4\pi} \frac{\partial \mathcal{P}_{\mu\nu}}{\partial k^\lambda} = V_{\mu\lambda\nu}, \quad (\text{XI.102})$$

where $ie^2 V_{\mu\lambda\nu}$ is the sum of the internal parts of all “photon three-tails” obtained in this manner.

For the second derivative of the polarisation operator, differentiating (XI.102) once more:

$$\frac{1}{4\pi} \frac{\partial^2 \mathcal{P}_{\mu\nu}}{\partial k^\rho \partial k^\lambda} = \mathcal{G}_{\mu\rho\lambda\nu} + \mathcal{G}_{\mu\lambda\rho\nu}, \quad (\text{XI.103})$$

where $ie^2 \mathcal{G}$ is the sum of the internal parts of all “photon four-tails”:

(XI.104)

(including diagrams with fictitious three-photon vertices (XI.101)).

§ 109. Electron propagator in an external field

If the system is in a given external field $A^{(e)}(x)$, then the exact electron propagator is defined by the same formula (XI.48), but the Hamiltonian $\hat{H} = \hat{H}_0 + \hat{V}$ now includes, together with the interaction between particles, also the interaction with the external field:

$$\hat{V} = e \int \hat{A}_\mu \hat{j}^\mu d^3x + e \int A_\mu^{(e)} \hat{j}^\mu d^3x. \quad (\text{XI.105})$$

Since the external field violates the homogeneity of space-time, the propagator $\mathcal{G}(x, x')$ depends on x and x' separately, not merely on the difference $x - x'$.

Passage to the interaction representation leads to the standard diagrammatic technique, with the difference that in the diagrams there appear, alongside the virtual photon lines,

also lines of the external field. Such a technique is inconvenient in those cases where the external field cannot be treated as a small perturbation — when particles may be in bound states in the field. The electron propagator in an external field is needed precisely for the purpose of studying the properties of bound states when radiative corrections are taken into account.

For the construction of such a propagator, it is natural to start from a representation of the operators in which the external field is already taken into account in the zeroth approximation, while only the electron–photon interaction itself is treated as a perturbation (W. H. Furry, 1951).

We shall assume the external field to be stationary (time-independent).

The required representation of the ψ -operators is given by the formulae of second quantisation in the external field:

$$\begin{aligned}\hat{\psi}^{(e)}(t, \mathbf{r}) &= \sum_n \left\{ \hat{a}_n \psi_n^{(+)}(\mathbf{r}) e^{-i\varepsilon_n^{(+)}t} + \hat{b}_n^\dagger \psi_n^{(-)}(\mathbf{r}) e^{i\varepsilon_n^{(-)}t} \right\}, \\ \hat{\bar{\psi}}^{(e)}(t, \mathbf{r}) &= \sum_n \left\{ \hat{a}_n^\dagger \bar{\psi}_n^{(+)}(\mathbf{r}) e^{i\varepsilon_n^{(+)}t} + \hat{b}_n \bar{\psi}_n^{(-)}(\mathbf{r}) e^{-i\varepsilon_n^{(-)}t} \right\},\end{aligned}\quad (\text{XI.106})$$

where $\psi_n^{(\pm)}(\mathbf{r})$ and $\varepsilon_n^{(\pm)}$ are the wave functions and energy levels of the electron and the positron — solutions of the single-particle Dirac equation in the field.

It is easy to see that the operators (XI.106) are the ψ -operators in a certain representation (Furry representation), intermediate between the Heisenberg and interaction representations:

$$\begin{aligned}\hat{\psi}^{(e)}(t, \mathbf{r}) &= e^{i\hat{H}_1 t} \hat{\psi}(\mathbf{r}) e^{-i\hat{H}_1 t}, \\ \hat{\bar{\psi}}^{(e)}(t, \mathbf{r}) &= e^{i\hat{H}_1 t} \hat{\bar{\psi}}(\mathbf{r}) e^{-i\hat{H}_1 t},\end{aligned}\quad (\text{XI.107})$$

where

$$\hat{H}_1 = \hat{H}_0 + e \int A_\mu^{(e)}(x) \hat{j}^\mu(x) d^3x.$$

The electromagnetic field operator $\hat{A}_\mu$ commutes with the second term in $\hat{H}_1$, so that for this operator the Furry representation coincides with the interaction representation.

The zeroth-approximation electron propagator in the new representation is:

$$G_{ik}^{(e)}(x, x') = -i \langle 0 | T \hat{\psi}_i^{(e)}(x) \hat{\bar{\psi}}_k^{(e)}(x') | 0 \rangle. \quad (\text{XI.108})$$

The operator $\hat{\psi}^{(e)}(t, \mathbf{r})$ satisfies the Dirac equation in the external field:

$$[\gamma \hat{p} - e \gamma A^{(e)}(x) - m] \hat{\psi}^{(e)}(t, \mathbf{r}) = 0, \quad (\text{XI.109})$$

and correspondingly for $G^{(e)}$:

$$[\gamma \hat{p} - e \gamma A^{(e)}(x) - m] G^{(e)}(x, x') = \delta^4(x - x') \quad (\text{XI.110})$$

(cf. the derivation of (107.5)).

The diagrammatic technique for the expansion of the exact propagator $\mathcal{G}$ in a series in powers of e^2 is constructed — by passing from the Heisenberg to the Furry representation

and then to the interaction representation — in exactly the same way as the usual technique. It leads to diagrams of the same type, with the sole difference that the solid lines in the diagrams now correspond to the factor $iG^{(e)}$ (instead of iG).

A slight difference arises in the analytic rules corresponding to the diagrams, since $G^{(e)}$ is a function not only of the difference $x - x'$, but of x and x' separately. In a constant external field the homogeneity of time is preserved, so that t and t' enter only through the difference $\tau = t - t'$. Thus $G^{(e)} = G^{(e)}(\tau, \mathbf{r}, \mathbf{r}')$.

Performing the Fourier transform over each of its arguments, we have:

$$G^{(e)}(\tau, \mathbf{r}, \mathbf{r}') = \int e^{i(\varepsilon\tau - \mathbf{p}_2 \cdot \mathbf{r} + \mathbf{p}_1 \cdot \mathbf{r}')} G^{(e)}(\varepsilon, \mathbf{p}_2, \mathbf{p}_1) \frac{d\varepsilon}{2\pi} \frac{d^3 p_1}{(2\pi)^3} \frac{d^3 p_2}{(2\pi)^3}. \quad (\text{XI.111})$$

Each solid line now carries the factor $iG^{(e)}(\varepsilon, \mathbf{p}_2, \mathbf{p}_1)$, with one energy value ε but two momentum values — the initial $\mathbf{p}_1$ and the final $\mathbf{p}_2$:

$$iG^{(e)}(\varepsilon, \mathbf{p}_2, \mathbf{p}_1). \quad (\text{XI.112})$$

Integration is performed over $d\varepsilon/(2\pi)$ for each independent loop, and over $d^3 p_1/(2\pi)^3$ and $d^3 p_2/(2\pi)^3$ independently, with conservation of momentum at each vertex. For example:

$$= e^2 \int G^{(e)}(\varepsilon, \mathbf{p}_2, \mathbf{p}'') \gamma^\mu G^{(e)}(\varepsilon - \omega, \mathbf{p}'' - \mathbf{k}, \mathbf{p}' - \mathbf{k}) \gamma^\nu G^{(e)}(\varepsilon, \mathbf{p}', \mathbf{p}_1) \\ \times D_{\mu\nu}(\omega, \mathbf{k}) \frac{d^4 k}{(2\pi)^4} \frac{d^3 p'}{(2\pi)^3} \frac{d^3 p''}{(2\pi)^3}. \quad (\text{XI.113})$$

It is important to note that in this technique one must also include diagrams with “self-closed” electron lines (loops), which in the usual technique are discarded as corresponding to a “vacuum current.” In the presence of an external field this current need not vanish, owing to the “polarisation of the vacuum.” For example, in the diagram:

$$(\text{XI.114})$$

the upper loop carries the factor

$$i \int G^{(e)}(\omega, \mathbf{p} + \mathbf{k}, \mathbf{p}) \frac{d^3 p}{(2\pi)^3} \frac{d\omega}{2\pi}. \quad (\text{XI.115})$$

Here one must clarify the meaning of the integral over ω . The point is that integrating the Fourier component of $G^{(e)}(\tau)$ over ω amounts to taking the value of this function at $\tau = 0$; but $G^{(e)}(\tau)$ is discontinuous at this point, and one must specify which of its two limiting values should be taken. To clarify this, we note that the integral (XI.115) arises from the contraction of ψ -operators at the same point: $\hat{j}^\mu = \hat{\psi}^{(e)}(t, \mathbf{r}) \gamma^\mu \hat{\psi}^{(e)}(t, \mathbf{r})$, where $\hat{\psi}^{(e)}$ stands to the left. According to the definition of the propagator (XI.108), this ordering of the factors corresponds to $t = t'$ if t' is understood as $t' = t + 0$, i.e. one

must take the limiting value of $G^{(e)}(t - t')$ at $t - t' \rightarrow -0$. In other words, the integral over $d\omega/2\pi$ in (XI.115) should be understood as

$$\int \dots e^{-i\omega\tau} \frac{d\omega}{2\pi}, \quad \tau \rightarrow -0. \quad (\text{XI.116})$$

The mass operator in the external field is defined in the same manner as in § 105: $-i\mathcal{M}$ is the sum of all compact self-energy blocks. It is now a function of the energy ε and of the momenta $\mathbf{p}_1$ and $\mathbf{p}_2$ on the external lines entering and leaving the block:

$$-i\mathcal{M}(\varepsilon, \mathbf{p}_2, \mathbf{p}_1). \quad (\text{XI.117})$$

Proceeding in exactly the same way as in deriving (XI.53):

$$\begin{aligned} \mathcal{G}(\varepsilon, \mathbf{p}_2, \mathbf{p}_1) - G^{(e)}(\varepsilon, \mathbf{p}_2, \mathbf{p}_1) \\ = \int \int G^{(e)}(\varepsilon, \mathbf{p}_2, \mathbf{p}'') \mathcal{M}(\varepsilon, \mathbf{p}'', \mathbf{p}') \mathcal{G}(\varepsilon, \mathbf{p}', \mathbf{p}_1) \frac{d^3 p'}{(2\pi)^3} \frac{d^3 p''}{(2\pi)^3}. \end{aligned} \quad (\text{XI.118})$$

It is more natural to write this equation in the coordinate representation:

$$\mathcal{G}(\varepsilon, \mathbf{r}, \mathbf{r}') = \int \int \mathcal{G}(\varepsilon, \mathbf{p}_2, \mathbf{p}_1) e^{i(\mathbf{p}_2 \cdot \mathbf{r} - \mathbf{p}_1 \cdot \mathbf{r}')} \frac{d^3 p_1 d^3 p_2}{(2\pi)^6}, \quad (\text{XI.119})$$

and analogously for the other quantities. Performing the inverse Fourier transform in (XI.118):

$$\mathcal{G}(\varepsilon, \mathbf{r}, \mathbf{r}') - G^{(e)}(\varepsilon, \mathbf{r}, \mathbf{r}') = \int \int G^{(e)}(\varepsilon, \mathbf{r}, \mathbf{r}_2) \mathcal{M}(\varepsilon, \mathbf{r}_2, \mathbf{r}_1) \mathcal{G}(\varepsilon, \mathbf{r}_1, \mathbf{r}') d^3 x_1 d^3 x_2.$$

We apply to both sides of this equation the operator $\gamma^0 \varepsilon - \gamma \hat{\mathbf{p}} - e\gamma A^{(e)}(\mathbf{r})$ (where ε is a number, and $\hat{\mathbf{p}} = -i\nabla$ differentiates with respect to $\mathbf{r}$). According to (XI.110):

$$[\gamma^0 \varepsilon - \gamma \hat{\mathbf{p}} - e\gamma A^{(e)}(\mathbf{r})] G^{(e)}(\varepsilon, \mathbf{r}, \mathbf{r}') = \delta(\mathbf{r} - \mathbf{r}'). \quad (\text{XI.120})$$

The result is:

$$[\gamma^0 \varepsilon - \gamma \hat{\mathbf{p}} - e\gamma A^{(e)}(\mathbf{r})] \mathcal{G}(\varepsilon, \mathbf{r}, \mathbf{r}') - \int \mathcal{M}(\varepsilon, \mathbf{r}, \mathbf{r}_1) \mathcal{G}(\varepsilon, \mathbf{r}_1, \mathbf{r}') d^3 x_1 = \delta(\mathbf{r} - \mathbf{r}'). \quad (\text{XI.121})$$

The particular value of the function $\mathcal{G}(\varepsilon, \mathbf{r}, \mathbf{r}')$ is that its poles as a function of ε determine the energy levels of the electron in the external field.

Let us show this first for the approximate function $G^{(e)}$. Substituting the operators (XI.106) into the definition of the propagator (XI.108), we get, in analogy with the formulae (75.12) for the free-particle propagator:

$$G_{ik}^{(e)}(t - t', \mathbf{r}, \mathbf{r}') = \begin{cases} -i \sum_n \psi_{ni}^{(+)}(\mathbf{r}) \bar{\psi}_{nk}^{(+)}(\mathbf{r}') e^{-i\varepsilon_n^{(+)}(t-t')}, & t > t', \\ i \sum_n \psi_{ni}^{(-)}(\mathbf{r}) \bar{\psi}_{nk}^{(-)}(\mathbf{r}') e^{i\varepsilon_n^{(-)}(t-t')}, & t < t', \end{cases} \quad (\text{XI.122})$$

and after Fourier transformation:

$$G_{ik}^{(e)}(\varepsilon, \mathbf{r}, \mathbf{r}') = \sum_n \left\{ \frac{\psi_{ni}^{(+)}(\mathbf{r}) \bar{\psi}_{nk}^{(+)}(\mathbf{r}')}{\varepsilon - \varepsilon_n^{(+)} + i0} + \frac{\psi_{ni}^{(-)}(\mathbf{r}) \bar{\psi}_{nk}^{(-)}(\mathbf{r}')}{\varepsilon + \varepsilon_n^{(-)} - i0} \right\}. \quad (\text{XI.123})$$

The function $G^{(e)}(\varepsilon, \mathbf{r}, \mathbf{r}')$, considered as an analytic function of ε , has poles on the positive real half-axis at the electron energy levels, and on the negative half-axis at the positron levels. The values $\varepsilon_n^{(\pm)} > m$ form a continuous spectrum, and the corresponding poles merge into two branch cuts: from $-\infty$ to $-m$ and from m to $+\infty$. On the interval $|\varepsilon| < m$ lie the poles that determine the discrete energy levels.

For the exact propagator $\mathcal{G}(\varepsilon, \mathbf{r}, \mathbf{r}')$ one obtains an analogous expansion by expressing it through matrix elements of the Schrödinger–Heisenberg operators:

$$\langle m | \hat{\psi}(t, \mathbf{r}) | n \rangle = \langle m | \hat{\psi}(\mathbf{r}) | n \rangle e^{-i(E_n - E_m)t}. \quad (\text{XI.124})$$

Here E_n are the exact (with all radiative corrections) energy levels of the system in the external field. The operator $\hat{\psi}$ increases the charge by unity, and $\hat{\bar{\psi}}$ decreases it by unity. In the matrix elements $\langle n | \hat{\psi} | 0 \rangle$ and $\langle 0 | \hat{\bar{\psi}} | n \rangle$, the states $|n\rangle$ must correspond to charge -1 — one positron plus electron–positron pairs and photons; we denote the energies $E_n^{(-)}$. Similarly, in $\langle 0 | \hat{\bar{\psi}} | n \rangle$ and $\langle n | \hat{\psi} | 0 \rangle$, the states have charge $+1$ — one electron plus pairs and photons; energies $E_n^{(+)}$.

We obtain:¹⁶

$$\mathcal{G}_{ik}(t - t', \mathbf{r}, \mathbf{r}') = \begin{cases} -i \sum_n \langle 0 | \hat{\psi}_i(\mathbf{r}) | n \rangle \langle n | \hat{\bar{\psi}}_k(\mathbf{r}') | 0 \rangle e^{-iE_n^{(+)}(t-t')}, & t > t', \\ i \sum_n \langle 0 | \hat{\bar{\psi}}_k(\mathbf{r}') | n \rangle \langle n | \hat{\psi}_i(\mathbf{r}) | 0 \rangle e^{iE_n^{(-)}(t-t')}, & t < t', \end{cases} \quad (\text{XI.125})$$

and hence:

$$\mathcal{G}_{ik}(\varepsilon, \mathbf{r}, \mathbf{r}') = \sum_n \left\{ \frac{\langle 0 | \hat{\psi}_i(\mathbf{r}) | n \rangle \langle n | \hat{\bar{\psi}}_k(\mathbf{r}') | 0 \rangle}{\varepsilon - E_n^{(+)} + i0} + \frac{\langle 0 | \hat{\bar{\psi}}_k(\mathbf{r}') | n \rangle \langle n | \hat{\psi}_i(\mathbf{r}) | 0 \rangle}{\varepsilon + E_n^{(-)} - i0} \right\}. \quad (\text{XI.126})$$

Let ε be close to one of the discrete energy levels $E_n^{(+)}$ (or $E_n^{(-)}$). Then in the sum (XI.126) only the single pole term survives. Substituting into (XI.121), the factors depending on the second argument $\mathbf{r}'$ drop out (when $\mathbf{r} \neq \mathbf{r}'$), giving a homogeneous integro-differential equation for the function $\langle 0 | \hat{\psi}(\mathbf{r}) | n \rangle$ (or $\langle n | \hat{\bar{\psi}}(\mathbf{r}) | 0 \rangle$), which we denote $\Psi_n(\mathbf{r})$ for brevity.¹⁷ Dropping the index n :

$$[\gamma^0 \varepsilon + i\gamma \nabla - e\gamma A^{(e)}(\mathbf{r})]_{ik} \Psi_k(\mathbf{r}) - \int \mathcal{M}_{ik}(\varepsilon, \mathbf{r}, \mathbf{r}_1) \Psi_k(\mathbf{r}_1) d^3x_1 = 0 \quad (\text{XI.127})$$

(J. Schwinger, 1951). The discrete energy levels E_n appear as eigenvalues of this equation, providing the regular procedure for determining these levels.

¹⁶It is assumed that the external field vanishes at spatial infinity.

¹⁷Neglecting radiative corrections, $\Psi_n(\mathbf{r})$ coincides (for states containing one electron or one positron) with the wave functions $\psi_n^{(\pm)}$ — solutions of the Dirac equation.

Let us express, for example, the first-order correction to a discrete electron energy level ε_n , obtained from the Dirac equation:

$$[\gamma^0 \varepsilon_n + i\boldsymbol{\gamma} \nabla - e\gamma A^{(e)}(\mathbf{r})]\psi_n(\mathbf{r}) = 0, \quad (\text{XI.128})$$

with normalisation

$$\int \psi_n^* \psi_n d^3x = 1. \quad (\text{XI.129})$$

We write the eigenfunction in the form

$$\Psi_n(\mathbf{r}) = \psi_n(\mathbf{r}) + \psi_n^{(1)}(\mathbf{r}), \quad (\text{XI.130})$$

where $\psi_n^{(1)}$ is the correction. Substituting (XI.130) into equation (XI.127), multiplying from the left by $\bar{\psi}_n(\mathbf{r})$, and integrating over d^3x ,¹⁸ we obtain:

$$E_n - \varepsilon_n \approx \iint \bar{\psi}_{ni}(\mathbf{r}) \mathcal{M}_{ik}(\varepsilon_n, \mathbf{r}, \mathbf{r}_1) \psi_{nk}(\mathbf{r}_1) d^3x d^3x_1. \quad (\text{XI.131})$$

§ 110. Physical conditions of renormalisation

The theory presented so far in this chapter has been to a significant extent formal in character. We operated with all quantities as if they were finite, intentionally leaving aside the infinities arising in the theory. However, in an actual computation of the functions $\mathcal{D}$, $\mathcal{G}$, Γ by perturbation theory, divergent integrals appear which cannot, without additional considerations, be ascribed any definite value. The emergence of such divergences is a manifestation of the logical incompleteness of quantum electrodynamics. We shall see, however, that in this theory one can establish certain prescriptions that allow an unambiguous “subtraction of infinities” and the obtaining of finite values for all quantities possessing direct physical meaning. At the basis of these prescriptions lie obvious physical requirements: that the photon mass be zero and that the charge and mass of the electron equal their observed values.

Let us begin with the conditions on the photon propagator.

Consider scattering through single-particle intermediate states containing one virtual photon. The amplitude must have a pole at $P^2 = 0$ (the square of the total 4-momentum of the initial particles P coincides with the square of the real photon mass). As we saw in §79, this follows from the unitarity condition. The pole term arises from diagrams of the type (79.1):

$$(\text{XI.132})$$

with radiative corrections, both parts of the diagram being connected by a bold dashed line (exact photon propagator). Therefore $\mathcal{D}(k^2)$ must have a pole at $k^2 = 0$:

$$\mathcal{D} \rightarrow \frac{4\pi Z}{k^2} \quad \text{when} \quad k^2 \rightarrow 0, \quad (\text{XI.133})$$

¹⁸In the integration one must use the self-adjointness of the differential operator of equation (XI.128) for transferring its action from $\psi_n^{(1)}$ to $\bar{\psi}_n$.

where Z is a constant. For the polarisation operator this means, according to (103.21):

$$\mathcal{P}(0) = 0. \quad (\text{XI.134})$$

The coefficient $1/Z = 1 - \mathcal{P}(k^2)/k^2|_{k^2 \rightarrow 0}$.

Further restrictions on $\mathcal{P}(k^2)$ follow from the physical definition of electric charge. Two classical (arbitrarily heavy) particles at rest at a large distance from one another must interact according to the Coulomb law $U = e^2/r$ ($r \gg 1/m$). On the other hand, this interaction is expressed by the diagram:

$$(\text{XI.135})$$

where the upper and lower lines correspond to the classical particles. The photon self-energy corrections are included in the virtual photon line. Other corrections, involving the heavy-particle lines, vanish in the limit $M \rightarrow \infty$: adding internal lines to the diagram (XI.135) (for example, connecting lines a and c , or a and b , by a photon line) leads to virtual heavy-particle propagators containing the mass M in the denominator and vanishing as $M \rightarrow \infty$.

From the diagram (XI.135), the factor $e^2 \mathcal{D}(k^2)$ must represent (up to sign) the Fourier transform of the interaction potential. Stationarity of the particles means the virtual photon frequency $\omega = 0$; large distances mean small wave vectors $\mathbf{k}$. The Fourier transform of the Coulomb potential is $4\pi e^2/\mathbf{k}^2$. Since $\mathcal{D}$ depends only on $k^2 = \omega^2 - \mathbf{k}^2$, we obtain:

$$\mathcal{D} \rightarrow \frac{4\pi}{k^2}, \quad k^2 \rightarrow 0. \quad (\text{XI.136})$$

Thus the coefficient Z in (XI.133) must equal unity: $Z = 1$ (the sign in (XI.136) is obvious: $\mathcal{D}(k^2)$ tends to the free photon propagator $D(k^2)$). For the polarisation operator this means:

$$\frac{\mathcal{P}(k^2)}{k^2} \rightarrow 0, \quad k^2 \rightarrow 0. \quad (\text{XI.137})$$

Besides condition (XI.134), this also gives:

$$\mathcal{P}'(0) = 0. \quad (\text{XI.138})$$

In §103 it was noted that the effective external photon line factor (103.15), with account of (103.16) and (103.20), is:

$$\left[1 + \frac{\mathcal{P}(0) \mathcal{D}(0)}{4\pi} \right] e^\mu.$$

Now, with (XI.136) and (XI.137), the correction term vanishes. In other words, in external photon lines one need not include radiative corrections at all.

Thus natural physical requirements lead to the establishment of definite (zero) values of $\mathcal{P}(0)$ and $\mathcal{P}'(0)$. Computing these quantities from perturbation-theory diagrams would lead to divergent integrals. The method of eliminating infinities consists in ascribing to the divergent expressions the values prescribed by the physical requirements. This procedure is called *renormalisation* of the corresponding quantities.¹⁹

¹⁹The idea of this approach was first expressed by Kramers (H. Kramers, 1947). Systematic application of the renormalisation method in quantum electrodynamics was carried out in the works of Dyson, Tomonaga (S. Tomonaga), Feynman and Schwinger.

The procedure can also be formulated differently. For charge renormalisation one introduces a non-physical “bare” charge e_0 as the parameter in the electromagnetic interaction Hamiltonian in the formal perturbation theory. After this, the renormalisation condition is formulated as: $e_0^2 \mathcal{D}(k^2) \rightarrow 4\pi e^2/k^2$ (at $k^2 \rightarrow 0$), where e is the true physical charge. From this $e_0^2 Z = e^2$, and together with Z the non-physical quantity e_0 is eliminated. By requiring $Z = 1$ from the outset, we perform the renormalisation “on the fly” and avoid introducing fictitious quantities even in intermediate calculations.

Now let us turn to the conditions on the electron propagator. Consider scattering through a single-particle intermediate state containing one virtual electron. The amplitude must have a pole when the square of the total 4-momentum of the initial particles P_i coincides with the square of the real electron mass: $P_i^2 = m^2$. The pole term in the amplitude arises from diagrams of the type

(XI.139)

where, with radiative corrections taken into account, the bold solid line is the exact electron propagator. This means that the function $\mathcal{G}(p)$ must have a pole at $p^2 = m^2$, i.e. must have the limiting form

$$\mathcal{G}(p) \approx Z_1 \frac{\gamma p + m}{p^2 - m^2 + i0} + g(p), \quad p^2 \rightarrow m^2, \quad (\text{XI.140})$$

where Z_1 is a scalar constant, and $g(p)$ remains finite as $p^2 \rightarrow m^2$. The matrix structure of the pole term in (XI.140) (proportionality to $\gamma p + m$) is a consequence of the same unitarity condition from which the requirement of the existence of the pole itself arose. Let us demonstrate this, simultaneously clarifying an important question about the renormalisation conditions for external electron lines.

If $\mathcal{G}(P)$ has the limiting form (XI.140), the inverse matrix

$$\mathcal{G}^{-1}(p) \approx \frac{1}{Z_1} (\gamma p - m) - (\gamma p - m) g (\gamma p - m), \quad p^2 \rightarrow m^2. \quad (\text{XI.141})$$

The mass operator

$$\mathcal{M} = G^{-1} - \mathcal{G}^{-1} \approx \left(1 - \frac{1}{Z_1}\right) (\gamma p - m) + (\gamma p - m) g (\gamma p - m), \quad p^2 \rightarrow m^2. \quad (\text{XI.142})$$

The effective external (say, incoming) electron line carries the factor (cf. (103.15))

$$\mathcal{U}(p) = u(p) + \mathcal{G}(p) \mathcal{M}(p) u(p), \quad (\text{XI.143})$$

where $u(p)$ is the ordinary spinor amplitude of the electron, satisfying the Dirac equation $(\gamma p - m)u = 0$. By virtue of the requirements of relativistic invariance ($\mathcal{U}$, like u , is a bispinor), the limiting value of $\mathcal{U}(p)$ as $p^2 \rightarrow m^2$ can differ from $u(p)$ only by a constant scalar factor:

$$\mathcal{U}(p) = Z' u(p). \quad (\text{XI.144})$$

This factor Z' is connected in a definite way with Z_1 , but finding this connection by direct substitution of (XI.141), (XI.142) into (XI.143) is not possible, owing to the

indeterminacy that arises: the result depends on the order in which the limit is taken for the various factors in (XI.143).

One can, however, avoid the question about the correct way to take the limit, by appealing instead to the unitarity condition applied to the reaction depicted by diagram (XI.139). The unitarity relation applies, generally speaking, not to individual diagrams but to amplitudes of processes as a whole. But when $p^2 \rightarrow m^2$ the pole diagram (XI.139) gives the dominant contribution to the corresponding amplitude M_{fi} , so that the other diagrams belonging to the same reaction can be disregarded.

By virtue of the unitarity requirements, as was shown in § 79, the single-particle intermediate state leads to the appearance in the imaginary part of the reaction amplitude of a term with a δ -function:

$$i\pi\delta(p^2 - m^2) \sum_{\text{pol}} M_{fn} M_{in}^*, \quad (\text{XI.145})$$

where in the present case the index n refers to a state containing one real electron, and the summation over polarisations is carried out as required by the symmetrisation of both sides of the unitarity relation over the helicities of the initial and final particles (as in § 79; then $M_{fi} = M_{if}$). The amplitude M_{fn} corresponds to the process depicted by the diagram and has the form

$$M_{fn} = (M'_{fn} \mathcal{U}) = Z' (M'_{fn} u),$$

where M'_{fn} is a quantity with one free bispinor index.²⁰ Analogously the amplitude M_{in}^* has the structure

$$M_{in}^* = (\overline{\mathcal{U}} M'^*_{in}) = Z' (\bar{u} M'^*_{in}).$$

The summation over electron polarisations replaces the product $(M'_{fn} u)(\bar{u} M'^*_{in})$ by $M'_{fn}(\gamma p + m) M'^*_{in}$, so that the term (XI.145) in the amplitude M'_{fi} takes the form

$$Z'^2 i\pi\delta(p^2 - m^2) [M'_{fn}(\gamma p + m) M'^*_{in}].$$

On the other hand, computing this same amplitude directly from diagram (XI.139) gives

$$iM_{fi} = iM'_{fn} \cdot i\mathcal{G}(p) \cdot iM'^*_{in}.$$

Comparing the two expressions confirms the limiting form for $\mathcal{G}(p)$ (the first term in (XI.140)), with

$$Z' = \sqrt{Z_1}. \quad (\text{XI.146})$$

Let us now show that once the required limiting form of the electron propagator has been established, there is no need for any further conditions on the vertex operator.

Consider the diagram

²⁰Strictly speaking, a clarification is needed here. An electron as a stable particle cannot actually decay into a different set of real particles. One can, however, formally regard as “final particles” certain fictitious particles with masses that would permit such a decay. The resulting relation should then be understood in the sense of analytic continuation to the real values of the masses.

(XI.147)

describing electron scattering in an external field $A^{(e)}(k)$ (in the first Born approximation), with all radiative corrections taken into account. In the limit $k \rightarrow 0$, $p_2 \rightarrow p_1 \equiv p$, the self-energy corrections to the external field line vanish (recall that these corrections vanish for any k when $k^2 = 0$). The diagram then corresponds to the amplitude

$$M_{fi} = -e \bar{\mathcal{U}}(p) \Gamma(p, p; 0) \mathcal{U}(p) A^{(e)}(k \rightarrow 0) \quad (\text{XI.148})$$

— the product of $A^{(e)}$ with the electron transition current $\bar{\mathcal{U}} \Gamma \mathcal{U}$. But when $k \rightarrow 0$ the potential $A^{(e)}(x)$ reduces to one independent of coordinates and time. Such a potential does not correspond to any physical field (it is a particular case of a gauge transformation), so that it cannot cause any change in the electron current. In other words, in the limit under consideration the transition current $\bar{\mathcal{U}} \Gamma \mathcal{U}$ must simply coincide with the free current $\bar{u} \gamma^\mu u$:

$$\bar{\mathcal{U}}(p) \Gamma^\mu(p, p; 0) \mathcal{U}(p) = Z_1 \bar{u}(p) \Gamma^\mu u(p) = \bar{u}(p) \gamma^\mu u(p). \quad (\text{XI.149})$$

This is, in essence, also an expression of the definition of the physical charge of the electron. It is easy to see that it is automatically satisfied regardless of the value of Z_1 . Indeed, substituting $\mathcal{G}^{-1}(p)$ from (XI.141) into the Ward identity (XI.97), we find

$$\Gamma^\mu(p, p; 0) = Z_1^{-1} \gamma^\mu - \gamma^\mu g(p) (\gamma p - m) - (\gamma p - m) g(p) \gamma^\mu,$$

and equality (XI.149) is satisfied by virtue of the equations $(\gamma p - m)u = 0$, $\bar{u}(\gamma p - m) = 0$.

We see that in constructing the amplitude of a physical process the “renormalisation constant” Z_1 drops out altogether. Moreover, taking advantage of the indeterminacy arising from the divergences in computing Γ , one can simply require that

$$\bar{u}(p) \Gamma^\mu(p, p; 0) u(p) = \bar{u}(p) \gamma^\mu u(p), \quad p^2 = m^2, \quad (\text{XI.150})$$

i.e. set $Z_1 = 1$. The convenience of this definition is that the need for introducing corrections to external electron lines disappears: we simply have

$$\mathcal{U}(p) = u(p).$$

One can also verify this directly: noting that when $Z_1 = 1$ the mass operator (XI.142) becomes

$$\mathcal{M} = (\gamma p - m) g (\gamma p - m) \quad (\text{XI.151})$$

and the second term in (XI.143) obviously vanishes. Thus, external lines — of both photons and electrons — will not require “renormalisation” corrections.²¹

²¹In renormalising the photon propagator the condition $Z = 1$ arose as a necessary physical requirement, and after this the disappearance of corrections to external photon lines occurs automatically. From a formal viewpoint, however, the situation for photon and electron external lines is analogous: if $Z \neq 1$ then the photon wave amplitude e_μ of a real photon would be multiplied by $\sqrt{Z}$.

§ 111. Analytic properties of the photon propagator

The investigation of the analytic properties of the photon propagator is conveniently begun by studying the properties of the function $\Pi(k^2)$. Direct use of the definition (103.1) is complicated by the gauge ambiguity of the operators $\hat{A}^\mu(x)$.

Starting from the expression for the photon self-energy function in terms of matrix elements of the gauge-invariant current operator, the integral representation of $\Pi(k^2)$ (104.11) was obtained in § 104. Denoting k^2 by t ,²² consider the properties of $\Pi(t)$ in the complex t -plane. From the integral representation:

$$\Pi(t) = \int_0^\infty \frac{\rho(t') dt'}{t - t' + i0}. \quad (\text{XI.152})$$

On the negative real half-axis, $\Pi(t)$ is real, and on the entire remaining plane satisfies:

$$\Pi(t^*) = \Pi^*(t). \quad (\text{XI.153})$$

$\Pi(t)$ can have singularities only at points of the function $\rho(t)$. These lie at values $t = k^2$ which are thresholds for production of various collections of real particles by the virtual photon. At these values new types of intermediate states “come into play” in the sum (104.9). The contribution of these states is zero below threshold and nonzero above, leading to singularities at the threshold points. These threshold values are real and non-negative.²³

Therefore the singular points of $\Pi(t)$ lie on the positive real half-axis of t . Making a cut along this half-axis, $\Pi(t)$ is analytic on the entire cut plane.

The $+i0$ in the denominator of (XI.152) shows that the pole $t' = t$ must be bypassed from below. Thus by the “value of $\Pi(t)$ ” on the real axis we mean its value on the upper edge of the cut. Using the rule (75.18):

$$\frac{1}{x \pm i0} = \mathcal{P} \frac{1}{x} \mp i\pi\delta(x), \quad (\text{XI.154})$$

we find for real t :

$$\text{Im } \Pi(t) \equiv \text{Im } \Pi(t + i0) = -\pi\rho(t). \quad (\text{XI.155})$$

On the lower edge of the cut the imaginary part has opposite sign, while $\text{Re } \Pi$ is the same. Hence the jump across the cut:

$$\Pi(t + i0) - \Pi(t - i0) = -2\pi i \rho(t). \quad (\text{XI.156})$$

The integral representation (XI.152) can itself be viewed as a Cauchy formula for the analytic function $\Pi(t)$. Indeed, applying the Cauchy formula:

$$\Pi(t) = \frac{1}{2\pi i} \oint_C \frac{\Pi(t') dt'}{t' - t} \quad (\text{XI.157})$$

to the contour

²²Not to be confused with the time designation!

²³For example, $k^2 = 0$ is the threshold for three (or more odd number of) real photons, $k^2 = 4m^2$ is the threshold for one electron-positron pair, etc.

(XI.158)

wrapping around the cut. Under the assumption of sufficiently fast decrease of $\Pi(t)$ at infinity, the integral over the large circle vanishes, and the integrals along the edges of the cut give the *dispersion relation*:

$$\Pi(t) = \frac{1}{\pi} \int_0^\infty \frac{\text{Im } \Pi(t' + i0)}{t' - t} dt' = \frac{1}{\pi} \int_0^\infty \frac{\text{Im } \Pi(t')}{t' - t - i0} dt'. \quad (\text{XI.159})$$

Substituting (XI.155), we recover (XI.152).²⁴

The analytic properties of $\mathcal{P}(t)$ and $\mathcal{D}(t)$ coincide with those of $\Pi(t)$, through which they are expressed by simple formulas (104.2) and (103.21). For $\mathcal{D}(t)$:

$$\mathcal{D}(t) = \frac{4\pi}{t(1 + \Pi(t)/t)}. \quad (\text{XI.160})$$

On the positive real half-axis ($t > 0$), t should be understood as $t + i0$. The imaginary part of $\mathcal{D}(t)$ can be computed using (XI.154) and (XI.155), taking into account that according to (XI.137) $\Pi(t)/t \rightarrow 0$ as $t \rightarrow 0$. Then:

$$\text{Im } \mathcal{D}(t) = -4\pi^2 \delta(t) + \frac{4\pi}{t^2} \text{Im } \Pi(t) = -4\pi^2 \delta(t) - \frac{4\pi^2}{t^2} \rho(t). \quad (\text{XI.161})$$

Applying the dispersion relation of type (XI.159) to $\mathcal{D}(t)$:

$$\mathcal{D}(t) = \frac{4\pi}{t + i0} + 4\pi \int_0^\infty \frac{\rho(t')}{t'^2} \frac{dt'}{t - t' + i0}. \quad (\text{XI.162})$$

This formula is called the *Källén–Lehmann representation* (G. Källén, 1952; H. Lehmann, 1954).

There is a close connection between the position of the cut for $\mathcal{D}(t)$ (and its imaginary part on the cut) and the unitarity condition for the amplitude of the process $a + b \rightarrow c + d$ depicted by diagram (110.4).

The unitarity condition (71.2):

$$T_{fi} - T_{if}^* = i(2\pi)^4 \sum_n T_{fn} T_{in}^* \delta^{(4)}(P_f - P_i).^{25} \quad (\text{XI.163})$$

The summation on the right is over all physical intermediate states n , which are evidently the states of systems of real pairs and photons — the same states that appear in the matrix elements in the definition of $\rho(k^2)$ (104.9). The amplitudes M_{fi} and M_{if} contain factors $\mathcal{D}(k^2)$ and $\mathcal{D}^*(k^2)$ respectively, and their difference contains the imaginary part $\text{Im } \mathcal{D}(k^2)$. We see that the connection (from (XI.155)) between $\text{Im } \mathcal{D}$ and the existence of the indicated intermediate states is a consequence of the necessary requirements of unitarity.

²⁴Dispersion relations were introduced into quantum field theory by M. Gell-Mann, M. L. Goldberger, and W. E. Thirring (1954).

²⁵Recall that amplitudes T_{fi} differ from M_{fi} only by factors (see (64.10)).

In subsequent work we shall see that in actual perturbation-theory computations of $\mathcal{D}(t)$ (or equivalently $\mathcal{P}(t)$) it is convenient to begin by computing $\text{Im } \mathcal{P}$, which involves no divergent expressions; the real part is then obtained from the dispersion formula of type (XI.159), and the integral may turn out to be divergent, requiring additional subtraction operations to satisfy the conditions $\mathcal{P}(0) = 0$ and $\mathcal{P}'(0) = 0$. This subtraction can be performed without explicit manipulation of divergent integrals. To do this, one applies the dispersion relation (XI.159) not to $\mathcal{P}(t)$ itself but to $\mathcal{P}(t)/t^2$. Then $\mathcal{P}(t)$ takes the form:

$$\mathcal{P}(t) = \frac{t^2}{\pi} \int_0^\infty \frac{\text{Im } \mathcal{P}(t')}{t'^2(t' - t - i0)} dt'. \quad (\text{XI.164})$$

This integral converges, and the function $\mathcal{P}(t)$ obtained automatically satisfies the required conditions.

Equation (XI.164) is called a dispersion relation “with two subtractions.” The meaning of the passage to $\mathcal{P}(t)/t^2$ becomes particularly clear if one writes (XI.164) in the form:

$$\mathcal{P}(t) = \frac{1}{\pi} \int_0^\infty \frac{\text{Im } \mathcal{P}(t')}{t' - t - i0} dt' - \frac{1}{\pi} \int_0^\infty \frac{\text{Im } \mathcal{P}(t')}{t'} dt' - \frac{t}{\pi} \int_0^\infty \frac{\text{Im } \mathcal{P}(t')}{t'^2} dt'. \quad (\text{XI.165})$$

If one denotes the first (“unregularised”) integral as $\tilde{\mathcal{P}}(t)$, then the entire expression on the right equals

$$\tilde{\mathcal{P}}(t) - \tilde{\mathcal{P}}(0) - t \tilde{\mathcal{P}}'(0).$$

§ 112. Regularisation of Feynman integrals

The physical renormalisation conditions considered in § 110 allow, in principle, obtaining unambiguously finite values for the amplitude of any electrodynamic process when computed in any approximation of perturbation theory.

Let us first clarify the character of the divergences in Feynman integrals. Consider an n -th order diagram (n vertices) with N_e external electron lines and N_γ external photon lines.

The number N_e is always even; the electron lines in the diagram form $N_e/2$ continuous sequences, each beginning and ending at an external end. The number of internal electron lines in each such sequence is one less than the number of vertices lying on it; therefore the total number of internal electron lines in the diagram equals $n - N_e/2$.

Each internal photon line connects two vertices; their total number is $(n - N_\gamma)/2$.

Each internal photon line carries a propagator $D(k) \sim k^{-2}$. Each internal electron line carries a propagator $G(p) \sim p^{-1}$. The total power of 4-momenta in the denominator of the integrand is thus

$$2 \frac{n - N_\gamma}{2} + \frac{n - N_e/2}{1} = 2n - \frac{1}{2}N_e - N_\gamma.$$

The number of independent integrations (over d^4p or d^4k) equals the total number of internal lines minus the $n - 1$ additional constraints imposed by 4-momentum conservation

at the vertices (one conservation law relates external momenta and does not constrain internal ones). Thus the total number of integrations over all components of 4-momenta is

$$4 \left[\frac{n - N_\gamma}{2} + n - \frac{N_e}{2} - (n - 1) \right] = 2(n - N_e - N_\gamma + 2).$$

The degree of divergence r of the diagram is the difference between this number and the power of momenta in the denominator:

$$r = 4 - \frac{3}{2} N_e - N_\gamma. \quad (\text{XI.166})$$

Remarkably, this number does not depend on the order n of the diagram.

The condition $r < 0$ for the diagram as a whole is, however, insufficient for convergence of the integral: every internal block of the diagram must likewise have a negative degree of divergence r' . The presence of blocks with $r' > 0$ can increase the degree of divergence of the diagram as a whole.

Since N_e and N_γ are positive integers, only a few pairs of values (N_e, N_γ) give $r \geq 0$. Let us enumerate these cases, immediately excluding $N_e = N_\gamma = 0$ (vacuum closed loops, devoid of physical meaning) and $N_e = 0, N_\gamma = 1$ (a “vacuum current,” to be discarded according to § 103). The remaining cases are:

$$(\text{XI.167})$$

In the first case the divergence is quadratic; in all the others ($r = 0$ or $r = 1$) it is logarithmic.

Diagram d) is the first correction to the vertex operator. It must satisfy condition (XI.150), which we write in the form

$$\bar{u}(p) \Lambda^\mu(p, p; 0) u(p) = 0, \quad p^2 = m^2, \quad (\text{XI.168})$$

where

$$\Lambda^\mu = \Gamma^\mu - \gamma^\mu. \quad (\text{XI.169})$$

Denote the Feynman integral written directly from the diagram rules by $\bar{\Lambda}^\mu(p_2, p_1; k)$. This integral diverges logarithmically and does not by itself satisfy (XI.168). However, a quantity satisfying this condition is obtained by forming the difference

$$\Lambda^\mu(p_2, p_1; k) = \bar{\Lambda}^\mu(p_2, p_1; k) - \bar{\Lambda}^\mu(p_1, p_1; 0) \Big|_{p_1^2 = m^2}. \quad (\text{XI.170})$$

The leading divergent part of the integral $\bar{\Lambda}^\mu$ is obtained by regarding the virtual photon 4-momentum f as arbitrarily large in the integrand. It has the form²⁶

$$-4\pi i e^2 \int \frac{\gamma^\nu(\gamma f) \gamma^\mu(\gamma f) \gamma_\nu}{f^2 \cdot f^2 \cdot f^2} \frac{d^4 f}{(2\pi)^4}$$

and does not depend on the values of the 4-momenta of the external lines. Therefore in the difference (XI.170) the divergence cancels and a finite quantity is obtained. This

²⁶The full expression of the integral is given in § 117 (see (117.2)).

operation of eliminating the divergence by subtraction is called *regularisation* of the integral.

Note that regularisability by a single subtraction is ensured by the fact that in this case the divergence is only logarithmic—the weakest possible. If the integral contained divergences of various orders, one subtraction might be insufficient.

Having determined the first correction Γ^μ (the first term of the expansion of Λ^μ), the first correction to the electron propagator (diagram (XI.167,c)) can be computed using the Ward identity (XI.97), which may also be written as

$$\frac{\partial \mathcal{M}(p)}{\partial p_\mu} = -\Lambda^\mu(p, p; 0), \quad (\text{XI.171})$$

introducing the mass operator $\mathcal{M}$ in place of $\mathcal{G}$ and Λ^μ in place of Γ^μ . This equation must be integrated with the boundary condition

$$\bar{u}(p) \mathcal{M}(p) u(p) = 0, \quad p^2 = m^2, \quad (\text{XI.172})$$

which follows from (110.20).

Finally, for the first term in the expansion of the polarisation operator, we use identity (108.14); after contraction over pairs of indices it gives

$$\frac{3}{4\pi} \frac{\partial^2 \mathcal{P}}{\partial k_\rho \partial k^\sigma} = 2\mathcal{G},$$

relating the scalar functions $\mathcal{P} = \frac{1}{3} \mathcal{P}^\mu{}_\mu$ and $\mathcal{G} = \mathcal{G}^\mu{}_\rho{}^{\rho\mu}$. Both depend only on the scalar k^2 :

$$2k^2 \mathcal{P}''(k^2) + \mathcal{P}'(k^2) = \frac{4}{3} \mathcal{P}(k^2), \quad (\text{XI.173})$$

where primes denote differentiation with respect to k^2 . Together with condition $\mathcal{P}'(0) = 0$, this also gives

$$\mathcal{P}(0) = 0. \quad (\text{XI.174})$$

In the first approximation, $\mathcal{P}(k^2)$ is determined by diagram (XI.167,e) (with 4-momenta of the ends $k, k, 0, 0$). The corresponding Feynman integral (denote it $\bar{\mathcal{G}}(k^2)$) diverges logarithmically, and its regularisation is accomplished by a single subtraction using condition (XI.174):

$$\mathcal{G}(k^2) = \bar{\mathcal{G}}(k^2) - \bar{\mathcal{G}}(0).$$

Then $\mathcal{P}(k^2)$ is determined as the solution of equation (XI.173) with boundary conditions $\mathcal{P}(0) = 0, \mathcal{P}'(0) = 0$.

In the next approximation, the correction to the vertex operator ($\Lambda_\mu^{(2)}$) is determined by diagrams (XI.73,d–i). The irreducible diagram (XI.73,d) is computed by the same regularisation (XI.170) as in the computation of the first-approximation $\Lambda_\mu^{(1)}$. In the reducible diagrams, the internal self-energy and vertex parts of lower order are replaced by the already-known (regularised) first-approximation quantities ($\mathcal{P}^{(1)}, \mathcal{M}^{(1)}, \Lambda_\mu^{(1)}$), after which the integrals are once again regularised using (XI.170).²⁷

²⁷In diagrams of still higher approximations it may also be necessary to replace “four-tails” by the regularised values of $\mathcal{G}$.

The systematic procedure described gives, in principle, finite expressions for $\mathcal{P}$, $\mathcal{M}$, and Λ_μ in any approximation of perturbation theory. Thereby it becomes possible to compute the amplitudes of physical scattering processes, whose diagrams contain the blocks $\mathcal{P}$, $\mathcal{M}$, Λ_μ as constituent parts.

We see that the physical conditions established in §§ 110 and 111 turn out to be sufficient for the unambiguous regularisation of all Feynman integrals. This is by no means a trivial property of quantum electrodynamics, and is called *renormalisability*.²⁸

For practical computation of radiative corrections the described procedure may not be the simplest or most rational. In the next chapter we shall see that it is often more convenient to begin by computing the imaginary parts of the corresponding quantities; these are given by integrals that do not diverge and contain no divergent integrals. The full quantity is then determined by analytic continuation using dispersion relations. This avoids the need for direct regularisation by subtraction.

²⁸Another approach to the theory of renormalisation in quantum electrodynamics is given in: N. N. Bogolyubov, D. V. Shirkov, *Introduction to the Theory of Quantised Fields*, Moscow: Nauka, 1984.

Radiative Corrections

§ 113. Computation of the polarisation operator

Proceeding to the actual computation of radiative corrections, we begin with the computation of the polarisation operator (J. Schwinger, 1949; R. P. Feynman, 1949).

In the first approximation of perturbation theory, it is given by the loop in the diagram

As was already noted, the task is simplified by starting the computation with $\text{Im } \mathcal{P}$. This computation is most conveniently carried out with the help of the unitarity relation. The virtual photon lines are regarded as a “real” particle—a vector boson of mass $M^2 = k^2$, interacting with the electron by the same law as the photon. Thus (113.1) becomes a diagram of a “real” process, which justifies the application of unitarity.

The crosses on diagram (113.1) show where the cut is to be made: the intermediate state contains an electron with 4-momentum $p_- = p$ and a positron with $p_+ = -(p - k)$.

The unitarity condition (71.2) with a two-particle intermediate state (71.4) gives:

$$2 \text{Im } M_{ii} = \frac{|\mathbf{p}|}{(4\pi)^2 \varepsilon} \sum_{\text{pol}} \int |M_{ni}|^2 d\omega. \quad (\text{XII.1})$$

Here the amplitude M_{ii} is constructed from diagram (113.1):

$$iM_{ii} = \sqrt{4\pi} e_\mu^* \cdot \sqrt{4\pi} e_\nu \cdot \frac{i\mathcal{P}^{\mu\nu}}{4\pi}, \quad (\text{XII.2})$$

where e_μ is the 4-vector of boson polarisation; according to (14.13) it satisfies the condition $e_\mu k^\mu = 0$.

The amplitude M_{ni} corresponds to the decay diagram:

$$M_{ni} = -e\sqrt{4\pi} e_\mu j^\mu, \quad j^\mu = \bar{u}(p_-)\gamma^\mu u(-p_+). \quad (\text{XII.3})$$

Substituting (XII.2), (XII.3) into (XII.1):

$$2e_\mu^* e_\nu \text{Im } \mathcal{P}^{\mu\nu} = \frac{e^2 |\mathbf{p}|}{4\pi \varepsilon} \sum_{\text{pol}} \int j^{\mu*} j^\nu e_\mu^* e_\nu d\omega. \quad (\text{XII.4})$$

Here $\mathbf{p} = \mathbf{p}_- = -\mathbf{p}_+$ and $\varepsilon = \varepsilon_- + \varepsilon_+ = 2\varepsilon_+$ are the momenta and total energy of the pair in the centre-of-inertia frame; the integration is over the directions of $\mathbf{p}$, and the summation is over the polarisations of both particles.

We average both sides over the boson polarisations. The averaging is performed by the substitution:

$$e_\mu^* e_\nu \rightarrow -\frac{1}{3} \left(g_{\mu\nu} - \frac{k_\mu k_\nu}{k^2} \right) \quad (\text{XII.5})$$

(cf. (14.5)). Taking into account the transversality of $\mathcal{P}^{\mu\nu}$ and j^μ ($\mathcal{P}^{\mu\nu}k_\nu = 0$, $j^\mu k_\mu = 0$) and using $\mathcal{P}^\mu{}_\mu = 3\mathcal{P}$:

$$2 \operatorname{Im} \mathcal{P} = \frac{1}{12\pi} \frac{|\mathbf{p}|}{\varepsilon} \sum_{\text{pol}} \int (j j^*) d\omega. \quad (\text{XII.6})$$

Summation over polarisations by the usual method, integration over $d\omega$ gives 4π , resulting in:

$$2 \operatorname{Im} \mathcal{P} = e^2 \operatorname{Sp} \gamma_\mu (\gamma p_- + m) \gamma^\mu (\gamma p_+ - m) \cdot \frac{8|\mathbf{p}|}{3\varepsilon} = -e^2 \frac{8|\mathbf{p}|}{3\varepsilon} (p_+ p_- + 2m^2).$$

We introduce the variable:

$$t = k^2 = (p_+ + p_-)^2 = 2(m^2 + p_+ p_-). \quad (\text{XII.7})$$

Then $\varepsilon^2 = t$, $\mathbf{p}^2 = t/4 - m^2$, and the final formula is:

$$\operatorname{Im} \mathcal{P}(t) = -\frac{\alpha}{3} \sqrt{\frac{t - 4m^2}{t}} (t + 2m^2), \quad t \geq 4m^2. \quad (\text{XII.8})$$

The value $t = 4m^2$ is the threshold for production of one electron-positron pair by a virtual photon (cf. footnote on p. 550); in the approximation considered ($\sim e^2$) a state with one pair is the only possible intermediate state in the unitarity condition (XII.1). In the same approximation, for $t < 4m^2$ the right side of (XII.1) is zero:

$$\operatorname{Im} \mathcal{P}(t) = 0, \quad t < 4m^2. \quad (\text{XII.9})$$

For the same reason, in the approximation considered, the cut for $\mathcal{P}(t)$ in the complex t plane extends only from $t = 4m^2$, and this point serves as the lower limit in the dispersion integral (11.13). Thus:

$$\mathcal{P}(t) = -\frac{\alpha}{3\pi} t^2 \int_{4m^2}^{\infty} \frac{dt'}{t'^2 (t' - t - i0)} \sqrt{\frac{t' - 4m^2}{t'}} (t' + 2m^2). \quad (\text{XII.10})$$

To formulate the result it is convenient to introduce, instead of t , another variable, defining it by:

$$\frac{t}{m^2} = -\frac{(1 - \xi)^2}{\xi}. \quad (\text{XII.11})$$

This transformation maps the upper half of the complex t plane onto the upper semicircle of unit radius in the upper half of the complex ξ plane, as shown in Fig. 19 (identically hatched segments correspond to each other in both planes). The non-physical region ($0 \leq t/m^2 \leq 4$) corresponds to the arc, and the physical regions ($t < 0$ and $t/m^2 > 4$) correspond to the right and left real radii.

The integral (XII.10) is most easily evaluated using the substitution $t'/(4m^2) = 1/(1 - x^2)$, considering first the case $t < 0$ (then the denominator does not vanish and the $i0$ can be omitted). Expressed in terms of the variable ξ , the result of integration is:

$$\mathcal{P}(\xi) = \frac{\alpha m^2}{3\pi} \left\{ -\frac{22}{3} + \frac{5}{3} \left(\xi + \frac{1}{\xi} \right) + \left(\xi + \frac{1}{\xi} - 4 \right) \frac{1 + \xi}{1 - \xi} \ln \xi \right\}. \quad (\text{XII.12})$$

The analytic continuation to $t > 4m^2$ is obtained by setting $\xi \rightarrow \xi e^{i\pi}$ (with the logarithm giving $\ln \xi = \ln |\xi| + i\pi$).¹ For the non-physical region one sets $\xi = e^{i\varphi}$:

$$\mathcal{P}(t) = \frac{2\alpha m^2}{3\pi} \left\{ -\frac{10}{3} \sin^2 \frac{\varphi}{2} - 2 + \left(1 + 2 \sin^2 \frac{\varphi}{2} \right) \varphi \cot \frac{\varphi}{2} \right\}, \quad \frac{t}{4m^2} = \sin^2 \frac{\varphi}{2}. \quad (\text{XII.13})$$

In the limit of small $|t|$ ($\xi \rightarrow 1$):

$$\mathcal{P}(t) = -\frac{\alpha}{15\pi} \frac{t}{m^2}, \quad |t| \ll 4m^2. \quad (\text{XII.14})$$

In the opposite limit of large $|t|$ ($\xi \rightarrow 0$):

$$\mathcal{P}(t) = \begin{cases} -\frac{\alpha}{3\pi} |t| \ln \frac{|t|}{m^2}, & -t \gg 4m^2; \\ \frac{\alpha}{3\pi} t \left(\ln \frac{t}{m^2} - i\pi \right), & t \gg 4m^2. \end{cases} \quad (\text{XII.15})$$

By the meaning of perturbation theory, the obtained formulas are valid as long as $\mathcal{P}/(4\pi) \ll D^{-1} = t/(4\pi)$. Therefore the condition for applicability of (XII.15) is:

$$\frac{\alpha}{3\pi} \ln \frac{|t|}{m^2} \ll 1. \quad (\text{XII.16})$$

Radiative corrections containing $\alpha \ln(|t|/m)$ are called *logarithmic*.

§ 114. Radiative corrections to Coulomb's law

Let us investigate, on the basis of the obtained formulas, the radiative corrections to Coulomb's law. These corrections can be described visually as the result of *polarisation of the vacuum* around a point charge.

Without corrections, the field of a stationary centre (with charge e_1) is given by the Coulomb scalar potential $\Phi \equiv A_0^{(e)} = e_1/r$. The components of its three-dimensional Fourier expansion:

$$\Phi(\mathbf{k}) \equiv A_0^{(e)}(\mathbf{k}) = \frac{4\pi e_1}{\mathbf{k}^2}.$$

With the inclusion of radiative corrections, this field is replaced by an "effective field":

$$A_0^{(e)} = A_0^{(e)} + D_{0\rho} \frac{\mathcal{P}^{\rho\lambda}}{4\pi} D A_\lambda^{(e)} = A_0^{(e)} + \frac{1}{4\pi} \mathcal{P} D A_0^{(e)} \quad (\text{XII.17})$$

(cf. (XI.19)). The second term gives the sought correction to the scalar potential. In the first approximation of perturbation theory, for $\mathcal{P}(k^2)$ one must take the expression obtained in the preceding section, and for the function $D(k^2)$ its zeroth approximation:

$$\mathcal{D}(k^2) \approx D(k^2) = -4\pi/k^2.$$

¹The analytic continuation so performed is, as required, continuation to the upper edge of the cut, since the semicircle in the ξ -plane corresponds to the upper half of the t -plane.

Thus, the radiative correction to the potential of the field:

$$\delta\Phi(\mathbf{k}) = -\frac{4\pi e_1}{(\mathbf{k}^2)^2} \mathcal{P}(-\mathbf{k}^2). \quad (\text{XII.18})$$

To determine the form of this correction in coordinate representation, one must carry out the inverse Fourier transform:

$$\delta\Phi(\mathbf{r}) = \int e^{i\mathbf{k}\cdot\mathbf{r}} \delta\Phi(\mathbf{k}) \frac{d^3k}{(2\pi)^3}. \quad (\text{XII.19})$$

Since $\delta\Phi(\mathbf{k})$ is a function only of $t = -\mathbf{k}^2$, performing the integration over angles, we obtain

$$\delta\Phi(r) = \frac{1}{4\pi^2} \int_0^\infty \delta\Phi(t) \frac{\sin(r\sqrt{-t})}{r} d(-t) = \frac{1}{4\pi^2 r} \text{Im} \int_{-\infty}^\infty \delta\Phi(-y^2) e^{iry} y dy$$

$\mathcal{P}(-y^2)$ (Fig. 20). This cut begins at the point $2im$ and runs upward along the imaginary axis (the left bank of the cut corresponding to the physical sheet). Introducing, in place of y , a new variable according to $y = ix$, we find

$$\delta\Phi(r) = \frac{1}{2\pi^2 r} \int_{2m}^\infty \text{Im} \delta\Phi(x^2) e^{-rx} x dx.$$

Finally, returning to integration over $t = x^2$, we have definitively:

$$\delta\Phi(r) = \frac{1}{2\pi^2 r} \int_{4m^2}^\infty \text{Im} \delta\Phi(t) e^{-r\sqrt{t}} dt. \quad (\text{XII.20})$$

The imaginary part

$$\text{Im} \delta\Phi = -\frac{4\pi e}{t^2} \text{Im} \mathcal{P}(t)$$

we take from (XII.7), and after an obvious change of variable we find

$$\Phi(r) + \delta\Phi(r) = \frac{e_1}{r} \left\{ 1 + \frac{2\alpha}{3\pi} \int_1^\infty e^{-2mr\zeta} \left(1 + \frac{1}{2\zeta^2} \right) \frac{\sqrt{\zeta^2 - 1}}{\zeta^2} d\zeta \right\} \quad (\text{XII.21})$$

(*E. Uehling, R. Serber, 1935*).

The integral entering here can be evaluated in two limiting cases.

Consider first small r ($mr \ll 1$). We split the integral from the first term in the round brackets into two:

$$I = \int_1^\infty e^{-2mr\zeta} \frac{\sqrt{\zeta^2 - 1}}{\zeta^2} d\zeta = \int_1^{\zeta_1} \dots d\zeta + \int_{\zeta_1}^\infty \dots d\zeta \equiv I_1 + I_2,$$

where ζ_1 is chosen so that $1/(mr) \gg \zeta_1 \gg 1$. By virtue of this, in the first integral one can set $r = 0$, giving

$$I_1 \approx \int_1^{\zeta_1} \frac{\sqrt{\zeta^2 - 1}}{\zeta^2} d\zeta \approx \ln 2\zeta_1 - 1.$$

In I_2 one can, on the contrary, neglect the unity under the square root:

$$I_2 \approx \int_{\zeta_1}^{\infty} e^{-2mr\zeta} \frac{d\zeta}{\zeta} = -\ln \zeta_1 \cdot e^{-2mr\zeta_1} + 2mr \int_{\zeta_1}^{\infty} e^{-2mr\zeta} \ln \zeta d\zeta.$$

In the exponent and the lower limit of the integral one can set $\zeta_1 = 0$. Making after this the substitution $2mr\zeta = x$, we obtain

$$I_2 = -\ln(2\zeta_1) + \ln \frac{1}{mr} + \int_0^{\infty} e^{-x} \ln x dx = -\ln(2\zeta_1) + \ln \frac{1}{mr} - C,$$

where $C = 0.577 \dots$ is the Euler constant. The integral from the second term in (XII.21) can be immediately evaluated at $r = 0$:

$$I_3 \approx \frac{1}{2} \int_1^{\infty} \frac{\sqrt{\zeta^2 - 1}}{\zeta^4} d\zeta = \frac{1}{6}.$$

Adding all three integrals (the auxiliary quantity ζ_1 cancels), we obtain

$$\Phi(r) = \frac{e_1}{r} \left[1 + \frac{2\alpha}{3\pi} \left(\ln \frac{1}{mr} - C - \frac{5}{6} \right) \right], \quad r \ll \frac{1}{m}. \quad (\text{XII.22})$$

For $mr \gg 1$, the essential region in the integral is $\zeta - 1 \sim 1/(mr) \ll 1$. Substituting $\zeta = 1 + \xi$ and with the corresponding simplifications, it reduces to the integral

$$e^{-2mr} \int_0^{\infty} e^{-2mr\xi} \cdot \frac{3}{2} \sqrt{2\xi} d\xi = \frac{3}{2} \sqrt{2} \cdot \frac{\sqrt{\pi}}{2(2mr)^{3/2}} e^{-2mr} = \frac{3}{8} \frac{\sqrt{\pi}}{(mr)^{3/2}} e^{-2mr}.$$

Thus, in this case²

$$\Phi(r) = \frac{e_1}{r} \left(1 + \frac{\alpha}{4\sqrt{\pi}} \frac{e^{-2mr}}{(mr)^{3/2}} \right), \quad r \gg \frac{1}{m}. \quad (\text{XII.23})$$

We see that vacuum polarisation distorts the Coulomb field of a point charge in the region $r \sim 1/m$ ($= \hbar/(mc)$), where m is the electron mass. Outside this region the distortion of the field falls off exponentially.

Let us make one more remark of a general character. We have been assuming up to now that the radiative corrections arise from the interaction of the photon field with the electron–positron vacuum. But the photon also interacts with the fields of other particles; the interaction with the “vacua” of these fields is described by the same self-energy diagrams, in which the internal electron lines are replaced by lines of the corresponding particles.

²The origin of the factor e^{-2mr} in $\delta\Phi(r)$ is clear already from the form of the original integral (XII.20): for large r the essential values of t in it lie near the lower limit, and the exponent of the exponential factor is determined by the position of the first singularity of the function $\delta\Phi(e)$.

§ 115. Computation of the imaginary part of the polarisation operator from the Feynman integral

By direct computation from the diagram (the loop in diagram (113.1)) the polarisation operator in the first approximation of perturbation theory would be given by the integral

$$i \frac{\mathcal{P}^{\mu\nu}}{4\pi} \rightarrow -e^2 \int \text{Sp } \gamma^\mu G(p) \gamma^\nu G(p-k) \frac{d^4 p}{(2\pi)^4}. \quad (\text{XII.24})$$

However, this integral, taken over all of four-dimensional p -space, diverges quadratically, and to obtain a finite result it must be regularised according to the rules described in § 112.

We shall not reproduce this derivation in full here, but shall show how one can compute from the integral (XII.24) the imaginary part of the polarisation operator (which was determined by us in § 113 using the unitarity condition); this derivation contains a number of instructive points.

The imaginary part of the integral (XII.24) does not contain divergences and therefore does not require regularisation. For the scalar function

$\text{Im } \mathcal{P} = \frac{1}{3} \text{Im } \mathcal{P}_\mu^\mu$ we have

$$\text{Im } \mathcal{P} = \text{Im} \left\{ i \frac{4\pi e^2}{3(2\pi)^4} \int \frac{\text{Sp } \gamma^\mu (\gamma p + m) \gamma_\mu (\gamma p + \gamma k + m)}{(p^2 - m^2 + i0)[(p-k)^2 - m^2 + i0]} d^4 p \right\}.$$

After computing the trace the integral takes the form

$$\begin{aligned} \text{Im } \mathcal{P}(k^2) &= \text{Im} \int \frac{i\varphi(p) d^4 p}{(p^2 - m^2 + i0)[(p-k)^2 - m^2 + i0]}, \\ \varphi(p) &= \frac{2e^2}{3\pi^3} (2m^2 + pk - p^2). \end{aligned} \quad (\text{XII.25})$$

Let $k^2 > 0$. We pass to the rest frame, in which $k = (k_0, 0)$. In this frame

$$(p-k)^2 = (p_0 - k_0)^2 - \mathbf{p}^2.$$

Introducing also the notation $\varepsilon = \sqrt{\mathbf{p}^2 + m^2}$ (ε does not coincide with the “energy” of the virtual electron p_0 !), we rewrite (XII.25) in the form

$$\begin{aligned} \text{Im } \mathcal{P}(k^2) &= \text{Im} \int d^3 p \int_{-\infty}^{\infty} dp_0 \frac{i\varphi(p_0, \mathbf{p})}{(p_0^2 - \varepsilon^2 + i0)[(p_0 - k_0)^2 - \varepsilon^2 + i0]}, \\ \varphi(p_0, \mathbf{p}) &= \frac{2e^2}{3\pi^3} (m^2 + \varepsilon^2 + p_0 k_0 - p_0^2). \end{aligned} \quad (\text{XII.26})$$

The integrand has four poles with respect to the variable p_0 :

$$\begin{aligned} a) \quad p_0 &= \varepsilon - i0, & a') \quad p_0 &= -\varepsilon - i0, \\ b) \quad p_0 &= k_0 - \varepsilon + i0, & b') \quad p_0 &= k_0 + \varepsilon - i0. \end{aligned}$$

The arrangement of these poles is shown in Fig. 21; for definiteness we shall assume that $k_0 > 0$ (the final answer is a function of k_0^2 and does not depend on the sign of k_0).

We compute the jump of the function $\mathcal{P}(t)$, which it undergoes on the cut in the plane of the complex variable $t = k^2 = k_0^2$, or, what is the same thing, on the real axis in the plane of complex k_0 . The real part of the function $\mathcal{P}(t)$ is continuous on the cut, so that the jump

$$\Delta\mathcal{P}(t) = 2i \operatorname{Im} \mathcal{P}(t). \quad (\text{XII.27})$$

First of all we shall show how one can already determine the position of the cut from the form of the integral. We denote in (XII.26) the integral over p_0 as $I(\mathbf{p}, k_0)$. As long as the upper and lower

poles in Fig. 21 are at finite distances from each other, the path of integration over p_0 can be moved away from the poles (dashed line in the figure). In this case the integral $I(\mathbf{p}, k_0)$ does not change upon an infinitesimally small displacement of the poles b and b' downward and upward from the real axis, i.e. upon the replacement $k_0 \rightarrow k_0 + i\delta$, $\delta \rightarrow 0$. In other words, the values $I(\mathbf{p}, k_0)$ as k_0 approaches its real value from above and from below will be equal, so that $I(\mathbf{p}, k_0)$ will not contribute to the jump $\Delta\mathcal{P}$. The situation will change only if two poles (when $k_0 > 0$ these can only be the poles a and b) end up one directly below the other, so that the integration contour will be “pinched” between them and cannot be moved away. Thus, the jump $\Delta\mathcal{P} \neq 0$ only if somewhere in the region of integration over d^3p the condition $k_0 - \varepsilon = \varepsilon$ is fulfilled, i.e. $k_0 = 2\varepsilon = 2\sqrt{\mathbf{p}^2 + m^2}$. For this, obviously, it is necessary that $k_0 \geq 2m$, i.e. $t \geq 4m^2$.³

We rewrite the integral $I(\mathbf{p}, k_0)$ in the form

$$I(\mathbf{p}, k_0) = \int_G \frac{i\varphi(p_0, \mathbf{p}) dp_0}{(p_0^2 - \varepsilon^2)[(p_0 - k_0)^2 - \varepsilon^2]}, \quad (\text{XII.28})$$

omitting the $i0$ terms in the denominator and correspondingly changing the contour C of integration, as shown in Fig. 22. We see that the emergence of the jump $\Delta\mathcal{P}(t)$ is connected with the impossibility of removing

the contour from pole a (when the contour is pinched between a and b). With this in mind, we replace the contour C by a contour C' passing below point a , adding correspondingly the integral over a small circle C'' around this point. After this the contour C' can be freely moved away from the poles, so that integration along it contributes only to the regular part of the function $\mathcal{P}(t)$. For the determination of the desired jump it therefore suffices to consider only the integral around the circle C'' , which reduces to taking the residue at pole a . This operation can be effected by the replacement in the integrand:

$$\frac{1}{p_0^2 - \varepsilon^2} \rightarrow -2\pi i \delta(p_0^2 - \varepsilon^2) \quad (\text{XII.29})$$

(the sign “ $-$ ” is connected with the fact that the circle around the pole is traversed in the negative direction). In doing so one must take into account in the argument of

³One similarly verifies the absence of the cut when $t = k^2 < 0$. Choosing in this case the rest frame in which $\mathbf{k} = (0, \mathbf{k})$, we find that the poles of the integrand lie at

$$p_0 = \pm(\varepsilon - i0), \quad p_0 = \pm(\sqrt{(\mathbf{p} - \mathbf{k})^2 + m^2} - i0).$$

Both lower poles always lie in the right half-plane, and both upper ones in the left half-plane of p_0 , so that no pair of them can end up one below the other.

the δ -function only the root $p_0 = +\varepsilon$ (the pole a , but not a'); this condition will be automatically satisfied if one stipulates that the integration is to be carried out only over the half of the impulse 4-space with $p_0 > 0$.

After the replacement (XII.29) the jump of the integral $I(\mathbf{p}, k_0)$ is computed directly:

$$\Delta I = \{I(\mathbf{p}, k_0 + i\delta) - I(\mathbf{p}, k_0 - i\delta)\}_{\delta \rightarrow +0} = -2\pi i \times \int_0^\infty \delta(p_0^2 - \varepsilon^2) i\varphi(p_0, \mathbf{p}) \left[\frac{1}{(k_0 - p_0)^2 - \varepsilon^2 + i\delta} - \frac{1}{(k_0 - p_0)^2 - \varepsilon^2 - i\delta} \right] dp_0.$$

Using the identity

$$\frac{1}{(k_0 - p_0)^2 - \varepsilon^2 \pm i\delta} = P \frac{1}{(k_0 - p_0)^2 - \varepsilon^2} \mp i\pi \delta[(k_0 - p_0)^2 - \varepsilon^2]$$

(cf. (111.3)), we obtain

$$\Delta I = i(2\pi i)^2 \int_0^\infty \delta(p_0^2 - \varepsilon^2) \delta[(k_0 - p_0)^2 - \varepsilon^2] \varphi(p_0, \mathbf{p}) dp_0.$$

The arguments of the δ -functions can be rewritten in invariant form, subtracting and adding to them $\mathbf{p}^2$:

$$p_0^2 - \varepsilon^2 = p^2 - m^2, \quad (k_0 - p_0)^2 - \varepsilon^2 = (k - p)^2 - m^2.$$

After this we find finally

$$\Delta \mathcal{P}(k^2) = i(2\pi i)^2 \int d^4 p \varphi(p) \delta(p^2 - m^2) \delta[(p - k)^2 - m^2]. \quad (\text{XII.30})$$

$$p_0 > 0$$

By virtue of the presence of the δ -functions the integration is in fact carried out only over the region of intersection of the hypersurfaces

$$p^2 = m^2, \quad (p - k)^2 = m^2. \quad (\text{XII.31})$$

Since in this region all 4-vectors p are time-like, the condition of integration over $p_0 > 0$ has an invariant character (the upper sheet of the cone $p^2 = m^2$).

Comparing (XII.30) with the original formula (XII.25), we see that the jump of the function $\mathcal{P}(t)$ on the cut in the t -plane can be obtained if in the original Feynman integral one makes the replacement

$$\frac{1}{p^2 - m^2 + i0} \rightarrow -2\pi i \delta(p^2 - m^2) \quad (\text{XII.32})$$

in the propagators corresponding to the lines cut by the diagram (113.1) (*S. Mandelstam*, 1958, *R. Cutkosky*, 1960).

Let us note that the conditions (XII.31) single out that region of impulse space in which the lines of virtual particles in the diagram correspond to real particles (or, as one says, the 4-momenta p and $p - k$ lie on the *mass shell*). Here the connection with the

method of the unitarity relation, in which these same lines were replaced by lines of real particles of the intermediate state, is clearly visible.

We also see the mathematical reason for the absence of the divergence in the imaginary part of the diagram: it is determined by integration over the finite region of the mass surface instead of integration over all of unbounded impulse 4-space in the original Feynman integral.

To obtain now from (XII.30) the formula derived in § 113, we return to the rest frame, in which $\mathbf{k} = 0$, and carry out the integration over

$$d^4p = |\mathbf{p}| \varepsilon d\varepsilon dp_0 do.$$

The integration reduces to the removal of the δ -functions. In doing so

$$\delta(p^2 - m^2) dp_0 = \delta(p_0^2 - \varepsilon^2) dp_0 \rightarrow \frac{1}{2\varepsilon} \delta(p_0 - \varepsilon) dp_0$$

and then

$$\begin{aligned} \delta[(p - k)^2 - m^2] |d\varepsilon| &= \delta[(p_0 - k_0)^2 - \varepsilon^2] |d\varepsilon| = \\ &= \delta(-2\varepsilon k_0 + k_0^2) d\varepsilon \rightarrow \frac{1}{2k_0} \delta\left(\varepsilon - \frac{k_0}{2}\right) d\varepsilon. \end{aligned}$$

As a result we obtain

$$\Delta\mathcal{P}(t) = -\frac{i\pi^2}{2} \sqrt{\frac{t - 4m^2}{t}} \int \varphi(\varepsilon, \mathbf{p}) do, \quad (\text{XII.33})$$

where $t = k^2 = k_0^2$, and the value of the function φ is taken at

$$p_0 = \varepsilon = k_0/2, \quad \mathbf{p}^2 = \varepsilon^2 - m^2 = k_0^2/4 - m^2,$$

i.e. is equal to

$$\varphi(\varepsilon, \mathbf{p}) = \frac{e^2}{3\pi^3} (2m^2 + t)$$

and does not depend on the angle. Therefore the integration over do reduces to multiplication by 4π , and we return to (113.8).

In the derivation presented only the fact that the diagram is cut into two parts by crossing just two lines is essential. Therefore the rule formulated remains valid also for diagrams composed of any two blocks connected by two (electron or photon) lines. The integral, computed by means of the replacement (XII.32), determines in this case that contribution to the imaginary part of the diagram which, in the method of the unitarity relation, is connected with the corresponding two-particle intermediate state.

§ 116. Electromagnetic form factors of the electron

In the derivation presented, only the fact is essential that the diagram is cut into two parts by the intersection of just two lines. Therefore the formulated rule remains valid also for diagrams composed of any two blocks connected by two (electron or photon) lines.

The integral, computed by substitution (115.9), then determines the contribution to the imaginary part of the diagram which, in the unitarity-relation method, is associated with the corresponding two-particle intermediate state.

Consider the vertex operator $\Gamma^\mu = \Gamma^\mu(p_2, p_1; k)$ in the case when the two electron lines are external and the photon line is internal. The electron external lines correspond to the spinor factors $u_1 = u(p_1)$ and $\bar{u}_2 = \bar{u}(p_2)$, so that Γ enters the expression for the diagram in the form of the product

$$j_{fi}^\mu = \bar{u}_2 \Gamma^\mu u_1. \quad (\text{XII.34})$$

As was already noted in § 111, this quantity represents the electron transition current with the inclusion of radiative corrections. The requirements of relativistic and gauge invariance make it possible to establish the general form of the matrix structure of this current.

The operator of electromagnetic interaction $\hat{V} = e(\hat{j}\hat{A})$ is a true scalar (and not a pseudoscalar), which expresses the conservation of spatial parity in these interactions. Therefore the transition current j_{fi} is a true 4-vector (and not a pseudovector). It can consequently be expressed only through true 4-vectors, constructed from the available two 4-momenta p_1 and p_2 (the third $k = p_2 - p_1$) and the bispinors u_1 and u_2 . Such independent 4-vectors, bilinear in $\bar{u}_2$ and u_1 , are three in total:

$$\bar{u}_2 \gamma u_1, \quad (\bar{u}_2 u_1) p_1, \quad (\bar{u}_2 u_1) p_2,$$

or, equivalently,

$$\bar{u}_2 \gamma^\mu u_1, \quad (\bar{u}_2 u_1) P^\mu, \quad (\bar{u}_2 u_1) k^\mu, \quad (\text{XII.35})$$

where $P = p_1 + p_2$. But the condition of gauge invariance requires transversality of the transition current with respect to the 4-momentum of the photon k :

$$j_{fi} k = 0. \quad (\text{XII.36})$$

This condition is satisfied by the first two of the 4-vectors (XII.35): the first by virtue of the Dirac equations

$$(\gamma p_1 - m)u_1 = 0, \quad \bar{u}_2(\gamma p_2 - m) = 0, \quad (\text{XII.37})$$

and the second because $Pk = 0$. The current j_{fi} is represented by a linear combination of these two 4-vectors:

$$j_{fi}^\mu = f_1(\bar{u}_2 u_1) P^\mu + f_2(\bar{u}_2 \gamma^\mu u_1),$$

where f_1, f_2 are invariant functions; they are called the *electromagnetic form factors* of the electron.

Since the 4-momenta p_1 and p_2 refer to a free electron, $p_1^2 = p_2^2 = m^2$, and from the three 4-vectors p_1, p_2, k (related by $k = p_2 - p_1$) one can form only one independent scalar variable, which we choose to be k^2 . The form factors are then functions of k^2 .

The expression for the current can also be presented in other forms, with a different choice of two independent terms. Using equations (XII.37) and the commutation rules of the matrices γ , it is easy to verify that

$$(\bar{u}_2 \sigma^{\mu\nu} u_1) k_\nu = -2m(\bar{u}_2 \gamma^\mu u_1) + (\bar{u}_2 u_1) P^\mu, \quad (\text{XII.38})$$

where $\sigma^{\mu\nu} = \frac{1}{2}(\gamma^\mu \gamma^\nu - \gamma^\nu \gamma^\mu)$. The coefficient of this term has, as we shall see, an important physical meaning, so we shall write

$$\Gamma^\mu = \gamma^\mu f(k^2) - \frac{1}{2m} g(k^2) \sigma^{\mu\nu} k_\nu, \quad (\text{XII.39})$$

where f, g are two other form factors; the meaning of separating the factor $1/(2m)$ will become clear below⁴. For brevity we write the vertex operator in place of the transition current, it being understood that it should be taken “between the spinors” $\bar{u}_2 \dots u_1$.

To elucidate the properties of the form factors, let us consider the diagram (110.16) of the process of electron interaction with an external field. The corresponding scattering amplitude is

$$M_{fi} = -e j_{fi}^\mu A_\mu^{(e)}(k), \quad (\text{XII.40})$$

where $A^{(e)}$ is the effective (with the inclusion of vacuum polarisation) external field. The amplitude (XII.40) describes two reaction channels. In the scattering channel the invariant variable

$$t = k^2 = (p_2 - p_1)^2 \leq 0.$$

Replacing $p_2 \rightarrow p_-, p_1 \rightarrow -p_+$, we pass to the annihilation channel, corresponding to the production of a pair with 4-momenta p_- and p_+ . In this channel

$$t = (p_- + p_+)^2 \geq 4m^2.$$

The region of values $0 < t < 4m^2$ is unphysical.

Let us turn to the unitarity condition (111.12). In the scattering channel ($t < 0$) there are in the given case no physical intermediate states: a free electron can neither change its momentum nor produce any other particles. There are none, of course, in the unphysical region either. Therefore for $t < 4m^2$ the right-hand side of the equality (111.12) vanishes, so that the matrix T_{fi} (or, equivalently, M_{fi}) is hermitian:

$$M_{fi} = M_{if}^*.$$

Interchanging the initial and final states means interchanging p_2 and p_1 , and thereby the substitution $k \rightarrow -k$. Representing M_{fi} in the form (XII.40), we obtain therefore

$$j_{fi}^\mu A_\mu^{(e)}(k) = j_{if}^{\mu*} A_\mu^{(e)*}(-k).$$

But $A^{(e)}(-k) = A^{(e)*}(k)$, so that it follows that the matrix of transition currents is also hermitian:

$$j_{fi} = j_{if}^* \quad \text{for } t < 4m^2. \quad (\text{XII.41})$$

⁴To avoid misunderstanding we recall: in the definition (XII.39) it is assumed that the 4-momentum k enters the vertex along the photon line; for an outgoing line the sign of the second term would be reversed.

Using the properties of the matrices γ (21.7), it is easy to verify that

$$(\bar{u}_2 \gamma^\mu u_1) = (\bar{u}_1 \gamma^\mu u_2)^*, \quad (\bar{u}_2 \sigma^{\mu\nu} u_1) = -(\bar{u}_1 \sigma^{\mu\nu} u_2)^*.$$

Therefore j_{if}^* differs from j_{fi} only by the replacement of the functions $f(t)$ and $g(t)$ by their complex conjugates. From equality (XII.41) it follows then that these functions are real:

$$\text{Im } f(t) = \text{Im } g(t) = 0, \quad t < 4m^2. \quad (\text{XII.42})$$

In the annihilation channel ($t > 4m^2$) the state is a pair, which can convert into a pair again with the same momenta (elastic scattering) or into some other more complex system. Therefore the right-hand side of the unitarity condition is different from zero, the matrix M_{fi} (and with it j_{fi}) is not hermitian, and therefore the form factors are complex.

The analytic properties of the functions $f(t)$ and $g(t)$ are entirely analogous to those of the function $\mathcal{P}(t)$ considered in § 111 (although it would be more difficult to prove this equally directly). These functions are analytic in the complex t -plane, cut along the positive real axis for $t > 4m^2$, with

$$f^*(t) = f(t^*), \quad g^*(t) = g(t^*).$$

The renormalisation condition (110.19), applied to the vertex operator (XII.39), leads to the requirement

$$f(0) = 1. \quad (\text{XII.43})$$

In order to satisfy this condition automatically (when computing the function $f(t)$ from its imaginary part), one must apply the dispersion relation of the type (111.8) not to the function $f(t)$ itself, but to $(f - 1)/t$. One then obtains a dispersion relation “with one subtraction”:

$$f(t) - 1 = \frac{t}{\pi} \int_{4m^2}^{\infty} \frac{\text{Im } f(t')}{t'(t' - t - i0)} dt'. \quad (\text{XII.44})$$

For the form factor $g(t)$, on the other hand, no values are prescribed in advance by physical requirements. Therefore for it the dispersion relation is written “without subtractions”:

$$g(t) = \frac{1}{\pi} \int_{4m^2}^{\infty} \frac{\text{Im } g(t')}{t' - t - i0} dt'. \quad (\text{XII.45})$$

The value $g(0)$ has an important physical meaning: it gives a correction to the magnetic moment of the electron. To see this, let us consider the scattering of a non-relativistic electron in a constant, slowly varying magnetic field in space.

The term in the scattering amplitude (XII.40) associated with the form factor $g(k^2)$ has the form

$$\delta M_{fi} = \frac{e}{2m} g(k^2) (\bar{u}_2 \sigma^{\mu\nu} u_1) k_\nu A_\mu^{(e)}(k). \quad (\text{XII.46})$$

For a purely magnetic field $A^{(e)\mu} = (0, \mathbf{A})$; constancy of the field in time means that the 4-vector $k^\mu = (0, \mathbf{k})$, and slow variation of the field in space corresponds to small $\mathbf{k}$ (having in mind the further passage to the limit $\mathbf{k} \rightarrow 0$, we write at once in (XII.46) $A^{(e)}$)

in place of the effective $A^{(e)}$. Expanding the expression (XII.46) and expressing it in terms of three-dimensional quantities, we obtain

$$\delta M_{fi} = \frac{e}{2m} g(-\mathbf{k}^2) (\bar{u}_2 \Sigma u_1) i[\mathbf{k} \mathbf{A}_{\mathbf{k}}],$$

where Σ is the matrix (21.21). We replace the product $i[\mathbf{k} \mathbf{A}_{\mathbf{k}}]$ by the magnetic field intensity $\mathbf{H}_{\mathbf{k}}$, after which one can pass to the limit $\mathbf{k} \rightarrow 0$. Finally, introducing the non-relativistic spinor amplitudes w_1, w_2 , according to (23.12),

$$\bar{u}_2 = \sqrt{2m} (w_2^* 0), \quad u_1 = \sqrt{2m} \begin{pmatrix} w_1 \\ 0 \end{pmatrix},$$

we find ultimately

$$\delta M_{fi} = \frac{e}{2m} g(0) \mathbf{H}_{\mathbf{k}} \cdot 2m (w_2^* \boldsymbol{\sigma} w_1). \quad (\text{XII.47})$$

Comparing this expression with the amplitude of scattering in a constant electric field with scalar potential $\Phi_{\mathbf{k}}$:

$$M_{fi} = -e(\bar{u}_2 \gamma^0 u_1) \Phi_{\mathbf{k}} \approx -e \Phi_{\mathbf{k}} \cdot 2m (w_2^* w_1).$$

We see that to the electron in a magnetic field one can ascribe an additional potential energy

$$-\frac{e}{2m} g(0) \boldsymbol{\sigma} \mathbf{H}_{\mathbf{k}}.$$

This means that the electron possesses an “anomalous” magnetic moment

$$\mu' = \frac{e\hbar}{2mc} g(0) \quad (\text{XII.48})$$

(ordinary units) in addition to the “normal” Dirac magnetic moment $e\hbar/(2mc)$.

§ 117. Computation of the electron form factors

Let us turn to the actual computation of the electron form factors (*J. Schwinger*, 1949).

In the zeroth approximation of perturbation theory, the vertex operator $\Gamma^\mu = \gamma^\mu$, i.e. the electron form factors are

$$f = 1, \quad g = 0.$$

The first radiative correction to the form factors is determined by the vertex diagram

$$(\text{XII.49})$$

(with two real electron ends and one virtual photon end). We begin with the computation of the imaginary parts of the form factors. As was shown in the preceding section, they differ from zero only in the annihilation channel ($k^2 > 4m^2$); in accordance with this, the 4-momenta of the electron ends in diagram (117.1) correspond to the electron and positron being produced, and are denoted by p_- and $-p_+$. The analytic expression for diagram (117.1):

$$-ie\bar{u}(p_-)\Gamma^\mu u(-p_+) = (-ie)^3 \bar{u}(p_-)\gamma^\nu i \int G(p)\gamma^\mu G(p-k)\gamma^\lambda D_{\lambda\nu}(f) \frac{d^4 p}{(2\pi)^4}, \quad (\text{XII.50})$$

or, in expanded form,

$$\gamma^\mu f(k^2) - \frac{1}{2m} \sigma^{\mu\nu} k_\nu = \frac{i\varphi^\mu(p) d^4p}{(p^2 - m^2)[(p - k)^2 - m^2]}, \quad (\text{XII.51})$$

where the notation is

$$\varphi^\mu(p) = -e^2 \frac{\gamma^\nu (\gamma p + m) \gamma^\mu (\gamma p - \gamma k + m) \gamma_\nu}{4\pi^3 (p_- - p)^2} \quad (\text{XII.52})$$

and for brevity the factors $\bar{u}(p_-) \dots u(-p_+)$ are omitted; everywhere below it is understood that both sides of the equations are taken between these “covers”.

The horizontal dashed line drawn across diagram (117.1) divides it into two parts, in order to display the intermediate state which would appear in the computation of the imaginary part of the form factor from the unitarity condition: this is the state of an electron–positron pair with momenta distinct from p_- , p_+ . This same division shows that in the integral (XII.50) the replacement of the pole factors should have been carried out, if in (XII.51) these factors are separated out in the subintegral expression, the computation is to be performed by the rule (115.9).

The integral in (XII.51) is of the same form as in (115.2). Therefore we can write the result directly in the form (115.10), bypassing the intermediate steps:

$$2\gamma^\mu \text{Im } f(t) - \frac{1}{2m} \sigma^{\mu\nu} k_\nu \text{Im } g(t) = -\frac{\pi^2}{2} \sqrt{\frac{t - 4m^2}{t}} \int \varphi^\mu(p) d\mathbf{p}, \quad (\text{XII.53})$$

where $t = k^2$, the integration is performed over the direction of the vector $\mathbf{p}$, and the 4-vectors $p'_- \equiv p$ and $p'_+ \equiv k - p$ in the definition of the function $\varphi^\mu(p)$ (cf. (XII.52)) become 4-momenta of real (and not virtual) particles. Expression (XII.53) refers to the centre-of-mass system, in which $\mathbf{k} = 0$; this is the system of the centre of inertia of the pair being produced, p_- , p_+ (and at the same time the “intermediate” pair p'_- , p'_+). In this system, therefore,

$$k = (k_0, 0), \quad p_- = \left(\frac{k_0}{2}, \mathbf{p}_-\right), \quad p_+ = \left(\frac{k_0}{2}, -\mathbf{p}_-\right), \quad p = \left(\frac{k_0}{2}, \mathbf{p}\right),$$

and it is easy to verify that

$$f^2 = (p - p_-)^2 = -2\mathbf{p}^2(1 - \cos \theta) = -\frac{t - 4m^2}{2}(1 - \cos \theta), \quad (\text{XII.54})$$

where θ is the angle between $\mathbf{p}$ and $\mathbf{p}_-$ (and $\mathbf{p}^2 = \mathbf{p}_-^2$). Substituting now (XII.52) into (XII.53) and eliminating in the subintegral expression the matrices $\gamma^\nu \dots \gamma_\nu$ with the aid of formulas (22.6), we obtain

$$\begin{aligned} \gamma^\mu \text{Im } f(t) - \frac{1}{2m} \sigma^{\mu\nu} k_\nu \text{Im } g(t) &= \\ &= -\frac{e^2}{4\sqrt{t(t - 4m^2)}} \int \frac{d\mathbf{p}}{2\pi(1 - \cos \theta)} \gamma^\nu (\gamma p + m) \gamma^\mu (\gamma p - \gamma k + m) \gamma_\nu \\ &= -\frac{e^2}{4\sqrt{t(t - 4m^2)}} \int \frac{d\mathbf{p}}{2\pi(1 - \cos \theta)} \left[-2m^2 \gamma^{\mu(\text{XII.55})} + 4m(p^\mu + 2f^\mu) + \right. \\ &\quad \left. + 2(\gamma p_+ - \gamma f) \gamma^\mu (\gamma p_- + \gamma f) \right], \end{aligned}$$

where the 4-vectors

$$f = p - p_- = (0, \mathbf{f}), \quad P = p_- - p_+ = (0, 2\mathbf{p}_-) \quad (\text{XII.56})$$

have been introduced. The integration now reduces to the computation of the integrals

$$(I, I^\mu, I^{\mu\nu}) = \int \frac{(1, f^\mu, f^\mu f^\nu)}{1 - \cos \theta} \frac{d\omega}{2\pi} \quad (\text{XII.57})$$

with each of the three listed numerators.

The integral I diverges logarithmically as $\theta \rightarrow 0$. Rewriting it as

$$I = \int_0^{t-4m^2} \frac{d(\mathbf{f}^2)}{\mathbf{f}^2} = \int_0^{-(t-4m^2)} \frac{d(f^2)}{f^2},$$

we see that the divergence corresponds to small “masses” of the virtual photon. Thus, this is an *infrared* divergence. We shall defer its detailed discussion to § 122. Here we note only that it is fictitious, in the sense that upon a proper accounting for all physical effects such divergences cancel out and disappear. Therefore we may in an arbitrary manner “cut off” the integral from below, and in what follows, when computing real physical phenomena, take the cut-off limit to zero.

Here it will be simplest to perform the cut-off in a relativistically invariant manner. To do this we ascribe to the virtual photon f a small but finite mass λ ($\lambda \ll m$), i.e. we replace in the photon propagator $D(f^2)$ in (XII.50)

$$f^2 \rightarrow f^2 - \lambda^2. \quad (\text{XII.58})$$

After this

$$I = \int_0^{-(t-4m^2)} \frac{d(f^2)}{f^2 - \lambda^2} = \ln \frac{t - 4m^2}{\lambda^2}. \quad (\text{XII.59})$$

The integral I^μ , in which f^μ is a spacelike 4-vector, must be expressed through the 4-vector P^μ (of the two 4-vectors at our disposal, P^μ and k^μ , only P^μ is spacelike for arbitrary p_+ , p_-). Therefore $I^\mu = AP^\mu$. Multiplying this equation by P_μ and computing the integral $P_\mu I^\mu$ in the centre-of-inertia system of the pair (the components of the 4-vectors f and P are given by (XII.56)), we find

$$A = \frac{1}{2\mathbf{p}^2} \int \frac{\mathbf{f} \cdot d \cos \theta}{1 - \cos \theta} = -\frac{1}{2} \int_{-1}^1 d \cos \theta = -1.$$

Thus,

$$I^\mu = -P^\mu. \quad (\text{XII.60})$$

In an analogous manner the integral $I^{\mu\nu}$ is computed:

$$I^{\mu\nu} = \frac{1}{4} P^2 \left(g^{\mu\nu} - \frac{P^\mu P^\nu}{P^2} \right) + \frac{1}{4} P^\mu P^\nu \quad (\text{XII.61})$$

(to determine the coefficients in this expression it is sufficient to compute the integrals $I^\mu{}_\mu$ and $I^{\mu\nu}P_\mu P_\nu$).

The further computation proceeds as follows. Substituting (XII.59)–(XII.61) into (XII.55), we obtain between the “covers” $\bar{u}(p_-) \dots u(-p_+)$ a sum of a series of terms. In each of them we “push through” (using the commutation rules for the matrix γ^μ) the factor γp_+ to the right and γp_- to the left; after this one can make the replacements $\gamma p_- \rightarrow m$, $\gamma p_+ \rightarrow -m$, since

$$\bar{u}(p_-)\gamma p_- = m\bar{u}(p_-), \quad \gamma p_+u(-p_+) = -mu(-p_+).$$

In the resulting sum

$$-4(p_+p_-)I\gamma^\mu + 2mP^\mu - 3P^2\gamma^\mu$$

one can further replace P^μ by the expression equivalent to it (between the covers!)

$$P^\mu \rightarrow 2m\gamma^\mu + \sigma^{\mu\nu}k_\nu$$

(cf. (116.5)). Finally, expressing all quantities through the invariant $t = k^2$ ($2p_+p_- = t - 2m^2$, $P^2 = 4m^2 - t$) and comparing both sides of equation (XII.55), we obtain the following formulas for the imaginary parts of the form factors:

$$\text{Im } g(t) = \frac{\alpha m^2}{\sqrt{t(t - 4m^2)}}, \quad (\text{XII.62})$$

$$\text{Im } f(t) = \frac{\alpha}{4\sqrt{t(t - 4m^2)}} \left[-3t + 8m^2 + 2(t - 2m^2) \ln \frac{t - 4m^2}{\lambda^2} \right]. \quad (\text{XII.63})$$

The infrared divergence is present only in $\text{Im } f(t)$.

The functions $f(t)$ and $g(t)$ themselves are computed from their imaginary parts with the help of formulas (116.11), (116.12). The integration in these formulas is conveniently performed using the same substitutions that were used in § 113 in computing $\mathcal{P}(t)$. Expressed through the variable ξ (113.11), the form factors are given by the formulas

$$g(\xi) = \frac{\alpha}{\pi} \frac{\xi \ln \xi}{\xi^2 - 1}, \quad (\text{XII.64})$$

$$\begin{aligned} f(\xi) - 1 = \frac{\alpha}{2\pi} \left\{ 2 \left(1 + \frac{1 + \xi^2}{1 - \xi^2} \ln \xi \right) \ln \frac{m}{\lambda} - \frac{3(1 + \xi^2) + 2\xi}{2(1 - \xi^2)} \ln \xi + \right. \\ \left. + \frac{1 + \xi^2}{1 - \xi^2} \left[\frac{\pi^2}{6} - \frac{1}{2} \ln^2 \xi - 2F(\xi) + 2 \ln \xi \ln(1 + \xi) \right] \right\}, \end{aligned} \quad (\text{XII.65})$$

where $F(\xi)$ is the Spence function, defined according to (131.19).

In the non-physical region ($0 < t/m^2 < 4$) one must set $\xi = e^{i\varphi}$. Then the expressions for the form factors can be brought to the form

$$f(\varphi) - 1 = \frac{\alpha}{\pi} \left\{ \left(1 - \frac{\varphi}{\tan \varphi} \right) \ln \frac{m}{\lambda} + \frac{3 \cos \varphi + 1}{2 \sin \varphi} \varphi + \frac{2}{\tan \varphi} \int_0^{\varphi/2} x \tan x \, dx \right\}, \quad (\text{XII.66})$$

$$g(\varphi) = \frac{\alpha}{2\pi} \frac{\varphi}{\sin \varphi}. \quad (\text{XII.67})$$

Finally, let us write down the limiting formulas for small $|t|$:

$$f(t) - 1 = \frac{\alpha t}{3\pi m^2} \left(\ln \frac{m}{\lambda} - \frac{3}{8} \right), \quad g(t) = \frac{\alpha}{2\pi}, \quad |t| \ll 4m^2, \quad (\text{XII.68})$$

and for large $|t|$:

$$f(t) - 1 = -\frac{\alpha}{\pi} \left(\frac{1}{2} \ln^2 \frac{|t|}{m^2} + 2 \ln \frac{m}{\lambda} \ln \frac{|t|}{m^2} \right) + \begin{cases} i \frac{\alpha}{2} \ln \frac{t}{\lambda^2}, & t \gg 4m^2, \\ 0, & -t \gg 4m^2, \end{cases} \quad (\text{XII.69})$$

$$g(t) = -\frac{\alpha m^2}{\pi t} \ln \frac{|t|}{m^2} + \begin{cases} i \frac{\alpha m^2}{t}, & t \gg 4m^2, \\ 0, & -t \gg 4m^2. \end{cases} \quad (\text{XII.70})$$

Formula (XII.69) is valid (with respect to $\text{Re } f$), as one says, with double logarithmic accuracy, i.e. to within the squares of large logarithms.⁵

§ 118. Anomalous magnetic moment of the electron

As was already indicated in § 116, the value $g(0)$ determines the radiative correction to the magnetic moment of the electron. If one sets as the goal the computation of only this quantity, then the computation of the entire function $g(t)$ is, of course, not obligatory. With the help of (117.14) and (116.12) we have

$$g(0) = \frac{1}{\pi} \int_{4m^2}^{\infty} \frac{\text{Im } g(t')}{t'} dt' = \frac{\alpha}{4\pi} \int_1^{\infty} \frac{dx}{x^{3/2} \sqrt{x-1}} = \frac{\alpha}{2\pi}. \quad (\text{XII.71})$$

Taking into account this correction, the magnetic moment of the electron is

$$\mu = \frac{e\hbar}{2mc} \left(1 + \frac{\alpha}{2\pi} \right). \quad (\text{XII.72})$$

This formula was first obtained by *Schwinger* (1949).

In the next approximation ($\sim \alpha^2$) the radiative corrections in form-factor diagrams are represented by seven diagrams (106.10, $a-i$). The determination of even just the value $g(0)$ in this approximation requires very complex computations. Referring for the details of the computations to the original articles, we give only the final value of the second-approximation correction⁶:

$$g^{(2)}(0) = \left(\frac{\alpha}{\pi} \right)^2 \left(\frac{197}{144} + \frac{\pi^2}{12} - \frac{\pi^2}{2} \ln 2 + \frac{3}{4} \zeta(3) \right) = -0.328 \frac{\alpha^2}{\pi^2}, \quad (\text{XII.73})$$

⁵The expression for the vertex operator in the case of one virtual and one real electron end and a real photon end, see: *A. I. Akhiezer, V. B. Berestetskii*, Quantum Electrodynamics. Moscow: Nauka, 1981. P. 5.1.3. P. 330.

⁶The computation by the unitarity method was carried out by *M. V. Terent'ev*, ZhETF **43**, 619 (1962).

so that the magnetic moment of the electron is

$$\mu = \frac{e\hbar}{2mc} \left(1 + \frac{\alpha}{2\pi} - 0.328 \frac{\alpha^2}{\pi^2} \right) \quad (\text{XII.74})$$

(C. Sommerfield, 1957; A. Petermann, 1957).

Let us dwell in particular on the contribution of vacuum polarisation to the correction $g^{(2)}(0)$. This is the diagram

$$(\text{XII.75})$$

containing a photon self-energy part. It differs from the diagram (117.1) of the first approximation only in that in place of the photon propagator $D(f^2) = 4\pi/f^2$, there now stands the product

$$D(f^2) \frac{\mathcal{P}(f^2)}{4\pi} D(f^2) = \frac{4\pi}{f^2} \frac{\mathcal{P}(f^2)}{f^2},$$

where $\mathcal{P}(f^2)$ is the polarisation operator computed in § 113 in the first ($\sim \alpha$) approximation. Partially repeating (with this modification) the computations carried out in the preceding section, we obtain for the “polarisation part” correction

$$\text{Im } g_{\text{polar}}^{(2)}(t) = \frac{\alpha m^2}{\sqrt{t(t-4m^2)}} \int_{-1}^1 \frac{\mathcal{P}(f^2)}{f^2} \frac{1+3\cos\theta}{2} d\cos\theta, \quad (\text{XII.76})$$

where

$$f^2 = \frac{t-4m^2}{2} (1-\cos\theta) \quad (\text{XII.77})$$

(see (117.6)). The computation of this integral, and then of the integral

$$g_{\text{polar}}^{(2)}(0) = \frac{1}{\pi} \int_{4m^2}^{\infty} \text{Im } g_{\text{polar}}^{(2)}(t') \frac{dt'}{t'} \quad (\text{XII.78})$$

leads to the value

$$g_{\text{polar}}^{(2)}(0) = \frac{\alpha^2}{\pi^2} \left(\frac{119}{36} - \frac{\pi^2}{3} \right) = 0.016 \frac{\alpha^2}{\pi^2}; \quad (\text{XII.79})$$

it constitutes $\sim 5\%$ of the entire value (118.3).

We have already noted (at the end of § 114) that a definite contribution to the radiative corrections can also be made by effects of vacuum polarisation of other particles. The contribution of muonic vacuum to the anomalous magnetic moment of the electron is obtained by the same formulae (118.6)–(118.8), in which (including in the definition of the variable f^2) m is as before the mass of the electron (m_e), but as the parameter m entering the expression for $\mathcal{P}(f^2)$, the muon mass (m_μ) must be taken. The quantity $\mathcal{P}(f^2)/f^2$ is a function only of the ratio f^2/m_μ^2 . In the integral (118.8) the essential region of values of t (and therefore also f^2) is of order m_e^2 ; so that the ratio $f^2/m_\mu^2 \sim (m_e/m_\mu)^2 \ll 1$, and for the evaluation of the integrals one can use the limiting formula (113.14), according to which

$$\frac{\mathcal{P}(f^2)}{f^2} = -\frac{\alpha}{15\pi} \frac{f^2}{m_\mu^2}.$$

From this it is clear that the contribution to $g^{(2)}(0)$ due to muonic vacuum polarisation has an additional small factor $(m_e/m_\mu)^2$.

The reverse situation arises, however, when finding corrections to the magnetic moment of the muon. Since in (118.3) the mass of the particle does not enter, this value of $g^{(2)}(0)$ pertains also to the muon, with the contribution of muonic vacuum polarisation taken into account. But the contribution of vacuum polarisation of other particles—electrons—turns out in this case to be considerably larger. It is computed by formulae (118.6)–(118.8), in which one must now replace $m \rightarrow m_\mu$, and as $\mathcal{P}(t)$ substitute the electronic polarisation operator. In contrast to the preceding case, the essential region of values will now be $f^2/m_e^2 \sim (m_\mu/m_e)^2 \gg 1$, and as $\mathcal{P}(f^2)$ one must take the limiting expression (113.15):

$$\frac{\mathcal{P}(f^2)}{f^2} = \frac{\alpha}{3\pi} \ln \frac{|f^2|}{m_e^2}.$$

The computation of the integrals leads to the value

$$[g^{(2)}(0)]_{\text{e.p.}} = \left(\frac{\alpha}{\pi}\right)^2 \left(\frac{1}{3} \ln \frac{m_\mu}{m_e} - \frac{25}{36}\right) = 1.09 \frac{\alpha^2}{\pi^2} \quad (\text{XII.80})$$

(*H. Suura, E. H. Wichmann, 1957; A. Petermann, 1957*).

Adding (XII.80) to (118.3), we obtain for the magnetic moment of the muon

$$\mu_{\text{muon}} = \frac{e\hbar}{2m_\mu c} \left(1 + \frac{\alpha}{2\pi} + 0.76 \frac{\alpha^2}{\pi^2}\right). \quad (\text{XII.81})$$

Note that the contribution of muonic vacuum polarisation (XII.79) constitutes $\sim 2\%$ of the entire value of $g^{(2)}(0)$. A contribution of the same order would be given by pionic vacuum polarisation, which (owing to the closeness of the masses) cannot in general be computed exactly. For this reason it would already be pointless to compute corrections $\sim \alpha^3$ to the magnetic moment of the muon.

§ 119. Computation of the mass operator

Using the example of the computation of the mass operator, we shall demonstrate the method of direct regularisation of Feynman integrals.

In the first non-vanishing approximation, the mass operator is represented by the loop in the diagram

$$(\text{XII.82})$$

The corresponding integral is

$$-i\bar{M}(p) = (-ie)^2 \int \gamma^\mu G(p-k) \gamma^\nu D_{\mu\nu}(k) \frac{d^4k}{(2\pi)^4};$$

substituting the propagators and bringing together the factors $\gamma^\mu \dots \gamma_\mu$ with the help of the formulas (22.6), we obtain

$$\bar{M}(p) = -\frac{8\pi i}{(2\pi)^4} e^2 \int \frac{2m - (\gamma p) + (\gamma k)}{[(p-k)^2 - m^2](k^2 - \lambda^2)} d^4k \quad (\text{XII.83})$$

(by the bar over $\mathcal{M}$ we denote the unregularised value of the integral). A fictitious “photon mass” λ has been introduced into the photon propagator in order to eliminate the infrared divergence (as in § 117).

We transform the integral with the help of formula (131.4), taking the two factors in the denominator of (XII.83) as a_1 and a_2 . After a simple regrouping of the terms in the denominator of the new integral, we obtain

$$\bar{\mathcal{M}}(p) = -\frac{8\pi i}{(2\pi)^4} e^2 \int d^4 k \int_0^1 dx \frac{2m - (\gamma p) + (\gamma k)}{[(k - px)^2 - a^2]^2}, \quad (\text{XII.84})$$

where

$$a^2 = m^2 x^2 - (p^2 - m^2)x(1 - x) + \lambda^2(1 - x). \quad (\text{XII.85})$$

The substitution $k \rightarrow k + px$ brings the integrand in (XII.84) to a form in which the denominator depends only on k^2 . In doing so, however, according to (131.17), (131.18), an additive constant is added to the integral:

$$\bar{\mathcal{M}}(p) = -\frac{8\pi i}{(2\pi)^4} e^2 \left\{ \int d^4 k \int_0^1 dx \frac{2m(1 - x) - \frac{i\pi^2}{4}(\gamma p)}{(k^2 - a^2)^2} \right\}, \quad (\text{XII.86})$$

the term with (γk) in the numerator being now omitted as it vanishes upon integration over the directions of the 4-vector k (cf. (131.8)).

Regularisation of this integral consists in such subtractions as would reduce it to an expression of the form (110.20). The latter vanishes upon multiplication by the wave amplitude $u(p)$, if p is the 4-momentum of a real electron. Without introducing $u(p)$ explicitly, one can formulate this condition as the requirement that $\mathcal{M}(p)$ vanish upon the substitution

$$(\gamma p) \rightarrow m, \quad p^2 \rightarrow m^2. \quad (\text{XII.87})$$

The form of the integral (XII.86) is convenient in that the 4-vector p enters it only in the form γp and p^2 (and terms of the form $k p$ are absent).

Subtracting from (XII.86) the same expression with the substitution (XII.87), we obtain

$$\begin{aligned} & -\frac{8\pi i e^2}{(2\pi)^4} \left\{ \int d^4 k \int_0^1 dx \cdot [2m - (\gamma p)(1 - x)] \left[\frac{1}{(k^2 - a^2)^2} - \frac{1}{(k^2 - a_0^2)^2} \right] \right. \\ & \quad \left. - \int d^4 k \int_0^1 dx \frac{1 - x}{(k^2 - a_0^2)^2} (\gamma p - m) - \frac{i\pi^2}{4} (\gamma p - m) \right\}, \quad (\text{XII.88}) \end{aligned}$$

where

$$a_0^2 = m^2 x^2 + \lambda^2(1 - x).$$

For the final regularisation, however, one more subtraction must be performed: according to (110.20), upon the substitution (XII.87) not only $\mathcal{M}(p)$ as a whole should vanish, but so should the same expression without one of the factors $\gamma p - m$. The corresponding subtractions eliminate entirely the second and third terms in the curly

brackets of (XII.88)⁷. The first integral is preliminarily transformed by introducing one more auxiliary integration with the help of formula (131.5), putting in it $n = 2$ and taking $k^2 - a^2$ and $k^2 - a_0^2$ as a and b . Then expression (XII.88) takes the form

$$(\gamma p - m) \frac{16\pi i}{(2\pi)^4} e^2 \int d^4 k \int_0^1 dx \int_0^1 dz \frac{[2m - (\gamma p)(1-x)]x(1-x)}{[k^2 - a_0^2 + (p^2 - m^2)x(1-x)z]^3}$$

(here we have also used the identity $p^2 - m^2 = (\gamma p - m)(\gamma p + m)$). We immediately perform the integration over $d^4 k$. Assuming that $p^2 - m^2 < 0$, and using (131.14), we obtain

$$(\gamma p - m) \frac{e^2}{2\pi} \int_0^1 dx \int_0^1 dz \frac{(\gamma p + m)[2m - (\gamma p)(1-x)]x(1-x)}{m^2 x^2 + \lambda^2(1-x) + (m^2 - p^2)x(1-x)z}.$$

It now remains to subtract the factor $(\gamma p - m)$, temporarily omitted, and to perform the same integral with the substitution (XII.87); after straightforward manipulations we obtain

$$\begin{aligned} \mathcal{M}(p) &= (\gamma p - m) \frac{e^2}{2\pi} \times \\ &\times \int_0^1 dx \int_0^1 dz \frac{m(1-x^2) - (\gamma p)(1-x)^2}{m^2 x + (m^2 - p^2)(1-x)z} \left[1 - \frac{2x(1+x)}{x^2 + (\lambda/m)^2} \right] \end{aligned} \quad (\text{XII.90})$$

(in the common denominator the term with λ^2 is omitted, since this does not lead to divergences here; in another place $\lambda^2(1-x)$ is replaced by λ^2 , since the infrared divergence will correspond to the divergence at $x \rightarrow 0$).

Integration in (XII.90) (first over z , then over x) is rather lengthy, but elementary and leads to the following final result:

$$\begin{aligned} \mathcal{M}(p) &= \frac{\alpha}{2\pi m} (\gamma p - m)^2 \left\{ \frac{1}{2(1-\rho)} \left(1 - \frac{2-3\rho}{1-\rho} \ln \rho \right) - \right. \\ &\quad \left. - \frac{\gamma p + m}{m\rho} \left[\frac{1}{2(1-\rho)} \left(2 - \rho + \frac{\rho^2 + 4\rho - 4}{1-\rho} \ln \rho \right) + 1 + 2 \ln \frac{\lambda}{m} \right] \right\}, \end{aligned} \quad (\text{XII.91})$$

where we have denoted

$$\rho = \frac{m^2 - p^2}{m^2}.$$

(*R. Karplus, N. M. Kroll, 1950*). The integral is computed under the assumption $\rho > 0$, and moreover $\rho \gg \lambda/m$. In accordance with the rule for going around poles, the analytic

⁷In doing so, during the process of “renormalisation on the fly” (cf. p. 545), we drop corrections to the renormalisation constant Z_1 (see § 110). The corresponding integrals diverge logarithmically. If one introduces a “cut-off parameter” $\Lambda^2 \gg m^2, p^2$, restricting the region of integration over $d^4 k$ by the condition $k^2 \leq \Lambda^2$, this correction can be computed explicitly. The computation gives the result

$$Z_1 = 1 + Z_1^{(1)}, \quad Z_1^{(1)} = -\frac{\alpha}{2\pi} \left[\frac{1}{2} \ln \frac{\Lambda^2}{m^2} + \ln \frac{\lambda^2}{m^2} + \frac{9}{4} \right]. \quad (\text{XII.89})$$

continuation of expression (XII.91) into the region $\rho < 0$ is defined by the substitution $m \rightarrow m - i0$; in doing so $\rho \rightarrow \rho - i0$, so that $\ln \rho$ for $\rho < 0$ must be understood as

$$\ln \rho = \ln |\rho| - i\pi, \quad \rho < 0. \quad (\text{XII.92})$$

Let us examine the behaviour of the mass operator for $p^2 \gg m^2$. We then have $-\rho \approx p^2/m^2 \gg 1$, and with logarithmic accuracy

$$\mathcal{M}(p) = -[\mathcal{G}^{-1}(p) - G^{-1}(p)] \approx \frac{\alpha}{4\pi} (\gamma p) \ln \frac{p^2}{m^2}. \quad (\text{XII.93})$$

As in the case of the photon propagator (cf. formulas (113.15), (113.16) for the polarisation operator), the correction to G^{-1} turns out to be small only at not too large energies, namely when

$$\frac{\alpha}{4\pi} \ln \frac{p^2}{m^2} \ll 1.$$

In the present case, however, the logarithmic growth is in a certain sense fictitious: it can be eliminated by an appropriate choice of gauge, i.e. of the function $D^{(l)}$ in the photon propagator (*L. D. Landau, A. A. Abrikosov, I. M. Khalatnikov*, 1954). Namely, one must set (in the notation of § 103)

$$D^{(l)} = 0, \quad (\text{XII.94})$$

whereas formula (XII.91) was obtained in the gauge

$$D^{(l)} = D. \quad (\text{XII.95})$$

This property of the gauge (XII.94) makes it especially convenient for investigating the behaviour of the theory at $p^2 \gg m^2$, which will be used below, in § 132.

To prove the assertion made, we note that if we are interested only in terms $\sim e^2$, then the transformation from the gauge (XII.95) to the gauge (XII.94) can be regarded as infinitesimally small. Correspondingly, one can directly use formula (105.14), putting in it

$$d^{(l)}(q) = -\frac{D}{q^2} = -\frac{4\pi}{(q^2)^2},$$

and also replacing, with the required accuracy, the function $\mathcal{G}$ in the integrand by G . In the integral over d^4q , the essential region is $q \gg p$; in this case $G(p - q)$ in the integrand is much smaller than $G(p)$, and it can be neglected. Then

$$\delta \mathcal{G}^{-1} = -\mathcal{G}^{-2}(p) \delta \mathcal{G}(p) = -ie^2 \mathcal{G}^{-1}(p) \int d^{(l)}(q) \frac{d^4q}{(2\pi)^4}.$$

Finally, applying the transformation (113.11), (113.12), we obtain

$$\delta \mathcal{G}^{-1} = -\frac{e^2}{4\pi} \mathcal{G}^{-1}(p) \int \frac{d(-q^2)}{-q^2} \approx -\frac{e^2}{4\pi} (\gamma p) \ln \frac{\Lambda^2}{p^2},$$

where Λ is an auxiliary upper limit, the divergence at which is removed by renormalisation. The latter consists in subtracting the same expression evaluated at $p^2 \approx m^2$, so that we finally have

$$\delta\mathcal{G}^{-1} = \frac{e^2}{4\pi} (\gamma p) \ln \frac{p^2}{m^2}.$$

This expression exactly cancels the difference $\mathcal{G}^{-1} - G^{-1}$ from (XII.93).

Finally, let us address the question of the reasons necessitating the introduction of a finite “photon mass” λ for the regularisation of the integral (XII.83), closely connected with its behaviour at $p^2 \rightarrow m^2$.

First of all, let us note that this integral is by itself finite at $\lambda = 0$ (for the elimination of the inessential divergence in this respect at large k , we assume that the integral is taken over a large but finite region of k -space). The need for introducing λ arises from subtracting the renormalisation integral, which would otherwise diverge at $p^2 = m^2$. Let us therefore clarify how the unregularised mass operator behaves as $p^2 \rightarrow m^2$. Since this behaviour depends significantly on the choice of gauge (whereas the integral (XII.83) was written already for a specific choice—(XII.95)), let us consider the general case of an arbitrary gauge.

We again make use of the transformation (105.14). Representing $d^{(l)}$ in the form

$$d^{(l)}(q) = -\frac{\delta D^{(l)}}{q^2} = -\frac{4\pi}{(q^2)^2} \delta a(q^2), \quad (\text{XII.96})$$

we shall consider δa to be a variation of the function $a(q^2)$, substantially changing only over intervals $q^2 \sim m^2$ and finite at $q^2 \approx m^2$. In the integrand on the right side of (105.14), in the difference $\mathcal{G}(p) - \mathcal{G}(p - q)$, for small q both terms are close and the integral converges. Since for small q

$$\mathcal{G}(p - q) \sim \frac{1}{p^2 - m^2 - 2pq},$$

$\mathcal{G}(p - q)$ can be omitted in comparison with $\mathcal{G}(p)$ when $q \gg (p^2 - m^2)/m$. The integral

$$\delta\mathcal{G}(p) = ie^2 \mathcal{G}(p) \int d^{(l)}(q) \frac{d^4 q}{(2\pi)^4} = -\frac{e^2}{4\pi} \mathcal{G}(p) \int \delta a(q^2) \frac{d(-q^2)}{-q^2}$$

diverges logarithmically in the region

$$(p^2 - m^2)/m^2 \ll q^2 \ll m^2.$$

With logarithmic accuracy we therefore have

$$\frac{\delta\mathcal{G}}{\mathcal{G}} = -\frac{e^2}{4\pi} \delta a(m^2) \ln \frac{m^2}{p^2 - m^2}.$$

This equation can be integrated. Noting that when $\alpha^2 \equiv e^2 \rightarrow 0$ the exact propagator $\mathcal{G}$ must coincide with the propagator of free particles G , we obtain

$$\mathcal{G}(p) = \frac{1}{\gamma p - m} \left(\frac{m^2}{p^2 - m^2} \right)^{\frac{\alpha}{2\pi} (C - a_0)}, \quad (\text{XII.97})$$

where $a_0 = a(m^2)$, C is a certain constant. To determine the latter, we compare the expression

$$\mathcal{G}^{-1}(p) = (\gamma p - m) \left[1 + \frac{\alpha}{2\pi} (C - a_0) \ln \rho \right], \quad (\text{XII.98})$$

obtained from (XII.97) in the first approximation with respect to α , with the analogous expression obtained from the integral (XII.83) at $\lambda = 0^8$:

$$\mathcal{G}^{-1}(p) = (\gamma p - m) \left[1 + \frac{\alpha}{\pi} \ln \rho \right]. \quad (\text{XII.99})$$

According to the definition (XII.96), the function $a(q^2)$ coincides with the ratio $D^{(l)}/D$. Therefore the gauge (XII.95), to which (XII.99) pertains, corresponds to $a = a_0 = 1$. Requiring the coincidence of (XII.98) and (XII.99) at this value of a_0 , we find $C = 3$.

Thus we finally find the following limiting expression (*infrared asymptotic*) of the unrenormalised electron propagator as $p^2 \rightarrow m^2$:

$$\mathcal{G}(p) = \frac{\gamma p + m}{p^2 - m^2} \left(\frac{m^2}{p^2 - m^2} \right)^{\frac{\alpha}{2\pi}(3-a_0)}. \quad (\text{XII.100})$$

(A. A. Abrikosov, 1955). Let us stress that the validity of this formula is subject to the inequalities $\alpha \ll 1$, $|\ln \rho| \gg 1$, whereas the formulas of perturbation theory would require the condition $\alpha |\ln \rho| / 2\pi \ll 1$ as well. Let us also note that the sign of the difference $p^2 - m^2$ is immaterial here, since the imaginary part of expression (XII.100) would in any case be beyond its limits of accuracy.

The renormalised propagator must have a simple pole at $p^2 = m^2$. We see that (XII.100) satisfies this requirement only in the gauge in which

$$D^{(l)} = 3D. \quad (\text{XII.101})$$

(so that $a_0 = 3$). In this case the regularisation of the Feynman integral (having the purpose of eliminating its divergence at the upper limits) will not require the introduction of a finite “photon mass”. In other gauges, however, a vanishing photon mass leads to the appearance of a branch point at $p^2 = m^2$ instead of a simple pole, and the removal of this “defect” requires the introduction of a finite parameter λ .

§ 120. Emission of soft photons with non-zero mass

When computing the electron form factors in § 117 we encountered the divergence of integrals at low frequencies of virtual photons. This divergence is closely connected with the infrared catastrophe discussed already in § 98. There it was shown that the cross section of any process involving charged particles (including the scattering of an electron by an external field, represented by the diagram of type (117.1)) has meaning not by itself, but only when the simultaneous emission of any number of soft photons is taken

⁸To obtain (XII.98), there is no need to redo the computation from scratch. The term $\sim \ln \rho$ in (XII.91) was in fact obtained under the assumption $\rho \gg \lambda$, which permits the passage to $\lambda \rightarrow 0$. The term $\sim \ln(\lambda/m)$ arises from the subtraction of the renormalisation integral and is absent in the original integral (XII.83).

into account. As will be explained in detail below (see § 122), in the total cross section, which takes into account the emission of soft photons, all the divergences cancel. In this case, of course, to obtain the correct result the preliminary “cut-off” of the divergent integrals in all the contributing cross sections must be carried out in a uniform manner.

In § 117 this cut-off was accomplished by introducing a fictitious finite mass λ of the virtual photon. We must now accordingly modify the formulae obtained in § 98, so that they describe the emission of soft “photons” with non-zero mass.

From a formal point of view, such a photon belongs to “vector” particles of spin 1, the free field of which was considered in § 14. It is described by the 4-vector ψ -operator

$$\hat{\psi}_\mu = \sqrt{4\pi} \sum_{\mathbf{k}\alpha} \frac{1}{\sqrt{2\omega}} (\hat{c}_{\mathbf{k}\alpha} e_\mu^{(\alpha)} e^{-ikx} + \hat{c}_{\mathbf{k}\alpha}^+ e_\mu^{(\alpha)*} e^{ikx}), \quad \alpha = 1, 2, 3 \quad (\text{XII.102})$$

(here the notation and normalisation have been modified compared to (14.16) to bring them into correspondence with the photon case).

The interaction of the “photons” (XII.102) with electrons must be described by a Lagrangian of the same form as for real photons:

$$-e \hat{j}^\mu \hat{\psi}_\mu \quad (\text{XII.103})$$

(with the replacement of the potential operators $\hat{A}_\mu$ by $\hat{\psi}_\mu$). Then the amplitudes of emission processes for photons of finite mass will be given by the usual formulae of the diagrammatic technique, with the sole difference that

$$k^2 = \lambda^2. \quad (\text{XII.104})$$

The summation over the polarisations of the emitted photon must now be performed over the three independent polarisations (two transverse and one longitudinal) instead of the two for an ordinary photon. This is equivalent to averaging over the density matrix of unpolarised particles

$$\rho_{\mu\nu} = -\frac{1}{3} \left(g_{\mu\nu} - \frac{k_\mu k_\nu}{\lambda^2} \right) \quad (\text{XII.105})$$

(cf. (14.15)) with subsequent multiplication by 3.

The propagator for “photons” with non-zero mass is

$$D_{\mu\nu} = \frac{4\pi}{k^2 - \lambda^2} \left(g_{\mu\nu} - \frac{k_\mu k_\nu}{\lambda^2} \right)$$

(cf. (76.18)). However, owing to gauge invariance the amplitudes of real scattering processes do not depend on the longitudinal part of the photon propagator, and this property is not connected with the specific form of its transverse part. Therefore the second term in the parentheses effectively drops out, and there remains an expression of the same type as for ordinary photons:

$$D_{\mu\nu} = \frac{4\pi}{k^2 - \lambda^2} g_{\mu\nu} \quad (\text{XII.106})$$

(which we have already used in §§ 117, 119).

Let us now turn to the study of soft (in the sense explained in § 98) photons.

The derivation carried out in § 98 of formulae (98.5), (98.6) carries over to the present case with only the modification that, upon expanding the squares $(p \pm k)^2$ in the denominators of the electron propagators, a term $k^2 = \lambda^2$ is added. As a result, instead of (98.6) we obtain

$$d\sigma = d\sigma_{\text{el}} \cdot e^2 \left| \frac{p'e}{(p'k) + \lambda^2/2} - \frac{pe}{(pk) - \lambda^2/2} \right|^2 \frac{d^3k}{4\pi^2\omega}, \quad (\text{XII.107})$$

where $d\sigma_{\text{el}}$ is the cross section for the same process without emission of a soft photon (which we conventionally call the “elastic” process). In what follows, when integrating over d^3k , the important values of $|\mathbf{k}|$ will be $\sim \lambda$. In this case $p'k \sim pk \gg \lambda^2$, so that the terms λ^2 in the denominators may be neglected. The summation over photon polarisations is carried out, as indicated, with the help of (XII.105). After the neglect made, the second term in (XII.105) does not contribute to the cross section⁹, and there remains

$$d\sigma = -d\sigma_{\text{el}} \cdot e^2 \left(\frac{p'}{(p'k)} - \frac{p}{(pk)} \right)^2 \frac{d^3k}{4\pi^2\omega}. \quad (\text{XII.108})$$

Thus we return to formula (98.7), in which, however, one must now understand ω as

$$\omega = \sqrt{\mathbf{k}^2 + \lambda^2}. \quad (\text{XII.109})$$

Formula (XII.108) has a completely general character. It is applicable both for elastic and for inelastic scattering, and even when the species of particles changes. The result of the subsequent integration over d^3k depends on the 4-vectors p and p' , in other words, on the character of the underlying scattering process.

Let us consider the case of elastic scattering, when

$$|\mathbf{p}| = |\mathbf{p}'|, \quad \varepsilon = \varepsilon',$$

and determine the total probability of emission of photons with frequency less than some ω_{max} ; it is assumed here that

$$\omega_{\text{max}} \gg \lambda, \quad (\text{XII.110})$$

while from above the value of ω_{max} is bounded by the conditions of applicability of the theory of soft photon emission (98.9), (98.10).

Let us first compute the integral over d^3k in the non-relativistic limit. When $|\mathbf{p}| = |\mathbf{p}'| \ll m$ we have

$$\left(\frac{p'}{(p'k)} - \frac{p}{(pk)} \right)^2 \approx \frac{(\mathbf{qk})^2}{m^2\omega^4} - \frac{\mathbf{q}^2}{m^2\omega^2}$$

($\mathbf{q} = \mathbf{p}' - \mathbf{p}$). Integration of this expression over the directions of $\mathbf{k}$ gives

$$\frac{4\pi\mathbf{q}^2}{m^2\omega^2} \left(\frac{\mathbf{k}^2}{3\omega^2} - 1 \right).$$

⁹At first sight one might doubt the admissibility of neglecting λ^2 in the averaging, in view of the presence of λ^2 in the denominator of the second term in (XII.105). However, it is easy to verify directly that this term, upon averaging, gives a contribution $\sim \lambda^4 \cdot \lambda^{-2}$ which can be neglected.

After this, from (XII.108) we have

$$d\sigma = d\sigma_{\text{el}} \cdot \frac{e^2 \mathbf{q}^2}{\pi m^2} \int_0^{\omega=\omega_{\text{max}}} \left[1 - \frac{\mathbf{k}^2}{3(\mathbf{k}^2 + \lambda^2)} \right] \frac{\mathbf{k}^2 d|\mathbf{k}|}{(\mathbf{k}^2 + \lambda^2)^{3/2}},$$

or, carrying out the integration under the assumption $\omega_{\text{max}}/\lambda \gg 1$,

$$d\sigma = d\sigma_{\text{el}} \cdot \frac{2\alpha}{3\pi} \left(\ln \frac{2\omega_{\text{max}}}{\lambda} - \frac{5}{6} \right), \quad \mathbf{q}^2 \ll m^2. \quad (\text{XII.111})$$

In the general relativistic case, to evaluate the integral we make use of formula (131.4). With its help we have for the angular integral

$$I = \int \frac{d\mathbf{o}_{\mathbf{k}}}{(pk)(p'k)} = \int_0^1 dx \int \frac{d\mathbf{o}_{\mathbf{k}}}{[(pk)_x + (p'k)(1-x)]^2},$$

or, expanding the scalar products with $p = (\varepsilon, \mathbf{p})$, $p' = (\varepsilon, \mathbf{p}')$,

$$I = \int_0^1 dx \int \frac{d\mathbf{o}_{\mathbf{k}}}{\{\varepsilon\omega - \mathbf{k}[\mathbf{p}x + \mathbf{p}'(1-x)]\}^2}.$$

The integral $d\mathbf{o}_{\mathbf{k}}$ is easily evaluated in spherical coordinates with the polar axis along the vector $\mathbf{p}x + \mathbf{p}'(1-x)$, after which

$$I = \int_0^1 \frac{4\pi dx}{(\varepsilon\omega)^2 - [\mathbf{p}x + \mathbf{p}'(1-x)]^2 \mathbf{k}^2} = \int_0^1 \frac{4\pi dx}{m^2 + \mathbf{q}^2 x(1-x) \mathbf{k}^2 + \varepsilon^2 \lambda^2}.$$

The two other integrals (with $(pk)^2$ and $(p'k)^2$ in the denominators) are obtained from this at $\mathbf{q} = 0$. Noting also that

$$pp' = \varepsilon^2 - \mathbf{p}\mathbf{p}' = m^2 + \mathbf{q}^2/2,$$

we obtain

$$d\sigma = \frac{2e^2}{\pi} \int_0^1 dx \int_0^{\omega_{\text{max}}} \frac{\mathbf{k}^2 d|\mathbf{k}|}{\sqrt{\mathbf{k}^2 + \lambda^2}} \left\{ \frac{m^2 + \mathbf{q}^2/2}{[m^2 + \mathbf{q}^2 x(1-x)] \mathbf{k}^2 + \varepsilon^2 \lambda^2} - \frac{m^2}{m^2 \mathbf{k}^2 + \varepsilon^2 \lambda^2} \right\} d\sigma_{\text{el}}. \quad (\text{XII.112})$$

The integration over $d|\mathbf{k}|$ reduces to evaluating integrals of the form

$$\int_0^{\omega_{\text{max}}} \frac{\mathbf{k}^2 d|\mathbf{k}|}{(a\mathbf{k}^2 + \lambda^2)\sqrt{\mathbf{k}^2 + \lambda^2}} = \frac{1}{a} \int_0^{\omega_{\text{max}}} \frac{d|\mathbf{k}|}{\sqrt{\mathbf{k}^2 + \lambda^2}} - \frac{\lambda^2}{a} \int_0^{\omega_{\text{max}}} \frac{d|\mathbf{k}|}{(a\mathbf{k}^2 + \lambda^2)\sqrt{\mathbf{k}^2 + \lambda^2}} \approx \frac{1}{a} \ln \frac{2\omega_{\text{max}}}{\lambda}$$

In the second integral the substitution $|\mathbf{k}| \rightarrow \lambda z$ has been made and the upper limit $(\omega_{\text{max}}/\lambda)$ replaced by ∞ , which is permissible since the integral converges.

The integrals over x arising then in (XII.112) cannot be expressed in terms of elementary functions in full generality. The result may be written in the form

$$d\sigma = \alpha \left[F\left(\frac{|\mathbf{q}|}{2m}\right) \ln \frac{2\omega_{\text{max}}}{\lambda} + F_1 \right] d\sigma_{\text{el}}, \quad (\text{XII.113})$$

where¹⁰

$$f(\xi) = \frac{2}{\pi} \left[\frac{2\xi^2 + 1}{\xi\sqrt{\xi^2 + 1}} \ln(\xi + \sqrt{\xi^2 + 1}) - 1 \right], \quad (\text{XII.115})$$

$$F = \frac{2\varepsilon}{\pi|\mathbf{p}|} \ln \frac{\varepsilon + |\mathbf{p}|}{m} - \frac{2m^2 + \mathbf{q}^2}{\pi\varepsilon^2} \int_0^1 \frac{dx}{a\sqrt{1-a}} \ln \frac{1 + \sqrt{1-a}}{\sqrt{a}}, \quad (\text{XII.116})$$

$$a = \frac{1}{\varepsilon^2} [m^2 + \mathbf{q}^2 x(1-x)].$$

Let us find the asymptotic expression for the cross section in the ultra-relativistic case. It is assumed here that not only $\varepsilon \gg m$, but also $|\mathbf{q}| \gg m$, i.e. the scattering angle is not too small. Under these conditions, in the integral (XII.116) the essential region of values of x is that in which $a \ll 1$; after the corresponding approximations

$$F_1 \approx \frac{\mathbf{q}^2}{2\pi\varepsilon^2} \ln \frac{\ln a}{a} da \approx \frac{1}{2\pi} \int \frac{\ln(q^2/\varepsilon^2) + \ln x + \ln(1-x)}{x(1-x)} dx.$$

The integral must be cut off at $a \sim m^2/\varepsilon^2$, i.e. at $x \sim m^2/\mathbf{q}^2$ from below, and at $1-x \sim m^2/\mathbf{q}^2$ from above. Then

$$F_1 \approx \frac{1}{2\pi} \left[2 \ln \frac{\mathbf{q}^2}{\varepsilon^2} \ln \frac{\mathbf{q}^2}{m^2} - \ln^2 \frac{\mathbf{q}^2}{m^2} \right] = \frac{1}{2\pi} \left[\ln^2 \frac{\mathbf{q}^2}{m^2} - 4 \ln \frac{\varepsilon}{m} \ln \frac{\mathbf{q}^2}{m^2} \right].$$

This formula is valid to the accuracy of squares of logarithms, or as one says, with *doubly logarithmic* accuracy. With this accuracy it suffices to set in the first term of (XII.113)

$$F(\xi) \approx \frac{4}{\pi} \ln \xi, \quad \xi \gg 1.$$

Finally,

$$d\sigma = \frac{2\alpha}{\pi} \left[\ln \frac{\mathbf{q}^2}{m^2} \ln \frac{\omega_{\max}}{\lambda} - \ln \frac{\varepsilon}{m} \ln \frac{\mathbf{q}^2}{m^2} + \frac{1}{4} \ln^2 \frac{\mathbf{q}^2}{m^2} \right] d\sigma_{\text{el}}, \quad \mathbf{q}^2 \gg m^2. \quad (\text{XII.117})$$

§ 121. Electron scattering in an external field in the second Born approximation

In the first two approximations with respect to the external field, electron scattering is represented by the diagrams

(XII.118)

The first of them corresponds to the amplitude $M^{(1)} \sim Ze^2$, considered in § 80. The amplitude of the second approximation $M^{(2)} \sim (Ze^2)^2$.

¹⁰The function $F(\xi)$ was already encountered in the problems to § 98. This is not surprising, since with logarithmic accuracy (XII.113) can be obtained by integrating the cross section for emission of photons of zero mass (98.8) over ω in the limits from λ to $\omega_{\max}$. If one introduces the variable θ instead of ξ according to $\xi = \sinh(\theta/2)$, then

$$F(\theta) = \frac{2}{\pi} (\theta \coth \theta - 1). \quad (\text{XII.114})$$

It is easy to see that terms of the same order also arise from radiative corrections. In the third order of perturbation theory, radiative corrections to the scattering amplitude are represented by the diagrams

$$(XII.119)$$

In this case $M^{(3)} \sim Ze^2 \cdot e^2$, and if $Z \sim 1$, then $M^{(3)} \sim M^{(2)}$. According to (64.26) the scattering cross section is

$$d\sigma = |M_{fi}^{(1)} + M_{fi}^{(2)} + M_{fi}^{(3)}|^2 \frac{d\sigma'}{16\pi^2}. \quad (XII.120)$$

In the square of the amplitude appearing here, alongside $|M_{fi}^{(1)}|^2$, we are entitled to retain also the interference terms between $M_{fi}^{(1)}$ and $M_{fi}^{(2)}$ and between $M_{fi}^{(1)}$ and $M_{fi}^{(3)}$. Thus, to accuracy $\sim e^6$ the cross section is represented by the sum

$$d\sigma = d\sigma^{(1)} + d\sigma^{(2)} + d\sigma_{\text{rad}}, \quad (XII.121)$$

where $d\sigma^{(1)}$ is the cross section in the first Born approximation (see § 80), and the corrections to it are

$$\begin{aligned} d\sigma^{(2)} &= 2 \operatorname{Re} M_{fi}^{(1)} M_{fi}^{(2)*} \frac{d\sigma'}{16\pi^2}, \\ d\sigma_{\text{rad}} &= 2 \operatorname{Re} M_{fi}^{(1)} M_{fi}^{(3)*} \frac{d\sigma'}{16\pi^2}. \end{aligned} \quad (XII.122)$$

Recall (see § 80) that

$$M_{fi}^{(1)} = |e| (\bar{u}' \gamma^0 u) \Phi(\mathbf{q}), \quad (XII.123)$$

where $\Phi(\mathbf{q})$ is the Fourier component of the scalar potential of the constant external field ($\Phi \equiv A_0^{(e)}$) and it is taken into account that the charge of the electron is $e = -|e|$.

The two expressions (XII.122) can, of course, be computed independently. The first will be considered in this section, and the second in the following one.

The amplitude of the second approximation, constructed according to the diagram (XII.118), is (after integration¹¹)

$$M_{fi}^{(2)} = -e^2 \int \left\{ \bar{u}(p') \gamma^0 \frac{\gamma f + m}{f^2 - m^2 + i0} \gamma^0 u(p) \right\} \Phi(\mathbf{p}' - \mathbf{f}) \Phi(\mathbf{f} - \mathbf{p}) \frac{d^3 f}{(2\pi)^3}, \quad (XII.124)$$

the “4-impulses” of the constant external field $q_1 = f - p$ and $q_2 = p' - f$ have no time components. Therefore

$$f_0 = \varepsilon = \varepsilon', \quad (XII.125)$$

where ε and ε' are the initial and final energies of the electron, which coincide with each other for elastic scattering.

¹¹We recall that here one must use the rules of diagrammatic technique pertaining to a constant external field; see the rule formulated in § 77, rule 8.

In a purely Coulomb field of a stationary charge $Z|e|$:

$$\Phi(\mathbf{q}) = \frac{4\pi Z|e|}{\mathbf{q}^2}.$$

For such a potential the integral (XII.124) diverges logarithmically (when $\mathbf{f} \approx \mathbf{p}$ and $\mathbf{f} \approx \mathbf{p}'$). This divergence is specific to the Coulomb field and is connected with the slowness of its decrease at large distances. Its origin is most easily clarified using the example of the non-relativistic case. According to Vol. III, (135.8), the coefficient in the spherical wave $\exp(i|\mathbf{p}|r)/r$ in the asymptotic expression of the wave function of the electron in a Coulomb field has the form

$$f(\theta) \exp\left(-i \frac{Z\alpha m}{|\mathbf{p}|} \ln |\mathbf{p}|r\right).$$

But this coefficient is the scattering amplitude of the electron in the field, and we see that its phase contains a divergence (when $r \rightarrow \infty$). When the scattering amplitude is expanded in powers of Z , this term leads to a divergence of all terms of the expansion, starting from the second (since the function $f(\theta)$ itself is proportional to $Z\alpha$). The situation in the relativistic case is, of course, of an analogous character.

These arguments show that at the same time the divergent terms must cancel when computing the scattering cross section, in which the phase of the amplitude is inessential. The simplest way to carry out the computation correctly is to consider first the scattering in a screened Coulomb field, i.e. to set

$$\Phi(q) = \frac{4\pi Z|e|}{\mathbf{q}^2 + \delta^2} \quad (\text{XII.126})$$

with a small screening constant δ ($\delta \ll |\mathbf{p}|$). In this way the divergence is removed in the scattering amplitude, and in the final expression for the cross section one can already set $\delta = 0$.

Substituting (XII.126) into (XII.124), we obtain

$$M_{fi}^{(2)} = -\frac{2}{\pi} Z^2 \alpha^2 \bar{u}(p') [(\gamma^0 \varepsilon + m)J_1 + \boldsymbol{\gamma} \mathbf{J}] u(p),$$

where the following notation has been introduced:

$$J_1 = \int \frac{d^3 f}{[(\mathbf{p}' - \mathbf{f})^2 + \delta^2][(\mathbf{f} - \mathbf{p})^2 + \delta^2][\mathbf{f}^2 - \mathbf{p}^2 - i0]}, \quad (\text{XII.127})$$

$$\mathbf{J} = \int \frac{\mathbf{f} d^3 f}{[(\mathbf{p}' - \mathbf{f})^2 + \delta^2][(\mathbf{f} - \mathbf{p})^2 + \delta^2][\mathbf{f}^2 - \mathbf{p}^2 - i0]} \equiv \frac{\mathbf{p} + \mathbf{p}'}{2} J_2.$$

Here $\mathbf{p}^2 = \varepsilon^2 - m^2 = \mathbf{p}'^2$ and the integral $\mathbf{J}$ is symmetric with respect to $\mathbf{p}$ and $\mathbf{p}'$; from considerations of vector symmetry it is obvious that the vector $\mathbf{J}$ must be directed along $\mathbf{p} + \mathbf{p}'$. Eliminating now the matrices $\boldsymbol{\gamma}$ with the aid of the identities

$$\boldsymbol{\gamma} \mathbf{p} u = (\gamma^0 \varepsilon - m)u, \quad \bar{u}' \boldsymbol{\gamma} \mathbf{p}' = \bar{u}' (\gamma^0 \varepsilon - m),$$

we obtain

$$M_{fi}^{(2)} = -\frac{2}{\pi} Z^2 \alpha^2 \bar{u}(p') [\gamma^0 \varepsilon (J_1 + J_2) + m (J_1 - J_2)] u(p). \quad (\text{XII.128})$$

For carrying out further computations we pass (as in § 80) from the bispinor amplitudes u and u' to the corresponding three-dimensional 3-spinors w and w' (according to (23.9) and (23.11)). By direct multiplication we find

$$\begin{aligned} \bar{u}' u &= w'^* \{ (\varepsilon + m) - (\varepsilon - m) \cos \theta + i \mathbf{v} \boldsymbol{\sigma} (\varepsilon - m) \sin \theta \} w, \\ \bar{u}' \gamma^0 u &= w'^* \{ (\varepsilon + m) - (\varepsilon - m) \cos \theta - i \mathbf{v} \boldsymbol{\sigma} (\varepsilon - m) \sin \theta \} w, \end{aligned}$$

where

$$\mathbf{v} = \frac{[\mathbf{nn}']}{\sin \theta}, \quad \mathbf{n} = \frac{\mathbf{p}}{|\mathbf{p}|}, \quad \mathbf{n}' = \frac{\mathbf{p}'}{|\mathbf{p}'|}, \quad \cos \theta = \mathbf{nn}'.$$

After this the amplitude (XII.128) takes the form¹²

$$M_{fi}^{(2)} = 4\pi w'^* (A^{(2)} + B^{(2)} \mathbf{v} \boldsymbol{\sigma}) w, \quad (\text{XII.129})$$

$$\begin{aligned} A^{(2)} &= -\frac{1}{2\pi^2} Z^2 \alpha^2 \{ [(\varepsilon + m) + (\varepsilon - m) \cos \theta] \varepsilon (J_1 + J_2) + \\ &\quad + [(\varepsilon + m) - (\varepsilon - m) \cos \theta] m (J_1 - J_2) \}, \end{aligned} \quad (\text{XII.130})$$

$$B^{(2)} = \frac{i}{2\pi^2} Z^2 \alpha^2 (\varepsilon - m) \sin \theta [\varepsilon (J_1 + J_2) - m (J_1 - J_2)].$$

The scattering amplitude in the first Born approximation, in analogous notation, has the form

$$\begin{aligned} M_{fi}^{(1)} &= 4\pi w'^* (A^{(1)} + B^{(1)} \mathbf{v} \boldsymbol{\sigma}) w, \\ A^{(1)} &= \frac{Z\alpha}{\mathbf{q}^2} [(\varepsilon + m) + (\varepsilon - m) \cos \theta], \\ B^{(1)} &= -i \frac{Z\alpha}{\mathbf{q}^2} (\varepsilon - m) \sin \theta, \end{aligned}$$

where $\mathbf{q} = \mathbf{p}' - \mathbf{p}$.

The scattering cross section and polarisation effects are expressed in terms of the quantities $A = A^{(1)} + A^{(2)}$ and $B = B^{(1)} + B^{(2)}$ by the formulae obtained in Vol. III, § 140. Thus, the cross section for scattering of unpolarised electrons is:

$$d\sigma = (|A|^2 + |B|^2) d\sigma' \approx d\sigma^{(1)} + 2(A^{(1)} \operatorname{Re} A^{(2)} - i B^{(1)} \operatorname{Im} B^{(2)}) d\sigma'.$$

After substitution of (XII.129), (XII.130) a straightforward computation gives

$$d\sigma^{(2)} = -d\sigma' \frac{Z^2 \alpha^3 \varepsilon^3}{\pi^2 \mathbf{p}^2 \sin^2(\theta/2)} \left[\left(1 - v^2 \sin^2 \frac{\theta}{2} \right) \operatorname{Re}(J_1 + J_2) + \frac{m^2}{\varepsilon^2} \operatorname{Re}(J_1 - J_2) \right], \quad (\text{XII.131})$$

¹²The definition of the quantities A and B here corresponds to the definition in § 37 and in Vol. III, § 140, and differs by a factor from the definition in § 80.

where $\mathbf{v} = |\mathbf{p}|/\varepsilon$ is the velocity of the electron and θ is the scattering angle. As a result of the scattering, the electrons become polarised; the polarisation vector of the final electrons is

$$\zeta' = \frac{2 \operatorname{Re}(AB^*)}{|A|^2 + |B|^2} \mathbf{v} \approx \frac{2(A^{(1)} \operatorname{Re} B^{(2)} - iB^{(1)} \operatorname{Im} A^{(2)})}{|A|^2 + |B|^2} \mathbf{v}$$

or, after substitution of (XII.129), (XII.130),

$$\zeta' = \frac{4Z\alpha m |\mathbf{p}|^4 \sin^3(\theta/2) \cos(\theta/2)}{\pi^2 \varepsilon^2 [1 - v^2 \sin^2(\theta/2)]} \operatorname{Im}(J_1 - J_2) \mathbf{v}. \quad (\text{XII.132})$$

We now turn to the computation of the integrals J_1 and J_2 . It is facilitated by application of the method of Feynman parametrisation according to the formula (131.2). The integral J_1 takes the form

$$J_1 = -2 \iiint_0^1 \frac{d^3 f d\xi_1 d\xi_2 d\xi_3 \cdot \delta(1 - \xi_1 - \xi_2 - \xi_3)}{\{[(\mathbf{p}' - \mathbf{f})^2 + \delta^2]\xi_1 + [(\mathbf{p} - \mathbf{f})^2 + \delta^2]\xi_2 + [\mathbf{f}^2 - \mathbf{p}^2 - i0]\xi_3\}^3}.$$

Integration over ξ_3 eliminates the δ -function; combining like terms in the denominator, we obtain

$$J_1 = -2 \int_0^1 \int_0^{1-\xi_2} \int \frac{d^3 f d\xi_1 d\xi_2}{\{\delta^2(\xi_1 + \xi_2) + \mathbf{p}^2(2\xi_1 + 2\xi_2 - 1) - 2\mathbf{f}(\xi_1 \mathbf{p}' + \xi_2 \mathbf{p}) + \mathbf{f}^2 - i0\}^3}.$$

Introducing in place of $\mathbf{f}$ a new variable $\mathbf{k} = \mathbf{f} - \xi_1 \mathbf{p}' - \xi_2 \mathbf{p}$, we reduce the integration over $d^3 f$ to an integral of the form

$$\int \frac{d^3 k}{(\mathbf{k}^2 - a^2 - i0)^3} = i \frac{\pi^2}{4a^3},$$

so that

$$J_1 = -\frac{i\pi^2}{2} \int_0^1 \int_0^{1-\xi_2} \frac{d\xi_1 d\xi_2}{\{\mathbf{p}^2(\xi_1^2 + \xi_2^2 - 2\xi_1 - 2\xi_2 + 1) + 2\xi_1 \xi_2 \mathbf{p} \mathbf{p}' - \delta^2(\xi_1 + \xi_2) - i0\}^{3/2}}.$$

In place of ξ_1 and ξ_2 we introduce the symmetric combinations: $x = \xi_1 + \xi_2$, $y = \xi_1 - \xi_2$. Integration over y (in the limits from 0 to x) is elementary and gives

$$J_1 = -\frac{i\pi^2}{2|\mathbf{p}|^3} \int_0^1 \frac{x dx}{\left[bx^2 - 2x + 1 - \frac{\delta^2}{\mathbf{p}^2} x - i0 \right]^{1/2} \left[(1-x)^2 - \frac{\delta^2}{\mathbf{p}^2} x - i0 \right]^{1/2}},$$

where $b = \frac{\mathbf{p}^2 + \mathbf{p} \mathbf{p}'}{2\mathbf{p}^2} = \cos^2 \frac{\theta}{2}$.

To compute the integral over x when $\delta \rightarrow 0$ we split the domain of integration into two parts:

$$\int_0^1 \dots dx = \int_0^{1-\delta_1} \dots dx + \int_{1-\delta_1}^1 \dots dx, \quad 1 \gg \delta_1 \gg \frac{\delta}{|\mathbf{p}|}.$$

In the first integral one can set $\delta = 0$; then¹³

$$\int_0^{1-\delta_1} \dots dx = \frac{1}{2(1-b)} \ln \frac{(1-x)^2}{bx^2 - 2x + 1 - i0} \Big|_0^{1-\delta_1} = \frac{1}{2(1-b)} \left[\ln \frac{\delta_1^2}{1-b} + i\pi \right].$$

In the second integral one can set $x = 1$ everywhere except in the term $(1-x)^2$, and also set $\delta = 0$ in the first bracket of the denominator. Then¹⁴

$$\begin{aligned} \int_{1-\delta_1}^1 \dots dx &= \frac{1}{1-b} \int_0^{\delta_1} \frac{dx'}{(x'^2 - \delta^2/\mathbf{p}^2 - i0)^{1/2}} = \\ &= -\frac{1}{1-b} \left[\int_0^{\delta_1} \frac{dx'}{(x'^2 - \delta^2/\mathbf{p}^2)^{1/2}} + i \int_0^{\delta/|\mathbf{p}|} \frac{dx'}{(\delta^2/\mathbf{p}^2 - x'^2)^{1/2}} \right] \\ &= -\frac{1}{1-b} \left[\ln \frac{2|\mathbf{p}|\delta_1}{\delta} + i \frac{\pi}{2} \right]. \end{aligned}$$

Upon adding the two integrals, the quantity δ_1 , as should have been expected, drops out, and one obtains

$$J_1 = \frac{i\pi^2}{2|\mathbf{p}|^3 \sin^2(\theta/2)} \left[-\ln \frac{2|\mathbf{p}|}{\delta} \sin \frac{\theta}{2} \right]. \quad (\text{XII.133})$$

The integral J_2 is computed in an analogous fashion and equals

$$J_2 = J_1 - \frac{\pi^3 [1 - \sin(\theta/2)]}{4|\mathbf{p}|^3 \cos^2 \frac{\theta}{2} \sin \frac{\theta}{2}} - \frac{i\pi^2}{2|\mathbf{p}|^3 \cos^2 \frac{\theta}{2}} \ln \sin \frac{\theta}{2}. \quad (\text{XII.134})$$

It remains to substitute these expressions into (XII.131), (XII.132), and we obtain the final results:

$$d\sigma^{(2)} = \frac{\pi(Z\alpha)^3 \varepsilon}{4|\mathbf{p}|^3 \sin \frac{\theta}{2}} \left(1 - \sin \frac{\theta}{2} \right) d\theta', \quad (\text{XII.135})$$

¹³The bypass rule (the term $i0$) allows one to determine the change in argument of the expression under the logarithm sign upon passing from 0 to $1 - \delta_1$: upon traversal from below the branch point, the argument changes from 0 to $-\pi$.

¹⁴Here too the bypass rule determines the sign of the square root upon passing from positive to negative values of the expression under the radical.

$$\zeta' = \frac{2Z\alpha m|\mathbf{p}|}{\varepsilon^2} \left| \frac{\sin^3 \frac{\theta}{2} \ln \sin \frac{\theta}{2}}{\left(1 - v^2 \sin^2 \frac{\theta}{2}\right) \cos \frac{\theta}{2}} \right| \mathbf{v} \quad (\text{XII.136})$$

(W. A. McKinley, H. Feshbach, 1948; R. H. Dalitz, 1950).

In the first Born approximation the cross sections for scattering of an electron and a positron (in one and the same external field) are identical. In the second approximation this symmetry disappears. For the scattering of a positron (charge $+|e|$) the first-order amplitude (XII.123) has the opposite sign, whereas $M_{fi}^{(1)}$ itself does not change. Therefore the cross section $d\sigma^{(2)}$, which is the interference term between $M_{fi}^{(1)}$ and $M_{fi}^{(2)}$, changes sign. The same happens to expression (XII.136) for the polarisation vector. In general, the transition from the formulae for electron scattering to the formulae for positron scattering can be effected by the formal substitution $Z \rightarrow -Z$.

§ 122. Radiative corrections to scattering

Let us proceed to the computation of radiative corrections to the scattering of an electron in an external field (J. Schwinger, 1949).

The corresponding part of the scattering amplitude is represented by two diagrams (121.2). Diagram *a* gives a contribution to the amplitude

$$-(\bar{u}'\gamma^0 u) \frac{\mathcal{P}(-\mathbf{q}^2)}{4\pi} D(-\mathbf{q}^2) \cdot e\Phi(\mathbf{q}),$$

where $\mathcal{P}(-\mathbf{q}^2)$ is the polarisation operator corresponding to the loop in the diagram. The contribution of diagram *b*:

$$-(\bar{u}'\Lambda^0 u)e\Phi(\mathbf{q}),$$

where Λ^0 is the correction term in the vertex operator ($\Gamma^\mu = \gamma^\mu + \Lambda^\mu$); according to (116.6)

$$\Lambda^0 = \gamma^0 [f(-\mathbf{q}^2) - 1] - \frac{1}{2m} \sigma^{0\nu} q_\nu g(-\mathbf{q}^2).$$

Adding both contributions, we obtain¹⁵

$$M_{fi}^{(3)} = -(\bar{u}'\gamma^0 Q_{\text{rad}} u)e\Phi(\mathbf{q}), \quad Q_{\text{rad}}(\mathbf{q}) = f(-\mathbf{q}^2) - 1 - \frac{1}{\mathbf{q}^2} \mathcal{P}(-\mathbf{q}^2) + \frac{1}{2m} g(-\mathbf{q}^2) \mathbf{q}\gamma. \quad (\text{XII.137})$$

Let us discuss first of all the question of the *infrared divergence* contained in the form factor $f(-\mathbf{q}^2)$ and thereby also in the scattering amplitude (XII.137).

It was already pointed out (see § 98) that the exact amplitude of purely elastic scattering is in itself equal to zero, i.e. has no meaning. Physical meaning is possessed only by the scattering amplitude defined as a process in which any number of soft photons may be emitted, each with energy less than some given value ω_{max} satisfying

¹⁵In this transformation one must remember that if $q^\mu = (0, \mathbf{q})$, then $q_\mu = (0, -\mathbf{q})$! Therefore $\sigma^{0\nu} q_\nu = -\mathbf{q}^0 \mathbf{q}$.

the conditions for applicability of the theory of soft-photon emission. In other words, only the sum has meaning

$$d\sigma = d\sigma_{\text{el}} + d\sigma_{\text{el}} \int_0^{\omega_{\text{max}}} dw_{\omega} + d\sigma_{\text{el}} \frac{1}{2!} \int_0^{\omega_{\text{max}}} dw_{\omega_1} \int_0^{\omega_{\text{max}}} dw_{\omega_2} + \dots, \quad (\text{XII.138})$$

where $d\sigma_{\text{el}}$ is the scattering cross section without photon emission, dw_{ω} is the differential probability of emission of a photon of frequency ω by the electron. It is here assumed that $d\sigma_{\text{el}}$ is itself computed by perturbation theory, i.e. as a series expansion in powers of α^2 . Then upon bringing together terms of each order in α from all the summands in (XII.138) we obtain $d\sigma$ in the form of a series in α , each term of which will be finite.

In the first Born approximation $d\sigma_{\text{el}} \sim \alpha^2$. This term, naturally, has meaning in its own right. If we wish to take into account the next correction to $d\sigma_{\text{el}}$ (a term $\sim \alpha^3$), then alongside it one must also take the second term in the sum (XII.138): since $dw_{\omega} \sim \alpha$, upon multiplication by $d\sigma_{\text{el}} \sim \alpha^2$ a quantity of order $\sim \alpha^3$ also arises from here.¹⁶ $\sim \alpha^3$. Let us show that upon adding these two quantities the infrared divergence is eliminated.

The divergent term in the form factor f according to (117.17) has the form¹⁷

$$-\frac{\alpha}{2} F \left(\frac{|\mathbf{q}|}{2m} \right) \ln \frac{m}{\lambda}.$$

The corresponding term in the amplitude (XII.137):

$$\frac{\alpha}{2} F \ln \frac{m}{\lambda} \cdot (\bar{u}' \gamma^0 u) e \Phi(\mathbf{q}),$$

and in the scattering cross section (121.5):

$$d\sigma^{\text{infra}} = -\alpha F \ln \frac{m}{\lambda} \cdot |\bar{u}' \gamma^0 u|^2 |e \Phi(\mathbf{q})|^2 \frac{d\omega'}{16\pi^2}.$$

Comparing this with the Born cross section

$$d\sigma^{(1)} = |\bar{u}' \gamma^0 u|^2 |e \Phi(\mathbf{q})|^2 \frac{d\omega'}{16\pi^2},$$

we find that

$$d\sigma^{\text{infra}} = -\alpha F \ln \frac{m}{\lambda} \cdot d\sigma^{(1)}. \quad (\text{XII.139})$$

On the other hand, the second term in (XII.138) with $\int dw_{\omega}$ from (120.11) gives

$$d\sigma_{\text{el}} \int_0^{\omega_{\text{max}}} dw_{\omega} = \alpha F \ln \frac{2\omega_{\text{max}}}{\lambda} \cdot d\sigma^{(1)}. \quad (\text{XII.140})$$

¹⁶As regards the probability dw_{ω} , the necessity of accounting for radiative corrections in it depends on ω_{max} ; the limit $\omega \rightarrow 0$ corresponds to the classical case, in which radiative corrections vanish; therefore by choosing a sufficiently small ω_{max} one can always make them small.

¹⁷This is easily verified using the relation

$$\frac{|\mathbf{q}|}{m} = \frac{1 - \xi}{\sqrt{\xi}}$$

between $|\mathbf{q}|$ and the variable ξ with the help of which formula (117.17) was written.

Finally, adding (XII.139) and (XII.140), we obtain

$$-d\sigma^{(1)} \cdot \alpha F\left(\frac{|\mathbf{q}|}{2m}\right) \ln \frac{m}{2\omega_{\max}}. \quad (\text{XII.141})$$

We see that the divergent contribution from soft ($|\mathbf{k}| \sim \lambda$) virtual photons is indeed cancelled by the contribution from the emission of equally soft real photons. The same situation takes place in any other scattering process.

At the same time, there appears a dependence of the scattering cross section on $\omega_{\max}$. This dependence is a consequence of the fact that the quantity $\omega_{\max}$ enters into the very definition of scattering as a process in which any number of soft photons may be emitted. Naturally,

the cross section of such a process will be smaller, the lower the limit $\omega_{\max}$ of photon frequencies whose emission we still attribute to the given scattering process.

Let us now find the full expression for the radiative correction to the scattering cross section. Proceeding by the standard rules (see (65.7)), we find for the cross section, averaged over the polarisations of the initial and summed over the polarisations of the final electrons:

$$d\sigma = d\sigma^{(1)} + d\sigma_{\text{rad}} = |e\Phi(\mathbf{q})|^2 \text{Sp}\{(\gamma p' + m)(\gamma^0 + \gamma^0 Q_{\text{rad}})(\gamma p + m)(\gamma^0 + \overline{Q}_{\text{rad}}\gamma^0)\} \frac{do'}{32\pi^2}. \quad (\text{XII.142})$$

According to (XII.137)

$$Q_{\text{rad}} = a + b\gamma\mathbf{q}, \quad \overline{Q}_{\text{rad}} = \gamma^0 Q_{\text{rad}}^+ \gamma^0 = a + b\gamma\mathbf{q},$$

$$a = f(-\mathbf{q}^2) - 1 - \frac{1}{\mathbf{q}^2} \mathcal{P}(-\mathbf{q}^2), \quad b = \frac{1}{2m} g(-\mathbf{q}^2).$$

To the accuracy of terms linear in a and b , the trace in (XII.142) equals

$$\frac{1}{4} \text{Sp}\{\dots\} = 2\left(\varepsilon^2 - \frac{\mathbf{q}^2}{4}\right) (1 + 2a) - 2bm\mathbf{q}^2.$$

Therefore

$$d\sigma_{\text{rad}} = 2\left\{f_{\lambda}(-\mathbf{q}^2) - 1 - \frac{1}{\mathbf{q}^2} \mathcal{P}(-\mathbf{q}^2) - \frac{\mathbf{q}^2}{4\varepsilon^2 - \mathbf{q}^2} g(-\mathbf{q}^2)\right\} d\sigma^{(1)}, \quad (\text{XII.143})$$

where $d\sigma^{(1)}$ is the Born cross section for scattering of unpolarised electrons (80.5); the form factor f has been given the index λ as a reminder that it is “cut off by the photon mass λ ”.

It remains to add to (XII.143) the cross section for emission of soft photons. If we represent f_{λ} in the form

$$f_{\lambda}(-\mathbf{q}^2) = 1 - \frac{\alpha}{2} F\left(\frac{|\mathbf{q}|}{2m}\right) \ln \frac{m}{\lambda} + \alpha F_2, \quad (\text{XII.144})$$

then according to (120.11) this addition reduces to the replacement of f_{λ} in (XII.143) by

$$f_{\omega_{\max}} = 1 - \frac{\alpha}{2} F\left(\frac{|\mathbf{q}|}{2m}\right) \ln \frac{m}{2\omega_{\max}} + \frac{\alpha}{2} F_1 + \alpha F_2. \quad (\text{XII.145})$$

With this replacement (XII.143) gives the final answer.

Let us note that in the non-relativistic limit¹⁸

$$f_{\omega_{\max}} = 1 - \frac{\alpha \mathbf{q}^2}{3\pi m^2} \left(\ln \frac{m}{2\omega_{\max}} + \frac{11}{24} \right), \quad \mathbf{q}^2 \ll m^2. \quad (\text{XII.146})$$

Let us draw attention to the fact that the specifics of the external field enter into the radiative correction to the cross section only through $d\sigma^{(1)}$; the multiplier in the curly brackets in (XII.143) has a universal character. In the non-relativistic approximation

$$d\sigma_{\text{rad}} = -d\sigma^{(1)} \frac{2\alpha}{3\pi} \frac{\mathbf{q}^2}{m^2} \left(\ln \frac{m}{2\omega_{\max}} + \frac{19}{30} \right), \quad \mathbf{q}^2 \ll m^2 \quad (\text{XII.147})$$

(here the contributions from all terms in the curly brackets of (XII.143) are included). In the opposite, ultra-relativistic, case the main contribution comes only from the term with $f_{\omega_{\max}} - 1$ and one obtains

$$d\sigma_{\text{rad}} = -d\sigma^{(1)} \cdot \frac{2\alpha}{\pi} \ln \frac{\mathbf{q}^2}{m^2} \ln \frac{\varepsilon}{\omega_{\max}}, \quad \mathbf{q}^2 \gg m^2. \quad (\text{XII.148})$$

Let us note in conclusion that the radiative corrections considered here do not lead to the appearance of any polarisation effects absent in the first Born approximation (in contrast to the corrections of the second Born approximation, considered in § 121). The point is that the specifics of the first Born approximation in the final count are connected with the hermiticity of the S -matrix. This property is, however, preserved also when the radiative corrections considered here are taken into account, since in this approximation there are no real intermediate states in the scattering channel (so that the right-hand side of the unitarity relation vanishes).¹⁹

§ 123. Radiative shift of atomic levels

Radiative corrections lead to a shift of the energy levels of bound states of the electron in an external field (the so-called *Lamb shift*). The most interesting case of this kind is the shift of levels of the hydrogen atom (or a hydrogen-like ion)²⁰.

A systematic method for computing corrections to the energy levels is based on the use of the exact electron propagator in the external field (see § 109). But if

¹⁸This expression differs from the non-relativistic (117.20) by the replacement $\ln \lambda \rightarrow \ln 2\omega_{\max} - 5/6$.

¹⁹The computation of radiative corrections to processes appearing only in the second approximation of perturbation theory is considerably more cumbersome and will not be reproduced in this book. We limit ourselves to some literature references: radiative corrections to photon scattering by an electron — *Brown L. M., Feynman R. P.*//Phys. Rev. 1952. V. 85. P. 231; to two-photon annihilation of a pair — *Harris J., Brown L. M.*//Phys. Rev. 1957. V. 105. P. 1656; to scattering of an electron by an electron and a positron — *Redhead M.*//Proc. Roy. Soc. 1953. V. A220. P. 219; *Polovin R. V.*//ZhETF. 1956. V. 31. P. 449; to bremsstrahlung — *Fomin P. I.*//ZhETF. 1958. V. 35. P. 707.

²⁰The shift of hydrogen levels was first computed by *Bethe* (H. A. Bethe, 1947) with logarithmic accuracy on the basis of a non-relativistic treatment; this calculation served as the impetus for the entire subsequent development of quantum electrodynamics. The difference of the levels $2s_{1/2}$ and $2p_{1/2}$ (in the first non-vanishing approximation of perturbation theory) was computed exactly by *Kroll and Lamb* (N. M. Kroll, W. E. Lamb, 1949), and the full formula for the shift of levels was found by *Weisskopf and French* (V. Weisskopf, J. B. French, 1949).

$$Z\alpha \ll 1, \quad (\text{XII.149})$$

one can make use of a simpler method, in which the external field is treated as a perturbation.

In the first approximation in the external field, the radiative correction to the interaction of the electron with a constant electric field is described by the same two diagrams (121.2) that we already considered in connection with the problem of the scattering of an electron in such a field; the transition from one problem to the other requires only a simple reformulation (see below).

It is easy to understand, however, that in this way one can find only that part of the level shift which is caused by interaction with virtual photons of sufficiently high frequencies. Indeed, consider, for instance, the following (according to the external field) radiative correction to the amplitude of scattering of the electron:

$$(\text{XII.150})$$

(in contrast to (121.2,*b*) this diagram contains two vertices of the external field). In the region of integration over d^4k where k_0 is sufficiently large, this correction contains an extra power of $Z\alpha$ and is therefore inessential. But on introducing into the diagram a second vertex of the external field, one also introduces into it an additional electron propagator $G(f)$. For small k (and non-relativistic external ends p and p') the essential impulses of the virtual electrons f , close to the pole of the propagator $G(f)$, turn out to be significant. The small denominator that appears as a result compensates the extra small factor $Z\alpha$. The same applies, obviously, also to corrections of all orders in the external field. In other words, in the region of small frequencies of virtual photons the external field must be taken into account exactly—not only with respect to the electron itself, but also for all intermediate states. Under condition (XII.149) the applicability regions of both methods of calculation overlap, which allows one to carry out a rigorous “joining” of both parts of the correction to the level.

Let us decompose the desired level shift δE_s ²¹ into two parts:

$$\delta E_s = \delta E_s^{(\text{I})} + \delta E_s^{(\text{II})}, \quad (\text{XII.151})$$

arising respectively from interactions with virtual photons of frequencies in the regions I) $k_0 > \kappa$, II) $k_0 < \kappa$. We choose κ so that

$$(Z\alpha)^2 m \ll \kappa \ll m \quad (\text{XII.152})$$

($Z^2\alpha^2 m$ is of the order of the binding energy of the electron in the atom). Then in region I it suffices to take the field of the nucleus into account only in the first approximation. In region II as well (by virtue of the condition $\kappa \ll m$) one can solve the problem in the non-relativistic approximation—not only with respect to the electron itself, but also for all intermediate states. Under condition (XII.152) the applicability regions of both

²¹In this section E_s denotes the energy of the electron in the atom, not including its rest energy. The index s stands for the totality of quantum numbers determining the state of the atom.

methods of calculation overlap, which allows one to carry out a rigorous “joining” of both parts of the correction to the level.

High-frequency part of the shift. Let us consider first region I. In it one can make use of the correction to the scattering amplitude (122.1), from which it is necessary to preliminary exclude the contribution of virtual photons belonging to region II. Such photons make only a small contribution to the form factor g , which therefore does not need modification. In the function f , however, virtual photons of small frequencies make a large contribution due to the infrared divergence. Therefore in place of f in (122.1) one should substitute the function f_{κ} , from which the region $k_0 < \kappa$ has already been excluded.

Such an exclusion could be carried out by a direct method, by computing from f the integral over the region $k_0 < \kappa$. The required result can, however, be obtained without new computations, using the results of § 122.

For this we note that the exclusion of frequencies $k_0 < \kappa$ can be regarded as one of the possible methods of infrared cutoff. The result for the correction to the scattering cross section cannot, of course, depend on the method of cutoff, provided that in the same way one also cuts off the probability of emission of real soft photons, i.e. in the concept of “elastic” scattering one includes the emission with frequencies only from κ to a given $\omega_{\max}$. If one chooses $\omega_{\max} = \kappa$, then the explicit account of photon emission becomes superfluous. It is then clear that f_{κ} is obtained from the function $f_{\omega_{\max}}$ defined in § 122 simply by the replacement of $\omega_{\max}$ by κ . In particular, in the non-relativistic case

$$f_{\kappa} - 1 = -\frac{\alpha \mathbf{q}^2}{3\pi m^2} \left(\ln \frac{m}{2\kappa} + \frac{11}{24} \right). \quad (\text{XII.153})$$

Let us now transform the correction (122.1) to the scattering amplitude, presenting it as the result of a corresponding correction to the effective potential energy of the electron in the field. Comparing the amplitude (122.1)

$$-e(\bar{u}^* Q_{\text{rad}} \Phi u)$$

with the Born scattering amplitude (121.6)

$$-e(\bar{u}^* \Phi u),$$

we see that the role of the correction is played (in the momentum representation) by the function

$$e \delta \Phi(\mathbf{q}) = e Q_{\text{rad}}(\mathbf{q}) \Phi(\mathbf{q}). \quad (\text{XII.154})$$

In the non-relativistic case, taking $\mathcal{P}$ and g from (113.14) and (117.20), and for f substituting f_{κ} from (XII.153), we obtain

$$\delta \Phi(\mathbf{q}) = \left\{ -\frac{\alpha \mathbf{q}^2}{3\pi m^2} \left(\ln \frac{m}{2\kappa} + \frac{11}{24} - \frac{1}{5} \right) + \frac{\alpha}{4\pi m} \mathbf{q} \gamma \right\} \Phi(\mathbf{q}). \quad (\text{XII.155})$$

The corresponding function $\delta\Phi(\mathbf{r})$ in the coordinate representation²²:

$$\delta\Phi(\mathbf{r}) = \frac{\alpha}{3\pi m^2} \left(\ln \frac{m}{2\kappa} + \frac{11}{24} - \frac{1}{5} \right) \Delta\Phi(\mathbf{r}) - i \frac{\alpha}{4\pi m} \boldsymbol{\gamma} \nabla\Phi(\mathbf{r}). \quad (\text{XII.156})$$

The level shift $\delta E_s^{(1)}$ is obtained by averaging $e\delta\Phi(\mathbf{r})$ over the wave function of the unperturbed state of the electron in the atom, i.e. as the corresponding diagonal matrix element²³:

$$\delta E_s^{(1)} = \frac{e\alpha}{3\pi m^2} \left(\ln \frac{m}{2\kappa} + \frac{11}{24} - \frac{1}{5} \right) \langle s | \Delta\Phi | s \rangle - i \frac{e\alpha}{4\pi m} \langle s | \boldsymbol{\gamma} \nabla\Phi | s \rangle. \quad (\text{XII.157})$$

In the first term it suffices to use the non-relativistic wave function of the electron when averaging. In the second term, however, such an approximation is insufficient: the zeroth approximation with respect to non-relativistic wave functions vanishes because of the absence of diagonal elements of the matrix $\boldsymbol{\gamma}$. Therefore one must use here the approximate relativistic wave function found in § 33, $\psi = \begin{pmatrix} \varphi \\ \chi \end{pmatrix}$, retaining in it the small (in the standard representation) component χ . We have

$$\psi^* \boldsymbol{\gamma} \psi = \varphi^* \boldsymbol{\sigma} \chi - \chi^* \boldsymbol{\sigma} \varphi$$

and, substituting from (33.4)

$$\chi = \frac{1}{2m} \boldsymbol{\sigma} \hat{\mathbf{p}} \varphi = -\frac{i}{2m} \boldsymbol{\sigma} \nabla \varphi,$$

we obtain

$$\begin{aligned} \langle s | \boldsymbol{\gamma} \nabla\Phi | s \rangle &= -\frac{i}{2m} \int \{ \varphi^* (\boldsymbol{\sigma} \nabla\Phi) (\boldsymbol{\sigma} \nabla\varphi) + (\nabla\varphi^* \cdot \boldsymbol{\sigma}) (\boldsymbol{\sigma} \nabla\Phi) \varphi \} d^3x = \\ &= \frac{i}{2m} \int \{ \varphi^* \Delta\Phi \cdot \varphi - 2i\sigma\varphi^* [\nabla\Phi \cdot \nabla\varphi] \} d^3x \end{aligned}$$

(in the transformation of the integral the identity (33.5) was used and integration by parts was performed). Since $\Phi = \Phi(r)$,

$$\nabla\Phi = \frac{\mathbf{r}}{r} \frac{d\Phi}{dr},$$

and therefore

$$-i\boldsymbol{\sigma} [\nabla\Phi \cdot \nabla] = \frac{1}{r} \frac{d\Phi}{dr} \boldsymbol{\sigma} \hat{\mathbf{r}},$$

²²Let us emphasise the difference of this correction to the potential from the correction considered in § 114. The latter included in itself only the effect of vacuum polarisation (diagram (121.2,a)) for the Coulomb field as such. The correction (XII.156), however, pertains already to the interaction of the field with the electron and includes in itself also the effect of the change in the motion of the electron (diagram (121.2,b)).

²³Strictly speaking, the form factors defined in § 117 referred to the vertex operator with two free electron ends ($p^2 = p'^2 = m^2$). For an electron in an atom with energy E_s , the level is in no way connected with $\mathbf{p}$. This distinction can, however, be neglected in region I.

where $\hat{\mathbf{l}} = -i[\mathbf{r}\nabla]$ is the orbital angular momentum operator. Finally, collecting the expressions obtained and substituting in (XII.157), we find

$$\delta E_s^{(1)} = \frac{e^3}{3\pi m^2} \left(\ln \frac{m}{2\kappa} + \frac{19}{30} \right) \langle s | \Delta \Phi | s \rangle + \frac{e^3}{4\pi m^2} \left\langle s \left| \boldsymbol{\sigma} \frac{1}{r} \frac{d\Phi}{dr} \right| s \right\rangle, \quad (\text{XII.158})$$

where already in both terms the averaging is carried out over the non-relativistic wave function.

Low-frequency part of the shift. For the computation of the second part of the level shift we use a method based ultimately on the unitarity condition.

By virtue of the possibility of photon emission, the excited state of the atom is quasistationary (and not strictly stationary). To such a state one can assign a complex value of the energy, with its imaginary part equal to $-w/2$, where w is the probability of decay of the state, i.e. in the present case the total probability of photon emission (see III, § 134). In the non-relativistic approximation the radiation is dipolar, and according to (45.7) we have

$$\text{Im } \delta E_s = -\frac{1}{2} w_s = -\frac{2}{3} \sum_{s'} |\mathbf{d}_{ss'}|^2 (E_s - E_{s'})^3 \quad (\text{XII.159})$$

(where the summation is carried out over all lower-lying levels, $E_{s'} < E_s$), or in equivalent form:

$$\text{Im } \delta E_s = -\frac{2}{3} \int_0^\infty d\omega \cdot \sum_{s'} |\mathbf{d}_{ss'}|^2 (E_s - E_{s'})^3 \delta(E_s - E_{s'} - \omega).$$

To find the real part of δE_s , one should regard E_s as a complex variable and carry out the analytic continuation. This can be done by treating the δ -function as arising from the poles. The rule for going around the poles is given, as usual, by adding a negative imaginary part to the masses of virtual particles, which in the present case are the masses $m_{s'}$ of the electron in intermediate states of the atom. The role of these masses is played by $m_{s'} = m + E_{s'}$, so that one must set

$$E_{s'} \rightarrow E_{s'} - i0,$$

whence it follows that the substitution

$$\delta(E_s - E_{s'} - \omega) = -\frac{1}{\pi} \text{Im} \frac{1}{E_s - E_{s'} - \omega + i0} \quad (\text{XII.160})$$

(cf. (111.3)).

Substituting (XII.160) in (XII.159), we find

$$\text{Im } \delta E_s = \text{Im} \frac{2}{3\pi} \int_0^\infty d\omega \cdot \sum_{s'} |\mathbf{d}_{ss'}|^2 \frac{(E_s - E_{s'})^3}{E_s - E_{s'} - \omega + i0}.$$

The desired analytic continuation is then obtained simply by dropping Im on both sides of the equality. We need, however, to extract from δE_s only that part which is associated

with the contribution of frequencies in region II: $\omega < \kappa$. For this it suffices to replace the upper limit of the integral by κ . Carrying out the integration, we obtain

$$\delta E_s^{(\text{II})} = \frac{2}{3\pi} \sum_{s'} |\mathbf{d}_{ss'}|^2 (E_s - E_{s'})^3 \ln \frac{\kappa}{|E_{s'} - E_s| + i0}. \quad (\text{XII.161})$$

(by virtue of inequality (XII.152) we have neglected at the upper limit the difference $E_s - E_{s'}$ in comparison with κ). In what follows we shall be interested only in the real part of the level; it is obtained by replacing in (XII.161) the argument of the logarithm by $\kappa/|E_{s'} - E_s|$.

In expression (XII.161) we transform the term with $\ln \kappa$, replacing the matrix elements of the dipole moment $\mathbf{d} = e\mathbf{r}$ by the matrix elements of the impulse $\mathbf{p} = m\mathbf{v}$ and its derivative $\dot{\mathbf{p}}$:

$$\begin{aligned} \sum_{s'} |\mathbf{d}_{ss'}|^2 (E_s - E_{s'})^3 &= -\frac{e^2}{m^2} \sum_{s'} |\mathbf{p}_{ss'}|^2 (E_s - E_{s'}) = \\ &= \frac{ie^2}{2m^2} \sum_{s'} \{(\dot{\mathbf{p}})_{ss'} \mathbf{p}_{s's} - \mathbf{p}_{ss'} (\dot{\mathbf{p}})_{s's}\}. \end{aligned}$$

Replacing now $\dot{\mathbf{p}}$ according to the operator equation of motion of the electron $\dot{\mathbf{p}} = -e\nabla\Phi$, we obtain

$$\begin{aligned} \sum_{s'} |\mathbf{d}_{ss'}|^2 (E_s - E_{s'})^3 &= -\frac{ie^3}{2m^2} \sum_{s'} \{(\nabla\Phi)_{ss'} \mathbf{p}_{s's} - \mathbf{p}_{ss'} (\nabla\Phi)_{s's}\} = \\ &= \frac{ie^3}{2m^2} \langle s | \mathbf{p} \nabla\Phi - \nabla\Phi \mathbf{p} | s \rangle = \frac{e^3}{2m^2} \langle s | \Delta\Phi | s \rangle. \quad (\text{XII.162}) \end{aligned}$$

We can therefore rewrite (XII.161) in the form

$$\delta E_s^{(\text{II})} = \frac{e^3}{3\pi m^2} \langle s | \Delta\Phi | s \rangle \ln \frac{2\kappa}{m} + \frac{2e^2}{3\pi} \sum_{s'} |\mathbf{r}_{ss'}|^2 (E_{s'} - E_s)^3 \ln \frac{m}{2|E_{s'} - E_s|}. \quad (\text{XII.163})$$

The full shift. Finally, combining both parts, we find the following final formula for the shift of the level:

$$\begin{aligned} \delta E_s &= \frac{2e^2}{3\pi} \sum_{s'} |\mathbf{r}_{ss'}|^2 (E_{s'} - E_s)^3 \ln \frac{m}{2|E_{s'} - E_s|} + \\ &+ \frac{e^3}{3\pi m^2} \frac{19}{30} \langle s | \Delta\Phi | s \rangle + \frac{e^3}{4\pi m^2} \left\langle s \left| \boldsymbol{\sigma} \frac{1}{r} \frac{d\Phi}{dr} \right| s \right\rangle \quad (\text{XII.164}) \end{aligned}$$

(as should be, the auxiliary quantity κ has dropped out)²⁴.

All matrix elements in (XII.164) are taken with respect to the non-relativistic wave functions of the electron in the atom. For the hydrogen atom (or hydrogen-like ion) these

²⁴The determination of corrections of the next order in the shift of levels requires very complex computations. A most complete summary and systematic derivation of such corrections (together with the corresponding bibliography) can be found in the articles: *Erickson G. W., Yennie D. R.* // Ann. of Phys. 1965. V. 35. P. 271, 447.

functions depend on only three quantum numbers: the principal quantum number n , the orbital angular momentum l and its projection m (but not on the total angular momentum j); correspondingly the energy levels depend only on n ²⁵. Let us introduce the notation

$$L_{nl} = \frac{n^3}{2m(Ze^2)^4} \sum_{n'l'm'} |\langle n'l'm' | \mathbf{r} | nlm \rangle|^2 (E_{n'} - E_n)^3 \ln \frac{m(Ze^2)^2}{2|E_{n'} - E_n|}. \quad (\text{XII.165})$$

The energy levels are proportional to $(Ze^2)^2$, and the characteristic size of the atom is proportional to Ze^2 ; therefore, as defined by (XII.165), the quantities L_{nl} do not depend on Z . These quantities can be numerically determined.

Let us consider separately the cases $l = 0$ and $l \neq 0$. When $l = 0$ the last term in (XII.164) vanishes. In the second term we make use of the equation

$$e \Delta \Phi = 4\pi Ze^2 \delta(\mathbf{r}),$$

which is satisfied by the Coulomb field of the nucleus. Hence

$$\langle nlm | \Delta \Phi | nlm \rangle = \begin{cases} 4m^3 (Ze^2)^4 n^{-3}, & l = 0, \\ 0, & l \neq 0 \end{cases}$$

(see (34.3)). In the first term we also introduce the notation (XII.165) and once more use the equality (XII.162):

$$\sum_{n'l'm'} |\langle n'l'm' | \mathbf{r} | n00 \rangle|^2 (E_{n'} - E_n)^3 = \frac{e}{2m^2} \langle n00 | \Delta \Phi | n00 \rangle = \frac{2m(Ze^2)^4}{n^3}.$$

As a result we obtain the following expression for the shift of s -terms:

$$\delta E_{n0} = \frac{4mc^2 Z^4 \alpha^5}{3\pi n^3} \left[\ln \frac{1}{(Z\alpha)^2} + L_{n0} + \frac{19}{30} \right] \quad (\text{XII.166})$$

(ordinary units). Numerical values of several quantities L_{n0} :

$n =$	1	2	3	4	...	∞
$L_{n0} =$	-2.984	-2.812	-2.768	-2.750	...	-2.721

The unperturbed levels $E_n = -mc^2(Z\alpha)^2/(2n^2)$; therefore the relative magnitude of the radiative shift is

$$\left| \frac{\delta E_{n0}}{E_{n0}} \right| \sim Z^2 \alpha^3 \ln \frac{1}{Z\alpha}. \quad (\text{XII.167})$$

In the case $l \neq 0$ in (XII.164) the second term vanishes. The third is computed with the help of the formulae given in § 34. In this term there is also a dependence on the number j . As a result we obtain

$$\delta E_{nlj} = \frac{4mc^2 Z^4 \alpha^5}{3\pi n^3} \left[L_{nl} + \frac{3}{8} \frac{j(j+1) - l(l+1) - 3/4}{l(l+1)(2l+1)} \right], \quad l \neq 0. \quad (\text{XII.168})$$

²⁵The matrix elements of $\mathbf{r}$ are diagonal in the numbers j and do not depend on j ; therefore the summation over s in (XII.164) reduces to summation over n, l, m . By virtue of spatial isotropy the sum (XII.165) does not depend, of course, also on the quantum number m .

Thus the radiative shift removes the last remaining degeneracy, left after accounting for the spin-orbit interaction,—the degeneracy of levels with the same values of n and j , but different $l = j \pm 1/2$. Thus, the numerical value $L_{21} = +0.030$ and from (XII.166)–(XII.168) one obtains the following value for the difference of the levels $2s_{1/2}$ and $2p_{1/2}$ of the hydrogen atom:

$$E_{20\frac{1}{2}} - E_{21\frac{1}{2}} = 0.41 mc^2 \alpha^5 \approx 1050 \text{ MHz}$$

(this difference corresponds to a frequency of 1050 MHz).

§ 124. Radiative shift of levels of mesoatoms

At the end of § 118 it was established that the effect of polarisation of the electronic vacuum plays an essential role in the radiative correction (of the second approximation) to the magnetic moment of the muon. Even more important is this effect for the radiative shift (already in the first approximation) of the levels of μ -mesohydrogen — a hydrogen-like system consisting of a proton and a μ -meson (*A. D. Galanin, I. Ya. Pomeranchuk, 1952*).

In the computation carried out in the preceding section of the shift of levels of an ordinary atom, the effect of polarisation of the electronic vacuum was taken into account, in particular, by the electron loop in diagram (121.2, *a*). If, in an analogous manner, for a mesoatom one takes into account the effect of polarisation of the muonic vacuum, then the entire calculation carries over completely to this case as well, with the sole replacement of the electron mass $m = m_e$ everywhere by the muon mass m_μ . Since the relative shift of levels turned out not to depend on the electron mass (see (123.19)), for mesohydrogen one would obtain exactly the same result.

It is easy to see, however, that a significantly larger contribution to the shift of the levels of a mesoatom comes from the effect of polarisation of the electronic vacuum. Indeed, replacing the muon loop in the electron diagram by an electronic one means replacing the muonic polarisation operator by the electronic one. But the polarisation operator $\mathcal{P}(q^2)$ is inversely proportional to the square of the particle mass (at non-relativistic values of q^2). It is clear, therefore, that the indicated replacement will lead to an increase of the effect by a factor of $(m_\mu/m_e)^2$. It is precisely this contribution that will

determine the order of magnitude of the level shift, which will be

$$\frac{\delta E}{|E|} \sim \alpha^3 \left(\frac{m_\mu}{m_e} \right)^2,$$

i.e. four orders of magnitude greater than for an ordinary atom of hydrogen²⁶.

More visually, the origin of this effect can be understood by recalling that the distortion of the Coulomb potential by polarisation of the electronic vacuum extends over distances $\sim 1/m_e$ (§ 114). In an ordinary atom of hydrogen, the electron is located at distances $\sim 1/(m_e \alpha)$, i.e. outside the main region of distortion of the field; whereas in

²⁶For an analogous reason, the contribution of polarisation of the muonic vacuum to the shift of levels of an ordinary atom of hydrogen will be, on the contrary, negligibly small.

mesohydrogen the muon is located at distances $\sim 1/(m_\mu\alpha)$, which just fall within this region.

For an exact computation of the shift of the levels of a mesoatom, one cannot, however, use the approximate non-relativistic expression for the polarisation operator, as was done in formula (123.7) used for computing the shift of levels of an ordinary atom. The point is that the characteristic momenta of the muon in the atom of mesohydrogen are $|\mathbf{p}| \sim \alpha m_\mu$. For the muon, such momenta are non-relativistic, but with respect to the electron they are already relativistic.

We must therefore make use of the full relativistic expression (114.5) for the effective potential of the nuclear field, distorted by polarisation of the electronic vacuum. The shift of levels is determined by averaging over the wave function of the muon in the atom:

$$\delta E_{nl} = -|e| \int |\psi_{nl}|^2 \delta\Phi(r) d^3x = -|e| \int R_{nl}^2(r) \delta\Phi(r) r^2 dr, \quad (\text{XII.169})$$

where R_{nl} is the radial part of the Coulomb (non-relativistic) wave function. For a hydrogen-like ion with nuclear charge $Z|e|$, the functions $R_{nl}(r)$ depend on r only through the dimensionless combination $\rho = Z\alpha m_\mu r$ (the distance measured in Coulomb units). Taking this into account and substituting $\delta\Phi(r)$ from (114.5) (with charge $Z|e|$ in place of e_1), we bring the integral (124.1) to the form

$$\delta E_{nl} = -\frac{2}{3\pi} \alpha^3 m_\mu Z Q_{nl} \left(\frac{m_e}{Z\alpha m_\mu} \right), \quad (\text{XII.170})$$

where

$$Q_{nl}(x) = \int_0^\infty \rho d\rho \int_1^\infty R_{nl}^2(\rho) e^{-2x\rho\zeta} \left(1 + \frac{1}{2\zeta^2} \right) \frac{\sqrt{\zeta^2 - 1}}{\zeta^2} d\zeta.$$

Thus, for several of the first levels of mesohydrogen, the numerical calculation gives the following values of the relative shift:

$$\frac{\delta E_{10}}{|E_{10}|} = -6.4 \cdot 10^{-3}, \quad \frac{\delta E_{20}}{|E_{20}|} = -2.8 \cdot 10^{-4}, \quad \frac{\delta E_{21}}{|E_{21}|} = -2.0 \cdot 10^{-5}.$$

§ 125. Relativistic equation for bound states

The method employed in the preceding sections for computing radiative corrections to atomic energy levels is inapplicable for solving the problem of determining the corrections to the levels of positronium—a system of two identical particles, neither of which can be treated as fixed relative to the other as a source of external field.

A systematic method for solving this problem is based on the fact that the energy levels of bound states are poles of the exact amplitude of mutual scattering of two particles (the amplitude considered as a function of the total energy of the particles in the centre-of-inertia system). Indeed, positronium in each of its discrete states can be regarded as an “intermediate particle” of definite mass, through whose formation the process of electron–positron scattering can proceed; each such “single-particle” intermediate state corresponds to a pole of the scattering amplitude (these poles lie, of course, in the non-physical region of 4-momenta of the scattering particles).

According to (106.17), the exact scattering amplitude is constructed from the exact “four-tailed” vertex part $\Gamma_{ik,lm}$ and the polarisation amplitudes u of the particles. The latter, evidently, have no relation to the pole singularities, and therefore it is more convenient not to consider them at all, speaking instead of the poles of the vertex part itself, i.e. of the function

$$\Gamma_{ik,lm}(p'_-, -p_+; p_-, -p'_+), \quad (\text{XII.171})$$

where the designations of the 4-momenta of the external lines of the diagram (106.12) correspond to the scattering of a positron on an electron.

Let us emphasise that the assertion about the existence of poles refers precisely to the exact scattering amplitude or to the exact vertex part; in each separate term of the perturbation-theory series the required pole is absent. The latter is already obvious from the fact that in the Feynman diagrams of every approximation only electron (and photon) lines appear, but not lines of the “composite particle”—positronium as a whole. From this it follows in turn that the computation of the scattering amplitude near its poles requires the summation of an infinite sequence of diagrams. Let us elucidate which diagrams specifically enter into this sequence.

In the first non-vanishing (first in α) approximation of perturbation theory for the vertex part (XII.171), there correspond two second-order diagrams:

$$(\text{XII.172})$$

or in analytic form:

$$\Gamma_{ik,lm} = -e^2 \gamma_{il}^\mu \gamma_{km}^\nu D_{\mu\nu}(p_- - p'_-) + e^2 \gamma_{im}^\mu \gamma_{kl}^\nu D_{\mu\nu}(p_- + p_+). \quad (\text{XII.173})$$

In the next (second in α) approximation there are already 10 fourth-order diagrams:

$$(\text{XII.174})$$

and five more diagrams differing by the interchange $p_- \leftrightarrow -p'_+$. All these diagrams have, compared with diagrams (XII.172), an extra power of smallness $e^2 = \alpha$. We shall show, however, that in diagram (125.4,a) this extra power of smallness is compensated by a small (for small momenta of the electron and positron) denominator.

We shall consider all quantities in the centre-of-inertia system. Since, however, the 4-momenta of the external lines of the diagrams are not assumed to be physical (i.e. $p^2 \neq m^2$), although in this system $\mathbf{p}_+ = -\mathbf{p}_-$, we have $\varepsilon_+ \neq \varepsilon_-$. Thus, the 4-momenta of the external lines are

$$\begin{aligned} p_- &= (\varepsilon_-, \mathbf{p}), & p_+ &= (\varepsilon_+, -\mathbf{p}), \\ p'_- &= (\varepsilon'_-, \mathbf{p}'), & p'_+ &= (\varepsilon'_+, -\mathbf{p}'), \\ \varepsilon_- + \varepsilon_+ &= \varepsilon'_- + \varepsilon'_+. \end{aligned} \quad (\text{XII.175})$$

The binding energy of the electron and positron in positronium is $\sim m\alpha^2$. Therefore, in the neighbourhood of the poles of the scattering amplitude that is of interest to us,

$$|\mathbf{p}| \sim |\mathbf{p}'| \sim m\alpha \ll m, \quad |\varepsilon_- - m| \sim |\varepsilon_+ - m| \sim \mathbf{p}^2/m \sim m\alpha^2, \dots \quad (\text{XII.176})$$

The contribution to the vertex part from diagram (125.4,a) is

$$\Gamma_{ik,lm}^{(1a)} = -ie^4 \int (\gamma^\lambda G(q) \gamma^\mu)_{il} (\gamma^\nu G(q-p_- - p_+) \gamma^\rho)_{km} \times D_{\lambda\rho}(q-p'_-) D_{\mu\nu}(p_- - q) \frac{d^4 q}{(2\pi)^4}. \quad (\text{XII.177})$$

In the integral (XII.177) the essential region of values of $q^\mu = (q_0, \mathbf{q})$ is that near the poles of both electron Green's functions G simultaneously. In this region $|\mathbf{q}|$ and $|q_0 - m|$ are small, and the electron propagators are

$$G(q) = \frac{\gamma^0 q_0 - \boldsymbol{\gamma} \mathbf{q} + m}{(q_0 + m)(q_0 - m) - \mathbf{q}^2 + i0} \approx \frac{\gamma^0 + 1}{2} \left[q_0 - m - \frac{\mathbf{q}^2}{2m} + i0 \right]^{-1},$$

$$G(q - p_- - p_+) \approx \frac{\gamma^0 - 1}{2} \left[q_0 - \varepsilon_- - \varepsilon_+ + m + \frac{\mathbf{q}^2}{2m} - i0 \right]^{-1}. \quad (\text{XII.178})$$

The poles of these two expressions lie on opposite sides of the real axis in the plane of the complex variable q_0 ; closing the integration contour along this axis, say, in the upper half-plane, we compute the integral over q_0 by the residue at the corresponding pole²⁷. As a result we find that

$$\Gamma^{(1a)} \sim e^4 \int \frac{d^3 q}{(q - p'_-)^2 (p_- - q)^2 (2m - \varepsilon_- - \varepsilon_+ + \mathbf{q}^2/m)}$$

and from this, taking into account (XII.176), the estimate

$$\Gamma^{(1a)} \sim \alpha^2 \frac{(m\alpha)^3}{(m\alpha)^4 m\alpha^2} = \frac{1}{m^2 \alpha}.$$

The contribution to Γ from the second-order diagram (125.2,a) (the first term in (XII.173)) is of the same order of magnitude, which proves the assertion made above regarding the smallness of diagram (125.4,a). An analogous situation occurs in all subsequent approximations of perturbation theory.

Thus, the computation of the vertex part of interest to us near its poles requires the summation of the infinite sequence of “anomalously large” diagrams with intermediate states of the type of the internal lines of diagram (125.4,a). It is characteristic of these diagrams that they can be cut between the external lines p_- , $-p_+$ and p'_- , $-p'_+$ into parts connected to each other only by two electron lines²⁸. The totality of all diagrams not satisfying this condition we shall call the *compact vertex part* and denote by $\tilde{\Gamma}_{ik,lm}$; since anomalously large diagrams are not included in it, this quantity can be computed by ordinary perturbation theory. Thus, in the first approximation $\tilde{\Gamma}$ is determined by both second-order diagrams (XII.172), and in the second—by the family of fourth-order diagrams (all diagrams, except for (125.4,a,b)).

Classifying the non-compact vertex parts by the number of their “double connections”, one can represent the full Γ in the form of an infinite series:

²⁷For diagram (125.4,e), which differs from (125.4,a) only in the mutual direction of the electron lines, both poles would lie on the same side of the real axis, so that after the approximations made the integral would vanish.

²⁸Such a definition includes in itself all anomalously large diagrams, but alongside them also some “normal” ones, for instance diagram (125.4,b).

(XII.179)

where all the internal thick solid lines are exact propagators $\mathcal{G}$ (a series of this type is often called *ladder diagrams*). To sum this series, we “multiply” it from the left by one more $\tilde{\Gamma}$ ²⁹:

Comparing now this series with the original series (XII.179), we see that

(XII.180)

This graphical equation is equivalent to the following integral equation:

$$\begin{aligned} i\Gamma_{ik,lm}(p'_-, -p_+; p_-, -p'_+) &= i\tilde{\Gamma}_{ik,lm}(p'_-, -p_+; p_-, -p'_+) + \\ &+ \int \tilde{\Gamma}_{ir,sm}(p'_-, q - p'_+ - p'_-; q, -p'_+) \mathcal{G}_{st}(q) \mathcal{G}_{nr}(q - p'_+ - p'_-) \times \\ &\times \Gamma_{lk,ln}(q, -p_+; p_-, -p'_+) \frac{d^4q}{(2\pi)^4}. \end{aligned} \quad (\text{XII.181})$$

The functions $\tilde{\Gamma}$ and $\mathcal{G}$ are computed by perturbation theory, after which equation (XII.181) gives, in principle, the possibility of computing Γ to any required accuracy.

For the determination of energy levels, however, it suffices to know only the position of the poles of the function Γ . Near the poles $\Gamma \gg \tilde{\Gamma}$, so that the first term on the right-hand side of (XII.181) (the second diagram on the right in (XII.180)) can be neglected, and the equation becomes homogeneous in Γ . In this equation the variables p_+ , p_- , and also the indices k, l become parameters, the dependence on which remains arbitrary (is not determined by the equation itself). Dropping these parameters (and with them the primes on the remaining variables p'_+, p'_-), we obtain the equation

$$i\Gamma_{i,m}(p_-; -p_+) = \int \tilde{\Gamma}_{ir,sm}(p_-, q - p_+ - p_-; q, -p_+) \mathcal{G}_{st}(q) \times \mathcal{G}_{nr}(q - p_+ - p_-) \Gamma_{l,n}(q; q - p_+ - p_-) \quad (\text{XII.182})$$

(E. E. Salpeter, H. A. Bethe, 1951).

Written in the centre-of-inertia system ($\mathbf{p}_+ + \mathbf{p}_- = 0$), equation (XII.182) has solutions only for certain values of $\varepsilon_+ + \varepsilon_-$, which give the energy levels of positronium. The function $\Gamma_{i,m}$ plays only an auxiliary role. Instead of it, it is more convenient to introduce another function:

$$\chi_{sr}(p_1, p_2) = \mathcal{G}_{st}(p_1) \Gamma_{t,n}(p_1; p_2) \mathcal{G}_{nr}(p_2). \quad (\text{XII.183})$$

Then equation (XII.182) takes the form

$$i[\mathcal{G}^{-1}(p_-) \chi(p_-, -p_+) \mathcal{G}^{-1}(-p_+)]_{im} = \int \tilde{\Gamma}_{ir,sm}(p_-, q - p_+ - p_-; q, -p_+) \chi_{sr}(q, q - p_+ - p_-) \frac{d^4q}{(2\pi)^4} \quad (\text{XII.184})$$

²⁹That is, we multiply all terms of the series by $\tilde{\Gamma}$ and two $\mathcal{G}$, and carry out the corresponding integration over 4-momenta of the new internal connections.

in which $\tilde{\Gamma}$ appears as the kernel of an integral operator. As was already mentioned, $\tilde{\Gamma}$ can be computed by perturbation theory; the same applies, of course, also to the function $\mathcal{G}^{-1}$.

Let us show that in the first (in α) approximation of perturbation theory equation (XII.184) reduces, as one should expect, to the non-relativistic Schrödinger equation for positronium.

In the first non-relativistic approximation $\tilde{\Gamma}$ is determined by diagram (125.2,a) alone (the annihilation diagram (125.2,b) vanishes in this approximation)³⁰. As for the analogous reason in § 83, it is convenient to choose the photon propagator in the Coulomb gauge (76.12), (76.13), keeping in it only the component D_{00} . Then

$$\begin{aligned}\tilde{\Gamma}_{ir, sm}(p_-, q - p + p_-; q, -p_+) &= -e^2 \gamma_{is}^0 \gamma_{rm}^0 D_{00}(q - p_-) = \\ &= -U(\mathbf{q} - \mathbf{p}_-) \gamma_{is}^0 \gamma_{rm}^0,\end{aligned}$$

where

$$U(\mathbf{q}) = -4\pi e^2 / \mathbf{q}^2$$

is the Fourier component of the potential energy of Coulomb interaction of the positron and electron. Equation (XII.184) takes the form

$$i\chi_{im}(p_-, -p_+) = \left[G(p_-) \gamma^0 \int U(\mathbf{q} - \mathbf{p}_-) \chi(q, q - p_+ - p_-) \frac{d^4 q}{(2\pi)^4} \cdot \gamma^0 G(-p_+) \right]_{im}, \quad (\text{XII.185})$$

where the exact propagators $\mathcal{G}$ have also been replaced by the free-electron propagators G . For the latter we have the approximate expressions (cf. (XII.178))

$$G(p_-) \approx \frac{1 + \gamma^0}{2} g(p_-), \quad G(-p_+) \approx \frac{1 - \gamma^0}{2} g(p_+),$$

where the matrix factors have been separated out, and $g(p)$ is a scalar function:

$$g(p) = [\varepsilon - m - \mathbf{p}^2/(2m) + i0]^{-1}. \quad (\text{XII.186})$$

Upon substituting these expressions into (XII.185), we notice that all matrix elements different from zero of

$$\left[\frac{1 + \gamma^0}{2} \gamma^0 \chi \gamma^0 \frac{1 - \gamma^0}{2} \right]_{im} = \left[\frac{\gamma^0 + 1}{2} \chi \frac{\gamma^0 - 1}{2} \right]_{im}$$

coincide with the elements $-\chi_{im}$. Therefore the matrix equation (XII.185) is equivalent to an equation for the scalar function:

$$i\chi(p_-, -p_+) = -g(p_-) g(p_+) \int U(\mathbf{q} - \mathbf{p}_-) \chi(q, q - p_+ - p_-) \frac{d^4 q}{(2\pi)^4}. \quad (\text{XII.187})$$

Let us now introduce, in place of p_+ , p_- , the variables

$$p \equiv (\varepsilon, \mathbf{p}) = \frac{p_- - p_+}{2}, \quad P = p_- + p_+$$

³⁰We recall that the velocities of the particles in positronium $v/c \sim \alpha$. In this sense the expansions in α and in $1/c$ are interrelated.

(4-momenta of relative motion of the particles and of positronium as a whole). In the centre-of-inertia system

$$P = (E + 2m, \mathbf{0}),$$

where the total energy is denoted by $E + 2m$, i.e. E is the energy level measured from the rest mass. Expressing things in terms of these variables, we rewrite (XII.187) in the form

$$\begin{aligned} i\chi(p, P) &= \\ &= -g\left(p + \frac{P}{2}\right)g\left(-p + \frac{P}{2}\right)\int U(\mathbf{q} - \mathbf{p}_-) \chi\left(q - \frac{P}{2}, P\right) \frac{d^4q}{(2\pi)^4} = \\ &= -g\left(p + \frac{P}{2}\right)g\left(-p + \frac{P}{2}\right)\int U(\mathbf{q}' - \mathbf{p}) \chi(q', P) \frac{d^4q'}{(2\pi)^4}. \end{aligned}$$

In this equation P enters only as a parameter, and the function χ enters the right-hand side only in the form of the integral

$$\psi(\mathbf{q}) = \int_{-\infty}^{\infty} \chi(q, P) dq_0.$$

Integrating both sides of the equation over $d\varepsilon$, we obtain a closed equation for ψ :

$$\psi(\mathbf{p}) = -\frac{1}{2\pi i} \int_{-\infty}^{\infty} g\left(p + \frac{P}{2}\right)g\left(-p + \frac{P}{2}\right)d\varepsilon \int U(\mathbf{q} - \mathbf{p}) \psi(\mathbf{q}) \frac{d^3q}{(2\pi)^3},$$

where

$$g\left(\pm p + \frac{P}{2}\right) = \left[\pm\varepsilon + \frac{E}{2} - \frac{\mathbf{p}^2}{2m} + i0\right]^{-1}.$$

Closing the integration contour over $d\varepsilon$, say, in the upper half-plane of complex ε , we compute the integral by the residue at the corresponding pole, and finally obtain

$$\left(\frac{\mathbf{p}^2}{2m} - E\right)\psi(\mathbf{p}) + \int U(\mathbf{q} - \mathbf{p}) \psi(\mathbf{q}) \frac{d^3q}{(2\pi)^3} = 0. \quad (\text{XII.188})$$

This is indeed the Schrödinger equation for positronium in the momentum representation (see III, (130.4)).

If we had restricted ourselves to diagrams (XII.172) for $\tilde{\Gamma}$ but had included in them (and also in $\mathcal{G}$) the next terms of the expansion in $1/c$, we would have obtained the Breit equation (see § 83). The inclusion of diagrams from (XII.174) (together with subsequent terms of the expansion in $1/c$) gives radiative corrections to the levels of positronium; however, the computations become very involved.

We present the result of the computation with these corrections of the difference of the ground levels of ortho- and parapositronium³¹:

$$E(^3S_1) - E(^1S_0) = \alpha^2 \frac{me^4}{2\hbar^2} \left\{ \frac{7}{6} - \left(\frac{16}{9} + \ln 2 \right) \frac{\alpha}{\pi} - i \frac{\alpha}{2} \right\}. \quad (\text{XII.189})$$

³¹Karplus R., Klein A. // Phys. Rev. 1952. V. 87. P. 848.

The first term in the curly brackets is the fine-structure splitting (see Problem 2, § 84). The second term is the radiative correction to the splitting of levels. The imaginary part of the difference is related to the annihilation probability of parapositronium (see (89.14)), i.e. to the complexity of the level 1S_0 ; for parapositronium the width of the level turns out to be of the same order of magnitude as the radiative correction to its real part.

§ 126. Double dispersion relation

The next in order of complexity beyond the vertex part with three external lines is a block with four tails. In quantum electrodynamics, three such simplest diagrams are possible:

(XII.190)

The first of these describes photon-photon scattering. The remaining two represent individual terms of the radiative corrections — to the scattering of a photon on an electron (diagram *b*) and to the scattering of an electron on an electron (diagram *c*).

This section is devoted to the study of certain general properties of diagrams of this kind. But for simplicity and concreteness we shall conduct the exposition as applied to a specific diagram — (126.1,*a*).

The momenta of the lines of such a diagram are denoted as follows:

(XII.191)

The 4-momenta k_1, k_2, k_3, k_4 correspond to real photons, so their squares are zero.

Having separated the dependence on the photon polarisations, the amplitude M_{fi} corresponding to diagram (XII.191) can be expressed through several scalar functions of the 4-momenta. These are the *invariant amplitudes*, discussed in § 70; the concrete extraction of them for the scattering of a photon on a photon will be carried out in the following section. Being scalar, they depend only on the scalar variables, as which one can choose, for example, any two of the quantities

$$s = (k_1 + k_2)^2, \quad t = (k_1 - k_3)^2, \quad u = (k_1 - k_4)^2, \quad s + t + u = 0; \quad (\text{XII.192})$$

below we choose s and t as the independent variables.

Each of the invariant amplitudes (which we denote here by the same letter M) can be represented as an integral of the form

$$M = \int \frac{iB d^4q}{[q^2 - m^2][(q - k_4)^2 - m^2][(q - k_1 - k_2)^2 - m^2][(q - k_2)^2 - m^2]}, \quad m^2 \rightarrow m^2 - i0, \quad (\text{XII.193})$$

where B is some function of all the 4-momenta; the factors in the denominator come from the propagators of the four virtual electrons.

At sufficiently small s and t the amplitudes M are real (more precisely, can be made so by an appropriate choice of the phase factor). Indeed, the smallness of s ensures the impossibility of the production of real particles (an electron–positron pair) by two photons in the s -channel, and the smallness of t ensures the same impossibility in the

t -channel³². In other words, in both channels there are no real intermediate states which could, according to the unitarity condition, give rise to an imaginary part of the amplitude.

Let us now increase s at fixed (small) values of t . At $s \geq 4m^2$ the amplitude M acquires an imaginary part, associated with the possibility of the production of a pair by two photons in the s -channel. Therefore for M one can write a dispersion relation “in the variable s ”:

$$M(s, t) = \frac{1}{\pi} \int_{4m^2}^{\infty} \frac{A_{1s}(s', t)}{s' - s - i0} ds', \quad (\text{XII.194})$$

where $A_{1s}(s, t)$ denotes the imaginary part of $M(s, t)$.

As for any diagram of the form

$A_{1s}(s, t)$ is computed by the rule (115.9), replacing in the integral (XII.193) the corresponding pole factors by δ -functions:

$$2iA_{1s}(s, t) = (2\pi i)^2 \int \frac{iB \delta(q^2 - m^2) \delta[(q - k_1 - k_2)^2 - m^2]}{[(q - k_4)^2 - m^2][(q - k_2)^2 - m^2]} d^4q, \quad (\text{XII.195})$$

where the integration is performed over the half of q -space in which $q^0 > 0$.

We can make an essential further step by noting that the integral (XII.195) has the structure (in the sense of its pole factors) of an amplitude of a reaction of the same type as that depicted by the diagram

Therefore the analytic properties of $A_{1s}(s, t)$ as a function of t are similar to the analytic properties of this amplitude. In particular, the function $A_{1s}(s, t)$ may acquire (upon increasing t) an imaginary part only when both factors in the denominator vanish simultaneously — which would not happen immediately upon reaching the value $t = 4m^2$, the threshold for pair production in the t -channel. The point is that the presence of δ -functions in the integrand constrains the region of integration in q -space, which may turn out to be incompatible with the value $t = 4m^2$. The extent of the integration region depends on the values of s (the arguments of the δ -functions contain k_1 and k_2). Therefore both s and the boundary value $t = t_c(s)$, at which the function $A_{1s}(s, t)$ becomes complex, depend on s .

Similarly to the way in which the function $M(s, t)$ is expressed through its imaginary part $A_{1s}(s, t)$ by formula (XII.194), the function $A_{1s}(s, t)$ is in turn expressed through $A_2(s, t) = \text{Im } A_{1s}(s, t)$ by a dispersion relation “in the variable t ”:

$$A_{1s}(s, t) = \frac{1}{\pi} \int_{t_c(s)}^{\infty} \frac{A_2(s, t')}{t' - t - i0} dt'. \quad (\text{XII.196})$$

Substituting now (XII.196) into (XII.194), we obtain the *double dispersion relation*, or *Mandelstam representation*, for the amplitude $M(s, t)$:

$$M(s, t) = \frac{1}{\pi^2} \int_{4m^2}^{\infty} \int_{t_c(s')}^{\infty} \frac{A_2(s', t')}{(s' - s - i0)(t' - t - i0)} dt' ds', \quad (\text{XII.197})$$

(S. Mandelstam, 1958).

³²In the t -channel, the incoming lines must be lines 1 and 3, so that the initial photon 4-momenta would be k_1 and $-k_3$. The physical regions for photon-photon scattering in the variables s, t are the hatched sectors in Fig. 8 (p. 299). Thus, the s -channel corresponds to the region in which $s > 0, t < 0, u < 0$.

The function $A_2(s, t)$ is called the *double spectral density* of the function $M(s, t)$. It can be obtained from the integral (XII.195) by a repeated application of the replacement rule (115.9):

$$l_1 = q, \quad l_2 = q - k_1, \quad l_3 = q - k_2, \quad l_4 = q - k_1 - k_2, \quad (\text{XII.198})$$

denoting for brevity we obtain

$$(2i)^2 A_2(s, t) = (2\pi i)^4 \int iB \delta(l_1^2 - m^2) \delta(l_2^2 - m^2) \delta(l_3^2 - m^2) \delta(l_4^2 - m^2) d^4 q, \quad (\text{XII.199})$$

where the integration is performed over the region $q^0 > 0$.

It should be borne in mind, however, that formula (XII.199) has only a symbolic meaning. The point is that the region $s > 0, t > 0$ is unphysical. Correspondingly, in this region the quantities $l_1, l_2, \dots$ for real q turn out to be, generally speaking, complex; the concept of a δ -function for complex values of its argument is not fully defined. It would be more precise to speak directly of taking the residues at the corresponding poles of the original integral (XII.193). In our case, however, this is not important. The condition that four denominators in (XII.193) vanish, or four arguments of δ -functions, fully determines the components of the 4-vector q . Passing to integration over $l_1^2, l_2^2, \dots$ (see below) and formally operating with the integral (XII.199) by the usual rules, we shall find (up to sign) the expression for A_2 .

For further calculations we choose the centre-of-inertia frame (in the s -channel). Then

$$k_1 = (\omega, \mathbf{k}), \quad k_2 = (\omega, -\mathbf{k}), \quad k_3 = (\omega, \mathbf{k}'), \quad k_4 = (\omega, -\mathbf{k}'), \quad (\text{XII.200})$$

$$s = 4\omega^2, \quad t = -(\mathbf{k} - \mathbf{k}')^2 = -4\omega^2 \sin^2(\theta/2), \quad u = -(\mathbf{k} + \mathbf{k}')^2 = -4\omega^2 \cos^2(\theta/2), \quad (\text{XII.201})$$

where θ is the angle between $\mathbf{k}$ and $\mathbf{k}'$ (the scattering angle). The x axis of the spatial Cartesian coordinates is directed along the vector $\mathbf{k} + \mathbf{k}'$, and the y axis along $\mathbf{k} - \mathbf{k}'$ ³³.

Let us now transform the integral (XII.199), choosing as the squares $l_1^2, l_2^2, \dots$ the new variables of integration (instead of the four components of q). We have

$$\frac{\partial(l_i^2)}{\partial q^\mu} = 2l_{i\mu}, \quad \dots,$$

so the Jacobian of the transformation is

$$\frac{\partial(l_1^2, l_2^2, l_3^2, l_4^2)}{\partial(q^0, q_x, q_y, q_z)} = 16D,$$

where D is the determinant composed of the 16 components of the 4-vectors l_1, l_2, l_3, l_4 . The integration in (XII.199) reduces simply to replacing the functions B and D in the

³³For $t > 0$ the quantity $(\mathbf{k} - \mathbf{k}')^2 < 0$, i.e. the vector $\mathbf{k} - \mathbf{k}'$ is imaginary. This difficulty can, however, be easily circumvented by expanding all vectorial expressions for $t < 0$ and subsequently performing the analytic continuation to $t > 0$.

integrand by their values at ³⁴

$$l_1^2 = l_2^2 = l_3^2 = l_4^2 = m^2. \quad (\text{XII.202})$$

From the conditions $l_1^2 = l_4^2 = m^2$ we obtain, as in § 115,

$$q^0 = \omega, \quad \mathbf{q}^2 = \omega^2 - m^2. \quad (\text{XII.203})$$

The remaining two conditions give

$$(q - k_1)^2 - m^2 = -2qk_4 = -2\omega^2 - 2\mathbf{q}\mathbf{k}' = 0,$$

$$(q - k_2)^2 - m^2 = -2\omega^2 - 2\mathbf{q}\mathbf{k} = 0,$$

so that

$$\mathbf{q}\mathbf{k} = \mathbf{q}\mathbf{k}' = -s/4,$$

or in components:

$$q^0 = \omega, \quad q_x = -\frac{s}{2(s+t)}, \quad q_y = 0, \quad q_z = \pm \sqrt{\omega^2 - m^2 - q_x^2} = \pm \left[\frac{st - 4m^2(s+t)}{4(s+t)} \right]^{1/2}. \quad (\text{XII.204})$$

Thus the integral (XII.199) equals

$$A_2(s, t) = \frac{\pi^4}{4D} \sum (-iB), \quad (\text{XII.205})$$

where the summation is performed over the two values of $\mathbf{q}$ from (XII.204).

The determinant D can be written with the help of the unit antisymmetric tensor:

$$D = e_{\mu\nu\rho\sigma} l_1^\mu l_2^\nu l_3^\rho l_4^\sigma = -e_{\mu\nu\rho\sigma} q^\mu k_4^\nu k_2^\rho k_1^\sigma = -e_{\mu\nu\rho\sigma} (q-k_1)^\mu (k_4-k_1)^\nu (k_2-k_1)^\rho k_1^\sigma$$

(in the transformation the antisymmetry of $e_{\mu\nu\rho\sigma}$ has been used). Noting that of the four factors only the temporal component of q contributes, we find

$$D = -\omega \mathbf{q} \cdot [(\mathbf{k} + \mathbf{k}')(\mathbf{k} - \mathbf{k}')].$$

Expanding this expression for $t < 0$ and then continuing analytically to $t > 0$, we obtain

$$D = -\omega q_z \sqrt{s+t} \sqrt{-t} \rightarrow \pm \frac{i}{4} \{st[st - 4m^2(s+t)]\}^{1/2}. \quad (\text{XII.206})$$

The choice of sign in this expression can be made on the basis of the following considerations. We set $B = 1$ for simplicity. Then it is clear that in the physical region ($s > 0, t < 0$) we have $A_{1s}(s, t) < 0$ ³⁵. Indeed, both denominators in the integrand of (XII.195) have the same (negative) sign:

$$(q - k_1)^2 - m^2 = -2\omega^2 - 2\mathbf{q}\mathbf{k}' < -2\omega(\omega - |\mathbf{q}|) < 0,$$

³⁴This method of integration automatically takes into account only one of the zeros of the arguments of the δ -functions.

³⁵This is, of course, not accidental. The negativity of A_{1s} follows in fact from the unitarity condition, as is especially clear at $t = 0$, when A_{1s} determines the total cross section.

$$(q - k_2)^2 - m^2 = -2\omega^2 - 2\mathbf{q}\mathbf{k} < -2\omega(\omega - |\mathbf{q}|) < 0$$

(here we have used the fact that, by virtue of two δ -functions in the numerator, the relation (XII.203) holds, and therefore $|\mathbf{q}| < \omega$). From (XII.196)

it is then clear that the function $A_2(s, t)$ must also be negative for $s > 0, t > 0$ (taking into account that, according to (XII.205), this function is of constant sign). This means that in (XII.206) one must choose the upper sign, so that finally

$$A_2 = -\pi^2 \frac{\sum B}{\{st[st - 4m^2(s + t)]\}^{1/2}}. \quad (\text{XII.207})$$

Since by its very meaning the function $A_2(s, t)$ must be real, in addition to the positivity of s and t there is also the condition of positivity of the expression in the square brackets in the denominator:

$$st - 4m^2(s + t) \geq 0, \quad s > 0, \quad t > 0. \quad (\text{XII.208})$$

These inequalities define the region over which the integration in the double dispersion integral (XII.197) should be carried out (hatched in Fig. 23). Its boundary is the curve

$$st - 4m^2(s + t) = 0$$

with asymptotes $s = 4m^2$ and $t = 4m^2$.

The dispersion relations in the form (XII.194) and (XII.197) do not yet take into account the conditions of renormalisation, and upon their literal application the integrals would diverge and would require regularisation. The renormalisation condition for the amplitude $M(s, t)$ consists of the requirement

$$M(0, 0) = 0. \quad (\text{XII.209})$$

Indeed, the amplitude of the scattering of a photon on a photon must vanish when $k_1 = k_2 = k_3 = k_4 = 0$ (and therefore $s = t = 0$), since $k = 0$ signifies a potential constant in time and space, to which no physical field corresponds (we shall discuss this condition in more detail in the following section).

To account for this condition automatically, one must write a *subtracted* dispersion relation (similarly to the passage from (111.8) to (111.13)). We arrive at such a relation naturally by first performing an identical transformation of (XII.197) with the help of the identity

$$\frac{1}{(s' - s)(t' - t)} = \frac{st}{(s' - s)(t' - t)s't'} + \frac{s}{(s' - s)s't'} + \frac{t}{(t' - t)s't'} + \frac{1}{s't'}.$$

Substituting it into the integrand of (XII.197), we obtain

$$M(s, t) = \frac{st}{\pi^2} \iint \frac{A_2(s', t') ds' dt'}{(s' - s)(t' - t)s't'} + \frac{s}{\pi} \int \frac{f(s') ds'}{(s' - s)s'} + \frac{t}{\pi} \int \frac{g(t') dt'}{(t' - t)t'} + C,$$

where

$$f(s) = \frac{1}{\pi} \int \frac{A_2(s, t')}{t'} dt', \quad g(t) = \frac{1}{\pi} \int \frac{A_2(s', t)}{s'} ds',$$

$$C = \frac{1}{\pi^2} \iint \frac{A_2(s', t')}{s' t'} ds' dt'.$$

The last equalities, however, would make sense only under the condition that all integrals converge. Otherwise, the functions $f(s)$, $g(t)$ and the constant C should be assigned definite values, prescribed by the renormalisation condition. Namely, one must set

$$C = 0, \quad f(s) = A_{1s}(s, 0), \quad g(t) = A_{1t}(0, t),$$

where A_{1t} is the imaginary part appearing upon increasing t at fixed small s , similarly to A_{1s} , the imaginary part appearing upon increasing s at fixed small t . The first of these equalities is obvious: $C = M(0, 0) = 0$. The second (and analogously the third) follows from comparison of the equality

$$M(s, 0) = \frac{s}{\pi} \int \frac{f(s') ds'}{(s' - s)s'}$$

with the single-variable dispersion relation (XII.194), written “with subtraction” in accordance with the condition (XII.209):

$$M(s, 0) = \frac{s}{\pi} \int \frac{A_{1s}(s', t)}{(s' - s)s'} ds'. \quad (\text{XII.210})$$

Thus, the final *subtracted* double dispersion relation is:

$$M(s, t) = \frac{st}{\pi^2} \iint \frac{A_2(s', t')}{(s' - s)(t' - t)s' t'} ds' dt' + \frac{s}{\pi} \int \frac{A_{1s}(s', 0)}{(s' - s)s'} ds' + \frac{t}{\pi} \int \frac{A_{1t}(0, t')}{(t' - t)t'} dt'. \quad (\text{XII.211})$$

If the values of s , t themselves lie in the region of integration of the integrals (XII.210), (XII.211), then as always one must understand as the limit

$$s \rightarrow s + i0, \quad t \rightarrow t + i0. \quad (\text{XII.212})$$

§ 127. Photon–photon scattering

The scattering of light by light (in vacuum) is a specifically quantum-electrodynamic process; in classical electrodynamics it is absent, owing to the linearity of Maxwell’s equations³⁶.

In quantum electrodynamics, photon–photon scattering is described as the result of the creation by two initial photons of a virtual electron–positron pair and its subsequent annihilation into the final quanta. The amplitude of this process (in the first non-vanishing approximation) is represented by six “box” diagrams with all possible relative arrangements of their four external legs. These diagrams are

(XII.213)

³⁶In the limiting case of low frequencies, this process was first considered by *H. Euler* (1936), and in the ultra-relativistic case by *A. I. Akhiezer* (1937). The complete solution of the problem was given by *R. Karplus* and *M. Neumann* (1951).

and three further diagrams differing from these only in the reversal of the direction of traversal of the internal electron loop. The contribution of the latter coincides with the contribution of diagrams (127.1), and therefore the full scattering amplitude is

$$M_{fi} = 2(M^{(a)} + M^{(b)} + M^{(c)}), \quad (\text{XII.214})$$

where $M^{(a)}$, $M^{(b)}$, $M^{(c)}$ are the contributions of diagrams a , b , c .

According to (64.19) the scattering cross section is

$$d\sigma = \frac{1}{64\pi^2} |M_{fi}|^2 \frac{do'}{(2\omega)^2}, \quad (\text{XII.215})$$

where do' is an element of solid angle for the direction of $\mathbf{k}'$ in the centre-of-mass frame. The scattering angle in this frame will be denoted by θ .

Invariant amplitudes. Separating out the polarisation factors of the four photons, we write M_{fi} in the form

$$M_{fi} = e_1^\lambda e_2^\mu e_3^{*\nu} e_4^{*\rho} M_{\lambda\mu\nu\rho}; \quad (\text{XII.216})$$

the 4-tensor $M_{\lambda\mu\nu\rho}$ (which is called the *photon-photon scattering tensor*) is a function of the 4-momenta of all the photons. If one writes the arguments of the functions with signs corresponding to identical directions of the external legs of diagrams (127.1), then by the symmetry of the set of diagrams it is obvious that the tensor

$$M_{\lambda\mu\nu\rho}(k_1, k_2, -k_3, -k_4)$$

is symmetric with respect to any permutation of its four arguments together with a simultaneous permutation of the same four indices. By virtue of gauge invariance of the amplitude (XII.216), it must not change under the substitution $e \rightarrow e + \text{const} \cdot k$. In other words, one must have

$$k_1^\lambda M_{\lambda\mu\nu\rho} = k_2^\mu M_{\lambda\mu\nu\rho} = \dots = 0. \quad (\text{XII.217})$$

As is easy to see, it follows from this, in particular, that the expansion of the scattering tensor in powers of the 4-momenta $k_1, k_2, \dots$ must begin with terms containing fourth-order products of their components. Thus in any case

$$M_{\lambda\mu\nu\rho}(0, 0, 0, 0) = 0. \quad (\text{XII.218})$$

For the purpose of explicitly isolating the invariant amplitudes, it is convenient from the outset to choose a definite gauge for the polarisation 4-vectors e — the gauge in which

$$e_1^\mu = (0, \mathbf{e}_1), \quad e_2^\mu = (0, \mathbf{e}_2), \quad \dots \quad (\text{XII.219})$$

Then

$$M_{fi} = M_{iklm} e_{1i} e_{2k} e_{3l}^* e_{4m}^*, \quad (\text{XII.220})$$

where M_{iklm} is a three-dimensional tensor.

As the two independent polarisations, we choose for each photon circular polarisations with opposite directions of rotation, i.e. two helicity states with helicities $\lambda = \pm 1$. After this, the tensor M_{iklm} can be written in the form

$$M_{iklm} = \sum_{\lambda_1 \lambda_2 \lambda_3 \lambda_4} M_{\lambda_1 \lambda_2 \lambda_3 \lambda_4} e_{1i}^{(\lambda_1)*} e_{2k}^{(\lambda_2)*} e_{3l}^{(\lambda_3)} e_{4m}^{(\lambda_4)}; \quad (\text{XII.221})$$

the 16 quantities $M_{\lambda_1 \lambda_2 \lambda_3 \lambda_4}$ are functions of s, t, u and play the role of invariant amplitudes; they are not all independent, however.

The quantities $M_{\lambda_1 \lambda_2 \lambda_3 \lambda_4}$ are three-dimensional scalars. Spatial inversion reverses the sign of the helicities; the invariant variables s, t, u remain unchanged. Therefore the requirement

of P -invariance leads to the relations

$$M_{\lambda_1 \lambda_2 \lambda_3 \lambda_4}(s, t, u) = M_{-\lambda_1 -\lambda_2 -\lambda_3 -\lambda_4}(s, t, u). \quad (\text{XII.222})$$

Time reversal interchanges the initial and final photons without changing their helicities; the variables s, t, u again remain unchanged. Therefore the requirement of T -invariance leads to the equality

$$M_{\lambda_1 \lambda_2 \lambda_3 \lambda_4}(s, t, u) = M_{\lambda_3 \lambda_4 \lambda_1 \lambda_2}(s, t, u). \quad (\text{XII.223})$$

Finally, there is one more relation arising from the invariance of the amplitude M_{fi} with respect to the interchange of two initial or two final photons. If one performs both interchanges simultaneously ($k_1 \leftrightarrow k_2, k_3 \leftrightarrow k_4$), the variables s, t, u do not change, and the rearrangement in the polarisation indices leads to the relation

$$M_{\lambda_1 \lambda_2 \lambda_3 \lambda_4}(s, t, u) = M_{\lambda_2 \lambda_1 \lambda_4 \lambda_3}(s, t, u). \quad (\text{XII.224})$$

It is easy to verify that, by virtue of the symmetry properties (XII.222)–(XII.224), the number of independent invariant amplitudes reduces to five; as such one can choose, for example,

$$M_{++++}, \quad M_{++--}, \quad M_{+-+-}, \quad M_{+--+}, \quad M_{++++-}$$

(the indices “+” and “−” denote helicities +1 and −1).

If one substitutes in (XII.215) in place of M_{fi} one of the amplitudes $M_{\lambda_1 \lambda_2 \lambda_3 \lambda_4}$, one obtains the scattering cross section with specified polarisations of the initial and final photons. If one instead sums over final and averages over initial polarisations, one obtains the replacement

$$|M_{fi}|^2 \rightarrow \frac{1}{4} \{ 2|M_{++++}|^2 + 2|M_{++--}|^2 + 2|M_{+-+-}|^2 + 2|M_{+--+}|^2 + 8|M_{++++-}|^2 \}. \quad (\text{XII.225})$$

The symmetry relations (XII.222)–(XII.224) relate to one another the various invariant amplitudes as functions of one and the same set of variables. Further functional relations arise as a consequence of crossing symmetry (see § 78), if one takes into account that the amplitude M_{fi} in all channels describes one and the same reaction (the mutual

scattering of two photons) and therefore must not change upon passing from one channel to another.

The transition from the s -channel (corresponding to the direction of the arrows on the diagrams (127.1)) to the t -channel is effected by the interchange of 4-momenta k_2 and $-k_3$ (i.e. by the substitution of variables $s \leftrightarrow t$) and the interchange of helicity indices $\lambda_2 \leftrightarrow -\lambda_3$. Analogously, the transition from the s - to the u -channel is effected by the interchange of k_2 and $-k_4$ (whereby $s \leftrightarrow u$) and the substitution $\lambda_2 \leftrightarrow -\lambda_4$. This leads to the relations

$$\begin{aligned} M_{+--+}(s, t, u) &= M_{++++}(u, t, s), \\ M_{+--+}(s, t, u) &= M_{++++}(t, s, u), \\ M_{++++}(s, t, u) &= M_{++++}(s, u, t), \end{aligned} \quad (\text{XII.226})$$

$M_{+--+}(s, t, u)$ and M_{++++} are completely symmetric in the variables s, t, u ³⁷. It is therefore sufficient to compute only 3 of the 16 amplitudes, for example,

$$M_{++++}, \quad M_{+--+}, \quad M_{++++}.$$

The relations (XII.222)–(XII.224), (XII.226) pertain to the full amplitudes — sums of the contributions of all three diagrams (127.1). But the individual contributions are themselves related to one another by relations that are obvious from comparison of the diagrams. Thus, diagram b is obtained from diagram a by the substitution $k_2 \leftrightarrow -k_4$, $\mathbf{e}_2 \leftrightarrow \mathbf{e}_4^*$, and therefore their contributions to the invariant amplitudes are obtained from each other by the substitution of the variables $s \leftrightarrow u$ and of the indices $\lambda_2 \leftrightarrow -\lambda_4$; analogously, the contribution of diagram c is obtained from a by the substitution $t \leftrightarrow u$, $\lambda_3 \leftrightarrow -\lambda_4$.

Computation of the amplitudes. The integral $M_{fi}^{(a)}$, corresponding to diagram (127.1, a), has the form (126.4), with

$$\begin{aligned} B^{(a)} &= \frac{e^4}{\pi^2} \text{Sp}\{(\gamma e_1)(\gamma q - \gamma k_2 + m)(\gamma e_2)(\gamma q + m) \times \\ &\quad \times (\gamma e_3^*)(\gamma q - \gamma k_4 + m)(\gamma e_4^*)(\gamma q - \gamma k_1 - \gamma k_2 + m)\}. \end{aligned} \quad (\text{XII.227})$$

The integrals (126.4) diverge logarithmically. In accordance with the condition (XII.218), their regularisation is effected by subtracting the values at $k_1 = k_2 = \dots = 0$ ³⁸. The computation of the regularised integrals, however, is exceedingly cumbersome.

³⁷Here the symmetry with respect to the pair of final photons is also taken into account. Since the three variables s, t, u are not independent, it would suffice to write two arguments (for example, the first two); we retain all three solely with the aim of displaying the symmetry of their permutations more clearly.

³⁸We note that when the contributions of all diagrams are summed, the divergent parts of the integrals cancel. This is easy to verify by observing that the asymptotic form (when $q \rightarrow \infty$) of the integral is

$$M_{\lambda\mu\nu\rho}^{(a)} \propto \int \text{Sp}\{\gamma_\lambda(\gamma q)\gamma_\mu(\gamma q)\gamma_\nu(\gamma q)\gamma_\rho(\gamma q)\} \frac{d^4 q}{(q^2)^4}.$$

Summation over the diagrams means symmetrisation of this expression with respect to the indices λ, μ, ν, ρ , as a result of which it vanishes. We emphasise, however, that this cancellation is of a fortuitous character and does not remove the necessity for regularisation, although it reduces the problem to the computation of a finite quantity.

The most natural approach for computing the amplitudes of photon–photon scattering is based on the use of a double dispersion relation (*B. De Tollis*, 1964). This method accounts most fully for the symmetry of the diagrams and almost entirely eliminates the difficulties of integration³⁹.

The function $A_{1s}^{(a)}(s, t)$ (and analogously $A_{1t}^{(a)}$) for each given set of helicities $\lambda_1, \lambda_2, \lambda_3, \lambda_4$ is computed according to (126.6). In view of the presence of two δ -functions under the integral, we need only the value of $B^{(a)}$ at

$$l_1^2 \equiv q^2 = m^2, \quad l_4^2 \equiv (q - k_1 - k_2)^2 = m^2; \quad (\text{XII.228})$$

these equalities can be taken into account already when computing the trace (XII.227). But for the subsequent substitution into (126.22), we actually need only the value of $A_{1s}^{(a)}$ at $t = 0$. (This equality means that $\mathbf{k} = \mathbf{k}'$ and $k_2 = k_1$.) The integral (126.6) then takes the form

$$A_{1s}^{(a)}(s, 0) = -\frac{\pi^2}{4} \sqrt{\frac{s - 4m^2}{s}} \int \frac{B^{(a)} d\mathbf{q}}{[(q - k_2)^2 - m^2]^2}, \quad (\text{XII.229})$$

(cf. the derivation (115.10)). Introducing the angle ϑ between $\mathbf{q}$ and $\mathbf{k}$, we obtain

$$(q - k_2)^2 - m^2 = -2\omega(1 - |\mathbf{q}| \cos \vartheta) = -\sqrt{s} \left[1 - \frac{1}{2} \sqrt{s - 4m^2} \cos \vartheta \right].$$

The integrals (XII.229) are effectively expressed through elementary functions. The computation of the function $A_2^{(a)}(s, t)$ according to its definition (126.18) does not require integration at all; the expression for $B^{(a)}$ must be taken from (XII.227) for the values of q satisfying (XII.228), and also the conditions $(q - k_2)^2 = m^2$, $(q - k_1)^2 = m^2$.

After computing the functions A_{1s} , A_{1t} , A_2 , the dispersion relation (126.22) gives the amplitude directly in the form of single and double definite integrals.

We present here the final result for three of the invariant amplitudes, sufficient, according to what was said above, for

³⁹Some details of the transformation of integrals, various representations of the transcendental functions B , T , I and their limiting expressions, can be found in *De Tollis B.*//Nuovo Cimento. 1964. V. 32. P. 757; 1965. V. 35. P. 1182; *Costantini V., De Tollis B., Pistoni G.*//Nuovo Cimento. 1971. V. 2A. P. 733.

the determination also of all the remaining amplitudes:

$$\begin{aligned} \frac{1}{8\alpha^2} M_{++++} = & -1 - \left(2 + \frac{4t}{s}\right) B(t) - \left(2 + \frac{4u}{s}\right) B(u) - \\ & - \left[\frac{2(t^2 + u^2)}{s^2} - \frac{8}{s} \right] [T(t) + T(u)] + \frac{4}{t} \left(1 - \frac{2}{s}\right) I(s, t) + \\ & + \frac{4}{u} \left(1 - \frac{2}{s}\right) I(s, u) + \left[\frac{2(t^2 + u^2)}{s^2} - \frac{16}{s} - \frac{4}{t} - \frac{4}{u} - \frac{8}{tu} \right] I(t, u), \end{aligned} \quad (\text{XII.230})$$

$$\begin{aligned} \frac{1}{8\alpha^2} M_{++--} = & 1 + \left(\frac{1}{s} + \frac{1}{t} + \frac{1}{u}\right) [T(s) + T(t) + T(u)] - \\ & - 4\left(\frac{1}{u} + \frac{2}{st}\right) I(s, t) - 4\left(\frac{1}{t} + \frac{2}{su}\right) I(s, u) - 4\left(\frac{1}{s} + \frac{2}{tu}\right) I(t, u), \\ \frac{1}{8\alpha^2} M_{+-+-} = & 1 - \frac{8}{st} I(s, t) - \frac{8}{su} I(s, u) - \frac{8}{tu} I(t, u). \end{aligned}$$

Here the following notation has been introduced:

$$\begin{aligned} B(s) &= \sqrt{1 - \frac{4}{s}} \operatorname{Arsh} \frac{\sqrt{-s}}{2} - 1, \quad s < 0, \\ T(s) &= \left(\operatorname{Arsh} \frac{\sqrt{-s}}{2} \right)^2, \quad s < 0, \\ I(s, t) &= \frac{1}{4} \int_0^1 \frac{dy}{y(1-y) - \frac{s+t}{st}} \left\{ \ln[1 - i0 - sy(1-y)] + \right. \\ &\quad \left. + \ln[1 - i0 - ty(1-y)] \right\}, \end{aligned} \quad (\text{XII.231})$$

and the expressions in the regions $0 < s < 4$ and $s > 4$ are obtained from (XII.231) by analytic continuation according to the rule $s \rightarrow s + i0$, i.e. through the upper half-plane of the complex variable. (For brevity of notation in formulae (XII.230), (XII.231), and only in them, the letters s and t denote the ratios $s/m^2, t/m^2$.)

Cross section of scattering. In the limiting case of low frequencies ($\omega \ll m$), the first terms of the expansion of the invariant amplitudes in these variables are:

$$\begin{aligned} M_{++++} &\approx \frac{11e^4}{45m^4} s^2, \quad M_{+---} \approx \frac{11e^4}{45m^4} t^2, \quad M_{+-+-} \approx \frac{11e^4}{45m^4} u^2, \\ M_{++--} &\approx -\frac{e^4}{15m^4} (s^2 + t^2 + u^2), \quad M_{+++-} \approx 0. \end{aligned} \quad (\text{XII.232})$$

Substituting these expressions into formula (XII.215), we obtain the cross sections for scattering of polarised photons. The differential cross section for scattering of unpolarised photons is computed according to (XII.225) and equals (in ordinary units)

$$d\sigma = \frac{139}{4\pi^2(90)^2} \alpha^2 r_e^2 \left(\frac{\hbar\omega}{mc^2} \right)^6 (3 + \cos^2 \theta) d\theta', \quad (\text{XII.233})$$

and the total cross section⁴⁰ is

$$\sigma = \frac{973}{10125\pi} \alpha^2 r_e^2 \left(\frac{\hbar\omega}{mc^2} \right)^6 = 0.031 \alpha^2 r_e^2 \left(\frac{\hbar\omega}{mc^2} \right)^6, \quad \hbar\omega \ll mc^2. \quad (\text{XII.234})$$

In the opposite, ultra-relativistic, case the total cross section for scattering of unpolarised photons⁴¹ is

$$\sigma = 4.7 \alpha^4 \left(\frac{c}{\omega} \right)^2, \quad \hbar\omega \gg mc^2. \quad (\text{XII.235})$$

Finally, let us present the differential cross section for scattering at small angles in the ultra-relativistic case:

$$d\sigma = \frac{\alpha^4 c^2}{\pi^2 \omega^2} \ln^4 \frac{1}{\theta} d\theta, \quad \frac{mc^2}{\hbar\omega} \ll \theta \ll 1. \quad (\text{XII.236})$$

This expression is valid with logarithmic accuracy — the next term of the expansion contains a power of the large logarithm one less. For passing to the limit $\theta = 0$ (forward scattering), formula (XII.236) is unsuitable. In its place, we have here

$$d\sigma = \frac{\alpha^4 c^2}{\pi^2 \omega^2} \ln^4 \frac{\hbar\omega}{mc^2} d\theta, \quad \theta \ll \frac{mc^2}{\hbar\omega}. \quad (\text{XII.237})$$

Expression (XII.237) is easy to obtain with the help of the general formulae (XII.230), by putting $t = 0$ in them and noting that when $s \gg 1$ only the function T contains the highest (second) power of the large logarithm:

$$T\left(\frac{s}{m^2}\right) \approx \frac{1}{4} \ln^2 \frac{s}{m^2} \approx \ln^2 \frac{\omega}{m}.$$

To this accuracy only the amplitudes

$$M_{++++} = M_{----} = M_{+-+-} = -16e^4 \ln^2(\omega/m)$$

are different from zero. We see, in particular, that in this case the polarisation of the photon does not change upon scattering.

The case of small frequencies. In the case of small frequencies ($\omega \ll m$) the amplitude of photon-photon scattering can also be obtained in an entirely different way, using the correction terms in the Lagrangian of the weak electromagnetic field (see below, § 129).

The small correction to the interaction Hamiltonian $\hat{V}'$ differs from the small correction to the Lagrangian only by sign. According to (129.21) we have

$$\hat{V}' = -\frac{e^4}{45 \cdot 8\pi^2 m^4} \int \{(\hat{\mathbf{E}}^2 - \hat{\mathbf{H}}^2)^2 + 7(\hat{\mathbf{E}}\hat{\mathbf{H}})^2\} d^3x. \quad (\text{XII.238})$$

⁴⁰In passing from $d\sigma$ to σ one must introduce the factor $1/2$, accounting for the identity of the two final photons.

⁴¹We shall return to the dependence of σ on ω at the end of § 134.

Since this operator is of fourth order with respect to the field, it has matrix elements for the transition of interest to us already in first approximation.

For the calculation one must substitute in (XII.238)

$$\hat{\mathbf{E}} = -\frac{\partial \hat{\mathbf{A}}}{\partial t}, \quad \hat{\mathbf{H}} = \text{rot } \hat{\mathbf{A}}, \quad \hat{\mathbf{A}} = \sqrt{4\pi} \sum_{\mathbf{k}\lambda} (\hat{c}_{\mathbf{k}\lambda} \mathbf{e}_{\mathbf{k}\lambda} e^{-ikx} + \hat{c}_{\mathbf{k}\lambda}^+ \mathbf{e}_{\mathbf{k}\lambda}^* e^{ikx}), \quad \alpha = 1, 2, 3 \quad (\text{XII.239})$$

(λ is the polarisation number), after which the S -matrix element is computed as

$$S_{fi} = -i \langle f | \int V' dt | i \rangle = -i \langle 0 | c_{\mathbf{k}_3\lambda_3} c_{\mathbf{k}_4\lambda_4} \int V' dt c_{\mathbf{k}_1\lambda_1}^+ c_{\mathbf{k}_2\lambda_2}^+ | 0 \rangle \quad (\text{XII.240})$$

(cf. §§ 72, 77). With the normalisation of $\hat{\mathbf{A}}$ as in (XII.239), the amplitude of scattering M_{fi} is directly determined from S_{fi} according to

$$S_{fi} = i(2\pi)^4 \delta^{(4)}(k_3 + k_4 - k_1 - k_2) M_{fi} \quad (\text{XII.241})$$

(cf. § 64). The average value in (XII.240) is computed by Wick's theorem (see (77.3)), and in the contractions only the “external” operators $\hat{c}_{\mathbf{k}\lambda}$, $\hat{c}_{\mathbf{k}\lambda}^+$ are to be contracted with the internal $\hat{\mathbf{A}}$.

§ 128. Coherent scattering of a photon in the field of a nucleus

Other (alongside scattering of a photon by a photon) non-linear effects, described by quadratic diagrams of the type (127.1), include the decay of a single photon into two photons in an external field (and the reverse process of “merging” of two photons into one) and the scattering of a photon in an external field. The first of these processes corresponds to diagrams in which one of the four external photon lines is replaced by a line of the external field. The second process corresponds to diagrams with two external lines of real photons and two — of virtual photons of the field.

The latter category includes, in particular, coherent (elastic) scattering of a photon in the constant electric field of a stationary nucleus. In the general case, the calculation leads to very cumbersome formulae (containing multiple quadratures)⁴². We shall confine ourselves here to only some estimates.

By virtue of the requirement of gauge invariance, the scattering amplitude at $\omega \rightarrow 0$ must contain products of the components of the 4-momenta of the initial (k) and final (k') photons (just as the decomposition of the amplitude of scattering of a photon by a photon starts with the fourth power of the products of the components of the 4-momenta of all photons). In other words, the amplitude of scattering of a photon of small frequency is proportional to ω^2 . Taking into account also that this amplitude contains the external field (the field of the nucleus with charge Ze) in the second order, we conclude that the scattering cross section

$$d\sigma \approx Z^4 \alpha^4 r_e^2 \left(\frac{\omega}{m} \right)^4 d\omega, \quad \omega \ll m. \quad (\text{XII.242})$$

⁴²See V. Costantini, B. De Tollis, G. Pistoni, Nuovo Cimento **2A**, 733, 1971; B. De Tollis, M. Lusignoli, G. Pistoni, Nuovo Cimento **32A**, 227, 1976.

The dependence on frequency is, of course, in agreement with the general conclusions of § 59.

The coefficient in (XII.242) cannot be computed using the Lagrangian of a homogeneous electromagnetic field (as could be done for the scattering of light by light). The reason is that in this process the essential distances from the nucleus are $r \sim l/m$, at which the field of the nucleus cannot be regarded as homogeneous.

We present the result of an exact calculation:

$$\begin{aligned} d\sigma_{++} = d\sigma_{--} &= 1.004 \cdot 10^{-3} (Z\alpha)^4 r_e^2 \left(\frac{\omega}{m}\right)^4 \cos^4 \frac{\theta}{2} d\theta, \\ d\sigma_{+-} = d\sigma_{-+} &= 3.81 \cdot 10^{-4} (Z\alpha)^4 r_e^2 \left(\frac{\omega}{m}\right)^4 \sin^4 \frac{\theta}{2} d\theta. \end{aligned} \quad (\text{XII.243})$$

The subscripts “+” and “−” denote here (as in § 127) the helicities +1 or −1 of the final or initial photons; θ is the scattering angle in the rest frame of the nucleus (V. Costantini, B. De Tollis, G. Pistoni, 1971).

To estimate the cross section at high frequencies, we use the optical theorem (cf. § 71). The intermediate state appearing in the right-hand side of the unitarity relation is, in this case, an electron–positron pair state (it corresponds to the cutting of the diagrams along two internal electron lines between the photon ends). Therefore the optical theorem relates the amplitude of elastic scattering of the photon at zero angle to the total cross section σ_{pair} for pair production by a photon in the field of a nucleus. Defining the amplitude $f(\omega, \theta)$ of scattering at angle θ so that the cross section $d\sigma = |f|^2 d\theta$ (cf. (71.5)), we must have

$$\text{Im } f(\omega, 0) = \frac{\omega}{4\pi} \sigma_{\text{pair}}.$$

The cross section σ_{pair} differs from zero, of course, only for $\omega > 2m$. In the ultra-relativistic case, taking σ_{pair} from (94.6), we obtain

$$f''(\omega) \equiv \text{Im } f(\omega, 0) = \frac{7}{9\pi} (Z\alpha)^2 r_e \frac{\omega}{m} \left[\ln \frac{2\omega}{m} - \frac{109}{42} \right], \quad \omega \gg m. \quad (\text{XII.244})$$

The real part of the scattering amplitude is determined from the imaginary part by the dispersion relation. This relation should be written “with one subtraction”, i.e. it must be written for the function f/t (where $t = \omega^2$), since as $\omega \rightarrow 0$ the amplitude $f \propto \omega^2$ (cf. the relation “with two subtractions” (111.13)). Extracting the real part of the dispersion integral (it suffices to understand the integral in the principal-value sense) and changing from integration over $t' = \omega'^2$ to integration over ω' , we have

$$f'(\omega) \equiv \text{Re } f(\omega, 0) = \frac{2\omega^2}{\pi} \int_{2m}^{\infty} \frac{f''(\omega') d\omega'}{\omega'(\omega'^2 - \omega^2)}. \quad (\text{XII.245})$$

For $\omega \gg m$, the essential values in the integral are $\omega' \sim \omega \gg m$, so that the expression (XII.244) can be used for $f''(\omega')$; the lower limit of integration can then be replaced by zero. The principal value of the integral can be represented as the half-sum

of integrals along contours passing above and below the right-hand real axis in the plane of the complex variable ω' ; in turn, these contours can then be rotated in the ω' -plane so as to coincide with the upper and lower imaginary half-axes respectively. As a result, $f'(\omega)$ takes the form

$$f'(\omega) = -\frac{\omega^2}{\pi} \int_0^\infty \frac{f'(i\xi) + f'(-i\xi)}{\xi(\xi^2 + \omega^2)} d\xi = \frac{7}{9\pi} (Z\alpha)^2 r_e \frac{\omega^2}{m} \int_0^\infty \frac{d\xi}{\xi^2 + \omega^2}$$

and finally

$$\operatorname{Re} f(\omega, 0) = \frac{7}{18} (Z\alpha)^2 r_e \frac{\omega}{m}. \quad (\text{XII.246})$$

Note that the real part of the amplitude, unlike the imaginary part, does not contain a large logarithm.

The sum of the squares of the expressions (XII.244) and (XII.246) gives the cross section for scattering at zero angle:

$$d\sigma|_{\theta=0} = \frac{49}{81\pi^2} (Z\alpha)^4 r_e^2 \left(\frac{\omega}{m}\right)^2 \left\{ \ln^2 \frac{0.15\omega}{m} + \frac{\pi^2}{4} \right\} d\omega \quad (\text{XII.247})$$

(F. Rohrlich, R. L. Gluckstern, 1952).

The result obtained for forward scattering (XII.247) is valid also in a certain region of small angles. One can show that the condition of its applicability is $\theta \ll (m/\omega)^2$. This region, however, contributes only a small fraction to the total scattering cross section. The principal contribution to the total cross section comes from angles $\theta \lesssim m/\omega$; this is easy to understand on the basis of the general (not at zero angle) unitarity relation connecting the amplitude of elastic photon scattering with the amplitude of photon pair production. In this region, however, there is no logarithmic term, so that the total scattering cross section is

$$\sigma \sim (Z\alpha)^4 r_e^2 \left(\frac{\omega}{m}\right)^2 \theta^2 \sim (Z\alpha)^4 r_e^2 \quad (\text{XII.248})$$

(H. A. Bethe, F. Rohrlich, 1952). Thus, at large ω the cross section for coherent scattering tends to a constant limit.

§ 129. Radiative corrections to the equations of the electromagnetic field

When quantising the electron-positron field (see § 25) we saw that in the expression for the vacuum energy there appears an infinite constant, which can be written in the form⁴³

$$\mathcal{E}_0 = - \sum_{\mathbf{p}\sigma} \varepsilon_{\mathbf{p}\sigma}^{(-)}, \quad (\text{XII.249})$$

where $\varepsilon_{\mathbf{p}\sigma}^{(-)}$ are the negative frequencies of solutions of the Dirac equation. In itself this constant has no physical meaning, since the vacuum energy is by definition equal to zero.

⁴³We write here $\mathcal{E}$ instead of E to avoid confusion with the electric field strength.

On the other hand, in the presence of an electromagnetic field the energy levels $\varepsilon_{\mathbf{p}\sigma}^{(-)}$ will change. These changes are finite and have a definite physical meaning. They describe the dependence of the properties of space on the field and modify the equations of the electromagnetic field in vacuum.

The modification of the field equations is expressed as a change in its Lagrangian function. The density L of the Lagrangian function is a relativistic invariant and therefore can be a function only of the invariants $\mathbf{E}^2 - \mathbf{H}^2$ and $\mathbf{E}\mathbf{H}$. The usual expression

$$L_0 = \frac{1}{8\pi}(\mathbf{E}^2 - \mathbf{H}^2) \quad (\text{XII.250})$$

is the first term of the general expansion in powers of the invariants.

We shall find the Lagrangian function in the case when the fields $\mathbf{E}$ and $\mathbf{H}$ vary so slowly in space and time that they can be regarded as uniform and constant; then L does not contain derivatives of the fields. The formulation of the necessary conditions for this we shall discuss at the end of this section.

However, for the problem as posed to have meaning, it is also necessary to assume that the electric field is sufficiently weak. The point is that a uniform electric field can create pairs from the vacuum. Consideration of the field as a closed system in its own right is permissible only if the probability of pair creation is sufficiently small. Specifically, one must have

$$|\mathbf{E}| \ll \frac{m^2}{|e|} \quad \left(= \frac{m^2 c^3}{|e| \hbar} \right) \quad (\text{XII.251})$$

(the change in the energy of a charge e at a distance $\hbar/(mc)$ must be small compared with mc^2). We shall see below (see also Problem 2) that in this case the probability of pair creation is exponentially small.

If alongside the electric field there is also a magnetic field, then, generally speaking, one can choose a reference frame in which $\mathbf{E}$ and $\mathbf{H}$ are parallel. Then the magnetic field does not affect the motion of the charge in the direction of $\mathbf{E}$. It is precisely in this frame (the choice of which will be implied in the subsequent calculations) that condition (XII.251) must be satisfied.

The computation of the Lagrangian function begins with the determination of the change W' in the energy of the vacuum. The quantity W' is given by the change in the “zero-point energy” (XII.249). From this quantity, however, one must also subtract the average value of the potential energy of the electrons in “states” with negative energy. The latter subtraction means simply that the total charge of the vacuum is by definition equal to zero.

The zero-point energy in the presence of the field is:

$$\mathcal{E}_0 = - \sum_{\mathbf{p}\sigma} \varepsilon_{\mathbf{p}\sigma}^{(-)} = \sum \int \psi_{\mathbf{p}\sigma}^{(-)*} i \frac{\partial}{\partial t} \psi_{\mathbf{p}\sigma}^{(-)} d^3x, \quad (\text{XII.252})$$

where $\psi_{\mathbf{p}\sigma}^{(-)}$ are the negative-frequency solutions of the Dirac equation in the given field.

We shall assume that the integration is performed over unit volume, and that the wave functions are normalised to 1 in this volume; then $\mathcal{E}_0$ is the energy per unit volume.

In accordance with the above, one must subtract from $\mathcal{E}_0$ the quantity

$$U_0 = \sum_{\mathbf{p}\sigma} \int \psi_{\mathbf{p}\sigma}^{(-)*} e\varphi \psi_{\mathbf{p}\sigma}^{(-)} d^3x,$$

where $\varphi = -\mathbf{E}\mathbf{r}$ is the potential of the uniform field. But according to the theorem on differentiation of the operator with respect to a parameter (see III, (11.16))

$$U_0 \equiv \mathbf{E} \sum_{\mathbf{p}\sigma} \int \psi_{\mathbf{p}\sigma}^{(-)*} \frac{\partial \hat{H}}{\partial \mathbf{E}} \psi_{\mathbf{p}\sigma}^{(-)} d^3x = -\mathbf{E} \sum_{\mathbf{p}\sigma} \frac{\partial \varepsilon_{\mathbf{p}\sigma}^{(-)}}{\partial \mathbf{E}} = \mathbf{E} \frac{\partial \mathcal{E}_0}{\partial \mathbf{E}}.$$

Thus, the total change in the vacuum energy density is finally

$$W' = \left(\mathcal{E}_0 - \mathbf{E} \frac{\partial \mathcal{E}_0}{\partial \mathbf{E}} \right) - \left(\mathcal{E}_0 - \mathbf{E} \frac{\partial \mathcal{E}_0}{\partial \mathbf{E}} \right)_{\mathbf{E}=\mathbf{H}=0}. \quad (\text{XII.253})$$

Let us relate W' to the change in the Lagrangian density L' ($L = L_0 + L'$). For this we use the general formula

$$W = \sum_i \dot{q}_i \frac{\partial L}{\partial \dot{q}_i} - L,$$

where q are the “generalised coordinates” of the field (see II, § 32). For the electromagnetic field the role of the quantities q is played by the potentials $\mathbf{A}$ and φ . Since

$$\mathbf{E} = -\dot{\mathbf{A}} - \nabla\varphi, \quad \mathbf{H} = \text{rot } \mathbf{A}, \quad (\text{XII.254})$$

of the “velocities” $\dot{q}$, only $\dot{\mathbf{A}}$ enters L , and differentiation with respect to $\dot{\mathbf{A}}$ is equivalent to differentiation with respect to $\mathbf{E}$. Therefore

$$W' = \mathbf{E} \frac{\partial L'}{\partial \mathbf{E}} - L'. \quad (\text{XII.255})$$

Comparing (XII.253) and (XII.255), we find

$$L' = -[\mathcal{E}_0 - \mathcal{E}_0]_{\mathbf{E}=\mathbf{H}=0}. \quad (\text{XII.256})$$

Thus, the computation of L' reduces to the evaluation of the sum (XII.249).

Let us consider first the case when only a magnetic field is present. The “negative” energy levels of the electron (charge $e = -|e|$) in a constant uniform field $H_z = H$ are

$$-\varepsilon_{\mathbf{p}}^{(-)} = -\sqrt{m^2 + |e|H(2n-1+\sigma) + p_z^2}, \quad n = 0, 1, 2, \dots; \quad \sigma = \pm 1 \quad (\text{XII.257})$$

(see Problem to § 32). To evaluate the sum we note that the number of states in the interval dp_z is

$$\frac{|e|H}{2\pi} \frac{dp_z}{2\pi}$$

(see III, § 112); the first factor is the number of states with distinct values of p_x , on which the energy does not depend. Moreover, all levels except the level with $n = 0, \sigma = -1$ are doubly degenerate: the levels with $n, \sigma = +1$ and $n+1, \sigma = -1$ coincide. Therefore

$$-\mathcal{E}_0 = \frac{|e|H}{(2\pi)^2} \int_{-\infty}^{\infty} \left\{ \sqrt{m^2 + p_z^2} + 2 \sum_{n=1}^{\infty} \sqrt{m^2 + 2|e|Hn + p_z^2} \right\} dp_z. \quad (\text{XII.258})$$

The divergence of the integrals in (XII.258) is removed upon evaluating L' (XII.256) by subtracting the value of the sum at $H = 0$. To carry out this “renormalisation” it is convenient first to compute the convergent expression

$$\begin{aligned}\Phi &\equiv -\frac{\partial^2 \mathcal{E}_0}{(\partial m^2)^2} = -\frac{|e|H}{2(2\pi)^2} \times \\ &\times \int_0^\infty \left\{ (m^2 + p_z^2)^{-3/2} + 2 \sum_{n=1}^\infty (m^2 + 2|e|Hn + p_z^2)^{-3/2} \right\} dp_z = \\ &= -\frac{|e|H}{8\pi^2} \left\{ \frac{1}{m^2} + 2 \sum_{n=1}^\infty \frac{1}{m^2 + 2|e|Hn} \right\}.\end{aligned}$$

The summation in the curly brackets can be reduced to a summation of a geometric progression in the following way:

$$\begin{aligned}\Phi &= -\frac{|e|H}{8\pi^2} \int_0^\infty e^{-m^2 \eta} \left[2 \sum_{n=0}^\infty e^{-2|e|Hn\eta} - 1 \right] d\eta = -\frac{|e|H}{8\pi^2} \int_0^\infty e^{-m^2 \eta} \left[\frac{2}{1 - e^{-2|e|H\eta}} - 1 \right] d\eta \\ &\quad \text{(XII.259)} \\ &= -\frac{|e|H}{8\pi^2} \int_0^\infty e^{-m^2 \eta} \coth(|e|H\eta) d\eta.\end{aligned}$$

To find L' one must now integrate Φ twice with respect to m^2 and then subtract the value of the resulting quantity at $H = 0$. We find

$$L' = -\frac{1}{8\pi^2} \int_0^\infty \frac{e^{-m^2 \eta}}{\eta^3} \{ \eta |e|H \coth(|e|H\eta) - 1 + c_1 + c_2 m^2 \} d\eta, \quad \text{(XII.260)}$$

where c_1 and c_2 depend on H but not on m^2 .

From considerations of dimensionality and evenness with respect to $\mathbf{H}$ it is obvious that L' as a function of H and m must have the form

$$L' = m^4 f\left(\frac{H^2}{m^4}\right).$$

Therefore terms odd in m^2 cannot appear in L' at all, so that $c_2 = 0$. The coefficient c_1 is determined from the condition that the expansion of L' in powers of H^2 must begin with the term $\sim H^4$. Indeed, a term $\sim H^2$ in L' would simply mean a change in the coefficient of H^2 in the original Lagrangian $L_0 = -H^2/(8\pi)$. But this would amount, in essence, to a change in the definition of the field strength and charge. Therefore the elimination of terms $\sim H^2$ signifies a charge renormalisation. It is easy to verify that for this one must set

$$c_1 = \frac{H^2 e^2}{3 \cdot 8\pi^2} \int_0^\infty \frac{e^{-\eta}}{\eta} d\eta.$$

Finally, making in (XII.260) the substitution $m^2 \eta \rightarrow \eta$, we obtain finally

$$L'(H; E = 0) = \frac{m^4}{8\pi^2} \int_0^\infty \left\{ -\eta b \coth b\eta + 1 + \frac{b^2 \eta^2}{3} \right\} e^{-\eta} \frac{d\eta}{\eta^3}, \quad \text{(XII.261)}$$

where $b = |e|H/m^2$.

Let us turn to the general case, when alongside the magnetic field there is also an electric field $\mathbf{E}$ parallel to it, satisfying condition (XII.251).

For the computation of L' in this case it is not, however, necessary to solve anew the problem of determining the energy levels $\varepsilon_{\mathbf{p}}^{(-)}$ of the electron in the field. It suffices to note that if one seeks the wave function — the solution of the second-order equation (32.7) — in the form of a product

$$\psi = \psi_E(z) e^{ip_x x} \chi_{n\sigma}(y),$$

where $\chi_{n\sigma}$ is the wave function in a magnetic field with $\mathbf{E} = 0$ and $p_z = 0$, then the mass m and the field H enter the equation for $\psi_E(z)$ only in the combination

$$m^2 + |e|H(2n + 1 + \sigma).$$

If one now takes into account that the summation over p_x (on which the energy levels do not depend) still gives the factor $|e|H/(2\pi)$, then on dimensional grounds one can write

$$\Phi(H, E) \equiv \frac{\partial^2 L'}{(\partial m^2)^2}$$

in the form

$$\Phi(H, E) = -\frac{|e|H}{8\pi^2} \sum_{n=0}^{\infty} \sum_{\sigma=\pm 1} \frac{F\left(\frac{m^2 + |e|H(2n + 1 + \sigma)}{|e|H}\right)}{m^2 + |e|H(2n + 1 + \sigma)} = -\frac{b}{8\pi^2} \left\{ F\left(\frac{1}{a}\right) + 2 \sum_{n=1}^{\infty} \frac{F\left(\frac{1 + 2bn}{a}\right)}{1 + 2bn} \right\}, \quad (\text{XII.262})$$

(each term of this sum is the derivative $-\partial^2 \varepsilon_{\mathbf{p}}^{(-)}/(dm^2)^2$, summed over all quantum numbers except n). Here F is as yet unknown function, which we shall find from considerations of relativistic invariance.

Indeed, Φ must be a function of the scalars $H^2 - E^2$ and $(EH)^2 = (\mathbf{E}\mathbf{H})^2$:

$$\Phi(H, E) = f(H^2 - E^2, (EH)^2).$$

Therefore

$$\Phi(0, E) = f(-E^2, 0) = \Phi(iE, 0).$$

But the function $\Phi(iE, 0)$ is obtained from (XII.259) by the substitution $H \rightarrow iE$; after relabelling the integration variable we find

$$\Phi(iE, 0) = \frac{1}{8\pi^2} \int_0^{\infty} e^{-\eta/a} \cot \eta \, d\eta. \quad (\text{XII.263})$$

Comparing this expression with the limit $\Phi(H \rightarrow 0, E)$ computed from (XII.262), we can find the function F .

The passage to the limit $H \rightarrow 0$ in (XII.262) is effected by replacing the summation over n by integration over $dn = dx/(2b)$:

$$\Phi(0, E) = -\frac{1}{8\pi^2} \int_0^{\infty} F\left(\frac{1+x}{a}\right) \frac{dx}{1+x} = -\frac{1}{8\pi^2} \int_{1/a}^{\infty} \frac{F(y)}{y} dy. \quad (\text{XII.264})$$

Equating expressions (XII.263) and (XII.264) and differentiating the equality with respect to $1/a \equiv z$, we obtain

$$\frac{F(z)}{z} = - \int_0^\infty e^{-z\eta} \eta \cot \eta \, d\eta.$$

After this the summation in (XII.262) is again reduced to summation of a geometric progression, and the subsequent calculations are analogous to those carried out above: we express Φ in terms of m^2 , E and H , integrate twice with respect to m^2 , subtract the value at $E = H = 0$ and determine the constants of integration, as in the derivation of (XII.261). The final result is⁴⁴:

$$L' = -\frac{m^4}{8\pi^2} \int_0^\infty \frac{e^{-\eta}}{\eta^3} \left\{ -(\eta a \cot \eta a)(\eta b \coth \eta b) + 1 - \frac{\eta^2}{3}(a^2 - b^2) \right\} d\eta, \quad (\text{XII.265})$$

$$a = \frac{|e|E}{m^2} \quad \left(= \frac{|e|\hbar E}{m^2 c^3} \right), \quad b = \frac{|e|H}{m^2} \quad \left(= \frac{|e|\hbar H}{m^2 c^3} \right).$$

The parameters a and b can be written in invariant form as

$$a = -\frac{i|e|}{\sqrt{2}m^2} \{(\mathcal{F} + i\mathcal{G})^{1/2} - (\mathcal{F} - i\mathcal{G})^{1/2}\}, \quad b = -\frac{|e|}{\sqrt{2}m^2} \{(\mathcal{F} + i\mathcal{G})^{1/2} + (\mathcal{F} - i\mathcal{G})^{1/2}\}, \quad (\text{XII.266})$$

where $\mathcal{F}$ and $\mathcal{G}$ denote the invariants:

$$\mathcal{F} = \frac{1}{2}(\mathbf{H}^2 - \mathbf{E}^2), \quad \mathcal{G} = \mathbf{E}\mathbf{H}, \quad \mathcal{F} \pm i\mathcal{G} = \frac{1}{2}(\mathbf{H} \pm i\mathbf{E})^2. \quad (\text{XII.267})$$

Once formula (XII.265) is expressed through the invariants $\mathcal{F}$ and $\mathcal{G}$, it thereby becomes applicable in an arbitrary reference frame (and not only in the one where $\mathbf{E} \parallel \mathbf{H}$).

Let us note at once the somewhat conventional nature of writing formula (XII.265). It is suitable only if the condition of smallness of the electric field, $a \ll 1$ (XII.251) (not taken into account in (XII.265) in explicit form), is satisfied. This is manifested in the fact that the integrand in (XII.265) has poles at $\eta = n\pi/a$ ($n = 1, 2, \dots$), so that the integral as written strictly speaking has no meaning. Therefore (XII.265) can, essentially, serve only to obtain the terms of an asymptotic expansion (see below) in powers of a by means of a formal expansion of $\cot a$.

The integral (XII.265) can be given meaning mathematically by going around the poles in the plane of complex η . At the same time, L' and thereby also the energy density W' acquire an imaginary part. A complex energy means, as usual, the quasi-stationarity of the state⁴⁵. In this case stationarity is violated by the creation of pairs, and the quantity $-2 \operatorname{Im} W'$ is the probability w of pair creation per unit volume per unit time; since the small corrections W and L differ only in sign, the probability w , expressed in terms of E and H , is simply

$$w = 2 \operatorname{Im} L'. \quad (\text{XII.268})$$

⁴⁴This result was first obtained by *W. Heisenberg* and *H. Euler* (1935). In the calculations presented here, the ideas of the derivation proposed by *V. Weisskopf* (1936) have also been used.

⁴⁵The direction of going around the contour must be chosen so that $\operatorname{Im} W' < 0$. This requirement corresponds to the usual rule of mass replacement $m^2 \rightarrow m^2 - i0$ (in the present case $a \rightarrow a + i0$).

It is obvious that it is proportional to $e^{-\pi/a}$ (see below (129.22)). It is precisely because of the exponential smallness of $\text{Im } W'$ when $a \ll 1$ that the asymptotic expansion in powers of a makes sense, retaining any finite number of terms.

Let us consider the limiting cases of formula (XII.265). In weak fields ($a \ll 1$, $b \ll 1$) the first terms of the expansion are:

$$L' = \frac{m^4}{8\pi^2} \frac{(a^2 - b^2)^2 + 7(ab)^2}{45} = \frac{e^4}{45 \cdot 8\pi^2 m^4} (4\mathcal{F}^2 + 7\mathcal{G}^2). \quad (\text{XII.269})$$

In particular, for $b = 0$ the relative correction is

$$\frac{L'}{L_0} = \alpha \frac{a^2}{45\pi}.$$

The imaginary part of L' when $a \ll 1$ is obtained from the integral (XII.265) by taking the residue at the pole of the cotangent nearest to zero, i.e. at $\eta a = \pi - i0$. According to (XII.268), this gives the probability of pair creation by a weak electric field:

$$w = \frac{m^4}{4\pi^3} a^2 e^{-\pi/a},$$

or, in ordinary units:

$$w = \frac{1}{4\pi^3} \left(\frac{eEh}{m^2 c^3} \right)^2 \frac{mc^2}{h} \left(\frac{mc}{h} \right)^3 \exp \left(-\frac{\pi m^2 c^3}{|e|\hbar E} \right). \quad (\text{XII.270})$$

In a strong magnetic field ($a = 0$, $b \gg 1$) we start from formula (XII.261), written (with the substitution $b\eta \rightarrow \eta$) in the form

$$L' = \frac{m^4 b^2}{8\pi^2} \int_0^\infty \frac{e^{-\eta/b}}{\eta} \left[\frac{1}{3} - \frac{\eta \coth \eta - 1}{\eta^2} \right] d\eta.$$

When $b \gg 1$ the essential region in this integral is $1 \ll \eta \ll b$. In it one can neglect the second term in the square brackets, and cut off the integral at the limits $\eta \approx 1$ and $\eta \approx b$. Then

$$L' = \frac{m^4 b^2}{24\pi^2} \ln b \quad (\text{XII.271})$$

(a more exact calculation replaces $\ln b$ by $\ln b - 2.29$). In this case

$$\frac{L'}{L_0} \approx \frac{\alpha}{3\pi} \ln b.$$

It is evident that the radiative corrections to the equations of the field can achieve a relative magnitude of order unity only in exponentially large fields:

$$H \approx \frac{m^2}{|e|} e^{3\pi/\alpha}. \quad (\text{XII.272})$$

Nevertheless, the computed corrections have meaning: they violate the linearity of Maxwell's equations and thereby lead to observable, in principle, effects (for example, to scattering of light in vacuum or in an external field).

The relation between the field strengths $\mathbf{E}$ and $\mathbf{H}$ and the potentials $\mathbf{A}$ and φ remains, by definition, the former one — (XII.254). Therefore the first pair of Maxwell's equations is also unchanged:

$$\operatorname{div} \mathbf{H} = 0, \quad \operatorname{rot} \mathbf{E} = -\frac{\partial \mathbf{H}}{\partial t}. \quad (\text{XII.273})$$

The second pair of equations is obtained by varying the action

$$S = \int (L_0 + L') d^4x$$

with respect to $\mathbf{A}$ and φ . They can be written in the form

$$\operatorname{rot}(\mathbf{H} - 4\pi\mathbf{M}) = \frac{\partial}{\partial t}(\mathbf{E} + 4\pi\mathbf{P}), \quad \operatorname{div}(\mathbf{E} + 4\pi\mathbf{P}) = 0, \quad (\text{XII.274})$$

$$(\text{XII.275})$$

where the notations are introduced:

$$\mathbf{P} = \frac{\partial L'}{\partial \mathbf{E}}, \quad \mathbf{M} = \frac{\partial L'}{\partial \mathbf{H}}. \quad (\text{XII.276})$$

In form, equations (XII.273)–(XII.274) coincide with the macroscopic Maxwell equations for a field in a material medium⁴⁶. It follows from this that the quantities $\mathbf{P}$ and $\mathbf{M}$ have the meaning of vectors of electric and magnetic polarisation of the vacuum.

Let us note that $\mathbf{P}$ and $\mathbf{M}$ vanish for a plane wave, in which, as is well known, both invariants $\mathbf{E}^2 - \mathbf{H}^2$ and $\mathbf{E}\mathbf{H}$ are zero. In other words, for a plane wave the non-linear corrections in the vacuum are absent.

Let us dwell, finally, on the conditions of applicability of the formulae obtained. For the fields to be regarded as constant, their relative changes over distances and time intervals $\sim 1/m$ must be small; this is ensured by the smallness of the derivatives of the corrections to L_0 compared with L_0 itself. Thus, if the field depends only on time, this leads to the natural condition

$$\omega \ll m. \quad (\text{XII.277})$$

For the case of a weak field, however, there exists a more stringent condition. It arises from the requirement that the fourth-order term in (XII.269) be large compared with the derivative correction to L_0 ; otherwise this term would lose meaning. Thus, for an electric field depending only on time, this leads to the condition

$$\omega \ll m \frac{|e|E}{m^2}, \quad (\text{XII.278})$$

more stringent than (XII.277).

Condition (XII.278) does not arise, however, in the problem of photon-photon scattering considered in the preceding section (§ 127). There we were from the very beginning interested only in a four-photon process, described by members of the fourth

⁴⁶Cf. VIII, § 75. In comparison it should be borne in mind that in macroscopic electrodynamics the average value of the microscopic magnetic field strength is denoted by $\mathbf{B}$, and not $\mathbf{H}$, as here.

order in the Lagrangian, and the question of the relative value of other terms in L' was irrelevant. It was sufficient to require that merely condition (XII.277) be fulfilled.

Problems

1. Determine the correction to the field of a small stationary charge e_1 , associated with the non-linearity of Maxwell's equations.

Solution. Setting $\mathbf{H} = 0$, from (XII.269) we have

$$\mathbf{P} = \frac{\partial L'}{\partial \mathbf{E}} = \frac{\alpha^2}{90\pi^2 m^4} \mathbf{E} E^2.$$

In the centrally symmetric case, from (XII.274) we find

$$(E + 4\pi P) r^2 = \text{const} = e_1$$

(the constant is determined from the condition that at $r \rightarrow \infty$ the field coincides with the Coulomb field of charge e_1). Solving approximately equation (2), we obtain

$$E = \frac{e_1}{r^2} \left(1 - \frac{2\alpha^2 e_1^2}{45\pi m^4 r^4} \right),$$

or

$$\Phi = \frac{e_1}{r} \left(1 - \frac{2\alpha^2 e_1^2}{225\pi m^4 r^4} \right).$$

The non-linear correction in (3) should be distinguished from the linear correction in (114.6), which is connected in the final analysis with the non-uniformity of the Coulomb field. Correction (3) is of higher order in α , but falls off more slowly with increasing distance and grows faster with increasing e_1 .

2. Estimate directly the probability of pair creation by a weak uniform constant electric field in the quasi-classical approximation (*F. Sauter*, 1931).

Solution. Motion in a weak field $\mathbf{E}$ is quasi-classical (in the slowly varying potential $\varphi = -\mathbf{E}\mathbf{r} = -Ez$). Since the amplitude of the reaction contains the wave function of the final positron, pair creation can be regarded as the transition of an electron from a "negative-frequency" to a "positive-frequency" state. In the first of these, in the presence of the field, the quasi-classical impulse is determined by the equation

$$\varepsilon = -\sqrt{p^2(z) + m^2} + |e|Ez, \quad (1)$$

and in the second

$$\varepsilon = +\sqrt{p^2(z) + m^2} + |e|Ez. \quad (2)$$

The transition from the first state to the second involves passage through a potential barrier (the region of imaginary $p(z)$), separating the domains of applicability of (1) and (2). The boundaries of this barrier z_1 and z_2 lie at $p(z) = 0$, i.e.

$$\varepsilon = -m + |e|Ez_1, \quad \varepsilon = +m + |e|Ez_2.$$

The probability of passage through the quasi-classical barrier is

$$w \propto \exp\left(-2 \int_{z_2}^{z_1} |p(z)| dz\right) = \exp\left(-4 \frac{m^2}{eE} \int_0^1 \sqrt{1 - \xi^2} d\xi\right),$$

whence

$$w \propto \exp\left(-\frac{\pi m^2}{|e|E}\right),$$

in agreement with (XII.270).

§ 130. Photon splitting in a magnetic field

Non-linear corrections in the equations of the electromagnetic field give rise to a number of specific effects in the propagation of photons in external fields.

In order to give these equations a more conventional form (cf. footnote on p. 646), we shall in this section denote the electric and magnetic field strengths by the letters **E** and **B**; the letters **D** and **H** will denote the quantities

$$\mathbf{D} = \mathbf{E} + 4\pi\mathbf{P}, \quad \mathbf{H} = \mathbf{B} - 4\pi\mathbf{M}, \quad \mathbf{P} = \frac{\partial \mathbf{L}'}{\partial \mathbf{E}}, \quad \mathbf{M} = \frac{\partial \mathbf{L}'}{\partial \mathbf{B}}.$$

Then equations (129.25)–(129.27) take the form

$$\begin{aligned} \operatorname{div} \mathbf{B} &= 0, \quad \operatorname{rot} \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}, \\ \operatorname{div} \mathbf{D} &= 0, \quad \operatorname{rot} \mathbf{B} = \frac{\partial \mathbf{D}}{\partial t}. \end{aligned} \tag{XII.279}$$

Let us consider the propagation of a photon in a constant uniform magnetic field **B**₀. Denoting the quantities pertaining to the weak field of the electromagnetic wave by letters with a prime, we shall have for them the equations

$$\begin{aligned} [\mathbf{kH}'] &= -\omega \mathbf{D}', \quad [\mathbf{kE}'] = \omega \mathbf{B}', \\ \mathbf{kB}' &= 0, \quad \mathbf{kD}' = 0, \end{aligned} \tag{XII.280}$$

where

$$D'_i = \varepsilon_{ik} E'_k, \quad B'_i = \mu_{ik} H'_k; \tag{XII.281}$$

the dielectric and magnetic permeability tensors of the vacuum are functions of the external field **B**₀. Assuming this field to be weak in the sense $|e|B_0/m^2 \ll 1$, we find from the Lagrangian function (129.21):

$$\begin{aligned} \varepsilon_{ik} &= \delta_{ik} + \frac{2e^4}{45m^4} B_0^2 \left(-\delta_{ik} + \frac{7}{2} b_i b_k \right), \\ \mu_{ik} &= \delta_{ik} + \frac{e^4}{45m^4} B_0^2 (\delta_{ik} + 2b_i b_k), \end{aligned} \tag{XII.282}$$

where $\mathbf{b} = \mathbf{B}_0/B_0$.

Let us recall that the photon frequency is assumed to be small: $\omega \ll m$ (the condition (129.29)). We note, however, that the character of the structure of the tensors ε_{ik} and μ_{ik} is not connected with this assumption, but is already a consequence of the invariance of quantum electrodynamics with respect to spatial inversion and charge conjugation. Thus,

the first forbids the appearance in $\mathbf{D}'$ of terms of the form $\text{const} \cdot \mathbf{B}'$ and $\text{const} \cdot \mathbf{B}_0(\mathbf{B}_0 \mathbf{B}')$ (inversion changes the sign of $\mathbf{E}$ and $\mathbf{D}$ whilst leaving $\mathbf{H}$ and $\mathbf{B}$ unchanged), and the second forbids the appearance in ε_{ik} and μ_{ik} of terms antisymmetric and of odd degree in $\mathbf{B}_0$, i.e. terms of the form $e_{ikl} B_{0l}$ (charge conjugation changes the sign of all fields simultaneously).

Owing to the existence in the problem under consideration of a preferred plane — the plane passing through $\mathbf{k}$ and $\mathbf{b}$ — it is natural to choose as the two independent polarisations of the photon linear polarisations in this plane and perpendicular to it. We shall label with the subscripts $\perp$ and $\parallel$ the polarisations for which the vector $\mathbf{B}'$ is respectively perpendicular to the plane $\mathbf{k}, \mathbf{b}$ or lies in it.

In the case of perpendicular polarisation, together with the vector $\mathbf{B}'$ the vector $\mathbf{H}'$ is also perpendicular to the plane $\mathbf{k}, \mathbf{b}$:

$$\mathbf{B}' = \left(1 + \frac{2e^4}{45m^4} B_0^2\right) \mathbf{H}'.$$

The vectors $\mathbf{E}'$ and $\mathbf{D}'$, on the other hand, lie in the plane $\mathbf{k}, \mathbf{b}$. In this case, from equations (XII.280) we obtain the dispersion law for photons in the form $k = n_{\perp} \omega$ with a “refractive index” (ordinary units)

$$n_{\perp} = 1 + \frac{7e^4 \hbar}{90m_e^4 c^7} B_0^2 \sin^2 \theta, \quad (\text{XII.283})$$

where θ is the angle between $\mathbf{k}$ and $\mathbf{B}_0$ ⁴⁷.

In the second case the vectors $\mathbf{B}'$ and $\mathbf{H}'$ lie in the plane $\mathbf{k}, \mathbf{b}$, while the vectors $\mathbf{E}'$ and $\mathbf{D}'$ are perpendicular to it. For the refractive index one obtains

$$n_{\parallel} = 1 + \frac{2e^4 \hbar}{45m_e^4 c^7} B_0^2 \sin^2 \theta. \quad (\text{XII.284})$$

We note that $n_{\perp} \geq n_{\parallel}$. The equality sign is attained at $\theta = 0$, when $n_{\perp} = n_{\parallel} = 1$.

The most interesting manifestation of the non-linearity of Maxwell's equations with the inclusion of radiative corrections is the splitting of a photon into two photons in an external magnetic field (*S. L. Adler, J. N. Bahcall, C. G. Callan, M. N. Rosenbluth*, 1970).

In a constant and uniform field this process proceeds with conservation of energy and momentum⁴⁸. Upon the decay of a photon $\mathbf{k}$ into photons $\mathbf{k}_1$ and $\mathbf{k}_2$ we have

$$\omega(\mathbf{k}) = \omega(\mathbf{k}_1) + \omega(\mathbf{k}_2), \quad \mathbf{k}_1 + \mathbf{k}_2 = \mathbf{k}. \quad (\text{XII.285})$$

For photons in a vacuum in the absence of external fields $\omega = k$, and the equalities (XII.285) can be fulfilled only for three photons moving in the same direction. But even in such a case the decay

⁴⁷Expressing $\mathbf{B}'$ through $\mathbf{H}'$ from the second of equations (XII.280), substituting into it $\mathbf{H}'$ from the first equation, and then projecting the latter onto the direction $\mathbf{b}$. The product $\mathbf{kE}'$ is expressed through $\mathbf{bE}'$ from the equation $\mathbf{kD}' = 0$.

⁴⁸Conservation of momentum is connected with the spatial homogeneity of the field, but it takes place, of course, only for processes involving uncharged particles. The Lagrangian function of charged particles involves not only the field strengths but also the potentials of the field, which depend on the coordinates even in a uniform field.

is strictly forbidden by invariance with respect to charge conjugation — by virtue of Furry's theorem (cf. § 79) the sum of diagrams with three photon external ends vanishes. The presence of the external field makes the decay of a photon possible (it is depicted by diagrams with three photon external ends and one or more lines of the external field). But this possibility turns out to be connected with the character of polarisation of the photons. This connection can be established already from an analysis of the conservation laws (XII.285), taking into account the modification of the dispersion law of photons in a magnetic field.

Let us write the dispersion law in the form

$$\omega = k + \beta(\mathbf{k}), \quad (\text{XII.286})$$

where $\beta(\mathbf{k})$ is a small (in a weak field) addition. Its presence makes it possible, in principle, for the equalities (XII.285) to be fulfilled for momenta $\mathbf{k}_1, \mathbf{k}_2$ lying in a narrow cone in the neighbourhood of the direction $\mathbf{k}$. In view of the closeness of the directions of all three vectors $\mathbf{k}, \mathbf{k}_1, \mathbf{k}_2$, in the small terms $\beta(\mathbf{k})$ one may set them all directed along $\mathbf{k}$ and consider that $k_1 + k_2 = k$. Then the law of conservation of energy is written as

$$\beta(\kappa k) - \beta_1(\kappa k_1) - \beta_2(\kappa(k - k_1)) = k_1 + |\mathbf{k} - \mathbf{k}_1| - k$$

($\kappa = \mathbf{k}/k$); since the dispersion law depends on the polarisation of the photon, the functions β, β_1, β_2 may be different. Taking into account that

$$|\mathbf{k} - \mathbf{k}_1| = [(k - k_1)^2 + 2kk_1(1 - \cos \vartheta)]^{1/2} \approx k - k_1 + \frac{kk_1}{2(k - k_1)} \vartheta^2$$

(ϑ is the small angle between $\mathbf{k}$ and $\mathbf{k}_1$), we have

$$\beta(\kappa k) - \beta_1(\kappa k_1) - \beta_2(\kappa(k - k_1)) = \frac{kk_1\vartheta^2}{2(k - k_1)} > 0. \quad (\text{XII.287})$$

This inequality determines the necessary conditions for the properties of the dispersion law for decay to take place.

For frequencies $\omega \ll m$ the dispersion law is given by formulae (XII.283), (XII.284), so that $\beta(\mathbf{k}) \approx -k[n(\kappa) - 1]$, where the function $n(\kappa)$ depends on the direction but not on the magnitude of the vector $\mathbf{k}$. Then there must be

$$k_1 n_1(\kappa) + (k - k_1) n_2(\kappa) - k n(\kappa) > 0. \quad (\text{XII.288})$$

Since $n_\perp > n_\parallel$, this condition immediately excludes the decays

$$\gamma_\perp \rightarrow \gamma_\parallel + \gamma_\parallel, \quad \gamma_\perp \rightarrow \gamma_\perp + \gamma_\perp,$$

where the symbol γ denotes a photon, and the subscripts $\perp$ and $\parallel$ correspond to the two polarisations defined above⁴⁹.

For the decays

$$\gamma_\perp \rightarrow \gamma_\perp + \gamma_\perp, \quad \gamma_\parallel \rightarrow \gamma_\parallel + \gamma_\parallel$$

⁴⁹Numerical calculations show that the inequality $n_\perp > n_\parallel$ is valid not only at $\omega \ll m$ (when the expressions (XII.283), (XII.284) hold), but also at all $\omega < 2m$ (the threshold for pair production by a photon).

the left-hand side of the inequality (XII.288) vanishes, since the functions n , n_1 , n_2 are identical. In order to elucidate this question one must in this case take into account the dependence of the refractive index on k , which appears as ω increases. The required inequality is:

$$k_1 n(\kappa, k_1) + (k - k_1) n(\kappa, k - k_1) - k n(\kappa, k) > 0.$$

Already from general considerations one can assert that $n(\kappa, k)$ is an increasing function of k , and therefore this inequality cannot be fulfilled, so that the decays under consideration are also impossible (indeed, replacing $n(k - k_1)$ and $n(k_1)$ by $n(k)$, we deliberately increase the whole sum, whereas after the replacement it becomes exactly zero). The assertion made applies to any transparent medium, and is a consequence of the Kramers–Kronig formula for the refractive index (cf. VIII, § 84). In the present case the external field constitutes a “transparent medium” for photons of all frequencies $\omega < 2m$ — up to the threshold for pair production, i.e. for the appearance of photon absorption.

Thus, the only allowed decay processes turn out to be

$$\gamma_{\parallel} \rightarrow \gamma_{\perp} + \gamma_{\perp}, \quad (\text{XII.289})$$

$$\gamma_{\parallel} \rightarrow \gamma_{\parallel} + \gamma_{\perp}. \quad (\text{XII.290})$$

It was already noted that the momenta $\mathbf{k}_1$ and $\mathbf{k}_2$ are directed at small angles ϑ to the momentum of the initial photon $\mathbf{k}$. If one neglects these angles, i.e. considers the momenta of all photons to be parallel (we shall call this the *collinear approximation*), then the decay (XII.290) turns out to be impossible, as is clear from the following arguments.

Analogously to (127.14), let us represent the decay amplitude in the form

$$M_{fi} = M_{\lambda\mu\nu} e^{\lambda} e_1^{\mu*} e_2^{\nu*},$$

where e , e_1 , e_2 are the polarisation 4-vectors of the photons, defined, as usual, by their 4-potentials A . Choosing the three-dimensional gauge of potentials, $e = (0, \mathbf{e})$, we rewrite this expression as

$$M_{fi} = M_{ikl} e_i e_{1k}^* e_{2l}^*.$$

The two independent polarisations are defined by the unit vectors⁵⁰

$$\mathbf{e}_{\parallel} \parallel [\mathbf{k}\mathbf{b}], \quad \mathbf{e}_{\perp} \parallel [\mathbf{k}[\mathbf{k}\mathbf{b}]]. \quad (\text{XII.291})$$

It is easy to see that in the expansion

$$M_{ikl} = \sum_{\lambda\lambda_1\lambda_2} M_{\lambda\lambda_1\lambda_2} e_i^{(\lambda)*} e_k^{(\lambda_1)} e_l^{(\lambda_2)}$$

(the indices λ , λ_1 , λ_2 run through the values $\perp$, $\parallel$; cf. (127.9)) the vectors $\mathbf{e}_{\perp}$ must appear in each term an even (0 or 2) number of times. Indeed, the amplitude M_{fi} is invariant with respect to CP transformations, and since the potentials $\mathbf{A}$ (and with them $\mathbf{e}$) are

⁵⁰The subscripts $\parallel$ and $\perp$ correspond to the polarisations defined above. It must be remembered that the unit vectors $\mathbf{e}$ determine the direction of the vector potential $\mathbf{A}$ (and thereby of the field $\mathbf{E}'$) and are perpendicular to the direction $\mathbf{B}'$.

CP -invariant, the tensor M_{ikl} must also be CP -invariant. Under CP transformation $\mathbf{e}_{\parallel} \rightarrow \mathbf{e}_{\parallel}$, $\mathbf{e}_{\perp} \rightarrow -\mathbf{e}_{\perp}$ (charge conjugation changes the sign of $\mathbf{b}$, and inversion changes the sign of $\mathbf{k}$, leaving the axial vector $\mathbf{b}$ unchanged). Therefore if in any term of the expansion the vector $\mathbf{e}_{\perp}$ appears an odd number of times, the corresponding scalar $M_{\lambda\lambda_1\lambda_2}$ must be CP -odd. But from the only two (in the collinear approximation) vectors $\mathbf{k} = \mathbf{k}_1 = \mathbf{k}_2$ and $\mathbf{b}$, which both change sign under CP transformation, one cannot construct a CP -odd scalar, which proves the assertion made.

Thus, in the collinear approximation the decay (XII.290) is forbidden. A more detailed estimate shows that the ratio of the amplitude of this process to the amplitude of the decay (XII.289) allowed in the collinear approximation is

$$\frac{M_{\parallel\perp\parallel}}{M_{\perp\perp\parallel}} \sim \vartheta^2 \sim \alpha^2 \left(\frac{B_0}{B_{\text{cr}}} \right), \quad (\text{XII.292})$$

where

$$B_{\text{cr}} = \frac{m^2}{|e|} \left(= \frac{m^2 c^3}{|e| \hbar} = 4.4 \cdot 10^{13} \text{ G} \right)$$

(the angles ϑ are estimated from (XII.287) as $\vartheta^2 \sim n_{\perp} - n_{\parallel}$).

The fact that, of all the decays, only the decay $\gamma_{\parallel} \rightarrow \gamma_{\perp} + \gamma_{\perp}$ turns out to be possible (to leading order) means that in an unpolarised beam of photons propagating in a magnetic field, eventually perpendicular ($\perp$) polarisation is established.

Let us proceed to the calculation of the decay amplitude $M_{fi} \equiv M_{\perp\perp\parallel}$ by perturbation theory, i.e. under the assumption $B_0 \ll B_{\text{cr}}$.

The first (both in α and in the external field) non-vanishing Feynman diagrams have the form

$$(\text{XII.293})$$

(with all possible permutations of the external ends), where the three end-point lines correspond to photons and one to the external field. But in the collinear approximation the amplitude corresponding to these diagrams vanishes. Indeed, by virtue of gauge invariance the external field can enter into the amplitude of the process only in the form of the 4-tensor of its field strengths $F_{\mu\nu}$, and the 4-vectors of polarisation of the photons — only in antisymmetric combinations

$$f_{\mu\nu} = k_{\mu} e_{\nu} - k_{\nu} e_{\mu}$$

with the wave 4-vectors. The final expression for the amplitude is built from the tensor of the external field $F_{\mu\nu}$, the tensors $f_{\mu\nu}$, $f_{1\mu\nu}$, $f_{2\mu\nu}$ of the three photons, and their wave 4-vectors k_{μ} , $k_{1\mu}$, $k_{2\mu}$; moreover, it must be linear in each of the tensors $f_{\mu\nu}$, and for the diagrams (XII.293) — linear also in $F_{\mu\nu}$. In the collinear approximation the 4-vectors k_1 and k_2 reduce to k : $k_1 = k\omega_1/\omega$, $k_2 = k\omega_2/\omega$. Under these conditions every scalar product constructed in the indicated manner vanishes identically: it is easy to see that such a product will contain at least one factor equal to zero, namely k^2 or ke .

Thus, in the collinear approximation the first non-zero contribution to the decay amplitude arises only from six triangle diagrams of the form

(XII.294)

with three lines of the external field⁵¹. The amplitude corresponding to these diagrams is constructed already with three factors $F_{\mu\nu}$. Such scalar products may be different from zero. But all non-zero products contain the wave vectors of the photons only through the tensors $f_{\mu\nu}$; it is easy to see that adding further factors of k leads to the appearance of a factor equal to zero in the product, namely k^2 or ke . But the components of the tensor $f_{\mu\nu}$ coincide with the components of the field strengths $\mathbf{E}'$ and $\mathbf{B}'$ of the photon. This means that if the decay amplitude corresponding to the diagrams (XII.294) is represented as a matrix element of a certain operator, then this operator, being expressed through the operators of the photon field strengths, does not depend

on their frequencies. In turn, it follows from this that the calculation of the decay amplitude (corresponding to the diagram (XII.294)) with the help of the Lagrangian (129.17) will give the correct answer, not restricted by the condition $\omega \ll m$.

At the end of § 127 it was explained how the Hamiltonian of the interaction is obtained from the Lagrangian function L found in § 129. Now we are dealing with a process involving three photons, and the corresponding interaction operator is obtained from those terms of the expansion of L that contain triple products of the fields of the photons $\mathbf{E}'$, $\mathbf{B}'$. In doing so one need consider only terms of the form

$$(\mathbf{B}'\mathbf{B}_0)(\mathbf{E}'\mathbf{B}_0)^2, \quad (\text{XII.295})$$

in which each of the vectors $\mathbf{B}'$ and $\mathbf{E}'$ enters multiplied scalarly by $\mathbf{B}_0$. Indeed, the products $\mathbf{E}'^2$, $\mathbf{B}'^2$, $\mathbf{E}'\mathbf{B}'$ originate, in four-dimensional notation, from the scalars of the form $f_{\mu\nu}f^{\mu\nu}$, which in the collinear approximation identically vanish. The fact that the term with precisely one factor $\mathbf{B}'$ and two factors $\mathbf{E}'$ is chosen is connected with the fact that the process under consideration is one with a $\parallel$ -photon and two $\perp$ -photons; from the first photon, the component along $\mathbf{B}_0$ has the field $\mathbf{B}'$, and from the last two — the field $\mathbf{E}'$.

The Lagrangian function L is expressed through the invariants $\mathcal{F} = (\mathbf{B}^2 - \mathbf{E}^2)/2$ and $\mathcal{G} = \mathbf{E}\mathbf{B}$. We need the term in the expansion obtained from the term $\propto \mathcal{F}\mathcal{G}^2$. Computation with the help of (129.17) gives for the latter expression

$$-\frac{13e^6}{630\pi^2 m^8} \mathcal{F}\mathcal{G}^2.$$

Setting $\mathbf{B} = \mathbf{B}_0 + \mathbf{B}'$ and $\mathbf{E} = \mathbf{E}'$, and extracting from $\mathcal{F}$ the term $\mathbf{B}_0\mathbf{B}'$, and from $\mathcal{G}$ the term $\mathbf{B}_0\mathbf{E}'$, we obtain the required term of the expansion of the form (XII.295). Thus, the operator of the three-photon interaction leading to the decay $\gamma_{\parallel} \rightarrow \gamma_{1\perp} + \gamma_{2\perp}$ is given by the expression

$$\hat{V}^{(3)} = \frac{13e^6}{315\pi^2 m^8} \int (\mathbf{B}_0\hat{\mathbf{E}}'_1)(\mathbf{B}_0\hat{\mathbf{E}}'_2)(\mathbf{B}_0\hat{\mathbf{B}}') d^3x, \quad (\text{XII.296})$$

⁵¹ Corrections connected with taking into account the non-collinearity in the diagrams (XII.293) would give in the amplitude a contribution of the next order in α in comparison with the contribution from diagrams (XII.294).

where

$$\begin{aligned}\hat{\mathbf{B}}' &= i\sqrt{4\pi} [\mathbf{k}\mathbf{e}_{\parallel}] e^{i(\mathbf{k}\mathbf{r} - \omega t)} \hat{\mathbf{e}}_{\mathbf{k}\parallel}, \\ \hat{\mathbf{E}}'_1 &= -i\sqrt{4\pi\omega_1} \mathbf{e}_1 e^{-i(\mathbf{k}_1\mathbf{r} - \omega_1 t)} \hat{\mathbf{e}}_{\mathbf{k}_1\perp}^+\end{aligned}$$

and analogously for $\hat{\mathbf{E}}'_2$; cf. (127.26), (127.27) ⁵².

According to the rules set forth in § 64, the decay amplitude M_{fi} is computed by the definition

$$S_{fi} = -i\langle f | \int \hat{V}^{(3)} dt | i \rangle = i(2\pi)^4 \delta^{(4)}(k - k_1 - k_2) M_{fi}$$

and equals

$$M_{fi} = -i \frac{13e^6}{315\pi^2 m^8} (4\pi)^{3/2} \omega \omega_1 \omega_2 B_0^3 \sin^3 \theta$$

(θ is the angle between $\mathbf{k}$ and $\mathbf{B}_0$). The decay probability per unit time (see (64.11)):

$$dw = (2\pi)^4 \delta(\mathbf{k} - \mathbf{k}_1 - \mathbf{k}_2) \delta(\omega - \omega_1 - \omega_2) |M_{fi}|^2 \frac{d^3 k_1 d^3 k_2}{2 \cdot 2\omega \cdot 2\omega_1 \cdot 2\omega_2 \cdot (2\pi)^6}$$

(the extra factor 1/2 takes into account the reduction of the phase space volume on account of the identity of the two final photons). The first δ -function is eliminated by integration over $d^3 k_2$. For the elimination of the second δ -function we note that, on neglecting the dispersion,

$$\omega - \omega_1 - \omega_2 = k - k_1 - |\mathbf{k} - \mathbf{k}_1| \approx -\frac{kk_1}{k - k_1} (1 - \cos \vartheta_1)$$

and therefore ⁵³

$$\int_0^\omega \int_0^1 \omega_1 \omega_2 \delta(\omega - \omega_1 - \omega_2) d \cos \vartheta_1 \cdot 2\pi \omega_1^2 d\omega_1 = 2\pi \int_0^\omega \omega_1^2 (\omega - \omega_1)^2 d\omega_1 = \frac{\pi}{15} \omega^5.$$

We finally find the total decay probability of the photon per unit time (ordinary units):

$$\begin{aligned}w &= \frac{\alpha^3}{15\pi^2} \left(\frac{13}{315} \right)^2 \frac{mc^2}{\hbar} \left(\frac{\hbar\omega}{mc^2} \right)^5 \left(\frac{B_0 \sin \theta}{B_{\text{cr}}} \right)^6 \\ &= 0.18 \alpha^6 \frac{mc^2}{\hbar} \left(\frac{B_0^2 \sin^2 \theta}{8\pi mc^2} \right)^3 \left(\frac{\hbar}{mc} \right)^9 \left(\frac{\hbar\omega}{mc^2} \right)^5.\end{aligned}\tag{XII.297}$$

As was already mentioned, the applicability of this formula does not require the condition $\omega \ll m$. It is restricted only by the condition of smallness of the terms corresponding to diagrams of the eighth order. To estimate, we note that in the matrix element of the eighth order there may appear, for example, a term differing from that of the sixth order by the dimensionless invariant factor of the form $(eF^{\mu\nu}k_\nu/m^3)^2$. The condition that it be small leads to the very weak condition

$$\omega \ll m(m^2/(|e|B_0)).$$

⁵²The doubling of the coefficient in (XII.296) is due to the fact that $\mathbf{E}'_1$ and $\mathbf{E}'_2$ may be taken from each of the two factors $\mathbf{E}'$ in L .

⁵³It is here assumed that when the dispersion is taken into account, the argument of the δ -function does indeed vanish at some $\cos \vartheta_1 < 1$. Thus, dispersion of the required character is necessary in order for the decay to be possible, but the decay probability itself (if it is small) does not depend on the dispersion.

§ 131. Computation of integrals over four-dimensional regions

We shall present here certain rules and formulae useful for computing the integrals that arise in the theory of radiative corrections. The typical formula for an integral corresponding to a Feynman diagram is

$$\int \frac{f(k) d^4 k}{a_1 a_2 \dots a_n}, \quad (\text{XII.298})$$

where $a_1, a_2, \dots$ are polynomials of second degree in the 4-vector k , $f(k)$ is a polynomial of some degree n' , and the integration is taken over the entire four-dimensional k -space.

A convenient method of computing such integrals (due to *Feynman*, 1949) is based on a preliminary transformation (parametrisation) of the integrand by introducing additional integrations over auxiliary variables $\xi_1, \xi_2, \dots$ according to the formula

$$\frac{1}{a_1 a_2 \dots a_n} = (n-1)! \int_0^1 d\xi_1 \int_0^1 \dots \int_0^1 d\xi_n \frac{\delta(\xi_1 + \xi_2 + \dots + \xi_n - 1)}{(a_1 \xi_1 + a_2 \xi_2 + \dots + a_n \xi_n)^n}. \quad (\text{XII.299})$$

As a result of such a transformation, instead of n distinct quadratic polynomials in the denominator there arises a degree n of just a single polynomial of second degree.

Eliminating the δ -function by integrating over $d\xi_n$ and introducing new variables according to

$$\xi_1 = x_{n-1}, \quad \xi_2 = x_{n-2} - x_{n-1}, \quad \dots, \quad \xi_{n-1} = x_1 - x_2,$$

$$\xi_1 + \xi_2 + \dots + \xi_{n-1} = x_1,$$

we obtain formula (XII.299) in the equivalent form:

$$\frac{1}{a_1 a_2 \dots a_n} = (n-1)! \int_0^1 dx_1 \int_0^{x_1} dx_2 \dots \int_0^{x_{n-2}} dx_{n-1} \frac{1}{[a_1 x_{n-1} + a_2(x_{n-2} - x_{n-1}) + \dots + a_n(1 - x_1)]^n}. \quad (\text{XII.300})$$

For $n = 2$ this formula takes the form

$$\frac{1}{a_1 a_2} = \int_0^1 \frac{dx}{[a_1 x + a_2(1-x)]^2} \quad (\text{XII.301})$$

and is verified by direct computation. For arbitrary n the formula can be proved by induction from $n-1$ to n . Indeed, carrying out in (XII.300) the integration over dx_{n-1} , one obtains on the right-hand side the difference of two $(n-2)$ -fold integrals of the same form.

Assuming the formula to be valid for them, we get $\frac{1}{a_1 - a_2} \left[\frac{1}{a_2 a_3 \dots a_n} - \frac{1}{a_1 a_3 \dots a_n} \right]$, which coincides with the expression on the left-hand side of (XII.300).

By differentiating (XII.300) with respect to $a_1, a_2, \dots$ one can obtain analogous formulae serving for the parametrisation of integrals containing in the denominator any of the polynomials raised to powers higher than the first.

Regularisation of divergent integrals is effected by subtracting from them integrals of analogous form. For the computation of such a difference it may prove expedient to

carry out a preliminary transformation of the difference of the integrands (each of which has already been transformed by means of (XII.299)) with the aid of the formula

$$\frac{1}{a^n} - \frac{1}{b^n} = - \int_0^1 \frac{n(a-b) dz}{[(a-b)z + b]^{n+1}}. \quad (\text{XII.302})$$

After the transformation, according to (XII.300), the four-dimensional integration in (XII.298) reduces to the form

$$\int \frac{f(k) d^4 k}{[(k-l)^2 - \alpha^2]^n}, \quad (\text{XII.303})$$

where l is a 4-vector and α^2 a scalar, both depending on the parameters $x_1, \dots, x_{n-1}$; the scalar α^2 will be taken to be positive.

If the integral (XII.303) converges, then one can carry out the substitution $k-l \rightarrow k$ (shift of the origin of coordinates), after which it takes the form

$$\int \frac{f(k) d^4 k}{(k^2 - \alpha^2)^n} \quad (\text{XII.304})$$

(with a different function $f(k)$), so that the denominator contains only the square k^2 . As regards the numerator, it suffices to consider scalar functions $f = F(k^2)$. Indeed, for integrals of other types with respect to the numerator we have

$$\int \frac{k^\mu F(k^2) d^4 k}{(k^2 - \alpha^2)^n} = 0, \quad (\text{XII.305})$$

$$\int \frac{k^\mu k^\nu F(k^2) d^4 k}{(k^2 - \alpha^2)^n} = \frac{1}{4} g^{\mu\nu} \int \frac{k^2 F(k^2) d^4 k}{(k^2 - \alpha^2)^n}, \quad (\text{XII.306})$$

$$\begin{aligned} & \int \frac{k^\mu k^\nu k^\rho k^\sigma F(k^2) d^4 k}{(k^2 - \alpha^2)^n} \\ &= \frac{1}{24} (g^{\mu\nu} g^{\rho\sigma} + g^{\mu\rho} g^{\nu\sigma} + g^{\mu\sigma} g^{\nu\rho}) \int \frac{(k^2)^2 F(k^2) d^4 k}{(k^2 - \alpha^2)^n}, \end{aligned} \quad (\text{XII.307})$$

and so on, which is evident from symmetry considerations (upon integration over all directions of k).

In the original integral (XII.298), each of the factors $a_1, a_2, \dots$ in the denominator has (as a function of k_0) two zeros, which are dealt with upon integration over dk_0 according to the usual rule (see § 75). After the transformation to the form (XII.304), instead of $2n$ simple poles the integrand has only two poles of order n , which are dealt with by the same rule (contour C in Fig. 25). Rotating the contour of integration, as indicated by the arrows in the figure, one can bring it into coincidence with the imaginary axis in the k_0 -plane (C' in Fig. 25). In other words, the variable k_0 is replaced by $k_0 = ik'_0$ with a real variable k'_0 . Changing also the notation $\mathbf{k}$ to $\mathbf{k}'$, we have

$$k^2 = k_0^2 - \mathbf{k}^2 \rightarrow -(k_0'^2 + \mathbf{k}'^2) = -k'^2, \quad (\text{XII.308})$$

where k' is a 4-vector in Euclidean metric. In this case

$$d^4 k \rightarrow i d^4 k' = i k'^2 dk'^2 d\Omega,$$

where $d\Omega$ is an element of four-dimensional solid angle. Integration over $d\Omega$ gives $2\pi^2$ (see II, § 111), after which

$$d^4 k \rightarrow i\pi^2 k'^2 d(k'^2). \quad (\text{XII.309})$$

Denoting $k'^2 = z$, we obtain finally

$$\int \frac{F(k^2) d^4 k}{(k^2 - \alpha^2)^n} = (-1)^n i\pi^2 \int_0^\infty \frac{F(-z) z dz}{(z + \alpha^2)^n}. \quad (\text{XII.310})$$

In particular,

$$\int \frac{d^4 k}{(k^2 - \alpha^2)^n} = \frac{(-1)^n i\pi^2}{\alpha^{2(n-2)} (n-1)(n-2)}. \quad (\text{XII.311})$$

The logarithmically divergent part in integrals of the form (XII.304) can be separated out as

$$\int \frac{d^4 k}{[(k-l)^2 - \alpha^2]^2}. \quad (\text{XII.312})$$

It is easy to see that in such an integral the substitution $k \rightarrow k+l$ is permissible. Indeed, the difference of the original and the transformed integrals

$$\int \left\{ \frac{1}{[(k-l)^2 - \alpha^2]^2} - \frac{1}{(k^2 - \alpha^2)^2} \right\} d^4 k$$

represents a convergent integral, and hence the replacement $k \rightarrow k+l$ within it is permissible in any case. Carrying it out and then replacing $k \rightarrow -k$, we obtain the same quantity with the opposite sign, whence its vanishing follows.

A linearly divergent integral must have the form

$$\int \frac{k^\mu d^4 k}{[(k-l)^2 - \alpha^2]^2}, \quad (\text{XII.313})$$

but in fact such an integral diverges only logarithmically: the integrand asymptotically (as $k \rightarrow \infty$) equals $k^\mu / (k^2)^2$ and vanishes upon averaging over directions. The shift of the origin of coordinates, however, does not leave the integral (XII.313) unchanged, but adds to it an additive constant. We shall demonstrate this for the case of an infinitesimally small shift $k \rightarrow k + \delta l$, computing the difference

$$\Delta^\mu = \int \left\{ \frac{k^\mu}{[(k-\delta l)^2 - \alpha^2]^2} - \frac{k^\mu + \delta l^\mu}{(k^2 - \alpha^2)^2} \right\} d^4 k. \quad (\text{XII.314})$$

To the accuracy of terms of first order in δl ,

$$\Delta^\mu = \int \left\{ \frac{4k^\mu (k\delta l)}{(k^2 - \alpha^2)^3} - \frac{\delta l^\mu}{(k^2 - \alpha^2)^2} \right\} d^4 k.$$

In the first term the averaging over directions replaces the numerator by $k^2 \delta l^\mu$ (cf. (XII.306)), after which we find⁵⁴

$$\Delta^\mu = \alpha^2 \delta l^\mu \int \frac{d^4 k}{(k^2 - \alpha^2)^3} = -\frac{i\pi^2}{2} \delta l^\mu. \quad (\text{XII.315})$$

In the final expressions for radiative corrections there frequently appears a transcendental function defined by the integral

$$F(\xi) = \int_0^\xi \frac{\ln(1+x)}{x} dx \quad (\text{XII.316})$$

(it is sometimes called the *Spence function*). We shall note here for reference some of its properties:

$$F(\xi) + F\left(\frac{1}{\xi}\right) = \frac{\pi^2}{6} + \frac{1}{2} \ln^2 \xi, \quad (\text{XII.317})$$

$$F(-\xi) + F(-1 + \xi) = -\frac{\pi^2}{6} + \ln \xi \ln(1 - \xi), \quad (\text{XII.318})$$

$$F(1) = \frac{\pi^2}{12}, \quad F(-1) = -\frac{\pi^2}{6}. \quad (\text{XII.319})$$

Expansion for small ξ :

$$F(\xi) = \xi - \frac{\xi^2}{4} + \frac{\xi^3}{9} - \frac{\xi^4}{16} + \dots \quad (\text{XII.320})$$

⁵⁴A more laborious calculation leads to the same result also for finite L .

Asymptotic Formulae of Quantum Electrodynamics

§ 132. High-Momentum Asymptotics of the Photon Propagator

In § 113 we calculated the first term (in α) in the expansion of the polarization operator $\mathcal{P}(k^2)$. To logarithmic accuracy, its form for $|k^2| \gg m^2$ was found to be

$$\mathcal{P}(k^2) = \frac{\alpha}{3\pi} k^2 \ln \frac{|k^2|}{m^2}. \quad (132.1)$$

It was also noted there that the derivation of this formula, as a first-order correction to the propagator $4\pi D^{-1} = k^2$, presupposes the condition

$$\frac{\alpha}{3\pi} \ln \frac{|k^2|}{m^2} \ll 1. \quad (132.2)$$

This limits its range of applicability at large $|k^2|$. We shall now show that (132.1) in fact remains valid under the much weaker condition

$$\frac{\alpha}{3\pi} \ln \frac{|k^2|}{m^2} \lesssim 1. \quad (132.3)$$

The argument proceeds as follows.¹ First observe that, although terms of every order in α in the perturbation series can in principle contribute to $\mathcal{P}(k^2)$ under condition (132.3), at order n it is enough to retain only terms $\sim \alpha^n \ln^n(|k^2|/m^2)$, in which the large logarithm has the same power as α . Terms containing lower powers of the logarithm are necessarily small because $\alpha \ll 1$.

The perturbation series for $\mathcal{P}$ can next be reduced to those for G and Γ^μ by means of the Dyson equation

$$\mathcal{P}(k^2) = i \frac{4\pi\alpha}{3} \text{Tr} \int \gamma_\mu G(p+k) \Gamma^\mu(p+k, p; k) G(p) \frac{d^4 p}{(2\pi)^4} \quad (132.4)$$

(see (107.4)). Since $\mathcal{P}(k^2)$ is gauge invariant, any gauge may be chosen for G and Γ . The Landau gauge is especially convenient here; in this gauge the free-photon propagator has the form (76.11)

$$D_{\mu\nu}(k) = \frac{4\pi}{k^2} \left(g_{\mu\nu} - \frac{k_\mu k_\nu}{k^2} \right) \quad (132.5)$$

($D^{(l)} = 0$ in (103.17)). It turns out that, in this gauge, the perturbation expansions of G and Γ^μ contain no terms with the required powers of the logarithms. Thus their

¹This formulation of the problem and the results presented here are due to L. D. Landau, A. A. Abrikosov, and I. M. Khalatnikov (1954).

zeroth-order values, $G = G_0$ and $\Gamma^\mu = \gamma^\mu$, suffice in (132.4), which then becomes

$$\mathcal{P}(k^2) = i \frac{4\pi\alpha}{3} \text{Tr} \int \gamma_\mu G(p+k) \gamma^\mu G(p) \frac{d^4 p}{(2\pi)^4}. \quad (132.6)$$

This is the Feynman integral represented by the first-order (in α) diagram (113.1), and, after the appropriate renormalization, it gives (132.1).

To prove these statements, let us first trace the origin of the logarithm in (132.6). The logarithmic term plainly comes from the integration region

$$p^2 \gg |k^2| \quad \text{when} \quad |k^2| \gg m^2. \quad (132.7)$$

Indeed, a formal expansion of G in powers of $1/(\gamma p)$ gives

$$\begin{aligned} G(p) &\approx \frac{1}{\gamma p} = \frac{\gamma p}{p^2}, \\ G(p-k) &\approx \frac{1}{\gamma p - \gamma k} \approx \frac{1}{\gamma p} + \frac{1}{\gamma p} \gamma k \frac{1}{\gamma p} + \frac{1}{\gamma p} \gamma k \frac{1}{\gamma p} \gamma k \frac{1}{\gamma p} = \\ &= \frac{\gamma p}{p^2} + \frac{(\gamma p)(\gamma k)(\gamma p)}{(p^2)^2} + \frac{(\gamma p)(\gamma k)(\gamma p)(\gamma k)(\gamma p)}{(p^2)^3}. \end{aligned}$$

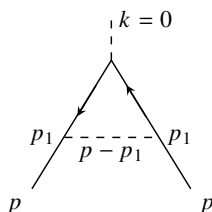
On substitution in (132.6), the first term, which is independent of k , is removed by regularization, consistently with the requirement $\mathcal{P}/k^2 \rightarrow 0$ as $k^2 \rightarrow 0$. The second term vanishes upon angular integration over p . The integral from the third term, however, diverges logarithmically in p^2 . Taking its limits from $p^2 \sim |k^2|$, the lower boundary of region (132.7), to an auxiliary “cutoff parameter” Λ^2 , we obtain

$$-\frac{\alpha}{3\pi} k^2 \ln \frac{\Lambda^2}{|k^2|}. \quad (132.8)$$

Regularization requires subtracting from $\mathcal{P}/k^2$ its value at $k^2 = 0$. Logarithmic accuracy, however, assumes $|k^2| \gg m^2$, so to this accuracy regularization amounts to subtracting the value at $|k^2| \sim m^2$. Consequently Λ^2 in the logarithm is replaced by m^2 , and (132.1) follows.

The corrections of interest in G and Γ^μ are logarithmic. When these are included, the exact functions therefore differ from G_0 and γ^μ only by slowly varying logarithmic factors. The same region (132.7) that matters in the approximate integral (132.6) is accordingly important in the exact integral (132.4). Nevertheless, one cannot simply set $k = 0$ in $\Gamma^\mu(p+k, p; k)$: because the integral is quadratically divergent, its regularization also requires the next two terms in the expansion of $\Gamma^\mu(p+k, p; k)$ in powers of k . Here, however, we shall discuss only the corrections to $\Gamma^\mu(p, p, 0)$. This already shows clearly both why the choice of gauge matters and how the integrals associated with diagrams of different types differ. There is no need to carry out an analogous investigation of G , since the corrections to Γ and G are related by the Ward identity (108.8).

The first-order correction (in α) to $\Gamma(p, p; 0)$ corresponds to the diagram



and hence to the integral²

$$\Gamma^{\mu(1)} = -i\alpha \int \gamma^\lambda G(p_1) \gamma^\mu G(p_1) \gamma^\nu D_{\lambda\nu}(p-p_1) \frac{d^4 p_1}{(2\pi)^4}. \quad (132.9)$$

In the usual gauge,

$$D_{\lambda\nu}(p-p_1) = g_{\lambda\nu} \frac{4\pi}{(p-p_1)^2},$$

and the relevant integration region is $p_1^2 \gg p^2$, where the integral diverges logarithmically. Evaluating

$$\Gamma^{\mu(1)} \approx -4\pi i\alpha \int \frac{\gamma^\lambda (\gamma p_1) \gamma^\mu (\gamma p_1) \gamma_\lambda}{(p_1^2)^3} \frac{d^4 p_1}{(2\pi)^4} \quad (132.10)$$

and regularizing the logarithm gives

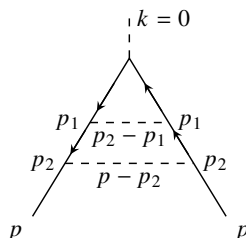
$$\Gamma^{\mu(1)} \approx -\frac{\alpha}{4\pi} \gamma^\mu \ln \frac{p^2}{m^2}.$$

In the Landau gauge, on the other hand, (132.10) is replaced by

$$\Gamma^{\mu(1)} \approx -4\pi i\alpha \int \left\{ \gamma^\lambda (\gamma p_1) \gamma^\mu (\gamma p_1) \gamma_\lambda - p_1^2 \gamma^\mu \right\} \frac{d^4 p_1}{(p_1^2)^3 (2\pi)^4}.$$

After averaging over the directions of p_1 and reducing the γ matrices, this integral is zero; the logarithmic term in $\Gamma^{\mu(1)}$ therefore disappears.³

Among the second-order corrections (in α), consider the diagram



²To avoid confusion in comparing with the results of § 117, recall that in § 117 both electron ends of the diagram were taken to be physical. Here, by contrast, $p^2 \gg |k^2| \gg m^2$, so neither line is physical.

³The corrections to G^{-1} in the two gauges, obtained from the correction $\Gamma^{(1)}$ by means of identity (108.8), agree, of course, with the results of § 119.

The corresponding integral is

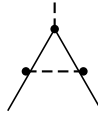
$$\Gamma^{\mu(2)} = -\alpha^2 \int \gamma^\lambda G(p_2) \gamma^\nu G(p_1) \gamma^\mu G(p_1) \gamma^\rho G(p_2) \gamma^\sigma \times \\ \times D_{\nu\rho}(p_2 - p_1) D_{\lambda\sigma}(p - p_2) \frac{d^4 p_1 d^4 p_2}{(2\pi)^8}.$$

With the usual gauge for the D functions, this integral contains a squared-logarithm term arising from the region

$$p_1^2 \gg p_2^2 \gg p^2. \quad (132.11)$$

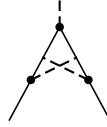
Indeed, after p_2 is neglected in the argument of $D_{\nu\rho}(p_2 - p_1)$, the integration over $d^4 p_1$ is of the same kind as in (132.9) and yields $\ln p_2^2$. The subsequent integration over $d^4 p_2$ is logarithmic once again and produces the square of $\ln(p^2/m^2)$. If the Landau gauge is used for the D functions, however, the logarithmic terms vanish in both integrations.

The same holds for every other diagram belonging to the skeleton diagram



(132.12)

Diagrams of other types, with crossing photon lines, such as those contained in the skeleton diagram



(132.13)

(cf. (106.11)), contain no terms with the required power of the logarithm in any gauge. For these diagrams there is no region of the integration variables in which the integral can be reduced to a succession of logarithmic integrations.

These arguments, together with their analogues for the following terms in the expansion of Γ in powers of k , confirm that in the Landau gauge no corrections to G and Γ arise with the required powers of the logarithm. Expression (132.1) is therefore valid under condition (132.3) as asserted.

The function $D(k^2)$ corresponding to the polarization operator (132.1) is

$$D(k^2) = \frac{4\pi}{k^2} \frac{1}{1 - \frac{\alpha}{3\pi} \ln \frac{|k^2|}{m^2}}. \quad (132.14)$$

Under condition (132.3), there is no need to expand this expression in powers of α .

§ 133. Relation between the “bare” and physical charges

At large $|k^2|$, however, formula (132.14) ceases to be applicable because its denominator becomes small. Indeed, its derivation rests on neglecting diagram (132.13) (as well as diagrams containing still more thick photon lines) relative to diagram (132.12). Yet the

introduction of each such line supplies the diagram with a factor $e^2 D$, where D is the exact propagator. Thus, in place of $\alpha = e^2$, the small parameter is

$$\frac{\alpha}{1 - \frac{\alpha}{3\pi} \ln \frac{|k^2|}{m^2}} \ll 1. \quad (133.1)$$

As $|k^2|$ grows and this quantity becomes of order unity, the theory in effect loses any small parameter at all.

The situation is seen more clearly if, in deriving (132.14), renormalization is not carried out along the way, but instead one first introduces a “bare” electron charge e_c . This charge is subsequently chosen so as to yield the correct observed value of the physical charge e (see § 110). If the integral is cut off, as above, at an auxiliary upper limit Λ^2 , the bare charge is a function of that limit, $e_c = e_c(\Lambda^2)$, and at the end one must take the limit $\Lambda \rightarrow \infty$.

With this procedure, the polarization operator is

$$\mathcal{P}(k^2) = -\frac{e_c^2}{3\pi} k^2 \ln \frac{\Lambda^2}{|k^2|}$$

(expression (132.8), with e_c in place of e), and hence

$$D(k^2) = \frac{4\pi}{k^2} \frac{1}{1 + \frac{e_c^2}{3\pi} \ln \frac{\Lambda^2}{|k^2|}}. \quad (133.2)$$

We now define the physical charge e by the condition

$$e_c^2 D(k^2) \rightarrow \frac{4\pi}{k^2} e^2, \quad k^2 \rightarrow \sim m^2.$$

This gives

$$e^2 = \frac{e_c^2}{1 + \frac{e_c^2}{3\pi} \ln \frac{\Lambda^2}{m^2}}, \quad (133.3)$$

or

$$e_c^2 = \frac{e^2}{1 - \frac{e^2}{3\pi} \ln \frac{\Lambda^2}{m^2}}. \quad (133.4)$$

If the limit $\Lambda \rightarrow \infty$ is taken formally in (133.3), then $e^2 \rightarrow 0$, irrespective of the form of the function $e_c^2(\Lambda)$. Such a “nullification” of the charge plainly signifies that renormalization cannot be performed rigorously. This limiting passage cannot, however, be made without violating the assumptions used in deriving (133.3). Equation (133.4) shows that, for a fixed value of e^2 , e_c^2 increases with Λ ; but as soon as $e_c^2 \sim 1$, the formulas no longer apply, since they were obtained under the assumption

$$e_c^2 \ll 1 \quad (133.5)$$

required for perturbation theory to be applicable to the “bare” interaction. The failure of inequality (133.5) as Λ increases is of considerable fundamental importance. It reveals that quantum electrodynamics is logically incomplete as a theory of weak interaction. In essence, it reveals the logical incompleteness of the available theory altogether. Its formalism depends precisely on the possibility of treating the electromagnetic interaction as a weak perturbation. Every calculable quantity is obtained in the theory as a power series in e_c^2 , and these series are in fact asymptotic. To assign a definite meaning to them when e_c^2 is not small would, at the very least, require additional considerations that do not follow from the general principles of the existing theory.

At the same time, it must be stressed that in quantum electrodynamics the difficulties just described may be of purely theoretical significance. They arise at fantastically high energies that are of no practical interest⁴. One may expect that, in reality, electromagnetic interactions become entangled with weak and strong interactions at incomparably lower energies, so that pure electrodynamics loses its meaning⁵.

To conclude this section, we show how formulas (133.3) and (133.4) can be obtained from simple arguments based on the meaning of renormalization and on dimensional considerations (*M. Gell-Mann, F. Low*, 1954).

Regard the square of the bare charge as a function of the cutoff parameter, $e_c^2(\Lambda^2)$, and introduce a function d that relates the values of e_c^2 at two different values of its argument: $e_c^2(\Lambda_2^2) = e_c^2(\Lambda_1^2)d$. When $\Lambda_1^2, \Lambda_2^2 \gg m^2$, the function d is independent of m ; being dimensionless, it can depend only on the likewise dimensionless quantities $e_c^2(\Lambda_1^2)$ and Λ_2^2/Λ_1^2 :

$$e_c^2(\Lambda_2^2) = e_c^2(\Lambda_1^2)d\left(e_c^2(\Lambda_1^2), \frac{\Lambda_2^2}{\Lambda_1^2}\right). \quad (133.6)$$

This functional relation may be converted into a differential equation. For this purpose, write equation (133.6) for infinitesimally close values of Λ_1^2 and Λ_2^2 . Denoting $\Lambda_1^2 \equiv \xi$ and setting $\Lambda_2^2 = \xi + d\xi$, we obtain the following differential equation for the function $\alpha_c(\xi) \equiv e_c^2(\Lambda_1^2)$:

$$d\alpha_c = \varphi(\alpha_c) \frac{d\xi}{\xi}. \quad (133.7)$$

Here we have introduced the notation

$$\varphi(\alpha_c) = \alpha_c \left[\frac{\partial d(\alpha_c, x)}{\partial x} \right]_{x=1} \quad (133.8)$$

and used the fact that, by definition (133.6), $d(\alpha_c, 1) \equiv 1$. Integrating equation (133.7) from $\xi = \Lambda_1^2$ to $\xi = \Lambda_2^2$, we find

$$\ln \frac{\Lambda_2^2}{\Lambda_1^2} = \int_{e_c^2(\Lambda_1^2)}^{e_c^2(\Lambda_2^2)} \frac{d\alpha}{\varphi(\alpha)}. \quad (133.9)$$

⁴For example, the equality $(\alpha/\pi) \ln(\varepsilon^2/m^2) = 1$ is reached at $\varepsilon \sim 10^{93}m$.

⁵The opposite situation occurs in theories where interaction between particles is mediated not by the electromagnetic field, but by the so-called Yang–Mills fields. In such theories the relation between the renormalized and bare charges is given by a formula of the type (133.4), but with the opposite sign in the denominator. Thus, at fixed e , the bare charge e_c decreases as Λ grows. This property of a theory is called *asymptotic freedom*. A theory with asymptotic freedom is, of course, fundamentally different from a theory with charge nullification.

Throughout the integration range, e_c^2 is small. We may therefore use for $\varphi(\alpha)$ the expression corresponding to the first approximation of perturbation theory. The correction to the bare charge e_c^2 is given by $e_c^2 k^2 \mathcal{P}(k^2)$. Taking the polarization operator in its first approximation (132.1), we obtain

$$d\left(\alpha_c, \frac{\Lambda_2^2}{\Lambda_1^2}\right) = 1 + \frac{\alpha_c}{3\pi} \ln \frac{\Lambda_2^2}{\Lambda_1^2}, \quad \varphi(\alpha) = \frac{\alpha^2}{3\pi},$$

after which integration in (133.9) yields

$$\frac{1}{3\pi} \ln \frac{\Lambda_2^2}{\Lambda_1^2} = \frac{1}{e_c^2(\Lambda_1^2)} - \frac{1}{e_c^2(\Lambda_2^2)}. \quad (133.10)$$

As $\Lambda_1^2 \rightarrow \sim m^2$, the bare charge $e_c(\Lambda_1^2)$ tends to the physical charge e , and (133.10) then agrees with (133.3) and (133.4)⁶.

§ 134. High-energy asymptotic behavior of scattering amplitudes

We shall examine the asymptotic behavior, at high energies, of the amplitudes and cross sections for two-particle scattering processes ($1 + 2 \rightarrow 3 + 4$). For the principal electrodynamic processes, the answer in the first nonvanishing order in α can be obtained from the explicit formulas derived in the preceding chapters, which hold at arbitrary energies. Here we instead approach the question more generally, so that these asymptotic forms can be found directly.

As in § 66, introduce the invariant variables

$$s = (p_1 + p_2)^2, \quad t = (p_1 - p_3)^2, \quad u = (p_1 - p_4)^2 \quad (134.1)$$

(with $p_1 + p_2 = p_3 + p_4$); this notation refers to reactions in the s -channel, which are the ones considered below. In the ultrarelativistic case, when the energies greatly exceed the particle masses, the two particle energies in the center-of-mass frame are approximately equal. If ε denotes the sum of the colliding-particle energies, then in this frame $p_1 = (\varepsilon/2, \mathbf{p}_1)$, $p_2 = (\varepsilon/2, -\mathbf{p}_1)$, $p_3 = (\varepsilon/2, \mathbf{p}_3)$, $p_4 = (\varepsilon/2, -\mathbf{p}_3)$, $\mathbf{p}_1^2 = \mathbf{p}_3^2 = \varepsilon^2/4$, and hence

$$s = \varepsilon^2, \quad t = -\frac{s}{2}(1 - \cos \theta), \quad u = -\frac{s}{2}(1 + \cos \theta), \quad (134.2)$$

where θ is the angle between $\mathbf{p}_1$ and $\mathbf{p}_2$.

First consider the asymptotic form of the reaction cross section at a fixed value of the scattering angle θ . All three variables s , t , and u are then proportional to one another and tend to infinity simultaneously. Particle masses cannot enter the result in the ultrarelativistic case, and the only quantity having the dimension of length is $1/\varepsilon$

⁶A systematic development of the method based on the functional properties of propagators and vertex parts (the so-called renormalization-group method) is given in: *N. N. Bogoliubov and D. V. Shirkov, Introduction to the Theory of Quantized Fields*. Moscow: Nauka, 1984.

($= \hbar c/\varepsilon$). Dimensional considerations alone therefore show that the differential cross section for a two-particle reaction decreases with energy according to the asymptotic law

$$d\sigma/d\omega \propto 1/s \quad \text{for} \quad s, |t|, |u| \rightarrow \infty. \quad (134.3)$$

If the cross section is referred to the differential dt rather than to the solid-angle element $d\omega$, then, since $d\omega \propto dt/s$,

$$d\sigma/dt \propto 1/s^2. \quad (134.4)$$

In the ultrarelativistic case the cross section is related to the scattering amplitude by $d\sigma/dt \propto |M_{fi}|^2/s$; see (64.22), (64.23). Thus (134.3) means that in the asymptotic limit the scattering amplitude has no dependence on s :

$$M_{fi} = \text{const.} \quad (134.5)$$

As is evident from the nature of the argument, these results apply not only in the first nonvanishing approximation, but also in higher orders of perturbation theory (that is, with radiative corrections included), provided logarithmic factors such as $\ln(s/m^2)$ are disregarded. Dependence on dimensionless logarithms plainly cannot be determined by dimensional reasoning⁷.

The situation differs when s is increased at fixed t , that is, at a fixed squared momentum transfer. In other words, one considers scattering through small angles that diminish as the energy rises:

$$s \rightarrow \infty, \quad |t| \sim s\theta^2 = \text{const}, \quad \theta \sim (|t|/s)^{1/2}. \quad (134.6)$$

In this case dimensional analysis tells us only that the total power of $1/s$ and $1/t$ in $d\sigma/dt$ is 2 (and that in the amplitude M_{fi} it is zero)⁸. To find the part of the cross section that decreases most slowly with increasing s , one must therefore isolate the factor with the highest power of $1/t$. Such factors arise only when a Feynman diagram can be divided into two parts, separating the ends 1, 3 from 2, 4, by cutting virtual-particle lines. The total four-momentum carried by these lines is $p_1 - p_3$, which is the origin of the factor's dependence on $t = (p_1 - p_3)^2$. Thus the asymptotic form of a diagram in the region (134.6) is governed by the possible cuts of the diagram in the t -channel.

Similarly, the asymptotic behavior in the region

$$s \rightarrow \infty, \quad |u| \sim s(\pi - \theta)^2 = \text{const}, \quad |\pi - \theta| \sim (|u|/s)^{1/2}, \quad (134.7)$$

which corresponds to scattering through angles near π , is determined by the possible cuts of the diagram in the u -channel (that is, between the ends 1, 4 and 2, 3).

The simplest example is electron-electron scattering, represented by the diagrams (73.13) and (73.14). Of these, the first admits a t -channel cut through the virtual-photon line; it consequently determines the asymptotic behavior of the scattering amplitude in

⁷Summing series that contain logarithmic corrections can produce an exponential dependence on the logarithms, thereby changing the exponent in the power law. This change is small, however, because α is small.

⁸Here the fixed value of $|t|$ is assumed to satisfy $|t| \gg m^2$. The results obtained in this way remain valid as regards their dependence on s (that is, on the energy) also when $|t| \sim m^2$.

the region (134.6). The D -function associated with the virtual-photon line is proportional to $1/t$. The asymptotic forms of the amplitude and differential scattering cross section are therefore

$$M_{fi} \propto s/t, \quad d\sigma \propto dt/t^2. \quad (134.8)$$

The asymptotic behavior in the limit (134.7), near the backward direction, is instead fixed by the “exchange” diagram (73.14); in this limit

$$M_{fi} \propto s/u, \quad d\sigma \propto du/u^2.$$

For mutual scattering of unlike particles (an electron and a muon), there is no exchange diagram. Its cross section for scattering through angles $\theta \approx \pi$ therefore decreases according to (134.3), (134.4)⁹.

We shall show that including radiative corrections does not alter these results for the asymptotic behavior of electron–electron scattering. To do so, consider corrections of different kinds to diagram (73.13).

We have already seen that diagrams representing corrections to the internal D -function (see (113.11)) or to the vertex parts (see (117.1)) produce only logarithmic corrections to the amplitude; they do not change the power dependence in (134.8). The same is true of a diagram that can be cut through two internal photon lines instead of one:

$$(134.9)$$

The scattering amplitude belonging to this diagram differs from that of diagram (73.13) by replacing the factor $1/t$ with

$$\frac{(\gamma(p_1 + q))(\gamma(p_2 - q))}{(p_1 + q)^2(p_2 - q)^2 q^2 (p_3 - p_1 - q)^2} d^4 q$$

and then integrating over $d^4 q$. The relevant integration region is the one that yields the smallest power of $1/s$. For this purpose q must in any event be small in comparison with p_1 and p_2 . Dropping terms that are small in this sense, as well as the terms $p_1^2 = p_2^2 = m^2$, we may rewrite this expression as

$$\frac{(\gamma p_1)(\gamma p_2)}{(p_1 q)(p_2 q) q^2 (p_3 - p_1 - q)^2} d^4 q. \quad (134.10)$$

The denominator contains no s if q_0 and q_x (the x axis lies along $\mathbf{p}_1 = -\mathbf{p}_2$) are proportional to $1/\sqrt{s}$, while the components q_y and q_z may be proportional to $1/\sqrt{|t|}$; the integration region is then proportional to $1/s$. The numerator, on the other hand, is of order $p_1 p_2 \propto s$. Consequently, replacing one internal photon line in the diagram by two does not change its dependence on s at fixed t ¹⁰. In other words, the contribution of

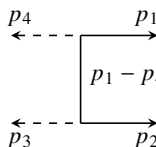
⁹These statements are, of course, consistent with the results of § 81; see (81.11) and problem 6.

¹⁰We emphasize once more that only power-law asymptotics are at issue, so logarithmic divergences in the integration may be disregarded. Diagrams of the type (134.9) will be studied in greater detail in § 137.

diagram (134.9) to the scattering amplitude obeys the same asymptotic law (134.8) as the basic diagram. The conclusion remains unchanged if further parallel internal photon lines are added to the diagram, or if corrections to the internal electron lines are inserted.

This result is general: every diagram that can be divided into two parts in the t -channel or the u -channel by cutting any number of internal photon lines contributes to the amplitude with the respective asymptotic behavior $M_{fi} \propto s/t$ for $t = \text{const}$ or s/u for $u = \text{const}$ (V. G. Gorshkov, V. N. Gribov, L. N. Lipatov, G. V. Frolov, 1967; H. Cheng, T. T. Wu, 1969).

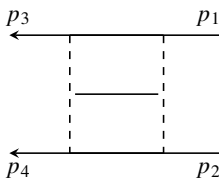
As another example, consider Compton scattering, which is represented by the two diagrams (74.14). These diagrams admit no cut in the t -channel, but the second can be cut in the u -channel through the internal electron line; in the notation of this section it is


(134.11)

Thus the scattering is concentrated chiefly near the backward direction, as was already noted at the end of § 86; see (86.20). To obtain the asymptotic behavior in this region, observe that the factor G associated with the internal line of diagram (134.11) is of order $1/\gamma(p_1 - p_4) \propto 1/\sqrt{|u|}$. Hence the scattering amplitude is $M_{fi} \propto \alpha(s/|u|)^{1/2}$; the factor α has been included because diagram (134.10) is of second order. It follows that the differential cross section is $d\sigma/du \propto \alpha^2/|u|s$. The integral of this expression over $|u|$ is governed by the region $|u| \ll s$. The resulting total cross section decreases with

increasing energy as $\sigma \propto \alpha^2/s$ (more precisely, $\sigma \propto (\alpha^2/s) \ln(s/m^2)$); compare (86.20)¹¹.

For this process, however, radiative corrections do alter the asymptotic behavior. The change comes from sixth-order diagrams of the form


(134.12)

They can be cut in the t -channel through two internal photon lines and therefore contribute to the amplitude with the asymptotic behavior $M_{fi} \propto \alpha^3 s/t$ (the factor α^3 corresponds to the diagram's sixth order). At sufficiently large s , this part of the amplitude becomes dominant, and the differential cross section is then

$$d\sigma/dt \propto \alpha^6/t^2.$$

The integral of this expression over t is governed by the region of small values $|t| \sim m^2$, that is, by scattering angles $\theta \sim m/\sqrt{s}$. Notice that the scattering now takes place chiefly

¹¹The exact dependence of the cross section on $|u|$ or $|t|$ when either is $\lesssim m^2$ cannot, of course, be determined from the considerations given here. It is understood that the integral over $|u|$ (or over $|t|$) converges at values of order m^2 . This is indeed the case for all processes except elastic scattering of charged particles.

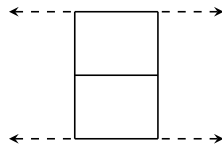
in the forward rather than the backward direction. As a result, the total cross section ceases to decrease with energy:

$$\sigma \propto \alpha^6/m^2 = \alpha^4/r_e^2. \quad (134.13)$$

The decreasing part of the cross section becomes comparable with its constant part at $\varepsilon = \sqrt{s} \propto m/\alpha^2$.

An analogous situation occurs in light-by-light scattering. In the first nonvanishing approximation it is represented by the “box” diagrams (127.1), which can be cut through two internal electron lines. The four-momentum of these lines is integrated over in the diagram, and the important momenta are of order $\sqrt{s}$; small values of t (or u) receive no special emphasis. Hence, for any fixed t (or fixed u), the asymptotic behavior of these diagrams is given by (134.5): $M_{fi} = \text{const} \propto \alpha^2$. The total cross section then decreases with increasing energy, $\sigma \propto \alpha^4/s$ (compare (127.23)); angles close to zero or to π are not singled out. In eighth order, however, diagrams appear that can be cut in the t - or u -channel through two

internal photon lines, for example,



$$(134.14)$$

These diagrams yield a constant asymptotic cross section: $\sigma \propto \alpha^8/m^2$ when $\sqrt{s} \gg m/\alpha^{212}$.

A constant asymptotic value of the total cross section is characteristic of scattering processes whose diagrams can be cut, in either the t - or the u -channel, through internal photon lines. The same property also holds when more than two particles are produced in the reaction's final state.

§ 135. Extraction of double-logarithmic terms in the vertex operator

Corrections of the form $(\alpha L)^n$ (where L is a large logarithm) can become appreciable, as noted at the end of § 133, only at fantastically high energies and are therefore of theoretical interest alone. Amplitudes of actual scattering processes, however, also contain much larger corrections, of the form $(\alpha L^2)^n$. Terms having two powers of a logarithm for every power of α are called double-logarithmic terms.

The characteristic expansion parameter for double-logarithmic corrections is

$$\frac{\alpha}{\pi} \ln^2 \frac{\varepsilon^2}{m^2}; \quad (135.1)$$

¹²The cross section for coherent photon scattering in a nuclear field already has a constant asymptotic form in the first nonvanishing approximation, represented by “box” diagrams two of whose ends are external-field lines (see (128.7)). In fact, however, these diagrams ought to be drawn in the form (134.12), with the upper solid line representing the nucleus. The external-field lines then become internal lines of the diagram, and the origin of the constant asymptotic behavior is evident.

here ε denotes an energy occurring in the problem (for example, the total energy of the colliding particles in their center-of-mass frame). Perturbation theory is applicable only if this quantity is small; this condition fails at energies

$$\varepsilon \sim m \exp\left(\frac{1}{2}\sqrt{\frac{\pi}{\alpha}}\right) \sim 3 \cdot 10^4 m. \quad (135.2)$$

Our aim will be to dispense with this restriction and obtain formulas valid under the condition

$$\frac{\alpha}{\pi} \ln^2 \frac{\varepsilon^2}{m^2} \lesssim 1. \quad (135.3)$$

This plainly requires summing the infinite series of corrections of every order $(\alpha L^2)^n$.

Double-logarithmic corrections occur in two classes of situations. The first comprises scattering processes through a fixed finite angle; as we saw in the preceding section, their cross sections always decrease in the high-energy asymptotic region. Here the double-logarithmic corrections are closely connected with infrared divergence. One example is elastic electron scattering by an external Coulomb field, whose first double-logarithmic correction to the cross section was found in § 122. This section and the next are devoted to determining these corrections completely subject to (135.3).

The second class comprises reaction cross sections that decrease with increasing energy when the square of the momentum transfer is fixed, that is, at angles asymptotically tending to zero or to π . As shown in the preceding section, this behavior occurs in processes whose diagrams cannot be cut in the t - or u -channel across internal photon lines. In this case the double-logarithmic corrections are unrelated to infrared divergence. Backward electron–muon scattering, for which $u = \text{const}$, will be considered as an example in § 137.

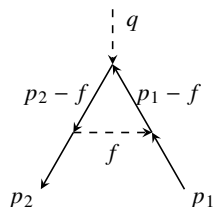
We first observe that under condition (135.3) the single-logarithmic corrections satisfy

$$\sim \frac{\alpha}{\pi} \ln \frac{\varepsilon^2}{m^2} \lesssim \sqrt{\frac{\alpha}{\pi}} \ll 1$$

and may therefore be omitted. Since G and D have no double-logarithmic corrections at all, these functions may now simply be replaced by their unperturbed values G and D .

The calculation of the vertex operator Γ , by contrast, requires the summation of double-logarithmic terms generated by an infinite series of diagrams. That problem is addressed in the next section. We shall first describe a method for extracting the double-logarithmic terms from individual Feynman integrals before actually performing the integrations over all their variables (V. V. Sudakov, 1956).

Consider the first-order (in α) correction to the vertex operator represented by diagram (117.1). Relabeling the variables, it is convenient to draw it here as



$$(135.4)$$

or, analytically,

$$\begin{aligned}\Gamma^{\mu(1)}(p_2, p_1; q) &= \\ &= -\frac{ie^2}{4\pi^3} \int \frac{\gamma^\nu(\gamma p_2 - \gamma f + m)\gamma^\mu(\gamma p_1 - \gamma f + m)\gamma_\nu d^4 f}{[(p_2 - f)^2 - m^2 - i0][(p_1 - f)^2 - m^2 + i0][f^2 + i0]}.\end{aligned}\quad (135.5)$$

We shall assume that

$$|q^2| \gg p_1^2, p_2^2, m^2, \quad (135.6)$$

where the legs p_1 and p_2 may be either physical or virtual. It follows from (135.6) that

$$|(p_1 p_2)| \approx \frac{1}{2}|q^2| \gg p_1^2, p_2^2, m^2, \quad (135.7)$$

so that the 4-vectors p_1 and p_2 have large components while their squares are small, a situation made possible by the pseudo-Euclidean character of the four-dimensional metric. Double-logarithmic terms arise precisely under conditions (135.6).

We shall see below that, in the integration over $d^4 f$, comparatively small values of f are important. Thus f may be neglected in the numerator of the integrand, after which $\Gamma^{(1)}$ becomes

$$\Gamma^{\mu(1)} = -\frac{ie^2}{4\pi^3} \gamma^\nu(\gamma p_2 + m)\gamma^\mu(\gamma p_1 + m)\gamma_\nu I_1, \quad (135.8)$$

where

$$I_1 = \int \frac{d^4 f}{[(p_2 - f)^2 - m^2 + i0][(p_1 - f)^2 - m^2 + i0][f^2 + i0]}. \quad (135.9)$$

The matrix factor in (135.8) can be simplified by noting that, in the diagrams where Γ occurs, it is in effect always multiplied by the matrices $(\gamma p_2 + m)$ and $(\gamma p_1 + m)$:

$$(\gamma p_2 + m)\Gamma(\gamma p_1 + m). \quad (135.10)$$

Indeed, for virtual lines p_1 and p_2 these factors come from $G(p_1)$ and $G(p_2)$; for lines corresponding to real

electrons, Γ is multiplied by $\bar{u}_2$ and u_1 , and the Dirac equations give

$$\bar{u}_2 = \bar{u}_2 \frac{\gamma p_2 + m}{2m}, \quad u_1 = \frac{\gamma p_1 + m}{2m} u_1.$$

Reordering the matrix factors and, at each step, using (135.7) to neglect the resulting squares p_1^2 , p_2^2 , and m^2 in comparison with $(p_1 p_2)$, we obtain

$$(\gamma p_2 + m)\Gamma^{\mu(1)}(\gamma p_1 + m) \approx -\frac{ie^2}{\pi^3} (p_1 p_2)(\gamma p_2 + m)\gamma^\mu(\gamma p_1 + m)I_1.$$

Consequently, $\Gamma^{(1)}$ can finally be written as

$$\Gamma^{\mu(1)} = \frac{ie^2}{2\pi^3} \gamma^\mu t I_1, \quad (135.11)$$

where

$$t = q^2 \approx -2(p_1 p_2). \quad (135.12)$$

Notice that the integral I_1 converges at large f and therefore needs no regularization.

The essential step in the calculation that follows is to introduce new, more convenient integration variables.

Resolve f into components tangent and normal to the plane of p_1 and p_2 :

$$f = u p_1 + v p_2 + f_\perp \equiv f_\parallel + f_\perp, \quad (135.13)$$

$$f_\perp p_1 = f_\perp p_2 = 0. \quad (135.14)$$

As the new variables we choose the coefficients u , v , and the quantity

$$\rho = -f_\perp^2. \quad (135.15)$$

Conditions (135.7) show that the metric in the $p_1 p_2$ plane is pseudo-Euclidean. The time axis can consequently be chosen within this plane, making $f_\perp$ a spacelike 4-vector and hence $\rho > 0$.

Temporarily denote by the indices 0, x the components of 4-vectors in the $p_1 p_2$ plane, and by y , z their components in the normal plane. To transform the 4-volume element $d^4 f = d^2 f_\perp d^2 f_\parallel$ to the new variables, write

$$d^2 f_\perp = |f_\perp| d|f_\perp| d\varphi = \frac{1}{2} d\rho d\varphi \rightarrow \pi d\rho$$

(the integrand in (135.9) being independent of the angle φ). Furthermore,

$$d^2 f_\parallel = \left| \frac{\partial(f_0, f_x)}{\partial(u, v)} \right| du dv = |p_{10} p_{2x} - p_{20} p_{1x}| du dv \approx \frac{1}{2} |q^2| du dv.$$

Indeed, since the square p_2^2 is small, $p_{2x}^2 \approx p_{20}^2$, and therefore

$$(p_{10} p_{2x} - p_{20} p_{1x})^2 \approx (p_{10} p_{20} - p_{2x} p_{1x})^2 = (p_1 p_2)^2 = (q^2/2)^2.$$

Thus

$$d^4 f = \frac{1}{2} |t| du dv d^2 f_\perp \rightarrow \frac{\pi}{2} |t| du dv d\rho. \quad (135.16)$$

The subsequent calculation depends on the relation among p_1^2 , p_2^2 , and m^2 . We consider two cases.

Virtual electron lines. Let the momenta p_1 and p_2 correspond to virtual electrons and satisfy

$$|p_1^2|, |p_2^2| \gg m^2. \quad (135.17)$$

We shall see that in this case the principal integration region producing the double-logarithmic expression is specified by

$$0 < \rho \ll |tu|, |tv|; \quad \left| \frac{p_1^2}{t} \right| \ll |v| \ll 1; \quad \left| \frac{p_2^2}{t} \right| \ll |u| \ll 1. \quad (135.18)$$

Accordingly, in the denominator of the integrand in (135.9), m^2 , p_1^2 , p_2^2 , and f^2 may be neglected compared with $(p_1 f)$ or $(p_2 f)$, giving

$$I_1 = \int \frac{d^4 f}{2(p_2 f) \cdot 2(p_1 f)(f^2 + i0)}. \quad (135.19)$$

For $(p_1 f)$, $(p_2 f)$, and f^2 we have

$$\begin{aligned} f^2 &= (up_1 + vp_2)^2 - \rho \approx -tuv - \rho, \\ 2(p_1 f) &= 2p_1(up_1 + vp_2) \approx -tv, \\ 2(p_2 f) &\approx -tu. \end{aligned}$$

It follows that

$$I_1 = -\frac{\pi}{2|t|} \int \frac{d\rho}{\rho + tuv - i0} \frac{du}{u} \frac{dv}{v}. \quad (135.20)$$

Under conditions (135.18), the ρ integration extends from 0 to the smaller of $|tv|$ and $|tu|$, and yields

$$\int_0^{\min\{|tu|, |tv|\}} \frac{d\rho}{\rho + tuv - i0} = \ln \min \left\{ \frac{1}{|u|}, \frac{1}{|v|} \right\} + \begin{cases} i\pi, & tuv < 0, \\ 0, & tuv > 0. \end{cases} \quad (135.21)$$

The logarithmic integration over v is carried out from -1 to $-|p_1^2/t|$ and from $|p_1^2/t|$ to 1, and similarly for u . When (135.21) is substituted in (135.20), the $du dv$ integral of the first term vanishes because its integrand is odd. For the second term, the integration runs over intervals where u and v have the same sign if $t < 0$, and opposite signs if $t > 0$. In either case the regions $v > 0$ and $v < 0$ make equal contributions after the integration over u . We thereby find (the sign of the integral is the sign of t)

$$I_1 = \frac{i\pi^2}{2t} \cdot 2 \int_{|p_2^2/t|}^1 \frac{du}{u} \int_{|p_1^2/t|}^1 \frac{dv}{v} = \frac{i\pi^2}{t} \ln \left| \frac{t}{p_1^2} \right| \ln \left| \frac{t}{p_2^2} \right|. \quad (135.22)$$

Finally, substituting I_1 into (135.11), we obtain

$$\Gamma^{\mu(1)}(p_2, p_1; q) = -\frac{\alpha}{2\pi} \gamma^\mu \ln \left| \frac{q^2}{p_1^2} \right| \ln \left| \frac{q^2}{p_2^2} \right|, \quad (135.23)$$

$$|q^2| \gg |p_1^2|, |p_2^2| \gg m^2.$$

Physical electron legs. Now let the momenta p_1 and p_2 correspond to real electrons, so that

$$p_1^2 = p_2^2 = m^2. \quad (135.24)$$

The relevant integration region is then

$$0 < \rho \ll |tu|, |tv|; \quad 0 < |v|, |u| \ll 1. \quad (135.25)$$

Since $p_1^2 - m^2 = p_2^2 - m^2 = 0$, neglecting p_1^2 and p_2^2 compared with $(p_1 f)$ or $(p_2 f)$ again reduces the integral (135.9) to the form (135.19). To remove the infrared divergence

that now appears, however, a finite photon mass $\lambda \ll m$ must also be introduced in the photon propagator (cf. § 117):

$$I_1 = \int \frac{d^4 f}{2(p_1 f) \cdot 2(p_2 f)(f^2 - \lambda^2 + i0)}. \quad (135.26)$$

We next have

$$f^2 \approx -tuv - \rho, \quad 2(p_1 f) \approx -tv + 2m^2 u, \quad 2(p_2 f) \approx -tu + 2m^2 v,$$

and hence

$$I_1 = -\frac{\pi}{2|t|} \int \frac{d\rho}{\rho + tuv + \lambda^2 - i0} \frac{du}{u - \tau v} \frac{dv}{v - \tau u}, \quad \tau = \frac{2m^2}{t} \ll 1. \quad (135.27)$$

After the ρ integration, carried out as in (135.21), we find

$$I_1 = -\frac{i\pi^2}{2|t|} \iint \frac{du}{u - \tau v} \frac{dv}{v - \tau u},$$

where the integration is subject to $tuv + \lambda^2 < 0$. The regions $v > 0$ and $v < 0$ once again give equal contributions, and integration over u gives

$$\begin{aligned} I_1 &= \frac{i\pi^2}{t} \int_0^1 dv \int_{\delta/v}^1 \frac{du}{(u - \tau v)(v - \tau u)} \\ &= \frac{i\pi^2}{t} \int_0^1 \ln \left| \frac{\tau\delta - v^2}{(\delta - \tau v^2)(\tau - v)} \right| \frac{dv}{v}, \end{aligned} \quad (135.28)$$

where $\delta = \lambda^2/|t|$, $|\delta| \ll |\tau|$, and $|\tau| \ll 1$ has been used.

In the integral (135.28), three regions of v give double-logarithmic expressions:

$$\text{I) } |\tau| \ll v \ll 1, \quad \text{II) } \sqrt{\delta/\tau} \ll v \ll |\tau|, \quad \text{III) } \sqrt{\tau\delta} \ll v \ll \sqrt{\delta/\tau}.$$

(For definiteness we suppose that $\sqrt{\delta/\tau} \ll |\tau|$. The result does not depend on this assumption.) Making the appropriate approximations in each region, we obtain

$$I_1 = \frac{i\pi^2}{2t} \left(\ln^2 \frac{|t|}{m^2} + 4 \ln \frac{|t|}{m^2} \ln \frac{m}{\lambda} \right). \quad (135.29)$$

Finally, substitution in (135.11) gives

$$\Gamma^{\mu(1)}(p_2, p_1; q) = -\frac{\alpha}{4\pi} \gamma^\mu \left(\ln^2 \frac{|q^2|}{m^2} + 4 \ln \frac{|q^2|}{m^2} \ln \frac{m}{\lambda} \right), \quad (135.30)$$

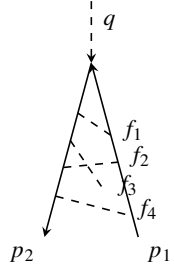
$$|q^2| \gg |p_1^2| = |p_2^2| = m^2,$$

in agreement with (117.21).

§ 136. Double-logarithmic asymptotics of the vertex operator

Once the corrections $\Gamma^{(1)}$ calculated in the preceding section become of order unity, evaluating the vertex operator calls for summing the entire infinite sequence of double-logarithmic terms of every order in α . This problem can be solved because terms of this kind are produced only by diagrams of a particular type, while simple relations connect the contributions from diagrams of different orders.

More precisely, as will be shown below, double-logarithmic terms arise from all diagrams of the form


(136.1)

and so forth, in which every photon line joins the right-hand electron line to the left-hand one; these photon lines may cross one another in any fashion.

Number the photon momenta $f_1, f_2, \dots$ according to the order of, say, the right-hand endpoints of their lines. Different diagrams of the same order then differ by a permutation of the left-hand endpoints of the photon lines. In each Feynman integral, make approximations in the numerator and denominator analogous to those made in integral (135.5), and subsequently transform the numerator by the same procedure used in deriving (135.11). The sum of all diagrams containing n photon lines, which supplies the term $\sim \alpha^n$ in Γ , consequently takes the form

$$\Gamma^{\mu(n)} = \gamma^{\mu} \left(\frac{i\alpha}{2\pi^3} t \right)^n I_n, \quad (136.2)$$

$$\begin{aligned} I_n = & \sum_{\text{perm}} \int_{\lambda^2} d^4 f_1 \dots d^4 f_n \times \\ & \times \frac{1}{2(p_1 f_1) \cdot 2(p_1 f_1 + p_1 f_2) \dots 2(p_1 f_1 + \dots + p_1 f_n)} \times \\ & \times \frac{1}{2(p_2 f_1) \dots 2(p_2 f_1 + \dots + p_2 f_n)} \times \frac{1}{f_1^2 f_2^2 \dots f_n^2}, \end{aligned} \quad (136.3)$$

where the sum extends over every permutation of the indices on the momenta f_k in the products $(p_2 f_k)$; for brevity, the $i0$ and λ^2 terms in the denominators have not been displayed.

Clearly, permuting in any way the indices of the factors f_k in the products $(p_1 f_k)$ occurring in the sum (136.3) merely renames the momenta and therefore leaves I_n unchanged. We may thus extend the summation in (136.3) to every permutation of the factors f_k both in the products $(p_2 f_k)$ and in $(p_1 f_k)$, and then divide the result by $n!$.

We now use the important identity

$$\sum_{\text{perm}} \frac{1}{a_1(a_1 + a_2) \dots (a_1 + a_2 + \dots + a_n)} = \frac{1}{a_1} \frac{1}{a_2} \dots \frac{1}{a_n}, \quad (136.4)$$

in which the summation is over permutations of the indices $1, 2, \dots, n$ ¹³. Applying this identity twice reduces the sum of integrals to a product of n identical integrals of the form (135.19) (or (135.26)), and hence

$$I_n = I_1^n / n!. \quad (136.5)$$

Substitution into (136.2), followed by summation of $\Gamma^{(n)}$ over $n = 0, 1, 2, \dots$, finally gives

$$\Gamma^\mu(p_2, p_1; q) = \gamma^\mu \exp \left(\frac{ie^2}{2\pi^3} t I_1 \right). \quad (136.6)$$

In particular, using I_1 from (135.22) yields the double-logarithmic asymptotic form of the vertex operator with virtual electron endpoints:

$$\Gamma^\mu(p_2, p_1; q) = \gamma^\mu \exp \left\{ -\frac{\alpha}{2\pi} \ln \frac{|q^2|}{|p_1^2|} \ln \frac{|q^2|}{|p_2^2|} \right\}, \quad |q^2| \gg |p_1^2|, |p_2^2| \gg m^2 \quad (136.7)$$

(V. V. Sudakov, 1956).

Using instead I_1 from (135.29), we obtain the asymptotic form of the vertex operator for real electron endpoints:

$$\Gamma^\mu(p_2, p_1; q) = \gamma^\mu \exp \left\{ -\frac{\alpha}{4\pi} \left(\ln^2 \frac{|q^2|}{m^2} + 4 \ln \frac{|q^2|}{m^2} \ln \frac{m}{\lambda} \right) \right\}, \quad |q^2| \gg p_1^2 = p_2^2 = m^2 \quad (136.8)$$

The factor by which Γ^μ differs from its unperturbed value γ^μ also specifies the departure of the electron-scattering amplitude in an external field from its Born value. The scattering cross section is therefore

$$d\sigma = d\sigma_B \exp \left\{ -\frac{\alpha}{2\pi} \left(\ln^2 \frac{|q^2|}{m^2} + 4 \ln \frac{|q^2|}{m^2} \ln \frac{m}{\lambda} \right) \right\}. \quad (136.9)$$

Eliminating the infrared divergence, however, also requires multiplication of this expression by the sum of the probabilities for emitting different numbers of soft photons whose energies do not exceed a certain small $\omega_{\max}$; that is, it must be multiplied by (see (122.2))

$$1 + \int_0^{\omega_{\max}} dw_\omega + \frac{1}{2!} \int_0^{\omega_{\max}} dw_{\omega_1} \int_0^{\omega_{\max}} dw_{\omega_2} + \dots = \exp \left\{ \int_0^{\omega_{\max}} dw_\omega \right\}. \quad (136.10)$$

For the integral in the exponent we use (120.14), specifically the expression multiplying $d\sigma_{\text{el}}$. The final result is the following asymptotic formula for scattering of an electron of energy ε when the momentum transfer is large:

$$d\sigma = d\sigma_B \exp \left\{ -\frac{2\alpha}{\pi} \ln \frac{|q^2|}{m^2} \ln \frac{\varepsilon}{\omega_{\max}} \right\}, \quad |q^2| \gg m^2, \quad \frac{\alpha}{2\pi} \ln^2 \frac{\varepsilon}{m} \sim 1 \quad (136.11)$$

¹³For $n = 2$ the identity is immediate, and its generalization follows readily by induction from n to $n + 1$.

(A. A. Abrikosov, 1956). As expected, the first-order term in α in the expansion of this expression agrees with (122.12).

Notice that on taking $\omega_{\max} \sim \varepsilon$, one of the logarithms in (136.11) becomes of order unity. Thus the double-logarithmic corrections cancel when the cross section includes simultaneous emission of photons with arbitrary energies¹⁴. In the approximation being used, the exponential factor in (136.11) then becomes unity, leaving the cross section equal to its Born value, in accordance with the general statement at the end of § 98.

§ 137. Double-logarithmic asymptotic form of the electron–muon scattering amplitude

As an example of a different kind, consider the scattering of an electron by a negative muon, restricting the discussion to exact backward scattering, i.e. to the angle $\theta = \pi$ (V. G. Gorshkov, B. N. Gribov, L. N. Lipatov, G. V. Frolov, 1967). This process is the simplest one in two respects. First, since the two particles are not identical, exchange diagrams are absent. Second, emission of soft photons is strongly suppressed in backward scattering, so no infrared divergence arises. Indeed, according to (98.8), the cross-section for soft-photon emission is

$$d\sigma = \alpha \left[\left(\frac{\mathbf{v}'_e}{1 - \mathbf{v}'_e \mathbf{n}} + \frac{\mathbf{v}'_\mu}{1 - \mathbf{v}'_\mu \mathbf{n}} - \frac{\mathbf{v}_e}{1 - \mathbf{v}_e \mathbf{n}} - \frac{\mathbf{v}_\mu}{1 - \mathbf{v}_\mu \mathbf{n}} \right) \mathbf{n} \right]^2 \frac{d\omega d\mathbf{n}}{4\pi^2\omega} d\sigma_{\text{el}}, \quad (137.1)$$

where $\mathbf{v}_e$, $\mathbf{v}_\mu$ and $\mathbf{v}'_e$, $\mathbf{v}'_\mu$ are the particle velocities before and after the collision. In the ultrarelativistic case, however, equal momenta imply equal velocities, and to this accuracy, in the centre-of-mass

system for backward scattering, $\mathbf{v}_e = -\mathbf{v}_\mu = -\mathbf{v}'_e = \mathbf{v}'_\mu$. Expression (137.1) therefore vanishes.

If the scattering process under consideration belongs to the s channel of the reaction, in the t channel it becomes the conversion of an electron–positron pair into a $\mu^+\mu^-$ pair. In this channel, the condition $\theta = \pi$ means that the directions of motion of e^- and μ^- coincide (as do those of e^+ and μ^+). Here the suppression of bremsstrahlung has a particularly transparent meaning: the direction of motion of a charge of either sign does not change at all.

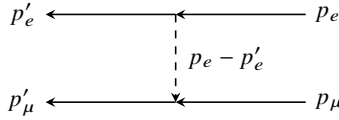
The cancellation of the leading terms in the radiation cross-section means that its asymptotic form contains no double-logarithmic corrections. Correspondingly, to the same double-logarithmic accuracy, integration over the virtual-photon momenta in the scattering amplitude produces no infrared divergence either.

Describing the process in terms of the invariant variables $s = (p_e + p_\mu)^2$, $t = (p_e - p'_e)^2$, $u = (p_e - p'_\mu)^2$, backward scattering in the ultrarelativistic case corresponds to

$$s = -t \gg m_\mu^2, \quad u = 0. \quad (137.2)$$

¹⁴For scattering through a finite angle, the condition stated in § 98 for a photon to be soft requires only $\omega_{\max} \ll \varepsilon$. To logarithmic accuracy, the formulas obtained here may consequently also be used when $\omega_{\max} \sim \varepsilon$.

In the first-order approximation (in α) of perturbation theory, electron–muon scattering is represented by the diagram



$$(137.3)$$

The corresponding amplitude is

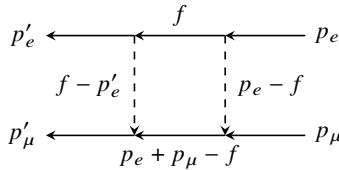
$$M_{fi}^{(1)} = \frac{4\pi\alpha}{t} (\bar{u}^{(\mu)'} \gamma^\nu u^{(\mu)}) (\bar{u}^{(e)'} \gamma_\nu u^{(e)}). \quad (137.4)$$

The limiting case (137.2) is obtained in this expression by replacing the matrix 4-vector γ^ν by its “projection” $\gamma_\perp^\nu$ onto the plane normal to the plane of p_e, p'_e (or, equivalently, to the plane of p_μ, p'_μ , since in ultrarelativistic backward scattering $p_e \approx p'_\mu, p'_e \approx p_\mu$). Indeed, the matrix components parallel to the plane of p_e, p'_e are

$$\frac{1}{\sqrt{s}}(\gamma p_e + \gamma p'_e), \quad \frac{1}{\sqrt{s}}(\gamma p_e - \gamma p'_e)$$

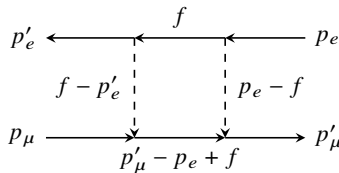
(the first is γ^0 , while the second equals $\mathbf{n}_e \boldsymbol{\gamma}$, where $\mathbf{n}_e$ is the unit vector along $\mathbf{p}_e$). The Dirac equations for the bispinors $u^{(e)}$ and $u^{(\mu)}$ show that $(\bar{u}^{(\mu)'} \gamma^\nu u^{(\mu)}) (\bar{u}^{(e)'} \gamma_\nu u^{(e)}) \sim 1/s$, so these terms may be omitted.

At the next order one must add the diagram



$$(137.5)$$

and also the diagram with “crossed” photon lines. It is conveniently drawn in a form differing from (137.5) only in the direction of one solid line:



$$(137.6)$$

Examination of the corresponding integrals shows that both diagrams acquire double-logarithmic contributions from regions of soft virtual photons: $|(f - p_e)^2| \ll m_e^2$ or $|(f - p'_e)^2| \ll m_e^2$. These contributions are associated with infrared divergences of the integrals and, by the argument above, must certainly cancel in the present case. Diagram (137.6), however, has a further double-logarithmic contribution from the region of large momenta: $|f^2| \gg m_\mu^2$. It is this contribution that must be evaluated.

The integral corresponding to diagram (137.6) is

$$M_{fi}^{(2)} = -i \frac{\alpha^2}{\pi^2} \int \frac{(\bar{u}^{(e)})' \gamma^\nu (\gamma f + m_e) \gamma^\lambda u^{(e)})(\bar{u}^{(\mu)})' \gamma^\lambda (\gamma f + m_\mu) \gamma_\nu u^{(\mu)}}{(p'_e - f)^2 (f^2 - m_e^2) (f^2 - m_\mu^2) (p_e - f)^2} d^4 f, \quad (137.7)$$

where $p_e \approx p'_\mu$ has already been used. Put once more

$$f = up_e + vp'_e + f_\perp \quad (137.8)$$

(cf. (137.13)). The double-logarithmic contribution comes from the region specified by

$$|su|, |sv| \gg \rho \gg m_\mu^2; \quad m_\mu^2/s \ll |u|, |v| \ll 1, \quad (137.9)$$

where $\rho = -f_\perp^2$. In (137.8), the 4-vector $f_\perp$ is defined by $f_\perp p_e = f_\perp p'_e = 0$; here (for backward scattering) it follows that $f_\perp^0 = 0$ in the centre-of-mass system, and hence $\rho = \mathbf{f}_\perp^2$.

In the numerator of the integral (137.7), m_e, m_μ , and all terms containing u or v may be neglected; factors u or v in the numerator

cancel the corresponding poles in the denominator (see below), preventing the required squared logarithms from arising. Noting that $(p'_e - f)^2 \approx tu \approx -su$, $(p_e - f)^2 \approx -sv$, $f^2 \approx suv - \rho$, and transforming the integration element $d^4 f$ according to (135.16), we rewrite (137.7) as

$$M_{fi}^{(2)} = -i \frac{\alpha^2}{2\pi^2} \int \frac{(\bar{u}^{(e)})' \gamma^\nu (\gamma f_\perp) \gamma^\lambda u^{(e)})(\bar{u}^{(\mu)})' \gamma_\lambda (\gamma f_\perp) \gamma_\nu u^{(\mu)}}{su \cdot sv (suv - \rho + i0)} s du dv d^2 f_\perp.$$

The numerator of the integrand is then transformed by averaging over the direction of $f_\perp$ and replacing $\gamma^\nu, \gamma^\lambda$ by $\gamma_\perp^\nu, \gamma_\perp^\lambda$ (for the same reasons as in (137.4)). Straightforward transformations give

$$M_{fi}^{(2)} = M_{fi}^{(1)} J^{(1)}, \quad J^{(1)} = -i \frac{\alpha}{4\pi^2} \int \frac{\rho du dv d\rho}{uv(suv - \rho + i0)^2}. \quad (137.10)$$

Finally, using the identity $\rho = (\rho - suv) + suv$ in the numerator, the second term may be dropped: it would cancel the simple poles and therefore yield no double-logarithmic contribution. Thus

$$J^{(1)} = -i \frac{\alpha}{4\pi^2} \int \frac{du dv d\rho}{uv(\rho - suv - i0)}. \quad (137.11)$$

This integral has the same form as (135.20), and the integration over ρ is consequently performed in the same way. Since now $\rho \gg m_\mu^2$, however, the condition becomes $suv \gg m_\mu^2$ (rather than $suv > 0$). We then obtain

$$J^{(1)} = \frac{\alpha}{2\pi} \int \frac{du dv}{uv}, \quad (137.12)$$

with the integration region bounded by

$$m_\mu^2/s < u, \quad v < 1, \quad suv > m_\mu^2$$

(in a calculation to logarithmic accuracy, the strong inequalities $\gg$ are replaced by ordinary inequalities $>$). Direct evaluation gives

$$J^{(1)} = \frac{\alpha}{4\pi} \ln^2 \frac{s}{m_\mu^2}. \quad (137.13)$$

At higher orders of perturbation theory, the contributions $\sim \alpha^n \ln^{2n} s$ of interest arise from “ladder” diagrams analogous to (137.6), with more “rungs”. The full double-logarithmic asymptotic form

of the scattering amplitude is therefore given by the infinite sum

$$iM_{fi} = \begin{array}{c} p'_e \leftarrow \leftarrow \leftarrow P_e \\ \vdots \quad \vdots \quad \vdots \\ p_\mu \rightarrow \rightarrow \rightarrow p'_\mu \end{array} + \begin{array}{c} \leftarrow \leftarrow \leftarrow \\ \vdots \quad \vdots \quad \vdots \\ \rightarrow \rightarrow \rightarrow \end{array} + \begin{array}{c} \leftarrow \leftarrow \leftarrow \\ \vdots \quad \vdots \quad \vdots \\ \rightarrow \rightarrow \rightarrow \end{array} + \dots \quad (137.14)$$

To establish the general form of the terms in this sum, consider also the third-order diagram (the third term of the series (137.14)). Its integral may be reduced to

$$M_{fi}^{(3)} = M_{fi}^{(1)} J^{(2)}, \quad J^{(2)} = \left(\frac{\alpha}{2\pi} \right)^2 \int \frac{du_1 dv_1 du_2 dv_2}{u_1 v_1 (u_1 + u_2)(v_1 + v_2)} \quad (137.15)$$

over the region

$$m_\mu^2/s < u_{1,2}, \quad v_{1,2} < 1, \quad su_1 v_1, \quad su_2 v_2 > m_\mu^2.$$

The double-logarithmic part of this integral can be isolated by imposing the further conditions

$$v_2 \gg v_1, \quad u_2 \gg u_1. \quad (137.16)$$

Then

$$J^{(2)} = \left(\frac{\alpha}{2\pi} \right)^2 \int \frac{du_1 dv_1 du_2 dv_2}{u_1 u_2 v_1 v_2} = \left(\frac{\alpha}{2\pi} \right)^2 \int d\xi_1 d\eta_1 d\xi_2 d\eta_2,$$

where $\xi_i = \ln(su_i/m_\mu^2)$, $\eta_i = -\ln v_i$, while the integration region is defined by

$$\xi_1 > \eta_1; \quad \xi_2 > \eta_2; \quad \sigma > \xi_2, \quad \eta_2 > 0; \quad \sigma = \ln(s/m_\mu^2).$$

Similarly, the n th term of the series may be written as $M_{fi}^{(n)} = M_{fi}^{(1)} J^{(n)}$, where

$$J^{(n)}(\sigma) = \left(\frac{\alpha}{2\pi} \right)^2 \int d\xi_1 d\eta_1 \dots d\xi_n d\eta_n, \quad (137.17)$$

and the integration region is

$$\xi_i > \eta_i \quad (i = 1, 2, \dots, n), \quad \sigma > \xi_n, \quad \eta_n > 0. \quad (137.18)$$

The complete scattering amplitude equals

$$M_{fi} = M_{fi}^{(1)} \left[1 + \sum_{n=1}^{\infty} J^{(n)}(\sigma) \right]. \quad (137.19)$$

To evaluate this sum, introduce auxiliary functions $A^{(n)}(\xi, \eta)$ given by the same integrals (137.17), but with the integration regions

$$\xi_i > \eta_i \quad (i = 1, 2, \dots, n), \quad \xi > \xi_n > 0, \quad \eta > \eta_n > 0 \quad (137.20)$$

(different upper limits for the integrations over ξ_n and η_n , instead of the equal ones in (137.18)). Clearly $M_{fi} = M_{fi}^{(1)} A(\sigma, \sigma)$, where

$$A(\xi, \eta) = \sum_{n=0}^{\infty} A^{(n)}(\xi, \eta), \quad A^{(0)} = 1. \quad (137.21)$$

The definition of $A^{(n)}(\xi, \eta)$ shows that these functions obey the recurrence relations

$$A^{(n)}(\xi, \eta) = \frac{\alpha}{2\pi} \int d\xi_1 d\eta_1 A^{(n-1)}(\xi_1, \eta_1).$$

Summing these relations over n from 1 to ∞ gives the integral equation determining $A(\xi, \eta)$:

$$A(\xi, \eta) = 1 + \frac{\alpha}{2\pi} \int A(\xi_1, \eta_1) d\xi_1 d\eta_1, \quad (137.22)$$

$$\xi_1 > \eta_1, \quad \xi > \xi_1 > 0, \quad \eta > \eta_1 > 0.$$

For what follows it is enough to consider $A(\xi, \eta)$ in the region $\xi > \eta$. Equation (137.22) can then be written

$$A(\xi, \eta) = 1 + \frac{\alpha}{2\pi} \int_0^\eta d\eta_1 \int_{\eta_1}^\xi A(\xi_1, \eta_1) d\xi_1. \quad (137.23)$$

Differentiating this equality with respect to η gives

$$\frac{\partial A(\xi, \eta)}{\partial \eta} = \frac{\alpha}{2\pi} \int_\eta^\xi A(\xi_1, \eta) d\xi_1, \quad (137.24)$$

and a further differentiation with respect to ξ yields the differential equation for $A(\xi, \eta)$,

$$\frac{\partial^2 A}{\partial \eta \partial \xi} - \frac{\alpha}{2\pi} A = 0. \quad (137.25)$$

This equation must be solved subject to the boundary conditions

$$A(\xi, 0) = 1, \quad \left. \frac{\partial A}{\partial \eta} \right|_{\xi=\eta} = 0, \quad (137.26)$$

which follow directly from (137.23), (137.24).

The solution can be found by applying a Laplace transform in ξ :

$$A(\xi, \eta) = \frac{1}{2\pi i} \int_C e^{p\xi} Q(p, \eta) dp, \quad (137.27)$$

where the contour C in the complex p plane is a closed curve enclosing $p = 0$. Substitution of (137.27) into (137.25), followed by setting the integrand to zero, gives

$$p \frac{\partial Q}{\partial \eta} = \frac{\alpha}{2\pi} Q, \quad Q = \varphi(p) \exp \frac{\alpha \eta}{2\pi p},$$

where $\varphi(p)$ is arbitrary. The first boundary condition in (137.26) now gives $\varphi(p) = 1/p + \psi(p)$, where $\psi(p)$ is analytic and has no singularities within C . The second condition in (137.26) is satisfied by choosing $\psi(p) = -2\pi p/\alpha$; indeed, then

$$\left. \frac{\partial A}{\partial \eta} \right|_{\xi=\eta} = -\frac{1}{2\pi i \xi} \int_C \frac{d}{dp} \exp \xi \left(p + \frac{\alpha}{2\pi p} \right) dp = 0.$$

Collecting the results and setting $\xi = \eta = \sigma$, we find

$$A(\sigma, \sigma) = -\frac{1}{2\pi i} \frac{2\pi}{\alpha \sigma} \int_C p \frac{d}{dp} \exp \left[\sigma \left(p + \frac{\alpha}{2\pi p} \right) \right] dp.$$

Finally, integrating by parts and using the known formula

$$I_1(z) = \frac{1}{2\pi i} \int_C \exp \left[\frac{z}{2} \left(p + \frac{1}{p} \right) \right] dp$$

($I_1(z) = -iJ_1(iz)$ is the Bessel function of imaginary argument), the scattering amplitude is ultimately

$$M_{fi} = M_{fi}^{(1)} \sqrt{\frac{2\pi}{\alpha \sigma^2}} I_1 \left(\sqrt{\frac{2\alpha}{\pi}} \sigma \right). \quad (137.28)$$

The scattering cross-section (for the angle $\theta = \pi$) is accordingly

$$d\sigma = d\sigma^{(1)} \frac{2\pi}{\alpha \ln^2(s/m_\mu^2)} I_1^2 \left(\sqrt{\frac{2\alpha}{\pi}} \ln \frac{s}{m_\mu^2} \right), \quad d\sigma^{(1)} = \frac{2\pi\alpha^2}{s^2} dt, \quad (137.29)$$

where $d\sigma^{(1)}$ is the Born-approximation cross-section in the ultrarelativistic case (see Problem 6, § 81)¹⁵.

¹⁵Further references to work on double-logarithmic asymptotics can be found in the review article: Gorshkov V. G.//UFN.—1973.—Vol. 110.—P. 45.

Electrodynamics of Hadrons

§ 138. Electromagnetic Form Factors of Hadrons

Up to this point, the quantum electrodynamics discussed in this book has concerned particles that do not undergo strong interactions: electrons, positrons, and muons. There is also a large number of particles that do participate in strong interactions; these are called hadrons¹.

Examples of hadrons include the spin-1/2 proton and neutron, the spin-0 π mesons, and other particles. Atomic nuclei are, of course, hadrons as well, since they consist of protons and neutrons.

Within the existing theory, it is impossible to construct a complete electrodynamics of hadrons. Evidently, equations determining the electromagnetic interactions of hadrons cannot be formulated without taking into account their much more intense strong interactions.

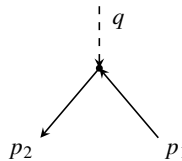
In particular, if the strong interactions are omitted, even the explicit form of the hadronic current that must describe quantum-electrodynamic interactions cannot be found.

Under these circumstances the hadronic current is introduced phenomenologically. Its structure is fixed only by general kinematic requirements, independently of any assumptions about interaction dynamics². The electromagnetic-interaction operator retains the form

$$e(\widehat{J}\widehat{A}), \quad (138.1)$$

where the current is now denoted by the capital letter J (to distinguish it from the electron current j). Since the scale of this interaction is set by the same elementary charge e , perturbation theory remains applicable³.

We shall determine the transition current between two states of a freely moving hadron (with no transformation of the hadron itself). This current occurs in the three-leg vertex



$$(138.2)$$

which may itself form part of a more complicated diagram (for example, one for elastic electron-hadron scattering). The dashed line in (138.2) represents a virtual photon. It

¹From the Greek word "hadros," meaning large or massive.

²This book does not discuss those aspects of hadron electrodynamics that are connected with the quark model.

³In this chapter e denotes the elementary charge ($e > 0$).

cannot correspond to a real photon, since a free particle cannot absorb (or emit) such a photon. Here $q^2 = (p_2 - p_1)^2 < 0$.

Consider first a spin-0 hadron. Let u_1 and u_2 be the wave amplitudes of its initial and final states, in which its 4-momenta are p_1 and p_2 ; for a spin-0 particle these amplitudes are scalars (or pseudoscalars)⁴. The hadronic transition current J_{fi} between these states must be bilinear in u_1 and u_2^* . Write it as

$$J_{fi} = u_2^* \Gamma u_1, \quad (138.3)$$

where the 4-vector Γ is the unknown vertex operator (the circle in diagram (138.2)). Setting $u_1 = u_2 = 1$ gives simply $J_{fi} = \Gamma$.

A universal property of an electrodynamic current, following from gauge invariance of the theory, is its conservation. In momentum space this is expressed by the orthogonality of the transition current to the photon 4-momentum $q = p_2 - p_1$:

$$q J_{fi} = 0. \quad (138.4)$$

In the present case this requires the operator Γ to have the form

$$\Gamma = P F(q^2), \quad (138.5)$$

where $P = p_1 + p_2$, while $F(q^2)$ is a scalar function of the sole independent invariant variable, the square q^2 . The species of hadron is unchanged in the transition, so $p_1^2 = p_2^2 = M^2$ (M is the hadron mass), and consequently $Pq = 0$.

The matrix elements (138.3), with Γ given by (138.5), are polar 4-vectors, as is the operator J itself. Hence the interaction operator (138.1) is a scalar. Thus the electromagnetic interaction of spin-0 hadrons is automatically P -invariant. It is also T -invariant. Indeed, time reversal first interchanges the initial and final 4-momenta, leaving their sum $P = p_1 + p_2$ unchanged. Second, time reversal reverses the spatial components of the 4-momenta without changing their time components; the components of the 4-potential A transform in the same way, and hence the product $\widetilde{J}A$ is unchanged.

The invariant function $F(q^2)$ is called the hadron's *electromagnetic form factor*. Its form plainly cannot be determined within a phenomenological theory. It can nevertheless be stated that this function is real (in the region $q^2 < 0$ considered here). This follows from the same argument used in § 116 for the electron form factors: when $q^2 < 0$, there are in any event no intermediate states that could occur on the right-hand side of the unitarity relation. The matrix M_{fi} , and with it J_{fi} , is therefore Hermitian.

At $q = 0$ the initial and final states coincide, and J_{fi} becomes a diagonal matrix element. In particular, $e(J^0)_{ii}/2\varepsilon_i = eF(0)$ is the charge density; with normalization to one particle per unit volume, this is equal to the particle's total charge Ze .

For an electrically neutral particle, $F(0) = 0$. This by no means implies, however, that the particle is genuinely neutral. If it is genuinely neutral and has definite charge

⁴Recall that a plane wave is written as $\psi = \frac{u}{\sqrt{2\varepsilon}} e^{-ipx}$. Normalization to one particle per unit volume corresponds, for spin-0 particles, to the scalar normalization $u^*u = 1$; one may then simply set $u = 1$ (see § 10). Below we define the transition current with the amplitudes u_1, u_2 displayed, in accordance with the notation adopted in § 64.

parity, then $F(q^2) = 0$ for every q^2 : since the current operator is odd under charge conjugation (see § 13), its matrix elements between two states of the same hadron vanish⁵.

We turn now to spin-1/2 hadrons. In this case the wave amplitudes u_1, u_2 are bispinors, and the hadronic current is

$$J_{fi} = \bar{u}_2 \Gamma u_1. \quad (138.6)$$

From the bilinear combinations of $\bar{u}_2$ and u_1 together with the 4-vectors p_1, p_2 , one can construct both polar 4-vectors and pseudovectors (satisfying (138.4)). Consequently, P -invariance of the interaction is no longer automatic and must be imposed as an additional condition⁶. As shown in § 116, under this condition the vertex operator contains two independent form factors, both real for $q^2 < 0$. We now write it in the form

$$\begin{aligned} \Gamma^\mu &= 2M(F_e - F_m) \frac{P^\mu}{P^2} + F_m \gamma^\mu = 2M \left(F_e - \frac{q^2}{4M^2} F_m \right) \frac{P^\mu}{P^2} - \frac{F_m}{2M} \sigma^{\mu\nu} q_\nu = \\ &= (4M^2 F_e - q^2 F_m) \frac{\gamma^\mu}{P^2} + \frac{2M}{P^2} (F_e - F_m) \sigma^{\mu\nu} q_\nu, \end{aligned} \quad (138.7)$$

where $F_e(q^2)$ and $F_m(q^2)$ are invariant form factors (M is the hadron mass). The equivalence of the three displayed expressions follows immediately from $P^2 + q^2 = 4M^2$ and (116.5)⁷.

Electromagnetic form factors belong to the class of invariant amplitudes introduced in § 70. They may be regarded as amplitudes for a “reaction” that, in its annihilation channel, is the decay of a virtual photon into a hadron and an antihadron. A virtual photon is a spin-1 “particle.” That its decay into two spin-1/2 particles must be described by two independent amplitudes can also be verified simply by counting the corresponding helicity amplitudes $\langle \lambda_b \lambda_c | S^J | \lambda_a \rangle$ (see § 69). Indeed, by P -invariance, the four nonzero S -matrix elements are equal in pairs:

$$\begin{aligned} \langle \tfrac{1}{2} \tfrac{1}{2} | S^1 | 1 \rangle &= \langle -\tfrac{1}{2} -\tfrac{1}{2} | S^1 | -1 \rangle, \\ \langle \tfrac{1}{2} -\tfrac{1}{2} | S^1 | 0 \rangle &= \langle -\tfrac{1}{2} \tfrac{1}{2} | S^1 | 0 \rangle. \end{aligned}$$

⁵This does not, of course, mean that such a hadron has no interaction at all with the electromagnetic field. The product of two current operators, $\hat{J}(x), \hat{J}(x')$, is even under charge conjugation, and its matrix elements for transitions between states of equal charge parity do not vanish. A genuinely neutral hadron can therefore scatter a photon and can also emit two photons simultaneously, i.e. it can participate in processes of higher order in α .

⁶We do not consider possible parity nonconservation in electromagnetic interactions arising from virtual weak interactions.

⁷The reason for defining the form factors as in (138.7) (*R. Sackhs*, 1962) will become apparent below. The form factors F_1, F_2 are also used in the literature; they are defined in analogy with f and g in (116.6), that is, by

$$\Gamma^\mu = F_1 \gamma^\mu - \frac{F_2}{2M} \sigma^{\mu\nu} q_\nu.$$

They are related to F_e, F_m by

$$F_e = F_1 + F_2 \frac{q^2}{4M^2}, \quad F_m = F_1 + F_2.$$

The requirement of T -invariance (or of C -invariance in the annihilation channel) supplies no further relations among these elements. This is connected with the fact that the interaction described by the vertex operator (138.7) is automatically T -invariant as well (this is no longer the case, however, for particles of higher spin).

At $q \rightarrow 0$, the terms of zeroth and first order in q in (138.7) give

$$\Gamma^\mu = F_e(0)\gamma^\mu - \frac{1}{2M}[F_m(0) - F_e(0)]\sigma^{\mu\nu}q_\nu. \quad (138.8)$$

It follows (see § 116) that $F_e(0) \equiv Z$ is the particle's electric charge in units of e , while $F_m(0) - F_e(0)$ is its anomalous magnetic moment in units of $e/2M$ ⁸.

So far we have used only form factors in momentum space. This is, of course, sufficient for describing observable phenomena. For illustrative purposes alone, however, the form factors may also be given a somewhat more pictorial interpretation by regarding them as Fourier transforms of certain functions of the coordinates.

For this purpose it is convenient to choose a frame in which $\mathbf{P} = \mathbf{p}_1 + \mathbf{p}_2 = 0$ (the so-called *Breit frame*); this is always possible because $P^2 > 4M^2 > 0$. In this frame $\varepsilon_1 = \varepsilon_2 \equiv \varepsilon$, so that $P^0 = 2\varepsilon$, while the components of the 4-vector q are $q^0 = 0$, $\mathbf{q} = 2\mathbf{p}_2 = -2\mathbf{p}_1$.

For a spin-0 hadron the transition current has a particularly simple form in the Breit frame:

$$\frac{J_{fi}^0}{2\varepsilon} = F(-\mathbf{q}^2), \quad \mathbf{J} = 0.$$

It follows that $F(-\mathbf{q}^2)$ may be interpreted as the Fourier transform of a static charge distribution with density

$$e\rho(\mathbf{r}) = e \frac{1}{(2\pi)^3} \int F(-\mathbf{q}^2) e^{i\mathbf{q}\mathbf{r}} d^3q. \quad (138.9)$$

In this sense one speaks of the particle's spatial electromagnetic structure: if $F = \text{const} = Z$, then $\rho(\mathbf{r}) = Z\delta(\mathbf{r})$; dependence of the form factor on $\mathbf{q}$ is interpreted as a departure of the charge distribution from a point distribution. This interpretation must not, however, be taken literally. The function $\rho(\mathbf{r})$ does not in general refer to any single frame, since a different frame corresponds to each value of $\mathbf{q}$.

Only in the nonrelativistic limit $\mathbf{q}^2 \ll M^2$, where the change in the particle's energy during scattering can be neglected, does the Breit frame coincide with the particle's rest frame and become independent of $\mathbf{q}$. In this approximation the initial and final states of the particle are the same, so the transition current becomes a diagonal matrix element and $\rho(\mathbf{r})$ acquires the actual meaning of a spatial charge distribution. For elementary particles, however, the characteristic values of $|\mathbf{q}|$ over which the form factors change appreciably are only slightly smaller than M . Thus, in the nonrelativistic limit one may replace $F(-\mathbf{q}^2)$ by $F(0)$ altogether and regard such a particle as pointlike. The situation is different for nuclei. The nuclear mass M is proportional to the number A of nucleons it contains, whereas the characteristic value $|\mathbf{q}| \sim 1/R$ is proportional to $A^{-1/3}$ (R is the

⁸For the proton, $F_e(0) = 1$, $F_m(0) - F_e(0) = 1.79$. For the neutron, $F_e(0) = 0$, $F_m(0) = -1.91$ (its magnetic moment is wholly "anomalous").

nuclear radius). For sufficiently heavy nuclei, therefore, the characteristic $q^2 \ll M^2$, and a nonrelativistic treatment is valid throughout the relevant range; the concept of the electromagnetic structure of a nucleus then has a well-defined meaning.

For a spin-1/2 particle, (138.7) gives in the Breit frame

$$J_{fi}^0 = (F_e - F_m) \frac{M}{\varepsilon} (\bar{u}_2 u_1) + F_m (\bar{u}_2 \gamma^0 u_1) = F_e (\bar{u}_2 \gamma^0 u_1), \quad (138.10)$$

$$\mathbf{J}_{fi} = -\frac{1}{2M} F_m i\mathbf{q} \times (\bar{u}_2 \boldsymbol{\Sigma} u_1), \quad (138.11)$$

where $\boldsymbol{\Sigma}$ is the three-dimensional spin operator (matrix) (21.21). In (138.10) we have used the relation $\varepsilon(\bar{u}_2 \gamma^0 u_1) = M(\bar{u}_2 u_1)$, which is readily verified from the Dirac equations for u_1 and $\bar{u}_2$ when $\mathbf{p}_1 = -\mathbf{p}_2$.

The time component of the transition current (138.10) differs from the expression for a “point particle,” the electron, by the factor $F_e(-q^2)$. Thus the form factor F_e (called the *charge form factor*) may be said to describe the “spatial charge distribution” according to (138.9).

In the same way, the three-vector (138.11) may be associated with a “spatial distribution” of current density $e\mathbf{j}(\mathbf{r}) = \nabla \times \boldsymbol{\mu}(\mathbf{r})$, where

$$\boldsymbol{\mu}(\mathbf{r}) = \frac{e}{2M} \boldsymbol{\Sigma} \int F_m(-q^2) e^{i\mathbf{q}\mathbf{r}} d^3q$$

is a “magnetic-moment density.” Thus the form factor F_m (called the *magnetic form factor*) may be interpreted as specifying the spatial distribution of magnetic moment, subject, of course, to the same qualifications made above for the charge distribution. Here F_m includes both the “normal” Dirac magnetic moment and the “anomalous” moment specific to the hadron; the difference $F_m - F_e$ corresponds to the “density” of the latter.

It is natural to suppose that the singularities of the hadronic electromagnetic form factors, like those of the electron form factors, lie at real positive values of the argument $t = q^2 = -q^2$. This permits definite conclusions about the asymptotic behavior of the distributions $\rho(r)$ (and $\boldsymbol{\mu}(r)$) as $r \rightarrow \infty$. In fact, applying to the integral (138.9) the same transformation that was used in § 114 to pass from (114.3) to (114.4), one finds at large r that $\rho(r) \propto e^{-\kappa_0 r}$, where κ_0^2 is the abscissa of the first singularity of the form factor $F(q^2)$ (compare also the note on p. 564). If the nearest singularity is the threshold for the virtual photon’s creation of a hadron pair (each of mass M_0), then $\kappa_0 = 2M_0$.

§ 139. Electron scattering by hadrons

We apply the formulas obtained in the preceding section to elastic scattering of an electron by a hadron. Let p_h and p'_h denote the initial and final 4-momenta of the hadron, and let p_e and p'_e be those of the electron; then

$$p_e + p_h = p'_e + p'_h. \quad (139.1)$$

The process under consideration is represented by the diagram

$$(139.2)$$

Emission of the virtual photon by the electron is described by the usual vertex operator γ , while its absorption by the hadron is described by the operator Γ .

Consider the most interesting case, a hadron of spin 1/2 (for example, electron scattering by a proton or neutron). The scattering amplitude associated with diagram (139.2) is

$$M_{fi} = -4\pi e^2 \frac{1}{q^2} (\bar{u}'_e \gamma^\mu u_e) (\bar{u}'_h \Gamma_\mu u_h) \quad (139.3)$$

(throughout this chapter the electron charge is $-e$!). Computing the cross section from this amplitude involves no difference of principle from the calculations in § 81; for this purpose it is convenient to write the operator Γ in the first of the forms (138.7).

For unpolarized particles the result is

$$d\sigma = \frac{\pi\alpha^2 dt}{[s - (M + m)^2][s - (M - m)^2]t^2(1 - t/4M^2)} \times \\ \times \left\{ F_e^2 [(s - u)^2 + (4M^2 - t)t] - \frac{t}{4M^2} F_m^2 [(s - u)^2 - (4M^2 - t)(4m^2 + t)] \right\}. \quad (139.4)$$

Here M is the hadron mass, m is the electron mass, and

$$s = (p_e + p_h)^2, \quad t = q^2 = (p_e - p'_e)^2, \quad u = (p_e - p'_h)^2, \\ s + t + u = 2m^2 + 2M^2.$$

We shall examine several limiting cases.

For electrons scattered by a heavy nucleus, the case of interest is one in which the momentum $|\mathbf{q}|$ transferred from the electron to the nucleus is small relative to the nuclear mass, but not relative to $1/R$ (where R is the nuclear radius), so that the nucleus cannot be regarded as pointlike. The centre-of-mass system then approximately coincides with the rest frame of the nucleus, its recoil may be neglected, and the electron energy remains unchanged. Thus

$$-t = \mathbf{q}^2 \ll M^2, \quad \pi|dt| = \mathbf{p}_e^2 d\theta'_e, \quad s - M^2 \approx M^2 - u \approx 2M\varepsilon_e$$

and formula (139.4) becomes

$$d\sigma = \frac{\alpha^2 d\theta'_e}{q^4} (4\varepsilon_e^2 - \mathbf{q}^2) F_e^2(-\mathbf{q}^2). \quad (139.5)$$

In this approximation, only the term containing the electric form factor remains in the cross section, and (139.5) agrees with formula (80.5) for electron scattering by a static charge distribution.

For an electron scattered by a neutron at rest in the same limiting case $\varepsilon_e \ll M$ (where M is the neutron mass), the form factors may be replaced by their values at $\mathbf{q} = 0$. Indeed, as noted already, the characteristic “radius” of the charge distribution of a single nucleon is comparable with $1/M^9$. Since the neutron is electrically neutral, $F_e(0) = 0$, and the cross section takes the form

$$d\sigma = \alpha\mu^2 \left[\frac{4(\varepsilon_e^2 - m^2)}{\mathbf{q}^2} + 1 \right] do'_e = \alpha\mu^2 \left(\frac{1}{\sin^2(\vartheta/2)} + 1 \right) do'_e, \quad (139.6)$$

where $\mu = \frac{e}{2M}F_m(0)$ is the neutron magnetic moment and ϑ is the scattering angle. This formula describes scattering of an electron by a stationary point magnetic moment.

Finally, let us give the cross section for an ultrarelativistic electron scattered by a nucleon when $|\mathbf{q}| \gg m$. As before, $\mathbf{q}^2$ denotes the square of the momentum transfer in the centre-of-mass system, so that the invariant is $t = -\mathbf{q}^2$. In the rest frame of the initial nucleon (the laboratory frame), however,

$$-t \approx 2(p_e p'_e) = 2\varepsilon_e \varepsilon'_e (1 - \cos \vartheta),$$

where ε_e and ε'_e are the initial and final electron energies, while ϑ is the scattering angle in this frame. In the ultrarelativistic case, ε'_e is related to ϑ by the same formula as in photon scattering (cf. (86.8)):

$$\frac{1}{\varepsilon'_e} - \frac{1}{\varepsilon_e} = \frac{1}{M}(1 - \cos \vartheta).$$

Consequently,

$$-t = \frac{4\varepsilon_e^2 \sin^2(\vartheta/2)}{1 + \frac{2\varepsilon_e}{M} \sin^2(\vartheta/2)}, \quad (139.7)$$

$$\pi d|t| = \frac{\varepsilon_e^2 do'_e}{\left(1 + \frac{2\varepsilon_e}{M} \sin^2(\vartheta/2)\right)^2}, \quad (139.8)$$

where $do'_e = 2\pi \sin \vartheta d\vartheta$. The electron mass m may be omitted throughout formula (139.4); expressing all quantities in terms of t and $s - M^2 = 2M\varepsilon_e$, we obtain

$$d\sigma = \frac{\pi\alpha^2 d|t|}{\varepsilon_e^2 t^2} \left\{ F_e^2(t) \left[\frac{(4M\varepsilon_e + t)^2}{4M^2 - t} + t \right] - \frac{t}{4M^2} F_m^2(t) \left[\frac{(4M\varepsilon_e + t)^2}{4M^2 - t} - t \right] \right\}, \quad (139.9)$$

⁹The empirical root-mean-square nucleon “radius” is $\approx 3.5/M \approx 1/(2m_\pi)$ (where m_π is the pion mass).

or, on using (139.7), (139.8),

$$d\sigma = do'_e \frac{\alpha^2}{4\epsilon_e^2} \frac{\cos^2(\vartheta/2)}{\sin^4(\vartheta/2)} \frac{1}{1 + \frac{2\epsilon_e}{M} \sin^2(\vartheta/2)} \times \\ \times \left\{ \frac{F_e^2 - \frac{t}{4M^2} F_m^2}{1 - \frac{t}{4M^2}} - \frac{t}{2M^2} F_m^2 \tan^2 \frac{\vartheta}{2} \right\}. \quad (139.10)$$

(*M. Rosenbluth*, 1950).

Notice that F_e and F_m contribute independently to the cross section: there are no interference terms. This confirms the usefulness of the chosen form factors.

PROBLEM

Find the cross section for electron scattering by a spin-zero hadron.

SOLUTION. Using (138.5), in place of (139.3) we have

$$M_{fi} = -\frac{4\pi e^2}{q^2} (\bar{u}'_e(\gamma P)u_e) F(q^2).$$

The cross section is therefore

$$d\sigma = \frac{\pi \alpha^2 dt [(s-u)^2 + (4M^2 - t)t]}{[s - (M+m)^2][s - (M-m)^2] t^2} F^2(t)$$

(the notation is the same as in (139.4)). For $|t| \gg m^2$,

$$d\sigma = do'_e \frac{\alpha^2}{4\epsilon_e^2} \frac{\cos^2(\vartheta/2)}{\sin^4(\vartheta/2)} \frac{F^2(t)}{1 + \frac{2\epsilon_e}{M} \sin^2(\vartheta/2)}$$

(the notation is the same as in (139.10)).

§ 140. The low-energy theorem for bremsstrahlung

In § 98 we examined photon emission during a collision of particles in the limit where the photon frequency tends to zero. The amplitude of that process was found to be inversely proportional to ω and to have a simple expression in terms of the amplitude for the same collision without emission of the soft photon (we shall again refer to the latter, by convention, as the “elastic” scattering amplitude and denote it by $M_{fi}^{(\text{el})}$). At the next order in ω we have

$$M_{fi} = M_{fi}^{(-1)} + M_{fi}^{(0)}, \quad (140.1)$$

where the leading term ($\propto \omega^{-1}$) is supplemented by a correction independent of ω ($\propto \omega^0$). We shall show that this correction, like the leading term, can be expressed through $M_{fi}^{(\text{el})}$ without reference to the details of the hadron’s electromagnetic structure. This statement is called the *low-energy theorem* for bremsstrahlung (*F. E. Low*, 1958).

We saw in § 98 that the main contribution to the amplitude for emission of a soft photon (the contribution corresponding to the first term in (140.1)) comes from diagrams in which the initial or final particle emits the photon directly. These diagrams have the form

$$(140.2)$$

in contrast to diagrams of the form

$$(140.3)$$

in which the photon line emerges from an internal part of the diagram. A graph of type (140.2) can be divided into two pieces by cutting a single virtual-hadron line (that of the initial or the final hadron). In other words, such graphs display the property essential here: a one-particle intermediate state containing one hadron. As we saw in § 79, the unitarity requirement makes this property alone sufficient to produce a pole singularity in the amplitude.

For simplicity, suppose that only one of the two colliding hadrons (the first) carries electric charge and can therefore radiate, and that both hadrons have zero spin. The wave amplitudes u of these hadrons are scalars, which we set equal to 1.

The contribution from the pole part of diagram (140.2,a) is then

$$iM_{fi}^{(a)} = \sqrt{4\pi} e_{\mu}^* (2p_1^{\mu} - k^{\mu}) eF \frac{1}{(p_1 - k)^2 - M^2} i\Gamma. \quad (140.4)$$

The first factor belongs to the photon k (e_{μ} is its polarization 4-vector). The second represents the electromagnetic hadron vertex (the solid dot in the diagram); it is written in the form (138.5), with F the hadron form factor. The third factor is the propagator of the virtual hadron with momentum $p_1 - k$ (and mass M). Finally, the factor $i\Gamma$ denotes the entire remaining block. This block differs from the elastic-process amplitude

$$iM_{fi}^{(\text{el})} = \quad (140.5)$$

by having the real hadron p_1 replaced with the virtual hadron $p_1 - k$.

The first terms in the expansion of (140.4) in powers of k include: 1) terms inversely proportional to ω ; 2) terms independent of ω but dependent on the direction of $\mathbf{k}$; and

3) terms independent of ω and also wholly independent of $\mathbf{k}$. Terms of the third kind (and only that kind) are also contributed by the “nonsingular” diagrams of type (140.3), which contain no pole singularity, and by the non-pole parts of diagrams (140.2). We shall see that gauge invariance fixes all such terms together uniquely from those of the first two types, so no separate calculation of them is required.

The amplitude of the elastic process (140.5) depends on only two invariant variables:

$$s = (p_1 + p_2)^2 = (p'_1 + p'_2)^2, \quad t = (p'_2 - p_2)^2. \quad (140.6)$$

Replacing p_1 by $p_1 - k$ not only changes s to $(p_1 - k + p_2)^2$, but also introduces dependence on the new variable

$$(p_1 - k)^2 - M^2 = -2(p_1 k),$$

which characterizes the “unphysicality” of the momentum $p_1 - k$. The first term alone in the expansion in this new (small) variable, however, removes the singularity in the amplitude (140.4); it can consequently contribute only terms independent of k , which, as explained above, are not yet of interest to us. We therefore reach the important conclusion that Γ in (140.4) may be replaced by the physical amplitude $M_{fi}^{(el)}(s, t)$, provided that within this amplitude we make the replacement

$$s \rightarrow (p_1 + p_2 - k)^2 = s - 2k(p_1 + p_2). \quad (140.7)$$

The first terms of its expansion are

$$\Gamma \rightarrow M_{fi}^{(el)}(s, t) - 2(k p_1 + k p_2) \left(\frac{\partial M_{fi}^{(el)}}{\partial s} \right)_t.$$

For the same reason, it is immaterial that the electromagnetic form factor F here belongs to a vertex for which only one of its two hadron legs (p_1 and $p_1 - k$) is physical. It may therefore be replaced by the form factor considered in § 138 for a vertex with two physical legs. Since the photon k is real in the present case, $F(k^2) = F(0) = Z_1$, where eZ_1 is the hadron charge.

Thus, from (140.8) we find

$$\begin{aligned} M_{fi}^a = & Z_1 e \sqrt{4\pi} \frac{2(e^* p_1)}{-2(k p_1)} M_{fi}^{(el)} - \\ & - Z_1 e \sqrt{4\pi} 2(e^* p_1) \frac{1}{-2(k p_1)} \cdot 2(p_2 k) \frac{\partial M_{fi}^{(el)}}{\partial s} + \dots, \end{aligned} \quad (140.8)$$

where the ellipsis denotes terms wholly independent of k (whereas the second term in (140.8) depends on the direction of $\mathbf{k}$). In the same way, the contribution to M_{fi} from diagram (140.2,b)

is obtained from (140.8) by replacing p_1 , p_2 , and k with p'_1 , p'_2 , and $-k$. The leading term in the expansion then gives the expression already found earlier:

$$M_{fi}^{(-1)} = Z_1 e \sqrt{4\pi} \left(\frac{(p'_1 e^*)}{(p'_1 k)} - \frac{(p_1 e^*)}{(p_1 k)} \right) M_{fi}^{(el)} \quad (140.9)$$

(cf. (98.5)).

The terms independent of k can instead be determined by requiring gauge invariance of the complete amplitude. More precisely, it must be unchanged under $e^* \rightarrow e^* + \text{const} \cdot k$; that is, it must have the form $M_{fi} = e_\mu^* J^\mu$, with $k_\mu J^\mu = 0$. It is readily seen that this requires adding to (140.8) the k -independent term

$$-2Z_1 e \sqrt{4\pi} (p_2 e^*),$$

and likewise for diagram (140.2, b). The final result is therefore

$$M_{fi}^{(0)} = 2Z_1 e \sqrt{4\pi} e_\mu^* \left[p_1^\mu \frac{(p_2 k)}{(p_1 k)} - p_2^\mu + p_1'^\mu \frac{(p_2' k)}{(p_1' k)} - p_2'^\mu \right] \frac{\partial M_{fi}^{(\text{el})}}{\partial s}. \quad (140.10)$$

This formula supplies the required result. It can be put in a more compact form by making the identity substitution

$$2p_{2\nu} \left(\frac{\partial}{\partial s} \right)_t = \left(\frac{\partial}{\partial p_1^\nu} \right)_{p_1', p_2, p_2'}$$

(and similarly for $\partial/\partial p_1'^\nu$), and by introducing the differential operators

$$\widehat{d}_{1\mu} = \frac{p_{1\mu}}{(p_1 k)} k^\nu \frac{\partial}{\partial p_1^\nu} - \frac{\partial}{\partial p_1^\mu} \quad (140.11)$$

(with $\widehat{d}_{1'\mu}$ defined analogously). Then

$$M_{fi}^{(0)} = Z_1 e \sqrt{4\pi} e_\mu^* (\widehat{d}_1^\mu + \widehat{d}_{1'}^\mu) M_{fi}^{(\text{el})}. \quad (140.12)$$

To the accuracy required, the cross section is determined by the squared amplitude

$$|M_{fi}|^2 = |M_{fi}^{(-1)}|^2 + 2 \text{Re}(M_{fi}^{(-1)} M_{fi}^{(0)*}). \quad (140.13)$$

The second term gives the correction sought for the radiation cross section. On summing over the photon polarizations, this correction becomes

$$-4\pi (Z_1 e)^2 \left(\frac{p'^\mu}{(p' k)} - \frac{p^\mu}{(p k)} \right) (\widehat{d}_1 + \widehat{d}_{1'})_\mu |M_{fi}^{(\text{el})}|^2. \quad (140.14)$$

Thus the correction to the radiation cross section is expressed through the elastic-process cross section and its derivative with respect to s .

If the charged hadron has spin $1/2$, the essential reasoning of the calculation remains unchanged. Only the particular forms of the vertices and propagators are different. After averaging over the hadron and photon polarizations, formula (140.14) nevertheless remains valid (*T. N. Burnett, N. M. Kroll, 1968*).

§ 141. Low-energy theorem for photon scattering by a hadron

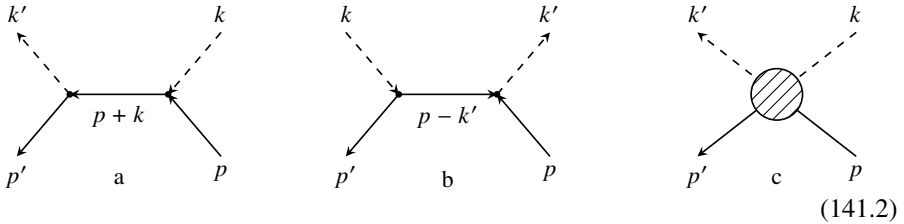
At low frequencies, the cross section for photon scattering by any stationary charged particle approaches the classical value given by the Thomson formula. This limit

corresponds to an amplitude independent of the photon frequency ω ; denote it by $M_{fi}^{(0)}$. It turns out, however, that in photon scattering (as in the bremsstrahlung discussed in the preceding section), not only this leading term but also the next term in the expansion of the amplitude in powers of ω is independent of the details of the hadron's electromagnetic structure:

$$M_{fi} = M_{fi}^{(0)} + M_{fi}^{(1)}, \quad (141.1)$$

where $M^{(1)} \sim \omega$ (F. E. Low, 1954; M. Gell-Mann, M. L. Goldberger, 1954).

The process is represented by three types of diagram:



The first two again contain a one-particle intermediate state and therefore have a pole singularity.

The reasoning and the essential features of the calculation are the same as in § 140. In fact, it suffices to calculate only the contribution of the pole parts of diagrams (141.2a–b), with their electromagnetic vertices expressed through the static form factors (the charge Ze and anomalous magnetic moment μ_{an}).

Unlike the bremsstrahlung case, however, the corrections to the Compton-scattering cross section that concern us here exist only for particles with spin. In bremsstrahlung, besides the spin-dependent corrections, there are corrections associated with the energy dependence of the amplitude of the “elastic” process. Here its role is played by the form factors, which reduce to constants on the “physical legs” and have no energy dependence. Thus, for photon scattering the corrections arise solely from the magnetic moment, which spinless particles do not possess. We shall consider below the scattering of a photon by a spin-1/2 hadron.

Let M_{fi} now mean the pole-diagram contribution to the scattering amplitude. We then have (cf. (86.3), (86.4))

$$M_{fi} = -4\pi(Ze)^2 e'_\mu e_\nu (\bar{u}' Q^{\mu\nu} u), \quad (141.3)$$

where

$$\begin{aligned} Q^{\mu\nu} = & (\gamma^\mu + S'^\mu) \frac{\gamma p + \gamma k + M}{s - M^2} (\gamma^\nu - S^\nu) + \\ & + (\gamma^\nu - S^\nu) \frac{\gamma p - \gamma k' + M}{u - M^2} (\gamma^\mu + S'^\mu), \end{aligned} \quad (141.4)$$

$$s = (p + k)^2 = (p' + k')^2, \quad u = (p - k')^2 = (p' - k)^2$$

and, for brevity, the notation

$$\mu_{\text{an}} \sigma^{\mu\lambda} k_\lambda = Ze S^\mu, \quad \mu_{\text{an}} \sigma^{\mu\lambda} k'_\lambda = Ze S'^\mu. \quad (141.5)$$

has been introduced. By moving the operators $\gamma p + M$ and using the equations

$$\bar{u}'(\gamma p' - M) = (\gamma p - M)u = 0,$$

expression (141.4) may be transformed into

$$\begin{aligned} Q^{\mu\nu} = & \left[(\gamma^\mu + S'^\mu) \frac{(\gamma k) \gamma^\nu + 2p^\nu}{2(pk)} + \frac{\gamma^\nu (\gamma k) - 2p'^\nu}{2(p'k)} (\gamma^\mu + S'^\mu) \right] - \\ & - \left[\frac{\gamma^\mu (\gamma k') + 2p'^\mu}{2(p'k')} S^\nu + S^\nu \frac{\gamma^\mu (\gamma k') - 2p'^\mu}{2(pk')} \right] - \\ & - \left[S'^\mu \frac{\gamma p + \gamma k + M}{2(pk)} S^\nu - S^\nu \frac{\gamma p - \gamma k' + M}{2(pk')} S'^\mu \right]. \end{aligned} \quad (141.6)$$

Written in this form (or in the analogous form with k and k' interchanged), the gauge invariance of expression (141.3) is manifest; its conditions are

$$k'_\mu (\bar{u}' Q^{\mu\nu} u) = (\bar{u}' Q^{\mu\nu} u) k_\nu = 0 \quad (141.7)$$

(in checking these relations, recall that $(\gamma k)(\gamma k) = 0$ and $kS = k'S' = 0$).

Since the pole part of the scattering amplitude is therefore gauge-invariant by itself, the regular part of the amplitude, which also includes the contribution from diagram (141.2c), must likewise be gauge-invariant separately. It follows in turn that the expansion of this part in powers of k and k' must begin with quadratic terms (compare the analogous observation concerning condition (127.5)). In other words, the regular part of the amplitude contains only terms starting at order $\omega\omega' \sim \omega^2$ and hence contributes nothing to the terms proportional to ω^0 and ω^1 that are of interest here. All such terms are consequently contained in expression (141.3).

For their explicit calculation, choose the laboratory frame in which the initial hadron is at rest. For the photons choose a three-dimensionally transverse gauge, in which $e_0 = e'_0 = 0$. Then $(pe) = 0$ and $(p'e'^*) \sim |\mathbf{p}'| \sim \omega$. It follows from (141.6) that the first terms in the expansion of M_{fi} are proportional to ω^0 , whereas the terms containing μ_{an} contribute only at order ω^1 .

To the required accuracy, the wave amplitudes of the initial and final hadrons in the laboratory frame are

$$u = \sqrt{2M} \begin{pmatrix} w \\ 0 \end{pmatrix}, \quad \bar{u}' = \sqrt{2M} \left(w'^*, -\frac{w'^*}{2M} (\mathbf{k} - \mathbf{k}') \boldsymbol{\sigma} \right),$$

where w and w' are 3-spinors.

Direct calculation gives

$$M_{fi}^{(0)} = -8\pi(Ze)^2 (\mathbf{e}'^* \mathbf{e}) (w'^* w), \quad (141.8)$$

$$\begin{aligned} M_{fi}^{(1)} = & -16\pi i M \mu_{\text{an}}^2 \omega (w'^* \boldsymbol{\sigma} w) (\mathbf{n}' \times \mathbf{e}'^*) (\mathbf{n} \times \mathbf{e}) - \\ & - 4\pi i Z e \mu_{\text{an}} \omega (w'^* \boldsymbol{\sigma} w) \left\{ \mathbf{n} ((\mathbf{n} \times \mathbf{e}) \mathbf{e}'^*) + (\mathbf{n} \times \mathbf{e}) (\mathbf{n} \mathbf{e}'^*) - \right. \\ & \left. - \mathbf{n}' ((\mathbf{n}' \times \mathbf{e}'^*) \mathbf{e}) - (\mathbf{n}' \times \mathbf{e}'^*) (\mathbf{n} \mathbf{e}) - 2(\mathbf{e}'^* \times \mathbf{e}) \right\}, \end{aligned} \quad (141.9)$$

where $\mathbf{n} = \mathbf{k}/\omega$ and $\mathbf{n}' = \mathbf{k}'/\omega'$.

The scattering cross section is

$$d\sigma = \frac{1}{64\pi^2} |M_{fi}|^2 \frac{\omega'^2}{M^2 \omega^2} d\omega' \quad (141.10)$$

(see (64.19)). For scattering by a charged particle, both $M_{fi}^{(1)}$ and $M_{fi}^{(0)}$ are nonzero. At the adopted accuracy, the terms $|M_{fi}^{(0)}|^2$ and $\text{Re}(M_{fi}^{(0)} M_{fi}^{(1)*})$ may therefore be retained in the square $|M_{fi}|^2$. The former gives the Thomson cross section. The latter, however, vanishes

after averaging over photon and hadron polarizations. Thus, in scattering by a charged hadron, the corrections under consideration appear only in polarization effects.

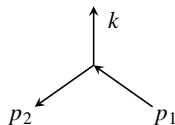
For scattering by an electrically neutral hadron, on the other hand, $M_{fi}^{(0)} = 0$, and the cross section is determined by $|M_{fi}^{(1)}|^2$. After averaging over the initial-particle polarizations and summing over those of the final particles, it is (in conventional units)

$$d\sigma = \frac{2\mu^4 \omega^2}{\hbar^2 c^4} (2 + \sin^2 \vartheta) d\omega', \quad (141.11)$$

where ϑ is the photon scattering angle and the anomalous magnetic moment is the same as the total moment μ . In its angular dependence, this cross section corresponds to antisymmetric scattering (see problem 2 in § 60).

§ 142. Multipole moments of hadrons

We now consider the transition current represented by a diagram of the same form as (138.2):



$$(142.1)$$

Here, however, the lines p_1 and p_2 correspond to different particles (with masses M_1 and M_2); it is more convenient to draw the photon line $k = p_1 - p_2$ as leaving the vertex. The photon may now be either virtual or real: the only restriction is $k^2 < (M_1 - M_2)^2$, so $k^2 = 0$ is allowed. The processes described by this diagram therefore include, in particular, photon emission accompanying transformations of particles, including nuclei (in the latter case the initial and final particles are the nucleus in different states).

Of greatest interest for the question at hand is the case in which the photon wavelength is large in comparison with the characteristic “dimensions” of the particle (that is, the dimensions entering its form factors; for a nucleus they of course coincide with its “radius”). The transition current can then be expanded in powers of k ¹⁰.

First, we note that

$$J_{fi} = 0 \quad \text{when} \quad k = 0. \quad (142.2)$$

¹⁰We follow below the method proposed by V. B. Berestetskii (1948).

Indeed, the limit $k \rightarrow 0$ corresponds to a potential constant in space and time. Such a potential, however, has no physical significance and cannot cause any real process. The same conclusion can also be reached more formally. The currents considered in § 138 were nonzero at $k = 0$ because of terms proportional to the 4-vector $P = p_1 + p_2$; when $M_1 \neq M_2$, however, $(Pk) \neq 0$, and the transversality condition on the current therefore forbids such terms.

In three-dimensional form, the transversality condition for the current $J_{fi} = (\rho_{fi}, \mathbf{J}_{fi})$ is

$$\mathbf{k} \mathbf{J}_{fi} = \omega \rho_{fi}. \quad (142.3)$$

It may be satisfied in either of two ways:

$$\mathbf{J}_{fi} = \omega \mathbf{v}(\mathbf{k}, \omega), \quad \rho_{fi} = \mathbf{k} \mathbf{v}(\mathbf{k}, \omega), \quad (142.4)$$

or

$$\mathbf{J}_{fi} = \mathbf{k} \times \mathbf{a}(\mathbf{k}, \omega), \quad \rho_{fi} = 0. \quad (142.5)$$

Here $\mathbf{v}$ is some polar vector, while $\mathbf{a}$ is an axial vector. The current is said to be of electric type in the first case and of magnetic type in the second. According to (142.2), $\mathbf{v}$ and $\mathbf{a}$ remain finite or tend to zero as $\mathbf{k}, \omega \rightarrow 0$.

Let the photon energy satisfy $\omega \ll M_1$. The recoil may then be neglected, and the final particle M_2 may also be regarded as being at rest (in the rest frame of the particle M_1); ω is then fixed: $\omega = M_1 - M_2$. The states of the stationary particles M_1 and M_2 are characterized by three-dimensional spinors w_1 and w_2 of ranks $2s_1$ and $2s_2$, where s_1 and s_2 are the particle spins. The transition current must be a bilinear combination of w_1 and w_2^* . From products of the components of these spinors one can construct irreducible tensors of ranks $l = s_1 + s_2, \dots, |s_1 - s_2|$ (for a specified l this will be a true tensor or a pseudotensor, according to the intrinsic parities of the particles M_1 and M_2). Apart from these tensors, the only quantity at our disposal is the vector $\mathbf{k}$. To construct the first term in the expansion of the current in powers of $\mathbf{k}$, these quantities must be used to form a vector of the lowest possible degree in $\mathbf{k}$. This is achieved by taking the tensor of lowest rank and contracting it scalarly $l - 1$ times with the vector $\mathbf{k}$. The result is the polar vector $\mathbf{v}$ or the axial vector $\mathbf{a}$.

Let Q_{lm} be the spherical components of the tensor formed from the particle wave amplitudes. The spherical components of the rank- $(l - 1)$ tensor made from the components of $\mathbf{k}$ are $|\mathbf{k}|^{l-1} Y_{l-1,m}(\mathbf{n})$, where $\mathbf{n} = \mathbf{k}/\omega$. By the general rule for coupling spherical tensors (see III, (107.3)), the spherical components

of the vector $\mathbf{v}$ may be written as

$$\begin{aligned} v_\lambda &= (-1)^{\lambda+1} i^l \frac{\sqrt{4\pi}}{(2l-1)!!} \sqrt{\frac{2l+1}{l}} |\mathbf{k}|^{l-1} \times \\ &\times \sum_m \begin{pmatrix} l-1 & 1 & l \\ \lambda+m & -\lambda & -m \end{pmatrix} Q_{l,-m} Y_{l-1,\lambda+m}(\mathbf{n}), \end{aligned}$$

where λ takes the values 0, ± 1 (the choice of the overall factor is discussed below).

Using formulas (7.16), we can express $\mathbf{v}$ in terms of vector spherical harmonics:

$$\mathbf{v} = i^l \frac{\sqrt{4\pi} |\mathbf{k}|^{l-1}}{(2l-1)!! \sqrt{l(2l+1)}} \sum_m (-1)^{l-m} Q_{l,-m} \times \\ \times \left[\sqrt{l+1} \mathbf{Y}_{lm}^{(e)}(\mathbf{n}) + \sqrt{l} \mathbf{Y}_{lm}^{(l)}(\mathbf{n}) \right]. \quad (142.6)$$

Substitution in (142.4) gives the El transition current:

$$\mathbf{J}_{fi} = i^l \frac{\sqrt{4\pi} |\omega| |\mathbf{k}|^{l-1}}{(2l-1)!! \sqrt{l(2l+1)}} \sum_m (-1)^{l-m} Q_{l,-m}^{(e)} \times \\ \times \left[\sqrt{l+1} \mathbf{Y}_{lm}^{(e)}(\mathbf{n}) + \sqrt{l} \mathbf{Y}_{lm}^{(l)}(\mathbf{n}) \right], \quad (142.7)$$

$$\rho_{fi} = i^l \frac{\sqrt{4\pi} |\mathbf{k}|^l}{(2l-1)!! \sqrt{l(2l+1)}} \sum_m (-1)^{l-m} Q_{l,-m}^{(e)} Y_{lm}(\mathbf{n}) \quad (142.8)$$

(we distinguish $|\mathbf{k}|$ from ω throughout, since the formulas may be applied to both real and virtual photons, for which these quantities do not coincide).

In (142.7), (142.8), the spherical tensor Q_{lm} (denoted there by $Q_{lm}^{(e)}$) is understood to be a true tensor. If it is instead a pseudotensor (which we then denote by $Q_{lm}^{(m)}$), formula (142.6) determines the pseudovector $\mathbf{a}$. Substitution in (142.5) then gives the Ml transition current:

$$\mathbf{J}_{fi} = i^l \frac{\sqrt{4\pi}}{(2l-1)!!} \sqrt{\frac{l+1}{l(2l+1)}} |\mathbf{k}|^l \sum_m (-1)^{l-m} Q_{l,-m}^{(m)} \mathbf{Y}_{lm}^{(m)}(\mathbf{n}), \quad (142.9)$$

$$\rho_{fi} = 0.$$

The quantities $Q_{lm}^{(e)}$ and $Q_{lm}^{(m)}$ are the electric and magnetic multipole transition moments of the hadron. Their role in hadron electrodynamics is entirely analogous to that played by the corresponding quantities in electron electrodynamics. However, whereas

for electronic systems these moments can, in principle, be calculated from wave functions (as matrix elements of the appropriate operators), in hadron electrodynamics they are phenomenological quantities whose values must be obtained experimentally.

The normalization adopted for these quantities in (142.7)–(142.9) agrees with their definition in § 46. This can be verified by treating the currents (142.7)–(142.9) as Fourier components of the transition current in coordinate representation. For example, expand the factor $e^{-i\mathbf{k}\mathbf{r}}$ in the integral

$$\rho_{fi}(\mathbf{k}) = \int \rho_{fi}(\mathbf{r}) e^{-i\mathbf{k}\mathbf{r}} d^3x \quad (142.10)$$

with formula (46.3). This gives

$$\rho_{fi}(\mathbf{k}) = 4\pi i^l \sum_{l,m} Y_{lm}(\mathbf{n}) \int \rho_{fi}(\mathbf{r}) Y_{lm}^* \left(\frac{\mathbf{r}}{r} \right) g_l(|\mathbf{k}|r) d^3x.$$

Retaining the term with the smallest l for which the integral is nonzero, and replacing $g_l(|\mathbf{k}|r)$ for $|\mathbf{k}|r \ll 1$ by the first term of its expansion (46.5), we recover formula (142.9), with

$$Q_{lm}^{(e)} = \sqrt{\frac{4\pi}{2l+1}} \int r^l \rho_{fi}(\mathbf{r}) Y_{lm}\left(\frac{\mathbf{r}}{r}\right) d^3x \quad (142.11)$$

in agreement with definition (46.7).

We shall also show that, when applied to emission of a real photon, the formulas obtained above lead back to results already known to us.

The amplitude for a transition accompanied by emission of a photon of momentum $\mathbf{k} = \omega \mathbf{n}$ and polarization $e = (0, \mathbf{e})$ is

$$M_{fi} = -e\sqrt{4\pi} \mathbf{e}^* \mathbf{J}_{fi}. \quad (142.12)$$

If the nucleus has definite values of the angular-momentum projection in the initial and final states (M_i and M_f), only one term remains in the sum over m in (142.7)–(142.9): $m = M_i - M_f$. Since, by (16.23), the products $\mathbf{Y}_{lm}^{(e)} \mathbf{e}^{(\lambda)*}$ or $\mathbf{Y}_{lm}^{(m)} \mathbf{e}^{(\lambda)*}$ ($\lambda = \pm 1$ is the photon helicity and $\mathbf{e}^{(\lambda)} \perp \mathbf{n}$) are proportional to $D_{\lambda m}^l$, we return to the formulas considered in § 48.

The differential radiation probability¹¹ is

$$dw = 2\pi\delta[\omega - (E_i - E_f)] |M_{fi}|^2 \frac{d^3k}{2\omega(2\pi)^3} \quad (142.13)$$

(E_i, E_f are the initial and final energies of the nucleus). The total probability is obtained by summing over polarizations and integrating over d^3k . Substituting (142.7) or (142.9) in (142.12), and then in (142.13), and carrying out these operations, we recover formula (46.9) (or (47.2)).

Formulas (142.7)–(142.9) include all cases that can arise in emission of a real photon. For virtual photons, however, one further case is possible which is not described by these formulas (*R. H. Fowler*, 1930).

If the initial and final states of the nucleus have the same spin and parity, their wave amplitudes can be used to form a scalar Q_0 , and hence a transition current of the form

$$\rho_{fi} = Q_0 \mathbf{k}^2, \quad \mathbf{J}_{fi} = Q_0 \omega \mathbf{k}. \quad (142.14)$$

The quantity Q_0 is called the *monopole* ($E0$) transition moment. The corresponding transition amplitude vanishes for emission of a real photon (because $\mathbf{e}^* \mathbf{k} = 0$). A monopole current can, however, produce transitions involving emission of a virtual photon. Moreover, it is the only such source when $s_1 = s_2 = 0$, for then all the multipole moments vanish.

The dependence of the monopole current (142.14) on ω and $\mathbf{k}$ is analogous to that of the electric quadrupole current. Accordingly, the moment Q_0 is of the same order as the quadrupole moment. The same conclusion follows by interpreting (142.14) as Fourier components of the current in coordinate representation. Expanding the factor

¹¹The factor $2\pi\delta$ in this formula, instead of $(2\pi)^4 \delta^{(4)}$ in (64.11), arises because neglecting nuclear recoil removes momentum conservation, leaving only conservation of energy.

$e^{-i\mathbf{k}\mathbf{r}}$ in (142.10) in powers of $\mathbf{k}\mathbf{r}$ and taking $\rho_{fi}(r)$ to be spherically symmetric, we obtain

$$\rho_{fi}(\mathbf{k}) = -\frac{1}{6}\mathbf{k}^2 \int \rho_{fi}(r)r^2 d^3x.$$

Comparison with (142.14) gives

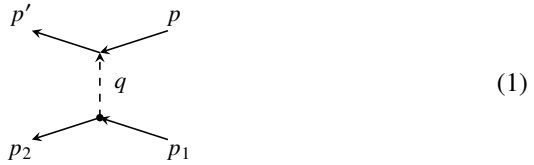
$$Q_0 = -\frac{1}{6} \int \rho_{fi}(r)r^2 d^3x. \quad (142.15)$$

The similarity between this quantity and the quadrupole moment is evident.

PROBLEM

1. Find the probability that an atom is ionized from its K shell by the nuclear excitation energy ω (the so-called internal conversion of γ radiation) in a nuclear Ml transition, neglecting both the electron binding energy in the atom and the effect of the nuclear field on its wave functions¹².

SOLUTION. The process is described by the diagram



where p_1 and p_2 refer to the stationary nucleus in different states, while $p = (m, 0)$ and $p' = (m + \omega, \mathbf{p}')$ are the 4-momenta of the initial and final electrons. The amplitude corresponding to this diagram is

$$M_{fi} = -e^2 \frac{4\pi}{q^2} \bar{u}(p')(\gamma J_{fi})u(p),$$

where J_{fi} is the nuclear transition current. Summing over the final electron polarizations and averaging over the initial ones gives

$$\frac{1}{2} \sum_{\text{pol}} |M_{fi}|^2 = e^4 \frac{16\pi^2}{(q^2)^2} \left\{ q^2 (J_{fi} J_{fi}^*) + 4(J_{fi} p)(J_{fi}^* p) \right\}$$

(we used $J_{fi} q = 0$, and hence $J_{fi} p = J_{fi} p'$). The conversion probability is calculated as

$$dw_{\text{conv}} = 2|\psi_i(0)|^2 \left(\frac{|\mathbf{p}|}{m} d\sigma \right)_{\mathbf{p} \rightarrow 0},$$

¹²This approximation requires a small nuclear charge and sufficiently large excitation energies ω (while $1/\omega$ is still assumed large compared with the dimensions of the nucleus). In fact, the approximation is not very satisfactory, and a more accurate calculation requires the nuclear Coulomb field to be taken into account.

where $d\sigma$ is the cross section for the scattering represented by diagram (1), with $p = (\varepsilon, \mathbf{p})$, and ψ_i is the wave function of the atomic electron; for a K electron,

$$|\psi_i(0)|^2 = (Z\alpha m)^3 / \pi.$$

The factor 2 accounts for the two electrons in the atom's K shell. The cross section $d\sigma$ is calculated from

$$d\sigma = 2\pi\delta(\varepsilon + \omega - \varepsilon') |M_{fi}|^2 \frac{d^3p'}{2|\mathbf{p}| 2\varepsilon' (2\pi)^3}$$

(cf. the note on p. 710).

For an MI transition, J_{fi} must be taken from (142.9). Integration of $d w_{\text{conv}}$ over $d\varepsilon'$ removes the δ function, while integration over $d\mathbf{o}'$ turns $|\mathbf{Y}_{lm}^{(m)}|^2$ into 1. The conversion probability is consequently expressed through $|Q_{l,-m}^{(m)}|^2$. According to (46.9), however, the probability w_γ of spontaneous photon emission in the same nuclear transition is expressed through this same quantity. The final result is

$$\frac{w_{\text{conv}}}{w_\gamma} = 2\alpha(Z\alpha)^3 \frac{m}{\omega} \left(1 + \frac{2m}{\omega}\right)^{l+1/2}$$

(this ratio is called the *conversion coefficient*).

2. The same for a nuclear El transition.

SOLUTION. The same procedure, using the transition current from (142.7), (142.8), gives

$$\frac{w_{\text{conv}}}{w_\gamma} = 2\alpha(Z\alpha)^3 \left(1 + \frac{4l}{l+1} \frac{m^2}{\omega^2}\right)^{l-1/2} \frac{m}{\omega}.$$

3. The same for a monopole transition of the nucleus.

SOLUTION. Using the transition current from (142.14), we obtain

$$w_{\text{conv}} = 4\alpha^2(Z\alpha)^3 m^3 \omega^2 \left(1 + \frac{2m}{\omega}\right)^{3/2} |Q_0|^2.$$

Since monopole photon emission is impossible, $|Q_0|^2$ cannot be eliminated from this expression.

§ 143. Inelastic scattering of electrons by hadrons

Elastic scattering of electrons by hadrons was considered in § 139. The problem of inelastic scattering can be formulated in a similar way. The difference is that the final hadronic state now corresponds either to another hadron or to a collection of hadrons. The momentum-conservation law (139.1) remains valid if p'_h is understood as the 4-momentum of the final hadron, or as the total 4-momentum of the complete collection of hadrons produced in the scattering process. Thus, we now have $p_h'^2 \neq p_h^2 \neq M^2$, where M is the mass of the initial hadron.

With this difference, the inelastic-scattering process is described by the same diagram (139.2). We denote the lower vertex of this diagram by J_{fi} , as in § 138. Unlike (138.3) or (138.6), however, we shall not express the transition current in terms of a vertex operator and state amplitudes, since that would fix the nature of the final hadronic state in advance.

The scattering amplitude can now be written in a form analogous to (139.3):

$$M_{fi} = -\frac{4\pi e^2}{(p_e - p'_e)^2} (\bar{u}'_e \gamma_\mu u_e) J_{fi}^\mu \quad (143.1)$$

(an amplitude of this kind was already used in Problem 1 of § 142, where energy transfer to an electron was considered; the amplitude in the problem of nuclear excitation by electrons has a similar structure).

We shall take the initial electron energy to be high enough that many hadrons can be produced in the final state. Our interest will be in the so-called *inclusive* cross section: in the final state only the electron momentum is specified, while a sum is taken over all hadronic states.

In accordance with the formulas of § 64, this differential cross section is

$$d\sigma = \frac{d^3 p'_e}{4I(2\pi)^3 2\varepsilon'_e} \sum_f (2\pi)^4 \delta^{(4)}(p_h + p'_h - p_e - p'_e) |M_{fi}|^2. \quad (143.2)$$

The inclusive cross section can depend on only three kinematic invariants that can be determined by measurements made solely on the electrons. There are three such invariants:

$$t = q^2 \equiv (p_e - p'_e)^2, \quad s = (p_e + p_h)^2 \quad (143.3)$$

and $p_h'^2$. The third invariant is required because, unlike in elastic scattering, $p_h'^2$ —the “mass” of the final hadronic state—is not fixed. It is more convenient, however, to replace $p_h'^2$ by the invariant

$$\nu = qp_h. \quad (143.4)$$

The relation between ν and $p_h'^2$ follows from $p_h' = p_h + q$:

$$p_h'^2 = M^2 + t + 2\nu. \quad (143.5)$$

If the initial hadron is stable (a proton, for example), the rest energy of the final state is greater than M , that is, $p_h'^2 \gg M^2$; since $t < 0$, it follows from (143.5) that

$$\nu \geq |t|/2 \quad (143.6)$$

(equality corresponds to elastic scattering).

The kinematic invariants can be expressed through the initial and final electron energies ε_e and ε'_e , and the scattering angle θ . Below, we shall regard the electron as ultrarelativistic ($\varepsilon_e \gg m$, $\varepsilon'_e \gg m$) and neglect its mass. In the rest frame of the initial hadron (the laboratory frame), we then obtain

$$t = -4\varepsilon_e \varepsilon'_e \sin^2(\theta/2), \quad \nu = M(\varepsilon_e - \varepsilon'_e), \quad s - M^2 = 2M\varepsilon. \quad (143.7)$$

Substituting (143.1) in (143.2) and carrying out the sum over electron polarizations in the usual way, we obtain the cross section for scattering of unpolarized electrons. We write it as

$$d\sigma = \frac{\alpha^2}{(q^2)^2} \frac{d^3 p'_e}{(2\pi)^3 \cdot 8M\varepsilon_e \varepsilon'_e} w_{\mu\nu} W^{\mu\nu}, \quad (143.8)$$

or

$$d\sigma = \frac{\alpha^2}{(q^2)^2} \frac{dt dv}{4(p_h p_e)^2} w_{\mu\nu} W^{\mu\nu}, \quad (143.9)$$

where

$$w_{\mu\nu} = 4p_{e\mu} p'_{e\nu} - 2(p_{e\mu} q_\nu + p_{e\nu} q_\mu) + q^2 g_{\mu\nu}, \quad (143.10)$$

$$W^{\mu\nu} = \sum_f (2\pi)^4 \delta^{(4)}(p'_h - p_h - q) J_{fi}^\mu J_{fi}^{\nu*}. \quad (143.11)$$

The tensor $W^{\mu\nu}$ naturally depends essentially on the properties of the hadronic currents, and in the general case we can only pose the problem of its phenomenological structure, analogous to the problem of hadron form factors. First, the tensor structure of $W^{\mu\nu}$ must be determined solely by the 4-vectors associated with the lower vertex of diagram (139.2), namely p_h and q . Together with the metric tensor $g_{\mu\nu}$, these can form only five independent tensors. Invariance under time reversal requires the tensor to be symmetric; four such tensors can be constructed. Finally, current conservation,

$$W^{\mu\nu} q_\nu = 0, \quad W^{\mu\nu} q_\mu = 0,$$

reduces the number of independent tensors to two. They may be chosen as

$$\tau_{\mu\nu}^{(1)} = \frac{q_\mu q_\nu}{q^2} - g_{\mu\nu}, \quad \tau_{\mu\nu}^{(2)} = \left(p_{h\mu} - \frac{\nu}{t} q_\mu\right) \left(p_{h\nu} - \frac{\nu}{t} q_\nu\right) \quad (143.12)$$

and $W_{\mu\nu}$ can be written as

$$W_{\mu\nu} = 4\pi M W_1 \tau_{\mu\nu}^{(1)} + \frac{4\pi}{M} W_2 \tau_{\mu\nu}^{(2)}. \quad (143.13)$$

Substituting (143.10) and (143.13) in (143.8), we put the cross section in the form

$$d\sigma = \left(W_2 + 2W_1 \tan^2 \frac{\theta}{2}\right) d\varepsilon'_e d\sigma_{\text{el}}, \quad (143.14)$$

where

$$d\sigma_{\text{el}} = \frac{\alpha^2}{4\varepsilon_e^2} \frac{\cos^2(\vartheta/2)}{\sin^4(\vartheta/2)} d\vartheta'$$

is the cross section for scattering of an ultrarelativistic electron in a Coulomb field (cf. (80.7)).

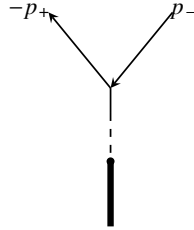
We see that the cross section is determined by two structure functions, each depending on the two invariants t and ν . If at high energies hadron physics contains no characteristic quantities with the dimensions of mass (the *scale-invariance hypothesis*), one may expect the structure functions at high energies to depend on the single dimensionless parameter t/ν . The functions W_1 and W_2 must then be functions of one variable, of the form

$$W_1 = \frac{M}{\nu} F_1\left(\frac{t}{\nu}\right), \quad W_2 = \frac{M}{\nu} F_2\left(\frac{t}{\nu}\right) \quad (143.15)$$

(notice that the ratio M/ν is independent of M).

§ 144. Conversion of an electron–positron pair into hadrons

Consider now the conversion of an electron–positron pair into hadrons. Denote the electron and positron 4-momenta by p_- and p_+ , and the (total) 4-momentum of the set of hadrons produced by p_h ; thus $p_- + p_+ = p_h$. The process is represented by the diagram


(144.1)

The lower vertex of this diagram corresponds to the current for a transition from the vacuum to some hadronic state $|n\rangle$; as in § 104, we denote it by $\langle n|J|0\rangle$.

The amplitude corresponding to diagram (144.1) is

$$M_n = -\frac{4\pi\alpha}{q^2} \bar{u}(-p_+) \gamma_\mu u(p_-) \langle n|J^\mu|0\rangle. \quad (144.2)$$

We shall be interested in the total cross section σ_h for annihilation into hadrons, that is, we sum over all final states $|n\rangle$. Then, in accordance with (64.18),

$$\sigma_h = \frac{1}{4I} \sum_n |M_n|^2 (2\pi)^4 \delta^{(4)}(p_h - q), \quad (144.3)$$

where $q = p_- + p_+$. We shall henceforth neglect the electron mass; then $q^2 = 2(p_- p_+)$ and $I = q^2/2$.

Proceeding as in § 143, write the cross section as

$$\sigma_h = \frac{(4\pi)^2}{2t^3} w^{\mu\nu} W_{\mu\nu}, \quad (144.4)$$

where

$$w^{\mu\nu} = \alpha \left(p_-^\mu q^\nu + p_-^\nu q^\mu - 2p_-^\mu p_-^\nu - \frac{1}{2} q^2 g^{\mu\nu} \right), \quad (144.5)$$

$$W_{\mu\nu} = \alpha \sum_n (2\pi)^4 \delta^{(4)}(p_h - q) \langle 0|J_\nu|n\rangle \langle n|J_\mu|0\rangle \quad (144.6)$$

and $t = q^2 > 0$.

Observe that t is the only kinematic invariant in the problem under consideration (the “three-tailed” diagram (144.1)), and q is the sole 4-vector on which $W_{\mu\nu}$ can depend. Taking current conservation into account, the tensor $W_{\mu\nu}$ may therefore be written

$$W_{\mu\nu} = \frac{1}{2} \rho_h(t) \left(\frac{q_\mu q_\nu}{q^2} - g_{\mu\nu} \right), \quad (144.7)$$

where $\rho_h(t)$ is the single invariant function that depends on the properties of the hadronic current and determines the annihilation cross section. Substitution of (144.5)–(144.7) into (144.4) gives

$$\sigma_h = \frac{4\pi^2\alpha}{t^2}\rho_h(t). \quad (144.8)$$

Notice that the function $\rho_h(t) = -2W_\mu^\mu/3$ is exactly the same as the function $\rho(t)$ defined in (104.9), provided the currents in that formula are understood to be hadronic currents. Recall also that $\rho(t)$ is the spectral density of the photon self-energy function $\Pi(t)$: $\text{Im } \Pi(t) = -\pi\rho(t)$. In the lowest-order approximation in α considered here, Π coincides with the polarization operator $\mathcal{P}$. Consequently, in this approximation $\rho_h(t)$ is also the spectral density of the hadronic contribution to the polarization operator:

$$\text{Im } \mathcal{P}_h(t) = -\pi\rho_h(t). \quad (144.9)$$

Using the dispersion relation (111.13) and expressing ρ_h through σ_h according to (144.8), we obtain

$$\mathcal{P}_h(t) = -\frac{t^2}{4\pi^2\alpha} \int_0^\infty \frac{\sigma_h(t') dt'}{t' - t - i0}, \quad (144.10)$$

which expresses the hadronic contribution to vacuum polarization in terms of the experimentally measurable annihilation cross section into hadrons.

The problem of electron-positron annihilation into a muon pair could be solved in precisely the same way (to first order in α , only one such pair can be produced). Analogously to (144.8), we would find

$$\sigma_\mu = \frac{4\pi^2\alpha}{t^2}\rho_\mu(t), \quad (144.11)$$

where $\rho_\mu(t)$ is the spectral density of muon vacuum polarization. It differs from the electron polarization only by replacement of the electron mass m with the muon mass μ and, according to (113.8), is given by

$$\rho_\mu(t) = \frac{\alpha}{3\pi}(t + 2\mu^2)\sqrt{1 - \frac{4\mu^2}{t}}.$$

Substitution into (144.11) reproduces the result already obtained in problem 8 of § 81.

[UNCLEAR: Scholarly edition]

Berestetskii Vladimir Borisovich
Lifshitz Evgenii Mikhailovich
Pitaevskii Lev Petrovich

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